AlgebraicJacobian

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Chapter 1

Sheaf condition along the structure-sheaf forget composite

The structure sheaf of a scheme is naturally a sheaf of commutative rings. Its underlying additive groups also form a sheaf, because the relevant forgetful functor preserves the limits that express the sheaf condition.

1.1 Statement

We record this preservation along the forgetful composite CommRing $\to$ Ring $\to$ Ab.

Theorem 1 (Sheaf condition transports along the structure-sheaf forget composite)**.** *Let X be a topological space. Then the Grothendieck topology of opens of X transports sheaves along the forgetful composite from commutative rings to abelian groups.*

Proof. Both forgetful functors in the composite CommRing $\to$ Ring $\to$ Ab create (in particular preserve) all small limits. The sheaf condition for the topology of opens is expressed as preservation of certain multiequalizer limits. Composing two limit-preserving functors gives a limit-preserving functor, so post-composition with the forget composite preserves the sheaf condition. $\square$

Chapter 2

Structure sheaf as a sheaf of abelian groups, sheafification and Ext

Let X be a topological space. Abelian-group-valued presheaves on the topology of opens admit sheafification, and their sheaf category admits Ext. Together with 1, these facts allow the structure sheaf of a scheme to be regarded as a sheaf of abelian groups.

1. Sheafification for abelian-group-valued presheaves on opens.
2. Ext in the resulting sheaf category.
3. The underlying abelian-group-valued structure sheaf $\mathcal{O}_C$.

2.1 Sheafification of abelian-group-valued sheaves on opens

Theorem 2 (Sheafification on the topology of opens, abelian-group valued)**.** *Let X be a topological space. Then the Grothendieck topology of opens of X admits sheafification for sheaves valued in the category of abelian groups.*

Proof. The category of abelian groups is a Grothendieck abelian category. Hence abelian-group-valued presheaves on the site of opens admit the usual sheafification, obtained from the plus construction, and the sheafification functor is left exact. □

2.2 Ext for sheaves on opens

Theorem 3 (Ext on the topology of opens, abelian-group valued)**.**

Let X be a topological space. Then the category of sheaves of abelian groups on the topology of opens of X admits Ext *groups.*

Proof. By 2, sheaves of abelian groups form a Grothendieck abelian category. In particular, this category has enough injectives, so the right derived functors of Hom define its Ext groups. □

2.3 Structure sheaf as a sheaf of abelian groups

Definition 4 (Structure sheaf as a sheaf of abelian groups)**.**

Let C be a scheme. The structure sheaf of C, viewed as a sheaf of abelian groups on the topology of opens of the underlying topological space of C, is obtained by post-composing the commutative-ring-valued structure sheaf with the forgetful composite $\mathrm{CommRing} \to \mathrm{Ring} \to \mathrm{Ab}$ of 1.

2.4 Cohomology of the structure sheaf

The preceding constructions define the abelian-group-valued cohomology of the structure sheaf:

$$H^n(C, \mathcal{O}_C) \ := \ \mathrm{toAbSheaf}(C).H\,n,$$

where H denotes sheaf cohomology. Each $H^n(C, \mathcal{O}_C)$ is therefore canonically an abelian group.

Chapter 3

Sheaves of k-modules: sheafification, Ext, and the structure sheaf as a sheaf of k-modules

Let C be a scheme over $\operatorname{Spec} k$. The structure morphism makes every ring of sections $\Gamma(C, U)$ a k-algebra, compatibly with restriction. Thus $\mathcal{O}_C$ is a sheaf of k-modules. Since the corresponding sheaf category is k-linear and abelian, its Ext groups carry canonical k-module structures.

Sections 3.1–3.2 establish sheafification and Ext for sheaves of k-modules. Section 3.3 constructs the k-module-valued structure sheaf from the algebra maps induced by the structure morphism.

3.1 Sheafification of k-module-valued sheaves on opens

Theorem 5 (Sheafification on the topology of opens, k-module valued)**.** *Let k be a commutative ring and X a topological space. Then the Grothendieck topology of opens of X admits sheafification for sheaves valued in the category of k-modules.*

Proof. The category of k-modules is Grothendieck abelian. Therefore module-valued presheaves on the site of opens admit the usual sheafification, whose functor is left exact. □

3.2 Ext for k-module-valued sheaves on opens

Theorem 6 (Ext on the topology of opens, k-module valued)**.**

Let k be a commutative ring and X a topological space. Then the category of sheaves of k-modules on the topology of opens of X admits Ext *groups.*

Proof. By 5, sheaves of k-modules form a Grothendieck abelian category. It has enough injectives, so the right derived functors of Hom define Ext. □

3.3 The structure sheaf of a $\operatorname{Spec} k$-scheme as a sheaf of k-modules

For C a scheme over $\operatorname{Spec} k$, the structure morphism $C \to \operatorname{Spec} k$ induces an algebra map $k \to \Gamma(C, U)$ for every open $U \subseteq C$. These maps equip the structure sheaf with the structure of a sheaf of k-modules.

3.3.1 The algebra map from the structure morphism

For each open $U \subseteq C$, the structure morphism induces a chain of ring maps

$$k \xrightarrow{\cong} \Gamma(\operatorname{Spec} k, \top) \xrightarrow{C.\mathrm{hom}} \Gamma(C, C.\mathrm{hom}^{-1}\top) \xrightarrow{\rho} \Gamma(C, U),$$

where ρ is the restriction along $U \le C.\mathrm{hom}^{-1}\top = \top$.

Definition 7 (Algebra map on a section)**.** Let k be a commutative ring and C a scheme over $\operatorname{Spec} k$. For any open U of C, define the ring map $k \to \Gamma(C, U)$ to be the composition of the inverse of the canonical isomorphism $\Gamma(\operatorname{Spec} k, \top) \cong k$, the structure morphism map on global sections, and the restriction from $\top$ to U.

Definition 8 (Algebra structure on a section)**.**
The ring map of 7 exhibits $\Gamma(C, U)$ as a k-algebra.

Lemma 9 (Defining identity for the algebra map)**.**
Under the algebra structure of 8, the structure map $k \to \Gamma(C, U)$ equals the ring map of 7.

Proof. The algebra structure of 8 is defined through the ring homomorphism of 7; its structure map is therefore that homomorphism. □

3.3.2 Naturality of the algebra map

Lemma 10 (Restriction is compatible with the algebra map)**.**
For any inclusion $V \le U$ of opens in C, the algebra map commutes with restriction: the composition $k \to \Gamma(C, U) \to \Gamma(C, V)$ equals the direct map $k \to \Gamma(C, V)$.

Proof. Both sides factor through the structure morphism map on global sections. The remaining triangle commutes by functoriality of the presheaf on composable inclusions. □

Lemma 11 (Algebra-map naturality)**.**
For any inclusion $V \le U$ of opens in C and any $r \in k$, the restriction map sends the image of r in $\Gamma(C, U)$ to the image of r in $\Gamma(C, V)$.

Proof. Replace the two algebra structure maps by the ring maps of 7 using 9. Their compatibility with restriction is exactly 10. □

3.3.3 The presheaf of k-modules and its sheaf condition

Definition 12 (Presheaf of k-modules from $\mathcal{O}_C$)**.**
Define the presheaf of k-modules on C by sending each open U to the k-module $\Gamma(C, U)$, with the k-action given by the algebra structure of 8, and on an inclusion $V \le U$ by the restriction map viewed as a k-linear map. The identity and composition laws are those of the structure sheaf.

Theorem 13 (Sheaf condition)**.**
The presheaf of 12 is a sheaf for the Grothendieck topology of opens of C.

Proof. The forgetful functor from k-modules to sets creates limits and reflects the sheaf condition. The underlying presheaf of sets is the underlying presheaf of the structure sheaf, which is a sheaf because $\mathcal{O}_C$ is a sheaf of commutative rings. □

3.3.4 The structure sheaf as a sheaf of k-modules

Definition 14 (Structure sheaf as a sheaf of k-modules)**.**
Bundle 12 and 13 into the sheaf of k-modules on C.

3.3.5 Forget-and-recover: bridging k-module and abelian-group values

Definition 15 (Forget-and-recover natural iso for the k-module structure sheaf)**.**
Let C be a scheme over $\operatorname{Spec} k$. Then forgetting the k-module structure from the sheaf of 14 yields a sheaf of abelian groups canonically isomorphic to the abelian-group-valued structure sheaf of 4.

Proof. At every open U, both sheaves have the abelian group underlying $\Gamma(C, U)$, and both restriction morphisms are the underlying additive maps of the usual restrictions. These identifications commute with restriction and hence define the claimed natural isomorphism. □

Lemma 16 (Object-evaluation lemma for the presheaf of k-modules)**.**
For every open U of C, the value of the presheaf of 12 at U is the k-module with carrier $\Gamma(C, U)$.

Proof. This is precisely the object assignment in 12. □

3.4 Sheaf cohomology with k-module coefficients

In a k-linear abelian category, Ext groups carry a canonical k-module structure. Applying this to sheaves of k-modules gives k-linear sheaf cohomology.

Definition 17 (Sheaf cohomology with k-module coefficients)**.**
Let k be a field, C a Grothendieck site admitting sheafification of k-modules and Ext. For a sheaf F of k-modules and $n \in \mathbb{N}$, define the k-module-valued sheaf cohomology

$$H^n_{\text{Module}}(F) \;:=\; \operatorname{Ext}^n(\text{constant sheaf at } k,\; F).$$

This automatically carries a k-module structure.

3.5 The H^0 algebraic bridge

After 17 the cohomology groups $H^n_{\text{Module}}(F)$ are well-defined and carry the canonical k-module structure. The next algebraic preliminary is the identification of the degree-zero piece $H^0_{\text{Module}}(F)$ with a Hom group: the bridge from sheaf cohomology to representable Hom-functors.

Definition 18 (H^0 as a k-linear Hom group)**.**
Let k be a field, C a Grothendieck site admitting sheafification of k-modules and Ext, and F a sheaf of k-modules. Then there is a canonical k-linear equivalence

$$H^0_{\text{Module}}(F) \;\simeq_k\; \operatorname{Hom}(\text{constant sheaf at } k,\; F).$$

Proof. In any abelian category, $\mathrm{Ext}^0(A, F)$ is canonically $\mathrm{Hom}(A, F)$. This identification respects scalar multiplication in a k-linear category; take A to be the constant sheaf at k. $\square$

3.6 Čech cohomology of the structure sheaf

The Čech complex of a sheaf with respect to an indexed open cover is the standard alternating-coface complex; its cohomology in each degree is a k-module.

Definition 19 (Čech complex of the structure sheaf)**.** Let C be a scheme over $\operatorname{Spec} k$ and $\mathcal{U}$ an indexed family of opens of C. Define the Čech cochain complex of the structure sheaf with respect to $\mathcal{U}$ to be the standard Čech complex applied to the underlying presheaf of the k-module-valued structure sheaf. This is a cochain complex valued in k-modules, indexed by $\mathbb{N}$.

Definition 20 (Čech cohomology of the structure sheaf)**.** With $(C, \mathcal{U})$ as in 19, define the n-th Čech cohomology to be the n-th homology of the Čech cochain complex. The result lives in the category of k-modules.

3.7 Cohomology of an open in k-module flavor

For an object X of the site, cohomology over X is represented by the sheafification of the free k-module on the representable presheaf at X.

Definition 21 (Cohomology of an open with k-module coefficients)**.**

Let k be a field, C a Grothendieck site admitting sheafification of k-modules and Ext, and F a sheaf of k-modules. For $n \in \mathbb{N}$ and X an object of C:

$$H'^n(X, F) \ := \ \mathrm{Ext}^n(\text{sheafified representable at } X,\ F) \ \in \ \mathrm{Type}.$$

The underlying type carries a k-module structure through the same Ext mechanism that powers H_{Module}.

3.8 The H^0 algebraic bridge for cohomology of an open

Parallel to 18, we identify the degree-zero piece of cohomology of an open with a Hom group.

Definition 22 (H^0 as a k-linear Hom group, open flavor)**.**

Let k be a field, C a Grothendieck site, X an object of C, and F a sheaf of k-modules. Then there is a canonical k-linear equivalence

$$H'^0(X, F) \ \simeq_k \ \mathrm{Hom}(\text{sheafified representable at } X,\ F).$$

Proof. Apply the canonical k-linear identification $\mathrm{Ext}^0(A, F) \cong \mathrm{Hom}(A, F)$ with A the sheafified representable at X. $\square$

Together with 18, this identifies degree-zero global and open-local cohomology with their corresponding representable Hom functors.

3.9 k-finite transport companions for the H^0 bridges

After the H^0 bridges identify cohomology with Hom groups, the immediate companion is transport of finite-dimensionality: if the Hom group is finite-dimensional over k, so is the cohomology group.

Theorem 23 (k-finite transport for global H^0)**.**

Let k be a field, C a Grothendieck site admitting sheafification of k-modules and Ext*, and F a sheaf of k-modules. If the k-linear Hom group* Hom(*constant sheaf at* k, F) *is finite-dimensional over k, then $H^0_{\mathrm{Module}}(F)$ is also finite-dimensional over k.*

Proof. Transport finite-dimensionality across the symmetric form of the H^0 bridge. ☐

Theorem 24 (k-finite transport for open H^0)**.** *Let k be a field, C a Grothendieck site, X an object of C, and F a sheaf of k-modules. If the k-linear Hom group* Hom(*sheafified representable at* X, F) *is finite-dimensional over k, then $H'^0(X, F)$ is also finite-dimensional over k.*

Proof. Transport finite-dimensionality across the symmetric form of the open H^0 bridge. ☐

3.10 Curve specialisations of the H^0 finite-dimensionality transports

Theorem 25 (Curve specialisation of global H^0 finite-dimensionality)**.** *Let k be a field and C a scheme over* Spec k*. If the k-linear Hom group from the constant sheaf at k to the k-module-valued structure sheaf is finite-dimensional over k, then $H^0_{\mathrm{Module}}(\mathcal{O}_C)$ is finite-dimensional over k.*

Theorem 26 (Curve specialisation of open H^0 finite-dimensionality)**.** *Let k be a field, C a scheme over* Spec k*, and U an open of C. If the k-linear Hom group from the sheafified representable at U to the k-module-valued structure sheaf is finite-dimensional over k, then $H'^0(U, \mathcal{O}_C)$ is finite-dimensional over k.*

3.11 Affine cohomology vanishing

For a Mayer–Vietoris argument on a two-affine cover, one uses Serre's theorem that the positive-degree cohomology of a quasicoherent sheaf on each affine piece vanishes. We isolate the corresponding vanishing property for a sheaf of k-modules.

Definition 27 (Affine cohomology vanishing)**.** Let k be a field, C a scheme over Spec k, and F a sheaf of k-modules on C. We say that F has affine cohomology vanishing if, for every affine open U of C and every degree $i > 0$, the group $H'^i(U, F)$ is zero.

Theorem 28 (Finite-dimensionality from affine vanishing)**.** *Let k be a field, C a scheme over* Spec k*, and F a sheaf of k-modules. Assume affine cohomology vanishing holds for F. For any affine open U of C and any $i > 0$, the cohomology group $H'^i(U, F)$ is finite-dimensional over k.*

Proof. A subsingleton k-module is generated by the empty set, hence finitely generated and finitely presented, hence finite-dimensional. ☐

3.12 Global H^0 Hom-finiteness

Global sections of a proper scheme can be finite-dimensional even though sections on its proper affine opens are not; for example, $\Gamma(\mathbb{A}^1_k, \mathcal{O}) = k[t]$. Accordingly, the relevant degree-zero finiteness condition concerns only the global Hom group.

Definition 29 (Global H^0 Hom-finiteness)**.** Let k be a field, C a scheme over $\operatorname{Spec} k$, and F a sheaf of k-modules on C. We say that F has global H^0 Hom-finiteness if Hom(constant sheaf at k, F) is finite-dimensional over k.

Theorem 30 (Finite-dimensionality of global H^0 from global Hom-finiteness)**.** *Let k be a field, C a scheme over* $\operatorname{Spec} k$*, and F a sheaf of k-modules. Assume global H^0 Hom-finiteness holds for F. Then $H^0_{\mathrm{Module}}(F)$ is finite-dimensional over k.*

Theorem 31 (Curve specialisation)**.** *Let k be a field and C a scheme over* $\operatorname{Spec} k$*. If the global morphism group from the constant sheaf at k to $\mathcal{O}_C$, viewed as a sheaf of k-modules, is finite-dimensional, then $H^0_{\mathrm{Module}}(\mathcal{O}_C)$ is finite-dimensional over k.*

Proof. Apply 30 to the k-module-valued structure sheaf. □

Remark 32. On a proper geometrically connected curve, Stein factorization gives $\Gamma(C, \mathcal{O}_C) = k$, which is finite-dimensional over k.

3.13 Stein finiteness of global sections

The geometric input for global Hom-finiteness is Stein's theorem: on a proper integral scheme over a field, global sections are finite-dimensional.

Theorem 33. *[Module-finiteness of the top-section map of a universally closed morphism]*

Let K be a commutative ring and X an integral scheme equipped with a universally closed, locally-of-finite-type morphism $f : X \to \operatorname{Spec} K$. Then the ring map on top sections $f^\sharp_\top$ is module-finite.

Theorem 34 (Stein finiteness of global sections)**.**

Let k be a field and C an integral scheme over $\operatorname{Spec} k$ *with proper structure morphism. Then $\Gamma(C, \mathcal{O}_C)$ is finite-dimensional over k.*

Proof. Apply the theorem that for an integral scheme with a universally closed, locally finite-type morphism to $\operatorname{Spec} k$, the structure-morphism ring map on global sections is module-finite. Transfer the base ring from $\Gamma(\operatorname{Spec} k, \top)$ to k along the canonical ring isomorphism. □

3.14 Global-sections evaluation isomorphism

To bridge from global sections (a concrete k-module) to the abstract Hom-from-constant-sheaf form, we lift the natural isomorphism between global sections and evaluation at the terminal object to a k-linear equivalence.

Theorem 35 (Global-sections-to-top linear equivalence)**.** *Let k be a field, X a topological space, and F a sheaf of k-modules on X. Then there is a k-linear equivalence between the global-sections object of F and the evaluation of F at the top open $\top$.*

Theorem 36 (Finite-dimensionality on global-sections object)**.** *Let k be a field and C an integral scheme over* $\operatorname{Spec} k$ *with proper structure morphism. Then the global-sections object of the k-module-valued structure sheaf is finite-dimensional over k.*

Proof. Combine 34 with the linear equivalence of 35, transporting finite-dimensionality across the symmetric form. ☐

3.15 Global Hom-finiteness for the structure sheaf

The constant-sheaf/global-sections adjunction and evaluation at $1 \in k$ identify the global Hom group with global sections. Stein finiteness then proves global H^0 Hom-finiteness for the structure sheaf of a proper integral scheme.

Lemma 37 (Additivity of the constant functor)**.** *Let C be a category and D a preadditive category. Then the constant functor* Functor.*const* $C : D \to (C \to D)$ *is additive.*

Proof. For morphisms f, g in D, the components of the constant natural transformation associated to $f+g$ are all $f+g$, which are also the components of the sum of the constant transformations associated to f and g. The same argument applies to zero, so the constant functor is additive. ☐

Lemma 38 (R-linearity of the constant functor)**.** *Let C be a category, R a semiring, and D an R-linear preadditive category. Then the constant functor* Functor.*const* $C : D \to (C \to D)$ *is R-linear.*

Proof. For $r \in R$ and a morphism f of D, both the constant transformation associated to $r \cdot f$ and r times the constant transformation associated to f have component $r \cdot f$ at every object. Equality of natural transformations is componentwise, so the constant functor is R-linear. ☐

Lemma 39 (Linear lifting of adjoint functors)**.** *Let $F \dashv G$ be an adjunction between R-linear preadditive categories. If G is R-linear, then F is R-linear.*

Proof. For a morphism $f : X \to X'$, the adjunction sends $h : FX \to FX'$ to $G(h) \circ \eta_X$. Naturality of the unit identifies the images of $F(r \cdot f)$ and $r \cdot F(f)$, because G preserves scalar multiplication. Since the adjunction map is injective, these morphisms agree. ☐

Lemma 40 (Right-adjoint version)**.** *Dual of 39: if F is R-linear, then G is R-linear.*

Proof. For $f : Y \to Y'$, the inverse adjunction map sends $h : GY \to GY'$ to $\varepsilon_{Y'} \circ F(h)$. Naturality of the counit identifies the images of $G(r \cdot f)$ and $r \cdot G(f)$, because F preserves scalar multiplication. Injectivity of the inverse adjunction map gives the result. ☐

Theorem 41 (Linear hom-equivalence from an adjunction)**.** *Let $F \dashv G$ be an adjunction between R-linear preadditive categories with G additive and R-linear. Then for all objects X and Y, the additive equivalence of the adjunction upgrades to an R-linear equivalence between* $\operatorname{Hom}(FX, Y)$ *and* $\operatorname{Hom}(X, GY)$*.*

Proof. The adjunction sends $f : FX \to Y$ to $G(f) \circ \eta_X$. Preadditivity and the R-linearity of G show that this map preserves addition and scalar multiplication. The unit argument shows that F is R-linear, so the inverse map $g \mapsto \varepsilon_Y \circ F(g)$ is also linear. Thus the adjunction bijection is an R-linear equivalence. ☐

Theorem 42 (Constant-sheaf / global-sections linear equivalence)**.** *Let k be a field, C a Grothendieck site admitting sheafification of k-modules, and F a sheaf of k-modules. For any k-module X, there is a k-linear equivalence between morphisms from the constant sheaf at X to F, and k-linear maps from X to the global sections of F.*

Proof. The constant-sheaf functor is left adjoint to global sections. The constant functor is k-linear by 38, and linearity passes across the adjunction by 39 and 40. The adjunction bijection is therefore k-linear by 41. □

Theorem 43 (Hom-from-k evaluation)**.** *For any field k and k-module M, there is a canonical k-linear equivalence between $\mathrm{Hom}_k(k, M)$ and M given by evaluation at 1.*

Proof. Evaluation sends f to $f(1)$. Its inverse sends $m \in M$ to the linear map $r \mapsto r \cdot m$. Linearity and the two inverse identities follow from $f(r) = r \cdot f(1)$. □

Theorem 44 (Global Hom-finiteness of the structure sheaf)**.** *Let k be a field and C an integral scheme over $\operatorname{Spec} k$ with proper structure morphism. Then global H^0 Hom-finiteness holds for the k-module-valued structure sheaf of C.*

Proof. Compose the constant-sheaf/global-sections linear equivalence (specialised to $X = k$) with the Hom-from-k evaluation to obtain a linear equivalence between the global Hom group and the global-sections object. 36 provides finite-dimensionality on the latter; transport across the symmetric form of the equivalence. □

3.16 Čech complexes for arbitrary sheaves

The Čech construction applies to an arbitrary sheaf of k-modules; the structure-sheaf construction is its specialization at $F = \mathcal{O}_C$.

Definition 45 (Parameterised Čech complex)**.** Let C be a scheme over $\operatorname{Spec} k$, F a sheaf of k-modules on C, and $\mathcal{U}$ an indexed family of opens. Define the Čech cochain complex of F with respect to $\mathcal{U}$ to be the standard Čech complex applied to the underlying presheaf of F. This is a cochain complex valued in k-modules.

Definition 46 (Parameterised Čech cohomology)**.** With $(C, F, \mathcal{U})$ as in 45, define the n-th Čech cohomology to be the n-th homology of the Čech cochain complex.

Theorem 47 (Čech complex bridge to structure-sheaf specialisation)**.** *The structure-sheaf Čech complex equals the parameterised Čech complex at $F = \mathcal{O}_C$.*

Proof. Both sides are the standard Čech complex of the same k-module-valued structure sheaf with respect to the same indexed family of opens. □

Theorem 48 (Čech cohomology bridge to structure-sheaf specialisation)**.** *The structure-sheaf Čech cohomology equals the parameterised Čech cohomology at $F = \mathcal{O}_C$.*

Proof. Both sides are the degree-n cohomology of the same structure-sheaf Čech complex, so they agree. □

3.17 Čech acyclicity

Positive-degree vanishing of Čech cohomology transports to derived cohomology whenever the two theories are related by a comparison isomorphism.

Definition 49 (Čech acyclicity)**.** Let k be a field, C a scheme over $\operatorname{Spec} k$, F a sheaf of k-modules on C, and $\mathcal{U}$ an indexed family of opens. The cover $\mathcal{U}$ is Čech-acyclic for F if the Čech cohomology vanishes in positive degrees: for every $n \geq 1$, the n-th Čech cohomology group is a subsingleton.

Theorem 50 (Derived vanishing from Čech acyclicity)**.** *Let k be a field, C a scheme over $\operatorname{Spec} k$, F a sheaf of k-modules, and $\mathcal{U}$ a Čech-acyclic cover for F. Suppose there is a k-linear comparison isomorphism between Čech cohomology and derived cohomology of the cover supremum for every degree. Then for every $n \geq 1$, the derived cohomology of the cover supremum is a subsingleton.*

Proof. Transport the subsingleton structure along the symmetric form of the comparison isomorphism. □

Theorem 51 (Curve specialisation)**.** *Let C be a scheme over $\operatorname{Spec} k$ and $\mathcal{U}$ a Čech-acyclic cover for $\mathcal{O}_C$. Suppose Čech and derived cohomology of the cover supremum are k-linearly isomorphic in every degree. Then, for every $n \geq 1$,*

$$H'^n\left(\bigsqcup_i \mathcal{U}_i, \mathcal{O}_C\right) = 0.$$

Proof. The positive-degree Čech group is zero by acyclicity. Its image under the assumed linear isomorphism is therefore the zero derived-cohomology group of the cover supremum. □

3.18 Cohomological consequences

The constructions above give the following compatible descriptions of structure-sheaf cohomology:

- For C a scheme over $\operatorname{Spec} k$, the cohomology $H^n_{\text{Module}}(\mathcal{O}_C)$ carries a canonical k-module structure.
- Forgetting the k-module structure recovers the abelian-group cohomology of 2; their underlying abelian groups agree.
- Finite-dimensionality can therefore be expressed as $\dim_k H^n_{\text{Module}}(\mathcal{O}_C) < \infty$, and in degree zero it is transported across the Hom and global-sections equivalences above.

Chapter 4

Mayer–Vietoris long exact sequence for sheaf cohomology with k-module coefficients

This chapter constructs the Mayer–Vietoris long exact sequence for sheaf cohomology with k-module coefficients and relates Čech acyclicity on affine covers to vanishing of derived cohomology. The groups $H^n(X, \mathcal{F})$ and $H'^n(U, \mathcal{F})$ are defined in 3.

4.1 Functoriality of cohomology in the open variable

The cohomology group $H'^n(X, \mathcal{F})$ is contravariantly functorial in the open variable X. Assembling this functoriality yields a presheaf

$$X \longmapsto H'^n(X, \mathcal{F}),$$

and, varying $\mathcal{F}$ as well, a bifunctor from sheaves of k-modules to presheaves of abelian groups. Each fiber also carries the k-module structure induced by the k-linear structure on Ext.

Definition 52 (Cohomology presheaf functor, H' flavor)**.**

Let k be a field, C a category equipped with a Grothendieck topology J, and assume sheafification and Ext exist for sheaves of k-modules on J. For $n \in \mathbb{N}$, define the cohomology presheaf functor

$$\mathcal{H}'^n(k, J) \;:\; \mathrm{Sheaf}\ J\,(\mathrm{ModuleCat}\,k) \longrightarrow (C^{\mathrm{op}} \longrightarrow \mathrm{AddCommGrpCat})$$

by the standard composition of the Yoneda embedding, the free-k-module functor, sheafification, and the n-th Ext functor.

Proof. Each morphism of C induces, by Yoneda and sheafification, a morphism between the corresponding source sheaves. Contravariance of Ext in its first variable gives the restriction maps, and functoriality of these constructions gives the stated bifunctor. □

Definition 53 (Cohomology presheaf, H' flavor)**.**

In the setting of 52, let $\mathcal{F}$ be a sheaf of k-modules. For $n \in \mathbb{N}$, define the cohomology presheaf

$$\mathcal{H}'^n(\mathcal{F}) \;:\; C^{\mathrm{op}} \longrightarrow \mathrm{AddCommGrpCat}$$

to be the evaluation of the bifunctor at $\mathcal{F}$.

Proof. Evaluating the bifunctor of 52 at $\mathcal{F}$ preserves identities and composition, and therefore gives the asserted presheaf. $\square$

Remark 54. For every object X of C, the value $\mathcal{H}'^n(\mathcal{F})(X)$ is canonically the abelian group underlying $H'^n(X, \mathcal{F})$.

4.2 Mayer–Vietoris long exact sequence

A Mayer–Vietoris square for a Grothendieck topology J on a category C is a pullback square

$$\begin{array}{ccc} X_1 & \longrightarrow & X_2 \\ \downarrow & & \downarrow \\ X_3 & \longrightarrow & X_4 \end{array}$$

that is a J-cover: the pair of morphisms $X_2 \to X_4$ and $X_3 \to X_4$ is a covering family, and X_1 is the pullback. For a sheaf $\mathcal{F}$, the cohomology groups of the four corners are linked by a long exact sequence whose building blocks are restriction maps, biproduct maps, and a connecting homomorphism.

4.2.1 The restriction maps

Definition 55 (Sum of two restriction maps)**.**
Let S be a Mayer–Vietoris square, $\mathcal{F}$ a sheaf of k-modules, and $n \in \mathbb{N}$. Define the morphism

$$\text{toBiprod} \;:\; H'^n(X_4, \mathcal{F}) \longrightarrow H'^n(X_2, \mathcal{F}) \oplus H'^n(X_3, \mathcal{F})$$

to be the biproduct lift of the two restriction maps induced by $X_2 \to X_4$ and $X_3 \to X_4$.

Proof. The biproduct exists because the category of abelian groups has finite biproducts. The morphism is built from the contravariant functoriality of the cohomology presheaf. $\square$

Definition 56 (Difference of two restriction maps)**.**
In the same setting, define the morphism

$$\text{fromBiprod} \;:\; H'^n(X_2, \mathcal{F}) \oplus H'^n(X_3, \mathcal{F}) \longrightarrow H'^n(X_1, \mathcal{F})$$

to be the biproduct desc of the restriction map $X_1 \to X_2$ and the additive inverse of the restriction map $X_1 \to X_3$.

Proof. The negation is taken in the preadditive morphism group of the codomain category. The sign convention encodes the alternating-sum structure of the Čech complex. $\square$

Lemma 57 (Composition vanishes)**.**
In the setting above,

$$\text{fromBiprod} \circ \text{toBiprod} \;=\; 0.$$

Proof. Expanding the two biproduct maps gives the difference of the restrictions along the two composites $X_1 \to X_2 \to X_4$ and $X_1 \to X_3 \to X_4$. These composites agree because the square commutes, so their difference is zero. $\square$

4.2.2 The short exact sequence of sheaves

The connecting homomorphism of the Mayer–Vietoris sequence is built from a short exact sequence of sheaves of k-modules obtained by sheafifying the Yoneda embedding composed with the free-k-module functor.

Definition 58 (Free-k-module functor is left adjoint)**.** Let k be a field. The free-k-module functor $\mathrm{Type} \to \mathrm{ModuleCat}\, k$ is a left adjoint.

Proof. The usual natural bijection

$$\mathrm{Hom}_k(k^{(S)}, M) \cong \mathrm{Hom}_{\mathrm{Set}}(S, |M|)$$

exhibits the free-k-module functor as left adjoint to the forgetful functor. ☐

Lemma 59 (Pushout of free sheaves)**.**
Let S be a Mayer–Vietoris square. The image of S under the composite functor

$$\mathit{Yoneda} \circ \mathit{whiskering\ with}\ (\mathrm{ModuleCat.\,free}\ k) \circ \mathrm{presheafToSheaf}$$

is a pushout square in the category of sheaves of k-modules.

Proof. The original square is a pushout. Post-composition with the sheafify-compose functor preserves pushouts because the free-k-module functor is a left adjoint. A canonical isomorphism translates the resulting square into the desired shape. ☐

Definition 60 (Short complex of free sheaves)**.**
In the setting of 59, define the short complex of sheaves of k-modules

$$\mathcal{F}_1 \xrightarrow{f} \mathcal{F}_2 \oplus \mathcal{F}_3 \xrightarrow{g} \mathcal{F}_4,$$

where f is the biproduct lift of the two maps induced by $X_1 \to X_2$ and $-X_1 \to X_3$, and g is the biproduct desc of the two maps induced by $X_2 \to X_4$ and $X_3 \to X_4$. The composition $g \circ f = 0$ follows from the cokernel-cofork condition of the pushout.

Proof. The commutativity relation in the pushout square says that the two composites from $\mathcal{F}_1$ to $\mathcal{F}_4$ agree. With the chosen sign on the second component of f, their sum is therefore zero, so $g \circ f = 0$. ☐

Definition 61 (Free-k-module functor preserves monomorphisms)**.** Let k be a field. The free-k-module functor preserves monomorphisms.

Proof. An injection of bases induces an injection of the corresponding free modules: a nonzero finite linear combination remains nonzero after relabeling its basis elements. Since monomorphisms of k-modules are precisely injective linear maps, the free-module functor preserves monomorphisms. ☐

Lemma 62 (The first map is mono)**.**
In the short complex of 60, the first map f is a monomorphism.

Proof. The map f followed by the second projection is the free-k-module functor applied to the monomorphism $X_1 \to X_3$, which is mono because the free functor preserves monomorphisms. Hence f itself is mono. ☐

Lemma 63 (The second map is epi)**.**
In the short complex of 60, the second map g is an epimorphism.

Proof. By the pushout lemma, g is the cokernel of f, and every cokernel is epi. ☐

Lemma 64 (Exactness at the middle term)**.**
In the short complex of 60, exactness holds at the middle term: the image of f equals the kernel of g.

Proof. By the pushout lemma, g is a cokernel of f. In an abelian category the kernel of a cokernel is the image of the original morphism, so $\mathrm{im}(f) = \ker(g)$. ☐

Lemma 65 (The short complex is short exact)**.**
The short complex of 60 is short exact: its first map is mono, its second map is epi, and exactness holds at the middle term.

Proof. Combine 62, 63, and 64. ☐

4.2.3 The connecting homomorphism and the long exact sequence

Definition 66 (Connecting homomorphism)**.**
Let S be a Mayer–Vietoris square, $\mathcal{F}$ a sheaf of k-modules, and $n_0, n_1 \in \mathbb{N}$ with $n_0 + 1 = n_1$. The connecting homomorphism

$$\delta \;:\; H'^{n_0}(X_1, \mathcal{F}) \longrightarrow H'^{n_1}(X_4, \mathcal{F})$$

is obtained by precomposing the extension class of the short exact sequence of 65 with the canonical degree-zero map.

Proof. The extension class lives in Ext^1 of the short exact sequence; precomposing with a map of degree zero and transporting through the additive structure yields the connecting homomorphism as a morphism of abelian groups. ☐

Definition 67 (Mayer–Vietoris long exact sequence)**.**
In the setting of 66, define the five-term Mayer–Vietoris sequence

$$H'^{n_0}(X_4, \mathcal{F}) \xrightarrow{\text{toBiprod}} H'^{n_0}(X_2, \mathcal{F}) \oplus H'^{n_0}(X_3, \mathcal{F}) \xrightarrow{\text{fromBiprod}} H'^{n_0}(X_1, \mathcal{F}) \xrightarrow{\delta} H'^{n_1}(X_4, \mathcal{F}) \xrightarrow{\text{toBiprod}} H'^{n_1}(X_2, \mathcal{F}) \oplus$$

Proof. The sources and targets of the five displayed morphisms match in succession, so they form the stated sequence of abelian groups. ☐

4.2.4 Comparison with the Ext contravariant sequence

The following four lemmas bridge the biproduct structure on the abelian-group side with the Ext biproduct additivity, preparing the comparison isomorphism.

Lemma 68 (Elementwise formula for toBiprod)**.**
For $y \in H'^n(X_4, \mathcal{F})$,

$$\text{toBiprod}(y) \;=\; (\text{restriction}_{X_4 \to X_2}(y),\ \text{restriction}_{X_4 \to X_3}(y)).$$

Proof. Follows from the definition of toBiprod as a biproduct lift and the naturality of the biproduct isomorphism with the product. ☐

Lemma 69 (Elementwise formula for fromBiprod)**.**
For $y_2 \in H'^n(X_2, \mathcal{F})$ *and* $y_3 \in H'^n(X_3, \mathcal{F})$*,*

$$\mathrm{fromBiprod}(y_2, y_3) \;=\; \mathrm{restriction}_{X_2 \to X_1}(y_2) \;-\; \mathrm{restriction}_{X_3 \to X_1}(y_3).$$

Proof. Follows from the definition of fromBiprod as a biproduct desc and the concrete description of the biproduct isomorphism. □

Lemma 70 (Ext biproduct bridge for toBiprod)**.**
For $x \in H'^n(X_4, \mathcal{F})$*, transport* toBiprod$(x)$ *from the abelian-group biproduct to the* Ext *biproduct. The result is the Yoneda product of* x *with the degree-zero class induced by the second map* g *of 60.*

Proof. Under the Ext biproduct isomorphism, the two components of the transported class are the pullbacks of x along the two components of g. This is also the componentwise description of its Yoneda product with the degree-zero class of g, so the classes agree. □

Lemma 71 (Ext biproduct bridge for fromBiprod)**.**
For a class x *in the biproduct of the* X_2*- and* X_3*-cohomology groups, its Yoneda product with the degree-zero class induced by the first map* f *of 60 equals* fromBiprod(x)*.*

Proof. Write $x = (x_2, x_3)$ using the biproduct isomorphism. The degree-zero class of f has components given by the two restriction maps, with a minus sign on the second. Bilinearity of the Yoneda product therefore gives the difference of the restrictions of x_2 and x_3, which is fromBiprod(x). □

Definition 72 (Comparison isomorphism)**.**
The Mayer–Vietoris sequence of 67 is canonically isomorphic to the Ext contravariant sequence applied to the short exact sequence of 65.

Proof. Use the identity on the non-biproduct terms and the canonical additive isomorphism between a biproduct of Ext groups and Ext from a biproduct on the other terms. The compatibility squares for the restriction maps are 70 and 71; the middle square is the definition of the connecting homomorphism. □

4.2.5 Exactness

Theorem 73 (Mayer–Vietoris exactness)**.**
The Mayer–Vietoris sequence of 67 is exact.

Proof. Transport exactness along the comparison isomorphism of 72. The Ext contravariant sequence is exact by the general exactness theorem for Ext applied to a short exact sequence. □

Lemma 74 (Connecting homomorphism annihilates toBiprod)**.**
In the setting above,

$$\delta \circ \mathrm{toBiprod} \;=\; 0.$$

Proof. Immediate from exactness at the third term of the sequence. □

Lemma 75 (fromBiprod annihilates the connecting homomorphism)**.**
In the setting above,

$$\mathrm{fromBiprod} \circ \delta \;=\; 0.$$

Proof. Immediate from exactness at the second term of the sequence. □

4.3 Two-affine open covers

Let X be a scheme equipped with two affine opens U_1, U_2 that cover X and whose intersection is affine. These opens determine a Mayer–Vietoris square and hence a long exact sequence.

Definition 76 (Two-affine Mayer–Vietoris square)**.** Let X be a scheme. A two-affine Mayer–Vietoris square on X consists of two open subschemes U_1, U_2 of X such that U_1, U_2, and $U_1 \cap U_2$ are affine, together with the hypothesis $U_1 \cup U_2 = X$.

Definition 77 (Underlying Mayer–Vietoris square)**.**
Given a two-affine cover S of X, the underlying abstract Mayer–Vietoris square has corners $X_1 = U_1 \cap U_2$, $X_2 = U_1$, $X_3 = U_2$, $X_4 = U_1 \cup U_2$.

Lemma 78 (Corner identification, X_1)**.**
$S.\text{toMayerVietorisSquare}.X_1 = S.U_1 \cap S.U_2$.

Proof. The first corner of the underlying square is defined to be $U_1 \cap U_2$. □

Lemma 79 (Corner identification, X_2)**.**
$S.\text{toMayerVietorisSquare}.X_2 = S.U_1$.

Proof. The second corner of the underlying square is U_1. □

Lemma 80 (Corner identification, X_3)**.**
$S.\text{toMayerVietorisSquare}.X_3 = S.U_2$.

Proof. The third corner of the underlying square is U_2. □

Lemma 81 (Cover-totality identification, X_4)**.**
$S.\text{toMayerVietorisSquare}.X_4 = \top$.

Proof. The fourth corner equals $U_1 \cup U_2$ by construction, and $U_1 \cup U_2 = X = \top$ by the cover hypothesis. □

Definition 82 (Mayer–Vietoris sequence on a two-affine cover)**.**
Let S be a two-affine cover of X and $\mathcal{F}$ a sheaf of k-modules. The five-term Mayer–Vietoris sequence on S is the specialisation of the abstract sequence to the underlying Mayer–Vietoris square of S.

Theorem 83 (Exactness on a two-affine cover)**.**
The sequence of 82 is exact.

Proof. By 73 applied to the underlying Mayer–Vietoris square. □

Definition 84 (Curve specialisation)**.**
Let C be a scheme over $\operatorname{Spec} k$ and S a two-affine cover of C. The Mayer–Vietoris sequence on S for the structure sheaf $\mathcal{O}_C$ is the specialisation of the sheaf-parameterised sequence at $\mathcal{F} = \mathcal{O}_C$.

Theorem 85 (Exactness for the structure sheaf on a curve)**.**
The sequence of 84 is exact.

Proof. By 83 applied to the structure sheaf. □

4.3.1 Serre-finiteness consequence

For a proper geometrically integral curve C over k, a two-affine cover with affine intersection always exists. Its Mayer–Vietoris sequence gives the following finiteness argument:

1. The Mayer–Vietoris LES applied to S and $\mathcal{O}_C$ relates H^0 and H^1 of C, U_1, U_2, and $U_1 \cap U_2$.
2. The cover hypothesis identifies the fourth corner with the whole curve.
3. Affineness of the pieces collapses $H^{>0}$ on those pieces, reducing the LES to a three-term sequence.
4. Finiteness of each H^0 piece then implies finiteness of $H^1(C, \mathcal{O}_C)$.

4.4 Cohomology of the whole space

At a terminal object T, the sheafified free k-module on the representable presheaf is the constant sheaf at k. Consequently, cohomology of the open T agrees with whole-space cohomology.

4.4.1 Source-object identification

Definition 86 (Source-object isomorphism for a terminal object)**.**
Let T be a terminal object of the site. There is a natural isomorphism in the sheaf category between the sheafified representable at T (composed with the free-k-module functor) and the constant sheaf at the one-dimensional k-module.

Proof. Compose three isomorphisms: the terminal-collapse of Yoneda, the constancy isomorphism after whiskering with the free functor, and the identification of the free module on a singleton with k itself. Then apply sheafification. □

4.4.2 Ext **transport**

Definition 87 (Ext transport for whole-space cohomology)**.**
Let T be a terminal object. There is a k-linear isomorphism

$$\mathrm{Ext}^n(\text{sheafified representable at } T,\ \mathcal{F}) \ \cong\ H^n(X, \mathcal{F}).$$

Proof. Pre- and post-compose with the degree-zero maps induced by the source-object isomorphism and its inverse. The round-trip equations collapse by associativity, composition of degree-zero maps, and the inverse laws of the isomorphism. □

Definition 88 (Ext transport for cohomology of an open)**.**
Let T be a terminal object. There is a k-linear isomorphism

$$H'^n(T, \mathcal{F}) \ \cong\ \mathrm{Ext}^n(\text{constant sheaf at } k,\ \mathcal{F}).$$

Proof. Precomposition with the source-object isomorphism of 86 gives the displayed map on Ext; precomposition with its inverse gives the inverse map. Associativity of the Yoneda product and the inverse identities prove that the two composites are identities. □

4.4.3 Change of universe and the full bridge

Changing the ambient universe does not change an Ext group. The canonical change-of-universe equivalence respects both its additive structure and, in an R-linear category, its R-module structure.

Lemma 89 (Universe change is additive on Ext)**.** *Let C be an abelian category for which* Ext *exists at two universes w and w'. For $a, b \in \mathrm{Ext}^n(X,Y)$ computed at universe w, the universe-change map satisfies*

$$\mathrm{chgUniv}(a+b) \;=\; \mathrm{chgUniv}(a) + \mathrm{chgUniv}(b).$$

Proof. Identify $\mathrm{Ext}^n(X,Y)$ with the corresponding morphism group in the derived category. Under this identification, change of universe commutes with the canonical embedding of morphisms, which is additive. Hence it carries the sum of a and b to the sum of their images. □

Lemma 90 (Universe change is R-linear on Ext)**.** *Let R be a ring and C an R-linear abelian category for which* Ext *exists at two universes w and w'. For $r \in R$ and $a \in \mathrm{Ext}^n(X,Y)$ computed at universe w, the universe-change map satisfies*

$$\mathrm{chgUniv}(r \cdot a) \;=\; r \cdot \mathrm{chgUniv}(a).$$

Proof. Under the identification of Ext with a derived-category morphism group, the R-action is the usual scalar action on morphisms. Change of universe commutes with this identification and with scalar multiplication, which gives the stated equality. □

Definition 91 (Linear change of universe)**.** Let R be a ring and C an R-linear abelian category. For any two universes at which Ext exists, there is an R-linear isomorphism between the two Ext groups, obtained by upgrading the universe-change bijection with its additivity and R-linearity.

Proof. The underlying bijection is the standard universe-change equivalence. Additivity and R-linearity follow from the fact that the universe change commutes with the homomorphism equivalence to the derived category, and the additive and R-linear structures are pulled back through that equivalence. □

Definition 92 (Full cover-totality bridge)**.**
Let T be a terminal object. There is a k-linear isomorphism

$$H'^n(T, \mathcal{F}) \;\cong\; H^n(X, \mathcal{F}).$$

Proof. Compose the open-cohomology identification of 88, the change-of-universe linear isomorphism of 91, and the whole-space identification of 87. □

4.4.4 Specialisation to the X_4 corner of a two-affine cover

Definition 93 (X_4 corner bridge, abstract sheaf form)**.**
Let S be a two-affine cover of X and $\mathcal{F}$ a sheaf of k-modules. There is a k-linear isomorphism

$$H'^n(S.\mathrm{toMayerVietorisSquare}.X_4, \mathcal{F}) \;\cong\; H^n(X, \mathcal{F}).$$

Proof. The fourth corner X_4 equals the top open $\top$, which is terminal; apply the full bridge. □

Definition 94 (X_4 corner bridge, curve form)**.**
Let C be a scheme over $\mathrm{Spec}\, k$ and S a two-affine cover of C. There is a k-linear isomorphism

$$H'^n(S.\mathrm{toMayerVietorisSquare}.X_4, \mathcal{O}_C) \;\cong\; H^n(C, \mathcal{O}_C).$$

Proof. Specialize 93 to the k-module-valued structure sheaf $\mathcal{O}_C$. □

4.4.5 Consequences for finite dimensionality

Theorem 95 (Finite rank via the X_4 corner, abstract sheaf form)**.**
Let S be a two-affine cover of X and $\mathcal{F}$ a sheaf of k-modules. Then

$$\dim_k H^n(X, \mathcal{F}) \;=\; \dim_k H'^n(S.\mathrm{toMayerVietorisSquare}.X_4, \mathcal{F}).$$

Proof. A linear isomorphism preserves finite rank. □

Theorem 96 (Finite rank via the X_4 corner, curve form)**.**
Let C be a scheme over $\operatorname{Spec} k$ *and S a two-affine cover of C. Then*

$$\dim_k H^n(C, \mathcal{O}_C) \;=\; \dim_k H'^n(S.\mathrm{toMayerVietorisSquare}.X_4, \mathcal{O}_C).$$

Proof. Apply 95 to $\mathcal{F} = \mathcal{O}_C$. □

Theorem 97 (Finite-dimensionality via the X_4 corner, abstract sheaf form)**.**
Let S be a two-affine cover of X and $\mathcal{F}$ a sheaf of k-modules. If $H'^n(S.\mathrm{toMayerVietorisSquare}.X_4, \mathcal{F})$ is finite-dimensional over k, then so is $H^n(X, \mathcal{F})$.

Proof. A linear isomorphism transports the property of being a finite module. □

Theorem 98 (Finite-dimensionality via the X_4 corner, curve form)**.**
Let C be a scheme over $\operatorname{Spec} k$ *and S a two-affine cover of C. If $H'^n(S.\mathrm{toMayerVietorisSquare}.X_4, \mathcal{O}_C)$ is finite-dimensional over k, then so is $H^n(C, \mathcal{O}_C)$.*

Proof. Apply 97 to $\mathcal{F} = \mathcal{O}_C$. □

4.5 Čech acyclicity and vanishing on affines

A Čech-acyclic cover whose Čech cohomology agrees with derived cohomology gives vanishing of $H'^n(U, \mathcal{F})$ in positive degrees. Applying this to covers of affine opens yields affine cohomology vanishing.

4.5.1 Top-supremum transport

When a cover $\mathfrak{U}$ is Čech-acyclic and its supremum is the whole space, the vanishing of Čech cohomology transports to the vanishing of whole-space cohomology.

Theorem 99 (Top-supremum transport, abstract sheaf form)**.**
Let $\mathfrak{U}$ be a Čech-acyclic cover with $\bigsqcup_i \mathfrak{U}_i = \top$, and assume a comparison isomorphism between Čech and derived cohomology. For $n \geq 1$, the type $H^n(X, \mathcal{F})$ is a subsingleton.

Proof. Acyclicity and the comparison isomorphism make $H'^n(\bigsqcup_i \mathfrak{U}_i, \mathcal{F})$ zero. The cover hypothesis identifies its open with $\top$, and 92 transports this vanishing to $H^n(X, \mathcal{F})$. □

Theorem 100 (Top-supremum transport, curve form)**.**
Let C be a scheme over $\operatorname{Spec} k$ *and $\mathfrak{U}$ a Čech-acyclic cover for $\mathcal{O}_C$ with $\bigsqcup_i \mathfrak{U}_i = \top$. If Čech and derived cohomology of the cover supremum are k-linearly isomorphic in every degree, then $H^n(C, \mathcal{O}_C) = 0$ for every $n \geq 1$.*

Proof. Apply 99 to the k-module-valued structure sheaf. □

4.5.2 Comparison data

A Čech-to-derived comparison consists of a linear isomorphism in every degree.

Definition 101 (Čech-to-derived comparison data)**.**
Let $\mathfrak{U}$ be an indexed open cover. The property `HasCechToHModuleIso` consists of a k-linear isomorphism between Čech cohomology and H' cohomology of the supremum in every degree.

Definition 102 (The comparison isomorphism)**.**
Given `HasCechToHModuleIso`, denote its degree-n comparison isomorphism by comp_n.

Theorem 103 (Top-supremum transport from comparison data, abstract sheaf form)**.**
If $\mathfrak{U}$ is Čech-acyclic for $\mathcal{F}$, has Čech-to-derived comparison data, and satisfies $\bigsqcup_i \mathfrak{U}_i = \top$, then $H^n(X, \mathcal{F}) = 0$ for $n \geq 1$.

Proof. Apply 99 using the comparison isomorphisms of 102. □

Theorem 104 (Top-supremum transport from comparison data, curve form)**.**
Let C be a scheme over $\operatorname{Spec} k$ and $\mathfrak{U}$ a Čech-acyclic cover for $\mathcal{O}_C$ with comparison data and $\bigsqcup_i \mathfrak{U}_i = \top$. Then $H^n(C, \mathcal{O}_C) = 0$ for $n \geq 1$.

Proof. Apply 103 to the k-module-valued structure sheaf. □

Theorem 105 (Supremum transport for H', abstract sheaf form)**.**
If $\mathfrak{U}$ is Čech-acyclic for $\mathcal{F}$ and has comparison data, then $H'^n(\bigsqcup_i \mathfrak{U}_i, \mathcal{F}) = 0$ for $n \geq 1$.

Proof. Apply 50 using the comparison isomorphisms of 102. □

Theorem 106 (Supremum transport for H', curve form)**.**
Let C be a scheme over $\operatorname{Spec} k$ and $\mathfrak{U}$ a Čech-acyclic cover for $\mathcal{O}_C$ with comparison data. Then $H'^n(\bigsqcup_i \mathfrak{U}_i, \mathcal{O}_C) = 0$ for $n \geq 1$.

Proof. Apply 105 to the k-module-valued structure sheaf. □

Theorem 107 (Vanishing on the covered open, abstract sheaf form)**.**
If $\mathfrak{U}$ is Čech-acyclic for $\mathcal{F}$, has comparison data, and satisfies $\bigsqcup_i \mathfrak{U}_i = U$, then $H'^n(U, \mathcal{F}) = 0$ for $n \geq 1$.

Proof. Identify U with $\bigsqcup_i \mathfrak{U}_i$ and apply 105. □

Theorem 108 (Vanishing on the covered open, curve form)**.**
Let C be a scheme over $\operatorname{Spec} k$, let $U \subseteq C$ be open, and let $\mathfrak{U}$ be a Čech-acyclic cover for $\mathcal{O}_C$ with comparison data and $\bigsqcup_i \mathfrak{U}_i = U$. Then $H'^n(U, \mathcal{O}_C) = 0$ for $n \geq 1$.

Proof. Apply 107 to the k-module-valued structure sheaf. □

4.5.3 Acyclic covers of affine opens

Suppose every affine open admits a Čech-acyclic cover equipped with comparison data. Then positive-degree cohomology vanishes on every affine open.

Definition 109 (Acyclic covers of affine opens)**.**
For every affine open U, there exists an indexed cover $\mathfrak{U}$ with $\bigsqcup \mathfrak{U}_i = U$ that is Čech-acyclic and carries a comparison isomorphism.

Theorem 110 (Affine vanishing from acyclic covers)**.**
If `HasAffineCechAcyclicCover` *holds, then* `IsAffineHModuleVanishing` *holds.*

Proof. For an affine open U, choose the cover supplied by 109. Its acyclicity and comparison isomorphisms give $H'^n(U, \mathcal{F}) = 0$ for $n \geq 1$ by 107. □

4.6 Finite-dimensionality consequence

Assume every affine open of C has a Čech-acyclic cover with Čech-to-derived comparison data. Then 110 gives positive-degree vanishing on affine opens. For a two-affine cover, the Mayer–Vietoris sequence therefore reduces in degrees zero and one to a short exact segment. If the relevant degree-zero terms are finite-dimensional, exactness implies that $H^1(C, \mathcal{O}_C)$ is finite-dimensional over k.

Chapter 5

Flat base change for the pushforward of a quasi-coherent sheaf ($i = 0$)

5.1 Setup and motivation

This chapter establishes the $i = 0$ (i.e. H^0 / direct-image) case of flat base change: the formation of the pushforward $f_*\mathcal{F}$ of a quasi-coherent sheaf commutes with flat base change on the target. It is the foundational, fully self-contained slice of the cohomology-and-base-change package that ultimately underlies relative representability statements; the higher derived cases $R^i f_*$ for $i \geq 1$ are treated separately and are out of scope here.

Throughout, $f : X \to S$ is a morphism of schemes that is quasi-compact and quasi-separated, and $\mathcal{F}$ is a quasi-coherent $\mathcal{O}_X$-module. A base change of f along an arbitrary morphism $g : S' \to S$ is recorded by the cartesian square obtained from the fibre product $X' = X \times_S S'$:

$$
\begin{array}{ccc}
X' & \xrightarrow{g'} & X \\
{\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\
S' & \xrightarrow{g} & S
\end{array}
$$

Here $f' : X' \to S'$ is the base change of f, and $g' : X' \to X$ is the first projection. We write $\mathcal{F}' = (g')^*\mathcal{F}$ for the pullback of $\mathcal{F}$ to X'.

5.2 Two routes to base change, and which one is active

There are two distinct ways to express "the pushforward commutes with flat base change" at the level of $\mathcal{O}_{S'}$-modules. This chapter records both, but commits to one as the load-bearing construction.

The canonical-mate route (Stacks-faithful, superseded). One fixes the canonical base-change morphism $\beta : g^*(f_*\mathcal{F}) \to f'_*((g')^*\mathcal{F})$ of Definition 111 — the adjoint mate of the $((g')^*, (g')_*)$-unit — and proves IsIso β directly, affine-locally. This is the route the Stacks Project takes, and it remains the mathematically canonical formulation, but its closure requires a 2-categorical mate-unwinding (identifying the abstract mate, transported through the four affine dictionaries, with the concrete cancellation isomorphism cancelBaseChange) that has no known direct proof. Only the definition of β is retained; the route's obligation lemmas and its two theorem statements were removed as superseded once the concrete-tilde route below closed the downstream consumers.

The concrete-tilde construction (active, consumed downstream). Rather than prove that the *canonical* map is an isomorphism, one *constructs a fresh* natural isomorphism

$$e : g^* \circ f_* \;\cong\; f'_* \circ (g')^*$$

directly, by gluing the affine-local isomorphisms of Lemma 129 (each assembled from the two tilde dictionaries and Mathlib's cancelBaseChange) along the affine basis of S'. This construction **never forms the adjoint mate**, so it entirely sidesteps the mate $\leftrightarrow$ cancelBaseChange coherence obligation of the superseded canonical route: the only remaining content is the naturality of the cancellation chain across affine restrictions of S' (Lemma 141), a concrete localization-and-tensor compatibility rather than a mate identification.

Why the active route suffices. Everything downstream of this chapter — the Čech tensorial base-change isomorphism (the cosimplicial residual e of the Čech chapter) and, through it, the Kleiman-style representability inputs — consumes *only* a base-change isomorphism (a Nonempty / $\cong$ datum) that commutes with the restriction maps, *not* the assertion that the canonical map β is that isomorphism. The fresh e supplies exactly such an isomorphism, and by construction it commutes with restriction (it is glued from restriction-compatible affine pieces). The canonical map pushforwardBaseChangeMap is not a protected declaration, so re-plumbing the downstream consumers to receive e is permitted. Consequently the key constituent of this chapter is Lemma 129 and the gluing chain (§5.5) built on it; the canonical-mate nodes are retained for faithfulness but are off the critical path.

5.3 The canonical base-change map

Definition 111 (Base-change map for the pushforward)**.** *Source: Stacks Project, Cohomology of Schemes, § Cohomology and base change, I.* Let $f : X \to S$ be a morphism of schemes, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$-module, and $g : S' \to S$ an arbitrary morphism, giving the cartesian square above with projection $g' : X' \to X$ and $\mathcal{F}' = (g')^*\mathcal{F}$. The *base-change map for the pushforward* is the canonical morphism of $\mathcal{O}_{S'}$-modules

$$g^*(f_*\mathcal{F}) \longrightarrow f'_*((g')^*\mathcal{F}) = f'_*\mathcal{F}'.$$

It is adjoint to the morphism $f_*\mathcal{F} \to g_*\, f'_*\mathcal{F}' = f_*\,(g')_*\mathcal{F}'$ obtained by applying f_* to the unit $\mathcal{F} \to (g')_*(g')^*\mathcal{F}$ of the $((g')^*, (g')_*)$-adjunction, using the commutativity $g \circ f' = f \circ g'$ of the square.

5.4 The affine case

5.4.1 Locality of isomorphisms for `Scheme.Modules` morphisms

Before treating the affine computation we record three project-local supplements to Mathlib. Mathlib provides the per-open criterion (IsIso $\varphi \iff \forall U,$ IsIso(φ_U)) and a stalkwise isomorphism criterion for TopCat.Sheaf-valued morphisms, but it does not package the stalk-local or basis-local criterion at the level of morphisms of $\mathcal{O}_X$-modules. The following lemmas bridge that gap; they are exactly the locality tools used in the first reduction of the affine base-change lemma below, where one checks the base-change map after restricting to the affine opens of the base. As bespoke infrastructure they carry no external citation and stand on their proof sketches.

Lemma 112 (Stalk-local criterion for isomorphisms of $\mathcal{O}_X$-modules)**.** *Let X be a scheme and $\varphi : M \to N$ a morphism of sheaves of $\mathcal{O}_X$-modules. Then φ is an isomorphism if and only if the underlying morphism of abelian-group presheaves induces an isomorphism on the stalk at every point $x \in X$.*

Proof. If φ is an isomorphism, then so is its image under the forgetful functor to abelian-group presheaves, and the stalk functor at each x, being a functor, carries isomorphisms to isomorphisms. Conversely, assume the stalk map is an isomorphism at every x. Package the underlying abelian-group presheaves of M and N as sheaves of abelian groups, and φ as a morphism of these sheaves; the stalkwise-isomorphism hypothesis lets us apply the TopCat.Sheaf-level stalk-local isomorphism criterion to conclude that this morphism of sheaves of abelian groups is an isomorphism. Pushing it through the forgetful functor to presheaves yields an isomorphism of the underlying abelian-group presheaf morphisms, and since the forgetful functor from $\mathcal{O}_X$-modules to abelian-group presheaves reflects isomorphisms, φ itself is an isomorphism. □

Lemma 113 (Basis-local criterion for isomorphisms of $\mathcal{O}_X$-modules)**.** *Let X be a scheme, let $\{B_i\}_{i\in\iota}$ be a family of opens whose range is a basis of the topology of X, and let $\varphi : M \to N$ be a morphism of sheaves of $\mathcal{O}_X$-modules. If φ restricts to an isomorphism on the sections over every basic open B_i, then φ is an isomorphism.*

Proof. By Lemma 112 it suffices to prove that the induced stalk map is bijective at each point $x \in X$. For injectivity, a germ that vanishes at x already vanishes on some open neighbourhood, which may be shrunk to a basic open B_i; on B_i the section map φ_{B_i} is injective, so the original section is zero, giving injectivity of the stalk map from injectivity over the basis. For surjectivity, an arbitrary germ at x is represented by a section of N over some basic open B_i containing x; since φ_{B_i} is surjective, this section lifts to a section of M over B_i, whose germ at x maps to the given germ. Thus the stalk map is bijective, and φ is an isomorphism. □

Lemma 114 (Affine-open locality criterion for isomorphisms of $\mathcal{O}_X$-modules)**.** *Let X be a scheme and $\varphi : M \to N$ a morphism of sheaves of $\mathcal{O}_X$-modules. Then φ is an isomorphism if and only if it restricts to an isomorphism on the sections over every affine open of X.*

Proof. The forward implication is immediate, since the section functor over a fixed open preserves isomorphisms. For the converse, the affine opens of a scheme form a basis of its topology, so the claim is the special case of Lemma 113 in which the basis is the family of affine opens. □

5.4.2 The affine computation

The whole result rests on the following elementary computation in the affine setting, where the base-change map is visibly an isomorphism because tensoring is associative. Before stating it we isolate the one ingredient not yet available as a ready-made result on which the affine reduction turns: the description of the pushforward of a tilde-module along a morphism of affine schemes.

We first record three bespoke Γ-fragment supports that compute the global sections of an affine pushforward; they carry no external citation and stand on their proof sketches.

Lemma 115 (Naturality of the global-sections comparison square)**.** *Let $\varphi : R \to R'$ be a homomorphism of commutative rings, and write $\mathrm{gs}_R : \Gamma(\operatorname{Spec} R, \top) \cong R$ and $\mathrm{gs}_{R'} : \Gamma(\operatorname{Spec} R', \top) \cong R'$ for the global-sections isomorphisms of the structure sheaves. Then the ring square*

$$\mathrm{gs}_R \circ (\operatorname{Spec}\varphi)^\sharp_\top = \varphi \circ \mathrm{gs}_{R'}$$

commutes, where $(\operatorname{Spec}\varphi)^\sharp_\top$ is the top-open component of the structure-sheaf comparison map underlying $\operatorname{Spec}\varphi$*.*

Proof. This is the naturality of the unit/counit of the $\Gamma \dashv \mathrm{Spec}$ adjunction read off at the top open: the global-sections map of $\mathrm{Spec}\,\varphi$ is, under the global-sections isomorphisms on each side, conjugate to φ itself. Transposing the naturality square of the Spec unit gives exactly the asserted ring equation. ☐

Lemma 116 (Global sections of an affine pushforward)**.** *Let $\varphi : R \to R'$ be a homomorphism of commutative rings and let N be an* $\mathrm{Spec}(R')$*-module. Then there is a canonical isomorphism of R-modules*

$$\Gamma\big((\mathrm{Spec}\,\varphi)_*N\big) \;\cong\; \mathrm{restr}_\varphi(\Gamma N),$$

identifying the R-module of global sections of the pushforward with the restriction of scalars along φ of the R'-module of global sections of N.

Proof. Because $(\mathrm{Spec}\,\varphi)^{-1}(\top) = \top$, the global sections of the pushforward $(\mathrm{Spec}\,\varphi)_*N$ are, on underlying abelian groups, the global sections of N with no transport along the open-set identification. The two sides therefore carry the same underlying abelian group; both unwind to nested restriction-of-scalars towers along the structure-sheaf global-sections maps. These towers are reconciled by applying the restriction-of-scalars composition identity twice and substituting the ring equation of Lemma 115, which shows the resulting R-action is exactly restriction of scalars along φ. ☐

Lemma 117 (Global sections of an affine pushforward of a tilde-module)**.** *Let $\varphi : R \to R'$ be a homomorphism of commutative rings and let M be an R'-module. Then there is a canonical isomorphism of R-modules*

$$\Gamma\big((\mathrm{Spec}\,\varphi)_*\widetilde{M}\big) \;\cong\; \mathrm{restr}_\varphi\, M.$$

Proof. Specialise Lemma 116 to $N = \widetilde{M}$ and compose with the tilde-unit isomorphism $\Gamma(\widetilde{M}) \cong M$ of R'-modules (which restricts to an isomorphism of R-modules after applying restr_φ). ☐

Lemma 118 (Sections of an affine pushforward over an arbitrary open)**.** *Let $\varphi : R \to R'$ be a homomorphism of commutative rings, let N be an* $\mathrm{Spec}(R')$*-module, and let U be an open of* $\mathrm{Spec}(R)$*, with preimage $V = (\mathrm{Spec}\,\varphi)^{-1}(U)$ in* $\mathrm{Spec}(R')$*. Then there is a canonical isomorphism of R-modules*

$$\Gamma\big((\mathrm{Spec}\,\varphi)_*N,\, U\big) \;\cong\; \mathrm{restr}_\varphi\big(\Gamma(N,\, V)\big).$$

This is the open-indexed generalization of Lemma 116, which is the case $U = \top$ (so that $V = \top$ as well). The family $\{e_U\}_U$ of these isomorphisms is moreover natural in the open argument: the structure-sheaf restriction maps on the two sides commute with the isomorphisms, so the family is a natural isomorphism of the two presheaves of R-modules in the open variable (this naturality is established in the proof below). For $U = D(a)$ it is exactly the linear equivalence $e_{D(a)}$ of movement (1) in the proof of Lemma 123.

Proof. The construction copies that of Lemma 116 verbatim, with the top open $\top$ replaced throughout by the arbitrary open U (and its preimage V). The point that makes this replacement free is that the R-module structure on the sections is supplied *uniformly* by restriction of scalars along the global-sections ring map at the top open, independently of the evaluation open; consequently both sides unwind to the same nested restriction-of-scalars towers, reconciled by applying the restriction-of-scalars composition identity twice and substituting the $\top$-level ring equation of Lemma 115. Naturality in the open follows because every constituent of the construction is either a structure-sheaf restriction map or conjugation by a fixed ring map, and all of these commute with further restriction along an inclusion of opens.

Naturality of the family $\{e_U\}_U$ in the open (remark). The family is in fact a natural isomorphism between two presheaves of R-modules on $\mathrm{Spec}(R)$: the source presheaf $U \mapsto \Gamma((\mathrm{Spec}\,\varphi)_* N, U)$, with the structure-sheaf restriction maps of the pushforward, and the target presheaf $U \mapsto \mathrm{restr}_\varphi(\Gamma(N, (\mathrm{Spec}\,\varphi)^{-1}U))$, with the structure-sheaf restriction maps of N transported through the preimage operation $(\mathrm{Spec}\,\varphi)^{-1}$ and read as R-linear maps by restriction of scalars along φ. Concretely, for any inclusion of opens $U' \subseteq U$, with $V = (\mathrm{Spec}\,\varphi)^{-1}U$ and $V' = (\mathrm{Spec}\,\varphi)^{-1}U'$, the square with horizontal maps $e_U, e_{U'}$ and vertical maps the respective structure-sheaf restrictions commutes. The reason is structural: each e_U is a composite whose constituents are of exactly two kinds — structure-sheaf restriction maps (of N, respectively of $(\mathrm{Spec}\,\varphi)_* N$) and identity-on-carrier restrictScalars repackagings (conjugation by the *fixed*, open-independent ring map supplied by the restriction-of-scalars reconciliation and by the $\top$-level ring equation of Lemma 115). Restriction maps commute with further restriction by the presheaf functoriality axiom (using $V' = (\mathrm{Spec}\,\varphi)^{-1}U' \subseteq (\mathrm{Spec}\,\varphi)^{-1}U = V$, as $(\mathrm{Spec}\,\varphi)^{-1}$ is monotone), and conjugation by a fixed ring map commutes with restriction because the ring map does not depend on U; the naturality square is the paste of these component squares, and each e_U being an isomorphism makes the family a natural isomorphism. This open-naturality is exactly the compatibility invoked, at the inclusion $D(a) \subseteq \top$, in the proof of Lemma 123. ☐

Lemma 119 (The tilde restriction from the top open to a basic open is a localization)**.** *Let M be an R'-module and $b \in R'$. Then the structure-sheaf restriction map*

$$\Gamma(\widetilde{M}, \top) \longrightarrow \Gamma(\widetilde{M}, D(b)),$$

read as a map of R'-modules, exhibits $\Gamma(\widetilde{M}, D(b))$ as the localization of $\Gamma(\widetilde{M}, \top)$ at $\mathrm{powers}(b)$*. This is the R'-side localization fact transported in movement (3) of the proof of Lemma 123.*

Proof. By the defining property of $\widetilde{(-)}$, the tilde-localization map $M \to \Gamma(\widetilde{M}, D(b))$ exhibits its target as the localization of M at $\mathrm{powers}(b)$. The global-sections map $M \to \Gamma(\widetilde{M}, \top)$ is the localization of M at the trivial submonoid $\mathrm{powers}(1)$ — since $D(1) = \top$ — and is therefore bijective. The triangle identity relating these two localization maps to the structure-sheaf restriction $\Gamma(\widetilde{M}, \top) \to \Gamma(\widetilde{M}, D(b))$ then exhibits the restriction itself, post-composed with the inverse of the bijection $M \cong \Gamma(\widetilde{M}, \top)$, as the localization of $\Gamma(\widetilde{M}, \top)$ at $\mathrm{powers}(b)$. ☐

We record three further bespoke supports that drive the basic-open verification of the affine-pushforward isomorphism below. They carry no external citation and stand on their proof sketches.

Lemma 120 (Restriction of scalars of a localized module)**.** *Let A be a commutative R-algebra, $S \subseteq R$ a submonoid, and $f : M \to N$ an A-linear map that exhibits N as the localization of M at the image submonoid* $\mathrm{algMap}_A(S) \subseteq A$*. Then the underlying R-linear map* $\mathrm{restr}_R\, f$ *exhibits N as the localization of M at S itself.*

Proof. This is the exact converse of the standard fact that a localization of M at a submonoid S of R remains a localization after enlarging the base to an R-algebra A along the image submonoid $\mathrm{algMap}_A(S)$. One checks the three defining conditions of a localized module directly. Each $s \in S$ acts invertibly on N because its image $\mathrm{algMap}_A(s)$ does and the two actions coincide by the scalar tower $R \to A \to N$. Surjectivity of f up to denominators and the equality-witness condition transport along the map $s \mapsto \mathrm{algMap}_A(s)$ from S onto $\mathrm{algMap}_A(S)$, again using $\mathrm{algMap}_A(s) \cdot = s \cdot$. In the application $A = R'$, $S = \mathrm{powers}(a)$, one has $\mathrm{algMap}_{R'}(\mathrm{powers}(a)) = \mathrm{powers}(\varphi a)$, so this identifies the localization of $\mathrm{restr}_\varphi M$ at a with the localization of M at φa; it is the ring-change core of the basic-open computation below. ☐

Lemma 121 (Section-level counit isomorphism from a localization hypothesis)**.** *Let N be an* $\operatorname{Spec}(R)$*-module and $a \in R$. Suppose the structure-sheaf restriction map $\Gamma(N, \top) \to \Gamma(N, D(a))$, read as an R-linear map of the global-section modules, exhibits $\Gamma(N, D(a))$ as the localization of $\Gamma(N, \top)$ at* powers(a)*. Then the tilde–Γ counit* fromTilde$\Gamma\, N : \widetilde{\Gamma(N, \top)} \to N$ *restricts to an isomorphism on the sections over the basic open $D(a)$.*

Proof. Over $D(a)$ the section map L of the counit sits in the triangle identity $L \circ j = \rho$, where $j : \Gamma(N, \top) \to \Gamma(\widetilde{\Gamma(N, \top)}, D(a))$ is the tilde-localization map and $\rho : \Gamma(N, \top) \to \Gamma(N, D(a))$ is the structure-sheaf restriction. Both j and ρ exhibit their targets as the localization of $\Gamma(N, \top)$ at powers(a) — j by the defining property of $\widetilde{(-)}$, and ρ by hypothesis. By the uniqueness of localized modules there is a unique R-linear isomorphism e with $e \circ j = \rho$; the triangle identity then forces $L = e$, so L is an isomorphism. (This is the per-basic-open input feeding the basis-local criterion Lemma 113.) ∎

Lemma 122 (Affine pushforward of a tilde-module, conditional form)**.** *Let $\varphi : R \to R'$ be a homomorphism of commutative rings and M an R'-module. Suppose that for every $a \in R$ the structure-sheaf restriction of the pushforward $N := (\operatorname{Spec}\varphi)_* \widetilde{M}$ exhibits its sections over $D(a)$ as the localization of its global sections at* powers(a)*. Then there is a canonical isomorphism of* $\operatorname{Spec}(R)$*-modules*

$$(\operatorname{Spec}\varphi)_* \widetilde{M} \;\cong\; \widetilde{(\operatorname{restr}_\varphi M)}.$$

Proof. Write $N := (\operatorname{Spec}\varphi)_* \widetilde{M}$. Under the hypothesis, Lemma 121 applies for each $a \in R$, so the counit fromTilde$\Gamma\, N$ restricts to an isomorphism on the sections over every basic open $D(a)$. Since the basic opens form a basis of $\operatorname{Spec}(R)$, the basis-local criterion Lemma 113 upgrades this to: the counit fromTilde$\Gamma\, N$ is an isomorphism, i.e. N lies in the essential image of $\widetilde{(-)}$. Composing the inverse counit with the tilde of the global-sections comparison Lemma 117 yields the asserted object isomorphism. ∎

Lemma 123 (Affine pushforward of a tilde-module)**.** *Let $\varphi : R \to R'$ be a homomorphism of commutative rings and let M be an R'-module. Then there is a canonical isomorphism of* $\operatorname{Spec}(R)$*-modules*

$$(\operatorname{Spec}\varphi)_* \widetilde{M} \;\cong\; \widetilde{(\operatorname{restr}_\varphi M)},$$

where $\operatorname{Spec}\varphi : \operatorname{Spec}(R') \to \operatorname{Spec}(R)$ is the induced morphism of affine schemes, $\widetilde{(-)}$ denotes the quasi-coherent sheaf associated to a module, and $\operatorname{restr}_\varphi M$ is the R-module obtained from M by restriction of scalars along φ. That is, pushing the quasi-coherent sheaf $\widetilde{M}$ forward along $\operatorname{Spec}\varphi$ is, up to this canonical identification, the tilde of the restriction of scalars of M.

This is the classical fact that the direct image of a quasi-coherent sheaf along a morphism of affine schemes is computed on modules by restriction of scalars; it is the affine companion of the pullback formula (the affine case of the pushforward analogue of Stacks "$\widetilde{M}$ and pullback"). It is recorded here as bespoke project infrastructure because the affine-pushforward identification does not appear as a ready-made result in the existing quasi-coherent sheaf theory.

Proof. We build the isomorphism *directly on a basis of basic opens*. This is the non-circular route: checking the comparison map on basic opens yields the object isomorphism, and quasi-coherence of the pushforward is then a corollary, not a prerequisite. (One must not reverse this order: the global-sections comparison $\Gamma((\operatorname{Spec}\varphi)_* \widetilde{M}) \cong \operatorname{restr}_\varphi M$ of Lemma 117 alone would only upgrade to an object isomorphism once the pushforward is already known to be quasi-coherent, so quasi-coherence cannot be deduced from an object isomorphism built that way.)

The comparison morphism. There is a canonical morphism of Spec(R)-modules

$$\alpha : \widetilde{(\mathrm{restr}_\varphi M)} \longrightarrow (\operatorname{Spec}\varphi)_* \widetilde{M},$$

namely the tilde-adjoint of the global-sections identification of Lemma 117. We show α is an isomorphism.

Reduction to basic opens. The basic opens $D(a)$, for $a \in R$, form a basis of the topology of Spec(R). By the basis-local criterion Lemma 113 it suffices to check that α restricts to an isomorphism on the sections over each basic open $D(a)$.

The computation over $D(a)$. On the source, the sections of $\widetilde{(\mathrm{restr}_\varphi M)}$ over $D(a)$ are the localization $(\mathrm{restr}_\varphi M)[1/a]$ as an $R[1/a]$-module. On the target, since $(\operatorname{Spec}\varphi)^{-1}D(a) = D(\varphi a)$, the sections of $(\operatorname{Spec}\varphi)_* \widetilde{M}$ over $D(a)$ are the sections of $\widetilde{M}$ over $D(\varphi a)$, i.e. the localization $M[1/\varphi a]$, regarded as an $R[1/a]$-module by restriction of scalars along the induced map $R[1/a] \to R'[1/\varphi a]$. These two localizations agree: the element a acts on $\mathrm{restr}_\varphi M$ through φa, so localizing $\mathrm{restr}_\varphi M$ at the multiplicative set generated by a is the same module as localizing M at the multiplicative set generated by φa. Both are the universal localization of M inverting φa, so the comparison map α is an isomorphism on the sections over $D(a)$. As this holds for every basic open, α is an isomorphism.

The explicit formalization route over $D(a)$. The agreement of the two localizations just described is the entire content, but it must be realized through the global ring R, because the R-action on the pushforward sections over $D(a)$ is built from the global-sections module of $N := (\operatorname{Spec}\varphi)_*\widetilde{M}$ rather than directly from $R[1/a]$. The reduction therefore proceeds exactly as the conditional assembly Lemma 122 prescribes: it suffices to supply, for each $a \in R$, the hypothesis hloc(a) that the structure-sheaf restriction $\Gamma(N, \top) \to \Gamma(N, D(a))$ exhibits $\Gamma(N, D(a))$ as the localization of $\Gamma(N, \top)$ at powers(a). The R-action on $\Gamma(N, D(a))$ is the honest restriction-of-scalars action carried by the scalar tower $R \to R' \to -$ determined by φ; against that structure the ring-change converse Lemma 120 is applicable, so the carrier of the R-module is exactly the one to which the localization claim refers, and no recursively-defined or ad-hoc section-level scalar action intervenes. We obtain hloc(a) by transporting the already-known R'-side localization (Lemma 119), in element-free fashion, across the open-indexed section comparison of Lemma 118 *using its naturality in the open* (the naturality remark in the proof of Lemma 118). This proceeds in three movements.

(1) The $D(a)$-level linear equivalence. This is the component at $U = D(a)$ of the open-indexed family of Lemma 118: the R-linear isomorphism

$$e_{D(a)} : \Gamma((\operatorname{Spec}\varphi)_*\widetilde{M},\, D(a)) \;\cong\; \mathrm{restr}_\varphi(\Gamma(\widetilde{M},\, D(\varphi a))),$$

using $(\operatorname{Spec}\varphi)^{-1}D(a) = D(\varphi a)$, which is definitional. No fresh $D(a)$-level construction is needed: this is precisely the open-indexed isomorphism already supplied by Lemma 118 evaluated at the basic open $D(a)$.

(2) The naturality square for $D(a) \subseteq \top$. The compatibility that relates $e_{D(a)}$ to the $\top$-level isomorphism $e_\top$ (the global-sections comparison of Lemma 116) is precisely the open-naturality square of the family $\{e_U\}_U$ (established in the proof of Lemma 118), instantiated at the inclusion $D(a) \subseteq \top$: the square

$$\begin{array}{ccc}
\Gamma((\operatorname{Spec}\varphi)_*\widetilde{M},\, \top) & \xrightarrow{e_\top} & \mathrm{restr}_\varphi(\Gamma(\widetilde{M},\, \top)) \\
{\scriptstyle \mathrm{res}_{\top\to D(a)}}\downarrow & & \downarrow{\scriptstyle \mathrm{res}_{\top\to D(\varphi a)}} \\
\Gamma((\operatorname{Spec}\varphi)_*\widetilde{M},\, D(a)) & \xrightarrow{e_{D(a)}} & \mathrm{restr}_\varphi(\Gamma(\widetilde{M},\, D(\varphi a)))
\end{array}$$

commutes, using $(\operatorname{Spec}\varphi)^{-1}D(a) = D(\varphi a)$, which is definitional. Because the family $\{e_U\}_U$ is a natural isomorphism, this single square is available uniformly in a; there is no per-a section-level identity to re-prove by hand. It intertwines the source restriction $\mathrm{res}_{\top\to D(a)}$ on the pushforward with the (restriction-of-scalars reading of the) target restriction $\mathrm{res}_{\top\to D(\varphi a)}$ on $\widetilde{M}$.

(3) Discharging $\mathrm{hloc}(a)$ *by transport.* The target restriction $\mathrm{res}_{\top\to D(\varphi a)} : \Gamma(\widetilde{M},\top) \to \Gamma(\widetilde{M}, D(\varphi a))$ exhibits $\Gamma(\widetilde{M}, D(\varphi a)) = M[1/\varphi a]$ as the powers(φa)-localization of $M = \Gamma(\widetilde{M},\top)$ over R'; this is exactly Lemma 119. Apply the ring-change converse Lemma 120 (with R-algebra $A = R'$ and submonoid powers(a), so that $\mathrm{algMap}_{R'}(\mathrm{powers}(a)) = \mathrm{powers}(\varphi a)$ — its $\mathrm{Algebra}\,R\,R'$ and scalar-tower hypotheses being supplied by the restriction-of-scalars structure along φ — to read this same map as the powers(a)-localization over R. The commuting naturality square of movement (2) intertwines this target restriction with the source restriction $\mathrm{res}_{\top\to D(a)} : \Gamma(N,\top) \to \Gamma(N, D(a))$ by way of the isomorphisms $e_\top$ and $e_{D(a)}$; transporting the localization property across that square — via the transport-along-an-equivalence principle for localized modules — yields that $\mathrm{res}_{\top\to D(a)}$ is the powers(a)-localization on $\Gamma(N, D(a))$ over R, i.e. $\mathrm{hloc}(a)$. No section-level computation specific to the individual a is performed: the only a-dependent input is the instantiation of the single natural-isomorphism square at $D(a)$. Feeding the family $\{\mathrm{hloc}(a)\}_{a\in R}$ to the conditional assembly Lemma 122 produces the unconditional object isomorphism α.

Naturality in the open drives the transport. The comparison assembled here, $(\operatorname{Spec}\varphi)_* \circ \widetilde{(-)} \cong \widetilde{(-)} \circ \mathrm{restr}_\varphi$, is natural in *both* arguments — in the module M and in the open U. The naturality in the module is the usual functoriality; the naturality in the open is the naturality remark in the proof of Lemma 118: the family $\{e_U\}_U$ of Lemma 118 commutes with the structure-sheaf restriction maps on both sides. It is precisely this open-naturality that drives the localization transport *uniformly*: the per-a localization fact $\mathrm{hloc}(a)$ follows from the $\top$-level localization Lemma 119 together with that naturality square for the inclusion $D(a) \hookrightarrow \top$, rather than from a section-level naturality square re-proved by hand for each individual a. This is the single remaining obligation of the construction; the carrier of the R-action on the pushforward sections is the restriction-of-scalars structure along φ, against which Lemma 120 applies directly.

The R-module structure on the R'-side sections to which the ring-change converse Lemma 120 is applied is the honest restriction-of-scalars structure: R acts through φ, i.e. via the R-algebra structure on R' determined by φ and the associated scalar tower $R \to R' \to M[1/\varphi a]$. It is *not* an ad-hoc, recursively-defined scalar action. Lemma 120 is stated against exactly this restriction-of-scalars / scalar-tower structure, and it is under that structure that localizing $\mathrm{restr}_\varphi M$ at a coincides with localizing M at φa.

Quasi-coherence as a corollary. The object isomorphism just constructed exhibits $(\operatorname{Spec}\varphi)_*\widetilde{M} \cong \widetilde{(\mathrm{restr}_\varphi M)}$ as the tilde of a module. Tilde modules are quasi-coherent and quasi-coherence of $\operatorname{Spec}(R)$-modules is closed under isomorphism, so the pushforward $(\operatorname{Spec}\varphi)_*\widetilde{M}$ is quasi-coherent. This conclusion is stated *after* the object isomorphism, of which it is a consequence — no separate "pushforward preserves quasi-coherent" input is needed for the affine lane. □

Remark 124. This single isomorphism discharges both inputs that the affine reduction below requires of the pushforward of a tilde-module.

1. *The global-sections comparison.* Applying the global-sections (module-of-sections) functor to the isomorphism identifies the R-module of sections of $(\operatorname{Spec}\varphi)_*\widetilde{M}$ with $\mathrm{restr}_\varphi M$; this is exactly the Γ-fragment comparison used to recognise the section-level base-change map.

2. *Quasi-coherence of the pushforward.* The tilde of any module is quasi-coherent, and quasi-coherence of $\mathrm{Spec}(R)$-modules is closed under isomorphism; hence $(\mathrm{Spec}\,\varphi)_* \widetilde{M}$ is quasi-coherent. In particular no separate "pushforward preserves quasi-coherent" theorem is needed for the affine lane.

Because both inputs the affine reduction asks of the pushforward are corollaries of this single object isomorphism — the section-level module identification and the quasi-coherence of the pushforward — this lemma is the sole bespoke ingredient the affine lane requires.

5.4.3 The pullback companion

The affine reduction below requires, alongside the pushforward dictionary, the *pullback* companion: the affine description of the inverse image of a tilde-module. It is the affine case of the classical fact that the pullback of a quasi-coherent sheaf is computed on modules by base change of scalars.

The pullback companion is built by uniqueness of left adjoints, which is driven by the following packaging of the per-object Γ-fragment comparison as a natural isomorphism of right-adjoint functors. It is bespoke project infrastructure and carries no external citation.

Lemma 125 (The pushforward–global-sections natural isomorphism)**.** *Let $\varphi : R \to R'$ be a homomorphism of commutative rings. Then the per-object global-sections comparison of Lemma 116 assembles into a natural isomorphism of functors* $(\mathrm{Spec}\,R').\mathrm{Modules} \to \mathrm{ModuleCat}(R)$

$$(\mathrm{Spec}\,\varphi)_* \circ \Gamma_R \;\cong\; \Gamma_{R'} \circ \mathrm{restr}_\varphi,$$

whose component at an $\mathrm{Spec}(R')$*-module N is the isomorphism* $\Gamma((\mathrm{Spec}\,\varphi)_* N) \cong \mathrm{restr}_\varphi(\Gamma N)$ *of Lemma 116.*

Proof. The components are the isomorphisms supplied by Lemma 116. Naturality in the module argument N holds on the nose: every constituent of the comparison (the restriction-of-scalars composition repackagings and the restriction-of-scalars congruence) is the identity on underlying elements, merely re-presenting the module structure on an unchanged carrier, so each naturality square is a pointwise identity. This is the right-adjoint natural isomorphism that drives the uniqueness-of-left-adjoints argument for the affine pullback dictionary Lemma 126: identifying the two right adjoints $(\mathrm{Spec}\,\varphi)_* \circ \Gamma_R$ and $\Gamma_{R'} \circ \mathrm{restr}_\varphi$ by this isomorphism forces the corresponding left adjoints to be isomorphic. ∎

Lemma 126 (Affine pullback of a tilde-module)**.** *Source: Stacks Project, Schemes, Lemma "$\widetilde{M}$ and pullback" (Tag 01I9), part (1). Let $\varphi : R \to R'$ be a homomorphism of commutative rings and let M be an R-module. Then there is a canonical isomorphism of* $\mathrm{Spec}(R')$*-modules*

$$(\mathrm{Spec}\,\varphi)^* \widetilde{M} \;\cong\; (\widetilde{R' \otimes_R M}),$$

where $\mathrm{Spec}\,\varphi : \mathrm{Spec}(R') \to \mathrm{Spec}(R)$ *is the induced morphism of affine schemes, $\widetilde{(-)}$ denotes the quasi-coherent sheaf associated to a module, and $R' \otimes_R M$ is the base change of M along φ — the R'-module obtained by extension of scalars. The isomorphism is natural in M. That is, pulling the quasi-coherent sheaf $\widetilde{M}$ back along* $\mathrm{Spec}\,\varphi$ *is, up to this canonical identification, the tilde of the base change of M. It is the pullback companion of the affine pushforward dictionary Lemma 123, and is part (1) of the same Stacks lemma of which the pushforward formula is part (2).*

Proof. Pullback along $\operatorname{Spec}\varphi$ is, on quasi-coherent sheaves, the extension-of-scalars operation $-\otimes_R R'$ on modules transported through the $\widetilde{(-)}$ / moduleSpec Γ equivalence between modules and quasi-coherent sheaves. Concretely, the global sections of $(\operatorname{Spec}\varphi)^*\widetilde{M}$ are the base change $R'\otimes_R M$: the pullback functor $(\operatorname{Spec}\varphi)^*$ is left adjoint to the pushforward $(\operatorname{Spec}\varphi)_*$, and under the global-sections identification the latter is restriction of scalars along φ (Lemma 123); the left adjoint of restriction of scalars is extension of scalars $M \mapsto R'\otimes_R M$. The comparison map is the unit of the base-change adjunction, and the universal property of base change identifies it with the canonical isomorphism $(\operatorname{Spec}\varphi)^*\widetilde{M} \cong (\widetilde{R'\otimes_R M})$. Naturality in M is the naturality of that unit. (One may equally read this off the standard-open description: on $D(\varphi a)$ the localization $(R'\otimes_R M)[1/a]$ is $R'[1/\varphi a]\otimes_{R[1/a]} M[1/a]$, which is the base change of the sections of $\widetilde{M}$ over $D(a)$.) ☐

5.4.4 The concrete-tilde affine base-change brick (active route)

The key affine input for this construction is the following isomorphism. It is assembled entirely from the two tilde dictionaries (Lemmas 123 and 126) and Mathlib's cancellation isomorphism, and — crucially — it never forms the adjoint mate of Definition 111, so it does not re-incur the mate $\leftrightarrow$ cancelBaseChange obligation of the superseded canonical route.

Lemma 127 (Cancellation isomorphism for iterated base change)**.** *Provided by Mathlib. Let $R \to R'$ and $R \to A$ be commutative-ring homomorphisms and M an A-module. There is a canonical isomorphism of R'-modules*

$$\mathrm{cancelBaseChange} : (R'\otimes_R A)\otimes_A M \xrightarrow{\sim} R'\otimes_R M, \qquad (r'\otimes a)\otimes m \mapsto r'\otimes(a\cdot m),$$

with no flatness hypothesis whatsoever.

Lemma 128 (Module-level base-change cancellation iso)**.** *Let $\varphi : R \to A$, $\psi : R \to R'$, $\rho : A \to B$, $\sigma : R' \to B$ be commutative-ring homomorphisms forming a pushout square in* CommRing *(so that B is, up to canonical isomorphism, the tensor product $A\otimes_R R'$), and let M be an A-module. Then there is a canonical isomorphism of R'-modules*

$$\mathrm{extendScalars}_\psi(\mathrm{restr}_\varphi M) \;\cong\; \mathrm{restr}_\sigma(\mathrm{extendScalars}_\rho M),$$

i.e. $R'\otimes_R M \cong \mathrm{restr}_\sigma(B\otimes_A M)$, realising the inverse $\mathrm{cancelBaseChange}^{-1}$ *of Lemma 127 as a map between the extension-of-scalars objects attached to the two legs of the square.*

Proof. The pushout hypothesis exhibits (R, R', A, B) as a commutative-algebra pushout. For such a square the cancellation isomorphism Lemma 127 is exactly the R'-linear isomorphism $B\otimes_A M \;\cong\; R'\otimes_R M$; its inverse, read through the definitional identifications $R'\otimes_R M = \mathrm{extendScalars}_\psi(\mathrm{restr}_\varphi M)$ and $B\otimes_A M = \mathrm{extendScalars}_\rho M$, is the displayed map. It is an isomorphism with no flatness hypothesis. ☐

Lemma 129 (Affine termwise base change, abstract-B framing)**.** *Source: Stacks Project, Cohomology of Schemes, Tag 02KG (`lemma-affine-base-change`). Let $\varphi : R \to A$, $\psi : R \to R'$, $\rho : A \to B$, $\sigma : R' \to B$ be commutative-ring homomorphisms forming a pushout square in* CommRing*,*

$$\begin{array}{ccc} R & \xrightarrow{\varphi} & A \\ {\scriptstyle\psi}\downarrow & & \downarrow{\scriptstyle\rho} \\ R' & \xrightarrow{\sigma} & B \end{array}$$

equivalently — applying Spec*, which carries a pushout of rings to a cartesian (pullback) square of affine schemes — the cartesian square*

$$\begin{array}{ccc} \operatorname{Spec} B & \xrightarrow{\operatorname{Spec}\rho} & \operatorname{Spec} A \\ {\scriptstyle \operatorname{Spec}\sigma}\big\downarrow & & \big\downarrow{\scriptstyle \operatorname{Spec}\varphi} \\ \operatorname{Spec} R' & \xrightarrow{\operatorname{Spec}\psi} & \operatorname{Spec} R. \end{array}$$

Write $S = \operatorname{Spec} R$*,* $S' = \operatorname{Spec} R'$*,* $X = \operatorname{Spec} A$*,* $X' = \operatorname{Spec} B$*, with* $f = \operatorname{Spec}\varphi$*,* $g = \operatorname{Spec}\psi$*,* $f' = \operatorname{Spec}\sigma$*,* $g' = \operatorname{Spec}\rho$*; let* M *be an* A*-module and* $\mathcal{F} = \widetilde{M}$*. Then there is an isomorphism of* $\operatorname{Spec}(R')$*-modules*

$$g^*(f_*\widetilde{M}) \;\cong\; f'_*((g')^*\widetilde{M}),$$

natural in the module M*, built directly from the tilde primitives and the cancellation isomorphism, and* never *identified with the canonical adjoint-mate base-change map.*

Proof. The two sides are transported to modules through the affine dictionaries and then compared by the cancellation isomorphism, in three composable moves; each move is an isomorphism, so the composite is.

(1) The source $g^*(f_*\widetilde{M})$*.* The pushforward dictionary Lemma 123 identifies $f_*\widetilde{M}$ over $\operatorname{Spec}(R)$ with $\widetilde{\operatorname{restr}_\varphi M}$. The pullback dictionary Lemma 126 then identifies its pullback $g^*(\widetilde{\operatorname{restr}_\varphi M})$ with $(\widetilde{R' \otimes_R M})$, the tilde of the R'-module $\operatorname{extendScalars}_\psi(\operatorname{restr}_\varphi M) = R' \otimes_R M$.

(2) The target $f'_*((g')^*\widetilde{M})$*.* The pullback dictionary Lemma 126 (applied to $g' = \operatorname{Spec}\rho$) identifies $(g')^*\widetilde{M}$ with $(\widetilde{\operatorname{extendScalars}_\rho M}) = (\widetilde{B \otimes_A M})$. The pushforward dictionary Lemma 123 (applied to $f' = \operatorname{Spec}\sigma$) identifies the pushforward of this tilde with $(\operatorname{restr}_\sigma\widetilde{(B \otimes_A M)})$, the tilde of $B \otimes_A M$ read as an R'-module by restriction of scalars along σ.

(3) Cancellation. The module-level core Lemma 128 supplies the R'-linear isomorphism $\operatorname{extendScalars}_\psi(\operatorname{restr}_\varphi M) \cong \operatorname{restr}_\sigma(\operatorname{extendScalars}_\rho M)$, i.e. $R' \otimes_R M \cong \operatorname{restr}_\sigma(B \otimes_A M)$; it is the realization of $\operatorname{cancelBaseChange}^{-1}$ (Lemma 127) for the square. Applying $\widetilde{(-)}$ and composing the inverse of (1), this iso, and (2) yields the asserted isomorphism $g^*(f_*\widetilde{M}) \cong f'_*((g')^*\widetilde{M})$.

Realization as a chain of isomorphisms. The three moves are assembled as a five-step chain of isomorphisms of $\operatorname{Spec}(R')$-modules

$$g^*(f_*\widetilde{M}) \xrightarrow{\sim} (\widetilde{R' \otimes_R M}) \xrightarrow{\widetilde{(\cdot)}\text{ of core}} (\operatorname{restr}_\sigma\widetilde{(B \otimes_A M)}) \xrightarrow{\sim} f'_*((g')^*\widetilde{M}),$$

in which the first arrow is move (1) (the pushforward dictionary Lemma 123 followed by the pullback dictionary Lemma 126), the middle arrow is $\widetilde{(-)}$ applied to the module-level core Lemma 128, and the last arrow is the inverse of move (2) (the pullback then pushforward dictionaries for g' and f'). The only step carrying mathematical content beyond the tilde dictionaries is the middle one: keeping B abstract ensures the cancellation Lemma 128 applies from the pushout hypothesis directly, with no adjoint mate formed.

Naturality in M is the naturality of each constituent: the two tilde dictionaries are natural in the module argument, and cancelBaseChange (hence the module core Lemma 128) is natural in M by Mathlib. No adjoint mate is formed at any point, so the mate $\leftrightarrow$ cancelBaseChange coherence obligation of the canonical route never arises. $\square$

5.5 Gluing the affine base-change isomorphisms to a global natural iso

The affine brick Lemma 129 produces, over each affine open $\mathrm{Spec}(R') \subseteq S'$ (with a chosen affine $\mathrm{Spec}(R) \subseteq S$ over its g-image), an isomorphism $e_{R'}$. We glue these into the global natural isomorphism $e : g^* \circ f_* \cong f'_* \circ (g')^*$ of the active route. The construction mirrors the worked sheaf-of-modules descent already carried out in the project's line-bundle dual, and proceeds in four steps: an affine-restriction naturality lemma (the one genuinely new statement), a gluing to a sheaf of abelian groups, the re-imposition of $\mathcal{O}_{S'}$-linearity, and the upgrade to an isomorphism packaged in $\mathcal{F}$. It rests on the following Mathlib anchors.

Lemma 130 (Morphism into a sheaf from its restriction to a basis)**.** *Provided by Mathlib. Let F be a presheaf and F' a sheaf (valued in a category) on a space X, and let $\{B_i\}_i$ be opens whose range is a basis of the topology of X. Then restriction along the basis is a bijection*

$$\left(B^{\mathrm{op}*}F \longrightarrow B^{\mathrm{op}*}F'\right) \;\simeq\; \left(F \longrightarrow F'\right),$$

so a morphism into the sheaf F' is determined by, and constructible from, a compatible family of restrictions to the basis opens (with extend_hom_app *recovering the component at each B_i).*

Lemma 131 (Re-imposing module linearity on a sheaf morphism)**.** *Provided by Mathlib. A morphism φ of the underlying abelian-group presheaves of two presheaves of R-modules M_1, M_2, satisfying the sectionwise linearity equation $\varphi(r \cdot m) = r \cdot \varphi(m)$ for all opens, all scalars $r \in R(U)$ and all sections m, assembles into a morphism $M_1 \to M_2$ of presheaves of R-modules.*

Lemma 132 (Packaging components into a natural isomorphism)**.** *Provided by Mathlib. A family of isomorphisms $\{\eta_X : G_1(X) \cong G_2(X)\}_X$, natural in X, assembles into a natural isomorphism $G_1 \cong G_2$ of functors.*

Lemma 133 (Compatibility of cancellation with localizing the outer base)**.** *Source: bespoke project infrastructure; no external citation. Let $R \to S \to S'$ be a tower of commutative-ring homomorphisms ($S \to S'$* arbitrary *— no localization hypothesis), let $R \to A$ be a further homomorphism, and let M be an A-module. Realizing the pushouts as the concrete tensor products $A \otimes_R S$ and $A \otimes_R S'$ (A the first factor), the cancellation isomorphisms* cancelBaseChange *(Lemma 127) for the S-tower and the S'-tower are intertwined by the outer base-change $S \to S'$: the square*

$$\begin{array}{ccc}
(A \otimes_R S) \otimes_A M & \xrightarrow{\text{cancelBaseChange}_S} & S \otimes_R M \\
\downarrow{\scriptstyle \text{rTensor}\,(\mathrm{id}_A \otimes (S \to S'))} & & \downarrow{\scriptstyle \text{rTensor}\,(S \to S')} \\
(A \otimes_R S') \otimes_A M & \xrightarrow{\text{cancelBaseChange}_{S'}} & S' \otimes_R M
\end{array}$$

commutes. Equivalently cancelBaseChange *commutes with changing the outer base $S \to S'$. This is pure* ModuleCat*-level commutative algebra, carrying no scheme scaffolding; the localization instance $S' = S[1/b]$ (with the standard* `Algebra`/`IsScalarTower` *data) is the case consumed by the affine-restriction naturality square (Lemma 141).*

Proof. Both cancelBaseChange maps are given on simple tensors by the shared formula $(a \otimes s) \otimes m \mapsto s \otimes (a \cdot m)$ (Lemma 127). The square is checked pointwise by a double `TensorProduct.induction_on` (outer over $(A \otimes_R S) \otimes_A M$, inner over $A \otimes_R S$); the `tmul` leaf reduces, on both routes around the square, to $(\text{algebraMap}\, S\, S')(s) \otimes (a \cdot m)$ by the simple-tensor formula together with rTensor functoriality and the A-algebra map $\mathrm{id}_A \otimes (S \to S')$. No `IsLocalizedModule` datum or universal property of localized modules is required: the identity holds for the *arbitrary* ring map

$S \to S'$. (This is *stronger* than the localization-only framing originally planned, and removes `tildeRestriction_isLocalizedModule` from this lemma's cone.) □

Lemma 134 (Cancellation–base-change compatibility, inverse form)**.** *Source: bespoke project infrastructure; no external citation. In the setting of Lemma 133, the inverse square commutes as well:* $\mathrm{rTensor}\,(S \to S') \circ \mathrm{cancelBaseChange}_S^{-1} = \mathrm{cancelBaseChange}_{S'}^{-1} \circ \mathrm{rTensor}\,(\mathrm{id}_A \otimes (S \to S'))$*. This is the form consumed by the affine brick, since* `baseChangeCancelModuleIso` $=$ $\mathrm{cancelBaseChange}^{-1}$*.*

Proof. Apply $\mathrm{cancelBaseChange}_{S'}$ (an isomorphism, hence injective) to both sides and rewrite by Lemma 133 and `LinearEquiv.apply_symm_apply` (twice); both sides become equal. □

5.5.1 The chart-tower indexing datum (scaffolding)

The affine-restriction naturality lemma compares the two affine bricks attached to an inclusion $\mathrm{Spec}(R'') \subseteq \mathrm{Spec}(R')$ of affine opens over a common chart $\mathrm{Spec}(R) \subseteq S$. We package the algebraic datum of such an inclusion — four rings R, A, R', R'', the chart map $\varphi : R \to A$, the base map $\psi : R \to R'$, the tower map $j : R' \to R''$, and the two affine pushout squares realizing the charts $B' = A \otimes_R R'$, $B'' = A \otimes_R R''$ — as a single structure, so the naturality statement is phrased against it.

Definition 135 (Base-change chart tower)**.** *Source: bespoke project infrastructure; no external citation.* A *base-change chart tower* is the datum of commutative rings R, A, R', R'', ring maps $\varphi : R \to A$, $\psi : R \to R'$, $j : R' \to R''$, chart rings B', B'' with their pushout legs $\rho', \sigma', \rho'', \sigma''$, and two cocartesian-square witnesses $h' : \mathrm{IsPushout}\,\varphi\,\psi\,\rho'\,\sigma'$ and $h'' : \mathrm{IsPushout}\,\varphi\,(\psi \circ j)\,\rho''\,\sigma''$. It indexes the affine bricks of Lemma 129 over the inclusion $\mathrm{Spec}(R'') \subseteq \mathrm{Spec}(R')$.

Definition 136 (Chart comparison of a tower)**.** *Source: bespoke project infrastructure; no external citation.* The canonical ring map $B' \to B''$ obtained from the universal property of the pushout B' applied to the cocone $(\rho'', j \circ \sigma'')$; it is the affine model of the inclusion $X_{\mathrm{Spec}(R'')} \subseteq X_{\mathrm{Spec}(R')}$.

Lemma 137 (Leg compatibilities of the chart comparison)**.** *Source: bespoke project infrastructure; no external citation. The chart comparison satisfies* $\rho' \circ \mathrm{connect} = \rho''$ *(and, by Lemma 138,* $\sigma' \circ \mathrm{connect} = j \circ \sigma''$*); both are immediate from the pushout universal property (*$\mathrm{inl\_desc}$*,* $\mathrm{inr\_desc}$*).*

Lemma 138 (Second leg compatibility of the chart comparison)**.** *Source: bespoke project infrastructure; no external citation.* $\sigma' \circ \mathrm{connect} = j \circ \sigma''$*, from the pushout universal property.*

Definition 139 (The two bricks of a tower)**.** *Source: bespoke project infrastructure; no external citation.* The brick $e_{R'}$ is affinePushforwardPullbackBaseChange over $\mathrm{Spec}(R')$ (base map ψ); the companion brick $e_{R''}$ (brickR$''$) is the same over $\mathrm{Spec}(R'')$ with the *composite* base map $\psi \circ j$. These are exactly the two isomorphisms the restriction-naturality square of Lemma 141 compares.

Definition 140 (The second brick of a tower)**.** *Source: bespoke project infrastructure; no external citation.* The brick $e_{R''} =$ affinePushforwardPullbackBaseChange over $\mathrm{Spec}(R'')$ with composite base map $\psi \circ j$; paired with brickR$'$ (Definition 139) in the restriction-naturality square.

Lemma 141 (Affine-restriction naturality of the cancellation chain)**.** *Let* $\mathrm{Spec}(R'') \subseteq \mathrm{Spec}(R')$ *be an inclusion of affine opens of the base* S' *(with the same chosen affine* $\mathrm{Spec}(R) \subseteq S$ *over the g-image, so that* $R \to R' \to R''$ *is a tower). Then the affine base-change isomorphisms* $e_{R'}$ *and* $e_{R''}$ *of Lemma 129 are intertwined by the structure-sheaf restriction maps: the square with* *horizontal maps $e_{R'}, e_{R''}$ and vertical maps the restrictions* $\mathrm{Spec}(R') \to \mathrm{Spec}(R'')$ *on* $g^*(f_*\widetilde{M})$ *and on* $f'_*((g')^*\widetilde{M})$ *commutes. Equivalently, the family* $\{e_{R'}\}$ *is a morphism of the restricted presheaves over the affine basis. This is the one genuinely new lemma of the active route: a concrete localization compatibility, replacing the intractable mate coherence.*

Proof. Each $e_{R'}$ is the three-move composite of Lemma 129: two tilde-dictionary isomorphisms and $\widetilde{\text{cancelBaseChange}}$. The restriction square factors as the paste of one component square per move, and each commutes for a concrete reason. The crucial structural point is that the two bricks live over *different base maps*: $e_{R'}$ is assembled from the dictionaries for the base map $\psi : R \to R'$, whereas $e_{R''}$ is assembled from the dictionaries for the *composite* base map $\psi \circ j$, with $j : R' \to R''$ the chart inclusion. The vertical restriction maps realize the passage $\psi \rightsquigarrow \psi \circ j$, so the relevant commutation is governed not by fixed-base open-naturality but by the *base-map composition cocycle* of the dictionaries.

- *Tilde-dictionary moves.* Restricting from $\mathrm{Spec}(R')$ to $\mathrm{Spec}(R'')$ replaces each dictionary factor for the base map ψ by the corresponding factor for the composite $\psi \circ j$. That these transitions commute with the dictionary isomorphisms is exactly the base-map composition cocycle of the affine dictionaries, instantiated at the tower $R \xrightarrow{\psi} R' \xrightarrow{j} R''$: on the pushforward/$\Gamma$ side the global-sections comparison of Lemma 116 factors through the comparisons for ψ and j separately (the source pushforward cocycle absorbed into Spec-functoriality and the target restriction-of-scalars cocycle folded in), and on the pullback side the corresponding factorization of the pullback dictionary of Lemma 126. (The fixed-base open-naturality of the section comparison, recorded in the proof of Lemma 118, supplies the compatibility of each individual brick with the structure-sheaf restriction maps internal to a single base; it is the base-map composition cocycle that reconciles the *two distinct base maps* ψ and $\psi \circ j$ the square actually compares.)
- *Cancellation move.* Localizing the base R' to R'' carries $(R' \otimes_R A) \otimes_A M$ to $(R'' \otimes_R A) \otimes_A M = R'' \otimes_{R'} ((R' \otimes_R A) \otimes_A M)$ and $R' \otimes_R M$ to $R'' \otimes_R M = R'' \otimes_{R'} (R' \otimes_R M)$; under these identifications cancelBaseChange for the tower over R'' is the localization of cancelBaseChange for the tower over R'. This is exactly the cancellation-localization compatibility Lemma 133 (whose R'-side localization datum is Lemma 119).

Pasting the three component squares yields the restriction square. As the inclusion was an arbitrary basic affine inclusion, the family $\{e_{R'}\}$ is restriction- compatible on the affine basis. $\square$

Lemma 142 (Gluing to a global sheaf-of-abelian-groups morphism)**.** *There is a morphism of sheaves of abelian groups*

$$\underline{e} : \mathrm{toSheaf}(g^*(f_*\mathcal{F})) \longrightarrow \mathrm{toSheaf}(f'_*((g')^*\mathcal{F}))$$

on S' whose restriction to each affine open $\mathrm{Spec}(R') \subseteq S'$ *of the basis is the underlying abelian-group morphism of the affine isomorphism* $e_{R'}$*.*

Proof. The affine opens of S' form a basis of its topology (X. isBasis_affineOpens). Forgetting the module structure of both $g^*(f_*\mathcal{F})$ and $f'_*((g')^*\mathcal{F})$ through SheafOfModules. toSheaf gives a presheaf source and a *sheaf* target of abelian groups. By the restriction naturality Lemma 141, the affine-local components $\{e_{R'}\}$ (Lemma 129) assemble into a morphism of the basis-restricted presheaves $B^{\mathrm{op}*}F \to B^{\mathrm{op}*}F'$. Mathlib's Lemma 130 (basis form) promotes this basis-local family to a global morphism of sheaves of abelian groups $\underline{e}$, with the component at each basis open $\mathrm{Spec}(R')$ recovered as the abelian-group morphism of $e_{R'}$; the corresponding extension property hom_ext makes $\underline{e}$ the unique such global morphism. $\square$

Lemma 143 (Re-imposing $\mathcal{O}_{S'}$-linearity)**.** *The abelian-group sheaf morphism $\underline{e}$ of Lemma 142 underlies a morphism of $\mathcal{O}_{S'}$-modules*

$$e^{\mathrm{lin}} : g^*(f_*\mathcal{F}) \longrightarrow f'_*((g')^*\mathcal{F}).$$

Proof. The linearity equation $\underline{e}(r \cdot m) = r \cdot \underline{e}(m)$ is a sectionwise identity. It holds on each affine open $\mathrm{Spec}(R') \subseteq S'$ of the basis, because there $\underline{e}$ agrees with the underlying map of $e_{R'}$, which is $\mathcal{O}_{S'}(\mathrm{Spec}(R'))$-linear by construction (Lemma 129). Since the affine opens form a basis and both sides of the equation are sections of a sheaf, the identity that holds on the basis holds on every open by separatedness. Mathlib's Lemma 131 packages the abelian-group morphism $\underline{e}$ together with this sectionwise linearity witness into the morphism e^{lin} of $\mathcal{O}_{S'}$-modules. ☐

Lemma 144 (The global base-change natural isomorphism e)**.** *The morphism e^{lin} of Lemma 143 is an isomorphism, and the family over quasi-coherent $\mathcal{F}$ assembles into a natural isomorphism of functors*

$$e : g^* \circ f_* \;\cong\; f'_* \circ (g')^*.$$

This e is the active-route deliverable consumed downstream (it commutes with the restriction maps by construction); it is built without ever forming the canonical adjoint mate.

Proof. By the affine-open locality criterion Lemma 114, e^{lin} is an isomorphism if and only if it restricts to an isomorphism on the sections over every affine open of S'. On each such affine open $\mathrm{Spec}(R')$ it agrees with $e_{R'}$, which is an isomorphism by Lemma 129; hence e^{lin} is an isomorphism. Naturality in $\mathcal{F}$ is checked affine-locally: the brick Lemma 129 is natural in the module M, so the components are natural in $\mathcal{F}$, and this naturality propagates through the gluing (the global morphism is the unique extension of natural affine data). Finally Mathlib's Lemma 132 packages the per-$\mathcal{F}$ isomorphisms into the natural isomorphism e. ☐

5.6 Mathlib stubs imported for the Grassmannian/Quot descent

Lemma 145 (The pullback–pushforward conjugate sends the pullback coherence to the pushforward coherence)**.** *Provided by Mathlib. Under the conjugation (mate) bijection between the composite pullback–pushforward adjunction $(g^* \dashv g_*).(f^* \dashv f_*)$ and the adjunction for the composite $((f \circ g)^* \dashv (f \circ g)_*)$, the inverse pullback-composition coherence is carried to the pushforward-composition coherence.*

Lemma 146 (Sheaf gluing: a compatible family has a unique amalgamation)**.** *Provided by Mathlib. Let $\mathcal{G}$ be a sheaf on X, $U : \iota \to \mathrm{Opens}(X)$ a family covering an open V, and $(s_i)_i$ a compatible family of sections. Then there is a unique $s \in \mathcal{G}(V)$ restricting to each s_i — the gluing half of the sheaf condition.*

Chapter 6

Higher direct images of quasi-coherent sheaves in positive degree

6.1 Setup and motivation

This chapter treats the higher derived direct images $R^i f_* \mathcal{F}$ for $i \geq 1$, complementing the $i = 0$ (direct-image) case of Chapter 5. These higher direct images are the single deepest foundation of the cohomology-and-base-change package: the Cohen–Macaulay regularity bound, semicontinuity of fibre cohomology, and the m-regularity estimate of FGA all rest on the basic finiteness and base-change properties of $R^i f_*$ recorded here.

We use the classical cohomology-sheaf formulation (not the derived category) and a Čech-complex approach, parallel to the $i = 0$ treatment of Chapter 5.

Throughout, $f : X \to S$ is a morphism of schemes that is quasi-compact and quasi-separated, and $\mathcal{F}$ is a quasi-coherent $\mathcal{O}_X$-module. A base change of f along a morphism $g : S' \to S$ is recorded by the cartesian square obtained from the fibre product $X' = X \times_S S'$:

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow{g} & S \end{array}$$

Here $f' : X' \to S'$ is the base change of f, $g' : X' \to X$ is the first projection, and $\mathcal{F}' = (g')^* \mathcal{F}$ is the pullback of $\mathcal{F}$ to X'.

6.2 Definition of the higher direct images

Definition 147 (Higher direct image)**.** Let $f : X \to S$ be a morphism of schemes and let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. For $i \geq 0$ the *i-th higher direct image* $R^i f_* \mathcal{F}$ is the i-th right derived functor of the pushforward functor f_* applied to $\mathcal{F}$. Concretely, $R^i f_* \mathcal{F}$ is the $\mathcal{O}_S$-module obtained as the sheafification of the presheaf

$$V \longmapsto H^i(f^{-1}(V), \mathcal{F}|_{f^{-1}(V)}), \qquad V \subset S \text{ open},$$

where $H^i(-,-)$ denotes sheaf cohomology. For $i = 0$ this recovers the ordinary pushforward $R^0 f_* \mathcal{F} = f_* \mathcal{F}$.

6.3 Quasi-coherence of the higher direct images

Lemma 148 (Quasi-coherence of higher direct images)**.** *Let $f : X \to S$ be a quasi-compact and quasi-separated morphism of schemes. Then for every quasi-coherent $\mathcal{O}_X$-module $\mathcal{F}$ and every $p \geq 0$, the higher direct image $R^p f_* \mathcal{F}$ is a quasi-coherent $\mathcal{O}_S$-module.*

Proof. Quasi-coherence is local on S, and forming higher direct images commutes with restriction to open subschemes, so we may assume S is affine; then X is quasi-compact and quasi-separated. We argue by the induction principle for quasi-compact opens of a quasi-compact quasi-separated scheme: for a quasi-compact open $U \subset X$ with $a = f|_U$, let $P(U)$ be the property that $R^p a_* \mathcal{G}$ is quasi-coherent on S for every quasi-coherent module $\mathcal{G}$ on U and every $p \geq 0$; the desired statement is $P(X)$.

Affine base. If U is affine then a is an affine morphism, so $R^p a_* \mathcal{G} = 0$ for $p \geq 1$ by the relative affine vanishing of Lemma 149, while $R^0 a_* \mathcal{G} = a_* \mathcal{G}$ is quasi-coherent because the pushforward of a quasi-coherent sheaf along a quasi-compact quasi-separated morphism is quasi-coherent. Hence $P(U)$ holds.

Inductive step. Let $U \subset X$ be quasi-compact open, $V \subset X$ affine open, and suppose $P(U)$, $P(V)$, $P(U \cap V)$ hold. Writing $a = f|_U$, $b = f|_V$, $c = f|_{U \cap V}$, $h = f|_{U \cup V}$, the relative Mayer–Vietoris sequence

$$0 \to h_* \mathcal{G} \to a_*(\mathcal{G}|_U) \oplus b_*(\mathcal{G}|_V) \to c_*(\mathcal{G}|_{U \cap V}) \to R^1 h_* \mathcal{G} \to \cdots$$

exhibits each $R^p h_* \mathcal{G}$ as an extension built from quasi-coherent sheaves $R^p a_*(\mathcal{G}|_U)$, $R^p b_*(\mathcal{G}|_V)$, $R^p c_*(\mathcal{G}|_{U \cap V})$: it sits between the cokernel of a map of quasi-coherent sheaves and the kernel of a map of quasi-coherent sheaves, hence is quasi-coherent. Thus $P(U \cup V)$ holds. By the induction principle $P(X)$ holds, proving the lemma. ☐

6.4 Vanishing of higher direct images along an affine morphism

Lemma 149 (Relative affine vanishing)**.** *Let $f : X \to S$ be a morphism of schemes and $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$-module. If f is affine, then $R^i f_* \mathcal{F} = 0$ for all $i \geq 1$.*

Proof. By Definition 147, $R^i f_* \mathcal{F}$ is the sheafification of the presheaf $V \mapsto H^i(f^{-1}(V), \mathcal{F}|_{f^{-1}(V)})$. Since f is affine, the preimage $f^{-1}(V)$ of any affine open $V \subset S$ is affine. The higher cohomology of a quasi-coherent sheaf on an affine scheme vanishes, so $H^i(f^{-1}(V), \mathcal{F}|_{f^{-1}(V)}) = 0$ for all $i \geq 1$ and all affine V. As the affine opens form a basis of S, the sheafification of a presheaf that vanishes on a basis is zero; hence $R^i f_* \mathcal{F} = 0$ for all $i \geq 1$. ☐

6.5 Flat base change for the higher direct images

Theorem 150 (Flat base change, $i \geq 1$ case)**.** *Consider the cartesian square*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow{g} & S \end{array}$$

with $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$-module and $\mathcal{F}' = (g')^*\mathcal{F}$. Assume that g is flat and that f is quasi-compact and quasi-separated. Then for every $i \geq 1$ the canonical base-change map (the higher-degree analogue of the pushforward base-change map of Definition 111) is an isomorphism

$$g^*(R^i f_* \mathcal{F}) \xrightarrow{\sim} R^i f'_*((g')^*\mathcal{F}) = R^i f'_* \mathcal{F}'.$$

Equivalently, in the affine situation $S = \operatorname{Spec}(A)$, $S' = \operatorname{Spec}(B)$ with $A \to B$ flat, the comparison map of B-modules

$$H^i(X, \mathcal{F}) \otimes_A B \xrightarrow{\sim} H^i(X_B, \mathcal{F}_B)$$

is an isomorphism, where $X_B = X \times_{\operatorname{Spec}(A)} \operatorname{Spec}(B)$ and $\mathcal{F}_B$ is the pullback of $\mathcal{F}$.

Proof. Reduction to the affine target. The assertion is local on S', so we may assume $S = \operatorname{Spec}(A)$ and $S' = \operatorname{Spec}(B)$ with $A \to B$ flat. By quasi-coherence of the higher direct images (Lemma 148), $R^i f_* \mathcal{F}$ is the quasi-coherent $\mathcal{O}_S$-module associated to the A-module $H^0(S, R^i f_* \mathcal{F}) = H^i(X, \mathcal{F})$, and likewise $R^i f'_* \mathcal{F}'$ is associated to the B-module $H^i(X', \mathcal{F}')$. Since pullback along g corresponds to $- \otimes_A B$ on modules, the base-change map is identified with the comparison map of B-modules

$$H^i(X, \mathcal{F}) \otimes_A B \longrightarrow H^i(X_B, \mathcal{F}_B), \qquad X_B = X \times_{\operatorname{Spec}(A)} \operatorname{Spec}(B),$$

and it suffices to prove that this map is an isomorphism.

The separated case via Čech complexes. Suppose first that X is separated over A. Choose a finite affine open cover $\mathcal{U} : X = U_1 \cup \cdots \cup U_t$; by separatedness every finite intersection $U_{i_0 \cdots i_p}$ is again affine, so Čech cohomology computes sheaf cohomology: $\check{H}^p(\mathcal{U}, \mathcal{F}) = H^p(X, \mathcal{F})$ for all p. Base changing the cover yields an affine open cover $\mathcal{U}_B : X_B = (U_1)_B \cup \cdots \cup (U_t)_B$ of the again-separated X_B, with the same property $\check{H}^p(\mathcal{U}_B, \mathcal{F}_B) = H^p(X_B, \mathcal{F}_B)$. The affine case identifies each term of the base-changed Čech complex with the original term tensored over A with B; that is, there is an isomorphism of cochain complexes

$$\check{\mathcal{C}}^\bullet(\mathcal{U}_B, \mathcal{F}_B) \;\cong\; \check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F}) \otimes_A B.$$

Because $A \to B$ is flat, the functor $- \otimes_A B$ is exact and hence commutes with the formation of cohomology of a complex; taking H^i in any degree $i \geq 1$ gives the desired isomorphism $H^i(X, \mathcal{F}) \otimes_A B \cong H^i(X_B, \mathcal{F}_B)$.

The quasi-separated case. In general X need only be quasi-separated. Choose a finite affine open cover $\mathcal{U} : X = U_1 \cup \cdots \cup U_t$. Now the intersections $U_{i_0 \cdots i_p}$ are quasi-compact and separated rather than affine, but they fall under the separated case already treated, so flat base change holds for each of them. Comparing the two Čech-to-cohomology spectral sequences with E_2-pages $\check{H}^p(\mathcal{U}, \underline{H}^q(\mathcal{F}))$ and $\check{H}^p(\mathcal{U}_B, \underline{H}^q(\mathcal{F}_B))$, converging to $H^{p+q}(X, \mathcal{F})$ and $H^{p+q}(X_B, \mathcal{F}_B)$ respectively, the separated case provides the term-by-term identification $\check{H}^p(\mathcal{U}_B, \underline{H}^q(\mathcal{F}_B)) = \check{H}^p(\mathcal{U}, \underline{H}^q(\mathcal{F})) \otimes_A B$ on the E_2-pages; flatness of $A \to B$ propagates this isomorphism through the spectral sequences to the abutment, giving $H^i(X, \mathcal{F}) \otimes_A B \cong H^i(X_B, \mathcal{F}_B)$ in every degree $i \geq 1$. Translating back to sheaves on S' yields the asserted isomorphism $g^*(R^i f_* \mathcal{F}) \xrightarrow{\sim} R^i f'_* \mathcal{F}'$. $\square$

Chapter 7

Acyclic resolutions compute right-derived functors

7.1 Setup and motivation

This chapter isolates the one abstract, scheme-independent piece of homological algebra on which the Čech computation of the higher direct images rests: an *acyclic resolution computes the right-derived functor*. The companion chapter 8 builds, from a separated quasi-compact morphism $f : X \to S$ and an affine open cover, an explicit complex of $\mathcal{O}_S$-modules — the relative Čech complex $\check{\mathcal{C}}^\bullet(\mathfrak{U}, \mathcal{F})$ — whose terms are pushforwards of $\mathcal{F}$ over the affine intersections. To identify the cohomology of that complex with $R^i f_* \mathcal{F}$, one observes that the Čech complex is a resolution of $\mathcal{F}$ by objects on which the higher direct images vanish (the affine acyclicity of Lemma 194), and then invokes the present theorem with $G = f_*$. This single homological lemma thereby replaces *both* the Čech-to-cohomology spectral sequence and the Leray spectral sequence.

Throughout, $\mathcal{A}$ and $\mathcal{B}$ are abelian categories, $\mathcal{A}$ has enough injectives (so that injective resolutions exist and the right-derived functors below are everywhere defined), and $G : \mathcal{A} \to \mathcal{B}$ is an additive functor. For $n \in \mathbb{N}$ we write $R^n G$ for the n-th right-derived functor of G (a functor $\mathcal{A} \to \mathcal{B}$), constructed from injective resolutions: for an injective resolution $X \to I^\bullet$ one has $(R^n G)(X) \cong H^n(G(I^\bullet))$ (Lemma 151 below). When in addition G is left exact — equivalently, G preserves finite limits, which holds for any G that is a right adjoint, such as a pushforward f_* — one has $R^0 G \cong G$ (Lemma 153).

7.2 Known results

The following three well-known results underpin the injective-resolution construction of $R^n G$ and are the foundational inputs to the argument.

Lemma 151 (Right-derived functor via an injective resolution)**.** *Provided by Mathlib. Let $\mathcal{A}$ be an abelian category with enough injectives, $\mathcal{B}$ abelian, and $G : \mathcal{A} \to \mathcal{B}$ additive. For an object X of $\mathcal{A}$, a choice of injective resolution I of X (an exact complex $0 \to X \to I^0 \to I^1 \to \cdots$ with each I^n injective), and every $n \in \mathbb{N}$, there is a canonical isomorphism in $\mathcal{B}$*

$$(R^n G)(X) \;\cong\; H^n(G(I^\bullet)),$$

where $G(I^\bullet)$ is the cochain complex obtained by applying G degreewise to the resolution complex $I^\bullet$ and H^n is its n-th cohomology object. This is the defining computation of the right-derived functor and the seed from which the acyclic-resolution generalization is bootstrapped.

Lemma 152 (Higher derived functors vanish on injectives)**.** *Provided by Mathlib. With $\mathcal{A}, \mathcal{B}, G$ as above, if J is an injective object of $\mathcal{A}$ then $(R^{n+1}G)(J) = 0$ for every $n \in \mathbb{N}$; that is, the higher right-derived functors vanish on injective objects. In particular every injective object is right-G-acyclic in the sense of Definition 155.*

Lemma 153 (The zeroth derived functor of a left-exact functor)**.** *Provided by Mathlib. With $\mathcal{A}, \mathcal{B}, G$ as above, if G is left exact (preserves finite limits) then the natural transformation $G \to R^0G$ is an isomorphism,*

$$R^0G \;\cong\; G.$$

Lemma 154 (Long exact homology sequence of a short exact sequence of complexes)**.** *Provided by Mathlib. Let $\mathcal{C}$ be an abelian category and let*

$$0 \longrightarrow S.X_1 \longrightarrow S.X_2 \longrightarrow S.X_3 \longrightarrow 0$$

be a short exact sequence of cochain complexes in $\mathcal{C}$ — that is, a short complex $S = (S.X_1 \to S.X_2 \to S.X_3)$ of objects of the category of $\mathbb{Z}$-indexed cochain complexes HomologicalComplex $\mathcal{C}$ *which is short exact (degreewise mono, epi and exact in the middle). Then for every degree n there is a connecting morphism*

$$\delta^n \;:\; H^n(S.X_3) \longrightarrow H^{n+1}(S.X_1)$$

(the Mathlib map `ShortComplex.ShortExact.δ`), and these assemble into a long exact sequence of homology objects in $\mathcal{C}$

$$\cdots \longrightarrow H^n(S.X_2) \longrightarrow H^n(S.X_3) \xrightarrow{\delta^n} H^{n+1}(S.X_1) \longrightarrow H^{n+1}(S.X_2) \longrightarrow \cdots.$$

This is the complex-level homology long exact sequence and its connecting homomorphism. The derived-functor long exact sequence used in the sequel is constructed from this complex-level result via a horseshoe lift; see Lemma 166.

7.3 Right-acyclic objects

Definition 155 (Right-G-acyclic object)**.** *Source: Stacks Project, Derived Categories, Tag 0157 (`definition-derived-functor`) and Tag 015C (`lemma-F-acyclic`).* Let $G : \mathcal{A} \to \mathcal{B}$ be an additive functor between abelian categories. An object J of $\mathcal{A}$ is *right-G-acyclic* when the higher right-derived functors of G vanish at J:

$$(R^kG)(J) = 0 \qquad \text{for all } k \geq 1.$$

Equivalently, writing the condition with the index shifted, $(R^{k+1}G)(J) = 0$ for all $k \in \mathbb{N}$. The Stacks Project defines a right-G-acyclic object intrinsically (as one for which the one-term complex $J[0]$ computes the derived functor RG); for an additive — in our applications, left exact — functor G that intrinsic condition is equivalent to the vanishing of all $(R^kG)(J)$ with $k \geq 1$, which is the form we adopt here and which is directly checkable against the injective-resolution computation of Lemma 151.

Remark (alternative quantifier shapes). Two equivalent formulations are available: "$(R^kG)(J) = 0$ for all $k \geq 1$" or the index-shifted "$(R^{k+1}G)(J) = 0$ for all $k \in \mathbb{N}$"; the latter avoids the inequality side-condition and aligns with the vanishing statement of Lemma 152.

7.4 Lifting a short exact sequence to injective resolutions

The passage from the complex-level homology long exact sequence (Lemma 154) to a long exact sequence of right-derived functors requires one geometric input: a short exact sequence of objects must be lifted to a short exact sequence of their injective resolutions which is, moreover, *degreewise split*. This is the dual Horseshoe Lemma. It is the single genuinely new homological-algebra construction the argument requires; individual injective resolutions (provided by enough injectives) furnish the building blocks, and the horseshoe construction assembles them into the required lift.

7.4.1 The horseshoe construction, step by step

The construction proceeds by induction on degree; we isolate its independent moving parts as the sub-lemmas below, from which the horseshoe lemma is assembled. Throughout, I_A and I_C are fixed injective resolutions of A and C (they exist because $\mathcal{A}$ has enough injectives), and we write d_A, d_C for their differentials and I^n_A, I^n_C for their degree-n terms. The middle resolution is built degreewise as the biproduct $I^n_B := I^n_A \oplus I^n_C$. All seven statements are structural pieces of the dual Horseshoe Lemma (dual of Weibel, *An Introduction to Homological Algebra*, Lemma 2.2.8); no new source is introduced.

Lemma 156 (Degree term: a finite biproduct of injectives is injective)**.** *Provided by Mathlib. In an abelian category, if P and Q are injective objects then their biproduct $P \oplus Q$ is injective. Consequently each degree term $I^n_B := I^n_A \oplus I^n_C$ of the middle complex is injective, being the biproduct of the injective terms I^n_A and I^n_C of the two chosen resolutions.*

Lemma 157 (Degree term: the biproduct short complex is split exact)**.** *Provided by Mathlib. For objects P, Q of an abelian category the short complex*

$$P \xrightarrow{\iota_P} P \oplus Q \xrightarrow{\pi_Q} Q$$

formed from the biproduct coprojection ι_P and projection π_Q carries a canonical `ShortComplex.Splitting`*; in particular it is a split short exact sequence. Applied in each degree n with $P = I^n_A$, $Q = I^n_C$, this exhibits every degree*

$$0 \longrightarrow I^n_A \longrightarrow I^n_B \longrightarrow I^n_C \longrightarrow 0$$

as split, furnishing the degreewise-splitting data required for the construction.

Lemma 158 (Per-stage monomorphism into the biproduct)**.** *Let $0 \to S.X_1 \xrightarrow{S.f} S.X_2 \xrightarrow{S.g} S.X_3 \to 0$ be an exact short complex with $S.f$ a monomorphism, and let $\alpha : S.X_1 \to P$, $\gamma : S.X_3 \to Q$ be monomorphisms with P injective. Then the map*

$$\langle\, \mathrm{factorThru}(\alpha, S.f)\,,\ S.g \circ \gamma \rangle \ :\ S.X_2 \longrightarrow P \oplus Q,$$

whose first component is the injective extension of α along the mono $S.f$ and whose second component is $S.X_2 \twoheadrightarrow S.X_3 \hookrightarrow Q$, is a monomorphism. This is the kernel/cokernel step driving each stage of the recursion: applied to the n-th cosyzygy short exact sequence with $P = I^{n+1}_A$, $Q = I^{n+1}_C$ it produces the monomorphism of the next cosyzygy into the next biproduct term, whose cokernel feeds the following stage (and, at the base stage with the original sequence $0 \to A \to B \to C \to 0$, the augmentation $\beta : B \hookrightarrow I^0_B$).

Proof. Write $u := \langle \mathrm{factorThru}(\alpha, S.f), S.g \circ \gamma \rangle$. To see u is mono it suffices, in an abelian category, to show $x \circ u = 0 \implies x = 0$ for any x. Composing $x \circ u$ with the two biproduct projections gives $x \circ \mathrm{factorThru}(\alpha, S.f) = 0$ and $x \circ S.g \circ \gamma = 0$; since γ is mono the latter yields $x \circ S.g = 0$. Exactness of S at $S.X_2$ (with $S.f$ mono) lets us lift x through $S.f$: there is y with $y \circ S.f = x$. Then $y \circ \alpha = y \circ S.f \circ \mathrm{factorThru}(\alpha, S.f) = x \circ \mathrm{factorThru}(\alpha, S.f) = 0$, using the defining property $S.f \circ \mathrm{factorThru}(\alpha, S.f) = \alpha$; as α is mono, $y = 0$, hence $x = y \circ S.f = 0$. □

Lemma 159 (Off-diagonal twist and augmentation lifts)**.** *There exist an augmentation $\beta : B \to I_B^0$ and a family of morphisms*

$$\tau^n : I_C^n \longrightarrow I_A^{n+1} \qquad (n \in \mathbb{N})$$

— the off-diagonal components of the middle differential — satisfying the cocycle identity

$$\tau^{n+1} \circ d_C^n \;=\; -\,d_A^{n+1} \circ \tau^n \qquad (n \in \mathbb{N}),$$

together with the compatibility of β with the augmentations $A \to I_A^0$, $C \to I_C^0$ and the maps of the original short exact sequence. Each τ^n (and β) is produced by the universal lifting property of injectives: the target summand I_A^{n+1} is injective and the relevant source map is a monomorphism (Lemma 158).

Proof. The lifts are constructed by recursion on n. For the augmentation, the first component of β extends the composite $A \to I_A^0$ along the monomorphism $A \hookrightarrow B$ using injectivity of I_A^0; the second component is $B \twoheadrightarrow C \to I_C^0$. At the inductive step we have τ^n (with $\tau^n : I_C^n \to I_A^{n+1}$) and seek $\tau^{n+1} : I_C^{n+1} \to I_A^{n+2}$ with $\tau^{n+1} \circ d_C^n = -\,d_A^{n+1} \circ \tau^n$. The right-hand side $-\,d_A^{n+1} \circ \tau^n : I_C^n \to I_A^{n+2}$ annihilates $\ker d_C^n = \operatorname{im} d_C^{n-1}$ — the equality $\ker d_C^n = \operatorname{im} d_C^{n-1}$ being the exactness of the injective resolution I_C in the positive degree n, which is part of the defining data of an injective resolution. Indeed, precomposing with d_C^{n-1} and using the previous-stage identity $\tau^n \circ d_C^{n-1} = -\,d_A^n \circ \tau^{n-1}$ gives $-\,d_A^{n+1} \circ \tau^n \circ d_C^{n-1} = d_A^{n+1} \circ d_A^n \circ \tau^{n-1} = 0$, where the final equality is the complex identity $d_A^{n+1} \circ d_A^n = 0$ for the cochain complex I_A. Hence $-\,d_A^{n+1} \circ \tau^n$ factors through the image of d_C^n, i.e. through the monomorphism $\operatorname{im} d_C^n \hookrightarrow I_C^{n+1}$; injectivity of the target I_A^{n+2} extends that factorization along this monomorphism to a map $\tau^{n+1} : I_C^{n+1} \to I_A^{n+2}$ defined on all of I_C^{n+1} and satisfying $\tau^{n+1} \circ d_C^n = -\,d_A^{n+1} \circ \tau^n$ by construction. This is exactly the dual of the inductive lift in the Horseshoe Lemma. □

Lemma 160 (The assembled middle differential squares to zero)**.** *With the twist family τ of Lemma 159, define the middle differential degreewise by the matrix*

$$d_B^n \;=\; \begin{pmatrix} d_A^n & \tau^n \\ 0 & d_C^n \end{pmatrix} \;:\; I_A^n \oplus I_C^n \longrightarrow I_A^{n+1} \oplus I_C^{n+1}.$$

Then $d_B^{n+1} \circ d_B^n = 0$ for every n, so $(I_B^\bullet, d_B)$ is a cochain complex.

Proof. Multiplying the two matrices, the composite $d_B^{n+1} \circ d_B^n$ has diagonal entries $d_A^{n+1} \circ d_A^n = 0$ and $d_C^{n+1} \circ d_C^n = 0$ (the complexes I_A, I_C being complexes), lower-left entry 0, and upper-right entry

$$d_A^{n+1} \circ \tau^n + \tau^{n+1} \circ d_C^n,$$

which vanishes precisely by the cocycle identity $\tau^{n+1} \circ d_C^n = -\,d_A^{n+1} \circ \tau^n$ of Lemma 159. Hence $d_B^{n+1} \circ d_B^n = 0$. □

Lemma 161 (The inclusion and projection are chain maps)**.** *With d_B as in Lemma 160, the degreewise biproduct coprojection $\iota^n = \iota_{I_A^n} : I_A^n \to I_B^n$ and projection $\pi^n = \pi_{I_C^n} : I_B^n \to I_C^n$ assemble into chain maps $\iota : I_A^\bullet \to I_B^\bullet$ and $\pi : I_B^\bullet \to I_C^\bullet$; that is, $d_B^n \circ \iota^n = \iota^{n+1} \circ d_A^n$ and $\pi^{n+1} \circ d_B^n = d_C^n \circ \pi^n$. Moreover ι, π are compatible with the augmentations and form, in each degree, the split short exact sequence $0 \to I_A^n \to I_B^n \to I_C^n \to 0$ of Lemma 157.*

Proof. These are direct biproduct computations from the matrix form of d_B. Pre-composing d_B^n with the coprojection ι^n of the first summand selects the first column $\begin{pmatrix} d_A^n \\ 0 \end{pmatrix}$, which equals $\iota^{n+1} \circ d_A^n$; so ι is a chain map. Post-composing d_B^n with the projection π^{n+1} onto the second summand selects the second row $\begin{pmatrix} 0 & d_C^n \end{pmatrix}$, which equals $d_C^n \circ \pi^n$; so π is a chain map. Compatibility with the augmentations is the corresponding statement in degree $-1 \to 0$, immediate from the construction of β in Lemma 159. In each degree $\iota^n = \iota_{I_A^n}$ and $\pi^n = \pi_{I_C^n}$ are the structure maps of the biproduct, so the degree is split exact by Lemma 157. $\square$

Lemma 162 (Middle-term quasi-isomorphism transfer)**.** *Project-local supplement (the τ_2 companion of Mathlib's τ_3 transfer* `quasiIso_` *). Let $\varphi : S_1 \to S_2$ be a morphism of short exact sequences of cochain complexes, with components $\varphi.\tau_1, \varphi.\tau_2, \varphi.\tau_3$ on the three columns. Suppose $\varphi.\tau_1$ and $\varphi.\tau_3$ are quasi-isomorphisms, and that the middle component $\varphi.\tau_2$ is a monomorphism in every degree i that has no incoming differential, and an epimorphism in every degree i that has no outgoing differential. Then $\varphi.\tau_2$ is a quasi-isomorphism.*

Proof. This is the homology four-lemma. Fix a degree i; we show $H^i(\varphi.\tau_2)$ is an isomorphism by proving it is both epimorphic and monomorphic and concluding that a map which is both is an isomorphism. *Epi:* when i has an outgoing differential to some j, the two short exact sequences yield a commuting ladder whose rows are the five-term exact windows of the homology long exact sequence (Lemma 154) about degree i; the four-lemma (epi form), using that $H^i(\varphi.\tau_1)$, $H^i(\varphi.\tau_3)$ are epi and $H^j(\varphi.\tau_1)$ is mono, forces $H^i(\varphi.\tau_2)$ epi. When i has no outgoing differential, $\varphi.\tau_2$ is epi in degree i by hypothesis, and an epimorphism of complexes that is epi in a degree with no outgoing differential is epi on homology there. *Mono:* dually, when i has an incoming differential from some k, the four-lemma (mono form), using that $H^k(\varphi.\tau_3)$ is epi and $H^i(\varphi.\tau_1), H^i(\varphi.\tau_3)$ are mono, forces $H^i(\varphi.\tau_2)$ mono; when i has no incoming differential it is mono in that degree by hypothesis, hence mono on homology. Being epi and mono in every degree, $H^i(\varphi.\tau_2)$ is an isomorphism for all i, so $\varphi.\tau_2$ is a quasi-isomorphism. $\square$

Lemma 163 (The middle complex resolves B)**.** *The cochain complex $(I_B^\bullet, d_B)$ of Lemma 160, with the augmentation $\beta : B \to I_B^0$ of Lemma 159, is an injective resolution of B: its terms are injective (Lemma 156), it is exact in every positive degree, and $H^0 \cong B$ via β.*

Proof. By Lemma 161 the maps ι, π form, degreewise, the split short exact sequence $0 \to I_A^n \to I_B^n \to I_C^n \to 0$; assembling these splittings (Lemma 157) gives a short exact sequence of cochain complexes

$$0 \longrightarrow I_A^\bullet \longrightarrow I_B^\bullet \longrightarrow I_C^\bullet \longrightarrow 0.$$

Apply the complex-level homology long exact sequence (Lemma 154). The outer complexes $I_A^\bullet$, $I_C^\bullet$ are injective resolutions, hence exact in every positive degree ($H^k(I_A) = H^k(I_C) = 0$ for $k \geq 1$) with $H^0(I_A) \cong A$, $H^0(I_C) \cong C$. In the long exact sequence the terms flanking $H^k(I_B)$ for $k \geq 1$ vanish, forcing $H^k(I_B) = 0$; in degree 0 the initial segment $0 \to H^0(I_A) \to H^0(I_B) \to H^0(I_C) \to H^1(I_A) = 0$ reads $0 \to A \to H^0(I_B) \to C \to 0$, which together with the original sequence $0 \to A \to B \to C \to 0$ and the augmentation β identifies $H^0(I_B) \cong B$. Hence $I_B^\bullet$ is exact in positive degrees and resolves B; with injective terms it is an injective resolution of B. $\square$

Lemma 164 (Horseshoe lift of a short exact sequence to injective resolutions)**.** *Let $\mathcal{A}$ be an abelian category with enough injectives and let*

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$$

be a short exact sequence in $\mathcal{A}$. Then there exist injective resolutions I_A of A, I_B of B, I_C of C, together with a short exact sequence of cochain complexes

$$0 \longrightarrow I_A \longrightarrow I_B \longrightarrow I_C \longrightarrow 0$$

which is compatible with the augmentations $A \to I_A^0$, $B \to I_B^0$, $C \to I_C^0$ and which is degreewise split*: for every n the sequence*

$$0 \longrightarrow I_A^n \longrightarrow I_B^n \longrightarrow I_C^n \longrightarrow 0$$

is a split short exact sequence of $\mathcal{A}$, so that $I_B^n \cong I_A^n \oplus I_C^n$.

Proof. This is the (dual) Horseshoe Lemma (dual of Weibel, *An Introduction to Homological Algebra*, Lemma 2.2.8), assembled from the sub-lemmas of §"The horseshoe construction, step by step". Take I_A, I_C and set $I_B^n := I_A^n \oplus I_C^n$: injective by Lemma 156, degreewise split by Lemma 157. The twist τ and augmentation exist with the cocycle identity (Lemma 159, via Lemma 158); the matrix differential then squares to zero (Lemma 160) with coprojection and projection chain maps (Lemma 161); and $I_B^\bullet$ resolves B (Lemma 163). These data are exactly the asserted degreewise-split short exact sequence $0 \to I_A \to I_B \to I_C \to 0$ of cochain complexes compatible with the augmentations. □

7.5 The dimension-shift step

The induction driving the comparison theorem rests on a single short-exact-sequence calculation: cutting a resolution into short exact sequences and reading off the long exact sequence of right-derived functors. The latter is not available off the shelf at the level of the derived-functor objects (R^kG); we build it from the complex-level homology long exact sequence (Lemma 154) via the horseshoe lift (Lemma 164). We first record the shift at the *resolution* level — where the horseshoe lift directly supplies the degreewise-split short exact sequence of injective resolutions — and then specialize to objects.

Lemma 165 (Resolution-level dimension shift)**.** *Let $G : \mathcal{A} \to \mathcal{B}$ be additive and let $0 \to A \to J \to Z \to 0$ be a short exact sequence with J right-G-acyclic, lifted to a* degreewise-split *short exact sequence of injective resolutions $0 \to I_A \to I_J \to I_Z \to 0$ (chain maps φ, ψ with $\psi \circ \varphi = 0$ and a splitting of $0 \to I_A^n \to I_J^n \to I_Z^n \to 0$ in each degree n). Then for every $k \geq 0$ the connecting map of the homology long exact sequence of $G(I_\bullet)$ is an isomorphism*

$$(R^{k+1}G)(Z) \;\cong\; (R^{k+2}G)(A).$$

Proof. Applying G degreewise to the degreewise-split sequence preserves each split degree, yielding a short exact sequence of cochain complexes $0 \to G(I_A) \to G(I_J) \to G(I_Z) \to 0$ in $\mathcal{B}$. Its homology long exact sequence (Lemma 154) has connecting maps $\delta^m : H^m(G(I_Z)) \to H^{m+1}(G(I_A))$; transport these across the injective-resolution computation $H^m(G(I_\_)) \cong (R^mG)(-)$ (Lemma 151). For $m = k+1 \geq 1$ the two homology objects flanking δ^m in the long exact sequence both sit on the middle complex $G(I_J)$, namely $H^{k+1}(G(I_J))$ and $H^{k+2}(G(I_J))$; both vanish because J is right-G-acyclic (Definition 155). With both neighbours zero, δ^{k+1} is an isomorphism, giving $(R^{k+1}G)(Z) \cong (R^{k+2}G)(A)$. □

Lemma 166 (Dimension shift across an acyclic short exact sequence)**.** *Source: Stacks Project, Derived Categories, Tag 015D (`lemma-F-acyclic-ses`). Let $G : \mathcal{A} \to \mathcal{B}$ be an additive left-exact functor, and let*

$$0 \longrightarrow A \longrightarrow J \longrightarrow Z \longrightarrow 0$$

be a short exact sequence in $\mathcal{A}$ with the middle term J right-G-acyclic. Then the connecting maps of the long exact sequence of right-derived functors furnish isomorphisms

$$(R^kG)(Z) \xrightarrow{\cong} (R^{k+1}G)(A) \qquad \textit{for all } k \geq 1.$$

(The companion lowest-degree statement $(R^1G)(A) \cong \operatorname{coker}(G(J) \to G(Z))$ is isolated as the sibling Lemma 167.)

Proof. We first construct the long exact sequence of right-derived functors associated to $0 \to A \to J \to Z \to 0$. In the classical treatment this sequence is supplied by the δ-functor structure of $\{R^kG\}$ on the derived category; here it is built directly from the complex-level homology long exact sequence via a horseshoe lift.

Building the long exact sequence. By the horseshoe lift (Lemma 164) choose injective resolutions I_A, I_J, I_Z of A, J, Z fitting into a short exact sequence of cochain complexes

$$0 \longrightarrow I_A \longrightarrow I_J \longrightarrow I_Z \longrightarrow 0$$

that is degreewise split. Apply G degreewise. Because each degree $0 \to I_A^n \to I_J^n \to I_Z^n \to 0$ is a *split* short exact sequence (so $I_J^n \cong I_A^n \oplus I_Z^n$ compatibly with the maps), the additive functor G carries it to a split short exact sequence $0 \to G(I_A^n) \to G(I_J^n) \to G(I_Z^n) \to 0$ — this is where degreewise splitness is essential, since G is not assumed exact and would not in general preserve a non-split short exact sequence. Hence

$$0 \longrightarrow G(I_A) \longrightarrow G(I_J) \longrightarrow G(I_Z) \longrightarrow 0$$

is a short exact sequence of cochain complexes in $\mathcal{B}$. The complex-level homology long exact sequence (Lemma 154) applied to it reads

$$\cdots \to H^k(G(I_J)) \to H^k(G(I_Z)) \xrightarrow{\delta^k} H^{k+1}(G(I_A)) \to H^{k+1}(G(I_J)) \to \cdots.$$

By Lemma 151 the homology of each applied resolution computes the right-derived functor: $H^k(G(I_A)) \cong (R^kG)(A)$, $H^k(G(I_J)) \cong (R^kG)(J)$ and $H^k(G(I_Z)) \cong (R^kG)(Z)$, naturally in the maps of the sequence. Transporting the homology long exact sequence across these isomorphisms gives the long exact sequence of right-derived functors

$$\cdots \to (R^kG)(J) \to (R^kG)(Z) \xrightarrow{\delta^k} (R^{k+1}G)(A) \to (R^{k+1}G)(J) \to \cdots,$$

and, using $R^0G \cong G$ (Lemma 153) at the bottom, its low-degree initial segment

$$0 \to G(A) \to G(J) \to G(Z) \xrightarrow{\delta^0} (R^1G)(A) \to (R^1G)(J) \to \cdots.$$

Killing the acyclic terms. Because J is right-G-acyclic (Definition 155), every term $(R^kG)(J)$ with $k \geq 1$ vanishes. For $k \geq 1$ the segment around $(R^kG)(J)$,

$$\underbrace{(R^kG)(J)}_{=0} \to (R^kG)(Z) \xrightarrow{\delta^k} (R^{k+1}G)(A) \to \underbrace{(R^{k+1}G)(J)}_{=0},$$

has both outer terms zero, so the connecting map $\delta^k : (R^kG)(Z) \to (R^{k+1}G)(A)$ is an isomorphism, which is the asserted dimension shift. (The lowest-degree segment of the same long exact sequence yields the cokernel description of $(R^1G)(A)$; that reading is recorded separately in Lemma 167.) $\square$

Lemma 167 (Lowest-degree cokernel description across an acyclic short exact sequence)**.** *Source: Stacks Project, Derived Categories, Tag 015D (`lemma-F-acyclic-ses`). Let $G : \mathcal{A} \to \mathcal{B}$ be an additive left-exact functor, and let*

$$0 \longrightarrow A \longrightarrow J \longrightarrow Z \longrightarrow 0$$

be a short exact sequence in $\mathcal{A}$ with the middle term J right-G-acyclic. Then there is an isomorphism in $\mathcal{B}$

$$(R^1G)(A) \;\cong\; \operatorname{coker}(G(J) \to G(Z)),$$

i.e. $(R^1G)(A)$ is the cokernel of the map $G(J) \to G(Z)$ induced by the surjection $J \twoheadrightarrow Z$.

Proof. This reads off the lowest-degree segment of the very long exact sequence of right-derived functors built, from the same short exact sequence $0 \to A \to J \to Z \to 0$, in the proof of Lemma 166; here we use only its initial terms, so we recall the construction briefly and then specialize. By the horseshoe lift (Lemma 164) choose injective resolutions I_A, I_J, I_Z of A, J, Z fitting into a degreewise-split short exact sequence of cochain complexes $0 \to I_A \to I_J \to I_Z \to 0$. Applying the additive functor G degreewise preserves each split degree, giving a short exact sequence of complexes $0 \to G(I_A) \to G(I_J) \to G(I_Z) \to 0$ in $\mathcal{B}$; its complex-level homology long exact sequence (Lemma 154), transported across the injective-resolution computation $H^k(G(I_\bullet)) \cong (R^kG)(-)$ (Lemma 151) and using $R^0G \cong G$ (Lemma 153), has low-degree initial segment

$$G(J) \to G(Z) \xrightarrow{\delta^0} (R^1G)(A) \to (R^1G)(J).$$

Because J is right-G-acyclic (Definition 155), the right end $(R^1G)(J)$ vanishes. Hence the segment $G(J) \to G(Z) \xrightarrow{\delta^0} (R^1G)(A) \to 0$ is exact with vanishing right end, so δ^0 is an epimorphism whose kernel is the image of $G(J) \to G(Z)$. Therefore δ^0 factors through an isomorphism $\operatorname{coker}(G(J) \to G(Z)) \xrightarrow{\cong} (R^1G)(A)$, which is the assertion. ☐

7.6 Cosyzygies of an acyclic resolution

The comparison theorem decomposes an acyclic resolution into the short exact sequences cut out by its cosyzygies, and reads the cohomology of the applied complex off those sequences using left-exactness of G. We isolate the two ingredients here.

Lemma 168 (Cosyzygy short exact sequences of an acyclic resolution)**.** *Source: Stacks Project, Derived Categories, Tag 05TA (`proposition-enough-acyclics`). Let $G : \mathcal{A} \to \mathcal{B}$ be additive and let*

$$0 \longrightarrow A \longrightarrow J^0 \longrightarrow J^1 \longrightarrow J^2 \longrightarrow \cdots$$

be an exact resolution of A with every J^n right-G-acyclic. Define the cosyzygies $Z^0 := A$ and, for $m \geq 1$,

$$Z^m \;:=\; \ker(J^m \to J^{m+1}) \;=\; \operatorname{im}(J^{m-1} \to J^m).$$

Then for every $m \geq 0$ the sequence

$$0 \longrightarrow Z^m \longrightarrow J^m \longrightarrow Z^{m+1} \longrightarrow 0 \qquad (S_m)$$

is short exact, with the inclusion $Z^m \hookrightarrow J^m$ the canonical one, the surjection $J^m \twoheadrightarrow Z^{m+1}$ the corestriction of $J^m \to J^{m+1}$ onto its image, and the middle term J^m right-G-acyclic. In particular $Z^1 = \operatorname{im}(J^0 \to J^1) = \operatorname{coker}(A \to J^0)$.

Proof. Fix $m \geq 0$. Mono of $Z^m \hookrightarrow J^m$ is immediate: Z^m is by definition a subobject of J^m (for $m = 0$ via the injection $A \hookrightarrow J^0$ of the resolution, for $m \geq 1$ as the kernel $\ker(J^m \to J^{m+1})$). The map $J^m \to Z^{m+1}$ is the corestriction of the differential $d^m : J^m \to J^{m+1}$ onto its image $Z^{m+1} = \operatorname{im}(d^m)$, hence an epimorphism. Exactness in the middle is exactness of the resolution at J^m: the image of $Z^m \hookrightarrow J^m$ equals $\ker(d^m)$ — for $m \geq 1$ by the definition $Z^m = \ker(J^m \to J^{m+1})$, and for $m = 0$ because exactness of $0 \to A \to J^0 \to J^1$ makes $A \hookrightarrow J^0$ identify A with $\ker(d^0)$ — while the kernel of the epimorphism $J^m \twoheadrightarrow Z^{m+1}$ is exactly $\ker(d^m)$. Thus $\operatorname{im}(Z^m \to J^m) = \ker(J^m \to Z^{m+1})$, so (S_m) is short exact. The middle term J^m is right-G-acyclic by hypothesis. The final identity $Z^1 = \operatorname{im}(J^0 \to J^1) = \operatorname{coker}(A \to J^0)$ is the case $m = 0$: $\operatorname{coker}(A \to J^0) = \operatorname{coker}(Z^0 \to J^0) \cong Z^1$ by (S_0). ☐

Lemma 169 (Applied cosyzygy is the cocycle object of the applied complex)**.** *Source: Stacks Project, Derived Categories, Tag 015D (`lemma-F-acyclic-ses`), left-exact initial segment. With G additive left exact and the acyclic resolution and cosyzygies of Lemma 168, for every $n \geq 0$ the morphism G applies to the inclusion $Z^n \hookrightarrow J^n$ to give an isomorphism*

$$G(Z^n) \xrightarrow{\cong} \ker(G(J^n) \to G(J^{n+1})),$$

identifying $G(Z^n)$, as a subobject of $G(J^n)$, with the n-th cocycle object of the applied complex $G(J^\bullet)$. For $n = 0$ this reads $G(A) \cong \ker(G(J^0) \to G(J^1)) = H^0(G(J^\bullet))$.

Proof. Fix $n \geq 0$. Apply the left-exact functor G to the cosyzygy short exact sequence (S_n), $0 \to Z^n \to J^n \to Z^{n+1} \to 0$ of Lemma 168; left-exactness yields an exact sequence

$$0 \longrightarrow G(Z^n) \longrightarrow G(J^n) \longrightarrow G(Z^{n+1}),$$

so $G(Z^n) \hookrightarrow G(J^n)$ is the kernel of $G(J^n) \to G(Z^{n+1})$. Next, the differential $G(J^n) \to G(J^{n+1})$ of the applied complex factors as $G(J^n) \xrightarrow{G(J^n \twoheadrightarrow Z^{n+1})} G(Z^{n+1}) \xrightarrow{G(Z^{n+1} \hookrightarrow J^{n+1})} G(J^{n+1})$, because the original differential $J^n \to J^{n+1}$ factors through its image Z^{n+1} as $J^n \twoheadrightarrow Z^{n+1} \hookrightarrow J^{n+1}$ and G preserves composition. The second factor $G(Z^{n+1}) \to G(J^{n+1})$ is a monomorphism (the image of the left-exact application of G to (S_{n+1}), whose first map $G(Z^{n+1}) \hookrightarrow G(J^{n+1})$ is mono). A monomorphism does not change kernels under precomposition, so $\ker(G(J^n) \to G(J^{n+1})) = \ker(G(J^n) \to G(Z^{n+1})) = G(Z^n)$. Hence $G(Z^n) \xrightarrow{\cong} \ker(G(J^n) \to G(J^{n+1}))$ as subobjects of $G(J^n)$. For $n = 0$, $Z^0 = A$ gives $G(A) \cong \ker(G(J^0) \to G(J^1)) = H^0(G(J^\bullet))$, the last equality being the definition of the zeroth cohomology of a cochain complex starting in degree 0 (no incoming differential). The use of $R^0G \cong G$ (Lemma 153) is what licenses reading $G(A)$ as $(R^0G)(A)$ when this identification feeds the comparison theorem. ☐

Lemma 170 (Cohomology of the applied resolution as a cosyzygy cokernel)**.** *Source: Stacks Project, Derived Categories, Tag 05TA (`proposition-enough-acyclics`) and Tag 015D (`lemma-F-acyclic-ses`). With G additive left exact and the acyclic resolution and cosyzygies of Lemma 168, the cohomology of the applied complex $G(J^\bullet) = (G(J^0) \to G(J^1) \to \cdots)$ is computed by*

$$H^0(G(J^\bullet)) \;\cong\; G(A), \qquad H^n(G(J^\bullet)) \;\cong\; \operatorname{coker}(G(J^{n-1}) \to G(Z^n)) \quad (n \geq 1),$$

where $G(J^{n-1}) \to G(Z^n)$ is G applied to the surjection $J^{n-1} \twoheadrightarrow Z^n$ of (S_{n-1}).

Proof. The degree-zero statement is the $n = 0$ case of Lemma 169: $H^0(G(J^\bullet)) = \ker(G(J^0) \to G(J^1)) \cong G(Z^0) = G(A)$. Fix now $n \geq 1$. By definition of the cohomology of the cochain complex $G(J^\bullet)$ in positive degree,

$$H^n(G(J^\bullet)) \;=\; \ker(G(J^n) \to G(J^{n+1}))\big/\operatorname{im}(G(J^{n-1}) \to G(J^n)).$$

By Lemma 169 the numerator is $\ker(G(J^n) \to G(J^{n+1})) \cong G(Z^n)$, realized as the subobject $G(Z^n) \hookrightarrow G(J^n)$. Under this identification the differential $G(J^{n-1}) \to G(J^n)$ lands inside $G(Z^n)$: it factors as $G(J^{n-1}) \xrightarrow{G(J^{n-1} \twoheadrightarrow Z^n)} G(Z^n) \hookrightarrow G(J^n)$ by the factorization of $J^{n-1} \to J^n$ through its image Z^n, so its image equals the image of $G(J^{n-1}) \to G(Z^n)$ inside the subobject $G(Z^n)$. Therefore the quotient $\ker(G(J^n) \to G(J^{n+1}))/\operatorname{im}(G(J^{n-1}) \to G(J^n))$ is exactly $G(Z^n)/\operatorname{im}(G(J^{n-1}) \to G(Z^n)) = \operatorname{coker}(G(J^{n-1}) \to G(Z^n))$. Hence $H^n(G(J^\bullet)) \cong \operatorname{coker}(G(J^{n-1}) \to G(Z^n))$. ☐

7.7 The acyclic-resolution comparison theorem

Theorem 171 (An acyclic resolution computes the right-derived functor). *Source: Stacks Project, Derived Categories, Tag 015E (Leray's acyclicity lemma) and Tag 05TA (`proposition-enough-acyclics`). Let $\mathcal{A}$ be an abelian category with enough injectives, $\mathcal{B}$ abelian, and $G : \mathcal{A} \to \mathcal{B}$ an additive left-exact functor. Let*

$$0 \longrightarrow A \longrightarrow J^0 \longrightarrow J^1 \longrightarrow J^2 \longrightarrow \cdots$$

be a resolution of A (an exact sequence) in which every object J^n is right-G-acyclic in the sense of Definition 155. Then for every $n \in \mathbb{N}$ the n-th right-derived functor of G at A is computed by the resolution: there is an isomorphism in $\mathcal{B}$

$$(R^nG)(A) \;\cong\; H^n(G(J^\bullet)),$$

where $G(J^\bullet)$ is the cochain complex $G(J^0) \to G(J^1) \to G(J^2) \to \cdots$ obtained by applying G degreewise to the augmentation-dropped resolution and H^n is its n-th cohomology object.

Proof. The argument is the classical dimension shift, now assembled from the cosyzygy infrastructure and the dimension-shift step proved above. Cut the resolution into its cosyzygy short exact sequences $(S_m) : 0 \to Z^m \to J^m \to Z^{m+1} \to 0$ with $Z^0 = A$ and acyclic middle terms, as furnished by Lemma 168.

Degree zero. By Lemma 170, $H^0(G(J^\bullet)) \cong G(A)$, and by $R^0G \cong G$ (Lemma 153) the right-hand side is $(R^0G)(A)$. This is the case $n = 0$.

Positive degrees. Fix $n \geq 1$. The sequences $(S_0), \ldots, (S_{n-2})$ all have acyclic middle terms, so the dimension-shift Lemma 166 applied to them gives, for each $k \geq 1$, isomorphisms $(R^kG)(Z^{m+1}) \cong (R^{k+1}G)(Z^m)$ for $0 \leq m \leq n-2$ (with $Z^0 = A$). Composing them down the staircase

$$(R^nG)(A) \;\cong\; (R^{n-1}G)(Z^1) \;\cong\; (R^{n-2}G)(Z^2) \;\cong\; \cdots \;\cong\; (R^1G)(Z^{n-1})$$

reduces $(R^nG)(A)$ to a single first-derived functor. (For $n = 1$ the staircase is empty and $(R^1G)(A) = (R^1G)(Z^0)$.) Now apply the lowest-degree cokernel description (Lemma 167) to (S_{n-1}), $0 \to Z^{n-1} \to J^{n-1} \to Z^n \to 0$, whose middle term J^{n-1} is acyclic:

$$(R^1G)(Z^{n-1}) \;\cong\; \operatorname{coker}(G(J^{n-1}) \to G(Z^n)).$$

Finally Lemma 170 identifies this cokernel with the cohomology of the applied complex, $\operatorname{coker}(G(J^{n-1}) \to G(Z^n)) \cong H^n(G(J^\bullet))$. Composing the staircase, the cokernel description, and this identification gives $(R^nG)(A) \cong H^n(G(J^\bullet))$ for every $n \geq 1$; together with the degree-zero case this proves the theorem for every $n \in \mathbb{N}$.

Every long exact sequence consumed above is the one produced by the dimension-shift Lemma 166 and its lowest-degree companion Lemma 167, and is therefore grounded — through the horseshoe lift and the complex-level homology long exact sequence on which those lemmas rest — on the injective-resolution computation of R^nG. The injective objects supplied by "enough injectives" are themselves right-G-acyclic (Lemma 152), which is what makes an injective resolution a special case of an acyclic resolution and reconciles the present formula with the defining computation $(R^nG)(A) \cong H^n(G(I^\bullet))$ for an injective resolution $I^\bullet$. $\square$

Chapter 8

Čech computation of higher direct images $R^i f_*$ (unconditional)

8.1 Setup and motivation

This chapter constructs the higher derived direct images $R^i f_* \mathcal{F}$ for $i \geq 1$ *without appealing to injective resolutions in the category of sheaves of modules.* The companion development of Chapter 6 defines $R^i f_*$ as a right derived functor via injective resolutions, which requires the ambient category of $\mathcal{O}_X$-modules to have enough injectives. The Čech approach developed here sidesteps this requirement: it computes $R^i f_* \mathcal{F}$ as the cohomology of an explicit complex built from the pushforwards of $\mathcal{F}$ over the finite intersections of an affine open cover, and so produces an *unconditional* construction of $R^i f_*$ for quasi-coherent $\mathcal{F}$ and separated quasi-compact f.

Throughout, $f : X \to S$ is a morphism of schemes that is quasi-compact and separated (so that all finite intersections of an affine open cover of X are again affine), and $\mathcal{F}$ is a quasi-coherent $\mathcal{O}_X$-module. We write $\mathrm{QCoh}(X)$ for the category of quasi-coherent $\mathcal{O}_X$-modules. A base change of f along a morphism $g : S' \to S$ is recorded by the cartesian square obtained from the fibre product $X' = X \times_S S'$:

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow{g} & S \end{array}$$

Here $f' : X' \to S'$ is the base change of f, $g' : X' \to X$ is the first projection, and $\mathcal{F}' = (g')^* \mathcal{F}$ is the pullback of $\mathcal{F}$ to X'.

8.2 The Čech nerve and Čech complex of an affine cover

Definition 172 (Čech nerve of an affine open cover)**.** *Source: Stacks Project, Cohomology of Schemes, §Čech cohomology of quasi-coherent sheaves.* Let X be a scheme and let $\mathcal{U} : X = \bigcup_{i \in I} U_i$ be a finite open cover of X by affine opens, with index set I. For a finite nonempty tuple $i_0, \dots, i_p \in I$ write $U_{i_0 \cdots i_p} = U_{i_0} \cap \cdots \cap U_{i_p}$ for the corresponding intersection. The *Čech nerve* of $\mathcal{U}$ is the augmented *cosimplicial* object in $\mathrm{QCoh}(X)$ whose object in cosimplicial degree p is the direct image, along the open immersions $j_{i_0 \cdots i_p} : U_{i_0 \cdots i_p} \hookrightarrow X$, of the restriction of $\mathcal{O}_X$ (or of any chosen quasi-coherent sheaf) to the $(p+1)$-fold intersections,

$$[p] \longmapsto \prod_{(i_0,\dots,i_p)\in I^{p+1}} (j_{i_0\cdots i_p})_* (\,\cdot\,|_{U_{i_0\cdots i_p}}),$$

with cofaces given by the restriction maps that omit one index and codegeneracies by the maps that repeat one index; the augmentation in degree -1 is the object on all of X. The variance is essential: the geometric nerve of the cover (the iterated fibre powers of $\coprod_i U_i$ over X, introduced in §8.2.1 below) is a *simplicial* object in *schemes*, but applying the covariant direct-image functor turns it into a *cosimplicial* object in $\mathrm{QCoh}(X)$; consequently the associated alternating-coface complex is a *cochain* complex (differentials raising degree), not a chain complex. When X is separated each intersection $U_{i_0\cdots i_p}$ is itself affine. The local affine model is the *standard open covering* $U = \bigcup_{i=1}^n D(f_i)$ of an affine open $U = \mathrm{Spec}(A)$ by basic opens, where $f_1,\dots,f_n \in A$ generate the unit ideal; on such a cover the intersections are again basic opens $D(f_{i_0}\cdots f_{i_p})$.

Definition 173 (Čech complex of a quasi-coherent sheaf)**.** *Source: Stacks Project, Cohomology of Schemes, `lemma-cech-cohomology-quasi-coherent-trivial` (standard-cover Čech vanishing).* Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module and let $\mathcal{U} : X = \bigcup_{i\in I} U_i$ be a finite affine open cover with all intersections $U_{i_0\cdots i_p}$ affine (e.g. X separated). The *Čech complex* $\check{\mathcal{C}}^\bullet(\mathcal{U},\mathcal{F})$ is the cochain complex obtained by applying the global-sections functor to the Čech nerve of Definition 172 evaluated at $\mathcal{F}$; in degree p it is

$$\check{\mathcal{C}}^p(\mathcal{U},\mathcal{F}) = \prod_{(i_0,\dots,i_p)\in I^{p+1}} \mathcal{F}(U_{i_0\cdots i_p}),$$

with differential $d : \check{\mathcal{C}}^p \to \check{\mathcal{C}}^{p+1}$ the alternating sum $(ds)_{i_0\cdots i_{p+1}} = \sum_{j=0}^{p+1}(-1)^j\, s_{i_0\cdots\widehat{i_j}\cdots i_{p+1}}|_{U_{i_0\cdots i_{p+1}}}$ of the restriction maps. Over an affine $U = \mathrm{Spec}(A)$ with $\mathcal{F}|_U = \widetilde{M}$ and a standard cover by the $D(f_i)$, this is the complex $\prod_{i_0} M_{f_{i_0}} \to \prod_{i_0 i_1} M_{f_{i_0} f_{i_1}} \to \cdots$ of localisations. The *relative Čech complex* $\check{\mathcal{C}}^\bullet(\mathfrak{U},\mathcal{F})$ for $f : X \to S$ is the analogous complex in $\mathrm{QCoh}(S)$ obtained by applying f_* over each finite intersection instead of global sections; its terms are $\prod_{i_0\cdots i_p} (f|_{U_{i_0\cdots i_p}})_*(\mathcal{F}|_{U_{i_0\cdots i_p}})$, and it is a complex in $\mathrm{QCoh}(S)$ because each $U_{i_0\cdots i_p}$ is affine and the pushforward of a quasi-coherent sheaf along a quasi-compact quasi-separated morphism is quasi-coherent.

8.2.1 The three-part construction of the relative Čech complex

It is convenient to factor the construction of Definitions 172–173 into three independent pieces. Two of them are purely formal and unconditional; the sole non-trivial mathematical content is isolated in the middle piece.

(1) Geometric backbone — the augmented Čech nerve of an arrow. Package the affine cover as a single morphism $\coprod_{i\in I} U_i \longrightarrow X$ (the canonical map out of the coproduct induced by the inclusions $U_i \hookrightarrow X$). The augmented Čech nerve of this arrow is an augmented *simplicial scheme* over X: its object in simplicial degree p is the $(p+1)$-fold fibre power

$$\Bigl(\coprod_i U_i\Bigr)^{\times_X (p+1)} = \coprod_{(i_0,\dots,i_p)\in I^{p+1}} U_{i_0} \times_X \cdots \times_X U_{i_p},$$

with the augmentation the cover map to X. Because the U_i are open subschemes of X, each fibre product $U_{i_0} \times_X \cdots \times_X U_{i_p}$ is canonically the intersection $U_{i_0\cdots i_p}$. This backbone exists *unconditionally*: it uses only that the category of schemes has coproducts and finite limits (hence the wide pullbacks underlying the Čech nerve of an arrow). It carries no sheaf-theoretic data yet — it is the simplicial diagram of spaces over which the construction is indexed.

(2) Push–pull functor — lifting the backbone to modules. The geometric backbone is turned into a cosimplicial $\mathcal{O}_X$-module by the *relative-direct-image functor on the over-category*

$$G : (X/\mathbf{Sch})^{\mathrm{op}} \longrightarrow \mathrm{QCoh}(X), \qquad (Y \xrightarrow{p} X) \;\longmapsto\; p_* \, p^* \mathcal{F},$$

sending an X-scheme $p : Y \to X$ to the pushforward along p of the pullback $p^*\mathcal{F}$. Composing G with the (opposite of the) simplicial backbone of (1) produces exactly the augmented cosimplicial object of Definition 172: on the degree-p piece $p : U_{i_0 \cdots i_p} \hookrightarrow X$ one has $p_*p^*\mathcal{F} = (j_{i_0\cdots i_p})_*(\mathcal{F}|_{U_{i_0\cdots i_p}})$, and a morphism of X-schemes $h : (Y_1, p_1) \to (Y_2, p_2)$ (so $h \circ p_2 = p_1$, reading p as the structure map) induces the restriction comparison $G(p_2) \to G(p_1)$ via the unit/counit of the pull-push adjunction together with the over-triangle identifying the two structure maps. This functor G is the *one non-formal step* in the whole construction. Its functoriality — that G respects identities and composition of morphisms over X — is precisely a statement about the coherence of the comparison isomorphisms $(p \circ q)_* \cong p_*q_*$ and $(p \circ q)^* \cong q^*p^*$ for pushforward and pullback of sheaves of modules, together with their compatibility with the adjunction units. Concretely, pullback and pushforward of sheaves of modules are organised as a pseudofunctor $\mathrm{LocallyDiscrete}(\mathbf{Sch}^{\mathrm{op}}) \to \mathbf{Adj\,Cat}$ (sending a morphism to the adjoint pair $p^* \dashv p_*$), and the comparison isomorphisms $(p \circ q)^* \cong q^*p^*$, $(p \circ q)_* \cong p_*q_*$ are its structure 2-cells. The functoriality of G needs exactly three facts: the mate identity expressing the conjugate of the inverse pullback comparison as the pushforward comparison, its normalisation at the identity morphism, and the pseudofunctor unitor and pentagon (associativity) coherences. The laws are nonetheless not formal trivialities: each is a multi-step mate calculation that transports the comparison isomorphisms along the over-triangle $g \circ p_2 = p_1$ of structure maps and past the adjunction unit, transporting along the over-triangle equality; the obstacle is the bookkeeping of those transports, not a missing geometric ingredient.

(3) Coherence-free plumbing — from the nerve to the cochain complex. The passage from the augmented cosimplicial $\mathcal{O}_X$-module of (1)–(2) to the relative Čech complex in $\mathrm{QCoh}(S)$ is again purely formal and uses *no* pushforward/pullback coherence. One forgets the augmentation, pushes the cosimplicial object forward along $f : X \to S$ (a covariant operation applied degreewise), and takes the alternating-coface-map cochain complex of the resulting cosimplicial object of $\mathcal{O}_S$-modules. This step relies only on the preadditivity of $\mathrm{QCoh}(S)$: the alternating sum $\sum_j (-1)^j d^j$ of the cofaces is defined in any preadditive category and squares to zero by the cosimplicial identities. Thus the relative Čech complex $\check{\mathcal{C}}^\bullet(\mathfrak{U}, \mathcal{F})$ of Definition 173 is genuinely *defined* in terms of the Čech nerve of Definition 172: once the nerve is constructed, the complex follows with no further input. In summary, of the three pieces only the push–pull functor (2) requires content beyond formal category theory, and that content is the pseudofunctor coherence of pushforward and pullback: the comparison 2-cells of the pseudofunctor $\mathrm{LocallyDiscrete}(\mathbf{Sch}^{\mathrm{op}}) \to \mathbf{Adj\,Cat}$ and their unitor/pentagon identities.

Formal bricks of the three-part construction. The geometric backbone (1), the push–pull functor (2), and the plumbing (3) are recorded as the following declarations. The backbone and plumbing are unconditional; the push–pull object and morphism maps are likewise unconditional bricks, and the functor laws assembling them are the lone coherence statement.

Definition 174 (Arrow of an affine cover)**.** Let $\mathcal{U} : X = \bigcup_{i \in I} U_i$ be an affine open cover. Its *cover arrow* is the single morphism $\coprod_{i \in I} U_i \longrightarrow X$ obtained as the universal map out of the coproduct induced by the open immersions $U_i \hookrightarrow X$. Packaging the cover as one arrow lets the generic Čech-nerve-of-an-arrow machinery apply.

Definition 175 (Augmented Čech nerve of a cover)**.** The augmented Čech nerve of the cover arrow of Definition 174: the augmented *simplicial scheme* over X whose degree-p object is the $(p+1)$-fold fibre power

$$\Big(\coprod_i U_i\Big)^{\times_X (p+1)} = \coprod_{(i_0,\dots,i_p) \in I^{p+1}} U_{i_0} \times_X \cdots \times_X U_{i_p},$$

with augmentation the cover map to X. It exists unconditionally because the category of schemes has all finite limits (hence the wide pullbacks underlying the Čech nerve of an arrow). This is the geometric backbone (1).

Definition 176 (Push–pull object map)**.** The object map of the push–pull functor G: to an X-scheme $p : Y \to X$ it assigns

$$G(Y, p) \;=\; p_* \, p^* \mathcal{F},$$

the pushforward along p of the pullback $p^*\mathcal{F}$.

Definition 177 (Push–pull comparison map)**.** The morphism map of G. A morphism $g : (Y_2, p_2) \to (Y_1, p_1)$ of X-schemes has underlying map $\bar{g} : Y_2 \to Y_1$ satisfying the over-triangle $p_1 \circ \bar{g} = p_2$; the contravariant functor produces the restriction comparison $G(Y_1, p_1) \to G(Y_2, p_2)$ as the five-step composite

$$p_{1*}\, p_1^* \mathcal{F} \xrightarrow{p_{1*}(\eta)} p_{1*}\, \bar{g}_*\, \bar{g}^*\, p_1^* \mathcal{F} \xrightarrow{\sim} (p_1 \circ \bar{g})_*\, \bar{g}^*\, p_1^* \mathcal{F} \;=\; p_{2*}\, \bar{g}^*\, p_1^* \mathcal{F} \xrightarrow{p_{2*}(\sim)} p_{2*}\, p_2^* \mathcal{F},$$

where η is the unit of the adjunction $\bar{g}^* \dashv \bar{g}_*$, the second arrow is the pushforward comparison $p_{1*}\bar{g}_* \cong (p_1\bar{g})_*$, the equality is the substitution $p_2 = p_1 \circ \bar{g}$ of the over-triangle, and the last arrow applies p_{2*} to the pullback comparison $\bar{g}^* p_1^* \cong (p_1\bar{g})^* = p_2^*$. The two over-triangle substitutions are realised as coercions along $p_1 \circ \bar{g} = p_2$.

Lemma 178 (Push–pull functor — identity law)**.** *The morphism map of Definition 177 respects identities: $G(\mathrm{id}_Y) = \mathrm{id}_{G(Y)}$ for every X-scheme Y.*

Proof. A direct calculation with comparison isomorphisms, with no sectionwise content. Rewrite the head $p_*(\eta)$ followed by the pushforward comparison as the conjugate-mate of the inverse pullback comparison; the identity leg then collapses by the identity normalisation of the pseudo-functor together with right unitality, and the two over-triangle transport coercions cancel against the unitality coercion. ☐

Lemma 179 (Push–pull functor — composition law)**.** *The morphism map of Definition 177 is contravariantly functorial: for composable morphisms $h : Y_3 \to Y_2$, $g : Y_2 \to Y_1$ of X-schemes,*

$$G(g \circ h) = G(h) \circ G(g).$$

Together with Lemma 178 this assembles Definitions 176–177 into a functor $G : (X/\mathbf{Sch})^{\mathrm{op}} \to \mathrm{QCoh}(X)$; composing G with the backbone of Definition 175 yields the Čech nerve of Definition 172.

Proof. The composition identity $G(g \circ h) = G(h) \circ G(g)$ follows from three coherence facts. First, the composite adjunction unit $\eta^{g \circ h}$ decomposes into the iterated units η^g, η^h via the mate identity of Lemma 180 and the naturality of the adjunction unit. Second, the resulting pullback-side identity is the *pentagon coherence* of the pullback pseudofunctor — the associativity 2-cell relating the two bracketings of the comparison isomorphism $(p \circ q \circ r)^* \cong r^* q^* p^*$. Third, the over-triangle transport coherences are absorbed by Lemma 181. □

Lemma 180 (Base-change unit (mate) identity for the push–pull head)**.** *For composable scheme morphisms $f : A \to B$, $p : B \to Z$ and a $\mathcal{O}_Z$-module N, the adjunction unit η^p_N of $p^* \dashv p_*$ at N, followed by p_* of the unit η^f for $f^* \dashv f_*$ at p^*N, followed by the pushforward comparison $p_* f_* \cong (f \circ p)_*$, equals the unit $\eta^{f \circ p}_N$ for the composite followed by $(f \circ p)_*$ applied to the inverse pullback comparison $(f \circ p)^* \cong f^* p^*$:*

$$\eta^p_N \;\circ\; p_*(\eta^f_{p^*N}) \;\circ\; (p_* f_* \cong (f \circ p)_*) \;=\; \eta^{f \circ p}_N \;\circ\; (f \circ p)_*((f \circ p)^* \cong f^* p^*)^{-1}.$$

This is the mate-calculus core that converts the single-morphism unit $\eta^{f \circ p}$ into the iterated units η^p, η^f; it is the reusable ingredient consumed by the functoriality (pentagon) law of the push–pull comparison map of Definition 177.

Proof. A direct mate-calculus computation: unfold both sides using the definition of the adjunction unit for the composite and the pseudofunctor structure maps, then verify the equality by the mate identity for the pushforward/pullback comparison. □

Lemma 181 (Over-triangle transport cancellation for the push–pull tail)**.** *Let $\bar{g} : Y_2 \to Y_1$, $p_1 : Y_1 \to X$, $p_2 : Y_2 \to X$ satisfy the over-triangle $\bar{g} \circ p_1 = p_2$, and let $\mathcal{F}$ be a $\mathcal{O}_X$-module. Then the pullback-comparison leg of the push–pull map, conjugated by the two* eqToHom *coercions along the over-triangle, collapses to the transport-light form*

$$e_1 \;\circ\; p_{2*}((\bar{g}^* p_1^* \cong (\bar{g} p_1)^*)_{\mathcal{F}}) \;\circ\; e_2 \;=\; (\bar{g} p_1)_*((\bar{g}^* p_1^* \cong (\bar{g} p_1)^*)_{\mathcal{F}}) \;\circ\; e_3,$$

where e_1, e_2, e_3 are the eqToHom *coercions induced by $\bar{g} \circ p_1 = p_2$. It states the over-triangle cancellation in the push–pull comparison map of Definition 177 with the over-triangle equality as a free hypothesis, so a single substitution turns the transports into identities; it is the reusable pre-coherence brick for the composition (pentagon) law.*

Proof. Substitute the over-triangle equality $\bar{g} \circ p_1 = p_2$ to replace transport coercions by identities, then verify the cancellation by direct calculation with the pushforward comparison. □

Definition 182 (Push–pull functor)**.** The *push–pull functor*

$$G : (X/\mathbf{Sch})^{\mathrm{op}} \longrightarrow \mathrm{QCoh}(X),$$

assembling the object map $p \mapsto p_* p^* \mathcal{F}$ of Definition 176 and the comparison morphism map of Definition 177 into a genuine functor. Functoriality is exactly the two laws established above: the identity law Lemma 178 and the composition law Lemma 179. The functor is contravariant in the over-category $(X/\mathbf{Sch})$; it is the lift of the geometric backbone (1) to $\mathcal{O}_X$-modules described as step (2) above.

Definition 183 (Čech nerve cosimplicial object)**.** The (non-augmented) cosimplicial $\mathcal{O}_X$-module obtained by transporting the geometric Čech backbone in $(X/\mathbf{Sch})$ of Definition 175 through the push–pull functor G of Definition 182: degreewise, the p-th term is the product over $(i_0, \ldots, i_p) \in I^{p+1}$ of the $(p+1)$-fold-intersection pushforwards $(j_{i_0 \cdots i_p})_*(\mathcal{F}|_{U_{i_0 \cdots i_p}})$. This is the underlying cosimplicial object that the Čech nerve of Definition 172 augments by the structure map $\mathcal{F} \to \mathcal{C}^0$.

Definition 184 (Relative Čech complex from a nerve)**.** Given $f : X \to S$ and an augmented cosimplicial $\mathcal{O}_X$-module N, the *relative Čech cochain complex* in $\mathrm{QCoh}(S)$ obtained by forgetting the augmentation of N, pushing the resulting cosimplicial object forward degreewise along f, and taking the alternating-coface-map cochain complex. This passage uses only the preadditivity of $\mathrm{QCoh}(S)$ and *no* pushforward/pullback coherence; it is the coherence-free plumbing (3) that turns the nerve of Definition 172 into the Čech complex.

8.3 Affine acyclicity of the Čech complex

Definition 185 (Standard affine cover from a spanning family)**.** *Provided by Mathlib (`Mathlib.AlgebraicGeometry.C`*
Let R be a commutative ring (so $R = \Gamma(\operatorname{Spec} R, \mathcal{O})$) and let $s : \iota \to R$ be a family of elements that generate the unit ideal, $\mathrm{Ideal.span}(\mathrm{range}\, s) = \top$. The *standard affine open cover* of $\operatorname{Spec} R$ associated with s is the affine open cover whose index set is ι, whose i-th member is the basic open $D(s_i)$, and whose i-th structure morphism is the map of affine schemes induced by the localisation homomorphism $R \to R_{s_i}$; that is, in Mathlib notation the defining equation $(\mathrm{affineOpenCoverOfSpanRangeEqTop})_f\, i = \mathrm{Spec.\,map}(\mathrm{algebraMap}\ R\ (\mathrm{Localization.\,Away}(s_i)))$, so that the sections over the i-th piece are the *away* localisation $R_{s_i} = \mathrm{Localization.\,Away}(s_i)$ and the geometric piece is $D(s_i)$. This is exactly the standard open covering $\operatorname{Spec} R = \bigcup_i D(s_i)$ by basic opens, packaged as a single cover object. It is the spanning-family bundle (s, hs) that the affine Čech vanishing of Lemma 194 is stated over.

8.3.1 The categorical–module bridge (L1)

The affine vanishing of Lemma 194 concerns the *section* Čech complex $\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F})$, a cochain complex of abelian groups, while its contracting homotopy is most naturally described on a concrete complex of localised R-modules. The passage between them has two parts: identify sections and restriction maps with away localisations and their canonical maps, then translate vanishing of homology into exactness of the resulting module complex. Definition 186 and Lemma 187 give these two parts.

Definition 186. [Quasi-coherent sections over a basic open are an away localisation] *Source: Stacks Project, Schemes, `lemma-spec-sheaves` (Tag 01HV), items (4)–(5).* Let R be a commutative ring, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_{\operatorname{Spec} R}$-module, and write $M = \Gamma(\operatorname{Spec} R, \mathcal{F})$ for its module of global sections. For every $g \in R$, the restriction-of-sections map $M = \mathcal{F}(\operatorname{Spec} R) \to \mathcal{F}(D(g))$ exhibits the section group $\mathcal{F}(D(g))$ as the away localisation M_g of M; that is, this map is an IsLocalizedModule for the multiplicative submonoid $\mathrm{Submonoid.\,powers}\, g$ of powers of g. Moreover, for $D(g_2) \subseteq D(g_1)$ the restriction map $\mathcal{F}(D(g_1)) \to \mathcal{F}(D(g_2))$ is, under these identifications, the canonical localisation map $M_{g_1} \to M_{g_2}$ (the unique R-linear map compatible with the localisation units). Item (4) of Tag 01HV gives the first assertion for $\mathcal{F} = \widetilde{M}$ ($\Gamma(D(g), \widetilde{M}) = M_g$ as an R_g-module) and item (5) gives the second (the restriction along $D(g) \subset D(f)$ is the localisation map $M_f \to M_g$).

For a standard sheaf $\widetilde{M}$, the distinguished-open comparison identifies $\Gamma(D(g), \widetilde{M})$ with M_g, and its naturality identifies restriction maps with localization maps. For arbitrary quasi-coherent $\mathcal{F}$, the affine equivalence $\mathcal{F} \cong \Gamma(\widetilde{\operatorname{Spec} R, \mathcal{F}})$ reduces the assertion to this standard-sheaf case.

Two combinatorial–geometric ingredients package the identification into the form consumed by Lemma 187. First, the multi-index intersection identity $\left(\bigcap_k D(s_{\sigma(k)})\right) = D(\prod_k s_{\sigma(k)})$ in $(\operatorname{Spec} R).\mathrm{Opens}$ (basicOpen_sprod, proved by induction on the index length using the infimum-splitting helper iInf_fin_succ and PrimeSpectrum. basicOpen_mul): it rewrites the $(p+1)$-fold intersection over a multi-index σ as the single basic open $D(s_\sigma)$ of the product $s_\sigma = \prod_k s_{\sigma(k)}$, so the section group there is the away localisation M_{s_σ}. Second, the restriction compatibility of item (5) is made explicit: for a basic-open inclusion $D(b) \subseteq D(a)$ the presheaf restriction map of $\widetilde{M}$ between the two section groups *is* the away-localisation comparison AwayComparison. comparison of the two localisation units (qcohRestriction_eq_comparison, by the uniqueness AwayComparison. comparison_uniq of a map of localised modules); this is the single differential brick that the Čech coface match of Lemma 187 feeds on.

Consequently the canonical localization isomorphisms commute with every face restriction. They identify the section Čech differential with the alternating sum of the corresponding localization maps.

Lemma 187 (Section Čech homology vanishes iff the localised complex is exact)**.** *Source: Stacks Project, Cohomology of Schemes,* `lemma-cech-cohomology-quasi-coherent-trivial` *(identification with the complex of localisations). Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_{\operatorname{Spec} R}$-module and $\mathcal{U}$ the standard cover associated with a spanning family $s : \iota \to R$. Under the section identification of Definition 186, the section Čech complex $\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F})$ of Definition 196 is isomorphic, as a cochain complex, to the concrete complex of localised R-modules*

$$D^\bullet : \quad \prod_{\sigma:\{0\}\to\iota} M_{s_\sigma} \longrightarrow \prod_{\sigma:\{0,1\}\to\iota} M_{s_\sigma} \longrightarrow \cdots, \qquad s_\sigma = \prod_k s_{\sigma(k)},$$

whose differential is the alternating sum of the canonical localisation maps. Its positive-degree homology vanishes:

$$1 \le p \implies \operatorname{IsZero}(\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F}).\mathrm{homology}\, p),$$

This follows from positive-degree exactness of the underlying group homomorphisms, Function. Exact$(D^{p-1} \to D^p \to D^{p+1})$*, which is Lemma 193.*

The degree-p object of $\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F})$ is the categorical product $\prod^c_{\sigma:\mathrm{Fin}(p+1)\to\iota} \mathcal{F}(D(s_\sigma))$ *in the category* Ab *of abelian groups (the alternating-coface-map complex of the* Ab*-valued cosimplicial object* sectionCechCosimplicial*), not an R-module object, so the appropriate exactness criterion is therefore the one in* Ab*. Vanishing of homology is equivalent to exactness at degree p, which in* Ab *is the underlying-group condition $\forall x_2,\ g(x_2) = 0 \Rightarrow \exists x_1,\ f(x_1) = x_2$ (Lemma 191). The categorical product is identified with per-σ localised modules by the element-level product equivalence* $\operatorname{ToType}(\prod^c F_\sigma) \simeq \prod_\sigma \operatorname{ToType}(F_\sigma)$ *(Lemma 188), under which the Čech coface differential matches the localisation differential* dDiff *of $D^\bullet$ (the abstract cosimplicial unfold Lemma 189 followed by the tilde-bridge identification Lemma 190). Transporting the exactness of Lemma 193 across the product equivalence then proves the group-level exactness.*

Proof. The term identification is the degreewise statement of Definition 186: the degree-p section group $\prod_\sigma \mathcal{F}(D(s_\sigma))$ is $\prod_\sigma M_{s_\sigma}$, since each $(p+1)$-fold intersection $D(s_{\sigma(0)}) \cap \cdots \cap D(s_{\sigma(p)})$ is the basic open $D(s_\sigma)$ of the product $s_\sigma = \prod_k s_{\sigma(k)}$ and the sections there are the away localisation M_{s_σ}. The differential identification is item (5) of Definition 186: each face restriction $\mathcal{F}(D(s_{\sigma\circ d_j})) \to \mathcal{F}(D(s_\sigma))$ is the canonical localisation map $M_{s_{\sigma\circ d_j}} \to M_{s_\sigma}$, so the alternating sum of restrictions is the alternating sum of localisation maps, the differential of $D^\bullet$ (the coface match of Lemma 190). The homology vanishing then follows in three steps. First, exactAt_iff_isZero_homology turns IsZero of the homology at p into ExactAt of the short complex at p. Second, because the degree-p objects are *categorical products in* Ab rather than R-modules, the abelian analogue ShortComplex. ab_exact_iff (Lemma 191) — not the module-category criterion — reduces ExactAt to the underlying-group exactness $\forall x_2,\ g(x_2) = 0 \Rightarrow \exists x_1,\ f(x_1) = x_2$. Third, the element-level product equivalence $\operatorname{ToType}(\prod^c F_\sigma) \simeq \prod_\sigma \operatorname{ToType}(F_\sigma)$

of Lemma 188 identifies each product group with the per-σ localised module D^p, so the group-level exactness is exactly $\mathrm{Function.Exact}(D^{p-1} \to D^p \to D^{p+1})$, supplied by dDiff_exact (Lemma 193).

This argument uses the equivalence between localized modules and its essential image: global sections is an inverse there and hence reflects zero homology. It does not require an a priori assertion that global sections preserves homology. ☐

Lemma 188. *[Element-level product equivalence for the section Čech terms] Let $\{F_\sigma\}_{\sigma:\mathrm{Fin}(p+1)\to\iota}$ be the finite family of abelian groups $F_\sigma = \mathcal{F}(D(s_\sigma))$ appearing in degree p of the section Čech complex of Definition 196. There is an equivalence of underlying types*

$$\mathrm{ToType}\Big(\prod_\sigma^c F_\sigma\Big) \simeq \prod_\sigma \mathrm{ToType}(F_\sigma),$$

between the elements of the categorical product $\prod_\sigma^c F_\sigma$ in Ab *and the dependent functions $\sigma \mapsto$ (element of F_σ). It is supplied by the Mathlib concrete-limit comparison* $\mathrm{CategoryTheory.Limits.Concrete.productEquiv}$, *which holds because the forgetful functor* forget Ab *preserves the discrete product (*$\mathrm{PreservesLimit}(\mathrm{Discrete.functor}\,F)$*). Under this equivalence each coordinate* $\mathrm{ToType}(\mathcal{F}(D(s_\sigma)))$ *is identified with the away localisation M_{s_σ} =* dCoeff *of Definition 192 by the* IsLocalizedModule *uniqueness of Definition 186 (both $\mathcal{F}(D(s_\sigma))$ and M_{s_σ} localise M at s_σ, so the comparison is the canonical isomorphism). This is the move between the categorical product of* Ab *and the per-σ localised modules. Its packaging as an additive isomorphism — the degreewise vertical map of the ladder transport in Lemma 191 — is recorded as* sectionProdAddEquiv, *the* AddEquiv *underlying the coordinatewise comparison. The coordinate projection of the equivalence is recorded separately as the definitional restatement* sectionCechProductEquiv_apply*: the σ-component of the image of y under the equivalence is the underlying group map of the categorical projection π_σ applied to y.*

Proof. The type-level equivalence is Concrete. productEquiv applied to the family $\sigma \mapsto F_\sigma$; its hypothesis is that forget Ab preserves the discrete-diagram limit, which is the standard instance for Ab. The per-coordinate identification $\mathrm{ToType}(\mathcal{F}(D(s_\sigma))) \cong M_{s_\sigma}$ is the uniqueness clause of IsLocalizedModule: by Definition 186 the restriction map $M \to \mathcal{F}(D(s_\sigma))$ and the localisation unit $M \to M_{s_\sigma}$ are both localisations of M at $\mathrm{Submonoid.powers}(s_\sigma)$ (qcohSectionsAwayLocalized), hence canonically isomorphic over M. Assembling the coordinatewise isomorphisms after the product equivalence gives the stated identification. ☐

Lemma 189 (Abstract cosimplicial unfold of the section Čech differential)**.** *Let $\mathcal{F}$ be a presheaf of $\mathcal{O}_X$-modules and $\mathcal{U} : X = \bigcup_{i\in I} U_i$ an open cover. For a multi-index $\sigma : \mathrm{Fin}(q+2) \to I$ and a face index $i \in \mathrm{Fin}(q+2)$, write*

$$\mathrm{sectionCechFaceRestr}(\sigma, i) : \mathcal{F}\Big(\bigcap_l U_{(\sigma\circ d_i)(l)}\Big) \longrightarrow \mathcal{F}\Big(\bigcap_k U_{\sigma(k)}\Big)$$

for the i-th presheaf face restriction — the value of $\mathcal{F}$ on the inclusion of opens $\bigcap_k U_{\sigma(k)} \subseteq \bigcap_l U_{(\sigma\circ d_i)(l)}$ obtained from the order-preserving coface d_i of the simplex category. Then, read through the product equivalence sectionCechProductEquiv *of Lemma 188, the underlying group action of the degree-q differential* objD q *of the alternating-coface-map complex* AlgebraicTopology. alternatingCofaceM *is the alternating sum of the face restrictions applied to the deleted-face coordinates:*

$$\big(\mathrm{sectionCechProductEquiv}_{q+1}\,(\mathrm{objD}\,q\,t)\big)_\sigma = \sum_{i=0}^{q+1}(-1)^i\ \mathrm{sectionCechFaceRestr}(\sigma, i)\Big(\big(\mathrm{sectionCechProductEquiv}_q\ t\big)_{\sigma\circ}$$

This is purely cosimplicial*: it records the shape of the section Čech differential as an alternating sum of presheaf restrictions, with* no *sheaf-theoretic or localisation identification of those restrictions. It is the abstract shape that the localisation differential* dDiff *of Definition 192 matches once the tilde-bridge identification of Lemma 190 is supplied.*

Proof. Unfold the degree-q map of alternatingCofaceMapComplex on sectionCechCosimplicial: by construction objD q is the alternating sum $\sum_i (-1)^i\, d^i$ of the cosimplicial cofaces, each coface d^i being the Pi. lift whose σ-component projects to the coordinate $\sigma \circ d_i$ and post-composes with the presheaf restriction sectionCechFaceRestr(σ, i). Reading the σ-coordinate through the product equivalence (the coordinate-projection restatement sectionCechProductEquiv_apply) commutes the projection past the categorical sum: applying the sum of Ab-morphisms to an element distributes over the sum (the finite-sum-application identity ab_hom_finsetSum_apply), and each summand becomes $(-1)^i$ times the face restriction of the $(\sigma \circ d_i)$-coordinate. The per-coordinate projection of the cosimplicial coface is the Pi. lift computation Pi. lift_, which identifies the projected coface with sectionCechFaceRestr(σ, i) precomposed with the projection to $\sigma \circ d_i$. Assembling the signs yields the stated alternating sum. □

Lemma 190. *[Coface match: the section Čech differential is the localisation differential] Take* $X = \operatorname{Spec} R$*, the standard affine cover* $\mathcal{U}$ *associated with a spanning family* $s : \iota \to R$*, and* $\mathcal{F} = \widetilde{M}$ *the quasi-coherent sheaf associated with an* R*-module* M *(for a general quasi-coherent* $\mathcal{F}$*, see the note following the statement). In the situation of Lemma 188, the degree-*p *differential of the section Čech complex* $\check{C}^\bullet(\mathcal{U}, \mathcal{F})$*, read through the product equivalence, is exactly the localisation differential* dDiff *of the module complex* $D^\bullet$ *of Definition 192. Concretely, under the per-coordinate identifications each section group* $\mathcal{F}(D(s_\sigma))$ *is the away localisation* M_{s_σ} *and each face restriction* sectionCechFaceRestr(σ, i) *is the away-localisation coface* dCoface*, so the alternating sum of face restrictions of Lemma 189 becomes the alternating sum of localisation cofaces, i.e.* dDiff*.*

Proof. The proof layers in two steps.

(i) Abstract unfold. By Lemma 189, the σ-coordinate of the degree-p differential, read through the product equivalence, is the alternating sum $\sum_i (-1)^i$ sectionCechFaceRestr(σ, i) of presheaf face restrictions applied to the deleted-face coordinates. This fixes the shape of the differential without any sheaf identification of the restrictions.

(ii) Tilde $\mathcal{F}$-bridge. For $\mathcal{F} = \widetilde{M}$ each face restriction is identified with the localisation coface SectionCechModule. dCoface by transporting along a per-coordinate isomorphism. The section group $\mathcal{F}(D(s_\sigma))$ and the localised module M_{s_σ} = LocalizedModule(Submonoid. powers s_σ) M both localise M at the submonoid of powers of $s_\sigma = \prod_k s_{\sigma(k)}$: the former by qcohSectionsAwayLocalized together with the multi-index intersection identity basicOpen_sprod (which rewrites $\bigcap_k D(s_{\sigma(k)})$ as the single basic open $D(s_\sigma)$), the latter by construction. Hence there is a canonical R-linear isomorphism

$$\varphi_\sigma : \mathrm{ToType}(\mathcal{F}(D(s_\sigma))) \xrightarrow{\sim} M_{s_\sigma},$$

the comparison of the two IsLocalizedModule structures, unique by AwayComparison. comparison_unique and supplied by IsLocalizedModule. iso. The naturality square

$$\varphi_\sigma \circ \mathrm{sectionCechFaceRestr}(\sigma, i) = \mathrm{dCoface} \circ \varphi_{\sigma \circ d_i}$$

is exactly qcohRestriction_eq_comparison of Definition 186: the presheaf restriction of $\widetilde{M}$ between the two section groups is the away-localisation comparison of the two localisation units. This face-naturality is recorded in two forms: at the bare localisation level as phiL_naturality and at the sheaf level (with φ_σ = phi) as phi_naturality. Substituting (ii) into (i) and summing with signs identifies the section Čech differential with dDiff.

The isomorphism φ_σ is obtained by matching the two descriptions of the sections of $\widetilde{M}$ over $D(s_\sigma)$: the cosimplicial functor provides one description and the localisation property of Definition 186 provides the other. Their agreement, together with the naturality square, is the content of this bridge. $\square$

Lemma 191 (Abelian-group exactness criterion for the section Čech complex)**.** *For $1 \le p$ the homology of the section Čech complex $\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F})$ of Definition 196 vanishes,* $\mathrm{IsZero}(\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F}).\mathrm{homology}\, p)$. *The argument splits into a homological reduction and a ladder transport:*

- Reduction. *By standard homological algebra, vanishing of the homology at degree $q+1$ is equivalent to exactness of the short complex around that node, and hence to function exactness of the two consecutive coface differentials d^q, d^{q+1} as maps of abelian groups.*
- Ladder transport. *The degreewise additive isomorphism (Lemma 188) between the section complex and the localised-module complex of Lemma 193 transports the R-module exactness across an additive ladder, yielding function exactness for the group-level coface differentials.*

Proof. Fix $p = q + 1 \ge 1$. By standard homological algebra it suffices to show that the two consecutive coface differentials d^q, d^{q+1} of the section complex are exact as maps of abelian groups (exactness at a node iff vanishing homology, via the short complex around it).

The function exactness is established by an additive ladder transport. The vertical maps of the ladder are the degreewise additive isomorphisms e_p: the product equivalence of Lemma 188 composed with the coordinatewise comparison φ_σ that identifies each section group with the localised module M_{s_σ}. The two commuting squares of the ladder follow from Lemma 189 (the differential unfolds as an alternating sum of face restrictions) and Lemma 190 (each face restriction equals the localisation coface), giving $e_{p+1} \circ d^p = d^p_M \circ e_p$. The horizontal exactness is the R-module exactness of Lemma 193; transporting it across the additive ladder yields function exactness for the coface differentials of the section complex. $\square$

Definition 192 (The un-localised section Čech module complex $D^\bullet$)**.** Let R be a commutative ring, $M = \Gamma(\operatorname{Spec} R, \mathcal{F})$ the global sections of a quasi-coherent $\mathcal{O}_{\operatorname{Spec} R}$-module $\mathcal{F}$, and $s : \iota \to R$ (ι finite) a spanning family of the unit ideal, $\mathrm{Ideal.span}(\mathrm{range}\, s) = \top$ (Definition 185). The *un-localised section Čech module complex* $D^\bullet$ is the cochain complex of R-modules whose degree-p term is the product over multi-indices $\sigma : \{0, \dots, p\} \to \iota$ of the away localisations

$$D^p = \prod_\sigma M_{s_\sigma}, \qquad s_\sigma = \prod_k s_{\sigma(k)}, \qquad M_{s_\sigma} = \mathrm{LocalizedModule}(\mathrm{Submonoid.powers}(s_\sigma))\, M,$$

with differential $d^p : D^p \to D^{p+1}$ the alternating sum $\sum_j (-1)^j \delta^j$ of the coface comparison maps δ^j, each of which is the canonical away-localisation comparison $M_{s_{\sigma \circ d_j}} \to M_{s_\sigma}$ sending a fraction to its image under the further localisation at the omitted factor. The single-coefficient comparison map and its localisation property are recorded by the `dCoface` and `dCoeff` data and the differential by `dDiff` (with action `dDiff_apply`); the auxiliary `spanIdx`/`spanIdx_spec` select a witnessing index of the spanning relation.

The complex $D^\bullet$ is tied to the section Čech complex of Definition 196 and to its localisations by three comparison families. First, the maps `dToCech` identify each term D^p with the degree-p section group $\prod_\sigma \mathcal{F}(D(s_\sigma))$ of Definition 186 as a localised module (`dToCech_isLocalizedModule`), intertwining the differentials (`cechCoface_dToCech`, `dToCech_comm`); the section-side cofaces are `cechCofaceLin` with action `cechCoface_apply`. Second, localising $D^\bullet$ at a spanning element s_r reproduces the *localised* Čech complex: the localised differential `locDiff` (action `locDiff_apply`) agrees, via `locDiff_eq_depDiff`, with the dependent-coefficient combinatorial differential, and its exactness is recorded as `locDiff_exact`. The localisation comparison maps `fLoc` (action `fLoc_apply`) exhibit each localised term as a localised module (`fLoc_isLocalizedModule`) intertwining `dDiff` with `locDiff` (`locDiff_fLoc`, `map_dDiff_eq_locDiff`). The key algebraic fact underlying all three families is the localisation transitivity $M_a[1/b] = M_{ab}$: localising the away module M_{s_σ} further at s_r gives $M_{s_r s_\sigma}$. This is the keystone `AwayComparison.comparison_isLocal` (the composite of two away localisations is again the away localisation at the product), with the cancellation bookkeeping `AwayComparison.Inverts.smul_pow_cancel`. Thus localising $D^\bullet$ at s_r (an IsLocalizedModule. Away) reproduces the localised Čech complex of Lemma 187, and exactness of $D^\bullet$ can be checked one localisation at a time.

Lemma 193 (Positive-degree exactness of the section Čech module complex)**.** *With the notation of Definition 192, the un-localised section Čech module complex $D^\bullet$ is exact in positive degrees: for every p,*

$$\mathrm{Function.Exact}(d^{p+1} : D^{p+1} \to D^{p+2},\ d^{p+2} : D^{p+2} \to D^{p+3}).$$

The homology vanishing of the section Čech complex follows from this exactness by Lemma 187.

Proof. By the local-to-global criterion for a spanning family, it suffices to prove exactness after localizing at each s_r. Localization transitivity identifies the localized differential with the Čech differential whose coefficients are all localized at s_r. Since s_r is then invertible, prepending the fixed index r defines a contracting homotopy and gives $dh + hd = \mathrm{id}$. Thus the complex is exact after every localization in the spanning family, and hence exact before localization. □

Lemma 194. *[Standard-cover Čech-complex vanishing on affines] Source: Stacks Project, Cohomology of Schemes, `lemma-cech-cohomology-quasi-coherent-trivial` (standard-cover Čech vanishing). Let R be a commutative ring, $U = \mathrm{Spec}(R)$ the associated affine scheme, and $\mathcal{F}$ a quasi-coherent $\mathcal{O}_U$-module ($\mathcal{F} : (\mathrm{Spec}\, R).\mathrm{Modules}$). Let $s : \iota \to R$ (ι finite) be a spanning family of the unit ideal, $\mathrm{Ideal.span}(\mathrm{range}\, s) = \top$, and let $\mathcal{U} : U = \bigcup_{i\in\iota} D(s_i)$ be the associated standard affine cover of Definition 185; the index set and the affine pieces $D(s_i)$ with sections R_{s_i} are read off directly from the spanning family. The statement is* intrinsic *to* $\mathrm{Spec}\, R$*: there is no relative morphism f and no pushforward in the conclusion. Then the* section Čech complex $\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F})$ *of Definition 196 — the cochain complex of abelian groups whose degree-p term is $\prod_\sigma \mathcal{F}(D(s_\sigma))$ — has vanishing cohomology in all positive degrees:*

$$\check{H}^p(\mathcal{U}, \mathcal{F}) = 0 \qquad \text{for all } p > 0,$$

where $\check{H}^p$ denotes the cohomology of the section complex $\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F})$. Equivalently, the extended complex $0 \to M \to \prod_{i_0} M_{s_{i_0}} \to \prod_{i_0 i_1} M_{s_{i_0} s_{i_1}} \to \cdots$ of localisations of $M = \Gamma(\mathrm{Spec}\, R, \mathcal{F})$ is exact in positive degrees. This is the standard-cover Čech vanishing only; the sheaf-cohomology Serre vanishing $H^p(U, \mathcal{F}) = 0$ is the separate Lemma 228 below, obtained from this complex vanishing by the basis-comparison Lemma 328.

Proof. Write R for the ring, $M = \Gamma(\mathrm{Spec}\, R, \mathcal{F})$ for the global sections, and $(s : \iota \to R,\ hs : \mathrm{Ideal.span}(\mathrm{range}\, s) = \top)$ for the spanning family of Definition 185.

Reduction to the localised complex. By the section identification of Definition 186, each $(p+1)$-fold intersection $D(s_{\sigma(0)}) \cap \cdots \cap D(s_{\sigma(p)})$ is the basic open $D(s_\sigma)$ of the product $s_\sigma = \prod_k s_{\sigma(k)}$, and its sections are the away localisation $\mathcal{F}(D(s_\sigma)) = M_{s_\sigma}$; under these identifications each face restriction is the canonical localisation map. Hence Lemma 187 identifies the section Čech complex $\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F})$ of Definition 196 with the concrete complex of localised R-modules

$$\prod_{i_0} M_{s_{i_0}} \to \prod_{i_0 i_1} M_{s_{i_0} s_{i_1}} \to \prod_{i_0 i_1 i_2} M_{s_{i_0} s_{i_1} s_{i_2}} \to \cdots,$$

and reduces the vanishing IsZero($\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F})$.homology p) for $p > 0$ to positive-degree Function. Exact of the underlying localisation maps. Equivalently, the extended complex $0 \to M \to \prod_{i_0} M_{s_{i_0}} \to \cdots$ (whose truncation is studied in Algebra, Lemma `algebra-lemma-cover-module`) is exact in positive degrees. For $\mathcal{F} = \widetilde{M}$, the coface identification is Lemma 190. The general case follows from the affine equivalence $\mathcal{F} \cong \widetilde{\Gamma\mathcal{F}}$.

Local exactness. By the local-to-global exactness criterion for the spanning family (s, hs), it remains to prove exactness after localization at each s_r. In that localization, s_r is invertible. Prepending the fixed index r therefore defines a homotopy h satisfying $dh + hd = \mathrm{id}$: the term involving the first coface is the identity, while the remaining terms cancel in pairs after shifting the omitted index. Hence every localized complex is contractible in positive degrees. The local-to-global criterion proves exactness of the original complex, and Lemma 187 gives the asserted vanishing. □

8.3.2 Presheaf-level Čech machinery

The Čech-acyclicity of injective $\mathcal{O}_X$-modules (Lemma 214 below) is not a statement about a single affine cover: it is proved on the abelian category $\mathrm{PMod}(\mathcal{O}_X)$ of presheaves of $\mathcal{O}_X$-modules, where the Čech complex of sections is the Hom-complex of an explicit free resolution. The argument rests on two presheaf-level facts, recorded here, and one transfer of injectivity. First, a free complex $K(\mathcal{U})_\bullet$ of presheaves of modules whose Hom-complex into $\mathcal{F}$ *is* the Čech complex of sections (Definition 195, Definition 196 and Lemma 197). Second, the resolution property of that complex: $K(\mathcal{U})_\bullet$ resolves $\mathcal{O}_\mathcal{U}$ concentrated in degree 0 (Lemma 208). Together these give: an injective $\mathcal{O}_X$-module is injective as a presheaf (Lemma 214 via Lemma 209 applied to the sheafification adjunction Lemma 210), so $\mathrm{Hom}(-, \mathcal{I})$ is exact and carries the augmented resolution to an exact section complex, i.e. $\check{H}^p(\mathcal{U}, \mathcal{I}) = 0$ for $p > 0$. This direct route uses neither enough-injectives in $\mathrm{PMod}(\mathcal{O}_X)$ nor a right-derived δ-functor identification.

Definition 195. [Free presheaf complex of a cover] *Source: Stacks Project, Cohomology,* `lemma-cech-map-into` *(the free complex $K(\mathcal{U})_\bullet$).* Let X be a ringed space and $\mathcal{U} : U = \bigcup_{i \in I} U_i$ an open covering, with $j_{i_0 \cdots i_p} : U_{i_0 \cdots i_p} \hookrightarrow X$ the open immersion of each finite intersection. The Stacks formulation uses, for each intersection $U_{i_0 \cdots i_p}$, the *extension-by-zero* presheaf $(j_{i_0 \cdots i_p})_! \mathcal{O}_X|_{U_{i_0 \cdots i_p}}$, the presheaf of modules sending an open W to $\mathcal{O}_X(W)$ when $W \subset U_{i_0 \cdots i_p}$ and to 0 otherwise. This extension-by-zero presheaf is, canonically, the *free presheaf of modules on the representable* $\mathrm{free}(\mathbf{y}\, U_{i_0 \cdots i_p})$: writing $\mathbf{y}\, V = \mathrm{Hom}(-, V)$ for the Yoneda presheaf of the open V on the site of opens of X, the free presheaf of modules $\mathrm{free}(\mathbf{y}\, V)$ sends W to the free $\mathcal{O}_X(W)$-module on the set $\mathrm{Hom}(W, V)$, which is $\mathcal{O}_X(W)$ if $W \subset V$ and 0 otherwise — exactly extension-by-zero of $\mathcal{O}_X|_V$. Its defining universal property

$$\mathrm{Hom}_{\mathrm{PMod}(\mathcal{O}_X)}(\mathrm{free}(\mathbf{y}\, V), \mathcal{F}) \;\cong\; \mathcal{F}(V)$$

is the free–forgetful adjunction for presheaves of modules composed with the Yoneda isomorphism $\mathrm{Hom}(\mathbf{y}\, V, G) \cong G(V)$. The *free presheaf complex* of $\mathcal{U}$ is the chain complex $K(\mathcal{U})_\bullet$ in the abelian category $\mathrm{PMod}(\mathcal{O}_X)$ of presheaves of $\mathcal{O}_X$-modules whose term in degree p is the direct sum

$$K(\mathcal{U})_p = \bigoplus_{(i_0,\dots,i_p)\in I^{p+1}} \mathrm{free}(\mathbf{y}\, U_{i_0\cdots i_p}),$$

with $K(\mathcal{U})_p$ placed in degree $p \geq 0$ (and 0 in negative degrees), and whose differential $K(\mathcal{U})_{p+1} \to K(\mathcal{U})_p$ is the alternating sum $\sum_{j=0}^{p+1}(-1)^j$ of the maps $\mathrm{free}(\mathbf{y}\, U_{i_0\cdots i_{p+1}}) \to \mathrm{free}(\mathbf{y}\, U_{i_0\cdots\widehat{i_j}\cdots i_{p+1}})$ obtained by applying free to the representable index-dropping maps $\mathbf{y}\, U_{i_0\cdots i_{p+1}} \to \mathbf{y}\, U_{i_0\cdots\widehat{i_j}\cdots i_{p+1}}$ induced by the inclusions of opens $U_{i_0\cdots i_{p+1}} \subset U_{i_0\cdots\widehat{i_j}\cdots i_{p+1}}$ that drop the index i_j. Each summand is a *projective* object of $\mathrm{PMod}(\mathcal{O}_X)$, because the universal property above identifies $\mathrm{Hom}_{\mathcal{O}_X}(\mathrm{free}(\mathbf{y}\, U_{i_0\cdots i_p}), -)$ with the exact "sections over $U_{i_0\cdots i_p}$" functor $(-)(U_{i_0\cdots i_p})$; this is the structural feature exploited below.

We assume the index set I is *finite* (so the required coproducts exist in $\mathrm{PMod}(\mathcal{O}_X)$; this matches the finiteness hypothesis of the comparison target). Here the direct sum $\bigoplus$ is the categorical coproduct $\coprod$ (the indexed Sigma colimit), which for a finite index set coincides with the finite biproduct.

The parallel `...Fam` declarations pinned above are the cover-agnostic raw-finite-family mirror of this construction — indexed by an arbitrary finite family $(U_i)_{i\in I}$ of opens carrying *no* covering hypothesis — and feed the family form of the Čech acyclicity lemma (injective_cech_acyclicFam).

Definition 196. [Section Čech complex of a presheaf of modules] *Source: Stacks Project, Cohomology, §Čech cohomology of presheaves.* Let X be a ringed space and $\mathcal{U} : U = \bigcup_{i\in I} U_i$ an open covering. For a presheaf of $\mathcal{O}_X$-modules $\mathcal{F}$, the *section Čech complex* $\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F})$ is the cochain complex of *abelian groups* whose term in degree p is the product of sections over the $(p+1)$-fold intersections,

$$\check{\mathcal{C}}^p(\mathcal{U}, \mathcal{F}) = \prod_{(i_0,\dots,i_p)\in I^{p+1}} \mathcal{F}(U_{i_0\cdots i_p}),$$

each factor being the value of the "sections over $U_{i_0\cdots i_p}$" functor (the evaluation of $\mathcal{F}$ at the open $U_{i_0\cdots i_p}$), with differential the alternating sum of restriction maps

$$d(s)_{i_0\cdots i_{p+1}} = \sum_{j=0}^{p+1} (-1)^j\, s_{i_0\cdots\widehat{i_j}\cdots i_{p+1}}\big|_{U_{i_0\cdots i_{p+1}}}.$$

This object is *distinct* from the relative Čech complex $\check{\mathcal{C}}^\bullet(\mathfrak{U}, \mathcal{F})$ of Definition 173: the latter is the pushforward complex with terms $\prod (f|_{U_{i_0\cdots i_p}})_*(\mathcal{F}|_{U_{i_0\cdots i_p}})$ in $\mathrm{QCoh}(S)$, an object of relative sheaves over the base, whereas the section Čech complex here is the simpler complex of *global sections* of $\mathcal{F}$ over the intersections, an object of abelian groups (the underlying additive structure of the sections). The two must not be conflated: the presheaf-level acyclicity argument below runs entirely on this section complex.

Lemma 197 (The free complex represents the Čech complex)**.** *Source: Stacks Project, Cohomology,* `lemma-cech-map-into`. *Let X be a ringed space and $\mathcal{U} : U = \bigcup_{i\in I} U_i$ an open covering. For every presheaf of $\mathcal{O}_X$-modules $\mathcal{F}$ there is an isomorphism of cochain complexes of* abelian groups

$$\mathrm{Hom}_{\mathcal{O}_X}(K(\mathcal{U})_\bullet, \mathcal{F}) \;=\; \check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F}),$$

natural in $\mathcal{F} \in \mathrm{PMod}(\mathcal{O}_X)$. Here the left side is the cochain complex obtained by applying the contravariant $\mathrm{Hom}_{\mathcal{O}_X}(-, \mathcal{F})$ to the chain complex $K(\mathcal{U})_\bullet$ of Definition 195, and the right side is the section Čech complex of Definition 196.

Proof. The free–forgetful adjunction for presheaves of modules gives, composed with the Yoneda isomorphism $\mathrm{Hom}(\mathbf{y}\,V, \mathcal{F}) \cong \mathcal{F}(V)$, the natural identification, for each multi-index,

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathrm{free}(\mathbf{y}\,U_{i_0\cdots i_p}),\, \mathcal{F}) \;\cong\; \mathrm{Hom}(\mathbf{y}\,U_{i_0\cdots i_p},\, \mathcal{F}) \;\cong\; \mathcal{F}(U_{i_0\cdots i_p}),$$

where the last group is the evaluation of $\mathcal{F}$ at the open $U_{i_0\cdots i_p}$. Taking the product over multi-indices, $\mathrm{Hom}_{\mathcal{O}_X}(K(\mathcal{U})_p, \mathcal{F}) = \prod_{i_0\cdots i_p} \mathcal{F}(U_{i_0\cdots i_p}) = \check{\mathcal{C}}^p(\mathcal{U}, \mathcal{F})$; the direct sum $K(\mathcal{U})_p$ therefore represents the functor $\mathcal{F} \mapsto \prod_{i_0\cdots i_p} \mathcal{F}(U_{i_0\cdots i_p})$. Under this identification the $(-1)^j$ maps $\mathrm{free}(\mathbf{y}\,U_{i_0\cdots i_{p+1}}) \to \mathrm{free}(\mathbf{y}\,U_{i_0\cdots \widehat{i_j}\cdots i_{p+1}})$ defining the differential of $K(\mathcal{U})_\bullet$ become exactly the alternating-sum restriction differential of $\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F})$, so the two cochain complexes coincide; naturality in $\mathcal{F}$ is immediate from the naturality of the free–forgetful adjunction and of Yoneda. The per-multi-index bijection $\mathrm{Hom}_{\mathcal{O}_X}(\mathrm{free}(\mathbf{y}\,U_{i_0\cdots i_p}), \mathcal{F}) \cong \mathcal{F}(U_{i_0\cdots i_p})$ is additive, so it is promoted to an isomorphism of abelian groups (an *additive* equivalence); taking the product over multi-indices and assembling these degreewise additive isomorphisms — which commute with the alternating-sum differentials — yields the complex isomorphism by matching components degree by degree. ☐

Definition 198. [The cover structure presheaf $\mathcal{O}_{\mathcal{U}}$] *Source: Stacks Project, Cohomology,* `lemma-homology-complex`. Let X be a ringed space and $\mathcal{U} : U = \bigcup_{i\in I} U_i$ an open covering. The *cover structure presheaf* $\mathcal{O}_{\mathcal{U}} : X.\mathrm{PresheafOfModules}$ is the image presheaf of the augmentation

$$\coprod_i \mathrm{free}(\mathbf{y}\,U_i) \;\longrightarrow\; \mathbf{1},$$

where the target $\mathbf{1} = \mathrm{PresheafOfModules.\,unit}(X.\mathrm{ringCatSheaf.obj})$ is the structure presheaf $\mathcal{O}_X$ regarded as a presheaf of modules over itself. Concretely $\mathcal{O}_{\mathcal{U}}(W) = \mathcal{O}_X(W)$ when W is contained in some U_i, and 0 otherwise. The degree-0 term $K(\mathcal{U})_0 = \coprod_i \mathrm{free}(\mathbf{y}\,U_i)$ of the free presheaf complex of Definition 195 carries the augmentation map $K(\mathcal{U})_0 \to \mathcal{O}_{\mathcal{U}}$ onto this image presheaf.

The parallel `...Fam` declarations pinned above are the cover-agnostic raw-finite-family mirror of this construction (an arbitrary finite family of opens, no covering hypothesis), feeding the family form of the Čech acyclicity lemma.

Lemma 199 (Objectwise detection of quasi-isomorphisms of presheaf-of-module complexes)**.** *Source: Stacks Project, Cohomology,* `lemma-homology-complex` *(sectionwise reduction). Let R be a presheaf of rings on a category C and let $\varphi : K \to L$ be a morphism of chain complexes in the abelian category* $\mathrm{PMod}(R)$ *of presheaves of R-modules. Suppose that for* every *object V the evaluated morphism — the image of φ under the functor* $(\mathrm{PresheafOfModules.\,evaluation}\,R\,V).\mathrm{mapHomologicalComplex}$*, a morphism of complexes of $R(V)$-modules — is a quasi-isomorphism. Then φ is a quasi-isomorphism. This is the objectwise-reduction step that turns the resolution claim of Lemma 208 into a family of sectionwise (combinatorial) statements: homology in* $\mathrm{PMod}(R)$ *is computed open-by-open, because the evaluation functors* $\mathrm{PresheafOfModules.\,evaluation}\,R\,V$ *are exact (hence preserve homology) and jointly conservative: a morphism in* $\mathrm{PMod}(R)$ *is an isomorphism iff each of its evaluations is an isomorphism, and each evaluation functor preserves homology.*

Proof. A morphism of complexes is a quasi-isomorphism iff it induces an isomorphism on homology in every degree (quasiIso_iff, quasiIsoAt_iff_isIso_homologyMap). Fix a degree i. The homology map $H_i(\varphi)$ is an isomorphism in $\mathrm{PMod}(R)$ iff each of its evaluations is an isomorphism, since the underlying presheaf-of-abelian-groups functor reflects isomorphisms and a morphism of presheaves is an isomorphism iff it is so objectwise (joint conservativity). For each V, evaluation preserves homology, so the evaluation of $H_i(\varphi)$ is the homology map of the evaluated complex morphism; the latter is an isomorphism because the evaluated morphism is a quasi-isomorphism by hypothesis. Hence $H_i(\varphi)$ is an isomorphism for all i, i.e. φ is a quasi-isomorphism. ☐

Lemma 200. *[Sectionwise description of the evaluated free complex] Source: Stacks Project, Cohomology,* `lemma-homology-complex` *(description of the evaluated complex). Fix an open $V \subset X$ and write $I_1 = I_1(V) = \{ i \in I : V \le U_i \}$ for the set of cover members containing V. Evaluating the free presheaf complex $K(\mathcal{U})_\bullet$ of Definition 195 at V gives, degreewise,*

$$K(\mathcal{U})_p(V) = \bigoplus_{\sigma:\mathrm{Fin}(p+1)\to I} \mathrm{free}(\mathbf{y}\, U_\sigma)(V) = \bigoplus_{\substack{\sigma:\mathrm{Fin}(p+1)\to I \\ V\le U_\sigma}} \mathcal{O}_X(V) = \bigoplus_{\sigma:\mathrm{Fin}(p+1)\to I_1} \mathcal{O}_X(V),$$

because $\mathrm{free}(\mathbf{y}\, W)(V) = \mathcal{O}_X(V)$ *when $V \le W$ and 0 otherwise (poset hom-set is a subsingleton), and $V \le U_\sigma = U_{\sigma(0)} \cap \cdots \cap U_{\sigma(p)}$ holds iff σ takes all its values in I_1. Under this identification the differential of $K(\mathcal{U})_\bullet(V)$ is the alternating-sum combinatorial Čech differential of the full simplex on the index set I_1 with coefficients in the single module $\mathcal{O}_X(V)$. Moreover the evaluated augmentation target is $\mathcal{O}_\mathcal{U}(V) = \mathcal{O}_X(V)$ when $I_1 \ne \emptyset$ and $\mathcal{O}_\mathcal{U}(V) = 0$ when $I_1 = \emptyset$. Thus the evaluation at V of the augmented complex $K(\mathcal{U})_\bullet \to \mathcal{O}_\mathcal{U}[0]$ is precisely the augmented combinatorial Čech complex of the simplex on I_1 with coefficients $\mathcal{O}_X(V)$.*

Proof. The degreewise identity is the value at V of the explicit description of the free presheaf $\mathrm{free}(\mathbf{y}\, W)$ recorded in Definition 195: it sends V to the free $\mathcal{O}_X(V)$-module on $\mathrm{Hom}(V, W)$, which is $\mathcal{O}_X(V)$ if $V \le W$ and 0 otherwise. Applying PresheafOfModules. evaluation V to the coproduct decomposition cechFreePresheafComplex_X commutes with the direct sum and collapses each summand to $\mathcal{O}_X(V)$ or 0 according to whether $V \le U_\sigma$, i.e. whether σ factors through I_1; the surviving summands are exactly those indexed by $\sigma : \mathrm{Fin}(p+1) \to I_1$. The evaluated faces are the index-dropping restriction maps, which on the surviving summands are identities of $\mathcal{O}_X(V)$, so the alternating sum is the combinatorial Čech differential. The augmentation value follows from $\mathcal{O}_\mathcal{U}(V) = \mathcal{O}_X(V)$ exactly when some U_i contains V ($I_1 \ne \emptyset$), recorded in Definition 198. ☐

Lemma 201 (Constant-coefficient combinatorial Čech contracting-homotopy engine)**.** *Source: Stacks Project, Cohomology,* `lemma-homology-complex` *(differential and contracting homotopy of the evaluated complex). Fix a nonempty index set J (in the application $J = I_1(V)$), a chosen index $i_{\mathrm{fix}} \in J$, and a single coefficient module M. The* constant-coefficient combinatorial Čech engine *consists of: the alternating-sum Čech differential* combDifferential *on the full simplex over J with coefficients M,*

$$(\mathrm{combDifferential}\, t)_{i_0\cdots i_{p+1}} = \sum_{j=0}^{p+1} (-1)^j\, t_{i_0\cdots\widehat{i_j}\cdots i_{p+1}},$$

which squares to zero, combDifferential ∘ combDifferential = 0 *(*combDifferential_comp*); the* prepend-i_{fix} *homotopy* combHomotopy, $(\mathrm{combHomotopy}\; i_{\mathrm{fix}}\, u)_{i_0\cdots i_{p+1}} = u_{i_1\dots i_{p+1}}$ *when $i_0 = i_{\mathrm{fix}}$ and 0 otherwise; and the contracting identity $d \circ h + h \circ d = \mathrm{id}$ (*combHomotopy_spec*), which yields positive-degree exactness* combDifferential_exact *and the cocycle-splitting form* combDifferential_eq_of_cocycle*. The remaining declarations are sign- and index-bookkeeping consumed by the contracting identity: the succ-shift reindexing* cons_comp_succAbove_succ *and the alternating-sign flip* combSign_flip*, together with the degenerate-case simplification* combHomotopy_zero*. The generic category-theory helper* isZero_sigma_of_forall_isZero *(a coproduct of zero objects is a zero object) is recorded here as well, used to collapse the empty-index direct sums of Lemma 204. This engine is self-contained combinatorics on* Fin. cons, Fin. succAbove, Fin. predAbove, *and* Finset. sum_involution*; it mirrors the contracting homotopy of the Stacks* `lemma-homology-complex` *proof.*

Lemma 202 (The constant-coefficient Čech engine *chain* complex $C_\bullet$)**.** *Source: Stacks Project, Cohomology,* `lemma-homology-complex` *(the evaluated complex $K^{ext}_\bullet(W)$ and its differential).*

*Fix an open V with $I_1(V) = \{ i : V \le U_i \} \ne \emptyset$, and let $\mathcal{O}_X(V)$ be the single coefficient module (*coverSectionModule*). The* engine chain complex $C_\bullet = \mathrm{cechEngineComplex}\,\mathcal{U}\,V$ *is the chain complex of $\mathcal{O}_X(V)$-modules with*

$$C_p \;=\; \coprod_{\sigma:\mathrm{Fin}(p+1)\to I_1(V)} \mathcal{O}_X(V) \qquad (\mathrm{cechEngineX}),$$

the coproduct taken in $\mathrm{ModuleCat}(\mathcal{O}_X(V))$*, and with differential* $d = \mathrm{cechEngineD}$ *given on the injection ι_σ of the index $\sigma : \mathrm{Fin}(p+1) \to I_1(V)$ by the alternating sum of index-drop reindexings,*

$$d \circ \iota_\sigma \;=\; \sum_{i=0}^{p} (-1)^i\, \iota_{\sigma\circ\mathrm{Fin.succAbove}\, i} \qquad (\mathrm{cechEngineD\_}),$$

*i.e. the chain transpose of the index-*raise *differential* FreeCechEngine. combDifferential *of Lemma 201. Then:*

- $d \circ d = 0$ *(*cechEngineD_comp*), so the family assembles into the chain complex* $C_\bullet = \mathrm{ChainComplex.\,of}$ *(*cechEngineComplex*).*
- *The* prepend-i_{fix} *map* $s \mapsto \iota_{\mathrm{Fin.cons}\; i_{\mathrm{fix}}\,\sigma}$ (cechEnginePrepend*, with its injection computation* cechEnginePrepend_*) is a contracting homotopy in positive degree:* $s \circ d + d \circ s = \mathrm{id}$ *(*cechEnginePrepend_spec*).*
- *Hence $C_\bullet$ is exact in positive degree,* $\mathrm{Function.\,Exact}(\mathrm{cechEngineD}(n+1), \mathrm{cechEngineD}\,n)$ *(*cechEngineD_exact*).*

Four object-half bricks realise the degreewise identification $(\mathrm{eval}\,V)(K(\mathcal{U})_p) \cong C_p$ *consumed by Lemma 203: evaluation commutes with the coproduct; the non-surviving summands (those σ with $V \not\le U_\sigma$) drop to the zero object (*cechFreeEvalDropZeros, cechFreeEvalEngine_X*); and the surviving index type is reindexed by the equivalence* $\mathrm{survivingEquiv} : (\mathrm{Fin}(p+1) \to I_1(V)) \simeq \{ \sigma \mathrel{//} V \le \mathrm{coverInterOpen}\,\mathcal{U}\,\sigma \}$*, where the membership criterion* $V \le \mathrm{coverInterOpen}\,\mathcal{U}\,\sigma \iff \forall k,\ V \le U_{\sigma(k)}$ *is* le_coverInterOpen_iff.

The parallel `...Fam` *engine declarations pinned above are the cover-agnostic raw-finite-family mirror of this construction (an arbitrary finite family of opens, no covering hypothesis), feeding the family form of the Čech acyclicity lemma.*

Proof. The complex $C_\bullet$ is the chain/coproduct dual of the constant-coefficient core of Lemma 201. The differential cechEngineD is defined on coproduct injections by the alternating-sign index-drop $\sigma \mapsto \sigma\circ\mathrm{Fin.\,succAbove}\,i$; both $d\circ d = 0$ and the contracting identity $s\circ d + d\circ s = \mathrm{id}$ are proved by extending along the injections with Limits. Sigma. hom_ext and reducing each injection-component to the corresponding combinatorial cancellation of FreeCechEngine. combDifferential_comp respectively FreeCechEngine. combHomotopy_spec (the prepend cechEnginePrepend is the coproduct dual of combHomotopy). Positive-degree exactness cechEngineD_exact reads off the contracting identity. The four object-half bricks are immediate: evaluation preserves $\coprod$ and the surviving summands are picked out by survivingEquiv under the criterion le_coverInterOpen_iff, the rest being zero objects (cechFreeEvalDropZeros). □

Lemma 203. *[Degreewise identification of the evaluated free complex with the engine complex] Source: Stacks Project, Cohomology,* `lemma-homology-complex` *(the evaluated complex and its differential). Fix an open V with $I_1(V) = \{ i : V \le U_i \} \ne \emptyset$. There is an isomorphism of chain complexes of $\mathcal{O}_X(V)$-modules*

$$((\mathrm{eval}\,V).\,\mathrm{mapHomologicalComplex})(\mathrm{cechFreePresheafComplex}\,\mathcal{U}) \;\cong\; C_\bullet, \qquad C_p = \coprod_{\sigma:\mathrm{Fin}(p+1)\to I_1(V)} \mathcal{O}_X(V),$$

where the coproduct is taken in $\mathrm{ModuleCat}(\mathcal{O}_X(V))$ *and the differential of* $C_\bullet$ *is the constant-coefficient Čech differential* FreeCechEngine. combDifferential *on the index set* $J = I_1(V)$ *(Lemma 201). The isomorphism is* degreewise *and* intertwines the evaluated alternating-face differential with combDifferential*. It is the chain/coproduct dual of the section-side cochain identification of Lemma 187; isolating it here gives the empty and nonempty cases (Lemmas 204 and 207) a common, ready degreewise model.*

The parallel `...Fam` *declarations pinned above are the cover-agnostic raw-finite-family mirror of this degreewise identification (an arbitrary finite family of opens, no covering hypothesis), feeding the family form of the Čech acyclicity lemma.*

Proof. Degreewise. By cechFreeEval_X of Lemma 200 evaluation at V commutes with the coproduct, $(\mathrm{eval}\, V)(K(\mathcal{U})_p) \cong \coprod_{\sigma:\mathrm{Fin}(p+1)\to\mathcal{U}.I_0} (\mathrm{eval}\, V)(\mathrm{free}(\mathbf{y}\, U_\sigma))$. Split the index set into the σ landing in $I_1(V)$ and the rest. Each summand with σ valued in $I_1(V)$ (so $V \le U_\sigma$) is $\cong \mathcal{O}_X(V)$ by freeYonedaEval_iso_of_le; each remaining summand is the zero object by freeYonedaEval_isZero_of_not_le. Hence the coproduct collapses to $\coprod_{\sigma:\mathrm{Fin}(p+1)\to I_1(V)} \mathcal{O}_X(V) = C_p$.

Differential match. The evaluated differential is $(\mathrm{eval}\, V)$ applied to the alternating face sum AlternatingFaceMapComplex. objD(cechFreeSimplicial $\mathcal{U}$). The face δ_i reindexes the σ-summand by $\sigma \mapsto \sigma \circ \mathrm{Fin.\,succAbove}\, i$ (the face action of cechFreeSimplicial); restricted to the I_1-summands this is exactly the index-drop that the engine encodes, and the sign $(-1)^i$ is supplied by objD. The match is verified on the coproduct injections by Limits. Sigma. hom_ext, reusing the naturality of Limits. PreservesCoproduct. iso; summing with signs identifies the evaluated differential with FreeCechEngine. combDifferential. This is the hard content of the identification and the chain/coproduct dual of the cochain coface match in Lemma 187.

The three layers commuted past. The degreewise iso is the 3-fold composite

$$(\mathrm{eval}\, V)(K(\mathcal{U})_p) \xrightarrow{\text{cechFreeEval\_X}} \coprod_{\sigma:\mathrm{Fin}(p+1)\to\mathcal{U}.I_0} (\mathrm{eval}\, V)(\mathrm{free}(\mathbf{y}\, U_\sigma)) \xrightarrow{\text{cechFreeEvalDropZeros}} \coprod_{V\le U_\sigma} \mathcal{O}_X(V) \xrightarrow{\text{(Limits.Sigma.wh}}$$

whose layers are: (1) cechFreeEval_X = the comparison Limits. PreservesCoproduct. iso expressing that evaluation at V commutes with the coproduct $\coprod$; (2) cechFreeEvalDropZeros, which kills the summands with $V \not\le U_\sigma$ (these evaluate to zero objects by freeYonedaEval_isZero_of_not_le); (3) (Limits. Sigma. whiskerEquiv survivingEquiv (freeYonedaEval_iso_of_le ...))$^{-1}$, which reindexes the surviving summands by survivingEquiv of Lemma 202 and identifies each evaluated summand with the single coefficient $\mathcal{O}_X(V)$ via freeYonedaEval_iso_of_le.

Naturality past each layer. The reindex survivingEquiv is natural in the face map $\sigma \mapsto \sigma \circ \mathrm{Fin.\,succAbove}\, i$: a surviving index σ (i.e. $V \le U_\sigma$, equivalently $V \le \mathrm{coverInterOpen}\ \mathcal{U}\ \sigma$ by le_coverInterOpen_iff) is carried to a surviving $\sigma \circ \mathrm{Fin.\,succAbove}\, i$ (still $V \le U_{\sigma\circ\mathrm{succAbove}\ i}$, since dropping a coordinate only weakens the meet), so the drop-zeros layer (2) and the whisker layer (3) commute with every face. On a surviving summand the evaluated transition freeYoneda .map(homOfLE ...) collapses to the identity under freeYonedaEval_iso_of_le; hence each face δ_i matches the engine reindexing $\sigma \mapsto \sigma \circ \mathrm{Fin.\,succAbove}\, i$ summand-for-summand, and summing with the objD signs $(-1)^i$ identifies the evaluated alternating-face differential with cechEngineD (= the chain transpose of combDifferential).

The genuine difficulty: variance. The free *chain* differential *lowers* Fin-arity ($\sigma \mapsto \sigma \circ \mathrm{Fin.\,succAbove}\, i$, faces), whereas FreeCechEngine. combDifferential *raises* it (it inserts an index, summing over $i \in I_1$). The two are reconciled because on the finite biproduct $\coprod = \prod$ the insertion/deletion adjunction makes the index-drop chain differential the transpose of the index-raise cochain differential; this transpose is precisely what cechEngineD of Lemma 202 packages, and verifying the commutation amounts to the injectionwise check Limits. Sigma. hom_ext outlined above. ☐

Lemma 204. *[Evaluated free complex over an open meeting no cover member] Source: Stacks Project, Cohomology,* `lemma-homology-complex` *(empty case). With the notation of Lemma 200, suppose $I_1(V) = \emptyset$, i.e. V is contained in no cover member U_i. Then the evaluated augmented complex is identically zero: every degree-p term $\bigoplus_{\sigma:\mathrm{Fin}(p+1)\to I_1} \mathcal{O}_X(V)$ is the empty direct sum 0 (there is no map $\mathrm{Fin}(p + 1) \to \emptyset$) and the augmentation target is $\mathcal{O}_{\mathcal{U}}(V) = 0$. Consequently the evaluation at V of* cechFreeComplexAug *is a morphism of zero complexes, hence a quasi-isomorphism. The same argument applies to an arbitrary finite family of opens, without a covering hypothesis.*

Proof. By Lemma 200 each evaluated term is $\bigoplus_{\sigma:\mathrm{Fin}(p+1)\to I_1} \mathcal{O}_X(V)$; when $I_1 = \emptyset$ the index type $\mathrm{Fin}(p + 1) \to I_1$ is empty, so the term is the zero object, and likewise $\mathcal{O}_{\mathcal{U}}(V) = 0$. A morphism whose source and target complexes are both zero has isomorphisms on all homology objects and is therefore a quasi-isomorphism. ☐

Lemma 205 (Prepend homotopy on the evaluated free complex, by transport)**.** *Source: Stacks Project, Cohomology,* `lemma-homology-complex` *(the contracting map h). With the notation of Lemma 200, suppose $I_1(V) \neq \emptyset$ and fix an index $i_{\mathrm{fix}} \in I_1$. The contracting homotopy is built at the engine-complex level as* cechEnginePrepend *of Lemma 202; its evaluated-complex avatar is obtained by transporting along the degreewise iso* cechFreeEvalEngineIso *of Lemma 203. Concretely the transported* prepend homotopy *is the family of $\mathcal{O}_X(V)$-linear maps*

$$h_p : K(\mathcal{U})_p(V) \longrightarrow K(\mathcal{U})_{p+1}(V), \qquad h_p(s)_{i_0\dots i_{p+1}} = \begin{cases} s_{i_1\dots i_{p+1}} & i_0 = i_{\mathrm{fix}}, \\ 0 & i_0 \neq i_{\mathrm{fix}}, \end{cases}$$

defined on the I_1-indexed direct sum of Lemma 200 as the Sigma. desc *of the per-summand prepend-i_{fix} maps. This is exactly the combinatorial content of* FreeCechEngine. combHomotopy *of Lemma 201, here with the single coefficient module $\mathcal{O}_X(V)$, transported across the degreewise identification of Lemma 203. The maps h_p (together with the degree $(-1) \to 0$ map onto $\mathcal{O}_{\mathcal{U}}(V) = \mathcal{O}_X(V)$) are the homotopy hom maps assembled into a* HomologicalComplex. Homotopy *in Lemma 207.*

Proof. Under the identification of Lemma 200 each evaluated term is a finite direct sum of copies of $\mathcal{O}_X(V)$ indexed by $\sigma : \mathrm{Fin}(p+1) \to I_1$. The prepend-$i_{\mathrm{fix}}$ prescription sends the basis component at $(i_1, \dots, i_{p+1})$ to the component at $(i_{\mathrm{fix}}, i_1, \dots, i_{p+1})$ and is $\mathcal{O}_X(V)$-linear; it is packaged as the Sigma. desc of these per-summand maps, exactly as FreeCechEngine. combHomotopy is defined. This is a definition, so there is nothing to prove beyond well-definedness, which is the linearity of Sigma. desc. ☐

Lemma 206 (The prepend homotopy contracts the augmented evaluated complex, by transport)**.** *Source: Stacks Project, Cohomology,* `lemma-homology-complex` *(the contracting identity). In the situation of Lemma 205 ($I_1(V) \neq \emptyset$, $i_{\mathrm{fix}} \in I_1$ fixed), the transported prepend homotopy h satisfies the contracting identity*

$$d \circ h + h \circ d = \mathrm{id}$$

on the augmented evaluated complex $\cdots \to K(\mathcal{U})_1(V) \to K(\mathcal{U})_0(V) \to \mathcal{O}_{\mathcal{U}}(V) \to 0$ (including the degree-0/degree-(-1) augmentation term). Here d is the alternating-sum Čech differential of Lemma 200.

Proof. This is the same alternating-sum cancellation as FreeCechEngine. combHomotopy_spec (combDifferential(combHomotopy $r\,t$)+combHomotopy r (combDifferential t) $= t$), specialised to coefficients $\mathcal{O}_X(V)$. Expanding $(dh+hd)(s)_{i_0\cdots i_p}$: the $j = 0$ term of dh prepends and then removes i_{fix}, returning $s_{i_0\cdots i_p}$; every remaining term of dh cancels in pairs against the corresponding term of hd (the prepend-i_{fix} of the differential), because removing an index $j \geq 1$ after prepending i_{fix} equals prepending i_{fix} after removing index $j-1$, with opposite signs. The net result is $s_{i_0\cdots i_p}$, i.e. $dh + hd = \text{id}$; the index bookkeeping is the FreeCechEngine. cons_comp_succAbove_succ / FreeCechEngine. combSign_flip content of Lemma 201. ☐

Lemma 207. *[Evaluated free complex over an open meeting a cover member] Source: Stacks Project, Cohomology,* `lemma-homology-complex` *(homotopy equivalence to zero). With the notation of Lemma 200, suppose $I_1(V) \neq \emptyset$. Then the evaluation at V of* cechFreeComplexAug *is a quasi-isomorphism: the augmented evaluated complex is contractible, hence its homology is concentrated in degree 0 where it equals $\mathcal{O}_{\mathcal{U}}(V) = \mathcal{O}_X(V)$.*

The parallel `...Fam` *declarations pinned above are the cover-agnostic raw-finite-family mirror of this nonempty-index case (an arbitrary finite family of opens, no covering hypothesis), feeding the family form of the Čech acyclicity lemma.*

Proof. By Lemmas 205 and 206 the degreewise prepend maps h_p satisfy $dh + hd = \text{id}$ on the augmented evaluated complex, so they assemble into a HomologicalComplex. Homotopy between the identity and the zero endomorphism of that complex. A complex whose identity is homotopic to zero is contractible; equivalently the augmented evaluated complex is a HomotopyEquiv of $K(\mathcal{U})_\bullet(V)$ with $\mathcal{O}_{\mathcal{U}}(V)[0] = \mathcal{O}_X(V)[0]$. A homotopy equivalence is a quasi-isomorphism (Homotopy. toQuasiIso, or HomotopyEquiv. toQuasiIso / QuasiIso. ofHomotopyEquiv), so the evaluated augmentation map is a quasi-isomorphism. ☐

Lemma 208. *[The free complex resolves $\mathcal{O}_U$] Source: Stacks Project, Cohomology,* `lemma-homology-complex`. *Let X be a ringed space and $\mathcal{U} : U = \bigcup_{i\in I} U_i$ an open covering. Let $\mathcal{O}_{\mathcal{U}}$ be the cover structure presheaf of Definition 198, the image presheaf of the augmentation $\bigoplus_i (j_i)_! \mathcal{O}_X|_{U_i} \to \mathcal{O}_X$ (so $\mathcal{O}_{\mathcal{U}}(W) = \mathcal{O}_X(W)$ when W is contained in some U_i, and 0 otherwise). Then the homology presheaves of the free presheaf complex $K(\mathcal{U})_\bullet$ of Definition 195 vanish away from degree 0 and equal $\mathcal{O}_{\mathcal{U}}$ in degree 0:*

$$H_i(K(\mathcal{U})_\bullet) = \begin{cases} \mathcal{O}_{\mathcal{U}} & i = 0, \\ 0 & i \neq 0. \end{cases}$$

Equivalently, the augmented complex $\cdots \to K(\mathcal{U})_1 \to K(\mathcal{U})_0 \to \mathcal{O}_{\mathcal{U}} \to 0$ is exact as a complex of presheaves; that is, $K(\mathcal{U})_\bullet$ is a (projective) resolution of $\mathcal{O}_{\mathcal{U}}[0]$, so the augmentation $K(\mathcal{U})_\bullet \to \mathcal{O}_{\mathcal{U}}[0]$ is a quasi-isomorphism — $K(\mathcal{U})_\bullet$ is quasi-isomorphic to $\mathcal{O}_{\mathcal{U}}$ concentrated in degree 0.

The companion cechFreeComplex_quasiIsoFam *is the cover-agnostic raw-finite-family mirror of this resolution statement — for an arbitrary finite family $(U_i)_{i\in I}$ of opens with* no *covering hypothesis — and is the quasi-isomorphism consumed by the family form of the Čech acyclicity lemma.*

Proof. The claim is that the augmentation chain map $K(\mathcal{U})_\bullet \to \mathcal{O}_{\mathcal{U}}[0]$ (cechFreeComplexAug) is a quasi-isomorphism. By the objectwise-reduction Lemma 199 (quasiIso_of_evaluation) it suffices to show that for every open $V \subset X$ the evaluation ((PresheafOfModules. evaluation V). mapHomologicalComplex) is a quasi-isomorphism of complexes of $\mathcal{O}_X(V)$-modules.

Fix V and set $I_1 = I_1(V) = \{\, i : V \leq U_i \,\}$. By Lemma 200 the evaluated augmented complex is the augmented combinatorial Čech complex of the simplex on I_1 with coefficients

$\mathcal{O}_X(V)$, with augmentation target $\mathcal{O}_{\mathcal{U}}(V)$. Split into two cases on I_1. If $I_1 = \emptyset$, Lemma 204 shows both sides vanish and the evaluated augmentation is a quasi-isomorphism. If $I_1 \neq \emptyset$, Lemma 207 provides the prepend-i_{fix} contracting homotopy, exhibiting the augmented evaluated complex as contractible and hence the evaluated augmentation as a quasi-isomorphism. In either case the hypothesis of Lemma 199 holds for V, so cechFreeComplexAug is a quasi-isomorphism; equivalently the homology presheaves of $K(\mathcal{U})_\bullet$ vanish away from degree 0 and equal $\mathcal{O}_{\mathcal{U}}$ in degree 0.

The contracting homotopy is built directly at the $\mathcal{O}_X(V)$-module (section) level as the prepend-i_{fix} map — the same combinatorial content as FreeCechEngine. combHomotopy / FreeCechEngine. combHom of Lemma 201. ☐

Lemma 209 (Right adjoint of a mono-preserving left adjoint preserves injectives)**.** *Provided by Mathlib (`Mathlib.CategoryTheory.Preadditive.Injective.Basic`). Let $L \dashv R$ be an adjunction between abelian (or, more generally, suitable) categories such that the left adjoint L preserves monomorphisms. Then for every injective object J of the target, $R(J)$ is an injective object of the source.*

Lemma 210 (Sheafification is left adjoint to the inclusion of sheaves of modules)**.** *Provided by Mathlib (`Mathlib.Algebra.Category.ModuleCat.Presheaf.Sheafification`). For the identity ring map $\alpha = \mathbf{1}$ over a scheme X, sheafification is left adjoint to the inclusion* $\mathrm{Mod}(\mathcal{O}_X) \hookrightarrow \mathrm{PMod}(\mathcal{O}_X)$ *of sheaves of $\mathcal{O}_X$-modules into presheaves of $\mathcal{O}_X$-modules:*

$$\text{sheafification} \dashv \iota, \qquad \iota = \textit{the inclusion } X.\text{Modules} \hookrightarrow X.\text{PresheafOfModules}.$$

The left adjoint sheafification is exact, in particular it preserves monomorphisms.

Lemma 211 (Opposite-transport identification of the hom-Čech complex)**.** *Let $\mathcal{U} : U = \bigcup_{i \in I} U_i$ be a finite open covering and $\mathcal{F}$ a presheaf of $\mathcal{O}_X$-modules. There is an isomorphism of cochain complexes of abelian groups*

$$\text{homCechComplex}\,\mathcal{U}\,\mathcal{F} \;\cong\; ((\text{preadditiveYoneda}\,.\text{obj}\,\mathcal{F}).\,\text{mapHomologicalComplex}(\text{up}\,\mathbb{N})).\text{obj}(\text{HomologicalComplex}.$$

exhibiting the Čech hom-complex $\mathrm{Hom}_{\mathcal{O}_X}(K(\mathcal{U})_\bullet, \mathcal{F})$ *of Lemma 197 as the image of the* opposite *of the free presheaf complex (Definition 195) under the covariant* $\mathrm{Hom}(-, \mathcal{F})$ *functor* preadditiveYoneda .obj $\mathcal{F}$*.*

Lemma 212. *[Opposite-transport identification of the section Čech complex] With the notation above, composing the opposite-transport iso* homCechComplexMapOpIso *of Lemma 211 with the Čech hom-identification* cechComplex_hom_identification *of Lemma 197 gives an isomorphism of cochain complexes*

$$((\text{preadditiveYoneda}\,.\text{obj}\,\mathcal{F}).\,\text{mapHomologicalComplex}).\text{obj}(\text{HomologicalComplex}.\,\text{op}(\text{cechFreePresheafComplex}\,\mathcal{U}))$$

i.e. sectionCechComplexMapOpIso *is* $\text{homCechComplexMapOpIso}^{-1}$ *followed by* cechComplex_hom_identification*. This is the bridge that lets the mapped-opposite of the free resolution be transported onto the section Čech complex of Definition 196.*

The companion sectionCechComplexMapOpIsoFam *is the cover-agnostic raw-finite-family mirror of this transport, stated over an arbitrary finite family $(U_i)_{i \in \iota}$ of opens with* no *covering hypothesis.*

Lemma 213 (Hom into an injective preserves exactness and quasi-isomorphisms)**.** *Let $\mathcal{A}$ be an abelian category and I an injective object. Then the contravariant hom functor* $\mathrm{Hom}(-, I)$:

$\mathcal{A}^{\mathrm{op}} \to \mathrm{Ab}$ *is exact: it is automatically left exact (preserves finite limits), and injectivity of* I *makes it carry epimorphisms to epimorphisms, so it preserves finite colimits and hence preserves homology. Consequently* $\mathrm{Hom}(-, I)$ *carries quasi-isomorphisms of chain complexes in* $\mathcal{A}^{\mathrm{op}}$ *to quasi-isomorphisms of the* Hom*-cochain complexes. Both statements are for a general abelian category, the right generality for reuse in Lemma 214. The mechanism — a right adjoint / exact contravariant functor preserves the relevant exactness — is the same homological principle behind Lemma 209.*

Proof. For injective I, injectivity gives that $\mathrm{Hom}(-, I)$ carries epimorphisms to epimorphisms. Combined with the automatic preservation of finite limits (left exactness), the short-exact-sequence criterion shows it preserves finite colimits; hence, being also left exact, it preserves homology. A functor that preserves homology sends quasi-isomorphisms of chain complexes to quasi-isomorphisms degreewise, which is the second statement. □

Lemma 214 (Injective modules are Čech-acyclic)**.** *Source: Stacks Project, Cohomology,* `lemma-injective-trivial-cech`*. Let* X *be a ringed space, let* $\mathcal{U} : U = \bigcup_{i \in I} U_i$ *be any open covering, and let* $\mathcal{J}$ *be an* injective $\mathcal{O}_X$*-module. Then the Čech cohomology of* $\mathcal{J}$ *is concentrated in degree zero:*

$$\check{H}^p(\mathcal{U}, \mathcal{J}) = \begin{cases} \mathcal{J}(U) & p = 0, \\ 0 & p > 0. \end{cases}$$

The positive-degree vanishing holds over an arbitrary finite family of opens $\{U_i\}_{i \in \iota}$*, with no covering hypothesis: the augmented free Čech resolution is exact* stalkwise *— at a point* x *it is the augmented simplicial chain complex of the full simplex on* $\{i \mid x \in U_i\}$ *with coefficients* $\mathcal{O}_{X,x}$*, which is contractible — and a covering hypothesis only fixes the identity of the augmentation target* $\mathcal{O}_{\mathcal{U}}$*, never the exactness. Consequently the lemma applies to any standard covering data of the affine cover system, with no restriction to coverings of the whole space.*

This cover-agnostic form is recorded as injective_cech_acyclicFam*, parameterized by a raw finite family* $\{U_i\}_{i \in \iota}$ *of opens (*ι *finite) carrying* no *covering hypothesis; it consumes the family-form resolution quasi-isomorphism* cechFreeComplex_quasiIsoFam *and is the form invoked over the standard covers of an arbitrary distinguished open* $D(f)$ *by the cover-agnostic affine vanishing statement (Tag 01XB).*

Proof. The argument has two independent parts.

Part 1: an injective sheaf is injective as a presheaf. Sheafification is left adjoint to the inclusion $\iota : \mathrm{Mod}(\mathcal{O}_X) \hookrightarrow \mathrm{PMod}(\mathcal{O}_X)$ of sheaves of $\mathcal{O}_X$-modules into presheaves of $\mathcal{O}_X$-modules (Lemma 210), and its left adjoint sheafification is exact, hence preserves monomorphisms. By Lemma 209 the right adjoint ι carries injective objects to injective objects; thus an injective $\mathcal{O}_X$-module $\mathcal{J}$ is injective as an object of $\mathrm{PMod}(\mathcal{O}_X)$. In particular the contravariant functor $\mathrm{Hom}_{\mathcal{O}_X}(-, \mathcal{J})$ is exact on $\mathrm{PMod}(\mathcal{O}_X)$.

Part 2: vanishing. By Lemma 208 the free presheaf complex $K(\mathcal{U})_\bullet$ of Definition 195 is a resolution of $\mathcal{O}_{\mathcal{U}}$ concentrated in degree 0; equivalently the augmented complex $\cdots \to K(\mathcal{U})_1 \to K(\mathcal{U})_0 \to \mathcal{O}_{\mathcal{U}} \to 0$ is exact in $\mathrm{PMod}(\mathcal{O}_X)$. Applying the exact functor $\mathrm{Hom}_{\mathcal{O}_X}(-, \mathcal{J})$ of Part 1 to this exact augmented complex yields an exact complex; by the identification $\mathrm{Hom}_{\mathcal{O}_X}(K(\mathcal{U})_\bullet, \mathcal{J}) = \check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{J})$ of Lemma 197 (with the section Čech complex of Definition 196), this is precisely the statement that the section Čech complex of $\mathcal{J}$ is exact in positive degrees, with $\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{O}_{\mathcal{U}}, \mathcal{J}) = \mathcal{J}(U)$ supplying degree 0.

Categorical assembly. Concretely, the exactness is produced by the opposite-transport bridge. The free Čech augmentation is a quasi-isomorphism (Lemma 208); forming its opposite complex and applying $\mathrm{Hom}(-, \mathcal{J})$ keeps it a quasi-isomorphism, since $\mathcal{J}$ is injective as a presheaf of modules

(Part 1) and $\mathrm{Hom}(-, \mathcal{I})$ then preserves homology (Lemma 213). Transporting source and target of this opposite quasi-isomorphism across the opposite-transport isomorphisms (Lemma 211 and Lemma 212) identifies it with the augmentation of the section Čech complex $\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{I})$ onto the degree-0-concentrated target $\mathcal{O}_{\mathcal{U}}[0]$. The latter has vanishing homology in positive degrees, so the section Čech complex does too. Hence $\check{H}^p(\mathcal{U}, \mathcal{I}) = 0$ for all $p > 0$ and $\check{H}^0(\mathcal{U}, \mathcal{I}) = \mathcal{I}(U)$. This direct route uses neither enough-injectives in $\mathrm{PMod}(\mathcal{O}_X)$ nor a right-derived δ-functor identification. ☐

Lemma 215. *[Surjectivity on sections from cofinal Čech-H^1 vanishing] Source: Stacks Project, Cohomology,* `lemma-ses-cech-h1`*. Let X be a ringed space and let*

$$0 \to \mathcal{F} \to \mathcal{G} \to \mathcal{H} \to 0$$

be a short exact sequence of $\mathcal{O}_X$-modules. Let $U \subset X$ be open. If there is a cofinal system of open coverings $\mathcal{U}$ of U with $\check{H}^1(\mathcal{U}, \mathcal{F}) = 0$, then the map on sections $\mathcal{G}(U) \to \mathcal{H}(U)$ is surjective. Equivalently, the short exact sequence is exact on sections over such U.

Proof. Let $s \in \mathcal{H}(U)$. Choose an open covering $\mathcal{U} : U = \bigcup_{i \in I} U_i$ fine enough that (a) $\check{H}^1(\mathcal{U}, \mathcal{F}) = 0$ and (b) each restriction $s|_{U_i}$ lifts to a section $s_i \in \mathcal{G}(U_i)$; both can be arranged simultaneously because the coverings with property (a) are cofinal and (b) holds on any sufficiently fine cover (surjectivity of $\mathcal{G} \to \mathcal{H}$ is a local condition). On the double overlaps form

$$s_{i_0 i_1} = s_{i_1}|_{U_{i_0 i_1}} - s_{i_0}|_{U_{i_0 i_1}}.$$

Because the s_i all lift the single section s, the difference $s_{i_0 i_1}$ maps to 0 in $\mathcal{H}$, hence lies in the subsheaf $\mathcal{F}(U_{i_0 i_1})$; the family $(s_{i_0 i_1})$ is a Čech 1-cocycle for $\mathcal{F}$. By $\check{H}^1(\mathcal{U}, \mathcal{F}) = 0$ it is a coboundary: there are $t_i \in \mathcal{F}(U_i)$ with $s_{i_0 i_1} = t_{i_1}|_{U_{i_0 i_1}} - t_{i_0}|_{U_{i_0 i_1}}$. Then the corrected sections $s_i - t_i \in \mathcal{G}(U_i)$ agree on overlaps, so by the sheaf condition they glue to a global section of $\mathcal{G}$ over U whose image in $\mathcal{H}$ is s. Hence $\mathcal{G}(U) \to \mathcal{H}(U)$ is surjective. ☐

8.4 Absolute sheaf cohomology as Ext of the corepresenting object

The two vanishing lemmas below speak about the *absolute* sheaf cohomology $H^p(U, \mathcal{F})$ of an $\mathcal{O}_X$-module over an open U. Their dimension-shift proof needs three structural facts about $H^p(U, -)$: a degree-zero identification with global sections, vanishing on injective modules, and a long exact sequence attached to a short exact sequence of modules. We realize $H^p(U, -)$ as the *Ext of a corepresenting object* in the single abelian category $X.\mathrm{Modules} = \mathrm{Mod}(\mathcal{O}_X)$: there is an $\mathcal{O}_X$-module $j_!\mathcal{O}_U$ corepresenting the sections-over-U functor $\mathcal{F} \mapsto \Gamma(U, \mathcal{F})$, and we set $H^p(U, \mathcal{F}) = \mathrm{Ext}^p_{\mathrm{Mod}(\mathcal{O}_X)}(j_!\mathcal{O}_U, \mathcal{F})$. The corepresenting object is built *without* the extension-by-zero functor — it is the sheafification of the free module on the representable presheaf $\mathbf{y}\,U$ — and the realization keeps the injective module in the *second* Ext argument, where Mathlib supplies all three facts off the shelf with *no* appeal to restriction of module sheaves. The corepresenting object and its corepresentability are fixed first, then the following definition fixes the realization, and the Mathlib dependency anchors after it pin the exact results it rests on.

Definition 216 (Extension-by-zero structure sheaf as a corepresenting object)**.** *Corepresenting object for sections over an open, built without the extension-by-zero functor.* Let X be a scheme and $U \subseteq X$ an open subscheme, regarded through the representable presheaf $\mathbf{y}\,U$ on the site of opens of X. The *extension-by-zero structure sheaf* $j_!\mathcal{O}_U$ is the object of X.Modules = $\mathrm{Mod}(\mathcal{O}_X)$ obtained by sheafifying the free presheaf of modules on $\mathbf{y}\,U$:

$$j_!\mathcal{O}_U \;:=\; \mathrm{sheafification}(\mathrm{free}(\mathbf{y}\,U)) \;\in\; X.\mathrm{Modules},$$

where free is the free presheaf-of-modules functor on the representable (the degree-zero generator of the free presheaf complex of Definition 195) and sheafification $\dashv \iota$ is the sheafification adjunction of Lemma 210. Up to isomorphism this is the classical extension-by-zero $j_!\mathcal{O}_U$ of the structure sheaf of U along the open immersion $j : U \hookrightarrow X$; crucially, we build only this *object* as a sheafified free module, never the extension-by-zero *functor* $j_!$. The object exists unconditionally: the free presheaf of modules on a representable and the sheafification of presheaves of $\mathcal{O}_X$-modules are both supplied by Mathlib.

Lemma 217. *[Corepresentability of global sections over U] The object $j_!\mathcal{O}_U$ corepresents the sections-over-U functor. For every $\mathcal{O}_X$-module $\mathcal{F}$ there is a natural additive isomorphism*

$$\mathrm{Hom}_{X.\mathrm{Modules}}(j_!\mathcal{O}_U,\, \mathcal{F}) \;\cong\; \Gamma(U, \mathcal{F}) \;=\; \mathcal{F}(U),$$

natural in $\mathcal{F}$. Thus $j_!\mathcal{O}_U$ corepresents the global-sections functor $\mathcal{F} \mapsto \Gamma(U, \mathcal{F})$ on X.Modules.

Proof. Compose two natural additive isomorphisms. The sheafification adjunction sheafification $\dashv$ ι of Lemma 210 transports corepresentability through sheafification:

$$\mathrm{Hom}_{X.\mathrm{Modules}}(\mathrm{sheafification}(\mathrm{free}(\mathbf{y}\,U)),\, \mathcal{F}) \;\cong\; \mathrm{Hom}_{\mathrm{PMod}(\mathcal{O}_X)}(\mathrm{free}(\mathbf{y}\,U),\, \iota\mathcal{F}).$$

The free–Yoneda additive equivalence freeYonedaHomAddEquiv of Lemma 197 then identifies the right-hand presheaf Hom-group with the evaluation $(\iota\mathcal{F})(U) = \mathcal{F}(U) = \Gamma(U, \mathcal{F})$:

$$\mathrm{Hom}_{\mathrm{PMod}(\mathcal{O}_X)}(\mathrm{free}(\mathbf{y}\,U),\, \iota\mathcal{F}) \;\cong\; \mathcal{F}(U).$$

Both isomorphisms are additive and natural in $\mathcal{F}$, so their composite is the asserted natural additive isomorphism. ☐

Definition 218 (Absolute sheaf cohomology as Ext of the corepresenting object)**.** *Realization of absolute sheaf cohomology in the module-sheaf category.* Let X be a scheme, $\mathcal{F}$ an $\mathcal{O}_X$-module, and $U \subseteq X$ an open subscheme with open immersion $j : U \hookrightarrow X$. Let $j_!\mathcal{O}_U \in X$.Modules be the corepresenting object of Definition 216. The *absolute sheaf cohomology of $\mathcal{F}$ over U* is defined to be the Ext of this corepresenting object in the single abelian category X.Modules = $\mathrm{Mod}(\mathcal{O}_X)$ of sheaves of $\mathcal{O}_X$-modules:

$$H^p(U, \mathcal{F}) \;:=\; \mathrm{Ext}^p_{\mathrm{Mod}(\mathcal{O}_X)}(j_!\mathcal{O}_U,\, \mathcal{F}), \qquad p \geq 0.$$

Both Ext arguments live in X.Modules; no restriction of $\mathcal{F}$ along j is taken. Here Ext is Mathlib's Ext bifunctor (Lemma 219) on X.Modules; the HasExt structure it requires is available unconditionally by Lemma 220. The three structural facts the downstream dimension-shift arguments consume are then exactly the corresponding Mathlib results:

1. **Degree zero is global sections.** In degree 0, Ext is the Hom-group, $\mathrm{Ext}^0(j_!\mathcal{O}_U, \mathcal{F}) \cong \mathrm{Hom}_{X.\mathrm{Modules}}(j_!\mathcal{O}_U, \mathcal{F})$, by the degree-zero Ext comparison (Lemma 221); composing with the corepresentability isomorphism $\mathrm{Hom}_{X.\mathrm{Modules}}(j_!\mathcal{O}_U, \mathcal{F}) \cong \Gamma(U, \mathcal{F})$ of Lemma 217 gives $H^0(U, \mathcal{F}) \cong \Gamma(U, \mathcal{F})$.

2. **Injective vanishing.** When the *second* Ext argument is an injective $\mathcal{O}_X$-module $\mathcal{I}$, one has $\mathrm{Ext}^{n+1}(-, \mathcal{I}) = 0$ by Lemma 222. Since $\mathcal{I}$ is exactly the second argument of $\mathrm{Ext}^{n+1}(j_!\mathcal{O}_U, \mathcal{I})$, this yields $H^{n+1}(U, \mathcal{I}) = 0$ directly: the short exact sequence stays in X.Modules, no restriction is taken, and restriction-preserves-injectives is not needed.

3. **Long exact sequence.** For a short exact sequence $0 \to \mathcal{F} \to \mathcal{I} \to \mathcal{Q} \to 0$ in X.Modules and the *fixed* first argument $j_!\mathcal{O}_U$, the covariant Ext long exact sequence (Lemma 223) is the long exact sequence of the groups $H^p(U, -)$, with connecting maps given by composition with the Ext class of the extension.

Definition 219 (Ext bifunctor in an abelian category). *Provided by Mathlib (`Mathlib.Algebra.Homology.DerivedC`* For an abelian category C with a HasExt structure and objects $X, Y \in C$ and $n \in \mathbb{N}$, Mathlib provides the Ext group $\mathrm{Ext}^n_C(X, Y)$, functorial (contravariantly in X, covariantly in Y) and equipped with a composition pairing.

Lemma 220 (Unconditional availability of HasExt). *Provided by Mathlib (`Mathlib.Algebra.Homology.DerivedCat` Every abelian category C carries a* HasExt *structure; in particular the module-sheaf category* U.Modules *admits the Ext bifunctor of Definition 219 unconditionally.*

Lemma 221 (Degree-zero Ext is the Hom-group). *Provided by Mathlib (`Mathlib.Algebra.Homology.DerivedCatego` For objects A, Y of an abelian category with* HasExt*, there is a natural (additive) isomorphism* $\mathrm{Ext}^0(A, Y) \cong (A \to Y)$ *between degree-zero Ext and the Hom-group. Applied to the first argument* $A = j_!\mathcal{O}_U$*, this identifies* $\mathrm{Ext}^0(j_!\mathcal{O}_U, \mathcal{F})$ *with* $\mathrm{Hom}_{X.\mathrm{Modules}}(j_!\mathcal{O}_U, \mathcal{F})$*, which the corepresentability isomorphism of Lemma 217 in turn identifies with* $\Gamma(U, \mathcal{F})$*.*

Lemma 222 (Ext vanishes on injectives in positive degree). *Provided by Mathlib (`Mathlib.Algebra.Homology.Deriv` Let C be an abelian category with* HasExt *and $I \in C$ an injective object. Then every element of* $\mathrm{Ext}^{n+1}(X, I)$ *is zero, i.e.* $\mathrm{Ext}^{n+1}(X, I) = 0$ *for all X and all $n \geq 0$. The vanishing requires only the* second *argument to be injective.*

Lemma 223 (Covariant Ext long exact sequence of a short exact sequence). *Provided by Mathlib (`Mathlib.Algebra.Homology.DerivedCategory.Ext.ExactSequences`). Let C be an abelian category with* HasExt*, let $0 \to S_1 \to S_2 \to S_3 \to 0$ be a short exact sequence in C, and fix a first argument $X \in C$. Then the groups* $\mathrm{Ext}^n(X, S_i)$ *assemble into a long exact sequence, with connecting homomorphism* $\mathrm{Ext}^{n_0}(X, S_3) \to \mathrm{Ext}^{n_1}(X, S_1)$ *(for $n_0 + 1 = n_1$) given by composition with the Ext class of the extension. Mathlib packages the exactness both as the five-term* ComposableArrows *exactness* covariantSequence_exact *and as the three step-lemmas three individual step-exactness lemmas.*

8.4.1 Project wrappers around the Ext realization

The following are the thin project-side specializations of the Mathlib Ext anchors above to the fixed first argument $j_!\mathcal{O}_U$; they are the declarations the dimension-shift sub-lemmas of Lemma 328 actually invoke.

Lemma 224 (Degree-zero absolute cohomology is global sections). *Project specialization of Lemma 221 composed with corepresentability. For every $\mathcal{O}_X$-module $\mathcal{F}$ and open $U \subseteq X$ there is an additive isomorphism*

$$H^0(U, \mathcal{F}) \;=\; \mathrm{Ext}^0(j_!\mathcal{O}_U, \mathcal{F}) \;\cong\; \Gamma(U, \mathcal{F}).$$

Proof. Compose the degree-zero Ext comparison $\mathrm{Ext}^0(j_!\mathcal{O}_U, \mathcal{F}) \cong \mathrm{Hom}_{X.\mathrm{Modules}}(j_!\mathcal{O}_U, \mathcal{F})$ (Lemma 221) with the corepresentability isomorphism $\mathrm{Hom}_{X.\mathrm{Modules}}(j_!\mathcal{O}_U, \mathcal{F}) \cong \Gamma(U, \mathcal{F})$ of Lemma 217. Both are additive, so is the composite. □

Lemma 225. *[Naturality of $H^0 \cong \Gamma$ in the coefficient sheaf] Project-bespoke obligation. The isomorphism $H^0(U,-) \cong \Gamma(U,-)$ of Lemma 224 is natural in the coefficient sheaf: for a morphism $g : \mathcal{F}_1 \to \mathcal{F}_2$ in X.Modules the square*

$$\begin{array}{ccc} H^0(U,\mathcal{F}_1) & \longrightarrow & H^0(U,\mathcal{F}_2) \\ \cong\downarrow & & \downarrow\cong \\ \Gamma(U,\mathcal{F}_1) & \xrightarrow{g_U} & \Gamma(U,\mathcal{F}_2) \end{array}$$

commutes, where the top arrow is $H^0(U,g)$, the functorial action of g on Ext^0 in the second variable, and the bottom arrow is the sections map $g_U = g(U)$.

Proof. Under the degree-zero Ext comparison, post-composition by the degree-zero Ext class of g corresponds to post-composition by g in $\mathrm{Hom}_{X.\mathrm{Modules}}(j_!\mathcal{O}_U,-)$; under the corepresentability isomorphism of Lemma 217 (which is natural in the second variable, being a composite of the sheafification adjunction and the free–Yoneda evaluation, both natural), this in turn corresponds to applying g on sections over U, i.e. g_U. Hence the square commutes. In particular, when g_U is surjective so is $H^0(U,g)$ — the transfer used by the surjectivity step of Lemma 326. □

Lemma 226 (Injective vanishing for absolute cohomology)**.** *Project specialization of Lemma 222. For an injective $\mathcal{O}_X$-module $\mathcal{I}$ and any $n \geq 0$,*

$$H^{n+1}(U,\mathcal{I}) \;=\; \mathrm{Ext}^{n+1}(j_!\mathcal{O}_U,\mathcal{I}) \;=\; 0.$$

The injective $\mathcal{I}$ is the second *Ext argument, so no restriction is taken.*

Proof. Immediate from Lemma 222 with $X = j_!\mathcal{O}_U$ and $I = \mathcal{I}$. □

Lemma 227 (Covariant long exact sequence for absolute cohomology)**.** *Project specialization of Lemma 223 at fixed first argument $j_!\mathcal{O}_U$. For a short exact sequence $0 \to \mathcal{F}_1 \to \mathcal{F}_2 \to \mathcal{F}_3 \to 0$ in X.Modules, the groups $H^n(U,\mathcal{F}_i) = \mathrm{Ext}^n(j_!\mathcal{O}_U,\mathcal{F}_i)$ assemble into the covariant long exact sequence, recorded in three step-exactness lemmas (for positions 1, 2, and 3 of the sequence).*

Proof. Each wrapper fixes the first Ext argument to $j_!\mathcal{O}_U$ and threads the short exact sequence and degree hypotheses unchanged into the corresponding step-exactness lemma of Lemma 223. □

Lemma 228 (Serre vanishing on affines)**.** *Source: Stacks Project, Cohomology of Schemes, Tag 01XB (`lemma-quasi-coherent-affine-cohomology-zero`, Serre vanishing on affines). Let U be an affine scheme and $\mathcal{F}$ a quasi-coherent $\mathcal{O}_U$-module. Then the sheaf cohomology of $\mathcal{F}$ vanishes in all positive degrees:*

$$H^p(U,\mathcal{F}) = 0 \qquad \text{for all } p > 0.$$

The formal statement carries [EnoughInjectives X.Modules] *as an explicit instance hypothesis, inherited from Lemma 328.*

Proof. The proof instantiates the basis-comparison criterion (Lemma 328) at the affine cover system. Assemble the affine cover system s of Definition 242: its basis $\mathcal{B}$ is the standard (distinguished) opens of the affine U and its admissible coverings Cov are the standard open covers, the three nontrivial fields being discharged by Lemmas 229, 231 and 241.

For the quasi-coherent coefficient $\mathcal{F}$, Lemma 307 supplies the seed: $\mathcal{F}$ has vanishing higher Čech cohomology for s (Definition 320), i.e. condition (3) of Lemma 328 holds for $\mathcal{F}$. Applying the basis-comparison lemma to s and $\mathcal{F}$ yields, for every standard open $V \in \mathcal{B}$ and every $p > 0$, the vanishing $H^p(V,\mathcal{F}) = 0$. Taking $V = U$ (the whole affine, itself a standard open) gives $H^p(U,\mathcal{F}) = 0$ for all $p > 0$. Throughout, the sheaf cohomology $H^p(U,-) = \mathrm{Ext}^p(j_!\mathcal{O}_U,-)$ is the Ext realization of Definition 218. □

8.4.2 The affine instantiation (Tag 01XB)

The Serre vanishing of Lemma 228 is obtained by feeding a concrete *affine* cover system into the basis-comparison criterion. We construct that cover system (Definition 242) and discharge its three nontrivial fields, then supply the quasi-coherent seed that meets condition (3). The sub-lemmas below are the formalize-ready cut points of this instantiation.

Lemma 229 (Distinguished opens are closed under finite intersection)**.** *Source: Stacks Project, Schemes, `lemma-standard-open` (standard opens). Let R be a commutative ring and* $\operatorname{Spec} R$ *the associated affine scheme. The distinguished opens $D(f)$, $f \in R$, form a basis closed under intersection: $D(f) \cap D(g) = D(fg)$. Consequently, for any standard open cover $\mathcal{U} : V = \bigcup_{i\in\iota} D(g_i)$ of a distinguished open V, every finite intersection $D(g_{i_0}) \cap \cdots \cap D(g_{i_p}) = D(g_{i_0} \cdots g_{i_p})$ of its members is again a distinguished open. This is the* faces_mem *field (condition (1)) of the affine cover system.*

Proof. The identity $D(f) \cap D(g) = D(fg)$ holds because a prime $\mathfrak{p}$ avoids both f and g iff it avoids the product fg. Iterating over the indices $i_0, \dots, i_p$ of a face gives $\bigcap_k D(g_{i_k}) = D(\prod_k g_{i_k})$, a distinguished open; the base set $V = D(f)$ is distinguished by hypothesis. Hence every face of a standard cover lies in the basis. □

Lemma 230 (Standard covers are cofinal among open covers of a distinguished open)**.** *Source: Stacks Project, Schemes, `lemma-standard-open` (2), and Sheaves on Spaces, Tag 009L, `lemma-cofinal-systems-coverings-standard-case`. Let $V = D(f)$ be a distinguished open of* $\operatorname{Spec} R$*, and let $\{W_\alpha\}_{\alpha\in A}$ be an arbitrary open covering of V (indexed by a type A, with each W_α an open of* $\operatorname{Spec} R$ *contained in V). Then there exist a finite index $n \in \mathbb{N}$, a family $g : \{1, \dots, n\} \to R$ of ring elements, and a reindexing $\varphi : \{1, \dots, n\} \to A$ such that*

$$D(f) = \bigcup_{i=1}^{n} D(g_i) \qquad \textit{and} \qquad D(g_i) \subseteq W_{\varphi(i)} \quad (1 \le i \le n).$$

That is, every open covering of V is refined *by a finite covering by distinguished opens, each member of which sits inside a member of the original covering. This concrete indexed-cover/refinement statement is exactly the cofinality input the sheaf-on-a-basis principle (Tag 009L) requires; together with the closure of the basis under intersection (Lemma 229) it makes the standard covers a cofinal system among the open coverings of V. It is the refinement step invoked in Lemma 231: a covering of V produced by local surjectivity is refined to a standard cover $\mathcal{U} \in$* Cov *carrying the local lifts, with the reindexing φ transporting the local sections.*

Proof. The basic opens $D(g)$, $g \in R$, form a topological basis of $\operatorname{Spec} R$ (Lemma 239), so each member W_α of the given covering is a union of basic opens contained in it; choosing one basic open inside each W_α through each point of V yields a basic-open covering of $D(f)$ refining $\{W_\alpha\}$. The distinguished open $D(f)$ is quasi-compact, hence a finite subfamily $D(g_1), \dots, D(g_n)$ already covers $D(f)$; recording for each i the index $\varphi(i) \in A$ of a member containing $D(g_i)$ gives the asserted finite refinement $D(f) = \bigcup_{i=1}^{n} D(g_i)$ with $D(g_i) \subseteq W_{\varphi(i)}$. The basis is closed under the intersections required by Tag 009L by Lemma 229, so the standard covers form a cofinal system in the sense of the sheaf-on-a-basis cofinality principle. □

Lemma 231 (Section surjectivity for the affine cover system)**.** *Source: Stacks Project, Cohomology, `lemma-ses-cech-h1`. Let V be a distinguished open of the affine and let $S : 0 \to S_1 \to S_2 \to S_3 \to 0$ be a short exact sequence of $\mathcal{O}_X$-modules whose left term S_1 has vanishing positive Čech cohomology over the standard covers of V, i.e. $\check{H}^p(\mathcal{U}, S_1) = 0$ for every standard cover $\mathcal{U}$ of V and every $p > 0$. Then the section map $S_2(V) \to S_3(V)$ is surjective. This discharges the* surj_of_vanishing *field of the affine cover system.*

Proof. Fix a distinguished open $V \in \mathcal{B}$ and a section $t \in S_3(V)$; we produce a global lift of t over V.

Step 1 (local surjectivity of the section map). The map $g : S_2 \to S_3$ is an epimorphism in the category of $\mathcal{O}_X$-modules. By the gap-fill Lemma 235 the underlying-abelian-sheaf functor preserves this epimorphism, so the induced map of abelian sheaves is an epimorphism; by Lemma 237 an epimorphism of sheaves is locally surjective, hence g is a locally surjective morphism of presheaves (Lemma 236).

Step 2 (refine to a standard cover carrying local lifts). Local surjectivity at t produces an open covering $\{W_\alpha\}$ of V together with sections $t_\alpha \in S_2(W_\alpha)$ mapping to $t|_{W_\alpha}$. Since the affine opens form a basis (Lemma 238) and standard covers are cofinal among open coverings of V (Lemma 230), refine $\{W_\alpha\}$ to a standard cover $\mathcal{U} = \{D(g_i)\} \in \mathrm{Cov}$ of V; restricting the t_α along the refinement yields local lifts $s_i^{\mathrm{loc}} \in S_2(D(g_i))$ with $g(s_i^{\mathrm{loc}}) = t|_{D(g_i)}$.

Step 3 (glue via $\check{H}^1 = 0$). The left term S_1 has vanishing positive Čech cohomology over every covering of V in Cov (the field hypothesis); in particular $\check{H}^1(\mathcal{U}, S_1) = 0$. Feeding the cover $\mathcal{U}$, the local lifts s^{loc}, and this vanishing to the Čech $\check{H}^1$-surjectivity criterion (Lemma 215, which is stated over a raw finite family of opens) produces a global section $s \in S_2(V)$ with $g(s) = t$. Hence $S_2(V) \to S_3(V)$ is surjective. □

Lemma 232 (The sheafification–toSheaf square)**.** *Provided by Mathlib. Let R be a sheaf of rings for the Grothendieck topology J, and write $L := \mathrm{sheafification}(\mathrm{id}_{R_0}) : \mathrm{PreMod}(R_0) \to \mathrm{Mod}(R)$ for the sheafification functor from presheaves of R_0-modules to sheaves of R-modules (a left adjoint and a localization, whose counit is an isomorphism). Then there is a natural isomorphism of functors*

$$L \circ \mathrm{toSheaf}\, R \;\cong\; \mathrm{toPresheaf}\, R_0 \circ \mathrm{sheafify}_J^{\mathrm{Ab}},$$

where $\mathrm{toPresheaf}\, R_0$ forgets a presheaf of modules to its underlying presheaf of abelian groups and $\mathrm{sheafify}_J^{\mathrm{Ab}} = \mathrm{presheafToSheaf}\, J\, \mathbf{Ab}$ is sheafification of presheaves of abelian groups. This is the commuting square relating module-sheafification to abelian-group-sheafification through the forgetful functors.

Lemma 233 (Colimit preservation gives epimorphism preservation)**.** *Provided by Mathlib. Let $F : \mathcal{C} \to \mathcal{D}$ be a functor that preserves colimits of shape* WalkingSpan *(equivalently, preserves pushouts). Then F preserves epimorphisms: if f is an epimorphism in $\mathcal{C}$ then $F(f)$ is an epimorphism in $\mathcal{D}$. In particular a functor preserving finite colimits preserves epimorphisms, since* WalkingSpan *is a finite shape.*

Lemma 234 (The underlying-abelian-sheaf functor preserves finite colimits)**.** *Project-bespoke gap-fill (the missing right-exact dual of* toSheaf*). The forgetful functor $\mathrm{toSheaf}\, R : \mathrm{Mod}(R) \to \mathrm{Sh}(J, \mathbf{Ab})$ from sheaves of $\mathcal{O}_X$-modules to the underlying sheaves of abelian groups preserves finite colimits. This is the right-exact counterpart of the limit-side fact that $\mathrm{toSheaf}\, R$ preserves finite limits; Mathlib supplies the latter but not the former, so it is built here.*

Proof. Write $L := \mathrm{sheafification}(\mathrm{id}_{R_0}) : \mathrm{PreMod}(R_0) \to \mathrm{Mod}(R)$ for the module-sheafification functor; L is a left adjoint whose counit is an isomorphism, so it is a localization that preserves all colimits. Throughout, $\circ$ denotes diagrammatic composition: $F \circ G$ means "apply F, then G".

Step 1 (the composite $L \circ \mathrm{toSheaf}$ preserves finite colimits). By the square isomorphism (Lemma 232), $L \circ \mathrm{toSheaf}\, R \cong \mathrm{toPresheaf}\, R_0 \circ \mathrm{sheafify}_J^{\mathrm{Ab}}$. The forgetful functor $\mathrm{toPresheaf}\, R_0$ on presheaves of modules preserves finite colimits (colimits of presheaves of modules are computed objectwise on the underlying abelian groups), and abelian-group sheafification $\mathrm{sheafify}_J^{\mathrm{Ab}}$ is a left adjoint, hence preserves all colimits. Therefore their composite, and so the isomorphic functor $L \circ \mathrm{toSheaf}\, R$, preserves finite colimits.

Step 2 (descend from the composite to toSheaf R *alone).* Let $\iota : \mathrm{Mod}(R) \hookrightarrow \mathrm{PreMod}(R_0)$ be the inclusion of sheaves of modules into presheaves of modules, so that $L \dashv \iota$ is the sheafification adjunction (Lemma 210). The descent proceeds in three clauses.

(i) The counit is an isomorphism. Every sheaf of modules is already a sheaf, so its sheafification recovers it: the counit of $L \dashv \iota$ is a natural isomorphism

$$\varepsilon : \iota \circ L \xrightarrow{\cong} \mathrm{id}_{\mathrm{Mod}(R)}.$$

(ii) toSheaf R *is a retract of* $L \circ$ toSheaf R*.* Whiskering the counit isomorphism ε with toSheaf R yields a natural isomorphism

$$\mathrm{toSheaf}\,R \;\cong\; \iota \circ (L \circ \mathrm{toSheaf}\,R),$$

which exhibits toSheaf R as a retract of the composite $L \circ$toSheaf R: the section is precomposition with ι, and the retraction is ε whiskered with toSheaf R. By Step 1 the composite $L \circ$ toSheaf R preserves finite colimits.

(iii) A retract of a finite-colimit-preserving functor preserves finite colimits. If G sends a finite colimit cocone to a colimit cocone and F is a retract of G, then the colimit comparison map for F is a retract of the (isomorphism) comparison map for G, hence itself an isomorphism, so F preserves that colimit. Applying this with $F =$ toSheaf R and $G = L \circ$ toSheaf R, we conclude that toSheaf R preserves finite colimits. ☐

Lemma 235 (The underlying-abelian-sheaf functor preserves epimorphisms)**.** *Project-bespoke gap-fill (instance). The forgetful functor from sheaves of $\mathcal{O}_X$-modules to the underlying sheaves of abelian groups preserves epimorphisms: if $g : S_2 \to S_3$ is an epimorphism of $\mathcal{O}_X$-modules then the induced morphism of underlying abelian sheaves is an epimorphism. This is the one small instance unlocking the passage from a module epimorphism to local surjectivity in Lemma 231.*

Proof. The functor toSheaf R preserves finite colimits (Lemma 234); in particular it preserves colimits of shape WalkingSpan, which is a finite shape. A functor that preserves pushouts preserves epimorphisms (Lemma 233), so toSheaf R carries the epimorphism g to an epimorphism of abelian sheaves. ☐

Lemma 236 (Locally surjective morphisms of presheaves)**.** *Provided by Mathlib. For a Grothendieck topology J, a morphism $\alpha : \mathcal{G} \to \mathcal{H}$ of presheaves is* locally surjective *when every section of $\mathcal{H}$ over an object is, locally on a covering sieve, in the image of α. On a topological space this is the predicate* TopCat. Presheaf. IsLocallySurjective*: for each open V and each section $t \in \mathcal{H}(V)$ there is an open covering $\{W_\alpha\}$ of V with $t|_{W_\alpha}$ in the image of α_{W_α}.*

Lemma 237 (An epimorphism of sheaves is locally surjective)**.** *Provided by Mathlib. A morphism of sheaves (of abelian groups, valued in a suitable category) is an epimorphism in the category of sheaves if and only if it is locally surjective. In particular an epimorphism of abelian sheaves on the space X is locally surjective, supplying the covering with local lifts used in Step 2 of Lemma 231.*

Lemma 238 (Affine opens form a basis)**.** *Provided by Mathlib. The affine opens of a scheme X form a basis for its topology. On* Spec R *this refines any open covering of a distinguished open to one by basic/affine opens, the input that turns a local-surjectivity covering into a standard cover lying in* Cov *(Lemma 230).*

Lemma 239 (Basic opens form a topological basis of the prime spectrum)**.** *Provided by Mathlib. For a commutative (semi)ring R, the distinguished/basic opens $D(r)$, $r \in R$, form a topological* *basis of* $\operatorname{Spec} R$*: the sets* $\{D(r) : r \in R\}$ *are open and every open of* $\operatorname{Spec} R$ *is a union of them. This is the basis input used to refine an arbitrary open covering of a distinguished open to one by basic opens (Lemma 230).*

Lemma 240 (Standard-cover opens are the distinguished opens $D(s_i)$)**.** *Project-bespoke compatibility lemma. Let* $\mathcal{U}$ *be the standard affine cover associated to a spanning family* $s : \iota \to R$ *with* $\operatorname{span}(\operatorname{range} s) = \top$ *(Definition 185). For each index* i *the* i*-th open of* $\mathcal{U}$ *is exactly the distinguished open* $D(s_i)$*:*

$$\operatorname{coverOpen} \mathcal{U}\, i \;=\; D(s_i) \quad \textit{in } \operatorname{Opens}(\operatorname{Spec} R).$$

This identifies the opens of a standard cover with distinguished opens, which is the form in which a standard cover is recognised inside the basis $\mathcal{B}$ *of the affine cover system.*

Proof. By construction the i-th member of the standard affine cover attached to s is the basic open $D(s_i)$ of $\operatorname{Spec} R$; the spanning hypothesis $\operatorname{span}(\operatorname{range} s) = \top$ guarantees these distinguished opens cover $\operatorname{Spec} R$, and the open carried at index i is $D(s_i)$ on the nose. □

Lemma 241 (Injective acyclicity for the affine cover system)**.** *Source: Stacks Project, Cohomology,* `lemma-injective-trivial-cech`*. Every injective* $\mathcal{O}_X$*-module* $\mathcal{I}$ *has vanishing positive-degree Čech cohomology over a standard cover* of the whole affine*: if* $s : \iota \to R$ *spans,* $\operatorname{span}(\operatorname{range} s) = \top$*, so that the distinguished opens* $D(s_i)$ *cover* $\operatorname{Spec} R$*, then* $\check{H}^p(\mathcal{U}, \mathcal{I}) = 0$ *for every* $p > 0$*. This is the special case of the family-form injective Čech-acyclicity (Lemma 214) at the spanning family* s*; the general* injective_acyclic *field of the affine cover system is discharged by that family-form lemma directly.*

Proof. The opens of the standard cover $\mathcal{U}$ attached to the spanning family s are the distinguished opens $D(s_i)$ (Lemma 240). Applying the family-form injective Čech-acyclicity (Lemma 214) to the finite family $i \mapsto D(s_i)$ gives $\check{H}^p(\mathcal{U}, \mathcal{I}) = 0$ for all $p > 0$. □

Definition 242 (The affine cover system)**.** *Source: Stacks Project, Cohomology of Schemes, Tag 01XB (*`lemma-quasi-coherent-affine-cohomology-zero`*, proof).* The *affine cover system* for an affine scheme $X = \operatorname{Spec} R$ (or an affine open of a scheme) is the instance of Definition 319 whose basis $\mathcal{B}$ is the distinguished opens $D(f)$, $f \in R$, and whose admissible coverings Cov are the standard open covers (finite coverings of a distinguished open by distinguished opens). Beyond the basis $\mathcal{B}$ and the covering set Cov just described, it carries three proof fields: the faces_mem field by Lemma 229; the surj_of_vanishing field by Lemma 231; and the injective_acyclic field by the family-form injective Čech-acyclicity Lemma 214 (applied to the distinguished opens of each standard cover).

Lemma 243 (Sections–tilde equivalence from an invertible counit)**.** *Source: Stacks Project, Schemes, Tag 01HV,* `lemma-spec-sheaves`*. Let* $X = \operatorname{Spec} R$ *and let* $\mathcal{F}$ *be an* $\mathcal{O}_X$*-module for which the tilde–global-sections counit*

$$\operatorname{fromTilde} : \widetilde{\Gamma(X, \mathcal{F})} \longrightarrow \mathcal{F}$$

is an isomorphism. Then its inverse gives a canonical isomorphism $\mathcal{F} \cong \widetilde{\Gamma(X, \mathcal{F})}$*; the forward map is the inverse of the counit and the reverse map is the counit itself.*

Proof. Take the inverse of fromTilde. The inverse laws give the two isomorphism identities, and identify its inverse component with the original counit. □

Lemma 244 (Affine structure theorem from a global presentation)**.** *Source: Stacks Project, Schemes, Tag 01I8,* `lemma-quasi-coherent-affine`*. Let $X = \operatorname{Spec} R$ and let $\mathcal{F}$ be an $\mathcal{O}_X$-module that admits a* global *presentation (a global exact sequence $\mathcal{O}_X^{(J)} \to \mathcal{O}_X^{(I)} \to \mathcal{F} \to 0$, the data* $\mathcal{F}$.Presentation*). Then $\mathcal{F}$ is isomorphic to the sheaf $\Gamma\widetilde{(X, \mathcal{F})}$ associated to its module of global sections. This is the presentation-driven discharge of the conditional hypothesis* IsIso(fromTilde) *of Lemma 243.*

Proof. A global presentation of $\mathcal{F}$ makes the tilde–Γ counit fromTilde an isomorphism (Lemma 245): Γ and $\widetilde{(-)}$ are exact enough to carry the presentation of $\mathcal{F}$ to a presentation of $\Gamma\widetilde{(X, \mathcal{F})}$, and the counit is then an isomorphism by the five lemma on the two presentations. Apply 243 to this invertible counit. □

Lemma 245 (Counit is iso from a presentation)**.** *Provided by Mathlib. For R a ring and $\mathcal{F}$ an $\mathcal{O}_{\operatorname{Spec} R}$-module equipped with a global presentation* $\mathcal{F}$.Presentation*, the tilde–Γ counit* fromTilde : $\Gamma(\widetilde{\operatorname{Spec} R, \mathcal{F}}) \to \mathcal{F}$ *is an isomorphism.*

8.5 The 01I8 counit isomorphism via section localization

Let $X = \operatorname{Spec} R$ and let $\mathcal{F}$ be quasi-coherent. The distinguished-open component of the counit fromTilde : $\Gamma\widetilde{(X, \mathcal{F})} \to \mathcal{F}$ is the canonical map between two localizations of $\Gamma(X, \mathcal{F})$ at the powers of f. Thus it is an isomorphism once the restriction map $\Gamma(X, \mathcal{F}) \to \Gamma(D(f), \mathcal{F})$ is known to have the localization property. Lemma 287 establishes this property; checking it on the basis of distinguished opens then proves the counit is an isomorphism.

Lemma 246 (The tilde–Γ counit)**.** *Provided by Mathlib. For a ring R and an $\mathcal{O}_{\operatorname{Spec} R}$-module $\mathcal{F}$, the tilde–Γ counit* fromTilde : $\Gamma(\widetilde{\operatorname{Spec} R, \mathcal{F}}) \to \mathcal{F}$ *is the natural morphism of sheaves of modules whose component over a distinguished open $D(f)$ is the localization lift* IsLocalizedModule.lift *of the section-restriction map $\Gamma(\operatorname{Spec} R, \mathcal{F}) \to \Gamma(D(f), \mathcal{F})$ along the structure-sheaf localization* $\widetilde{(-)}$.toOpen *(the latter exhibiting $\Gamma(D(f), \widetilde{M}) = M_f$ as a localization at the powers of f).*

Lemma 247 (Criterion for the counit to be an isomorphism)**.** *Provided by Mathlib. For an $\mathcal{O}_{\operatorname{Spec} R}$-module $\mathcal{F}$, the counit* fromTilde *is an isomorphism if and only if $\mathcal{F}$ lies in the essential image of the tilde functor $M \mapsto \widetilde{M}$; equivalently $\mathcal{F} \cong \widetilde{M}$ for some R-module M.*

Lemma 248 (The forgetful functor reflects isomorphisms)**.** *Provided by Mathlib. The forgetful functor* forget *from sheaves of $\mathcal{O}_X$-modules to the underlying sheaves of abelian groups is fully faithful, hence reflects isomorphisms; the same holds for* toSheaf*. Consequently a morphism of sheaves of modules is an isomorphism as soon as its underlying morphism of sheaves of abelian groups is, and — since a morphism of sheaves which is an isomorphism on every member of a basis of the topology is an isomorphism — it suffices to check* fromTilde *on the basis of distinguished opens $\{D(f)\}$.*

Lemma 249 (Two localizations at the same set are canonically isomorphic)**.** *Provided by Mathlib. Let $S \subseteq R$ be a submonoid of a commutative ring and let $\varphi : M \to M'$ and $\psi : M \to M''$ be R-linear maps that each exhibit their target as the localization of M at S (i.e.* IsLocalizedModule $S\,\varphi$ *and* IsLocalizedModule $S\,\psi$*). Then there is a canonical R-linear isomorphism $M' \cong M''$ compatible with φ, ψ. In particular the localization lift* IsLocalizedModule.lift $S\,\varphi\,\psi$ *is an isomorphism whenever both φ and ψ are localizations of M at S.*

Lemma 250 (The structure-sheaf localization of $\widetilde{M}$ over $D(f)$)**.** *Provided by Mathlib. For M an R-module and $f \in R$, Mathlib's map* toOpen : $M \to \Gamma(D(f), \widetilde{M})$ *exhibits* $\Gamma(D(f), \widetilde{M}) = M_f$ *as the localization of M at the powers of f: it carries an* IsLocalizedModule(powers f) *instance. Moreover the global component* toOpen$(M, \top)$ *is an isomorphism (Mathlib's* isIso_toOpen_top*), identifying* $\Gamma(\operatorname{Spec} R, \widetilde{M}) = M$.

Lemma 251 (Transport of a presentation along a colimit-preserving functor)**.** *Provided by Mathlib. Let F :* SheafOfModules $R \to$ SheafOfModules S *be a functor between categories of sheaves of modules (possibly over different sites) that preserves colimits and carries the unit object to the unit object, $F(\mathcal{O}_R) \cong \mathcal{O}_S$. Then any presentation of a sheaf of modules $\mathcal{M}$ — a right-exact sequence $\mathcal{O}_R^{(J)} \to \mathcal{O}_R^{(I)} \to \mathcal{M} \to 0$ of generators and relations — is carried by F to a presentation of $F(\mathcal{M})$.*

Lemma 252 (Transport of a presentation across an isomorphism)**.** *Provided by Mathlib. Let $\phi : \mathcal{M} \to \mathcal{N}$ be an isomorphism of sheaves of modules. A presentation of $\mathcal{M}$ induces a presentation of $\mathcal{N}$, by composing its generators and relations with ϕ.*

Lemma 253 (Site-equivalence transport of sheaves of modules)**.** *Provided by Mathlib. Let $e : \mathcal{C} \simeq \mathcal{D}$ be an equivalence of sites whose functor and inverse are continuous. If $\mathcal{O}_{\mathcal{C}}$ and $\mathcal{O}_{\mathcal{D}}$ are sheaves of rings related by mutually inverse comparison maps along the two pushforwards, then e induces an equivalence of categories of sheaves of modules* SheafOfModules $\mathcal{O}_{\mathcal{C}} \simeq$ SheafOfModules $\mathcal{O}_{\mathcal{D}}$*, compatible with those pushforwards. This is the engine that bridges two pictures of "restriction to an open".*

Lemma 254 (The over-site of a topological space is equivalent to its subspace site)**.** *Provided by Mathlib. For a topological space T and an open $U \subseteq T$, the over-category* (Opens T). over U *of opens contained in U is equivalent to the category* Opens U *of opens of the subspace U.*

Lemma 255 (Continuity of the over-site equivalence)**.** *Both functors of the over-site equivalence* (Opens T). over $U \simeq$ Opens U *of Lemma 254 are cover-preserving and continuous for the respective Grothendieck topologies of pointwise covers. The same holds after identifying the subspace $D(g)$ with its associated open subscheme.*

Proof. The opens Grothendieck topology is the pointwise-cover predicate; the over-topology unfolds via the membership criterion for over-topologies, and the subspace-versus-image point correspondence is the obvious $\langle x', hx' \rangle$ / underlying-value bookkeeping, giving cover-preservation of each functor. Continuity then follows from cover-preservation together with the cover-density, local-fullness and local-faithfulness that hold automatically for the fully-faithful, essentially surjective functors of an equivalence. $\square$

Lemma 256 (Quasi-coherence datum: a cover with a presentation on each member)**.** *Provided by Mathlib. A* quasi-coherence datum *for a sheaf of modules $\mathcal{F}$ consists of a family of objects U_i of the site that covers the terminal object, together with, for each i, a presentation of the over-picture restriction* $\mathcal{F}$. over U_i. *A sheaf of modules is quasi-coherent precisely when such a datum exists.*

Lemma 257 (Sections of a restricted module are sections over the image open)**.** *Provided by Mathlib. For an $\mathcal{O}_Y$-module $\mathcal{M}$, an open immersion $\iota : X \to Y$, and an open $U \subseteq X$, the sections of the restricted module over U coincide definitionally with the sections of $\mathcal{M}$ over the image open $\iota(U)$:* $\Gamma(\mathcal{M}.\text{restrict}\,\iota, U) = \Gamma(\mathcal{M}, \iota(U))$*, and likewise for the restriction maps. The section comparison is therefore an equality, not merely an isomorphism.*

Lemma 258 (Section restriction of $\widetilde{M}$ localizes)**.** *Let M be an R-module and $f \in R$. The section-restriction map $\Gamma(\operatorname{Spec} R, \widetilde{M}) \to \Gamma(D(f), \widetilde{M})$ of the associated sheaf $\widetilde{M}$ exhibits its target as the localization of its source at the powers of f: one has* IsLocalizedModule(powers f) *for it.*

Proof. By Lemma 250 the distinguished-open component toOpen : $M \to \Gamma(D(f), \widetilde{M})$ is a localization at the powers of f, and the global component $M \to \Gamma(\operatorname{Spec} R, \widetilde{M})$ is an isomorphism. Up to this global isomorphism, the section-restriction map is exactly toOpen over $D(f)$; transporting the localization property along the global-sections isomorphism that identifies M with $\Gamma(\operatorname{Spec} R, \widetilde{M})$ on the source shows that the section-restriction map is itself a localization at the powers of f. □

Lemma 259 (Section restriction localizes when the counit is an isomorphism)**.** *Let $\mathcal{F}$ be an $\mathcal{O}_{\operatorname{Spec} R}$-module for which the tilde–Γ counit* fromTilde *is an isomorphism, and let $f \in R$. Then the section-restriction map $\Gamma(\operatorname{Spec} R, \mathcal{F}) \to \Gamma(D(f), \mathcal{F})$ is a localization at the powers of f.*

Proof. When fromTilde is an isomorphism, $\mathcal{F} \cong \widetilde{M}$ with $M = \Gamma(\operatorname{Spec} R, \mathcal{F})$ (Lemma 243). The counit isomorphism, pushed through the forgetful functor, identifies the section-restriction map of $\mathcal{F}$ with that of $\widetilde{M}$ — compatibly on source and target by naturality. The latter is a localization by Lemma 258; conjugating by the isomorphism transports the localization property to the section-restriction map of $\mathcal{F}$. □

Lemma 260 (Section restriction localizes for a globally presented module)**.** *Let $\mathcal{F}$ be an $\mathcal{O}_{\operatorname{Spec} R}$-module that admits a global presentation, and let $f \in R$. Then the section-restriction map $\Gamma(\operatorname{Spec} R, \mathcal{F}) \to \Gamma(D(f), \mathcal{F})$ is a localization at the powers of f.*

Proof. A global presentation makes the counit fromTilde an isomorphism (Lemma 245); now apply Lemma 259. □

Lemma 261 (Finite presentation cover from quasi-coherence)**.** *Let $X = \operatorname{Spec} R$ and let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Then there exist a finite family $g_1, \dots, g_n \in R$ with* $\operatorname{span}\{g_1, \dots, g_n\} = R$ *and, for each j, a member $U_{\varphi(j)}$ of the quasi-coherence cover with $D(g_j) \subseteq U_{\varphi(j)}$ carrying a presentation of the over-picture restriction $\mathcal{F}.\operatorname{over} U_{\varphi(j)}$.*

Proof. Quasi-coherence supplies a quasi-coherence datum (Lemma 256): a cover $(U_i)_i$ of X together with a presentation of $\mathcal{F}.\operatorname{over} U_i$ for each i. The cover-of-terminal-object condition translates to $\bigcup_i U_i = \top$; feeding this to the finite basic-open refinement (Lemma 291) yields finitely many g_j with $D(g_j) \subseteq U_{\varphi(j)}$ and $\operatorname{span}\{g_j\} = R$. The required presentation of $\mathcal{F}.\operatorname{over} U_{\varphi(j)}$ is the one attached to $U_{\varphi(j)}$ by the datum. □

Lemma 262 (Restricting a presentation to a basic open)**.** *Let $\mathcal{M}$ be an $\mathcal{O}_{\operatorname{Spec} R}$-module, let U be an open carrying a presentation of the over-restriction $\mathcal{M}.\operatorname{over} U$, and let $g \in R$ with $D(g) \subseteq U$. Then $\mathcal{M}.\operatorname{over} D(g)$ admits a presentation.*

Proof. The further over-restriction from U to the smaller open $D(g)$ is a left adjoint, hence preserves colimits, and carries the unit object to the unit object; the nested over-restriction instances make it a functor between the corresponding categories of sheaves of modules. Transport the given presentation of $\mathcal{M}.\operatorname{over} U$ along this functor by Lemma 251 to obtain a presentation of $\mathcal{M}.\operatorname{over} D(g)$. Concretely the transport is realized by first building the relevant site equivalence via Lemma 253, transporting the presentation through its inverse with Presentation.map, and closing along the residual isomorphism with Presentation.ofIsIso (Lemma 252). □

Definition 263 (Module equivalence over a basic open)**.** For $g \in R$, there is an equivalence of categories of sheaves of modules

$$\widehat{D(g)}.\operatorname{toScheme}.\mathrm{Modules} \;\simeq\; \mathrm{SheafOfModules}((\operatorname{Spec} R).\operatorname{ringCatSheaf}.\operatorname{over} D(g))$$

obtained by feeding the structure-sheaf comparison data $\varphi = \mathrm{overBasicOpenRingHom}\, g$, $\psi = \mathrm{overBasicOpenRingInvHom}\, g$ and their coherence witnesses into Lemma 253 along the now-continuous site equivalence of Lemma 255. The comparison morphisms φ, ψ are built by whiskering the structure sheaf along the functor isomorphism $\mathrm{overForgetIso}\, g : \mathrm{Over}.\operatorname{forget} D(g) \cong \mathrm{overEquivalence}.\operatorname{functor} \circ \iota.\operatorname{opensFunctor}$ (whose object component is the image–preimage identity $\iota(\iota^{-1}V) = V$ for opens $V \le D(g)$) and its definitional inverse $\mathrm{overForgetInvIso}\, g$. Lemma 264 gives its object-level comparison with restriction to the open subscheme.

Lemma 264 (Compatibility between over-restriction and open-subscheme restriction)**.** *Let $\mathcal{F}$ be an $\mathcal{O}_{\operatorname{Spec} R}$-module and $g \in R$; write $\iota : \widehat{D(g)} \hookrightarrow \operatorname{Spec} R$ for the open immersion of the basic open and $\mathrm{e} = \mathrm{modulesOverBasicOpenEquivalence}\, g$ (Definition 263) for this equivalence. Then its inverse carries the over-picture object $\mathcal{F}.\operatorname{over} D(g)$ — a sheaf of modules over the localized site $(\mathrm{Opens}\operatorname{Spec} R).\operatorname{over} D(g)$, where the quasi-coherence presentation lives — to the honest subspace restriction $\mathcal{F}.\operatorname{restrict}\iota$:*

$$\mathrm{e}.\operatorname{inverse}.\operatorname{obj}(\mathcal{F}.\operatorname{over} D(g)) \;\cong\; \mathcal{F}.\operatorname{restrict}\iota \qquad \text{in } \widehat{D(g)}.\mathrm{Modules}.$$

Thus the over-picture restriction agrees with restriction to the corresponding open subscheme.

Proof. Both objects describe $\mathcal{F}$ restricted to $D(g)$, in two equivalent sites. The equivalence

$$(\mathrm{Opens}\operatorname{Spec} R).\operatorname{over} D(g) \simeq \mathrm{Opens}(\widehat{D(g)})$$

is continuous by Lemma 255. The structure sheaf on the over-site and the structure sheaf of the open subscheme agree under this equivalence: both assign to an open $W \subseteq \widehat{D(g)}$ the sections over its image $\iota(W)$. Lemma 253 therefore induces an equivalence of their module categories. Under it, Lemma 257 identifies the subscheme restriction $\mathcal{F}|_{\widehat{D(g)}}$ with the over-picture object $\mathcal{F}.\operatorname{over} D(g)$. Reading this identification in the inverse direction gives the asserted isomorphism. ☐

Lemma 265 (Presentation of the affine restriction)**.** *Let $\mathcal{F}$ be an $\mathcal{O}_{\operatorname{Spec} R}$-module, let U be an open carrying a presentation of $\mathcal{F}.\operatorname{over} U$, and let $g \in R$ with $D(g) \subseteq U$. Then the affine restriction $\mathcal{F}_{(g)} = \mathrm{modulesRestrictBasicOpen}\, g\, \mathcal{F}$, as an $(\operatorname{Spec} R_g).\mathrm{Modules}$-object, admits a global presentation.*

Proof. By Lemma 262, $\mathcal{F}.\operatorname{over} D(g)$ admits a presentation. Carry this presentation through the inverse equivalence $\mathrm{e}.\operatorname{inverse}$ (Definition 263) and across the isomorphism $\mathrm{e}.\operatorname{inverse}.\operatorname{obj}(\mathcal{F}.\operatorname{over} D(g)) \cong \mathcal{F}.\operatorname{restrict}\iota$ (Lemma 264) using Lemma 252, obtaining a presentation of the subspace restriction $\mathcal{F}.\operatorname{restrict}\iota$ on $\widehat{D(g)}$. Finally transport along the canonical affine identification $D(g) \cong \operatorname{Spec} R_g$: restriction along this isomorphism preserves colimits and carries the structure-sheaf unit of $\widehat{D(g)}$ to the unit of $\operatorname{Spec} R_g$ — this unit compatibility holds because the identification is an isomorphism of schemes. The transported presentation, under the comparison of Lemma 292, is a presentation of $\mathcal{F}_{(g)}$. ☐

Lemma 266 (Degree-0/1 sheaf-axiom equalizer of a sheaf on a finite standard cover)**.** *Let $X = \operatorname{Spec} R$, let $\mathcal{F}$ be an $\mathcal{O}_X$-module (a sheaf of modules on the spectral site), let $W \subseteq X$ be an* *open, and let $g_1, \dots, g_n \in R$ be a finite family with $W = \bigcup_{j=1}^n (W \cap D(g_j))$. Then the augmented two-term Čech sequence*

$$0 \longrightarrow \Gamma(W, \mathcal{F}) \xrightarrow{\rho} \prod_j \Gamma(W \cap D(g_j), \mathcal{F}) \xrightarrow{\delta} \prod_{j,k} \Gamma(W \cap D(g_j g_k), \mathcal{F})$$

is exact at both nonzero terms: ρ — the product of the restriction maps — is injective, and a family $(s_j)_j$ lies in the image of ρ iff $\delta((s_j)_j) = 0$, where δ is the difference of the two overlap-restriction maps and $W \cap D(g_j g_k) = W \cap D(g_j) \cap D(g_k)$.

Proof. This is nothing but the sheaf axiom for $\mathcal{F}$, specialised to the finite open cover $\{W \cap D(g_j)\}_j$ of W and read off in degrees 0 and 1. The sheaf of modules $\mathcal{F}$ on $\operatorname{Spec} R$ is a sheaf, so its underlying presheaf satisfies the gluing axiom for every open cover. For the cover $\{W \cap D(g_j)\}_j$ of W the gluing axiom says exactly two things. First, a section of $\mathcal{F}$ over W is determined by its restrictions to the $W \cap D(g_j)$: this is the injectivity of ρ. Second, a *matching family* — a tuple $(s_j)_j$ whose members agree on the pairwise intersections $W \cap D(g_j) \cap D(g_k)$, which for basic opens are again basic opens $W \cap D(g_j g_k)$ — glues to a (necessarily unique) section over W: this is the surjectivity of ρ onto $\ker \delta$. Reading the two restriction-to-overlap maps as the two cofaces of the Čech nerve and δ as their difference turns "matching family" into the equation $\delta((s_j)_j) = 0$, giving the asserted exactness. Only the sheaf condition on the cover $\{W \cap D(g_j)\}_j$ is used; no positive-degree Čech vanishing enters. □

Lemma 267 (Localisation preserves exactness)**.** *Provided by Mathlib. Let $S \subseteq R$ be a submonoid of a commutative ring and let $M_0 \xrightarrow{a} M_1 \xrightarrow{b} M_2$ be R-linear maps forming an exact sequence at M_1 (the image of a equals the kernel of b). Fix, for each i, a localisation map $\ell_i : M_i \to M_i'$ at S (so each M_i' is the localised module $S^{-1}M_i$). Then the induced maps $S^{-1}a : M_0' \to M_1'$ and $S^{-1}b : M_1' \to M_2'$ again form an exact sequence at M_1'. In other words, localisation at S is an exact functor; in particular it carries short exact sequences to short exact sequences and commutes with the formation of kernels.*

Lemma 268 (Base-ring descent of a localisation along an algebra map)**.** *Let A be an R-algebra, let M and N be A-modules that are also R-modules compatibly with the algebra structure ($R \to A \to M$ and $R \to A \to N$ are scalar towers), and let $\varphi : M \to N$ be A-linear. Fix $f \in R$ and write f_A for its image in A under the algebra structure map $R \to A$. If φ exhibits N as the localisation of M at the powers of f_A (as A-modules), then φ, viewed as an R-linear map via restriction of scalars along $R \to A$, exhibits N as the localisation of M at the powers of f (as R-modules).*

Proof. The three defining clauses of being a localised module transfer directly. Surjectivity of quotients and injectivity-up-to-torsion are statements about the underlying additive maps and are unchanged by restricting scalars. For the unit clause one must see that multiplication by f^k on N is bijective as an R-endomorphism: by the scalar tower $f^k \cdot x = f_A^k \cdot x$ for every x, the R-endomorphism $x \mapsto f^k \cdot x$ coincides as a function with the A-endomorphism $x \mapsto f_A^k \cdot x$, and bijectivity is a property of the underlying map independent of the choice of base ring. Hence all three clauses hold over R. □

Lemma 269 (Image opens of the affine tile identification)**.** *Let $g, f \in R$, let $R_g = R[g^{-1}]$ be the localisation of R away from g, and let $\iota : \operatorname{Spec} R_g \xrightarrow{\sim} D(g) \hookrightarrow X$ be the open immersion identifying $\operatorname{Spec} R_g$ with the basic open $D(g) \subseteq X = \operatorname{Spec} R$ (the composite of the canonical homeomorphism*

$\operatorname{Spec} R_g \xrightarrow{\sim} D(g)$ *with the open inclusion). Then the image opens of* ι *of the two relevant opens of* $\operatorname{Spec} R_g$ *are*

$$\iota(\operatorname{Spec} R_g) = D(g), \qquad \iota(D(\bar{f})) = D(gf) = D(g) \cap D(f),$$

where $\bar{f}$ *is the image of* f *under the localisation map* $R \to R_g$*.*

Proof. The canonical homeomorphism $\operatorname{Spec} R_g \xrightarrow{\sim} D(g)$ carries $\operatorname{Spec} R_g$ isomorphically onto $D(g)$, so the first identity is just the image of the whole space under that homeomorphism followed by the open inclusion. For the second, $\bar{f}$ is the image of f under $R \to R_g$; its basic open $D(\bar{f}) \subseteq \operatorname{Spec} R_g$ maps under the homeomorphism to $D(g) \cap D(f)$, since restricting to $D(g)$ and then taking the locus where f is invertible is the locus where both g and f are invertible. Finally $D(g) \cap D(f) = D(gf)$ since the basic open of a product is the intersection of the basic opens. These are equalities of opens, proved set-theoretically; they are not definitional. □

Lemma 270 (Native action of the global-ring section functor is the structure-sheaf action)**.** *Let* $\mathcal{F}$ *be an* $\mathcal{O}_{\operatorname{Spec} R}$*-module, let* $W \subseteq \operatorname{Spec} R$ *be open, let* $r \in R$*, and let* x *be a section over* W *of the global-ring section functor* $\Gamma_R(-, \mathcal{F})$*. Then the native* R*-action on* x *coincides with the structure-sheaf action of the restricted global-sections image of* r*:*

$$r \cdot x \;=\; c_R \cdot x_{\mathcal{F}}, \qquad c_R \;=\; \rho^W(\theta_R(r)),$$

where $\theta_R : R \xrightarrow{\sim} \Gamma(\operatorname{Spec} R, \mathcal{O})$ *is the global-sections identification,* ρ^W *is restriction of sections from* $\operatorname{Spec} R$ *to* W*, and* $x_{\mathcal{F}}$ *denotes* x *regarded as a section of* $\mathcal{F}$ *over* W *for its structure-sheaf module structure (the underlying carrier is unchanged).*

Proof. The action of the global-ring section functor $\Gamma_R(-, \mathcal{F})$ is, by definition, the restriction of scalars of $\mathcal{F}$'s structure-sheaf module action along the global-sections identification θ_R. Unfolding that definition, the R-scalar r acts as the structure-sheaf scalar $\rho^W(\theta_R(r))$ on the unchanged underlying section; the equality therefore holds by definitional unfolding. □

Lemma 271 (Tile action transports definitionally to the ambient structure-sheaf action)**.** *Let* $\mathcal{F}$ *be an* $\mathcal{O}_{\operatorname{Spec} R}$*-module and let* $g \in R$*. Write* $\mathcal{F}_{(g)}$ *for the restricted module (tile) over* $\operatorname{Spec} R_g$ *obtained as the iterated restriction of* $\mathcal{F}$ *along the affine identification* $\operatorname{Spec} R_g \cong D(g)$ *followed by the open inclusion* $D(g) \hookrightarrow \operatorname{Spec} R$*. Let* $c \in \Gamma(\top, \mathcal{O}_{\operatorname{Spec} R_g})$ *and let* m *be a section of* $\mathcal{F}_{(g)}$ *over* $\top$*. Then the tile action of* c *on* m *transports to the ambient structure-sheaf action of* $\mathcal{F}$*:*

$$c \cdot m \;=\; (\alpha_g^{-1}(\beta_g^{-1}(c))) \cdot m_{\mathcal{F}},$$

where β_g *and* α_g *are the ring isomorphisms on global sections induced respectively by the affine identification* $\operatorname{Spec} R_g \cong D(g)$ *and by the open inclusion* $D(g) \hookrightarrow \operatorname{Spec} R$*, and* $m_{\mathcal{F}}$ *denotes* m *regarded as a section of* $\mathcal{F}$ *over the iterated-image open* $W \subseteq \operatorname{Spec} R$ *(which equals* $D(g)$ *by Lemma 269).*

Proof. Restriction of a module along an open immersion is the pushforward whose scalar action is reindexed by the inverse of the open immersion's global-sections ring map. The tile $\mathcal{F}_{(g)}$ is the composite of two such restrictions — along the affine identification $\operatorname{Spec} R_g \cong D(g)$ and along the open inclusion $D(g) \hookrightarrow \operatorname{Spec} R$ — so its action of c is, by definition, the ambient $\mathcal{F}$-action of $\alpha_g^{-1}(\beta_g^{-1}(c))$ on the unchanged underlying section. The equality holds by definitional unfolding; keeping the $\mathcal{F}$-side open in the iterated-image form W (rather than rewriting it to $D(g)$) is what makes the two carriers coincide on the nose. □

Lemma 272 (Tile action transports to the ambient action over an arbitrary open)**.** *The general-open companion of Lemma 271. Let $\mathcal{F}$ be an $\mathcal{O}_{\operatorname{Spec} R}$-module, let $g \in R$, and let $V \subseteq \operatorname{Spec} R_g$ be an* arbitrary *open (the sibling treats only $V = \top$). For $c \in \Gamma(V, \mathcal{O}_{\operatorname{Spec} R_g})$ and a section m of the tile $\mathcal{F}_{(g)}$ over V, the tile action of c on m transports to the ambient structure-sheaf action of $\mathcal{F}$ over the iterated-image open $\iota(V) \subseteq \operatorname{Spec} R$:*

$$c \cdot m \;=\; \left(\alpha_g^{-1}(\beta_g^{-1}(c))\right) \cdot m_{\mathcal{F}},$$

where β_g and α_g are the section ring isomorphisms induced by the affine identification $\operatorname{Spec} R_g \cong D(g)$ and by the open inclusion $D(g) \hookrightarrow \operatorname{Spec} R$ (now read at the open V rather than at $\top$), and $m_{\mathcal{F}}$ denotes m regarded as a section of $\mathcal{F}$ over the iterated-image open $\iota(V)$.

Proof. As in Lemma 271, the tile $\mathcal{F}_{(g)}$ is the composite of the two open-immersion restrictions, whose scalar action is by definition reindexed by the inverses of the two section ring maps. At an arbitrary open V this reindexing is identical, so the equality holds by definitional unfolding; keeping the $\mathcal{F}$-side open in the iterated-image form $\iota(V)$ (rather than rewriting it to a basic open) is what makes the two carriers coincide on the nose. □

Lemma 273 (Open-immersion section iso reads as a restriction)**.** *Let $f : X \to Y$ be an open immersion of schemes. On global sections f induces the comparison ring map $f^\sharp_\top : \Gamma(Y, \mathcal{O}_Y) \to \Gamma(X, \mathcal{O}_X)$, and the open-immersion section isomorphism $\beta_f : \Gamma(X, \mathcal{O}_X) \xrightarrow{\sim} \Gamma(f(\top), \mathcal{O}_Y)$ identifies the global sections of X with the sections of Y over the image open $f(\top) \subseteq Y$. Then the composite of $f^\sharp_\top$ with β_f^{-1} is the structure-sheaf restriction of Y from $\top$ down to the image open:*

$$\beta_f^{-1} \circ f^\sharp_\top \;=\; \rho_Y^{f(\top)}.$$

Equivalently, under the section identification β_f the global-sections comparison map of f is nothing but restriction along $f(\top) \hookrightarrow Y$.

Proof. The forward map of β_f is, by construction, the comparison map $f^\sharp$ read on the image open. Rewriting the identity in the form $f^\sharp_\top = \rho_Y^{f(\top)} \circ \beta_f$ and applying the naturality of the comparison maps $f^\sharp$ with respect to the restriction maps of $\mathcal{O}_X$ and $\mathcal{O}_Y$, the right-hand side reduces to $f^\sharp_\top$ precomposed with the identity restriction of $\mathcal{O}_X$ on $\top$; the source open $\top$ and its image are matched by the thin-category uniqueness of open inclusions. This leaves exactly $\rho_Y^{f(\top)}$. □

Lemma 274 (Γ–Spec naturality of the localisation immersion, section form)**.** *Let $g \in R$, write $R_g = R[g^{-1}]$, let $\lambda : R \to R_g$ be the localisation map, and let $\iota = \operatorname{Spec}(\lambda) : \operatorname{Spec} R_g \to \operatorname{Spec} R$ be the induced affine morphism (it is the open immersion identifying $\operatorname{Spec} R_g$ with $D(g) = \iota(\top)$). Then the restriction to the image open $D(g)$ of the global-sections identification $\theta_R : R \xrightarrow{\sim} \Gamma(\operatorname{Spec} R, \mathcal{O})$ factors as the localisation map followed by the global-sections identification of $\operatorname{Spec} R_g$ and the (inverse) section isomorphism of ι:*

$$\rho^{D(g)} \circ \theta_R \;=\; \beta_\iota^{-1} \circ \theta_{R_g} \circ \lambda,$$

where $\theta_{R_g} : R_g \xrightarrow{\sim} \Gamma(\operatorname{Spec} R_g, \mathcal{O})$ is the global-sections identification of $\operatorname{Spec} R_g$ and β_ι is the open-immersion section isomorphism of ι at $\top$.

Proof. This is the naturality of the Spec-to-global-sections comparison applied to the ring map λ. Concretely, the Mathlib naturality square for the global-sections identification (the Γ-Spec adjunction unit, `ΓSpecIso` naturality) gives λ followed by θ_{R_g} equals θ_R followed by the comparison map $\iota^\sharp_\top$ of $\iota = \operatorname{Spec}(\lambda)$. Substituting this into the right-hand side and reading the composite of $\iota^\sharp_\top$ with β_ι^{-1} as the restriction to the image open $D(g)$ by Lemma 273 yields $\rho^{D(g)} \circ \theta_R$. $\square$

Lemma 275 (Composition of the two tile section isomorphisms)**.** *In the situation of Lemma 274 the localisation immersion ι factors as the composite of the affine identification* $\operatorname{Spec} R_g \xrightarrow{\sim} D(g)$ *(of schemes) with the open inclusion $D(g) \hookrightarrow \operatorname{Spec} R$. The (inverse) open-immersion section isomorphisms of these two factors compose into the single (inverse) section isomorphism β_ι^{-1} of ι at $\top$, up to a structure-sheaf transport along the equality of image opens $\iota(\top) = D(g)$:*

$$\beta_{\hookrightarrow}^{-1} \circ \beta_{\mathrm{aff}}^{-1} \;=\; \tau \circ \beta_\iota^{-1},$$

where β_{aff} and $\beta_{\hookrightarrow}$ are the section isomorphisms of the affine identification and of the open inclusion respectively, and τ is the structure-sheaf transport induced by the equality of the composed image open with $D(g)$.

Proof. The functoriality of the open-immersion section isomorphism under composition of open immersions (`comp_appIso`) expresses β_ι as the composite of the two factor isomorphisms, with the section isomorphism of the open inclusion being the identity. Folding the resulting transport through the equality of image opens $\iota(\top) = D(g)$ gives the displayed identity, with τ the structure-sheaf map of that equality of opens. $\square$

Lemma 276 (Structure-sheaf ring identity of the tile comparison, morphism form)**.** *In the situation of Lemma 274, the restriction to the tile image open $D(g)$ of the global-sections identification θ_R of* $\operatorname{Spec} R$ *equals the localisation map followed by the global-sections identification of* $\operatorname{Spec} R_g$ *and the two (inverse) open-immersion section isomorphisms of the tile:*

$$\rho^{D(g)} \circ \theta_R \;=\; \beta_{\hookrightarrow}^{-1} \circ \beta_{\mathrm{aff}}^{-1} \circ \theta_{R_g} \circ \lambda.$$

This is the morphism-level (ring-map) form of the residual ring identity of Lemma 280.

Proof. Substitute the Γ-Spec naturality identity of Lemma 274, which rewrites $\rho^{D(g)} \circ \theta_R$ as $\beta_\iota^{-1} \circ \theta_{R_g} \circ \lambda$, and then replace the single section isomorphism β_ι^{-1} by the composite of the two tile factor isomorphisms using Lemma 275. The structure-sheaf transport τ of Lemma 275 merges with the restriction maps in the thin category of opens (functoriality of the structure presheaf), and the two displayed sides coincide. $\square$

Lemma 277 (Structure-sheaf ring identity of the tile comparison at an arbitrary open)**.** *The general-open companion of Lemma 276. For an* arbitrary *open $V \subseteq$* $\operatorname{Spec} R_g$*, the restriction to the iterated-image open $\iota(V)$ of the global-sections identification θ_R of* $\operatorname{Spec} R$ *equals the localisation map followed by the global-sections identification of* $\operatorname{Spec} R_g$*, the restriction to V, and the two (inverse) open-immersion section isomorphisms of the tile read at V:*

$$\rho^{\iota(V)} \circ \theta_R \;=\; \beta_{\hookrightarrow,V}^{-1} \circ \beta_{\mathrm{aff},V}^{-1} \circ \rho^V_{R_g} \circ \theta_{R_g} \circ \lambda.$$

This supplies the ring identity at the target open $V = D(\bar f)$ consumed by the general-open scalar reconciliation Lemma 279.

Proof. Derive from the $V = \top$ case (Lemma 276) by post-composing both sides with the structure-sheaf restriction $\iota(V) \le \iota(\top)$ and pushing that restriction through the two open-immersion section isomorphisms. The push-through is the section-restriction form of the open-immersion section-iso naturality: for an open immersion f and opens $U' \le U$, the inverse section iso at U followed by the Y-restriction $f(U') \le f(U)$ equals the X-restriction $U' \le U$ followed by the inverse section iso at U'. The residual restriction $\rho^{\top}_{R_g}$ on $\operatorname{Spec} R_g$ is the identity by uniqueness of morphisms in the thin category of opens. $\square$

Lemma 278 (Scalar compatibility of the tile comparison at the source open)**.** *Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_{\operatorname{Spec} R}$-module, let $g, r \in R$, and let x be a section of the affine tile $\mathcal{F}_{(g)}$ over $\top$ — equivalently, regarding the underlying carrier, a section of $\mathcal{F}$ over the tile image open $D(g)$. Then the native R-action of r on x (through $\Gamma_R(-,\mathcal{F})$) coincides with the R_g-action of the localised scalar $\lambda(r) \in R_g$ on x (through the tile $\Gamma_{R_g}(-,\mathcal{F}_{(g)})$):*

$$r \cdot x \;=\; \lambda(r) \cdot x, \qquad \lambda : R \to R_g \text{ the localisation map.}$$

This is the scalar core, at the source open $V = \top$, of the section comparison of Lemma 280; it is the scalar-tower compatibility $R \to R_g$ on the common underlying section carrier. It is not *the full natural isomorphism, only the equality of the two scalar actions on the single open $V = \top$.*

Proof. By Lemma 270 the native R-action of r on x is the structure-sheaf action of $\rho^{D(g)}(\theta_R(r))$, and by Lemma 271 the R_g-action of $\lambda(r)$ on x (as a tile section) transports to the structure-sheaf action of $\beta_{\hookrightarrow}^{-1}(\beta_{\mathrm{aff}}^{-1}(\theta_{R_g}(\lambda(r))))$ on the same underlying carrier. Both actions therefore reduce to a structure-sheaf scalar action, and the two scalars agree by the morphism-level ring identity of Lemma 276 evaluated at r. $\square$

Lemma 279 (Scalar compatibility of the tile comparison at an arbitrary open)**.** *The general-open companion of Lemma 278. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_{\operatorname{Spec} R}$-module, let $g, r \in R$, let $V \subseteq \operatorname{Spec} R_g$ be an* arbitrary *open, and let x be a section of the affine tile $\mathcal{F}_{(g)}$ over V — equivalently, on the underlying carrier, a section of $\mathcal{F}$ over the iterated-image open $\iota(V)$. Then the native R-action of r on x coincides with the R_g-action of the localised scalar $\lambda(r) \in R_g$ on x:*

$$r \cdot x \;=\; \lambda(r) \cdot x, \qquad \lambda : R \to R_g \text{ the localisation map.}$$

The instance at $V = D(\bar{f})$ is the scalar-tower compatibility $R \to R_g$ at the target *open of the per-tile section localization (Lemma 283); the $V = \top$ case is Lemma 278.*

Proof. The same argument as Lemma 278, now at an arbitrary open V. By Lemma 270 the native R-action of r on x reduces to a structure-sheaf scalar action, and by Lemma 272 the R_g-action of $\lambda(r)$ on x (as a tile section over V) reduces to a structure-sheaf scalar action on the same underlying carrier. The two scalars then agree by the morphism-level ring identity of Lemma 277 evaluated at r. $\square$

Lemma 280 (Natural section comparison between the tile and the ambient sheaf)**.** *In the situation of Lemma 269, with $\mathcal{F}_{(g)}$ the restriction of the quasi-coherent module $\mathcal{F}$ along $\iota : \operatorname{Spec} R_g \cong D(g) \hookrightarrow X$, there is an R_g-linear isomorphism, natural in the open $V \subseteq \operatorname{Spec} R_g$ and intertwining the restriction maps of the two sheaves,*

$$\Gamma_{R_g}(V, \mathcal{F}_{(g)}) \;\cong\; \Gamma_R(\iota(V), \mathcal{F}).$$

Here $\Gamma_{R_g}(-,\mathcal{F}_{(g)})$ is the global-ring section functor attached to the presented tile over R_g (valued in R_g-modules), and $\Gamma_R(-,\mathcal{F})$ the corresponding functor for $\mathcal{F}$ over R.

Proof. By Lemma 257, both sides have the same underlying additive group, and these identifications commute with restriction maps. It remains to compare scalar multiplication. For $r \in R$, naturality of global sections for the localization morphism gives

$$\rho^{D(g)}(\theta_R(r)) \;=\; \beta_g^{-1}(\theta_{R_g}(\bar{r})), \qquad \bar{r} \;=\; \text{image of } r \text{ in } R_g,$$

where θ_R and θ_{R_g} are the affine global-section identifications and β_g is induced by $\operatorname{Spec} R_g \cong D(g)$. This is precisely the compatibility of $\Gamma(D(g), \mathcal{O}) \cong R_g$ with the localization map $R \to R_g$, as recorded in Lemma 276. Together with Lemmas 270 and 271, it shows that the additive comparison respects the R_g-action. Hence it is an R_g-linear isomorphism, and its compatibility with restrictions makes it natural in V. $\square$

Lemma 281 (Scalar tower on a bundled restriction-of-scalars module)**.** *Let $R \to S$ be a ring map and let M be an S-module. Then the R-module obtained from M by restriction of scalars along $R \to S$, written $(\mathrm{restrictScalars}_{R\to S})\, M$, carries a scalar-tower structure: the restricted R-action factors through the S-action via the algebra map $R \to S$.*

Proof. By definition of restriction of scalars, the R-action on M is given by $r \cdot m = \phi(r) \cdot m$ where $\phi : R \to S$ is the algebra map. This is precisely the scalar-tower identity: the R-action factors through the S-action via ϕ. Hence R, S, and ϕ form a scalar tower on $(\mathrm{restrictScalars}_{R\to S})\, M$. $\square$

Lemma 282 (Reconciliation equivalence for the tile section comparison)**.** *Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_{\operatorname{Spec} R}$-module, let $g \in R$, write R_g for the localisation of R away from g, and let $V \subseteq \operatorname{Spec} R_g$ be an open. Let $\iota : \operatorname{Spec} R_g \xrightarrow{\sim} D(g) \hookrightarrow \operatorname{Spec} R$ be the localisation open immersion, so that $\iota(V)$ is the iterated image of V under the affine identification $\operatorname{Spec} R_g \cong D(g)$ followed by the open inclusion $D(g) \hookrightarrow \operatorname{Spec} R$. Then there is an R-linear isomorphism*

$$(\mathrm{restrictScalars}_{R\to R_g})\, \Gamma_{R_g}(V, \mathcal{F}_{(g)}) \xrightarrow{\ \sim\ } \Gamma_R(\iota(V),\ \mathcal{F}),$$

which is the identity on underlying elements. The forward and inverse maps are each the identity on elements, so the only content is the reconciliation of the two module structures carried by the common section type.

Proof. The two section modules share one underlying carrier: the tile sections over V coincide as sets with the sections of $\mathcal{F}$ over the image open $\iota(V)$, by construction of the restriction functor. The underlying map is therefore the identity, and it is additive. Its R-linearity is exactly the general-open scalar-tower compatibility of Lemma 279: the native R-action on the $\mathcal{F}$-side (described by Lemma 270) coincides with the R_g-action read through the localisation map $R \to R_g$, i.e. the restriction-of-scalars action carried by the tile side. The identity map is thus R-linear with R-linear inverse, hence an R-linear isomorphism. $\square$

Lemma 283. *[Per-tile section localisation at f] Let $X = \operatorname{Spec} R$, let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module, let $f \in R$, and let $g \in R$ be either a cover element $g = g_j$ or an overlap product $g = g_j g_k$ of the finite presentation cover of Lemma 261, so that the restriction $\mathcal{F}_{(g)}$ of $\mathcal{F}$ to $\operatorname{Spec} R_g$ admits a global presentation. Then the restriction-of-sections map $\Gamma(D(g), \mathcal{F}) \to \Gamma(D(gf), \mathcal{F})$ exhibits its target as the localisation of its source at the powers of f; equivalently, there is a canonical R-linear isomorphism $\Gamma(D(g), \mathcal{F})_f \cong \Gamma(D(gf), \mathcal{F})$.*

Proof. Let $\iota : \operatorname{Spec} R_g \xrightarrow{\sim} D(g) \hookrightarrow X$ be the open immersion of Lemma 269, with $\bar{f}$ the image of f under the localisation map $R \to R_g$. We compare the section maps after restriction of scalars from R_g to R.

Step 1 (global presentation of the tile). Since $D(g)$ lies in a presentation member of the cover, Lemma 265 presents the tile $\mathcal{F}_{(g)}$ globally over $\operatorname{Spec} R_g$.

Step 2 (localisation over R_g). Apply Lemma 260 to the globally presented $\mathcal{F}_{(g)}$ and $\bar{f}$: the section-restriction map $\Gamma_{R_g}(\operatorname{Spec} R_g, \mathcal{F}_{(g)}) \to \Gamma_{R_g}(D(\bar{f}), \mathcal{F}_{(g)})$ exhibits its target as a localisation at the powers of $\bar{f}$, *as R_g-modules.* Note this is over R_g, not yet over R.

Step 3 (opens identities). By Lemma 269 the two image opens of ι are $\iota(\operatorname{Spec} R_g) = D(g)$ and $\iota(D(\bar{f})) = D(gf) = D(g) \cap D(f)$.

Step 4 (the section comparison, as a restriction of scalars). The localisation of Step 2 lives over R_g; it must be related to the section-restriction map of $\mathcal{F}$ over R. The two maps share the same underlying map of section sets: this is the carrier identification of Lemma 257, now read as an equality of R-linear maps (with the $\mathcal{F}$-side open kept in its iterated-image form $W = \iota(V)$). The one genuine difference is the base ring: the tile sections carry an R_g-module structure, while the $\mathcal{F}$-sections carry an R-module structure. The descent that reconciles them is precisely the *restriction of scalars* along the localisation map $R \to R_g$: each tile section module, viewed through restriction of scalars, becomes an R-module, and the Step-2 map so viewed is the section-restriction map $\Gamma_R(D(g), \mathcal{F}) \to \Gamma_R(D(gf), \mathcal{F})$.

For this restriction of scalars to be meaningful, the native R-action on each section must agree with its R_g-action read through $R \to R_g$; this is the scalar-tower compatibility. At the source open $V = \top$ (image $D(g)$) it is exactly Lemma 278, whose two scalar prescriptions agree by the scalar-reduction identities (Lemmas 270 and 271). At the target open $V = D(\bar{f})$ (image $D(gf)$) it is the general-open companion Lemma 279, the $V = D(\bar{f})$ instance of the same ring-naturality content. Both opens of the Step-2 map are thus covered. In the descent these two scalar reconciliations are packaged uniformly as the R-linear reconciliation isomorphism of Lemma 282 (instantiated at the source and target opens), whose R-linearity is supplied by the general-open compatibility Lemma 279; the descent transports the Step-2 localisation through these isomorphisms.

Reading Step 2's localisation-at-$\bar{f}$ property through restriction of scalars along $R \to R_g$, and matching source and target opens by Step 3, yields: the map $\Gamma_R(D(g), \mathcal{F}) \to \Gamma_R(D(gf), \mathcal{F})$ exhibits its target as a localisation at the powers of $\bar{f}$, *still as R_g-modules.*

Step 5 (base-ring descent to R). Finally descend the base ring from R_g to R by Lemma 268, applied with $A = R_g$ and the scalar tower $R \to R_g$ on the two section modules (supplied structurally on the bundled restriction-of-scalars carriers by Lemma 281): since $\bar{f}$ is the image of f under $R \to R_g$, the powers of $\bar{f}$ descend to the powers of f, and the restriction of scalars (from R_g to R) of the Step-4 map exhibits $\Gamma_R(D(gf), \mathcal{F})$ as the localisation of $\Gamma_R(D(g), \mathcal{F})$ at the powers of f *over* R. Equivalently there is a canonical R-linear isomorphism $\Gamma(D(g), \mathcal{F})_f \cong \Gamma(D(gf), \mathcal{F})$.

Crucially this localisation is carried out entirely on the tile $\mathcal{F}_{(g)}$, where $\mathcal{F}$ is associated to a module (the tile is globally presented, hence of the form $\widetilde{M_g}$), and never on the global object $\mathcal{F}$. This is exactly the structural feature that keeps the keystone Lemma 287 non-circular: the keystone-at-g statement is *not* invoked; only the presented-tile lemma is. ☐

Lemma 284 (Per-overlap section localisation at f)**.** *Let $X = \operatorname{Spec} R$, let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module, let $f \in R$, and let $a, b \in R$ be such that $D(a)$ lies in a cover member of Lemma 261 carrying a global presentation (so the product ab is an overlap of that cover). Then the section-restriction map $\Gamma(D(a) \cap D(b), \mathcal{F}) \to \Gamma(D(af) \cap D(bf), \mathcal{F})$ exhibits its target as the localisation of its source at the powers of f; equivalently, there is a canonical R-linear isomorphism $\Gamma(D(a) \cap D(b), \mathcal{F})_f \cong \Gamma(D(af) \cap D(bf), \mathcal{F})$.*

Proof. This is the per-tile localisation Lemma 283 taken at the single element $g = ab$, transported along the two basic-open lattice identities $D(ab) = D(a) \cap D(b)$ and $D((ab)f) = D(af) \cap D(bf)$. Applying the per-tile lemma to $g = ab$ gives that the restriction $\Gamma(D(ab), \mathcal{F}) \to \Gamma(D(abf), \mathcal{F})$ localises at the powers of f. The source identity rewrites $D(ab)$ as $D(a) \cap D(b)$; the target identity rewrites $D(abf) = D((ab)f)$ as $D(af) \cap D(bf)$ (both are the closed-under-products meet of the corresponding basic opens). Transporting the localisation map along the induced isomorphisms of section modules — which fold the three intervening restriction maps into the single restriction along $D(af) \cap D(bf) \subseteq D(a) \cap D(b)$ — preserves the IsLocalizedModule property, yielding the claim. □

Lemma 285 (Kernel comparison: a left-exact ladder localises on the left)**.** *Let $S \subseteq R$ be a submonoid of a commutative ring, and consider a commutative ladder of R-modules*

$$\begin{array}{ccccc} A & \xrightarrow{i} & B & \xrightarrow{p} & C \\ \downarrow a & & \downarrow b & & \downarrow c \\ A' & \xrightarrow{i'} & B' & \xrightarrow{p'} & C' \end{array}$$

*in which both rows are left-exact (i and i' injective, with $\operatorname{im} i = \ker p$ and $\operatorname{im} i' = \ker p'$). Suppose the two right-hand verticals $b : B \to B'$ and $c : C \to C'$ are localisation maps at S (*IsLocalizedModule $S\, b$ *and* IsLocalizedModule $S\, c$*). Then the left vertical $a : A \to A'$ is a localisation map at S (*IsLocalizedModule $S\, a$*). This is the converse direction of the Mathlib fact that localisation preserves exactness (Lemma 267): there, the verticals being localisations forces the rows to stay exact; here, the rows being exact lets a localisation on the two right columns be transported to the left column.*

Proof. We verify the three defining clauses of IsLocalizedModule $S\, a$ by a diagram chase, using only the left-exactness of the two rows and the localisation hypotheses on b and c.

(i) Each $s \in S$ acts invertibly on A'. Injectivity of $s \cdot -$ on A': if $s \cdot x = s \cdot y$ then $s \cdot i'(x) = s \cdot i'(y)$, so $i'(x) = i'(y)$ because $s \cdot -$ is bijective on B'; injectivity of i' gives $x = y$. Surjectivity of $s \cdot -$ onto A': given $x \in A'$, pick $w \in B'$ with $s \cdot w = i'(x)$ (bijectivity of $s \cdot -$ on B'); then $s \cdot p'(w) = p'(i'(x)) = 0$, and since $s \cdot -$ is injective on C' we get $p'(w) = 0$, so left-exactness of the bottom row yields $x' \in A'$ with $i'(x') = w$, whence $i'(s \cdot x') = s \cdot w = i'(x)$ and $s \cdot x' = x$. Thus $s \cdot -$ is bijective on A'.

(ii) Every $y \in A'$ is $s^{-1} \cdot a(x)$ for some $x \in A$, $s \in S$. Map y into B' via i'; the surjectivity clause of b gives $w \in B$ and $s \in S$ with $b(w) = s \cdot i'(y)$. Then $c(p(w)) = p'(b(w)) = s \cdot p'(i'(y)) = 0 = c(0)$, so since c is a localisation map there exists $t \in S$ with $t \cdot p(w) = 0$, i.e. $p(t \cdot w) = 0$. Left-exactness of the top row yields $x \in A$ with $i(x) = t \cdot w$. Then $i'(a(x)) = b(i(x)) = t \cdot b(w) = (ts) \cdot i'(y) = i'((ts) \cdot y)$, and injectivity of i' gives $a(x) = (ts) \cdot y$, i.e. $y = (ts)^{-1} \cdot a(x)$.

(iii) If $a(x) = a(x')$ then $s \cdot x = s \cdot x'$ for some $s \in S$. From $a(x) = a(x')$ we get $b(i(x)) = i'(a(x)) = i'(a(x')) = b(i(x'))$; since b is a localisation map there exists $s \in S$ with $s \cdot i(x) = s \cdot i(x')$, i.e. $i(s \cdot x) = i(s \cdot x')$, and injectivity of i gives $s \cdot x = s \cdot x'$.

The three clauses are exactly IsLocalizedModule $S\, a$. □

Lemma 286 (Kernel comparison: the localised global equalizer matches the $D(f)$-cover equalizer)**.** *Let $X = \operatorname{Spec} R$, let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module, let $f \in R$, and let $g_1, \dots, g_n$ be the finite presentation cover of Lemma 261, with* $\operatorname{span}\{g_j\} = R$*. Then the canonical R-linear map $\Gamma(X, \mathcal{F})_f \to \Gamma(D(f), \mathcal{F})$ induced by $\rho_f : \Gamma(X, \mathcal{F}) \to \Gamma(D(f), \mathcal{F})$ (the localisation lift of ρ_f at the powers of f) is an isomorphism.*

Proof. By Lemma 287, the section-restriction map $\rho_f : \Gamma(X, \mathcal{F}) \to \Gamma(D(f), \mathcal{F})$ is shown to exhibit $\Gamma(D(f), \mathcal{F})$ as the localisation of $\Gamma(X, \mathcal{F})$ at the powers of f. Packaging this localisation map as a linear equivalence (Lemma 249) yields the canonical R-linear isomorphism $\Gamma(X, \mathcal{F})_f \cong \Gamma(D(f), \mathcal{F})$; its underlying map is exactly the localisation lift of ρ_f, so that lift is an isomorphism. □

Lemma 287 (Sections over a basic open localize the global sections)**.** *Source: Stacks Project, Schemes, Tag 01HV,* `lemma-spec-sheaves` *(4) and* `lemma-widetilde-pullback`*. Let $X = \operatorname{Spec} R$, let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module, and let $f \in R$. Then the restriction-of-sections* *map $\rho_f : \Gamma(X,\mathcal{F}) \to \Gamma(D(f),\mathcal{F})$ exhibits $\Gamma(D(f),\mathcal{F})$ as the localization of $\Gamma(X,\mathcal{F})$ at the powers of f. This is the generalization of Tag 01HV (4) ($\Gamma(D(f),\widetilde{M}) = M_f$) from $\widetilde{M}$ to an arbitrary quasi-coherent $\mathcal{F}$.*

Proof. By Lemma 261, refine the quasi-coherence cover of $\mathcal{F}$ to a finite standard cover $\operatorname{Spec} R = \bigcup_{j=1}^{n} D(g_j)$ with $\operatorname{span}\{g_1,\dots,g_n\} = R$, where each $D(g_j)$ — and hence each overlap $D(g_jg_k)$ — lies in a cover member carrying a presentation of the over-picture restriction, so that the tiles $\mathcal{F}_{(g_j)}$ and $\mathcal{F}_{(g_jg_k)}$ are globally presented.

Write the degree-0/1 sheaf-axiom equalizer (Lemma 266) twice: once for $\mathcal{F}$ on the standard cover $X = \bigcup_j D(g_j)$ (take $W = X$), and once for the induced cover $D(f) = \bigcup_j (D(f) \cap D(g_j))$ of $D(f)$ (take $W = D(f)$, using $D(f) \cap D(g_j) = D(g_jf)$):

$$0 \to \Gamma(X,\mathcal{F}) \xrightarrow{\rho} \prod_j \Gamma(D(g_j),\mathcal{F}) \xrightarrow{\delta} \prod_{j,k} \Gamma(D(g_jg_k),\mathcal{F}),$$

$$0 \to \Gamma(D(f),\mathcal{F}) \xrightarrow{\rho'} \prod_j \Gamma(D(g_jf),\mathcal{F}) \xrightarrow{\delta'} \prod_{j,k} \Gamma(D(g_jg_kf),\mathcal{F}).$$

Both rows are left-exact (the equalizer presents the first map as the kernel of the second). They assemble into a commutative ladder

$$\begin{array}{ccccc}
\Gamma(X,\mathcal{F}) & \xrightarrow{\rho} & \prod_j \Gamma(D(g_j),\mathcal{F}) & \xrightarrow{\delta} & \prod_{j,k} \Gamma(D(g_jg_k),\mathcal{F}) \\
\downarrow \rho_f & & \downarrow b & & \downarrow c \\
\Gamma(D(f),\mathcal{F}) & \xrightarrow{\rho'} & \prod_j \Gamma(D(g_jf),\mathcal{F}) & \xrightarrow{\delta'} & \prod_{j,k} \Gamma(D(g_jg_kf),\mathcal{F})
\end{array}$$

whose three verticals are the section-restriction maps along the inclusions $D(f) \subseteq X$, $D(g_jf) \subseteq D(g_j)$, and $D(g_jg_kf) \subseteq D(g_jg_k)$. Both squares commute by functoriality of restriction: restricting a global section first to $D(g_j)$ and then to $D(g_jf)$ equals restricting first to $D(f)$ and then to $D(g_jf)$, and likewise on the overlaps.

The two right-hand verticals are localisations at the powers of f. Componentwise, the per-tile localisation (Lemma 283) exhibits $\Gamma(D(g_jf),\mathcal{F})$ as the localisation of $\Gamma(D(g_j),\mathcal{F})$ at the powers of f, and the per-overlap localisation (Lemma 284) — the transport of Lemma 283 along $D(g_jg_k) = D(g_j) \cap D(g_k)$ and $D(g_jg_kf) = D(g_jf) \cap D(g_kf)$ — exhibits $\Gamma(D(g_jg_kf),\mathcal{F})$ as the localisation of $\Gamma(D(g_jg_k),\mathcal{F})$. Since a finite product of localisation maps at the same set is again a localisation, the product maps b and c are IsLocalizedModule at the powers of f.

Apply the abstract left-exact-ladder kernel comparison Lemma 285 to this ladder, with $S =$ powers f, the two left-exact rows, and the two localised right-hand verticals b, c. It concludes that the left vertical $\rho_f : \Gamma(X,\mathcal{F}) \to \Gamma(D(f),\mathcal{F})$ is itself a localisation map at the powers of f; equivalently, its canonical lift $\Gamma(X,\mathcal{F})_f \cong \Gamma(D(f),\mathcal{F})$ is an isomorphism. (The exactness of localisation, Lemma 267, is the converse statement: it certifies that the localised X-row is again left-exact, the consistency under which the kernel comparison operates.)

Crucial non-circularity point. The only "sections-localise" inputs are the per-tile localisations (Lemma 283), which live on the tiles $\mathcal{F}_{(g_j)}$ and $\mathcal{F}_{(g_jg_k)}$. There $\mathcal{F}$ is associated to a module — the tile is globally presented, hence $\mathcal{F}_{(g)} \cong \widetilde{M_g}$ (since $\widetilde{(-)}$ preserves colimits), **not** by computing $\Gamma(D(g),-)$ of an abstract cokernel — and the localisation is given by Lemma 260 via the canonical section equality $\Gamma(\mathcal{F}_{(g)},V) = \Gamma(\mathcal{F},\iota(V))$ (Lemma 257). The global statement for $\mathcal{F}$ is then recovered from the tiles by the sheaf-axiom equalizer and the exactness of localisation, **not** by identifying $\Gamma(X,\mathcal{F})$ with an abstract localised module of its restrictions. In particular no instance of the keystone at the g_j is invoked, so the regress that an algebraic span-cover descent on the global sections would create (reducing the keystone-at-f to the keystone-at-g_j, the same statement) is avoided. □

Lemma 288 (Quasi-coherence implies the counit is an isomorphism, via section-localization (01I8))**.** *Source: Stacks Project, Schemes, Tag 01I8,* `lemma-quasi-coherent-affine`*. Let $X = \operatorname{Spec} R$ and let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Then the tilde–Γ counit* fromTilde : $\widetilde{\Gamma(X, \mathcal{F})} \to \mathcal{F}$ *is an isomorphism. This is the affine structure theorem 01I8, proved here by checking the counit on the basis of distinguished opens. It is the unconditional instance that discharges the* IsIso(fromTilde) *hypothesis of Lemma 243.*

Proof. The forgetful functor reflects isomorphisms and the distinguished opens $\{D(f)\}$ form a basis of $\operatorname{Spec} R$ (Lemma 248), so it suffices to show that the component of fromTilde over each $D(f)$ is an isomorphism. By Lemma 246 that component is the localization lift IsLocalizedModule. lift(powers f) of the section-restriction map $\rho_f : \Gamma(X, \mathcal{F}) \to \Gamma(D(f), \mathcal{F})$ along the structure-sheaf localization $\widetilde{(-)}$. toOpen, the latter being an IsLocalizedModule at the powers of f. By the keystone Lemma 287, ρ_f is itself an IsLocalizedModule at the powers of f; hence both the source and the target of the component are localizations of $\Gamma(X, \mathcal{F})$ at the powers of f, and the lift between two such localizations is an isomorphism (Lemma 249). Thus fromTilde is an isomorphism on every $D(f)$, and therefore an isomorphism. (Equivalently, this exhibits $\mathcal{F}$ in the essential image of $\widetilde{(-)}$, the criterion of Lemma 247.) □

Remark 289 (The 01I8 decomposition: quasi-coherence $\Rightarrow$ counit is iso)**.** *Source: Stacks Project, Schemes, Tag 01I8,* `lemma-quasi-coherent-affine` *(proof).* Let $X = \operatorname{Spec} R$ and let $\mathcal{F}$ be quasi-coherent. Choose a finite basic-open cover $\{D(g_j)\}$ on which $\mathcal{F}$ admits presentations. The sheaf axiom presents both $\Gamma(X, \mathcal{F})$ and $\Gamma(D(f), \mathcal{F})$ as equalizers for the covers $\{D(g_j)\}$ and $\{D(fg_j)\}$. Exactness of localization, together with the section-localization isomorphisms on each tile and overlap, identifies these equalizers after localizing at f. Hence

$$\Gamma(D(f), \mathcal{F}) \cong \Gamma(X, \mathcal{F})_f.$$

The component of fromTilde on $D(f)$ is the canonical map between these two localizations, so it is an isomorphism. Since distinguished opens form a basis, the counit is an isomorphism and $\mathcal{F} \cong \widetilde{\Gamma(X, \mathcal{F})}$.

Alternatively, one may choose global generators for $\mathcal{F}$, prove that the kernel of the resulting epimorphism from a free module is quasi-coherent, and compare the two resulting presentations after applying $\widetilde{(-)}$. The lemmas below record this global-generation argument.

We now give the global-generation argument. It uses quasi-coherence of free modules and finite basic-open refinements of covers, as in the Stacks Project section "Quasi-coherent sheaves on affines".

Lemma 290 (Free $\mathcal{O}$-modules are quasi-coherent)**.** *Source: Stacks Project, Schemes,* `lemma-colimit-quasi-coherent`*. Let $X = \operatorname{Spec} R$. For any index type ι the free $\mathcal{O}_X$-module $\mathcal{O}_X^{(\iota)}$ on ι generators is quasi-coherent. Indeed it is the sheaf $\widetilde{R^{(\iota)}}$ associated to the free R-module $R^{(\iota)} = \bigoplus_\iota R$ (the finitely-supported functions $\iota \to R$): the tilde functor carries $R^{(\iota)}$ to a direct sum of copies of the structure sheaf $\mathcal{O}_X = \widetilde{R}$, and every sheaf of the form $\widetilde{M}$ is quasi-coherent. Quasi-coherence is preserved under the isomorphism $\mathcal{O}_X^{(\iota)} \cong \widetilde{R^{(\iota)}}$; concretely it is obtained from the identification of the free $\mathcal{O}_X$-module with the tilde of the free R-module.*

Lemma 291 (Finite basic-open refinement of a cover of $\operatorname{Spec} R$)**.** *Source: Stacks Project, Schemes,* `lemma-standard-open` *and* `lemma-quasi-coherent-affine` *(proof). Let R be a ring and $(U_i)_{i\in\iota}$ a family of opens of* $\operatorname{Spec} R$ *with* $\bigsqcup_i U_i = \operatorname{Spec} R$ *(i.e.* $\bigcup_i U_i = \top$*). Then there exist finitely many elements $f_1, \dots, f_n \in R$ and indices $\varphi : \{1, \dots, n\} \to \iota$ such that each basic open satisfies $D(f_j) \subseteq U_{\varphi(j)}$ and the f_j generate the unit ideal,* $\operatorname{span}\{f_1, \dots, f_n\} = R$ *(equivalently* $\bigcup_j D(f_j) = \operatorname{Spec} R$*).*

Proof. The basic opens $D(f)$, $f \in R$, form a basis of the Zariski topology on $\operatorname{Spec} R$; hence every U_i is a union of basic opens, and refining the cover (U_i) yields a cover of $\operatorname{Spec} R$ by basic opens $D(f)$, each contained in some U_φ. The space $\operatorname{Spec} R$ is quasi-compact, so finitely many of these basic opens $D(f_1), \dots, D(f_n)$ already cover $\operatorname{Spec} R$, with $D(f_j) \subseteq U_{\varphi(j)}$ by construction. Finally a finite family of basic opens covers all of $\operatorname{Spec} R$ if and only if the f_j generate the unit ideal: $\bigcup_j D(f_j) = \operatorname{Spec} R \iff \operatorname{span}\{f_j\} = R$, since the closed complement $V(\operatorname{span}\{f_j\})$ is empty exactly when the ideal is the whole ring. ☐

Lemma 292 (Restriction of an $\mathcal{O}_{\operatorname{Spec} R}$-module to $D(f)$ as an $\mathcal{O}_{\operatorname{Spec} R_f}$-module)**.** *Source: Stacks Project, Schemes,* `lemma-widetilde-pullback`*, Tag 01I8. Let $X = \operatorname{Spec} R$, let $\mathcal{F}$ be an $\mathcal{O}_X$-module (an object of* $(\operatorname{Spec} R).\mathrm{Modules}$*), and let $f \in R$. Then, transporting along the canonical isomorphism of affine schemes $D(f) \cong \operatorname{Spec} R_f$, the restriction $\mathcal{F}|_{D(f)}$ is naturally an $\mathcal{O}_{\operatorname{Spec} R_f}$-module: there is an object* $\mathcal{F}_{(f)} \in (\operatorname{Spec} R_f).\mathrm{Modules}$ *together with an isomorphism $\mathcal{F}|_{D(f)} \cong \mathcal{F}_{(f)}$ of sheaves of modules over the identification $D(f) \cong \operatorname{Spec} R_f$. This assignment is functorial in $\mathcal{F}$.*

Proof. The basic open $D(f) \subseteq \operatorname{Spec} R$ is an affine scheme, and the canonical comparison $D(f) \cong \operatorname{Spec} R_f$ is an isomorphism of affine schemes (it corresponds to the localisation ring map $R \to R_f$). Pulling the restricted module $\mathcal{F}|_{D(f)}$ back along the inverse of this isomorphism transports it to a sheaf of $\mathcal{O}_{\operatorname{Spec} R_f}$-modules $\mathcal{F}_{(f)}$; the unit of the transport supplies the comparison isomorphism. Restriction of sheaves of modules and pullback along an isomorphism are both functorial, so the assignment $\mathcal{F} \mapsto \mathcal{F}_{(f)}$ is functorial in $\mathcal{F}$. ☐

Definition 293 (The distinguished open $D(f)$ as an open of $\operatorname{Spec} R$)**.** Let R be a commutative ring and $f \in R$. The *distinguished open* $D(f)$ is the open subset $\mathrm{specBasicOpen}\, f = \{\mathfrak{p} : f \notin \mathfrak{p}\}$ of $\operatorname{Spec} R$, packaged as an object of the frame of opens $(\operatorname{Spec} R).\mathrm{Opens}$; it is the basic open $\mathrm{PrimeSpectrum.basicOpen}\, f$.

Lemma 294 (The localisation morphism $\operatorname{Spec} R_f \to \operatorname{Spec} R$)**.** *Let R be a commutative ring and $f \in R$. The morphism of schemes* $\mathrm{specAwayToSpec}\, f : \operatorname{Spec} R_f \to \operatorname{Spec} R$ *is the composite of the inverse of the comparison isomorphism $D(f) \cong \operatorname{Spec} R_f$* ($\mathrm{basicOpenIsoSpecAway}\, f$) *with the open immersion $D(f) \hookrightarrow \operatorname{Spec} R$ of Definition 293; it is the geometric morphism underlying the restriction transport of Lemma 292.*

Lemma 295 (The localisation morphism is the spectrum of $R \to R_f$)**.** *With notation as in Lemma 294, the morphism* $\mathrm{specAwayToSpec}\, f$ *equals the spectrum of the localisation ring map, i.e.* $\mathrm{specAwayToSpec}\, f = \mathrm{Spec.map}(\mathrm{algebraMap}\, R\, (R_f))$ *where* $R_f = \mathrm{Localization.Away}\, f$*. Hence the restriction-to-$D(f)$ transport is pullback along* Spec *of the localisation $R \to R_f$.*

Proof. By definition $\mathrm{specAwayToSpec}\, f$ is $(\mathrm{basicOpenIsoSpecAway}\, f)^{-1} \circ \iota_{D(f)}$; rewriting with Iso.inv_comp_eq reduces the claim to the defining property of the comparison isomorphism $\mathrm{basicOpenIsoSpecAway}\, f$, which identifies $D(f)$ with $\operatorname{Spec} R_f$ via the localisation map $R \to R_f$. ☐

Lemma 296 (Restricting $\widetilde{M}$ to $D(f)$ gives $\widetilde{M_f}$). *Source: Stacks Project, Schemes,* `lemma-widetilde-pullback`, *Tag 01I8. Let R be a ring, M an R-module, and $f \in R$. Under the identification $D(f) \cong \operatorname{Spec} R_f$ of Lemma 292, the restriction of the associated sheaf $\widetilde{M}$ to the standard open $D(f)$ is isomorphic to the associated sheaf of the away-localised module: $\widetilde{M}|_{D(f)} \cong \widetilde{M_f}$ as $\mathcal{O}_{\operatorname{Spec} R_f}$-modules, where $M_f = (\operatorname{powers} f)^{-1}M$.*

Proof. Transport $\widetilde{M}|_{D(f)}$ to an $\mathcal{O}_{\operatorname{Spec} R_f}$-module by Lemma 292. On a standard open $D(g) \subseteq D(f)$ the sections of $\widetilde{M}|_{D(f)}$ are the localisation M_{gf}, which is exactly the localisation of M_f at (the image of) g; since $\psi^{-1}(D(f)) = D(\psi^\sharp(f))$ for the localisation map $\psi^\sharp : R \to R_f$, the value on $D(f)$ itself is M_f. These identifications are compatible with restriction, hence patch to an isomorphism of sheaves $\widetilde{M}|_{D(f)} \cong \widetilde{M_f}$ over $\operatorname{Spec} R_f$. □

Lemma 297 (Presentations restrict along an open immersion of basic opens). *Source: Stacks Project, Schemes,* `lemma-quasi-coherent-affine` *(proof), Tag 01I8. Let $X = \operatorname{Spec} R$, let $\mathcal{F}$ be an $\mathcal{O}_X$-module, let $U \subseteq X$ be open, and suppose $\mathcal{F}|_U$ admits a presentation $\mathcal{O}_U^{(J)} \to \mathcal{O}_U^{(I)} \to \mathcal{F}|_U \to 0$. Then for every basic open $D(g) \subseteq U$, the restricted module $\mathcal{F}|_{D(g)}$ — viewed as an $\mathcal{O}_{\operatorname{Spec} R_g}$-module via Lemma 292 — admits a presentation $\mathcal{O}^{(J)} \to \mathcal{O}^{(I)} \to \mathcal{F}|_{D(g)} \to 0$ over $\operatorname{Spec} R_g$; in particular $\mathcal{F}|_{D(g)}$ is quasi-coherent over R_g.*

Proof. Restriction of sheaves of modules along the open immersion $D(g) \hookrightarrow U$ is an additive functor that preserves the structure sheaf, hence preserves cokernels and arbitrary direct sums of $\mathcal{O}$; applying it to the given right-exact sequence yields a right-exact sequence $\mathcal{O}_{D(g)}^{(J)} \to \mathcal{O}_{D(g)}^{(I)} \to \mathcal{F}|_{D(g)} \to 0$. Transporting along $D(g) \cong \operatorname{Spec} R_g$ (Lemma 292) turns the free modules $\mathcal{O}_{D(g)}^{(I)}$, $\mathcal{O}_{D(g)}^{(J)}$ into the free $\mathcal{O}_{\operatorname{Spec} R_g}$-modules $\widetilde{R_g}^{(I)}$, $\widetilde{R_g}^{(J)}$ (the structure sheaf restricts to the structure sheaf, and free modules restrict to free modules by Lemma 296). This is a presentation of $\mathcal{F}|_{D(g)}$ over $\operatorname{Spec} R_g$, exhibiting it as quasi-coherent. □

Lemma 298 (Restriction of a quasi-coherent sheaf to a basic open is quasi-coherent). *Source: Stacks Project, Schemes,* `lemma-quasi-coherent-affine` *(proof), Tag 01I8. Let $X = \operatorname{Spec} R$, let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module, and let $f \in R$. Then, transporting along the canonical isomorphism of affine schemes $D(f) \cong \operatorname{Spec} R_f$ (Lemma 292), the restriction $\mathcal{F}|_{D(f)}$ is a quasi-coherent $\mathcal{O}_{\operatorname{Spec} R_f}$-module, and in particular it admits a presentation $\mathcal{O}^{(J)} \to \mathcal{O}^{(I)} \to \mathcal{F}|_{D(f)} \to 0$ over $\operatorname{Spec} R_f$. Consequently $\mathcal{F}|_{D(f)} \cong \widetilde{M}$ for the R_f-module $M = \Gamma(D(f), \mathcal{F})$.*

Proof. Refine the quasi-coherence cover of $\mathcal{F}$ to a finite standard-open cover $\operatorname{Spec} R = \bigcup_j D(g_j)$, $D(g_j) \subseteq U_{\varphi(j)}$, by Lemma 291. Transport $\mathcal{F}|_{D(f)}$ to an $\mathcal{O}_{\operatorname{Spec} R_f}$-module via Lemma 292. The basic opens $D(g_j f) = D(g_j) \cap D(f)$ cover $D(f) = \operatorname{Spec} R_f$; on each, Lemma 297 restricts the presentation over $U_{\varphi(j)}$ to one of $\mathcal{F}|_{D(g_j f)}$, and Lemma 296 identifies the restriction with the associated sheaf of an away-localised module. Hence $\mathcal{F}|_{D(f)}$ is locally of the form $\widetilde{(-)}$, and the finitely many local presentations assemble into a presentation of $\mathcal{F}|_{D(f)}$ over $\operatorname{Spec} R_f$. Feeding it to Lemma 244 yields $\mathcal{F}|_{D(f)} \cong \Gamma(\widetilde{D(f)}, \mathcal{F})$ as an R_f-module. □

Lemma 299 (IsLocalizedModule is local on a finite spanning cover). *Let R be a commutative ring, let M and N be R-modules, let $g : M \to N$ be an R-linear map, let $f \in R$, and let $s : \{1, \dots, n\} \to R$ be a finite family with $\operatorname{span}\{s_1, \dots, s_n\} = R$ (the unit ideal). For each j write $M_{s_j} = (\operatorname{powers} s_j)^{-1}M$ and $N_{s_j} = (\operatorname{powers} s_j)^{-1}N$ for the localisations at the powers of s_j, and* *let $g_{s_j} : M_{s_j} \to N_{s_j}$ be the induced R-linear (equivalently R_{s_j}-linear) map. Suppose that for every j the localised map g_{s_j} exhibits N_{s_j} as the localisation of M_{s_j} at the powers of (the image of) f, i.e.* $\mathrm{IsLocalizedModule}(\mathrm{powers}\, f)\, g_{s_j}$. *Then g itself exhibits N as the localisation of M at the powers of f:* $\mathrm{IsLocalizedModule}(\mathrm{powers}\, f)\, g$.

Proof. This is a purely commutative-algebra statement: the predicate $\mathrm{IsLocalizedModule}(\mathrm{powers}\, f)\, g$ is the conjunction of its three defining conditions, and each is verified by descent along the spanning cover. Throughout we use the standard partition of unity: since $\mathrm{span}\{s_j\} = R$, for any chosen exponents N_j the powers $s_j^{N_j}$ again span the unit ideal, so there are $r_j \in R$ with $\sum_j r_j s_j^{N_j} = 1$; and an element of an R-module is zero iff its image in M_{s_j} vanishes for every j (locality of being zero on a spanning cover, the s_j-torsion form of the same partition of unity).

f acts invertibly on N. For each j, the hypothesis $\mathrm{IsLocalizedModule}(\mathrm{powers}\, f)\, g_{s_j}$ makes multiplication by f on N_{s_j} bijective. Multiplication by f on N is an isomorphism iff it is so after localising at each s_j (an R-linear endomorphism is bijective iff it is bijective on the cover $\{D(s_j)\}$, since injectivity and surjectivity are both local on a spanning cover); hence $f \cdot (-) : N \to N$ is bijective, and likewise every power f^k.

Every $y \in N$ satisfies $f^k y = g(x)$ for some $x \in M$, k. Fix $y \in N$. For each j, surjectivity of g_{s_j} as a localisation at the powers of f yields $m_j \in M$, exponents a_j (clearing the s_j-denominator) and k_j with $s_j^{a_j} f^{k_j} y = g(m_j)$ holding after localising at s_j; clearing the remaining s_j-torsion (enlarging a_j) we may take this as an honest equality $s_j^{a_j} f^{k_j} y = g(m_j)$ in N. Choose a common $k = \max_j k_j$ (replacing m_j by $f^{k-k_j} m_j$) so that $s_j^{a_j} f^k y = g(m_j)$ for all j. Pick, by the partition of unity, r_j with $\sum_j r_j s_j^{a_j} = 1$ (possible because the $s_j^{a_j}$ span the unit ideal). Then $f^k y = \sum_j r_j s_j^{a_j} f^k y = \sum_j r_j\, g(m_j) = g(\sum_j r_j m_j)$, so $x = \sum_j r_j m_j$ and the exponent k work.

If $g(x) = g(x')$ then $f^k(x - x') = 0$ for some k. Put $z = x - x'$, so $g(z) = 0$. For each j, the equaliser clause of $\mathrm{IsLocalizedModule}(\mathrm{powers}\, f)\, g_{s_j}$ gives, from the vanishing of $g_{s_j}(z) = 0$ in N_{s_j}, an exponent k_j with $f^{k_j} z = 0$ after localising at s_j; clearing the s_j-denominator there is b_j with $s_j^{b_j} f^{k_j} z = 0$ in M. Take $k = \max_j k_j$, so $s_j^{b_j} f^k z = 0$ for all j. With r_j chosen so that $\sum_j r_j s_j^{b_j} = 1$, $f^k z = \sum_j r_j s_j^{b_j} f^k z = 0$. Hence $f^k(x - x') = 0$.

The three clauses are exactly the data of $\mathrm{IsLocalizedModule}(\mathrm{powers}\, f)\, g$. ☐

Lemma 300 (Localised sections of a quasi-coherent sheaf). *Source: Stacks Project, Schemes, `lemma-widetilde-pullback`. Let $X = \mathrm{Spec}\, R$, let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module, and let $f \in R$. Then the restriction of sections $\Gamma(X, \mathcal{F}) \to \Gamma(D(f), \mathcal{F})$ exhibits $\Gamma(D(f), \mathcal{F})$ as the localisation of $\Gamma(X, \mathcal{F})$ at the powers of f: it is an* IsLocalizedModule *for the submonoid $\{1, f, f^2, \ldots\}$. In particular every section over $D(f)$ is, after multiplication by a suitable power f^N, the restriction of a global section, and a global section restricting to* 0 *on $D(f)$ is annihilated by a power of f.*

Proof. Choose, by Lemma 291, a finite basic-open cover $\mathrm{Spec}\, R = \bigcup_{j=1}^n D(s_j)$ with $\mathrm{span}\{s_1, \ldots, s_n\} = R$. On each member $D(s_j) = \mathrm{Spec}\, R_{s_j}$ the restriction $\mathcal{F}|_{D(s_j)}$ is, by Lemma 298, a quasi-coherent $\mathcal{O}_{\mathrm{Spec}\, R_{s_j}}$-module admitting a presentation, hence (by the affine structure theorem) $\mathcal{F}|_{D(s_j)} \cong \widetilde{M_j}$ for an R_{s_j}-module $M_j = \Gamma(D(s_j), \mathcal{F})$. For the standard sheaf $\widetilde{M_j}$ the sections over a distinguished open $D(g) \subseteq D(s_j)$ are the away localisation $(M_j)_g$ (`lemma-widetilde-pullback`: $\Gamma(D(g), \widetilde{M}) = M_g$). Applying this with g the image of f shows that, after localising at s_j, the restriction map $\Gamma(X, \mathcal{F}) \to \Gamma(D(f), \mathcal{F})$ becomes the localisation map $(M_j) \to (M_j)_f$, i.e. the localised map $g_{s_j} : \Gamma(X, \mathcal{F})_{s_j} \to \Gamma(D(f), \mathcal{F})_{s_j}$ is an IsLocalizedModule for the powers of f.

Now apply Lemma 299 with $M = \Gamma(X, \mathcal{F})$, $N = \Gamma(D(f), \mathcal{F})$, the restriction map g, the element f, and the spanning family $s_1, \ldots, s_n$: since g_{s_j} is an IsLocalizedModule for the powers of f for every j and the s_j generate the unit ideal, the patching primitive concludes that g itself is an IsLocalizedModule for $\{1, f, f^2, \ldots\}$. Concretely this packages the two clauses: every section over $D(f)$ is, after multiplication by some f^N, the restriction of a global section, and a global section vanishing on $D(f)$ is annihilated by a power of f. □

Lemma 301 (Global generation of a quasi-coherent sheaf on an affine)**.** *Source: Stacks Project, Schemes, Tag 01I8, `lemma-quasi-coherent-affine` (proof); cf. Hartshorne II.5.16–17. Let $X = \operatorname{Spec} R$ and $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$-module. Then $\mathcal{F}$ is generated by its global sections: there is an index set I of global sections and an epimorphism of $\mathcal{O}_X$-modules $\mathcal{O}_X^{(I)} \twoheadrightarrow \mathcal{F}$. Equivalently, $\mathcal{F}$ carries a global generating family (the data $\mathcal{F}$.*GeneratingSections*).*

Proof. By Lemma 291 pick a finite basic-open cover $\operatorname{Spec} R = \bigcup_{j=1}^n D(f_j)$ on each of which $\mathcal{F}|_{D(f_j)} \cong \widetilde{M_j}$. Each M_j is generated, as an R_{f_j}-module, by local sections $t \in \Gamma(D(f_j), \mathcal{F})$. By the surjectivity half of Lemma 300, after multiplying by a suitable power f_j^N each such local section is the restriction of a global section $s \in \Gamma(X, \mathcal{F})$; since f_j is a unit on $D(f_j)$, these global sections still generate $\mathcal{F}$ over $D(f_j)$. Collecting the finitely many resulting global sections over all j gives a family that generates $\mathcal{F}$ on every member of the cover, hence everywhere: the induced map $\mathcal{O}_X^{(I)} \to \mathcal{F}$ is surjective on each $D(f_j)$ and so is an epimorphism of sheaves. □

Lemma 302 (The tilde functor preserves kernels)**.** *Source: Stacks Project, Schemes, `lemma-kernel-cokernel-quasi-coherent` (and Tag 01HV, exactness of $\widetilde{(-)}$). Let $X = \operatorname{Spec} R$. The functor $M \mapsto \widetilde{M}$ from R-modules to $\mathcal{O}_X$-modules is exact; in particular it preserves kernels: for an R-module map $\varphi : M \to N$ the induced $\widetilde{\varphi} : \widetilde{M} \to \widetilde{N}$ has $\operatorname{Ker}(\widetilde{\varphi}) = \widetilde{\operatorname{Ker}(\varphi)}$, and more generally $\widetilde{(-)}$ carries finite limits of R-modules to the corresponding limits of associated sheaves.*

Proof. Source: Stacks Project, Schemes, `lemma-spec-sheaves` (item (7) and proof), Tag 01HV. Kernels in the category of sheaves of $\mathcal{O}_X$-modules are computed objectwise and sheafified, and isomorphisms of sheaves may be checked on stalks: a morphism of $\mathcal{O}_X$-modules is an isomorphism if and only if it induces an isomorphism on every stalk. So it suffices to compare stalks. By the stalk computation for the tilde functor (`lemma-spec-sheaves` item (7)) the stalk of $\widetilde{M}$ at a point $x \in \operatorname{Spec} R$ corresponding to the prime $\mathfrak{p}$ is the localisation $\widetilde{M}_x = M_{\mathfrak{p}}$, naturally in M: the stalk functor at x applied to $\widetilde{(-)}$ is the localisation functor $M \mapsto M_{\mathfrak{p}}$.

Now let $\varphi : M \to N$ be an R-module map with kernel $K = \operatorname{Ker}(\varphi)$, giving the exact sequence $0 \to K \to M \xrightarrow{\varphi} N$. Localisation $R \to R_{\mathfrak{p}}$ is flat, hence the functor $M \mapsto M_{\mathfrak{p}}$ is exact and in particular preserves kernels: $K_{\mathfrak{p}} = \operatorname{Ker}(\varphi_{\mathfrak{p}} : M_{\mathfrak{p}} \to N_{\mathfrak{p}})$. Translating through the stalk identification, the natural comparison map $\widetilde{K} \to \operatorname{Ker}(\widetilde{\varphi})$ induces, at every point $x \leftrightarrow \mathfrak{p}$, the isomorphism $K_{\mathfrak{p}} \xrightarrow{\sim} \operatorname{Ker}(\varphi_{\mathfrak{p}})$ (using that the stalk functor, being a filtered colimit, is itself exact and commutes with forming the kernel). Being a stalkwise isomorphism, the comparison map is an isomorphism of sheaves, so $\operatorname{Ker}(\widetilde{\varphi}) = \widetilde{\operatorname{Ker}(\varphi)}$. The same stalkwise-flatness argument applied to a finite diagram of R-modules shows that $\widetilde{(-)}$ carries finite limits to finite limits; equivalently $\widetilde{(-)}$ is left exact, i.e. preserves finite limits (kernels).

□

Lemma 303 (The kernel of a generating epimorphism is quasi-coherent)**.** *Source: Stacks Project, Schemes,* `lemma-kernel-cokernel-quasi-coherent`*. Let $X = \operatorname{Spec} R$, let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module, and let $\sigma : \mathcal{F}.\mathrm{GeneratingSections}$ be the generating family of Lemma 301, with epimorphism $\sigma.\pi : \mathcal{O}_X^{(I)} \twoheadrightarrow \mathcal{F}$. Then $\operatorname{Ker}(\sigma.\pi)$ is again quasi-coherent on $\operatorname{Spec} R$, and therefore — by Lemma 301 applied to it — globally generated, yielding a second generating family $\tau : (\operatorname{Ker} \sigma.\pi).\mathrm{GeneratingSections}$.*

Proof. The free sheaf $\mathcal{O}_X^{(I)}$ is quasi-coherent (Lemma 290) and $\mathcal{F}$ is quasi-coherent by hypothesis. On each member $D(f_j)$ of a finite trivialising cover both source and target are of the form $\widetilde{(-)}$, and the map $\sigma.\pi$ corresponds to an R_{f_j}-module map; since $\widetilde{(-)}$ preserves kernels (Lemma 302), $\operatorname{Ker}(\sigma.\pi)|_{D(f_j)}$ is the associated sheaf of the kernel of that module map, hence quasi-coherent locally on the cover. Quasi-coherence is local, so $\operatorname{Ker}(\sigma.\pi)$ is quasi-coherent on all of $\operatorname{Spec} R$. Applying Lemma 301 to the quasi-coherent $\operatorname{Ker}(\sigma.\pi)$ produces the generating family τ. ☐

Lemma 304 (Counit is iso from two global generating families)**.** *Let $X = \operatorname{Spec} R$ and $\mathcal{F}$ an $\mathcal{O}_X$-module. Suppose given a global generating family $\sigma : \mathcal{F}.\mathrm{GeneratingSections}$ (an epimorphism $\sigma.\pi : \mathcal{O}_X^{(I)} \twoheadrightarrow \mathcal{F}$) and a global generating family $\tau : (\operatorname{Ker} \sigma.\pi).\mathrm{GeneratingSections}$ of its kernel. Then the tilde–Γ counit $\mathrm{fromTilde} : \widetilde{\Gamma(X, \mathcal{F})} \to \mathcal{F}$ is an isomorphism.*

Proof. The two generating families assemble into a global presentation of $\mathcal{F}$: the composite $\mathcal{O}_X^{(J)} \xrightarrow{\tau.\pi} \operatorname{Ker} \sigma.\pi \hookrightarrow \mathcal{O}_X^{(I)} \xrightarrow{\sigma.\pi} \mathcal{F}$ is a global exact sequence $\mathcal{O}_X^{(J)} \to \mathcal{O}_X^{(I)} \to \mathcal{F} \to 0$, i.e. the data $\mathcal{F}.\mathrm{Presentation}$. Feeding this presentation to Lemma 245 makes fromTilde an isomorphism. ☐

Lemma 305 (Affine structure theorem from two generating families)**.** *Let $X = \operatorname{Spec} R$ and $\mathcal{F}$ an $\mathcal{O}_X$-module equipped with a global generating family σ and a global generating family τ of $\operatorname{Ker} \sigma.\pi$. Then $\mathcal{F} \cong \widetilde{\Gamma(X, \mathcal{F})}$, naturally identifying $\Gamma(D(f), \mathcal{F})$ with $\Gamma(X, \mathcal{F})_f$ for every $f \in R$.*

Proof. By Lemma 304 the counit fromTilde is an isomorphism, which is exactly the conditional hypothesis of Lemma 243; its conclusion gives the isomorphism $\mathcal{F} \cong \widetilde{\Gamma(X, \mathcal{F})}$ and the identification of distinguished-open sections with away localisations. ☐

Lemma 306 (Quasi-coherence implies the counit is an isomorphism by global generation)**.** *Source: Stacks Project, Schemes, Tag 01I8,* `lemma-quasi-coherent-affine`*. Let $X = \operatorname{Spec} R$ and $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$-module. Then the tilde–Γ counit $\mathrm{fromTilde} : \widetilde{\Gamma(X, \mathcal{F})} \to \mathcal{F}$ is an isomorphism. This is the affine global-generation instance ($[\,\mathrm{IsQuasicoherent}\ \mathcal{F}\,] \Rightarrow \mathrm{IsIso}(\mathrm{fromTilde})$, Stacks Tag 01I8) that discharges the conditional hypothesis of Lemma 243.*

Proof. Quasi-coherence of $\mathcal{F}$ gives, by Lemma 301, a global generating family σ. Its kernel $\operatorname{Ker} \sigma.\pi$ is again quasi-coherent and hence globally generated (Lemma 303), yielding a generating family τ. Feeding the pair (σ, τ) to Lemma 304 makes fromTilde an isomorphism. ☐

Lemma 307 (Standard-cover Čech vanishing for quasi-coherent coefficients)**.** *Source: Stacks Project, Cohomology of Schemes, Tag 01XB (*`lemma-quasi-coherent-affine-cohomology-zero`*, proof). Let $X = \operatorname{Spec} R$ be an affine scheme and $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$-module. Then $\mathcal{F}$ has vanishing higher Čech cohomology for the affine cover system (Definition 242, Definition 320): for every standard cover $\mathcal{U}$ and every $p > 0$ one has $\check{H}^p(\mathcal{U}, \mathcal{F}) = 0$. This supplies condition (3) of the basis-comparison Lemma 328, via the tilde-case residual Lemma 314.*

Proof. Step 1 (reduction to the tilde case). By Lemma 243 the quasi-coherent $\mathcal{F}$ is isomorphic to $\widetilde{M}$ for $M = \Gamma(X, \mathcal{F})$, with sections over each face $D(g_\sigma)$ the away localisation M_{g_σ}. Since

Čech cohomology transports along an isomorphism of coefficient presheaves (Lemma 308), it suffices to prove the vanishing for the tilde sheaf $\widetilde{M}$: for every standard cover $\mathcal{U}$ and every $p > 0$, $\check{H}^p(\mathcal{U}, \widetilde{M}) = 0$.

Step 2 (the tilde residual). A standard cover of the affine $X = \operatorname{Spec} R$ in the affine cover system is a cover of a distinguished open $D(f) = \bigsqcup_i D(g_i)$ by smaller distinguished opens; for a proper $D(f)$ the family $\{g_i\}$ does *not* span the unit ideal of R, so the whole-space standard-cover vanishing of Lemma 194 (stated under $\operatorname{span}\{g_i\} = R$) does not apply directly. The required positive-degree vanishing $\check{H}^p(\{D(g_i)\}, \widetilde{M}) = 0$ over such a proper cover is exactly Lemma 314, proved by change of base to $R_f = \text{Localization.Away}\, f$ (where the cover *does* span the unit ideal). Combining Steps 1 and 2 gives $\check{H}^p(\mathcal{U}, \mathcal{F}) = 0$ for all $p > 0$ and every standard cover $\mathcal{U}$; hence $\mathcal{F}$ has vanishing higher Čech cohomology for the affine cover system. ☐

Lemma 308 (Čech cohomology transports along an isomorphism of coefficients)**.** *Project-local transport lemma. Let $U : \iota \to \operatorname{Opens}(X)$ be an index family and let $e : \mathcal{F} \cong \mathcal{G}$ be an isomorphism of presheaves of $\mathcal{O}_X$-modules. If $\check{H}^p(\mathcal{U}, \mathcal{F}) = 0$ for some p, then $\check{H}^p(\mathcal{U}, \mathcal{G}) = 0$.*

Proof. The section Čech complex is functorial in the coefficient presheaf (Definition 316), so e induces an isomorphism of section Čech complexes $\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F}) \cong \check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{G})$, and applying the degree-p homology functor yields an isomorphism $\check{H}^p(\mathcal{U}, \mathcal{F}) \cong \check{H}^p(\mathcal{U}, \mathcal{G})$. A group isomorphic to a zero object is itself zero. ☐

Lemma 309 (A finite family of distinguished opens covers $\operatorname{Spec} R$ iff its elements span the unit ideal)**.** *Provided by Mathlib. For a family $s : \iota \to R$, one has $\bigsqcup_i D(s_i) = \operatorname{Spec} R$ if and only if* $\operatorname{span}(\{s_i : i \in \iota\}) = R$.

Lemma 310 (A standard cover of $D(f)$ spans the unit ideal of R_f)**.** *Project-local infrastructure. Let R be a ring, $f \in R$, and $g : \iota \to R$ a finite family with $D(f) = \bigsqcup_i D(g_i)$ in* $\operatorname{Spec} R$*. Then the images* $\text{algebraMap}\, R\, R_f\, (g_i)$ *span the unit ideal of $R_f = \text{Localization.Away}\, f$:*

$$\operatorname{span}(\{\, g_i/1 : i \in \iota \,\}) = R_f.$$

Proof. Pull the equality $D(f) = \bigsqcup_i D(g_i)$ back along the canonical open immersion $\operatorname{Spec} R_f \cong D(f) \hookrightarrow \operatorname{Spec} R$, whose underlying continuous map is the comap of $\text{algebraMap}\, R\, R_f$. Comap commutes with distinguished opens and with suprema, so the pullback of $\bigsqcup_i D(g_i)$ is $\bigsqcup_i D(g_i/1)$, while the pullback of $D(f)$ is all of $\operatorname{Spec} R_f$ because f is a unit in R_f. Thus $\bigsqcup_i D(g_i/1) = \operatorname{Spec} R_f$, which by the span criterion (a family of distinguished opens covers the whole spectrum iff the elements generate the unit ideal) is equivalent to $\operatorname{span}\{g_i/1\} = R_f$. ☐

Lemma 311 (Localisation transitivity for away modules)**.** *Project-local infrastructure. Let R be a commutative ring, M an R-module, and $a, b \in R$. Write $M_a = \text{LocalizedModule}(\text{Submonoid.powers}\, a)\, M$ for the away localisation of M at a. Then the canonical transitivity comparison $M_a \to M_{ab}$ exhibits M_{ab} as the localisation of M_a at (the image of) b:*

$$M_a[1/b] \;=\; M_{ab},$$

i.e. the composite of the two away-localisation units is again the away localisation at the product ab. In particular, it yields the degreewise identification $M_{g_\sigma} \cong (M_f)_{g_\sigma}$ used in Lemma 314: when f is invertible in M_{g_σ} (because $g_\sigma \in \sqrt{(f)}$), localising M away from g_σ and localising M_f away from g_σ present the same module.

Lemma 312 (Two-step away localisation localises at the product factor)**.** *Project-local infrastructure. Let $R \to R_f$ be the away localisation of R at f, let M be an R-module, and let $\mathrm{mk}_f : M \to M_f$ exhibit M_f as the localisation of M at the powers of f. Suppose $a \in R$ lies in the radical of (f), say $a^j = f \cdot h$ for some $h \in R$ and $j \geq 1$ (equivalently $f \mid a^j$), and let $g : M_f \to N$ be an R_f-linear map exhibiting N as the localisation of M_f at the powers of the image* $\mathrm{algebraMap}\, R\, R_f\, (a)$*. Then the R-linear composite*

$$g \circ \mathrm{mk}_f \;:\; M \longrightarrow N$$

exhibits N as the localisation of M at the powers of a; that is, $N = M[1/a]$.

Proof. Verify the three defining clauses of a localisation module directly. *Units act invertibly:* a power a^n acts on N through its image $\mathrm{algebraMap}\, R\, R_f\, (a)^n$, which is invertible because it is the localisation base for g. *Surjectivity:* any $y \in N$ is $g(m_0)$ divided by a power of a for some $m_0 \in M_f$, and m_0 is $\mathrm{mk}_f(m)$ divided by a power f^l of f; writing $a^{jl} = (fh)^l = f^l h^l$ converts the f^l in the denominator into a power of a, so y is the image of $h^l \cdot m$ up to a power of a. *Cancellation:* if $g(\mathrm{mk}_f(m)) = g(\mathrm{mk}_f(m'))$, chaining the cancellation property of g and then of mk_f produces a single power of a annihilating $m - m'$. This route avoids the converse of "a localisation survives restriction of scalars". □

Lemma 313 (Section Čech module exactness from a localisation-away spanning datum)**.** *Project-local infrastructure. Let M be an R-module, $s : \iota \to R$ a finite family, and $f \in R$. Assume*

1. *each s_i lies in the radical of (f) (so $f \mid s_i^{k_i}$ for some k_i), and*
2. *the images* $\{\mathrm{algebraMap}\, R\, R_f\, (s_i)\}$ *span the unit ideal of* $R_f = \mathrm{Localization.Away}\, f$.

Then the un-localised section Čech module complex $D^\bullet(s, M)$ of Definition 192 is exact in every positive degree — without *assuming that s spans the unit ideal of R itself.*

Proof. Run the computation over R_f. Instantiate the polymorphic positive-degree exactness of Lemma 193 over the base ring R_f, with module $M_f = \mathrm{LocalizedModule}(\mathrm{Submonoid.powers}\, f)\, M$ and the family $s/1 : \iota \to R_f$; hypothesis (2) is exactly the spanning datum that lemma requires, so the R_f-side complex $D^\bullet(s/1, M_f)$ is exact in positive degrees. Transport this exactness back to the R-side along the degreewise R-linear isomorphisms $M_{s_\sigma} \cong (M_f)_{s_\sigma}$ of Lemma 312 — valid since each $s_\sigma = \prod_k s_{\sigma k}$ lies in the radical of (f) by (1). These isomorphisms are the vertical maps of an isomorphism of cochain complexes, commuting with the alternating-sum cofaces by naturality of the localisation comparison, and exactness transfers across an isomorphism of complexes. □

Lemma 314 (Standard subcover Čech vanishing for the tilde sheaf (the residual))**.** *Source: Stacks Project, Cohomology of Schemes, `lemma-cech-cohomology-quasi-coherent-trivial`. Let R be a ring, M an R-module, $f \in R$, and $g : \{1, \dots, n\} \to R$ a finite family with $D(f) = \bigsqcup_i D(g_i)$ in $\operatorname{Spec} R$. Then for every $p > 0$,*

$$\check{H}^p(\{D(g_i)\}, \widetilde{M}) = 0.$$

Proof. Set $R_f = \mathrm{Localization.Away}\, f$. Following the Stacks proof's "Write $U = \mathrm{Spec}(A)$" step, we run the whole computation over the ring R_f, where the cover *does* span the unit ideal. The two algebraic inputs this needs are read off from the geometry of the cover.

(a) Each g_i lies in the radical of (f). From $D(f) = \bigsqcup_i D(g_i)$ we have in particular $D(g_i) \subseteq D(f)$ for every i, and the inclusion of basic opens $D(g_i) \subseteq D(f)$ is equivalent to $g_i \in \sqrt{(f)}$ (a

power of g_i is a multiple of f). *(b) The cover spans over R_f.* Because f is a unit in R_f, the equality $D(f) = \bigsqcup_i D(g_i)$ pulls back to $\operatorname{Spec} R_f = \bigsqcup_i D(g_i/1)$, so the images $g_i/1$ span the unit ideal of R_f (Lemma 310).

With (a) and (b) in hand, Lemma 313 supplies the positive-degree exactness of the unlocalised section Čech module complex $D^\bullet(g, M)$: internally it instantiates the polymorphic exactness of Lemma 193 over R_f with the module $M_f = \text{LocalizedModule}(\text{Submonoid.powers}\, f)\, M$, and transports the result back along the degreewise isomorphisms $M_{g_\sigma} \cong (M_f)_{g_\sigma}$ of Lemma 312 (for a tuple σ, $g_\sigma = \prod_k g_{\sigma k} \in \sqrt{(f)}$, so both sides present M localised away from $g_\sigma f$).

Finally, the tilde-bridge ladder identifies the section Čech complex of $\widetilde{M}$ over $\{D(g_i)\}$ with this localised-module complex — via the same section-to-module identification and coface matching used in Lemma 187 — and wraps positive-degree exactness as the vanishing of the degree-p homology, giving $\check{H}^p(\{D(g_i)\}, \widetilde{M}) = 0$ for all $p > 0$. □

The Čech-to-cohomology comparison on a basis (Lemma 328) is proved by a dimension shift on the coefficient module, decomposed below into four sub-lemmas: the short exact sequence of Čech complexes coming from a basis (Lemma 322), the resulting stability of vanishing higher Čech cohomology under the injective quotient (Lemma 325), the sheaf-cohomology base case $H^1(U, \mathcal{F}) = 0$ (Lemma 326), and the dimension-shift induction $H^p(U, \mathcal{F}) = 0$ for $p > 0$ (Lemma 327).

8.5.1 Section Čech infrastructure and the cover-system encoding

The sub-lemmas below have a *cover-local*, presheaf-level form: each lemma fixes a single index family $U : \iota \to \text{Opens}(X)$ and works with the section Čech complexes of that one covering, taking the geometric inputs (section-level exactness on faces, positive-degree Čech vanishing) as hypotheses. The cover-global $(\mathcal{B}, \text{Cov})$ data of Lemma 328 re-enters only in the inductive step (Lemma 327), where it is packaged by the cover-system encoding (Definition 319, Definition 320) and discharges those hypotheses via the per-face short-exact-sequence derivation (Lemma 323). The supporting definitions and the degreewise product engine are collected first.

Definition 315 (Čech cohomology accessor)**.** *Project-local accessor.* For an index family $U : \iota \to \text{Opens}(X)$ (giving the covering $\mathcal{U}$), a presheaf of $\mathcal{O}_X$-modules $\mathcal{F}$, and $p \geq 0$, the *Čech cohomology* $\check{H}^p(\mathcal{U}, \mathcal{F})$ is the degree-p homology of the section Čech complex of Definition 196,

$$\check{H}^p(\mathcal{U}, \mathcal{F}) \;:=\; H^p(\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F})),$$

taken as an object of Ab. This is the named wrapper that the 01EO chain refers to uniformly.

Definition 316 (Functoriality of the section Čech complex)**.** *Project-local.* A morphism $\varphi : \mathcal{F} \to \mathcal{G}$ of presheaves of $\mathcal{O}_X$-modules acts on the section Čech data of Definition 196 coordinatewise: on the basic-open section group of each intersection it is the map $\varphi(U_{i_0 \dots i_p}) : \mathcal{F}(U_{i_0 \dots i_p}) \to \mathcal{G}(U_{i_0 \dots i_p})$. These assemble into a cosimplicial morphism (the *section Čech cosimplicial map*), functorial in φ (the *section Čech cosimplicial functor*). Composing with the alternating-coface-map complex functor packages the section Čech complex as a functor in the coefficient presheaf (the *section Čech complex functor*) and yields, for each φ, the induced chain map of section Čech complexes $\check{\mathcal{C}}^\bullet(\mathcal{U}, \varphi)$. These are the maps used to build the short complex of Definition 318.

Definition 317 (Per-face section short complex)**.** Let $U : \iota \to \text{Opens}(X)$ be an index family and let $P : \mathcal{P}_1 \to \mathcal{P}_2 \to \mathcal{P}_3$ be a short complex of presheaves of $\mathcal{O}_X$-modules. For a degree p and a tuple $\sigma : \{0, \dots, p\} \to \iota$ write $V_\sigma = \bigcap_k U(\sigma k)$ for the corresponding basic open. The *per-face* *short complex* face(U, P, σ) is the short complex of abelian groups obtained by evaluating P on sections over V_σ,

$$\mathcal{P}_1(V_\sigma) \longrightarrow \mathcal{P}_2(V_\sigma) \longrightarrow \mathcal{P}_3(V_\sigma),$$

the maps being the section-level actions over V_σ of the two maps of P. It is the (p, σ)-indexed factor whose product over all tuples σ is the degree-p term of the section Čech complex.

Definition 318 (Short complex of section Čech complexes)**.** *Project-local.* For an index family U and a short complex $P : \mathcal{P}_1 \to \mathcal{P}_2 \to \mathcal{P}_3$ of presheaves of $\mathcal{O}_X$-modules, the *short complex of section Čech complexes* is

$$0 \to \check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{P}_1) \to \check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{P}_2) \to \check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{P}_3) \to 0,$$

a short complex in the category of cochain complexes of abelian groups, with the two maps the chain maps induced by the maps of P through the functoriality of Definition 316. Its short-exactness is the conclusion of Lemma 322 and the input to Lemma 325.

Definition 319. [Cover system on a basis] *Project-bespoke encoding of the hypotheses of Stacks 01EO.* A *cover system on a basis* for a scheme X is a record with five fields. A *covering datum* (the helper CovDatum) is a pair (ι, U) of an index type ι together with an index family $U : \iota \to \mathrm{Opens}(X)$; it is the raw shape $\Sigma\, \iota,\ \iota \to \mathrm{Opens}(X)$ over which the section Čech complex is built. The cover system bundles:

1. a basis $\mathcal{B} \subseteq \mathrm{Opens}(X)$ for the topology;
2. a set Cov of admissible coverings, each a CovDatum;
3. *faces-in-basis* (faces_mem): for every covering in Cov every finite intersection $U_{i_0 \cdots i_p}$ of its opens lies in $\mathcal{B}$ (condition (1) of Lemma 328);
4. *section surjectivity* (surj_of_vanishing): for every $V \in \mathcal{B}$ and every short exact sequence $S : 0 \to S_1 \to S_2 \to S_3 \to 0$ of $\mathcal{O}_X$-modules whose left term S_1 has vanishing positive Čech cohomology over the coverings of V in Cov, the section map $S_2(V) \to S_3(V)$ is surjective;
5. *injective acyclicity* (injective_acyclic): for every injective $\mathcal{O}_X$-module $\mathcal{I}$ and every covering $\mathcal{U} \in$ Cov the positive-degree Čech cohomology vanishes, $\check{H}^p(\mathcal{U}, \mathcal{I}) = 0$ for all $p > 0$.

Unlike a bare combinatorial record, the structure carries two sheaf-theoretic (Čech-cohomology) fields. The field surj_of_vanishing is *not* the raw cofinality datum: it is the section-surjectivity *output* that cofinality together with Lemma 215 produces, packaged directly as the consequence the 01EO chain consumes. The field injective_acyclic is the injective Čech-acyclicity of Lemma 214 for the system's own coverings. These are exactly the data that the dimension-shift induction Lemma 327 consumes.

Definition 320 (Vanishing higher Čech cohomology for a cover system)**.** *Project-bespoke inductive predicate (condition (3) of Stacks 01EO, per module).* Given a cover system s on a basis (Definition 319), an $\mathcal{O}_X$-module $\mathcal{F}$ *has vanishing higher Čech cohomology for* s when

$$\check{H}^p(\mathcal{U}, \mathcal{F}) = 0 \qquad \text{for every } \mathcal{U} \in s.\mathrm{Cov} \text{ and every } p > 0,$$

with $\check{H}^p$ the accessor of Definition 315. This is the per-module form of condition (3). It is deliberately stated for an *arbitrary* $\mathcal{O}_X$-module: the dimension-shift induction of Lemma 327 feeds the quotient $\mathcal{Q} = \mathcal{I}/\mathcal{F}$ back as the coefficient, and $\mathcal{Q}$ need not be quasi-coherent even when $\mathcal{F}$ is. The inductive class is therefore "$\mathcal{O}_X$-modules with vanishing higher Čech cohomology for s", a quasi-coherent $\mathcal{F}$ entering only as the seed at the affine (Tag 01XB) instantiation; the predicate must *not* be specialized to quasi-coherent modules.

Lemma 321 (Product of short exact sequences in Ab (AB4*))**.** *Project AB4* lemma (Archon-original infrastructure). Let* $(S_j)_{j\in J}$ *be a family, indexed by an arbitrary type* J*, of short exact sequences of abelian groups,* $S_j : 0 \to A_j \xrightarrow{f_j} B_j \xrightarrow{g_j} C_j \to 0$*. Then the short complex assembled from the product maps,*

$$0 \to \prod_{j\in J} A_j \xrightarrow{\prod_j f_j} \prod_{j\in J} B_j \xrightarrow{\prod_j g_j} \prod_{j\in J} C_j \to 0,$$

is again short exact. This is the AB4 exactness-of-products property of* Ab*; it is the degreewise engine of Lemma 322, since each Čech term is a product of basic-open section groups (Definition 173).*

Proof. Monomorphy of the product map $\prod_j f_j$ is automatic, as products preserve monomorphisms. Exactness at the middle is checked componentwise: an element of $\prod_j B_j$ annihilated by $\prod_j g_j$ is annihilated in each coordinate j, hence lies in the image of f_j by exactness of S_j, and the resulting coordinatewise preimages assemble to a preimage in $\prod_j A_j$. The one nontrivial point is epimorphy of $\prod_j g_j$, which is *not* an automatic instance in Ab: given a target tuple in $\prod_j C_j$, surjectivity of each g_j supplies a coordinatewise preimage, and these assemble — through the canonical identification of an Ab-valued product with the set-theoretic product of its components — to a preimage of the whole tuple. Hence $\prod_j g_j$ is an epimorphism and the displayed sequence is short exact. □

Lemma 322 (Short exact sequence of Čech complexes from a basis)**.** *Source: Stacks Project, Cohomology, Tag 01EO,* `lemma-cech-vanish-basis` *(proof). Fix an index family* $U : \iota \to \mathrm{Opens}(X)$ *(giving the covering* $\mathcal{U}$*) and a short complex*

$$P : \mathcal{P}_1 \to \mathcal{P}_2 \to \mathcal{P}_3$$

of presheaves of $\mathcal{O}_X$*-modules. Assume that for every degree* p *and every tuple* $\sigma : \{0,\dots,p\} \to \iota$ *the per-face short complex of Definition 317 — the section sequence*

$$\mathcal{P}_1(V_\sigma) \longrightarrow \mathcal{P}_2(V_\sigma) \longrightarrow \mathcal{P}_3(V_\sigma), \qquad V_\sigma = \bigcap_k U(\sigma k)$$

— is short exact. Then the induced short complex of section Čech complexes of Definition 318,

$$0 \to \check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{P}_1) \to \check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{P}_2) \to \check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{P}_3) \to 0,$$

is short exact (as a short complex of cochain complexes of abelian groups).

Remark. *In the cover-global* $(\mathcal{B}, \mathrm{Cov})$ *setting of Lemma 328, where* P *is the section short complex of a short exact sequence* $0 \to \mathcal{F} \to \mathcal{I} \to \mathcal{Q} \to 0$ *of sheaves, the per-face hypothesis is discharged by the per-face short-exact-sequence derivation of Lemma 323: left-exactness of the section sequence over* V_σ *is automatic, and surjectivity* $\mathcal{I}(V_\sigma) \to \mathcal{Q}(V_\sigma)$ *is the output of Lemma 215 at the basic open* V_σ *(condition (1) placing* $V_\sigma \in \mathcal{B}$*).*

Proof. Short-exactness of a short complex of cochain complexes is checked degreewise: it suffices to verify short-exactness in each cohomological degree p. In degree p the term of the section Čech complex is the product $\prod_\sigma (-)(V_\sigma)$ over all tuples $\sigma : \{0,\dots,p\} \to \iota$ of section groups over the basic opens V_σ, and the two maps are the products of the corresponding per-face maps; thus the degree-p short complex is the product over σ of the per-face short complexes (Definition 317). Each of these is short exact by hypothesis, so their product is short exact by the AB4* product-of-short-exact-sequences lemma (Lemma 321). As this holds in every degree, the short complex of section Čech complexes (Definition 318) is short exact. □

Lemma 323. *[Per-face short exact sequence from a sheaf short exact sequence] Source: Stacks Project, Cohomology, Tag 01EO,* `lemma-cech-vanish-basis` *(proof). Let $S : 0 \to \mathcal{F} \to \mathcal{I} \to \mathcal{Q} \to 0$ be a short exact sequence of sheaves of $\mathcal{O}_X$-modules, let $U : \iota \to \mathrm{Opens}(X)$ be an index family, and suppose that on every face $V_\sigma = \bigcap_k U(\sigma k)$ the section map $\mathcal{I}(V_\sigma) \to \mathcal{Q}(V_\sigma)$ is surjective — the basis-surjectivity datum supplied by Lemma 215. Writing P for the section short complex of S regarded as a short complex of presheaves of modules, the per-face short complex of Definition 317 is then short exact for every degree p and tuple σ. This lemma produces the per-face hypothesis consumed by Lemma 322, and is the bridge by which Lemma 327 instantiates the cover-local lemmas from a genuine sheaf short exact sequence.*

Proof. Fix p and σ, and put $V = V_\sigma = \bigcap_k U(\sigma k)$. Monomorphy of $\mathcal{F}(V) \to \mathcal{I}(V)$ and exactness at the middle of $\mathcal{F}(V) \to \mathcal{I}(V) \to \mathcal{Q}(V)$ are the left-exactness of the sections functor applied to the short exact sequence of sheaves S: the sections functor $\mathcal{G} \mapsto \mathcal{G}(V)$ is the composite of the inclusion of sheaves of $\mathcal{O}_X$-modules into presheaves of modules (a right adjoint, hence preserving finite limits) with evaluation at V (which preserves limits), so it is left exact and carries the exactness of S to exactness of the section sequence at $\mathcal{I}(V)$. The remaining map $\mathcal{I}(V) \to \mathcal{Q}(V)$ is surjective by hypothesis, i.e. by the output of Lemma 215 at the basic open V. Hence the per-face short complex is short exact. ☐

Lemma 324 (Quotient preserves vanishing positive homology (homological core))**.** *Abstract homological core of Lemma 325. Let $T : 0 \to T_1 \to T_2 \to T_3 \to 0$ be a short exact sequence of cochain complexes (cohomological $\mathbb{N}$-grading) of abelian groups, and suppose the middle term T_2 and the left term T_1 both have vanishing positive homology: $H^p(T_2) = 0$ and $H^p(T_1) = 0$ for all $p > 0$. Then the right term T_3 also has vanishing positive homology, $H^p(T_3) = 0$ for all $p > 0$. This is the section-Čech-free homological core consumed by Lemma 325.*

Proof. Fix $p > 0$. In the homology long exact sequence of T (Lemma 154) the two homology groups of the middle term adjacent to the connecting map at T_3 in degree p, namely $H^p(T_2)$ and $H^{p+1}(T_2)$, both vanish (positive degrees, by the hypothesis on T_2); hence the connecting homomorphism is an isomorphism $H^p(T_3) \cong H^{p+1}(T_1)$. The right-hand side vanishes since $p + 1 > 0$ and T_1 has vanishing positive homology. Therefore $H^p(T_3) = 0$. ☐

Lemma 325 (Quotient preserves vanishing higher Čech cohomology)**.** *Source: Stacks Project, Cohomology, Tag 01EO,* `lemma-cech-vanish-basis` *(proof). Fix an index family $U : \iota \to \mathrm{Opens}(X)$ (giving the covering $\mathcal{U}$) and a short complex $P : \mathcal{P}_1 \to \mathcal{P}_2 \to \mathcal{P}_3$ of presheaves of $\mathcal{O}_X$-modules, as in Lemma 322. Suppose:*

1. *the short complex of section Čech complexes of P is short exact (the conclusion of Lemma 322);*
2. *the middle term $\mathcal{P}_2$ has vanishing positive Čech cohomology, $\check{H}^p(\mathcal{U}, \mathcal{P}_2) = 0$ for all $p > 0$; and*
3. *the left term $\mathcal{P}_1$ has vanishing positive Čech cohomology, $\check{H}^p(\mathcal{U}, \mathcal{P}_1) = 0$ for all $p > 0$.*

Then the right term $\mathcal{P}_3$ (the quotient) also has vanishing positive Čech cohomology: $\check{H}^p(\mathcal{U}, \mathcal{P}_3) = 0$ for all $p > 0$. Here $\check{H}^p$ is the accessor of Definition 315. At instantiation hypothesis (2) is supplied for $\mathcal{P}_2 = \mathcal{I}$ injective by Lemma 214, and hypothesis (3) for $\mathcal{P}_1 = \mathcal{F}$ by condition (3) of Lemma 328.

Proof. Apply the abstract homological core (Lemma 324) to the short exact sequence T of cochain complexes furnished by hypothesis (1) — the short complex of section Čech complexes of P. Its homology in degree p is by definition $\check{H}^p(\mathcal{U}, -)$ (Definition 315), so hypotheses (2) and (3) are exactly the vanishing of the positive homology of the middle and left terms of T. The core lemma then yields the vanishing of the positive homology of the right term, that is $\check{H}^p(\mathcal{U}, \mathcal{P}_3) = 0$ for all $p > 0$. Concretely, the connecting isomorphism of the homology long exact sequence realizes the dimension shift $\check{H}^p(\mathcal{U}, \mathcal{P}_3) \cong \check{H}^{p+1}(\mathcal{U}, \mathcal{P}_1)$, the step used in the 01EO proof; at instantiation $\check{H}^{>0}(\mathcal{U}, \mathcal{P}_2) = 0$ is the injective acyclicity of Lemma 214. □

Lemma 326 (Sheaf-cohomology base case $H^1(U, \mathcal{F}) = 0$)**.** *Source: Stacks Project, Cohomology, Tag 01EO, `lemma-cech-vanish-basis` (proof). Let $U \subseteq X$ be an open, and let $S : 0 \to \mathcal{F} \to \mathcal{I} \to \mathcal{Q} \to 0$ be a short exact sequence in X.*Modules *with $\mathcal{I}$ injective. Assume that the section map $\mathcal{I}(U) \to \mathcal{Q}(U)$ — the action on sections over U of the surjection of S — is surjective. Then the first absolute cohomology vanishes,*

$$H^1(U, \mathcal{F}) = 0,$$

where $H^p(U, -)$ is the Ext realization of Definition 218. The only geometric input is the section-surjectivity hypothesis; at instantiation it is supplied, at a basic open $U \in \mathcal{B}$, by the per-face surjectivity of Lemma 215 (through Lemma 322).

Proof. Run the covariant Ext long exact sequence (Lemma 223) of $0 \to \mathcal{F} \to \mathcal{I} \to \mathcal{Q} \to 0$ at the fixed first argument $j_!\mathcal{O}_U$, giving the exact strand

$$H^0(U, \mathcal{I}) \xrightarrow{g} H^0(U, \mathcal{Q}) \xrightarrow{\delta} H^1(U, \mathcal{F}) \to H^1(U, \mathcal{I}).$$

By Lemma 222 (the injective $\mathcal{I}$ is the *second* Ext argument, Lemma 226) the term $H^1(U, \mathcal{I}) = \operatorname{Ext}^1(j_!\mathcal{O}_U, \mathcal{I}) = 0$. In degree 0 the identification $H^0(U, -) = \Gamma(U, -)$ of Lemma 224 is natural in the coefficient sheaf (Lemma 225), so it carries $g = H^0(U, \mathcal{I} \to \mathcal{Q})$ to the section map $\mathcal{I}(U) \to \mathcal{Q}(U)$; the latter is surjective by hypothesis, hence so is g. Surjectivity of g makes the connecting map $\delta = 0$; since also $H^1(U, \mathcal{I}) = 0$, the strand reads $0 \to H^1(U, \mathcal{F}) \to 0$, whence $H^1(U, \mathcal{F}) = 0$. □

Lemma 327. *[Dimension-shift induction $H^p(U, \mathcal{F}) = 0$ for $p > 0$] Source: Stacks Project, Cohomology, Tag 01EO, `lemma-cech-vanish-basis` (proof). Let s be a cover system on a basis for X (Definition 319). For every $\mathcal{O}_X$-module $\mathcal{F}$ that has vanishing higher Čech cohomology for s (Definition 320), every basic open $U \in s.\mathcal{B}$, and every $p > 0$, the absolute cohomology vanishes,*

$$H^p(U, \mathcal{F}) = 0,$$

where $H^p(U, -)$ is the Ext realization of Definition 218. The formal statement carries [EnoughInjectives X.Modules] *as an explicit instance hypothesis: enough injectives in the category of sheaves of $\mathcal{O}_X$-modules is needed to embed $\mathcal{F}$ into an injective $\mathcal{I}$, and that instance is not currently available in the ambient library for sheaves of modules (it would follow from* IsGrothendieckAbelian(SheafOfModules R)*).*

Proof. Induct on $p \geq 1$ with the statement quantified over *all* $\mathcal{O}_X$-modules $\mathcal{F}$ having vanishing higher Čech cohomology for s (Definition 320) simultaneously, so that the quotient may take the role of $\mathcal{F}$ at the next stage. Fix such an $\mathcal{F}$, embed $\mathcal{F} \hookrightarrow \mathcal{I}$ into an injective $\mathcal{O}_X$-module, and put $\mathcal{Q} = \mathcal{I}/\mathcal{F}$, giving a short exact sequence $S : 0 \to \mathcal{F} \to \mathcal{I} \to \mathcal{Q} \to 0$.

We first record the geometric input that both the quotient-closure and the base case need. For every $\mathcal{U} \in s.\mathrm{Cov}$, the faces-in-basis field faces_mem (condition (1)) places every face V_σ in $s.\mathcal{B}$; the section-surjectivity field surj_of_vanishing of the cover system, applied to the short exact sequence S at V_σ, then gives section surjectivity $\mathcal{I}(V_\sigma) \to \mathcal{Q}(V_\sigma)$. Its left-term Čech-vanishing hypothesis $\check{H}^{>0}(-, \mathcal{F}) = 0$ over the coverings of V_σ is supplied by $\mathcal{F}$'s vanishing higher Čech cohomology (Definition 320); mathematically this field packages Lemma 215 (cofinality plus the Čech $\check{H}^1$-surjectivity criterion). By the per-face derivation (Lemma 323) the per-face short complexes of S are short exact, so Lemma 322 gives the short exact sequence of section Čech complexes for $\mathcal{U}$; in particular the section surjectivity $\mathcal{I}(U) \to \mathcal{Q}(U)$ holds at every basic open $U \in s.\mathcal{B}$ (take $\mathcal{U}$ a covering of U in $s.\mathrm{Cov}$ and the length-one face). Feeding the Čech short exact sequence into Lemma 325 — with $\check{H}^{>0}(\mathcal{U}, \mathcal{I}) = 0$ from injective acyclicity and $\check{H}^{>0}(\mathcal{U}, \mathcal{F}) = 0$ from $\mathcal{F}$'s hypothesis — shows $\check{H}^{>0}(\mathcal{U}, \mathcal{Q}) = 0$ for every $\mathcal{U} \in s.\mathrm{Cov}$; that is, $\mathcal{Q}$ again has vanishing higher Čech cohomology for s, so it is itself a valid instance of the inductive statement.

Base case $p = 1$: apply Lemma 326 to S at the basic open $U \in s.\mathcal{B}$, its section-surjectivity hypothesis discharged by the paragraph above.

Step $p \Rightarrow p+1$: the covariant Ext long exact sequence (Lemma 223) of $0 \to \mathcal{F} \to \mathcal{I} \to \mathcal{Q} \to 0$ at $j_!\mathcal{O}_U$ gives the exact strand

$$H^p(U, \mathcal{Q}) \xrightarrow{\delta} H^{p+1}(U, \mathcal{F}) \to H^{p+1}(U, \mathcal{I}).$$

The right term vanishes since $\mathcal{I}$ is injective (Lemma 222), and the left term $H^p(U, \mathcal{Q}) = 0$ by the inductive hypothesis applied to $\mathcal{Q}$ (admissible by the previous paragraph). Exactness forces $H^{p+1}(U, \mathcal{F}) = 0$. This completes the induction, giving $H^p(U, \mathcal{F}) = 0$ for all $p > 0$ and all $U \in s.\mathcal{B}$. ☐

Lemma 328 (Čech-to-cohomology comparison on a basis)**.** *Source: Stacks Project, Cohomology, Tag 01EO, Lemma `lemma-cech-vanish-basis`. Let X be a ringed space, let $\mathcal{B}$ be a basis for its topology, let $\mathcal{F}$ be an $\mathcal{O}_X$-module, and suppose there is a set* Cov *of open coverings such that: (1) every $\mathcal{U} : U = \bigcup_{i \in I} U_i$ in* Cov *has $U, U_i \in \mathcal{B}$ and all intersections $U_{i_0 \cdots i_p} \in \mathcal{B}$; (2) for each $U \in \mathcal{B}$ the coverings of U occurring in* Cov *are cofinal among all open coverings of U; and (3) for every $\mathcal{U} \in$* Cov *the Čech cohomology vanishes, $\check{H}^p(\mathcal{U}, \mathcal{F}) = 0$ for all $p > 0$. Then*

$$H^p(U, \mathcal{F}) = 0 \qquad \textit{for all } p > 0 \textit{ and any } U \in \mathcal{B}.$$

This is the basis-comparison vehicle that upgrades the standard-cover Čech vanishing of Lemma 194 to the Serre vanishing of Lemma 228: instantiating $\mathcal{B}$ as the affine opens and Cov *as the standard affine covers, condition (3) is exactly Lemma 194. The formal target of this lemma is the* vanishing *conclusion $H^p(U, \mathcal{F}) = 0$ for $p > 0$ under hypotheses (1)–(3), not an explicit isomorphism $\check{H}^p(\mathcal{U}, \mathcal{F}) \cong H^p(U, \mathcal{F})$ of Čech and sheaf cohomology; only the positive-degree vanishing is needed downstream, and it is all the statement establishes. The formal statement carries* [EnoughInjectives X.Modules] *as an explicit instance hypothesis, inherited from the dimension-shift induction (Lemma 327): it is needed to embed the coefficient module into an injective, and that instance is not currently available in the ambient library for sheaves of $\mathcal{O}_X$-modules (it would follow from* IsGrothendieckAbelian(SheafOfModules R)*).*

Proof. Here $H^p(U, -)$ denotes the Ext realization of absolute sheaf cohomology fixed in Definition 218. The proof is the thin assembly of the sub-lemmas through the cover-system encoding. Conditions (1)–(2) of the hypotheses are exactly the faces-in-basis and cofinality data of a cover system on a basis, so $\mathcal{B}$ and Cov assemble into an instance s of Definition 319 with $s.\mathcal{B} = \mathcal{B}$ and $s.\mathrm{Cov} = \mathrm{Cov}$; condition (3) is exactly the assertion that $\mathcal{F}$ has vanishing higher Čech cohomology for s (Definition 320).

The conclusion is then the dimension-shift induction Lemma 327 applied to this instance: for every basic open $U \in s.\mathcal{B} = \mathcal{B}$ and every $p > 0$ one has $H^p(U, \mathcal{F}) = 0$. That induction internally embeds $\mathcal{F}$ into an injective $\mathcal{I}$, forms the quotient $\mathcal{Q} = \mathcal{I}/\mathcal{F}$, keeps $\mathcal{Q}$ inside the inductive class via the Čech short exact sequence (Lemma 322) and its quotient-stability (Lemma 325), and bottoms out at the base case (Lemma 326).

The argument never uses affine sheaf-cohomology vanishing (Lemma 228); its only Čech-vanishing input is condition (3), which on the affine instantiation is supplied by the standard-cover Čech vanishing of Lemma 194. Thus the dependency affine Serre vanishing → this lemma is one-directional and the graph is acyclic. □

8.6 The Čech complex computes $R^i f_*$

The comparison is proved by the *acyclic-resolution* (Cartan–Leray) route. The augmented Čech complex is a resolution of $\mathcal{F}$ whose terms are acyclic for f_*; the abstract homological-algebra Lemma 171 then identifies the derived functor with the cohomology of the complex obtained by applying f_*. The two geometric inputs are isolated in the following sub-lemmas.

8.6.1 Object layer for the augmented Čech complex

Before stating the resolution lemma we name the object it is about: the augmented Čech complex $0 \to \mathcal{F} \to \mathcal{C}^0 \to \mathcal{C}^1 \to \cdots$ as a cochain complex in the category X.Modules of sheaves of $\mathcal{O}_X$-modules, together with its augmentation map and the identity $\varepsilon \cdot d^0 = 0$ that makes the augmentation legitimate. This object layer is purely formal — it uses only the Čech nerve of Definition 172 and the unitors of the push–pull adjunction — and is independent of the exactness (acyclicity) statement.

Definition 329 (Un-augmented Čech cochain complex on X)**.** *Project-local.* Given an open cover $\mathcal{U}$ of X and a sheaf of $\mathcal{O}_X$-modules $\mathcal{F}$, the *un-augmented Čech cochain complex*

$$\mathcal{C}^0 \to \mathcal{C}^1 \to \mathcal{C}^2 \to \cdots$$

on X is the alternating-coface-map cochain complex of the cosimplicial $\mathcal{O}_X$-module obtained by forgetting the augmentation of the Čech nerve of Definition 172 (its Augmented.drop). Its degree-p term is $\mathcal{C}^p = \prod_{(i_0,\ldots,i_p)} (j_{i_0\cdots i_p})_*(\mathcal{F}|_{U_{i_0\cdots i_p}})$, exactly as in the relative Čech complex of Definition 184 specialised to $f = \mathrm{id}_X$.

Definition 330 (Augmentation point isomorphism)**.** *Project-local.* The augmentation point of the Čech nerve of Definition 172 is the value of the nerve at the initial augmentation object, namely $(\mathrm{id}_X)_*(\mathrm{id}_X)^*\mathcal{F}$. The *augmentation point isomorphism* is the canonical isomorphism

$$(\mathrm{CechNerve}\ \mathcal{U}\ \mathcal{F}).\mathrm{left}\ \cong\ \mathcal{F}$$

obtained by composing the pullback unitor $(\mathrm{id}_X)^*\mathcal{F} \cong \mathcal{F}$ with the pushforward unitor $(\mathrm{id}_X)_*\mathcal{G} \cong \mathcal{G}$ of the push–pull adjunction.

Definition 331 (Čech augmentation map)**.** *Project-local.* The *Čech augmentation* is the morphism $\varepsilon : \mathcal{F} \to \mathcal{C}^0$ of sheaves of $\mathcal{O}_X$-modules obtained by composing the inverse of the augmentation point isomorphism of Definition 330 with the degree-0 component of the nerve's augmentation natural transformation (which lands in the degree-0 term $\mathcal{C}^0$ of the complex of Definition 329).

Lemma 332 (The augmentation kills the first differential)**.** *Project-local. The composite of the augmentation $\varepsilon : \mathcal{F} \to \mathcal{C}^0$ of Definition 331 with the first Čech differential vanishes,*

$$\varepsilon \cdot d^0\ =\ 0.$$

Proof. This is the standard augmented-cosimplicial identity. The differential $d^0 = \delta^0 - \delta^1$ is the alternating sum of the two cofaces, and naturality of the augmentation natural transformation of the cosimplicial object intertwines ε with δ^0 and with δ^1 in the same way; the two contributions therefore cancel. □

Definition 333 (Augmented Čech complex)**.** *Project-local.* The *augmented Čech complex* is the cochain complex

$$0 \to \mathcal{F} \to \mathcal{C}^0 \to \mathcal{C}^1 \to \cdots$$

in X.Modules obtained by prepending the coefficient sheaf $\mathcal{F}$ in degree 0 to the complex of Definition 329 along the augmentation ε of Definition 331, using the augmentation identity $\varepsilon \cdot d^0 = 0$ of Lemma 332. Concretely its degree-0 term is $\mathcal{F}$ and its degree-$(p+1)$ term is $\mathcal{C}^p$. This is the object whose exactness is the content of Lemma 337.

Lemma 334 (Sheafification kills a presheaf that is locally bijective to a locally-zero one)**.** *Project-bespoke site-theory lemma (pure Mathlib site theory). Let J be a Grothendieck topology on a site $\mathcal{C}$ and let $J.W$ denote the class of* locally bijective *maps of presheaves — the local-isomorphism class for J, i.e. the maps inverted by sheafification. Then:*

1. *Sheafification* presheafToSheaf *carries every map in $J.W$ to an isomorphism; consequently it sends a presheaf that is locally bijective to a presheaf with zero sheafification to a presheaf with zero sheafification. Precisely, if $\varphi : P \to Q$ lies in $J.W$ and the sheafification of Q is the zero sheaf, then the sheafification of P is the zero sheaf.*

2. *In particular, if P admits a locally bijective map to the zero presheaf — for instance if P is locally zero, i.e. vanishes on a basis of the topology — then the sheafification of P is the zero sheaf.*

3. *For an abelian-valued presheaf the locally-bijective hypothesis is the conjunction of local injectivity and local surjectivity of the comparison map; a map that is both locally injective and locally surjective lies in $J.W$.*

Proof. Sheafification presheafToSheaf : $\widehat{\mathcal{C}} \to \mathrm{Sh}(J)$ inverts exactly the maps of the class $J.W$: by definition, $\varphi \in J.W$ if and only if presheafToSheaf(φ) is an isomorphism. Hence for $\varphi : P \to Q$ in $J.W$ the induced map sheafify$(P) \to$ sheafify(Q) is an isomorphism, and being a zero object transports along an isomorphism: if sheafify(Q) is zero then so is sheafify(P). This proves (1); part (2) is the special case $Q = 0$ (the zero presheaf, whose sheafification is the zero sheaf), and a presheaf that vanishes on a basis admits such a locally bijective map to 0. For (3), sheafification is additive, and a map of abelian presheaves that is simultaneously locally injective and locally surjective is a local isomorphism, i.e. lies in $J.W$; applying (1) then collapses its sheafification. □

Lemma 335. *[A faithful zero-preserving functor reflects zero objects] Let $F : \mathcal{C} \to \mathcal{D}$ be a functor between categories with zero morphisms that is* faithful *and* preserves zero morphisms*. If the image $F(Y)$ of an object Y is a zero object of $\mathcal{D}$, then Y is a zero object of $\mathcal{C}$.*

Proof. An object Y is a zero object if and only if its identity morphism equals its zero endomorphism, $\mathrm{id}_Y = 0$. Applying this characterization to $F(Y)$, the hypothesis gives $\mathrm{id}_{F(Y)} = 0$. Since F preserves identities and zero morphisms, $F(\mathrm{id}_Y) = \mathrm{id}_{F(Y)} = 0 = F(0)$; faithfulness of F then makes the map on hom-sets injective, so $\mathrm{id}_Y = 0$, whence Y is a zero object. □

Lemma 336 (Sheafification of a presheaf that is locally a zero object vanishes)**.** *Let J be a Grothendieck topology on a site $\mathcal{C}$ and let Q be a presheaf of abelian groups on $\mathcal{C}$. Suppose that for every object U there is a covering sieve $S \in J(U)$ all of whose members $g : V \to U$ land on an object V where $Q(V)$ is a zero object. Then the sheafification of Q is the zero sheaf.*

Proof. This is the "locally zero" packaging of Lemma 334. Let Z be the constant presheaf with value the zero group; its sheafification is the zero sheaf. The zero map $0 : Q \to Z$ is locally injective — on each covering sieve furnished by the hypothesis the source sections $Q(V)$ are subsingletons, so the comparison is injective locally — and it is locally surjective because the target is objectwise a zero object. Hence $0 : Q \to Z$ is locally bijective, i.e. lies in the local-isomorphism class $J.W$, and part (3) followed by part (1) of Lemma 334 collapses the sheafification of Q to the zero sheaf. □

Lemma 337 (The augmented Čech complex is a resolution)**.** *Source: Stacks Project, Cohomology of Schemes,* `lemma-cech-cohomology-quasi-coherent-trivial` *(proof). Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module and let $\mathfrak{U} : X = \bigcup_{i\in I} U_i$ be a finite affine open cover with all intersections affine. Form the augmented Čech complex on X,*

$$0 \to \mathcal{F} \to \mathcal{C}^0 \to \mathcal{C}^1 \to \mathcal{C}^2 \to \cdots, \qquad \mathcal{C}^p = \prod_{(i_0,\dots,i_p)\in I^{p+1}} (j_{i_0\cdots i_p})_*(\mathcal{F}|_{U_{i_0\cdots i_p}}),$$

the degree-p term being the degree-p object of the Čech nerve of Definition 172, with augmentation the canonical map $\mathcal{F} \to \mathcal{C}^0$. This augmented complex is exact; that is, the Čech nerve is a resolution of $\mathcal{F}$ in $\mathrm{QCoh}(X)$*.*

Proof. Source: Stacks Project, Cohomology of Schemes, `lemma-cech-cohomology-quasi-coherent-trivial` *(proof).* We prove exactness of the augmented Čech complex of Definition 333 in X.Modules by showing that each homology object $\mathcal{H}^p$ ($p \geq 1$, together with the augmentation node) is the zero object. The argument works entirely with sections and sheafification, never with stalks.

Step 1: reflect through the forgetful functor to abelian sheaves. The reflection is carried out through the forgetful functor SheafOfModules.toSheaf : X.Modules $\to \mathrm{Sh}(X, \mathbf{Ab})$ from sheaves of $\mathcal{O}_X$-modules to sheaves of abelian groups. This functor is faithful and additive, hence preserves zero morphisms; a faithful additive functor reflects the property of an object being zero. It therefore suffices to prove that the image SheafOfModules.toSheaf($\mathcal{H}^p$) of each homology object is zero in the category of abelian sheaves.

Step 2: the homology sheaf is the sheafification of the presheaf homology. By the sheafification-commutes-with-homology engine of Lemma 407, the degree-p homology of the complex of abelian sheaves is the sheafification of the degree-p homology of the underlying complex of *presheaves*. The presheaf homology evaluated at an open V is the homology of the section complex $\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F})(V)$ — the Čech cochain complex of $\mathcal{F}$ over the cover restricted to V — so the presheaf in question is $V \mapsto \check{H}^p(V, \mathcal{F})$.

Step 3: the presheaf homology vanishes locally, via the prepend-i_{fix} homotopy. This presheaf is not zero as a presheaf, but it vanishes on a basis. The relevant basis is the family of opens V contained in some single cover element U_i; since $\{U_i\}$ covers X, every open is covered by such V (intersect with the U_i), so this family is cofinal among all opens. Fix such a V with $V \subseteq U_{i_{\mathrm{fix}}}$. The section complex $\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F})(V)$ is the Čech cochain complex of $\mathcal{F}$ over the *restricted* cover $\{U_s \cap V\}_s$ of V; this restricted cover has the member $U_{i_{\mathrm{fix}}} \cap V = V$ equal to the whole of V. A cover possessing a member equal to the total space carries a *contracting homotopy*: the prepend-i_{fix} map $(h\,s)_{i_0\cdots i_p} = s_{i_{\mathrm{fix}}\,i_0\cdots i_p}$ (with the evident restriction along $U_{i_0\cdots i_p} \cap V \subseteq U_{i_{\mathrm{fix}}\,i_0\cdots i_p} \cap V$) satisfies $d \circ h + h \circ d = \mathrm{id}$. This is the section-level objectwise analogue of the combinatorial homotopy combHomotopy / combHomotopy_spec of Lemma 205; it is *cover-agnostic and coefficient-agnostic* — it uses neither quasi-coherence of $\mathcal{F}$ nor affineness of the cover, only that V sits inside a cover member. Hence the section complex over every such V is contractible, so its homology vanishes in every degree. Consequently the canonical map

$0 \to$ (presheaf homology) is a local isomorphism on this basis — it is locally bijective. (This is a different mechanism from the affine Čech vanishing of Lemma 314, which concerns the *standard* cover $\{D(f_i)\}$ of an affine and feeds the 01XB Serre vanishing, not the present local check.)

To convert this into a vanishing statement about the abelian-sheaf homology of Step 1, the module-level sheafification of the homology (the comparison isomorphism homologyIsoSheafify underlying Lemma 407) is transported to the abelian-sheaf side along the *sheafification square*

$$\mathrm{SheafOfModules.toSheaf} \circ \mathrm{sheafify} \;\cong\; \mathrm{presheafToSheaf} \circ \mathrm{forget},$$

the natural isomorphism expressing that sheafifying a presheaf of modules and then forgetting to abelian sheaves agrees with forgetting to abelian presheaves and then sheafifying. Under this square the image $\mathrm{SheafOfModules.toSheaf}(\mathscr{H}^p)$ becomes the abelian sheafification presheafToSheaf of the abelian presheaf homology $V \mapsto \check{H}^p(V, \mathscr{F})$. Lemma 334 now applies: the locally bijective comparison $0 \to$ (presheaf homology) is inverted by presheafToSheaf (in the abelian setting it is simultaneously locally injective and locally surjective, hence in the local-isomorphism class), so its abelian sheafification is the zero sheaf. Combined with Steps 1–2 this gives $\mathscr{H}^p = 0$ for all $p \geq 1$.

Step 3, formalization mechanism (from the section homotopy to the vanishing of homology). The prose above describes the contracting homotopy mathematically; we now record the precise mechanism by which it produces the required vanishing of the section-complex homology, since this is the step a formalization must instantiate.

(a) After Steps 1–2 and the sheafification/site reduction, the residual obligation is exactly that, for every $V \leq U_{i_{\mathrm{fix}}}$ and every degree p, the homology of the *section cochain complex* $\Gamma(V, \mathscr{C}^\bullet_{\mathrm{aug}}(\mathfrak{U}, \mathscr{F}))$ — viewed as a complex of abelian groups — is a zero object. In the categorical packaging this is the homology of the complex obtained by applying the evaluation-at-V functor (through toPresheaf) to the augmented Čech complex.

(b) This abstract evaluated complex is identified with the *concrete* Čech section cochain complex $D^\bullet$ whose degree-p term is the product $\prod_\sigma \Gamma(U_\sigma \cap V, \mathscr{F})$ with the alternating coface differential. The identification is the naturality of the map-of-homological-complexes functor together with a per-degree isomorphism of the product-of-sections term; it is the same categorical-to-combinatorial section-complex identification used for the free evaluated complex (Lemmas 200 and 203). This per-degree identification is carried out in Lemma 392: its degreewise object isomorphism $\Gamma(V, \mathrm{pushPullObj}\, \mathscr{F}\, Y_p) \cong \prod_\sigma \Gamma(U_\sigma \cap V, \mathscr{F})$ is Lemma 373 (which rests on the coproduct-to-product decomposition of the backbone push–pull object, Lemma 371, the geometric backbone identification Lemma 350, and the per-leg section calculation Lemma 372), and its differential match reuses Lemma 189.

(c) Because $V \leq U_{i_{\mathrm{fix}}}$, prepending the index i_{fix} is the *identity* on each section term — $U_{i_{\mathrm{fix}}\sigma} \cap V = U_\sigma \cap V$ since $V \subseteq U_{i_{\mathrm{fix}}}$ — so the prepend map gives a contracting homotopy from the identity of $D^\bullet$ to the zero map, i.e. a homotopy $\mathrm{id}_{D^\bullet} \simeq 0$, whose component maps are the combinatorial prepend maps combHomotopy i_{fix} and whose homotopy relation $d \circ h + h \circ d = \mathrm{id}$ is exactly combHomotopy_spec (Lemma 205 and its specification Lemma 206). On the concrete section complex $D'^\bullet$ this contracting homotopy is supplied by Lemma 393: since $V \leq U_{i_{\mathrm{fix}}}$ makes $U_{i_{\mathrm{fix}}} \cap U_\sigma \cap V = U_\sigma \cap V$, prepending i_{fix} is the identity on each coefficient and the dependent combinatorial engine yields $\mathrm{id}_{D'^\bullet} \simeq 0$.

(d) A homotopy $\mathrm{id}_{D^\bullet} \simeq 0$ forces the homology to vanish in every degree: by homotopy-invariance the map induced on degree-p homology by the identity equals the map induced by the zero map; the former is the identity of $D^\bullet$'s degree-p homology and the latter is zero, so that identity equals zero, which is the criterion for the homology to be a zero object. (This composes the homotopy-invariance of the induced homology map with the identity/zero computations and the "identity equals zero" characterization of a zero object — the same characterization used in Lemma 335, packaged once and for all as Lemma 338.) Items (a)–(d) together are exactly the content of Lemma 394, which is the single residual obligation discharging this step.

(e) One route is recorded as a dead end: the homotopy must be built directly as a homotopy of homological complexes, following the construction of Lemma 207. It cannot be obtained from an extra-degeneracy package, because the only such package available is the *simplicial* one producing a homotopy equivalence of *chain* complexes (face maps), whereas the augmented Čech complex here is a *cochain* complex (coface maps); there is no cosimplicial/cochain dual to invoke.

Step 4: the augmentation node. The augmentation $\varepsilon : \mathcal{F} \to \mathcal{C}^0$ is handled by the *same* prepend-i_{fix} homotopy of Step 3. Over an open $V \subseteq U_{i_{\text{fix}}}$, the contracting homotopy extends to the augmentation term: the degree-(-1)/degree-0 component of h sends a section $s \in \prod_s \mathcal{F}(U_s \cap V)$ to its i_{fix}-component $s_{i_{\text{fix}}} \in \mathcal{F}(U_{i_{\text{fix}}} \cap V) = \mathcal{F}(V)$, so the identity $d \circ h + h \circ d = \text{id}$ holds at the augmentation node as well. In the formalization mechanism above this is the degree-0 instance: the same per-degree identification (b) and the same homotopy (c) cover the augmentation term, the i_{fix}-component map $s \mapsto s_{i_{\text{fix}}}$ landing in $\mathcal{F}(U_{i_{\text{fix}}} \cap V) = \mathcal{F}(V)$. Thus the augmented section complex over each such V is exact at degree 0 (and the augmentation is the kernel inclusion there). Sheafifying this local exactness as in Step 3 makes the augmentation-node homology zero. Together with Steps 1–3, every homology object of the augmented complex is zero, so the complex is exact and the Čech nerve of Definition 172 is a resolution of $\mathcal{F}$ in X.Modules. In particular the statement holds for an arbitrary $\mathcal{O}_X$-module $\mathcal{F}$ and an arbitrary open cover; quasi-coherence and affineness are needed only by the downstream consumers (termwise f_*-acyclicity), not by this exactness.

Remark. An alternative proof uses the stalk-at-prime criterion directly: the complex is exact if and only if it is exact after localizing at each prime $\mathfrak{p}$, and for each $\mathfrak{p}$ one fixes an index i_{fix} with $f_{i_{\text{fix}}} \notin \mathfrak{p}$ and defines a contracting homotopy of the localized complex by $h(s)_{i_0 \cdots i_p} = s_{i_{\text{fix}} i_0 \cdots i_p}$, as in the Stacks source proof above. This approach is equivalent to the sections-and-sheafification argument of Steps 1–4. □

Lemma 338 (A contracting homotopy of the identity makes homology vanish)**.** *Let $\mathcal{A}$ be a preadditive category and $D^\bullet$ a cochain complex in $\mathcal{A}$. If there is a homotopy $\text{id}_{D^\bullet} \simeq 0$ between the identity endomorphism of $D^\bullet$ and the zero endomorphism, then for every degree p the homology object $H^p(D^\bullet)$ is a zero object.*

Proof. The map induced on homology is a homotopy invariant: homotopic chain maps induce equal maps on each homology object. Applying this to the homotopy $\text{id}_{D^\bullet} \simeq 0$, the map induced by $\text{id}_{D^\bullet}$ on $H^p(D^\bullet)$ equals the map induced by 0. The former is $\text{id}_{H^p(D^\bullet)}$ (homology is a functor) and the latter is the zero endomorphism of $H^p(D^\bullet)$; hence $\text{id}_{H^p(D^\bullet)} = 0$. An object whose identity morphism equals its zero endomorphism is a zero object, so $H^p(D^\bullet)$ is zero. □

Lemma 339 (Homology vanishes after transporting a null-homotopic complex across an isomorphism)**.** *Let $\mathcal{A}$ be a preadditive category with homology and let $D^\bullet, D'^\bullet$ be cochain complexes in $\mathcal{A}$. If $D^\bullet \cong D'^\bullet$ and $D'^\bullet$ admits a contracting homotopy $\text{id}_{D'^\bullet} \simeq 0$, then for every degree p the homology $H^p(D^\bullet)$ is a zero object. This is the consumer glue that turns the contractibility of the model complex $D'^\bullet$ into the vanishing of the homology of the isomorphic complex $D^\bullet$ actually occurring in Step 3 of Lemma 337.*

Proof. By Lemma 338 the null-homotopy $\mathrm{id}_{D'^\bullet} \simeq 0$ forces $H^p(D'^\bullet)$ to be a zero object in every degree. Homology is a functor, so the isomorphism $D^\bullet \cong D'^\bullet$ induces an isomorphism $H^p(D^\bullet) \cong H^p(D'^\bullet)$; an object isomorphic to a zero object is itself a zero object, whence $H^p(D^\bullet)$ is zero. $\square$

Lemma 340 (The degree-p Čech-nerve object is a wide pullback)**.** *Let $\mathcal{U} : X = \bigcup_{i\in I} U_i$ be an open cover with cover arrow $q = \mathrm{Sigma.\,desc}\,\mathcal{U}.f : \coprod_{i\in I} U_i \to X$. For every p, the underlying scheme of the degree-p object* $(\mathrm{coverCechNerveOver}\,\mathcal{U}).\mathrm{obj}\,[p]$ *of the Čech nerve of Definition 175 is the wide pullback*

$$\mathrm{widePullback}\,X\;(\,\lambda\,k : \mathrm{Fin}(p+1).\;\textstyle\coprod_i U_i\,)\;(\,\lambda\,k.\;q\,),$$

the $(p+1)$-fold fibre power of q over X, with structure map to X the canonical $\mathrm{WidePullback.\,base}$*. Equivalently, in* $\mathrm{Over}\,X$ *the object is* $\mathrm{Over.\,mk}(\mathrm{WidePullback.\,base})$.

Proof. Pure unfolding of the definitions. By Definition 175, $\mathrm{coverCechNerveOver}\,\mathcal{U}$ is the lift into $\mathrm{Over}\,X$ of $\mathrm{coverCechNerve}\,\mathcal{U} = (\mathrm{coverArrow}\,\mathcal{U}).\mathrm{augmentedCechNerve}$ along its augmentation. The augmented Čech nerve of an arrow f has, in degree n, the object $\mathrm{Arrow.\,cechNerve}\,f.\mathrm{obj}\,n = \mathrm{widePullback}\,f.\mathrm{right}\,(\lambda\,k : \mathrm{Fin}(n+1).\;f.\mathrm{left})\,(\lambda\,k.\;f.\mathrm{hom})$. Here $\mathrm{coverArrow}\,\mathcal{U} = \mathrm{Arrow.\,mk}(q)$ with right $= X$, left $= \coprod_i U_i$ and hom $= q$, so the degree-p object is exactly the displayed wide pullback, and the Over. lift records its structure map to X as $\mathrm{WidePullback.\,base}$. $\square$

Lemma 341. *[Wide fibre power over a base is the iterated product in the slice] Let $\mathcal{C}$ be a category with pullbacks, $S \in \mathcal{C}$, and $(Z_k \to S)_{k\in\mathrm{Fin}(n)}$ a finite family over S. Then the wide pullback over S,*

$$\mathrm{widePullback}\,S\;(\lambda\,k.\;Z_k)\;(\lambda\,k.\;Z_k \to S),$$

is the n-fold product $\prod_k Z_k$ of the same family taken in the slice category $\mathrm{Over}\,S$ *— i.e. the n-fold fibre power over S is the n-fold product in* $\mathrm{Over}\,S$*. In particular the fibre power has the clean recursion $\prod_{k:\mathrm{Fin}(p+2)} Z_k \cong Z_0 \times_S \prod_{k:\mathrm{Fin}(p+1)} Z_{k+1}$ along* $\mathrm{Fin}(p+2) \cong \mathrm{Fin}(1) \oplus \mathrm{Fin}(p+1)$.

Proof. In the slice $\mathrm{Over}\,S$ the object $\mathrm{Over.\,mk}(\mathrm{id}_S)$ is terminal by Lemma 405, and the legs of the wide-pullback diagram all land in this terminal base. By Lemma 404 a limiting fan of the legs is then a limit of the wide-pullback diagram, so the wide pullback over S coincides with the product of the legs computed in $\mathrm{Over}\,S$. The recursion is the standard splitting of a finite product along $\mathrm{Fin}(p+2) \cong \mathrm{Fin}(1) \oplus \mathrm{Fin}(p+1)$ (a binary product with the head factor). $\square$

Lemma 342 (Base case of the wide-fibre-power decomposition)**.** *Let $\mathcal{C}$ be a category with pullbacks, ι a finite index type, $S \in \mathcal{C}$, and $(X_i \to S)_{i\in\iota}$ a finite family over S. Then the 1-fold wide fibre power of $\coprod_i X_i \to S$ over S decomposes as*

$$\left(\textstyle\coprod_i X_i\right)^{\times_S 1} \;\cong\; \textstyle\coprod_{\sigma:\mathrm{Fin}(1)\to\iota} X_{\sigma(0)},$$

the σ-component being the 1-fold fibre power $X_{\sigma(0)}$.

Proof. By Lemma 341 the 1-fold fibre power is the 1-fold product in $\mathrm{Over}\,S$, which is the object itself: $(\coprod_i X_i)^{\times_S 1} \cong \coprod_i X_i$. On the right, the evaluation-at-0 bijection $\mathrm{Fin}(1) \to \iota \;\cong\; \iota$ identifies $\coprod_{\sigma:\mathrm{Fin}(1)\to\iota} X_{\sigma(0)}$ with $\coprod_i X_i$ (reindexing a coproduct along a bijection of index types). Composing the two gives the claim — the trivial reindexing. $\square$

Lemma 343. *[One-sided binary distributivity in the slice] Let $\mathcal{C}$ be a finitary pre-extensive category with pullbacks, $S \in \mathcal{C}$, and fix objects over S. For a finite family $(Y_\tau \to S)_\tau$ and a single $A \to S$ the fibre product distributes over the coproduct on each side:*

$$A \times_S (\textstyle\coprod_\tau Y_\tau) \;\cong\; \textstyle\coprod_\tau (A \times_S Y_\tau), \qquad (\textstyle\coprod_i X_i) \times_S B \;\cong\; \textstyle\coprod_i (X_i \times_S B),$$

with each isomorphism the Sigma. desc *of the pullback maps, hence compatible with the projections to S.*

Proof. This is the binary distributivity instance Lemma 401 ($\coprod_{(i,j)}(X_i \times_S Y_j) \cong (\coprod_i X_i) \times_S (\coprod_j Y_j)$) specialised by taking one of the two families to be a singleton: with the first family the singleton $\{A\}$ one gets $A \times_S \coprod_\tau Y_\tau \cong \coprod_\tau (A \times_S Y_\tau)$, and with the second family the singleton $\{B\}$ one gets $(\coprod_i X_i) \times_S B \cong \coprod_i (X_i \times_S B)$. □

Lemma 344 (One-sided distributivity in Over S binary-product form)**.** *Let $\mathcal{C}$ be a finitary pre-extensive category with pullbacks and $S \in \mathcal{C}$. For a finite index type ι :* Type 0*, a family $(A_i)_{i \in \iota}$ of objects of* Over S*, and a single $B \in$* Over S*, the* binary product in the slice *distributes over the coproduct:*

$$(\textstyle\coprod_i A_i) \times B \;\cong\; \textstyle\coprod_i (A_i \times B) \qquad \textit{in } \mathrm{Over}\, S,$$

where $\times$ is the categorical binary product of Over S*. This is the form in which the inductive step of Lemma 348 consumes distributivity: the recursion lives entirely in* Over S*, so it needs the slice binary product, not the $\mathcal{C}$-level fibre product of Lemma 343.*

Proof. The forgetful functor Over $S \to \mathcal{C}$ reflects isomorphisms: a morphism of the slice is an isomorphism iff its underlying $\mathcal{C}$-morphism (its .left) is. It therefore suffices to identify the underlying objects. The coproduct in Over S is computed on underlying objects, $(\coprod_i A_i).\mathrm{left} = \coprod_i A_i.\mathrm{left}$, and the underlying object of a binary product in Over S is the fibre product of the two structure maps, $(Y \times Z).\mathrm{left} \cong \mathrm{pullback}\, Y.\mathrm{hom}\, Z.\mathrm{hom}$, via Over. prodLeftIsoPullback (Lemma 346). Under these identifications both sides become the $\mathcal{C}$-level objects of the one-sided distributivity Lemma 343: $(\coprod_i A_i.\mathrm{left}) \times_S B.\mathrm{left}$ on the left and $\coprod_i (A_i.\mathrm{left} \times_S B.\mathrm{left})$ on the right. Matching .left with that $\mathcal{C}$-level isomorphism, and noting all comparison maps commute with the projection to S, gives the slice isomorphism. □

Lemma 345 (One-sided distributivity in Over S, product-on-the-right form)**.** *Let $\mathcal{C}$ be a finitary pre-extensive category with pullbacks and $S \in \mathcal{C}$. For a finite index type ι :* Type 0*, a single $A \in$* Over S *and a family $(Y_i)_{i \in \iota}$ of objects of* Over S*, the slice binary product distributes over the coproduct in the second argument:*

$$A \times (\textstyle\coprod_i Y_i) \;\cong\; \textstyle\coprod_i (A \times Y_i) \qquad \textit{in } \mathrm{Over}\, S.$$

This is the braiding-twin of Lemma 344; the inductive step of Lemma 348 uses it to distribute the head factor over the coproduct of multi-indexed fibre powers.

Proof. Braid the slice binary product, $A \times (\coprod_i Y_i) \cong (\coprod_i Y_i) \times A$, apply the left-handed distributivity (Lemma 344) to get $\coprod_i (Y_i \times A)$, and braid each summand back to $A \times Y_i$. All three steps are isomorphisms of Over S, so the composite is the asserted distributivity. □

Lemma 346 (The underlying object of a slice binary product is the pullback)**.** *Provided by Mathlib. For $S \in \mathcal{C}$ and two objects $Y, Z \in$* Over S *(with $\mathcal{C}$ having the relevant pullback), the underlying object of their binary product in the slice is the fibre product of their structure maps: $(Y \times Z).\mathrm{left} \cong \mathrm{pullback}\, Y.\mathrm{hom}\, Z.\mathrm{hom}$, compatibly with the projections.*

Lemma 347 (Nested coproduct flattens and reindexes to the next fibre power)**.** *Let ι be a finite index type, $S \in \mathcal{C}$, $(X_i \to S)_i$ a family over S, and for $\tau : \mathrm{Fin}(p+1) \to \iota$ write $Y_\tau = X_{\tau(0)} \times_S \cdots \times_S X_{\tau(p)}$. Then the nested coproduct flattens and reindexes to a coproduct over $(p+2)$-fold multi-indices:*

$$\coprod_i \coprod_\tau (X_i \times_S Y_\tau) \;\cong\; \coprod_{(i,\tau)} (X_i \times_S Y_\tau) \;\cong\; \coprod_{\sigma : \mathrm{Fin}(p+2) \to \iota} (X_{\sigma(0)} \times_S \cdots \times_S X_{\sigma(p+1)}),$$

where the second isomorphism is the reindexing $(i, \tau) \mapsto \sigma = \mathrm{Fin.cons}\, i\, \tau$, a bijection $\iota \times (\mathrm{Fin}(p+1) \to \iota) \cong (\mathrm{Fin}(p+2) \to \iota)$, under which $X_i \times_S Y_\tau$ is exactly the $(p+2)$-fold fibre power $X_{\sigma(0)} \times_S \cdots \times_S X_{\sigma(p+1)}$.

Proof. The first isomorphism is sigmaSigmaIso of Lemma 406, collapsing the doubly-indexed coproduct $\coprod_i \coprod_\tau$ into a coproduct over pairs (i, τ). The Fin. cons bijection $\iota \times (\mathrm{Fin}(p+1) \to \iota) \cong (\mathrm{Fin}(p+2) \to \iota)$, $(i, \tau) \mapsto \mathrm{Fin.cons}\, i\, \tau$, reindexes that coproduct over σ; and since $\sigma(0) = i$ and $\sigma(k+1) = \tau(k)$, the component $X_i \times_S Y_\tau = X_i \times_S (X_{\tau(0)} \times_S \cdots)$ is term-for-term the $(p+2)$-fold fibre power $X_{\sigma(0)} \times_S \cdots \times_S X_{\sigma(p+1)}$. □

Lemma 348 (Coproduct distributes over the wide fibre power)**.** *Let $\mathcal{C}$ be a category with pullbacks that is finitary pre-extensive (*FinitaryPreExtensive $\mathcal{C}$*), let ι be a finite index type, $S \in \mathcal{C}$, and $(X_i \to S)_{i \in \iota}$ a finite family over S. Then for every p the $(p+1)$-fold wide fibre power of the coproduct map $\coprod_i X_i \to S$ over S decomposes as a coproduct over multi-indices:*

$$(\textstyle\coprod_i X_i)^{\times_S (p+1)} \;\cong\; \textstyle\coprod_{\sigma : \mathrm{Fin}(p+1) \to \iota} (X_{\sigma(0)} \times_S \cdots \times_S X_{\sigma(p)}),$$

the σ-component being the $(p+1)$-fold fibre power of the legs $X_{\sigma(k)} \to S$ over S, and the structure map of the coproduct being Sigma. desc *of the component structure maps.*

Slice-product normal form. *The proof carries out the recursion in* Over S*, so the σ-component is presented as the iterated* slice product $\prod^{\mathrm{c}}_k \mathrm{Over.mk}(f(\sigma k))$ *over* Over S *rather than as the wide fibre power directly; the two are identified by Lemma 341 (wide fibre power over S equals the iterated product in* Over S*). The result is stated abstractly for* FinitaryPreExtensive $\mathcal{C}$ *(reusable) and then instantiated at $\mathcal{C} =$* Scheme*, which is finitary extensive by Lemma 400.*

Universe constraint. *This statement requires the index type ι :* Type 0*: the binary-distributivity primitive it rests on (Lemma 402) does not hold for index types in higher universes. When the cover index is finite but lives in a higher universe, one first reindexes along an equivalence to* Fin(n) *before applying this result; see Lemma 350.*

Proof. Induction on p. Throughout, Lemma 341 lets us treat the $(p+1)$-fold fibre power over S as the $(p+1)$-fold product in Over S, so the induction has the clean recursion $P^{\times_S (p+2)} \cong P \times_S P^{\times_S (p+1)}$ with $P = \coprod_i X_i$.

Base step $p = 0$ is Lemma 342: the 1-fold fibre power is $\coprod_i X_i$ itself, indexed by $\sigma : \mathrm{Fin}(1) \to \iota$.

Inductive step: assume $(\coprod_i X_i)^{\times_S (p+1)} \cong \coprod_{\tau : \mathrm{Fin}(p+1) \to \iota} Y_\tau$ with $Y_\tau = X_{\tau(0)} \times_S \cdots \times_S X_{\tau(p)}$. By the recursion of Lemma 341 and the inductive hypothesis,

$$(\textstyle\coprod_i X_i)^{\times_S (p+2)} \;\cong\; (\textstyle\coprod_i X_i) \times_S (\textstyle\coprod_\tau Y_\tau).$$

Applying the one-sided distributivity of Lemma 343 on each side turns the right-hand side into $\coprod_i \coprod_\tau (X_i \times_S Y_\tau)$. Since the recursion lives in Over S, this distributivity is consumed in its Over S binary-product form (Lemma 344), which expresses $(\coprod_\tau Y_\tau) \times B \cong \coprod_\tau (Y_\tau \times B)$ for the binary product in the slice rather than the $\mathcal{C}$-level pullback. Finally Lemma 347 flattens this nested coproduct (via sigmaSigmaIso) and reindexes the pairs (i, τ) as $\sigma = \mathrm{Fin.cons}\, i\, \tau : \mathrm{Fin}(p+2) \to \iota$, identifying $X_i \times_S Y_\tau$ with $X_{\sigma(0)} \times_S \cdots \times_S X_{\sigma(p+1)}$. The structure maps to S agree because each constituent isomorphism is built from Sigma. desc / product comparisons that commute with the projection to S by construction. Instantiating at $\mathcal{C} = \mathrm{Scheme}$ (Lemma 400) gives the scheme-level statement. □

Lemma 349. *[The wide fibre power of open immersions is the intersection open] Let $\mathcal{U} : X = \bigcup_{i\in I} U_i$ be an open cover and $\sigma : \mathrm{Fin}(p+1) \to I$ a multi-index. The $(p+1)$-fold wide fibre power over X of the open immersions $(U_{\sigma(k)} \hookrightarrow X)_k$ is the intersection open:*

$$\mathrm{widePullback}\, X\, (\lambda\, k.\, U_{\sigma(k)})\, (\lambda\, k.\, (U_{\sigma(k)}).\iota) \;\cong\; (\mathrm{coverInterOpen}\, \mathcal{U}\, \sigma).\mathrm{toScheme} \;=\; (\textstyle\bigcap_k U_{\sigma(k)}).\mathrm{toScheme},$$

over X, and the structure map is the open immersion $j_\sigma : \mathrm{coverInterOpen}\, \mathcal{U}\, \sigma \hookrightarrow X$ of Definition 195.

Proof. Induction on the family, iterating the binary case. The pullback over X of two open immersions $U.\iota, V.\iota$ is the intersection open $(U \sqcap V).\mathrm{toScheme}$ with the two homOfLE inclusions — this is isPullback_opens_inf of Lemma 403. The wide pullback of the family $(U_{\sigma(k)})_k$ is the iterated binary pullback over X along $\mathrm{Fin}(p+1)$, so iterating the binary intersection gives $\bigcap_k U_{\sigma(k)}$, which is the indexed infimum $\mathrm{coverInterOpen}\, \mathcal{U}\, \sigma = \bigcap_k \mathrm{coverOpen}\, \mathcal{U}\, (\sigma(k))$ of Definition 195 (recall $\mathrm{coverOpen}\, \mathcal{U}\, i = (\mathcal{U}.f\, i).\mathrm{opensRange}$ is the image open U_i). The structure map to X is the inclusion of the intersection, namely j_σ. □

Lemma 350. *[The degree-p Čech backbone is the coproduct of intersection opens] Let $\mathcal{U} : X = \bigcup_{i\in I} U_i$ be an open cover with cover arrow $q : \coprod_{i\in I} U_i \to X$, and let Y_p denote the degree-p object of the Čech nerve* coverCechNerveOver $\mathcal{U}$ *of Definition 175 — the $(p+1)$-fold fibre power of q over X. Then Y_p is canonically the coproduct*

$$Y_p \;\cong\; \coprod_{\sigma:\mathrm{Fin}(p+1)\to I} U_\sigma, \qquad U_\sigma \;=\; \mathrm{coverInterOpen}\, \mathcal{U}\, \sigma \;=\; \bigcap_k U_{\sigma(k)},$$

indexed by the multi-indices σ, and the structure map $q_p : Y_p \to X$ is $\mathrm{Sigma.desc}(\sigma \mapsto j_\sigma)$, where $j_\sigma : U_\sigma \hookrightarrow X$ is the open immersion of the intersection open U_σ of Definition 195. In words: the fibre power of the open immersions $U_i \hookrightarrow X$ is the intersection open, and the disjoint union over σ assembles the Čech backbone.

Proof. Assemble the chain in Over X. By Lemma 340 the degree-p object Y_p is the wide pullback $(\coprod_i U_i)^{\times_X (p+1)}$, the $(p+1)$-fold fibre power of q over X. By Lemma 348 (instantiated at $\mathcal{C} = \mathrm{Scheme}$, with $S = X$ and the family $U_i \hookrightarrow X$) this fibre power is the coproduct $\coprod_{\sigma:\mathrm{Fin}(p+1)\to I}(U_{\sigma(0)} \times_X \cdots \times_X U_{\sigma(p)})$ over the multi-indices. This decomposition delivers each σ-component in its slice-product normal form, namely the iterated product $\prod^{\mathrm{c}}_k \mathrm{Over.mk}(U_{\sigma(k)} \hookrightarrow X)$ in Over X; before it can be identified as an intersection open it must be converted back to the wide-pullback form, which is exactly Lemma 341 (the iterated product over X in Over X equals the wide fibre power over X). With the component thus in wide-pullback form, Lemma 349 applies: each σ-component, being the wide fibre power over X of the open immersions $U_{\sigma(k)} \hookrightarrow X$, is the intersection open $U_\sigma = \mathrm{coverInterOpen}\, \mathcal{U}\, \sigma = \bigcap_k U_{\sigma(k)}$, with structure map the open immersion j_σ. The per-σ component identification packaging this slice-product→intersection-open isomorphism is coverInterProdIso, and the transport of a wide fibre power along an equality of its base object (used to align the two wide-pullback forms over X) is widePullbackBaseCongr. Composing these isomorphisms gives $Y_p \cong \coprod_\sigma U_\sigma$; the structure-map agreement is automatic, since each step's isomorphism commutes with the projection to X, so the universal map out of the coproduct is $\mathrm{Sigma.desc}(\sigma \mapsto j_\sigma)$.

Universe reduction. The distributivity Lemma 348 and the binary-distributivity leaf it rests on (Lemma 402) require the index type to lie in Type 0; the cover index I of $\mathcal{U}$ is a finite type that need not live in Type 0. The reduction is therefore performed here: from finiteness of I choose an equivalence $e : I \cong \mathrm{Fin}(n)$ (Mathlib's Finite. equivFin, or Fintype. equivFin once a Fintype instance is fixed), and reindex the cover family $(U_i \hookrightarrow X)$ and every coproduct- and wide-pullback-index along it to land at $\mathrm{Fin}(n) : \mathrm{Type}\,0$. Two index transports are involved. On the left, the source coproduct is reindexed along e,

$$\coprod_{i\in I} \mathcal{U}.X \;\cong\; \coprod_{j\in \mathrm{Fin}(n)} (\mathcal{U}.X \circ e^{-1}),$$

the coproduct comparison induced by the index equivalence (Sigma. whiskerEquiv). On the right, the multi-index type is reindexed on its codomain,

$$(\mathrm{Fin}(p+1) \to \mathrm{Fin}(n)) \;\cong\; (\mathrm{Fin}(p+1) \to I),$$

the arrow-congruence Equiv. arrowCongr applied to e^{-1} in the codomain, under which the coproduct over σ-components of the $\mathrm{Fin}(n)$-decomposition matches the coproduct over the original multi-indices. Apply the Type 0 decomposition of Lemma 348 at $\mathrm{Fin}(n)$, then transport the resulting isomorphism back along these two equivalences. Reindexing a coproduct (resp. a wide fibre power) along a bijection of index types is an isomorphism compatible with the structure maps, so the transported isomorphism is again $\mathrm{Sigma.desc}(\sigma \mapsto j_\sigma)$ over X. □

Lemma 351. *[The cover arrow is the coproduct of its member arrows in the slice] Let $\mathcal{U} : X = \bigcup_{i\in I} U_i$ be an open cover with cover arrow $q = \mathrm{Sigma.desc}\,\mathcal{U}.f : \coprod_{i\in I} U_i \to X$. Then in the slice category* Over X *the object* $\mathrm{Over.mk}(q)$ *carrying the cover arrow is the coproduct of the member arrows* $\mathrm{Over.mk}(\mathcal{U}.f\,i)$*:*

$$\mathrm{Over.mk}(\mathrm{Sigma.desc}\,\mathcal{U}.f) \;\cong\; \coprod_{i\in I} \mathrm{Over.mk}(\mathcal{U}.f\,i) \qquad \textit{in}\ \ \mathrm{Over}\,X.$$

This is the degree-0 instance of the Čech backbone identification (Lemma 350): the source of the cover arrow is the disjoint union of the cover members, structurally over X.

Proof. The object $\mathrm{Over.mk}(q)$ is the coproduct of the family $(\mathrm{Over.mk}(\mathcal{U}.f\,i))_i$ in Over X. The i-th coproduct inclusion is the slice morphism lifting the inclusion $\iota_i : U_i \hookrightarrow \coprod_{j\in I} U_j$; it is well-defined as a morphism over X because $q \circ \iota_i = \mathcal{U}.f\,i$. The universal property holds: any family of slice morphisms $\mathrm{Over.mk}(\mathcal{U}.f\,i) \to Z$ into a common slice object Z factors uniquely through $\mathrm{Over.mk}(q)$ via the universal map of the underlying coproduct $\coprod_{j\in I} U_j$, lifted to the slice since the forgetful functor $\mathrm{Over}\,X \to \mathrm{Scheme}$ reflects the required equalities. Comparing with the standard coproduct in Over X yields the asserted isomorphism. □

Lemma 352 (Reflect an isomorphism of module sheaves to the underlying presheaf of abelian groups)**.** *Let $\varphi : \mathcal{M} \to \mathcal{N}$ be a morphism in X.Modules. If the image of φ under the forgetful functor* Scheme. Modules. toPresheaf $X : X.\mathrm{Modules} \to \mathrm{Presheaf(Ab)}$ *to the underlying presheaf of abelian groups is an isomorphism, then φ is an isomorphism in X.Modules.*

Proof. The forgetful functor from sheaves of $\mathcal{O}_X$-modules to the underlying (pre)sheaf of abelian groups is fully faithful, hence reflects isomorphisms (Lemma 248); the inclusion of sheaves of abelian groups into presheaves of abelian groups is likewise fully faithful, so the composite to presheaves of abelian groups reflects isomorphisms as well. A functor reflecting isomorphisms turns the hypothesis that the image of φ is an isomorphism into the conclusion that φ itself is one. $\square$

Lemma 353 (Restrict–stalk comparison along an open immersion)**.** *Provided by Mathlib (`Mathlib.AlgebraicGeomet`* *Let $f : U \hookrightarrow X$ be an open immersion of schemes and $y \in U$ a point with image $x = f(y)$. For a module sheaf $\mathcal{M} \in X.\mathrm{Modules}$, restricting $\mathcal{M}$ along f and then taking the stalk of the underlying presheaf of abelian groups at y is naturally isomorphic to the stalk of $\mathcal{M}$ (as a presheaf of abelian groups) at x; i.e. there is a natural isomorphism between the functors $\mathcal{M} \mapsto (\mathrm{restrict}\, f\, \mathcal{M})_y$ and $\mathcal{M} \mapsto \mathcal{M}_x$.*

Lemma 354 (Stalkwise conservativity for sheaves of abelian groups)**.** *Provided by Mathlib (`Mathlib.Topology.Sheaves.Stalks`). A morphism φ of sheaves of abelian groups on a topological space X is an isomorphism if and only if for every point $x \in X$ the induced map on stalks φ_x is an isomorphism. (Stalk functors are jointly conservative on $\mathrm{Sheaf}(\mathrm{Ab}, X)$.)*

Lemma 355 (Underlying presheaf of a module sheaf is a sheaf)**.** *Provided by Mathlib (`Mathlib.AlgebraicGeometry.M`* *For $\mathcal{M} \in X.\mathrm{Modules}$, the underlying presheaf of abelian groups $\mathcal{M}.\mathrm{presheaf}$ satisfies the sheaf condition, so it lifts to an object of $\mathrm{Sheaf}(\mathrm{Ab}, X)$.*

Lemma 356 (A module-sheaf morphism is an isomorphism iff it is one on every member of an open cover)**.** *Let X be a scheme, $\varphi : \mathcal{M} \to \mathcal{N}$ a morphism in $X.\mathrm{Modules}$, and $\mathfrak{U}$ an open cover of X. Then φ is an isomorphism if and only if, for every index j, its restriction along the open immersion $\mathfrak{U}.\mathrm{map}\, j$ to the cover member $\mathfrak{U}.\mathrm{obj}\, j$ is an isomorphism.*

Proof. ($\Rightarrow$) Functors preserve isomorphisms: if φ is an isomorphism then each restriction functor $\mathrm{restrictFunctor}(\mathfrak{U}.\mathrm{map}\, j)$ sends it to an isomorphism.

($\Leftarrow$) Assume every restriction is an isomorphism. It suffices to show that the image of φ under the forgetful functor $\mathrm{Scheme.Modules.toPresheaf}\, X$ to the underlying presheaf of abelian groups is an isomorphism, because that functor reflects isomorphisms (352). The underlying presheaves are sheaves of abelian groups by 355; by stalk conservativity (354), it suffices to show that φ is an isomorphism on every stalk.

Fix $x \in X$ and choose an index j such that x lies in the image of the cover map $\mathfrak{U}.\mathrm{map}\, j$ (the covering property); choose y in $\mathfrak{U}.\mathrm{obj}\, j$ mapping to x. The restrict-versus-stalk comparison $\mathrm{restrictStalkNatIso}(\mathfrak{U}.\mathrm{map}\, j)\, y$ (353) — a natural isomorphism between "restrict along the open immersion $\mathfrak{U}.\mathrm{map}\, j$ then take the stalk at y" and "take the stalk at $x = (\mathfrak{U}.\mathrm{map}\, j)\, y$" — conjugates the stalk at y of the restricted morphism $(\mathrm{restrictFunctor}(\mathfrak{U}.\mathrm{map}\, j)).\mathrm{map}\, \varphi$ with the stalk of φ at x. Since the restricted morphism is an isomorphism by hypothesis, its stalk at y is an isomorphism, hence so is the stalk of φ at x. As x was arbitrary, φ is a stalkwise isomorphism, hence an isomorphism. $\square$

Lemma 357 (Push–pull on a binary coproduct of two legs is the product of the two leg push–pulls)**.** *Let $Y_0, Y_1 : \mathrm{Over}\, X$ be two objects of the slice category over a scheme X, and form the binary coproduct of their underlying schemes $Y_0.\mathrm{left} \amalg Y_1.\mathrm{left}$ with structure map $\mathrm{coprod.desc}\, Y_0.\mathrm{hom}\, Y_1.\mathrm{hom} : Y_0.\mathrm{left} \amalg Y_1.\mathrm{left} \to X$. Then the push–pull object on this binary coproduct is the binary product of the two per-leg push–pull objects:*

$$\mathrm{pushPullObj}\, \mathcal{F}\, \Big(\mathrm{Over.mk}(\mathrm{coprod.desc}\, Y_0.\mathrm{hom}\, Y_1.\mathrm{hom})\Big) \cong \mathrm{pushPullObj}\, \mathcal{F}\, Y_0 \times \mathrm{pushPullObj}\, \mathcal{F}\, Y_1 \qquad \textit{in } X.\mathrm{Modul}$$

Proof. Write $q = \mathrm{coprod.desc}\, Y_0.\mathrm{hom}\, Y_1.\mathrm{hom}$ and $M = q^*\mathcal{F}$; let $\mathrm{inl} : Y_0.\mathrm{left} \to Y_0.\mathrm{left} \amalg Y_1.\mathrm{left}$ and $\mathrm{inr} : Y_1.\mathrm{left} \to Y_0.\mathrm{left} \amalg Y_1.\mathrm{left}$ be the coproduct inclusions, each an arrow over X (since $q \circ \mathrm{inl} = Y_0.\mathrm{hom}$ and $q \circ \mathrm{inr} = Y_1.\mathrm{hom}$), and write $\overline{\mathrm{inl}}, \overline{\mathrm{inr}}$ for these over-arrows.

The canonical comparison (mandatory framing). The asserted isomorphism is taken to be the *canonical* map

$$\mathrm{prod.\,lift}(\mathrm{pushPullMap}\,\mathcal{F}\,\overline{\mathrm{inl}},\ \mathrm{pushPullMap}\,\mathcal{F}\,\overline{\mathrm{inr}}) : \mathrm{pushPullObj}\,\mathcal{F}\,(\mathrm{Over.\,mk}\,q) \longrightarrow \mathrm{pushPullObj}\,\mathcal{F}\,Y_0 \times \mathrm{pushPullObj}$$

the product-lift of the two push–pull maps of the inclusions (Definition 177). This framing is mandatory, not cosmetic: the downstream section-identification and Čech-contractibility steps require the forward map of this isomorphism to be *literally* this prod. lift of push–pull maps; a non-canonical chain isomorphism with the same source and target would not serve them. So the task is precisely to prove this canonical map is an isomorphism.

The reference chain. Compare the canonical map against a manifestly invertible composite built from the binary disjoint-union decomposition. Write $P = \mathrm{inl}_*(M|_{\mathrm{inl}})$ and $Q = \mathrm{inr}_*(M|_{\mathrm{inr}})$ for the two restriction-to-component pushforwards, and let $\mathrm{coprodDecompMap}\,M : M \to P \times Q$ be their product-lift of restriction-adjunction units. Then

$$(\mathrm{pushforward}\,q).\mathrm{obj}\,M \xrightarrow{(\mathrm{pushforward}\,q)(\mathrm{coprodDecompMap}\,M)} (\mathrm{pushforward}\,q).\mathrm{obj}(P\times Q) \xrightarrow{\sim} (\mathrm{pushforward}\,q).\mathrm{obj}\,P\times(\mathrm{pushf}$$

is a chain of isomorphisms: the first arrow is q_* applied to the decomposition iso (an isomorphism by the leaf below); the second is that $q_* = \mathrm{pushforward}\,q$ preserves the binary product $P \times Q$; the third is $\mathrm{prod.\,mapIso}\ \mathrm{idiso}_0\ \mathrm{idiso}_1$, the product of the two per-leg identifications $\mathrm{idiso}_0 : (\mathrm{pushforward}\,q).\mathrm{obj}\,P \cong \mathrm{pushPullObj}\,\mathcal{F}\,Y_0$ and its inr-twin. Each idiso_0 is assembled by pushing forward along q_* the inner isomorphism that chains the identification of restriction with open-immersion pullback (Lemma 396), the pullback-composition comparison for $\mathrm{inl} \circ q$, and the over-triangle reindexing $q \circ \mathrm{inl} = Y_0.\mathrm{hom}$, together with a transport of the codomain along this same triangle. That codomain transport is almost entirely definitional: pushforward-composition is identity-on-objects, so $q_*(\mathrm{inl}_*\,N) = (\mathrm{inl} \circ q)_* N$ holds by definition and only the over-triangle relabelling remains.

The leaf — binary disjoint-union decomposition. The decomposition comparison $\mathrm{coprodDecompMap}\,M$ is an isomorphism. By Lemma 352 it suffices to check this on sections over each open $V \subseteq Y_0.\mathrm{left} \amalg Y_1.\mathrm{left}$. There the open splits as the disjoint union of its two component traces $\mathrm{inl}^{-1}\,V$ and $\mathrm{inr}^{-1}\,V$ (disjoint because the coproduct summands are), so the binary disjoint-union sheaf decomposition — sections over a binary coproduct scheme are the product of the sections over the two preimages, Lemma 398, equivalently $F(U \sqcup V') \cong F(U) \times F(V')$ for disjoint opens, Lemma 399 — exhibits $\Gamma(V, M) \cong \Gamma(\mathrm{inl}^{-1}\,V, M|_{\mathrm{inl}}) \times \Gamma(\mathrm{inr}^{-1}\,V, M|_{\mathrm{inr}})$, and this identification is exactly the pair of restriction-adjunction units making up $\mathrm{coprodDecompMap}\,M$. (Evaluation at V preserves the binary product, Lemma 397, so the right-hand sections are the corresponding product.) Hence $\mathrm{coprodDecompMap}\,M$ is an isomorphism, and the reference chain above is an isomorphism.

Matching the canonical map to the chain. It remains to see the canonical prod. lift of push–pull maps coincides with the (invertible) reference composite, whence it too is an isomorphism. Two maps into a binary product agree once they agree after composing with each projection, so it suffices to match the two legs. The inl-leg of the reference composite is $(\mathrm{pushforward}\,q).\mathrm{map}\,u_0$ followed by idiso_0, where u_0 is the inl-component of $\mathrm{coprodDecompMap}\,M$; the per-leg coherence identity (Lemma 359) says exactly that this equals $\mathrm{pushPullMap}\,\mathcal{F}\,\overline{\mathrm{inl}}$, the inl-leg of the canonical map, and symmetrically for inr. Therefore the canonical map equals the reference isomorphism and is itself an isomorphism in X.Modules. □

Lemma 358 (Unit of a left-adjoint-uniqueness isomorphism)**.** *Provided by Mathlib. If F, F' are both left adjoint to G (adjunctions $\mathrm{adj}_1, \mathrm{adj}_2$), the uniqueness isomorphism $\mathrm{leftAdjointUniq}\,\mathrm{adj}_1\,\mathrm{adj}_2 : F \cong F'$ intertwines the units: composing the unit of adj_1 with G applied to the isomorphism recovers the unit of adj_2, i.e. at each object $\mathrm{adj}_2.\mathrm{unit} = \mathrm{adj}_1.\mathrm{unit} \circ G((\mathrm{leftAdjointUniq}\,\mathrm{adj}_1\,\mathrm{adj}_2).\mathrm{hom})$.*

Lemma 359 (Per-leg coherence of the binary push–pull comparison)**.** *In the setting of Lemma 357, with* $q = \mathrm{coprod.desc}\ Y_0.\mathrm{hom}\ Y_1.\mathrm{hom}$ *and* $M = q^*\mathcal{F}$*, write* u_0 *for the* inl*-component of* coprodDecompMap M *(the unit of the restriction adjunction along* inl *at* M*) and* $\mathrm{idiso}_0 : (\mathrm{pushforward}\, q).\mathrm{obj}(\mathrm{inl}_*(M|_{\mathrm{inl}})) \cong \mathrm{pushPullObj}\, \mathcal{F}\, Y_0$ *for the per-leg identification of the reference chain. Then the* inl*-leg of the canonical comparison factors through these:*

$$\mathrm{pushPullMap}\, \mathcal{F}\, \overline{\mathrm{inl}} \;=\; (\mathrm{pushforward}\, q).\mathrm{map}\, u_0 \;\circ\; \mathrm{idiso}_0.\mathrm{hom},$$

and symmetrically for the inr*-leg.*

Proof. The two legs are symmetric; treat inl. Unfold the push–pull map on the left-hand side to its raw five-step form (Definition 177): $\mathrm{pushPullMap}\, \mathcal{F}\, \overline{\mathrm{inl}}$ is q_* applied to the composite of the restriction-adjunction unit η^{inl} at M with inl_* of the pullback-composition comparison for $\mathrm{inl} \circ q$ acting on $\mathcal{F}$, followed by the over-triangle transport that idiso_0 encodes. Now the open immersion inl presents its restriction functor as a left adjoint, and the comparison restrictFunctorIsoPullback (Lemma 396) is precisely the uniqueness isomorphism between this restriction functor and the open-immersion pullback. The unit-intertwining identity (Lemma 358) rewrites the adjunction unit as $\eta^{\mathrm{inl}} = u_0 \circ \mathrm{inl}_*((\mathrm{restrictFunctorIsoPullback}\, \mathrm{inl}).\mathrm{hom})$. Substituting and cancelling the comparison isomorphism against the matching inverse factor inside idiso_0, the residual identity is an equality of two canonical equality-transport morphisms along the single open-set equality $q \circ \mathrm{inl} = Y_0.\mathrm{hom}$; these agree by proof-irrelevance of that equality. Hence $\mathrm{pushPullMap}\, \mathcal{F}\, \overline{\mathrm{inl}} = (\mathrm{pushforward}\, q).\mathrm{map}\, u_0 \circ \mathrm{idiso}_0.\mathrm{hom}$, as claimed. □

Lemma 360 (Canonical leg identification of the binary push–pull chain)**.** *In the setting of Lemma 357, let* c *be one of the coproduct inclusions* $\mathrm{inl}, \mathrm{inr} : C \to Y_0.\mathrm{left} \amalg Y_1.\mathrm{left}$*, with structure map* $p_C : C \to X$ *and over-triangle* $w_C : c \circ q = p_C$*. Writing* $M = q^*\mathcal{F}$*, there is a canonical isomorphism identifying the corresponding factor of* $q_*(P \times Q)$ *with the per-leg push–pull object:*

$$(\mathrm{pushforward}\, q).\mathrm{obj}\Big((\mathrm{pushforward}\, c).\mathrm{obj}((q^*\mathcal{F}).\mathrm{restrict}\, c)\Big) \;\cong\; \mathrm{pushPullObj}\, \mathcal{F}\, (\mathrm{Over.mk}\, p_C).$$

Proof. Push forward along q_* the inner isomorphism chaining the identification of restriction with open-immersion pullback (Lemma 396), the pullback-composition comparison for $c \circ q$, and the over-triangle reindexing $c \circ q = p_C$; then transport the codomain along the same triangle by an equality-of-objects rewrite. This is the per-leg identification idiso of the reference chain in Lemma 357. □

Lemma 361 (Splitting an Option-indexed coproduct)**.** *Let* $\mathcal{C}$ *be a category and* $Z : \mathrm{Option}\, \alpha \to \mathcal{C}$ *a family for which the relevant coproducts exist (the total coproduct* $\coprod Z$*, the coproduct* $\coprod_{a:\alpha} Z(\mathrm{some}\, a)$ *over the proper part, and their binary coproduct). Then the total coproduct splits off its* none *summand:*

$$\coprod_{o:\mathrm{Option}\,\alpha} Z(o) \;\cong\; Z(\mathrm{none}) \amalg \Big(\coprod_{a:\alpha} Z(\mathrm{some}\, a)\Big).$$

Proof. The forward map is Sigma.desc of the cocone sending the none component to the left binary inclusion and each some a component to the composite of the a-th inclusion into $\coprod_a Z(\mathrm{some}\, a)$ with the right binary inclusion. The inverse is coprod.desc of the none inclusion and Sigma.desc of the some inclusions. The two round-trip identities follow from the universal properties of the coproducts, checked componentwise on each none and some term. (Mathlib provides the associativity isomorphism for nested coproducts but no Option-split, so this is supplied directly.) □

Lemma 362 (Push–pull objects transport along a slice isomorphism)**.** *Let* $\mathcal{F} : X.\mathrm{Modules}$ *and let* $e : Y \cong Y'$ *be an isomorphism in* $\mathrm{Over}\, X$*. Then the push–pull objects on* Y *and* Y' *are isomorphic,*

$$\mathrm{pushPullObj}\, \mathcal{F}\, Y \;\cong\; \mathrm{pushPullObj}\, \mathcal{F}\, Y'.$$

Proof. The push–pull object is a contravariant functor of its slice argument through the push–pull map (Definition 177). Take the forward map to be $\mathrm{pushPullMap}\, \mathcal{F}\, e.\mathrm{inv}$ and the backward map $\mathrm{pushPullMap}\, \mathcal{F}\, e.\mathrm{hom}$ (contravariance: the hom direction uses $e.\mathrm{inv}$). The two composites are identities by the functoriality of the push–pull map under composition and identity of over-morphisms. □

Lemma 363 (Slice lift of the Option-coproduct splitting)**.** *Let* $\mathrm{legs} : \mathrm{Option}\, \alpha \to \mathrm{Over}\, X$ *be a family of objects of the slice over* X*. The splitting of Lemma 361 lifts to an isomorphism in* $\mathrm{Over}\, X$ *between the coproduct slice object and the binary-split slice object:*

$$\mathrm{Over.\,mk}(\mathrm{Sigma.\,desc}(o \mapsto (\mathrm{legs}\, o).\mathrm{hom})) \;\cong\; \mathrm{Over.\,mk}(\mathrm{coprod.\,desc}\,(\mathrm{legs\ none}).\mathrm{hom}\,(\mathrm{Sigma.\,desc}(a \mapsto (\mathrm{legs}\,(\mathrm{some}\, a)).$$

Proof. Build the slice isomorphism as $\mathrm{Over.\,isoMk}$ of the underlying-scheme isomorphism $\mathrm{sigmaOptionIso}(o \mapsto (\mathrm{legs}\, o).\mathrm{left})$ (Lemma 361), together with the verification that its forward map commutes with the two structure arrows. The latter is checked componentwise: on each none and $\mathrm{some}\, a$ component both descent maps reduce to $(\mathrm{legs}\, o).\mathrm{hom}$ by the universal properties of each coproduct. □

Lemma 364 (Splitting an Option-indexed product)**.** *Dual to Lemma 361: let* $W : \mathrm{Option}\, \alpha \to \mathcal{C}$ *with the relevant products existing. Then the total product splits off its* none *factor:*

$$\prod_{o:\mathrm{Option}\, \alpha} W(o) \;\cong\; W(\mathrm{none}) \times \Big(\prod_{a:\alpha} W(\mathrm{some}\, a)\Big).$$

Proof. The same construction as Lemma 361 with all arrows reversed: the forward map is $\mathrm{prod.\,lift}$ of the projection onto $W(\mathrm{none})$ and the $\mathrm{Pi.\,lift}$ of the some projections; the inverse is $\mathrm{Pi.\,lift}$ of the matching projections out of the binary product. The round-trip identities follow from the universal properties of the products. □

Lemma 365 (Push–pull on the empty coproduct is terminal)**.** *For the empty index type* $\iota = \mathrm{PEmpty}$ *and the empty family of legs, the coproduct* $\coprod_i \mathrm{legs}\, i$ *is the initial scheme* $\bot$*; the push–pull object on its unique structure map is terminal in* $X.\mathrm{Modules}$*, hence isomorphic to the empty product* $\prod_{i \in \mathrm{PEmpty}} \mathrm{pushPullObj}\, \mathcal{F}\, (\mathrm{legs}\, i)$*, which is likewise terminal:*

$$\mathrm{pushPullObj}\, \mathcal{F}\, \Big(\mathrm{Over.\,mk}(\mathrm{Sigma.\,desc}(i \mapsto (\mathrm{legs}\, i).\mathrm{hom}))\Big) \;\cong\; \prod_{i \in \mathrm{PEmpty}} \mathrm{pushPullObj}\, \mathcal{F}\, (\mathrm{legs}\, i).$$

Proof. The coproduct over the empty index type is the initial scheme $\bot$: the structure map $\mathrm{Sigma.\,desc}$ out of $\coprod_{i \in \mathrm{PEmpty}} (\mathrm{legs}\, i).\mathrm{left}$ is the unique morphism $q : \bot \to X$ out of the initial object. Every sheaf of modules over the empty scheme is the zero object: the only open of $\bot$ is the empty open, over which the structure sheaf has subsingleton (hence zero) sections, so every $\mathcal{O}_\bot$-module has zero sections over every open and is therefore the zero module sheaf. In particular the pulled-back module $(\mathrm{pullback}\, q).\mathrm{obj}\, \mathcal{F}$ is the zero object of $\bot.\mathrm{Modules}$.

The push–pull object is $\mathrm{pushPullObj}\, \mathcal{F}\, (\mathrm{Over.\,mk}\, q) = (\mathrm{pushforward}\, q).\mathrm{obj}\big((\mathrm{pullback}\, q).\mathrm{obj}\, \mathcal{F}\big)$. The module pushforward is additive, so it sends the zero object to a zero object, which is terminal. The empty product $\prod_{i \in \mathrm{PEmpty}} \mathrm{pushPullObj}\, \mathcal{F}\, (\mathrm{legs}\, i)$ is likewise terminal. A morphism between terminal objects is an isomorphism, so the comparison map $\mathrm{coprodToProdMap}\, \mathcal{F}\, \mathrm{legs}$ (Definition 367) is an isomorphism.

The single remaining formal obligation is that the pulled-back module is zero, namely $\mathrm{IsZero}((\mathrm{pullback}\, q).\mathrm{obj}\, \mathcal{F})$ for q the map out of the empty scheme. This is a presheaf-of-modules IsZero statement established pointwise through the faithful sections-to-presheaf functor, using the subsingleton sections of the structure sheaf over the empty scheme. ☐

Definition 366 (The i-th coproduct inclusion as an over-morphism)**.** Let $(\mathrm{legs} : \iota \to \mathrm{Over}\, X)$ be a family of objects of the slice over a scheme X, and suppose the coproduct $\coprod_i (\mathrm{legs}\, i).\mathrm{left}$ exists. For each index i, the canonical coproduct inclusion $\iota_i : (\mathrm{legs}\, i).\mathrm{left} \to \coprod_i (\mathrm{legs}\, i).\mathrm{left}$ is promoted to a morphism of the slice category $\mathrm{Over}\, X$,

$$\overline{\iota_i} : \mathrm{legs}\, i \longrightarrow \mathrm{Over.\,mk}(\mathrm{Sigma.\,desc}(i \mapsto (\mathrm{legs}\, i).\mathrm{hom})),$$

whose underlying scheme map is ι_i; the over-triangle commutes because $\iota_i \circ \mathrm{Sigma.\,desc}(i \mapsto (\mathrm{legs}\, i).\mathrm{hom}) = (\mathrm{legs}\, i).\mathrm{hom}$ by the defining property of Sigma. desc.

Definition 367 (Canonical coproduct-to-product comparison of push–pull objects)**.** In the setting of Definition 366, assume in addition that the product $\prod_i \mathrm{pushPullObj}\, \mathcal{F}\, (\mathrm{legs}\, i)$ exists. The *canonical comparison map*

$$\mathrm{coprodToProdMap}\, \mathcal{F}\, \mathrm{legs} : \mathrm{pushPullObj}\, \mathcal{F}\, (\mathrm{Over.\,mk}(\mathrm{Sigma.\,desc}(i \mapsto (\mathrm{legs}\, i).\mathrm{hom}))) \longrightarrow \prod_i \mathrm{pushPullObj}\, \mathcal{F}\, (\mathrm{legs}\, i)$$

is the Pi. lift whose i-th component is the push–pull map $\mathrm{pushPullMap}\, \mathcal{F}\, \overline{\iota_i}$ (Definition 177) of the i-th coproduct inclusion (Definition 366). Equivalently, it is the unique map whose composite with the i-th product projection equals $\mathrm{pushPullMap}\, \mathcal{F}\, \overline{\iota_i}$ for every i. This is the canonical framing kept invariant throughout the finite induction of Lemma 370.

Lemma 368 (The comparison map is invariant under index reindexing)**.** *Let $e : \alpha \simeq \beta$ be an equivalence of finite index types and let $\mathrm{legs} : \beta \to \mathrm{Over}\, X$ be a family of slice objects. If the canonical comparison map $\mathrm{coprodToProdMap}\, \mathcal{F}\, (\mathrm{legs} \circ e)$ for the reindexed family is an isomorphism, then so is $\mathrm{coprodToProdMap}\, \mathcal{F}\, \mathrm{legs}$. In other words, the predicate "$\mathrm{coprodToProdMap}\, \mathcal{F}\, (\,\cdot\,)$ is an isomorphism" is stable under reindexing the leg family along an index equivalence.*

Proof. The induction hypothesis supplies that $\mathrm{coprodToProdMap}\, \mathcal{F}\, (\mathrm{legs} \circ e)$ is an isomorphism; it suffices to exhibit $\mathrm{coprodToProdMap}\, \mathcal{F}\, \mathrm{legs}$ as the conjugate of that map by two reindexing isomorphisms, since conjugating an isomorphism by isomorphisms again yields an isomorphism.

Transporting the source. Reindexing the coproduct along e (the whiskered reindexing $\mathrm{Sigma.\,whiskerEquiv}\, e\, (i \mapsto \mathrm{Iso.\,refl})$) together with the identification of descent objects gives an isomorphism in $\mathrm{Over}\, X$

$$\mathrm{Over.\,mk}(\mathrm{Sigma.\,desc}(o \mapsto (\mathrm{legs}\, o).\mathrm{hom})) \;\cong\; \mathrm{Over.\,mk}(\mathrm{Sigma.\,desc}(a \mapsto ((\mathrm{legs} \circ e)\, a).\mathrm{hom})).$$

Carrying the push–pull object across this slice isomorphism by Lemma 362 identifies the source of $\mathrm{coprodToProdMap}\, \mathcal{F}\, \mathrm{legs}$ with the source of $\mathrm{coprodToProdMap}\, \mathcal{F}\, (\mathrm{legs} \circ e)$.

Transporting the target. Reindexing the product along e (the whiskered reindexing $\mathrm{Pi.\,whiskerEquiv}\, e\, (a \mapsto \mathrm{Iso.\,refl})$, equivalently the product reindexing isomorphism Pi. mapIso along e) identifies $\prod_{o:\beta} \mathrm{pushPullObj}\, \mathcal{F}\, (\mathrm{legs}\, o)$ with $\prod_{a:\alpha} \mathrm{pushPullObj}\, \mathcal{F}\, ((\mathrm{legs} \circ e)\, a)$.

Matching the canonical form. It remains to check that conjugating $\mathrm{coprodToProdMap}\, \mathcal{F}\, (\mathrm{legs} \circ e)$ by these two reindexing isomorphisms reproduces $\mathrm{coprodToProdMap}\, \mathcal{F}\, \mathrm{legs}$. Since both maps are determined by their composites with the product projections, the check is performed projection by projection: by the defining projection law of the comparison map ($\mathrm{coprodToProdMap}\, \mathcal{F}\, \mathrm{legs} \circ \pi_i = \mathrm{pushPullMap}\, \mathcal{F}\, \overline{\iota_i}$, Definition 367), each projection of either side is the push–pull map of a coproduct inclusion, and the reindexing isomorphisms carry the i-th inclusion $\overline{\iota_i}$ of the β-coproduct to the $e^{-1}(i)$-th inclusion of the α-coproduct. The naturality of pushPullMap $\mathcal{F}$ under this reindexing of the inclusions makes the two projections agree. Hence the conjugate equals coprodToProdMap $\mathcal{F}$ legs, and the induction hypothesis yields IsIso(coprodToProdMap $\mathcal{F}$ legs). ☐

Lemma 369 (Adjoining one index to the comparison-map induction)**.** *Suppose that for every family* legs : $\alpha \to$ Over X *the canonical comparison map* coprodToProdMap $\mathcal{F}$ legs *is an isomorphism. Then for every family* legs : Option $\alpha \to$ Over X *the comparison map* coprodToProdMap $\mathcal{F}$ legs *is an isomorphism.*

Proof. Write ls = legs∘some for the restriction to the proper part. The Option α-coproduct splits as the binary coproduct of the none leg and the α-coproduct (Lemma 363); transporting the push–pull object across this slice isomorphism (Lemma 362) and applying the binary decomposition (Lemma 357) together with the induction hypothesis on ls presents the source as a binary product, which the inverse of the Option-product splitting (Lemma 364) reassembles into the total product $\prod_{o:\mathrm{Option}\,\alpha}$ pushPullObj $\mathcal{F}$ (legs o). A per-Option-case projection chase then matches this composite reference isomorphism with the canonical comparison map coprodToProdMap $\mathcal{F}$ legs (Definition 367), whence the latter is an isomorphism. ☐

Lemma 370 (Push–pull on a finite coproduct of legs is the product of the leg push–pulls)**.** *Let* (legs : $\iota \to$ Over X) *be a finite family of objects of the slice category over a scheme X (ι finite), and form the coproduct* $\coprod_{i\in\iota}$ legs i *with structure map* Sigma. desc($i \mapsto$ (legs i).hom) : $\coprod_i$ legs $i \to X$. *Then the push–pull object on this coproduct is the product of the per-leg push–pull objects:*

$$\mathrm{pushPullObj}\,\mathcal{F}\left(\mathrm{Over.\,mk}(\mathrm{Sigma.\,desc}(i \mapsto (\mathrm{legs}\,i).\mathrm{hom}))\right) \cong \prod_{i\in\iota} \mathrm{pushPullObj}\,\mathcal{F}\,(\mathrm{legs}\,i) \qquad \textit{in } X.\mathrm{Modules}.$$

This is the general indexed-coproduct→product disjoint-union decomposition of a module sheaf: the legs may have overlapping images in X, but they are disjoint inside the coproduct space $\coprod_i$ legs i*, so push–pull splits componentwise.*

Proof. Argue by induction on the finite index type ι, using the finite-index recursion principle: a property of finite types that is stable under index equivalence, holds for the empty type, and is preserved by adjoining one element (passing from α to Option α) holds for every finite type. Throughout, the isomorphism is kept in its *canonical* form as the product of push–pull maps $\prod_i$ pushPullMap $\mathcal{F}$ $\overline{\iota_i}$, where $\overline{\iota_i}$ denotes the i-th coproduct inclusion viewed as an over-morphism into Over. mk(Sigma. desc($i \mapsto$ (legs i).hom)).

Empty base. For ι = PEmpty the claim is Lemma 365: both sides are terminal.

Reindexing. The property is stable under index equivalence by Lemma 368.

Adjoining one index. Suppose the claim holds for α; the passage to Option α is Lemma 369: the Option-coproduct splits as the binary coproduct of the none leg and the α-coproduct (Lemma 363), the binary decomposition (Lemma 357) and the induction hypothesis present the source as a binary product, the inverse product splitting (Lemma 364) reassembles it into the total product, and a per-Option-case projection chase matches the result with the canonical comparison map. (All isomorphisms produced live in X.Modules, verified through the underlying presheaf by Lemma 352 where needed.) ☐

Lemma 371. *[Push–pull over the Čech backbone is the product over intersection opens] With the notation of Lemma 350, the push–pull object on the degree-p backbone decomposes as a product* *over the multi-indices:*

$$\mathrm{pushPullObj}\,\mathcal{F}\,Y_p \;\cong\; \prod_{\sigma:\mathrm{Fin}(p+1)\to I} \mathrm{pushPullObj}\,\mathcal{F}\,(\mathrm{Over.\,mk}\,j_\sigma) \qquad \textit{in } X.\mathrm{Modules},$$

equivalently $(q_p)_*\,(q_p)^*\mathcal{F} \cong \prod_\sigma (j_\sigma)_*\,(j_\sigma)^*\mathcal{F}$.

Proof. By Lemma 350 the backbone Y_p is the coproduct $\coprod_\sigma U_\sigma$ with structure map $q_p = \mathrm{Sigma.\,desc}(\sigma \mapsto j_\sigma)$. Apply Lemma 370 to this coproduct family, with the finite multi-index type $\sigma : \mathrm{Fin}(p+1) \to I$ as the index and the intersection-open legs $\mathrm{Over.\,mk}\,j_\sigma$, to obtain the asserted product decomposition. □

Lemma 372 (Sections of one push–pull leg over an open)**.** *Let $j_\sigma : U_\sigma \hookrightarrow X$ be the open immersion of the intersection open $U_\sigma = \mathrm{coverInterOpen}\,\mathcal{U}\,\sigma$, and let $V \subseteq X$ be an open. Then there is a natural isomorphism of section groups*

$$\Gamma(V,\,\mathrm{pushPullObj}\,\mathcal{F}\,(\mathrm{Over.\,mk}\,j_\sigma)) \;\cong\; \Gamma(U_\sigma \cap V,\,\mathcal{F}).$$

Proof. This follows from three standard identifications. First, the leg is $(j_\sigma)_*\,(j_\sigma)^*\mathcal{F}$; push-forward sections are computed by preimage, $\Gamma(V,(j_\sigma)_*N) = \Gamma(j_\sigma^{-1}(V),N)$ (Lemma 395, an equality), applied with $N = (j_\sigma)^*\mathcal{F}$. Second, pullback along the open immersion j_σ is the restriction functor: $(j_\sigma)^*\mathcal{F} \cong \mathcal{F}.\,\mathrm{restrict}\,j_\sigma$ by Lemma 396. Third, the sections of a restricted module over $j_\sigma^{-1}(V)$ are the sections of $\mathcal{F}$ over the image open (Lemma 257, an equality): $\Gamma(j_\sigma^{-1}(V),\mathcal{F}.\,\mathrm{restrict}\,j_\sigma) = \Gamma(j_\sigma(j_\sigma^{-1}(V)),\mathcal{F}) = \Gamma(U_\sigma \cap V,\mathcal{F})$, the last equality because the image of the preimage of V under the open immersion $j_\sigma : U_\sigma \hookrightarrow X$ is exactly $U_\sigma \cap V$. Composing the three gives the stated isomorphism. □

Lemma 373 (Degreewise section identification of the Čech backbone)**.** *For an open $V \subseteq X$ and every degree p there is an isomorphism*

$$\Gamma(V,\,\mathrm{pushPullObj}\,\mathcal{F}\,Y_p) \;\cong\; \prod_{\sigma:\mathrm{Fin}(p+1)\to I} \Gamma(\mathrm{coverInterOpen}\,\mathcal{U}\,\sigma \cap V,\,\mathcal{F}).$$

Proof. Combine the product decomposition of the backbone push–pull object (Lemma 371), $\mathrm{pushPullObj}\,\mathcal{F}\,Y_p \cong \prod_\sigma \mathrm{pushPullObj}\,\mathcal{F}\,(\mathrm{Over.\,mk}\,j_\sigma)$, with the fact that the section (evaluation-at-V) functor on X.Modules preserves products (Lemma 397): $\Gamma(V,\prod_\sigma N_\sigma) \cong \prod_\sigma \Gamma(V,N_\sigma)$. This reduces the left-hand side to $\prod_\sigma \Gamma(V,\mathrm{pushPullObj}\,\mathcal{F}\,(\mathrm{Over.\,mk}\,j_\sigma))$, and the per-leg section identification (Lemma 372) rewrites each factor as $\Gamma(U_\sigma \cap V,\mathcal{F})$. □

Lemma 374 (An additive functor commutes with augmentation, degreewise the identity)**.** *Let $\Phi : \mathcal{A} \to \mathcal{B}$ be an additive functor between preadditive categories, $C^\bullet$ a cochain complex in $\mathcal{A}$, and $f : Y \to C^0$ an augmentation map satisfying $f \cdot d^0 = 0$. Then applying Φ degreewise commutes with the augmentation construction: there is an isomorphism of cochain complexes*

$$\Phi(C^\bullet.\,\mathrm{augment}\,f) \;\cong\; (\Phi C^\bullet).\,\mathrm{augment}(\Phi f),$$

every component of which is the identity. (Here Φf satisfies its own augmentation identity $\Phi(f) \cdot d^0 = 0$ by Lemma 375.)

Proof. The two augmented complexes have, by construction, the same underlying graded object: in degree 0 both read ΦY, and in degree $n+1$ both read ΦC^n. Define the isomorphism by taking every degreewise component to be the identity id. The differential-compatibility square in degree 0 and in each degree $n+1$ then closes by unfolding the definition of augment: the augmented differentials of $\Phi(C^\bullet.\,\mathrm{augment}\,f)$ and of $(\Phi C^\bullet).\,\mathrm{augment}(\Phi f)$ are literally the same map (Φf in the bottom slot, $\Phi(d^n)$ above), so each square commutes on the nose. □

Lemma 375 (Additive functors preserve the augmentation condition)**.** *Let $\Phi : \mathcal{A} \to \mathcal{B}$ be an additive functor between preadditive categories, $C^\bullet$ a cochain complex in $\mathcal{A}$, and $f : Y \to C^0$ a map with $f \cdot d^0 = 0$. Then the image map $\Phi f : \Phi Y \to \Phi C^0$ again satisfies the augmentation condition $\Phi(f) \cdot d^0 = 0$, where d^0 is the lowest differential of the degreewise image complex $\Phi C^\bullet$.*

Proof. The lowest differential of $\Phi C^\bullet$ is, by definitional reshaping, $\Phi(d^0)$ applied to the original lowest differential of $C^\bullet$. Hence $\Phi(f) \cdot \Phi(d^0) = \Phi(f \cdot d^0) = \Phi(0) = 0$ by functoriality of Φ on composites and its preservation of zero. □

Lemma 376 (Gluing an iso of augmented complexes from base, node, and compatibility data)**.** *Let $C^\bullet, C'^\bullet$ be cochain complexes in a preadditive category, with augmentation maps $f : Y \to C^0$ ($f \cdot d^0 = 0$) and $f' : Y' \to C'^0$ ($f' \cdot d'^0 = 0$). Given an isomorphism $\varphi : C^\bullet \cong C'^\bullet$ of the base complexes, an isomorphism $e_Y : Y \cong Y'$ of the augmentation objects, and a compatibility square $f \cdot \varphi_0 = e_Y \cdot f'$ intertwining the augmentation maps through e_Y and the degree-0 component φ_0 of φ, there is an isomorphism of augmented cochain complexes*

$$C^\bullet.\,\mathrm{augment}\, f \;\cong\; C'^\bullet.\,\mathrm{augment}\, f'.$$

Proof. Define the isomorphism degreewise: take the degree-0 component to be e_Y and the degree-$n+1$ component to be φ_n, the n-th component of φ. The degree-0 differential square is exactly the supplied compatibility square $f \cdot \varphi_0 = e_Y \cdot f'$. For each degree $n+1$, the augmented differential is the original differential of the base complex (the augmentation only modifies the bottom slot), so the square reduces to the commutativity condition of φ at degree n. Hence all squares commute and the components assemble to an isomorphism. □

Lemma 377 (Open-meet identity: intersection with V distributes over the cover meet)**.** *Let $\mathcal{U}$ be an open cover of X, let $V \subseteq X$ be an open, and let $\sigma : \mathrm{Fin}(p+1) \to I$ be a multi-index. Then, as opens of X,*

$$\mathrm{coverInterOpen}\, \mathcal{U}\, \sigma \cap V \;=\; \bigcap_k (\mathrm{coverOpen}\, \mathcal{U}\, (\sigma k) \cap V).$$

Proof. By definition $\mathrm{coverInterOpen}\, \mathcal{U}\, \sigma = \bigcap_k \mathrm{coverOpen}\, \mathcal{U}\, (\sigma k)$, the $(p+1)$-fold meet of the cover-member opens. The index type $\mathrm{Fin}(p+1)$ is nonempty, and in the frame of opens a binary meet distributes over a *nonempty* indexed meet: $(\bigcap_k A_k) \cap V = \bigcap_k (A_k \cap V)$. (Nonemptiness is essential: over the empty index the left side is $\top \cap V = V$ while the right side is $\top$.) Taking $A_k = \mathrm{coverOpen}\, \mathcal{U}\, (\sigma k)$ gives the claim. □

Lemma 378 (Degreewise object iso of the non-augmented core comparison)**.** *Fix an open $V \subseteq X$ and a degree p. Write $\Psi = \mathrm{forget} \cdot \mathrm{restrictScalars}(\mathrm{id})$ for the push–pull adapter and $G_V = \mathrm{toPresheaf} \cdot \mathrm{ev}_V$ for the section-at-V functor. Then the degree-p terms of the evaluated non-augmented backbone complex and of the concrete restricted section Čech complex are isomorphic as abelian groups:*

$$((G_V \circ \Psi)\, \check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F})).X\, p \;\cong\; (\check{\mathcal{C}}^\bullet(\mathcal{U}', \mathcal{F})).X\, p,$$

i.e. $\Gamma(V, \mathrm{pushPullObj}\, \mathcal{F}\, Y_p) \cong \prod_{\sigma:\mathrm{Fin}(p+1)\to I} \Gamma(\bigcap_k (\mathrm{coverOpen}\, \mathcal{U}\, (\sigma k) \cap V),\, \mathcal{F})$, where $\mathcal{U}'$ is the restricted family $U'_i = \mathrm{coverOpen}\, \mathcal{U}\, i \cap V$.

Proof. Lemma 373 supplies the evaluated push–pull product iso $\Gamma(V, \mathrm{pushPullObj}\, \mathcal{F}\, Y_p) \cong \prod_\sigma \Gamma(\mathrm{coverInterOpen}\, \mathcal{U}\, \sigma \cap V, \mathcal{F})$. Post-compose with the product reindexing $\prod_\sigma \Gamma(\cdot, \mathcal{F})$ along the open-meet identity $\mathrm{coverInterOpen}\, \mathcal{U}\, \sigma \cap V = \bigcap_k (\mathrm{coverOpen}\, \mathcal{U}\, (\sigma k) \cap V)$ of Lemma 377 — factorwise the eqToIso on the section group $\Gamma(\cdot, \mathcal{F})$ obtained by applying $\Gamma(\cdot, \mathcal{F})$ to that equality of opens. The composite lands on the degree-p term $\prod_\sigma \Gamma(\bigcap_k (\mathrm{coverOpen}\, \mathcal{U}\, (\sigma k) \cap V), \mathcal{F})$ of

$\check{\mathcal{C}}^\bullet(\mathcal{U}', \mathcal{F})$, since for the restricted family $U'_{\sigma k} = \mathrm{coverOpen}\,\mathcal{U}\,(\sigma k) \cap V$ the section Čech term at σ is exactly $\Gamma(\bigcap_k U'_{\sigma k}, \mathcal{F})$. ☐

Lemma 379 (Per-leg coherence of the push–pull map of an open-immersion over-morphism)**.** *Let $q : A \to X$ be a morphism of schemes, let $c : C' \hookrightarrow A$ be an open immersion, and let $pC : C' \to X$ satisfy $c \cdot q = pC$. For a sheaf of $\mathcal{O}_X$-modules $\mathcal{F}$, the push–pull map of the over-X morphism* $\mathrm{Over.\,mk}\,c : \mathrm{Over.\,mk}\,pC \to \mathrm{Over.\,mk}\,q$ *factors, through a canonical leg isomorphism, as the pushforward of the restriction-adjunction unit:*

$$\mathrm{pushPullMap}\,\mathcal{F}\,(\mathrm{Over.\,mk}\,c) \;=\; q_*(\eta^c_{q^*\mathcal{F}}) \cdot (\mathrm{pushPullLegIso}\,q\,c\,pC\,\mathcal{F}).\mathrm{hom},$$

where η^c is the unit of the restriction adjunction along the open immersion c, evaluated at the pullback $q^*\mathcal{F}$, and pushPullLegIso *is the canonical leg isomorphism assembled from the restriction-versus-pullback comparison isomorphism* restrictFunctorIsoPullback *(Lemma 396), the pullback-composition comparison* pullbackComp *for the composite $c \cdot q$, and the reindexing* pullbackCongr *along the defining equality $c \cdot q = pC$.*

Proof. Unfold the raw description of the push–pull map of an open-immersion over-morphism. The key mate-calculus identity is that the restriction-adjunction unit η^c, post-composed with the pushforward of the comparison isomorphism restrictFunctorIsoPullback, coincides with the unit of the pullback–pushforward adjunction along c (Lemma 452); this is the general uniqueness of a left adjoint applied to the two presentations of pullback as restriction and as the genuine inverse image. Substituting the defining equality $c \cdot q = pC$ collapses the reindexing comparison pullbackCongr to the identity, and the surviving comparison isomorphisms compose to exactly the leg isomorphism pushPullLegIso. Hence the push–pull map equals the pushforward of the restriction unit followed by that leg isomorphism. ☐

Lemma 380 (Per-leg restriction naturality of the push–pull face)**.** *Fix the finite cover $\mathcal{U}$, a sheaf of $\mathcal{O}_X$-modules $\mathcal{F}$, an open $V \subseteq X$, a degree p, a multi-index $\sigma' : \mathrm{Fin}(p{+}2) \to I$, and a coface index k. Write $U_{\sigma'} = \mathrm{coverInterOpen}\,\mathcal{U}\,\sigma'$ and $j_{\sigma'} : U_{\sigma'} \hookrightarrow X$ for the intersection-open immersion. Let* $\mathrm{interLegHom}\,\mathcal{U}\,\sigma'\,k : \mathrm{Over.\,mk}\,j_{\sigma'} \to \mathrm{Over.\,mk}\,j_{\sigma' \circ d_k}$ *be the over-X morphism whose underlying map is the open immersion $c : U_{\sigma'} \hookrightarrow U_{\sigma' \circ d_k}$ of intersection opens (deleting the k-th index can only shrink the meet). Through the per-leg section identifications* pls = pushPull_leg_sections *(Lemma 372) at source and target, the section over V of the push–pull map of* interLegHom *equals the plain $\mathcal{F}$-restriction along $U_{\sigma'} \cap V \subseteq U_{\sigma' \circ d_k} \cap V$:*

$$G_V(\Psi(\mathrm{interLegHom}\,\mathcal{U}\,\sigma'\,k)) \cdot \mathrm{pls}(\sigma').\mathrm{hom} \;=\; \mathrm{pls}(\sigma' \circ d_k).\mathrm{hom} \cdot \mathcal{F}(\mathrm{homOfLE}\,(U_{\sigma'} \cap V \subseteq U_{\sigma' \circ d_k} \cap V)).$$

Proof. The argument has four steps.

(a) Per-leg coherence. By Lemma 379, the push–pull map of an over-morphism $\mathrm{Over.\,mk}\,c$ with c an open immersion is, through the canonical leg isomorphism, the pushforward of the restriction-adjunction unit η^c. Thus the only nontrivial content of $\mathrm{pushPullMap}\,\mathcal{F}\,(\mathrm{interLegHom})$ is the restriction unit along the geometric face c; the remaining factors are canonical comparison isomorphisms.

(b) Unit versus pullback comparison. The restriction-adjunction unit, composed with the pushforward of the comparison isomorphism restrictFunctorIsoPullback (the uniqueness isomorphism between the restriction adjunction and the genuine pullback–pushforward adjunction), is itself the unit of the pullback–pushforward adjunction. This replaces the restriction unit by the inverse-image unit without changing the map.

(c) Composition law along c and q. The unit of the composite restriction/pullback factors through the units of the two pieces via the comparison isomorphisms pullbackComp and the

analogous restriction-composition comparison: their naturality squares chain the inner comparison along the defining equality $c \cdot q = pC$. This identifies $\mathrm{pushPullMap}\, \mathcal{F}\, (\mathrm{interLegHom})$ postcomposed with the inverse of the target leg isomorphism with the pushforward of the restriction unit, transported by the reindexing isomorphisms at $c \cdot q = pC$.

(d) On sections over V. At the level of sections the restriction-adjunction unit acts as the plain presheaf restriction $\mathcal{F}(\mathrm{homOfLE} \cdots)$ — the value of $\mathcal{F}$ on the inclusion of opens induced by c. The leg isomorphism $\mathrm{pls}.\mathrm{hom}$ reads, on sections, as the section-over-V of the pushforward of the comparison isomorphism followed by a canonical equality isomorphism. Composing the contributions of (a)–(c), every comparison and transport factor is a morphism between section groups indexed by opens; since the lattice of opens is a thin category, all such reindexing morphisms are uniquely determined and compose to a single restriction. The whole composite therefore collapses to the single $\mathcal{F}$-restriction along $U_{\sigma'} \cap V \subseteq U_{\sigma' \circ d_k} \cap V$, which is exactly the claimed homOfLE map. ☐

Lemma 381 (Backbone inclusion projects to the canonical component map)**.** *Fix a degree p and a multi-index $\tau : \mathrm{Fin}(p+1) \to I$. The backbone inclusion* $\mathrm{backboneIncl}\, \mathcal{U}\, p\, \tau : \mathrm{Over.mk}\, j_\tau \to Y_p$ *of the τ-summand, followed by the l-th wide-pullback projection* $\mathrm{backboneProj}\, \mathcal{U}\, p\, l$*, equals the canonical component map* $\mathrm{interProj}\, \mathcal{U}\, \tau\, l$ *— the over-X inclusion of the intersection open U_τ into the (τl)-th cover member:*

$$\mathrm{backboneIncl}\, \mathcal{U}\, p\, \tau \cdot \mathrm{backboneProj}\, \mathcal{U}\, p\, l \;=\; \mathrm{interProj}\, \mathcal{U}\, \tau l.$$

Proof. The backbone inclusion is assembled, through the coproduct decomposition of the backbone (Lemma 350), from several canonical comparison isomorphisms: the distributivity isomorphisms reassociating the iterated fibre product, a canonical summand reindexing, and the projection of the wide-pullback comparison. Composing with the l-th projection and reassociating these layers reduces the left-hand side to a single lift into the (τl)-th cover member $\mathcal{U}.f(\tau l)$. Because that member arrow is an open immersion, hence a monomorphism, the lift is rigid: any two over-X maps agreeing after post-composition with it are equal. The canonical component map interProj is, by its defining factorization (coverInterToMember_fac), exactly such a lift, so the two composites coincide. ☐

Lemma 382 (Backbone inclusion intertwines the nerve coface with the geometric face)**.** *For a multi-index $\sigma' : \mathrm{Fin}(p+2) \to I$ and a coface index k, the backbone inclusion intertwines the k-th Čech-nerve coface $\delta_k^{\mathrm{nerve}}$ with the geometric face* interLegHom*:*

$$\mathrm{backboneIncl}\, \mathcal{U}\, (p+1)\, \sigma' \cdot \delta_k^{\mathrm{nerve}} \;=\; \mathrm{interLegHom}\, \mathcal{U}\, \sigma'\, k \cdot \mathrm{backboneIncl}\, \mathcal{U}\, p\, (\sigma' \circ d_k).$$

Proof. Both sides are over-X maps into the degree-p backbone term Y_p. By the wide-pullback extensionality principle backbone_hom_ext, it suffices to check equality after post-composition with each component projection $\mathrm{backboneProj}\, \mathcal{U}\, p\, l$. On the left, reassociating and applying nerve_backboneProj (the nerve coface followed by a projection is again the projection at the dropped index) and then Lemma 381 yields the canonical component map $\mathrm{interProj}\, \mathcal{U}\, \sigma'\, (d_k\, l)$. On the right, applying Lemma 381 and then interLegHom_interProj (the geometric face intertwines the canonical component maps) yields the same map. Hence the two composites agree after every projection, so they are equal. ☐

Lemma 383 (Coordinate formula for the degreewise object iso)**.** *With the notation of Lemma 378, fix a degree p, an open $V \subseteq X$, and a multi-index $\tau : \mathrm{Fin}(p+1) \to I$. The τ-projection of the degreewise object isomorphism* $(\mathrm{objIso}\, p).\mathrm{hom}$ *is the evaluated push–pull map of the backbone* *inclusion* backboneIncl $\mathcal{U}\, p\, \tau$*, followed by the per-leg section identification* $\mathrm{pls}(\tau)$ *(Lemma 372) and the open-meet reindex of Lemma 377:*

$$(\mathrm{objIso}\, p).\mathrm{hom}\cdot\pi_\tau \;=\; G_V(\mathrm{pushPullMap}\, \mathcal{F}\, (\mathrm{backboneIncl}\, \mathcal{U}\, p\, \tau))\cdot \mathrm{pls}(\tau).\mathrm{hom}\cdot(\textit{open-meet reindex on } \Gamma(\cdot, \mathcal{F})).$$

Proof. Unfold $(\mathrm{objIso}\, p)$.hom as in Lemma 378: it is the evaluated push–pull product isomorphism of Lemma 373, reindexed factorwise by the open-meet identity of Lemma 377. Projecting onto the τ-factor, the product comparison of the section functor composed with the τ-projection of the product isomorphism is, by the leg of the coproduct-to-product decomposition (Lemma 371), the push–pull map of the backbone inclusion of the τ-summand. Reading off the three surviving layers term by term: the product-decomposition leg supplies $G_V(\mathrm{pushPullMap}\, \mathcal{F}\, (\mathrm{backboneIncl}\, \mathcal{U}\, p\, \tau))$, the per-leg section identification supplies $\mathrm{pls}(\tau)$.hom, and the factorwise object-equality isomorphism supplies the open-meet reindex. Composing these gives the asserted coordinate formula. ☐

Lemma 384 (Per-leg naturality of the core comparison coface)**.** *With the notation of Lemma 378, fix a degree p, a coface index $k \le p{+}1$, and a multi-index $\sigma' : \mathrm{Fin}(p{+}2) \to I$. Read the degree-$p$ and degree-$(p+1)$ terms of both complexes through the product equivalence* sectionCechProductEquiv *(Lemma 188). Then the σ'-coordinate of the evaluated push–pull coface followed by the object iso,* $G_V(\Psi(\delta_k^{\mathrm{nerve}}))\cdot(\mathrm{objIso}\,(p{+}1))$.hom*, equals the presheaf face restriction* $\mathrm{sectionCechFaceRestr}(\sigma', k)$ *— the value of $\mathcal{F}$ on the inclusion of intersection opens*

$$\bigcap_j (\mathrm{coverOpen}\, \mathcal{U}\, (\sigma' j) \cap V) \;\subseteq\; \bigcap_l (\mathrm{coverOpen}\, \mathcal{U}\, ((\sigma' \circ d_k) l) \cap V)$$

— applied to the $(\sigma' \circ d_k)$-coordinate of the image of the source under $(\mathrm{objIso}\, p)$.hom*. In words: the projection of the evaluated push–pull coface onto the σ'-component is exactly the $\mathcal{F}$-restriction along the k-th face inclusion.*

Proof. Unwind $(\mathrm{objIso}\,(p+1))$.hom through its construction in Lemma 378: it is the evaluated push–pull product iso pushPull_eval_prod_iso (Lemma 373), reindexed factorwise by the open-meet identity (Lemma 377). The product iso factors through the coproduct-to-product decomposition of the backbone push–pull object (Lemma 371), the preservation of products by the section functor, and the per-leg section identification $\Gamma(V, \mathrm{pushPullObj}\, \mathcal{F}\, (\mathrm{Over.mk}\, j_{\sigma'})) \cong \Gamma(U_{\sigma'} \cap V, \mathcal{F})$ (Lemma 372). Projecting onto the σ'-leg, the evaluated Čech-nerve coface $\delta_k^{\mathrm{nerve}}$ acts on sections as the restriction of $\mathcal{F}$ along the inclusion of the σ'-intersection open into the $(\sigma' \circ d_k)$-intersection open — precisely the coface induced by deleting the k-th index. The open-meet identity (Lemma 377) rewrites both intersection opens as the meets $\bigcap_j (\mathrm{coverOpen}\, \mathcal{U}\, (\sigma' j) \cap V)$ and $\bigcap_l (\mathrm{coverOpen}\, \mathcal{U}\, ((\sigma' \circ d_k) l) \cap V)$, identifying this restriction with $\mathrm{sectionCechFaceRestr}(\sigma', k)$ on the nose. ☐

Lemma 385 (Per-coface square of the core comparison)**.** *With the notation of Lemma 378, for each degree p and each coface index $k \le p+1$ the individual cofaces are intertwined by the object isomorphisms:*

$$(\mathrm{objIso}\, p).\mathrm{hom}\cdot \delta_k^{\mathrm{sec}} \;=\; G_V(\Psi(\delta_k^{\mathrm{nerve}}))\cdot(\mathrm{objIso}\,(p+1)).\mathrm{hom},$$

where δ_k^{sec} is the k-th section-Čech coface and $\delta_k^{\mathrm{nerve}}$ the k-th Čech-nerve coface of the backbone, both maps into the degree-$(p+1)$ term.

Proof. Both sides are maps into the degree-$(p+1)$ term, which is the product over multi-indices $\sigma' : \mathrm{Fin}(p+2) \to I$; by the product equivalence sectionCechProductEquiv (Lemma 188) it suffices to check equality coordinate by coordinate. Fix σ'. The σ'-coordinate of the left-hand side is, by the per-coface coordinate computation underlying Lemma 189, the presheaf face restriction sectionCechFaceRestr(σ', k) applied to the $(\sigma' \circ d_k)$- coordinate of $(\mathrm{objIso}\, p).\mathrm{hom}$. The σ'-coordinate of the right-hand side is the same expression by Lemma 384. Hence all coordinates agree and the two cofaces are equal. ☐

Lemma 386 (Alternating-sum assembly of the core comparison square)**.** *With the notation of Lemma 378, for each degree p the full alternating-coface differentials are intertwined by the object isomorphisms:*

$$(\mathrm{objIso}\, p).\mathrm{hom} \cdot d^p_{\check{\mathcal{C}}(\mathcal{U}')} \;=\; d^p_{(G_V \circ \Psi)\check{\mathcal{C}}(\mathcal{U})} \cdot (\mathrm{objIso}\,(p+1)).\mathrm{hom}.$$

Proof. Both differentials are the alternating coface sums $d^p = \sum_{k=0}^{p+1} (-1)^k \delta_k$ (the objD of the alternating-coface-map complex). Work elementwise: by Lemma 189, read through the product equivalence (Lemma 188), each side — evaluated at an element of the source and at a coordinate σ' — is the pointwise finite sum $\sum_k (-1)^k (\cdots)_k$ of the same shape. The two sums agree summand by summand: the k-th summands carry the same sign $(-1)^k$ and, by the per-coface square of Lemma 385, the same coface contribution. A termwise comparison of the two alternating sums therefore yields equality of the differentials in every degree. ☐

Lemma 387 (The core comparison intertwines the Čech differentials)**.** *With the notation of Lemma 378, for each p the degreewise object isomorphisms* $\mathrm{objIso}\, p$ *intertwine the two differentials:*

$$(\mathrm{objIso}\, p).\mathrm{hom} \cdot d^p_{\check{\mathcal{C}}(\mathcal{U}')} \;=\; d^p_{(G_V \circ \Psi)\check{\mathcal{C}}(\mathcal{U})} \cdot (\mathrm{objIso}\,(p+1)).\mathrm{hom}.$$

Consequently $\mathrm{coreIso} := \mathrm{isoOfComponents}(\mathrm{objIso}, \cdot)$ *is an isomorphism of cochain complexes* $(G_V \circ \Psi)\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F}) \cong \check{\mathcal{C}}^\bullet(\mathcal{U}', \mathcal{F})$.

Proof. The intertwining square for each p is exactly the alternating-sum assembly of Lemma 386: both Čech differentials are the alternating coface sums $\sum_k (-1)^k \delta_k$, and that lemma matches them by summing the per-coface squares of Lemma 385 termwise. Holding for every p, the degreewise object isomorphisms $\mathrm{objIso}\, p$ together with these commuting squares assemble $\mathrm{coreIso} := \mathrm{isoOfComponents}(\mathrm{objIso}, \cdot)$ into an isomorphism of cochain complexes. ☐

Definition 388 (Augmentation of the section Čech complex over V)**.** Fix an open $V \subseteq X$. The augmentation of the concrete section Čech complex for the restricted family $U_i' = U_i \cap V$ is the product of the restriction maps

$$\varepsilon_V : \Gamma(V, \mathcal{F}) \longrightarrow \prod_{\sigma : \mathrm{Fin}\, 1 \to I} \Gamma\left(\bigcap_k (U_{\sigma(k)} \cap V), \mathcal{F} \right), \qquad t \longmapsto \left(t|_{\cap_k (U_{\sigma(k)} \cap V)} \right)_\sigma.$$

Lemma 389 (Coordinatewise value of the section augmentation)**.** *Fix an open $V \subseteq X$ and a one-element multi-index $\sigma : \mathrm{Fin}\, 1 \to I$. The σ-coordinate of the augmentation ε_V of Definition 388, namely its composite with the projection onto the σ-factor of $\prod_\tau \Gamma(\bigcap_l (U_{\tau l} \cap V), \mathcal{F})$, is the plain presheaf restriction of $\mathcal{F}$ along the inclusion of intersection opens $\bigcap_l (U_{\sigma l} \cap V) \subseteq V$:*

$$\mathrm{pr}_\sigma \circ \varepsilon_V \;=\; \mathcal{F}\big(\textstyle\bigcap_l (U_{\sigma l} \cap V) \subseteq V \big) \;:\; \Gamma(V, \mathcal{F}) \longrightarrow \Gamma\big(\textstyle\bigcap_l (U_{\sigma l} \cap V), \mathcal{F}\big).$$

Proof. This is the defining property of a map into a product: composing the product of the restrictions with its σ-projection gives the restriction indexed by σ. ☐

Lemma 390 (The section augmentation kills the first differential)**.** *For the augmentation ε_V of Definition 388,*

$$\varepsilon_V \cdot d^0 \;=\; 0.$$

Proof. It suffices to check every coordinate indexed by $\sigma : \mathrm{Fin}\,2 \to I$. The alternating differential has two terms. By Lemma 389, each is the composite restriction from V, first to one of $U_{\sigma(0)} \cap V$ and $U_{\sigma(1)} \cap V$, and then to their intersection. Transitivity of restriction identifies both composites with the restriction from V to $(U_{\sigma(0)} \cap V) \cap (U_{\sigma(1)} \cap V)$. They occur with opposite signs, so their sum is zero. □

Lemma 391 (Comparison of the evaluated and concrete section augmentations)**.** *Let* $\Psi = \mathrm{forget} \cdot \mathrm{restrictScalars(id)}$*, let* $G_V = \mathrm{toPresheaf} \cdot \mathrm{ev}_V$*, and let* $\mathrm{objIso}\,0$ *be the degree-zero isomorphism of Lemma 378. Then*

$$G_V(\Psi(\mathrm{cechAugmentation})) \cdot (\mathrm{objIso}\,0).\mathrm{hom} = \varepsilon_V.$$

Proof. Maps into the product are equal if all their coordinates are equal. Fix $\sigma : \mathrm{Fin}\,1 \to I$. The σ-projection of $\mathrm{objIso}\,0$ identifies the corresponding push–pull term with sections over $U_{\sigma(0)} \cap V$. The augmentation followed by this leg is the unit for the open immersion $U_{\sigma(0)} \hookrightarrow X$; under the section identification, that unit is restriction from V to $U_{\sigma(0)} \cap V$. This is the σ-coordinate of ε_V by Lemma 389. □

Lemma 392. *[The evaluated augmented Čech complex is the concrete section Čech complex] Fix an open $V \subseteq X$. The augmented Čech complex of Definition 333 evaluated at V (through* toPresheaf*) — call it $D^\bullet$, the cochain complex of abelian groups whose augmentation node is $\Gamma(V, \mathcal{F})$ and whose Čech degree-p term is $\Gamma(V, \mathrm{pushPullObj}\,\mathcal{F}\,Y_p)$ — is isomorphic, as a cochain complex, to the* augmented *concrete section Čech complex*

$$D'^{\bullet}_{\mathrm{aug}} \;:=\; (\check{\mathcal{C}}^\bullet(\mathcal{U}', \mathcal{F})).\,\mathrm{augment}\,\varepsilon\,h_\varepsilon$$

built from the concrete section Čech complex over the restricted family $U'_\sigma = \mathrm{coverInterOpen}\,\mathcal{U}\,\sigma \cap V$ (degree-p term $\prod_\sigma \Gamma(U_\sigma \cap V, \mathcal{F})$, alternating coface restriction differential) by prepending the augmentation node $\Gamma(V, \mathcal{F})$ along

- *the restriction product map $\varepsilon : \Gamma(V, \mathcal{F}) \to \prod_i \Gamma(U_i \cap V, \mathcal{F})$, $t \mapsto (t|_{U_i \cap V})_i$ of Definition 388; by Lemma 391, this is the evaluation at V of the Čech augmentation $\mathcal{F} \to \mathcal{C}^0$ of Definition 331; and*
- *the augmentation identity $h_\varepsilon : \varepsilon \cdot d^0 = 0$, the content of Lemma 390.*

Concretely: at the augmentation node both complexes carry $\Gamma(V, \mathcal{F})$, identified by the identity map; at Čech degree p the object isomorphism $\Gamma(V, \mathrm{pushPullObj}\,\mathcal{F}\,Y_p) \cong \prod_\sigma \Gamma(U_\sigma \cap V, \mathcal{F})$ is Lemma 373; and these degreewise maps intertwine the two differentials, including the augmentation differential ε.

Proof. The augmentation node of both complexes is $\Gamma(V, \mathcal{F})$, matched by the identity. At Čech degree p the object isomorphism between $\Gamma(V, \mathrm{pushPullObj}\,\mathcal{F}\,Y_p)$ and $\prod_\sigma \Gamma(U_\sigma \cap V, \mathcal{F})$ is Lemma 373. It remains to check the degreewise isomorphisms commute with the differentials.

Augmentation bookkeeping (peeling the node). Before matching differentials we strip the augmentation node off $D^\bullet$. Writing $\Psi = \mathrm{forget} \cdot \mathrm{restrictScalars(id)}$ for the push–pull adapter and $G_V = \mathrm{toPresheaf} \cdot \mathrm{ev}_V$ for the section functor, the evaluated complex is $D^\bullet = (G_V \circ \Psi)((\check{\mathcal{C}}^\bullet).\,\mathrm{augment}\,(\mathrm{cechAugmentatio}$
Applying Lemma 374 twice — once for Ψ and once for G_V, each an additive functor, with the

augmentation condition transported by Lemma 375 — commutes both functors past augment, so $D^\bullet \cong ((G_V \circ \Psi)\check{\mathcal{C}}^\bullet)$. augment $((G_V \circ \Psi)(\mathrm{cechAugmentation}))$ with identity components. Lemma 376 then glues an iso of augmented complexes from three pieces: the *non-augmented* core iso

$$\mathrm{coreIso} : (G_V \circ \Psi)(\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F})) \;\cong\; \check{\mathcal{C}}^\bullet(\mathcal{U}', \mathcal{F}),$$

the augmentation-object iso $\Gamma(V, \mathcal{F}) \cong \Gamma(V, \mathcal{F})$ (the identity e_Y), and the degree-0 compatibility square h_{compat}. This reduces the goal to the non-augmented coreIso, and the differential-match steps below apply to that reduced (non-augmented) goal — not to the full augmented $D^\bullet$, whose augmentation bookkeeping the three helper lemmas already discharge.

The non-augmented core iso coreIso*.* It is assembled componentwise from its degreewise object isos and a differential-match square. The degree-p object iso $\mathrm{objIso}\, p : \Gamma(V, \mathrm{pushPullObj}\, \mathcal{F}\, Y_p) \cong \prod_\sigma \Gamma(U_\sigma \cap V, \mathcal{F})$ is Lemma 378 (the evaluated push–pull product iso Lemma 373, reindexed by the open-meet identity Lemma 377), and the differential-match square is Lemma 387.

Čech differentials (degrees $p \geq 0$). This is the content of Lemma 387: the differential of the section complex is, read through the product equivalence sectionCechProductEquiv (Lemma 188), the alternating sum of presheaf face restrictions $\sum_i (-1)^i\, \mathrm{sectionCechFaceRestr}(\sigma, i)$ computed in Lemma 189. The differential of $D^\bullet$ is the evaluation at V of the Čech-nerve coface maps, assembled from the same push–pull restriction comparisons; under the degreewise identification each coface of $D^\bullet$ becomes the corresponding face restriction, so the alternating sums agree. Reusing the alternating-sum bookkeeping of Lemma 189 — rather than rebuilding it — matches these differentials.

Augmentation differential. The lowest differential of $D^\bullet$ is the evaluation at V of the Čech augmentation $\varepsilon : \mathcal{F} \to \mathcal{C}^0$ (Definition 331); read through the degree-0 instance of Lemma 373 it is the restriction product map $t \mapsto (t|_{U_i \cap V})_i$ by Lemma 391. This supplies the compatibility square h_{compat} for Lemma 376. The source augmentation is a complex differential by Lemma 332, while the concrete restriction product is one by Lemma 390. Hence the degreewise maps assemble to an isomorphism of cochain complexes $D^\bullet \cong D'^\bullet_{\mathrm{aug}}$. □

Lemma 393. *[The section Čech complex inside a cover member is contractible] Source: Stacks Project, Cohomology of Schemes,* `lemma-cech-cohomology-quasi-coherent-trivial` *(proof, the projection homotopy). Suppose $V \leq \mathrm{coverOpen}\, \mathcal{U}\, i_{\mathrm{fix}}$ for some fixed index i_{fix}, i.e. V lies inside one cover member. Then the* augmented *concrete section Čech complex $D'^\bullet_{\mathrm{aug}}$ of Lemma 392 — the section Čech complex over the family $U'_j = \mathrm{coverInterOpen}\, \mathcal{U}\, j \cap V$, augmented at $\Gamma(V, \mathcal{F})$ along the restriction product ε — admits a contracting homotopy $\mathrm{id}_{D'^\bullet_{\mathrm{aug}}} \simeq 0$.*

Proof. Because $V \leq U_{i_{\mathrm{fix}}}$, the restricted family $U'_j = U_j \cap V$ has a *maximum*: $U'_{i_{\mathrm{fix}}} = U_{i_{\mathrm{fix}}} \cap V = V$, and every $U'_j \leq V$. Consequently, for each multi-index σ, $U'_{i_{\mathrm{fix}}\sigma} = U_{i_{\mathrm{fix}}} \cap U_\sigma \cap V = U_\sigma \cap V = U'_\sigma$: prepending the index i_{fix} does not change the open, so the prepend map is the *identity* on each coefficient $\Gamma(U'_\sigma, \mathcal{F})$ (the membership criterion le_coverInterOpen_iff of Lemma 202). The contracting homotopy is the prepend-i_{fix} map, $h(s)_\sigma = s_{i_{\mathrm{fix}}\sigma}$, exactly the projection homotopy of the Stacks proof; it splits into two parts.

Positive Čech degrees ($p \geq 1$). On the section Čech part the dependent-coefficient combinatorial Čech engine — depDiff, depHomotopy, depHomotopy_spec, depDiff_exact of Lemma 194 — supplies the prepend-i_{fix} homotopy h with $d \circ h + h \circ d = \mathrm{id}$, since its unit, shift, and coface-commutation compatibilities all hold with the prepend map acting as the identity on each coefficient.

Augmentation node (the piece the engine does not supply). The engine governs only the Čech degrees; the contracting-homotopy identity at the augmentation node must be checked directly. The relevant homotopy component runs from the lowest Čech term $\prod_i \Gamma(U_i \cap V, \mathcal{F})$ down to the augmentation term $\Gamma(V, \mathcal{F})$, and is the i_{fix}-th coordinate projection $\pi_{i_{\text{fix}}} : (s_j)_j \mapsto s_{i_{\text{fix}}}$ — well defined because $U'_{i_{\text{fix}}} = V$, so $s_{i_{\text{fix}}} \in \Gamma(U_{i_{\text{fix}}} \cap V, \mathcal{F}) = \Gamma(V, \mathcal{F})$. We verify $\varepsilon \circ \pi_{i_{\text{fix}}} +$ (degree-0 engine term) $= \mathrm{id}$ on $s \in \prod_i \Gamma(U_i \cap V, \mathcal{F})$. The augmentation differential ε is the restriction product, so $\left(\varepsilon(\pi_{i_{\text{fix}}} s)\right)_j = s_{i_{\text{fix}}}|_{U_j \cap V}$; the degree-0 contribution of the Čech homotopy supplies the complementary term $s_j - s_{i_{\text{fix}}}|_{U_j \cap V}$ coming from the prepend-i_{fix} shift on the Čech part. Their sum is s_j, i.e. the identity. (This is the bottom node of the Stacks projection homotopy: h on $\prod_{i_0} M_{f_{i_0}} \to M$ is projection onto the factor $M_{f_{i_{\text{fix}}}} = M$.) Together the two parts give a homotopy $\mathrm{id}_{D'^{\bullet}_{\text{aug}}} \simeq 0$ of cochain complexes. ☐

Lemma 394 (Local vanishing of the evaluated augmented Čech homology)**.** *Let* $V \le \operatorname{coverOpen} \mathcal{U}\, i$ *for some index* i*. Then for every degree* p *the homology* $H^p(D^\bullet)$ *of the augmented Čech complex evaluated at* V *(the complex* $D^\bullet$ *of Lemma 392) is a zero object; equivalently* $D^\bullet$ *carries a contracting homotopy* $\mathrm{id}_{D^\bullet} \simeq 0$*. This is the residual obligation of Step 3 of Lemma 337.*

Proof. By Lemma 392 the evaluated complex $D^\bullet$ is isomorphic to the augmented concrete section Čech complex $D'^{\bullet}_{\text{aug}}$. The hypothesis $V \le \operatorname{coverOpen} \mathcal{U}\, i$ places V inside one cover member, so Lemma 393 gives a contracting homotopy $\mathrm{id}_{D'^{\bullet}_{\text{aug}}} \simeq 0$; transporting it across the complex isomorphism yields $\mathrm{id}_{D^\bullet} \simeq 0$. Lemma 338 then makes $H^p(D^\bullet)$ a zero object in every degree. ☐

Lemma 395 (Pushforward sections are sections over the preimage)**.** *Provided by Mathlib. For a morphism* $f : X \to Y$*, an* $\mathcal{O}_X$*-module* $\mathcal{M}$*, and an open* $U \subseteq Y$*, the sections of the pushforward* $f_*\mathcal{M}$ *over* U *coincide definitionally with the sections of* $\mathcal{M}$ *over the preimage:* $\Gamma(U, f_*\mathcal{M}) = \Gamma(f^{-1}(U), \mathcal{M})$*.*

Lemma 396 (Pullback along an open immersion is the restriction functor)**.** *Provided by Mathlib. For an open immersion* $f : X \to Y$*, the restriction functor* $\operatorname{restrictFunctor} f : Y.\text{Modules} \to X.\text{Modules}$ *is naturally isomorphic to the module pullback functor* $f^* = \operatorname{pullback} f$*. In particular* $f^*\mathcal{M} \cong \mathcal{M}.\operatorname{restrict} f$ *for every* $\mathcal{O}_Y$*-module* $\mathcal{M}$*.*

Lemma 397 (Evaluation of a sheaf of modules preserves products)**.** *Provided by Mathlib. For a sheaf of rings on a site and any object* U*, the evaluation (sections-at-*U*) functor* $\operatorname{evaluation} U : \text{SheafOfModules} \to \text{Ab}$ *preserves limits of every shape; in particular it preserves products, so* $\Gamma(U, \prod_\sigma N_\sigma) \cong \prod_\sigma \Gamma(U, N_\sigma)$*.*

Lemma 398 (Sections over a binary coproduct scheme are a product)**.** *Provided by Mathlib. For schemes* X, Y *and an open* U *of the coproduct* $X \amalg Y$*, the structure-sheaf sections over* U *decompose as the product of the sections over the two preimages:* $(X \amalg Y).\mathcal{O}(U) \cong X.\mathcal{O}(\operatorname{inl}^{-1} U) \times Y.\mathcal{O}(\operatorname{inr}^{-1} U)$*. This is the binary disjoint-union sheaf decomposition that Lemma 371 ports to the indexed,* SheafOfModules *setting.*

Lemma 399 (Sections of a sheaf over a disjoint union are a product)**.** *Provided by Mathlib. For a sheaf* F *(valued in a category with products) on a topological space and two opens* U, V *with* $U \sqcap V = \bot$ *(disjoint), the sections over the union are the product of the sections:* $F(U \sqcup V) \cong F(U) \times F(V)$*. Iterated over a finite family, this is the disjoint-union decomposition underlying Lemma 371.*

Lemma 400 (The category of schemes is finitary extensive)**.** *Provided by Mathlib. The category* Scheme *is finitary extensive: finite coproducts are disjoint and universal (stable under pullback). In particular it is finitary pre-extensive and has finite coproducts and pullbacks, so the abstract distributivity of Lemma 401 applies at* $\mathcal{C} =$ Scheme*.*

Lemma 401 (Coproducts distribute over the binary fibre product)**.** *Provided by Mathlib. In a finitary pre-extensive category with pullbacks, for finite index types ι, ι', a base S, and families $(X_i \to S)_i$, $(Y_j \to S)_j$, the canonical map $\coprod_{(i,j)}(X_i \times_S Y_j) \to (\coprod_i X_i) \times_S (\coprod_j Y_j)$,* Sigma. desc *of the pullback maps, is an isomorphism. Taking one family to be a singleton gives the one-sided distributivity $(\coprod_i X_i) \times_S Y \cong \coprod_i (X_i \times_S Y)$ used in the induction of Lemma 348.*

Lemma 402 (The coproduct-of-pullback-fst comparison is an isomorphism)**.** *Provided by Mathlib. In a finitary pre-extensive category, for a finite index type α in* Type 0*, a base S, a family $(X_a \to S)_{a\in\alpha}$ and a single $Y \to S$, the comparison map* Sigma. desc *of the pullback first-projections $\coprod_a (X_a \times_S Y) \to (\coprod_a X_a) \times_S Y$ is an isomorphism. This is the primitive on which the binary distributivity Lemma 401 and the one-sided form Lemma 343 rest.* It is stated in Mathlib only for $\alpha : \mathrm{Type}\,0$*, which is why the whole distributivity chain (Lemma 348) is confined to* Type 0 *and the cover index is reduced to* $\mathrm{Fin}(n)$ *at Lemma 350.*

Lemma 403 (Pullback of two open immersions is the intersection open)**.** *Provided by Mathlib. For two opens U, V of a scheme X, the square with vertex* $(U \sqcap V)$.toScheme *and the two* homOfLE *inclusions into U and V is a pullback of the open immersions $U.\iota, V.\iota$. Equivalently* pullback $U.\iota\, V.\iota \cong (U \sqcap V)$.toScheme*: the fibre product of open immersions over X is the intersection of their images.*

Lemma 404 (A wide pullback over a terminal base is the product of its legs)**.** *Provided by Mathlib. When the base of a wide pullback is a terminal object, a limiting fan of the legs is a limit of the wide-pullback diagram. Hence in* Over S *— where* Over. mk(id_S) *is terminal — the wide pullback over S of a family is the product of the legs in the slice, which is what lets the fibre power over S be treated as an iterated product in Lemma 348.*

Lemma 405 (The identity arrow is terminal in the slice category)**.** *Provided by Mathlib. For an object S of a category,* Over. mk(id_S) *is a terminal object of the slice category* Over S*. This is what makes the base of a wide pullback over S terminal in the slice, so that Lemma 404 exhibits the fibre power over S as the product in* Over S *in Lemma 341.*

Lemma 406 (Nested coproducts flatten to a coproduct over pairs)**.** *Provided by Mathlib. For a doubly-indexed family $(Z_{i,j})$, the nested coproduct is the coproduct over pairs: $\coprod_i \coprod_j Z_{i,j} \cong \coprod_{(i,j)} Z_{i,j}$. This collapses the nested coproduct produced by one inductive step of the binary distributivity into a single coproduct over the multi-indices σ in Lemma 348.*

Lemma 407 (Cohomology sheaf is the sheafification of the objectwise homology)**.** *Let K be a cochain complex of presheaves of modules over a site admitting sheafification. For every index i there is a natural isomorphism between the i-th homology of the sheafified complex and the sheafification of the i-th objectwise (presheaf-level) homology of K,*

$$H^i(\mathrm{sheafify}(K)) \;\cong\; \mathrm{sheafify}(H^i(K)).$$

Equivalently, sheafification commutes with the formation of homology. This is the reusable engine underlying the presheaf description of the higher direct images (Stacks Tag 01XJ): it is what turns the resolution-internal cohomology sheaf into a sheafification of an explicit cohomology presheaf.

Proof. Sheafification is exact: as the left adjoint of the inclusion of sheaves into presheaves it preserves all colimits, and on a site admitting sheafification it additionally preserves finite limits, so it preserves both the kernels and the cokernels through which homology is computed. An exact additive functor commutes with the homology functor of a cochain complex; chasing this through the counit complex isomorphism $\mathrm{sheafify}(\iota K) \cong K$ (the inclusion–sheafification adjunction counit, an isomorphism on sheaves) gives $H^i(K) \cong H^i(\mathrm{sheafify}\, K) \cong \mathrm{sheafify}(H^i(K))$. The supporting bricks are the additivity of sheafification, the counit complex isomorphism, and the complex-level transport of the short-complex map–homology isomorphism along an additive homology-preserving functor. □

Lemma 408 (Presheaf description of the higher direct images)**.** *Source: Stacks Project, Cohomology, Tag 01XJ (`lemma-describe-higher-direct-images`). Let $f : X \to S$ be a morphism of schemes and $\mathcal{G}$ an $\mathcal{O}_X$-module, and assume the category of $\mathcal{O}_X$-modules has enough injective objects, so that the derived pushforward $R^k f_*$ is defined. Choose an injective resolution $\mathcal{G} \to \mathcal{I}^\bullet$ in $\mathcal{O}_X$-modules and push it forward degreewise to obtain the cochain complex of presheaves of modules $f_*\mathcal{I}^\bullet$ on S. For every $k \geq 0$ the higher direct image $R^k f_*\mathcal{G}$ is isomorphic to the* sheafification of the presheaf of objectwise homology *of this pushed complex,*

$$R^k f_*\mathcal{G} \;\cong\; \mathrm{sheafify}\Big(V \longmapsto H^k((f_*\mathcal{I}^\bullet)(V)) \Big), \qquad V \subseteq S \text{ open},$$

the sheafification of the k-th presheaf-level homology of $f_*\mathcal{I}^\bullet$. As a closing remark, this gives the affine-local vanishing criterion consumed downstream (Lemmas 414 and 467): since the sheafification of a presheaf vanishes exactly when that presheaf vanishes on a basis of opens, $R^k f_*\mathcal{G} = 0$ iff $H^k((f_*\mathcal{I}^\bullet)(V)) = 0$ for every V in a basis of the topology of S; the affine opens form such a basis, so it suffices to check the vanishing on affine V.

The identification of the displayed presheaf homology with the absolute cohomology of the preimage,

$$H^k((f_*\mathcal{I}^\bullet)(V)) \;=\; H^k(f^{-1}(V), \mathcal{G}|_{f^{-1}(V)}),$$

which uses that $\mathcal{I}^\bullet|_{f^{-1}(V)}$ is again injective over $f^{-1}(V)$, is supplied at point of use; this lemma records the resolution-internal sheafify-of-objectwise-homology form.

Proof. Choose an injective resolution $\mathcal{G} \to \mathcal{I}^\bullet$ in $\mathcal{O}_X$-modules. By definition $R^k f_*\mathcal{G}$ is the k-th cohomology sheaf of the complex $f_*\mathcal{I}^\bullet$ of presheaves of modules over S. The comparison engine of Lemma 407 now applies: taking homology in $\mathcal{O}_S$-modules is the sheafification of the objectwise (presheaf-level) homology, because the sheafification/inclusion adjunction is exact (it preserves the finite limits and colimits computing kernels and cokernels). Hence

$$R^k f_*\mathcal{G} \;\cong\; \mathrm{sheafify}\Big(V \longmapsto H^k((f_*\mathcal{I}^\bullet)(V)) \Big),$$

which is the stated isomorphism. Since the sheafification of a presheaf vanishes exactly when the underlying presheaf vanishes on a basis of opens, this yields the affine-local vanishing criterion. □

Lemma 409 (Ext as the cohomology of a Hom complex into an injective resolution)**.** *Provided by Mathlib. Let $\mathcal{I}^\bullet$ be an injective resolution of Y in an abelian category with enough injectives. Then there is a natural identification, for every n,*

$$\mathrm{Ext}^n(X, Y) \;\cong\; H^n(\mathrm{Hom}(X[0], \mathcal{I}^\bullet)),$$

the composite of the Ext-to-cohomology-class equivalence extAddEquivCohomologyClass *with the cohomology-class-to-homology equivalence* homologyAddEquiv *of the Hom complex; the degreewise comparison* cochainComplexXIso *matches the $\mathbb{N}$- and $\mathbb{Z}$-indexed forms of the resolution. The companion right-derived-object comparison $(F.\mathrm{rightDerived}\, n)(X) \cong H^n(F \cdot \mathcal{I}^\bullet)$ for an additive functor F is recorded in Lemma 151.*

Lemma 410 (Ext-vanishing forces the coyoneda right-derived object to vanish)**.** *Let $\mathcal{C}$ be an abelian category with enough injectives, let $P, H \in \mathcal{C}$, and let $q \geq 1$. If every class $e \in \operatorname{Ext}^q(P, H)$ is zero, then the right-derived object of the covariant hom-functor vanishes at H:*

$$((\operatorname{Hom}(P, -)).\text{rightDerived}\, q)(H) \quad \textit{is a zero object.}$$

Proof. Choose an injective resolution $H \to \mathcal{I}^\bullet$. By the right-derived-object comparison (Lemma 151) the value $((\operatorname{Hom}(P, -)).\text{rightDerived}\, q)(H)$ is identified with the degree-q homology of the Hom-cochain complex $n \mapsto (P \to \mathcal{I}^n)$, whose differential is post-composition with the differential of $\mathcal{I}^\bullet$. A degree-q cocycle is a map $f : P \to \mathcal{I}^q$ with $f \circ d = 0$, and by the Ext–Hom-complex identification (Lemma 409) such an f is a coboundary exactly when its class $\text{extMk}\, f \in \operatorname{Ext}^q(P, H)$ vanishes. The hypothesis makes every such class zero, so every q-cocycle is a coboundary; the homology at q is therefore zero and the right-derived object is a zero object. □

Lemma 411 (An isomorphism in the first Ext-argument transfers subsingletons)**.** *Let $\mathcal{C}$ be an abelian category with an Ext-theory, $\varphi : A \cong B$ an isomorphism, $Y \in \mathcal{C}$ an object and q a degree. If $\operatorname{Ext}^q(B, Y)$ is a subsingleton (has at most one element), then so is $\operatorname{Ext}^q(A, Y)$.*

Proof. Ext is contravariant in its first argument. For any $z \in \operatorname{Ext}^q(A, Y)$, associativity of Yoneda composition with the degree-0 classes of φ gives

$$z = \text{mk}_0(\varphi.\text{hom}) \circ (\text{mk}_0(\varphi.\text{inv}) \circ z),$$

using $\varphi.\text{inv} \circ \varphi.\text{hom} = \text{id}$ and the identity law $\text{mk}_0(\text{id}) \circ z = z$. The inner factor $\text{mk}_0(\varphi.\text{inv}) \circ z$ lives in the subsingleton $\operatorname{Ext}^q(B, Y)$, hence equals 0; composing with 0 gives $z = 0$. Since every element is 0, $\operatorname{Ext}^q(A, Y)$ is a subsingleton. □

Lemma 412 (Injective resolutions provide enough injectives)**.** *An abelian category equipped with chosen injective resolutions has enough injectives: every object admits a monomorphism into an injective object.*

Proof. Given an object X, take its chosen injective resolution $X \to \mathcal{I}^\bullet$. The degree-0 term $\mathcal{I}^0$ is injective and the resolution's unit $X \to \mathcal{I}^0$ is a monomorphism, so it is an injective presentation of X. As every object has one, the category has enough injectives. □

Lemma 413 (Affine open immersions have vanishing higher direct images)**.** *Source: Stacks Project, Cohomology of Schemes, `lemma-relative-affine-vanishing`. Let X be separated and $j : U \hookrightarrow X$ the open immersion of an affine open U. Then j is an affine morphism (an open immersion of an affine open into a separated scheme), and its higher direct images vanish:*

$$R^q j_* \mathcal{H} = 0 \qquad \textit{for all } q \geq 1$$

and every quasi-coherent $\mathcal{O}_U$-module $\mathcal{H}$.

Proof. An open immersion of an affine open U into a separated scheme X is an affine morphism: separatedness makes the immersion j quasi-separated and the preimage of any affine open of X meets U in an affine, so j is affine. The relative affine vanishing for affine morphisms then gives $R^q j_* \mathcal{H} = 0$ for $q \geq 1$; concretely, by the presheaf description (Lemma 408) $R^q j_* \mathcal{H}$ is the sheafification of $V \mapsto H^q(j^{-1}(V), \mathcal{H})$, and for affine V the preimage $j^{-1}(V)$ is affine, on which the Serre vanishing of Lemma 228 kills H^q for $q \geq 1$. Since the affine opens form a basis of X, the presheaf $V \mapsto H^q(j^{-1}(V), \mathcal{H})$ is locally zero, so its sheafification $R^q j_* \mathcal{H}$ vanishes for $q \geq 1$.

Proof details. The argument rests on three identifications.

(1) The cohomology-presheaf identification. The objectwise homology in degree n of the pushed-forward resolution complex $j_*\mathcal{J}^\bullet$ (for a chosen injective resolution $\mathcal{H} \to \mathcal{J}^\bullet$ of the coefficient), evaluated at an open V, is identified with the absolute cohomology presheaf $V \mapsto H^n(j^{-1}(V), \mathcal{H}) = \mathrm{Ext}^n(j_!\mathcal{O}_{j^{-1}(V)}, \mathcal{H})$. Concretely: $\mathrm{Ext}^n(X, Y)$ over an injective resolution of the second argument Y is identified with the n-th cohomology of the Hom cochain complex $H^n(\mathrm{Hom}(X[0], \mathcal{J}^\bullet))$, via the composite of the Ext-to-cohomology-class equivalence and the cohomology-class-to-homology equivalence of the Hom complex (the equivalences of Lemma 409). Taking $X = j_!\mathcal{O}_{j^{-1}(V)}$ and $Y = \mathcal{H}$, the degree-k Hom group $\mathrm{Hom}(j_!\mathcal{O}_{j^{-1}(V)}, \mathcal{J}^k)$ is identified with $\Gamma(j^{-1}(V), \mathcal{J}^k) = (j_*\mathcal{J}^k)(V)$ by the corepresentability $j_!\mathcal{O}_{j^{-1}(V)} \dashv \Gamma(j^{-1}(V), -)$ of the absolute-cohomology setup, degreewise and naturally; the $\mathbb{N}/\mathbb{Z}$ indexing of the two cochain complexes is matched by the resolution's degreewise comparison isomorphism. The right-derived side is settled by the already-used identification of $R^n j_*$ with the homology of $j_*\mathcal{J}^\bullet$ (Lemma 408); no separate "right-derived equals Ext" comparison lemma is needed, since both compute the same homology over the same injective resolution and meet at corepresentability.

(2) Serre vanishing for a general affine open. The Serre vanishing of Lemma 228 is pinned to the *top* open $\top$ of a spectrum $\mathrm{Spec}\, R$: it kills $\mathrm{Ext}^q(j_!\mathcal{O}_\top, -)$ for $q \geq 1$. The cohomology that actually occurs here is over a *general* affine open — the preimage $j^{-1}(V)$ of an affine open V of X that may straddle the boundary of $j(U)$, so $j^{-1}(V)$ is a general, possibly non-distinguished, affine open of U. The residual is therefore $\mathrm{Ext}^q(j_!\mathcal{O}_{j^{-1}(V)}, \mathcal{H}) = 0$ over the abstract affine scheme U, for $q \geq 1$. We discharge it by splitting it into two independent transports.

(2a) The residual lives over an affine scheme. The immersion j is an affine morphism: for U affine and X separated, $j : U \hookrightarrow X$ is affine, so the preimage of every affine open of X is an affine open of U. In particular $j^{-1}(V) = U \cap V$ is an affine open of the affine scheme U. After the reduction to the right-derived sections functor — by the presheaf description (Lemma 408) and the corepresentability isomorphism of Lemma 418, upgraded to right derived functors by Lemma 419 — the residual reads $\mathrm{Ext}^q_U(j_!\mathcal{O}_{j^{-1}(V)}, \mathcal{H}) = 0$, an Ext vanishing in the module category of the abstract affine scheme U, for $q \geq 1$.

(2b) Transport along the whole-scheme spectrum isomorphism (Need #1). The abstract affine scheme U carries the canonical *whole-scheme* isomorphism $U \cong \mathrm{Spec}\, \Gamma(U, \mathcal{O}_U)$ (Scheme.isoSpec, Lemma 420), a genuine isomorphism of schemes because U is affine. By Lemma 463 it induces an equivalence of module categories U.Modules $\simeq$ ($\mathrm{Spec}\, \Gamma U$).Modules and transports Ext along it (via Ext.mapExactFunctor), carrying the residual to $\mathrm{Ext}^q_{\mathrm{Spec}\, \Gamma U}(j_!\mathcal{O}_{V'}, \mathcal{H}') = 0$, where V' is the image affine open of $j^{-1}(V)$ and $\mathcal{H}'$ the transported (still quasi-coherent) coefficient. Under the equivalence V' is in general a non-distinguished affine open of $\mathrm{Spec}\, \Gamma U$.

(2c) General-affine-open Serre vanishing (Need #2). Over $\mathrm{Spec}\, \Gamma U$ the open V' is a general affine open, and Lemma 434 gives $\mathrm{Ext}^q(j_!\mathcal{O}_{V'}, \mathcal{H}') = 0$ for $q \geq 1$ and every affine open V' and quasi-coherent $\mathcal{H}'$. This is read off from the basis-comparison criterion (Lemma 328) with the cover system's basis $\mathcal{B}$ enlarged from the distinguished opens to *all* affine opens; it requires no fresh cover system on the open V' and stays entirely inside ($\mathrm{Spec}\, \Gamma U$).Modules. Combining (2b) and (2c) kills the residual.

Remark (a blocked approach). One might try to make $j^{-1}(V)$ the *whole* space by transporting along the spectrum isomorphism of the *open subscheme* $j^{-1}(V) \cong \mathrm{Spec}\, \Gamma(j^{-1}(V), \mathcal{O})$. This is a dead end: the inclusion $j^{-1}(V) \hookrightarrow U$ is an open immersion, not an isomorphism, so there is no equivalence U.Modules $\simeq$ ($j^{-1}(V)$).Modules to transport Ext along; using the open-subscheme isomorphism forces a restriction functor $\mathcal{H} \mapsto \mathcal{H}|_{j^{-1}(V)}$, whose derived comparison $\mathrm{Ext}^q_U(j_!\mathcal{O}_{j^{-1}(V)}, \mathcal{H}) \cong H^q(j^{-1}(V), \mathcal{H}|_{j^{-1}(V)})$ is exactly "restriction preserves injectives" — infrastructure not available in the present formalism. The sound route therefore keeps the spectrum isomorphism at the level of the *whole* affine U as in (2b) and handles the general affine open ambiently via (2c).

(3) Locally zero implies zero sheafification. The passage from "$H^q(\cdots) = 0$ on a basis of affine opens" to "$R^q j_* = 0$" is the module-sheafification analogue of Lemma 334 (and of its packaged form Lemma 336). It is obtained by transporting the abelian-sheaf statement across the natural isomorphism $\mathrm{toSheaf} \circ \mathrm{sheafify} \cong \mathrm{presheafToSheaf} \circ \mathrm{forget}$ between module-sheafification followed by the forgetful functor and abelian sheafification; the "locally zero" hypothesis it consumes is exactly the basis-of-affines vanishing supplied by identifications (1) and (2) above. □

Lemma 414 (Pushing forward an affine open immersion degenerates the composite direct image)**.** *Let X be separated, $j : U \hookrightarrow X$ the open immersion of an affine open U, and $f : X \to S$ any morphism, with composite $g := f \circ j : U \to S$. Then for every quasi-coherent $\mathcal{O}_U$-module $\mathcal{H}$ and every $k \geq 0$,*

$$R^k f_*(j_*\mathcal{H}) \;\cong\; R^k g_*\mathcal{H}.$$

Proof. Choose an injective resolution $\mathcal{H} \to \mathcal{I}^\bullet$ in $\mathcal{O}_U$-modules and apply the pushforward j_* degreewise, forming the complex $K := j_*\mathcal{I}^\bullet$ of $\mathcal{O}_X$-modules. We verify the hypotheses of the acyclic-resolution comparison (Lemma 171) for the functor $G = f_*$ and the candidate f_*-acyclic resolution K of $j_*\mathcal{H}$, then transport along the pushforward-composition isomorphism. The argument is the formalization-friendly categorical route; it establishes the f_*-acyclicity of each $j_*\mathcal{I}^n$ by an adjoint-preserves-injectives argument and is an equivalent, cleaner alternative to the presheaf-description argument of the Stacks Project (which would compute cohomology over the opens $U \cap f^{-1}(V)$; these are general opens of the affine U and need not be affine, since $f^{-1}(V)$ is open but not affine in X, so that route re-enters the restriction-of-injectives obstruction we deliberately avoid).

(b) Each $j_*\mathcal{I}^n$ is f_*-acyclic. The pushforward $j_* = \mathrm{pushforward}\, j$ preserves injective objects. Indeed, it is the right adjoint in the adjunction $\mathrm{pullback}\, j \dashv \mathrm{pushforward}\, j$ (Lemma 453), and its left adjoint $\mathrm{pullback}\, j$ preserves monomorphisms: for an open immersion j the pullback functor is isomorphic to the restriction functor, $\mathrm{restrictFunctor}\, j \cong \mathrm{pullback}\, j$ (Lemma 396), and restriction of a sheaf of modules to an open subscheme is exact, hence preserves monos. By the criterion that the right adjoint of a mono-preserving left adjoint preserves injectives (Lemma 209), each $j_*\mathcal{I}^n$ is injective in X.Modules. An injective object is right-G-acyclic for *any* additive functor G (Lemma 152, in the sense of Definition 155), so $j_*\mathcal{I}^n$ is f_*-acyclic: $R^k f_*(j_*\mathcal{I}^n) = 0$ for all $k \geq 1$. No Serre vanishing on $U \cap f^{-1}(V)$ is needed, and the restriction-of-injectives wall is avoided.

(a) K is exact in positive degrees. The objectwise homology of the pushed-forward resolution $K = j_*\mathcal{I}^\bullet$ in degree k computes the right-derived functor of j_* along the injective resolution $\mathcal{I}^\bullet$: by Lemma 151,

$$H^k(j_*\mathcal{I}^\bullet) \;\cong\; (R^k j_*)(\mathcal{H}) \;=\; R^k j_*\mathcal{H}.$$

By Lemma 413 (j affine, $\mathcal{H}$ quasi-coherent on the affine U) this vanishes for $k \geq 1$; hence K is exact at every positive degree $n + 1$.

Augmentation: $j_*\mathcal{H} \cong K.\mathrm{cycles}\, 0$. The pushforward j_* is left exact, being a right adjoint (Lemma 453) and so finite-limit-preserving. For a left-exact functor the zeroth right-derived functor is the functor itself, $R^0 j_* \cong j_*$ (Lemma 153), and the degree-zero cycles of the applied resolution compute $R^0 j_*$; thus j_* carries the resolution augmentation $\mathcal{H} \cong (\mathcal{I}^\bullet).\mathrm{cycles}\, 0$ to

$$j_*\mathcal{H} \;\cong\; H^0(j_*\mathcal{I}^\bullet) \;=\; K.\mathrm{cycles}\, 0,$$

the augmentation that makes K a resolution of $j_*\mathcal{H}$.

By (a), (b) and the augmentation, $K = j_*\mathcal{I}^\bullet$ is an f_*-acyclic resolution of $j_*\mathcal{H}$. The acyclic-resolution comparison (Lemma 171) applied with $G = f_*$ gives, for every k, an isomorphism

$$R^k f_*(j_*\mathcal{H}) \;\cong\; H^k(f_*(K)) \;=\; H^k(f_*(j_*\mathcal{I}^\bullet)).$$

Transport: $H^k(f_*(K)) \cong R^k g_* \mathcal{H}$. The pushforward-composition isomorphism pushforwardComp $j\, f$: pushforward $j \circ$ pushforward $f \cong$ pushforward$(j \circ f)$ (Lemma 454) identifies $f_*(j_* \mathcal{I}^\bullet)$ with $(f \circ j)_* \mathcal{I}^\bullet = g_* \mathcal{I}^\bullet$; applying it degreewise and taking homology, while recognising $H^k(g_* \mathcal{I}^\bullet) \cong R^k g_* \mathcal{H}$ via the same injective-resolution identification (Lemma 151), yields

$$H^k(f_*(j_* \mathcal{I}^\bullet)) \;\cong\; R^k g_* \mathcal{H}.$$

Composing the acyclic-resolution comparison with this transport gives the stated isomorphism $R^k f_*(j_* \mathcal{H}) \cong R^k g_* \mathcal{H}$. □

Lemma 415 (The pushforward-sections functor and its additivity)**.** *Let $j : U \to X$ be a morphism and $W \subseteq X$ an open. The* pushforward-sections functor pushforwardSectionsFunctor $j\, W$: U.Modules $\to$ Ab *sends an $\mathcal{O}_U$-module M to its sections $\Gamma(W, j_* M) = \Gamma(j^{-1}(W), M)$ — the pushforward along j, forgotten to abelian groups, viewed as a presheaf and evaluated at W. It is an additive functor, and its q-th right derived functor applied to H computes the absolute cohomology $H^q(j^{-1}(W), H)$.*

Proof. The functor is the composite of pushforward pushforward j (additive), the forgetful functor to presheaves of abelian groups (additive), and evaluation at W (additive). A composite of additive functors is additive; the additivity follows from the additivity of each factor. The identification of its right derived functor with absolute cohomology is the presheaf description of Lemma 408. □

Lemma 416 (The presheaf-of-modules functor is additive)**.** *The functor* toPresheafOfModules X : X.Modules $\to$ PresheafOfModules$(\mathcal{O}_X)$, *sending a sheaf of $\mathcal{O}_X$-modules to its underlying presheaf of modules, is additive: it preserves finite biproducts and the additive structure on hom-groups.*

Proof. The functor is the composite of the forgetful functor SheafOfModules$(\mathcal{O}_X) \to$ PresheafOfModules$(\mathcal{O}_X)$ and the restriction of scalars along the identity restrictScalars$(\mathrm{id}_{\mathcal{O}_X})$. Both factors are additive, and a composite of additive functors is additive. □

Lemma 417 (The sections functor is additive)**.** *For an open $V \subseteq X$ the global-sections functor* sectionsFunctor $V = \Gamma(V, -) : X$.Modules $\to$ Ab *is additive.*

Proof. The functor factors as the additive functor to presheaves of modules (Lemma 416), followed by the underlying-presheaf functor and evaluation at V, each of which is additive; the composite is therefore additive. □

Lemma 418 (The sections functor is corepresented by $j_! \mathcal{O}_V$)**.** *For an open $V \subseteq X$ there is a natural isomorphism of additive functors*

$$\text{sectionsFunctor}\, V \;\cong\; \mathrm{Hom}_{X.\mathrm{Modules}}(j_! \mathcal{O}_V, -)$$

between $\Gamma(V, -)$ and the covariant additive hom-functor corepresented by $j_! \mathcal{O}_V$. It is the functorial upgrade of the corepresentability bijection of Lemma 217.

Proof. Assemble the natural transformation from the additive corepresentability isomorphism $\mathrm{Hom}(j_! \mathcal{O}_V, \mathcal{F}) \cong \Gamma(V, \mathcal{F})$ of Lemma 217, taken componentwise; its naturality square in $\mathcal{F}$ is exactly the naturality of that bijection in the coefficient sheaf. The components are isomorphisms, so the assembled transformation is a natural isomorphism. Both functors are additive (Lemma 417 and the additivity of a hom-functor). □

Lemma 419 (Right derived functors transport along a natural isomorphism)**.** *Let $\mathcal{A}$ be an abelian category with enough injectives, $\mathcal{B}$ abelian, and $F, G : \mathcal{A} \to \mathcal{B}$ additive functors. A natural isomorphism $F \cong G$ induces, for every n, a natural isomorphism of right derived functors*

$$F.\mathrm{rightDerived}\, n \;\cong\; G.\mathrm{rightDerived}\, n.$$

Proof. Right derivation is functorial in the functor argument: a natural transformation $F \to G$ induces a natural transformation $F.\mathrm{rightDerived}\, n \to G.\mathrm{rightDerived}\, n$, compatibly with composition and identities. Applying this to the hom and inverse of the given isomorphism $F \cong G$ yields two derived natural transformations, and the composition and identity compatibilities show they are mutually inverse; hence $F.\mathrm{rightDerived}\, n \cong G.\mathrm{rightDerived}\, n$. □

Lemma 420 (The canonical isomorphism of an affine scheme with its spectrum)**.** *Provided by Mathlib. Let U be a scheme with* [IsAffine U]*. Then there is a canonical* whole-scheme *isomorphism of schemes*

$$U \;\cong\; \operatorname{Spec}\Gamma(U, \mathcal{O}_U),$$

identifying U with the spectrum of its ring of global sections. This is a genuine isomorphism of schemes (not merely an open immersion), so it induces an equivalence of module categories.

Lemma 421 (Ext transports along an exact functor)**.** *Provided by Mathlib. Let $F : \mathcal{A} \to \mathcal{B}$ be an exact functor between abelian categories. Then for objects X, Y of $\mathcal{A}$ and every n there is an induced additive map*

$$\mathrm{Ext}^n_{\mathcal{A}}(X, Y) \;\longrightarrow\; \mathrm{Ext}^n_{\mathcal{B}}(FX, FY),$$

compatible with the additive structure, with degree-zero Hom*-action, and with Yoneda composition. When F is one half of an equivalence the maps for F and its inverse are mutually inverse, so* Ext *is carried isomorphically across the equivalence.*

Lemma 422 (Pushforward of sheaves of modules and its coherences)**.** *Provided by Mathlib. For a morphism of schemes $f : X \to Y$ there is an additive pushforward functor* $\mathrm{pushforward}\, f : X.\mathrm{Modules} \to Y.\mathrm{Modules}$*, together with the coherence isomorphisms* $\mathrm{pushforward}(\mathrm{id}_X) \cong \mathrm{id}$, $\mathrm{pushforward}\, f \circ \mathrm{pushforward}\, g \cong \mathrm{pushforward}(f \circ g)$*, and* $\mathrm{pushforward}\, f \cong \mathrm{pushforward}\, g$ *for $f = g$. When $\varphi : X \cong Y$ is an isomorphism, these assemble* $\mathrm{pushforward}\, \varphi.\mathrm{hom}$ *and* $\mathrm{pushforward}\, \varphi.\mathrm{inv}$ *into an equivalence* $X.\mathrm{Modules} \simeq Y.\mathrm{Modules}$*.*

8.6.2 The enlarged affine instantiation (general affine opens, Need #2)

The distinguished-open affine cover system (Definition 242) computes Serre vanishing only on distinguished opens $D(f)$. To reach *every* affine open V we enlarge the basis $\mathcal{B}$ to all affine opens, keeping the admissible coverings Cov the standard covers $i \mapsto D(g_i)$. The three cover-system fields and the quasi-coherent Čech-vanishing seed are re-discharged below in their general-affine-open form; the only genuinely new content is the seed (Lemma 430), whose change-of-ring discharge replaces the single localisation R_f of the distinguished case by the coordinate ring $\Gamma(V)$.

Lemma 423 (Distinguished opens of $\operatorname{Spec} R$ are affine)**.** *Project-local infrastructure. For a ring R and $r \in R$, the distinguished open $D(r) \subseteq \operatorname{Spec} R$ is an affine open: it is isomorphic, as a scheme, to $\operatorname{Spec} R[1/r]$. Consequently every distinguished open lies in the enlarged basis of all affine opens, which discharges the* faces_mem *field of the enlarged affine cover system (Definition 426).*

Proof. The canonical open immersion $D(r) \cong \operatorname{Spec} R[1/r] \hookrightarrow \operatorname{Spec} R$ (the basic-open / away-localisation isomorphism basicOpenIsoSpecAway) is an isomorphism onto $D(r)$; since its source is affine and affineness transports across a scheme isomorphism, $D(r)$ is affine. ☐

Lemma 424 (Standard covers are cofinal among open covers of a general affine open)**.** *Source: Stacks Project, Sheaves on Spaces, Tag 009L,* `lemma-cofinal-systems-coverings-standard-case`*. Let V be* any *affine open of* $\operatorname{Spec} R$*, and let $\{W_\alpha\}_{\alpha\in A}$ be an arbitrary open covering of V (each W_α an open of* $\operatorname{Spec} R$ *contained in V). Then there exist a finite n, a family $g : \{1,\dots,n\} \to R$, and a reindexing $\varphi : \{1,\dots,n\} \to A$ such that*

$$V = \bigcup_{i=1}^{n} D(g_i) \qquad \textit{and} \qquad D(g_i) \subseteq W_{\varphi(i)} \quad (1 \le i \le n).$$

That is, every open covering of an arbitrary affine open V is refined by a finite standard cover by distinguished opens, each sitting inside a member of the original covering. This is the general-affine-open companion of Lemma 230, the only change being the source of quasi-compactness: the compactness of the affine open V (every affine scheme is quasi-compact) in place of the compactness of a distinguished $D(f)$.

Proof. The basic opens $D(g)$ form a topological basis of $\operatorname{Spec} R$ (Lemma 239), so each W_α is a union of basic opens it contains; choosing one basic open inside a member through each point of V yields a basic-open covering of V refining $\{W_\alpha\}$. An affine open V is quasi-compact, so a finite subfamily $D(g_1),\dots,D(g_n)$ already covers V; recording for each i an index $\varphi(i)$ with $D(g_i) \subseteq W_{\varphi(i)}$ gives the asserted finite refinement $V = \bigcup_i D(g_i)$. The basis is closed under the intersections required by Tag 009L (Lemma 229), so the standard covers form a cofinal system among the open coverings of V. ☐

Lemma 425 (Section surjectivity for the enlarged affine cover system)**.** *Project-local infrastructure. Let V be* any *affine open of* $\operatorname{Spec} R$ *and let $S : 0 \to S_1 \to S_2 \to S_3 \to 0$ be a short exact sequence of $\mathcal{O}_X$-modules whose left term S_1 has vanishing positive Čech cohomology over every standard cover $i \mapsto D(g_i)$* whose union is affine*. Then the section map $S_2(V) \to S_3(V)$ is surjective. This is the* surj_of_vanishing *field of the enlarged affine cover system, the general-affine-open companion of Lemma 231.*

Proof. The proof of Lemma 231 carries over verbatim. A module epimorphism $S_2 \to S_3$ is locally surjective on sections (Lemmas 235, 237, 236); the resulting open covering of V carrying local lifts of a section $t \in S_3(V)$ is refined to a finite standard cover $i \mapsto D(g_i)$ of V by Lemma 424 (this is where the quasi-compactness of the affine open V replaces that of $D(f)$), whose union equals V and is therefore affine. The Čech H^1-gluing sequence (Lemma 215), fed the vanishing $\check{H}^1(\{D(g_i)\}, S_1) = 0$ for that affine-union standard cover, glues the local lifts to a global lift of t over V. ☐

Definition 426 (The enlarged affine cover system)**.** *Project-local: enlargement of Definition 242.* The *enlarged affine cover system* for $\operatorname{Spec} R$ is the instance of Definition 319 whose basis $\mathcal{B}$ is *all* affine opens of $\operatorname{Spec} R$ and whose admissible coverings Cov are the standard finite covers $i \mapsto D(g_i)$ whose union $\bigcup_i D(g_i)$ is affine. It generalises Definition 242, which uses only the distinguished opens $D(f)$ as its basis. Its three proof fields are discharged by: faces_mem — each face $\bigcap_k D(g_{\sigma k}) = D(\prod_k g_{\sigma k})$ is distinguished, hence affine (Lemma 423), so lies in the enlarged basis (Lemma 229); surj_of_vanishing by Lemma 425; and injective_acyclic by the cover-agnostic family-form injective Čech-acyclicity (Lemma 214).

Lemma 427 (Localised base change is a localisation away)**.** *Project-local infrastructure, assembled from Mathlib base-change primitives. Let $\varphi : R \to S$ be a ring map, M an R-module, and $g \in R$ with image $\bar{g} = \varphi(g) \in S$. The base change $M \otimes_R S$, further localised at the powers of $\bar{g}$, is a localisation of M: the composite*

$$M \longrightarrow M \otimes_R S \longrightarrow (M \otimes_R S)_{\bar{g}}$$

exhibits $(M \otimes_R S)_{\bar{g}}$ as the base change of M along the composite $R \to S \to S_{\bar{g}}$. In particular, whenever the localised ring $S_{\bar{g}}$ is identified with the localisation R_g — as happens when $D_S(\bar{g})$ and $D_R(g)$ name the same basic open — the composite presents $(M \otimes_R S)_{\bar{g}}$ as the away localisation M_g.

Proof. Tensoring with S exhibits $M \otimes_R S$ as the base change of M along $R \to S$ (TensorProduct. isBaseChange); localising at the powers of $\bar{g}$ exhibits $(M \otimes_R S)_{\bar{g}}$ as the base change of $M \otimes_R S$ along $S \to S_{\bar{g}}$ (isLocalizedModule_iff_isBaseChange). Composing the two base changes (IsBaseChange. comp) exhibits the composite $M \to (M \otimes_R S)_{\bar{g}}$ as the base change of M along $R \to S_{\bar{g}}$. Under the identification $S_{\bar{g}} = R_g$ this is exactly the away localisation of M at g; the two-step composite specialises to the localised module recipe already recorded in Lemma 312. ☐

Lemma 428 (Section Čech module exactness from an affine-cover base-change datum)**.** *Project-local infrastructure; mirror of Lemma 313. Let M be an R-module and $g : \iota \to R$ a finite family whose union $V = \bigcup_i D(g_i)$ is an affine open of $\operatorname{Spec} R$, with coordinate ring $S = \Gamma(V, \mathcal{O}_V)$ and restriction map $\varphi : R \to S$, $\bar{g}_i = \varphi(g_i)$. Since the $D_S(\bar{g}_i)$ cover $\operatorname{Spec} S = V$, the images $\bar{g}_i$ span the unit ideal of S. Then the un-localised section Čech module complex $D^\bullet(g, M)$ of Definition 192 is exact in every positive degree* — without *assuming that g spans the unit ideal of R itself.*

Proof. Run the computation over S. Instantiate the polymorphic positive-degree exactness of Lemma 193 over the base ring S, with module $M \otimes_R S$ and the spanning family $\bar{g} : \iota \to S$; spanning is exactly the affineness of the cover's union, so the S-side complex $D^\bullet(\bar{g}, M \otimes_R S)$ is exact in positive degrees. Transport this exactness back to the R-side along the degreewise R-linear isomorphisms $M_{g_\sigma} \cong (M \otimes_R S)_{\bar{g}_\sigma}$ of Lemma 427 (for a tuple σ, $g_\sigma = \prod_k g_{\sigma k}$ and $\bar{g}_\sigma = \varphi(g_\sigma)$, with $\Gamma(D(g_\sigma))$ the common localisation). Realised through IsLocalizedModule. iso, these comparisons are the vertical maps of a ladder, and exactness transfers across it by Function. Exact. of_ladder_addEquiv_of_exact, the cofaces commuting by naturality of the localisation comparison. ☐

Lemma 429 (Tilde Čech vanishing over an affine-union standard cover)**.** *Project-local infrastructure; mirror of Lemma 314. Let R be a ring, M an R-module, and $g : \iota \to R$ a finite family whose union $V = \bigcup_i D(g_i)$ is an affine open of $\operatorname{Spec} R$. Then for every $p > 0$ the homology of the section Čech complex of $\widetilde{M}$ over $\{D(g_i)\}$ vanishes,*

$$\operatorname{IsZero}(\check{\mathcal{C}}^\bullet(\{D(g_i)\}, \widetilde{M}).\text{homology}\, p).$$

Proof. Lemma 428 supplies positive-degree exactness of the un-localised section Čech module complex $D^\bullet(g, M)$ from the affineness of the cover's union. The abelian-group exactness criterion (the private sectionCechAbExact_affine packaging of Lemma 191) transports this module exactness across the degreewise additive ladder identifying the section complex with the localised-module complex (Lemma 187), and reads the resulting function exactness off as the vanishing of the degree-p homology via sectionCech_isZero_homology_of_objD_exact. ☐

Lemma 430. *[The general-affine-open change-of-ring Čech-vanishing seed] Source: Stacks Project, Schemes, Tag 01HV, `lemma-spec-sheaves` (4), and Cohomology of Schemes, `lemma-cech-cohomology-quasi-coherent-trivial`. Let R be a ring, M an R-module, and $g : \{1, \dots, n\} \to R$*

a finite family such that $V := \bigcup_i D(g_i)$ *is an affine open of* $\operatorname{Spec} R$ *(a general affine open,* not *assumed to be a single* $D(f)$*). Then for every* $p > 0$ *the section Čech complex of* $\widetilde{M}$ *over the cover* $\{D(g_i)\}$ *has vanishing positive homology:*

$$\check{H}^p(\{D(g_i)\}, \widetilde{M}) = 0.$$

Proof. Because V is a proper affine open, the family $\{g_i\}$ need not span the unit ideal of R, so the standard-cover vanishing of Lemma 194 does not apply over R. As in the distinguished case (Lemma 314) we run the computation over a ring where the cover *does* span the unit ideal — here the coordinate ring of V itself, rather than a single localisation R_f.

Change of ring to $S = \Gamma(V, \mathcal{O}_V)$*.* Set $S := \Gamma(V, \mathcal{O}_V)$, let $\varphi : R \to S$ be the restriction map and $\bar{g}_i := \varphi(g_i)$. Since V is affine, Scheme. isoSpec (Lemma 420) gives a scheme isomorphism $V \cong \operatorname{Spec} S$ under which each $D(g_i) \subseteq V$ (an affine open, by Lemma 423) corresponds to the distinguished open $D_S(\bar{g}_i)$ of $\operatorname{Spec} S$. The $D_S(\bar{g}_i)$ cover $\operatorname{Spec} S = V$, so the $\bar{g}_i$ *span the unit ideal of* S*.*

Identification of the two section complexes. The section Čech complex of $\widetilde{M}$ over $\{D(g_i)\}$ has degree-p term $\prod_\sigma \Gamma(D(g_\sigma), \widetilde{M}) = \prod_\sigma M_{g_\sigma}$ (Tag 01HV item (4), $g_\sigma = \prod_k g_{\sigma k}$). We identify it, degreewise, with the standard-cover (full-span) section Čech complex over $\operatorname{Spec} S$ of $\widetilde{M \otimes_R S}$, whose degree-$p$ term is $\prod_\sigma (M \otimes_R S)_{\bar{g}_\sigma}$. The per-$\sigma$ comparison $M_{g_\sigma} \cong (M \otimes_R S)_{\bar{g}_\sigma}$ is a localised base-change isomorphism: $\Gamma(D(g_\sigma)) = R_{g_\sigma}$ is simultaneously the localisation of R away from g_σ and the localisation of S away from $\bar{g}_\sigma$ (since $(\operatorname{Spec} R)$. basicOpen $\bar{g}_\sigma = D(g_\sigma)$), and $(M \otimes_R S)_{\bar{g}_\sigma}$ is the base change of M along the composite $R \to S \to S_{\bar{g}_\sigma}$ (transitivity of base change for localised modules). These comparisons are the vertical maps of an isomorphism of cochain complexes, commuting with the alternating-sum cofaces by naturality of the localisation comparison.

Conclusion. Over S the family $\bar{g}$ spans the unit ideal, so the full-span standard-cover vanishing — the polymorphic core of Lemma 194, re-instantiated over S with the module $M \otimes_R S$, exactly as it supplies the quasi-coherent seed of Lemma 307 on a whole spectrum — gives positive-degree exactness of the $\operatorname{Spec} S$ complex. This change-of-base computation and its transport back along the per-σ base-change isomorphisms $M_{g_\sigma} \cong (M \otimes_R S)_{\bar{g}_\sigma}$ (Lemma 427) is exactly Lemma 429, whose conclusion is the asserted vanishing $\check{H}^p(\{D(g_i)\}, \widetilde{M}) = 0$ for all $p > 0$. ☐

Lemma 431 (General-affine-open quasi-coherent Čech vanishing from the tilde seed)**.** *Project-local: enlarged-system analogue of the reduction inside Lemma 307. Let* $\mathcal{F}$ *be a quasi-coherent* $\mathcal{O}_{\operatorname{Spec} R}$*-module with* $M = \Gamma(\operatorname{Spec} R, \mathcal{F})$*. Then* $\mathcal{F}$ *has vanishing higher Čech cohomology for the enlarged affine cover system (Definition 426): for every standard cover* $i \mapsto D(g_i)$ *whose union is affine and every* $p > 0$*,* $\check{H}^p(\{D(g_i)\}, \mathcal{F}) = 0$*. The reduction is transport along the affine structure isomorphism* $\mathcal{F} \cong \widetilde{M}$ *(Lemma 243) to the tilde seed of Lemma 430.*

Proof. By Lemma 243 the quasi-coherent $\mathcal{F}$ is isomorphic to $\widetilde{M}$. Čech cohomology transports along an isomorphism of coefficients (Lemma 308), so it suffices to vanish $\check{H}^p(\{D(g_i)\}, \widetilde{M})$ for every standard cover whose union is affine and every $p > 0$ — which is exactly Lemma 430. ☐

Lemma 432 (General-affine Serre vanishing from the enlarged Čech seed)**.** *Project-local: enlarged-system instantiation of Lemma 328. Let* $\mathcal{F}$ *be a quasi-coherent* $\mathcal{O}_{\operatorname{Spec} R}$*-module which has vanishing higher Čech cohomology for the enlarged affine cover system (Definition 426). Then for every affine open* V *of* $\operatorname{Spec} R$ *and every* $p > 0$ *the absolute cohomology* $H^p(V, \mathcal{F}) = \operatorname{Ext}^p(j_! \mathcal{O}_V, \mathcal{F})$ *vanishes. The statement carries* [EnoughInjectives $(\operatorname{Spec} R)$.Modules]*, inherited from Lemma 328.*

Proof. Apply the basis-comparison criterion (Lemma 328) to the enlarged affine cover system (Definition 426), whose basis is all affine opens. Conditions (1) and (2) are the system's faces_mem and cofinality data, and condition (3) is the supplied seed. Reading off the conclusion at the member $V \in \mathcal{B}$ gives $H^p(V, \mathcal{F}) = 0$ for $p > 0$. $\square$

Lemma 433 (General-affine Serre vanishing from the tilde leaf)**.** *Project-local: the fully-reduced working form behind Lemma 434. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_{\operatorname{Spec} R}$-module and V any affine open of* Spec R*. Then* $\operatorname{Ext}^p(j_!\mathcal{O}_V, \mathcal{F}) = 0$ *for all $p > 0$. This composes the seed reduction (Lemma 431) with the basis-comparison instantiation (Lemma 432).*

Proof. Lemma 431 supplies the enlarged-system seed HasVanishingHigherCech(affineCoverSystemGeneral R) $\mathcal{F}$ (its tilde-leaf hypothesis being discharged by Lemma 430); feeding this seed to Lemma 432 yields $\operatorname{Ext}^p(j_!\mathcal{O}_V, \mathcal{F}) = 0$ for $p > 0$. $\square$

Lemma 434 (Serre vanishing for a general affine open (Need #2))**.** *Let R be a ring, V* any *affine open of* Spec R*, and $\mathcal{H}$ a quasi-coherent $\mathcal{O}_{\operatorname{Spec} R}$-module. Then for every $q \geq 1$ the absolute cohomology vanishes:*

$$H^q(V, \mathcal{H}) \;=\; \operatorname{Ext}^q(j_!\mathcal{O}_V, \mathcal{H}) \;=\; 0.$$

This is the general-affine-open companion of the distinguished-open Serre vanishing (Lemma 228): the basis $\mathcal{B}$ of the affine cover system is enlarged from the distinguished opens $D(f)$ to all affine opens, while the admissible coverings Cov *remain the standard covers $i \mapsto D(g_i)$.*

Proof. Instantiate the basis-comparison criterion (Lemma 328) at the enlarged affine cover system (Definition 426), whose basis $\mathcal{B}$ is *all* affine opens of Spec R and whose admissible coverings Cov are the standard covers of an affine open $V = \bigcup_i D(g_i)$. Its three cover-system fields are:

- faces_mem: the faces of a standard cover are the distinguished opens $\bigcap_k D(g_{\sigma k}) = D(\prod_k g_{\sigma k})$, still distinguished and a fortiori affine, so they lie in the enlarged $\mathcal{B}$ (Lemmas 229, 423).
- injective_acyclic: the family-form injective Čech-acyclicity (Lemma 214) is cover-agnostic and applies to the standard covers of any affine open verbatim.
- surj_of_vanishing: the general-affine-open section surjectivity (Lemma 425), whose refinement step (Lemma 424) uses the quasi-compactness of the affine open V in place of that of a distinguished $D(f)$.

With $\mathcal{F} = \mathcal{H}$ quasi-coherent, condition (3) of Lemma 328 (standard-cover Čech vanishing) is *not* the distinguished-open seed of Lemma 228: a standard cover here covers a general affine open $V = \bigcup_i D(g_i)$, whose coordinate ring $\Gamma(V)$ is not a single localisation R_f, so the $D(f)$ seed does not apply. Condition (3) is instead supplied by the enlarged-system seed HasVanishingHigherCech(affineCoverSystemGeneral R) $\mathcal{H}$ of Lemma 431, which transports along $\mathcal{H} \cong \widetilde{M}$ to the change-of-ring tilde seed (Lemma 430). Reading off the conclusion at the member $V \in \mathcal{B}$ gives $H^q(V, \mathcal{H}) = 0$ for $q \geq 1$; the identification $H^q(V, -) = \operatorname{Ext}^q(j_!\mathcal{O}_V, -)$ is the Ext realization of Definition 218. $\square$

Lemma 435 (A scheme isomorphism carries the corepresenting object along)**.** *Let $\varphi : X \cong Y$ be an isomorphism of schemes, $V \subseteq X$ an open, and $\Phi =$ pushforwardEquivOfIso $\varphi : X$.Modules $\simeq$ Y.Modules the induced equivalence. Then the pushforward carries the corepresenting object of Definition 216 along the isomorphism:*

$$\Phi.\text{functor.obj}(j_!\mathcal{O}_V) \;\cong\; j_!\mathcal{O}_{\varphi.\text{inv}^{-1} V},$$

where $\varphi.\text{inv}^{-1} V$ is the preimage of V under φ.inv, i.e. the image open of V in Y.

Proof. Both sides corepresent the functor sectionsFunctor$(\varphi.\mathrm{inv}^{-1}V) \circ$ forget: the object Φ.functor$(j_!\mathcal{O}_V)$ via the adjunction underlying the equivalence Φ, and the object $j_!\mathcal{O}_{\varphi.\mathrm{inv}^{-1}V}$ via the defining corepresentability of $\Gamma(\varphi.\mathrm{inv}^{-1}V, -)$ by $j_!\mathcal{O}$ (Definition 216). Both corepresent the same functor, so uniqueness of corepresenting objects identifies them.

In detail, the argument is a corepresentability transport; $j_!\mathcal{O}_V$ is never unfolded. Write $A = j_!\mathcal{O}_V$. We exhibit two corepresentations of one and the same Type-valued functor and invoke the uniqueness-up-to-isomorphism of a corepresenting object to obtain the stated isomorphism, with no adjunction-mate transposition or naturality square computed by hand. The common functor is built by chaining three identifications.

1. The adjunction underlying the equivalence Φ presents the functor corepresented by Φ.functor(A) as the composite of Φ.inverse with the functor corepresented by A.

2. The corepresentability isomorphism of Lemma 418, whiskered by the forgetful functor to types, identifies the functor corepresented by A with the sections functor: it is sectionsFunctor V as a Type-valued functor. This rests on the corepresentability of $\Gamma(V, -)$ by $j_!\mathcal{O}_V$ of Definition 216.

3. Sections of a pushforward are, by definition, sections over the preimage open: Φ.inverse $\circ$ sectionsFunctor V = sectionsFunctor$(\varphi.\mathrm{inv}^{-1}V)$, an equality of functors. Reading Lemma 418 at the open $\varphi.\mathrm{inv}^{-1}V$ then exhibits this same functor as the one corepresented by $j_!\mathcal{O}_{\varphi.\mathrm{inv}^{-1}V}$.

Thus Φ.functor(A) and $j_!\mathcal{O}_{\varphi.\mathrm{inv}^{-1}V}$ corepresent the same functor; the uniqueness of corepresenting objects up to isomorphism reflects this to the stated isomorphism of objects. The only geometric inputs are the two definitional equalities "corepresented functor = sections functor" and "sections of a pushforward = sections over the preimage". □

Lemma 436 (Quasi-coherence from a cover with quasi-coherent restrictions)**.** *Provided by Mathlib. Let $\mathcal{F}$ be a sheaf of modules and $(U_i)_{i\in I}$ a family of objects of the site covering the terminal object. If each over-picture restriction $\mathcal{F}$.over U_i is quasi-coherent, then $\mathcal{F}$ is quasi-coherent.*

Lemma 437 (Extraction of the local quasi-coherence datum)**.** *Provided by Mathlib. A quasi-coherent sheaf of modules $\mathcal{F}$ admits a quasi-coherence datum: there is a family $(U_i)_{i\in I}$ of objects of the site covering the terminal object together with, for each i, a presentation of $\mathcal{F}$.over U_i.*

Lemma 438 (Pushforward along an iso commutes with restriction to an open)**.** *Let $\varphi : X \cong Y$ be an isomorphism of schemes, Φ = pushforwardEquivOfIso φ the induced equivalence of module categories, and $V \subseteq X$ an open with image $V' = \varphi.\mathrm{inv}^{-1}\, V \subseteq Y$. The homeomorphism underlying φ restricts to a homeomorphism $V \cong V'$, hence to an equivalence of the opens-sites* Opens $V \simeq$ Opens V'*; let*

$$e : (X.\mathrm{Modules}\restriction V) \;\simeq\; (Y.\mathrm{Modules}\restriction V')$$

be the induced equivalence of restricted-module categories obtained from the site-equivalence transport of Lemma 253. Then for every $\mathcal{O}_X$-module H there is a comparison isomorphism

$$e.\mathrm{functor.obj}(H.\mathrm{over}\,V) \;\cong\; (\Phi.\mathrm{functor.obj}\,H).\mathrm{over}\,V',$$

identifying "restrict to V then push forward over V'" with "push forward then restrict to V'".

Proof. Both sides are computed open-by-open on the subspace site of V'. For an open $W \subseteq V'$ with preimage $W_0 = \varphi.\mathrm{hom}^{-1}\, W \subseteq V$, the sections of the right-hand restriction are the sections of Φ.functor.obj H over the image open W, which — pushforward acting on opens by the inverse-image homeomorphism — are the sections of H over W_0 (the restriction-of-objects identity of Lemma 257, applied to the open immersions $V \hookrightarrow X$ and $V' \hookrightarrow Y$). The same sections compute the left-hand side: the site-equivalence transport e of Lemma 253 relabels the open W of V' to the corresponding open W_0 of V and reads off $H.\operatorname{over} V$ there. The two presheaves of sections therefore agree, compatibly with restriction maps, and the comparison is a morphism of sheaves of modules over V'; being an isomorphism on every section group it is an isomorphism. This is the open-restriction analogue of the bind construction's per-piece transport, with the homeomorphism-induced opens-site equivalence in place of the iterated-slice equivalence. □

Lemma 439 (Quasi-coherence of an iso-pushforward reduces to its slices). *Let $\varphi : X \cong Y$ be an isomorphism of schemes, $\Phi = \mathrm{pushforwardEquivOfIso}\,\varphi$ the induced equivalence, $H : X.\mathrm{Modules}$, and $(U_i)_{i\in I}$ a family of opens covering X. Then quasi-coherence of the pushforward $\Phi.\mathrm{functor.obj}\,H$ reduces to quasi-coherence of its restrictions to the preimage-transported cover: if for every i the slice*

$$(\Phi.\mathrm{functor.obj}\,H).\operatorname{over}(\varphi.\mathrm{inv}^{-1}\,U_i)$$

is quasi-coherent, then $\Phi.\mathrm{functor.obj}\,H$ is quasi-coherent.

Proof. The preimages $V_i = \varphi.\mathrm{inv}^{-1}\,U_i$ again cover Y: the underlying map of $\varphi.\mathrm{inv}$ is a homeomorphism, so a covering family of X pulls back to a covering family of Y. Concretely, membership in $\mathrm{Opens.grothendieckTopology}$ is preserved under the iso by transporting the covering sieve $\mathrm{Sieve.ofObjects}$ along φ (the cover-transport step). With (V_i) covering Y and each slice $(\Phi.\mathrm{functor.obj}\,H).\operatorname{over} V_i$ quasi-coherent by hypothesis, quasi-coherence of $\Phi.\mathrm{functor.obj}\,H$ follows from the cover criterion (Lemma 436); the family (U_i) is supplied by the quasi-coherence datum of H (Lemma 437). □

Lemma 440 (The continuous-functor pullback of sheaves of modules). *Provided by Mathlib. Let $F : C \to D$ be a continuous functor of sites ($F.\mathrm{IsContinuous}\,J\,K$), let $S : \mathrm{Sheaf}\,J\,\mathrm{RingCat}$, $R : \mathrm{Sheaf}\,K\,\mathrm{RingCat}$, and let $\varphi : S \to (F.\mathrm{sheafPushforwardContinuous}\,\mathrm{RingCat}\,J\,K).\mathrm{obj}\,R$ be a morphism of sheaves of rings such that the induced pushforward $\mathrm{SheafOfModules.pushforward}\,\varphi$ admits a right adjoint. Then there is a pullback functor $\mathrm{pullback}\,\varphi : \mathrm{SheafOfModules}\,S \to \mathrm{SheafOfModules}\,R$, and it is a left adjoint (of the pushforward); in particular it preserves all colimits.*

Lemma 441 (The functor on sheaves of rings induced by a continuous functor). *Provided by Mathlib. A continuous functor of sites $F : C \to D$ ($F.\mathrm{IsContinuous}\,J\,K$) induces a pushforward functor $F.\mathrm{sheafPushforwardContinuous}\,A\,J\,K : \mathrm{Sheaf}\,K\,A \to \mathrm{Sheaf}\,J\,A$ on A-valued sheaves. Taking $A = \mathrm{RingCat}$, this is the receiving object $(F.\mathrm{sheafPushforwardContinuous}\,\mathrm{RingCat}\,J\,K).\mathrm{obj}\,R$ of the structure-sheaf comparison ψ_r below.*

Lemma 442 (Structure-sheaf comparison of a scheme morphism). *Provided by Mathlib. A morphism of schemes $f : X \to Y$ carries an associated morphism of sheaves of (commutative) rings comparing the structure sheaves along f; this is the sheaf-of-rings datum underlying the comorphism $f^\sharp$, packaged as a single morphism in the sheaves-of-rings category.*

Lemma 443 (Beck–Chevalley identity for the slice post-composition functor). *Provided by Mathlib. For a functor $F : T \to D$ and an object $X : T$, post-composition on slices commutes with the forgetful functors out of the slices: $\mathrm{Over.post}\,F \circ \mathrm{Over.forget} = \mathrm{Over.forget} \circ F$ as an equality of functors $(\mathrm{Over}\,X) \to D$.*

Lemma 444 (The slice structure-sheaf ring map ψ_r). *Let $\varphi : X \cong Y$ be an isomorphism of schemes, $U_i \subseteq X$ an (affine) open and $V_i = \varphi.\mathrm{inv}^{-1}\,U_i \subseteq Y$ its image. Let $F : \mathrm{Opens}\,U_i \to \mathrm{Opens}\,V_i$ be the continuous functor of opens-sites induced by the homeomorphism underlying φ restricted to U_i; since φ is an isomorphism, F is an equivalence, so it is continuous*

(F. IsContinuous $J\,K$) and final (F. Final). Let $\mathcal{O}_{U_i}$: Sheaf J RingCat and $\mathcal{O}_{V_i}$: Sheaf K RingCat be the structure sheaves of the two sliced opens-sites, regarded as sheaves of rings. Then there is a morphism of sheaves of rings

$$\psi_r : \mathcal{O}_{U_i} \longrightarrow (F.\,\mathrm{sheafPushforwardContinuous}\,\mathrm{RingCat}\,J\,K).\mathrm{obj}\,\mathcal{O}_{V_i},$$

carrying the instance (SheafOfModules. pushforward ψ_r). IsRightAdjoint *— the right-adjointness obligation that makes the pullback functor along ψ_r (Lemma 440) available. This single cross-ring slice structure-sheaf ring map is the datum from which the pullback functor of Lemma 460 is built.*

Proof. The functor F and the two structure sheaves $\mathcal{O}_{U_i}$, $\mathcal{O}_{V_i}$ arise by restricting the scheme isomorphism φ to U_i: the underlying homeomorphism restricts to a homeomorphism $U_i \cong V_i$, inducing the opens-site equivalence F. The map ψ_r is the structure-sheaf comparison of φ.hom — its associated morphism of sheaves of rings (Lemma 442) — read along the opens-equivalence F. The transport to the sliced sites uses the Beck–Chevalley identity that post-composition with a functor commutes with the forgetful functor out of the slice, Over. post $F \circ$ Over. forget $=$ Over. forget $\circ F$ (Lemma 443, an equality of functors holding by definition), so the comparison descends to the over-categories with no further coherence datum. The right-adjointness instance for SheafOfModules. pushforward ψ_r holds because F is an equivalence, whence its induced push-forward of sheaves of rings has a right adjoint. The result is the slice structure-sheaf ring map ψ_r. □

Lemma 445 (Uniqueness of left adjoints)**.** *Provided by Mathlib. If $F, F' : C \to D$ are both left adjoint to a single functor $G : D \to C$ (witnessed by adjunctions $\mathrm{adj}_1 : F \dashv G$ and $\mathrm{adj}_2 : F' \dashv G$), then F and F' are canonically naturally isomorphic,* leftAdjointUniq adj_1 adj_2 *: $F \cong F'$.*

Lemma 446 (Two pushforwards from an adjunction of sites are adjoint)**.** *Provided by Mathlib. Let $F : C \to D$ and $G : D \to C$ be continuous functors of sites carrying an adjunction* adj *: $F \dashv G$; let S :* Sheaf J RingCat, *R :* Sheaf K RingCat, *and let $\varphi : S \to$* (F. sheafPushforwardContinuous RingCat $J\,K$).obj R *and $\psi : R \to$* (G. sheafPushforwardContinuous RingCat $K\,J$).obj *be morphisms of sheaves of rings satisfying the two compatibility squares H_1 (relating ψ and φ to* adj*'s counit) and H_2 (the unit triangle). Then the induced module pushforwards are adjoint:* SheafOfModules. pushforward $\varphi \dashv$ SheafOfModules. pushforward ψ.

Lemma 447 (Post-composition equivalence on slices, with its correction term)**.** *Provided by Mathlib. An equivalence of categories $F : T \simeq D$ induces an equivalence of slice categories* Over. postEquiv $X\,F$: Over $X \simeq$ Over(F.functor.obj X)*. Its forward functor is* Over. post F.functor*; crucially its* inverse *is not the bare* Over. post F.inverse *but the corrected composite* Over. post F.inverse$\circ$ Over. map(F.unitIso.inv.app X)*. The* Over. map *factor repairs the fact that* F.inverse.obj(F.functor.obj X) *is only canonically isomorphic to X, not equal — the source of all the correction bookkeeping in the route below.*

Lemma 448 (Opens functor of a space isomorphism)**.** *Provided by Mathlib. An isomorphism $h : T \cong T'$ of topological spaces induces an equivalence of the lattices of opens* Opens. mapMapIso h : Opens $T \simeq$ Opens T'*, assembled from the* Opens. map *functors of the two components of h.*

Lemma 449 (Over-map functors are continuous)**.** *Provided by Mathlib. For a Grothendieck topology J on $\mathcal{C}$ and a morphism $f : X \to X'$ in $\mathcal{C}$, the slice functor* Over. map f : Over $X \to$ Over X' *is continuous for the induced over-topologies* J.over X *and* J.over X'*.*

Lemma 450 (Continuity is closed under composition)**.** *Provided by Mathlib. If $F : (\mathcal{C}, J) \to (\mathcal{D}, K)$ and $G : (\mathcal{D}, K) \to (\mathcal{E}, L)$ are continuous functors of sites, then their composite $F \circ G$ is continuous for J and L.*

Lemma 451 (Cover-preservation lifts to slices)**.** *Provided by Mathlib. If $F : \mathcal{C} \to \mathcal{D}$ is cover-preserving for topologies J, K, then for any $X : \mathcal{C}$ the post-composition functor* $\mathrm{Over.post}\, F$: $\mathrm{Over}\, X \to \mathrm{Over}(F.\mathrm{obj}\, X)$ *is cover-preserving for the induced over-topologies.*

Lemma 452 (Pullback is left adjoint to pushforward for sheaves of modules)**.** *Provided by Mathlib. For a morphism of sheaves of rings φ along a continuous functor of sites whose induced pushforward admits a right adjoint, the pullback functor of Lemma 440 is left adjoint to the pushforward:* $\mathrm{pullback}\, \varphi \dashv \mathrm{pushforward}\, \varphi$.

Lemma 453 (Pullback is left adjoint to pushforward for modules on schemes)**.** *Provided by Mathlib. For a morphism of schemes $f : X \to Y$, the pullback functor of sheaves of modules is left adjoint to the pushforward,*

$$\mathrm{pullback}\, f \;\dashv\; \mathrm{pushforward}\, f \qquad (\mathrm{pullback}\, f : Y.\mathrm{Modules} \to X.\mathrm{Modules},\ \mathrm{pushforward}\, f : X.\mathrm{Modules} \to Y.\mathrm{Modules}).$$

In particular $\mathrm{pushforward}\, f$ *is a right adjoint, hence preserves finite limits (is left exact).*

Lemma 454 (Pushforward of modules composes for a composite of schemes)**.** *Provided by Mathlib. For morphisms of schemes $f : X \to Y$ and $g : Y \to Z$, the composition of the two pushforward functors on sheaves of modules identifies with the pushforward along the composite:*

$$\mathrm{pushforward}\, f \circ \mathrm{pushforward}\, g \;\cong\; \mathrm{pushforward}(f \circ g).$$

Lemma 455 (Continuity of the slice equivalence of a scheme isomorphism)**.** *Let $\varphi : X \cong Y$ be an isomorphism of schemes, $U_i \subseteq X$ an open with image $V_i = \varphi.\mathrm{inv}^{-1}\, U_i$, and write J_X, J_Y for the opens-Grothendieck-topologies of X, Y. Then:*

- *the opens-lattice functor* $\mathrm{Opens.\,map}\, \varphi.\mathrm{hom.base}$ *is an equivalence;*
- *φ induces an equivalence of opens-sites* $\mathrm{opensEquivOfIso}\, \varphi : \mathrm{Opens}\, X \simeq \mathrm{Opens}\, Y$, *whose forward functor is definitionally* $\mathrm{Opens.\,map}\, \varphi.\mathrm{inv.base}$ *and whose inverse is* $\mathrm{Opens.\,map}\, \varphi.\mathrm{hom.base}$*;*
- *the slice equivalence* $\mathrm{sliceOversEquiv}\, \varphi\, U_i = \mathrm{Over.\,postEquiv}\, U_i\, (\mathrm{opensEquivOfIso}\, \varphi) : \mathrm{Over}\, U_i \simeq \mathrm{Over}\, V_i$ *has both its functor and its inverse continuous for the over-Grothendieck-topologies.*

Proof. The opens-equivalence is $\mathrm{opensEquivOfIso}\, \varphi = \mathrm{Opens.\,mapMapIso}(\mathrm{forgetToTop}\, .\mathrm{mapIso}\, \varphi).\mathrm{symm}$ (Lemma 448); its functor is definitionally $\mathrm{Opens.\,map}\, \varphi.\mathrm{inv.base}$ and its inverse $\mathrm{Opens.\,map}\, \varphi.\mathrm{hom.base}$, so the latter is an equivalence. Forward continuity of the slice functor is the cover-preservation of post-composition along the opens-map (Lemma 451, applied to $\mathrm{Opens.\,map}\, \varphi.\mathrm{inv.base}$, which is cover-preserving since the opens-map of a morphism is). For the inverse, read it through $\mathrm{Over.\,postEquiv}$'s inverse (Lemma 447): it is the composite of $\mathrm{Over.\,post}(\mathrm{Opens.\,map}\, \varphi.\mathrm{hom.base})$ — continuous by cover-preservation of the opens-map (Lemma 451) — with the correction $\mathrm{Over.\,map}(\mathrm{unitIso.inv})$, which is always continuous (Lemma 449). Continuity of a composite of continuous functors (Lemma 450) gives the inverse's continuity. □

Lemma 456 (The reverse slice ring map φ'')**.** *In the setting of Lemma 455, write* $\mathrm{eqv} = \mathrm{sliceOversEquiv}\, \varphi\, U_i$. *There is a morphism of sheaves of rings*

$$\varphi'' : (Y.\mathrm{ringCatSheaf.\,over}\, V_i) \longrightarrow (\mathrm{eqv.inverse.\,sheafPushforwardContinuous}\, \mathrm{RingCat}\, (J_Y.\mathrm{over}\, V_i)\, (J_X.\mathrm{over}\, U_i)).\mathrm{obj}\, (.$$

along the correction-carrying *inverse* $\mathrm{eqv.inverse} = \mathrm{Over.\,post}(\mathrm{Opens.\,map}\, \varphi.\mathrm{hom.base}) \circ \mathrm{Over.\,map}(\mathrm{unitIso.inv})$. *The map φ'' is* object-level correction-free*: over any $W : (\mathrm{Over}\, V_i)^{\mathrm{op}}$ the codomain section is the section of* $X.\mathrm{ringCatSheaf.\,over}\, U_i$ *over* $\mathrm{eqv.inverse.obj}\, W$, *whose underlying open is* $\varphi.\mathrm{hom}^{-1}\, W.\mathrm{left}$

— *because* Over. map *leaves the underlying open* $(\cdot)$.left *unchanged, only post-composing the structure arrow. Thus at the level of sections* φ'' *is exactly the slice restriction of the structure-sheaf comparison* φ.hom. toRingCatSheafHom*; the* Over. map *correction affects only the action of* eqv.inverse *on morphisms.*

Proof. Define φ'' to be the forward slice structure-sheaf comparison of Lemma 444 for the inverse isomorphism $\varphi^{-1} = \varphi.\mathrm{symm}$ over the open $\varphi.\mathrm{inv}^{-1}\, U_i = V_i$, that is

$$\varphi'' := \mathrm{sliceStructureSheafHom}(\varphi.\mathrm{symm})\,(\varphi.\mathrm{inv}^{-1}\, U_i).$$

This morphism is constructed exactly as the forward slice ring map, but for φ.symm; no separate codomain-bridge construction is required, because the codomain it naturally carries — along the *bare* composite Over. post(Opens. map φ.symm.inv.base) — is *definitionally equal* to the codomain demanded in the statement, namely the one along the correction-carrying inverse slice functor eqv.inverse = Over. post(Opens. map φ.hom.base) $\circ$ Over. map(unitIso.inv$_{U_i}$).

The reconciliation of the two codomains is absorbed by definitional equality and needs no explicit isomorphism. Two coherences make this so. First, the composition law for pushforward-continuous functors, Functor. sheafPushforwardContinuousComp for the two factors of eqv.inverse, is the identity isomorphism: composing pushforward-continuous functors along a composite of continuous functors agrees on the nose with the pushforward along the composite. Second, the relabelling Over. map$(g)\circ$forget = forget (Over. mapForget) holds by reflexivity, since Over. map leaves the underlying open $(\cdot)$.left unchanged and only post-composes the structure arrow. As both coherences are definitional, the would-be codomain bridge reduces to the identity, and φ'' is simply sliceStructureSheafHom(φ.symm) V_i retyped along the (definitionally equal) corrected-inverse codomain. At the level of sections φ'' is therefore exactly the slice restriction of the structure-sheaf comparison of φ^{-1}; the Over. map correction affects only the action on morphisms, which is what the counit and unit compatibility squares H_1, H_2 absorb. ☐

Lemma 457 (Counit compatibility square H_1 for the slice adjunction)**.** *In the setting of Lemma 456, let* adj = (sliceOversEquiv $\varphi\, U_i$).symm. toAdjunction *be the adjunction underlying the slice equivalence. Then the reverse ring map* φ'' *and the forward slice ring map* ψ_r *satisfy the counit-naturality compatibility square* H_1 *required by Mathlib's two-pushforward adjunction (Lemma 446), absorbing the* Over. map(unitIso.inv) *correction of the inverse functor.*

Proof. The square relates φ'', ψ_r and the counit of adj. Since φ.hom. toRingCatSheafHom and φ.inv. toRingCatSheafHom are mutually inverse structure-sheaf comparisons, and the two slice ring maps are their respective over-pullbacks, naturality collapses the square to an equality of canonical equality-transport isomorphisms along the open identity $\varphi.\mathrm{hom}^{-1}(\varphi.\mathrm{inv}^{-1}U_i) = U_i$, which holds by proof-irrelevance. ☐

Lemma 458 (Unit compatibility square H_2 for the slice adjunction)**.** *In the setting of Lemma 456, with* adj *as in Lemma 457, the ring maps* φ'' *and* ψ_r *satisfy the unit-triangle compatibility square* H_2 *required by Lemma 446.*

Proof. Identical in shape to H_1 (Lemma 457): the unit triangle reduces, via the mutually inverse structure-sheaf comparisons of φ.hom and φ.inv, to an equality of equality-transport morphisms along the same open identity, true by proof-irrelevance. ☐

Lemma 459 (The two-pushforward adjunction for the slice ring maps)**.** *Let* $\varphi : X \cong Y$, $U_i \subseteq X$, $V_i = \varphi.\mathrm{inv}^{-1}\, U_i$*. Let* ψ_r = sliceStructureSheafHom $\varphi\, U_i$ *be the slice structure-sheaf ring map of*

Lemma 444, and let φ'' be the reverse slice ring map of Lemma 456 — the object-level correction-free comparison along the corrected inverse slice functor (sliceOversEquiv $\varphi\, U_i$).inverse. *Then the two module pushforwards are adjoint:*

$$\mathrm{SheafOfModules.\,pushforward}\,\varphi'' \;\dashv\; \mathrm{SheafOfModules.\,pushforward}\,\psi_r.$$

Proof. Apply Mathlib's two-pushforward adjunction (Lemma 446) to the adjunction adj $=$ (sliceOversEquiv $\varphi\, U_i$).symm. toAdjunction underlying the slice equivalence (Lemma 447). Its left adjoint is the corrected inverse (sliceOversEquiv $\varphi\, U_i$).inverse $=$ Over. post(Opens. map φ.hom.base)$\circ$ Over. map(unitIso.inv), and its right adjoint is the forward functor Over. post(Opens. map φ.inv.base) along which ψ_r lives. The required continuity of both functors is Lemma 455. Feed Mathlib the two ring maps φ'' (Lemma 456) and ψ_r (Lemma 444), together with the two compatibility squares H_1 (Lemma 457) and H_2 (Lemma 458); the output is the asserted adjunction pushforward $\varphi'' \dashv$ pushforward ψ_r.

The substance isolated by this decomposition is the Over. map(unitIso.inv) correction carried by the inverse of Over. postEquiv (Lemma 447). Because the open identity $\varphi.\mathrm{hom}^{-1}(\varphi.\mathrm{inv}^{-1}U_i) = U_i$ does *not* hold definitionally, the bare functor Over. post(Opens. map φ.hom.base) lands in Over($\varphi.\mathrm{hom}^{-1}V_i$) rather than Over U_i, and the two slices are reconciled only up to the canonical isomorphism unitIso.inv. Consequently the reverse ring map must be taken along the *corrected* inverse slice functor (Lemma 456) — it is *not* sliceStructureSheafHom $\varphi^{-1}\, V_i$, which lives along the bare post-composition and lands in the wrong slice — and the same correction is absorbed by the two squares H_1, H_2, each reducing to a proof-irrelevant equality-transport identity. □

Lemma 460 (Pullback along ψ_r realizes the per-slice pushforward)**.** *Let $\varphi : X \cong Y$, $U_i \subseteq X$, $V_i = \varphi.\mathrm{inv}^{-1}\, U_i$, let ψ_r be the slice structure-sheaf ring map of Lemma 444, and write Φ = pushforwardEquivOfIso φ. For every $\mathcal{O}_X$-module H the pullback functor* pullback ψ_r : SheafOfModules $\mathcal{O}_{U_i} \to$ SheafOfModules $\mathcal{O}_{V_i}$ *applied directly to the slice* H. over U_i *(which already lies in its domain) is isomorphic to the restricted pushforward:*

$$(\mathrm{pullback}\,\psi_r).\mathrm{obj}(H.\,\mathrm{over}\,U_i) \;\cong\; (\Phi.\mathrm{functor.obj}\,H).\,\mathrm{over}\,V_i.$$

Proof. Assemble the isomorphism in two steps. The unit/free-module comparison pullbackObjUnitToUnit is *not* used: it identifies the pullback only on the unit (rank-one free) module, whereas H. over U_i is an arbitrary module. Instead we identify the whole functor pullback ψ_r by uniqueness of left adjoints, which is valid for every H at once.

Step 1 — identify the pullback functor with a pushforward. The pullback pullback ψ_r is left adjoint to pushforward ψ_r (Lemma 452). By Lemma 459 the pushforward pushforward φ'' along the reverse slice ring map φ'' of Lemma 456 is *also* left adjoint to the same pushforward ψ_r. Two left adjoints of one functor are canonically isomorphic (Lemma 445); applied to these two adjunctions it yields a natural isomorphism of functors

$$\mathrm{leftAdjointUniq}(\mathrm{pullbackPushforwardAdjunction}\,\psi_r)\,\mathrm{adj}_A : \mathrm{pullback}\,\psi_r \;\cong\; \mathrm{pushforward}\,\varphi'',$$

where adj_A is the two-pushforward adjunction of Lemma 459.

Step 2 — the section identity. Evaluate this functor isomorphism at H. over U_i:

$$(\mathrm{pullback}\,\psi_r).\mathrm{obj}(H.\,\mathrm{over}\,U_i) \;\cong\; (\mathrm{pushforward}\,\varphi'').\mathrm{obj}(H.\,\mathrm{over}\,U_i).$$

The right-hand object is (Φ.functor.obj H). over V_i by direct definitional unfolding: pushforward sections are computed by preimage (Lemma 395), so over an open W of the slice both objects have the very same sections $\Gamma(H, \varphi.\mathrm{hom}^{-1}\, W.\mathrm{left})$ — the slice pushforward along φ'' and the slice of the pushforward Φ.functor.obj H over V_i unfold to one and the same sections functor. Composing the functor isomorphism of Step 1, evaluated at $H.\mathrm{over}\, U_i$, with this definitional identification yields the stated isomorphism. $\square$

Lemma 461 (Pushforward along an iso preserves quasi-coherence)**.** *Source: Stacks Project, Schemes, "Functoriality for quasi-coherent modules". Let $\varphi : X \cong Y$ be an isomorphism of schemes and $H : X.\mathrm{Modules}$ a quasi-coherent module. Then the pushforward $\Phi.\mathrm{functor.obj}\, H$ along φ is again quasi-coherent.*

Proof. Quasi-coherence is a *local* datum: it transports across the equivalence Φ by moving H's local presentations to the image cover, member by member. We never seek a global presentation of H (it has none — H is quasi-coherent only on the abstract source); we move the local presentations one open at a time. The carrying functor need not be the full restricted-module equivalence: a single *left adjoint* pullback functor along one cross-ring slice structure-sheaf ring map suffices, which keeps the colimit-preservation that presentation-transport requires while collapsing the heavy equivalence bookkeeping to one ring map, one unit isomorphism, and one comparison isomorphism.

The cover. Extract H's quasi-coherence datum (Lemma 437): a family $(U_i)_{i\in I}$ of opens covering X together with, for each i, a presentation p_i of $H.\mathrm{over}\, U_i$. Form the image cover $V_i = \varphi.\mathrm{inv}^{-1}\, U_i$ of Y (the image of an affine open under the scheme isomorphism is again affine, Lemma 462, when one wants the cover to remain affine). By Lemma 439 it suffices to show that $(\Phi.\mathrm{functor.obj}\, H).\mathrm{over}\, V_i$ is quasi-coherent for every i.

Per-member transport. Fix i and write $W = U_i$, $V = V_i = \varphi.\mathrm{inv}^{-1}\, W$. Let ψ_r be the slice structure-sheaf ring map of Lemma 444, and consider the pullback functor pullback ψ_r. Being a left adjoint, it preserves colimits, so by Lemma 251 the presentation p_i of $H.\mathrm{over}\, W$ (transported across the opens-equivalence) transports to a presentation of $(\mathrm{pullback}\, \psi_r).\mathrm{obj}(H.\mathrm{over}\, W\ \mathrm{transported})$. The comparison isomorphism of Lemma 460,

$$(\mathrm{pullback}\, \psi_r).\mathrm{obj}(H.\mathrm{over}\, W\ \mathrm{transported}) \;\cong\; (\Phi.\mathrm{functor.obj}\, H).\mathrm{over}\, V,$$

then carries that presentation, via Lemma 252, to a presentation of $(\Phi.\mathrm{functor.obj}\, H).\mathrm{over}\, V$. A sheaf of modules with a (global) presentation is quasi-coherent, so each member of the image cover is quasi-coherent, and the claim follows by Lemma 439.

This is the isomorphism case of the Stacks functoriality remark ("it is in general not the case that the pushforward of a quasi-coherent module is quasi-coherent"; here it holds because φ is an isomorphism, equivalently $\varphi_* = (\varphi^{-1})^*$ is a pullback). $\square$

Lemma 462 (The image of an affine open under an open immersion is affine)**.** *Provided by Mathlib. Let $f : X \to Y$ be an open immersion of schemes (in particular an isomorphism) and U an open of X. Then the image open $f''\, U$ is affine if and only if U is affine. In particular the image V' of an affine open V under a scheme isomorphism is an affine open.*

Lemma 463 (Ext transport along the whole-scheme spectrum equivalence (Need #1))**.** *Let U be a scheme with $[\mathrm{IsAffine}\ U]$, let $V \subseteq U$ be an affine open, and let $\mathcal{H}$ be a quasi-coherent $\mathcal{O}_U$-module. The whole-scheme isomorphism $U \cong \mathrm{Spec}\,\Gamma(U, \mathcal{O}_U)$ (Lemma 420) induces, for every q, an isomorphism*

$$\mathrm{Ext}^q_U(j_!\mathcal{O}_V, \mathcal{H}) \;\cong\; \mathrm{Ext}^q_{\mathrm{Spec}\,\Gamma U}(j_!\mathcal{O}_{V'}, \mathcal{H}'),$$

where V' is the affine open image of V under the isomorphism and $\mathcal{H}'$ is the transported (still quasi-coherent) coefficient. In particular the residual over U vanishes iff its transport over $\mathrm{Spec}\,\Gamma U$ does.

Proof. Since U is affine, Scheme. isoSpec (Lemma 420) gives a genuine isomorphism of schemes $\varphi : U \cong \operatorname{Spec}\Gamma(U, \mathcal{O}_U)$. Assemble from the pushforward functors and their coherences (Lemma 422) the equivalence of module categories

$$\Phi = \text{pushforwardEquivOfIso}\,\varphi : U.\text{Modules} \;\simeq\; (\operatorname{Spec}\Gamma U).\text{Modules},$$

with Φ.functor the pushforward along φ.hom and its quasi-inverse the pushforward along φ.inv; the unit and counit are supplied by the composition and identity coherences. An equivalence is exact and additive, so Ext. mapExactFunctor (Lemma 421) applies and yields a map on Ext; since Φ.functor, being one half of an equivalence, preserves injective objects, its induced Ext-map is bijective (using enough injectives on U.Modules), giving an additive isomorphism. This gives the displayed isomorphism once the two arguments are identified. For the first argument, a scheme isomorphism carries the corepresenting object along (Lemma 435): $\Phi.\text{functor}(j_!\mathcal{O}_V) \cong j_!\mathcal{O}_{V'}$, where $V' = \varphi.\text{hom}\,''\,V$ is the image, an affine open of $\operatorname{Spec}\Gamma U$ by Lemma 462. For the second, $\Phi.\text{functor}(\mathcal{H}) = \mathcal{H}'$ is again quasi-coherent because pushforward along an isomorphism preserves quasi-coherence (Lemma 461). ☐

Lemma 464 (Serre Ext-vanishing over an affine scheme via the spectrum transport)**.** *Let U be a scheme with chosen injective resolutions on U.*Modules*, let $\varphi : U \cong \operatorname{Spec} R$ be an isomorphism of schemes, $\Phi =$* pushforwardEquivOfIso φ *the induced equivalence of module categories, $V \subseteq U$ an open and $\mathcal{H}$ an $\mathcal{O}_U$-module. Suppose $\Phi.\text{functor}(j_!\mathcal{O}_V) \cong j_!\mathcal{O}_{V'}$ for some* affine *open $V' \subseteq \operatorname{Spec} R$ (Lemma 435) and that $\Phi.\text{functor}(\mathcal{H})$ is quasi-coherent (Lemma 461). Then for every $q \geq 1$ every class $e \in \operatorname{Ext}^q_U(j_!\mathcal{O}_V, \mathcal{H})$ vanishes.*

Proof. The chosen injective resolutions provide enough injectives on U.Modules (Lemma 412), which transport across the equivalence Φ to enough injectives on $(\operatorname{Spec} R)$.Modules. Over $\operatorname{Spec} R$ the general-affine-open Serre vanishing (Lemma 434), applied to the affine open V' and the quasi-coherent coefficient $\Phi.\text{functor}(\mathcal{H})$, makes $\operatorname{Ext}^q(j_!\mathcal{O}_{V'}, \Phi.\text{functor}\,\mathcal{H})$ a subsingleton. Transferring along the transport isomorphism $\Phi.\text{functor}(j_!\mathcal{O}_V) \cong j_!\mathcal{O}_{V'}$ in the first Ext-argument (Lemma 411) shows $\operatorname{Ext}^q(\Phi.\text{functor}(j_!\mathcal{O}_V), \Phi.\text{functor}\,\mathcal{H})$ is a subsingleton as well. The spectrum Ext-transport additive isomorphism

$$\operatorname{Ext}^q_U(j_!\mathcal{O}_V, \mathcal{H}) \;\cong\; \operatorname{Ext}^q_{\operatorname{Spec} R}(\Phi.\text{functor}(j_!\mathcal{O}_V), \Phi.\text{functor}\,\mathcal{H})$$

(Lemma 463) is injective, so it carries e into a subsingleton and forces $e = 0$. ☐

Lemma 465 (Sheafification of a sectionwise-locally-zero presheaf vanishes)**.** *Let J be a Grothendieck topology on a site $\mathcal{C}$ and Q a presheaf of abelian groups. Suppose that for every object U and every section $s \in Q(U)$ there is a covering sieve $S \in J(U)$ on which s restricts to 0 — i.e. $Q(g)(s) = 0$ for every arrow $g : V \to U$ in S. Then the sheafification of Q is the zero sheaf.*

This is the sectionwise *(local-injectivity) strengthening of Lemma 336. That lemma requires the* objectwise *vanishing $Q(V) = 0$ for every member V of a single covering sieve, which forces the vanishing to hold on a downward-closed family. The present hypothesis only asks each individual section to die on* some *covering sieve, so the witnessing sieve may be generated by a cofinal* basis *of good opens (such as the affine opens, where the relevant section group already vanishes) even though that basis is not downward closed — exactly the situation of the relative-affine vanishing argument, where the affine opens are not downward closed in the opens topology.*

Proof. Compare Q with the constant zero presheaf Z via the zero map $0 : Q \to Z$. The hypothesis, applied to a difference of sections $x - y$, shows 0 is locally injective; local surjectivity is automatic because the target is objectwise a zero object. Hence $0 : Q \to Z$ is locally bijective, i.e. lies in the local-isomorphism class $J.W$, and Lemma 334 collapses the sheafification of Q to the zero sheaf. ☐

Lemma 466 (Pullback of a quasi-coherent module along an open immersion is quasi-coherent)**.** *Let $\iota : U \hookrightarrow X$ be the inclusion of an open subscheme and let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Then the pullback $\iota^*\mathcal{F}$ — equivalently, by Lemma 396, the restriction of $\mathcal{F}$ to U — is a quasi-coherent $\mathcal{O}_U$-module.*

Proof. This is the general-open-immersion analogue of Lemma 264 and its quasi-coherence corollary Lemma 298, where the affine open $D(g)$ is replaced by an arbitrary open $U \subseteq X$. Quasi-coherence is local on the base, so it suffices to verify it on the members of an open cover; we cover X by affine opens and check $\iota^*\mathcal{F}$ on each, then conclude by the cover-to-top criterion. On an affine chart, the site-theoretic equivalence between $\mathcal{O}_U$-modules and sheaves over the localized slice category $(\mathrm{Opens}\, X)/U$ allows the quasi-coherence presentation of $\mathcal{F}$ to be restricted: the induced over-presentation of $\mathcal{F}$ over U transports, slice by slice, to a presentation of $\mathcal{F}|_U$ over the chart, exhibiting it as quasi-coherent there. The canonical isomorphism between $\mathcal{F}|_U$ and the pullback $\iota^*\mathcal{F}$ supplied by Lemma 396, together with coherence of the relevant unit maps, makes the transport well-defined; transporting the quasi-coherence property across this isomorphism yields quasi-coherence of $\iota^*\mathcal{F}$. ☐

Lemma 467 (Each Čech term is f_*-acyclic)**.** *Source: Stacks Project, Cohomology of Schemes,* `lemma-relative-affine-vanishing` *(relative affine vanishing). Let $f : X \to S$ be a quasi-compact morphism of schemes with* both *the source X and the target S separated, let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module, and let $\mathfrak{U}$ be a finite affine open cover of X (so, by separatedness of X, every intersection $U_{i_0\cdots i_p}$ is again affine). Assume moreover that for every multi-index $\sigma : \{0, \dots, p\} \to I$ the category of $\mathcal{O}$-modules on the intersection subscheme $U_\sigma = U_{\sigma(0)\dots\sigma(p)}$ admits injective resolutions (hypothesis* hres*). Then for every $p \geq 0$ the degree-p Čech term $\mathcal{C}^p = \prod_{i_0\cdots i_p} (j_{i_0\cdots i_p})_*(\mathcal{F}|_{U_{i_0\cdots i_p}})$ is right $(\,\cdot\,)_*$-acyclic for the pushforward f_*:*

$$R^k f_*(\mathcal{C}^p) = 0 \qquad \text{for all } k \geq 1.$$

Two hypotheses beyond the bare "f separated and quasi-compact" are genuinely required. First, the target *S must be separated: the structure map $f \circ j_\sigma : U_\sigma \to S$ from the affine U_σ is an affine morphism only because a morphism from an affine scheme to a separated scheme is affine (Stacks Project, Tag 01SG). Over a non-separated base this fails — for the affine line with doubled origin S and $f = \mathrm{id}$, the inclusion of a standard affine chart is not an affine morphism, and termwise acyclicity breaks. Second, the higher direct images are taken in the derived-functor sense, which presupposes enough injectives; since this is not available for* QCoh *in general, the existence of injective resolutions on each intersection subscheme U_σ is carried as the explicit hypothesis* hres *(the relative analogue of asking that* $\mathrm{QCoh}(X)$ *have injective resolutions).*

Proof. The degree-p Čech term is identified, via the push–pull product decomposition of Lemma 371, with the finite product $\mathcal{C}^p \cong \prod_\sigma (j_\sigma)_*(\mathcal{F}|_{U_\sigma})$ over the multi-indices σ, with each U_σ affine. A finite product of right f_*-acyclic objects is right f_*-acyclic (Lemma 469), so it suffices to treat a single factor $(j_s)_*(\mathcal{F}|_{U_s})$, where $s = (i_0, \dots, i_p)$, $U_s = U_{i_0\cdots i_p}$ is affine, and $j_s : U_s \hookrightarrow X$ is the open immersion. Acyclicity for f_* is a local question on S: the derived pushforward $R^k f_* \mathcal{G}$ is the sheaf associated to the presheaf $V \mapsto H^k(f^{-1}(V), \mathcal{G}|_{f^{-1}(V)})$, so it vanishes for $k \geq 1$ iff that cohomology vanishes for all affine $V \subseteq S$. Restricting to such a $V = \mathrm{Spec}(A)$, the base change $f_V : f^{-1}(V) \to V$ is again separated and quasi-compact over the separated base V, hence $f^{-1}(V)$ is separated and the affine U_s meets it in the affine open $U_s \cap f^{-1}(V)$.

Now decompose the structure map $g_s := f \circ j_s : U_s \to S$. The open immersion $j_s : U_s \hookrightarrow X$ is the inclusion of an affine open into the separated scheme X, hence an affine morphism. Composing, g_s is a morphism from the affine scheme U_s to the separated scheme S, and a morphism from an affine scheme to a separated scheme is affine (Stacks Project, Tag 01SG); this is exactly the point at which separatedness of the *target* S enters, and it fails over a non-separated base. The single factor of the product is the pushforward $(j_s)_*(\mathcal{F}|_{U_s})$, whose restriction $\mathcal{F}|_{U_s}$ is quasi-coherent on U_s by Lemma 466; identifying its higher direct image under f_* with the higher direct image of the quasi-coherent sheaf $\mathcal{F}|_{U_s}$ along the affine morphism g_s, the relative affine vanishing gives

$$R^k f_*((j_s)_*(\mathcal{F}|_{U_s})) \;\cong\; R^k(g_s)_*(\mathcal{F}|_{U_s}) = 0 \qquad (k \geq 1).$$

That relative vanishing is itself a consequence of Serre vanishing (Lemma 228): $R^k(g_s)_*\mathcal{G}$ is the sheafification of $V \mapsto H^k(g_s^{-1}(V), \mathcal{G})$ (Lemma 408), and for affine V the preimage $g_s^{-1}(V)$ is affine — precisely because g_s is an affine morphism — on which higher cohomology of a quasi-coherent sheaf vanishes. The derived-functor formation of these higher direct images uses the injective resolutions supplied by the hypothesis hres on each intersection subscheme. Combining, $R^k f_*(\mathcal{C}^p) = 0$ for all $k \geq 1$. □

Lemma 468 (Čech complex computes the higher direct images)**.** *Source: Stacks Project, Cohomology of Schemes, Tag 02KE (`lemma-cech-cohomology-quasi-coherent`) and `lemma-quasi-coherence-higher-direct-images-application`. Let $f : X \to S$ be a quasi-compact* separated *morphism of schemes with both the target S and the source X separated, let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module, and let $\mathfrak{U} : X = \bigcup_{i \in I} U_i$ be a finite open cover all of whose members U_i are affine (the hypothesis* h$\mathfrak{U}$*; so, by separatedness of X, every intersection $U_{i_0 \cdots i_p}$ is affine). Assume the category of $\mathcal{O}_X$-modules admits injective resolutions, and that for every degree n and every multi-index $\sigma : \{0, \ldots, n\} \to I$ the category of $\mathcal{O}$-modules on the intersection subscheme U_σ admits injective resolutions (the hypothesis* hres *inherited from Lemma 467). Then the cohomology sheaves of the relative Čech complex $\check{\mathcal{C}}^\bullet(\mathfrak{U}, \mathcal{F})$ of Definition 173 compute the higher direct images: for every $i \geq 0$ there is a canonical isomorphism of $\mathcal{O}_S$-modules*

$$H^i(\check{\mathcal{C}}^\bullet(\mathfrak{U}, \mathcal{F})) \;\cong\; R^i f_*\mathcal{F}.$$

In particular, over an affine base $S = \operatorname{Spec}(A)$, taking global sections gives $H^i(X, \mathcal{F}) = \check{H}^i(\mathfrak{U}, \mathcal{F}) = H^0(S, R^i f_*\mathcal{F})$ as A-modules for all i.

Scope. *This is the X-separated specialization of Tag 02KE: routing termwise acyclicity through $R^q(j_\sigma)_* = 0$ for the* affine *morphisms j_σ genuinely needs every j_σ affine, i.e. X separated, which is strictly stronger than the literal Stacks hypothesis "all finite intersections affine". The separatedness of X is in fact a consequence of that of f and S (the structure morphism $X \to \operatorname{Spec} \mathbb{Z}$ factors through f and separatedness is stable under composition); the formalization nonetheless carries X separated as an explicit, redundant hypothesis to match the producer Lemma 467. Without separatedness and affineness the statement fails: a single-element cover $\mathfrak{U} = \{X\}$ with X non-affine already breaks termwise acyclicity (e.g. $X = \mathbb{P}^1$, $\mathcal{F} = \mathcal{O}(-2)$: $H^i(\check{\mathcal{C}}^\bullet) = 0$ but $R^1 f_*\mathcal{F} \neq 0$).*

Two points about the precise form of the comparison should be noted. First, the derived-functor higher direct image $R^i f_*\mathcal{F}$ on the right-hand side is the one obtained from injective resolutions, which is available only when the category of $\mathcal{O}_X$-modules has enough injectives; accordingly this comparison is asserted under the hypothesis that injective resolutions exist in $\operatorname{QCoh}(X)$*, and it compares the (always-defined) Čech higher direct image $H^i(\check{\mathcal{C}}^\bullet(\mathfrak{U}, \mathcal{F}))$ against the derived-functor one only in that case. Second, we assert the comparison in the weak form of the* existence *of an isomorphism, i.e. as $(H^i(\check{\mathcal{C}}^\bullet(\mathfrak{U}, \mathcal{F})) \cong R^i f_*\mathcal{F})$ without singling out a chosen natural isomorphism. This existence statement is all the downstream consumers (representability of the relative Picard functor, local triviality of the comparison) require: they use only that the unconditional Čech $R^i f_*$ and the derived-functor $R^i f_*$ agree as objects, never a specific identification, so committing* *to a canonical natural transformation would add coherence obligations that play no role in the applications.*

Proof. Apply the abstract Lemma 171 with the right-derivable additive functor $G = f_*$ (the pushforward $\mathrm{QCoh}(X) \to \mathrm{QCoh}(S)$), the object $A = \mathcal{F}$, and the resolution $J^\bullet = \mathcal{C}^\bullet$ given by the Čech nerve of Definition 172. The two hypotheses of the abstract lemma are exactly the geometric sub-lemmas established above:

1. By Lemma 337 the augmented Čech complex $0 \to \mathcal{F} \to \mathcal{C}^0 \to \mathcal{C}^1 \to \cdots$ is exact, i.e. $\mathcal{C}^\bullet$ is a resolution of $\mathcal{F}$ in $\mathrm{QCoh}(X)$.

2. By Lemma 467 each term $\mathcal{C}^p$ is right f_*-acyclic: $(f_*).\mathrm{rightDerived}\, k\, (\mathcal{C}^p) = 0$ for $k \geq 1$.

Lemma 171 then gives, for every i, an isomorphism between the i-th derived functor evaluated at $\mathcal{F}$ and the i-th cohomology of the complex obtained by applying f_* to the resolution,

$$(f_*).\mathrm{rightDerived}\, i\, (\mathcal{F}) \;\cong\; H^i(f_*\, \mathcal{C}^\bullet).$$

By construction (Definition 173) the complex $f_*\, \mathcal{C}^\bullet$ is precisely the relative Čech complex $\check{\mathcal{C}}^\bullet(\mathfrak{U}, \mathcal{F})$, so its i-th cohomology is $H^i(\check{\mathcal{C}}^\bullet(\mathfrak{U}, \mathcal{F}))$; and $(f_*).\mathrm{rightDerived}\, i\, (\mathcal{F})$ is the derived-functor higher direct image $R^i f_* \mathcal{F}$ of Definition 147. This yields the isomorphism $H^i(\check{\mathcal{C}}^\bullet(\mathfrak{U}, \mathcal{F})) \cong R^i f_* \mathcal{F}$ asserted by the lemma; taking it in the existence (Nonempty) form discards the choice of resolution, as required. Over an affine base $S = \mathrm{Spec}(A)$, passing to global sections recovers $H^i(X, \mathcal{F}) = \check{H}^i(\mathfrak{U}, \mathcal{F}) = H^0(S, R^i f_* \mathcal{F})$. ☐

Lemma 469 (A finite product of right-f_*-acyclic objects is right-f_*-acyclic)**.** *Project-local. Let $G : \mathcal{A} \to \mathcal{B}$ be an additive functor between abelian categories that admits a right-derived functor, and let $(X_s)_{s \in \Sigma}$ be a* finite *family of objects of $\mathcal{A}$, each right G-acyclic — $G.\mathrm{rightDerived}\, k\, (X_s) = 0$ for all $k \geq 1$ and all s. Then the product $\prod_{s \in \Sigma} X_s$ is right G-acyclic: $G.\mathrm{rightDerived}\, k\, (\prod_s X_s) = 0$ for all $k \geq 1$.*

Proof. A finite product in an abelian category is a finite biproduct, and an additive right-derived functor preserves finite biproducts; hence for every k there is an isomorphism

$$G.\mathrm{rightDerived}\, k \left(\prod_s X_s \right) \;\cong\; \prod_s G.\mathrm{rightDerived}\, k\, (X_s).$$

For $k \geq 1$ each factor on the right vanishes by the acyclicity hypothesis, so the product vanishes, and the isomorphism forces the left-hand side to vanish as well. ☐

Lemma 470 (Pushforward commutes with the Čech complex functor)**.** *Project-local. Let $f : X \to S$ be a morphism of schemes, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$-module and $\mathfrak{U}$ a finite affine open cover of X. There is a canonical isomorphism of cochain complexes in* $S.\mathrm{Modules}$

$$(f_*).\mathrm{mapHomologicalComplex}(\mathcal{C}^\bullet(\mathfrak{U}, \mathcal{F})) \;\cong\; \check{\mathcal{C}}^\bullet(\mathfrak{U}, \mathcal{F})$$

between the termwise image under the direct image f_* of the un-augmented Čech complex on X of Definition 329 and the relative Čech complex of Definition 173.

Proof. The direct image f_* is an additive functor, so it commutes with the functor that assembles a cosimplicial module into its alternating-coface cochain complex. Concretely, in each degree p both complexes have the same object, the image $f_*(\mathcal{C}^p)$ of the degree-p Čech term; the degree-p differential of the relative Čech complex is exactly the image under f_* of the degree-p differential of the complex on X. The latter differential is the alternating sum $\sum_j (-1)^j \delta_j$ of the cofaces δ_j, and additivity of f_* — its preservation of finite sums, of integer scalar multiples and of negation — gives

$$f_*\Big(\sum_j (-1)^j \delta_j\Big) \;=\; \sum_j (-1)^j\, f_*(\delta_j).$$

Assembling these degreewise data — the identity on objects together with the commuting squares expressing the differential identities — produces the asserted isomorphism of cochain complexes. ☐

Lemma 471 (From augmented exactness to the acyclic-resolution input data)**.** *Project-local. Suppose the augmented Čech complex of Definition 333 is exact (its homology vanishes in every degree; Lemma 337). Then the un-augmented complex $\mathcal{C}^\bullet$ on X of Definition 329 carries exactly the data required to feed the abstract acyclic-resolution lemma:*

1. *an isomorphism $e : \mathcal{F} \cong (\mathcal{C}^\bullet).\mathrm{cycles}\,0$ identifying $\mathcal{F}$ with the 0-cocycles, and*
2. *exactness in every positive degree, $(\mathcal{C}^\bullet).\mathrm{ExactAt}\,(n+1)$ for all n.*

Proof. The augmented complex is the complex obtained by prepending $\mathcal{F}$ to $\mathcal{C}^\bullet$ along the augmentation ε: its degree-0 term is $\mathcal{F}$ and its degree-$(p+1)$ term is $\mathcal{C}^p$. This augmentation shifts homological degree by one, so the homology of the augmented complex at degree $n+2$ coincides with the homology of the un-augmented complex $\mathcal{C}^\bullet$ at degree $n+1$. Vanishing of the augmented homology in these degrees therefore gives $(\mathcal{C}^\bullet).\mathrm{ExactAt}\,(n+1)$ for every n, which is (2).

For (1) consider the bottom of the augmented complex, $0 \to \mathcal{F} \xrightarrow{\varepsilon} \mathcal{C}^0 \xrightarrow{d^0} \mathcal{C}^1$. Vanishing of the augmented homology in degree 0 says that ε has trivial kernel, i.e. ε is a monomorphism; vanishing in degree 1 says that the image of ε is exactly the kernel of d^0, i.e. ε is an epimorphism onto the 0-cocycles $(\mathcal{C}^\bullet).\mathrm{cycles}\,0 = \ker d^0$. A monomorphism that is also an epimorphism onto cycles 0 is an isomorphism, giving the desired $e : \mathcal{F} \cong (\mathcal{C}^\bullet).\mathrm{cycles}\,0$. ☐

8.7 The unconditional higher direct image

Definition 472 (Unconditional higher direct image via Čech)**.** *Source: Stacks Project, Cohomology of Schemes, `lemma-quasi-coherence-higher-direct-images-application`; unconditional Čech packaging is Archon-original.* Let $f : X \to S$ be a separated quasi-compact morphism and $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$-module. Fix a finite affine open cover $\mathfrak{U}$ of X. The *(unconditional) i-th higher direct image* is defined, for each $i \geq 0$, as the i-th cohomology sheaf of the relative Čech complex,

$$R^i f_* \mathcal{F} \;:=\; H^i(\check{\mathcal{C}}^\bullet(\mathfrak{U}, \mathcal{F})) \in \mathrm{QCoh}(S).$$

This requires *no* hypothesis that the category of $\mathcal{O}_X$-modules have enough injectives: the right-hand side is the cohomology of an explicit complex of quasi-coherent sheaves. By Lemma 468 the result is canonically isomorphic to the derived-functor higher direct image wherever the latter is defined, and in particular is independent of the chosen affine cover $\mathfrak{U}$ up to canonical isomorphism. For $i = 0$ one recovers the ordinary pushforward $R^0 f_* \mathcal{F} = f_* \mathcal{F}$.

8.8 The relative Čech-to-derived-pushforward spectral sequence

This section assembles the relative Čech-to-derived-pushforward spectral sequence

$$E_2^{p,q} = \check{H}^p(\mathfrak{U}, R^q f_* \mathcal{F}) \implies R^{p+q} f_* \mathcal{F},$$

functorial in $\mathcal{F}$, for a separated quasi-compact morphism $f : X \to S$ and a finite affine open cover $\mathfrak{U}$ of X. It is the relative (over the base S) version of the absolute Čech-to-cohomology spectral sequence; the absolute statement is cited verbatim in 479, the relativization replacing the global sections functor by f_* and absolute cohomology H^q by the derived pushforward $R^q f_*$.

Mathlib supplies only the *abstract* spectral-sequence scaffold (the total-complex and spectral-object API recorded as Mathlib anchors below); it does *not* contain the Grothendieck spectral sequence as a black box. The construction is therefore decomposed into the bicomplex of a Čech complex of an injective resolution, its total complex, the two filtrations whose associated graded pages give respectively $R^n f_* \mathcal{F}$ and $\check{H}^p(\mathfrak{U}, R^q f_* \mathcal{F})$, and the final assembly through the Mathlib bridge.

8.8.1 Mathlib anchors for the abstract spectral-sequence scaffold

Lemma 473 (Total complex of a bicomplex)**.** *Provided by Mathlib (`Mathlib.Algebra.Homology.TotalComplex`). For a double complex K in a preadditive category with the requisite coproducts, Mathlib forms the associated total complex* $\mathrm{Tot}(K)$ *and, for an isomorphism $e : K \cong L$ of double complexes, a functorial isomorphism* $\mathrm{Tot}(K) \cong \mathrm{Tot}(L)$ *of the totals. This is the total-complex half of the abstract scaffold on which the spectral object below is built.*

Lemma 474 (Spectral sequence of a spectral object)**.** *Provided by Mathlib (`Mathlib.Algebra.Homology.SpectralOb` `Mathlib.Algebra.Homology.SpectralSequence.Basic`). A* spectral object *in an abelian category C (Mathlib's* SpectralObject*) packages the filtration data of a filtered cochain complex; from one, together with a page datum, Mathlib produces a genuine spectral sequence via* SpectralObject. spectralSequence, *and in the cohomological-quadrant indexing it is a* CohomologicalSpectralSequenceNat*. Applied to the two filtrations of a total complex this is exactly the abstract machine that turns a filtered total complex into a convergent first-quadrant spectral sequence; the project supplies the filtered total complex (476) and the page identifications (477, 478).*

8.8.2 The bicomplex and its total complex

Definition 475 (Čech bicomplex of an injective resolution)**.** *Source: Stacks Project, Cohomology, Tag 03OW (`lemma-cech-spectral-sequence`); the bicomplex is the explicit model of the Grothendieck spectral sequence used in its proof.* Choose an injective resolution $\mathcal{F} \to \mathcal{I}^\bullet$ in the category of $\mathcal{O}_X$-modules (it exists by 412). Applying the relative Čech-complex construction 184 of the cover $\mathfrak{U}$ to each $\mathcal{I}^q$ yields a double complex

$$C^{p,q} \; = \; \check{\mathcal{C}}^p(\mathfrak{U}, \mathcal{I}^q) \in \mathrm{QCoh}(S),$$

with horizontal differential the Čech (alternating-coface) differential and vertical differential induced by $\mathcal{I}^\bullet$. Each row $C^{\bullet,q}$ is the relative Čech complex of $\mathcal{I}^q$; each column $C^{p,\bullet}$ is f_* of the push–pull nerve term at level p applied to the resolution $\mathcal{I}^\bullet$.

Definition 476 (Total complex of the Čech bicomplex)**.** The total complex $\mathrm{Tot}^\bullet(C^{\bullet,\bullet})$ of the bicomplex 475, formed via Mathlib's $\texttt{HomologicalComplex}_2\texttt{.total}$ (473). Carrying both the row filtration (by the Čech degree p) and the column filtration (by the resolution degree q), it is the single filtered complex whose two filtrations produce the two pages of the spectral sequence.

8.8.3 The two filtrations

Lemma 477 (First filtration: degeneration onto $R^n f_* \mathcal{F}$)**.** *Source: Stacks Project, Cohomology,* `lemma-cech-spectral-sequence-application` *(the degeneration corollary). The filtration of* $\mathrm{Tot}^\bullet(C^{\bullet,\bullet})$ *by the resolution degree q has associated spectral object whose first page is computed by the columns. Because each Čech term is a finite product of pushforwards of injectives along affine intersections, it is f_*-acyclic (467); hence taking column cohomology in the q-direction is concentrated in the resolution direction and recovers $\mathcal{I}^\bullet$ up to homotopy, so the total complex computes the derived pushforward: $H^n(\mathrm{Tot}^\bullet) \cong R^n f_* \mathcal{F}$ with $R^n f_* \mathcal{F}$ as in 472. The exact-functor/homology interchange 482 supplies the termwise identifications.*

Lemma 478 (Second filtration: $E_2 = \check{H}^p(\mathfrak{U}, R^q f_* \mathcal{F})$)**.** *Source: Stacks Project, Cohomology,* `lemma-cech-cohomology-derived-presheaves` *and* `lemma-injective-trivial-cech`*. The filtration of* $\mathrm{Tot}^\bullet(C^{\bullet,\bullet})$ *by the Čech degree p has E_2 page*

$$E_2^{p,q} \;=\; \check{H}^p(\mathfrak{U}, R^q f_* \mathcal{F}).$$

Taking row cohomology first computes, in each column q, the Čech cohomology of the resolution sheaf $\mathcal{I}^q$; since injectives are Čech-acyclic (214, the Grothendieck-SS hypothesis), the presheaf-level Čech functor $\check{H}^0(\mathfrak{U}, -)$ is computed by the Čech complex 197, and its right derived functors are the Čech cohomology functors $\check{H}^p(\mathfrak{U}, -) = R^p \check{H}^0(\mathfrak{U}, -)$ on presheaves (using the exact inclusion i of 210). Applying this with the column cohomology $R^q f_* \mathcal{F}$ (472) in the coefficient slot identifies the second page with $\check{H}^p(\mathfrak{U}, R^q f_* \mathcal{F})$.

8.8.4 Assembly of the spectral sequence

Lemma 479 (Relative Čech-to-derived-pushforward spectral sequence)**.** *Source: Stacks Project, Cohomology, Tag 03OW (*`lemma-cech-spectral-sequence`*, "Čech-to-cohomology spectral sequence"); stated here in its relative form over the base S. Let $f : X \to S$ be separated and quasi-compact, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$-module, and $\mathfrak{U}$ a finite affine open cover of X. There is a first-quadrant cohomological spectral sequence, functorial in $\mathcal{F}$,*

$$E_2^{p,q} = \check{H}^p(\mathfrak{U}, R^q f_* \mathcal{F}) \implies R^{p+q} f_* \mathcal{F},$$

with $R^\bullet f_*$ the unconditional Čech higher direct images of 472.

Proof. Form the Čech bicomplex $C^{\bullet,\bullet}$ of an injective resolution $\mathcal{F} \to \mathcal{I}^\bullet$ (475) and its total complex $\mathrm{Tot}^\bullet$ (476). Feeding the two filtrations of $\mathrm{Tot}^\bullet$ into Mathlib's spectral-object machine (474) yields two spectral sequences with the same abutment $H^{p+q}(\mathrm{Tot}^\bullet)$. The column (resolution-degree) filtration degenerates at its second page onto $R^n f_* \mathcal{F}$ (477), identifying the abutment with $R^{p+q} f_* \mathcal{F}$. The row (Čech-degree) filtration has second page $\check{H}^p(\mathfrak{U}, R^q f_* \mathcal{F})$ (478). The resulting spectral sequence is the asserted one; functoriality in $\mathcal{F}$ is inherited from functoriality of the injective resolution, the Čech complex, and the total-complex/spectral-object constructions. This is the relative incarnation of the Grothendieck spectral sequence for the composite $\check{H}^0(\mathfrak{U}, -) \circ i$ recorded in the source. ∎

8.9 Flat base change for the Čech higher direct images

Lemma 480 (Flat base change of $R^i f_*$ via Čech)**.** *Source: Stacks Project, Cohomology of Schemes, Tag 02KH (`lemma-flat-base-change-cohomology`, "Flat base change"). Consider the cartesian square*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow{g} & S \end{array}$$

with $f : X \to S$ separated and quasi-compact, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$-module, and $\mathcal{F}' = (g')^*\mathcal{F}$. Assume g is flat. Fix a finite affine open cover $\mathfrak{U}$ of X; the cover $\mathfrak{U}'$ of X' is required to be the base change of $\mathfrak{U}$ along g' *(the cover by the preimages $(g')^{-1}(U_i)$) — the isomorphism is constructed only for this matched pair, not for an unrelated cover $\mathfrak{U}'$. Then for every $i \geq 0$ the canonical base-change map between the unconditional Čech higher direct images of Definition 472 is an isomorphism*

$$g^*(R^i f_* \mathcal{F}) \xrightarrow{\sim} R^i f'_*((g')^*\mathcal{F}) = R^i f'_* \mathcal{F}'.$$

Equivalently, when $S = \operatorname{Spec}(A)$ and $S' = \operatorname{Spec}(B)$ with $A \to B$ flat, the comparison map of B-modules $H^i(X, \mathcal{F}) \otimes_A B \to H^i(X', \mathcal{F}')$ is an isomorphism.

Proof. This is the *separated* case of Stacks Tag 02KH. For a separated quasi-compact f the relative Čech complex already computes $R^\bullet f_*$ on the nose, so the separated case needs *no* spectral sequence; the Čech-to-derived-pushforward spectral sequence (479) enters only in the separated $\to$ general quasi-separated promotion (see the remark below). Write $\check{\mathcal{C}}^\bullet = \check{\mathcal{C}}^\bullet(\mathfrak{U}, \mathcal{F})$ for the relative Čech complex in $\mathrm{QCoh}(S)$, so that $R^i f_* \mathcal{F} = H^i(\check{\mathcal{C}}^\bullet)$ by Definition 472. The proof is the two-step composite

$$g^*(H^i(\check{\mathcal{C}}^\bullet)) \xrightarrow[\sim]{(1)} H^i(g^*\check{\mathcal{C}}^\bullet) \xrightarrow[\sim]{(2)} H^i(\check{\mathcal{C}}^\bullet(\mathfrak{U}', \mathcal{F}')) = R^i f'_* \mathcal{F}'.$$

Step (1) is Lemma 494: since g is flat the pullback g^* is exact, hence commutes with taking the i-th cohomology of a complex. Step (2) is Definition 536: applying g^* degreewise to $\check{\mathcal{C}}^\bullet$ recovers the Čech complex of the base-changed data $(\mathfrak{U}', \mathcal{F}')$, the Čech complex transforming tensorially under base change (Stacks Tag 02KG). Functoriality of H^i (`HomologicalComplex.homologyMapIso`) turns the complex isomorphism of (2) into the homology isomorphism. Composing the two gives the claim. □

Remark 481 (Lift to the general quasi-separated case)**.** The separated case above suffices for the curve/Jacobian application, where the structure morphisms in play are separated. The general (quasi-compact, quasi-separated) case of Stacks Tag 02KH is obtained from the separated case by base change functoriality of the spectral sequence 479: the flat morphism g induces, by g^*-exactness, a comparison of the entire E_2 page $g^*\check{H}^p(\mathfrak{U}, R^q f_* \mathcal{F}) \to \check{H}^p(\mathfrak{U}', R^q f'_* \mathcal{F}')$, whose termwise input is the affine base change of 536; an isomorphism on E_2 yields an isomorphism on the abutment $R^{p+q} f_*$, promoting the separated conclusion to the general one. This downstream lift is recorded as a remark only; it is not needed for the present development.

8.10 Homology-side machinery for Čech flat base change

The Čech flat-base-change lemma factors through two ingredients: exactness of the flat pullback g^* (so it commutes with cohomology) and the tensorial base change of the Čech complex itself.

We record both here. The homology-side pieces are already proved; the genuine open content is isolated in the two clearly-flagged gaps below.

Lemma 482 (Exact functor commutes with complex homology)**.** *Let $F : \mathcal{C} \to \mathcal{D}$ be an additive functor between categories with homology that preserves homology (i.e. preserves kernels and cokernels). Then for every cochain complex K and index i there is a canonical isomorphism*

$$H^i((F\text{-applied degreewise to } K)) \;\cong\; F(H^i(K)).$$

Indeed, the short complex defining degree-i homology after applying F is the image under F of the short complex defining $H^i(K)$.

Lemma 483 (Flat pullback preserves finite colimits)**.** *For any morphism $g : S' \to S$ the pullback $g^* : \mathrm{QCoh}(S) \to \mathrm{QCoh}(S')$ preserves finite colimits, being a left adjoint.*

8.10.1 Left exactness of flat pullback

Let φ be the morphism of ring sheaves underlying g. Pullback of sheaves of modules factors through presheaves: one forgets the sheaf condition, applies presheaf pullback, and sheafifies. Left exactness can therefore be checked on these three factors.

Lemma 484 (Factorisation of the sheaf-of-modules pullback)**.** *Provided by Mathlib (`Mathlib.Algebra.Category.Mod`*
For a continuous morphism of ringed sites carried by a ring-sheaf hom φ, the pullback of sheaves of modules factors as the composite of three functors

$$\texttt{SheafOfModules.pullback}\,\varphi \;\cong\; \mathrm{forget} \circ (\texttt{PresheafOfModules.pullback}\,\varphi_{\mathrm{hom}}) \circ \texttt{PresheafOfModules.sheafific}$$

i.e. forget the sheaf condition, take the presheaf-level pullback, then re-sheafify.

Lemma 485 (Forgetful functor is left-exact)**.** *Provided by Mathlib. The forgetful functor* forget *from sheaves of modules to the underlying sheaves of abelian groups preserves finite limits.*

Lemma 486 (Sheafification of presheaves of modules is left-exact)**.** *Provided by Mathlib. Sheafification of presheaves of modules is a left-exact reflector, hence preserves finite limits.*

Lemma 487 (Flat extension of scalars is left-exact)**.** *Provided by Mathlib (`Mathlib.Algebra.Category.ModuleCat.D`*
If $A \to B$ is a flat ring homomorphism then the extension-of-scalars functor extendScalars : $\mathrm{Mod}_A \to \mathrm{Mod}_B$*, $M \mapsto B \otimes_A M$, preserves finite limits.*

The two outer factors are left exact by Lemmas 485 and 486. It remains to treat presheaf pullback.

Definition 488 (Inverse-image model of presheaf pullback)**.** The inverse-image model for presheaf-level pullback along φ_{hom} is the inverse-image functor g^{-1} on presheaves of modules followed by degreewise extension of scalars along the underlying flat ring map, i.e. the composite $g^{-1} \circ$ extendScalars, assembled sectionwise from the presheaf inverse image and extension of scalars.

Lemma 489 (The inverse-image model computes presheaf pullback)**.** *The model $g^{-1} \circ$extendScalars of Definition 488 is left adjoint to `PresheafOfModules.pushforward`φ_{hom}; by uniqueness of adjoints it is therefore canonically isomorphic to `PresheafOfModules.pullback`φ_{hom} = (`PresheafOfModules.pushfor`*
The unit and counit are obtained by composing, sectionwise, the inverse-image/direct-image adjunction with the extension-of-scalars/restriction-of-scalars adjunction. Their triangle identities follow from those of the constituent adjunctions, and uniqueness of left adjoints gives the stated isomorphism.

Lemma 490 (Presheaf inverse image is exact)**.** *The inverse-image functor g^{-1} on presheaves of modules is exact, hence preserves finite limits. It is computed by filtered colimits over the relevant comma categories. Filtered colimits of modules are exact, so they commute with the finite limits and colimits defining exact sequences.*

Lemma 491 (Presheaf pullback is left exact under flatness)**.** *Let φ_{hom} be the ring-sheaf hom underlying a flat morphism g. Then* `PresheafOfModules.pullback` φ_{hom} *preserves finite limits.*

Proof. By Lemma 489 it suffices to prove that the model $g^{-1} \circ \text{extendScalars}$ preserves finite limits. The inverse image g^{-1} is exact by Lemma 490, hence preserves finite limits; the extension of scalars along the flat ring map preserves finite limits by Lemma 487. A composite of two finite-limit-preserving functors preserves finite limits, and a functor isomorphic to such a composite does as well. □

Lemma 492 (Flat pullback is left exact)**.** *If g is flat then g^* preserves finite limits. The flatness-dependent content reduces, via the factorisation of Lemma 484, to the single middle factor (Lemma 491); the two outer factors are left-exact unconditionally.*

Proof. By Lemma 484, on the genuine scheme site g^* factors as the composite forget $\circ$(`PresheafOfModules.pullback` φ `PresheafOfModules.sheafification`. The outer two factors preserve finite limits unconditionally (Lemmas 485 and 486, both supplied by Mathlib on the scheme site). The middle factor preserves finite limits because g is flat (Lemma 491). A composite of finite-limit-preserving functors preserves finite limits, so g^* does. □

Lemma 493 (Flat pullback preserves homology)**.** *For g flat, g^* preserves homology. Derived from left-exactness and left-adjointness via* `Functor.preservesHomologyOfExact`*.*

Lemma 494 (Flat pullback commutes with Čech homology)**.** *For g flat, applying g^* to the i-th cohomology of a cochain complex of $\mathcal{O}_S$-modules agrees with the i-th cohomology of the degreewise pullback complex: $g^*(H^i(K)) \cong H^i(g^*K)$. A direct specialisation of Lemma 482 to g^*.*

8.10.2 The Čech base-change isomorphism via a cosimplicial natural iso

By Definition 184, $\check{\mathcal{C}}^\bullet(\mathfrak{U}, \mathcal{F}) = \text{alternatingCofaceMapComplex}((\text{pushforward } f) \circ \text{CechNerve}(\mathfrak{U}, \mathcal{F}))$, whose degree-$p$ term is $(\text{pushforward } f)$ of a push–pull nerve object and whose differentials are $(\text{pushforward } f)$ of cosimplicial coface maps. A natural isomorphism of the underlying cosimplicial objects therefore induces an isomorphism of Čech complexes by functoriality of alternatingCofaceMapComplex.

Lemma 495 (Additive functors commute with the alternating-coface differential)**.** *For an additive functor F between preadditive categories, F applied to the degree-p alternating-coface differential* objD *of a cosimplicial object C agrees with the alternating-coface differential of $F \circ C$: the differential is an alternating sum of coface maps, and F preserves finite sums and carries each coface map to the corresponding coface map of $F \circ C$, so the two differentials coincide termwise.*

Lemma 496 (Flat pullback commutes with the alternating-coface complex)**.** *For an additive functor F between preadditive categories, F applied degreewise commutes with the alternating-coface-map cochain complex functor: there is a natural isomorphism* $F(\text{alternatingCofaceMapComplex}(C)) \cong \text{alternatingCofaceMapComplex}(F \circ C)$*, natural in the cosimplicial object C. In particular this holds for $F = g^*$, which is additive. The degree-p terms agree canonically (F applied to the degree-p term of C); the differential is an alternating sum of coface maps, and since F is additive* *it commutes with finite sums and with each coface map, so the term-wise identifications assemble into a chain isomorphism.*

Lemma 497 (Degreewise affine reduction of the Čech base change)**.** *Fix a cosimplicial degree p. The $(p+1)$-fold fibre power $U_{i_0\cdots i_p} = U_{i_0} \times_X \cdots \times_X U_{i_p}$ appearing in the Čech nerve is, on the standard affine model of the cover,* Spec *of the finite affine intersection $A_{i_0} \otimes_R \cdots \otimes_R A_{i_p}$ of the coordinate rings of the members $U_{i_j} = \operatorname{Spec} A_{i_j}$. On that affine model the X-level cartesian square defining the base change along g' restricts to the affine pushout (tensor) square of rings, so the degreewise Beck–Chevalley comparison $g'^*(p_*p^*F) \cong p'_*p'^*(g'^*F)$ over $U_{i_0\cdots i_p}$ IS the affine termwise base change* affinePushforwardPullbackBaseChange *(129). These affine identifications are natural with respect to the index-omission maps that generate the cosimplicial structure of the nerve.*

Proof. Spec-reduction. On the standard affine model the members of the cover are $U_{i_j} = \operatorname{Spec} A_{i_j}$ over the affine base $\operatorname{Spec} R$, and the intersection $U_{i_0\cdots i_p}$ — a fibre power over X — is, by the affine description of fibre products, Spec of the iterated tensor product $A_{i_0} \otimes_R \cdots \otimes_R A_{i_p}$. The base change g' is induced by a ring map $R \to R'$; pulling back along g' replaces this affine intersection by Spec of $(A_{i_0} \otimes_R \cdots \otimes_R A_{i_p}) \otimes_R R'$. Thus the X-level cartesian square restricts, over each intersection, to the affine pushout square of rings, and the degreewise base-change isomorphism is by construction affinePushforwardPullbackBaseChange (129); the latter is assembled from the tilde dictionaries 123, 126 and the cancellation cancelBaseChange (127), *not* from the canonical adjoint mate.

Index-omission naturality. Each coface map of the Čech nerve (183) omits one index i_j, realised on rings as the inclusion $A_{i_0} \otimes_R \ldots \widehat{A_{i_j}} \cdots \otimes_R A_{i_p} \to A_{i_0} \otimes_R \cdots \otimes_R A_{i_p}$ (insertion of the missing tensor factor); codegeneracies repeat an index dually. The affine base-change isomorphism affinePushforwardPullbackBaseChange is natural in the ring under consideration, so the degreewise comparisons commute with these omission/insertion maps and assemble into a morphism of cosimplicial objects. This is exactly the per-degree data that both 518 and the twisted-nerve identification consume. $\square$

8.10.3 The abstract-to-affine bridge for the base-change leaves

The brick 497 speaks the affine $\widetilde{(-)}$-model over Spec: it compares $g^*((\operatorname{Spec}\varphi)_*\widetilde{N})$ with $(\operatorname{Spec}\varphi')_*(g'^*\widetilde{N})$ for an explicit module N. The two leaves below, however, present each degree as the *abstract* push–pull sheaf pushPullObj $\mathcal{F}\, Y_n = (p_n)_*\, p_n^*\mathcal{F}$ (176) over a fibre power Y_n, with no $\widetilde{(-)}$ in sight. The present subsection supplies the missing identification, at the two well-typed altitudes at which it can be stated, and it is what lets each leaf hand its degreewise square to the affine brick. The bridge rests on four already-available ingredients, recorded first: the affine structure theorem in counit form, preservation of quasi-coherence by pullback and by affine pushforward, and the affineness of fibre powers of an affine open-immersion cover of a separated scheme.

Lemma 498 (The counit-iso criterion in localizing form)**.** *Provided by Mathlib. For an $\mathcal{O}_{\operatorname{Spec} R}$-module $\mathcal{F}$, the tilde–Γ counit (the canonical map $\widetilde{\Gamma(\mathcal{F})} \to \mathcal{F}$) is an isomorphism if and only if $\mathcal{F}$ is* localizing*: for every $f \in R$ the section-restriction map $\Gamma(\operatorname{Spec} R, \mathcal{F}) \to \Gamma(D(f), \mathcal{F})$ exhibits its target as the localization of the global sections at the powers of f. Combined with the keystone 287, this yields the unconditional 01I8 instance 288 directly.*

Lemma 499 (Affine pushforward preserves the localizing property)**.** *Provided by Mathlib. Let $\varphi : R \to A$ be a ring homomorphism and let $\mathcal{G}$ be a localizing (equivalently, $\widetilde{(-)}$-essential-image)*

$\mathcal{O}_{\operatorname{Spec} A}$*-module. Then the pushforward* $(\operatorname{Spec}\varphi)_*\mathcal{G}$ *along the affine morphism* $\operatorname{Spec}\varphi : \operatorname{Spec} A \to \operatorname{Spec} R$ *is again localizing; equivalently its tilde–*Γ *counit is an isomorphism, so* $(\operatorname{Spec}\varphi)_*\widetilde{N} \cong \widetilde{N|_R}$ *with* $N|_R$ *the restriction of scalars of* N *along* φ.

Lemma 500 (Finite intersections of affine opens of a separated scheme are affine)**.** *Provided by Mathlib. Let* X *be a scheme whose diagonal is affine — in particular any* separated *scheme. Then any finite intersection* $U_{i_0} \cap \cdots \cap U_{i_p}$ *of affine open subschemes of* X *is again an affine open. Consequently, for an affine open-immersion cover* $\mathfrak{U} = \{U_i = \operatorname{Spec} A_i\}$ *of a separated scheme, every* $(p+1)$*-fold fibre power* $Y_n = U_{i_0} \times_X \cdots \times_X U_{i_p} = U_{i_0} \cap \cdots \cap U_{i_p}$ *of the Čech nerve is affine,* $Y_n = \operatorname{Spec} A_n$.

Lemma 501 (Čech intersection opens are affine, separated case)**.** *Let* $f : X \to S$ *be a* separated *morphism with affine base* $S = \operatorname{Spec} R$*, and let* $\mathfrak{U} = \{U_i = \operatorname{Spec} A_i\}$ *be an affine open-immersion cover of* X*. Then for every* finite, nonempty *index family* $\sigma : \kappa \to I$ *the Čech intersection open* $\operatorname{coverInterOpen} \mathfrak{U}\, \sigma = \bigcap_k (U_{\sigma(k)})$ *— the meet of the ranges of the cover members indexed by* σ *— is an affine open subscheme of* X.

Proof. Since $S = \operatorname{Spec} R$ is affine it is a separated scheme, so its structure morphism to the terminal scheme is separated. The structure morphism of X factors as f followed by that of S; a composite of separated morphisms is separated, so the structure morphism of X is separated, i.e. X is a separated scheme. Separatedness says the diagonal $\Delta_X : X \to X \times X$ is a closed immersion, in particular an affine morphism, so X has affine diagonal. Consequently any finite *nonempty* intersection of affine opens of X is again affine — this is the indexed-meet form of 500, where each successive meet is an affine open because it is cut out by the affine diagonal. Applying it to the nonempty finite family $\{U_{\sigma(k)}\}_k$ shows $\operatorname{coverInterOpen} \mathfrak{U}\, \sigma$ is affine. (Nonemptiness of κ is load-bearing: an empty meet would be all of X, which need not be affine.) ☐

Lemma 502 (Pullback preserves quasi-coherence (open-immersion case))**.** *Source: Stacks Project, Modules on Sites, Tag 01BG,* `lemma-pullback-quasi-coherent`*. Let* $\iota_V : V \hookrightarrow X$ *be the open immersion of an open subscheme* V *of* X*, and let* $\mathcal{F}$ *be a quasi-coherent* $\mathcal{O}_X$*-module. Then the restriction pullback* $\iota_V^*\mathcal{F}$ *is a quasi-coherent* $\mathcal{O}_V$*-module. In particular, each fibre power* $Y_n = U_{i_0} \cap \cdots \cap U_{i_p}$ *of the Čech nerve embeds in* X *by an open immersion* $p_n : Y_n \hookrightarrow X$*, so* $p_n^*\mathcal{F}$ *is quasi-coherent on* Y_n.

Proof. This is the open-immersion instance of 466: the preimages of a quasi-coherence cover of X cover V, and quasi-coherence descends along that cover. The Čech case follows since each fibre-power inclusion $p_n : Y_n \hookrightarrow X$ is an open immersion. ☐

Lemma 503 (The opens-preimage comma posets of a continuous map are filtered)**.** *Let* $\varphi : T \to T'$ *be a continuous map of topological spaces and let* d *be an open subset of* T*. The poset of open subsets* $U \subseteq T'$ *with* $d \subseteq \varphi^{-1}(U)$ *is filtered.*

Proof. The poset is nonempty: T' itself satisfies $d \subseteq \varphi^{-1}(T') = T$. It is directed: for U_1, U_2 in the poset the union $U_1 \cup U_2$ again lies in it, since $d \subseteq \varphi^{-1}(U_1) \subseteq \varphi^{-1}(U_1 \cup U_2)$, and it dominates both. A poset is thin, so any parallel pair of maps is already equal, and a nonempty directed thin category is filtered. ☐

Lemma 504 (The opens-preimage functor of a continuous map is final)**.** *For any continuous map* $\varphi : T \to T'$*, the preimage functor* $U \mapsto \varphi^{-1}(U)$*, from open subsets of* T' *to open subsets of* T*, is a final functor.*

Proof. A functor is final as soon as, for every object d of the target, the comma category of maps from d into its image is connected; by 503 these comma categories are filtered, and filtered categories are connected. $\square$

Lemma 505 (Pullback of the structure sheaf)**.** *For any morphism of schemes $g : Y \to X$, the canonical comparison morphism $g^*\mathcal{O}_X \to \mathcal{O}_Y$ of sheaves of $\mathcal{O}_Y$-modules is an isomorphism.*

Proof. By the pullback–pushforward adjunction, morphisms $g^*\mathcal{O}_X \to \mathcal{M}$ correspond to morphisms $\mathcal{O}_X \to g_*\mathcal{M}$, i.e. to global sections of $g_*\mathcal{M}$. Since the opens-preimage functor underlying g is final (504), global sections of $g_*\mathcal{M}$ coincide with global sections of $\mathcal{M}$, which correspond to morphisms $\mathcal{O}_Y \to \mathcal{M}$. Thus $g^*\mathcal{O}_X$ and $\mathcal{O}_Y$ corepresent the same functor, and the comparison morphism realizes this identification, hence is an isomorphism. $\square$

Lemma 506 (Per-slice presentation of a pullback)**.** *Let $g : Y \to X$ be a morphism of schemes, $\mathcal{F}$ an $\mathcal{O}_X$-module, and $A \subseteq X$ an open subset over which $\mathcal{F}$ admits a presentation $\bigoplus_\iota \mathcal{O} \to \bigoplus_\sigma \mathcal{O} \to \mathcal{F}|_A \to 0$. Then the pullback $g^*\mathcal{F}$ admits a presentation over the preimage open $W = g^{-1}(A)$.*

Proof. Let $g_A : W \to A$ denote the restriction of g, so that $\iota_W \circ g = g_A \circ \iota_A$ with ι_A, ι_W the open inclusions. Pseudofunctoriality of the module pullback applied to this square identifies $(g^*\mathcal{F})|_W \cong g_A^*(\mathcal{F}|_A)$. The functor g_A^* is a left adjoint, hence preserves colimits, and it carries the structure sheaf to the structure sheaf (505), hence free modules to free modules; applying it to the given presentation of $\mathcal{F}|_A$ therefore yields a presentation of $g_A^*(\mathcal{F}|_A) \cong (g^*\mathcal{F})|_W$. Carrying this presentation of the restriction back to the slice over W is the same restrict–over bridge used in 466. $\square$

Lemma 507 (Pullback preserves quasi-coherence (general morphism))**.** *Source: Stacks Project, Tag 01BG, `lemma-pullback-quasi-coherent`. Let $g : Y \to X$ be a morphism of schemes and $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$-module. Then $g^*\mathcal{F}$ is a quasi-coherent $\mathcal{O}_Y$-module.*

Proof. Choose an open cover $\{A_i\}$ of X over whose members $\mathcal{F}$ admits presentations. The preimages $g^{-1}(A_i)$ cover Y, and over each of them $g^*\mathcal{F}$ admits a presentation by 506. Quasi-coherence is local on the base, so $g^*\mathcal{F}$ is quasi-coherent. $\square$

Lemma 508 (Abstract push–pull is the affine tilde-model on a fibre power)**.** *Source: Stacks Project, Schemes, Tag 01I8, `lemma-quasi-coherent-affine`. Let $f : X \to S$ be separated with affine base $S = \operatorname{Spec} R$, let $\mathfrak{U} = \{U_i = \operatorname{Spec} A_i\}$ be an affine open-immersion cover of X, and fix a cosimplicial degree, writing Y_n for the corresponding fibre power with projection $p_n : Y_n \to X$. By 500 the separatedness makes $Y_n = \operatorname{Spec} A_n$ affine. There are two well-typed identifications of the abstract push–pull data with the affine $\widetilde{(-)}$-model.*

Altitude 1 (restriction level). *Since the formation of $\widetilde{(-)}$ requires a genuine* Spec*, and the open subscheme Y_n is not literally the spectrum of a ring, we transport along the whole-scheme affine identification $\sigma_n : Y_n \xrightarrow{\sim} \operatorname{Spec}\Gamma(X, Y_n)$. Pushing the restriction $p_n^*\mathcal{F}$ forward along σ_n onto the genuine affine $\operatorname{Spec}\Gamma(X, Y_n)$, and setting $N_n := \Gamma(\operatorname{Spec}\Gamma(X, Y_n),\ (\sigma_n)_* p_n^*\mathcal{F})$,*

$$(\sigma_n)_* p_n^*\mathcal{F} \;\cong\; \widetilde{N_n} \qquad (\text{over } \operatorname{Spec}\Gamma(X, Y_n)).$$

Altitude 2 (assembled pushed-forward level). *Over the affine base S, with $\varphi_n : R \to A_n$ the ring map presenting the composite $f \circ p_n$ as $\operatorname{Spec}\varphi_n$,*

$$(\text{pushforward } f).\text{obj}(\text{pushPullObj } \mathcal{F}\, Y_n) \;\cong\; (\text{pushforward}(\operatorname{Spec}\varphi_n)).\text{obj}(\widetilde{N_n}).$$

The left-hand side is exactly the term $f_*((p_n)_* p_n^* \mathcal{F})$ whose pullback along g appears in the degree-n Beck–Chevalley square; the right-hand side is the affine pushforward of a $\widetilde{(-)}$-module, the form to which 129 applies.

Proof. *Altitude 1.* The restriction $p_n^*\mathcal{F}$ is quasi-coherent on Y_n (502). Pushing it forward along the whole-scheme iso $\sigma_n : Y_n \cong \operatorname{Spec}\Gamma(X, Y_n)$ (420) lands it on a genuine affine scheme, where $\widetilde{(-)}$ is defined, and pushforward along an isomorphism preserves quasi-coherence (461). The affine structure theorem 01I8 then applies to the quasi-coherent $(\sigma_n)_*\, p_n^*\mathcal{F}$: the tilde–Γ counit is an isomorphism (288), so $(\sigma_n)_*\, p_n^*\mathcal{F} \cong \widetilde{N_n}$ with $N_n = \Gamma(\operatorname{Spec}\Gamma(X, Y_n),\ (\sigma_n)_*\, p_n^*\mathcal{F})$ by 243.

Altitude 2. Collapse $f_* \circ (p_n)_* \cong (f \circ p_n)_*$ by 454; thus $(\text{pushforward}\, f).\text{obj}(\text{pushPullObj}\, \mathcal{F}\, Y_n) = (f \circ p_n)_*\, p_n^*\mathcal{F}$. As Y_n is affine, the composite $f \circ p_n : Y_n \to S$ of schemes with affine source and target is presented as $\operatorname{Spec}\varphi_n$ for $\varphi_n = \Gamma(f \circ p_n) : R \to A_n$, using the whole-scheme identification $Y_n \cong \operatorname{Spec}\Gamma(Y_n)$ (420). Applying the pushforward of this presented morphism to the altitude-1 iso gives $(f \circ p_n)_*\, p_n^*\mathcal{F} \cong (\operatorname{Spec}\varphi_n)_*\widetilde{N_n}$, which is the asserted altitude-2 form; that the affine pushforward of a $\widetilde{(-)}$-module is again of $\widetilde{(-)}$-form is 499. ☐

Lemma 509 (Abstract push–pull tilde-model over an abstract affine base)**.** *Let $f : X \to S$ be* separated *with S an* abstract *affine scheme (*[IsAffine S]*, so S need not be a literal* Spec*), and write $e_S : S \xrightarrow{\sim} \operatorname{Spec}\Gamma(S, \mathcal{O}_S) = \operatorname{Spec} R$ for the canonical affine identification* Scheme. isoSpec *(420), $R = \Gamma(S, \mathcal{O}_S)$. Fix an affine open-immersion cover $\mathfrak{U} = \{U_i\}$ of X and a cosimplicial degree, writing Y_n for the corresponding fibre power and N_n, $\varphi_n : R \to A_n$ for the data of 508. Then the altitude-2 identification holds over the abstract base S, with the affine $\widetilde{(-)}$-form transported back along e_S so that it lands in $\mathcal{O}_S$-modules rather than $\mathcal{O}_{\operatorname{Spec} R}$-modules:*

$$(\text{pushforward}\, f).\text{obj}(\text{pushPullObj}\, \mathcal{F}\, Y_n) \ \cong\ (\text{pushforward}\, e_S^{-1}).\text{obj}\Big((\text{pushforward}(\operatorname{Spec}\varphi_n)).\text{obj}(\widetilde{N_n})\Big).$$

Proof. Let $e_S : S \cong \operatorname{Spec}\Gamma(S, \mathcal{O}_S)$ be the affine identification (420). The composite $e_S \circ f : X \to \operatorname{Spec}\Gamma(S, \mathcal{O}_S)$ is separated with a *literal* affine base $\operatorname{Spec} R$, $R = \Gamma(S, \mathcal{O}_S)$, so the literal-base bridge 508 at *altitude 2* applies to it, presenting $(e_S \circ f) \circ p_n$ as $\operatorname{Spec}\varphi_n$ and giving

$$(\text{pushforward}(e_S \circ f)).\text{obj}(\text{pushPullObj}\, \mathcal{F}\, Y_n) \ \cong\ (\text{pushforward}(\operatorname{Spec}\varphi_n)).\text{obj}(\widetilde{N_n}).$$

Collapsing the pushforward of the composite, $\text{pushforward}(e_S \circ f) \cong (\text{pushforward}\, e_S) \circ (\text{pushforward}\, f)$ (454). Applying pushforward e_S^{-1} to both sides and cancelling pushforward $e_S^{-1} \circ$ pushforward $e_S \cong \text{pushforward}(e_S \circ e_S^{-1}) = \text{pushforward id} = \text{id}$ on the left (again 454) yields the asserted isomorphism, with the right-hand side transported by pushforward e_S^{-1}. The single affine identification $e_S =$ Scheme. isoSpec is reused on both the f-side and the presented-morphism side. ☐

Lemma 510 (Spec-cartesian commutative squares are pushouts of rings)**.** *A commutative square of commutative rings $\varphi : A \to B$, $\psi : A \to C$, $\rho : B \to P$, $\sigma : C \to P$ whose* Spec*-square*

$$\begin{tikzcd}
\operatorname{Spec} P \arrow[r, "\operatorname{Spec}\rho"] \arrow[d, "\operatorname{Spec}\sigma"'] & \operatorname{Spec} B \arrow[d, "\operatorname{Spec}\varphi"] \\
\operatorname{Spec} C \arrow[r, "\operatorname{Spec}\psi"] & \operatorname{Spec} A
\end{tikzcd}$$

is cartesian in the category of schemes is a pushout (cocartesian) square in the category of commutative rings; equivalently, the corner ring is the tensor product $P = B \otimes_A C$ of the two legs over the base A. (This is the converse used by the ring-pushout carve; the forward direction is Mathlib's isPullback_SpecMap_of_isPushout*.)*

Proof. The functor Spec : $\mathrm{CommRing}^{\mathrm{op}} \to \mathrm{Sch}$ is fully faithful, and a fully faithful functor reflects limits: the cartesian scheme square of the four Spec-maps reflects to a pullback square in $\mathrm{CommRing}^{\mathrm{op}}$, i.e. the original square is a pushout in CommRing. $\square$

Lemma 511 (Restriction of the cartesian base-change square over an intersection open is cartesian)**.** *Let $g' : X' \to X$ be an arbitrary morphism of schemes and let $V = \mathrm{coverInterOpen}\,\mathfrak{U}\,\sigma = \bigcap_k U_{\sigma(k)}$ be a Čech intersection open of an affine open-immersion cover $\mathfrak{U}$ of X, with open immersion $j_\sigma : V \hookrightarrow X$. Form the fibre product $X' \times_X V$ and the cartesian square*

$$\begin{array}{ccc} X' \times_X V & \longrightarrow & V \\ \big\downarrow & & \big\downarrow{\scriptstyle j_\sigma} \\ X' & \xrightarrow{g'} & X \end{array}$$

in which the top map is the projection pullback. snd $g'\,j_\sigma$*, the left map is* pullback. fst $g'\,j_\sigma$*, the right map is the open immersion j_σ, and the bottom map is g'. This square is cartesian: it is a pullback square of schemes.*

Proof. The fibre product $X' \times_X V$ exists and its defining projections fit, by construction, into a cartesian square. Reading off that pullback square with the projections in the stated positions gives exactly the asserted cartesian square; no further hypotheses are required. $\square$

Lemma 512 (LHS abstract $\to$ tilde for a single intersection open)**.** *Let $f : X \to S$ be* separated *with affine base $S = \mathrm{Spec}\,R$, fix an affine open-immersion cover $\mathfrak{U} = \{U_i = \mathrm{Spec}\,A_i\}$ of X and a* finite, nonempty *multi-index $\sigma : \kappa \to I$, with intersection open $V = \mathrm{coverInterOpen}\,\mathfrak{U}\,\sigma = \bigcap_k U_{\sigma(k)}$ and open immersion $j_\sigma : V \hookrightarrow X$. By 501 the open $V = \mathrm{Spec}\,A_\sigma$ is affine; write $\varphi : R \to A_\sigma$ for the ring map presenting the composite $f \circ j_\sigma : V \to S$ as $\mathrm{Spec}\,\varphi$, and $N = \Gamma(V, (j_\sigma)^*\mathcal{F})$ for the A_σ-module of sections of the (quasi-coherent) restriction. Then there is an isomorphism of $\mathcal{O}_S$-modules*

$$f_*(\mathrm{pushPullObj}\,\mathcal{F}\,(\mathrm{Over.mk}\,j_\sigma)) \;=\; f_*((j_\sigma)_*(j_\sigma)^*\mathcal{F}) \;\cong\; (\mathrm{Spec}\,\varphi)_*\,\widetilde{N}.$$

Proof. The intersection open V is affine, $V = \mathrm{Spec}\,A_\sigma$, by 501 (the separated-diagonal instance). The single open immersion $j_\sigma : V \hookrightarrow X$ is the projection of the one-fold fibre power of the cover indexed by σ, so $\mathrm{Over.mk}\,j_\sigma$ is the degenerate fibre power Y to which 508 applies. Reading that bridge at *altitude 2* over the affine base S collapses $f_* \circ (j_\sigma)_* \cong (f \circ j_\sigma)_*$ and presents the composite as $\mathrm{Spec}\,\varphi$ with $\varphi = \Gamma(f \circ j_\sigma) : R \to A_\sigma$, yielding exactly $f_*((j_\sigma)_*(j_\sigma)^*\mathcal{F}) \cong (\mathrm{Spec}\,\varphi)_*\widetilde{N}$ with $N = \Gamma(V, (j_\sigma)^*\mathcal{F})$. $\square$

Lemma 513 (Base-change cancellation: the restricted corner module is the tensor base change of N)**.** *Let $\varphi : R \to A_\sigma$, $\rho : A_\sigma \to B$, $\psi : R \to R'$, $\sigma' : R' \to B$ be commutative-ring homomorphisms forming a pushout (cocartesian) square in* CommRing*, so that the corner ring is the tensor product $B = A_\sigma \otimes_R R'$ of the two legs over the base R (510). Let N be an arbitrary A_σ-module. Then there is a canonical isomorphism of R'-modules*

$$\mathrm{restr}_{\sigma'}(\mathrm{extendScalars}_\rho\,N) \;\cong\; \mathrm{extendScalars}_\psi(\mathrm{restr}_\varphi\,N), \qquad \textit{i.e.} \qquad \mathrm{restr}_{\sigma'}(B \otimes_{A_\sigma} N) \;\cong\; R' \otimes_R N.$$

This is the pure commutative-algebra cancellation that underlies the geometric corner identification of 514; it carries no sheaf, pullback, or Γ-content.

Proof. This is exactly the module-level base-change cancellation 128 read for the pushout square $(\varphi, \rho, \psi, \sigma')$ and the module N: its inverse realises the Mathlib cancellation cancelBaseChange

(127), $(R' \otimes_R A_\sigma) \otimes_{A_\sigma} N \cong R' \otimes_R N$, which after the identification $B \cong A_\sigma \otimes_R R'$ of the corner ring (510) reads as the asserted isomorphism $\mathrm{restr}_{\sigma'}(B \otimes_{A_\sigma} N) \cong R' \otimes_R N$. No flatness hypothesis is required. ☐

Lemma 514 (RHS abstract → tilde for a single intersection open (base-changed side))**.** *In the cartesian base-change setting of 517, let $f : X \to \operatorname{Spec} R$ be* separated *over the affine base $\operatorname{Spec} R$, let $g : \operatorname{Spec} R' \to \operatorname{Spec} R$ be the base-change morphism, and let $f' : X' \to \operatorname{Spec} R'$ be the* separated *([IsSeparated f']) pulled-back morphism with $g' : X' \to X$ the top map of the cartesian square. Assume the pulled-back affine open-immersion cover $\mathfrak{U}' = \{\mathfrak{U}'.X\,i\}$ of X' consists of affine opens ([$\forall i$, IsAffine$(\mathfrak{U}'.X\,i)$]). Fix the affine intersection open $V = \operatorname{Spec} A_\sigma$ (501) with $j_\sigma : V \hookrightarrow X$ and $N = \Gamma(V, (j_\sigma)^*\mathcal{F})$; let $j'_\sigma : V' \hookrightarrow X'$ be the pullback of j_σ along g', so that $V' = X' \times_X V = \operatorname{Spec}(A_\sigma \otimes_R R')$, and set*

$$N' \;=\; \Gamma(V', (j'_\sigma)^*((g')^*\mathcal{F}))$$

for the raw base-changed corner module. Write $\psi : R' \to A_\sigma \otimes_R R'$ for the ring map presenting the composite $f' \circ j'_\sigma : V' \to \operatorname{Spec} R'$ as $\operatorname{Spec}\psi$. Then there is an isomorphism of $\mathcal{O}_{\operatorname{Spec} R'}$-modules

$$f'_*\Big((g')^*(\mathrm{pushPullObj}\,\mathcal{F}\,(\mathrm{Over.mk}\,j_\sigma))\Big) \;\cong\; (\operatorname{Spec}\psi)_*\,\widetilde{N'}.$$

(The further identification $N' \cong N \otimes_R R'$ is 515.)

Proof. First reorganise the pulled-back data by the per-intersection-open X-level Beck–Chevalley isomorphism 531: since j_σ is an open immersion whose pullback along g' is the open immersion $j'_\sigma : V' \hookrightarrow X'$ onto the preimage $(g')^{-1}(V)$,

$$(g')^*(\mathrm{pushPullObj}\,\mathcal{F}\,(\mathrm{Over.mk}\,j_\sigma)) = (g')^*((j_\sigma)_*(j_\sigma)^*\mathcal{F}) \;\cong\; (j'_\sigma)_*(j'_\sigma)^*((g')^*\mathcal{F}) = \mathrm{pushPullObj}((g')^*\mathcal{F})\,(\mathrm{Over.mk}\,j'_\sigma$$

Pushing forward along f' reduces the left side of the claim to $f'_*(\mathrm{pushPullObj}((g')^*\mathcal{F})\,(\mathrm{Over.mk}\,j'_\sigma))$. The base-changed module $(g')^*\mathcal{F}$ is quasi-coherent by 507, and the intersection open $V' =$ coverInterOpen $\mathfrak{U}'\,\sigma$ of the base-changed cover is affine for the separated $f' : X' \to \operatorname{Spec} R'$ over the affine base $\operatorname{Spec} R'$ (501). Hence the LHS-side bridge 512, applied to the base-changed data $(f', \mathfrak{U}', (g')^*\mathcal{F})$ at the index tuple σ, gives

$$f'_*(\mathrm{pushPullObj}((g')^*\mathcal{F})\,(\mathrm{Over.mk}\,j'_\sigma)) \;\cong\; (\operatorname{Spec}\psi)_*\,\widetilde{N'}, \qquad N' = \Gamma(V', (j'_\sigma)^*((g')^*\mathcal{F})),$$

with $\psi = \Gamma(f' \circ j'_\sigma) : R' \to A_\sigma \otimes_R R'$. This is the asserted isomorphism. ☐

Lemma 515 (Corner identification of the base-changed section module)**.** *In the setting of 514, the raw base-changed corner module $N' = \Gamma(V', (j'_\sigma)^*((g')^*\mathcal{F}))$ satisfies*

$$N' \;\cong\; N \otimes_R R'$$

naturally in $\mathcal{F}$, so the right-hand side of 514 may equivalently be written $(\operatorname{Spec}\psi)_*\,\widetilde{N \otimes_R R'}$.

Proof. The restricted square $V' \to V$, $V' \to X'$, $g' : X' \to X$, j_σ is cartesian by 511, and supplies the canonical pullback identification $(j'_\sigma)^*((g')^*\mathcal{F}) \cong (g'_V)^*((j_\sigma)^*\mathcal{F})$ (the same comparison of the cartesian square used in 520), where $g'_V : V' \to V$ is the top map of that square. Over the affine $V = \operatorname{Spec} A_\sigma$ the restriction $(j_\sigma)^*\mathcal{F}$ is quasi-coherent, $\cong \widetilde{N}$. The top map g'_V is the base change of the base-change morphism g over the affine V, hence the affine morphism $\operatorname{Spec}\rho$ for the corner ring map $\rho : A_\sigma \to A_\sigma \otimes_R R'$; pulling $\widetilde{N}$ back along it is, by the pullback tilde dictionary 126, $(g'_V)^*\widetilde{N} \cong N \otimes_{A_\sigma} \widetilde{(A_\sigma \otimes_R R')} = N \widetilde{\otimes_R} R'$. Taking global sections over $V' = \operatorname{Spec}(A_\sigma \otimes_R R')$

gives $N' \cong N \otimes_{A_\sigma} (A_\sigma \otimes_R R') = N \otimes_R R'$, naturally in $\mathcal{F}$; the underlying R'-module cancellation $\mathrm{restr}_{\sigma'}(B \otimes_{A_\sigma} N) \cong R' \otimes_R N$ (with $B = A_\sigma \otimes_R R'$) is the pure algebraic core 513. A consumer rewrites the right-hand side $(\operatorname{Spec}\psi)_*\widetilde{N'}$ as $(\operatorname{Spec}\psi)_*\widetilde{N \otimes_R R'}$ along this identification. ☐

Lemma 516 (The restricted square carves into a ring pushout)**.** *In the setting of 517, write $V = \operatorname{Spec} A_\sigma$ for the affine intersection open (501), $\varphi : R \to A_\sigma$ for the ring map presenting $f \circ j_\sigma$, $\gamma : R \to R'$ for the ring map presenting g, $\rho : A_\sigma \to B$ for the corner ring map obtained by applying global sections to the top map $V' \to V$ of the restricted square, and $\psi : R' \to B$ for the ring map presenting $f' \circ j'_\sigma$, where $B = \Gamma(X', V'_\sigma)$. Then the square*

$$\begin{array}{ccc} R & \xrightarrow{\varphi} & A_\sigma \\ {\scriptstyle\gamma}\downarrow & & \downarrow{\scriptstyle\rho} \\ R' & \xrightarrow{\psi} & B \end{array}$$

is a pushout of commutative rings; in particular $B \cong A_\sigma \otimes_R R'$.

Proof. Pasting the restricted cartesian square of 511 vertically onto the ambient cartesian base-change square exhibits $X' \times_X V$ as the fibre product of $V \xrightarrow{j_\sigma} X \xrightarrow{f} S$ and $g : S' \to S$. The categorical pullback $X' \times_X V$ is identified with the base-changed intersection open $V' = V'_\sigma$: the first projection is an open immersion with range $(g')^{-1}(V)$, which equals V' because the base-changed cover's member opens are the g'-preimages of the original member opens. Under this identification and the affine identifications $V = \operatorname{Spec} A_\sigma$, $V' = \operatorname{Spec} B$, all four corners of the resulting cartesian square are affine and all four maps are Spec-maps of the stated ring maps; by 510 the ring square is a pushout, and the corner of a ring pushout is the tensor product $A_\sigma \otimes_R R'$. ☐

Lemma 517 (Per-intersection-open base change of the push–pull object)**.** *Let $f : X \to S$ be* separated *with affine base $S = \operatorname{Spec} R$, let $g : S' \to S$ be a base-change morphism with $S' = \operatorname{Spec} R'$ affine, and let*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow{g} & S \end{array}$$

be the associated cartesian square. Fix an affine open-immersion cover $\mathfrak{U} = \{U_i = \operatorname{Spec} A_i\}$ of X and a multi-index $\sigma : \kappa \to I$, with intersection open $V = \mathrm{coverInterOpen}\,\mathfrak{U}\,\sigma = \bigcap_k U_{\sigma(k)}$ and open immersion $j_\sigma : V \hookrightarrow X$. Then there is an isomorphism of $\mathcal{O}_{S'}$-modules

$$g^*\Big(f_*((j_\sigma)_*(j_\sigma)^*\mathcal{F})\Big) \;\cong\; f'_*\Big((g')^*((j_\sigma)_*(j_\sigma)^*\mathcal{F})\Big),$$

i.e. $g^*(f_*(\mathrm{pushPullObj}\,\mathcal{F}\,(\mathrm{Over.mk}\,j_\sigma))) \cong f'_*((g')^*(\mathrm{pushPullObj}\,\mathcal{F}\,(\mathrm{Over.mk}\,j_\sigma)))$. This is the per-intersection-open instance of the Beck–Chevalley comparison; it is the genuine open content that survives the coproduct/product decomposition of the degreewise Čech nerve.

Proof. The two sides of the comparison are routed independently to the common affine $(\operatorname{Spec}\cdot)_*\widetilde{(-)}$ form, where the affine termwise base change closes the gap between them.

1. *V is affine.* The intersection open $V = \bigcap_k U_{\sigma(k)}$ is a finite nonempty intersection of affine opens of the separated scheme X, hence affine by 501 (the separated-diagonal instance of 500), $V = \operatorname{Spec} A_\sigma$; write $\varphi : R \to A_\sigma$ for the ring map presenting $f \circ j_\sigma$ as $\operatorname{Spec}\varphi$ and $N = \Gamma(V, (j_\sigma)^*\mathcal{F})$.

2. *LHS → tilde.* By 512 the left abstract term is the affine pushforward of $\widetilde{N}$ over S,

$$f_*(\text{pushPullObj}\,\mathcal{F}\,(\text{Over.}\,\text{mk}\,j_\sigma)) \;\cong\; (\text{Spec}\,\varphi)_*\,\widetilde{N},$$

so applying g^* gives $g^*\big(f_*(\text{pushPullObj}\,\mathcal{F}\,(\text{Over.}\,\text{mk}\,j_\sigma))\big) \cong g^*\big((\text{Spec}\,\varphi)_*\widetilde{N}\big)$.

3. *RHS → tilde.* By 514 the right abstract term is likewise the affine pushforward of the base-changed module $N \otimes_R R'$ over S',

$$f'_*\Big((g')^*(\text{pushPullObj}\,\mathcal{F}\,(\text{Over.}\,\text{mk}\,j_\sigma))\Big) \;\cong\; (\text{Spec}\,\psi)_*\,\widetilde{N \otimes_R R'},$$

where $\psi : R' \to A_\sigma \otimes_R R'$ presents $f' \circ j'_\sigma$. Both comparison sides are now in $(\text{Spec}\,\cdot)_*\widetilde{(-)}$ form.

4. *The affine pushout square closes the gap.* Restrict the cartesian base-change square over $j_\sigma : V \hookrightarrow X$. By 511 the resulting square $X' \times_X V \to V$, $X' \times_X V \to X'$, g', j_σ is cartesian, and with $V = \text{Spec}\,A_\sigma$ (step 1), $S = \text{Spec}\,R$, $S' = \text{Spec}\,R'$ affine it is a cartesian square of affine schemes, $\text{Spec}(A_\sigma \otimes_R R') \to \text{Spec}\,A_\sigma$, $\text{Spec}(A_\sigma \otimes_R R') \to \text{Spec}\,R'$. Applying global sections, the affine-pullback ↔ ring-pushout equivalence 510 converts it into the cocartesian square of commutative rings

$$\begin{array}{ccc} R & \xrightarrow{\varphi} & A_\sigma \\ \downarrow & & \downarrow \\ R' & \xrightarrow{\psi} & A_\sigma \otimes_R R' \end{array}$$

with corner $A_\sigma \otimes_R R'$. For this pushout square the affine termwise base change 497, the per-open instance of 129, supplies

$$g^*\big((\text{Spec}\,\varphi)_*\widetilde{N}\big) \;\cong\; (\text{Spec}\,\psi)_*\,\widetilde{N \otimes_R R'}.$$

Composing step 2 (and g^*), this affine isomorphism, and the inverse of step 3 yields the asserted comparison $g^*\big(f_*(\text{pushPullObj}\,\mathcal{F}\,(\text{Over.}\,\text{mk}\,j_\sigma))\big) \cong f'_*\big((g')^*(\text{pushPullObj}\,\mathcal{F}\,(\text{Over.}\,\text{mk}\,j_\sigma))\big)$.

∎

Lemma 518 (Beck–Chevalley natural iso through the Čech nerve)**.** *Let $f : X \to S$ be* separated *with affine base $S = \text{Spec}\,R$, fix an affine open-immersion cover $\mathfrak{U} = \{U_i = \text{Spec}\,A_i\}$ of X (so that, by 500, every fibre power Y_n of the Čech nerve is affine). Whiskered through the Čech nerve, there is a natural isomorphism of cosimplicial $\mathcal{O}_{S'}$-modules*

$$g^* \circ (\text{pushforward}\,f) \;\cong\; (\text{pushforward}\,f') \circ (g')^*,$$

the Beck–Chevalley comparison for the cartesian square, valid at every cosimplicial degree.

Proof. The natural isomorphism is built degreewise. Fix a cosimplicial degree n and work over the affine base $S = \text{Spec}\,R$. The degree-n comparison is between $g^*\big(f_*(\text{pushPullObj}\,\mathcal{F}\,Y_n)\big)$ and $f'_*\big((g')^*(\text{pushPullObj}\,\mathcal{F}\,Y_n)\big)$, where Y_n is the degree-n fibre power of the Čech nerve.

1. *Product decomposition.* The degree-n fibre power is the coproduct $Y_n = \coprod_\sigma U_\sigma$ over index tuples $\sigma : \text{Fin}(n+1) \to I$ of the intersection opens $U_\sigma = \text{coverInterOpen}\,\mathfrak{U}\,\sigma$, so by 371 (which requires the cover to be finite) the push–pull object decomposes as the finite product $\text{pushPullObj}\,\mathcal{F}\,Y_n \cong \prod_\sigma \text{pushPullObj}\,\mathcal{F}\,(\text{Over.}\,\text{mk}\,j_\sigma)$. Both pushforward f and g^* (and likewise f'_*, $(g')^*$) preserve this finite product, so the degree-n comparison reduces termwise to the per-σ single-open base change.

2. *The per-σ factor.* Each summand is exactly the per-intersection-open base-change isomorphism $g^*(f_*((j_\sigma)_*(j_\sigma)^*\mathcal{F})) \cong f'_*((g')^*((j_\sigma)_*(j_\sigma)^*\mathcal{F}))$ of 517. There the intersection open $U_\sigma = U_{i_0} \cap \cdots \cap U_{i_p}$ is affine by separatedness (500); the abstract pushed-forward term is identified with the affine $\widetilde{(-)}$-form $(\operatorname{Spec}\varphi)_*\widetilde{N}$ over S by 508 (*altitude 2*); and on that affine model the comparison IS the affine termwise base change 129 (via 497), assembled from the concrete tilde dictionaries 123 and 126 with the commutative-algebra cancellation cancelBaseChange (127) — not the canonical adjoint mate.

3. *Naturality.* The naturality condition is the index-omission restriction: each coface map is a restriction along an inclusion of finite affine intersections, and the termwise isomorphisms are natural in the intersection; naturality in the base-change morphism g is inherited termwise from the affine base change.

$\square$

Lemma 519 (The base-changed cover intersection is the preimage of the intersection)**.** *Let $g' : X' \to X$ be the pullback of a base change $g : S' \to S$ along a separated $f : X \to S$, and let $\mathfrak{U}'$ be the base change along g' of the affine open-immersion cover $\mathfrak{U}$ of X. Then for every finite index family σ the Čech intersection open of the base-changed cover is the preimage of the intersection open of the original cover:*

$$\text{coverInterOpen}\,\mathfrak{U}'\,\sigma \;=\; (g')^{-1}(\text{coverInterOpen}\,\mathfrak{U}\,\sigma).$$

Proof. The base-changed cover member $\mathfrak{U}'.X\,i$ is by definition the pullback $(g')^{-1}(U_i)$, so its range in X' is the preimage of the range of U_i. Taking preimages of opens commutes with finite meets (the inverse-image map on the frame of opens preserves intersections), whence $\text{coverInterOpen}\,\mathfrak{U}'\,\sigma = \bigcap_k (g')^{-1}(U_{\sigma(k)}) = (g')^{-1}(\bigcap_k U_{\sigma(k)}) = (g')^{-1}(\text{coverInterOpen}\,\mathfrak{U}\,\sigma)$. $\square$

Lemma 520 (Open-immersion Beck–Chevalley over a restricted cartesian square)**.** *Let*

$$\begin{array}{ccc} V' & \xrightarrow{g'_V} & V \\ {\scriptstyle p'}\big\downarrow & & \big\downarrow{\scriptstyle p} \\ X' & \xrightarrow{g'} & X \end{array}$$

be the cartesian square obtained by restricting $g' : X' \to X$ over an open immersion $p : V \hookrightarrow X$ with V affine *(in Lean p is taken to be the inclusion $V_0.\iota$ of an affine open V_0 : X.*Opens, IsAffineOpen V_0 *— the form in which every consumer instantiates the square), so that $p' : V' \hookrightarrow X'$ is again an open immersion (the pullback of p along g', with image the preimage $(g')^{-1}(V)$). Assume in addition that X is* separated *(so that the affine V meets every affine open of X in an affine; in Lean this is a separatedness hypothesis on X — e.g.* [IsSeparated (terminal.from X)], *or any equivalent input guaranteeing affine intersections, at the prover's choice). Then for any* quasi-coherent $\mathcal{O}_X$*-module $\mathcal{F}$ (*`hF` : $\mathcal{F}$.IsQuasicoherent*) there is a Beck–Chevalley isomorphism*

$$(g')^*(p_*\,p^*\mathcal{F}) \;\cong\; p'_*\,p'^*((g')^*\mathcal{F}),$$

i.e. $(g')^*(\text{pushPullObj}\,\mathcal{F}\,(\text{Over.mk}\,p)) \cong \text{pushPullObj}((g')^*\mathcal{F})\,(\text{Over.mk}\,p')$*. This is the open-immersion (in particular flat) case of Beck–Chevalley. The affineness of V together with the quasi-coherence of $\mathcal{F}$ is what reduces it, after refining X' by an affine open cover, to the affine termwise base-change isomorphism 129. The separatedness hypothesis on X is what makes the affine V meet each affine open of X in an affine; in the application the consumer discharges it from its own separatedness hypothesis through 501.*

Proof. Throughout write $G := p^*\mathcal{F}$ for the restricted module on V. A warning on the geometry first: p being an open immersion does *not* make the *right*-adjoint direct image p_* extension-by-zero. Extension-by-zero is the *left* adjoint $p_!$; the right adjoint p_* has genuinely nonzero sections off V (over an open $U \subseteq X$ its sections are $\Gamma(U \cap V, p^*\mathcal{F})$, not 0 for $U \nsubseteq V$). So there is no "both sides vanish away from V'" support argument. The off-V' data is real and is exactly the content of the comparison; the correct tool is a sectionwise criterion. The orientation is the open-immersion (flat) special case of base change, Stacks Project Tag 02KH (`lemma-flat-base-change-cohomology`); the section-level input is supplied, after refining X' by an affine open cover and using that V is affine and $\mathcal{F}$ quasi-coherent, by the affine termwise base-change brick 129 (Stacks Project Tag 02KG), which already packages the affine tilde dictionaries and the cancellation cancelBaseChange.

Stage 1 (telescope; no affineness required). Since p and p' are open immersions, pullback along them *is* restriction: the restriction functor and the module pullback agree, restrict $p \cong p^*$ and restrict $p' \cong p'^*$ (396). Using the pseudofunctor structure of pullback — the comparison $(q \circ r)^* \cong r^* \circ q^*$ for composable morphisms, together with reindexing of $(-)^*$ along an equality of morphisms — and the commuting square $g'_V \circ p' = p \circ g'$ (in the sense $p' \cdot g' = g'_V \cdot p$ as maps $V' \to X$), one telescopes

$$p'^*\,(g')^* \;\cong\; (p' \cdot g')^* \;=\; (g'_V \cdot p)^* \;\cong\; (g'_V)^*\,p^*.$$

Applied to $\mathcal{F}$ this identifies $p'^*(g')^*\mathcal{F} \cong (g'_V)^*G$, hence the right-hand side rewrites as $p'_*\,p'^*(g')^*\mathcal{F} \cong p'_*\,(g'_V)^*G = (\text{pushforward}\,p')((g'_V)^*G)$, while the left-hand side is $(g')^*(p_*\,G) = (g')^*((\text{pushforward}\,p)G)$. The whole telescope is formal and uses no affineness. It reduces the lemma, with the single fixed coefficient $G = p^*\mathcal{F}$, to the assertion that the *bare base-change map*

$$\text{bareBC}\colon\quad (g')^*((\text{pushforward}\,p)\,G) \;\longrightarrow\; (\text{pushforward}\,p')((g'_V)^*G)$$

— the canonical mate of $\text{pushforward}\,p \circ (g')^* \to (g'_V)^* \circ \text{pushforward}\,p'$ attached to the cartesian square — is an isomorphism. (This canonical mate is the local comparison morphism; it is stated from first principles, independently of any relative-spectrum base-change theory.)

Stage 2 (the bare comparison is an isomorphism). It remains only to prove that bareBC is an isomorphism. This is exactly 526. By the mate formula, bareBC at G is the p'-unit at $M = (g')^*(p_*\,G)$ followed by isomorphisms (the telescope components and the p-counit, invertible for the open immersion p by 522); since p'_* is fully faithful, the unit is an isomorphism iff M lies in the essential image of p'_*, and that membership is cover-local (524). Check it on an affine open cover $\{W_j\}$ of X' refining $(g')^{-1}$(affine opens of X) (521): over each member $W_j = \operatorname{Spec} B$, with $g'(W_j) \subseteq U_j = \operatorname{Spec} A$, — because V is affine and X is separated — the intersection $V \cap U_j$ is affine, the restricted local square is a *fully-affine* pullback, and the affine termwise base-change isomorphism 129, applied with the open immersion $V \cap U_j \hookrightarrow U_j$ in the role of the affine map and presenting module $M' = \Gamma(V \cap U_j,\ p^*\mathcal{F})$ (quasi-coherent by `hF`), exhibits $M|_{W_j}$ as a pushforward from $W_j \cap (g')^{-1}(V)$ (525). Hence bareBC is an isomorphism. Combined with the Stage 1 reduction this proves the lemma.

Remark (the hypotheses are essential). The affineness of V, the separatedness of X, and the quasi-coherence of $\mathcal{F}$ are not cosmetic. Separatedness is what makes the affine V meet each affine open U_j of X in an *affine* intersection $V \cap U_j$; without it the local square over W_j need not be fully affine and the reduction to the affine brick breaks. The outer base pullback $(g')^*$ is the sheafification of the presheaf tensor $U \mapsto (p_*\,p^*\mathcal{F})(U) \otimes_{\mathcal{O}_X(U)} \mathcal{O}_{X'}(W)$; this tensor description agrees with the sections of $p'_*\,(g'_V)^*(p^*\mathcal{F})$ only on the *quasi-coherent* locus, so for a general $\mathcal{O}_X$-module the comparison fails — quasi-coherence of $\mathcal{F}$ is genuinely required. Affineness of V is what makes the local square over each W_j fully affine and hence reducible to the affine brick; in the application it is supplied by the consumer, where each fibre power U_σ of the Čech nerve is affine because the cover is affine and the structure morphism is separated (501). In particular the bare-mate route never forms the global push–pull $p_* p^* \mathcal{F}$, so no quasi-compactness of p is needed; there is no claim of a hypothesis-free open-immersion Beck–Chevalley. $\square$

Lemma 521 (Affine cover of X' refining preimages of affine opens of X)**.** *Let $g' : X' \to X$ be any morphism of schemes. Then there is an affine open cover $\{W_j\}$ of X' together with, for each j, an affine open $U_j \subseteq X$ such that $g'(W_j) \subseteq U_j$; equivalently the restricted map $g'|_{W_j} : W_j \to U_j$ is a map of affine schemes, hence a* Spec*-map $g'|_{W_j} = \operatorname{Spec} \varphi_j$ for a ring map $\varphi_j : \Gamma(U_j, \mathcal{O}_X) \to \Gamma(W_j, \mathcal{O}_{X'})$.*

Proof. Take the standard affine open cover $\mathfrak{A} = \{A_i = \operatorname{Spec} R_i\}$ of X and pull it back along g': the base-changed cover $g'^{-1}\mathfrak{A}$ is an open cover of X' whose member over index i maps into the affine A_i. Its members $g'^{-1}(A_i)$ need not be affine, so refine each of them by its own affine open cover and combine the results into a single cover of X'. The resulting cover $\{W_j\}$ is affine, and each W_j sits inside some $g'^{-1}(A_i)$, whence $g'(W_j) \subseteq A_i =: U_j$. A morphism between two affine schemes is induced by a ring map on global sections (420), so $g'|_{W_j} = \operatorname{Spec} \varphi_j$ with $\varphi_j = \Gamma(g'|_{W_j})$. $\square$

Lemma 522 (The geometric counit at an open immersion is invertible)**.** *Let $q : V \hookrightarrow X$ be an open immersion of schemes. Then for every $\mathcal{O}_V$-module c the counit $q^*(q_* c) \to c$ of the pullback–pushforward adjunction $q^* \dashv q_*$ is an isomorphism. (Equivalently: q_* is fully faithful for an open immersion q.)*

Proof. For an open immersion the module pullback q^* is naturally isomorphic to the restriction functor (396), and the counit of the restriction–pushforward adjunction is invertible: over an open $O \subseteq V$ it is the identification $\Gamma(O,\ \mathrm{restrict}_q(q_* c)) = \Gamma(q^{-1}(q(O)),\ c) = \Gamma(O,\ c)$, using that q is an open embedding so $q^{-1}(q(O)) = O$. Counits of adjunctions with a common right adjoint correspond under the canonical comparison of left adjoints, so the geometric counit is the composite of an isomorphism (the comparison, evaluated at $q_* c$) with an isomorphism (the restriction counit), hence an isomorphism. $\square$

Lemma 523 (Restriction maps of a pushforward-presented module are isomorphisms)**.** *Let W be a scheme, $O_0 \subseteq W$ an open with inclusion $\iota : O_0 \hookrightarrow W$, and N an $\mathcal{O}_W$-module lying in the essential image of ι_*, say $N \cong \iota_* K$. Then for every open $O \subseteq W$ the presheaf restriction map*

$$N(O) \longrightarrow N(O \cap O_0)$$

is an isomorphism.

Proof. Transport along the isomorphism $N \cong \iota_* K$ (naturality of its components in the open). On the pushforward side the map in question is the restriction $(\iota_* K)(O) = K(\iota^{-1}(O)) \to K(\iota^{-1}(O \cap O_0)) = (\iota_* K)(O \cap O_0)$, and $\iota^{-1}(O \cap O_0) = \iota^{-1}(O) \cap \iota^{-1}(O_0) = \iota^{-1}(O)$ since $\iota^{-1}(O_0)$ is everything; so it is K applied to an identity inclusion, hence an isomorphism. $\square$

Lemma 524 (Essential-image membership for p'_* is cover local)**.** *Let $p' : V' \hookrightarrow X'$ be an open immersion with open range $R = p'(V')$, let M be an $\mathcal{O}_{X'}$-module, and let $\{W_j\}$ be an open cover of X'. If for every j the restriction $M|_{W_j}$ lies in the essential image of the pushforward along the open inclusion $W_j \cap R \hookrightarrow W_j$, then M lies in the essential image of p'_*.*

Proof. Since p'_* is fully faithful (522), M lies in its essential image iff the unit $M \to p'_* p'^* M$ of $p'^* \dashv p'_*$ is an isomorphism. Under the identification of p'^* with restriction (396) the unit has, over an open $U \subseteq X'$, the plain presheaf restriction component $M(U) \to M(U \cap R)$. Invertibility of this morphism of sheaves is local on X' (356): it suffices that each restriction to W_j be invertible, and — because restriction along an open immersion has definitional components — the restricted morphism has, over an open $O \subseteq W_j$, again the plain component $M(w_j(O)) \to M(p'(p'^{-1}(w_j(O))))$. By the lattice identity $w_j(O \cap w_j^{-1}(R)) = w_j(O) \cap R = p'(p'^{-1}(w_j(O)))$ this component is, up to the induced identification of the target sections, the restriction map $M|_{W_j}(O) \to M|_{W_j}(O \cap w_j^{-1}(R))$, which is an isomorphism by 523 applied to the member datum. Hence the unit is an isomorphism and M lies in the essential image of p'_*. $\square$

Lemma 525 (Local essential-image comparison over an affine cover member)**.** *Keep the data of 520. Assume that V is an affine open of X, that X is separated, and that $\mathcal{F}$ is quasi-coherent. Write $M := (g')^*(p_* p^* \mathcal{F})$. Let $W_j = \operatorname{Spec} B \subseteq X'$, $U_j = \operatorname{Spec} A \subseteq X$ be as in 521, with $g'|_{W_j} = \operatorname{Spec} \varphi$, $\varphi : A \to B$. Since V is affine and X is separated, the intersection $V \cap U_j$ is affine, say $V \cap U_j = \operatorname{Spec} A'$, and the open immersion $V \cap U_j \hookrightarrow U_j$ is* Spec *of a ring map $A \to A'$. Then the restriction $M|_{W_j}$ lies in the essential image of the pushforward along the open inclusion $W_j \cap (g')^{-1}(V) \hookrightarrow W_j$.*

Proof. By the pullback pseudofunctor along $w_j \cdot g' = g'|_{W_j} \cdot (U_j \hookrightarrow X)$, the restriction $M|_{W_j}$ is the pullback along $g'|_{W_j} = \operatorname{Spec} \varphi$ of the restriction $(p_* p^* \mathcal{F})|_{U_j}$; the open-restriction identity $(p_* p^* \mathcal{F})|_{U_j} = (V \cap U_j \hookrightarrow U_j)_* (p^* \mathcal{F}|_{V \cap U_j})$ (both sides are site-level pushforwards along the same opens functor) presents that restriction as $f_* \widetilde{M'}$ for $f = \operatorname{Spec}(A \to A')$ and the A'-module $M' = \Gamma(V \cap U_j,\ p^* \mathcal{F})$. This M' exists because $p^* \mathcal{F}$ is quasi-coherent (the restriction p^* of the quasi-coherent $\mathcal{F}$, 502) and a quasi-coherent module over an affine is the $\widetilde{(-)}$ of its sections (126). Applying Spec to the ring pushout

$$\begin{array}{ccc} A & \xrightarrow{\varphi} & B \\ \downarrow & & \downarrow \\ A' & \longrightarrow & B \otimes_A A' \end{array}$$

(cut out by the affine map $\varphi : A \to B$ presenting $g'|_{W_j}$ and the affine open immersion $A \to A'$ presenting $V \cap U_j \hookrightarrow U_j$) gives a fully-affine cartesian square, and the affine termwise base-change isomorphism

$$g^*(f_* \widetilde{M'}) \ \cong \ f'_*((g')^* \widetilde{M'})$$

of 129 for this pushout square (with $g = \operatorname{Spec} \varphi$) exhibits $M|_{W_j}$ as the pushforward of $(g')^* \widetilde{M'}$ along $f' = \operatorname{Spec}(B \to B \otimes_A A')$, the base change of the open immersion $V \cap U_j \hookrightarrow U_j$; its range is $g'|_{W_j}^{-1}(V \cap U_j) = W_j \cap (g')^{-1}(V)$. Transporting the pushforward along this range identification, $M|_{W_j}$ lies in the essential image of the pushforward along $W_j \cap (g')^{-1}(V) \hookrightarrow W_j$, as claimed. $\square$

Lemma 526 (Stage 2, assembly: bareBC is an isomorphism)**.** *For the cartesian square $p' \cdot g' = g'_V \cdot p$ of 520 with p an open immersion, V affine (*[IsAffine V]*), X separated, and $\mathcal{F}$ a* quasi-coherent *$\mathcal{O}_X$-module (*`hF : F.IsQuasicoherent`*), the bare base-change map* bareBC *(529), evaluated at $G = p^* \mathcal{F}$, is an isomorphism. (These hypotheses are essential: the proof applies the affine member node 525, which needs each $V \cap U_j$ affine — hence V affine and X separated — and needs $p^* \mathcal{F}$ quasi-coherent — hence* `hF`*. The arbitrary-$\mathcal{F}$ form is false.)*

Proof. Write $M := (g')^*(p_* G)$. *Mate factorization.* By the componentwise mate formula, bareBC evaluated at G is the composite of the unit $M \to p'_* p'^* M$ of $p'^* \dashv p'_*$, followed by p'_* applied to the pullback telescope components (530, isomorphisms), followed by $p'_* (g'_V)^*$ applied to the counit $p^*(p_* G) \to G$ — an isomorphism because p is an open immersion (522). Hence bareBC is an isomorphism as soon as the leading unit is.

The unit is an isomorphism. Since p' is an open immersion (528), p'_* is fully faithful (522), so the unit at M is an isomorphism iff M lies in the essential image of p'_*. That membership is cover-local (524); check it on the affine open cover $\{W_j\}$ of X' refining $(g')^{-1}$(affine opens of X) supplied by 521: over each member the restriction $M|_{W_j}$ is a pushforward from $W_j \cap (g')^{-1}(V)$ by the member node 525. Hence M is in the essential image, the unit is an isomorphism, and bareBC is an isomorphism. □

Lemma 527 (Open immersions are stable under base change)**.** *Provided by Mathlib (`Mathlib.AlgebraicGeometry.O` The class of open immersions of schemes is stable under base change: in any pullback square, the base change of an open immersion is again an open immersion.*

Lemma 528 (Left base change of an open immersion is an open immersion)**.** *Given a cartesian square with maps g_V, p', p, g' satisfying $p' \cdot g_V = g' \cdot p$, if p is an open immersion then so is its base change p'.*

This supplies the open-immersion property of p' needed for the target-side restriction in Stage 2 of 520.

Proof. Open immersions are stable under base change (527); applying that stability to the given cartesian square turns the open immersion p into the open immersion p'. □

Lemma 529 (Bare Beck–Chevalley mate of the restricted square)**.** *For the cartesian square $p' \cdot g' = g'_V \cdot p$ of 520 there is a canonical natural transformation — the* bare base-change map *—*

$$\text{bareBC}\colon \quad \text{pushforward}\, p \circ (g')^* \implies (g'_V)^* \circ \text{pushforward}\, p',$$

namely the Beck–Chevalley mate*, under the two pullback–pushforward adjunctions $p^* \dashv p_*$ and $p'^* \dashv p'_*$, of the canonical pullback 2-isomorphism $(g')^* \circ p'^* \cong p^* \circ (g'_V)^*$ attached to the square. It exists for* any *morphisms forming the cartesian square — no flatness, no open-immersion hypothesis — and its being an isomorphism is the genuine Beck–Chevalley content discharged in Stage 2 of 520.*

Proof. Assemble the pullback 2-isomorphism $(g')^* p'^* \cong (p' \cdot g')^* = (g'_V \cdot p)^* \cong (g'_V)^* p^*$ by combining the pseudofunctor composition isomorphism for $p' \cdot g'$, the reindexing along the square equation $p' \cdot g' = g'_V \cdot p$, and the inverse composition isomorphism for $g'_V \cdot p$ (the content of 530). The mate construction for the adjunctions $p^* \dashv p_*$ and $p'^* \dashv p'_*$ turns this 2-isomorphism into the stated natural transformation. The construction is formal and uses no hypothesis on the maps. □

Lemma 530 (Pullback telescope across the restricted square)**.** *In the cartesian square of 520, the iterated pullback of any $\mathcal{O}_X$-module $\mathcal{F}$ telescopes:*

$$p'^*((g')^* \mathcal{F}) \;\cong\; (p' \cdot g')^* \mathcal{F} \;=\; (g'_V \cdot p)^* \mathcal{F} \;\cong\; (g'_V)^*(p^* \mathcal{F}),$$

using only the pseudofunctor comparison $(q \cdot r)^* \cong r^* \circ q^*$ and reindexing of $(-)^*$ along the square equation $p' \cdot g' = g'_V \cdot p$. No flatness and no affineness hypotheses are needed.

Proof. Compose the pseudofunctor composition isomorphism for $p' \cdot g'$, the reindexing isomorphism along $p' \cdot g' = g'_V \cdot p$, and the inverse composition isomorphism for $g'_V \cdot p$, then evaluate the resulting natural isomorphism at $\mathcal{F}$. □

Lemma 531 (Per-intersection-open X-level Beck–Chevalley for the twisted nerve)**.** *In the setting of 532, fix an index family σ with intersection open $U_\sigma = \mathrm{coverInterOpen}\,\mathfrak{U}\,\sigma$, $j_\sigma : U_\sigma \hookrightarrow X$, and write U'_σ, $j'_\sigma : U'_\sigma \hookrightarrow X'$ for the corresponding data of the base-changed cover $\mathfrak{U}'$. Then there is the per-intersection-open X-level Beck–Chevalley isomorphism*

$$(g')^*((j_\sigma)_*(j_\sigma)^*\mathscr{F}) \;\cong\; (j'_\sigma)_*(j'_\sigma)^*((g')^*\mathscr{F}),$$

over the base-changed intersection open U'_σ; this is the single-open factor of the twisted-nerve identification. The index family σ is required to be finite*, so that 519 applies to identify $(g')^{-1}(U_\sigma)$ with* $\mathrm{coverInterOpen}\,\mathfrak{U}'\,\sigma$ *(the preimage commutes with the cover intersection only for finite index families); in the Čech application σ ranges over the finite index sets of the nerve.*

Proof. The restriction $j_\sigma : U_\sigma \hookrightarrow X$ is an open immersion, and its pullback along g' is the open immersion $j'_\sigma : U'_\sigma \hookrightarrow X'$ onto the preimage $(g')^{-1}(U_\sigma)$, which by 519 is exactly the base-changed intersection open $U'_\sigma = \mathrm{coverInterOpen}\,\mathfrak{U}'\,\sigma$. Restricting the cartesian square $g' : X' \to X$ over j_σ is therefore the restricted cartesian square of 520, whose open-immersion Beck–Chevalley isomorphism is the asserted comparison. When the affine $\widetilde{(-)}$-model is needed downstream, both sides are identified with their affine forms by 508 at altitude 1. □

Lemma 532 (The base-changed nerve is the nerve of the base-changed data)**.** *Let $f : X \to S$ be* separated *and fix an affine open-immersion cover $\mathfrak{U} = \{U_i = \mathrm{Spec}\,A_i\}$ of X (so every fibre power is affine, 500). Applying $(g')^*$ to the Čech nerve of $(\mathfrak{U}, \mathscr{F})$ yields the Čech nerve of the base-changed data $(\mathfrak{U}', (g')^*\mathscr{F})$, where $\mathfrak{U}'$ is the base change of $\mathfrak{U}$ along g': a natural isomorphism* $(g')^* \circ \mathrm{CechNerve}(\mathfrak{U}, -) \cong \mathrm{CechNerve}(\mathfrak{U}', (g')^*(-))$.

Proof. Geometric backbone (first half). The cover Čech nerve coverCechNerve of $\mathfrak{U}$ base-changes to that of $\mathfrak{U}'$: the fibre powers $U_{i_0} \times_X \cdots \times_X U_{i_p}$ of the cover commute with the base change g' (pullback preserves fibre products), so the preimages $(g')^{-1}(U_{i_0\cdots i_p})$ are exactly the corresponding intersections $U'_{i_0\cdots i_p}$ of $\mathfrak{U}'$. This identifies the underlying diagrams of open subschemes. It does *not* by itself produce the nerve isomorphism: the Čech nerve is valued not in open subschemes but in the push–pull functor p_*p^* (for p the projection of the relevant fibre power), so the comparison carries genuine sheaf-theoretic content beyond the identification of supports.

Genuine content (second half, Beck–Chevalley). The degree-n component is the X-level square along g': the asserted identification is the sheaf-level Beck–Chevalley isomorphism $(g')^*(\mathrm{pushPullObj}\,\mathscr{F}\,Y_n) \cong \mathrm{pushPullObj}((g')^*\mathscr{F})\,Y'_n$ (that is, $g'^*(p_*p^*F) \cong p'_*p'^*(g'^*F)$), where Y_n, Y'_n are the degree-n fibre powers of the cover Čech nerves of $\mathfrak{U}$ and $\mathfrak{U}'$.

1. *Product decomposition (LHS).* The degree-n fibre power is the coproduct $Y_n = \coprod_\sigma U_\sigma$ over index tuples $\sigma : \mathrm{Fin}(n+1) \to I$, so by 371 (which requires the cover to be finite) the push–pull object decomposes as the finite product $\mathrm{pushPullObj}\,\mathscr{F}\,Y_n \cong \prod_\sigma \mathrm{pushPullObj}\,\mathscr{F}\,(\mathrm{Over.mk}\,j_\sigma)$, and $(g')^*$ preserves this finite product. The left-hand side thus reduces to $\prod_\sigma (g')^*((j_\sigma)_*(j_\sigma)^*\mathscr{F})$.

2. *Per-σ component.* Each factor is the per-intersection-open X-level Beck–Chevalley iso $(g')^*((j_\sigma)_*(j_\sigma)^*\mathscr{F}) \cong (j'_\sigma)_*(j'_\sigma)^*((g')^*\mathscr{F})$ of 531 (the X-level analogue of 517 for the restricted cartesian square over U_σ). Here each intersection $U_\sigma = U_{i_0} \cap \cdots \cap U_{i_p}$ is affine by separatedness (501); the abstract data is identified with the affine $\widetilde{(-)}$-form by 508 at *altitude 1* (the restriction level $p_n^*\mathscr{F} \cong \widetilde{N_n}$ over the affine Y_n), pushed forward along g'; and on that model the comparison IS the affine termwise base change 129 (via 497), assembled from the concrete tilde dictionaries 123 and 126 with the commutative-algebra cancellation cancelBaseChange (127), *not* the canonical adjoint mate.

3. *RHS reassembly (the residual cover-base-change obligation).* To recognise the product $\prod_\sigma (j'_\sigma)_*(j'_\sigma)^*((g')^*\mathcal{F})$ as the push–pull object pushPullObj$((g')^*\mathcal{F})\, Y'_n$ of the base-changed cover one applies 371 *in reverse* to $\mathfrak{U}'$. This requires the base-changed cover $\mathfrak{U}'$ to be finite with affine members, together with the cover-base-change identification coverInterOpen $\mathfrak{U}'\,\sigma = (g')^{-1}(\text{coverInterOpen}\,\mathfrak{U}\,\sigma)$ of 519 (the preimages of the intersections of $\mathfrak{U}$ are the intersections of $\mathfrak{U}'$). This geometric matching of the base-changed cover with the preimage cover is the Beck–Chevalley content of the argument.

4. *Naturality.* The naturality condition is verified by the index-omission restriction: each coface and codegeneracy is induced by an index omission, hence by an inclusion of finite affine intersections, and the termwise comparisons are natural in those inclusions.

□

Definition 533 (Degreewise pullback of the Čech complex as an alternating-coface complex)**.** Specialising 496 to $F = g^*$ and to the push–pull nerve underlying $\check{\mathcal{C}}^\bullet(\mathfrak{U}, \mathcal{F})$: applying g^* degreewise to the relative Čech complex is the alternating-coface complex of $g^* \circ (\text{pushforward}\, f) \circ \text{CechNerve}(\mathfrak{U}, \mathcal{F})$. This is step (1) of the base-change construction below.

Definition 534 (Reduction of the Čech base change to a cosimplicial iso)**.** Given a cosimplicial natural isomorphism

$$e : g^* \circ (\text{pushforward}\, f) \circ \text{CechNerve}(\mathfrak{U}, \mathcal{F}) \;\cong\; (\text{pushforward}\, f') \circ \text{CechNerve}(\mathfrak{U}', (g')^*\mathcal{F}),$$

the Čech base-change complex isomorphism follows mechanically: it is the composite

$$\text{pullback\_cechComplex\_alternatingIso} \,\circ\, \text{alternatingCofaceMapComplex}\,.\text{mapIso}(e),$$

i.e. the degreewise pullback identification (533, step (1)) followed by Functor. mapIso of e through the alternating-coface complex functor. No per-degree Čech differential square is checked by hand: functoriality of alternatingCofaceMapComplex supplies the differential compatibility automatically.

The hypothesis e is precisely the residual obligation. It is constructed in 535 as the whiskered composite of the Beck–Chevalley natural iso 518 with the twisted-nerve identification 532, both pushed through the Čech nerve. Supplying e is the sole remaining open obligation (Stacks Tag 02KG); 536 below obtains the complex iso by feeding e into the present reduction.

Definition 535 (The Čech base-change cosimplicial isomorphism e)**.** The cosimplicial natural isomorphism e required by 534 is assembled from the two base-change leaves. It is the composite

$$e \;=\; (518)\,;(\text{pushforward}\, f').\text{mapIso}(532),$$

i.e. first the Beck–Chevalley natural iso $g^* \circ (\text{pushforward}\, f) \cong (\text{pushforward}\, f') \circ (g')^*$ whiskered through the Čech nerve, followed by the image under the functor pushforward f' (via Functor. mapIso) of the twisted-nerve identification $(g')^* \circ \text{CechNerve}(\mathfrak{U}, -) \cong \text{CechNerve}(\mathfrak{U}', (g')^*(-))$. Both factors are isomorphisms of cosimplicial objects, so e is one as well; it is precisely the hypothesis consumed by 534.

Definition 536 (Tensorial base change of the Čech complex (Stacks 02KG, OPEN))**.** *Source: Stacks Project, Cohomology of Schemes, Tag 02KG (`lemma-affine-base-change`); the Čech-complex relation in the proof of Tag 02KH (`lemma-flat-base-change-cohomology`).* Applying g^* degreewise to the relative Čech complex $\check{\mathcal{C}}^\bullet(\mathfrak{U}, \mathcal{F})$ yields a Čech complex naturally isomorphic to the relative Čech complex $\check{\mathcal{C}}^\bullet(\mathfrak{U}', (g')^*\mathcal{F})$ of the base-changed data, with $\mathfrak{U}'$ the base change of $\mathfrak{U}$ along g'. Only the *existence* of such a natural isomorphism is asserted — the downstream assembly Lemma 480 concludes Nonempty of an isomorphism — so no identification with any canonical base-change map is required.

Proof. The construction factors through 534: it suffices to supply the cosimplicial iso e and feed it into that reduction. We bypass the canonical base-change map entirely. The sole genuine input is the *affine termwise* base change 129, which is assembled from the concrete tilde identifications 123 and 126 together with the commutative-algebra cancellation cancelBaseChange (127) — not the canonical adjoint mate pushforwardBaseChangeMap. Unwinding the reduction, the construction is the composite, pushed through the functor alternatingCofaceMapComplex:

1. By Lemma 496, applying g^* degreewise commutes with alternatingCofaceMapComplex, so g^* of the Čech complex is the alternating-coface complex of $g^* \circ (\mathrm{pushforward}\, f) \circ \mathrm{CechNerve}(\mathfrak{U}, \mathcal{F})$.

2. The Beck–Chevalley natural iso (Lemma 518) rewrites the cosimplicial object as $(\mathrm{pushforward}\, f')\circ (g')^* \circ \mathrm{CechNerve}(\mathfrak{U}, \mathcal{F})$; since this is a natural iso of cosimplicial objects, feeding it through the *functor* alternatingCofaceMapComplex via Functor. mapIso makes compatibility with the Čech differentials automatic — no per-degree naturality square need be checked by hand.

3. Lemma 532 identifies the $(g')^*$-twisted nerve with $\mathrm{CechNerve}(\mathfrak{U}', (g')^*\mathcal{F})$, so the alternating-coface complex of the rewritten object is exactly $\check{\mathcal{C}}^\bullet(\mathfrak{U}', (g')^*\mathcal{F})$.

Composing (1)–(3) yields the required isomorphism of complexes; the homology isomorphism that Lemma 480 consumes is then homologyMapIso of this complex iso, exactly as in that lemma's assembly. Following Stacks Tag 02KH, since g is flat the relation persists on taking cohomology; flatness is *not* needed for the present complex iso, the affine base change being flatness-free. □

Chapter 9

Genus of a smooth proper curve

Throughout this chapter, k denotes a field and C a smooth proper geometrically irreducible curve over k of relative dimension 1. In other words, $C \to \operatorname{Spec} k$ is a morphism of schemes carrying the geometric typeclasses that make it a curve in the modern sense.

9.1 Definition

Definition 537 (Genus of a smooth proper curve)**.**
The *genus* of a smooth proper curve C over k is the natural number

$$g(C) \;=\; \dim_k H^1(C,\, \mathcal{O}_C),$$

the k-dimension of the first sheaf-cohomology group of the structure sheaf of C.

Proof. The definition is realised by passing the structure sheaf through the sheaf-of-k-modules construction and applying the k-linear sheaf cohomology H^1. The result is a k-vector space; its dimension is extracted via the standard finiteness rank.

Remark 538 (Lean implementation)**.** The formal definition uses the project's ModuleCat$k$-flavoured sheaf cohomology of the structure sheaf:

$$\texttt{genus}\, C \;:=\; \text{Module.}\texttt{finrank}\, k\, (\text{Scheme. HModule } k\, (\text{Scheme. toModuleKSheaf } C)\, 1).$$

The carrier toModuleKSheaf C is the sheaf-of-k-modules constructed from $\mathcal{O}_C$ via the structure morphism; the wrapper HModule $kF1$ is the Module$k$-flavoured sheaf cohomology, given by Ext from the constant sheaf at ModuleCat.of $k\,k$, which carries a canonical Module$k$-structure inherited from the Linear$k$-enrichment of the sheaf category.

Mathlib's Module.`finrank` returns the k-dimension when the underlying module is finite-dimensional and zero otherwise. For a smooth proper geometrically irreducible curve, Serre's theorem on coherent cohomology of a proper k-scheme guarantees $\dim_k H^1(C, \mathcal{O}_C) < \infty$, so $g(C)$ is the geometric genus on the relevant case. The closure is therefore the one-liner above, confirmed to typecheck against current Mathlib. □

Equivalent reformulations available once Riemann–Roch and Serre duality are formalised:

- $g(C) = \dim_k H^0(C,\, \Omega^1_{C/k})$ (Serre duality).

- $g(C) = 1 - \chi(\mathcal{O}_C)$, where χ is the Euler characteristic.
- For a smooth projective curve over $\mathbb{C}$, $g(C)$ equals the topological genus of $C(\mathbb{C})$.

Each of these requires Mathlib infrastructure that is currently absent. The first reformulation in particular requires the sheaf of relative differentials of schemes (Phase B), and the others require Riemann–Roch / Serre duality / a topological-vs-algebraic geometry comparison.

9.2 Mathlib gap

The honest definition $\dim_k H^1(C, \mathcal{O}_C)$ requires a Module$k$-flavoured sheaf cohomology of $\mathcal{O}_C$ on the topology of opens of the underlying space of C. Through earlier iterations the project has built this:

- **Sheaf cohomology of** AddCommGrpCat**-valued sheaves** (Mathlib): defined as Ext from the constant sheaf at AddCommGrpCat.of (ULift $\mathbb{Z}$).
- **Sheaf cohomology of** ModuleCat$k$**-valued sheaves** (this project): the parallel Module$k$-valued construction with unit object ModuleCat.of $k\,k$ instead of ULift $\mathbb{Z}$, gluing through the Linear$k$-enrichment of the sheaf category to recover a Module$k$ structure on the cohomology groups.
- **Structure sheaf as a** ModuleCat$k$**-valued sheaf** (this project): for C over Spec k, the sheafification of the presheaf $U \mapsto$ ModuleCat.of $k\,(\Gamma(C, U))$ where the k-action is the algebra structure induced by the structure morphism.
- H^0 **algebraic bridge** (this project): $H^0(C, \mathcal{O}_C)_{\mathrm{Module}k} \simeq_k \mathrm{Hom}_{\mathrm{Sheaf}}(\mathrm{const}, \mathrm{toModuleKSheaf}\, C)$, via Mathlib's Ext linear equivalence. Used downstream for the geometric content $H^0(C, \mathcal{O}_C) \simeq k$ on a connected proper k-curve.

The remaining Mathlib gap for the *theorem* (as opposed to the *definition*) of genus:

- **Serre finiteness for $H^i(C, \mathcal{F})$ on a proper k-curve.** Mathlib's ModuleCat.`finite_ext` provides Module.`Finite` for module-Ext only, not sheaf-Ext. The Čech-cohomology scaffold is the recommended direction.

9.3 User authorisation of `noncomputable`

The original challenge file writes `def genus`, not `noncomputable def genus`. Because Module.`finrank`, Abelian. Ext .`instModule`, and the structure-sheaf-as-k-module construction are all noncomputable in Mathlib, the honest closure of the body requires the `noncomputable` modifier on the `def`. The user explicitly authorised agents to add this modifier as a non-signature change: names, argument types, and argument order remain verbatim.

9.4 What this iteration does and does not

This iteration *defines* the genus. It does not *compute* it on any specific curve, and it does not prove that the result equals the geometric genus in the classical sense (which requires Serre finiteness for non-vacuity and Riemann–Roch for verification on examples). The downstream theorem `smoothOfRelativeDimension_genus` in `Jacobian.lean` states that the Jacobian of C is smooth of relative dimension $g(C)$ — that theorem will require Serre finiteness on the way and is queued for the iteration that closes the Jacobian construction itself.

Chapter 10

The rigidity lemma and its corollaries

This chapter proves the rigidity lemma and the two elementary consequences used in the Albanese construction. The argument is characteristic-free and uses only properness, affine constancy, and separatedness. In particular, it does not use the theorem of the cube or coherent cohomology.

Throughout, $\bar{k}$ is an algebraically closed field. Products are fibre products over $\operatorname{Spec} \bar{k}$, and a point means a $\bar{k}$-point.

10.1 The rigidity lemma

Lemma 539 (Rigidity Lemma (Mumford, Form I))**.** *Let X be a complete variety, let Y and Z be varieties, fix a point $x_0 \in X$, and let $f\colon X \times Y \to Z$ be a morphism. Suppose that for some $y_0 \in Y$ the image $f(X \times \{y_0\})$ is the single point z_0. Then there is a morphism $g\colon Y \to Z$ such that*

$$f = g \circ p_2,$$

where $p_2\colon X \times Y \to Y$ is the second projection.

Proof. Define $g(y) = f(x_0, y)$. Let U_0 be an affine neighbourhood of z_0, put $F = Z \setminus U_0$, and set

$$G = p_2(f^{-1}(F)).$$

The projection p_2 is closed because X is complete, hence G is closed. The hypothesis on the fibre over y_0 gives $y_0 \notin G$, so $V = Y \setminus G$ is nonempty. For every $y \in V$, the complete variety $X \times \{y\}$ maps into the affine variety U_0, and therefore its image is a point. That point is $f(x_0, y)$. Thus f and $g \circ p_2$ agree on the nonempty open set $X \times V$. This open set is dense, and the target is separated, so the two morphisms agree everywhere. □

Remark 540. The collapse assumption is essential. For example, the first projection $X \times Y \to X$ does not agree on a nonempty open set with the map obtained by replacing the first coordinate by a fixed point, unless X is itself a point. The proof above uses neither the theorem of the cube nor cohomology.

Remark 541. The proof separates into four ingredients: the cartesian identity for the collapsed map, closedness of the second projection, constancy of a proper integral slice mapping to an affine scheme, and extension of equality from a dense set to a separated target. The following lemmas record these ingredients individually.

Lemma 542 (Cartesian collapse identity). *Fix $x_0 \in X$, and let $s_{x_0} : Y \to X \times Y$ be the section $y \mapsto (x_0, y)$ of p_2. Then $s_{x_0} \circ p_2$ is the endomorphism $(x, y) \mapsto (x_0, y)$ of $X \times Y$.*

Proof. The second component is the identity because $p_2 \circ s_{x_0} = \mathrm{id}_Y$. The first component is the constant map with value x_0. The product universal property therefore identifies the composite with $(x, y) \mapsto (x_0, y)$. □

Lemma 543 (Completeness makes the second projection closed). *If X is proper over $\bar{k}$, then the underlying topological map of $p_2 : X \times Y \to Y$ is closed.*

Proof. The projection is the base change of $X \to \operatorname{Spec} \bar{k}$ along $Y \to \operatorname{Spec} \bar{k}$. Proper morphisms are universally closed, and universal closedness is stable under base change. Hence p_2 is closed. □

Lemma 544 (Scheme-theoretic gluing core). *Suppose that X is proper, $X \times Y$ is reduced, geometrically irreducible, and locally of finite type over $\bar{k}$, and Z is separated. Fix points $x_0 \in X$, $y_0 \in Y$, and $z_0 \in Z$. If $f : X \times Y \to Z$ maps $X \times \{y_0\}$ to z_0, then f equals the morphism $(x, y) \mapsto f(x_0, y)$ on all of $X \times Y$.*

Proof. By 545, the collapse hypothesis gives a nonempty open set on which the two morphisms agree. Geometric irreducibility makes this open set dense. Two morphisms from a reduced scheme to a separated scheme that agree after a dominant open immersion are equal, so the equality holds everywhere. □

Lemma 545 (Dense-open agreement). *Let X be proper, suppose that $X \times Y$ is reduced, geometrically irreducible, and locally of finite type over $\bar{k}$, and let Z be separated. Fix points $x_0 \in X$, $y_0 \in Y$, and $z_0 \in Z$. If $f : X \times Y \to Z$ maps $X \times \{y_0\}$ to z_0, then there is a nonempty open subset on which*

$$f(x, y) = f(x_0, y).$$

Proof. Choose an affine open neighbourhood U_0 of z_0, set $F = Z \setminus U_0$, and let $G = p_2(f^{-1}(F))$. By 543, the set G is closed. The collapsed fibre is disjoint from $f^{-1}(F)$, so $y_0 \notin G$. Hence $V = Y \setminus G$ is nonempty and $U = p_2^{-1}(V)$ is a nonempty saturated open subset of $X \times Y$ with $f(U) \subseteq U_0$. Apply 546 to U. □

Lemma 546 (Constancy on a saturated open). *Let X be proper, suppose that $X \times Y$ is reduced, geometrically irreducible, and locally of finite type over $\bar{k}$, and let Z be separated. Fix $x_0 \in X$, and let $U = p_2^{-1}(V)$ be a saturated open subset of $X \times Y$. If $f : X \times Y \to Z$ maps U into an affine open subset $U_0 \subseteq Z$, then on U one has $f(x, y) = f(x_0, y)$.*

Proof. The open subscheme U is locally of finite type over the Jacobson scheme $\operatorname{Spec} \bar{k}$, hence its closed points are dense. At every closed point of U, the desired equality follows from 550. The source is reduced and the target is separated, so 547 promotes these pointwise equalities to equality of morphisms on U. □

Lemma 547 (Equality from closed points). *Let W be a reduced Jacobson scheme and let Z be separated. If two morphisms $g_1, g_2 : W \to Z$ agree after restriction along $\operatorname{Spec} \kappa(x) \to W$ for every closed point $x \in W$, then $g_1 = g_2$.*

Proof. Form the coproduct

$$P = \coprod_{x \in W_{\mathrm{cl}}} \operatorname{Spec} \kappa(x)$$

and its natural morphism $q\colon P \to W$. Its image contains every closed point of W, so it is dense because W is Jacobson. Thus q is dominant. The hypotheses and the coproduct universal property give $g_1 \circ q = g_2 \circ q$. Dominance of q, reducedness of W, and separatedness of Z then imply $g_1 = g_2$. $\square$

Lemma 548 (Proper integral schemes are constant on affine targets)**.** *Let W be an integral scheme, proper and locally of finite type over $\bar{k}$, and let V be affine. If $a, b\colon \operatorname{Spec} \bar{k} \to W$ are sections of the structure morphism and $g\colon W \to V$, then $g \circ a = g \circ b$.*

Proof. The ring $\Gamma(W, \mathcal{O}_W)$ is a field because W is integral and proper. It is finite over $\bar{k}$ because W is locally of finite type. Since $\bar{k}$ is algebraically closed, the structure map $\bar{k} \to \Gamma(W, \mathcal{O}_W)$ is an isomorphism. The two sections induce the same inverse on global sections. Morphisms into the affine scheme V are determined by their maps on global sections, so $g \circ a = g \circ b$. $\square$

Lemma 549 (Integrality descends to a retract)**.** *Let T be an integral scheme. If morphisms $r\colon S \to T$ and $p\colon T \to S$ satisfy $p \circ r = \mathrm{id}_S$, then S is integral.*

Proof. The map p is surjective because it has the section r. Therefore the continuous image of the irreducible space T is all of S, and S is irreducible. For each $x \in S$, functoriality of stalk maps and $p \circ r = \mathrm{id}_S$ show that

$$\mathcal{O}_{S,x} \longrightarrow \mathcal{O}_{T,r(x)}$$

is split injective. The target is reduced, so $\mathcal{O}_{S,x}$ is reduced. Hence every stalk of S is reduced, and S is reduced. Reducedness and irreducibility give integrality. $\square$

Lemma 550 (Constancy on a closed slice)**.** *In the setting of 546, let x be a closed point of U. Then f and the map $(x', y) \mapsto f(x_0, y)$ agree after restriction along the residue-field point $\operatorname{Spec} \kappa(x) \to U$.*

Proof. Since U is locally of finite type over the algebraically closed field $\bar{k}$, the residue field of its closed point x is $\bar{k}$. Thus x is a $\bar{k}$-point. Write $y = p_2(x)$; then y and (x_0, y) are also $\bar{k}$-points. Since U is saturated, the whole fibre X_y lies in U, and its image under f lies in the affine open U_0. Identifying X_y with X, the slice section and first projection exhibit X as a retract of the integral product $X \times Y$, so it is integral by 549; it is also proper and locally of finite type over $\bar{k}$. The two points of X_y represented by x and (x_0, y) therefore have the same image in U_0 by 548. Restricting this equality along $\operatorname{Spec} \kappa(x) \to U$ gives the claim. $\square$

10.2 Product decompositions and pointed maps

Remark 551. The results in this section are direct consequences of the rigidity lemma. The theorem of the cube enters the theory of abelian varieties later and is not needed here.

Lemma 552 (Canonical decomposition over a product)**.** *Let V be proper, suppose that $V \times W$ is a reduced geometrically irreducible variety, and let A be a proper group variety. Fix points $v_0 \in V$, $w_0 \in W$. If $h\colon V \times W \to A$ satisfies $h(v_0, w_0) = e$, define*

$$f(v) = h(v, w_0), \qquad g(w) = h(v_0, w).$$

Then

$$h(v, w) = f(v)g(w).$$

Equivalently, $h = (f \circ p)(g \circ q)$, where p and q are the two projections. Commutativity of A is not required.

Proof. Set $\varphi = h/((f \circ p)(g \circ q))$. On $V \times \{w_0\}$, the map φ is the identity element because $g(w_0) = h(v_0, w_0) = e$. By the rigidity lemma, φ factors through q. On the section $\{v_0\} \times W$, it is again the identity, so the factor through q is identically the identity. Hence $\varphi = e$ and the displayed decomposition follows. □

Corollary 553 (Unique additive decomposition (Milne Corollary 1.5))**.** *Let V and W be complete varieties with points v_0 and w_0, and let A be an abelian variety. Every morphism $h\colon V \times W \to A$ with $h(v_0, w_0) = 0$ has a unique expression*

$$h = f \circ p + g \circ q,$$

where $f(v_0) = 0$ and $g(w_0) = 0$.

Proof. Existence is 552. Restricting a decomposition to $V \times \{w_0\}$ recovers f, and restricting it to $\{v_0\} \times W$ recovers g. Thus the two summands are unique. □

Lemma 554 (A pointed regular map is a homomorphism)**.** *Let A and B be abelian varieties, and let $\alpha\colon A \to B$ be a regular map with $\alpha(0_A) = 0_B$. Then α is a group homomorphism.*

Proof. Consider

$$\varphi(a, a') = \alpha(a + a') - \alpha(a) - \alpha(a').$$

This is a regular map $A \times A \to B$, and it vanishes on both coordinate axes. Apply 552 to φ. Both canonical summands are zero, hence $\varphi = 0$. Therefore $\alpha(a + a') = \alpha(a) + \alpha(a')$ for all a, a', so α is a homomorphism. □

Corollary 555 (Regular maps are translations of homomorphisms (Milne Corollary 1.2))**.** *Every regular map $\alpha\colon A \to B$ of abelian varieties is the composite of a homomorphism with a translation.*

Proof. Put $b = \alpha(0_A)$ and define $\beta(a) = \alpha(a) - b$. Then $\beta(0_A) = 0_B$, so 554 shows that β is a homomorphism. Thus α is β followed by translation by b. □

Chapter 11

The Jacobian as an abelian variety

Let C be a smooth proper geometrically irreducible curve over a field k, with genus $g(C)$. The Jacobian is constructed uniformly as the identity component $\mathrm{Pic}^0_{C/k}$ of the Picard scheme. It is an abelian variety of dimension $g(C)$; when $g(C) = 0$, it is $\operatorname{Spec} k$.

11.1 Albanese objects

Definition 556 (Albanese universal property of a pointed curve)**.** Let $P \in C(k)$. An abelian variety J together with a pointed morphism $\iota_P\colon C \to J$ is an *Albanese object* for (C, P) if every pointed morphism $f\colon C \to A$ to an abelian variety factors uniquely through ι_P: there is a unique morphism of k-schemes $g\colon J \to A$ such that

$$P \circ \iota_P = \eta_J, \qquad f = \iota_P \circ g.$$

Remark 557. The phrase "abelian variety" includes the group structure, properness, smoothness, and geometric irreducibility of both J and every target A. These are hypotheses of the universal property, rather than extra conclusions inside it.

11.1.1 The universal morphism

Definition 558 (The Albanese universal morphism)**.** An Albanese structure on (C, P, J) determines a distinguished morphism $\iota_P\colon C \to J$, called its universal morphism.

Lemma 559 (Pointed condition)**.** *The universal morphism satisfies $P \circ \iota_P = \eta_J$.*

Lemma 560 (Universal factorisation)**.** *Every pointed morphism $f\colon C \to A$ to an abelian variety factors uniquely as $f = \iota_P \circ g$ through a morphism of k-schemes $g\colon J \to A$.*

Theorem 561 (Uniqueness of the Albanese object)**.** *If (J_1, ι_1) and (J_2, ι_2) are Albanese objects for (C, P), there is a unique morphism $e\colon J_1 \to J_2$ satisfying $\iota_2 = \iota_1 \circ e$.*

Proof. The universal property of J_1 applied to ι_2 gives $e\colon J_1 \to J_2$; that of J_2 applied to ι_1 gives $e' : J_2 \to J_1$. Both $e \circ e'$ and the identity of J_1 factor ι_1 through itself, so uniqueness gives $e \circ e' = \mathrm{id}_{J_1}$; similarly $e' \circ e = \mathrm{id}_{J_2}$. The original universal property gives the required uniqueness of e. $\square$

Remark 562. Consequently the Albanese object is unique up to a unique isomorphism compatible with the universal morphism.

11.2 A uniform Jacobian witness

Definition 563 (Jacobian witness)**.** A *Jacobian witness* for C consists of a single k-scheme J, an abelian-variety structure on J, a proof that J is smooth of relative dimension $g(C)$, and, for every $P \in C(k)$, an Albanese structure on (C, P, J).

Remark 564. The same J works for every marked point. Replacing P by another rational point translates the universal morphism by an automorphism of J, so it does not change the underlying abelian variety.

Remark 565. Smoothness is implied by smoothness of relative dimension $g(C)$. It is nevertheless useful to record both the abelian-variety property and the dimension assertion as parts of the witness.

11.2.1 The Picard witness $J = \mathrm{Pic}^0_{C/k}$

Theorem 566 (The Picard identity component is a Jacobian witness)**.** *The identity component* $\mathrm{Pic}^0_{C/k}$ *carries a Jacobian witness for* C*.*

Proof. Representability of the relative Picard functor gives $\mathrm{Pic}_{C/k}$, and its identity component $\mathrm{Pic}^0_{C/k}$ is a group scheme, smooth, proper and geometrically irreducible by 1646. Those four data are the first four fields of the witness; of the four, 1643 and 1644 are the ones whose own proofs are still open.

Three further steps are needed, and each is isolated as a lemma because it does not follow from the four above.

First, the hypotheses must be reconciled: $\mathrm{Pic}^0_{C/k}$ is constructed for a geometrically *integral* curve carrying a k-rational point, whereas a witness is required for every smooth proper geometrically irreducible curve. Geometric integrality is automatic (567); the rational point is not, and is the single genuine gap (570), a gap only over a general base field — over an algebraically closed field it is 571 and the witness is 573.

Second, the smoothness supplied by 1646 carries no relative dimension, while the witness records smoothness at the specific relative dimension $g(C)$; the tangent-space identification $T_0 \, \mathrm{Pic}^0_{C/k} \cong H^1(C, \mathcal{O}_C)$ of 1602 supplies the dimension, and 574 is the resulting refinement.

Third, the Albanese structure is required for *every* $P \in C(k)$ and for every genus, whereas 587 is proved over an algebraically closed base field and for $g(C) > 0$; the extension is 577. Together these seven data form the required witness. ☐

Lemma 567 (A smooth geometrically irreducible curve is geometrically integral)**.** *Let* C *be a smooth proper geometrically irreducible curve over a field* k*. Then* C *is geometrically integral over* k*.*

Proof. A smooth morphism is geometrically reduced (568), and a morphism that is both geometrically reduced and geometrically irreducible is geometrically integral: for every field extension K/k the base change C_K is then both reduced and irreducible, hence integral. ☐

Lemma 568 (A smooth morphism is geometrically reduced)**.** *A morphism of schemes* $f \colon X \to Y$ *that is smooth of some relative dimension is geometrically reduced: for every field* K *and every morphism* $\mathrm{Spec}\, K \to Y$*, the base change* $X \times_Y \mathrm{Spec}\, K$ *is a reduced scheme.*

Proof. Smoothness is stable under base change, so it suffices to show that a scheme X smooth over a field K is reduced. Reducedness is local, and X is covered by affine opens $\operatorname{Spec} S$ with S standard smooth over K; each is reduced by 569 applied to the domain K. □

Lemma 569 (A standard smooth algebra over a domain is reduced)**.** *Let B be a domain and S a standard smooth B-algebra. Then S is reduced.*

Proof. By the structure theorem for standard smooth algebras, S is étale over a polynomial ring $P = B[x_1, \dots, x_n]$, which is again a domain. So it suffices to prove that a flat, formally unramified P-algebra S essentially of finite type over a domain P is reduced, étale algebras being flat, formally unramified and essentially of finite type.

Let $F = \operatorname{Frac}(P)$. The algebra $F \otimes_P S$ is formally unramified and essentially of finite type over the field F, hence reduced. Since S is flat over P and $P \to F$ is injective, the induced map $S \cong P \otimes_P S \to F \otimes_P S$ is injective, so S embeds into a reduced ring and is therefore reduced. □

Lemma 570 (The curve has a k-rational point)**.** *A smooth proper geometrically irreducible curve C over a field k has a k-rational point.*

Proof. This assertion is *false* as stated. A smooth proper geometrically integral curve need not have a k-rational point: a conic over $\mathbb{Q}$ with no rational point is a genus-0 example, and there are genus-2 curves over $\mathbb{Q}$ with no rational point. Over an algebraically closed field a rational point always exists, since a curve there is a nonempty Jacobson space whose closed points are rational.

The lemma is therefore stated here only to name the gap between the hypotheses of the witness and those of the Picard construction, and it is not provable in this form. Closing it requires either weakening the claim — carry $C(k) \neq \emptyset$ as a hypothesis of the Jacobian itself, which proves something weaker than intended — or replacing the input by a representability statement that needs no section, namely representability of the étale sheafification (25.1). That choice is open. The algebraically closed case is not affected by it and is 571. □

Lemma 571 (A curve over an algebraically closed field has a rational point)**.** *Let C be a smooth proper geometrically irreducible curve over an algebraically closed field $\bar{k}$. Then C has a $\bar{k}$-rational point.*

Proof. Geometric irreducibility over the one-point base $\operatorname{Spec} \bar{k}$ makes the underlying space of C irreducible, in particular nonempty, and smoothness makes $C \to \operatorname{Spec} \bar{k}$ locally of finite type, so that space is a Jacobson space. A nonempty Jacobson space has a closed point x. Over an algebraically closed field the residue field at a closed point of a scheme locally of finite type is $\bar{k}$ itself, so x is the image of a section of $C \to \operatorname{Spec} \bar{k}$, which is the required rational point. □

Remark 572 (Which leaf is false and which is merely unproved)**.** 570 and 571 state the same conclusion under different hypotheses, and the pair delimits what is open. Over an arbitrary base field the statement is false, so it admits no proof and the witness built on it rests on an inconsistent hypothesis — its consequences are vacuously true. Over an algebraically closed field the same statement is a theorem. So the open question is not the existence of a rational point in general, but which of two formulations is claimed over an arbitrary k: representability of $\operatorname{Pic}^{\sharp}_{C/k}$ with $C(k) \neq \emptyset$ as a hypothesis, or representability of the étale sheafification $\operatorname{Pic}_{(C/k)\text{ét}}$, which needs no section (25.1). The remaining obligations of the witness — representability itself (1568), 1643, 1644, 574 and 577 — are of the other kind: true statements awaiting proofs.

Discharging the rational point does not reduce their number. The identity component is defined only once $\mathrm{Pic}_{C/k}$ exists, so over an algebraically closed field the discharge makes representability an obligation that must be met rather than one carried along inside the rational-point hypothesis. What changes is the kind of every remaining obligation, not the count.

Theorem 573 (A Jacobian witness over an algebraically closed field)**.** *Over an algebraically closed field $\bar{k}$, the identity component $\mathrm{Pic}^0_{C/\bar{k}}$ carries a Jacobian witness for every smooth proper geometrically irreducible curve C, with no rational-point hypothesis.*

Proof. The construction is that of 566, with the rational point supplied by 571 instead of assumed. Every other field of the witness is unchanged, so the open obligations are representability (1568), 1643, 1644, 574 and 577, all of them true statements (572). □

Lemma 574 (Relative dimension of the identity component)**.** *For a smooth proper geometrically integral curve C/k whose Picard functor is representable, $\mathrm{Pic}^0_{C/k}$ is smooth over k of relative dimension $g(C)$.*

Proof. $\mathrm{Pic}^0_{C/k}$ is a smooth k-group scheme by 1646. For a smooth group scheme over a field the relative dimension is constant and equals the dimension of the tangent space at the identity, which is $H^1(C, \mathcal{O}_C)$ by 1602. Its k-dimension is $g(C)$ by the definition of the genus (537). □

Lemma 575 (Tangent space of the identity component has dimension the genus)**.** *For a smooth proper geometrically integral curve C/k whose Picard functor is representable, the Zariski tangent space of $\mathrm{Pic}^0_{C/k}$ at the identity has dimension $g(C)$ over the residue field of the identity point.*

Proof. The tangent space at the identity is $H^1(C, \mathcal{O}_C)$ (1602), whose k-dimension is $g(C)$ by the definition of the genus (537). □

Remark 576 (What separates the tangent dimension from the relative dimension)**.** 575 is not 574. Two further ingredients separate them: smoothness of $\mathrm{Pic}^0_{C/k}$ over k, and the identification of the tangent-space dimension with the relative dimension of a smooth morphism. For the second, the relative dimension of a smooth morphism is computed by the rank of the module of Kähler differentials Ω, so the step is the duality, over a field, between that rank and the dimension of the tangent space at a rational point — together with the passage from that affine-local statement to a cover of $\mathrm{Pic}^0_{C/k}$. Translation by a rational point makes the dimension independent of the chosen point, which is what lets a computation at the identity determine the relative dimension everywhere.

Lemma 577 (Albanese property at every point, in every genus)**.** *Let C/k be a smooth proper geometrically integral curve whose Picard functor is representable. Then for every $P \in C(k)$, the scheme $\mathrm{Pic}^0_{C/k}$ satisfies the Albanese universal property for the pointed curve (C, P).*

Proof. For $g(C) > 0$ and k algebraically closed this is 578.

Three extensions are required. Over a general k, apply the algebraically closed case over k^s and descend: the universal morphism and the factoring homomorphism are unique, so they are Galois-equivariant and descend to k. For $g(C) = 0$ the tangent space $H^1(C, \mathcal{O}_C)$ vanishes, so $\mathrm{Pic}^0_{C/k}$ is a smooth proper group scheme of dimension 0, namely $\mathrm{Spec}\, k$; the universal property then asserts that every pointed morphism from C to an abelian variety is constant, which is the rigidity statement that a morphism from a genus-0 curve to an abelian variety is constant. The third is the basepoint condition $P \circ \iota_P = \eta$, a hypothesis of 578 rather than a restriction on it. □

Lemma 578 (Albanese property over an algebraically closed field)**.** *Let C be a smooth proper geometrically integral curve over an algebraically closed field $\bar{k}$, with representable Picard functor and genus $g(C) > 0$, and let $P \in C(\bar{k})$ satisfy $P \circ \iota_P = \eta$ for the Abel–Jacobi morphism ι_P of (C, P). Then $\mathrm{Pic}^0_{C/\bar{k}}$ satisfies the Albanese universal property for the pointed curve (C, P).*

Proof. The universal property of 556 has two clauses. The basepoint clause is the hypothesis. The factorisation clause — every pointed morphism $\varphi : C \to A$ into an abelian variety factors uniquely through ι_P — is 587 at the same object, the Albanese development's Jacobian scheme being $\mathrm{Pic}^0_{C/\bar{k}}$. □

Theorem 579 (Existence of a Jacobian witness)**.** *Every smooth proper geometrically irreducible curve has a Jacobian witness.*

Proof. Take the witness on $\mathrm{Pic}^0_{C/k}$ supplied by Theorem 566. □

Definition 580 (Chosen Jacobian witness)**.** Fix one Jacobian witness for C.

11.3 The Jacobian and its structure

Definition 581 (Jacobian)**.** The *Jacobian* $\mathrm{Jac}(C)$ is the underlying k-scheme of the chosen Jacobian witness. Equivalently, $\mathrm{Jac}(C) = \mathrm{Pic}^0_{C/k}$ up to the unique isomorphism of Albanese objects.

Theorem 582 (Group structure)**.** *The Jacobian is a group scheme over k.*

Proof. This is the group structure carried by the chosen Jacobian witness. □

Theorem 583 (Dimension and smoothness)**.** *The structural morphism $\mathrm{Jac}(C) \to \mathrm{Spec}\, k$ is smooth of relative dimension $g(C)$.*

Proof. This is the dimension assertion carried by the chosen Jacobian witness. □

Theorem 584 (Properness)**.** *The structural morphism $\mathrm{Jac}(C) \to \mathrm{Spec}\, k$ is proper.*

Proof. This is the properness assertion carried by the chosen Jacobian witness. □

Theorem 585 (Geometric irreducibility)**.** *The structural morphism $\mathrm{Jac}(C) \to \mathrm{Spec}\, k$ is geometrically irreducible.*

Proof. This is the geometric-irreducibility assertion carried by the chosen Jacobian witness. □

Remark 586. If $g(C) = 0$, then $\mathrm{Jac}(C) = \mathrm{Spec}\, k$. If $g(C) = 1$ and $C(k)$ is nonempty, the choice of a rational point identifies C with its Jacobian. For $g(C) \geq 2$, the Jacobian has dimension $g(C)$, whereas C has dimension one.

11.4 The Albanese property of Pic^0

Theorem 587 (Albanese universal property of $\mathrm{Pic}^0_{C/k}$)**.** *Fix $P \in C(k)$, put $J = \mathrm{Pic}^0_{C/k}$, and let $f^P(Q) = [\mathcal{O}_C(Q - P)]$. For every pointed morphism $\varphi \colon C \to A$ to an abelian variety there is a unique group-scheme homomorphism $\psi \colon J \to A$ such that $\varphi = f^P \circ \psi$.*

Proof. For $g = g(C) > 0$, the morphism

$$(Q_1, \dots, Q_g) \longmapsto \varphi(Q_1) + \cdots + \varphi(Q_g)$$

is symmetric and therefore descends to $C^{(g)}$. The Abel map $C^{(g)} \to J$ is birational onto J, so this descent defines a rational map $J \dashrightarrow A$. A rational map from a nonsingular variety to an abelian variety extends uniquely to a regular morphism; denote the extension by ψ. By construction $\varphi = f^P \circ \psi$, and $\psi(0) = 0$, so rigidity makes ψ a homomorphism. If ψ' is another such homomorphism, then ψ and ψ' agree on the sum of g copies of $f^P(C)$, which is all of J; hence $\psi = \psi'$. When $g = 0$, $J = \operatorname{Spec} k$ and the result follows from rigidity. $\square$

Chapter 12

The Abel–Jacobi map

Let C be a smooth proper geometrically irreducible curve over a field k, let $P \in C(k)$, and let $\mathrm{Jac}(C)$ be the Albanese variety constructed in Chapter 11. This chapter records the universal morphism and its two defining properties.

12.1 Definition

Definition 588 (Abel–Jacobi map)**.** The *Abel–Jacobi map at P* is the universal pointed morphism

$$\alpha_P \colon C \longrightarrow \mathrm{Jac}(C)$$

supplied by the Albanese property of $\mathrm{Jac}(C)$.

Remark 589 (Picard description)**.** Under the identification $\mathrm{Jac}(C) = \mathrm{Pic}^0_{C/k}$, the morphism is

$$\alpha_P(Q) = [\mathcal{O}_C(Q - P)].$$

Equivalently, it is classified by $\mathcal{O}_{C\times C}(\Delta - p_1^* P)$, regarded as a family of degree-zero line bundles over the second factor.

12.2 Pointed property

Lemma 590 (α_P sends P to the identity)**.** *The Abel–Jacobi map is pointed:*

$$P \circ \alpha_P = \eta_{\mathrm{Jac}(C)}.$$

Proof. This is the pointed condition in the defining Albanese property of $\mathrm{Jac}(C)$. In the Picard description it is the equality $[\mathcal{O}_C(P - P)] = [\mathcal{O}_C]$. $\square$

Remark 591 (Restriction of the universal line bundle)**.** Restricting $\mathcal{O}_{C\times C}(\Delta - p_1^* P)$ along the section (P, id_C) gives the trivial line bundle on C, which is the identity point of $\mathrm{Pic}^0_{C/k}$.

12.3 Universal property

Theorem 592 (Albanese property of the Jacobian)**.** *Let A be an abelian variety over k. For every morphism $f\colon C \to A$ satisfying $P \circ f = \eta_A$, there exists a unique morphism of k-schemes $g\colon \mathrm{Jac}(C) \to A$ such that*

$$f = \alpha_P \circ g.$$

Proof. By definition, $\mathrm{Jac}(C)$ is an Albanese object for (C, P), and α_P is its universal pointed morphism. Applying the universal factorisation property to f gives the required g, while the uniqueness clause gives uniqueness. □

Chapter 13

Relative Spec

13.1 Setup and motivation

The Route A construction of $J = \mathrm{Pic}^0_{C/k}$ as a g-dimensional abelian variety is organised around the relative Picard functor $\mathrm{Pic}^\sharp_{C/k}(T) = \mathrm{Pic}(C \times_k T)/\pi_T^* \mathrm{Pic}(T)$ on the category of k-schemes. Representing $\mathrm{Pic}^\sharp_{C/k}$ requires the line-bundle pullback construction on a relative curve $C \times_k T \to T$, which in turn rests on a *relative scheme* construction on a quasi-coherent sheaf of $\mathcal{O}_X$-algebras $\mathcal{A}$, the *relative spectrum* $\pi : \underline{\mathrm{Spec}}_X(\mathcal{A}) \to X$. This chapter packages the construction of $\underline{\mathrm{Spec}}_X(-)$ and its three core properties — universal property, affine base, base change — as the foundational A.1.a building block. The relative Picard functor itself (A.1.c), the line-bundle pullback functor on a product (A.1.b), and the identity-component / dimension theory of J (A.3) live in separate sibling chapters devoted to those constructions.

13.2 Quasi-coherent sheaves of algebras

Definition 593 (Quasi-coherent $\mathcal{O}_X$-algebra)**.** *Source: [Stacks Project], tag 01LL (situation-relative-spec).* Let X be a scheme. A *quasi-coherent sheaf of $\mathcal{O}_X$-algebras* is a sheaf $\mathcal{A}$ of $\mathcal{O}_X$-algebras (i.e. an $\mathcal{O}_X$-module equipped with $\mathcal{O}_X$-linear multiplication and unit morphisms satisfying associativity, commutativity, and the unit law as sheaf morphisms) whose underlying $\mathcal{O}_X$-module is quasi-coherent. We write $\mathrm{QcohAlg}(X)$ for the category of quasi-coherent sheaves of $\mathcal{O}_X$-algebras and $\mathcal{O}_X$-algebra morphisms.

13.3 The relative-spectrum scheme

The construction is given affine-locally: on every affine open $U = \mathrm{Spec}\, R \subseteq X$ one sets $\pi^{-1}(U) := \mathrm{Spec}(\mathcal{A}(U))$ regarded as an R-algebra, and the local pieces glue compatibly because $\mathcal{A}$ is quasi-coherent (the transition isomorphism $\mathcal{A}(U) \otimes_R S \xrightarrow{\sim} \mathcal{A}(V)$ for an inclusion $V = \mathrm{Spec}\, S \subseteq U$ gives an open immersion of the corresponding Specs).

Theorem 594 (Relative spectrum exists)**.** *Source: [Stacks Project], tag 01LQ (lemma-glue-relative-spec); cf. [Hartshorne], II Ex. 5.17(a). Let X be a scheme and $\mathcal{A}$ a quasi-coherent $\mathcal{O}_X$-algebra (593). There exists a scheme $\underline{\mathrm{Spec}}_X(\mathcal{A})$ together with an affine morphism $\pi : \underline{\mathrm{Spec}}_X(\mathcal{A}) \to X$ such that*

1. *for every affine open $U \subseteq X$, there is an isomorphism $i_U : \pi^{-1}(U) \xrightarrow{\sim} \mathrm{Spec}(\mathcal{A}(U))$ over U;*
2. *for $U \subseteq U' \subseteq X$ both affine open, the composite $\mathrm{Spec}(\mathcal{A}(U)) \xrightarrow{i_U^{-1}} \pi^{-1}(U) \hookrightarrow \pi^{-1}(U') \xrightarrow{i_{U'}} \mathrm{Spec}(\mathcal{A}(U'))$ coincides with the open immersion induced by the restriction $\mathcal{A}(U') \to \mathcal{A}(U)$.*

The pair $(\underline{\mathrm{Spec}}_X(\mathcal{A}), \pi)$ is unique up to unique isomorphism over X.

Proof. By the open-cover gluing principle for schemes, it suffices to specify affine local pieces $\mathrm{Spec}(\mathcal{A}(U))$ for each affine open $U \subseteq X$ and open-immersion transition data $\mathrm{Spec}(\mathcal{A}(U)) \hookrightarrow \mathrm{Spec}(\mathcal{A}(U'))$ for $U \subseteq U'$ satisfying the cocycle condition. The transition morphisms come from the ring maps $\mathcal{A}(U') \to \mathcal{A}(U)$; quasi-coherence yields the cartesian square $\mathrm{Spec}(\mathcal{A}(U)) = U \times_{U'} \mathrm{Spec}(\mathcal{A}(U'))$, so each transition is an open immersion. Transitivity (Stacks lemma-transitive-spec) gives the cocycle condition. Uniqueness of the glued scheme is the standard property of the gluing functor. □

Definition 595 (Structure morphism of the relative spectrum)**.** Let X be a scheme and $\mathcal{A}$ a quasi-coherent $\mathcal{O}_X$-algebra. The *structure morphism* $\pi : \underline{\mathrm{Spec}}_X(\mathcal{A}) \to X$ is the canonical projection of the relative spectrum to its base, obtained as the colimit descent of the affine-local structure morphisms $\mathrm{Spec}(\mathcal{A}(U)) \to U$ over the directed affine open cover of X. It is the morphism whose affineness and base-change behaviour are the subject of the universal-property and base-change results below.

Theorem 596 (Universal property of the relative spectrum)**.** *Source: [Stacks Project], tag 01LQ (lemma-spec + section-spec functor description); [Hartshorne], II Ex. 5.17(b). Let X be a scheme and $\mathcal{A}$ a quasi-coherent $\mathcal{O}_X$-algebra. Then $\underline{\mathrm{Spec}}_X(\mathcal{A})$ represents the functor $F : \mathrm{Sch}^{op} \to \mathrm{Set}$, $T \mapsto F(T) := \{(f, \varphi) : f : T \to X$ a morphism, $\varphi : f^*\mathcal{A} \to \mathcal{O}_T$ an $\mathcal{O}_T$-algebra map$\}$. Equivalently, for any X-scheme $g : T \to X$, there is a natural bijection*

$$\mathrm{Hom}_X(T, \underline{\mathrm{Spec}}_X(\mathcal{A})) \;\cong\; \mathrm{Hom}_{\mathcal{O}_X\text{-}alg}(\mathcal{A}, g_*\mathcal{O}_T)$$

(using the adjunction between g_* and g^* to convert between $\mathcal{A} \to g_*\mathcal{O}_T$ and $g^*\mathcal{A} \to \mathcal{O}_T$). The bijection is natural in T.

Proof. Following Stacks tag 01LQ (proof of lemma-spec): the functor F is a Zariski sheaf because morphisms of schemes and $\mathcal{O}$-algebra maps both glue under open covers. Choose an affine open cover $X = \bigcup U_i$. The subfunctor $F_i \subseteq F$ of pairs (f, φ) with $f(T) \subseteq U_i$ is representable by $\mathrm{Spec}(\mathcal{A}(U_i))$. Indeed, factoring f through U_i identifies $f^*\mathcal{A}$ with the pullback of $\mathcal{A}|_{U_i}$, and the affine-base case (597) represents the resulting algebra maps. Each $F_i \hookrightarrow F$ is an open subfunctor (pull back U_i along f). The F_i jointly cover F since the U_i cover X. The gluing of representable open subfunctors produces a representing scheme, which is canonically isomorphic to the glued scheme $\underline{\mathrm{Spec}}_X(\mathcal{A})$ of 594. □

13.4 Affine base

Theorem 597 (Relative spectrum over an affine base)**.** *Source: [Stacks Project], tag 01LO (lemma-spec-affine). Let $X = \mathrm{Spec}\, R$ be an affine scheme and let $\mathcal{A}$ be a quasi-coherent $\mathcal{O}_X$-algebra. Set $A := \Gamma(X, \mathcal{A})$, regarded as an R-algebra. Then there is a canonical isomorphism of X-schemes*

$$\underline{\mathrm{Spec}}_X(\mathcal{A}) \;\cong\; \mathrm{Spec}(A).$$

In particular, on an affine base the relative spectrum reduces to the absolute spectrum of the global sections.

Proof. Write $X = \operatorname{Spec} R$ and $A = \Gamma(X, \mathcal{A})$. Quasi-coherence of $\mathcal{A}$ gives $\mathcal{A} = \widetilde{A}$ as $\mathcal{O}_X$-modules (with the R-algebra structure on A inducing the algebra structure on $\widetilde{A}$). The ring map $R \to A$ gives a morphism $f_{\mathrm{univ}} : \operatorname{Spec} A \to \operatorname{Spec} R = X$ and a canonical $\mathcal{O}_{\operatorname{Spec} A}$-algebra map $\varphi_{\mathrm{univ}} : f_{\mathrm{univ}}^* \mathcal{A} = \widetilde{A \otimes_R A} \to \mathcal{O}_{\operatorname{Spec} A} = \widetilde{A}$ given by multiplication $a \otimes a' \mapsto aa'$. We verify $(\operatorname{Spec} A, f_{\mathrm{univ}}, \varphi_{\mathrm{univ}})$ represents F on the affine base: given a test scheme T and a pair (f, φ), the composite $A = \Gamma(X, \mathcal{A}) \xrightarrow{f^\sharp} \Gamma(T, f^*\mathcal{A}) \xrightarrow{\varphi} \Gamma(T, \mathcal{O}_T)$ is a ring map $A \to \Gamma(T, \mathcal{O}_T)$ and hence corresponds to a morphism $T \to \operatorname{Spec} A$ by the affine universal property (Mathlib `Scheme.toSpec_naturality` / `Scheme.Hom.toSpec`). One checks this assignment inverts the universal map $(f, \varphi) \mapsto a$ defined by $f = f_{\mathrm{univ}} \circ a$, $\varphi = a^* \varphi_{\mathrm{univ}}$. □

13.5 Base change

Theorem 598 (Base change of the relative spectrum)**.** *Source: [Stacks Project], tag 01LS (lemma-spec-base-change + lemma-spec-properties). Let $g : T \to X$ be a morphism of schemes and let $\mathcal{A}$ be a quasi-coherent $\mathcal{O}_X$-algebra. Then $g^*\mathcal{A}$ is a quasi-coherent $\mathcal{O}_T$-algebra and there is a canonical isomorphism of T-schemes*

$$T \times_X \underline{\operatorname{Spec}}_X(\mathcal{A}) \;\cong\; \underline{\operatorname{Spec}}_T(g^*\mathcal{A}).$$

Proof. By 596, $\underline{\operatorname{Spec}}_X(\mathcal{A})$ represents the functor F sending an X-scheme $h : S \to X$ to the set of $\mathcal{O}_S$-algebra maps $h^*\mathcal{A} \to \mathcal{O}_S$. A morphism $S \to T \times_X \underline{\operatorname{Spec}}_X(\mathcal{A})$ over T is the same as a pair of an X-morphism $S \to \underline{\operatorname{Spec}}_X(\mathcal{A})$ together with a T-factorisation of $S \to X$, i.e. a morphism $S \to T$ whose composition with g equals $S \to X$. Tracing through, this is a T-morphism $f' : S \to T$ together with an $\mathcal{O}_S$-algebra map $(f')^*(g^*\mathcal{A}) = f^*\mathcal{A} \to \mathcal{O}_S$. This is precisely $F'(S)$, the functor for $(T, g^*\mathcal{A})$, so the canonical map $T \times_X \underline{\operatorname{Spec}}_X(\mathcal{A}) \to \underline{\operatorname{Spec}}_T(g^*\mathcal{A})$ is an isomorphism by Yoneda. □

13.6 Pullback compatibility helpers

This section records the substrate lemmas and constructions underlying the base-change isomorphism (598). Throughout, $g : T \to X$ is a morphism of schemes, $\mathcal{A}$ is a quasi-coherent $\mathcal{O}_X$-algebra, and q denotes the first projection of the pullback $T \times_X \underline{\operatorname{Spec}}_X(\mathcal{A})$ along the structure morphism $\pi : \underline{\operatorname{Spec}}_X(\mathcal{A}) \to X$. Write $\mathcal{A}_g := q_* \mathcal{O}_{T \times_X \underline{\operatorname{Spec}}_X(\mathcal{A})}$ for the quasi-coherent algebra reconstructed from this affine pullback.

Lemma 599 (Affineness of the structural pullback projection)**.** *The first projection $q : T \times_X \underline{\operatorname{Spec}}_X(\mathcal{A}) \to T$ of the pullback of the structure morphism π along g is an affine morphism.*

Proof. The structure morphism π is affine by 594, and affineness of morphisms is stable under base change; hence its pullback q is affine. □

Lemma 600 (Quasi-coherence overlay for the pushforward along the affine projection)**.** *The pushforward $q_* \mathcal{O}_{T \times_X \underline{\operatorname{Spec}}_X(\mathcal{A})}$ along the affine projection q carries the Stacks 01LL quasi-coherence overlay: for every affine open $U \subseteq T$ and section $r \in \Gamma(T, U)$, the restriction of the pushforward to the basic open $D(r) \subseteq U$ is the localization away from r of its sections over U. This is the predicate making $\mathcal{A}_g$ a well-formed quasi-coherent $\mathcal{O}_T$-algebra.*

Proof. Since q is affine (599), the preimage $q^{-1}(U)$ of an affine open U is affine, and $q^{-1}(D(r))$ is the basic open of $q^{-1}(U)$ cut out by the pulled-back section; the localization-away identification then follows from the affine-open localization criterion. $\square$

Definition 601 (Algebra reconstructed from the base-changed spectrum)**.** Let $g : T \to X$ be a morphism of schemes and $\mathcal{A}$ a quasi-coherent $\mathcal{O}_X$-algebra (593). Define the reconstructed $\mathcal{O}_T$-algebra $\mathcal{A}_g$ as the pushforward $q_*\mathcal{O}_{T\times_X \underline{\mathrm{Spec}}_X(\mathcal{A})}$ along the first projection $q : T \times_X \underline{\mathrm{Spec}}_X(\mathcal{A}) \to T$ of the pullback of the structure morphism π along g. Concretely, the underlying $\mathcal{O}_T$-module is the topological pushforward $q_*\mathcal{O}$ and the unit is the comorphism $q^\sharp : \mathcal{O}_T \to q_*\mathcal{O}$; the Stacks 01LL quasi-coherence (coequifibered) overlay is supplied by 600, which itself rests on the affineness of q (599). Thus $\mathcal{A}_g$ is an object of $\mathrm{QcohAlg}(T)$.

Definition 602 (Per-affine-open piece of the base-change iso)**.** For an affine open $U \subseteq T$, the preimage $q^{-1}(U)$ is affine and is canonically isomorphic to Spec of the sections of $\mathcal{A}_g$ over U. This per-piece isomorphism feeds the global base-change iso construction.

Proof. As q is affine (599), $q^{-1}(U)$ is affine; its canonical isomorphism with $\operatorname{Spec}\Gamma(q^{-1}(U), \mathcal{O})$ matches the sections of the pushforward $\mathcal{A}_g$ over U by definitional unfolding. $\square$

Definition 603 (Pullback cocone on the relative-gluing functor)**.** A cocone on the relative-gluing-data functor of $\mathcal{A}_g$ whose apex is the pullback $T \times_X \underline{\mathrm{Spec}}_X(\mathcal{A})$; its component at an affine open U is the canonical map $\mathrm{Spec}(\mathcal{A}_g(U)) \to T \times_X \underline{\mathrm{Spec}}_X(\mathcal{A})$ into the affine piece $q^{-1}(U)$.

Proof. The components are the affine-open inclusion maps of the pulled-back pieces; their compatibility (naturality) follows from the functoriality of these inclusions under restriction along basic opens. $\square$

Lemma 604 (Descent of the pullback cocone covers the structure morphism)**.** *The colimit descent of the pullback cocone (603), as a morphism $\underline{\mathrm{Spec}}_T(\mathcal{A}_g) \to T \times_X \underline{\mathrm{Spec}}_X(\mathcal{A})$, composed with the projection q, equals the structure morphism $\underline{\mathrm{Spec}}_T(\mathcal{A}_g) \to T$.*

Proof. By the universal property of the colimit it suffices to check the identity on each cocone component, where it reduces to the compatibility of the affine-piece inclusion with the projection q. $\square$

Lemma 605 (Base-change descent is invertible)**.** *The colimit descent map $\underline{\mathrm{Spec}}_T(\mathcal{A}_g) \to T \times_X \underline{\mathrm{Spec}}_X(\mathcal{A})$ determined by the pullback cocone is an isomorphism of schemes.*

Proof. Being an isomorphism is local on the target, so it suffices to check it over each member $q^{-1}(U)$ of the affine open cover. There the descent restricts, via 604 and the range identification of the gluing inclusions, to the per-affine-open isomorphism (602). $\square$

Definition 606 (Base-change iso construction)**.** The canonical isomorphism of T-schemes $T \times_X \underline{\mathrm{Spec}}_X(\mathcal{A}) \cong \underline{\mathrm{Spec}}_T(\mathcal{A}_g)$, defined as the inverse of the colimit descent map of the pullback cocone.

Proof. The descent map is an isomorphism by 605, and the construction takes its inverse. $\square$

Lemma 607 (Existence of the base-change isomorphism)**.** *There exists a canonical isomorphism of T-schemes $T \times_X \underline{\mathrm{Spec}}_X(\mathcal{A}) \cong \underline{\mathrm{Spec}}_T(\mathcal{A}_g)$.*

Proof. Witnessed by the explicit construction 606. $\square$

13.7 Functoriality

Theorem 608 (Relative Spec functor)**.** *Source: [Stacks Project], tag 01LR; [Hartshorne], II Ex. 5.17(b). The construction $\mathcal{A} \mapsto \underline{\mathrm{Spec}}_X(\mathcal{A})$ assembles into a contravariant functor*

$$\underline{\mathrm{Spec}}_X : \mathrm{QcohAlg}(X)^{op} \longrightarrow \mathrm{AffSch}/X$$

from quasi-coherent $\mathcal{O}_X$-algebras to affine X-schemes: a morphism of $\mathcal{O}_X$-algebras $\psi : \mathcal{A} \to \mathcal{B}$ induces, by the universal property (596) applied to the pair $(\pi_{\mathcal{B}}, \psi \circ \pi_{\mathcal{B}}^*) : \pi_{\mathcal{B}}^*\mathcal{A} \to \pi_{\mathcal{B}}^*\mathcal{B} \to \mathcal{O}_{\mathrm{Spec}\,\mathcal{B}}$, an X-morphism $\underline{\mathrm{Spec}}_X(\psi) : \underline{\mathrm{Spec}}_X(\mathcal{B}) \to \underline{\mathrm{Spec}}_X(\mathcal{A})$. Functoriality (identity, composition) is direct from the universal property. The pushforward $\pi_*\mathcal{O}_{\underline{\mathrm{Spec}}_X(\mathcal{A})} \cong \mathcal{A}$ exhibits $\underline{\mathrm{Spec}}_X$ as an equivalence between $\mathrm{QcohAlg}(X)^{op}$ and the full subcategory of affine X-schemes, with inverse $f \mapsto f_*\mathcal{O}_Y$ (Stacks lemma-spec-properties part (3)).

Proof. Functoriality follows directly from 596: for $\psi : \mathcal{A} \to \mathcal{B}$, the pair $(\pi_{\mathcal{B}} : \underline{\mathrm{Spec}}_X(\mathcal{B}) \to X,\ \pi_{\mathcal{B}}^*(\psi))$ defines an X-morphism $\underline{\mathrm{Spec}}_X(\mathcal{B}) \to \underline{\mathrm{Spec}}_X(\mathcal{A})$ via the representing universal pair. Identity and composition follow from naturality of the representing bijection in $\mathcal{A}$. For the equivalence claim, $\underline{\mathrm{Spec}}_X$ is fully faithful because the isomorphism $\pi_*\mathcal{O}_{\underline{\mathrm{Spec}}_X(\mathcal{A})} \cong \mathcal{A}$ (locally identifying with $\Gamma(\mathrm{Spec}(\mathcal{A}(U)), \mathcal{O}) = \mathcal{A}(U)$ via 597) reconstructs $\mathcal{A}$ from $\underline{\mathrm{Spec}}_X(\mathcal{A})$. Essential surjectivity onto affine X-schemes uses the canonical morphism $X \to \underline{\mathrm{Spec}}_X(\pi_*\mathcal{O}_X)$ of Stacks lemma-canonical-morphism; on an affine X-scheme $\pi : Y \to X$, this canonical morphism is an isomorphism (affine-locally it is the identity $\mathrm{Spec}\,\Gamma(\pi^{-1}(U), \mathcal{O}_Y) \to \mathrm{Spec}\,\Gamma(\pi^{-1}(U), \mathcal{O}_Y)$). □

13.8 Construction by affine gluing

The relative spectrum is determined by the affine schemes $\mathrm{Spec}(\mathcal{A}(U))$ over the affine opens $U \subseteq X$. Quasi-coherence identifies their restrictions on overlaps, and the resulting cocycle glues both the schemes and their structure morphisms. The universal property shows that this glued object is independent of the chosen affine cover.

13.9 Further properties

Additional hypotheses on $\mathcal{A}$, such as finite presentation, flatness, or smoothness, impose corresponding finiteness and geometric properties on $\underline{\mathrm{Spec}}_X(\mathcal{A}) \to X$. The relative Picard construction uses the functoriality and base-change results above together with such properties in its own geometric setting.

Chapter 14

Line-bundle pullback on a relative curve

Let S be a scheme, let $C \to S$ be a morphism, and let $T \to S$ be a test scheme. Write $P = C \times_S T$ for their fibre product. When $S = \operatorname{Spec} k$ and C is a smooth proper geometrically integral curve, this is the relative curve used in the Picard construction. We first record the local description of line bundles needed for pullback.

14.1 Line bundles and local triviality

Definition 609 (Local triviality of a module)**.** An $\mathcal{O}_X$-module $\mathcal{M}$ on a scheme X is *locally trivial of rank one* if every point $x \in X$ has an affine open neighbourhood U and an isomorphism

$$\mathcal{M}|_U \;\cong\; \mathcal{O}_U.$$

Equivalently, in the category of sheaves of $\mathcal{O}_X$-modules,

$$\forall x : X,\ \exists U : X.\text{Opens},\quad x \in U \ \wedge\ U \text{ affine } \wedge\ \mathcal{M}|_U \cong \mathcal{O}_U.$$

Definition 610 (Line bundle on a product)**.** A *line bundle on* $P = C \times_S T$ is a locally trivial module of rank one on P. Thus the type of line bundles on the product is

$$\text{OnProduct}(C \to S, T \to S) := \{\mathcal{M} \in P.\text{Modules} \mid \mathcal{M} \text{ is locally trivial of rank one}\}.$$

Isomorphism classes of these objects are the usual elements of $\operatorname{Pic}(P)$; the tensor-product group structure is not needed for the constructions below.

Lemma 611 (Pullback preserves local triviality)**.** *Let $f : Y \to X$ be a morphism of schemes and let $\mathcal{M}$ be locally trivial of rank one on X. Then $f^*\mathcal{M}$ is locally trivial of rank one on Y.*

Proof. Fix $y \in Y$ and put $x = f(y)$. Choose an affine open U containing x and an isomorphism $\mathcal{M}|_U \cong \mathcal{O}_U$. The inverse image $f^{-1}(U)$ is an open neighbourhood of y; choose an affine open $V \subseteq f^{-1}(U)$ containing y. The map $V \to X$ factors through U, and restriction followed by pullback gives

$$(f^*\mathcal{M})|_V \cong (V \to U)^*(\mathcal{M}|_U) \cong (V \to U)^*\mathcal{O}_U \cong \mathcal{O}_V.$$

Hence every point of Y has an affine trivialising neighbourhood. $\square$

Definition 612 (Underlying module)**.** For a line bundle $\mathcal{L} \in \mathrm{OnProduct}(C \to S, T \to S)$, its *carrier* $\mathcal{L}.\mathrm{carrier}$ is the underlying $\mathcal{O}_P$-module obtained by forgetting the local-triviality witness.

Lemma 613 (A line bundle is locally trivial)**.** *If* $\mathcal{L} \in \mathrm{OnProduct}(C \to S, T \to S)$*, then* $\mathcal{L}.\mathrm{carrier}$ *is locally trivial of rank one.*

Proof. An element of OnProduct is a pair consisting of a module and a local-triviality witness. Projecting to the first component leaves exactly that witness, so the carrier has the asserted property. ☐

14.2 Pullback along the projection

Let $\pi_T : P \to T$ be the second projection. Pullback along π_T sends a line bundle on T to a line bundle on P.

Definition 614 (Pullback along the projection)**.** If $\mathcal{N}$ is a locally trivial $\mathcal{O}_T$-module of rank one, define

$$\pi_T^*\mathcal{N} := \big((\pi_T)^*\mathcal{N},\ \text{the local-triviality witness supplied by pullback preservation}\big) \in \mathrm{OnProduct}(C \to S, T \to S).$$

On isomorphism classes this is the usual pullback operation on line bundles.

14.3 Base change and composition

Lemma 615 (Composition of line-bundle pullbacks)**.** *Let* $g : T' \to T$ *be a morphism over* S*, so that* $\pi_{T'} = \pi_T \circ g$*. Let*

$$g_C : C \times_S T' \longrightarrow C \times_S T$$

be the induced base-change morphism. For every $\mathcal{O}_T$*-module* $\mathcal{N}$ *there exists an isomorphism*

$$g_C^*\pi_T^*\mathcal{N} \;\cong\; \pi_{T'}^*g^*\mathcal{N}.$$

Proof. The fibre-product projections satisfy $\pi_T \circ g_C = g \circ \pi_{T'}$. Pullback of modules is compatible with composition, so both sides are isomorphic to the pullback of $\mathcal{N}$ along this common composite. Composing these isomorphisms gives the result. ☐

14.4 Isomorphism classes and their naturality

The current line-bundle carrier records modules together with a local trivialisation. Before adding tensor products and quotienting by bundles from the base, we can already form its set of isomorphism classes.

Theorem 616 (Isomorphism classes of line bundles)**.** *For fixed* $C \to S$ *and* $T \to S$*, define*

$$\mathcal{L} \sim \mathcal{L}' \quad\Longleftrightarrow\quad \mathcal{L}.\mathrm{carrier} \cong \mathcal{L}'.\mathrm{carrier}.$$

This is an equivalence relation on $\mathrm{OnProduct}(C \to S, T \to S)$*, and the quotient set*

$$\mathrm{LB}(C \times_S T) := \mathrm{OnProduct}(C \to S, T \to S)/\!\sim$$

is the set of isomorphism classes of line bundles on $C \times_S T$*. The canonical projection onto this quotient is surjective.*

Proof. Reflexivity follows from the identity isomorphism, symmetry from taking the inverse of an isomorphism, and transitivity from composition. Thus $\sim$ is an equivalence relation, so its quotient exists; the quotient projection is surjective by definition. ☐

14.5 Naturality in the test scheme

Theorem 617 (Pullback descends to isomorphism classes)**.** *Let $g : T' \to T$ be a morphism over S and let g_C be its base change along $C \to S$. Pullback along g_C sends line bundles on $C \times_S T$ to line bundles on $C \times_S T'$ and sends isomorphic carriers to isomorphic carriers. Consequently it induces a unique map of quotient sets*

$$g^\sharp : \mathrm{LB}(C \times_S T) \longrightarrow \mathrm{LB}(C \times_S T'), \qquad [\mathcal{L}] \longmapsto [g_C^* \mathcal{L}].$$

Proof. Local triviality of $g_C^*\mathcal{L}$ follows from 611. If $e : \mathcal{L}.\mathrm{carrier} \cong \mathcal{L}'.\mathrm{carrier}$, the functoriality of pullback gives an isomorphism $g_C^*\mathcal{L}.\mathrm{carrier} \cong g_C^*\mathcal{L}'.\mathrm{carrier}$. Hence pullback respects $\sim$ and the universal property of the quotient in 616 gives the displayed map. $\square$

Chapter 15

The relative Picard functor and its étale sheafification

15.1 Setup and motivation

Fix a base field k and a smooth proper geometrically integral curve C over k. As in 14, write $\pi_T : C \times_k T \to T$ for the projection of a relative curve over a k-scheme T and $\pi_T^* : \mathrm{Pic}(T) \to \mathrm{Pic}(C \times_k T)$ for the induced pullback of isomorphism classes of line bundles (614).

The set-valued relative Picard presheaf

$$\mathrm{Pic}^{\sharp}_{C/k}(T) \;=\; \mathrm{Pic}(C \times_k T) \,/\, \pi_T^* \mathrm{Pic}(T)$$

of 616 was set up forgetting the abelian-group structure inherited from tensor product. This chapter restores that structure and then sheafifies. The first half (15.2 and 15.3) verifies that the quotient is canonically an abelian group and the functoriality of 617 respects that group structure; the result is a group-valued presheaf $\mathrm{Pic}^{\sharp}_{C/k} : (\mathrm{Sch}\,/k)^{op} \to \mathrm{Ab}$ on the category of k-schemes. The second half (15.4) takes the étale sheafification $\mathrm{Pic}^{\sharp}_{(C/k)\text{ét}} := (\mathrm{Pic}^{\sharp}_{C/k})^{\sim}$ in the étale topology on $(\mathrm{Sch}\,/k)^{op}$, and verifies that sheafification preserves the abelian-group target. The étale-sheafified functor is the object represented by the Picard scheme in 25.

15.2 Group structure on the relative Picard quotient

The Picard group $\mathrm{Pic}(X)$ of any scheme is canonically an abelian group under tensor product of line bundles (Stacks tag 01CR; the inverse of $[\mathcal{L}]$ is $[\mathcal{L}^{-1}] = [\mathcal{H}om_{\mathcal{O}_X}(\mathcal{L}, \mathcal{O}_X)]$). The pullback map $\pi_T^* : \mathrm{Pic}(T) \to \mathrm{Pic}(C \times_k T)$ of 614 respects this structure: pullback of $\mathcal{O}_X$-modules sends $\mathcal{O}_T$ to $\mathcal{O}_{C \times_k T}$ and is multiplicative on tensor products.

Lemma 618 (Group structure descends to the relative Picard quotient)**.**

Source: [Kleiman], "The Picard scheme", §2, Defs. df:aPf + df:Pfs (the Picard functors are abelian-group-valued, and the relative one is a quotient of abelian groups; cf. Stacks tag 01CR for the abelian-group structure on Pic*). Let C/k be a smooth proper geometrically integral curve and let T be a k-scheme. The Picard group* $\mathrm{Pic}(C \times_k T)$ *is an abelian group under tensor product of line bundles, and the pullback map*

$$\pi_T^* : \;\mathrm{Pic}(T) \longrightarrow \mathrm{Pic}(C \times_k T)$$

of 614 is a group homomorphism. Its image

$$H_T \;:=\; \pi_T^* \mathrm{Pic}(T)$$

is therefore a subgroup of $\mathrm{Pic}(C \times_k T)$*, and the quotient set of 616*

$$\mathrm{Pic}^\sharp_{C/k}(T) \;=\; \mathrm{Pic}(C \times_k T)\,/\,H_T$$

inherits a canonical abelian-group structure under which the quotient map $\mathrm{Pic}(C \times_k T) \twoheadrightarrow \mathrm{Pic}^\sharp_{C/k}(T)$ *is a surjective homomorphism with kernel exactly* H_T*.*

Proof.

The construction proceeds in four steps over the locally-trivial carrier

$$\mathrm{OnProduct}(\pi_C, \pi_T) \;=\; \{\, M : \text{a module on } C \times_k T \;\big|\; M \text{ is locally trivial}\,\}$$

of 610, on whose isomorphism classes the relative Picard quotient is taken. It is authored *in parallel* against two inputs being built concurrently: the locally-trivial pullback–tensor comparison isomorphism 696 (consumed in Step 2 to make π_T^* additive) and the tensor inverse 656 (consumed in Step 1 for group closure); both are themselves deferred targets.

Step 1 (abelian group on the carrier). On isomorphism classes of the carrier $\mathrm{OnProduct}(\pi_C, \pi_T)$, addition is the descent of the scheme-level tensor product of line bundles,

$$[L] + [L'] \;:=\; [\, L \otimes_{\mathcal{O}_{C\times_k T}} L'\,],$$

where $L \otimes L'$ is the locally-trivial descent of the scheme-level tensor product (657); the sum lands again in the carrier because the tensor product of two locally-trivial modules is locally trivial (648). The zero element is the class $[\mathcal{O}_{C\times_k T}]$ of the structure sheaf, the monoidal unit, which is locally trivial. Associativity, commutativity, and left/right unitality of $+$ descend from the scheme-level coherence isomorphisms — the associator (653), the braiding (655), and the unitors (654) — exactly as for the absolute Picard group on tensor-invertible isomorphism classes (664). The additive inverse is

$$-[L] \;:=\; [\, L^{\mathrm{inv}}\,],$$

where L^{inv} is the witness supplied by 656: it returns a *locally-trivial* module together with an isomorphism $L \otimes L^{\mathrm{inv}} \cong \mathcal{O}_{C\times_k T}$, so the inverse stays inside the loc-triv carrier and the group is closed under negation.

Step 2 (the pullback homomorphism π_T^*). Pullback along the projection π_T gives an additive map

$$\pi_T^* : \;\; \mathrm{Pic}(T) \;\longrightarrow\; \mathrm{OnProduct}(\pi_C, \pi_T),$$

whose underlying assignment of line bundles is 614. It is a homomorphism of abelian groups: it sends the unit to the unit, since $\pi_T^* \mathcal{O}_T \cong \mathcal{O}_{C\times_k T}$ by the unconditional pullback–unit isomorphism (671); and it is additive, since pullback commutes with tensor product on locally-trivial modules,

$$\pi_T^*(N \otimes_{\mathcal{O}_T} N') \;\cong\; \pi_T^* N \otimes_{\mathcal{O}_{C\times_k T}} \pi_T^* N',$$

by the locally-trivial comparison isomorphism 696 specialised to the projection π_T on the OnProduct carrier. This is the exact analogue, for the geometric pullback, of the additive monoid homomorphism on Picard groups induced by a ring map respecting tensor product, and is modelled on it field-for-field. Write $H_T := \mathrm{im}(\pi_T^*)$ for its image, a subgroup of the carrier's isomorphism-class group.

Step 3 (setoid reconciliation). The set-valued quotient of 616 is taken modulo the isomorphism-class equivalence that declares two line bundles related when their classes differ by an element of H_T. This relation coincides with the left-coset relation of the subgroup H_T inside the abelian group of Steps 1–2: both encode

$$L \sim L' \quad \Longleftrightarrow \quad [\, L^{\mathrm{inv}} \otimes L' \,] \in H_T.$$

Establishing this equivalence of relations is a concrete isomorphism-class argument over the carrier; it consumes the tensor inverse (Step 1) and the group law, but not the computational content of the comparison isomorphism.

Step 4 (transport). By Step 3 the quotient set underlying $\mathrm{Pic}^{\sharp}_{C/k}(T)$ is in canonical bijection with the group quotient $\mathrm{OnProduct}(\pi_C, \pi_T)/H_T$, which carries the canonical abelian-group structure of a quotient of an abelian group by a subgroup. Transporting this structure along the bijection of Step 3 endows $\mathrm{Pic}^{\sharp}_{C/k}(T)$ with its abelian-group structure, under which the canonical surjection $\mathrm{Pic}(C \times_k T) \twoheadrightarrow \mathrm{Pic}^{\sharp}_{C/k}(T)$ is a homomorphism with kernel exactly H_T. □

15.2.1 Helper constructions underlying the group law

The abelian-group instance of 618 is assembled from a handful of project-internal helper constructions on the locally-trivial carrier $\mathrm{OnProduct}(\pi_C, \pi_T)$: the tensor-product operation that realises addition, the local triviality of the unit (the zero element), the uniqueness of tensor inverses (which makes negation well defined), and the two descended operations on isomorphism classes. Each is recorded here for the one-to-one Lean correspondence; all are proved directly in Lean.

Definition 619 (Tensor product on relative line bundles)**.** For relative line bundles L, L' on the locally-trivial carrier $\mathrm{OnProduct}(\pi_C, \pi_T)$ of 610, their *tensor product* is

$$L \otimes L' \;:=\; L \otimes_{\mathcal{O}_{C \times_S T}} L',$$

the locally-trivial descent of the scheme-level tensor product (657). The result again lies in the carrier because the tensor product of two locally-trivial modules is locally trivial (648). This is the operation that realises addition $[L] + [L'] := [L \otimes L']$ on isomorphism classes.

Lemma 620 (The structure-sheaf unit is locally trivial)**.** *The structure sheaf $\mathcal{O}_{C \times_S T}$ (the monoidal unit) is locally trivial, hence defines an element of the carrier* $\mathrm{OnProduct}(\pi_C, \pi_T)$*. Its isomorphism class is the zero element of the group of 618.*

Proof. Proved directly in Lean. Local triviality is checked on an affine chart: for any point x, choose an affine open $W \ni x$; the restriction of the structure sheaf to W is its pullback along the inclusion $W \hookrightarrow C \times_S T$, and the pullback of the structure sheaf is again the structure sheaf by the unconditional pullback–unit isomorphism (671). Composing these two isomorphisms trivialises $(\mathcal{O}_{C \times_S T})|_W \cong \mathcal{O}_W$. □

Lemma 621 (Uniqueness of the tensor inverse)**.** *Source: [Stacks Project], Tag 0B8K (§17.25, invertible modules): any two $\otimes$-inverses of $\mathcal{L}$ are each isomorphic to $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{L}, \mathcal{O}_X)$, hence to one another. Let M, N, N' be relative line bundles, and suppose there are isomorphisms*

$$M \otimes N \;\cong\; \mathcal{O}_{C \times_S T} \qquad \text{and} \qquad M \otimes N' \;\cong\; \mathcal{O}_{C \times_S T}.$$

Then $N \cong N'$. In other words, the tensor inverse of a relative line bundle is unique up to isomorphism.

Proof. Proved directly in Lean. One transports N across the chain of coherence isomorphisms

$$N \;\cong\; N\otimes\mathcal{O} \;\cong\; N\otimes(M\otimes N') \;\cong\; (N\otimes M)\otimes N' \;\cong\; (M\otimes N)\otimes N' \;\cong\; \mathcal{O}\otimes N' \;\cong\; N',$$

using the right unitor and left unitor (654), the given trivialisations together with bifunctoriality of $\otimes$ (657), the associator (653), and the braiding (655). This is the relative-line-bundle analogue of the standard uniqueness of inverses in the Picard group. ☐

Definition 622 (Descended addition on the relative Picard quotient)**.** The addition on the relative Picard quotient $\mathrm{Pic}^\sharp_{C/k}(T) = \mathrm{Pic}(C\times_S T)/\pi_T^*\mathrm{Pic}(T)$ is the descent of the tensor product 619 to isomorphism classes:

$$[L]+[L'] \;:=\; [\,L\otimes L'\,].$$

Proof. Proved directly in Lean. Well-definedness on the quotient (616) is bifunctoriality of the tensor product: replacing L, L' by isomorphic representatives $L \cong M$, $L' \cong M'$ yields an isomorphism $L\otimes L' \cong M\otimes M'$ (657), so the class $[L\otimes L']$ is independent of the chosen representatives. ☐

Definition 623 (Descended negation on the relative Picard quotient)**.** The negation on the relative Picard quotient is

$$-[L] \;:=\; [\,L^{\mathrm{inv}}\,],$$

where L^{inv} is the locally-trivial tensor-inverse witness supplied by 656, together with its trivialisation $L\otimes L^{\mathrm{inv}} \cong \mathcal{O}_{C\times_S T}$.

Proof. Proved directly in Lean. Well-definedness on the quotient uses uniqueness of the tensor inverse (621): if $L \cong M$, then the chosen inverse witnesses L^{inv} and M^{inv} of 656 both trivialise the same class up to the isomorphism $L \cong M$, hence $L^{\mathrm{inv}} \cong M^{\mathrm{inv}}$, so $[L^{\mathrm{inv}}]$ is independent of the representative. ☐

The realized carrier of 618 is the locally-trivial iso-class set modulo the H_T-coset relation $\sim$, in its multiplicative form

$$L\sim L' \quad\Longleftrightarrow\quad \exists\, N \text{ locally trivial on } T,\ L\cong \pi_T^*N\otimes L'.$$

Keeping the tensor inverse out of the definition — it appears only in the symmetry and negation steps — confines the bridge locally trivial $\Rightarrow$ invertible to those two places. The following helpers assemble the quotient directly: the relation itself, the iso-class inclusion bridge, its three setoid axioms, the setoid it defines, the middle-four interchange, the well-definedness of addition, and the compatibility of negation. Each is recorded here for the one-to-one Lean correspondence.

Definition 624 (The H_T-coset relation on locally-trivial bundles)**.** On the locally-trivial carrier $\mathrm{OnProduct}(\pi_C, \pi_T)$ of 610, define, for locally-trivial L, L' on $C\times_k T$, the relation

$$L\sim L' \quad\Longleftrightarrow\quad \exists\, N \text{ locally trivial on } T,\ L \;\cong\; \pi_T^*N\otimes L',$$

where $\pi_T^*N = \texttt{pullbackAlongProjection}\,\pi_C\,\pi_T\,N$ is the pullback of 614. This *multiplicative form* is the left-coset relation of $H_T := \pi_T^*\mathrm{Pic}(T)$: it holds exactly when $[L]$ and $[L']$ differ by an element of H_T (equivalently $L\otimes L'^{-1} \cong \pi_T^*N$). It is coarser than the absolute iso-class relation `RelPicPresheaf.preimage_subgroup`.

Lemma 625 (Iso-class inclusion into the H_T-coset relation)**.** *If L and L' have isomorphic underlying bundles, then $L\sim L'$; in particular $\sim$ is coarser than the absolute iso-class relation, which is what lets the objectwise coherence isomorphisms descend to the quotient.*

Proof. Take $N = \mathcal{O}_T$. Then

$$\pi_T^*\mathcal{O}_T \otimes L' \cong \mathcal{O}_{C\times_k T} \otimes L' \cong L' \cong L,$$

the first isomorphism by the pullback–unit isomorphism 671, the second by the left unitor 654, and the last by hypothesis. Hence $L \cong \pi_T^*\mathcal{O}_T \otimes L'$, so $L \sim L'$. $\square$

Lemma 626 (Reflexivity of the H_T-relation)**.** *For every locally-trivial L on $C \times_k T$ one has $L \sim L$.*

Proof. Immediate from 625 applied to the identity isomorphism $L \cong L$. Concretely, taking $N = \mathcal{O}_T$ gives $\pi_T^*\mathcal{O}_T \otimes L \cong \mathcal{O}_{C\times_k T} \otimes L \cong L$, so $L \sim L$. $\square$

Lemma 627 (Symmetry of the H_T-relation)**.** *If $L \sim L'$ then $L' \sim L$.*

Proof. Suppose $L \cong \pi_T^*N \otimes L'$ with N locally trivial, and let N^{-1} be its tensor inverse on T (656), again locally trivial. Tensoring the hypothesis by $\pi_T^*(N^{-1})$ and multiplying the pullbacks through the locally-trivial comparison isomorphism 696,

$$\pi_T^*(N^{-1}) \otimes L \cong \pi_T^*(N^{-1}) \otimes \pi_T^*N \otimes L' \cong \pi_T^*(N^{-1} \otimes N) \otimes L' \cong \pi_T^*\mathcal{O}_T \otimes L' \cong L',$$

after reassociating and braiding with 653, 655, and 654 and applying the pullback–unit isomorphism 671. Hence $L' \cong \pi_T^*(N^{-1}) \otimes L$ with N^{-1} locally trivial, so $L' \sim L$. $\square$

Lemma 628 (Transitivity of the H_T-relation)**.** *If $L \sim L'$ and $L' \sim L''$ then $L \sim L''$.*

Proof. Suppose $L \cong \pi_T^*N \otimes L'$ and $L' \cong \pi_T^*N' \otimes L''$ with N, N' locally trivial. Substituting and reassociating with the associator 653,

$$L \cong \pi_T^*N \otimes (\pi_T^*N' \otimes L'') \cong (\pi_T^*N \otimes \pi_T^*N') \otimes L'' \cong \pi_T^*(N \otimes N') \otimes L'',$$

the last isomorphism by the locally-trivial comparison 696. Since $N \otimes N'$ is locally trivial (648), $L \sim L''$. $\square$

Lemma 629 (Middle-four interchange for $\otimes$)**.** *For arbitrary modules A, B, C, D on a scheme X there is a canonical isomorphism*

$$(A \otimes B) \otimes (C \otimes D) \;\cong\; (A \otimes C) \otimes (B \otimes D)$$

that interchanges the two inner factors. It is the coherence isomorphism used to show that the objectwise tensor product descends to a well-defined addition on the relative Picard quotient (630).

Proof. The isomorphism is assembled entirely from the unconditional associator 653 and the braiding 655, applied to a factor through the bifunctoriality of $\otimes$ (862); no coherence pentagon is invoked. Reassociating and braiding the middle pair,

$$(A\otimes B)\otimes(C\otimes D) \cong A\otimes(B\otimes(C\otimes D)) \cong A\otimes((B\otimes C)\otimes D) \cong A\otimes((C\otimes B)\otimes D) \cong A\otimes(C\otimes(B\otimes D)) \cong (A\otimes C)\otimes(B\otimes D),$$

where the first, third-to-last, and last isomorphisms are instances of the associator, the middle isomorphism braids $B \otimes C \cong C \otimes B$ inside the fixed outer factors, and each step past the first is transported through $\otimes$ by 862. $\square$

Lemma 630 (Addition is well defined on the H_T-quotient)**.** *If $L_1 \sim L_2$ and $L_1' \sim L_2'$ then $L_1 \otimes L_1' \sim L_2 \otimes L_2'$; hence the descended tensor product $+$ is well defined on the quotient* $\mathrm{OnProduct}(\pi_C, \pi_T)/\!\sim$.

Proof. Suppose $L_1 \cong \pi_T^*N \otimes L_2$ and $L_1' \cong \pi_T^*N' \otimes L_2'$ with N, N' locally trivial. By the middle-four interchange $(A \otimes B) \otimes (C \otimes D) \cong (A \otimes C) \otimes (B \otimes D)$ of 629, assembled from the associator and the braiding 653 and 655,

$$L_1 \otimes L_1' \cong (\pi_T^*N \otimes L_2) \otimes (\pi_T^*N' \otimes L_2') \cong (\pi_T^*N \otimes \pi_T^*N') \otimes (L_2 \otimes L_2') \cong \pi_T^*(N \otimes N') \otimes (L_2 \otimes L_2'),$$

the last step by the locally-trivial comparison 696. As $N \otimes N'$ is locally trivial (648), $L_1 \otimes L_1' \sim L_2 \otimes L_2'$. □

Definition 631 (The relative Picard carrier setoid)**.** The H_T-coset relation $\sim$ of 624 is an equivalence relation on $\mathrm{OnProduct}(\pi_C, \pi_T)$, by reflexivity 626, symmetry 627, and transitivity 628. The resulting setoid presents the relative Picard quotient $\mathrm{Pic}(C \times_k T)/H_T$, distinct from the absolute iso-class setoid `RelPicPresheaf.preimage_subgroup`.

Lemma 632 (Negation is compatible with the H_T-relation)**.** *If $L \sim M$, then their chosen tensor inverses satisfy $L^{-1} \sim M^{-1}$. Hence the descended negation $[L] \mapsto [L^{-1}]$ is well defined on the quotient* $\mathrm{OnProduct}(\pi_C, \pi_T)/\!\sim$*.*

Proof. Suppose $L \cong \pi_T^*N \otimes M$ with N locally trivial, and let N^{-1} be its tensor inverse (656), again locally trivial. Write L^{-1}, M^{-1} for the chosen tensor inverses of L, M. We show $\pi_T^*(N^{-1}) \otimes M^{-1}$ is a tensor inverse of L: tensoring the hypothesis with $\pi_T^*(N^{-1}) \otimes M^{-1}$, reassociating and braiding with 653, 655, and 654, and multiplying the pullbacks through the locally-trivial comparison isomorphism 696,

$$L \otimes (\pi_T^*(N^{-1}) \otimes M^{-1}) \cong \pi_T^*(N \otimes N^{-1}) \otimes (M \otimes M^{-1}) \cong \pi_T^*\mathcal{O}_T \otimes \mathcal{O}_{C \times_k T} \cong \mathcal{O}_{C \times_k T},$$

using $M \otimes M^{-1} \cong \mathcal{O}$ and the pullback–unit isomorphism 671. By uniqueness of tensor inverses (621), $L^{-1} \cong \pi_T^*(N^{-1}) \otimes M^{-1}$, i.e. $L^{-1} \sim M^{-1}$. □

15.3 The relative Picard presheaf as a group-valued presheaf

We now refine the set-valued functor of 617 into a group-valued presheaf.

Definition 633 (Relative Picard presheaf)**.**

Source: [Kleiman], "The Picard scheme", §2, Def. df:Pfs. Let C/k be a smooth proper geometrically integral curve. The *relative Picard presheaf* of C over k is the functor

$$\mathrm{Pic}^\sharp_{C/k} \;:\; (\mathrm{Sch}\,/k)^{op} \longrightarrow \mathrm{Ab}, \qquad T \longmapsto \mathrm{Pic}(C \times_k T)\,/\,\pi_T^*\mathrm{Pic}(T),$$

where the right-hand side is the quotient abelian group of 703 on the H_T-coset carrier of 631, and the structure morphisms are the pullback maps $[L] \mapsto [g_C^*L]$, which descend to the quotient by 636 and are group homomorphisms by 634 below. In Kleiman's notation with $X = C$ and $S = \mathrm{Spec}\,k$, this is exactly $\mathrm{Pic}_{X/S}$ of df:Pfs.

Lemma 634 (Naturality respects the group structure)**.**

Source: [Kleiman], "The Picard scheme", §2, Defs. df:aPf + df:Pfs (the target category is the category of abelian groups; the structure maps are group homomorphisms). For a morphism $g : T' \to T$ of k-schemes, the assignment $[L] \mapsto [g_C^*L]$ is a well-defined homomorphism of abelian groups $g^\sharp : \mathrm{Pic}^\sharp_{C/k}(T) \to \mathrm{Pic}^\sharp_{C/k}(T')$ with respect to the H_T-quotient structure of 703, and the identity and composition laws $\mathrm{id}^\sharp = \mathrm{id}$ and $(g \circ h)^\sharp = h^\sharp \circ g^\sharp$ hold as identities of group homomorphisms.

Proof. Well-definedness across H_T-cosets is 636. The map preserves zero because pullback preserves the structure sheaf (671) and preserves addition because pullback commutes with the tensor product of locally-trivial modules (696); both isomorphisms descend through 625. The identity law reduces to $\mathrm{id}_C \times_k \mathrm{id}_T = \mathrm{id}$ and the invariance of classes under pullback along the identity; the composition law to $\mathrm{id}_C \times_k (h \circ g) = (\mathrm{id}_C \times_k g) \circ (\mathrm{id}_C \times_k h)$ and pseudo-functoriality of pullback (615). □

Theorem 635 ($\mathrm{Pic}^\sharp_{C/k}$ is a group-valued presheaf)**.**
Source: [Kleiman], "The Picard scheme", §2, Defs. df:aPf + df:Pfs. The assignment $T \mapsto \mathrm{Pic}^\sharp_{C/k}(T)$ of 633, together with the structure maps $g \mapsto g^\sharp$ of 634, defines a presheaf of abelian groups

$$\mathrm{Pic}^\sharp_{C/k} \ : \ (\mathrm{Sch}\,/k)^{op} \longrightarrow \mathrm{AddCommGroup}$$

on the category of k-schemes.

Proof. The object map is well defined by 703: for each k-scheme T, the value $\mathrm{Pic}^\sharp_{C/k}(T)$ is an abelian group. The morphism map $g \mapsto g^\sharp$ is a group homomorphism by 634, and satisfies the identity and composition laws by the same lemma. These data assemble into a functor $(\mathrm{Sch}\,/k)^{op} \to \mathrm{AddCommGroup}$, i.e. a presheaf of abelian groups on $(\mathrm{Sch}\,/k)$. □

15.3.1 Well-definedness, the absolute functor, and the quotient comparison

The well-definedness lemma below lets the pullback action descend to the H_T-coset carrier of 631; it is the substrate of the headline 634. The absolute iso-class functor $T \mapsto \mathrm{Pic}(C \times_k T)$ (Kleiman's *absolute Picard functor* df:aPf) is retained as 637, with the natural componentwise quotient comparison 638 onto the relative functor of 633.

Lemma 636 (Pullback descends to the H_T-quotient)**.** *Source: [Kleiman], "The Picard scheme", §2, Def. df:Pfs (functoriality of the quotient). Let $g : T' \to T$ be a morphism of k-schemes and $g_C := \mathrm{id}_C \times_k g$. If $L \sim L'$ in the H_T-relation, then $g_C^* L \sim g_C^* L'$ in the $H_{T'}$-relation.*

Proof. Let N be a coset witness, so $L \cong \pi_T^* N \otimes L'$ with N locally trivial on T. Applying g_C^* and the locally-trivial comparison isomorphism (696), $g_C^* L \cong g_C^* \pi_T^* N \otimes g_C^* L'$. By the pullback square $\pi_T \circ g_C = g \circ \pi_{T'}$ (615), $g_C^* \pi_T^* N \cong \pi_{T'}^*(g^* N)$, and $g^* N$ is again locally trivial. Hence $g^* N$ witnesses $g_C^* L \sim g_C^* L'$. □

Definition 637 (Absolute Picard functor)**.** *Source: [Kleiman], "The Picard scheme", §2, Def. df:aPf (the absolute Picard functor).* The *absolute Picard functor* of C over k: the group-valued presheaf

$$(\mathrm{Sch}\,/k)^{op} \longrightarrow \mathrm{Ab}, \qquad T \longmapsto \mathrm{Pic}(C \times_k T),$$

on the iso-class carrier, with the tensor-product group law of 618 and the pullback structure maps of 617. In Kleiman's notation this is Pic_X of df:aPf (with $X = C \times_k -$ over the test scheme). The Lean declaration bundles the object/morphism data directly; the helpers `PicSharp.functorial` (group-hom form of the structure maps) and `PicSharp.presheaf` (a re-export of the bundle) belong to this node. It compares onto the relative functor of 633 via 638.

Lemma 638 (Quotient comparison from the absolute functor)**.** *Source: [Kleiman], "The Picard scheme", §2, Def. df:Pfs (the quotient defining $\mathrm{Pic}_{X/S}$). The componentwise coset-quotient maps $\mathrm{Pic}(C \times_k T) \to \mathrm{Pic}(C \times_k T)/H_T$ assemble into a natural transformation from the absolute functor of 637 to the relative functor of 633, and each component is a surjective group homomorphism.*

Proof. Each component is the canonical quotient map, well defined because the H_T-relation is coarser than isomorphism (625); it is a group homomorphism because both group laws descend the same tensor-product operations on representatives, and it is surjective as a quotient map. Naturality holds on representatives: both routes send $[L]$ to $[g_C^* L]$. $\square$

15.4 Étale sheafification

The presheaf $\mathrm{Pic}^\sharp_{C/k}$ is generally not an étale (or even Zariski) sheaf. The standard obstruction (Kleiman §2, L1292–L1302): for a k-scheme T carrying a non-trivial line bundle whose pullback to $C \times_k T$ becomes non-trivial after an open cover, the Zariski separated-presheaf condition fails. Therefore, to obtain a representable functor in the sense of 25, one must replace $\mathrm{Pic}^\sharp_{C/k}$ by its *associated étale sheaf*. Kleiman §2 names this sheaf $\mathrm{Pic}_{(X/S)\text{ét}}$ and singles it out as the functor that Kleiman's main existence theorem in §4 represents.

Definition 639 (Étale sheafification of the relative Picard presheaf)**.**
Source: [Kleiman], "The Picard scheme", §2, Def. df:Pfs (the étale-sheaf notation $\mathrm{Pic}_{(X/S)\text{ét}}$*) and the associated-sheaf paragraph immediately following df:Pfs.* Let C/k be a smooth proper geometrically integral curve. The *étale-sheafified relative Picard functor* is the sheafification of the presheaf $\mathrm{Pic}^\sharp_{C/k}$ (635) in the global étale topology on (Sch/k):

$$\mathrm{Pic}^\sharp_{(C/k)\text{ét}} \;:=\; \left(\mathrm{Pic}^\sharp_{C/k}\right)^{\sim_{\text{ét}}}.$$

Equivalently, $\mathrm{Pic}^\sharp_{(C/k)\text{ét}}$ is the unique (up to canonical isomorphism) sheaf of abelian groups on the étale site of Sch/k equipped with a presheaf morphism $\mathrm{Pic}^\sharp_{C/k} \to \mathrm{Pic}^\sharp_{(C/k)\text{ét}}$ which is universal among presheaf morphisms from $\mathrm{Pic}^\sharp_{C/k}$ to étale sheaves of abelian groups. In Kleiman's notation this is $\mathrm{Pic}_{(X/S)\text{ét}}$ with $X = C$ and $S = \mathrm{Spec}\, k$. When k is algebraically closed and $T = \mathrm{Spec}\, k'$ for an algebraically closed field extension k'/k, the natural map $\mathrm{Pic}^\sharp_{C/k}(T) \to \mathrm{Pic}^\sharp_{(C/k)\text{ét}}(T)$ is a bijection (Kleiman Exercise ex:Alr, L1357–L1361).

Theorem 640 (Étale sheafification preserves the abelian-group structure)**.**
Source: [Kleiman], "The Picard scheme", §2, Def. df:Pfs together with the associated-sheaf paragraph immediately following (the étale sheaf $\mathrm{Pic}_{(X/S)\text{ét}}$ *is named together with the abelian-group-valued relative Picard functor, meaning the sheaf is taken in the category of abelian groups, and is universal among such sheaves receiving a presheaf morphism from* $\mathrm{Pic}_{X/S}$*). This block records the bare existence of a morphism of presheaves of abelian groups*

$$\mathrm{Pic}^\sharp_{C/k} \;\longrightarrow\; \mathrm{Pic}^\sharp_{(C/k)\text{ét}}$$

inside the category of presheaves $(\mathrm{Sch}/k)^{op} \to \mathrm{AddCommGroup}$*; equivalently, the hom-set of presheaf morphisms from* $\mathrm{Pic}^\sharp_{C/k}$ *to* $\mathrm{Pic}^\sharp_{(C/k)\text{ét}}$ *is nonempty. The full canonical statement — that this morphism is the canonical sheafification unit and carries the universal property characterising* $\mathrm{Pic}^\sharp_{(C/k)\text{ét}}$ *as the initial étale sheaf of abelian groups receiving a presheaf morphism from* $\mathrm{Pic}^\sharp_{C/k}$*, so in particular that the abelian-group structure on* $\mathrm{Pic}^\sharp_{(C/k)\text{ét}}(T)$ *is uniquely determined — is recorded separately as 641 below.*

Proof. The canonical morphism from a presheaf to its associated sheaf in any Grothendieck topology (Stacks tag 00WI; the canonical sheafification unit) supplies, in particular, a morphism of presheaves $\mathrm{Pic}^\sharp_{C/k} \to \mathrm{Pic}^\sharp_{(C/k)\text{ét}}$; the hom-set of presheaf morphisms between these two objects is therefore nonempty. The full universal property of this canonical morphism — and consequently the claim that the abelian-group structure on the sheafification is uniquely pinned by it — is the content of the forward-looking 641 below; see its proof sketch for the canonical formulation. ☐

Theorem 641 (Canonical sheafification unit with universal property (forward-looking))**.**
Source: [Kleiman], "The Picard scheme", §2, Def. df:Pfs together with the associated-sheaf paragraph immediately following (the natural map from a presheaf to its associated sheaf, taken inside the category of abelian groups, is universal among presheaf morphisms from $\mathrm{Pic}_{X/S}$ *into étale sheaves of abelian groups). Let* C/k *be a smooth proper geometrically integral curve. There exists a canonical morphism of presheaves of abelian groups*

$$\eta : \ \mathrm{Pic}^\sharp_{C/k} \longrightarrow \mathrm{Pic}^\sharp_{(C/k)\text{ét}}$$

with the following universal property. For every étale sheaf $\mathcal{F}$ *of abelian groups on* (Sch/k) *and every morphism of presheaves of abelian groups* $\varphi : \mathrm{Pic}^\sharp_{C/k} \to \mathcal{F}$*, there is a unique morphism of étale sheaves of abelian groups* $\tilde{\varphi} : \mathrm{Pic}^\sharp_{(C/k)\text{ét}} \to \mathcal{F}$ *such that* $\tilde{\varphi} \circ \eta = \varphi$*. Equivalently, the pair* $(\mathrm{Pic}^\sharp_{(C/k)\text{ét}}, \eta)$ *is initial among pairs* $(\mathcal{F}, \mathrm{Pic}^\sharp_{C/k} \to \mathcal{F})$ *consisting of an étale sheaf of abelian groups on* (Sch/k) *and a morphism of presheaves of abelian groups from* $\mathrm{Pic}^\sharp_{C/k}$ *to* $\mathcal{F}$*. In particular, the abelian-group structure on* $\mathrm{Pic}^\sharp_{(C/k)\text{ét}}(T)$ *at each* k*-scheme* T *is uniquely determined (up to canonical isomorphism) by the abelian-group structure on* $\mathrm{Pic}^\sharp_{C/k}(T)$ *and the universal property of* η*.*

Proof. In any Grothendieck topology, every presheaf has a canonical associated sheaf, and the canonical morphism from a presheaf to its associated sheaf is universal among morphisms from the presheaf into sheaves (Stacks tag 00WI; Kleiman §2, paragraph following df:Pfs). The étale site of (Sch/k) is a Grothendieck topology, so this applies in particular to $\mathrm{Pic}^\sharp_{C/k}$ and yields the canonical set-valued sheafification $(\mathrm{Pic}^\sharp_{C/k})^{\sim_{\text{ét}}}$ together with the universal arrow out of $\mathrm{Pic}^\sharp_{C/k}$. The forgetful functor AddCommGroup $\to$ **Set** preserves and reflects the filtered colimits and finite limits used by the plus-construction, so performing the same plus-construction inside AddCommGroup yields a sheaf of abelian groups whose underlying set-valued sheaf coincides with the set-valued sheafification; the abelian-group structure on the sheafification is transported back through this identification. The resulting morphism η is then by construction the universal arrow in the abelian-group-valued sheafification, which is the universal property in the form stated. ☐

15.5 Lean encoding

The Lean target file is the new `AlgebraicJacobian/Picard/RelPicFunctor.lean`, sibling to `AlgebraicJacobian/Picard/LineBundlePullback.lean` (14) and `AlgebraicJacobian/Picard/RelativeSpec.lea` (13). The blueprint declarations correspond to Lean targets as follows.

- `AlgebraicGeometry.Scheme.PicSharp.addCommGroup` (618) is the abelian-group instance on the quotient set built in `AlgebraicGeometry.Scheme.RelPicPresheaf.preimage_subgroup` (616). Mathlib's `QuotientAddGroup` machinery on a normal subgroup of an abelian group is the backbone; the project-side work is to certify that π_T^* is a group homomorphism (one extra `LinearMap.map_mul`–style line on tensor product preservation).

- `AlgebraicGeometry.Scheme.PicSharp.relPresheaf` (633; run-0008 repin) is the group-valued presheaf $T \mapsto \mathrm{Pic}(C \times_k T)/\pi_T^*\mathrm{Pic}(T)$ on the honest H_T-coset carrier `Quotient (relPicSetoid)` with the group structure of 703. The ABSOLUTE iso-class functor `AlgebraicGeometry.Scheme.PicSharp` (637) compares onto it via `toRelPresheaf` (638).

- `AlgebraicGeometry.Scheme.PicSharp.relFunctorial` (634; run-0008 repin) packages the structure-map group homomorphism on the H_T-quotient, with well-definedness from `relPicRel_pullback` (636). The absolute-carrier helper is `PicSharp.functorial`.

- 635 carries no separate pin: the bundled functor is the same Lean declaration `relPresheaf` as 633 (the absolute-carrier bundle `PicSharp.presheaf` belongs to 637).

- `AlgebraicGeometry.Scheme.PicSharp.etSheaf` (639) is the étale sheafification of `PicSharp`. The Lean form is `CategoryTheory.presheafToSheaf` applied to the big étale topology on `Scheme` with target category `AddCommGrpCat`. **Lean API note.** The big étale Grothendieck topology on the category of schemes is `AlgebraicGeometry.Scheme.etaleTopology`, of type `CategoryTheory.GrothendieckTopology AlgebraicGeometry.Scheme` (module `Mathlib.AlgebraicGeo` an `abbrev` for the topology generated by the big étale pretopology `Scheme.etalePretopology`). The sheafification entry point with abelian-group target is `CategoryTheory.presheafToSheaf AlgebraicGeometry.Scheme.etaleTopology AddCommGrpCat`, of type `Functor (Scheme ⇒ AddCommGrpCat) (Sheaf etaleTopology AddCommGrpCat)` (module `Mathlib.CategoryTheory.Sites.She` it is available under a `CategoryTheory.HasWeakSheafify` instance for the pair `(etaleTopology, AddCommGrpCat)`. The slice $(\mathrm{Sch}\,/k)$ on the Lean side is captured by working over the global `Scheme` category and restricting along `Over (Spec k)` via Mathlib's `CategoryTheory.Sites.Over` / `GrothendieckTopology` pullback construction (the relative Picard functor's natural domain is the slice; Kleiman's notation $(\mathrm{Sch}\,/S)^{op}$ corresponds to `(Over (Spec k))`). If the `HasWeakSheafify` instance for the pair `(etaleTopology, AddCommGrpCat)` is missing at the pinned Mathlib commit, the project-side fallback combines the concrete set-valued sheafification `CategoryTheory.GrothendieckTopology.sheafify` (module `Mathlib.CategoryTheory.Sites.` with the colimit-preservation of the forgetful functor `AddCommGrpCat` → **Set** and the `plusPlusSheaf` plus-construction on the underlying set-valued presheaf, transported back through the `ConcreteCategory` instance on `AddCommGrpCat`.

- `AlgebraicGeometry.Scheme.PicSharp.etSheaf_group_structure` (640) packages the fact that the étale sheafification preserves the abelian-group target. On the Lean side it is the canonical `Functor (Scheme ⇒ AddCommGrpCat)` obtained by applying `CategoryTheory.presheafToSheaf AlgebraicGeometry.Scheme.etaleTopology AddCommGrpCat` to `PicSharp.presheaf` and then forgetting from the sheaf category back to a presheaf-of-abelian-groups via `(CategoryTheory.sheafToPres etaleTopology AddCommGrpCat)`. The proof is structural: it identifies `etSheaf` (typed as a sheaf of sets via the concrete plus-construction `CategoryTheory.GrothendieckTopology.sheafify`) with the value of `presheafToSheaf etaleTopology AddCommGrpCat PicSharp.presheaf` under the forgetful functor, using `CategoryTheory.GrothendieckTopology.sheafification_obj` (the equality `J.sheafification D`.obj P = `J.sheafify` P) and the fact that the forgetful functor `AddCommGrpCat` → **Set** preserves the multiequalizer / filtered-colimit data used by the plus-construction (`plusPlusSheaf` commutes with `forget` on a `ConcreteCategory` target).

The relationship with 25 is captured by the slogan: the FGA Picard scheme of A.2.c represents $\mathrm{Pic}^\sharp_{(C/k)\text{ét}}$ (Kleiman §4, the main existence theorem: "...represents $\mathrm{Pic}_{(X/S)\text{ét}}$"), *not* $\mathrm{Pic}^\sharp_{C/k}$ directly. The étale sheafification step of this chapter is precisely what makes the representability theorem in A.2.c well-posed.

15.6 Out of scope

The following constructions are downstream of this chapter and live in dedicated sibling chapters; this chapter does *not* touch them:

- **Representability of $\mathrm{Pic}^{\sharp}_{(C/k)\text{ét}}$ by a scheme** (A.2.c, 25) — the Grothendieck–Mumford–Kleiman existence theorem is the next step but is not part of this chapter.
- **Comparison theorem** $\mathrm{Pic}_{X/S} \hookrightarrow \mathrm{Pic}_{(X/S)\mathrm{Zar}} \hookrightarrow \mathrm{Pic}_{(X/S)\text{ét}} \hookrightarrow \mathrm{Pic}_{(X/S)\mathrm{fppf}}$ (Kleiman §2, the comparison theorem) — the chain of inclusions under $\mathcal{O}_S \cong f_*\mathcal{O}_X$; we use only the étale-sheafified functor here, not the full comparison chain.
- **Identity component** $\mathrm{Pic}^0_{C/k}$ (A.3) — the open subgroup scheme cut out by Hilbert polynomials, treated in a downstream chapter.
- **Albanese universal property** (A.4) — consumes the full Route A stack.
- **Group-scheme structure on the representing object** — once $\mathrm{Pic}^{\sharp}_{(C/k)\text{ét}}$ is representable (A.2.c), its representing scheme inherits a k-group-scheme structure from the abelian-group-valued target via the Yoneda embedding; that refinement is part of A.2.c, not of this chapter.

15.7 Internal-consistency check

The six Lean-pinned declaration blocks (plus the forward-looking 641, which carries no `\lean{…}` pin and so is not yet tracked by the deterministic `sync_leanok` phase) form a connected `\uses{…}` chain rooted in 14:

- 618 uses 610, 614, and 616 (all from A.1.b on disk) for the statement, and in its proof additionally consumes the tensor-substrate lemmas 657, 648, 653, 655, 654, 664, 656, and 671 together with the locally-trivial pullback–tensor comparison isomorphism 696 (the concurrently-built A.1.c.sub input feeding additivity of π_T^*).
- 633 (run-0008 repin: `relPresheaf`) uses 610, 614, 616, 631, 703, and 636.
- 634 (run-0008 repin: `relFunctorial`) uses 633, 615, 636, and 625.
- 635 (no separate pin; same Lean declaration as 633) uses 633, 634, and 703.
- 637 (the absolute functor, `Scheme.PicSharp`) uses 610, 618, and 617; 638 compares it onto 633.
- 639 uses 637 (run-0008: sheafifies the absolute bundle; no longer consumed by the FGA chapter).
- 640 uses 639 and 635.
- 641 uses 639, 635, and 640. This block records the canonical sheafification unit and its universal property; it carries no `\lean{…}` pin and is not yet tracked by the deterministic `sync_leanok` phase.

All `\uses` pointers resolve either inside this chapter or inside the already on-disk 14. No declaration cross-references a chapter that does not yet exist on disk; the reference to 25 is purely commentary in the section "Lean encoding" and "Out of scope", not a `\uses` target.

Chapter 16

Relative Picard sheaf — `Scheme.Modules.tensorObj` substrate (A.1.c.SubT)

16.1 Motivation: the abelian-group-valued relative Picard sheaf

Fix a base field k and a smooth proper geometrically integral curve C over k. In Kleiman's framework (15, `rel_pic_sharp`), the relative Picard functor

$$\mathrm{Pic}_{C/k} \;:\; (\mathrm{Sch}\,/k)^{op} \longrightarrow \mathrm{AddCommGrpCat}, \qquad T \longmapsto \mathrm{Pic}(C \times_k T)\,/\,\pi_T^*\mathrm{Pic}(T),$$

is, by definition, abelian-group-valued. The abelian-group structure on each fiber $\mathrm{Pic}(C \times_k T)$ comes from tensor product of line bundles (Stacks tag 01CR / Hartshorne II.§6), and the structure descends to the quotient because π_T^* is a group homomorphism.

Source: [Kleiman], "The Picard scheme", §2, Defs. df:aPf + df:Pfs. The absolute and relative Picard functors take values in the category of abelian groups; the relative functor is the quotient of one such group by another, and the structure morphisms of the resulting presheaf are group homomorphisms.

Concretely, the abelian-group law on $\mathrm{Pic}(C\times_k T)$ is the assignment $[\,\mathcal{L}\,]+[\,\mathcal{L}'\,] := [\,\mathcal{L}\otimes_{\mathcal{O}_{C\times_k T}} \mathcal{L}'\,]$, with neutral element the class of the structure sheaf $\mathcal{O}_{C\times_k T}$ and inverse $-[\,\mathcal{L}\,] := [\,\mathcal{L}^{-1}\,]$, where $\mathcal{L}^{-1} = \mathcal{H}om_{\mathcal{O}_{C\times_k T}}(\mathcal{L}, \mathcal{O}_{C\times_k T})$ is the dual invertible sheaf. Pullback of $\mathcal{O}$-modules along $\pi_T : C \times_k T \to T$ preserves tensor products and the structure sheaf, so the image $\pi_T^*\mathrm{Pic}(T) \subset \mathrm{Pic}(C \times_k T)$ is a subgroup, and the quotient $\mathrm{Pic}(C \times_k T)/\pi_T^*\mathrm{Pic}(T)$ inherits a canonical abelian-group structure. This is the content of 618 in chapter 15.

The mathematics is straightforward; the obstacle to formalisation is purely infrastructural, and — crucially — much smaller than a full monoidal-category construction. The group law lives on *isomorphism classes* of line bundles: each group axiom is a *proposition* asserting the existence of an isomorphism ($\mathrm{Nonempty}(\cdots \cong \cdots)$), and a proposition is insensitive to which isomorphism witnesses it. In particular the coherence data of a monoidal category — the pentagon, triangle and hexagon identities, and a fortiori any closed structure — is *never consumed* by a group law on iso-classes. What the law genuinely requires on the carrier is therefore only:

1. a binary tensor-product operation $\otimes$: Scheme.`Modules` $X \times$ Scheme.`Modules` $X \to$ Scheme.`Modules` X, well-defined on isomorphism classes;

2. the structure sheaf $\mathcal{O}_X$ as a designated unit object for $\otimes$;

3. the $\otimes$-invertibility predicate ($\exists N,\ M \otimes_X N \cong \mathcal{O}_X$) together with the three monoidal coherence isomorphisms of $\otimes_X$: associativity $(M \otimes N) \otimes P \cong M \otimes (N \otimes P)$, unit $\mathcal{O}_X \otimes M \cong M$ (and on the right), and commutativity $M \otimes N \cong N \otimes M$. The inverse is carried by the predicate, not by a separate lemma.

At Mathlib's pinned commit one might be tempted to obtain (3) by instantiating a *full* symmetric monoidal category on Scheme.`Modules` X and reading the three isomorphisms off it. Two Mathlib facts make such a route conceivable. First, the category `PresheafOfModules` R of presheaves of modules over a presheaf of commutative rings R is already a (symmetric) monoidal category in Mathlib (`PresheafOfModules.monoidalCategory`), with the sectionwise relative tensor $(M \otimes_R N)(U) = M(U) \otimes_{R(U)} N(U)$; this applies verbatim to the *varying* structure sheaf $R = \mathcal{O}_X$. Second, Mathlib's localization-of-monoidal-categories API (`CategoryTheory.Localization.Monoidal.LocalizedMonoid` takes a monoidal category C, a localization functor $L : C \to D$ at a morphism property W, and an instance W.`IsMonoidal`, and produces a *full* monoidal-category structure on the localization D; sheafification of presheaves of modules is the localization $L =$ `sheafification` at the locally-bijective class $W = J.W$, so a full monoidal structure on Scheme.`Modules` X would follow once $J.W$.`IsMonoidal` were supplied.

This full-monoidal-category route is, however, *not* the goal, and is not pursued. The goal is the commutative *group* on locally-trivial iso-classes (à la `CommRing.Pic`), and that group law consumes only propositions, so it never needs a coherent monoidal category — neither the pentagon/triangle/hexagon identities nor the $J.W$.`IsMonoidal` instance that the `LocalizedMonoidal` API requires. What it needs is exactly the *existence* of the three coherence isomorphisms of (3), and these are produced directly: the associator by gluing canonical local isomorphisms (653, via the single linchpin 646), the unitors by the cheap `mapIso` pattern (654), and the braiding likewise (655). With those three existence facts the group law assembles *by hand* the commutative group of $\otimes$-invertible $\otimes$-iso-classes — the Picard group (664) — each group-axiom field fed a single existence-of-isomorphism. This is the scheme-theoretic analogue of Mathlib's fixed-ring `CommRing.Pic` $R :=$ Skeleton(`Module.Invertible` R), whose `CommGroup` is read off the Skeleton of the coherent monoidal category `MonoidalCategory`(`Module.Invertible` R) that Mathlib supplies over a fixed ring; the present construction cannot take that Skeleton route, since the coherent monoidal category on Scheme.`Modules` X for the varying structure sheaf is exactly what it avoids. The $\otimes$-invertibility predicate (662, Mathlib's `Module.Invertible` idiom) carries each inverse existentially. The *full* `LocalizedMonoidal` / $J.W$.`IsMonoidal` coherent monoidal category is therefore not built for the group law and is off the critical path; what the group law does still need from this circle of ideas is the *existence* of the associator (653), realized as a sheafification transport whose single open left-whisker obligation is closed by the stalk–tensor commutation (d.2). The stalk apparatus (d.1/d.2) is accordingly *on* the critical path, while the coherent monoidal category itself is not.

16.2 Mathlib API survey

The relevant pieces of the Mathlib API at the pinned commit (`b80f227`) were originally assembled into a single *realization route* (route (e)): obtaining a full monoidal structure on the substrate by instantiating Mathlib's abstract localization-of-monoidal-categories API rather than hand-assembling the associator. That route is *superseded*. The group law on iso-classes never consumes a coherent monoidal category, and the associator is obtained by direct gluing through 646, so the $J.W$.`IsMonoidal` instance on which route (e) hinges is never required. The survey below is retained as a Mathlib-API reference and to record why the route is vestigial; the four Mathlib facts are all verified against the on-disk Mathlib source.

(i) The presheaf-of-modules monoidal category. Mathlib's `Mathlib.Algebra.Category.ModuleCat.Preshea` constructs a monoidal-category instance `PresheafOfModules.monoidalCategory` on `PresheafOfModules` ($R\circ$ $\texttt{forget}_2$ `CommRingCat RingCat`), with the sectionwise relative tensor

$$(M \otimes_R N)(U) \quad = \quad M(U) \otimes_{R(U)} N(U)$$

Source: [Mathlib], `PresheafOfModules.monoidalCategoryStruct` / `PresheafOfModules.monoidalCategory`*, in* `Mathlib/Algebra/Category/ModuleCat/Presheaf/Monoidal.lean`*, L104–L125.* together with associator, left/right unitors and braiding making it symmetric monoidal. The base presheaf of rings $R \circ \texttt{forget}_2$ `CommRingCat RingCat` is exactly the project's `Sheaf.val` X.`ringCatSheaf` carrier (equal by `rfl`), so this monoidal structure applies verbatim to the *varying* structure sheaf $\mathcal{O}_X$ — no project work. This is the presheaf-level analogue of Stacks tag 03DM (relative tensor product of $\mathcal{O}_X$-modules).

(ii) The localization-of-monoidal-categories API. Mathlib's `Mathlib.CategoryTheory.Localization.Monoi` provides the class `MorphismProperty.IsMonoidal` and the construction `Localization.Monoidal.LocalizedMonoidal` The class records exactly the whiskering-stability data: *Source: [Mathlib],* `MorphismProperty.IsMonoidal` *and* `Localization.Monoidal.LocalizedMonoidal`*, in* `Mathlib/CategoryTheory/Localization/Monoidal/Basic.` *L44–L440.* Given [MonoidalCategory C], [W.`IsMonoidal`], a localization functor $L : C \to D$ with [L.`IsLocalization` W] and an isomorphism $\varepsilon : L(\mathbf{1}_C) \cong$ `unit`, the localized category `LocalizedMonoidal` $L\,W\,\varepsilon$ is a MonoidalCategory with *all* coherence (associator, unitors, pentagon, triangle, naturalities) derived. There is *no* `MonoidalClosed` anywhere in the hypothesis chain.

(iii) The proof template for W-monoidality. Mathlib's `Mathlib.CategoryTheory.Sites.Point.IsMonoidalW` proves $(J.W(A := A))$.`IsMonoidal` *stalkwise* for presheaves valued in a *fixed* monoidal concrete category A (i.e. $C^{op} \to A$), via the conservative-family characterisation `hP.W_iff`, `Functor.Monoidal.map_tensor` and typeclass inference: *Source: [Mathlib],* `ObjectProperty.IsConservativeFamilyOfPoints.isMonoidal_W`*, in* `Mathlib/CategoryTheory/Sites/Point/IsMonoidalW.lean`*, L48–L57.* This is the *template* for the proof technique used in route (e), but it does *not* apply directly to `PresheafOfModules` R: modules over a *varying* ring are not of the form $C^{op} \to A$ for a fixed monoidal A.

The gap and route (e), superseded. The substrate Scheme.`Modules.tensorObj` of this chapter is the sheafification of `PresheafOfModules.Monoidal.tensorObj`, and sheafification of presheaves of modules is exactly the localization $L =$ `sheafification` at the locally-bijective class $W = J.W$. Route (e) proposed to obtain a *full* monoidal structure on Scheme.`Modules` X = `SheafOfModules` X.`ringCatSheaf` by applying `LocalizedMonoidal` (ii) to the already-monoidal category (i), which would require the instance $J.W$.`IsMonoidal` (its two whiskering-stability fields 652, with load-bearing local-injectivity half 651). This is *not* how the group law is realized. The group law consumes only the *existence* of the three coherence isomorphisms, and the associator is built directly by gluing canonical local isomorphisms through 646 (653); it never invokes the *full* $J.W$.`IsMonoidal` instance or a coherent monoidal category. The *full monoidal-category* route (e) is therefore off the critical path. The associator does, however, consume a *single* left-whisker existence lemma 659, closed by the d.2 stalk–tensor commutation; so the stalk apparatus (d.1/d.2) itself is on the critical path, even though the full coherent monoidal category built from it is not.

The stalk-infrastructure analysis that route (e) turned on is recorded here for reference only, since that route is superseded. The *stalk module structure* is already present in Mathlib: the file `Mathlib.Algebra.Category.ModuleCat.Stalk` supplies, for a topological space X, a presheaf of commutative rings R on X and a point $x \in X$, the R.`stalk` x-module structure on the stalk `stalk` M.`presheaf` x of a presheaf of modules M, together with the germ–scalar compatibility `germ_smul`; a linearity packaging of the induced stalk map of a morphism was built project-side (650, the *linearity half* of ingredient (d.1)). Route (e) additionally required a (d.1)-*bridge* between the site-level $J.W$ and the topological stalkwise-isomorphism criterion on `Opens` X, and (d.2) the commutation $(A \otimes^p B)_x \cong A_x \otimes_{R_x} B_x$ of the stalk with the relative module-presheaf tensor; there is moreover no monoidal `SheafOfModules` in Mathlib. The full $J.W$.`IsMonoidal` coherent monoidal category that route (e) would build is not pursued; but the (d.1)-linearity packaging and the (d.2) stalk–tensor commutation are *on* the critical path, since the associator's single open left-whisker obligation 659 is closed precisely by (d.2) (16.6). What is superseded is only the *pair*-of-whiskering-fields $J.W$.`IsMonoidal` packaging and the `LocalizedMonoidal` monoidal category, not the stalk infrastructure.

The project's policy on this gap is *project-side*: the substrate is built inside `AlgebraicJacobian/Picard/` (the file `AlgebraicJacobian/Picard/TensorObjSubstrate.lean` is its home) and the consumer instance is closed against it. A parallel Mathlib upstream PR with the same content is welcome but not on the critical path for the project; the consumer is closed against the project-side substrate independently of any upstream timing.

16.3 The substrate Scheme.`Modules.tensorObj`

Definition 642 (The substrate operation $\otimes$ on Scheme.`Modules` X)**.** Let X be a scheme. For $M, N \in$ Scheme.`Modules` X, define

$$M \otimes_X N \quad \in \quad \text{Scheme}.\mathtt{Modules}\, X$$

as follows. Choose an affine open cover $\{U_i = \operatorname{Spec} A_i\}_{i \in I}$ of X. On each U_i, the restriction $M|_{U_i}$ corresponds (via the equivalence between $\mathcal{O}_{\operatorname{Spec} A_i}$-modules and A_i-modules) to an A_i-module M_i, and similarly $N|_{U_i}$ corresponds to an A_i-module N_i. Define $(M \otimes_X N)|_{U_i}$ as the sheafification of the presheaf $U_i \mapsto M_i \otimes_{A_i} N_i$. On overlaps $U_i \cap U_j = \operatorname{Spec} A_{ij}$ the resulting modules agree via the canonical compatibility $(M_i \otimes_{A_i} N_i) \otimes_{A_i} A_{ij} \;\cong\; (M_j \otimes_{A_j} N_j) \otimes_{A_j} A_{ij}$, so the data $\{(M \otimes_X N)|_{U_i}\}$ glue to a single object $M \otimes_X N$ in Scheme.`Modules` X. The result is canonically independent of the affine cover chosen.

Equivalently: the underlying presheaf of $M \otimes_X N$ is the presheaf-of-modules tensor product `PresheafOfModules.Monoidal.tensorObj` of the underlying presheaves of M and N, composed with the sheafification functor on the small Zariski site of X. The substrate operation $\otimes_X$ is, by construction, a Scheme.`Modules` X object reflecting the relative tensor product of $\mathcal{O}_X$-modules described in Stacks tag 03DM.

Lemma 643 (Functoriality of $\otimes_X$)**.** *Let X be a scheme. The assignment $(M, N) \mapsto M \otimes_X N$ of 642 extends to a bifunctor*

$$\otimes_X \;:\; \text{Scheme}.\mathtt{Modules}\, X \times \text{Scheme}.\mathtt{Modules}\, X \;\longrightarrow\; \text{Scheme}.\mathtt{Modules}\, X$$

natural in both arguments: a pair of morphisms $f : M \to M'$ and $g : N \to N'$ determines a morphism $f \otimes g : M \otimes_X N \to M' \otimes_X N'$ compatible with identities and composition. This *bifunctorial action on morphisms is the only datum the present lemma provides. The further coherence data of a monoidal structure — the unitors*

$$\lambda_M : \mathcal{O}_X \otimes_X M \xrightarrow{\sim} M, \qquad \rho_M : M \otimes_X \mathcal{O}_X \xrightarrow{\sim} M,$$

the associator $\alpha_{M,N,P} : (M \otimes_X N) \otimes_X P \xrightarrow{\sim} M \otimes_X (N \otimes_X P)$*, and the braiding* $\beta_{M,N} : M \otimes_X N \xrightarrow{\sim} N \otimes_X M$ *— are* not *outputs of this lemma; they are off the critical path, recorded in 644.*

Proof. Pick an affine open cover $\{U_i = \operatorname{Spec} A_i\}$ of X. On each affine, the assignment is the standard tensor product of modules over a commutative ring A_i, which is bifunctorial. The bifunctorial action on morphisms glues across the cover because the underlying gluing data of 642 respect the canonical morphisms on overlaps; equivalently, the morphism action is inherited from `PresheafOfModules.Monoidal.tensorObj` (Mathlib module `Mathlib.CategoryTheory.Monoidal.PresheafOfModul` under sheafification. The unitor/associator/braiding coherence data are *not* constructed here; they are off the critical path (644). ☐

Remark 644 (The full monoidal category on all of Scheme.`Modules` X is off-path; the group law consumes only propositions)**.** One could ask for the bifunctor $\otimes_X$ of 643, together with $\mathcal{O}_X$, α, λ, ρ and β, to assemble into a full symmetric MonoidalCategory structure on *all* of Scheme.`Modules` X, discharging the pentagon, triangle and hexagon coherence identities. This is explicitly *off-path* and *not pursued*. One could in principle obtain it as the output of `Localization.Monoidal.LocalizedMonoidal` applied to the already-monoidal presheaf category `PresheafOfModules` $\mathcal{O}_X$ and the sheafification localizer $J.W$, supplying the instance $J.W$.`IsMonoidal` (652); but the group law never consumes a coherent monoidal category, so this construction is superseded and not carried out. The substrate provides only the *existence* of the three coherence isomorphisms that the group law actually consumes.

Were that superseded route pursued, the instance $J.W$.`IsMonoidal` would need *no* closed structure: its `whiskerLeft` field $W g \Rightarrow W(F \lhd g)$ for an *arbitrary* module F is the flatness-free stalkwise statement 651 (a $J.W$-morphism g is a stalkwise isomorphism, so $(F \lhd g)_x = \mathrm{id}_{F_x} \otimes g_x$ is again an isomorphism for any F), the same technique Mathlib uses for fixed-base presheaves in `CategoryTheory.Sites.Point.IsMonoidalW`. It would use neither `MonoidalClosed`(`PresheafOfModules` R) nor the sectionwise-flatness hypothesis of `flat_whisker_localizer` (which is false over a non-affine open and is likewise off the critical path). The *full* $J.W$.`IsMonoidal` instance and the `LocalizedMonoidal` monoidal category are not on the critical path: the group law is realized without them. The single left-whisker existence lemma and its underlying d.1/d.2 stalk apparatus, by contrast, *are* on the critical path, since the associator's sheafification transport consumes exactly that one whisker obligation (659, closed via d.2).

Independently of how the monoidal structure is built, the relative Picard group law of 618 is a structure on *isomorphism classes* of line bundles, and every group axiom there is a *proposition* of the form Nonempty($\cdots \cong \cdots$). Such a proposition is insensitive to which isomorphism witnesses it, so the coherence data of the monoidal category — pentagon, triangle, hexagon, and a fortiori any monoidal-closed structure — is never *consumed* by the group law. The group law needs only the existence of the three coherence isomorphisms 653, 654, and 655, supplied directly: the associator by gluing canonical local isomorphisms through the restriction-compatibility isomorphism 646 (653), and the unitors and braiding by the cheap `mapIso` pattern (654 and 655). The carrier — iso-classes of the substrate tensor $\otimes_X$ — is the scheme-theoretic analogue of Mathlib's Picard group `CommRing.Pic` $R :=$ Skeleton(`Module.Invertible` R) (`Mathlib.RingTheory.PicardGroup`), but it is built *differently*: Mathlib reads its `CommGroup` off the Skeleton of the coherent monoidal category `MonoidalCategory`(`Module.Invertible` R) over a fixed ring, whereas here — with no coherent monoidal category on Scheme.`Modules` X available for the varying structure sheaf — the group is assembled by hand from the existence-of-isomorphism facts.

16.4 The lift through `LineBundle.OnProduct`

The consumer in chapter 15 works not with Scheme.`Modules`$(C \times_k T)$ directly but with the subtype `LineBundle.OnProduct` $\pi_C\,\pi_T$ of locally-trivial modules on the fiber product $C \times_k T$, introduced in 14. This section records the *existence-of-isomorphism facts on line bundles* (associativity, unit, commutativity and inverse) that the relative Picard group law consumes. These facts are supplied *directly*: associativity by the direct-gluing associator 653 (through the linchpin 646), unit and commutativity by the cheap `mapIso` pattern (654 and 655), and the inverse existentially from ⊗-invertibility. They are *not* read off any MonoidalCategory structure: the route that would derive them from `jw_ismonoidal` (route (e)) is superseded and off-path. The section also records the closure of ⊗ on the locally-trivial `LineBundle.OnProduct` subtype (657) and the interaction with the pullback functor π_T^*. The open-immersion restriction-compatibility isomorphism 646 is on the critical path precisely because it is consumed by the direct-gluing associator 653; it is *not* tied to the superseded whiskering-stability lemma 651, which belongs to the off-path route (e).

Lemma 645 (Sectionwise lax-monoidal structure on `restrictScalars` φ)**.** *Let $\varphi : R \to S$ be a morphism of presheaves of* commutative *rings on a site (so that both R and S factor through* `CommRingCat` *and the category of presheaves of modules over each is monoidal). Then the presheaf-of-modules restriction-of-scalars functor* `PresheafOfModules.restrictScalars` φ *is lax monoidal. Consequently, composing with the already-monoidal Mathlib functor* `pushforward`$_0$`OfCommRingCat` *via* `Functor.LaxMonoidal.comp`*, the full pushforward*

$$\texttt{pushforward}\,\varphi \;:=\; \texttt{pushforward}_0\,F R \circ \texttt{restrictScalars}\,\varphi$$

is lax monoidal. This lemma is a self-contained `CommRingCat`*-level lax-monoidal supplement: it stands on its own and is retained for potential reuse, but it is* off the critical path *for the tensor group law. In particular it is* not *an ingredient of 646, whose proof identifies restriction along an open immersion with base change along a structure-sheaf isomorphism and consequently needs no lax-monoidal structure on the pushforward.*

Proof. This is a *sectionwise* lift of the existing fact that, for a ring map $f : R \to S$, the module restriction-of-scalars functor `ModuleCat.restrictScalars` f is lax monoidal (`Mathlib.Algebra.Category.ModuleCat` where it is obtained as the `rightAdjointLaxMonoidal` of the extension/restriction adjunction `extendRestrictScalarsAdj` — i.e. Mathlib already uses the same mate idiom one level down). The presheaf monoidal structure is computed sectionwise, $(M_1 \otimes M_2).\mathrm{obj}\,X = M_1.\mathrm{obj}\,X \otimes M_2.\mathrm{obj}\,X$, and the presheaf functor `restrictScalars` φ acts sectionwise as `ModuleCat.restrictScalars` $(\varphi.\mathrm{app}\,X)$. Hence the lax structure maps μ (laxity) and ε (unit) assemble from the per-section `ModuleCat` ones, the required naturality squares being the naturality of the sectionwise maps against the presheaf restriction morphisms. Packaging these as a `Functor.CoreLaxMonoidal` yields the lax-monoidal structure; no new diagram chase is needed beyond transporting the sectionwise coherence. The composite `pushforward` φ is then lax monoidal by `Functor.LaxMonoidal.comp` together with the monoidality of `pushforward`$_0$`OfCommRingCat` (Mathlib), modulo a definitional transport across the unfolding of `pushforward` φ.

The commutativity hypothesis is essential: the presheaf-of-modules monoidal category requires `CommRingCat`-valued presheaves of rings, so the instance must be stated over `CommRingCat`-factored R, S. In the project this holds: the relevant φ is (`Scheme.Hom.toRingCatSheafHom` f).hom, which is structure-sheaf-valued and hence commutative-ring-valued. ☐

Lemma 646 (Tensor product commutes with restriction along an open immersion)**.** *Let X be a scheme, let $M, N \in$* Scheme.`Modules` X *be* arbitrary *$\mathcal{O}_X$-modules, and let $f : U \hookrightarrow X$ be an* *open immersion, with $(-)|_f$ the restriction of $\mathcal{O}_X$-modules along f. The canonical comparison morphism*

$$(M \otimes_X N)\big|_f \to M\big|_f \otimes_U N\big|_f$$

is an isomorphism. No local-freeness, flatness, or line-bundle hypothesis on M, N is required: the statement holds for all $\mathcal{O}_X$-modules. The reason is structural to open immersions — restriction along f is base change along the structure-sheaf isomorphism *induced by f, not along a general ring map — and base change along a ring isomorphism is trivially invertible and commutes with tensor products.*

Proof. Write φ for the morphism of structure-sheaf-valued presheaves of rings induced by the open immersion f (concretely $\varphi = (\texttt{Scheme.Hom.toRingCatSheafHom}\, f)$.hom), so that restriction along f is computed by the inverse-image (pullback) functor on presheaves of modules. The proof exploits the defining feature of an open immersion: the comparison of structure-sheaf rings over the image is an *isomorphism*, not merely a ring map. For an open $V \subseteq U$ the section ring $\Gamma(X, f(V))$ of $\mathcal{O}_X$ over the image of V is identified with $\Gamma(U, V)$ by the canonical *ring isomorphism* $f.\texttt{appIso}\, V : \Gamma(X, f(V)) \xrightarrow{\sim} \Gamma(U, V)$ (the `CommRingCat` isomorphism `AlgebraicGeometry.Scheme.Hom.appIso`). Restriction along f is therefore extension of scalars along this ring isomorphism, and the proof needs no flatness anywhere.

Step 1 (reduce restriction to the abstract pullback). The restriction $(-)|_f$ of $\mathcal{O}_X$-modules along the open immersion f is naturally isomorphic to the abstract presheaf-of-modules pullback $(\texttt{PresheafOfModules.pullback}\, \varphi)$.obj$(-)$, via the present comparison `Scheme.Modules.restrictFunctorIsoPullbac`
It therefore suffices to produce the comparison and prove it an isomorphism for the abstract pullback functor.

Step 2 (move the pullback inside the sheafification). The substrate $\otimes_X$ is sheafification of the presheaf-of-modules tensor product (642). The pullback commutes with sheafification: there is a natural isomorphism $\texttt{sheafification} \circ \texttt{pullback}\, \varphi \cong \texttt{PresheafOfModules.pullback}\, \varphi \circ \texttt{sheafification}$ (the present `SheafOfModules.sheafificationCompPullback`). Hence the whole question descends to the presheaf level, where it is the base-change comparison

$$(\texttt{pullback}\, \varphi).\mathrm{obj}(A \otimes B) \longrightarrow (\texttt{pullback}\, \varphi).\mathrm{obj}\, A \otimes (\texttt{pullback}\, \varphi).\mathrm{obj}\, B$$

for the underlying presheaves of modules A, B of M and N.

Source: [Stacks Project], Modules on Ringed Spaces, Lemma `lemma-exactness-pushforward-pullback`: pullback and pushforward of $\mathcal{O}$-modules along a morphism form an adjoint pair (f^*, f_*), the abstract universal property on which the comparison below rests.

Step 3 (the genuine residual: H1 alone). Completing this comparison at the presheaf level requires a sectionwise handle on the abstract functor $\texttt{PresheafOfModules.pullback}\, \varphi$, which is defined as the left adjoint of pushforward (the presheaf-of-modules avatar of the $f^* \dashv f_*$ adjunction of the cited Stacks lemma) and admits no sectionwise formula in Mathlib at the pinned commit. The closure passes through two presheaf-level ingredients, of which exactly one now remains open.

The first ingredient — a strong-monoidal upgrade of `restrictScalars` along the structure-sheaf ring *isomorphism* f.appIso, so that base change along the isomorphism commutes with $\otimes$ — is *closed in this chapter* as 647 (`restrictScalarsRingIsoTensorEquiv`): for a ring isomorphism $e : R \simeq S$ the canonical map $a \otimes_R b \mapsto a \otimes_S b$ is an R-linear equivalence $\texttt{restrictScalars}_e(M) \otimes_R \texttt{restrictScalars}_e(N) \xrightarrow{\sim} \texttt{restrictScalars}_e(M \otimes_S N)$.

The *sole remaining residual* is therefore the presheaf-level identification

$$\texttt{pushforward}\, \beta \cong \texttt{pullback}\, \varphi$$

via `Adjunction.leftAdjointUniq`. One exhibits a presheaf-level adjunction `pushforward` $\beta \dashv$ `pushforward` φ and concludes that `pushforward` β, being a left adjoint of `pushforward` φ, is canonically isomorphic to the abstract left adjoint `pullback` φ. Here β is the structure map of the restriction functor, so that $(M|_f)$.val $=$ (`pushforward` β).obj M.val definitionally; composing the resulting comparison with 647 yields the asserted isomorphism.

This presheaf-level adjunction is the presheaf analogue of the sheaf-level `SheafOfModules.pushforwardPushforwar` and must be built from presheaf-level `pushforwardNatTrans` and `pushforwardCongr`. These two building blocks are *Mathlib-absent* at the pinned commit: only the *sheaf*-level versions exist (`Mathlib/.../SheafOfModules/Sheaf/PushforwardContinuous.lean`), whereas `pushforwardId` and `pushforwardComp` do already exist at the presheaf level. This residual is being constructed *directly*, mirroring the existing sheaf-level template app-for-app at the presheaf level (a Mathlib-gradient build, not a deferral to an upstream PR).

This lemma is the single substrate linchpin of the relative Picard group law for *arbitrary* $\mathcal{O}_X$-modules M, N: the associator 653 consumes the comparison $(M \otimes_X N)|_f \cong M|_f \otimes_U N|_f$ on its sole consumer 648, where the restriction is commuted past the *sheafified* tensor $M \otimes_X N$ *before* any local trivialisation of M, N is introduced — so the comparison is required for the global, untrivialised modules, and no free-cover specialisation can stand in for it. Consequently the H1 residual `pushforward` $\beta \cong$ `pullback` φ is genuinely on the critical path, and closing it is the *single open substrate obligation* of this chapter. The superseded route (e) $(J.W)$.`IsMonoidal` $\to$ `Sheaf.monoidalCategory` (`jw_ismonoidal`) remains named only as a last-resort fallback should the presheaf-level adjunction itself bottom out; it is not the plan. ☐

Lemma 647 (Base change along a ring isomorphism commutes with $\otimes$)**.** *Let $e : R \simeq S$ be an isomorphism of commutative rings, and let M, N be S-modules. Restriction of scalars along e commutes with the tensor product: the canonical map*

$$\texttt{restrictScalars}_e(M) \otimes_R \texttt{restrictScalars}_e(N) \xrightarrow{\sim} \texttt{restrictScalars}_e(M \otimes_S N), \qquad a \otimes_R b \longmapsto a \otimes_S b,$$

is an R-linear equivalence. Equivalently, `restrictScalars`$_e$ *is* strong *monoidal, not merely lax, when e is a ring* isomorphism*: base change along an isomorphism $R \simeq S$ identifies the two ground rings, so the lax structure map of* `ModuleCat.restrictScalars` *is invertible. The forward map is R-linear, but its inverse $a \otimes_S b \mapsto a \otimes_R b$ is only* additive *(it is not S-linear), so the inverse is constructed as a homomorphism of additive groups and then verified to be the two-sided inverse of the forward R-linear map. This closes the strong-monoidal ingredient of 646, leaving the presheaf-level pushforward adjunction as the sole open residual there.*

Lemma 648 (Tensor product of locally-trivial modules is locally trivial)**.** *Let X be a scheme, and let $L, L' \in$ Scheme.*`Modules` *X be locally-trivial modules of rank one (i.e. line bundles). The tensor product $L \otimes_X L'$ of 642 is again locally trivial of rank one. Consequently the bifunctor $\otimes_X$ restricts to a bifunctor on the full subcategory* `LineBundle` *$X \subset$ Scheme.*`Modules` *X of locally-trivial rank-one modules.*

Proof. Pick a common affine open cover $\{U_i\}$ of X on which both $L|_{U_i}$ and $L'|_{U_i}$ are free of rank one. On each U_i there are trivialisations $L|_{U_i} \cong \mathcal{O}_{U_i}$ and $L'|_{U_i} \cong \mathcal{O}_{U_i}$. The tensor product of two free rank-one modules over a commutative ring is again free of rank one; hence $(L \otimes_X L')|_{U_i} \cong \mathcal{O}_{U_i}$. So $L \otimes_X L'$ is locally trivial of rank one, i.e. an object of `LineBundle` X. ☐

Closed standalone result — off the critical path. The following lemma *is* formalized and sorry-free; it is not, however, consumed by the relative Picard group law. The associator 653 is realized *unconditionally* through the varying-ring stalk–tensor commutation 659 (16.6), which closes the single left-whiskering obligation for *all* modules with no flatness and no local-triviality hypothesis. The sectionwise-flatness whiskering statement below is therefore neither the live whiskering-stability route nor on the critical path: it is a valid, self-contained result about flat whiskering of sheafification-localizer morphisms, retained for potential reuse. (Unlike the genuinely abandoned 652, it is a closed lemma, not a stub to be skipped.)

Superseded route — off path, not to be formalized. The following lemma belongs to the abandoned coherent-monoidal-category / sheafification-whiskering approach to the associator. The group law on iso-classes consumes only the *existence* of the coherence isomorphisms (664), and the associator is realized as the sheafification transport of the presheaf associator (653), whose single open left-whisker obligation is closed by the d.2 route of 659. The *full* $(J.W)$.`IsMonoidal` / `LocalizedMonoidal` monoidal-category packaging is not required. This block is retained for the historical record only and must *not* be formalized.

Lemma 649. *[Sheafification inverts a localizer morphism of presheaves of modules] Let R_0 be a presheaf of commutative rings on a site carrying a Grothendieck topology J, let R_{sh} be its sheafification, and let $\alpha : R_0 \to R_{sh}.\mathrm{obj}$ be the (locally bijective) comparison map of presheaves of rings. Let $f : A \to B$ be a morphism of presheaves of R_0-modules whose underlying morphism of presheaves of abelian groups* (`toPresheaf` R_0)$.\mathrm{map}\, f$ *lies in the sheafification localizer class $J.W$ — equivalently, is locally bijective for J. Then the presheaf-of-modules sheafification functor* `PresheafOfModules.sheafification` α *sends f to an isomorphism:*

$$(\texttt{sheafification}\,\alpha).\mathrm{map}\, f \quad \textit{is an isomorphism.}$$

This is a closed go/no-go bridge — it converts "a morphism lies in $J.W$" into "its sheafification is invertible". Under route (e) the associator is the API-derived one (`jw_ismonoidal`), so this bridge is retained as a supplement rather than as the associator's load-bearing step.

Proof. This is a one-morphism reading of the structural identity that the sheafification of presheaves of R_0-modules *is* the localization of `PresheafOfModules` R_0 at the class $J.W$.`inverseImage` (`toPresheaf` R_0) of morphisms whose underlying map of presheaves of abelian groups is locally bijective. Concretely, the underlying `AddCommGrp`-valued sheafification inverts exactly the locally bijective morphisms, and Mathlib's `PresheafOfModules.inverseImage_W_toPresheaf_eq_inverseImage_isomorphisms` identifies the preimage of $J.W$ under `toPresheaf` R_0 with the preimage of the class of isomorphisms under the module-level sheafification functor. Reading that equality of morphism classes at the single morphism f: the hypothesis $J.W$((`toPresheaf` R_0)$.\mathrm{map}\, f$) places f in that preimage, hence (`sheafification` α)$.\mathrm{map}\, f$ is an isomorphism. No new construction is required beyond instantiating the existing Mathlib identity at one morphism; this lemma is a closed thin wrapper, retained as a supplement under route (e). □

16.5 Route (e): the $J.W$-monoidality instance and the localized monoidal structure

Full-monoidal-category construction — off path, not to be formalized. The construction below realizes a *full* symmetric monoidal category on all of Scheme.`Modules` X by instantiating Mathlib's localization-of-monoidal-categories API. It was the original plan for supplying the associator. The relative Picard group law, however, is a structure on *isomorphism classes* of invertible sheaves and consumes only the *existence* of the associator, unitor and braiding isomorphisms (664, built by hand as the scheme-theoretic analogue of Mathlib's `CommRing.Pic`, not off a coherent monoidal category's `Skeleton`). The associator is obtained as the sheafification transport of the presheaf associator (653), whose *single* open left-whisker obligation is closed by the d.2 stalk–tensor commutation (659 and 658); it uses *no full* $(J.W)$.`IsMonoidal` instance and *no* `LocalizedMonoidal`

monoidal category. Accordingly the *full monoidal-category* target of this section — the *pair* of whiskering-stability fields (652) and the `LocalizedMonoidal` instance (`jw_ismonoidal`) — is off path and must not be re-attempted; it is kept for the historical record. The d.1/d.2 stalk apparatus it shares with the live single whisker obligation is *not* vestigial, however: it is on the critical path through 659.

The monoidal structure on the substrate is realized by *instantiating* Mathlib's localization-of-monoidal-categories API (16.2): apply `Localization.Monoidal.LocalizedMonoidal` to the already-monoidal presheaf category `PresheafOfModules` $\mathcal{O}_X$ and the sheafification localizer $J.W$. The single new input the API requires is the instance $J.W$.`IsMonoidal`, whose two fields are the whiskering-stability lemmas below.

What the group law consumes. The group law is assembled directly on isomorphism classes of *locally trivial* (invertible) sheaves (664), mirroring Mathlib's ring-level Picard group `Module.Invertible` / `CommRing.Pic` (`Mathlib/RingTheory/PicardGroup.lean`). It needs only the *existence* of the associator 653, the unitors 654 and the braiding 655, together with $\otimes$-invertibility supplying inverses; it uses *no* $J.W$.`IsMonoidal` instance, *no* `LocalizedMonoidal` monoidal category, and *no* `Skeleton`. It does, however, consume the associator 653, which is realized as the sheafification transport of the presheaf associator and whose *single* open left-whisker obligation 659 is closed by the stalk–tensor commutation (d.2, 658). Hence the stalk ingredients (d.1)/(d.2) are *on* the critical path of the group law — through the associator's single whisker obligation — even though the full $J.W$.`IsMonoidal` $\to$ `LocalizedMonoidal` monoidal category on all of `SheafOfModules` (the subject of the present section) is *not* built: that full monoidal generality, with its *pair* of whiskering-stability fields (652) and all derived coherence, is what the group law does not consume. The single open obligation of the present section's full-monoidal target is thus the local-injectivity half packaged for an arbitrary whisker object (651), itself a consequence of the same d.2 commutation.

The following lemma is the R_x-linear stalk-map packaging (the "d.1" ingredient). It is consumed by the stalk–tensor commutation of 16.6 (658 and 659) and is therefore a live ingredient of the unconditional associator route.

Lemma 650. *[The* R.`stalk` x*-linear stalk map of a morphism of presheaves of modules] Let* X *be a topological space,* R *a presheaf of commutative rings on* X*, and* $x \in X$ *a point. For a morphism* $g : M \to N$ *of presheaves of* R*-modules, the induced map on stalks*

$$g_x \;:\; M_x \longrightarrow N_x$$

is R.`stalk` x*-linear (*`stalkLinearMap`*), and is characterised on germs by* $g_x(\mathrm{germ}_x\, s) = \mathrm{germ}_x(g_U\, s)$ *(*`stalkLinearMap_germ`*). When the underlying map of* `Ab`*-valued stalks is an isomorphism,* g_x *is bijective (*`stalkLinearMap_bijective_of_isIso`*); in that case it bundles to an* R.`stalk` x*-linear equivalence* $M_x \xrightarrow{\sim} N_x$ *(*`stalkLinearEquivOfIsIso`*).*

These four declarations are the (d.1)*-partial implementation — the* R_x*-linearity packaging of stalk maps — consumed by the* $\mathrm{id} \otimes g_x$ *step in the proofs of 658 and 659. They build on the stalk module structure of* `Mathlib.Algebra.Category.ModuleCat.Stalk`*; the linearity packaging itself is project-original.*

Proof. The stalk $M_x =$ `stalk` M.`presheaf` x carries its R.`stalk` x-module structure from `Mathlib.Algebra.Category`
The underlying additive map g_x on stalks is the filtered colimit of the section maps g_U; since each section map is $R(U)$-linear and the scalar action on the stalk is the colimit of the section actions, g_x respects the R.`stalk` x-action, which gives the linear map together with its germ characterisation. Bijectivity is read off the underlying `Ab`-stalk isomorphism (a bijective linear map is a linear equivalence), and bundling the inverse yields the linear equivalence. ∎

Off-path duplicate — the full-generality packaging, not to be formalized. The following lemma is the load-bearing whiskering-stability field of the *full* $(J.W)$.`IsMonoidal` / `LocalizedMonoidal` monoidal-category route to the associator. That *full monoidal category* is not built: the group law on iso-classes consumes only the *existence* of the associator (664). The associator instead arises as the sheafification transport whose single open left-whisker obligation is closed by the live lemma 659. The stalk ingredients (d.1)/(d.2) themselves — the R_x-linear stalk map 650 and the stalk–tensor commutation 658 — are *on* the critical path through that live lemma; what is off path is only the *full-generality* $J.W$.`IsMonoidal` packaging of the following duplicate, retained for the historical record.

Lemma 651. *[Local injectivity of a whiskered localizer morphism (stalkwise, flatness-free)] Let R be a presheaf of commutative rings on a site carrying a Grothendieck topology J, and let $J.W$ be the sheafification localizer (the locally bijective morphisms). Concretely the site carries the* `WEqualsLocallyBijective` *property for* `Ab`*, so $J.W$ is exactly the class of J-locally bijective morphisms; this is recorded as a typeclass hypothesis* $[J.$`WEqualsLocallyBijective Ab`$]$ *on the statement. Let F be an* arbitrary *presheaf of R-modules, and let $g : M \to N$ be a morphism of presheaves of R-modules whose underlying morphism of presheaves of abelian groups* (`toPresheaf` R).map g *lies in $J.W$. Then the left-whiskered morphism*

$$F \lhd g \;=\; \mathrm{id}_F \otimes g$$

is J-locally injective. *No flatness and no local-triviality hypothesis on F is required.*

This is the `whiskerLeft`*-field content (the injectivity half) of the* full *instance* $J.W$.`IsMonoidal` *(`jw_ismonoidal`), the load-bearing field of the off-path* `LocalizedMonoidal` *monoidal-category path (arbitrary F). Its content for an arbitrary F is exactly the live lemma 659, which the associator's sheafification transport consumes and which is closed* unconditionally *by the d.2 stalk–tensor commutation (658); the d.1/d.2 ingredients are therefore on the critical path. The two proofs recorded below — a sheaf-level argument specialised to locally trivial F, and the general stalkwise argument via (d.1)/(d.2) — are retained for the historical record; the live group law closes this obligation in its general (d.2) form through 659.*

Proof.

PRIMARY route (F locally trivial — the consumer's case; sheaf-level, no stalks, no (d.2)). Suppose F is locally trivial (`LineBundle.IsLocallyTrivial` F), which is exactly the hypothesis its sole consumer — the associator 653 — supplies. J-local injectivity of a morphism is *local on a covering family*: it suffices to produce a cover of X on each member of which $F \lhd g$ restricts to a locally injective morphism. Take a trivialising open cover $\{V\}$ of X for F, so that $F|_V \cong \mathcal{O}_V$ on each V. On such a V:

- the substrate–restriction compatibility 646 gives a canonical isomorphism

  $$(F \lhd g)\big|_V \;\cong\; (F|_V) \lhd (g|_V)$$

  — restriction along the open immersion $V \hookrightarrow X$ commutes with $\otimes$, and the whiskering $F \lhd g = \mathrm{id}_F \otimes g$ is $\otimes$ with an identity, so it restricts to $\mathrm{id}_{F|_V} \otimes (g|_V) = (F|_V) \lhd (g|_V)$;

- the trivialisation $F|_V \cong \mathcal{O}_V$ together with the left unitor 654 ($\mathcal{O}_V \otimes_V (-) \cong (-)$) identifies $(F|_V) \lhd (g|_V) \cong g|_V$.

Composing, $(F \lhd g)|_V$ is isomorphic to $g|_V$. Now $g \in J.W$, and membership in $J.W$ (local bijectivity) is stable under restriction along an open immersion, so $g|_V \in J.W$; in particular $g|_V$ is J-locally injective. An isomorphism preserves local injectivity, hence $(F \lhd g)|_V$ is locally injective on V. As the $\{V\}$ cover X and local injectivity is local on the cover, $F \lhd g$ is J-locally injective. *No stalks and no (d.2) stalk–tensor commutation enter*; the single comparison lemma 646 (a presheaf-level comparison, decomposed into its two open-immersion base-change halves) carries the argument, and thereby moves *onto* the critical path.

This is *not* the section-level route that tensors a bare section map $g(V)$ — only injective — and stalls on a Tor_1/flatness obligation. Here the argument is sheaf-level and uses that g is locally *bijective* (a $J.W$ morphism), reduced through the unitor to $g|_V$ (again a $J.W$ morphism); a mere injection is never tensored, so no flatness is needed.

FALLBACK route (arbitrary F; stalkwise, required only for the full-generality route (e) `LocalizedMonoidal` *path).* For an *arbitrary* (not necessarily locally trivial) F, local injectivity is proved stalkwise, by the argument Mathlib uses for fixed-base presheaves in `CategoryTheory.Sites.Point.IsMonoidal` (16.2), ported to `PresheafOfModules`. A morphism in $J.W$ is, on the topological site of X, a *stalkwise isomorphism*: at each point x the induced map on stalks g_x is an isomorphism. The presheaf-of-modules tensor is sectionwise, so it commutes with the stalk (a filtered colimit), giving

$$(F \lhd g)_x \;=\; \mathrm{id}_{F_x} \otimes_{R_x} g_x \;:\; F_x \otimes_{R_x} M_x \;\longrightarrow\; F_x \otimes_{R_x} N_x.$$

Tensoring an isomorphism g_x by the identity id_{F_x} yields an isomorphism — no flatness is needed, since g_x is invertible, not merely injective. Hence $F \lhd g$ is a stalkwise isomorphism, in particular locally bijective, in particular locally injective, for *any* F.

The stalkwise computation rests on three ingredients with distinct status at the pinned commit:

- (d.1-done) the R.`stalk` x-linear stalk-map packaging (`stalkLinearMap` and its companions, 650): a morphism of presheaves of modules induces an R.`stalk` x-linear map on stalks, characterised on germs, that becomes a linear equivalence once the underlying `Ab`-stalk map is an isomorphism. This is built project-side, axiom-clean, and furnishes the $\mathrm{id} \otimes g_x$ step above.
- (d.1-bridge) (remaining) the site-$J.W \iff$ stalkwise-iso characterisation on `Opens` X: that a morphism lies in $J.W$ iff its stalk maps are isomorphisms at every point. The concrete Mathlib assembly route uses the `WEqualsLocallyBijective` structure together with the topological stalk criteria — local injectivity $\iff$ injectivity on stalks and local surjectivity $\iff$ surjectivity on stalks — bridging the site-level `Presheaf.IsLocallyInjective` / `IsLocallySurjective` to the `TopCat.Presheaf` stalk versions. This half is still Mathlib-absent at the `PresheafOfModules` level.
- (d.2) (remaining) the commutation $(A \otimes^p B)_x \cong A_x \otimes_{R_x} B_x$ of the stalk (a filtered colimit) with the relative module-presheaf tensor, under which $(F \lhd g)_x$ is identified with lTensor $F_x\,(g_x) = \mathrm{id}_{F_x} \otimes g_x$ (using that `tensorLeft` / `tensorRight` preserve filtered colimits over a module category). This is genuinely Mathlib-absent over a *varying* ring and is the largest remaining piece.

Mathlib supplies the analogues for presheaves valued in a fixed monoidal concrete category, but not for modules over a varying ring; porting (d.1-bridge) and (d.2) to `PresheafOfModules` is the remaining work. Once (d.2) lands, the conclusion is immediate and flatness-free: by `stalkLinearMap_bijective_of_isIso` the stalk map g_x is a linear equivalence, and the linear-equivalence lTensor by F_x carries it to an isomorphism, finishing the proof.

This stalkwise fallback is required *only* if the full-generality $J.W$.`IsMonoidal` / `LocalizedMonoidal` route (e) — the monoidal category on *all* of Scheme.`Modules` X, for an arbitrary whisker object F — is later wanted. For the relative Picard group law the whisker objects are locally trivial, and the PRIMARY route above suffices, trading the Mathlib-absent (d.2) for the single comparison 646. ∎

Superseded route — off path, not to be formalized. The following lemma supplies the two whiskering-stability fields of the abandoned $(J.W)$.`IsMonoidal` instance (`jw_ismonoidal`). The group law needs no such instance: the associator is built directly by gluing canonical local isomorphisms (653, via 646). This block is retained for the historical record only and must *not* be formalized.

Lemma 652. *[Flatness-free whiskering preserves the sheafification localizer] With notation as in 651: for an* arbitrary *presheaf of R-modules F and a morphism g whose underlying additive-presheaf map lies in $J.W$, both whiskerings again lie in $J.W$:*

$$g \in J.W \implies F \lhd g \in J.W, \qquad g \in J.W \implies g \rhd F \in J.W.$$

These two statements are precisely the `whiskerLeft` *and* `whiskerRight` *data required by the class* `MorphismProperty.IsMonoidal` *(16.2) for the morphism property $J.W$; together they furnish the instance $J.W$.*`IsMonoidal` *of* `jw_ismonoidal`*. No flatness and no local triviality is used — they are the flatness-free replacements for the flat variants of* `flat_whisker_localizer`*.*

Proof. By the locally-bijective characterisation of $J.W$, it suffices to show $F \lhd g$ is both locally injective and locally surjective. *Local surjectivity* is free and needs no hypothesis on F: the presheaf tensor is right exact, so $F \otimes^p -$ carries the locally-zero cokernel of g to the cokernel of $F \lhd g$, which is therefore locally zero (Mathlib `isLocallySurjective_whiskerLeft`). *Local injectivity* is 651. The two together give $F \lhd g \in J.W$. The right-whiskered statement $g \rhd F$ follows by conjugating $F \lhd g$ with the braiding of the symmetric presheaf monoidal structure, which is an isomorphism and hence preserves membership in $J.W$. □

Superseded route — off path, not to be formalized. The following lemma is the route (e) target: a full MonoidalCategory structure on all of Scheme.`Modules` X from `LocalizedMonoidal`. It is not needed. The group law consumes only the existence of the associator, unitor and braiding isomorphisms (664, assembled by hand rather than off a coherent monoidal category's `Skeleton`), and the associator is now built directly by gluing (653, via 646). This block is retained for the historical record only and must *not* be formalized.

Lemma 653 (Associator for $\otimes_X$ (unconditional))**.** *Let X be a scheme and let $M, N, P \in$* Scheme.`Modules` X*. There is an isomorphism*

$$(M \otimes_X N) \otimes_X P \xrightarrow{\sim} M \otimes_X (N \otimes_X P)$$

in Scheme.`Modules` X*, natural in M, N, P. The group law on $\otimes$-iso-classes (664) consumes only the* existence *of this isomorphism, the proposition* $\mathrm{Nonempty}((M \otimes_X N) \otimes_X P \cong M \otimes_X (N \otimes_X P))$*; no pentagon, triangle, or naturality coherence of it is ever required.*

The Lean pin is unconditional: it holds for arbitrary $\mathcal{O}_X$-modules M, N, P, with no local-triviality or invertibility hypothesis. (Earlier drafts carried vestigial `IsLocallyTrivial` *hypotheses to match the carrier; these were dropped once the construction below was seen to use none of them.)*

Proof. **Realization (sheafification transport, unconditional via d.2).** The associator is the sheafification transport of the presheaf-of-modules associator `PresheafOfModules.monoidalCategoryStruct.asso` which exists for *all* presheaves of modules over the varying ring. The transport requires the single left-whisker $F \lhd$ `toSheafify` to lie in the sheafification localizer $J.W$ for the relevant F; that single open left-whiskering obligation `isLocallyInjective_whiskerLeft_of_W` is closed *unconditionally* — for an arbitrary whiskering object, with no flatness and no local triviality — by the d.2 stalk–tensor commutation 659 (16.6): a $J.W$-morphism g is a stalkwise isomorphism, and by the d.2 commutation $\left(F \lhd g\right)_x \cong \mathrm{id}_{F_x} \otimes g_x$ is again an isomorphism, so $F \lhd g \in J.W$. Hence the associator is itself unconditional and axiom-clean once d.2 (658) is formalized; it does *not* require the full $J.W$.`IsMonoidal` instance or the `LocalizedMonoidal` monoidal category — only the one left-whisker lemma — and no sectionwise flatness enters. The local-gluing description below records the equivalent concrete local realization through the restriction-compatibility comparison 646.

Concretely, the same isomorphism is described by gluing canonical local isomorphisms through the restriction-compatibility comparison 646. Fix a common open cover $\{U\}$ of X. On each member U form the canonical composite

$$((M \otimes_X N) \otimes_X P)\big|_U \xrightarrow{\sim} (M|_U \otimes_U N|_U) \otimes_U P|_U \xrightarrow{\sim} M|_U \otimes_U (N|_U \otimes_U P|_U) \xrightarrow{\sim} (M \otimes_X (N \otimes_X P))\big|_U,$$

in which:

- the first arrow is two applications of the restriction-compatibility isomorphism 646 along the open immersion $U \hookrightarrow X$: first $(M \otimes_X N)|_U \cong M|_U \otimes_U N|_U$ tensored with $P|_U$, then $((M \otimes_X N) \otimes_X P)|_U \cong (M \otimes_X N)|_U \otimes_U P|_U$;
- the middle arrow is the *canonical presheaf-level associator* α of the (symmetric) monoidal category `PresheafOfModules` $\mathcal{O}_U$ (`PresheafOfModules.monoidalCategoryStruct.associator`, 16.2(i)), read on the restricted modules;
- the third arrow is again two applications of 646, in the opposite direction.

Each of the three arrows is a *canonical* isomorphism: the restriction-compatibility comparison 646 is induced by the open immersion with no choices, hence natural in its module arguments, and the presheaf associator α is a fixed natural transformation. Therefore, on a double overlap $U \cap U'$, restricting the U-composite and the U'-composite to $U \cap U'$ yields the same morphism: each of the three arrows commutes with the further restriction to $U \cap U'$ by the naturality of 646 and of the presheaf associator, so the two local composites agree on the overlap.

Because the internal hom $\mathcal{H}om(-,-)$ of sheaves of $\mathcal{O}_X$-modules is itself a sheaf on X, a family of morphisms defined on the members of an open cover and agreeing on the overlaps glues to a single global morphism. Applying this both to the local composites above and to their local inverses produces a global morphism $(M \otimes_X N) \otimes_X P \to M \otimes_X (N \otimes_X P)$ whose restriction to each U is the local composite, together with a two-sided inverse glued from the local inverses; hence the glued morphism is a global isomorphism — the associator. The single non-formal input of this re-route is the restriction-compatibility comparison 646; the presheaf associator (16.2(i)) and the sheaf-of-modules internal-hom gluing are already available.

Note that, unlike the pointwise-existence statement 648 — a *proposition* proved cover-pointwise with *no* overlap obligation — the associator is *global data*, a single isomorphism of sheaves on all of X. Assembling it from local pieces is therefore genuinely stronger than that pattern: the overlap-naturality compatibility of the local composites must be discharged before gluing, and is discharged above. The locally-trivial hypotheses of the statement are not consumed; the comparison 646 is applied to the global, untrivialised restrictions $(M \otimes_X N)|_U$, so no free-cover specialisation can replace it. What this proof does *not* require is the *full* $(J.W)$.`IsMonoidal` instance or the `LocalizedMonoidal` monoidal category (652): only the single left-whisker obligation 659, closed via the d.2 stalk–tensor commutation 658, is consumed by the sheafification transport. The group law on $\otimes$-iso-classes consumes only the *existence* of the resulting isomorphism, never its pentagon, triangle, or naturality coherence. $\square$

Lemma 654 (Left and right unitors for $\otimes_X$)**.** *Let X be a scheme and $M \in$ Scheme.*`Modules` *X an arbitrary $\mathcal{O}_X$-module. With $\mathcal{O}_X =$* `SheafOfModules.unit` *X.*`ringCatSheaf` *the structure-sheaf unit object, there are natural isomorphisms*

$$\mathcal{O}_X \otimes_X M \xrightarrow{\sim} M, \qquad M \otimes_X \mathcal{O}_X \xrightarrow{\sim} M.$$

Proof. The presheaf monoidal structure on `PresheafOfModules` R supplies the left and right unitors λ, ρ at the presheaf level. Their sheafifications `sheafification.mapIso`(λ), `sheafification.mapIso`(ρ), composed with the sheafification counit identifying the sheafified structure-sheaf presheaf with $\mathcal{O}_X$ (the step already used in the structure-sheaf-unit comparison `tensorObj_unit_iso` in the substrate file), give the two isomorphisms. This route is the cheap `mapIso` pattern and uses no abstract pullback. ☐

Lemma 655 (Braiding for $\otimes_X$)**.** *Let X be a scheme and $M, N \in$ Scheme.*`Modules` *X arbitrary $\mathcal{O}_X$-modules. There is a natural isomorphism*

$$M \otimes_X N \xrightarrow{\sim} N \otimes_X M.$$

Proof. The presheaf-of-modules monoidal category is symmetric, so it carries a braiding β at the presheaf level (`Mathlib.CategoryTheory.Monoidal.PresheafOfModules`). Its sheafification `sheafification.mapIso`(β) is the asserted isomorphism, again by the cheap `mapIso` pattern. ☐

Lemma 656 (Inverse of an invertible module)**.** *Source: [Stacks Project], Tag 01CR, Lemma* `lemma-constructions-invertible` *(item 2, the dual of an invertible is invertible and the contraction is an isomorphism) and Definition* `definition-pic` *(the Picard group is the abelian group of iso-classes of invertible $\mathcal{O}_X$-modules under tensor product). Let X be a scheme and let $L \in$* `LineBundle` *X be a line bundle. The dual*

$$L^{-1} \;:=\; \mathcal{H}om_{\mathcal{O}_X}(L, \mathcal{O}_X)$$

is again a line bundle on X, and there is a canonical natural isomorphism $\varepsilon_L : L \otimes_X L^{-1} \xrightarrow{\sim} \mathcal{O}_X$. Consequently every line bundle has a two-sided tensor inverse under $\otimes_X$, and the monoid (`LineBundle` *$X/\sim, \otimes_X, \mathcal{O}_X)$ of isomorphism classes is an abelian group — the Picard group* $\mathrm{Pic}(X)$ *(Stacks tag 01CR).*

Proof. The dual object and its evaluation are now supplied by the internal-hom infrastructure of 16.13. Take $L^{-1} := \mathcal{H}om_{\mathcal{O}_X}(L, \mathcal{O}_X)$ to be the sheaf-level dual 710 — a genuine object of Scheme.`Modules` X, assembled from the presheaf internal hom 706 by sheafification — and let $\varepsilon_L : L \otimes_X L^{-1} \to \mathcal{O}_X$ be the evaluation (contraction) morphism 709. With both in hand the argument is three steps.

Step 1 (L^{-1} is a line bundle). Local triviality of L (`LineBundle.IsLocallyTrivial` L) furnishes an open cover $\{U\}$ of X with trivialisations $L|_U \cong \mathcal{O}_U$. The internal hom commutes with restriction along the open immersion $U \hookrightarrow X$, so on each U

$$L^{-1}\big|_U \;\cong\; \mathcal{H}om_{\mathcal{O}_U}(L|_U, \mathcal{O}_U) \;\cong\; \mathcal{H}om_{\mathcal{O}_U}(\mathcal{O}_U, \mathcal{O}_U) \;\cong\; \mathcal{O}_U,$$

the last identification being the evaluation-at-1 isomorphism for the dual of a free rank-one module. Hence L^{-1} is locally trivial of rank one, i.e. an object of `LineBundle` X; this is exactly 739.

Step 2 (the contraction is a local isomorphism). The evaluation/contraction morphism

$$\varepsilon_L \;:\; L \otimes_X L^{-1} \;=\; L \otimes_X \mathcal{H}om_{\mathcal{O}_X}(L, \mathcal{O}_X) \;\longrightarrow\; \mathcal{O}_X, \qquad s \otimes \phi \longmapsto \phi(s),$$

is a morphism in Scheme.$\texttt{Modules}\,X$. Restricting to a trivialising open U, the restriction-compatibility isomorphism 646 commutes $\otimes_X$ past $(-)|_U$:

$$(L \otimes_X L^{-1})\big|_U \;\cong\; L|_U \otimes_U L^{-1}|_U \;\cong\; \mathcal{O}_U \otimes_U \mathcal{O}_U,$$

using the trivialisations of Step 1. Under these identifications $\varepsilon_L|_U$ becomes the contraction $\mathcal{O}_U \otimes_U \mathcal{O}_U \to \mathcal{O}_U$ of the free rank-one module with its dual, which is precisely the left unitor 654 ($\mathcal{O}_U \otimes_U (-) \cong (-)$) and hence an isomorphism. Thus $\varepsilon_L|_U$ is an isomorphism on each member of the cover.

Step 3 (glue to a global isomorphism). The local trivialisations are induced by the open immersions with no choices, so the local contraction isomorphisms $\varepsilon_L|_U$ agree on the overlaps $U \cap U'$ by the naturality of 646 and of the unitor under further restriction. As "being an isomorphism" is local on X — a morphism of sheaves of $\mathcal{O}_X$-modules whose restriction to every member of an open cover is an isomorphism is itself an isomorphism — the globally-defined ε_L is an isomorphism $L \otimes_X L^{-1} \xrightarrow{\sim} \mathcal{O}_X$. Equivalently, the local inverses glue (they agree on overlaps for the same reason) to a two-sided inverse of ε_L.

Consequently every line bundle L has the two-sided tensor inverse L^{-1} under $\otimes_X$, and the monoid of $\otimes$-iso-classes of line bundles is an abelian group — the Picard group $\mathrm{Pic}(X)$ (Stacks tag 01CR). The local-triviality of L^{-1} (Step 1) is 739, the dual object and its evaluation are 710 and 709, and the local-iso⇒global-iso passage (Step 3) is the already-closed 646.

Formal dependency. The proof of `exists_tensorObj_inverse` is *not* a one-line consequence: it is the gluing argument 748, which assembles the global contraction ε_L from the chart-local left-unitor contractions. Its load- bearing inputs are: the gluing of compatible local morphisms 746 together with its restriction-value connector 747; the local-iso⇒global-iso passage 740; the overlap cocycle equality, whose two hardest constituents are the trivialisation restriction-compatibility chain 841 and the unit self-duality collapse 846. The dual restriction commutation 737 is one input among several. All of these inputs are established, including the restriction-naturality of the trivialisation chain (841), itself a chain of per-constituent naturality squares. □

Lemma 657 (Closure of $\otimes$ on the locally-trivial carrier `LineBundle.OnProduct`)**.** *Let $C \to S$ and $T \to S$ be S-schemes. The carrier is the* locally-trivial *subtype*

$$\textit{\texttt{LineBundle.OnProduct}}\,\pi_C\,\pi_T \;=\; \bigl\{\, M \in \mathrm{Scheme}.\texttt{Modules}(C \times_S T) \;\big|\; \texttt{IsLocallyTrivial}\,M \,\bigr\}$$

introduced in 14. The bifunctor $\otimes_{C\times_S T}$ of 643 restricts to this subtype: if `IsLocallyTrivial` M *and* `IsLocallyTrivial` M' *then* `IsLocallyTrivial`$(M \otimes_{C\times_S T} M')$*. Hence $\otimes$ closes on* ***`LineBundle.OnProduct`*** $\pi_C\,\pi_T$*, supplying the binary operation the relative group law multiplies with.*

Proof. Closure is exactly 648: the tensor product of two locally-trivial rank-one modules is again locally trivial of rank one (`tensorObj_isLocallyTrivial`). The Lean construction `tensorObjOnProduct` applies this directly to a pair of objects of the `IsLocallyTrivial` subtype `LineBundle.OnProduct`, returning the tensor with its locally-trivial witness; no $\otimes$-invertibility predicate and no coherence isomorphism enters this closure statement. (The $\otimes$-invertibility carrier `IsInvertible` of 662 is a separate predicate, connected to `IsLocallyTrivial` only by an off-critical-path equivalence; the units-group packaging is 664, not this lemma.) □

16.6 The varying-ring stalk–tensor commutation (d.2) and the unconditional left-whiskering

This section supplies the one genuinely Mathlib-absent ingredient on which the *unconditional* associator 653 rests: the commutation of the stalk functor with the relative-ring tensor product of presheaves of modules ("d.2"). Together with the already-built R_x-linear stalk comparison (650, "d.1") it closes the single open left-whiskering obligation for an *arbitrary* whiskering object — with no flatness and no local-triviality hypothesis — which is exactly what makes the sheafification transport of the presheaf associator work for all modules. The construction is to be formalized in the dedicated Lean file `AlgebraicJacobian/Picard/TensorObjSubstrate/StalkTensor.lean`.

Lemma 658 (Stalk of a tensor product (d.2): varying-ring stalk–tensor commutation)**.** *Source: [Stacks Project], Modules on Ringed Spaces, Lemma* `lemma-stalk-tensor-product`*. Let* $R : (\mathtt{Opens}\,X)^{op} \to \mathtt{CommRingCat}$ *be the structure presheaf of a scheme* X*, and let* A, B *be presheaves of* R*-modules (objects of* $\mathtt{PresheafOfModules}\,(R \circ \mathtt{forget}_2\,\mathtt{CommRingCat}\,\mathtt{RingCat})$*). For each point* $x \in X$ *there is a canonical isomorphism of* $(R \circ \mathtt{forget}_2).\mathtt{stalk}\,x$*-modules*

$$(A \otimes^p B).\mathtt{stalk}\,x \xrightarrow{\sim} A.\mathtt{stalk}\,x \otimes_{R.\mathtt{stalk}\,x} B.\mathtt{stalk}\,x,$$

natural in A *and* B*, where* $\otimes^p$ *is the presheaf-of-modules tensor product* `PresheafOfModules.Monoidal.tensorObj`*.*

Proof. The stalk at x is the filtered colimit over the neighbourhoods $U \ni x$ of the sections functor, and sectionwise the tensor product is computed pointwise: $(A \otimes^p B)(U) = A(U) \otimes_{R(U)} B(U)$ (`PresheafOfModules.Monoidal.tensorObj_obj`). The neighbourhood system $\{U \ni x\}$ is cofiltered under inclusion, so the colimit is filtered. The isomorphism is assembled in five named stages. Stages (i)–(ii) construct the forward additive comparison map and its germ characterisation; stages (iii)–(v) upgrade it to an R_x-linear map, construct the reverse map, and check mutual inversion.

(i) Per-neighbourhood balanced bilinear map and its section-level descent (`stalkTensorBilin`, `stalkTensorDescU`). For a fixed neighbourhood $U \ni x$ the assignment

$$(a, b) \longmapsto \mathrm{germ}_x\, a \otimes \mathrm{germ}_x\, b, \qquad a \in A(U),\ b \in B(U),$$

is $R(U)$-balanced bilinear into $A_x \otimes_{R_x} B_x$: it is additive in each variable, and for $r \in R(U)$ the two ways of moving r across the tensor agree after passing to germs, since $\mathrm{germ}_x(r \cdot a) = (\mathrm{germ}_x r) \cdot \mathrm{germ}_x a$ and the stalk tensor is balanced over $R_x = R.\mathtt{stalk}\,x$ (the germ–scalar compatibility of 650). By the universal property of the relative tensor product it descends to an additive map on the section module

$$\mathtt{stalkTensorDescU}_U : (A \otimes^p B)(U) = A(U) \otimes_{R(U)} B(U) \longrightarrow A_x \otimes_{R_x} B_x, \qquad a \otimes b \longmapsto \mathrm{germ}_x a \otimes \mathrm{germ}_x b.$$

(ii) Colimit descent to the forward comparison map (`stalkTensorDesc`, `stalkTensorDesc__germ__tmul`). The maps $\mathtt{stalkTensorDescU}_U$ are compatible with the restriction maps of the presheaf $A \otimes^p B$ as U shrinks toward x (a smaller neighbourhood computes the same germs), so they form a cocone over the filtered diagram $\{(A \otimes^p B)(U)\}_{U \ni x}$. The induced map out of the colimit is the forward additive comparison map

$$\mathtt{stalkTensorDesc} : (A \otimes^p B).\mathtt{stalk}\,x \longrightarrow A_x \otimes_{R_x} B_x,$$

characterised on germs of simple tensors by $\mathtt{stalkTensorDesc}(\mathrm{germ}_x(a \otimes b)) = \mathrm{germ}_x a \otimes \mathrm{germ}_x b$ (`stalkTensorDesc__germ__tmul`). This is the forward half of the asserted isomorphism (`stalk_tensor_desc_forward`).

(iii) R_x*-linearity packaging* (`stalkTensorLinearMap`). The additive map of (ii) is upgraded to an $R.\mathtt{stalk}\,x$-linear map: on a germ $\mathrm{germ}_x r$ of a section $r \in R(U)$ one checks $\mathtt{stalkTensorDesc}((\mathrm{germ}_x r) \cdot \xi) = (\mathrm{germ}_x r) \cdot \mathtt{stalkTensorDesc}(\xi)$ on germs of simple tensors and extends by additivity. The one subtlety is a *carrier-duality obstacle*: the section-level tensor $A(U) \otimes_{R(U)} B(U)$ is a module over the `RingCat` carrier $(R \circ \mathtt{forget}_2)(U)$, whereas the natural scalar $r : R(U)$ lives in the

`CommRingCat` carrier $R(U)$; the two carriers are canonically identified but not definitionally equal, so the scalar-pulling lemma `TensorProduct.smul_tmul′` applies only after inserting a canonical `RingEquiv` bridge between the `CommRingCat` and `RingCat` carriers. Once that bridge is in place the linearity computation mirrors the germ-representative pattern used for the d.1 stalk map `stalkLinearMap` (650).

(iv) The reverse map. The map

$$A_x \otimes_{R_x} B_x \;\longrightarrow\; (A \otimes^p B).\mathtt{stalk}\,x$$

is built by the universal property of the tensor product over R_x from the R_x-bilinear map of the two stalks

$$(\mathrm{germ}_x a,\ \mathrm{germ}_x b) \;\longmapsto\; \mathrm{germ}_x(a|_{U\cap V} \otimes b|_{U\cap V}),$$

where $a \in A(U)$ and $b \in B(V)$ are chosen representatives; this is a *nested* colimit descent, descending first the A-variable out of the filtered colimit $\{A(U)\}_{U\ni x}$ and then the B-variable out of $\{B(V)\}_{V\ni x}$, passing to the common refinement $U \cap V$ to form the simple tensor and its germ. Well-definedness on germs (independence of the chosen representatives and of the neighbourhoods U, V) is exactly filteredness of the neighbourhood system.

It remains to check that the reverse map is R_x-bilinear, and in particular that its underlying bilinear map is *balanced* over R_x, i.e. that pulling a scalar out of the left variable agrees with pulling it out of the right. The key point is to establish this balancing *at the stalk level*, with every scalar living in the stalk ring $R_x = R.\mathtt{stalk}\,x$ throughout. Concretely, for a germ $\mathrm{germ}_x r$ of a section $r \in R(W)$ one verifies the identity

$$\mathtt{revBihom}((\mathrm{germ}_x r)\cdot \mathrm{germ}_x a,\ \mathrm{germ}_x b) \;=\; (\mathrm{germ}_x r)\cdot \mathtt{revBihom}(\mathrm{germ}_x a,\ \mathrm{germ}_x b),$$

and symmetrically on the right variable, where the scalar $\mathrm{germ}_x r \in R_x$ never leaves the stalk ring. This holds because of the germ–scalar compatibility $\mathrm{germ}_x(r\cdot s) = (\mathrm{germ}_x r)\cdot(\mathrm{germ}_x s)$ of the germ map $R(W) \to R_x$ (the Lean lemma `germ_smul`) — the same germ–scalar compatibility already used in stages (i) and (iii). Multiplying a representative $r \cdot a$ and then taking germs produces $(\mathrm{germ}_x r)\cdot \mathrm{germ}_x a$, so the action of r is transparently the R_x-action on the stalk and may be moved freely across the tensor of stalks, where balancedness over R_x is built into $A_x \otimes_{R_x} B_x$. Because the whole computation takes place over the single ring R_x, no carrier identification intervenes and the balancing follows directly.

One must *not* instead reduce this balancing to a section-level identity. If the scalar were pushed through the presheaf restriction to a common neighbourhood W and the balancing read off inside the section module $A(W) \otimes_{R(W)} B(W)$, then the scalar action appearing there is the one produced by the presheaf-of-modules structure of A via restriction of scalars: it lives over the `RingCat` carrier $(R \circ \mathtt{forget_2})(W)$, *not* over the `CommRingCat` carrier $R(W)$ that annotates the section tensor. This is precisely the `CommRingCat`-vs-`RingCat` carrier-duality obstacle already met in stage (iii), here compounded by the extra restriction-of-scalars wrapper, so the naive "move the scalar across the tensor" step (`TensorProduct.smul_tmul′`) does not apply at the section level without first inserting the carrier bridge. Working at the stalk level as above sidesteps this wrapper entirely.

Where a `CommRingCat`-vs-`RingCat` carrier identification is genuinely unavoidable, it is the *same* canonical bridge already invoked for the stage-(iii) R_x-linearity packaging (the `stalkTensorDescU_smul/stalkT` step, `stalk_tensor_linear_map`): the two carriers are canonically identified but not definitionally equal, and the same `RingEquiv` bridge resolves the obstruction here. Thus the stage-(iv) balancing reuses the technique already resolved in stage (iii) rather than re-deriving it.

(v) Mutual inversion on germ generators (`stalkTensorIso`). The two composites are checked to be identities on generators. The composite $A_x \otimes_{R_x} B_x \to (A \otimes^p B).\mathtt{stalk}\,x \to A_x \otimes_{R_x} B_x$ is the

identity on a simple tensor $\mathrm{germ}_x a \otimes \mathrm{germ}_x b$ by the germ characterisation `stalkTensorDesc_germ_tmul` of (ii), and extends to all of $A_x \otimes_{R_x} B_x$ by `TensorProduct.induction_on`. The opposite composite $(A \otimes^p B)$.`stalk` $x \to A_x \otimes_{R_x} B_x \to (A \otimes^p B)$.`stalk` x is the identity on a germ $\mathrm{germ}_x(a \otimes b)$ of a simple tensor by the same characterisation; since germs generate the stalk and the germ maps are jointly epimorphic (`stalk_hom_ext`), together with `TensorProduct.induction_on` applied to each section, the composite is the identity. Bundling the R_x-linear forward map of (iii) with the reverse map of (iv) and these two inversion identities yields the asserted isomorphism `stalkTensorIso`.

Naturality in A, B is inherited from naturality of the sectionwise tensor and of the germ maps. This is the presheaf-of-modules, varying-ring analogue of the Stacks lemma above; the only subtlety beyond the ringed-space statement is that the ground ring varies with U, handled by the "tensor of filtered colimits over the colimit ring" identification together with the carrier-duality bridge of (iii). Conceptually the tensor product of modules commutes with filtered colimits in each variable and the ground ring $R(U)$ itself colimits to the stalk ring R.`stalk` x, which is the reason the forward map (ii) and the reverse map (iv) are mutually inverse. ☐

The proof of the live whiskering lemma below factors through two named sub-lemmas: the *d.1-bridge* relating membership in the sheafification localizer $J.W$ to stalkwise isomorphism, and the *second-factor naturality* of the d.2 commutation isomorphism 658. We state both before the lemma that consumes them.

Lemma 659 (Unconditional left-whiskering of the localizer, via d.2)**.** *Let R, J be as above with J.*`WEqualsLocallyBijective Ab`*, let F be an* arbitrary *presheaf of R-modules, and let $g : M \to N$ be a morphism whose underlying morphism lies in $J.W$. Then the left-whiskered morphism $F \lhd g = \mathrm{id}_F \otimes g$ is J-locally injective (indeed lies in $J.W$). No flatness and no local-triviality hypothesis on F is required. This discharges the open left-whiskering obligation 651 that the unconditional associator 653 consumes.*

Proof. The proof runs in three movements: the d.1-bridge turns membership in $J.W$ into a stalkwise statement, the second-factor naturality of the d.2 isomorphism transports the whiskered stalk into a tensor of the stalk maps, and a flatness-free "tensoring by an equivalence" argument concludes.

(a) The d.1-bridge. Specialise to the topological site `Opens` X, where stalks exist. By the hypothesis $g \in J.W$ and the d.1-bridge `W_implies_stalkwise_iso`, the `Ab`-stalk map $g_x : M_x \to N_x$ is an isomorphism at every point $x \in X$. By 650 this stalk map carries an R_x-linear structure, and the bijectivity just obtained upgrades it to an R_x-linear equivalence `stalkLinearEquivOfIsIso` : $M_x \xrightarrow{\sim} N_x$.

(b) Identifying the whiskered stalk. By the second-factor naturality of the stalk–tensor comparison `stalk_tensor_commutation_naturality_right`, the isomorphism `stalkTensorIso` of 658 identifies the stalk $(F \lhd g)_x$ of the whiskered morphism with the map

$$\mathrm{id}_{F_x} \otimes g_x \;=\; \texttt{LinearMap.lTensor}\, F_x\, g_x \;:\; F_x \otimes_{R_x} M_x \longrightarrow F_x \otimes_{R_x} N_x.$$

(c) Conclusion. Tensoring the R_x-linear equivalence $g_x : M_x \xrightarrow{\sim} N_x$ of (a) on the left by the identity of F_x yields an R_x-linear equivalence `LinearMap.lTensor` $F_x\, g_x = \mathrm{id}_{F_x} \otimes g_x$ (this is `LinearEquiv.lTensor`; it needs *no* flatness of F_x, because lifting an isomorphism through $F_x \otimes_{R_x} (-)$ only uses the functoriality of the tensor product, and the inverse is supplied by the inverse equivalence). By (b), $(F \lhd g)_x$ is therefore an isomorphism of `Ab`-stalks for every x. Applying the converse half of the d.1-bridge `W_implies_stalkwise_iso` to the morphism $F \lhd g$ gives $F \lhd g \in J.W$, and in particular it is J-locally injective (`GrothendieckTopology.W.isLocallyInjective`). No flatness, no local triviality, and the whiskering object F is arbitrary.

The statement holds over any Grothendieck site J satisfying J.`WEqualsLocallyBijective Ab`, and is applied in particular at the topological site `Opens` X of a scheme, where the stalk arguments of (a)–(c) are available and through which the unconditional associator 653 is realized. □

16.6.1 Stalk–tensor comparison helper declarations

The following are the internal Lean helper declarations assembling the forward comparison map `stalk_tensor_desc_forward`, its R_x-linear packaging `stalk_tensor_linear_map`, the reverse map, and the mutual-inversion germ identities feeding the stalk–tensor isomorphism 658. Each is proved directly in Lean and carries the dependency edges of the construction.

Definition 660 (Forward stalk–tensor comparison map and its R_x-linear packaging)**.** *Source: [Stacks Project], Modules on Ringed Spaces, Lemma* `lemma-stalk-tensor-product`*.* The forward half of the stalk–tensor commutation 658 is assembled from four maps. For a neighbourhood $U \ni x$, the $R(U)$-balanced bilinear map `stalkTensorBilin`: $A(U) \times B(U) \to A_x \otimes_{R_x} B_x$, $(a, b) \mapsto \mathrm{germ}_x a \otimes \mathrm{germ}_x b$, descends through the neighbourhood colimit to the additive comparison map `stalkTensorDesc`: $(A \otimes^p B)$.`stalk` $x \to A_x \otimes_{R_x} B_x$; its R_x-linear repackaging is `stalkTensorLinearMap`, and the descent of the R_x-balanced dual bilinear map (661) is the candidate inverse `stalkTensorRev`. Together these are the underlying maps of the isomorphism 658.

Proof. Proved directly in Lean. `stalkTensorBilin` is bilinear and $R(U)$-balanced because $\mathrm{germ}_x(r \cdot a) = (\mathrm{germ}_x r) \cdot \mathrm{germ}_x a$ and the target tensor is balanced over R_x; the universal property of the relative tensor product and the colimit universal property produce `stalkTensorDesc`, and germ–scalar compatibility upgrades it to the R_x-linear `stalkTensorLinearMap`. The reverse map `stalkTensorRev` is the `TensorProduct.liftAddHom` of the dual bihom 661. □

Definition 661 (The dual (co)evaluation bihom of the reverse stalk–tensor map)**.** Project-local construction of the R_x-balanced additive bilinear map underlying the reverse map `stalkTensorRev` of 660. For a fixed A-section a over U, the inner leg `revInnerLeg` sends a B-section b over V to $\mathrm{germ}_x(a|_{U\cap V} \otimes b|_{U\cap V})$; descending it through the B-stalk colimit gives `revInner`, the outer leg `revOuterLeg` packages this additively in a, and descending through the A-stalk colimit gives the additive bilinear map `revBihom`: $A_x \times B_x \to (A \otimes^p B)_x$.

Proof. Proved directly in Lean. Additivity of each leg is checked on germ generators via bilinearity of the section tensor; well-definedness of the nested colimit descents is filteredness of the neighbourhood system, and passing to the common refinement $U \cap V$ forms the simple tensor whose germ is taken. This is the presheaf-of-modules, varying-ring analogue of the dual (co)evaluation pairing of an invertible module; no single external statement covers this bihom, so it stands on the sketch. □

16.7 Tensor invertibility and the commutative-monoid group-law engine

The relative Picard group law is assembled from ⊗-*invertibility* and the monoidal coherence isomorphisms of 653, 654, and 655, following Mathlib's Picard-group idiom (`Module.Invertible`, `CommRing.Pic` = Units(Skeleton(`ModuleCat` R)), `instCommGroupPic`; `Mathlib.RingTheory.PicardGroup`) as the design template. Under the route (e) fallback the inverse of an invertible object is carried by the predicate itself, so neither the open-immersion restriction-compatibility isomorphism 646 nor a separate tensor-inverse lemma is needed there. Under the PRIMARY locally-trivial route, by contrast, both 646 (via the PRIMARY proof of 651) and the tensor-inverse lemma 656 (supplying inverses to 664) are on the critical path.

Definition 662 (⊗-invertibility of an $\mathcal{O}_X$-module)**.** *Source: [Stacks Project], Tag 01CR (Modules on Ringed Spaces, "Invertible modules"), Definition 01CS and Lemma 0B8K. An invertible $\mathcal{O}_X$-module is one for which $\mathcal{L} \otimes_{\mathcal{O}_X} (-)$ is an equivalence; equivalently there exists $\mathcal{N}$ with $\mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{N} \cong \mathcal{O}_X$, and then $\mathcal{N} \cong \mathcal{H}om_{\mathcal{O}_X}(\mathcal{L}, \mathcal{O}_X)$.* Let X be a scheme and $M \in \text{Scheme.}\mathtt{Modules}\, X$. We say M is *⊗-invertible* when it admits a tensor inverse: there is an object $N \in \text{Scheme.}\mathtt{Modules}\, X$ together with an isomorphism

$$M \otimes_X N \;\cong\; \mathtt{SheafOfModules.unit}\, X.\mathtt{ringCatSheaf} \;=\; \mathcal{O}_X,$$

i.e. $\mathtt{IsInvertible}(M) :\equiv \exists\, N,\ \text{Nonempty}(M \otimes_X N \cong \mathtt{SheafOfModules.unit}\, X.\mathtt{ringCatSheaf})$, with $\otimes_X$ the substrate of 642. The predicate is a *proposition* carrying its inverse witness existentially; consequently the dual/inverse needed by the group law is definitional and no separate "a tensor inverse exists" lemma is consumed downstream. This is the scheme-level analogue of Mathlib's `Module.Invertible`, transported one categorical level up to $\text{Scheme.}\mathtt{Modules}\, X$ with the substrate $\otimes_X$.

Lemma 663 (The unit object is ⊗-invertible)**.** *Source: [Stacks Project], Modules on Ringed Spaces, Lemma 0B8K. The unit object $\mathcal{O}_X = \mathtt{SheafOfModules.unit}\, X.\mathtt{ringCatSheaf}$ is ⊗-invertible: taking $N = \mathcal{O}_X$, the substrate unitor gives $\mathcal{O}_X \otimes_X \mathcal{O}_X \cong \mathcal{O}_X$, so the $\exists N$ criterion 662 is met. This is the invertibility of the identity element of the relative Picard group law.*

Proof. Proved directly in Lean. The witness is $N = \mathcal{O}_X$ with the isomorphism $\mathcal{O}_X \otimes_X \mathcal{O}_X \cong \mathcal{O}_X$ supplied by the substrate unitor 654. □

Lemma 664 (The commutative monoid of ⊗-iso-classes and its units)**.** *Source: [Stacks Project], Tag 01CR (Modules on Ringed Spaces, "Invertible modules"), Definition 01CX and Lemma `lemma-constructions-invertible`. The isomorphism classes of invertible $\mathcal{O}_X$-modules form an abelian group* $\text{Pic}(X)$ *under ⊗, with unit $\mathcal{O}_X$ and inverse the dual; the tensor product of invertibles is invertible and the dual is a tensor inverse. Let X be a scheme. The isomorphism classes of ⊗-invertible objects of* $\text{Scheme.}\mathtt{Modules}\, X$ *(662) form a commutative monoid under the substrate tensor product $\otimes_X$ in which every element is a unit — equivalently, an abelian group — the Picard group* $\text{Pic}(X)$ *(Stacks tag 01CR).*

Primary construction (directly on locally-trivial iso-classes, à la `Module.Invertible`***).*** *The commutative group is built directly on isomorphism classes of* locally-trivial *(equivalently ⊗-invertible) sheaves, mirroring Mathlib's ring-level Picard group* `Module.Invertible` / `CommRing.Pic` *(`Mathlib/RingTheory/PicardGroup.lean`). The group operation is $[L] \cdot [M] := [L \otimes_X M]$, well-defined on iso-classes by the bifunctoriality of $\otimes_X$ (643) and again locally trivial by the closure lemma 657; the identity is $[\mathcal{O}_X]$; and the inverse of $[L]$ is the class of the dual $[L^{-1}]$ supplied by 656 (with $L \otimes_X L^{-1} \cong \mathcal{O}_X$).*

The four group-axiom fields are defined by hand*, each discharged by a* single *existence-of-isomorphism read as an equality of iso-classes — passage to iso-classes turns an isomorphism of sheaves into an equality of classes:*

$$\mathtt{one\_mul} \leftarrow \langle \lambda_X \rangle, \quad \mathtt{mul\_one} \leftarrow \langle \rho_X \rangle, \quad \mathtt{mul\_assoc} \leftarrow \langle \alpha_{X,Y,Z} \rangle, \quad \mathtt{mul\_comm} \leftarrow \langle \beta_{X,Y} \rangle,$$

where the unit laws are the unitors 654, associativity is the associator 653, and commutativity is the braiding 655, each consumed only as the proposition $\text{Nonempty}(\cdots \cong \cdots)$*; the pentagon, triangle and hexagon coherence carried by a* MonoidalCategory *typeclass is* never *invoked in any of the four axiom proofs.*

This by-hand assembly does not *go through* `CategoryTheory.monoidOfSkeletalMonoidal` *or* `Skeleton` *(`Mathlib/CategoryTheory/Monoidal/Skeleton.lean`): those constructions* require *a coherent* [`MonoidalCategory`] *(and, for the commutative law,* [`BraidedCategory`]*) on the carrier category — precisely the coherent monoidal category on* Scheme.`Modules` X *over the* varying *structure sheaf that this construction explicitly avoids. Consequently the group law requires* no $J.W$.`IsMonoidal`*,* no `LocalizedMonoidal` *and* no `Skeleton` *instance on* Scheme.`Modules` X*. The genuine precedent is Mathlib's fixed-ring Picard group* `CommRing.Pic` $R :=$ Skeleton(`Module.Invertible` R) *(`Mathlib/RingTheory/PicardGroup.lean`): there the* `CommGroup` is *read off the* Skeleton *of the coherent monoidal category* `MonoidalCategory`(`Module.Invertible` R) *that Mathlib supplies over a fixed ring. The present construction* cannot *take that* Skeleton *route — no coherent monoidal category on* Scheme.`Modules` X *for the varying structure sheaf is available — so its* `CommGroup` *is built directly from the existence-of-isomorphism propositions above, the construction ultimately resting, as* `CommRing.Pic` *does, on iso-classes of invertible modules. The pinned carrier* `tensorObjIsoclassCommMonoid` *realizes this commutative-group structure; the scheme-level construction is genuinely absent from Mathlib at the pinned commit.*

This is the lower-risk *path: every ingredient is either already proved (the unitors 654, the braiding 655) or locally-trivial-scoped and assembled (the associator 653). The only remaining obligations on this critical path are the restriction-compatibility comparison 646* and *the dual/inverse 656. The associator 653 is supplied by the unconditional route of 16.6 and transitively invokes the d.2 stalk–tensor commutation 658.*

Fallback (full route (e)). *Alternatively, once* `jw_ismonoidal` *lands,* Scheme.`Modules` $X =$ `SheafOfModules` X.`ringCatSheaf` *is a full (symmetric)* MonoidalCategory *via* `Localization.Monoidal.LocalizedMo` *and the Picard group is the units of its skeleton,* $\mathrm{Pic}(X) =$ Units(Skeleton(Scheme.`Modules` X))*, exactly as Mathlib builds* `CommRing.Pic` $R =$ Units(Skeleton(`ModuleCat` R)) *and transports* `instCommGroupPic`*. This fallback yields the group on* all *invertible modules and is retained for the full monoidal generality, but it is not required for the relative Picard group law, which the primary construction already supplies.*

Proof. Work directly on isomorphism classes of locally-trivial sheaves, as in Mathlib's `Module.Invertible` / `CommRing.Pic` (`Mathlib/RingTheory/PicardGroup.lean`). The binary operation $[L] \cdot [M] := [L \otimes_X M]$ is well-defined on iso-classes by the bifunctoriality of $\otimes_X$ (643) — an isomorphism in either argument tensors to an isomorphism — and its value is again locally trivial by the closure lemma 657, so the operation closes on the carrier. The identity element is $[\mathcal{O}_X]$. The four group-axiom fields are each defined *by hand* from a *single* existence-of-isomorphism, read as an equality of iso-classes,

$$\texttt{mul\_assoc} \leftarrow \langle\alpha\rangle, \quad \texttt{one\_mul} \leftarrow \langle\lambda\rangle, \quad \texttt{mul\_one} \leftarrow \langle\rho\rangle, \quad \texttt{mul\_comm} \leftarrow \langle\beta\rangle:$$

associativity is 653, the unit laws are the unitors 654, and commutativity is the braiding 655. No pentagon, triangle or hexagon identity is invoked, since each law asserts only Nonempty$(\cdots \cong \cdots)$. This is the same one-isomorphism-per-law shape that Mathlib's `monoidOfSkeletalMonoidal` (`Mathlib/CategoryTheory/Monoidal/Skeleton.lean`) uses on a skeletal monoidal category via its skeletality map `hC`, but we do *not* instantiate that construction: it requires a coherent [`MonoidalCategory`] (and [`BraidedCategory`]) on Scheme.`Modules` X, which is exactly the structure for the varying structure sheaf that is unavailable. Finally every element is invertible: the dual L^{-1} of 656 satisfies $L \otimes_X L^{-1} \cong \mathcal{O}_X$, so $[L^{-1}]$ is a two-sided inverse of $[L]$; equivalently, $[L]$ is a unit precisely when L is $\otimes$-invertible (662), the inverse witness carried existentially by `IsInvertible`. This assembles the abelian group — the Picard group of X (Stacks tag 01CR) — with *no* $J.W$.`IsMonoidal`, `LocalizedMonoidal` or `Skeleton`, exactly as `CommRing.Pic` carries its `CommGroup` on iso-classes of invertible modules (there off a fixed-ring `Skeleton`, here by hand).

Alternatively (fallback), once `jw_ismonoidal` provides the full MonoidalCategory structure on Scheme.`Modules` X via `LocalizedMonoidal`, the group is the units of the skeleton, Units(Skeleton(Scheme.`Modules` exactly the Mathlib `Units`(Skeleton($\cdots$)) template used for `CommRing.Pic`; this route is not needed for the group law. $\square$

16.8 Pullback-monoidality: invertibility under a general scheme morphism

The relative Picard consumer (15) carries its line bundles on the *local-triviality* predicate `IsLocallyTrivial` (609): its carrier `OnProduct` is the subobject $\{ L \in \text{Scheme.}\texttt{Modules}\,(C \times_S T) \mid \texttt{IsLocallyTrivial}\, L \}$ of locally-trivial $\mathcal{O}$-modules, and its functorial structure maps are *pullback* maps of $\mathcal{O}$-modules: the projection pullback π_T^* along $C \times_S T \to T$ and the base-change pullback along $C \times_S T' \to C \times_S T$. For these maps to descend to the carrier and to be *additive* (an `AddMonoidHom` of isomorphism classes) one needs the comparison isomorphism

$$f^*(L \otimes_X L') \;\cong\; f^*L \otimes_Y f^*L'$$

for *locally-trivial* L, L' and a *general* morphism of schemes $f : Y \to X$ — the projection and the base-change morphism are neither open immersions nor flat. This restricted comparison isomorphism, on locally-trivial pairs, is the load-bearing result of the section (696); the invertibility corollary `isinvertible_pullback` is then a three-line consequence. The comparison is needed *only* on locally-trivial pairs — the carrier never presents a general $\mathcal{O}$-module — so no comparison for arbitrary modules is required.

This is a genuinely different fact from the open-immersion compatibility 646, and not a corollary of it. That lemma handles restriction along an open immersion, and it does so by routing through the *lax right* adjoint `pushforward/restrictScalars`, which is strong monoidal there *only* because an open immersion's structure-sheaf ring map is a local *isomorphism* (`appIso` f). A general scheme morphism has no such isomorphism, so the right-adjoint route is unavailable.

The reduction to two facts about pullback. The corollary `isinvertible_pullback` is a three-line consequence of two facts about pullback, exactly as in the Stacks proof of `lemma-pullback-invertible`. The first is that pullback commutes with the tensor product on the *locally-trivial* pairs the consumer presents: the canonical comparison $f^*(L \otimes_X L') \cong f^*L \otimes_Y f^*L'$ is an isomorphism whenever L, L' are locally trivial, for every f (696). The second is that pullback preserves the structure sheaf, $f^*\mathcal{O}_X \cong \mathcal{O}_Y$ (671, already in hand). Given a locally-trivial inverse N with $L \otimes_X N \cong \mathcal{O}_X$, pulling back yields

$$f^*L \otimes_Y f^*N \;\cong\; f^*(L \otimes_X N) \;\cong\; f^*\mathcal{O}_X \;\cong\; \mathcal{O}_Y,$$

so f^*N is a tensor inverse of f^*L. The locally-trivial comparison 696 is the load-bearing input.

Why the apparent obstruction does not apply. It is tempting to argue that the comparison can never be an isomorphism because an (op)lax monoidal functor need not preserve invertible objects — global sections Γ is lax monoidal, yet $\Gamma(\mathbb{P}^1, \mathcal{O}(1)) = 0$ is not invertible. But that counterexample concerns the *lax* tensorator of *pushforward*: Γ is a right adjoint (global sections), and its comparison $\Gamma(\mathcal{F}) \otimes \Gamma(\mathcal{G}) \to \Gamma(\mathcal{F} \otimes \mathcal{G})$ genuinely fails to be an isomorphism. The comparison we need is the *oplax* δ of *pullback*, a *left* adjoint, running the other way, $f^*(M \otimes N) \to f^*M \otimes f^*N$. Pullback *provably preserves local triviality* (611): if M is trivialised by $\mathcal{O}_U$ on a cover $\{U_i\}$, then f^*M is trivialised by $\mathcal{O}_{f^{-1}U_i}$ on $\{f^{-1}U_i\}$ — using only $f^*\mathcal{O} \cong \mathcal{O}$ and the compatibility of pullback with restriction along open immersions. Consequently, on locally-trivial objects, both sides of δ are locally trivial, and δ restricts to a comparison between the structure sheaf and itself on each chart, where it is forced to be an isomorphism. The bare oplax structure supplies δ as a *morphism* for free (667); the entire remaining content is the chart-local check that, on a trivialising cover, δ is an isomorphism.

The construction route. The isomorphism is obtained by *upgrading the already-built oplax comparison map* δ_{sheaf} = `pullbackTensorMap` (667) to an isomorphism through a chart-chase — the same shape as the proven local-triviality results 611 and 648, and *not* via any concrete inverse-image model or filtered-colimit construction. The argument decomposes into four steps, carried out in the sub-lemmas cited by 696:

(D1′) *Naturality of δ in (M, N).* The sheaf-level comparison δ_{sheaf} is natural in both arguments, assembled from the naturality of the abstract oplax δ (Mathlib's `Functor.OplaxMonoidal.δ_natural`) and of the sheafification reconciliations used to build it (672). This naturality is what lets a chart replace $M|_{U_i}, N|_{U_i}$ by the trivial bundle $\mathcal{O}$ without changing whether δ is an isomorphism there.

(D2′) *δ on the unit pair $(\mathcal{O}, \mathcal{O})$ is an isomorphism.* On the trivial bundle the comparison $\delta_{(\mathcal{O},\mathcal{O})}$: $f^*(\mathcal{O} \otimes \mathcal{O}) \to f^*\mathcal{O} \otimes f^*\mathcal{O}$ is reduced, by the δ-wrapping `isIso_sheafifyDelta_unitPair_of_isIso_sheafifyE` (the presheaf oplax left-unitality factored through sheafification, using `W_of_isIso_sheafification` and the flatness-free right-whiskering `W_whiskerRight_of_W`), to the single *unit square* a_Y.map(ηF)∘`sheafifyUnitIso` = `pullbackValIso`∘`pullbackObjUnitToUnit` (`eta_bridge_unit_square`). That square, transposed across the sheaf pullback–pushforward adjunction, bottoms out in the structure-sheaf comparison `pullbackUnitIso` (671, an isomorphism for *all* f); hence $\delta_{(\mathcal{O},\mathcal{O})}$ is an isomorphism (675).

(D3′) *Composition coherence of δ* (the genuinely new ingredient). For composable scheme morphisms h, f the comparison δ for the composite $h \circ f$ factors through the comparisons for f and for h, conjugated by the pullback pseudofunctoriality isomorphism `pullbackComp` $h\,f$ (685). Specialising to $h := j_i'$ at an open U_i of X with preimage $f^{-1}U_i$ — with $g_i : f^{-1}U_i \to U_i$ the restriction of f and $j_i : U_i \hookrightarrow X$, $j_i' : f^{-1}U_i \hookrightarrow Y$ the open immersions, so $f \circ j_i' = j_i \circ g_i$ — this intertwines the restriction-to-$f^{-1}U_i$ of δ for f with the δ for g_i. Because `pullbackTensorMap` is a hand-built four-fold composite rather than an adjunction transpose, the unit-coherence mate calculus of 670 does *not* transfer; the coherence is instead established as a paste of four composition-coherence squares, decomposing δ of the composite pullback via Mathlib's `Functor.OplaxMonoidal.comp_δ` and the conjugation `conjugateEquiv_pullbackComp_inv` of the pseudofunctor isomorphisms.

(D4′) *Chart-chase assembly.* Choose a common affine cover $\{U_i\}$ of X trivialising both M and N (the common refinement, exactly as in 648). On each $f^{-1}U_i$, the base-change coherence (D3′) presents the restriction of δ for f as the δ for g_i *flanked* by the open-immersion comparisons $\delta^{j_i'}, \delta^{j_i}$; cancelling those flanking factors — legitimate only because each is invertible by the open-immersion comparison 695 (K1) — aligns the restriction with the δ for g_i; naturality (D1′) replaces the trivialised $M|_{U_i}, N|_{U_i}$ by $\mathcal{O}$; and (D2′) makes the resulting comparison on the unit pair an isomorphism. Thus δ restricts to an isomorphism on every member of the cover $\{f^{-1}U_i\}$ of Y, and the restriction-detects-isomorphism criterion 740 promotes it to a global isomorphism. The structure mirrors the chase of 611.

Two sub-steps carry genuinely new content: (D3′) the composition coherence of δ, and — as the realised chart-chase makes explicit — (K1) the open-immersion comparison 695, whose invertibility is what licenses cancelling the flanking factors $\delta^{j_i'}, \delta^{j_i}$ in (D4′) and reducing to the δ for

g_i. The remaining ingredients are lighter: (D1′) is Mathlib naturality, (D2′) is the unit coherence backed by `pullbackUnitIso`, and the cover assembly of (D4′) reuses the already-proven machinery (648, 740). No concrete inverse-image model, no filtered-colimit/⊗ interchange, and no general comparison for arbitrary modules is constructed.

Lemma 665 (The presheaf pushforward is lax monoidal)**.** *Let φ be a morphism of* `CommRingCat`*-valued ring presheaves, presenting a pushforward of presheaves of modules. Then the presheaf pushforward* `PresheafOfModules.pushforward` φ *is* lax monoidal*: it carries the comparison*

$$\mu : (\texttt{pushforward}\,\varphi).\mathrm{obj}\,A \otimes (\texttt{pushforward}\,\varphi).\mathrm{obj}\,B \longrightarrow (\texttt{pushforward}\,\varphi).\mathrm{obj}\,(A \otimes B),$$

obtained as the composite of the strong-monoidal pushforward of presheaves over a fixed ring presheaf ($\texttt{pushforward}_0\texttt{OfCommRingCat}$*) with the lax monoidal restriction of scalars* `restrictScalars` φ *along the ring-presheaf map. This is a statement at the* presheaf *level of $\mathcal{O}$-modules; it does not require a monoidal structure on* `SheafOfModules`*.*

Proof. Restriction of scalars along a homomorphism of ring presheaves is lax monoidal — the comparison sends $(a \otimes b)$ to itself under the identification of the restricted tensor product with the source tensor product — and the pushforward of presheaves of modules over a *fixed* ring presheaf is strong monoidal. The presheaf pushforward along φ factors as the composite of these two functors, so its lax monoidal structure is the composite of the strong-monoidal structure on the first factor with the lax structure on the second; the comparison μ is the whiskered composite of the two comparisons. □

Lemma 666 (The presheaf pullback is oplax monoidal; the comparison map δ)**.** *In the same setting, the presheaf pullback* `PresheafOfModules.pullback` φ *— the left adjoint of the lax-monoidal presheaf pushforward of 665 — is* oplax monoidal*. In particular it carries the canonical comparison* morphism

$$\delta : (\texttt{pullback}\,\varphi).\mathrm{obj}\,(A \otimes B) \longrightarrow (\texttt{pullback}\,\varphi).\mathrm{obj}\,A \otimes (\texttt{pullback}\,\varphi).\mathrm{obj}\,B,$$

produced by the doctrinal adjunction transfer `Adjunction.leftAdjointOplaxMonoidal`*: the oplax structure on a left adjoint is the adjunction mate of the lax structure on its right adjoint. The comparison δ is at the* presheaf *level, and it is only a* morphism*: upgrading δ to an isomorphism is the remaining content, and is exactly what the chart-chase of 696 supplies for a locally-trivial pair.*

Proof. The presheaf pullback is by construction the left adjoint of the presheaf pushforward, which is lax monoidal by 665. Mathlib's `Adjunction.leftAdjointOplaxMonoidal` turns the lax-monoidal structure on a right adjoint into an oplax-monoidal structure on the left adjoint, with the oplax comparison δ defined as the mate of the lax comparison μ under the adjunction. Applying this to the pullback–pushforward adjunction yields the asserted oplax structure and the comparison morphism δ. □

Lemma 667 (The sheaf-level pullback–tensor comparison map δ_{sheaf})**.** *Let $f : Y \to X$ be a morphism of schemes and let $M, N \in$* Scheme.`Modules` X *be* arbitrary *$\mathcal{O}_X$-modules. There is a canonical comparison* morphism

$$\delta_{\mathrm{sheaf}} : (\texttt{Scheme.Modules.pullback}\,f).\mathrm{obj}\,(M \otimes_X N) \longrightarrow (\texttt{Scheme.Modules.pullback}\,f).\mathrm{obj}\,M \otimes_Y (\texttt{Scheme.Modules.pu}$$

of $\mathcal{O}_Y$-modules, natural in M and N. It is a map only*: in general it is not asserted to be an isomorphism.*

Proof. The presheaf pullback is oplax monoidal (666), so at the presheaf level it carries the comparison morphism

$$\delta : (\mathtt{pullback}\,\varphi).\mathrm{obj}\,(A \otimes B) \;\longrightarrow\; (\mathtt{pullback}\,\varphi).\mathrm{obj}\,A \otimes (\mathtt{pullback}\,\varphi).\mathrm{obj}\,B,$$

where φ is the ring-presheaf map presenting f. One transports δ through sheafification to the sheaf level, using the same `sheafificationCompPullback` device that supplies Steps 1–2 of 646: the abstract pullback of $\mathcal{O}$-modules agrees, up to canonical natural isomorphism, with the sheafification of the presheaf pullback, and likewise the sheafified tensor product $\otimes_Y$ is the sheafification of the presheaf tensor. The right-hand side reconciliation of the unit and tensor factors with their sheafified forms is supplied by the sheafification-monoidality brick `sheafifyTensorUnitIso`. Whiskering and composing these canonical isomorphisms with the sheafification of δ produces the asserted sheaf-level morphism δ_{sheaf}, natural in M, N because each ingredient is. No claim of invertibility is made: δ_{sheaf} is the carrier on which the invertible-case isomorphism argument of `isinvertible_pullback` acts. □

Lemma 668 (Reduction of `pullbackTensorMap` iso-ness to the sheafified presheaf δ)**.** *Let $f : Y \to X$ be a morphism of schemes and let $M, N \in \mathrm{Scheme}.\mathtt{Modules}\,X$. Write*

$$\varphi' : (X.\mathtt{presheaf} \circ \mathtt{forget}_2) \;\longrightarrow\; (\mathtt{Opens.map}\,f.\mathtt{base})^{\mathrm{op}} \circ (Y.\mathtt{presheaf} \circ \mathtt{forget}_2)$$

for the induced presheaf-of-modules map presenting f, and let $a_Y = \mathtt{PresheafOfModules.sheafification}\,(\mathbf{1}_{Y.\mathtt{ringCatShe}}$ denote sheafification on Y. If the sheafified presheaf-level oplax comparison

$$a_Y.\mathrm{map}(\delta\,(\mathtt{PresheafOfModules.pullback}\,\varphi')\,M.\mathtt{val}\,N.\mathtt{val})$$

is an isomorphism, then the sheaf-level comparison $\mathtt{pullbackTensorMap}\,f\,M\,N : f^*(M \otimes_X N) \to f^*M \otimes_Y f^*N$ of 667 is an isomorphism.

Proof. Unfolding `pullbackTensorMap` (667) exhibits it as a four-fold composite

$$\mathtt{pullbackTensorMap}\,f\,M\,N \;=\; p_1 \;\;;\;\; a_Y.\mathrm{map}\,\delta \;\;;\;\; p_3 \;\;;\;\; p_4,$$

where: $p_1 = (\mathtt{sheafificationCompPullback}).\mathrm{hom}$, reconciling the abstract $\mathcal{O}$-module pullback with the sheafification of the presheaf pullback; $a_Y.\mathrm{map}\,\delta$ is the sheafification of the presheaf-level oplax comparison δ; $p_3 = (\mathtt{sheafifyTensorUnitIso}).\mathrm{hom}$, the sheafification-monoidality brick reconciling the sheafified tensor with the sheaf tensor; and $p_4 = a_Y.\mathrm{map}$ of the tensor $\otimes_{\mathrm{m}}$ of the two `pullbackValIso` comparisons (one for M, one for N) identifying the sheafified pullback value with the pullback.

The outer three factors are isomorphisms unconditionally: p_1 and p_3 are the .hom components of isomorphisms, and p_4 is the image under the additive functor $a_Y.\mathrm{map}$ of a tensor product of the two `pullbackValIso` *isomorphisms*, hence itself an isomorphism. The hypothesis supplies the one conditional middle factor $a_Y.\mathrm{map}\,\delta$. A composite of isomorphisms is an isomorphism, so $\mathtt{pullbackTensorMap}\,f\,M\,N$ is an isomorphism. □

Lemma 669 (Composition coherence of the pushforward unit comparison)**.** *Let $h : Z \to Y$ and $f : Y \to X$ be composable morphisms of schemes. Write* upu *for the pushforward (right-adjoint) unit comparison* `SheafOfModules.unitToPushforwardObjUnit`*, the canonical map $\mathcal{O} \to f_*(\mathcal{O})$ comparing the structure sheaf of the target with the pushforward of the structure sheaf of the source. Then* upu *satisfies the composition coherence*

$$\mathrm{upu}(h\,;f) \;=\; \mathrm{upu}\,f \;\;;\;\; (\mathtt{pushforward}\,f).\mathrm{map}(\mathrm{upu}\,h) \;\;;\;\; (\mathtt{pushforwardComp}\,h\,f).\mathrm{hom},$$

where `pushforwardComp` $h\,f : (h\,;f)_* \cong f_* \circ h_*$ *is the pseudofunctoriality isomorphism of the pushforward. Sectionwise both sides are nothing but the functoriality of the structure-sheaf ring maps, so the identity holds on the nose after the comparison of sections. This is the right-adjoint half of the unit composition coherence, and it is the input consumed by the mate transport that produces the pullback-side coherence 670.*

Proof. Both sides are morphisms of $\mathcal{O}$-modules; it suffices to compare them on sections over each open. Each comparison upu acts on sections as the structure-sheaf ring homomorphism induced by the relevant morphism of schemes, and the pseudofunctor coherence `pushforwardComp` acts as the identification of $(h\,;f)$-pushforward sections with iterated $f_*\,h_*$-pushforward sections. The composition coherence is therefore the functoriality identity $(h;f)^\sharp = h^\sharp \circ f^\sharp$ of the structure-sheaf ring maps, which holds definitionally; the two sides agree section by section. ☐

Lemma 670 (Composition coherence of the pullback unit comparison)**.** *Let* $h : Z \to Y$ *and* $f : Y \to X$ *be composable morphisms of schemes. Write* pbu *for the pullback (left-adjoint) unit comparison* `SheafOfModules.pullbackObjUnitToUnit`*, the canonical map* $f^*\mathcal{O} \to \mathcal{O}$*. Then* pbu *satisfies the composition coherence*

$$\mathrm{pbu}(h\,;f) \;=\; (\texttt{pullbackComp}\,h\,f).\mathrm{inv} \;\;;\;\; (\texttt{pullback}\,h).\mathrm{map}(\mathrm{pbu}\,f) \;\;;\;\; \mathrm{pbu}\,h,$$

where `pullbackComp` $h\,f : (h;f)^* \cong h^* \circ f^*$ *is the pseudofunctoriality isomorphism of the pullback. This is the genuinely-new ingredient for 671: the left-adjoint pullback is defined abstractly, with no sectionwise value, so this identity is* not *a sectionwise statement and cannot be checked on sections.*

Proof. The pullback f^* is the left adjoint of the pushforward f_*, and likewise for h and $h\,;f$; the comparison $\mathrm{pbu}\,f$ is the adjunction *mate* of the pushforward comparison $\mathrm{upu}\,f$ (669) across the pullback–pushforward adjunction, and similarly for h and $h\,;f$. Since the mate correspondence is a bijection, the asserted identity follows once both sides are transposed under the adjunction $(h\,;f)^* \dashv (h\,;f)_*$.

Under this transpose the left-hand side $\mathrm{pbu}(h\,;f)$ becomes the pushforward comparison $\mathrm{upu}(h\,;f)$. The right-hand side, a composite involving $(\texttt{pullbackComp}\,h\,f)$.inv and the mates of $\mathrm{pbu}\,f$ and $\mathrm{pbu}\,h$, transposes — using the conjugation of the pseudofunctoriality isomorphisms (`pullbackComp` $h\,f$ on the left adjoints corresponds to `pushforwardComp` $h\,f$ on the right adjoints) and the unit of the composite adjunction $(f^* \dashv f_*) \circ (h^* \dashv h_*)$ — exactly into the right-hand side of the pushforward coherence 669, namely $\mathrm{upu}\,f\,;f_*(\mathrm{upu}\,h)\,;(\texttt{pushforwardComp}\,h\,f)$.hom. Thus the two transposed sides coincide by 669, and the original identity follows by the injectivity of the transpose. ☐

Lemma 671 (Pullback preserves the structure sheaf)**.** *Let* $f : Y \to X$ *be a morphism of schemes. The pullback of the structure sheaf is the structure sheaf: there is a canonical isomorphism*

$$f^*(\texttt{SheafOfModules.unit}\,X.\texttt{ringCatSheaf}) \xrightarrow{\sim} \texttt{SheafOfModules.unit}\,Y.\texttt{ringCatSheaf}, \qquad \textit{i.e.} \qquad f^*\mathcal{O}_X \cong \mathcal{O}_Y,$$

where `SheafOfModules.unit` X.`ringCatSheaf` $= \mathcal{O}_X$ *is the monoidal unit of* Scheme.`Modules` X.

Proof. This is a one-step consequence of a Mathlib instance, with no chart-chase. The canonical comparison map is the genuine Mathlib morphism

$$\texttt{SheafOfModules.pullbackObjUnitToUnit}\,f : f^*\mathcal{O}_X \longrightarrow \mathcal{O}_Y,$$

the unit comparison η of f^*; abbreviate it $\mathrm{pbu}\,f$. Mathlib proves this map is an isomorphism whenever the comparison functor `Opens.map` f.`base` on opens is `Final`: the instance `SheafOfModules.instIsIsoPullbackObjUnitToUnitOfFinal`.

The point is that the finality hypothesis holds *unconditionally*, for every morphism of schemes f. The preimage functor `Opens.map` f`.base` : `Opens` $X \to$ `Opens` Y is a frame homomorphism: it preserves arbitrary unions and finite intersections, hence preserves all finite limits. A functor between small categories that preserves finite limits is representably flat, so `final_of_representablyFlat` gives (`Opens.map` f`.base`)`.Final` with no hypothesis on f whatsoever. (There is no need to pass to affine charts: finality is not a special feature of open immersions but a general property of preimage-on-opens.)

Consequently the instance `instIsIsoPullbackObjUnitToUnitOfFinal` fires directly, and pbu f is an isomorphism for every f. The asserted isomorphism $f^*\mathcal{O}_X \cong \mathcal{O}_Y$ is the bundled iso `pullbackUnitIso` f of this comparison. □

Lemma 672 (Naturality of the sheaf-level pullback–tensor comparison δ_{sheaf} (D1′))**.** *Let* $f : Y \to X$ *be a morphism of schemes. The sheaf-level comparison map* $\delta_{\text{sheaf}} =$ `pullbackTensorMap` *of* 667 *is* natural *in both module arguments: for morphisms* $a : M \to M'$ *and* $b : N \to N'$ *in* Scheme.`Modules` X *the square*

$$\begin{array}{ccc} f^*(M \otimes_X N) & \xrightarrow{\delta_{\text{sheaf}}} & f^*M \otimes_Y f^*N \\ \big\downarrow{\scriptstyle f^*(a\otimes b)} & & \big\downarrow{\scriptstyle f^*a \otimes f^*b} \\ f^*(M' \otimes_X N') & \xrightarrow{\delta_{\text{sheaf}}} & f^*M' \otimes_Y f^*N' \end{array}$$

commutes.

Proof. The map δ_{sheaf} is built (667) by transporting the abstract oplax comparison δ of the presheaf pullback (666) through the sheafification reconciliations. The asserted square is the pasting of four naturality squares, one for each ingredient of this composite; we record them in order, the only delicate one being the naturality of the oplax comparison itself.

Square 2: naturality of the oplax comparison δ. The comparison δ is the oplax-monoidal comparison of the presheaf-level pullback functor $F =$ `PresheafOfModules.pullback` φ', whose defining ring map $\varphi' =$ (`toRingCatSheafHom` f).hom has, as the source of its domain, the ring presheaf of X. For the monoidal structure on `PresheafOfModules` to be available on this domain the ring presheaf must be presented in its *canonical* form as the composite $\mathcal{O}_X \circ$ $\texttt{forget}_2$ `CommRingCat RingCat` — the presentation on which that monoidal structure is defined. When the comparison map's construction is unfolded, however, the domain ring re-presents in a form definitionally equal, but not explicitly identical, to the canonical one, and on that form the monoidal structure is not directly available. The naturality of δ,

$$\delta \circ F(a \otimes b) \;=\; (Fa \otimes Fb) \circ \delta,$$

which holds for any oplax monoidal functor, is therefore applied by re-establishing the canonical domain-ring spelling at the functor argument itself: one ascribes the defining ring map φ' explicitly to the canonical composite-functor source, so that the domain monoidal structure resolves, and the naturality identity then matches the goal definitionally. This is a structural device — it changes only the presentation of F's defining ring map. No restatement of `pullbackTensorMap` or of its helper isomorphisms is needed, and the unit-case lemma `pullbackTensorMap_unit_isIso` (D2′) is unaffected. After this step the comparison δ commutes past $F(a \otimes b)$, producing the factor $Fa \otimes Fb$.

Squares 1, 3, 4 and bifunctoriality. The remaining ingredients of δ_{sheaf} are honest natural transformations and contribute already-closed naturality squares: the sheafification, the tensor/unit reconciliation `sheafifyTensorUnitIso` (Square 3, 673), and the `pullbackValIso` reconciliation (Square 4, 674). Finally, the tensor of the two component morphisms factors as

$f^*a \otimes f^*b$ by the bifunctoriality of $\otimes_Y$ — the interchange law $(f_1 \otimes f_2) \circ (g_1 \otimes g_2) = (f_1 \circ g_1) \otimes (f_2 \circ g_2)$, here `tensorHom_comp_tensorHom`. Pasting the four squares together with this bifunctoriality identity yields the asserted commuting square. ☐

Lemma 673 (Section-level naturality of the tensor/unit reconciliation `sheafifyTensorUnitIso`). *The tensor/unit reconciliation isomorphism* `sheafifyTensorUnitIso` *entering the fourth naturality square of 672 is natural in both module arguments. Equivalently: after descending to sections over each open U and passing to the underlying module components, the naturality square is the bilinear, sectionwise naturality of the sheafification unit η, verified one pure tensor $p \otimes q$ at a time, using bilinearity.*

Proof. Write Φ for the natural transformation whose naturality square (in the two module arguments M, N, along morphisms $a : M \to M'$ and $b : N \to N'$) is to be shown to commute; Φ is assembled from the reconciliation isomorphism `sheafifyTensorUnitIso` (the intermediate construction inside 667), so its components are morphisms of presheaves of modules. The square is closed at the categorical (tensor-of-morphisms) level, without any descent to sections or to elements.

Step 1: a single tensor-of-morphisms presentation. The hom leg of `sheafifyTensorUnitIso` is rewritten as a *single* tensor of morphisms of the form $a.\mathrm{map}(\eta_P \otimes \eta_Q)$, where $\eta =$ (`sheafificationAdjunction`).unit is the unit $\mathbf{1} \Rightarrow (\text{sheafify}) \circ (\text{inclusion})$ of the sheafification adjunction and a is the sheafification functor. The essential point is that all factors are kept inside *one* monoidal structure on `PresheafOfModules` — the canonical presentation as a composite with the forgetful functor $\texttt{forget}_2$ — so that the bifunctoriality law of $\otimes$ can be applied consistently within one monoidal structure.

Step 2: merge the two functor-image factors. Both composites around the square are images, under the sheafification functor, of a tensor of morphisms. The two functor-image factors are merged into a single image of a composite by functoriality, $F(g) \circ F(h) = F(g \circ h)$.

Step 3: bifunctoriality and unit naturality. The resulting equality of two tensors of morphisms is closed by the bifunctoriality (interchange) law

$$(f_1 \otimes f_2) \circ (g_1 \otimes g_2) \;=\; (f_1 \circ g_1) \otimes (f_2 \circ g_2),$$

applied within the single chosen monoidal structure, together with the two single-component naturality squares of the sheafification unit η, one in each module argument:

$$p \circ \eta_{M'} = \eta_M \circ (f^*a), \qquad q \circ \eta_{N'} = \eta_N \circ (f^*b).$$

Each of these holds because η is a natural transformation. Combining the two single-component squares through the interchange law equates the two tensors of morphisms, which is exactly the naturality square of `sheafifyTensorUnitIso`. ☐

Lemma 674 (Naturality of the `pullbackValIso` comparison). *The* `pullbackValIso` *comparison — the isomorphism reconciling the underlying presheaf of a pullback sheaf of modules with the pullback of its underlying presheaf — is natural in its module argument. This is one of the (closed) input squares of the pasting that proves 672.*

Lemma 675 (The comparison on the unit pair $(\mathcal{O}, \mathcal{O})$ is an isomorphism (D2′)). *Let $f : Y \to X$ be a morphism of schemes. On the monoidal unit pair $(\mathcal{O}_X, \mathcal{O}_X)$ the sheaf-level comparison*

$$\delta_{(\mathcal{O},\mathcal{O})} : f^*(\mathcal{O}_X \otimes_X \mathcal{O}_X) \;\longrightarrow\; f^*\mathcal{O}_X \otimes_Y f^*\mathcal{O}_X$$

of 667 is an isomorphism.

Proof. By 668 applied on the unit pair, it suffices to verify that the sheafified presheaf comparison a_Y.map $(\delta\,(\texttt{pullback}\,\varphi')\,\mathbf{1}\,\mathbf{1})$ is an isomorphism; the underlying presheaf of the structure-sheaf unit coincides with the presheaf monoidal unit $\mathbf{1}$.

On the unit pair the presheaf-level oplax comparison δ is governed by Mathlib's oplax-monoidal *unit coherence*: `Functor.OplaxMonoidal.left_unitality_hom` expresses $\delta\,(\texttt{pullback}\,\varphi')\,\mathbf{1}\,\mathbf{1}$ as a composite of the functorial unitor F.map $(\lambda_{\mathbf{1}})$.hom, the target unitor $\lambda_{F.\mathrm{obj}\,\mathbf{1}}$, and the whiskered presheaf unit comparison $\eta\,(\texttt{pullback}\,\varphi') \triangleright F.\mathrm{obj}\,\mathbf{1}$. The unitors are isomorphisms, so after sheafification by a_Y the genuine remaining content is

$$\texttt{IsIso}\big(a_Y.\mathrm{map}\,(\eta\,(\texttt{pullback}\,\varphi'))\big),$$

the sheafified presheaf unit comparison (the δ-wrapping reduction). By `isiso_sheafifyeta_of_unitsquare` it suffices, in turn, to prove the *unit square*

$$(\texttt{pullbackValIso}\,f\,\mathcal{O}_X).\mathrm{inv};a_Y.\mathrm{map}\,(\eta\,(\texttt{pullback}\,\varphi'));\texttt{sheafifyUnitIso}.\mathrm{hom} \;=\; \texttt{pullbackObjUnitToUnit}\,\varphi,$$

which is `eta_bridge_unit_square`. Granting that square, `isiso_sheafifyeta_of_unitsquare` yields $\texttt{IsIso}\big(a_Y.\mathrm{map}\,(\eta\,(\texttt{pullback}\,\varphi'))\big)$ — the right-hand side $\texttt{pullbackObjUnitToUnit}\,\varphi = \texttt{pullbackUnitIso}\,f : f^*\mathcal{O}_X \xrightarrow{\sim} \mathcal{O}_Y$ being an isomorphism for *every* f (671) — and the δ-wrapping reduction above then delivers that $\delta_{(\mathcal{O},\mathcal{O})}$ is an isomorphism. ◻

16.8.1 The unit square (D2′): a mate-calculus telescope

The proof of 675 bottoms out in the single morphism equation labelled the *unit square* below. Its proof is a telescope across three nested adjunction layers (sheafification, presheaf pullback–pushforward, sheaf pullback–pushforward). This subsection isolates the load-bearing intermediate identities as named lemmas so each can be discharged independently, then assembles them.

We fix throughout a morphism of schemes $f : Y \to X$ and write $\varphi = f.\texttt{toRingCatSheafHom}$, $\varphi' = \varphi$.hom. Let a_X, a_Y denote sheafification of presheaves of modules on X, Y; $\mathbf{1}^{\mathrm{p}}$ the presheaf monoidal unit; $F = \texttt{pullback}\,\varphi'$ the presheaf pullback; $\eta F : F.\mathrm{obj}\,\mathbf{1}^{\mathrm{p}} \to \mathbf{1}^{\mathrm{p}}$ its oplax unit comparison; and $\varepsilon(G)$ the lax unit comparison of a lax monoidal functor G. The isomorphisms $\texttt{pullbackValIso}\,f\,\mathcal{O}_X$ and `sheafifyUnitIso` are the file's comparison morphisms relating the sheaf-level pullback value to the sheafification of the presheaf-level pullback. Write $\texttt{sheafCounit}_\bullet = (\texttt{sheafificationAdjunction}\,(\mathbf{1}_{\bullet.\texttt{ringCatSheaf}.\mathrm{val}})).\mathrm{counit}$.

Lemma 676 (Composite-adjunction homEquiv factorisation)**.** *Let* $\texttt{adj}_1 : L_1 \dashv R_1$ *and* $\texttt{adj}_2 : L_2 \dashv R_2$ *be composable adjunctions and* $A = \texttt{adj}_1.\mathrm{comp}\,\texttt{adj}_2 : L_1 \circ L_2 \dashv R_2 \circ R_1$ *their composite. Then the hom-set bijection of the composite factors as the composite of the two factor bijections: for every* g*,*

$$A.\mathrm{homEquiv}\,g \;=\; \texttt{adj}_1.\mathrm{homEquiv}(\texttt{adj}_2.\mathrm{homEquiv}\,g).$$

Consequently, for the composite adjunction A *of the unit square (left adjoint* $a_X \circ \texttt{pullback}_{\mathrm{sheaf}}\,\varphi$*), the head of step 2 of* `eta_bridge_unit_square` *equals* $(\texttt{sheafAdj}_X.\mathrm{homEquiv})^{-1}(A.\mathrm{homEquiv}\,g)$ *with* $g = (\texttt{sheafificationCompPullback}\,\varphi).\mathrm{hom.app}\,\mathbf{1}^{\mathrm{p}}$*.*

Proof. The composite adjunction `Adjunction.comp` is defined so that its unit is the pasting $\texttt{adj}_1.\mathrm{unit};R_1(\texttt{adj}_2.\mathrm{unit})_{L_1(-)}$ and its counit dually; equivalently its hom-set bijection A.homEquiv is the composite of the two factor bijections in the order $\texttt{adj}_2$.homEquiv then $\texttt{adj}_1$.homEquiv. This is the standard factorisation `Adjunction.comp_homEquiv` (the naturality of `homEquiv` under composition of adjunctions): a morphism $L_1 L_2\, c \to d$ is transposed first across $\texttt{adj}_1$ to $L_2\, c \to R_1\, d$ and then across $\texttt{adj}_2$ to $c \to R_2 R_1\, d$. Applying this with $\texttt{adj}_1 = \texttt{sheafificationAdjunction}\,(\mathbf{1}_X)$, $\texttt{adj}_2 = \texttt{pullbackPushforwardAdjunction}_{\mathrm{sheaf}}\,\varphi$ and transposing back along $\texttt{adj}_1$ (i.e. applying $(\texttt{sheafAdj}_X.\mathrm{homEquiv})^{-1}$) gives the stated rewrite of the head of step 2. ◻

Lemma 677 (The sheafify-then-pullback comparison is the left-adjoint-uniqueness isomorphism)**.** *Write A for the composite adjunction obtained by following the sheafification adjunction on X with the sheaf-level pullback–pushforward adjunction of φ, namely $A = (\texttt{sheafificationAdjunction}\,(\mathbf{1}_X)).\mathrm{comp}\,(\texttt{pullbac}$ whose left adjoint is $a_X \circ \texttt{pullback}_{\mathrm{sheaf}}\,\varphi$; and write B for the composite adjunction obtained by following the presheaf-level pullback–pushforward adjunction of φ' with the sheafification adjunction on Y, namely $B = (\texttt{pullbackPushforwardAdjunction}\,\varphi').\mathrm{comp}\,(\texttt{sheafificationAdjunction}\,(\mathbf{1}_Y))$, whose left adjoint is $F \circ a_Y$. These two left adjoints are canonically isomorphic, and the canonical "sheafify-then-pullback" comparison agrees on the nose with the left-adjoint-uniqueness isomorphism:*

$$\texttt{sheafificationCompPullback}\,\varphi \;=\; \texttt{Adjunction.leftAdjointUniq}\,A\,B.$$

Proof. The two composite adjunctions A and B share the same right adjoint: pushing forward along φ and then forgetting to presheaves of modules on X is, on the nose, the same as forgetting to presheaves on Y and then pushing forward along φ' — the pushforward commutes with the forgetful functor to presheaves by definition. Hence A and B are two adjunctions sharing the same right adjoint, so the left-adjoint-uniqueness isomorphism $\texttt{Adjunction.leftAdjointUniq}\,A\,B$ between their left adjoints is defined. Unfolding the canonical comparison $\texttt{sheafificationCompPullback}\,\varphi$ and unfolding $\texttt{Adjunction.leftAdjointUniq}\,A\,B$ produces the same morphism — both sides are the composite isomorphism assembled from the two factor adjunctions in the same order. The equality therefore holds by reflexivity. □

Lemma 678 (leftAdjointUniq transport of the composite unit)**.** *Let A, B be the two composite adjunctions of* ***`eta_bridge_unit_square`*** *(with left adjoints $a_X \circ \texttt{pullback}_{\mathrm{sheaf}}\,\varphi$ and $F \circ a_Y$ respectively, which agree up to the canonical iso $\texttt{sheafificationCompPullback}\,\varphi = \texttt{Adjunction.leftAdjointUniq}\,A\,B$), and let $g = (\texttt{sheafificationCompPullback}\,\varphi).\mathrm{hom.app}\,\mathbf{1}^{\mathrm{p}}$. Then*

$$A.\mathrm{homEquiv}\,g \;=\; B.\mathrm{unit.app}\,\mathbf{1}^{\mathrm{p}} \;=\; \texttt{presheafAdj}.\mathrm{unit.app}\,\mathbf{1}^{\mathrm{p}} \,;\, (\texttt{pushforward}\,\varphi').\mathrm{map}(\texttt{toSheafify}_Y(F.\mathrm{obj}\,\mathbf{1}^{\mathrm{p}})).$$

Proof. The comparison $\texttt{sheafificationCompPullback}\,\varphi$ is by definition $\texttt{Adjunction.leftAdjointUniq}\,A\,B$ (677), the canonical isomorphism between two left adjoints of the same functor. Mathlib's $\texttt{Adjunction.homEquiv\_leftAdjointUniq\_hom\_app}\,A\,B$ states that applying $A.\mathrm{homEquiv}$ to the component of this comparison at an object c returns $B.\mathrm{unit.app}\,c$; taking $c = \mathbf{1}^{\mathrm{p}}$ gives the first equality. For the second, expand $B.\mathrm{unit}$ by $\texttt{Adjunction.comp\_unit\_app}$ on $B = (\texttt{pullbackPushforwardAdjunction}\,\varphi').\mathrm{comp}\,(\texttt{sheafificationAdjunction}\,(\mathbf{1}_Y))$: the unit of a composite adjunction at $\mathbf{1}^{\mathrm{p}}$ is the presheaf unit $\texttt{presheafAdj}.\mathrm{unit.app}\,\mathbf{1}^{\mathrm{p}}$ followed by the pushforward of the sheafification unit $\texttt{toSheafify}_Y$ at $F.\mathrm{obj}\,\mathbf{1}^{\mathrm{p}}$, which is the displayed right-hand side. □

Lemma 679 (leftAdjointUniq transport of the composite unit (P-general form))**.** *This is the P-general twin of 678: the latter is the specialisation to the monoidal unit $P = \mathbf{1}^{\mathrm{p}}$, whereas the present form holds for an* arbitrary *presheaf P over X. Fix a morphism of schemes φ (with structure-ring map φ'), and let*

$$A_\varphi \;=\; (\texttt{PresheafPullbackPushforwardAdj}\,\varphi').\mathrm{comp}\,\texttt{sheafAdj}$$

be the two-layer composite adjunction (presheaf pullback–pushforward composed with the sheafification adjunction), and let B_φ denote its sheaf-level counterpart (the composite adjunction whose left adjoint is the sheaf-level pullback). Writing $\texttt{sheafCompPb} = \texttt{SheafOfModules.sheafificationCompPullback}$ for the canonical comparison, the lemma asserts

$$A_\varphi.\mathrm{homEquiv}\,P\_\,((\texttt{sheafCompPb}\,\varphi).\mathrm{hom.app}\,P) \;=\; B_\varphi.\mathrm{unit.app}\,P.$$

*It is the key **step-1** brick of the tail assembly of 685: applying A_φ.homEquiv to the comparison component recovers the composite-adjunction unit at P. Its proof is identical to that of 678 with $\mathbf{1}^{\mathrm{p}}$ replaced by P:* `sheafCompPb` φ *is by definition* `Adjunction.leftAdjointUniq` $A_\varphi\, B_\varphi$ *(677), so Mathlib's* `Adjunction.homEquiv_leftAdjointUniq_hom_app` *at P returns B_φ.unit.app P.*

Lemma 680 (Sheafification kills the `pullbackComp` hom–inv pair (D3′ residual brick))**.** *Let $h : Z \to Y$ and $f : Y \to X$ be composable, write* `PrPbComp` $=$ `PresheafOfModules.pullbackComp` $\varphi'_f\,\varphi'_h$ *for the presheaf pullback pseudofunctoriality isomorphism* `pullback` $\varphi'_f \circ$ `pullback` $\varphi'_h \cong$ `pullback` $\varphi'_{h\circ f}$*, and let a_Z be sheafification of presheaves of modules on Z. Then for every presheaf T over X the sheafified hom–inv pair cancels:*

$$a_Z.\mathrm{map}(\texttt{PrPbComp}.\mathrm{hom.app}\, T)\ ;\ a_Z.\mathrm{map}(\texttt{PrPbComp}.\mathrm{inv.app}\, T)\ =\ \mathrm{id}_{a_Z.\mathrm{map}\,(\cdots)}.$$

Proof. The two factors are a_Z.map of the components at T of the hom–inv pair of the isomorphism `PrPbComp`. By `Iso.hom_inv_id_app` the underlying composite `PrPbComp`.hom.app T; `PrPbComp`.inv.app T is the identity; applying a_Z.map (`congrArg`, then `Functor.map_comp` to split and `Functor.map_id` to collapse the identity) yields the claim. The cancellation is built as this *standalone* equation and then spliced into the D3′ paste by `erw`; a plain `rw` does not fire across the parenthesised group boundary because of the `Sheaf.val` Z carrier-spelling mismatch. ☐

Lemma 681 (Sq3 — composition coherence of `sheafifyTensorUnitIso` (D3′))**.** *With notation as in 685, write $F^{\mathrm{p}}_\bullet$ =* `pullback` $\varphi'_\bullet$ *for the presheaf-level pullback and $a_\bullet$ for sheafification. The tensor/unit reconciliation* `sheafifyTensorUnitIso` *of the* composite *$h \circ f$, evaluated on $F^{\mathrm{p}}_{h\circ f}M$.val and $F^{\mathrm{p}}_{h\circ f}N$.val, agrees — after transport through the presheaf pullback pseudofunctoriality identification* `PrPbComp` $=$ `PresheafOfModules.pullbackComp` $\varphi'_f\,\varphi'_h$ *(880 licenses the composite ring map) — with the f-instance carried by h^* followed by the h-instance on the pulled-back arguments. Concretely, the* `hom` *legs of the two presentations are equal as morphisms in* Scheme.`Modules` Z.

Proof. By 877 each `sheafifyTensorUnitIso`.hom is a single a.map of a `tensorHom` of sheafification-unit components $\eta_P \otimes \eta_Q$, kept inside one monoidal structure on `PresheafOfModules`. The composite-vs-interleaved discrepancy is therefore the image, under a_Z.map, of the naturality of the sheafification unit η against the presheaf identification `PrPbComp`, one tensor factor at a time; bifunctoriality of $\otimes$ (the interchange law) recombines the two single-factor squares into the asserted equality. ☐

Lemma 682 (Sq4 — per-leg presheaf composition coherence of `pullbackValIso` (D3′))**.** *Let $h : Z \to Y$, $f : Y \to X$ be composable and $W \in$ Scheme.*`Modules` *X. Write $F^{\mathrm{p}}_\bullet$ =* `pullback` $\varphi'_\bullet$ *for the presheaf-level pullback, $\eta^\bullet$ for the sheafification unit on $\bullet$,* `forget` *for the forgetful functor* `SheafOfModules` $\to$ `PresheafOfModules`, `pVI` $=$ `pullbackValIso` *(878),* `PrPbComp` $=$ `PresheafOfModules.pullbackComp` $\varphi'_f\,\varphi'_h$*, and* `pullbackComp` $h\,f$ *for the scheme-level pullback pseudofunctoriality isomorphism $(h \circ f)^* \cong h^* \circ f^*$, with f^*W = (*`pullback` *f).obj W. Then the following identity of morphisms in* `PresheafOfModules` *Z.*`ringCatSheaf`*.obj holds:*

$$\texttt{PrPbComp}.\mathrm{hom.app}\, W.\mathrm{val}\ ;\ \eta^{Z}_{F^{\mathrm{p}}_{h\circ f}W.\mathrm{val}}\ ;\ \texttt{forget}\,(\texttt{pVI}\,(h\circ f)\,W).\mathrm{hom}$$

$$=\ F^{\mathrm{p}}_h(\eta^{Y}_{F^{\mathrm{p}}_f W.\mathrm{val}}\,;\texttt{forget}\,(\texttt{pVI}\,f\,W).\mathrm{hom})\,;\eta^{Z}_{F^{\mathrm{p}}_h(f^*W).\mathrm{val}}\,;\texttt{forget}\,(\texttt{pVI}\,h\,(f^*W)).\mathrm{hom}\,;\texttt{forget}\,((\texttt{pullbackComp}\,h\,f).\mathrm{ap}$$

This is the per-leg identity actually consumed — once for M, once for N — by the final `tensorObjIsoOfIso`$\big($(`pullbackComp` $h\,f)_M$, (`pullbackComp` $h\,f)_N\big)$ *step of 685; it is the presheaf-level form of the composition coherence of the connecting comparison* `pullbackValIso`*, relating the $(h \circ f)$-comparison to the f- and h-comparisons conjugated by* `pullbackComp` $h\,f$.

Proof. Expand the connecting comparison on each of the three legs by its definition (878): for every g and V,

$$(\texttt{pVI}\, g\, V).\mathrm{hom} \;=\; \left((\texttt{sheafificationCompPullback}\, \varphi_g).\mathrm{app}\, V.\mathrm{val}\right)^{-1} \;;\; g^*(\varepsilon_V),$$

the inverse of the sheafify–pullback comparison `sheafCompPb` at V.val followed by the pullback of the sheafification counit ε_V. Applying `forget` and distributing it over the composite splits each `forget` (`pVI` $g\,V$).hom into a $\texttt{sheafCompPb}^{-1}$ factor and a pulled-back-counit factor; after this expansion both sides of the asserted identity are composites whose factors are exactly these two kinds, threaded by the sheafification units $\eta^\bullet$ and the presheaf comparison `PrPbComp`.

The two families of factors reassemble by the two sub-coherences. The $\texttt{sheafCompPb}^{-1}$ factors recombine by 683, the inverse of the Sq1 coherence 884. The pulled-back-counit factors recombine by 684, the naturality (pseudofunctoriality) of the sheafification counit across `pullbackComp` $h\, f$. Substituting both reassemblies and reassociating the intervening $\eta^\bullet$-units and the `PrPbComp` factor turns the left-hand composite into the right-hand one.

The two reassemblies are not independent: the f-leg counit $f^*(\varepsilon_W)$ and the h-leg counit ε_{f^*W} are bridged by the f-instance of `sheafCompPb`, so the counit reassembly of 684 interleaves with the $\texttt{sheafCompPb}^{-1}$ reassembly of 683 through that mediating comparison. □

Lemma 683 (Sq4a — the $\texttt{sheafCompPb}^{-1}$ reassembly (inverse of Sq1))**.** *Let $h : Z \to Y$, $f : Y \to X$ be composable and P a presheaf over X; write a_X, a_Z for the sheafifications, $F^{\mathrm{p}}_\bullet =$ `pullback` $\varphi'_\bullet$,* `sheafCompPb = SheafOfModules.sheafificationCompPullback`*, and* `PrPbComp = PresheafOfModules.pullbackCo` *Then the inverse of the composite sheafify–pullback comparison factors as*

$$\left((\texttt{sheafCompPb}\,(h\circ f)).\mathrm{app}\, P\right)^{-1} \;=\; a_Z.\mathrm{map}(\texttt{PrPbComp}.\mathrm{inv}_P) \;;\; \left((\texttt{sheafCompPb}\, h).\mathrm{app}\,(F^{\mathrm{p}}_f P)\right)^{-1} \;;\; h^*(((\texttt{sheafCompPb}\, f).$$

Proof. The Sq1 coherence 884 expresses the hom leg of the composite comparison as the ordered composite

$$((\texttt{sheafCompPb}\,(h\circ f)).\mathrm{app}\, P).\mathrm{hom} \;=\; (\texttt{pullbackComp}\, h\, f).\mathrm{inv}_{a_X P} \;;\; h^*(((\texttt{sheafCompPb}\, f).\mathrm{app}\, P).\mathrm{hom}) \;;\; ((\texttt{sheafCompP}$$

Taking inverses of both sides reverses the order of the four factors and inverts each. The inverse of a_Z.map(`PrPbComp`.hom_P) is a_Z.map(`PrPbComp`.inv_P), since functoriality of a_Z sends the inverse of an isomorphism to the inverse of its image; likewise the inverse of $h^*(\cdots.\mathrm{hom})$ is $h^*(\cdots.\mathrm{inv})$ by functoriality of h^*; and (`pullbackComp` $h\, f$).inv^{-1} = (`pullbackComp` $h\, f$).hom. Substituting these gives the displayed factorisation. □

Lemma 684 (Sq4b — the counit reassembly (counit pseudofunctoriality))**.** *Let $h : Z \to Y$, $f : Y \to X$ be composable and $W \in$ Scheme.*`Modules` *X; write ε for the sheafification counit (878) and* `pullbackComp` *$h\, f : (h \circ f)^* \cong h^* \circ f^*$ for the scheme-level pullback pseudofunctoriality isomorphism. The pulled-back counit of the composite reassembles through* `pullbackComp` *$h\, f$: naturality of* `pullbackComp` *$h\, f$ at the counit morphism $\varepsilon_W : a_X(W.\mathrm{val}) \to W$ gives*

$$(h \circ f)^*(\varepsilon_W) \;\;;\;\; (\texttt{pullbackComp}\, h\, f).\mathrm{hom}_W \;=\; (\texttt{pullbackComp}\, h\, f).\mathrm{hom}_{a_X(W.\mathrm{val})} \;\;;\;\; h^*(f^*(\varepsilon_W)).$$

This is the counit-pseudofunctoriality identity that, in the assembly of 682, moves the composite counit into the h-image of the f-counit, where it meets the h-leg counit ε_{f^*W}.

Proof. The displayed equality is the naturality square of the natural isomorphism `pullbackComp` $h\, f$: $(h \circ f)^* \Rightarrow h^* \circ f^*$ evaluated at the morphism $\varepsilon_W : a_X(W.\mathrm{val}) \to W$: for a natural transformation, the domain image followed by the codomain component equals the source component followed by the codomain image, and $(h^* \circ f^*)(\varepsilon_W) = h^*(f^*(\varepsilon_W))$ by functoriality of the composite. No further input is needed; the counit ε enters only as the morphism at which the natural isomorphism is evaluated. □

Lemma 685 (Composition coherence of the pullback tensorator δ (D3′))**.** *Let $h : Z \to Y$ and $f : Y \to X$ be composable morphisms of schemes and let $M, N \in$* Scheme.`Modules` X*. Write δ_{sheaf} =* `pullbackTensorMap` *for the sheaf-level pullback–tensor comparison of 667, $h^*(-)$ =* (`pullback` h).map *for the pullback functor, and* `pullbackComp` $h\,f$ *for the pullback pseudofunctoriality isomorphism $(h \circ f)^* \cong h^* \circ f^*$. Then the comparison for the composite $h \circ f$ factors through the comparisons for f and for h, conjugated by* `pullbackComp` $h\,f$*:*

$$\delta^{h\circ f}_{\mathrm{sheaf}}(M,N) \;=\; (\texttt{pullbackComp}\,h\,f).\mathrm{inv}_{M\otimes_X N} \;;\; h^*(\delta^{f}_{\mathrm{sheaf}}(M,N)) \;;\; \delta^{h}_{\mathrm{sheaf}}(f^*M,\, f^*N) \;;\; \texttt{tensorObjIsoOfIso}((\texttt{pullbac}$$

an equality of morphisms $(h \circ f)^*(M \otimes_X N) \to (h^* f^* M) \otimes_Z (h^* f^* N)$ in Scheme.`Modules` Z*, where* `tensorObjIsoOfIso` *is the functoriality of $\otimes$ in a pair of isomorphisms.*

Remark (the open-immersion base-change square). *The base-change form consumed downstream by 696 is the* specialisation *of this coherence. Given $f : Y \to X$, an open $U \subseteq X$ with preimage $f^{-1}U$, the restriction $g : f^{-1}U \to U$ and the open immersions $j : U \hookrightarrow X$, $j' : f^{-1}U \hookrightarrow Y$, the morphism $f^{-1}U \to X$ admits the two equal factorisations $j' \circ f = g \circ j$ (i.e. $j' ; f = g ; j$). Applying the displayed coherence to each factorisation and equating the two results identifies the restriction $(j')^*(\delta^f_{\mathrm{sheaf}}(M,N))$ with $\delta^g_{\mathrm{sheaf}}(j^*M, j^*N)$, through the pseudofunctoriality isomorphisms; in particular each is an isomorphism if and only if the other is. This is the form used in the chart-chase, where M, N are trivialised on U.*

Proof. Unfolding δ_{sheaf} = `pullbackTensorMap` (667) on both sides exposes each as the same four-fold composite

$$S_1 \;\;;\;\; a.\mathrm{map}\,\delta \;\;;\;\; S_3 \;\;;\;\; S_4,$$

where S_1 is the sheafification–pullback comparison `sheafificationCompPullback`, a.map δ is the sheafification of the presheaf-level oplax comparison δ of 666, S_3 is the monoidal-unit sheafification isomorphism `sheafifyTensorUnitIso`, and S_4 is the tensor of the connecting isomorphisms `pullbackValIso`. The identity to prove is therefore a paste of *four commuting squares*, conjugated throughout by the pullback pseudofunctoriality isomorphism `pullbackComp` $h\,f$. Unlike the *naturality* paste of 672 (D1′), this is a *composition*-coherence paste: the left-hand side is the composite-morphism $(h \circ f)$ instance, while the right-hand side interleaves the f-instance (transported by h^*) with the h-instance (on the pulled-back modules f^*M, f^*N).

We record *why* the unit-comparison mirror does not apply. The unit coherence 670 succeeds because `pullbackObjUnitToUnit` is, by definition, an adjunction transpose, so its composition coherence is obtained by transposing across the pullback–pushforward adjunction. The tensorator `pullbackTensorMap` is *not* a transpose — it is the four-fold composite above — and there is no analogous `homEquiv`-bridge, so the transpose-and-inject opening of the unit analog leaves an un-evaluable transpose of a concrete composite. The proof instead pastes the four squares directly.

Sq2 (the δ core). The presheaf-level oplax comparison δ of the *composite* pullback decomposes by the Mathlib oplax-monoidal coherence `Functor.OplaxMonoidal.comp_`δ, once the composite presheaf pullback `pullback` $\varphi'_{h\circ f}$ is identified with the composite functor `pullback` $\varphi'_f \circ$ `pullback` φ'_h through the Mathlib presheaf coherence `PresheafOfModules.pullbackComp`. This identification rests on the ring-map reconciliation

$$(\texttt{toRingCatSheafHom}\,(h \circ f)).\mathrm{hom} \;=\; \varphi'_f \;;\; (\texttt{Opens.map}\,f.\mathrm{base})^{\mathrm{op}}.\texttt{whiskerLeft}\,\varphi'_h,$$

which is *definitional*: although the two sides are presented in functor categories that nominally differ by the equality `Opens.map` (h.base $\circ$ f.base) = `Opens.map` f.base $\circ$ `Opens.map` h.base (`Opens.map_comp`), this equality and the underlying scheme-composition data already hold up to definitional equality at default transparency, so the reconciliation is definitional (`toRingCatSheafHom_comp_hom_rec`

The genuine content of Sq2 is therefore not the reconciliation but the monoidality of `pullbackComp` itself, isolated as Sq2b below.

Sq2b (monoidality of `pullbackComp` *— the genuinely new ingredient).* The decomposition in Sq2 reduces the δ-core to a single structural fact, absent from Mathlib: the connecting isomorphism

$$\texttt{pullbackComp}\,\varphi'_f\,\varphi'_h \;:\; \texttt{pullback}\,\varphi'_f \circ \texttt{pullback}\,\varphi'_h \;\cong\; \texttt{pullback}\,(\varphi'_f \,;\, (\texttt{Opens.map}\,f.\mathrm{base})^{\mathrm{op}}.\texttt{whiskerLeft}\,\varphi'_h)$$

is a *monoidal* natural isomorphism between the oplax-monoidal functors $(\texttt{pullback}\,\varphi'_f{\circ}\texttt{pullback}\,\varphi'_h,\ \texttt{comp\_}\delta)$ and $(\texttt{pullback}\,(\varphi'_f\,;\cdots),\ \delta)$. Writing δ for the oplax tensorator of the single composite pullback and $\texttt{comp\_}\delta = (\texttt{pullback}\,\varphi'_h).\mathrm{map}\,\delta^{\varphi'_f}\,;\delta^{\varphi'_h}$ for the composite tensorator of the two-step pullback, monoidality is the identity

$$\delta_{M,N} \;=\; (\texttt{pullbackComp}\,\varphi'_f\,\varphi'_h).\mathrm{inv}_{M\otimes N}\,;\,(\texttt{comp\_}\delta)_{M,N}\,;\,((\texttt{pullbackComp}\,\varphi'_f\,\varphi'_h).\mathrm{hom}_M \otimes (\texttt{pullbackComp}\,\varphi'_f\,\varphi'_h).\mathrm{hom}$$

This is precisely the δ-twin of the unit coherence 670, and it is proved by the *same* mate-calculus argument with the unit comparison η replaced by the tensorator δ. The pivot is that, for the canonical oplax structure on a left adjoint (`leftAdjointOplaxMonoidal`, here 666), δ is *itself* an adjunction transpose: $\delta_{M,N} = \texttt{homEquiv}^{-1}((\eta_M \otimes \eta_N)\,;\,\mu_{M,N})$, where μ is the lax tensorator of the `pushforward`. Consequently the transpose-and-inject opening that closes the unit coherence ports verbatim: applying `homEquiv`.injective for the pullback–pushforward adjunction and transporting `pullbackComp` to `pushforwardComp` across the adjunctions via `conjugateEquiv_pullbackComp_inv` (with `unit_conjugateEquiv` and `Adjunction.comp_unit_app`) reduces the goal to the lax-μ composition coherence of `PresheafOfModules.pushforward` across `pushforwardComp`. This last residual — isolated as the lemma $\texttt{pushforwardComp\_lax\_}\mu$ — is the *right-adjoint twin* of the δ-coherence: it is precisely the assertion that the pseudofunctoriality isomorphism `pushforwardComp` is *monoidal* for the lax tensorators μ, i.e. that comparing the lax-μ structure of the single composite `pushforward` with the composite of the two individual lax-μ structures agrees across `pushforwardComp`. Two points must be stated precisely.

First, the mate-calculus reduction proceeds as follows: the transpose-and-inject opening, the $\delta = \texttt{homEquiv}^{-1}((\eta\otimes\eta)\,;\,\mu)$ pivot, the `conjugateEquiv_pullbackComp_inv` transport across the adjunctions (with `unit_conjugateEquiv` and `Adjunction.comp_unit_app`), and the two-argument $\texttt{tensorHom}/\delta_{\mathrm{natural}}$ bookkeeping — which follows the template of Mathlib's `CategoryTheory.Adjunction.isMonoidal_comp` (the canonical "a composite of monoidal adjunctions is monoidal") — together reduce Sq2b to the single residual $\texttt{pushforwardComp\_lax\_}\mu$. This is the content of 670's δ-mirror.

Second, this residual $\texttt{pushforwardComp\_lax\_}\mu$ is discharged by a short *sectionwise pure-tensor collapse*, not by a change-of-rings calculation. On the strongly-monoidal pushforward objects the pushforward tensorator μ reduces *definitionally* to the lighter `restrictScalars` tensorator μ (the identity $\texttt{pushforward\_}\mu\texttt{\_eq}$, which holds by reflexivity); after this reduction, evaluated on a pure tensor $m\otimes n$ every `restrictScalars` tensorator is the identity (Mathlib's $\texttt{ModuleCat.restrictScalars\_}\mu\texttt{\_tmul}$), so both sides of the interchange collapse to the same pure tensor $m\otimes n$; no change-of-rings construction is required. This closes the Sq2b obligation.

Crucially, this sub-step is stated at the `PresheafOfModules` level — about $\texttt{pullback}\,\varphi'$ with φ' the ring map $(\mathcal{S}_0\,;\,\texttt{forget}_2) \to f^{\mathrm{op}}\,;\,(\mathcal{R}_0\,;\,\texttt{forget}_2)$ carried by `pullbackTensorMap` — rather than at the scheme level, where the composite factorisation is immediate from `pullbackComp` and the ring-map reconciliation is definitional.

The observation above — that `pullbackTensorMap` is not an adjunction transpose — concerns the *full* four-fold composite and does not constrain Sq2b: the inner tensorator δ genuinely *is* a transpose ($\texttt{homEquiv}^{-1}$ of $(\eta \otimes \eta)\,;\,\mu$), so for Sq2b specifically the $\eta \to \delta$ mirror of 670 is the correct route.

Sq1 (sheafification $\leftrightarrow$ pullback) — the sole open ingredient. The composition coherence of `SheafOfModules.sheafificationCompPullback` across $h \circ f$ — the analog of `pullbackComp` for the "sheafify-then-pullback" natural isomorphism — is *absent from Mathlib* and is supplied as a named, private project sub-lemma `sheafificationCompPullback_comp`. For a presheaf P over X, writing a_X, a_Z for the sheafifications and `sheafCompPb` = `SheafOfModules.sheafificationCompPullback`, it asserts

$$((\texttt{sheafCompPb}\,(h \circ f)).\mathrm{app}\,P).\mathrm{hom} \;=\; (\texttt{pullbackComp}\,h\,f).\mathrm{inv}_{a_X P}\,;\,h^*(((\texttt{sheafCompPb}\,f).\mathrm{app}\,P).\mathrm{hom})\,;\,((\texttt{sheafCompPb}$$

It is proved by the `leftAdjointUniq` mate calculus: both `sheafCompPb` isomorphisms are `leftAdjointUniq` of composite adjunctions, so transposing the identity under the composite adjunction for $h \circ f$ and evaluating the left-hand side by the mate identity `homEquiv_leftAdjointUniq_hom_app` reduces it to a unit-level identity; the two `pullbackComp` factors are then transported across the adjunction units — the δ-free twin of 670.

The reduced tail goal. After peeling the leading pseudofunctoriality factor $(\texttt{pullbackComp}\,h\,f).\mathrm{inv}$, the remaining obligation is a unit identity for the two-layer composite adjunction (presheaf pullback–pushforward adjunction composed with the sheafification adjunction), isolated as the named sub-lemma `sheafificationCompPullback_comp_tail`. Writing B_φ for the sheaf-level composite adjunction of 679 and $\texttt{PresheafPullback}_f$ for the presheaf-level pullback along f, the assembly proceeds in the following ordered steps, mirroring the already-closed 670 (whose proof is the δ-free analog of this one, exactly one sheafification layer down):

(a) *Strip the outer identity wrapper.* The tail goal carries an outer $\texttt{restrictScalars}(\mathbf{1})$ change-of-rings wrapper coming from the identity ring map; this restriction-of-scalars-along-identity factor is removed first, so that the genuine comparison factors are exposed.

(b) *Distribute the right-hand sheaf composite under* `forget`*.* The right-hand side is the sheaf-level composite carrying the two sub-comparison factors R_1 (the f-layer) and R_5 (the h-layer). Pushing the forgetful functor `forget` (sheaf $\to$ presheaf) through the composite separates these two factors *without* disturbing the left-hand composite unit $B_{h \circ f}.\mathrm{unit}$, which is left intact for the final reassembly.

(c) *Recover the sub-comparison units.* Apply 679 (`leftAdjointUniqUnitEta_app`, the P-general step-1 brick) to rewrite R_1 as the composite-adjunction unit $B_f.\mathrm{unit.app}\,P$ and R_5 as $B_h.\mathrm{unit.app}\,(\texttt{PresheafPullback}_f\,P)$. This is precisely the uniqueness-of-left-adjoints characterisation `homEquiv_leftAdjointUniq_hom_app`, now used in its P-general form.

(d) *Slide the* `pushforwardComp` *comparison past the recovered units.* The sheaf-level comparison $\texttt{pushforwardComp}\,h\,f$ is moved across the two recovered units $B_f.\mathrm{unit}$ and $B_h.\mathrm{unit}$ by its naturality.

(e) *Reassemble the composite unit.* Expanding $B_{h \circ f}.\mathrm{unit}$ by the composite-adjunction unit formula `comp_unit_app` and applying `unit_naturality` identifies the slid expression of step (d) with the left-hand composite unit, closing the tail.

The precise binding obligation. The one place where the assembly is not yet a mechanical paste is the bridge between the *presheaf*-level left-hand data (the presheaf pushforward $\texttt{pushforward}^{\mathrm{pre}}$ and the composite unit $B_{h \circ f}.\mathrm{unit}$) and the *sheaf*-level right-hand factors R_1, R_5 wrapped in `forget`. Crossing this boundary requires the compatibility

$$\texttt{forget} \circ \texttt{pushforward}^{\mathrm{sheaf}} \;=\; \texttt{pushforward}^{\mathrm{pre}} \circ \texttt{forget},$$

interacting with the recovered B_f- and B_h-units of step (c). This compatibility is the natural unit to isolate as its own named sub-lemma *before* the assembly: once it is in hand, steps (a)–(e) become a mechanical paste, whereas without it the presheaf- and sheaf-level pushforwards cannot be aligned at the recovered units. The remainder of the tail is bookkeeping around this single compatibility.

This is the sheafification-laden analog of 670; unlike that lemma it is not definitional sectionwise, because the adjunctions involved are composite rather than single transposes. The outer Sq1 lemma `sheafificationCompPullback_comp` *consumes* this tail: it peels the leading (`pullbackComp` $h\,f$).inv factor and then invokes `sheafificationCompPullback_comp_tail` for the residual unit identity. *Warning:* transposing the *whole* tail back under the composite adjunction — as opposed to recovering the two sub-units individually via step (c) and reassembling — is *circular* (verified): it returns the very identity one set out to prove, so the step-(c) recovery of the individual R_1, R_5 units is essential.

Sq3 (unit-iso transport). The monoidal-unit sheafification isomorphism `sheafifyTensorUnitIso` is carried through the same `pullbackComp` identification used for Sq2. This composition coherence is isolated as the named brick 681 (`sheafifyTensorUnitIso_comp`): via 877 the `hom` leg is a single a.map of $\eta \otimes \eta$, so the coherence reduces to the naturality of the sheafification unit η against `PrPbComp`, recombined by bifunctoriality.

Sq4 (the connecting iso). The scheme-level composition coherence of `pullbackValIso` is isolated as the named brick 682 (`pullbackValIso_comp`): it relates `pullbackValIso` $(h \circ f)\,M$ to h^*(`pullbackValIso` $f\,M$), `pullbackValIso` $h\,(f^*M)$, and the pseudofunctoriality component (`pullbackComp` $h\,f)_M$, and produces the final `tensorObjIsoOfIso`((`pullbackComp` $h\,f)_M$, (`pullbackComp` $h\,f)_N$) factor. It is *absent from Mathlib*; but `pullbackValIso` factors through `sheafCompPb` (as `sheafCompPb`$^{-1}$ followed by the pullback of a counit), so by 682 its composition coherence reduces to that of Sq1 (884, now closed) together with the pseudofunctoriality of the counit — i.e. Sq4 is a short corollary once Sq1 is in hand. It is the naturality across the connecting-object boundary where the abstract pullback's `.val` presentation meets the helper-`.obj` presentation. The prover treats it as its own brick (its own statement and connecting-object bookkeeping), not as a step inlined in the proof of Sq1.

State of the paste and the residual chain. Sq1 (`sheafificationCompPullback_comp`) and Sq2b (`pullbackComp_`δ with `pushforwardComp_lax_`μ) are *closed*, and the Sq2 ring-map reconciliation is definitional. In the Lean paste these are consumed by the committed splice of the equation `h`δ (the Sq2b decomposition of a_Z.map $\delta_{h\circ f}$ into a_Z.map(`PrPbComp`.inv) ; a_Z.map δ_{comp} ; a_Z.map `tcomp`, via 881 under `congrArg` a_Z.map) by `erw`. After that splice the residual is a three-brick chain:

(i) *The hom–inv cancellation* 680 (`sheafifyMap_pullbackComp_hom_inv_id`): the adjacent pair a_Z.map(`PrPbComp`.hom.app $(M \otimes N)$) ; a_Z.map(`PrPbComp`.inv.app $(M \otimes N)$) created by the `h`δ-splice collapses to the identity. The two cancelling factors are adjacent in the composite only after a reassociation; the intervening bracket joins through a `SheafOfModules` instance that is definitionally but not syntactically equal to the one in the ambient goal, so the cancellation is carried out via a private generic middle-cancellation lemma over an abstract category, invoked by definitional-equality unification. The helper carries no separate blueprint pin.

(ii) *The* δ_{comp} *split* via `Functor.OplaxMonoidal.comp_`δ (mechanical): δ_{comp} becomes (`pullback` φ'_h).map($\delta_f\, M$.val
$\delta_h\,(F^{\mathrm{p}}_f M.\mathrm{val})\,(F^{\mathrm{p}}_f N.\mathrm{val})$.

(iii) *Sq3 681 and Sq4 682*, discharged via the interleaved four-square merge below.

The adjunction-mate bridge `conjugateEquiv_pullbackComp_inv` (relating `pullbackComp` for the left adjoints to `pushforwardComp` for the right adjoints) is the device used inside Sq2b and Sq4 to transport the pseudofunctoriality coherence across the adjunctions.

The final assembly (the interleaved four-square merge). The four squares do *not* paste row-by-row, and step (iii) is *not* the D1′ paste applied verbatim. In D1′ the four ingredients are honest natural transformations whose squares close independently; here two distinct obstructions — a misaligned S_1^h argument and the `pullbackValIso` bridge between Sq3 and Sq4 — force the last three factors into a single combined chase.

The slide of S_1^h. After the δ_{comp} split (ii), the h-tensorator δ_h acts on the *sheaf*-pullback arguments $((\texttt{pullback}\, f).\mathrm{obj}\, M)\otimes((\texttt{pullback}\, f).\mathrm{obj}\, N)$, whereas S_1^h — the h-instance of `sheafificationCompPullback` (Sq1, 884, closed) — must be made to act instead on the *presheaf*-pullback arguments $F_f^{\mathrm{p}} M.\mathrm{val}\otimes F_f^{\mathrm{p}} N.\mathrm{val}$. This re-alignment is a naturality step in its own right, *not* the D1′ slide of δ past $F(a \otimes b)$: the connecting comparison `sheafificationCompPullback` h is a natural transformation, and its naturality square at the oplax-tensorator morphism

$$\delta_f(M, N) \;:\; F_f^{\mathrm{p}}(M.\mathrm{val} \otimes N.\mathrm{val}) \;\longrightarrow\; F_f^{\mathrm{p}} M.\mathrm{val} \otimes F_f^{\mathrm{p}} N.\mathrm{val}$$

slides S_1^h from the sheaf-pullback arguments to the presheaf-pullback arguments. It is this bespoke square — the naturality of the *connecting* transformation at δ_f, not the oplax naturality of a comparison — that effects the slide.

The merged Sq3/Sq4 chase. The two remaining bricks — Sq3 681 (the composition coherence of `sheafifyTensorUnitIso`) and Sq4 682 (the composition coherence of `pullbackValIso`) — do *not* discharge separately. Their arguments are bridged by the connecting isomorphism `pullbackValIso` f (878): one side presents the data through the sheaf-pullback $((\texttt{pullback}\, f).\mathrm{obj}\, M).\mathrm{val}$, the other through the presheaf-pullback $F_f^{\mathrm{p}} M.\mathrm{val}$, so a paste of Sq3 followed by Sq4 never lines up. They must be discharged *together*, as one chase, exactly as in the merge of 672 (D1′): there the two functor-image factors are folded into a single image by functoriality $F(g);F(h) = F(g;h)$, reducing the sheaf-level equality to a presheaf-level one. We reproduce that structure here: the composite $a.\mathrm{map}\,\delta;S_3;S_4$ on the left, and the matching tail on the right, are each folded into a single a.map of one presheaf morphism, collapsing the sheaf-level equation to a single presheaf-level identity between the two folded presheaf morphisms.

Closing the presheaf identity. That presheaf identity is closed by three facts already in hand: the naturality of the oplax tensorator δ (666); the single-`tensorHom` presentation `sheafifyTensorUnitIso`.hom $= a.\mathrm{map}(\eta\otimes\eta)$ of 877, which keeps every factor inside one monoidal instance; and the `pullbackValIso` factorisation of 878 (the inverse of `sheafificationCompPullback` followed by the pulled-back sheafification counit), which rewrites the connecting isomorphism into the same presheaf-level form on both sides. With the slide supplying the δ_h alignment and the closed Sq2b decomposition supplying the δ-core, the four squares assemble into the displayed identity. The base-change-square consequence (the Remark) is then the specialisation $h := j'$ obtained by equating the two factorisations $j' \,;\, f = g \,;\, j$; the iff-clause is immediate, an isomorphism being preserved and reflected by the pseudofunctoriality isomorphism it is conjugated by. □

16.8.2 The K1 monoidal-mate bridge on the multiplication side

The open-immersion comparison lemma K1 (695) reduces, by the same two-ingredient model as 646, to the statement that the canonical left-adjoint-uniqueness isomorphism $H_1 =$ `leftAdjointUniq` $: G_\beta \xrightarrow{\sim}$ `pullback` φ' is a *monoidal* natural isomorphism. Here $f : Y \to X$ is the open immersion, $\varphi' = f.$`toRingCatSheafHom`.hom is its presheaf-level structure map, β is the sectionwise-bijective structure-ring map of the restriction (sectionwise $f.$`appIso`$^{-1}$), and $G_\beta =$ $\texttt{pushforward}_0^{\mathrm{Comm}}$ $f.$`opensFunctor` $X.$preshe

$\texttt{restrictScalars}\,\beta'$ is the resulting *strong*-monoidal restriction functor (a left adjoint of $\texttt{pushforward}\,\varphi'$, so H_1 is defined). That H_1 is monoidal is two compatibilities — one on the unit comparison, one on the tensorator — and each is an open-immersion pure-tensor collapse of the bijective structure-ring map $f.\texttt{appIso}$. These are the μ/δ-side twins of the already-proved η/ε-side template 686, and the open-immersion analogues of the D3′ composition coherence 893; they use the same D3′ pure-tensor section calculus. We isolate them as the next two lemmas; granting them, K1's monoidal-mate obligation is discharged by the standard mate transport across H_1.

Lemma 686 (Presheaf-side mate identity for the η-bridge)**.** *For a scheme morphism $f : Y \to X$ with $\varphi' = f.\texttt{toRingCatSheafHom}.\mathrm{hom}$, the presheaf-of-modules adjunction unit at the monoidal unit, post-composed with the pushforward of the oplax unit comparison $\eta(\texttt{pullback}\,\varphi')$, recovers the lax unit comparison $\varepsilon(\texttt{pushforward}\,\varphi')$:*

$$(\texttt{pullbackPushforwardAdjunction}\,\varphi').\mathrm{unit.app}\,\mathbf{1}^{\mathrm{p}}\ ;\ (\texttt{pushforward}\,\varphi').\mathrm{map}(\eta(\texttt{pullback}\,\varphi'))\ =\ \varepsilon(\texttt{pushforward}\,\varphi').$$

This is the presheaf-level driver of the D2′ η-bridge, and the template the K1 μ-side bridge mirrors.

Proof. This is the mate identity $\texttt{Adjunction.unit\_app\_unit\_comp\_map\_}\eta$ for the adjunction $\texttt{pullbackPushforwardAdjunction}\,\varphi'$, whose $\texttt{Adjunction.IsMonoidal}$ structure is supplied by $\texttt{presheafPushforwardLaxMonoidal}$ (665) and $\texttt{presheafPullbackOplaxMonoidal}$ (666). ☐

Lemma 687 (K1 η-side collapse: H_1 respects the unit comparison)**.** *In the K1 setting above, the strong unit comparison $\eta\,G_\beta$ of the restriction functor agrees, under H_1, with the oplax unit comparison $\eta(\texttt{pullback}\,\varphi')$:*

$$\eta\,G_\beta\ =\ H_1.\mathrm{hom.app}\,\mathbf{1}^{\mathrm{p}}\ ;\ \eta(\texttt{pullback}\,\varphi').$$

Equivalently, H_1 carries the monoidal unit of G_β to that of $\texttt{pullback}\,\varphi'$.

Proof. This is the η/ε-side of "H_1 is a monoidal natural isomorphism", the open-immersion analogue of the D2′ template 686. It is verified sectionwise on an open U. On the unit module the strong unit comparison $\eta\,G_\beta$ of $\texttt{restrictScalars}\,\beta'$ is, sectionwise, the structure-ring identity of the *bijective* $f.\texttt{appIso}$ (the strong-monoidal upgrade of base change along a ring isomorphism, as in 646); the mate $\eta(\texttt{pullback}\,\varphi')$ is the same base-change value transported across H_1, so the two sides coincide. The collapse of the adjunction-unit composite to a sectionwise structure-ring map uses the presheaf-level value 688, after which both sides reduce to the ring-unit identity $1 \mapsto 1$. ☐

Lemma 688 (Presheaf-level value of the pushforward–pushforward adjunction unit)**.** *For the pushforward–pushforward adjunction $F \dashv G$ assembled from a morphism of presheaves of rings (the adjunction $\texttt{pushforwardPushforwardAdj}\,adj\,\varphi\,\psi\,H_1\,H_2$), the unit's sectionwise carrier action is precomposition by the pullback of the adjunction counit: for a module M, an open U and a section x,*

$$(((\texttt{pushforwardPushforwardAdj}\,adj\,\varphi\,\psi\,H_1\,H_2).\mathrm{unit.app}\,M).\mathrm{app}\,U).\mathrm{hom}\,x\ =\ (M.\mathrm{map}\,(adj.\mathrm{counit.app}\,U^{\mathrm{op}})^{\mathrm{op}}).\mathrm{hom}$$

Proof. Definitional. The constituent isomorphisms $\texttt{pushforwardId}$ and $\texttt{pushforwardComp}$ are $\texttt{Iso.refl}$, so the unit composite collapses to the $\texttt{pushforwardNatTrans}$ of the adjunction counit, whose carrier action is exactly the displayed precomposition; the equation holds by $\texttt{rfl}$ on the element-applied form. ☐

Lemma 689 (Presheaf-level value of the pushforward–pushforward adjunction counit)**.** *The exact dual of 688. For the same pushforward–pushforward adjunction $F \dashv G$ (`pushforwardPushforwardAdj` $adj\,\varphi\,\psi\,H_1\,H$ the* counit*'s sectionwise carrier action is precomposition by the pullback of the adjunction* unit*: for a module N, an open U and a section y,*

$$(((\texttt{pushforwardPushforwardAdj}\; adj\,\varphi\,\psi\,H_1\,H_2).\mathrm{counit}.\mathrm{app}\,N).\mathrm{app}\,U).\mathrm{hom}\,y \;=\; (N.\mathrm{map}\,(adj.\mathrm{unit}.\mathrm{app}\,U^{\mathrm{op}})^{\mathrm{op}}).\mathrm{hom}\,y$$

In the K1 setting this is the structure-ring action of the bijective f.`appIso` *on each tensor factor, which is what discharges the counit pair leg of the LHS mate telescope.*

Proof. Definitional, exactly as the unit dual. The constituent isomorphisms `pushforwardId` and `pushforwardComp` are `Iso.refl`, so the counit composite collapses to the `pushforwardNatTrans` of the adjunction unit, whose carrier action is the displayed precomposition; the equation holds by `rfl` on the element-applied form. □

Lemma 690 (Unit-preservation of the strong-monoidal `restrictScalars` oplax unit)**.** *Let $\alpha : R \to S$ be a morphism of presheaves of commutative rings whose every sectionwise carrier α_W is* bijective*, so that* `restrictScalars` α *is* strong *monoidal. Then its oplax-monoidal unit comparison η(*`restrictScalars` α*) preserves the section ring unit: for every object W,*

$$(\eta(\texttt{restrictScalars}\,\alpha)_W)(1_{S(W)}) = 1_{R(W)}.$$

Proof. Since α is sectionwise bijective, `restrictScalars` α carries a *strong* monoidal structure, in which the oplax unit η is the two-sided inverse of the lax unit ε. The lax unit sends $1 \mapsto 1$: on sections it is the change-of-rings unit, namely the ring map α_W applied to the unit, and $\alpha_W(1) = 1$. Composing ε with its inverse η yields the identity ($\varepsilon\,;\eta = \mathbf{1}$), so η too sends the unit to the unit. □

Lemma 691 (Sectionwise comparison of the LHS mate tensorator on a pure tensor)**.** *In the K1 setting the adjoint-transported tensorator μ(*`rightAdjointLaxMonoidal hadj′`*$)_W$ on the objects $G_\beta A$, $G_\beta B$ arises as an adjunction* mate*, whereas the directly-defined composition tensorator μ(*`presheafPushforwardLaxMonoidal` φ'*$)_W$ is given outright. This lemma is the per-section comparison of the two: on every open W, evaluated at a pure tensor $m \otimes_t n$, the mate tensorator agrees with the directly-defined tensorator. Concretely, expanding the mate to its explicit adjunction form*

$$\texttt{hadj}'.\mathrm{unit} \;;\; (\texttt{pushforward}\,\varphi').\mathrm{map}(\delta\,G_\beta\,A\,B \;;\; (\texttt{hadj}'.\mathrm{counit} \otimes_{\mathrm{m}} \texttt{hadj}'.\mathrm{counit})),$$

the comparison reduces to computing its value on $m \otimes_t n$; that value is the base-change value $m \otimes_t n$, matching the directly-defined tensorator.

Proof. The value lemma 688 characterises the adjunction unit in the canonical `pushforwardPushforwardAdjunction` form. Since the mate is expressed via the adjunction `hadj′`, we identify `hadj′` with `pushforwardPushforwardAdjuncti` and then evaluate the three legs of the mate sectionwise at W on the pure tensor $m \otimes_t n$. The adjunction unit acts by precomposition with the pullback of the counit, M.map (`hadj′`.counit$)^{\mathrm{op}}$, by 688. Reindexing the (`pushforward` φ').map leg to the section over $F^{\mathrm{op}}W$, the strong-monoidal oplax tensorator $\delta\,G_\beta$ of the composite $\texttt{pushforward}_0^{\mathrm{Comm}} \cdot \texttt{restrictScalars}\,\beta'$ is, on a pure tensor, the identity: its `restrictScalars` β' factor is the base-change tensorator, which is the identity on pure tensors by 887, 889 and 888, while its $\texttt{pushforward}_0^{\mathrm{Comm}}$ factor has identity tensorator. Finally the pair of counits `hadj′`.counit $\otimes_{\mathrm{m}}$ `hadj′`.counit acts as the structure-ring identity of the *bijective* f.`appIso` on each factor. The three legs compose to the identity, so the value is $m \otimes_t n$.

We spell out the three closing steps on the pure tensor explicitly. *Step 3 (the δ leg).* After reindexing the inner ε-leg, the comonoidal δ of the composite functor $G_\beta = \texttt{pushforward}_0^{\mathrm{Comm}} \cdot \texttt{restrictScalars}\,\beta'$ splits via `Functor.OplaxMonoidal.comp`: the $\texttt{pushforward}_0^{\mathrm{Comm}}$ factor's δ is the identity on a pure tensor, leaving only the $\texttt{restrictScalars}\,\beta'$ factor's δ, which is discharged on the pure tensor by 890. *Step 4 (the counit pair).* Split the $\texttt{tensorHom}\ \texttt{hadj}'.\mathrm{counit} \otimes_{\mathrm{m}} \texttt{hadj}'.\mathrm{counit}$ over the pure tensor into its two factors and collapse each counit leg by 689. *Closing (triangle cancellation).* The unit-restriction leg produced by the inner-unit discharge and the counit-restriction leg of Step 4 are mutually inverse on the image of the open immersion — this is the adjunction triangle identity read on opens — so the two legs cancel. What remains is $m \otimes_t n$, which matches the right-hand value by 692. □

Lemma 692 (Sectionwise value of the RHS composition tensorator on a pure tensor)**.** *In the K1 setting, the directly-defined composition tensorator* $\mu(\texttt{presheafPushforwardLaxMonoidal}\,\varphi')$ *on* $G_\beta A$, $G_\beta B$, *evaluated on an open* W *at a pure tensor* $m \otimes_t n$, *is the base-change value* $m \otimes_t n$.

Proof. By 892 the pushforward composition tensorator reduces to the $\texttt{restrictScalars}\,\varphi'$ tensorator on the $\texttt{pushforward}_0^{\mathrm{Comm}}$ reindexed objects. Exposing the `ModuleCat` tensorator by 887 and collapsing on the pure tensor by 889 and 888, with the outer $(\texttt{pushforward}\,\varphi').\mathrm{map}$ leg collapsed by 891, the value is $m \otimes_t n$. This reuses the same helper family as the composition coherence of the single pushforward. □

Lemma 693 (Bare tensorator comparison of the two lax structures on $\texttt{pushforward}\,\varphi'$)**.** *In the K1 setting above, the functor* $\texttt{pushforward}\,\varphi'$ *carries* two *lax monoidal structures: the adjoint-transported one* $\texttt{rightAdjointLaxMonoidal}\,(\texttt{hadj}')$, *induced from the strong-monoidal left adjoint* G_β *through the adjunction* $\texttt{hadj}' : G_\beta \dashv \texttt{pushforward}\,\varphi'$, *and the directly-defined* `presheafPushforwardLaxMonoidal` *(665). Their tensorators agree on the objects* $G_\beta A$, $G_\beta B$:

$$\mu(\texttt{rightAdjointLaxMonoidal}\,\texttt{hadj}')\,(G_\beta A)\,(G_\beta B) \;=\; \mu(\texttt{presheafPushforwardLaxMonoidal})\,(G_\beta A)\,(G_\beta B).$$

This is a bare *tensorator identity between two* different kinds *of lax structures on the same functor, with* no *appeal to any monoidal-adjunction structure on* $\texttt{hadj}'$*: that structure is precisely the monoidal adjunction being established, which in turn depends on this identity.*

Proof. Crucially the two sides are *not* symmetric. The left-hand tensorator $\mu(\texttt{rightAdjointLaxMonoidal}\,\texttt{hadj}')$ is an adjunction *mate* of the strong oplax tensorator of G_β, whereas the right-hand $\mu(\texttt{presheafPushforwardLaxMonoid$ is a *composition* structure. They carry no common normal form, and must be brought to a common value by two distinct routes. No appeal is made to a monoidal structure on the adjunction $\texttt{hadj}'$; such an appeal would be circular, since that structure is precisely the monoidal adjunction instance being established, which in turn consumes this identity.

We first reduce the right-hand side at the level of *morphisms*, replacing $\mu(\texttt{presheafPushforwardLaxMonoidal}\,\varphi')$ by the $\texttt{restrictScalars}\,\varphi'$ tensorator via 892 at the *morphism level*, before descending to sections — 892 is an identity of presheaf morphisms and must be applied there, prior to section evaluation. We then argue sectionwise: on an open W the domain $G_\beta.\mathrm{obj}\,(A \otimes B)$ is, sectionwise, an honest `ModuleCat` tensor, so pure-tensor extensionality reduces the goal to checking equality on each pure tensor $m \otimes_t n$. By 691 the left-hand mate evaluates to $m \otimes_t n$, and by 692 the right-hand composition tensorator evaluates to the same $m \otimes_t n$. The two sides therefore agree on every pure tensor, hence as presheaf morphisms. □

Lemma 694 (K1 μ/δ-side collapse: H_1 respects the tensorator)**.** *In the K1 setting above, for all presheaves of modules* A, B *the strong tensorator* $\delta\, G_\beta\, A\, B$ *of the restriction functor agrees with the* H_1*-conjugate of the oplax tensorator* $\delta(\texttt{pullback}\,\varphi')\,A\,B$:

$$\delta\, G_\beta\, A\, B \;=\; H_1.\mathrm{hom.app}\,(A \otimes B) \;;\; \delta(\texttt{pullback}\,\varphi')\,A\,B \;;\; (H_1.\mathrm{inv.app}\,A \otimes_{\mathrm{m}} H_1.\mathrm{inv.app}\,B).$$

Equivalently, H_1 is monoidal on tensorators. This is the genuine geometric content of K1 (the μ/δ-side twin of 686, and the open-immersion analogue of 893).

Proof. The genuine geometric content is the bare tensorator comparison 693; this proof is the formal reduction to it, and uses *no* mate machinery (`hadj'.IsMonoidal`), which would be circular. Four steps: (1) cancel the isomorphism H_1 = `leftAdjointUniq`, restating the goal on the `pullback` φ' domain; (2) transpose across `hadj'` : $G_\beta \dashv$ `pushforward` φ' by hom-equivalence injectivity, turning the left adjoint's oplax tensorator into the right adjoint's lax tensorator μ(`rightAdjointLaxMonoidal hadj'`); (3) replace $\delta\, G_\beta$ by $\mu^{-1}_{\mathrm{Iso},\beta}$; (4) reduce δ(`pullback` φ') to μ(`presheafPushforwardLaxMonoidal`) via 892. The goal is then exactly 693, which closes it. ☐

Lemma 695 (Pullback commutes with ⊗ along an open immersion (K1))**.** *Let $f : Y \to X$ be an* open immersion *of schemes and let $M, N \in$* Scheme.`Modules` X *be* arbitrary *$\mathcal{O}_X$-modules. Then the sheaf-level comparison map* `pullbackTensorMap` $f\,M\,N$ *of 667,*

$$f^*(M \otimes_X N) \xrightarrow{\ \sim\ } (f^*M) \otimes_Y (f^*N),$$

is an isomorphism. No local-triviality hypothesis on M, N is required: along an open immersion the pullback functor is *the restriction functor, which* is strong *monoidal, so the comparison is invertible for every pair of modules. Source: [Stacks Project], Modules on Ringed Spaces, Lemma* `lemma-tensor-product-pullback`*: pullback commutes with ⊗ functorially. Along an open immersion the pullback is restriction, base change along the structure-sheaf* isomorphism*, so the comparison is an isomorphism for all modules.*

Proof. Source: [Stacks Project], Modules on Ringed Spaces, Lemma `lemma-tensor-product-pullback`*.*

Unfold `pullbackTensorMap` $f\,M\,N$ by its four-fold factorisation (667, made explicit in 668). The three outer factors p_1, p_3, p_4 are isomorphisms unconditionally, so the comparison is an isomorphism as soon as the single middle factor — the sheafification a_Y.map δ of the presheaf-level oplax comparison δ(`pullback` φ')(M.val, N.val) — is an isomorphism, where φ' is the presheaf structure map presenting f. Sheafification is a functor and hence preserves isomorphisms, so it suffices to show the *presheaf*-level comparison δ(`pullback` φ') is an isomorphism.

For an open immersion this presheaf-level statement is exactly the strong-monoidality of pullback = restriction, and we establish it by the same two-ingredient model that closes 646.

H1 (the presheaf identification). Let β be the structure map of the restriction functor along the open immersion f. One exhibits the presheaf-level adjunction `pushforward` $\beta \dashv$ `pushforward` φ' and concludes, by `Adjunction.leftAdjointUniq`, a canonical isomorphism `pushforward` $\beta \cong$ `pullback` φ' between the two left adjoints of `pushforward` φ'. (This is the very same H1 identification that closes 646.)

H2 (the strong-monoidal tensorator). Along an open immersion the structure map β is sectionwise the bijective structure-ring isomorphism f.`appIso`, so `restrictScalars` β, and hence `pushforward` β, is *strong* monoidal — exactly the strong-monoidal upgrade of base change along a ring isomorphism used in 646. In particular `pushforward` β carries an *invertible* tensorator.

The two pure-tensor collapses — the unit compatibility 687 (with template 686) and the tensorator compatibility 694 — confirm that H_1 is a monoidal natural isomorphism. The tensorator compatibility in particular supplies the hypothesis of the abstract δ-conjugation 772, yielding that $\delta\, G_\beta$ is the H_1-conjugate of δ(`pullback` φ'). Since G_β is strong monoidal (**H2**), its oplax tensorator $\delta\, G_\beta$ is the inverse of its strong tensorator $\mu_{\mathrm{Iso}}\, G_\beta$ (771) and is therefore an isomorphism; 773 then concludes that δ(`pullback` φ') is an isomorphism.

Hence δ(`pullback` φ') is an isomorphism, so its sheafification a_Y.map δ is too, and therefore `pullbackTensorMap` $f\,M\,N$ is an isomorphism. ☐

Lemma 696 (Pullback commutes with $\otimes$ on locally-trivial pairs (D4′))**.** *Let $f : Y \to X$ be a morphism of schemes and let $M, N \in$ Scheme.*`Modules` *X be* locally trivial *(609). Then the sheaf-level comparison map $\delta_{\mathrm{sheaf}} =$* `pullbackTensorMap` *of 667 is an isomorphism:*

$$f^*(M \otimes_X N) \xrightarrow{\sim} (f^*M) \otimes_Y (f^*N),$$

natural in M and N. This is the comparison the relative Picard consumer needs; it is established by promoting the already-built map δ_{sheaf} to an isomorphism through a chart-chase, with no comparison for arbitrary modules and no filtered-colimit machinery. Source: [Stacks Project], Modules on Ringed Spaces, Lemmas `lemma-tensor-product-pullback` and `lemma-pullback-locally-free`: pullback commutes with $\otimes$ functorially, and the pullback of a locally free (locally trivial) module is again locally free; on locally-trivial pairs the comparison is therefore an isomorphism, checkable on a trivialising cover.

Proof. *Source: [Stacks Project], Lemmas `lemma-tensor-product-pullback` and `lemma-pullback-locally-free`.*

By local triviality of M and N (609) choose a common affine open cover $\{U_i\}$ of X on which both restrict to the trivial bundle, $M|_{U_i} \cong \mathcal{O}_{U_i}$ and $N|_{U_i} \cong \mathcal{O}_{U_i}$ — the common refinement of the two trivialising covers, exactly as in the proof that the tensor product of locally-trivial modules is locally trivial (648). The preimages $\{f^{-1}U_i\}$ form an open cover of Y. It suffices, by the restriction-detects-isomorphism criterion 740, to show that the canonical map $\delta_{\mathrm{sheaf}}(M, N)$ restricts to an isomorphism on each $f^{-1}U_i$; the canonical global map being shown locally-iso is exactly what lets us conclude (a non-canonical patchwork of local isomorphisms would not glue).

Fix i and write $g_i : f^{-1}U_i \to U_i$ for the restriction of f, with open immersions $j_i : U_i \hookrightarrow X$ and $j_i' : f^{-1}U_i \hookrightarrow Y$, so $f \circ j_i' = j_i \circ g_i$. The base-change coherence 685 expresses the restriction $(j_i')^*\delta^f_{\mathrm{sheaf}}(M, N)$ as the comparison $\delta^{g_i}_{\mathrm{sheaf}}(j_i^*M, j_i^*N)$ for g_i on the restricted modules, *flanked* by the comparisons $\delta^{j_i'}$ and δ^{j_i} along the two open immersions. Isolating the restriction factor δ^{g_i} from this four-fold composite is legitimate only because those flanking factors are *invertible*: by 695 (K1) the comparison along an open immersion is an isomorphism for arbitrary modules, so $\delta^{j_i'}$ and δ^{j_i} may be cancelled. The claim is thereby reduced to: $\delta^{g_i}_{\mathrm{sheaf}}(j_i^*M, j_i^*N)$ is an isomorphism. On U_i the restricted modules are trivial, $j_i^*M \cong \mathcal{O}_{U_i}$ and $j_i^*N \cong \mathcal{O}_{U_i}$; by naturality of δ_{sheaf} in both arguments (672) these isomorphisms transport $\delta^{g_i}_{\mathrm{sheaf}}(j_i^*M, j_i^*N)$ to the comparison on the unit pair $\delta^{g_i}_{\mathrm{sheaf}}(\mathcal{O}_{U_i}, \mathcal{O}_{U_i})$. That comparison is an isomorphism by 675 — the unit-pair case, where the structure-sheaf comparison `pullbackUnitIso` (an isomorphism for all g_i) and the unitors do the work. Hence $\delta^{g_i}_{\mathrm{sheaf}}(j_i^*M, j_i^*N)$ is an isomorphism, so is its base-change identification $(j_i')^*\delta^f_{\mathrm{sheaf}}(M, N)$, and thus $\delta^f_{\mathrm{sheaf}}(M, N)$ restricts to an isomorphism on each $f^{-1}U_i$. (Both sides are indeed locally trivial on this cover — $f^*M, f^*N, f^*(M \otimes_X N)$ are locally trivial by 611 and 648 — which is what makes the chart-local comparison a map between copies of the structure sheaf.) By 740, $\delta_{\mathrm{sheaf}}(M, N)$ is a global isomorphism, natural in M, N by 672. □

Lemma 697 (δ_{sheaf} is an isomorphism on a structure-sheaf pair (D4′ base case))**.** *Let $f : Y \to X$ be a morphism of schemes and let $P, Q \in$ Scheme.*`Modules` *X each be isomorphic to the structure sheaf, $P \cong \mathcal{O}_X$ and $Q \cong \mathcal{O}_X$. Then the comparison* `pullbackTensorMap` *$f\,P\,Q$ of 667 is an isomorphism.*

Proof. By naturality of δ_{sheaf} in both arguments (672), the chosen isomorphisms $P \cong \mathcal{O}_X$ and $Q \cong \mathcal{O}_X$ transport `pullbackTensorMap` $f\,P\,Q$ onto the comparison on the unit pair `pullbackTensorMap` $f\,\mathcal{O}_X\,\mathcal{O}_X$, which is an isomorphism by 675 (D2′). Transport of an isomorphism along isomorphisms is again an isomorphism, so `pullbackTensorMap` $f\,P\,Q$ is one. □

Lemma 698 (Per-chart isomorphism obligation for the D4′ chart-chase)**.** *In the situation of [696](#), fix a member $f^{-1}U_i$ of the trivialising cover, with restriction $g_i : f^{-1}U_i \to U_i$ of f and open immersions $j_i : U_i \hookrightarrow X$, $j'_i : f^{-1}U_i \hookrightarrow Y$. Then the restriction $(j'_i)^*\delta^f_{\mathrm{sheaf}}(M,N)$ of the comparison to that chart is an isomorphism.*

Proof. The base-change coherence 685, applied at the two factorisations $f \circ j'_i = j_i \circ g_i$, presents $(j'_i)^*\delta^f_{\mathrm{sheaf}}(M,N)$ as the comparison δ^{g_i} flanked by the two open-immersion comparisons $\delta^{j'_i}, \delta^{j_i}$; these flanking factors are isomorphisms by 695 (K1, applied twice) and cancel, while δ^{g_i} on the trivialised modules $j_i^*M \cong \mathcal{O}_{U_i}, j_i^*N \cong \mathcal{O}_{U_i}$ is an isomorphism by the structure-sheaf-pair case 697. (The restriction criterion 740 is what makes this chart-local invertibility the relevant per-cover obligation.) Hence $(j'_i)^*\delta^f_{\mathrm{sheaf}}(M,N)$ is an isomorphism. □

16.9 The invertibility-carrier Picard group

The group law of 664 is carried on the *locally-trivial* predicate `IsLocallyTrivial`, and on that carrier the existence of a tensor inverse for each class is an extra obligation (656): one must *manufacture* the dual object $L^{-1} = \mathcal{H}om_{\mathcal{O}_X}(L, \mathcal{O}_X)$ and prove the contraction $L \otimes_X L^{-1} \cong \mathcal{O}_X$, which is the whole content of the dual-infrastructure build of 16.13.

This section carries the group law on a *different* predicate for which that obligation evaporates: the ⊗-invertibility predicate `IsInvertible` of 662,

$$\texttt{IsInvertible}(M) \;:\equiv\; \exists\, N,\ \mathrm{Nonempty}(M \otimes_X N \cong \mathcal{O}_X).$$

On this carrier the inverse of a class $[M]$ is supplied by the *membership witness itself* — the very N whose existence `IsInvertible`(M) asserts — so no inverse object need be constructed and no separate tensor-inverse lemma is consumed. The cost is shifted to a routine well-definedness fact (702: the witness N is determined up to isomorphism), proved from the monoidal coherence isomorphisms alone. The carrier is the scheme-theoretic analogue of Mathlib's `CommRing.Pic`, whose elements are exactly the iso-classes of invertible modules (Stacks tag 01CX).

Lemma 699 (Associator for $\otimes_X$ on invertible objects (corollary of the unconditional associator))**.** *Let X be a scheme and let $M, N, P \in$ Scheme.`Modules` X be ⊗-invertible (662). There is an isomorphism*

$$(M \otimes_X N) \otimes_X P \xrightarrow{\sim} M \otimes_X (N \otimes_X P)$$

in Scheme.`Modules` X*. The invertibility-carrier group law (`pic_commgroup`) consumes only the* existence *of this isomorphism, the proposition* $\mathrm{Nonempty}((M \otimes_X N) \otimes_X P \cong M \otimes_X (N \otimes_X P))$*; no pentagon, triangle, or naturality coherence is ever required.*

Proof. Immediate specialisation of the *unconditional* associator 653 to the invertible arguments M, N, P; no hypothesis on the arguments is used. (The earlier route argued via "invertible ⇒ locally free ⇒ sectionwise flat" to feed the flat-whiskering lemma `flat_whisker_localizer`; that step is *false* — local freeness does not give global sectionwise flatness $M(U)$ over $\mathcal{O}_X(U)$ for a non-affine open U — and is no longer needed, since the general associator is itself unconditional, its single open left-whiskering obligation 651 being closed via the d.2 stalk–tensor commutation 659.) □

Definition 700 (The invertibility-carrier Picard group Pic X)**.** *Source: [Stacks Project], Tag 01CX, Definition `definition-pic`: the Picard group* $\mathrm{Pic}(X)$ *is the abelian group of isomorphism* *classes of invertible $\mathcal{O}_X$-modules, with addition the tensor product.* Let X be a scheme. Consider the subtype of $\otimes$-invertible objects

$$\texttt{Invertible}\,X \;:=\; \{\, M \in \mathrm{Scheme}.\texttt{Modules}\,X \;\big|\; \texttt{IsInvertible}\,M \,\}$$

of 662, and equip it with the setoid whose relation is "there exists an isomorphism", $M \sim M' :\equiv \mathrm{Nonempty}(M \cong M')$ (a proposition-valued equivalence relation: reflexive, symmetric and transitive because isomorphisms compose and invert). The *Picard group carrier*

$$\mathrm{Pic}\,X \;:=\; \texttt{Invertible}\,X\,/\!\sim$$

is the quotient type of isomorphism classes of $\otimes$-invertible $\mathcal{O}_X$-modules. We write $[M] \in \mathrm{Pic}\,X$ for the class of an invertible M. This is the scheme-theoretic analogue of Mathlib's $\texttt{CommRing.Pic}\,R$, carried on `IsInvertible` rather than on `IsLocallyTrivial`; on a scheme the two predicates agree (Stacks tag 0B8M), but only the invertibility form carries the inverse witness existentially.

Lemma 701 ($\otimes$-invertibility is closed under tensor product)**.** *Source: [Stacks Project], Tag 01CR, Lemma* `lemma-constructions-invertible` *(item 1): the tensor product of two invertible $\mathcal{O}_X$-modules is again invertible. Let X be a scheme and $M, M' \in \mathrm{Scheme}.\texttt{Modules}\,X$. If* $\texttt{IsInvertible}\,M$ *and* $\texttt{IsInvertible}\,M'$*, then* $\texttt{IsInvertible}(M \otimes_X M')$*.*

Proof. Let N be a tensor inverse of M (so $M \otimes_X N \cong \mathcal{O}_X$) and N' a tensor inverse of M' (so $M' \otimes_X N' \cong \mathcal{O}_X$), furnished by the two invertibility witnesses. We claim $N \otimes_X N'$ is a tensor inverse of $M \otimes_X M'$. Indeed, reassociating and commuting the four factors with the associator 699 and the braiding 655,

$$(M \otimes_X M') \otimes_X (N \otimes_X N') \;\cong\; (M \otimes_X N) \otimes_X (M' \otimes_X N') \;\cong\; \mathcal{O}_X \otimes_X \mathcal{O}_X \;\cong\; \mathcal{O}_X,$$

the middle step substituting the two witness isomorphisms and the last being the unitor 654. Hence $N \otimes_X N'$ witnesses $\texttt{IsInvertible}(M \otimes_X M')$. Every step is consumed only as the existence of an isomorphism, so no coherence identity is required. □

Lemma 702 (The tensor inverse is determined up to isomorphism)**.** *Source: [Stacks Project], Tag 01CR, Lemma* `lemma-invertible`*: when $\mathcal{L}$ is invertible the module $\mathcal{N}$ with $\mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{N} \cong \mathcal{O}_X$ is isomorphic to the dual $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{L}, \mathcal{O}_X)$, hence unique up to isomorphism. Let X be a scheme and $M, N, N' \in \mathrm{Scheme}.\texttt{Modules}\,X$. If $M \otimes_X N \cong \mathcal{O}_X$ and $M \otimes_X N' \cong \mathcal{O}_X$, then $N \cong N'$. Consequently the assignment $[M] \mapsto [N]$, sending an invertible class to the class of its tensor inverse, is well-defined on $\mathrm{Pic}\,X$.*

Proof. This is the standard "inverse-of-inverse" argument in a commutative monoid, run with isomorphisms in place of equalities. Chain the unitor 654, the two hypotheses, the associator 699 and the braiding 655:

$$N \;\cong\; N \otimes_X \mathcal{O}_X \;\cong\; N \otimes_X (M \otimes_X N') \;\cong\; (N \otimes_X M) \otimes_X N' \;\cong\; \mathcal{O}_X \otimes_X N' \;\cong\; N',$$

where the second step substitutes $\mathcal{O}_X \cong M \otimes_X N'$, the third is the associator, the fourth uses the braiding $N \otimes_X M \cong M \otimes_X N \cong \mathcal{O}_X$, and the outer two are the unitors. Hence $N \cong N'$. For well-definedness on classes, note that if $M \cong M'$ then a tensor inverse of M is also a tensor inverse of M' (tensor with the isomorphism), so the inverse class depends only on $[M]$. □

16.10 Consumer: the AddCommGroup instance for the relative Picard sheaf

The substrate of 16.3 and 16.4 supplies all the data missing from 618's Lean encoding. The consumer instance below packages the abelian-group law $[\,\mathcal{L}\,] + [\,\mathcal{L}'\,] := [\,\mathcal{L} \otimes_{C\times_S T} \mathcal{L}'\,]$ on the relative Picard quotient via the substrate.

Theorem 703 (Abelian-group instance on the relative Picard quotient via Scheme.`Modules.tensorObj`).

Source: [Kleiman], "The Picard scheme", §2, Defs. df:aPf + df:Pfs (the target category is the category of abelian groups; the relative Picard functor is a quotient of abelian groups). Let C/k be a smooth proper geometrically integral curve over a field k, let $\pi_C : C \to \operatorname{Spec} k$ be the structure morphism, and let T be a k-scheme with structure morphism $\pi_T : T \to \operatorname{Spec} k$. The set

$$\operatorname{Pic}^\sharp_{C/k}(T) \;\; := \;\; \operatorname{Pic}(C \times_k T)\,/\,\pi_T^*\operatorname{Pic}(T)$$

of 616 carries a canonical abelian-group structure with addition $[\,\mathcal{L}\,] + [\,\mathcal{L}'\,] := [\,\mathcal{L} \otimes_{C\times_k T} \mathcal{L}'\,]$, neutral element $[\,\mathcal{O}_{C\times_k T}\,]$, and inverse $-[\,\mathcal{L}\,] := [\,\mathcal{L}^{-1}\,]$, where $\otimes_{C\times_k T}$ and $(-)^{-1}$ are the operations supplied by the substrate of 642 restricted to the `LineBundle.OnProduct` $\pi_C\,\pi_T$ carrier via 657. With this structure the quotient map $\operatorname{Pic}(C \times_k T) \twoheadrightarrow \operatorname{Pic}^\sharp_{C/k}(T)$ is a surjective group homomorphism with kernel exactly $\pi_T^*\operatorname{Pic}(T)$, in agreement with 618.

Proof. Write $\operatorname{Pic}(C\times_k T)$ for the set of isomorphism classes of objects of `LineBundle.OnProduct` $\pi_C\,\pi_T$ (line bundles on $C \times_k T$), with the operation $[L] + [L'] := [L \otimes_{C\times_k T} L']$. This operation is well-defined on isomorphism classes: an isomorphism $L \cong L_0$ tensors to an isomorphism $L \otimes L' \cong L_0 \otimes L'$ by functoriality of $\otimes$ (643), and symmetrically in the second argument. The abelian-group structure on $\operatorname{Pic}(C \times_k T)$ is then exactly the *units group of the commutative monoid of $\otimes$-iso-classes* (664): multiplication $[L] \cdot [L'] = [L \otimes_{C\times_k T} L']$ is well-defined and commutative-monoidal by the coherence isomorphisms 653, 654, and 655, and an iso-class is a unit precisely when its carrier is $\otimes$-invertible (662), the inverse being carried by the predicate. Each axiom is a *proposition* — the existence of an isomorphism — so no monoidal-coherence datum (pentagon, triangle, hexagon) is invoked; the AddCommGroup on $\operatorname{Pic}(C \times_k T)$ is the additive form of this units group. This mirrors Mathlib's own Picard group: `CommRing.Pic` R carries its `CommGroup` (`instCommGroupPic`, `Mathlib.RingTheory.PicardGroup`) transported from Units(Skeleton(`ModuleCat` R)); here the same Units($\cdots$) shape is transported to the scheme-level $\otimes_X$. (The scheme-level Picard group is genuinely absent from Mathlib — `Mathlib.RingTheory.PicardGroup` records "connect to invertible sheaves" as an open task — so the project supplies it lightly.) The invertibility-predicate carrier is the project's `IsInvertible` (662), the scheme-level analogue of Mathlib's `Module.Invertible`; the geometric predicate `IsLocallyTrivial` is retained separately and is connected to `IsInvertible` only by an off-critical-path equivalence (only the geometric `IsInvertible` $\Leftarrow$ `IsLocallyTrivial`-style direction is nontrivial, and it is not needed here).

By 696 applied to the projection $\pi_T : C \times_k T \to T$, pullback along π_T commutes with the tensor product of locally-trivial Scheme.`Modules`: for line bundles $\mathcal{N}, \mathcal{N}'$ on T there is a natural isomorphism $\pi_T^*(\mathcal{N} \otimes_T \mathcal{N}') \xrightarrow{\sim} (\pi_T^*\mathcal{N}) \otimes_{C\times_k T} (\pi_T^*\mathcal{N}')$, and π_T^* sends the structure sheaf to the structure sheaf. Hence the pullback carries tensor products to tensor products and the unit to the unit, so the induced map on isomorphism classes $\pi_T^* : \operatorname{Pic}(T) \to \operatorname{Pic}(C \times_k T)$ is a group homomorphism, and its image is a subgroup.

By 616, the underlying set of $\operatorname{Pic}^\sharp_{C/k}(T)$ is the set of cosets of this subgroup. The quotient of an abelian group by a subgroup is itself an abelian group, via the standard `QuotientAddGroup` construction, and the quotient map is a surjective group homomorphism whose kernel is exactly the subgroup (Stacks tag 01CR for the Picard-group special case; the general quotient-of-abelian-groups statement is in any algebra textbook). Putting the pieces together gives the AddCommGroup instance on $\mathrm{Pic}^{\sharp}_{C/k}(T)$ of the statement, and identifies it with the structure of 618. □

16.11 Out of scope

The following are downstream of this chapter and live in dedicated sibling chapters; this chapter does *not* touch them:

- **The sheafified relative Picard functor and its abelian-group target** (`rel_pic_etale_sheafification`, `rel_pic_etale_sheaf_group_structure`) — once 703 is established, the AddCommGroup-valued presheaf, its étale sheafification, and the sheafification preserving the abelian-group target close against it (chapter 15); none of those declarations is the responsibility of this chapter.
- **Representability of $\mathrm{Pic}^{\sharp}_{(C/k)\text{ét}}$ by a scheme** (A.2.c, the FGA Picard-representability chapter (parent scope)) — consumes the full AddCommGroup-valued sheafified presheaf; representability is downstream of `rel_pic_etale_sheaf_group_structure`.
- **Identity component** $\mathrm{Pic}^{0}_{C/k}$ (A.3, the Picard identity-component chapter (parent scope)) — once representability is in hand, the identity component is cut out by Hilbert polynomials; that is independent of the tensor-product substrate of this chapter.
- **Albanese universal property** (A.4, the Albanese universal-property chapter (parent scope)) — consumes the full Route A stack downstream of A.3.
- **Mathlib upstream PR of** Scheme.`Modules.tensorObj` — the substrate of this chapter is project-side, intentionally; a parallel Mathlib upstream PR is welcome but not on the critical path. Once the substrate ships in Mathlib, the `TensorObjSubstrate.lean` file can be slimmed to a single import + namespace re-export, but no downstream Lean file changes.

16.12 Internal-consistency check

The Lean-pinned declaration blocks of this chapter form a connected `\uses{…}` chain whose leaves all resolve either inside this chapter, inside chapter 14 (A.1.b on disk), or inside chapter 15 (A.1.c on disk):

- 642 introduces the substrate operation and depends on no prior block.
- 643 uses 642.
- 644 uses 642, 643, and 652; it records that, under route (e), the full monoidal category is *cheap* (the output of `LocalizedMonoidal`) and that the group law still consumes only existence-of-isomorphism propositions.
- 645 uses 642; it is a self-contained `CommRingCat`-level supplement, off the critical path and *not* an ingredient of 646 — nothing downstream depends on it.

- 662 (the $\otimes$-invertibility predicate, carrying the inverse) uses 642.
- The route (e) whisker lemmas 651 (local injectivity of $F \triangleleft g$ for $g \in J.W$: the PRIMARY route assumes F locally trivial and proves it sheaf-level via 646 and the unitor, with no stalks; the FALLBACK route, for arbitrary F, is the stalkwise argument with residual ingredients d.1/d.2 — the same (d.2) stalk–tensor commutation 658 that the *live* associator obligation 659 consumes, hence on the critical path) and 652 (its closed left/right packaging) supply the two fields of the instance `jw_ismonoidal`. The latter applies Mathlib's `LocalizedMonoidal` to the already-monoidal `PresheafOfModules` $\mathcal{O}_X$ and the sheafification localizer to give the full MonoidalCategory structure on Scheme.`Modules` X; it is stated-but-unformalized, pending the sole open proof obligation.
- The coherence isomorphisms 654 (left/right unitors) and 655 (braiding) are the cheap `sheafification.mapIso` of the presheaf-level coherence data. The associator 653 is the sheafification transport of the presheaf-level associator, whose single open left-whiskering obligation 659 is closed *unconditionally* — for an arbitrary whiskering object, with no flatness and no local triviality — by the d.2 stalk–tensor commutation 658 (16.6). None uses `MonoidalClosed` or sectionwise flatness; the associator's pinned `IsLocallyTrivial` hypotheses are vestigial (the construction is unconditional).
- 649 (sheafification sends a morphism whose underlying map is in $J.W$ to an isomorphism) depends on no prior block; it is a closed go/no-go bridge, retained as a supplement (the route (e) associator is the API-derived one, so this bridge is no longer the associator's load-bearing step).
- `flat_whisker_localizer` (left/right whiskering of a sheafification-localizer morphism by a flat object stays in the localizer class) uses 642; it is a valid closed standalone result, superseded on the critical path by the d.2 route of 16.6 (659).
- 664 (the commutative group on locally-trivial $\otimes$-iso-classes) uses 642, 662, 643, 657, 656, 653, 654, and 655. This is the group-law engine: the PRIMARY construction is the commutative group on locally-trivial iso-classes à la `Module.Invertible` / `CommRing.Pic`, needing only 646 and 656 on top of the assembled locally-trivial associator/unitors/braiding; the Units(Skeleton(Scheme.`Modules` X)) presentation via `jw_ismonoidal` is retained as the route (e) fallback.
- Under the PRIMARY locally-trivial group law, 646 (consumed by the PRIMARY proof of 651) and 656 (in the `\uses` of 664, supplying inverses) are ON the critical path. The remaining supplements 645 and 648 use 642 and are auxiliary. The group law rests on 664 and 662.
- 657 restricts $\otimes$ to the locally-trivial carrier `LineBundle.OnProduct` on $C \times_S T$, using 642 and 648 (the Lean `tensorObjOnProduct` applies `tensorObj_isLocallyTrivial` directly).
- 703 uses 657, 696, 662, 664, 616, and 618; the last two are in 14 (A.1.b) and 15 (A.1.c) respectively.

All `\uses` targets resolve either in this chapter, in 14, or in 15. No declaration cross-references a chapter that does not yet exist on disk.

16.13 Sheaf internal-hom and tensor-inverse infrastructure

The PRIMARY group law (664) needs a $\otimes$-inverse for every line bundle, supplied by 656 as the dual $L^{-1} = \mathcal{H}om_{\mathcal{O}_X}(L, \mathcal{O}_X)$. The standard route constructs that dual as a special case of the *sheaf internal hom* of $\mathcal{O}_X$-modules; over a fixed ring the internal hom of R-modules is the linear-coyoneda module `ihom` $M\,N = (M \to N)$, with dual `ihom` $M\,\mathbf{1} = M^\vee$, but for the *varying* structure sheaf no such internal-hom / monoidal-closed structure exists at the `PresheafOfModules`, `SheafOfModules`, or general-categorical level. This section decomposes the construction of the missing object into individually-formalizable sub-steps, following the slice presentation of the internal hom; each block names the intended Lean declaration and its dependencies. The specialisation that closes 656 is recorded in 748; an alternative descent-theoretic route is noted in 847.

Status: deferred, off the group's critical path. The blocks of this section — the dual object `dual` and its restriction compatibility 737, the local-triviality of the dual 739, the gluing engine `sheafofmodules_hom_of_local_compat`, and the discharge remark 748 — together constitute the *deferred* `IsInvertible` $\Leftrightarrow$ `IsLocallyTrivial` bridge: on a scheme an invertible $\mathcal{O}_X$-module is locally free of rank one, and conversely (Stacks tag 0B8M), so the two carriers agree, and the locally-trivial carrier's missing inverse (656) is recovered precisely from the dual built here. However, the relative Picard *group law* is now carried on the $\otimes$-invertibility predicate `IsInvertible` (16.9, `pic_commgroup`), where the inverse is the membership witness and *no* dual object is constructed. This section is therefore needed only by a downstream consumer that specifically requires the locally-trivial form of an inverse — not by the group law. The blocks are retained intact for that future use.

A structural point governs the whole construction. The naive rule $U \mapsto \mathrm{Hom}_{\mathcal{O}_X(U)}(M(U), N(U))$ is *contravariant* in the restriction maps of M (a restriction $M(U) \to M(V)$ for $V \subseteq U$ induces precomposition $\mathrm{Hom}(M(V), N(V)) \to \mathrm{Hom}(M(U), \cdot)$ in the wrong direction), so it is *not* a presheaf in the covariant sense and cannot be sheafified like the tensor product 642. The remedy is the *slice* formula $U \mapsto (M|_U \to N|_U)$ — the module of morphisms of *restricted* modules over the restricted ring — whose presheaf functoriality (restrict a global-over-U morphism to a global-over-V morphism, for $V \subseteq U$) *is* covariant. Constructing that slice presheaf, and proving it satisfies the sheaf condition, is the substance of the build.

Definition 704 (The global-ring module on morphisms of presheaves of modules (value module))**.** *Source: [Stacks Project], "Modules on Ringed Spaces", §Internal Hom (tag area 01CM): the section ring acts on a morphism of restricted modules by post-composition with multiplication by the section.* Let R be a presheaf of commutative rings on a base category $\mathcal{C}$ that has a terminal object T, and let $M, N \in$ `PresheafOfModules` R. The abelian group of morphisms $M \longrightarrow N$ of presheaves of R-modules carries a canonical $R(T)$-module structure — the *morphism module* `homModule` $M\,N$. A global scalar $f \in R(T)$ acts on a morphism $\varphi : M \to N$ by post-composition with multiplication by f,

$$f \cdot \varphi \;:=\; m_f^N \circ \varphi,$$

where $m_f^N : N \to N$ is the *multiplication-by-f* endomorphism of N: the endomorphism that, on each object $U \in \mathcal{C}$, is multiplication by the restriction $f|_U \in R(U)$ of f along the unique morphism $U \to T$ to the terminal object. The module axioms hold because the assignment $f \mapsto m_f^N$ is a ring homomorphism $R(T) \to \mathrm{End}(N)$ into the endomorphism ring of N — a product $f \cdot g$ goes to the *composite* $m_f^N \circ m_g^N$ of the two multiplication endomorphisms, the order in which the factors compose being reversed by the fact that the scalar acts by *post*-composition — and composition of morphisms is biadditive in the preadditive category `PresheafOfModules` R. Distributivity and the associativity of the action then follow from bilinearity of composition together with the homomorphism property of $f \mapsto m_f^N$. This is the genuinely new core of the construction: Mathlib supplies the *fixed*-ring internal hom `ihom` $M\,N = (M \to N)$ over a single base ring, but nothing equipping the morphism group with the module structure for the *varying* structure presheaf at the `PresheafOfModules` level.

Definition 705 (Slice specialisation of the morphism module)**.** Specialise 704 to a slice category. For an object (open) U of $\mathcal{C} = \mathrm{Opens}X$, restricting along the forgetful functor `Over.forget` $U : \mathcal{C}/U \to \mathcal{C}$ — the open-immersion restriction $M \mapsto M|_U$ — carries M, N to presheaves of modules $M|_U, N|_U$ over the restricted structure ring on the slice $\mathcal{C}/U$. The slice $\mathcal{C}/U$ has the terminal object `Over.mk` (id_U), at which the restricted ring takes the value $R(U)$; hence 704 makes the morphism group $M|_U \longrightarrow N|_U$ an $R(U)$-module, the `internalHomObjModule`. This $R(U)$-module $(M|_U \to N|_U)$ is exactly the value $\mathcal{H}om(M,N)(U)$ of 706.

Definition 706 (Presheaf internal hom of $\mathcal{O}_X$-modules)**.** *Source: [Stacks Project], "Modules on Ringed Spaces", §Internal Hom (tag area 01CM).* Let R be a presheaf of commutative rings on $\mathrm{Opens}X$ and let $M, N \in$ `PresheafOfModules` R. The *presheaf internal hom* $\mathcal{H}om(M,N)$ is the presheaf of R-modules whose value on an open U is the module of morphisms of restricted modules

$$\mathcal{H}om(M,N)(U) \ := \ \texttt{ModuleCat.of}\ (R(U))\ (M|_U \longrightarrow N|_U),$$

the $R(U)$-module of $R|_U$-linear morphisms $M|_U \to N|_U$ of presheaves of modules on U, formed via the open-immersion restriction $U \hookrightarrow X$. This value module is the slice specialisation 705 of the morphism module 704.

Explicitly, $\mathcal{H}om(M,N)$ is the presheaf of R-modules assembled from three pieces. *(a) Value modules.* On each open U the value is the $R(U)$-module $(M|_U \to N|_U)$ of 704 / 705. *(b) Restriction maps.* For an inclusion $V \subseteq U$ the restriction map $\mathcal{H}om(M,N)(U) \to \mathcal{H}om(M,N)(V)$ is the further-restriction map 707, $\varphi \mapsto \varphi|_V$, which is additive and semilinear over the ring restriction $R(g) : R(U) \to R(V)$ induced by $g : V \to U$ — the compatibility datum that the presheaf-of-modules assembly (`PresheafOfModules.mk`, via `ofPresheaf`) consumes. *(c) Functoriality.* Restriction along the identity $U \to U$ is the identity map, and restriction along a composite $V \xrightarrow{g} U \xrightarrow{h} W$ of inclusions $V \subseteq U \subseteq W$ (the morphism $g : V \to U$ followed by $h : U \to W$ in $\mathcal{C} = \mathrm{Opens}X$) is the composite of the two restriction maps $\mathcal{H}om(M,N)(W) \to \mathcal{H}om(M,N)(U) \to \mathcal{H}om(M,N)(V)$; both identity and composition are inherited from the functoriality of further restriction, since `Over.map` is a functor. These three pieces make $\mathcal{H}om(M,N)$ a genuine, covariant presheaf of R-modules; this assembled presheaf is the construction target `PresheafOfModules.internalHom`.

The slice formula is forced by contravariance: the pointwise rule $U \mapsto \mathrm{Hom}_{R(U)}(M(U), N(U))$ varies *against* the restriction maps of M and so is not a covariant presheaf, whereas the module of morphisms of *restricted* objects $M|_U \to N|_U$ restricts covariantly. Thus, unlike the tensor product 642 — which is covariant in the restriction maps and sheafifies directly — the internal hom must be built through the slice presentation, and there is no presheaf-level dual to sheafify naively.

Lemma 707 (Restriction map of the presheaf internal hom)**.** *Source: [Stacks Project], "Modules on Ringed Spaces", §Internal Hom (tag area 01CM): the rule $U \mapsto \mathrm{Hom}(\mathcal{F}|_U, \mathcal{G}|_U)$, with its further-restriction maps, is a (pre)sheaf of abelian groups. Let $g : V \to U$ be a morphism of $\mathcal{C} = \mathrm{Opens}X$ (an inclusion $V \subseteq U$ of opens), and let $M, N \in$ `PresheafOfModules` R. There is a restriction map*

$$\mathcal{H}om(M,N)(U) \longrightarrow \mathcal{H}om(M,N)(V), \qquad \varphi \longmapsto \varphi|_V,$$

the `restrictionMap`*, sending a morphism of restricted modules $M|_U \to N|_U$ to its further restriction $M|_V \to N|_V$. Concretely, the functor* `Over.map` $g : \mathcal{C}/V \to \mathcal{C}/U$ *realises "restrict from*

U to V", and $\varphi|_V$ is φ pulled back along it — the further restriction of the same morphism of presheaves of modules. This map is additive, and it is semilinear *over the ring restriction $R(g) : R(U) \to R(V)$: for $f \in R(U)$,*

$$(f \cdot \varphi)\big|_V \;=\; R(g)(f) \cdot (\varphi|_V).$$

This semilinearity is the compatibility datum that the presheaf assembly of 706 (through `PresheafOfModules.ofPreshea` */* `mk`*) requires.*

Proof. Further restriction along `Over.map` g is functorial, so it carries a morphism $M|_U \to N|_U$ to a morphism $M|_V \to N|_V$ and respects identities and composites; in particular the induced map on morphism groups is additive. For the semilinearity, recall that the global-scalar action of 704 acts on φ by post-composition with the multiplication-by-f endomorphism m_f^N; restricting that endomorphism from U to V is multiplication by $f|_V = R(g)(f)$ on $N|_V$. Since further restriction commutes with composition of morphisms, restricting $f \cdot \varphi = m_f^N \circ \varphi$ yields $m^{N|_V}_{R(g)(f)} \circ (\varphi|_V) = R(g)(f) \cdot (\varphi|_V)$, which is the asserted compatibility. □

Definition 708 (Presheaf dual against the structure presheaf)**.** With R and M as in 706, the *presheaf dual* of M is the internal hom into the unit (structure) presheaf,

$$M^\vee \;:=\; \mathcal{H}om(M, R),$$

i.e. $M^\vee(U) =$ `ModuleCat.of` $(R(U))\,(M|_U \to R|_U)$, the $R(U)$-module of $R|_U$-linear functionals on $M|_U$. This is the presheaf-of-modules analogue of the linear dual `Module.Dual` $R\,M = (M \to R)$ that serves as the $\otimes$-inverse of an invertible module over a fixed ring; here the fixed-ring dual is taken open-by-open and assembled by the slice presentation of 706.

Lemma 709 (Evaluation/contraction of the internal hom)**.** *Source: [Stacks Project], "Modules on Ringed Spaces", §Internal Hom (tag area 01CM). With R and M as above, there is a canonical* evaluation *(contraction) morphism of presheaves of R-modules*

$$\mathrm{ev}_M \,:\; M \otimes_R M^\vee \;\longrightarrow\; R, \qquad s \otimes \phi \longmapsto \phi(s),$$

the counit of the tensor–hom adjunction $\mathcal{H}om(M \otimes_R -, R) \cong \mathcal{H}om(-, M^\vee)$ specialised at R. On each open U it is the open-by-open contraction $(M|_U) \otimes_{R|_U} (M|_U \to R|_U) \to R|_U$, $s \otimes \phi \mapsto \phi(s)$, and these are compatible with restriction, so they assemble to a morphism of presheaves of modules. The tensor product is the presheaf tensor underlying 642.

Proof. On a fixed open U, the value $M^\vee(U) = (M|_U \to R|_U)$ is the module of $R|_U$-linear functionals on $M|_U$, and the evaluation $s \otimes \phi \mapsto \phi(s)$ is the standard contraction $(M|_U) \otimes_{R(U)} (M|_U \to R|_U) \to R(U)$ bilinear over $R(U)$, hence a well-defined $R(U)$-linear map out of the tensor product. For $V \subseteq U$ the square relating the U- and V-contractions commutes because evaluating then restricting equals restricting both arguments then evaluating ($\phi(s)|_V = (\phi|_V)(s|_V)$); hence the open-by-open contractions are natural in the restriction maps and define a morphism of presheaves of R-modules. Concretely, the morphism is defined by specifying, on each open U, the per-open contraction $(M|_U) \otimes_{R(U)} (M|_U \to R|_U) \to R(U)$, $s \otimes \phi \mapsto \phi(s)$. The single remaining datum is the naturality square, which reduces to the evaluate/restrict-commutation identity $\phi(s)|_V = (\phi|_V)(s|_V)$ above: the restriction $\phi|_V$ of a functional inside the dual $M^\vee$ is its restriction to the open V, and restriction is functorial, so evaluating the restricted functional on the restricted section agrees with restricting the value of the evaluation. This is the same restriction-functoriality argument already used to make the internal-hom restriction maps functorial in 707. This is the counit of the tensor–hom adjunction of 706 specialised to the unit presheaf. □

Lemma 710 (The internal hom is a sheaf; the sheaf-level dual)**.** *Source: [Stacks Project], "Modules on Ringed Spaces", §Internal Hom (tag area 01CM). Let X be a scheme, $R = X.\texttt{ringCatSheaf}$, and let $M, N \in \text{Scheme}.\texttt{Modules}\,X$ with N a sheaf of modules. Then the presheaf internal hom $\mathcal{H}om(M, N)$ of 706 satisfies the sheaf condition: for an open cover $\{U_i\}$ of U, a family of morphisms $M|_{U_i} \to N|_{U_i}$ agreeing on the overlaps $U_i \cap U_j$ glues to a unique morphism $M|_U \to N|_U$, because N is a sheaf and morphisms of sheaves of modules are determined and gluable section-wise. Hence $\mathcal{H}om(M, N)$ descends to an object of* Scheme.$\texttt{Modules}\,X$. *Specialising to $N = \mathcal{O}_X = \texttt{SheafOfModules.unit}\,R$ gives the* sheaf-level dual

$$\texttt{dual}\,M \;:=\; \mathcal{H}om_{\mathcal{O}_X}(M, \mathcal{O}_X) \;\in\; \text{Scheme}.\texttt{Modules}\,X,$$

and the evaluation 709 descends to a morphism of sheaves of modules $M \otimes_X \texttt{dual}\,M \to \mathcal{O}_X$.

Proof. Fix an open U and an open cover $\{U_i\}$ of U. A morphism $M|_U \to N|_U$ is a section of $\mathcal{H}om(M, N)$ over U; restricting it to the U_i gives a compatible family. Conversely, given morphisms $\varphi_i : M|_{U_i} \to N|_{U_i}$ with $\varphi_i|_{U_i \cap U_j} = \varphi_j|_{U_i \cap U_j}$, section-wise gluing in the sheaf N produces, for each open $W \subseteq U$ and section $s \in M(W)$, a well-defined section of $N(W)$ by gluing the $\varphi_i(s|_{W \cap U_i})$ (these agree on overlaps by hypothesis, and N is a sheaf so the glued section is unique). The resulting assignment is $R(W)$-linear and natural in W, hence a morphism $M|_U \to N|_U$ restricting to the φ_i; uniqueness is the separatedness of N. This is exactly the sheaf condition for $\mathcal{H}om(M, N)$, so it is a sheaf of $\mathcal{O}_X$-modules. Taking $N = \mathcal{O}_X$ yields $\texttt{dual}\,M$, and the evaluation morphism of 709, being a morphism of presheaves into the sheaf $\mathcal{O}_X$, is a morphism of the corresponding sheaves of modules. ☐

Lemma 711 (Base change along a ring isomorphism commutes with the linear dual (H2′))**.** *Let $e : R \simeq S$ be an isomorphism of commutative rings, and let M be an S-module. Restriction of scalars along e commutes with the formation of the linear dual: the canonical map*

$$\texttt{restrictScalars}_e(M \to_S S) \xrightarrow{\;\sim\;} (\texttt{restrictScalars}_e M \to_R R), \qquad \varphi \longmapsto e^{-1} \circ \varphi,$$

is an R-linear equivalence, with inverse $\psi \mapsto e \circ \psi$. On both sides the R-module structure is the S-module structure of M pulled back along e, so that $r \cdot m = e(r) \cdot m$: the left-hand side is the S-linear dual $(M \to_S S)$ regarded as an R-module by restriction, and the right-hand side is the R-linear dual of the restricted module $\texttt{restrictScalars}_e M$ into the restricted ground ring. This is the dual analogue of 647: it is to the internal hom / dual exactly what that lemma is to the tensor product. It is the H2′ ingredient *of the restriction-compatibility isomorphism of the C-bridge 739 — the ring-iso/dual commutation that, composed with the open-immersion pushforward comparison H1, identifies the dual of a restriction with the restriction of the dual.*

Proof. The two assignments $\varphi \mapsto e^{-1} \circ \varphi$ and $\psi \mapsto e \circ \psi$ are mutually inverse, since $e \circ e^{-1} = \mathrm{id}_S$ and $e^{-1} \circ e = \mathrm{id}_R$ as ring homomorphisms, so the two composites are the identity on functionals. Each assignment is R-linear: additivity is immediate, and for a scalar $r \in R$ the defining relation $r \cdot m = e(r) \cdot m$ of the pulled-back module structure makes each linearity check a single rewrite — e^{-1} (resp. e) intertwines the R-action on the source with the R-action on the target precisely because e and e^{-1} are inverse ring homomorphisms identifying R with S. The forward map is therefore an R-linear bijection, i.e. an R-linear equivalence; packaging the two one-sided maps with the inverse identities gives `restrictScalarsRingIsoDualEquiv`. ☐

Definition 712 (Precomposition equivalence on dual sections)**.** Let R be a presheaf of commutative rings and let $e : M \xrightarrow{\sim} M'$ be an isomorphism of presheaves of R-modules. On a fixed open U, precomposition with e gives an $R(U)$-linear equivalence between the dual sections

$$\texttt{dualPrecompEquiv}\,e\,U : \;(M'|_U \to R|_U) \xrightarrow{\;\sim\;} (M|_U \to R|_U), \qquad \varphi \longmapsto \varphi \circ e|_U,$$

with inverse $\psi \mapsto \psi \circ (e^{-1})|_U$. This is the section-level core of the contravariant functoriality of the presheaf dual $M \mapsto M^\vee$ (708) in isomorphisms: precomposing a functional on M' with the comparison isomorphism produces a functional on M, $R(U)$-linearly and invertibly.

Definition 713 (The presheaf dual carries an iso $M \cong M'$ to an iso $M'^\vee \cong M^\vee$)**.** An isomorphism $e : M \xrightarrow{\sim} M'$ of presheaves of R-modules induces a canonical isomorphism of presheaf duals

$$\texttt{dualIsoOfIso}\, e : M'^\vee \xrightarrow{\sim} M^\vee$$

— note the reversal of direction, the dual being contravariant — assembled open-by-open from the precomposition equivalences $\texttt{dualPrecompEquiv}\, e\, U$ of 712. Naturality in U, i.e. compatibility of these sectionwise equivalences with restriction, is the routine check that precomposition by e commutes with restricting functionals.

Definition 714 (The sheaf-level dual carries an iso $M \cong M'$ to an iso $\texttt{dual}\, M' \cong \texttt{dual}\, M$)**.** Let X be a scheme and $e : M \xrightarrow{\sim} M'$ an isomorphism in Scheme.$\texttt{Modules}\, X$. Then there is a canonical isomorphism of sheaf-level duals

$$\texttt{dualIsoOfIso}\, e : \texttt{dual}\, M' \xrightarrow{\sim} \texttt{dual}\, M,$$

obtained by sheafifying the presheaf isomorphism $\texttt{dualIsoOfIso}\, e$ of 713 through the sheaf-level dual of 710. It is the dual (contravariant) analogue of the functoriality of the tensor product in isomorphisms: where an isomorphism of a tensor factor induces an isomorphism of tensor products, an isomorphism $M \cong M'$ induces an isomorphism of duals in the reverse direction.

Lemma 715 (The per-section dual transport $\texttt{sliceDualTransport}$)**.** *Let $f : Y \hookrightarrow X$ be an open immersion of schemes and let $M \in$ Scheme.$\texttt{Modules}\, X$. Write α for the open-immersion structure-presheaf natural transformation with components $\alpha_U = (f.\texttt{appIso}\, U).\mathrm{inv}$, a* $\texttt{CommRingCat}$*-level ring isomorphism, and $\beta = \texttt{whiskerRight}\, \alpha\, (\texttt{forget}_2\, \texttt{CommRingCat}\, \texttt{RingCat})$ for its* $\texttt{RingCat}$*-level shadow, with component $\beta_V \colon \mathcal{O}_Y(V) \xrightarrow{\sim} \mathcal{O}_X(fV)$ ($fV := f.\texttt{opensFunctor}(V)$). Then for every open $V \subseteq Y$ there is an $\mathcal{O}_Y(V)$-linear isomorphism*

$$((\texttt{pushforward}\, \beta).\texttt{obj}\, (\texttt{dual}\, M.\texttt{val}))(V) \xrightarrow{\sim} (\texttt{dual}\, ((\texttt{pushforward}\, \beta).\texttt{obj}\, M.\texttt{val}))(V),$$

realised as a single $\texttt{LinearEquiv.toModuleIso}$ *that packages* both *leg (A) — the slice-Hom base-change reindexing across $f.\texttt{opensFunctor}$ — and leg (B) — the unit codomain ring-iso swap. The naturality of the section family in W has* two *distinct parts: the underlying base-morphism uniqueness in $(\texttt{Over}\, V.\mathrm{unop})^{\mathrm{op}}$ is* $\texttt{Subsingleton.elim}$ *(a thin poset has at most one inclusion between objects), but the accompanying equation of $\mathcal{O}_Y(V)$-module maps is a genuine, separate obligation: it is the ε-naturality of* $\texttt{restrictScalars}$ *along the structure-ring iso β_W, which* $\texttt{Subsingleton.elim}$ *does* not *discharge.*

Proof. The isomorphism is produced from an $\mathcal{O}_Y(V)$-linear equivalence between the left-hand module — the $\mathcal{O}_Y(V)$-module inherited from $\texttt{pushforward}\, \beta$ — and the right-hand $\texttt{internalHomObjModule}$, assembled via $\texttt{LinearEquiv.toModuleIso}$, with both module structures identified explicitly.

Leg (A): the slice-Hom reindexing, built categorically. A dual section φ of the left-hand value is a morphism $\texttt{restr}_{fV} M.\texttt{val} \to \texttt{restr}_{fV} \mathbf{1}_X$ over $\texttt{Over}\,(fV)$. Its image is the dual section over $\texttt{Over}\, V$ whose component at $W \le V$ (with image $fW := f.\texttt{opensFunctor}(W)$) is the categorical image

$$(\texttt{restrictScalars}\, \beta_W).\mathrm{map}\, (\varphi_{(\texttt{Over.mk}\,(f.\texttt{opensFunctor}(W \le V)))}),$$

i.e. the component of φ at the f-image of W, reindexed across $f.\texttt{opensFunctor}$ by restriction of scalars along β_W, and then post-composed with the leg-(B) codomain swap below.

Leg (B): the unit codomain ring-iso swap, via the lax-monoidal unit ε. The codomain of the reindexed component is the restriction of scalars of the unit section, which must be carried to the unit section over Y. This is exactly the inverse of the lax-monoidal unit comparison of the restriction-of-scalars functor,

$$\texttt{codomainMap} := \texttt{inv}(\varepsilon(\texttt{restrictScalars}\,g)), \qquad g := (f.\texttt{appIso}\,W).\mathrm{inv.hom},$$

(719) where g is taken at the `CommRingCat` level. The unit ε is invertible because g is a ring isomorphism (716), so $\varepsilon(\texttt{restrictScalars}\,g)$ is an isomorphism. The presheaf-level identification underlying the unit handling is the dual-of-unit isomorphism for the unit section (726).

Inverse. The inverse reindexing depends on a set-theoretic down-set fact: because f is an open immersion, $fV \subseteq \mathrm{range}\, f$, so every open $W'' \leq fV$ already lies in the image of $f.\texttt{opensFunctor}$, namely $W'' = f.\texttt{opensFunctor}(f^{-1}W'')$. This is the open-image down-set coverage of an `IsOpenImmersion`: an open contained in the (open) image of the immersion is the functor-image of its own preimage. The project records it as the down-set identity $\iota_!(\iota^{-1}V) = V$ for $V \leq W$ (the helper `image_preimage_of_le` (741)), and it is exactly the lemma the inverse reindexing relies on to define the components on the whole down-set of fV.

With this in hand the inverse `PresheafOfModules.Hom` mirrors the forward component formula, with two changes: it uses $(f.\texttt{appIso}\,W'').\mathrm{hom}$ in place of its inverse, and its ε-codomain swap is $\texttt{dualUnitRingSwapHom} = \texttt{inv}(\varepsilon(\texttt{restrictScalars}\,(f.\texttt{appIso}\,W'').\mathrm{hom}))$ (721) — that is, the *inverse* of ε in the hom-direction (where ε is invertible by 717), *not* ε of the inv-direction β. (The reverse reindex turns an $\mathcal{O}_X(fW'')$-section map into an $\mathcal{O}_Y(W'')$-map, so it must restrict along the hom-direction ring iso, whose codomain swap is therefore $\texttt{inv}\,\varepsilon$ of *that* functor.) The whole family is reindexed by the inverse of the down-set bijection $W'' \leftrightarrow f^{-1}W''$ above. The round-trip identities `left_inv` and `right_inv` then close component-wise by `Iso.inv_hom_id` and `Iso.hom_inv_id` of $f.\texttt{appIso}$ (the forward/inverse β-swaps cancel) together with the cancellation of the down-set bijection $\iota_!(\iota^{-1}V) = V$ against its inverse.

For additivity and $\mathcal{O}_Y(V)$-linearity, one must be careful about *which* module structure carries the left-hand value. The additive and scalar structure on a dual section of the left-hand value $((\texttt{pushforward}\,\beta).\texttt{obj}\,(\texttt{dual}\,M.\texttt{val}))(V)$ is the $\mathcal{O}_Y(V)$-module structure on the internal-hom object (705) — the pointwise module structure in which the sum and scalar multiple of dual sections are formed at the terminal object id_{fV} of the slice over fV. This agrees with, but is *a priori* a distinct presentation of, the additive and scalar structure of the underlying presheaf morphism (the pointwise add and scalar action on morphisms of presheaves of modules). The verification of additivity and $\mathcal{O}_Y(V)$-linearity of the transport therefore proceeds in three distinct steps, the first of which is a required first move rather than an afterthought:

(i) *Identify the two module presentations.* The internal-hom module structure's addition and scalar action are identified with the underlying pointwise add and scalar action on presheaf morphisms. This is a definitional identification of the two $\mathcal{O}_Y(V)$-module presentations of the same group of dual sections: the internal-hom operations, formed at the terminal object of the slice, are precisely the pointwise morphism-level operations of the presheaf morphism they package. Until this identification is made, the additivity and scalar-compatibility obligations of the transport cannot be read off the change-of-rings compatibility, because their sums and scalar multiples are taken in the *internal-hom* structure, not the morphism structure.

(ii) *Match the X-side and Y-side scalars by the β-naturality ring identity.* The scalar acting on the X-side (the pushforward-restricted scalar s by which $\texttt{restrictScalars}\,\beta_W$ multiplies the reindexed component) and the scalar acting on the Y-side (the $\mathcal{O}_Y(V)$-scalar c of the internal-hom action) are *a priori* elements of different rings and cannot be compared directly. The bridge is the ring identity

$$s \;=\; (\beta.\mathtt{app}\, W').\mathrm{hom}\, c,$$

obtained from `InternalHom.termRingMap_naturality` — which records how the internal-hom term-ring map transports the scalar action across the slice — together with the naturality of β on the thin poset Opens Y. Without this identity the two sides of the scalar-compatibility equation are not even of matching type, so it must be supplied before the change-of-rings compatibility of step (iii) can be invoked. The restricted scalar action is made explicit by `ModuleCat.restrictScalars.smul_def′`, which exhibits the restricted smul as the original smul along β_W.

(iii) *Apply the change-of-rings compatibility on the morphism level.* Once the operations are expressed pointwise on the underlying morphisms and the scalars matched by (ii), additivity and $\mathcal{O}_Y(V)$-linearity follow from the post-composition compatibility of the change-of-rings reindexing: the functor `restrictScalars` β_W preserves sums and scalar multiples of morphisms, and the structure scalar is intertwined by the ring isomorphism β_W. This last is the presheaf-level shadow of 711.

Naturality of the section family in W. This splits into two obligations of different character, and it is a mistake to discharge the whole square by `Subsingleton.elim` alone. (a) The underlying *base*-morphism naturality — that the two routes around a square agree on the *indexing* morphisms of $(\mathtt{Over}\, V.\mathrm{unop})^{\mathrm{op}}$ — holds by `Subsingleton.elim`, since Opens Y is a thin poset and there is at most one inclusion between any two objects; this discharges *only* the uniqueness of the base morphisms. (b) The actual naturality obligation, however, is an equation between $\mathcal{O}_Y(V)$-*module* maps, and it is *not* settled by (a). Once the base morphisms are pinned by (a), the module-map equation reduces to the ε-naturality square of `restrictScalars` — the naturality of the lax-monoidal unit ε with respect to the restriction maps of *both* the source and the target presheaf, along the structure-ring isomorphism β_W — post-composed with the definition of the codomain swap `dualUnitRingSwap` = `inv` ε(`restrictScalars` β_W). This ε-naturality step is a genuine, separate obligation; the thin-poset argument of (a) does *not* discharge it. The required ε-commutativity-with-restriction-maps square is delivered by the extracted lemma `sliceDualTransport_naturality_apply`, which closes the module-map square *pointwise* rather than through a single natural-transformation field. Once the base morphisms are pinned by (a), it reduces step (b) to two genuine ingredients: the ring-level square `appIso_hom_naturality_apply` — that (f.`appIso`).hom intertwines the X- and Y-side restriction maps, this being the β_W-naturality of the structure-ring isomorphism — post-composed with the codomain swap `dualUnitRingSwap` = `inv` ε(`restrictScalars` β_W) evaluated through its proven action `dualUnitRingSwap_apply`. The thin-poset `Subsingleton.elim` of (a) settles only the base morphisms. □

16.13.1 Dual-inverse lane: the slice-transport helper declarations

These are the individual ingredients out of which 715 and its inverse leg are assembled: the leg-(B) unit-ring swaps, their invertibility and cancellation identities, and the presheaf-level dual-of-unit identification. Each is a self-contained internal construction.

Lemma 716 (Leg-(B) unit ε is invertible, inv-direction)**.** *Let $f : Y \hookrightarrow X$ be an open immersion and $W' \subseteq Y$ an open. The lax-monoidal unit ε of the change-of-rings functor* `restrictScalars` ((f.`appIso` W').inv.hom *taken along the* `CommRingCat`*-level structure-ring isomorphism, is an isomorphism.*

Proof. Proved directly in Lean — the underlying ring map $(f.\mathtt{appIso}\,W').\mathrm{inv}.\mathrm{hom}$ is an isomorphism, hence bijective, so its restriction-of-scalars unit ε (852) is invertible by the bijectivity criterion for ε. ☐

Lemma 717 (Leg-(B) unit ε is invertible, hom-direction)**.** *In the same setting, the lax-monoidal unit ε of* $\mathtt{restrictScalars}\,((f.\mathtt{appIso}\,W').\mathrm{hom}.\mathrm{hom})$ *— the* hom*-direction of the structure-ring iso — is likewise an isomorphism. This is the variant powering the reverse (*`invFun`*) leg, which reindexes a $\mathcal{O}_Y$-section map back to a $\mathcal{O}_X$-map.*

Proof. Proved directly in Lean — identical to 716 with the hom-direction (also bijective) ring map. ☐

Lemma 718 (Unit ε is invertible, section-relabel direction)**.** *Let X be a scheme and let $a = b$ be an equality of opens of X (taken in $(\mathrm{Opens}\,X)^{\mathrm{op}}$), inducing the* `eqToHom` *section-ring relabel $X.\mathtt{presheaf}.\mathrm{map}\,(\mathtt{eqToHom}\,e) \colon \mathcal{O}_X(b) \to \mathcal{O}_X(a)$. The lax-monoidal unit ε of the change-of-rings functor* $\mathtt{restrictScalars}\,((X.\mathtt{presheaf}.\mathrm{map}\,(\mathtt{eqToHom}\,e)).\mathrm{hom})$ *along that relabel is an isomorphism. This is the* `eqToHom`*-induced-ring-map variant of the ε-invertibility criterion, the section-relabel sibling of 716.*

Proof. Proved directly in Lean — the underlying ring map $(X.\mathtt{presheaf}.\mathrm{map}\,(\mathtt{eqToHom}\,e)).\mathrm{hom}$ is the image of an `eqToHom` under a functor, hence an isomorphism and bijective, so its restriction-of-scalars unit ε (852) is invertible by the bijectivity criterion for ε. ☐

Definition 719 (Leg-(B) codomain unit ring-iso swap, inv-direction)**.** The codomain unit ring-iso swap

$$\mathtt{dualUnitRingSwap}\,f\,W' \;:=\; \mathtt{inv}(\varepsilon(\mathtt{restrictScalars}\,(f.\mathtt{appIso}\,W').\mathrm{inv}.\mathrm{hom})) \;:\; (\mathtt{restrictScalars}\,(f.\mathtt{appIso}\,W')$$

the inverse of the (invertible, 716) lax-monoidal unit ε, carrying the restriction-of-scalars of the X-unit section to the Y-unit section. It is the leg-(B) ingredient of 715.

Proof. Proved directly in Lean — the categorical inverse of an isomorphism. ☐

Definition 720 (Leg-(B) unit ring-iso swap, reverse (ε itself))**.** The reverse map $\mathtt{dualUnitRingSwapInv}\,f\,W' :=$ $\varepsilon(\mathtt{restrictScalars}\,(f.\mathtt{appIso}\,W').\mathrm{inv}.\mathrm{hom})$ — the lax-monoidal unit ε *itself* (not its inverse) — from $\mathbf{1}_Y(W')$ back to the restriction-of-scalars of the X-unit section. It is the two-sided inverse of 719.

Proof. Proved directly in Lean — the unit ε, invertible by 716. ☐

Definition 721 (Reverse leg codomain unit ring-iso swap, hom-direction)**.** The hom-direction codomain swap

$$\mathtt{dualUnitRingSwapHom}\,f\,W' \;:=\; \mathtt{inv}(\varepsilon(\mathtt{restrictScalars}\,(f.\mathtt{appIso}\,W').\mathrm{hom}.\mathrm{hom})) \;:\; (\mathtt{restrictScalars}\,(f.\mathtt{appIso}$$

the inverse of the (invertible, 717) unit ε in the hom-direction. It is the leg-(B) ingredient of the reverse transport 728.

Proof. Proved directly in Lean — the categorical inverse of an isomorphism. ☐

Definition 722 (Unit-section relabel swap (the fourth leg of the reverse transport))**.** For an equality $a = b$ of opens of X (in $(\mathrm{Opens}\,X)^{\mathrm{op}}$), the unit-section relabel swap

$$\mathtt{unitRelabelSwap}\,e \;:=\; \mathtt{inv}(\varepsilon(\mathtt{restrictScalars}\,(X.\mathtt{presheaf}.\mathrm{map}\,(\mathtt{eqToHom}\,e)).\mathrm{hom})) \;:\; (\mathtt{restrictScalars}\,(X.\mathtt{pre}$$

the inverse of the (invertible, 718) unit ε of the section-ring relabel. It transports the codomain unit section $\mathbf{1}_X$ across the cross-fiber section-ring relabel $\mathcal{O}_X(b) \cong \mathcal{O}_X(a)$, and is the fourth leg of the reverse transport 728 — the unit-section mirror of 721, needed because the source relabel M.val(eqToHom) is semilinear and the in-fiber core must be conjugated back into the slice section's fiber.

Proof. Proved directly in Lean — the categorical inverse of an isomorphism. ☐

Lemma 723 (Round-trip of the swap pair, inv∘swap)**.** dualUnitRingSwap $f\,W'$∘dualUnitRingSwapInv $f\,W'$ = id*: the swap followed by its reverse is the identity.*

Proof. Proved directly in Lean — the Iso.inv_hom_id cancellation of ε with its inverse. ☐

Lemma 724 (Round-trip of the swap pair, swap∘inv)**.** dualUnitRingSwapInv $f\,W'$∘dualUnitRingSwap $f\,W'$ = id*: the reverse followed by the swap is the identity.*

Proof. Proved directly in Lean — the Iso.hom_inv_id cancellation of ε with its inverse. ☐

Definition 725 (Section equivalence for the dual of the unit)**.** For a presheaf of modules over a base ring presheaf R_0 and an object X, the $R_0(X)$-linear equivalence

$$(\mathtt{restr}_X\,\mathbf{1} \to \mathtt{restr}_X\,\mathbf{1}) \;\simeq_{R_0(X)}\; R_0(X)$$

identifying endomorphisms of the (restricted) monoidal unit with the ground ring, via evaluation at the global section 1; the inverse is multiplication by a global scalar (850).

Proof. Proved directly in Lean. The forward map is evalLin at 1 (851); the inverse is the global-scalar action globalSMul (850). The substantive field is left_inv: every endomorphism of the unit is multiplication by its value at 1, which follows from the φ-naturality of the section toward the terminal object of the slice. The module structure on the hom-space is the slice internal-hom structure (705). ☐

Definition 726 (Presheaf dual of the unit is the unit (general base))**.** The presheaf-level isomorphism dual $\mathbf{1} \cong \mathbf{1}$ of presheaves of modules over a general single-universe base ring presheaf R_0, assembled sectionwise from 725 with the evaluation-at-1 naturality.

Proof. Proved directly in Lean. The sectionwise components are the section equivalences 725 packaged by isoMk; the naturality square is the evaluation-at-1 naturality of the internal-hom evaluation (709) specialised at $M = \mathbf{1}$. ☐

Definition 727 (Scheme-level presheaf dual of the unit is the unit)**.** The scheme-level instance dual $\mathbf{1} \cong \mathbf{1}$ of the presheaf-of-modules dual-of-unit isomorphism, obtained by specialising 726 at $R_0 = Y$.presheaf. It is the presheaf core of the third leg dual_unit_iso (738).

Proof. Proved directly in Lean — a direct instance of 726. ☐

Lemma 728 (Reverse slice dual transport (the invFun of 715))**.** *Source: internal categorical construction; no external reference. In the setting of 715, with $f : Y \hookrightarrow X$ an open immersion, $M \in$ Scheme.Modules X, $V \subseteq Y$ an open and β the open-immersion structure-ring morphism, there is a reverse transport: a dual section ψ : $\mathtt{restr}_V((\mathtt{pushforward}\,\beta)\,M.\mathtt{val}) \to \mathtt{restr}_V \mathbf{1}_Y$ over Over V is sent to an X-slice dual section $\mathtt{restr}_{fV} M.\mathtt{val} \to \mathtt{restr}_{fV} \mathbf{1}_X$ over Over (fV), where $fV = f$.opensFunctor(V). This is the inverse leg of the forward equivalence 715.*

Proof. We build the underlying `PresheafOfModules.Hom` component-by-component over the X-slice `Over` (fV), mirroring the forward component formula of 715 with two changes.

The component at W''. Fix $W'' \leq fV$ and set $W' := W''.\mathrm{left}$ and $P := f^{-1}(W')$ for its preimage. Because f is an open immersion and $fV \subseteq \mathrm{range}\, f$, the open $W' \leq fV$ lies in the image of f.`opensFunctor`, so the down-set identity $f.\texttt{opensFunctor}(P) = W'$ holds — but only *propositionally* (741); this is the single set-theoretic fact the inverse reindexing depends on. The component of the reverse section at W'' is the X-slice mirror of the forward component:

$$M.\texttt{val}(\texttt{eqToHom}) \circ (\texttt{restrictScalars}\,(f.\texttt{appIso}\,P).\mathrm{hom}).\mathrm{map}(\psi_{(\texttt{Over.mk}\,(P \leq V))}) \circ \texttt{dualUnitRingSwapHom}\,f\,P \circ \texttt{unitRe}$$

conjugated on source and target by the `eqToHom` transports arising from the propositional equality $f.\texttt{opensFunctor}(P) = W'$ (the mirror of `homLocalSection`). Compared with the forward formula, this map uses $(f.\texttt{appIso}\,P).\mathrm{hom}$ in place of its inverse — because the reverse reindex turns a $\mathcal{O}_Y(P)$-section map into a $\mathcal{O}_X(fP)$-map — and its codomain swap is `dualUnitRingSwapHom` (721), the inverse of ε in the hom-direction (717). The fourth leg `unitRelabelSwap` $(\texttt{eqToHom}\,he.\mathrm{symm})$ transports the codomain unit $\mathbf{1}_X$ across the cross-fiber section-ring relabel $\mathcal{O}_X(W') \cong \mathcal{O}_X(fP)$: it is the $\texttt{inv}\,\varepsilon$ of that `eqToHom`-induced relabel, the unit-section mirror of `dualUnitRingSwapHom`, needed because the source relabel $M.\texttt{val}(\texttt{eqToHom})$ is semilinear and the in-fiber core must be conjugated back into the fiber $\mathcal{O}_X(W')$ of the slice section.

This reverse component is *not* available for an arbitrary β: collapsing the two stacked change-of-rings layers on $M.\texttt{val}(fP)$ into the identity — the step `restrictScalarsComp′App` that the in-fiber core depends on — requires the compatibility identity

$$h\beta : \forall P, \quad (\beta.\mathrm{app}\,(\texttt{op}\,P)).\mathrm{hom} \circ (f.\texttt{appIso}\,P).\mathrm{hom}.\mathrm{hom} = \mathrm{id},$$

which the lemma takes as an explicit hypothesis. The equality is *false* for a general structure morphism β; at the unique call site β is the open-immersion structure-ring morphism $(f.\texttt{appIso})^{-1}$, and the identity is discharged by `Iso.hom_inv_id`.

Additivity and $\mathcal{O}_Y(V)$-linearity follow exactly as in the forward leg (steps (i)–(iii) of 715): identify the two module presentations, match the X- and Y-side scalars by the β-naturality ring identity, and apply the change-of-rings compatibility, the module-level shadow of 711.

Naturality of the component family across morphisms of $(\texttt{Over}\,(fV))^{\mathrm{op}}$ reduces, after the thin-poset `Subsingleton.elim` on the indexing morphisms, to the ε-naturality of `restrictScalars` along the structure-ring iso, exactly as in the forward leg. The unit handling is the presheaf dual-of-unit identification 727. The whole family is reindexed by the inverse of the down-set bijection $W'' \leftrightarrow f^{-1}(W'')$ supplied by 741. □

Lemma 729 (Pointwise naturality square of the reverse slice transport)**.** *Source: internal categorical construction; no external reference. With notation as in 728, the reverse-transport component family satisfies the pointwise naturality square: for a morphism g of the slice* $(\texttt{Over}\,(fV))^{\mathrm{op}}$ *and a section z, the two ways of relabelling and transporting z agree, with the two* $\texttt{inv}\,\varepsilon$ *legs (*`unitRelabelSwap`*,* `dualUnitRingSwapHom`*) kept* shallow *(the down-set facts passed explicitly). It is the clean-statement extraction whose* `exact` *(by defeq) closes the* `naturality` *field of* `sliceDualTransportInv`*.*

Proof. After rewriting the two $\texttt{inv}\,\varepsilon$ legs into their shallow `_apply` forms (so no `whnf` of the deep $\texttt{inv}\,\varepsilon$ composite occurs) the goal is a concrete element equation, closed by three ingredients: the M-side coherence (both legs are $M.\texttt{val}$-restrictions of z, fused by $M.\texttt{val.map\_comp}$ and the thin-poset `Subsingleton.elim`); ψ-naturality (`PresheafOfModules.naturality_apply`) at the slice morphism; and `appIso`-naturality (`Scheme.Hom.appIso_inv_naturality`), the X-presheaf round-trip closing by a `CommRingCat` morphism identity. □

Lemma 730 (Clean pointwise form of the reverse-transport component)**.** *Source: internal categorical construction; no external reference. The* `app` *component of* `sliceDualTransportInv` *evaluated at a section equals its explicit reverse-transport formula (the leg reductions are definitional). This is the* `rfl` *bridge reused by the* `left_inv` *and* `right_inv` *round-trip proofs of 715 to expose the inverse leg in clean form before the* `appIso` *round-trip cancellation.*

Proof. Definitional: every leg reduction in the component formula is a `rfl`-identity, so the equation holds by `rfl` once the down-set proofs are supplied. □

Lemma 731 (Clean pointwise form of the forward-transport component)**.** *Source: internal categorical construction; no external reference. The* `app` *component of* `sliceDualTransport` $f\,M\,V$ *— the forward per-slice dual-transport family of 715 — evaluated at a dual section φ equals its explicit forward-transport formula. On a slice object W of $(\mathtt{Over}\,V)^{\mathrm{op}}$, writing $W' = W.\mathrm{left}$, the value at a section z is*

$$(\mathtt{sliceDualTransport}\,f\,M\,V\,\varphi).\mathtt{app}\,W(z) \;=\; (\mathtt{dualUnitRingSwap}\,f\,W').\mathrm{hom}\Big(\varphi.\mathtt{app}\big(\mathtt{Over.mk}\,(f.\mathtt{opensFunctor}\,W.\mathrm{ho}$$

i.e. the reindexed component $\varphi.\mathtt{app}$ *at the f-image of W followed by the codomain unit ring-swap* `dualUnitRingSwap` $f\,W'$ *(719, the* $(f.\mathtt{appIso}\,W').\mathrm{hom}$ *conjugation); the intervening* `restrictScalars` *leg is a definitional identity on carriers. This is the forward* `rfl` *bridge exposing the forward leg in clean form — the mirror of 730 — and is the sectionwise tool that closes the pushforward-flank residual of 799.*

Proof. Definitional: every leg reduction in the forward component formula is a `rfl`-identity, so the equation holds by `rfl` once the down-set proofs are supplied. □

Lemma 732 (Action of the codomain unit ring-swap)**.** *Source: internal categorical construction; no external reference. The underlying map of the codomain unit ring-swap* `dualUnitRingSwap` $f\,W'$ *(719) is the* hom*-direction of the open-immersion structure-ring isomorphism. For a section $x \in X.\mathrm{presheaf}(f.\mathtt{opensFunctor}\,W')$,*

$$(\mathtt{dualUnitRingSwap}\,f\,W').\mathrm{hom}\,(x) \;=\; (f.\mathtt{appIso}\,W').\mathrm{hom}.\mathrm{hom}\,(x).$$

This is the workhorse pointwise action exposing `dualUnitRingSwap` *as a plain change-of-rings ring map; it is the rewrite that drives the forward-transport sectionwise reductions.*

Proof. `dualUnitRingSwap` is the categorical inverse of the unit ε, and $\varepsilon(\mathtt{restrictScalars}\,(f.\mathtt{appIso}\,W')^{-1})$ acts on carriers as $(f.\mathtt{appIso}\,W').\mathrm{inv}$. Applying injectivity of $(f.\mathtt{appIso}\,W').\mathrm{inv}$ to the round-trip identity $\mathtt{dualUnitRingSwap} \circ \mathtt{dualUnitRingSwapInv} = \mathbf{1}$ collapses the left-hand carrier action to $(f.\mathtt{appIso}\,W').\mathrm{hom}$, which is the claim. □

Lemma 733 (Open-immersion structure-ring cocycle)**.** *Provided by Mathlib. For composable open immersions $f : X \hookrightarrow Y$ and $g : Y \hookrightarrow Z$ and an open $U \subseteq X$, the structure-ring isomorphism of the composite factors as the* `eqToIso`*-twisted product of the two factor isomorphisms,*

$$(f;g).\mathtt{appIso}\,U \;=\; Z.\mathrm{presheaf}.\mathrm{mapIso}\big(\mathtt{eqToIso}\,(\mathtt{comp\_image}\,f\,g\,U)\big)^{\mathrm{op}};g.\mathtt{appIso}\big(f.\mathtt{opensFunctor}\,U\big);f.\mathtt{appIso}\,U,$$

the cocycle reconciling the composite direct image $(f;g)''\,U$ with the iterated one $g''\,(f''\,U)$. This is the structure-ring half of the open-immersion pseudofunctor coherence.

Lemma 734 (Sectionwise split of the composite slice dual transport via the `appIso` cocycle)**.** *Source: internal categorical construction; dual of the proven tensor base-change 685 (statement anchor: internal-hom base-change, Stacks Modules ch. 17). Let $f : Y \hookrightarrow X$ and $h : Z \hookrightarrow Y$*

be composable open immersions, so that $h\,;f : Z \hookrightarrow X$*, let* $M \in \mathrm{Scheme}.\texttt{Modules}\,X$*, and let* $V \subseteq Z$ *be open. This is the structure-ring half of the pseudofunctoriality (composition) law for the forward slice dual transport (715): evaluated sectionwise, the composite unit ring-swap* $\texttt{dualUnitRingSwap}\,(h\,;f)$ *splits into the two single-immersion swaps of* f *and of* h*. Concretely, on a slice object* W *of* $(\texttt{Over}\,V)^{\mathrm{op}}$*, writing* $W' = W.\mathrm{left}$*, the value of* $\texttt{sliceDualTransport}\,(h\,;f)\,M\,V\,\varphi$ *at* W *and a section* z *is*

$$(\texttt{sliceDualTransport}\,(h;f)\,M\,V\varphi).\texttt{app}\,W(z) \;=\; (\texttt{dualUnitRingSwap}\,h\,W').\mathrm{hom}\Big((\texttt{dualUnitRingSwap}\,f\,(h.\texttt{opensFun}$$

where $\rho = X.\mathrm{presheaf.map}\big(\texttt{eqToHom}\,(\texttt{comp\_image}\,h\,f\,W')^{-1}\big)^{\mathrm{op}}$ *is the reindex along the direct-image composition equality* $(h\,;f).\texttt{opensFunctor}\,W' = f.\texttt{opensFunctor}(h.\texttt{opensFunctor}\,W')$ *(an equality of* $\mathrm{Opens}\,X$*, 733). In words: the reindexed component* $\varphi.\texttt{app}$ *at the* $(h\,;f)$*-image of* W*, retracted across* `comp_image` *and then conjugated successively by the codomain unit ring-swaps of* f *and of* h*. There is no separate* `dualIsoOfIso` *correction factor: the whole splitting is carried by the structure-ring cocycle* `comp_appIso` *(733) reindexed by* `comp_image`*.*

Proof. Evaluate both sides at W and the section z. The forward apply lemma 731 rewrites the left-hand component as a single $\texttt{dualUnitRingSwap}\,(h;f)$-conjugated reindex of φ, and its action 732 replaces that swap by the hom-direction of the composite structure-ring iso $(h;f).\texttt{appIso}\,W'$. The cocycle `comp_appIso` (733) factors that composite iso as the `eqToIso`-twisted product $(h\,;f).\texttt{appIso}\,W' = X.\mathrm{presheaf.mapIso}(\texttt{eqToIso}\,\texttt{comp\_image})^{\mathrm{op}}\,;f.\texttt{appIso}(h.\texttt{opensFunctor}\,W')\,;h.\texttt{appIso}\,W'$, and applying 732 once more to each of the f- and h-factors re-expresses the two right-hand swaps in hom form. Expanding the composite of homs — the opposite-presheaf map ρ of the `comp_image` equality followed by the two factor isos — the two sides coincide on the nose. The identity is thus purely the structure-ring cocycle reindexed by `comp_image`; no adjunction unit/counit layer enters.

Working sectionwise in the slice category — where 731 yields a definitional identity at each component — this composition law separates from the adjunction unit/counit layers of 799. □

Lemma 735 (Pointwise hom-naturality of the structure ring iso)**.** *Source: internal categorical construction; no external reference. Let* $f : Y \hookrightarrow X$ *be an open immersion,* $i : U \to V$ *an inclusion of opens of* Y*, and* w *a section of* $\mathcal{O}_X$ *over* $f.\texttt{opensFunctor}(U)$*. Then* $(f.\texttt{appIso}\,V).\mathrm{hom}$ *intertwines the* X*- and* Y*-side restriction maps:*

$$(f.\texttt{appIso}\,V).\mathrm{hom}((X.\texttt{presheaf}.\mathrm{map}\,(f.\texttt{opensFunctor}^{\mathrm{op}}\,i))\,w) \;=\; (Y.\texttt{presheaf}.\mathrm{map}\,i)((f.\texttt{appIso}\,U).\mathrm{hom}\,w).$$

This is the single ring-level square underlying the forward naturality and round-trip pastes of the slice transport (the hom*-direction* β_W*-naturality of the structure-ring isomorphism).*

Proof. Proved directly in Lean. It is the β_W-naturality of the structure-ring isomorphism, obtained by rotating the forward naturality square of $(f.\texttt{appIso}).\mathrm{inv}$ through the iso: $f.\texttt{appIso}\,V$ is an isomorphism, so its inv-leg is bijective; applying that injectivity and cancelling the hom/inv round-trips on both sides reduces the claim to the inv-direction naturality square, which holds. □

Lemma 736 (Pointwise naturality square of the forward slice transport)**.** *Source: internal categorical construction; no external reference. With notation as in 715, the forward-transport component family satisfies the pointwise naturality square: for a morphism* f_1 *of the slice* $(\texttt{Over}\,V)^{\mathrm{op}}$ *and a section* z*, the* `dualUnitRingSwap`*-conjugated reindex of a dual section* φ *commutes with the slice restriction maps. It is the* `toFun`*-side mirror of 729, serving as the naturality certificate for* `sliceDualTransport`*.*

Proof. After substituting the $\mathtt{inv}\,\varepsilon$ leg via `dualUnitRingSwap_apply`, the goal reduces to a concrete element equation closed by two ingredients: φ-naturality (`PresheafOfModules.naturality_apply`) at the f-image of f_1 in the X-slice (the indexing triangle pinned by the thin-poset `Subsingleton.elim`), and the ring-level square 735 — that $(f.\mathtt{appIso}).\mathrm{hom}$ intertwines the X- and Y-side restriction maps. ∎

Lemma 737 (Restriction commutes with the dual along an open immersion (the C-bridge))**.** *Source: [Stacks Project], "Modules on Ringed Spaces", §Internal Hom, Lemma* ***`lemma-pullback-internal-hom`*** *(tag area 01CM). Let $f : Y \hookrightarrow X$ be an open immersion of schemes, with image the open $U \subseteq X$, and let $M \in \mathrm{Scheme}.\mathtt{Modules}\,X$. Then there is a canonical isomorphism of $\mathcal{O}_Y$-modules*

$$(\mathtt{dual}\,M)\big|_f \xrightarrow{\sim} \mathtt{dual}(M|_f),$$

natural in M, between the restriction of the sheaf-level dual (710) and the dual of the restriction. (Only opens $V \subseteq U$ intervene downstream, where the comparison is needed.)

Proof. This is the classical internal-hom base-change isomorphism $f^*\mathcal{H}om_{\mathcal{O}_X}(M, \mathcal{O}_X) \cong \mathcal{H}om_{\mathcal{O}_Y}(f^*M, f^*\mathcal{O}_X)$ of Lemma `lemma-pullback-internal-hom`, specialised to the structure sheaf $\mathcal{G} = \mathcal{O}_X$. For an open immersion the morphism f is flat and restriction $(-)|_f$ is the exact pullback f^*, so the finite-presentation and flatness hypotheses of the general ringed-space statement are automatic; this is the *favorable* (open-immersion) case of the base-change comparison, in which the map is an isomorphism for *every* M, not only the finitely-presented ones.

We emphasise at the outset that this is a genuine *natural isomorphism*, not a definitional equality, and — crucially — that the dual is *not sectionwise*. Unlike the tensor product, whose value is computed fibrewise ($(M \otimes N)(V) = M(V) \otimes N(V)$), the dual evaluates to a Hom over an entire down-set: for an open $V \subseteq U$,

$$(\mathtt{dual}\,M)(V) \;=\; \mathcal{H}om(M|_V,\; \mathcal{O}|_V)\big|_{\mathtt{Over}\,V},$$

a module of morphisms of presheaves over the whole slice $\mathtt{Over}\,V \subseteq \mathrm{Opens}\,Y$, *not* a single fibre $M(V) \to \mathcal{O}(V)$. Consequently the assertion that restriction along the open immersion f commutes with the dual does *not* reduce to a single ground-ring reconciliation. The proof proceeds in two stages: a preliminary adjunction-uniqueness rewrite (stage H1) that re-presents the pulled-back dual in pushforward form, followed by the resolution of the resulting *residual* into *two* legs, carried out objectwise in the open V (with $V \subseteq U$) and assembled over the slice with poset-thin naturality.

Stage H1 — adjunction-uniqueness rewrite. The pulled-back dual is initially presented in the $\mathtt{pullback}\,\varphi$-form inherited from the abstract definition of the restriction functor, and no sectionwise comparison can begin against that opaque form. We first rewrite it into a $\mathtt{pushforward}\,\beta$-form: the restriction-along-an-open-immersion pullback and the pushforward along the structure-ring map β are both left adjoints to one and the same functor, so by uniqueness of left adjoints — the adjunction `pushforwardPushforwardAdj` composed with `leftAdjointUniq` — they are canonically isomorphic, and this isomorphism transports the pulled-back dual into pushforward form. After H1 the residual obligation is *exactly*

$$(\mathtt{pushforward}\,\beta).\mathtt{obj}\,(\mathtt{dual}\,M.\mathtt{val}) \xrightarrow{\sim} \mathtt{dual}((\mathtt{pushforward}\,\beta).\mathtt{obj}\,M.\mathtt{val}),$$

i.e. "$\mathtt{pushforward}\,\beta$ commutes with the dual". It is this residual — *not* the original Step-4 isomorphism — that legs (A) and (B) below resolve; legs (A) and (B) are the content remaining *after* the H1 rewrite, not a standalone two-leg split of the whole step.

Leg (A) — slice-site Hom base-change (Beck–Chevalley). The domain presheaf must first be transported across the open immersion. Because f is an open immersion, its direct-image functor on open sets is fully faithful with image the down-set $\{W : W \le U\}$; this carries the restricted domain $\texttt{restr}_V(f_*M)$ over $\texttt{Over}\,V \subseteq \text{Opens}\,Y$ to the restricted domain $\texttt{restr}_{fV}\,M$ over $\texttt{Over}\,(fV) \subseteq \text{Opens}\,X$, and hence identifies the two Hom-modules computed over these matched slices. This is a genuine slice-site Hom base-change — a Beck–Chevalley square on the internal hom across the fully-faithful open-immersion functor, restricted to the down-set of fV — *not* a fibrewise identity.

Leg (B) — ground-ring reconciliation. The codomain of the internal hom is in both cases the structure sheaf. Once leg (A) has matched the domains, the unit codomain is reconciled along the open-immersion ring isomorphism $\beta_V = (\texttt{toRingCatSheafHom}\,f).\text{hom}_V \colon \mathcal{O}_X(fV) \xrightarrow{\sim} \mathcal{O}_Y(V)$: restriction of scalars along the ring *isomorphism* β_V is an equivalence, and its ModuleCat-level shadow on duals is exactly 711, carrying the $\mathcal{O}_X(fV)$-linear dual onto the $\mathcal{O}_Y(V)$-linear dual, $R(V)$-linearly and invertibly. This is the dual analogue, at the module level, of the ring-isomorphism tensor equivalence ($\texttt{restrictscalars\_ringiso\_strongmonoidal}$).

The combined leg-(A)∘(B) atom $\texttt{sliceDualTransport}$. Both legs are realised together by a single $\mathcal{O}_Y(V)$-linear isomorphism $\texttt{sliceDualTransport}\,f\,M\,V$ (715), built *by hand* — it is *not* obtained by consuming a sheaf-level slice equivalence. This is essential: the equivalence of *sheaves of modules* over an open immersion compares the restriction functors and the units, but carries *no* internal-hom or dual content; the assertion realized by leg (A) — that the dual commutes with slice reindexing along the open-immersion direct-image functor $f_!$ on opens ($f.\texttt{opensFunctor}$) — cannot be supplied by such an object without a monoidal-closed structure on sheaves of modules, which this development deliberately does *not* carry (cf. 644). The atom must therefore be constructed directly.

Concretely (715), the forward component of $\texttt{sliceDualTransport}\,f\,M\,V$ at $W \le V$ is the categorical image $(\texttt{restrictScalars}\,\beta_W).\text{map}\,(\varphi_\bullet)$ of the dual-section component of φ at the f-image of W, reindexed across $f.\texttt{opensFunctor}$ by restriction of scalars along β_W, post-composed with the leg-(B) codomain swap $\texttt{inv}(\varepsilon(\texttt{restrictScalars}\,\beta_W))$. Leg (A) is built categorically via $(\texttt{restrictScalars}\,\beta_W).\text{map}$: a genuine change-of-rings functor acts here, so the reindexing must be applied as a functor map rather than pointwise. Leg (B)'s unit comparison ε is invertible because β_W is a ring isomorphism, as detailed in 715. Because Opens Y is a thin poset, the naturality squares commute by $\texttt{Subsingleton.elim}$. The forward and inverse maps identify the slice-Hom value over $\texttt{Over}\,(fV)$ with the slice-Hom value over $\texttt{Over}\,V$, $\mathcal{O}_Y(V)$-linearly.

The Step-4 residual is then $\texttt{PresheafOfModules.isoMk}$ applied *directly* to the family $V \mapsto \texttt{sliceDualTransport}\,f\,M\,V$ — which already packages *both* legs (A) and (B) — with the outer naturality square thin-poset-trivial. No separate leg-(B) step is interposed in $\texttt{isoMk}$: the per-section atom of 715 carries the full leg-(A)∘(B) content. The outer $\texttt{isoMk}$ naturality square follows from the per-section family naturality $\texttt{sliceDualTransport.naturality}$ of 715: the ε-naturality is established first at the level of individual sections, and the outer presheaf-morphism naturality of $\texttt{isoMk}$ then follows.

The residual of stage H1 ("$\texttt{pushforward}\,\beta$ commutes with the dual") is the sectionwise composite of leg (A) followed by leg (B), packaged over the slice with poset-thin naturality; the full Step-4 isomorphism is then the H1 isomorphism followed by this composite. The slice-reindexing coherence is *light*, because Opens is a thin poset and the relevant squares commute by $\texttt{Subsingleton.elim}$, whereas H1 and legs (A) and (B) carry the substantive content.

It is worth recording *why* the strong-monoidal restriction ($\texttt{presheaf\_pushforward\_adj\_substrate}$) has no direct dual analogue here. For the tensor product the transport *collapsed to leg (B) alone*: because ⊗ is sectionwise, pushing a tensor forward reduced, section by section, to a single ground-ring tensor equivalence, with no domain slice left to re-index. The dual's non-sectionwise nature removes that collapse and forces the additional leg (A): the domain $\texttt{restr}_V(f_*M)$ genuinely lives over a varying slice and must be transported before the ground ring can be reconciled.

We caution that the naive "fixed-value-category slice equivalence" of `open_immersion_slice_sheaf_equiv` is *not* applicable to leg (A). That instrument operates in a fixed value category over a fixed base ring, whereas the residual of leg (A) lives at the *presheaf* level, with a per-section ring $\mathcal{O}_Y(V)$ that varies with V and a finer slicing than the sheaf-level equivalence resolves; leg (A) must therefore be built directly at the presheaf-of-modules / varying-ring level.

Composing legs (A) and (B) gives, for each $V \subseteq U$, an $\mathcal{O}_Y(V)$-linear isomorphism between the two values; assembling these into a presheaf-of-modules natural isomorphism and sheafifying through the sheaf-level dual of 710 resolves the H1 residual. Precomposing with the stage-H1 rewrite then yields the stated isomorphism $(\texttt{dual}\,M)|_f \cong \texttt{dual}(M|_f)$.

The inverse-uniqueness shortcut is not viable. One might hope to derive the isomorphism from the already-established 646 by uniqueness of monoidal inverses — the dual being the $\otimes$-inverse of an invertible module and restriction commuting with $\otimes$ — sidestepping leg (A). The abstract statements of this idiom (`hasRightDualOfEquivalence`, `rightDualIso`) require, however, four structures the present setting lacks: a registered `MonoidalCategory` instance on `PresheafOfModules` (the monoidal structure here is carried entirely by hand), a *strong* (not merely lax/oplax) monoidal restriction functor, restriction being an *equivalence* (false for a non-surjective open immersion, which is a localization), and an exact pairing `ExactPairing` $M\,(\texttt{dual}\,M)$ (which exists only for dualizable M, whereas the lemma is stated for general M). Even restricted to the invertible consumer this would require building a coevaluation and verifying the zig-zag identities — strictly *more* infrastructure than leg (A) — so the construction commits to legs (A) and (B) above. ☐

Lemma 738 (Dual of the structure sheaf)**.** *Let Y be a scheme. The sheaf-level dual of the structure sheaf is canonically the structure sheaf itself,*

$$\texttt{dual}\,\mathcal{O}_Y \xrightarrow{\sim} \mathcal{O}_Y,$$

obtained by sheafifying the presheaf-level "evaluation at 1" isomorphism $\mathcal{H}om(\mathbf{1},\mathbf{1}) \cong \mathbf{1}$: every endomorphism of the unit presheaf is multiplication by its value at the global section 1, so evaluation at 1 identifies the internal hom of the unit into itself with the unit.

Proof. At the presheaf level the dual of the unit $\mathbf{1} = \mathcal{O}_Y$ is the internal hom $\mathcal{H}om(\mathbf{1},\mathbf{1})$, whose value on an open U is the module of $\mathcal{O}(U)$-linear endomorphisms of $\mathbf{1}|_U$. Each such endomorphism is determined by its value at the global section 1 and equals multiplication by that value; hence *evaluation at* 1 is itself the sectionwise isomorphism $\mathcal{H}om(\mathbf{1},\mathbf{1}) \cong \mathbf{1}$ of presheaves of modules, with inverse the scalar-action map $r \mapsto (\,\cdot\, r)$. (No left unitor is needed: the identification is the direct evaluation-at-1 / global-scalar-multiplication isomorphism, not the composite through $\mathbf{1}\otimes\mathbf{1} \cong \mathbf{1}$.) Sheafifying through the sheaf-level dual of 710 yields $\texttt{dual}\,\mathcal{O}_Y \cong \mathcal{O}_Y$. The scheme-level alias of this presheaf evaluation-at-1 isomorphism is `presheafDualUnitIso` (727). ☐

Lemma 739 (The dual of a line bundle is a line bundle)**.** *Source: [Stacks Project], Tag 01CR, Lemma `lemma-constructions-invertible` (item 2). Let X be a scheme and $L \in$* Scheme.`Modules` X *with* `LineBundle.IsLocallyTrivial` L*. Then the sheaf-level dual* `dual` L *of 710 is again locally trivial of rank one, i.e.* `LineBundle.IsLocallyTrivial` $(\texttt{dual}\,L)$.

Proof. Local triviality of L furnishes, around each point, an affine open U with a trivialisation $L|_U \cong \mathcal{O}_U$; writing $f : U \hookrightarrow X$ for the associated open immersion, it suffices to exhibit an isomorphism $(\texttt{dual}\,L)|_U \cong \mathcal{O}_U$. This is the following three-step chain:

$$(\texttt{dual}\,L)\big|_f \xrightarrow{\sim} \texttt{dual}(L|_f) \xrightarrow{\sim} \texttt{dual}\,\mathcal{O}_U \xrightarrow{\sim} \mathcal{O}_U.$$

1. The first isomorphism is the restriction-compatibility isomorphism 737: restriction along the open immersion f commutes with the formation of the dual. This is the C-bridge proper; its substantive content is the sectionwise ring-iso transport of the module structure (711), as detailed there.

2. The second isomorphism is the dual applied to the trivialisation $L|_U \cong \mathcal{O}_U$, via the contravariant functoriality of the sheaf-level dual in isomorphisms (`dualIsoOfIso`, 714): an isomorphism $L|_U \cong \mathcal{O}_U$ induces $\mathtt{dual}(L|_U) \cong \mathtt{dual}\,\mathcal{O}_U$.

3. The third isomorphism trivialises the dual of the unit, $\mathtt{dual}\,\mathcal{O}_U \cong \mathcal{O}_U$ (738): the dual of the structure sheaf is the structure sheaf itself, the comparison being evaluation at $1 \in \mathcal{O}_U$, under which $\mathcal{H}om_{\mathcal{O}_U}(\mathcal{O}_U, \mathcal{O}_U) \cong \mathcal{O}_U$.

Composing the three exhibits $(\mathtt{dual}\,L)|_U \cong \mathcal{O}_U$. Hence $\mathtt{dual}\,L$ is locally trivial of rank one, as required by item 2 of Stacks Lemma `lemma-constructions-invertible` (tag 01CR). ☐

Lemma 740 (Local isomorphisms of $\mathcal{O}_X$-modules glue to a global isomorphism)**.** *Let X be a scheme and $\varphi\colon M \to N$ a morphism in* Scheme.`Modules` X = `SheafOfModules` X.`ringCatSheaf`. *Suppose that for every point $x \in X$ there is an open neighbourhood U_x of x such that the restriction of φ to U_x — the image* (`restrictFunctor` (U_x).`i`).`map` φ *under the restriction functor along the open immersion* (U_x).`i`: $U_x \to X$ *— is an isomorphism in* Scheme.`Modules` U_x. *Then φ is an isomorphism.*

Proof. Being an isomorphism for a morphism of sheaves of $\mathcal{O}_X$-modules can be checked on the underlying morphism of sheaves of abelian groups, and there it can be checked stalkwise. Fix a point $x \in X$. Choose a preimage $x' \in U_x$ of x under the open immersion (U_x).`i`. By hypothesis the restriction (`restrictFunctor` (U_x).`i`).`map` φ is an isomorphism, hence so is its image under any functor — in particular under the stalk-at-x' functor on the underlying ab-sheaves. The natural isomorphism "restriction commutes with stalks" (`restrictStalkNatIso`) identifies this with the stalk of (the underlying ab-sheaf morphism of) φ at the image point x; transporting along it shows that stalk is an isomorphism. As this holds at every x, and a morphism of sheaves of abelian groups all of whose stalks are isomorphisms is itself an isomorphism (stalkwise iso-detection, `isIso_of_stalkFunctor_map_iso` on `TopCat.Sheaf`), the underlying ab-sheaf morphism of φ is an isomorphism. Finally the forgetful functor Scheme.`Modules.toPresheaf` reflects isomorphisms, so φ itself is an isomorphism. ☐

Lemma 741 (Down-set image identity $\iota_!(\iota^{-1}V) = V$)**.** *For opens $V \le W$ of a scheme X, the image under the open immersion $W.\iota$ of the preimage of V is V again: $W.\iota_!(W.\iota^{-1}(V)) = V$. This is the equality powering the* `eqToHom`*-conjugations of* `homLocalSection` *(`scheme_modules_hom_local_section`) and of the reverse slice transport (728).*

Proof. Proved directly in Lean — the image of the preimage is the intersection with the (open) range of the immersion, which equals V since $V \le W$ lies in that range. ☐

Definition 742 (Local section of the hom-sheaf from a compatible family)**.** For a family of open subsets $\{U_i\}_{i\in\iota}$ of a scheme X and a compatible family of morphisms $f_i : M|_{U_i} \to N|_{U_i}$ of $\mathcal{O}_X$-modules, the local section 'homLocalSection' packages the section of the hom-sheaf over U_i whose value on a smaller open $V \le U_i$ is obtained by conjugating the component of f_i along the canonical down-set identity $\iota_!(\iota^{-1}V) = V$. The naturality in the slice is automatic on the thin poset of opens once the down-set identity is in place.

Definition 743 (Top section to global hom)**.** A section of the hom-sheaf over the terminal open $\top$ determines a global morphism of presheaves $F \to G$ by the standard sections-to-morphisms equivalence: 'topSectionToHom' is the transport of that section through the canonical hom-sections equivalence, so that a compatible top section yields a global morphism.

Lemma 744 (Sectionwise value of 'topSectionToHom')**.** *Let s be a section over the terminal open. Then the component of the induced morphism at an open W is exactly the W-component of s, evaluated at the unique map $W \to \top$. In particular, the recovered morphism is sectionwise the same as the input section, just reindexed along the terminal inclusion.*

Definition 745 (Promote an $\mathcal{O}_X$-linear presheaf morphism to a module morphism)**.** A presheaf morphism of the underlying abelian-group sheaves that is sectionwise $\mathcal{O}_X$-linear packages as a morphism of $\mathcal{O}_X$-modules. This is the promotion step that turns the glued Ab-valued morphism into an actual module morphism.

Lemma 746 (Compatible local morphisms glue to a global morphism)**.** *Given a cover $\{U_i\}$ of X, a family of local morphisms $f_i : M|_{U_i} \to N|_{U_i}$ that agree on overlaps glues to a unique global $\mathcal{O}_X$-linear morphism $M \to N$. The proof glues the associated hom-sheaf sections using the sheaf property, converts the glued top section to a global morphism, and checks sectionwise $\mathcal{O}_X$-linearity on each open of the cover.*

Lemma 747. *[The glued morphism restricts to its local datum] In the situation of 746 — a cover $\{U_i\}$ of X and a compatible family $f_i : M|_{U_i} \to N|_{U_i}$ — the restriction of the glued global morphism back to a cover member recovers its local datum:*

$$(\texttt{restrictFunctor}\, U_i).\texttt{map}\,(\texttt{homOfLocalCompat}\, f) \;=\; f_i.$$

This is the companion gluing-value identity needed to apply the B-connector 740: when every f_i is an isomorphism, this identity transports that to the restriction of the glued morphism, so the global morphism is locally an isomorphism and hence an isomorphism.

Proof. By construction, `homOfLocalCompat` produces its global morphism by gluing the local sections of the hom-sheaf associated to each f_i along the cover; the sheaf gluing axiom states that the resulting global section restricts on U_i to the section corresponding to f_i. Converting the global section back to a morphism is sectionwise faithful (744), and the section-to-morphism construction preserves the underlying section, so the restriction of the global morphism to U_i is sectionwise equal to f_i — hence equal as a morphism. ☐

Remark 748 (How the dual infrastructure discharges the $\otimes$-inverse)**.** Once the dual package 739 and the A-bridge 746 are available, 656 is discharged by taking $L^{-1} := \texttt{dual}\, L$ (710), which is a line bundle by 739. The global contraction $\varepsilon_L : L \otimes_X L^{-1} \to \mathcal{O}_X$ is *not* obtained by sheafifying the presheaf-level evaluation; it is instead built by *gluing the canonical local contraction morphisms.* On each trivialising open U, the restriction-compatibility iso 646 carries $(L \otimes_X L^{-1})|_U$ to $\mathcal{O}_U \otimes_U \mathcal{O}_U$, and the canonical local contraction $\mathcal{O}_U \otimes_U \mathcal{O}_U \to \mathcal{O}_U$ of the free rank-one module with its dual is the left unitor 654, an isomorphism. These local contractions are canonical, hence agree on overlaps, and so glue — by 746 — to a single global morphism $\varepsilon_L : L \otimes_X L^{-1} \to \mathcal{O}_X$. Since "being an isomorphism" is local on X and ε_L restricts to an isomorphism on each U, it is a global isomorphism $L \otimes_X L^{-1} \xrightarrow{\sim} \mathcal{O}_X$ by the B-connector 740.

This descent construction avoids the "sheafify-the-evaluation" route, which would require commuting the sheafification unit past tensor-stalk whiskering; the gluing strategy computes no tensor stalk anywhere.

Two conditions make the gluing of 746 applicable. First, *overlap compatibility*: on each overlap $V \subseteq U_i \cap U_j$ the sectionwise restrictions of the two local contractions f_i and f_j to V must agree. They do because each f_x is the canonical evaluation $L \otimes L^{-1} \to \mathcal{O}$, independent of the chosen trivialisation: the dual trivialisation e_x^N is built from the primal e_x^M via 646, `tensorObjIsoOfIso`, `dualIsoOfIso`, and 654, so the transition isomorphism g_{ij} entering e_x^M and its inverse g_{ij}^{-1} entering e_x^N cancel through `tensorObjIsoOfIso`; this cancellation is a genuine computation of abelian-group homomorphisms. Concretely, the mechanism realising the cancellation is the *functoriality of the iso-transport operators*: bifunctoriality of `tensorObjIsoOfIso` (863, 864) and contravariant functoriality of `dualIsoOfIso` (869, 870). Pushing both legs to a pure tensor $a \otimes b$, these laws rewrite e_x^N so that the g_{ij} inside e_x^M meets $g_{ij}^{-1} =$ `dualIsoOfIso` (g_{ij}) inside e_x^N as adjacent factors, where `tensorObjIsoOfIso` g_{ij} (`dualIsoOfIso` g_{ij}) followed by `tensorObj_unit_iso` reduces to `tensorObj_unit_iso` alone (the unit self-duality evaluation collapse); the iso-level identities above thereby reduce sectionwise to the required cocycle equality. Second, the *local-iso* step: by the gluing-value identity 747, the restriction of ε_L to U_x equals the local isomorphism f_x, so 740 upgrades ε_L to a global isomorphism.

The two mechanism steps named in the overlap-compatibility argument of 748 are isolated below as formal target lemmas, so the cocycle obligation (residual A) of 656 has a named formalization target for each. The iso-transport operators `tensorObjIsoOfIso` and `dualIsoOfIso` and the unit comparison `tensorObj_unit_iso` are the existing helpers (863, 869, 654).

The five restriction-naturality squares constituting 841 are isolated here, *in chain order*, as named, independently-provable sub-lemmas; the target then telescopes them. Throughout, fix opens $V \subseteq U$ of X and write $j : V \hookrightarrow U$ for the open immersion Scheme.`Hom.resLE` $(\mathrm{id}_X)\, U\, V$, the chart morphism with $j \,;\, \iota_U = \iota_V$; restriction along j is the functor $(-)|_j =$ `restrict` j. The two a-priori-distinct opens $\iota_U^{-1}(V)$ and $\iota_V^{-1}(V)$ are identified as V only through the direct-image equality $\iota_U{}''(\iota_U^{-1} V) = V = \iota_V{}''(\iota_V^{-1} V)$ (741); write ϱ for the canonical equality-of-opens reindexing isomorphism it induces, sitting on *both* flanks of each constituent's square.

The four squares S2/S3/S4a/S4b are *not* four independent chases: each is a *base-change naturality* square for one and the same comparison morphism — the sheaf-level pullback–tensor map `pullbackTensorMap` f (667) — transported from the pullback world onto the restriction world along the comparison `restrictFunctorIsoPullback` f : `restrict` $f \cong$ `pullback` f (769) and reindexed by ϱ. For an open immersion f the comparison `pullbackTensorMap` f is invertible (695, a public witness), so every transported square is a genuine isomorphism square. Crucially there is *no* monoidal structure on the sheaf categories Scheme.`Modules` in play: the only monoidal data used lives one level down, on presheaves of modules, where the pullback is oplax monoidal with oplax tensorator $\delta =$ `pullbackTensorMap`, and the whole coherence is governed by uniqueness of left adjoints rather than by a sheaf-level `MonoidalCategory`.

Two supporting bridges organise the squares. *Bridge B1* (776) identifies `tensorObj_restrict_iso` f with the transported comparison: it is the *direct* conjugate factorisation `tensorObj_restrict_iso` $f =$ `restrictFunctorIsoPullback` f; `asIso`(`pullbackTensorMap` f) ; (per-leg reindex), proved without any sheaf-level monoidal transport — both sides share the `restrictFunctorIsoPullback` f; `sheafificationCompPullback` f prefix, and after cancelling it the residue is the presheaf-level identity that the sheafified oplax tensorator $\delta($`pullback` $\varphi')$ equals the left-adjoint-transported tensorator building the tail of `tensorObj_restrict_iso`, discharged by the δ-conjugation identity (694) together with uniqueness of left adjoints. *Bridge B2* (752) is the reindex coherence: `restrictFunctorIsoPullback` : `restrict` $f \cong$ `pullback` f (the left-adjoint uniqueness isomorphism) is pseudonatural across $j \,;\, \iota_U$, identifying $\varrho =$ `restrictCompReindex` j (749) with `pullbackComp` conjugated by `restrictFunctorIsoPullback` on each factor; its closed form is `leftAdjointUniq_trans` threaded by `mateEquiv_vcomp`. With B1 in hand each of

S2/S3/S4a/S4b reduces to pure base-change bookkeeping: substitute B1 to turn both the U- and V-built `tensorObj_restrict_iso` into `asIso`(`pullbackTensorMap`) conjugates, whereupon the morphism equation is the already-proven composition law `pullbackTensorMap_restrict` (685) together with naturality `pullbackTensorMap_natural` (672), transported back along `restrictFunctorIsoPullback` and reindexed by ϱ. The remaining global-unit square S4c rests on B2 alone, as the unit composition law `pullbackObjUnitToUnit_comp` (670) re-wrapped by the reindex coherence. The constituents are therefore landed in the dependency order

$$\mathrm{B2} \to \mathrm{B1} \to \mathrm{S4c} \to \mathrm{S2} \to \mathrm{S4b} \to \mathrm{S3} \to \mathrm{S4a}.$$

Definition 749 (Restriction-composite reindexing ϱ)**.** Let $j : V \hookrightarrow U$ be an open immersion of opens of X with $j ; \iota_U = \iota_V$, and let $A \in \mathrm{Scheme}.\mathtt{Modules}\, X$. The *restriction-composite reindexing* is the isomorphism

$$\varrho_A := \mathtt{restrictCompReindex}\, j\, A : A|_{\iota_V} \;\cong\; (A|_{\iota_U})\big|_j,$$

defined as the composite $(\mathtt{restrictFunctorCongr}\, h_{j\iota})^{-1}$;$\mathtt{restrictFunctorComp}\, j\, \iota_U$: the (Mathlib) restriction-composition isomorphism `restrictFunctorComp` precomposed with the congruence along $j ; \iota_U = \iota_V$. It is the ϱ appearing on both flanks of every square below.

Definition 750 (Unit-restriction identification u_ι)**.** For an open immersion $f : Y \hookrightarrow X$ the restriction of the global unit to Y is the structure sheaf of Y: the *unit-restriction identification* is

$$u_\iota(f) := (\mathtt{restrictFunctorIsoPullback}\, f).\mathrm{app}\, \mathcal{O}_X \,;\, \mathtt{pullbackUnitIso}\, f : \mathcal{O}_X|_f \;\cong\; \mathcal{O}_Y,$$

the comparison `restrictFunctorIsoPullback` into the pullback world followed by the structure-sheaf comparison `pullbackUnitIso` (671). It is the unit identification on the chart-scheme side of S4a/S4b and the final leg u_ι^{-1} of S4c.

Lemma 751 (Bridge B2, general literal composite: `restrictFunctorIsoPullback` across a composite of immersions)**.** *Source: internal categorical construction; no external reference. Let $f : Z \hookrightarrow Y$ and $g : Y \hookrightarrow X$ be composable open immersions (*general *— not tied to the chart j nor to any equality of opens) and $A \in$* Scheme.`Modules` X. *The restriction-to-pullback comparison* `restrictFunctorIsoPullback` *(769) is pseudonatural across the* literal *composite $f ; g$: it carries the restriction-world composition isomorphism* `restrictFunctorComp` $f\, g$ *(792) onto the pullback-world composition isomorphism* `pullbackComp` $f\, g$*. Concretely, the leading* `restrictFunctorIsoPullback` *factors of the two flanks intertwine* `restrictFunctorComp` $f\, g$ *with* `pullbackComp` $f\, g$*, the head being the* bare `restrictFunctorComp` $f\, g$ *(not the chart reindexing* `restrictCompReindex` *of 749), so that* no `pullbackCongr`/`restrictFunctorCongr` *congruence legs occur. The* hom*-component is recorded separately as* `restrictFunctorIsoPullback_comp_general_hom` *— the* .hom *of this iso identity.*

Proof. This is the general pseudofunctoriality coherence of the comparison `restrictFunctorIsoPullback`: both `restrictFunctorComp` $f\, g$ and `pullbackComp` $f\, g$ are the canonical isomorphisms comparing the base change along a composite with the composite of base changes, and `restrictFunctorIsoPullback` intertwines them by naturality of the comparison in each factor. Since $f;g$ is the literal composite — with no equality-of-opens substitution interposed — there is no congruence leg to thread, and the two composition isomorphisms correspond directly. This is the general (chart-free) form of the reindex coherence; the chart variant 752 is its specialisation along $j ; \iota_U = \iota_V$, where the additional congruence legs appear. ∎

Lemma 752 (Bridge B2: reindex coherence of `restrictFunctorIsoPullback`)**.** *Let $j : V \hookrightarrow U$ with $j \,;\, \iota_U = \iota_V$. The natural isomorphism* `restrictFunctorIsoPullback` $:$ `restrict` $f \cong$ `pullback` f *is pseudonatural across the composite $j \,;\, \iota_U$: the restriction-world reindexing $\varrho =$* `restrictCompReindex` j *(749) corresponds, under* `restrictFunctorIsoPullback` *applied on each factor, to the pullback-world reindexing* `pullbackComp` $j\, \iota_U$*, threaded by the equality-of-opens transport along $j \,;\, \iota_U = \iota_V$ (741).*

Proof. This is a general coherence of the pseudofunctor comparison `restrictFunctorIsoPullback`. Both `restrictFunctorComp` and `pullbackComp` are the canonical isomorphism comparing a composite base change with the composite of base changes, and `restrictFunctorIsoPullback` intertwines them by naturality of the comparison in each factor. The equality-of-opens transport of $j \,;\, \iota_U = \iota_V$ matches on the two sides because both reindexings are defined by transport along the *same* equality of opens. The isomorphism-existence shadow of this identity is already used in 698; the present statement promotes it to an equality of morphisms. In closed form this equality holds at the level of natural transformations: it is exactly the hom-level factorisation $(\texttt{restrictFunctorIsoPullback}\, \iota_V).\mathrm{hom.app}\, A = c_1 \,;\, c_2 \,;\, c_3 \,;\, c_4 \,;\, c_5 \,;\, c_6$ of 753, evaluated at A. That factorisation is obtained not from a horizontal-composite mate identity but by conjugating both sides onto the common right adjoint `pushforward` ι_V and distributing the conjugation *leg-by-leg* through `conjugateEquiv_comp` (762): every intermediate functor is a left adjoint X`.Modules` $\to V$`.Modules` of the common `pushforward` ι_V, each leg maps to a known pushforward-world morphism, and their reversed product telescopes to the identity. The app A statement of B2 then follows by transporting that natural-transformation identity across the unit triangle of `restrictFunctorIsoPullback` ι_V (764), together with the transitivity `leftAdjointUniq_trans` (760) aligning the two transports of the comparison. ☐

Lemma 753 (Bridge B2, hom level: leg-by-leg factorisation of `restrictFunctorIsoPullback`)**.** *Let $j : V \hookrightarrow U$ with $j \,;\, \iota_U = \iota_V$ and let $A \in \mathrm{Scheme}.\texttt{Modules}\, X$. Evaluated at A, the single restriction-to-pullback comparison map factors as the composite of the six reindex legs*

$$(\texttt{restrictFunctorIsoPullback}\, \iota_V).\mathrm{hom.app}\, A \;=\; c_1 \,;\, c_2 \,;\, c_3 \,;\, c_4 \,;\, c_5 \,;\, c_6,$$

where $c_1 = ((\texttt{restrictFunctorCongr}\, h_{j\iota})^{-1}).\mathrm{hom.app}\, A$, $c_2 = (\texttt{restrictFunctorComp}\, j\, \iota_U).\mathrm{hom.app}\, A$, $c_3 = (\texttt{restrict}\, j)((\texttt{restrictFunctorIsoPullback}\, \iota_U).\mathrm{hom.app}\, A)$, $c_4 = (\texttt{restrictFunctorIsoPullback}\, j).\mathrm{hom.app}$ $c_5 = (\texttt{pullbackComp}\, j\, \iota_U).\mathrm{hom.app}\, A$ *and* $c_6 = (\texttt{pullbackCongr}\, h_{j\iota}).\mathrm{hom.app}\, A$*. This is the genuine natural-transformation content to which Bridge B2 (752) reduces.*

Proof. It suffices to prove the underlying identity of natural transformations `restrictFunctor` $\iota_V \Rightarrow$ `pullback` ι_V; the app A statement follows by evaluating the whisker legs c_3, c_4 componentwise. Every functor in the right-hand chain is a functor X`.Modules` $\to V$`.Modules` and a *left* adjoint of the common right adjoint `pushforward` ι_V; hence both sides are determined by their image under the conjugation bijection `conjugateEquiv` (`pullbackPushforwardAdjunction` ι_V) (`restrictAdjunction` ι_V) onto natural transformations `pushforward` $\iota_V \Rightarrow$ `pushforward` ι_V, and it is enough to compare the two images.

The left-hand image is $\mathbf{1}_{\texttt{pushforward}\, \iota_V}$: the comparison `restrictFunctorIsoPullback` ι_V is itself the left-adjoint-uniqueness comparison, and the conjugate of such a hom is the identity. On the right-hand side the chain factors through the seven intermediate left adjoints of `pushforward` ι_V, $G_0 =$ `restrictAdjunction` ι_V, $G_1 =$ `restrictAdjunction` $(j \,;\, \iota_U)$, $G_2 =$ (`restrictAdjunction` ι_U).comp (`restrictAdjunction` j), $G_3 =$ (`pullbackPushforwardAdjunction` ι_U).comp (`restr` $G_4 =$ (`pullbackPushforwardAdjunction` ι_U).comp (`pullbackPushforwardAdjunction` j), $G_5 =$ `pullbackPushforwardAdjunction` $(j \,;\, \iota_U)$, $G_6 =$ `pullbackPushforwardAdjunction` ι_V, so the conjugation distributes *leg-by-leg* through the composition law `conjugateEquiv_comp` (762); no horizontal-composite mate identity is required. Each leg is a known pushforward-world map:

- $c_2 \mapsto (\texttt{pushforwardComp}\, j\, \iota_U)$.hom by the keystone 768;
- $c_5 \mapsto (\texttt{pushforwardComp}\, j\, \iota_U)$.inv by 754;
- $c_3, c_4 \mapsto$ the pushforward-whiskered units of ι_U and j by 756 and 757;
- $c_1, c_6 \mapsto$ the pushforward congruences along $j \,;\, \iota_U = \iota_V$ by 758.

The resulting reversed product of pushforward-world legs telescopes to $\mathbf{1}_{\texttt{pushforward}\,\iota_V}$ by 759: the two `pushforwardComp` legs are mutually inverse, and the unit and congruence legs cancel against the reindexing $j \,;\, \iota_U = \iota_V$ (741). Both conjugate images equal $\mathbf{1}_{\texttt{pushforward}\,\iota_V}$, so the two natural transformations agree. □

The telescope above is assembled from the following atomic per-leg conjugate computations, each a small fail-fast obligation shared with the Bridge B1 crux (774).

Lemma 754 (c_5: conjugate of the pullback-composition hom)**.** *Let $f : X \hookrightarrow Y$, $g : Y \hookrightarrow Z$ be open immersions. Under the conjugation between the composite pullback adjunction* $(\texttt{pullbackPushforwardAdjunction}\, g)$.c *and the pullback adjunction of the composite, the hom of the pullback-composition isomorphism corresponds to the* inverse *of the pushforward-composition isomorphism:*

$$\texttt{conjugateEquiv}\ \ldots\ (\texttt{pullbackComp}\, f\, g).\mathrm{hom} \;=\; (\texttt{pushforwardComp}\, f\, g).\mathrm{inv}.$$

Proof. The conjugation bijection is an additive-inverse-respecting equivalence on the relevant hom-groups only formally; concretely, apply it to the identity $(\texttt{pullbackComp}\, f\, g).\mathrm{hom}; (\texttt{pullbackComp}\, f\, g).\mathrm{inv} = \mathbf{1}$ and use that the conjugate of $(\texttt{pullbackComp}\, f\, g)$.inv is $(\texttt{pushforwardComp}\, f\, g)$.hom (763). Since `conjugateEquiv` sends the hom of an isomorphism to a map inverse to the image of its inverse, the image of $(\texttt{pullbackComp}\, f\, g)$.hom is the inverse of $(\texttt{pushforwardComp}\, f\, g)$.hom, namely $(\texttt{pushforwardComp}\, f\, g)$.inv. □

Lemma 755 (LHS-collapse: the conjugate of the comparison hom is the identity)**.** *For an open immersion f, the conjugate of the comparison hom under* $\texttt{conjugateEquiv}\, (\texttt{pullbackPushforwardAdjunction}\, f)$ (res *is the identity:*

$$\texttt{conjugateEquiv}\, ((\texttt{restrictFunctorIsoPullback}\, f).\mathrm{hom}) \;=\; \mathbf{1}_{\texttt{pushforward}\, f}.$$

This is the left-hand collapse used by the B2 telescope and by both whisker legs.

Proof. By definition $\texttt{restrictFunctorIsoPullback}\, f$ is the left-adjoint-uniqueness comparison isomorphism of $\texttt{restrictAdjunction}\, f$ against $\texttt{pullbackPushforwardAdjunction}\, f$. The conjugation bijection sends the hom of this uniqueness comparison to the identity natural transformation of the common right adjoint $\texttt{pushforward}\, f$, since conjugation transports the defining unit-triangle of the comparison to the identity. □

Lemma 756 (c_3: conjugate of the ι_U-comparison whiskered by $\texttt{restrict}\, j$)**.** *For the chart $j : V \hookrightarrow U$, the leg $c_3 = \texttt{whiskerRight}\, (\texttt{restrictFunctorIsoPullback}\, \iota_U).\mathrm{hom}\, (\texttt{restrict}\, j)$ maps, under the conjugation through the adjunctions $G_2 \to G_3$, to the j-pushforward of the unit-triangle datum of $\texttt{restrictFunctorIsoPullback}\, \iota_U$ — i.e. to the pushforward-world transport of the left-adjoint-uniqueness unit of the ι_U comparison.*

Proof. Since $\texttt{restrictFunctorIsoPullback}\, \iota_U$ is the left-adjoint-uniqueness comparison of $\texttt{restrictAdjunction}\, \iota_U$ against $\texttt{pullbackPushforwardAdjunction}\, \iota_U$, its conjugate is computed by the defining unit triangle `unit_leftAdjointUniq_hom_app` (764); the outer `whiskerRight` by $\texttt{restrict}\, j$ transports this through the second factor of G_2, G_3, yielding the displayed j-pushforward whisker of the unit. □

Lemma 757 (c_4: conjugate of the j-comparison whiskered into $\texttt{pullback}\,\iota_U$)**.** *For the chart $j : V \hookrightarrow U$, the leg $c_4 = \texttt{whiskerLeft}\,(\texttt{pullback}\,\iota_U)\,(\texttt{restrictFunctorIsoPullback}\,j).\mathrm{hom}$ maps, under the conjugation through the adjunctions $G_3 \to G_4$, to the unit-triangle datum of* $\texttt{restrictFunctorIsoPullback}\,j$ *precomposed with* $\texttt{pullback}\,\iota_U$ *— the pushforward-world transport of the left-adjoint-uniqueness unit of the j comparison.*

Proof. As in 756, the comparison $\texttt{restrictFunctorIsoPullback}\,j$ is a $\texttt{leftAdjointUniq}$ hom; its conjugate is given by the unit triangle $\texttt{unit\_leftAdjointUniq\_hom\_app}$ (764). The $\texttt{whiskerLeft}$ by $\texttt{pullback}\,\iota_U$ threads this datum through the first factor of G_3, G_4, giving the displayed map. □

Lemma 758 (c_1, c_6: conjugate of the reindex congruences along $j\,;\iota_U = \iota_V$)**.** *The two flanking congruence legs — c_1 the inverse hom of $\texttt{restrictFunctorCongr}\,h_{j\iota}$ (relating G_0 and G_1) and c_6 the hom of $\texttt{pullbackCongr}\,h_{j\iota}$ (relating G_5 and G_6) — both arise by transport along the single equality of opens $j\,;\iota_U = \iota_V$ (741). Under conjugation they map to the corresponding* $\texttt{pushforward}$ *congruences (*$\texttt{eqToHom}$*-transports of* $\texttt{pushforward}$ *along that same equality), inverse to one another modulo the reindex.*

Proof. Both congruences are $\texttt{eqToHom}/\texttt{eqToIso}$ transports along $j\,;\iota_U = \iota_V$; conjugation commutes with such equality-of-functor transports, sending each to the $\texttt{pushforward}$ congruence along the same equality. As the two endpoints G_0, G_6 of the chain are both $\texttt{pushforward}\,\iota_V$, the two congruence images are mutually inverse transports and cancel in the telescope. □

Lemma 759 (Telescope of the reindexed pushforward legs)**.** *With $j;\iota_U = \iota_V$, the reversed product of the six per-leg conjugate images — the two* $\texttt{pushforward}$ *congruences along $j\,;\iota_U = \iota_V$, the two whiskered units, and the pair $(\texttt{pushforwardComp}\,j\,\iota_U).\mathrm{hom}\,;(\texttt{pushforwardComp}\,j\,\iota_U).\mathrm{inv}$ — equals the identity natural transformation $\mathbf{1}_{\texttt{pushforward}\,\iota_V}$.*

Proof. The central pair $(\texttt{pushforwardComp}\,j\,\iota_U).\mathrm{hom}\,;(\texttt{pushforwardComp}\,j\,\iota_U).\mathrm{inv}$ cancels to the identity. The two whiskered unit legs are then absorbed by the pushforward triangle identities, leaving the two $\texttt{pushforward}$ congruences along $j\,;\iota_U = \iota_V$ (741), which are mutually inverse transports and cancel. The product is therefore $\mathbf{1}_{\texttt{pushforward}\,\iota_V}$. □

Bridge B2 (the reindex coherence of $\texttt{restrictFunctorIsoPullback}$) rests on the following Mathlib facts about uniqueness of left adjoints and the compositionality of mates, recorded here as dependency anchors.

Lemma 760 (Transitivity of the left-adjoint uniqueness isomorphism)**.** *Provided by Mathlib. For adjunctions $F \dashv G$, $F' \dashv G$, $F'' \dashv G$ sharing the right adjoint, the left-adjoint uniqueness comparisons compose:*

$$(\texttt{leftAdjointUniq}\,\mathrm{adj}_1\,\mathrm{adj}_2)_{\mathrm{hom}};(\texttt{leftAdjointUniq}\,\mathrm{adj}_2\,\mathrm{adj}_3)_{\mathrm{hom}} = (\texttt{leftAdjointUniq}\,\mathrm{adj}_1\,\mathrm{adj}_3)_{\mathrm{hom}}.$$

Lemma 761 (Mate of a horizontal composite is the vertical composite of mates)**.** *Provided by Mathlib. For adjunctions $\mathrm{adj}_1, \mathrm{adj}_2, \mathrm{adj}_3$ and two-squares α, β, the mate of the horizontal composite equals the vertical composite of the mates: $\texttt{mateEquiv}\,\mathrm{adj}_1\,\mathrm{adj}_3\,(\alpha \cdot_h \beta) = (\texttt{mateEquiv}\,\mathrm{adj}_1\,\mathrm{adj}_2\,\alpha) \cdot_v (\texttt{mateEquiv}\,\mathrm{adj}_2\,\mathrm{adj}_3\,\beta)$.*

Lemma 762 (Conjugation distributes over composition)**.** *Provided by Mathlib. For adjunctions $\mathrm{adj}_1 : L_1 \dashv R_1$, $\mathrm{adj}_2 : L_2 \dashv R_2$, $\mathrm{adj}_3 : L_3 \dashv R_3$ all between the same pair of categories, and natural transformations $\alpha : L_2 \Rightarrow L_1$, $\beta : L_3 \Rightarrow L_2$, the conjugation bijection takes a composite to the (reversed) composite of conjugates:*

$$\texttt{conjugateEquiv}\,\mathrm{adj}_1\,\mathrm{adj}_2\,\alpha\,;\texttt{conjugateEquiv}\,\mathrm{adj}_2\,\mathrm{adj}_3\,\beta \;=\; \texttt{conjugateEquiv}\,\mathrm{adj}_1\,\mathrm{adj}_3\,(\beta\,;\alpha).$$

Lemma 763 (Conjugate of the pullback-composition inverse)**.** *Provided by Mathlib. For open immersions f, g of schemes, under the conjugation between the composite pullback adjunction* ($\texttt{pullbackPushforwardAdjunction}\, g$).comp ($\texttt{pullbackPushforwardAdjunction}\, f$) *and the pullback adjunction of the composite, the* inverse *of the pullback-composition isomorphism corresponds to the hom of the pushforward-composition isomorphism:*

$$\texttt{conjugateEquiv}\ \ldots\ (\texttt{pullbackComp}\, f\, g).\mathrm{inv} \ = \ (\texttt{pushforwardComp}\, f\, g).\mathrm{hom}.$$

It holds on the nose because `pullbackComp` *is* defined *as the left-adjoint composition isomorphism* `leftAdjointCompIso`*.*

Lemma 764 (Unit triangle of the left-adjoint uniqueness comparison)**.** *Provided by Mathlib. For adjunctions* $\mathrm{adj}_1 : F \dashv G$ *and* $\mathrm{adj}_2 : F' \dashv G$ *sharing the right adjoint, the left-adjoint uniqueness comparison satisfies the unit triangle*

$$\mathrm{adj}_1.\mathrm{unit.app}\, x\, ;\, G((\texttt{leftAdjointUniq}\, \mathrm{adj}_1\, \mathrm{adj}_2).\mathrm{hom.app}\, x) \ = \ \mathrm{adj}_2.\mathrm{unit.app}\, x.$$

Lemma 765 (Hom-transpose of the left-adjoint uniqueness comparison is the second unit)**.** *Provided by Mathlib. For adjunctions* $\mathrm{adj}_1 : F \dashv G$ *and* $\mathrm{adj}_2 : F' \dashv G$ *sharing the right adjoint, applying* $\mathrm{adj}_1.\mathrm{homEquiv}$ *to the component of the left-adjoint uniqueness comparison* $\texttt{leftAdjointUniq}\, \mathrm{adj}_1\, \mathrm{adj}_2$ *at an object x returns the unit of* adj_2 *at x:*

$$\mathrm{adj}_1.\mathrm{homEquiv}((\texttt{leftAdjointUniq}\, \mathrm{adj}_1\, \mathrm{adj}_2).\mathrm{hom.app}\, x) \ = \ \mathrm{adj}_2.\mathrm{unit.app}\, x.$$

Lemma 766 (Counit triangle of the left-adjoint uniqueness comparison)**.** *Provided by Mathlib. For adjunctions* $\mathrm{adj}_1 : F \dashv G$ *and* $\mathrm{adj}_2 : F' \dashv G$ *sharing the right adjoint, the left-adjoint uniqueness comparison satisfies the counit triangle*

$$(\texttt{leftAdjointUniq}\, \mathrm{adj}_1\, \mathrm{adj}_2).\mathrm{hom.app}\, (G.\mathrm{obj}\, x)\, ;\, \mathrm{adj}_2.\mathrm{counit.app}\, x \ = \ \mathrm{adj}_1.\mathrm{counit.app}\, x.$$

Lemma 767 (Generic adjunction unit/counit collapse)**.** *Source: internal categorical construction; no external reference. Let* $\mathrm{adj} : L \dashv R$ *be an adjunction between categories C and D, let $P \in C$, $B \in D$, and let $g : L.\mathrm{obj}\, P \to B$. Then transporting g to its unit form and collapsing through the counit recovers g:*

$$L.\mathrm{map}(\mathrm{adj.unit.app}\, P ; R.\mathrm{map}\, g)\, ;\, \mathrm{adj.counit.app}\, B \ = \ g.$$

This is the round-trip identity of the hom-set adjunction equivalence ($\mathrm{adj.homEquiv}$*) read at the morphism g: the left factor is the unit-presentation* $\mathrm{adj.homEquiv}\, g$ *of g's transpose, and post-composing the counit is the inverse transpose, so the composite is the identity round trip. Stated abstractly over arbitrary* [`Category`] *so that its applications cross the* `SheafOfModules` *sheafification seam by definitional unification rather than a keyed rewrite.*

Proof. Pure category algebra. Rewrite the unit-form factor as the adjunction transpose $\mathrm{adj.unit.app}\, P; R.\mathrm{map}\, g = \mathrm{adj.homEquiv}\, P B\, g$ (the unit triangle of homEquiv); the left-hand side becomes the counit-form of $\mathrm{adj.homEquiv}\, P B\, g$, which is the inverse transpose $(\mathrm{adj.homEquiv})^{-1}$ applied to $\mathrm{adj.homEquiv}\, g$, i.e. g by the round-trip law of the equivalence. ☐

Lemma 768 (Restrict-side conjugate of the restriction-composition isomorphism)**.** *Let $f : X \hookrightarrow Y$ and $g : Y \hookrightarrow Z$ be open immersions. Under the conjugation equivalence between the composite restriction adjunction* ($\texttt{restrictAdjunction}\, g$) $\circ$ ($\texttt{restrictAdjunction}\, f$) *and the restriction* *adjunction of the composite* `restrictAdjunction` $(f\,;g)$*, the hom of the restriction-composition isomorphism corresponds to the hom of the pushforward-composition isomorphism:*

$$\texttt{conjugateEquiv}\ ((\texttt{restrictAdjunction}\,g).\mathrm{comp}\,(\texttt{restrictAdjunction}\,f))\ (\texttt{restrictAdjunction}\,(f;g))\ (\texttt{restrict}$$

This is the restriction-world mirror of Mathlib's `conjugateEquiv_pullbackComp_inv` *(which is available for the pullback world only because* `pullbackComp` *is* definitionally equal *to* `leftAdjointCompIso`*, so the pullback identity is immediate from the definition). It is the foundational lemma on which both Bridge B2 (752) and the Bridge B1 crux* `H1inv_app_eq_pullbackVal_restrict` *depend.*

Proof. Both `restrictFunctorComp` and `pushforwardComp` are sectionwise `eqToHom`/identity maps: on each open the restriction-composition hom is `presheaf.map` of an equality-of-opens `eqToHom`, and the pushforward-composition hom is the identity. The composite restriction adjunction's unit and counit are sectionwise the presheaf restriction maps `presheaf.map` (`homOfLE` $\cdots$) along the canonical inclusions of preimage opens. Evaluating the `conjugateEquiv` app-formula therefore reduces the claim to a composite endomorphism of each section $\Gamma(M', U)$ built entirely from `presheaf.map` of morphisms of the thin poset $\texttt{Opens}^{\mathrm{op}}$. By functoriality these collapse to `presheaf.map` of a single opens-morphism, which is forced to be the identity by thin-poset uniqueness (every parallel pair of opens-morphisms is equal), whence the composite is `presheaf.map` $\mathbf{1} = \mathbf{1}$. The computation is carried out at the level of `presheaf.map` — composing opens-morphisms before applying the functor, so the module carrier is never forced — which keeps it free of the `pushforward`-carrier definitional diamond. □

Definition 769 (Restriction–pullback comparison)**.** *Provided by Mathlib.* For an open immersion $f : Y \hookrightarrow X$, presented by the ring-presheaf map φ, the restriction functor `restrict` f on $\mathcal{O}$-modules is naturally isomorphic to the abstract presheaf-of-modules pullback,

$$\texttt{restrictFunctorIsoPullback}\,f : \texttt{restrict}\,f\ \cong\ \texttt{pullback}\,f.$$

It is the left-adjoint uniqueness comparison between the two left adjoints of the pushforward, and it is the comparison through which the pullback-world base-change data are transported onto the restriction world throughout this section.

The strong-monoidality of pullback-along-an-open-immersion, used both by K1 (695) and by Bridge B1 (776), rests on three pieces of machinery: the presheaf-level strong-monoidal upgrade **H2** of restriction of scalars, an abstract conjugation of one left adjoint's oplax tensorator into another's across uniqueness of left adjoints, and the resulting "read `IsIso` δ off the conjugation" principle. They are recorded here as load-bearing nodes.

Lemma 770 (H2: strong-monoidal restriction of scalars along a sectionwise iso)**.** *Let* $\alpha : R \to S$ *be a morphism of presheaves of* commutative *rings whose every section* α_U *is* bijective*. Then the restriction-of-scalars functor* `restrictScalars` α *on presheaves of modules is* strong *monoidal: its lax tensorator* μ *and lax unit* ε *(645) are assembled sectionwise from the* `ModuleCat`*-level structure maps, which are isomorphisms for a bijective ground-ring map; hence* μ *and* ε *are sectionwise isomorphisms, hence isomorphisms, and the lax structure upgrades to a strong (genuinely monoidal) one. This is the presheaf-level* **H2** *ingredient: along an open immersion the structure map of the restriction functor is sectionwise the bijective structure-ring isomorphism, so restriction is strong monoidal and carries an invertible tensorator.*

Lemma 771 (Strong-monoidal tensorator iso)**.** *Provided by Mathlib. A strong (genuinely) monoidal functor* F *carries an* invertible *tensorator* $\mu_{\mathrm{Iso}}\,F\,A\,B : F A \otimes F B \cong F(A \otimes B)$*; its inverse is the oplax tensorator* $\delta\,F\,A\,B$*. This is the abstract source of every "*δ *is an isomorphism" statement for a strong-monoidal functor.*

Lemma 772 (Abstract δ-conjugation across uniqueness of left adjoints)**.** *Let F_1, F_2 be two left adjoints of a common right adjoint G, with adjunctions $\mathrm{adj}_1 : F_1 \dashv G$ and $\mathrm{adj}_2 : F_2 \dashv G$; suppose F_1, F_2 are oplax monoidal, G is lax monoidal, and adj_2 is a* monoidal *adjunction. Write $H_1 := \texttt{leftAdjointUniq}\,\mathrm{adj}_1\,\mathrm{adj}_2 : F_1 \cong F_2$ for the canonical comparison of the two left adjoints. If the lax tensorator of G induced by adj_1 (its `rightAdjointLaxMonoidal` structure) agrees with the ambient lax tensorator of G on the image of F_1,*

$$\mu(\texttt{rightAdjointLaxMonoidal}\,\mathrm{adj}_1)\,(F_1 A)\,(F_1 B) \;=\; \mu\, G\,(F_1 A)\,(F_1 B),$$

then the oplax tensorator of F_1 is the H_1-conjugate of that of F_2:

$$\delta\, F_1\, A\, B \;=\; H_1.\mathrm{hom.app}\,(A \otimes B)\,;\, \delta\, F_2\, A\, B\,;\, (H_1.\mathrm{inv.app}\, A \otimes_{\mathrm{m}} H_1.\mathrm{inv.app}\, B).$$

Proof. Transpose the claimed identity across the hom-equivalence of adj_1. The oplax tensorator $\delta\, F_1$ transposes to the lax tensorator $\mu(\texttt{rightAdjointLaxMonoidal}\,\mathrm{adj}_1)$ by the adjoint-mate identity relating a left adjoint's oplax tensorator to its right adjoint's lax tensorator. On the conjugate side, the two legs $H_1.\mathrm{hom}$ and $H_1.\mathrm{inv}$ of the left-adjoint uniqueness comparison collapse against the units of adj_1 and adj_2 by the triangle identity characterising `leftAdjointUniq` (so that $\mathrm{adj}_2.\mathrm{unit}\,;\, G(H_1.\mathrm{inv}) = \mathrm{adj}_1.\mathrm{unit}$), while the lax tensorator of G coming from adj_2 is the ambient one because adj_2 is monoidal; mate compositionality (761) and transitivity of the uniqueness comparison (760) align the two transports. The residue is exactly the hypothesis comparing the two lax tensorators of G on the image of F_1, which closes the identity. □

Lemma 773 (Reading `IsIso` δ off a conjugation)**.** *Let F_1 be a strong monoidal functor and F_2 an oplax monoidal functor, and let $e : F_1 \cong F_2$ be a natural isomorphism. Suppose the oplax tensorator of F_1 is the e-conjugate of that of F_2,*

$$\delta\, F_1\, A\, B \;=\; e.\mathrm{hom.app}\,(A \otimes B)\,;\, \delta\, F_2\, A\, B\,;\, (e.\mathrm{inv.app}\, A \otimes_{\mathrm{m}} e.\mathrm{inv.app}\, B).$$

Then $\delta\, F_2\, A\, B$ is an isomorphism.

Proof. Solve the conjugation identity for $\delta\, F_2$: it equals

$$\texttt{inv}(e.\mathrm{hom.app}\,(A \otimes B))\,;\, \delta\, F_1\, A\, B\,;\, \texttt{inv}(e.\mathrm{inv.app}\, A \otimes_{\mathrm{m}} e.\mathrm{inv.app}\, B).$$

Each factor is an isomorphism: the two flanking legs are inverses of components of the natural isomorphism e (and the tensor of two isos is an iso), and the middle factor $\delta\, F_1$ is the inverse of the strong-monoidal tensorator $\mu_{\mathrm{Iso}}\, F_1$ (771) since F_1 is strong monoidal. A composite of isomorphisms is an isomorphism. □

Lemma 774 (Bridge B1 crux: sheafification intertwines the presheaf and sheaf adjoint-uniqueness comparisons)**.** *Let $f : Y \hookrightarrow X$ be an open immersion presented by the presheaf-ring map φ', and let $M \in \mathrm{Scheme}.\texttt{Modules}\, X$. Write H_1 for the presheaf-level left-adjoint-uniqueness comparison $\texttt{pushforward}\,\beta \cong \texttt{pullback}\,\varphi'$ (the linchpin of Step 4 of `tensorObj_restrict_iso`, where β is the structure-ring map of the inverse-image presentation). Then $H_1.\mathrm{inv}$, evaluated at the underlying presheaf M^{p} of M, factors — after the sheafification unit η — as the sheaf-level per-leg reindex*

$$H_1.\mathrm{inv.app}\, M^{\mathrm{p}} \;=\; \eta_{(\texttt{pullback}\,\varphi')\, M^{\mathrm{p}}} ; \texttt{forget}((\texttt{pullbackValIso}\, f\, M).\mathrm{hom}) ; \texttt{forget}(((\texttt{restrictFunctorIsoPullback}\, f).$$

where η is the unit of the sheafification adjunction on Y, `pullbackValIso` f (878) is the sheaf-level pullback comparison, and the maps are transported through the forgetful functor $\mathrm{Scheme}.\texttt{Modules}\, Y \to$ `PresheafOfModules`. *This is the per-leg sheafification bookkeeping behind the M- and N-legs of the residual `tensorHom` in Bridge B1 (776).*

Proof. Both sides are maps $(\texttt{pullback}\,\varphi').\mathrm{obj}\,M^{\mathrm{p}} \to (\texttt{pushforward}\,\beta).\mathrm{obj}\,M^{\mathrm{p}}$, and $H_1.\mathrm{inv}$ is itself a `leftAdjointUniq` hom of the presheaf adjunction onto $\texttt{pushforward}\,\varphi'$; so by `homEquiv`-injectivity it suffices to compare the two images under that adjunction's hom-equivalence. The left image is the defining unit datum (`unit_leftAdjointUniq_hom_app`, 764); the right image is expanded by the unit formula. Rewriting `pullbackValIso` (878) and `restrictFunctorIsoPullback` (769) by their `leftAdjointUniq` presentations, and identifying the sheafify-then-pullback comparison with the left-adjoint-uniqueness comparison (677), turns the obligation into a pure `leftAdjointUniq`-coherence: the sheafification unit leg is supplied by the composite-unit transport `leftAdjointUniqUnitEta` (678), and the remaining comparison reduces to the *same* leg-by-leg conjugate telescope as the Bridge B2 helper (753): the conjugation distributes through `conjugateEquiv_comp` (762) and transitivity `leftAdjointUniq_trans` (760), the `restrictFunctorComp` factor is discharged by the keystone (768), and the whisker, congruence and pullback-composition legs are the shared atomic computations (756, 757, 758, 754). In the realized formalization this comparison is split as a four-part `forget`-transport of the sheaf `unit_leftAdjointUniq` triangle, whose genuine sheafification-boundary content is isolated in Part III (775). ☐

Lemma 775 (Bridge B1 crux, Part III: the sheaf pullback unit, forget-transported, factors through sheafification)**.** *Let $f : Y \hookrightarrow X$ be an open immersion presented by the presheaf-ring map φ', and let $M \in \mathrm{Scheme}.\texttt{Modules}\,X$. The unit of the* sheaf*-level pullback–pushforward adjunction* $\texttt{pullbackPushforwardAdjunction}\,f$*, transported through the forgetful functor* $\texttt{forget} : \mathrm{Scheme}.\texttt{Modules}\,Y \to \texttt{PresheafOfModules}$*, equals the* presheaf*-level pullback–pushforward unit followed by the sheafification unit and the pullback comparison* `pullbackValIso` *(878):*

$$\texttt{forget}((\texttt{pullbackPushforwardAdjunction}\,f).\mathrm{unit.app}\,M) = (\texttt{pullbackPushforwardAdjunction}\,\varphi').\mathrm{unit.app}\,M^{\mathrm{p}};(\ldots$$

where η is the unit of the sheafification adjunction on Y and M^{p} is the underlying presheaf of M. This is the lone sheafification-boundary residual of the Bridge B1 crux H_1 (774); the other three parts of that crux are definitional `forget`/`pushforward` *functoriality identities.*

Proof. As shown in the Lean development the identity is, after computing the opaque sheaf unit via the concrete `PullbackConstruction.adjunction` (using Mathlib's $\texttt{pullbackIso}\,\varphi = \texttt{leftAdjointUniq}(\texttt{pullbackPushforwardAdjunction}\,\varphi)\,(\texttt{PullbackConstruction.adjunction}\,\varphi)$), equivalent to the single comparison identity

$$(\texttt{pullbackIso}\,\varphi).\mathrm{inv.app}\,M = (\texttt{pullbackValIso}\,f\,M).\mathrm{hom} = \texttt{sheafificationCompPullback}.\mathrm{inv.app}\,M^{\mathrm{p}};(\texttt{pullback}\ldots$$

i.e. the two `leftAdjointUniq` comparison isos — one over the right adjoint $\texttt{pushforward}\,\varphi$, the other (`sheafificationCompPullback`) over the composite right adjoint $\texttt{pushforward}\,\varphi \circ \texttt{forget}$ — agree on M. Transposing along the *whole composite* `homEquiv` is circular (it returns the parent goal), and `leftAdjointUniq_trans` does not apply (distinct right adjoints).

The identity is the `SheafOfModules` analogue of Mathlib's `Functor.toSheafify_pullbackSheafificationCompa…` (`Sites/CoverLifting.lean`: 362–382), which proves the literal twin — a `leftAdjointUniq` of two *composite* adjunctions, pushed through the forgetful functor `sheafToPresheaf` and matched against the sheafification unit `toSheafify`. Its proof skeleton ports directly:

1. Transpose along the *innermost* adjunction $\texttt{pullbackPushforwardAdjunction}\,\varphi'$ (the presheaf-level one whose `homEquiv` the crux H_1 already uses), via $((\texttt{pullbackPushforwardAdjunction}\,\varphi').\texttt{homEquiv}\,\_\,\_)\ldots$ — supplying the two objects explicitly — *not* the composite. This is the step that breaks circularity.

2. Feed the defining unit triangle of the composite comparison, $\texttt{hpin} := \texttt{unit\_leftAdjointUniq\_hom\_app}\,A\,B\,M^{\mathrm{p}}\ldots$ (764), where $A = \texttt{sheafAdj}_X\,.\,\texttt{comp}\,\texttt{pullbackPPAdj}_{\mathrm{sheaf}}$ and $B = \texttt{pullbackPPAdj}_{\mathrm{pre}}$.

comp sheafAdj$_Y$ (the two composite adjunctions whose left-adjoint comparison *is* sheafificationCompPullback
677).

3. Expand both composite units with Adjunction.comp_unit_app (applied twice), exposing the sheafification unit η as an explicit separate factor — the factor that is invisible from the composite homEquiv and the reason the naive route is circular.

4. Transpose back (homEquiv_unit/homEquiv_counit together with unit naturality), and discharge the residual sheafification–pullback leg with the X-counit naturality square hnat: $\varepsilon_M \,;\, \eta_M = \eta_{(\mathtt{a}_X)\,M^{\mathrm{p}}} \,;\, (\mathtt{pushforward}\,\varphi).\mathrm{map}((\mathtt{pullback}\,\varphi).\mathrm{map}\,\varepsilon_M)$.

Both non-circular inputs hpin (step 2) and hnat (step 4) are exactly the two inputs the Mathlib precedent uses, and are already typed and compiling in the file. The remaining three parts of the crux H_1 (774) are definitional forget/pushforward functoriality identities. ☐

Lemma 776 (Bridge B1: tensorObj_restrict_iso is the conjugated base-change comparison)**.** *Let $f : Y \hookrightarrow X$ be an open immersion and $M, N \in$ Scheme.Modules X. The tensor-restriction comparison is the* restrictFunctorIsoPullback*-conjugate of the base-change tensor map, i.e. the conjugate factorisation*

$$\mathtt{tensorObj\_restrict\_iso}\, f\, M\, N = (\mathtt{restrictFunctorIsoPullback}\, f).\mathrm{app}\,(M \otimes_X N) ; \mathtt{asIso}(\mathtt{pullbackTensorMap}\, f$$

where $\rho_M :=$ (restrictFunctorIsoPullback f).app M is the per-leg comparison and pullbackTensorMap f *(667) is an isomorphism for the open immersion f (695).*

Proof. The identity is proved *directly*, with no monoidal structure on the sheaf category. Both sides are built from the *same* presheaf-level comparison through the *same* transport prefix. By construction the tensor-restriction comparison tensorObj_restrict_iso f (646) factors as the shared prefix restrictFunctorIsoPullback f;sheafificationCompPullback f — reducing the restriction to the abstract pullback and moving it inside sheafification — followed by a tail assembled from the structure-sheaf reconciliation; the base-change tensor map pullbackTensorMap f (667) is built from the identical prefix followed by the sheafified oplax tensorator δ(PresheafOfModules.pullback φ') (668).

Cancelling the common prefix, the goal becomes a *presheaf-level* identity: the sheafified oplax tensorator δ(PresheafOfModules.pullback φ') equals the left-adjoint–transported strong tensorator μ-iso(pushforward$_0$OfCommRingCat;restrictScalars β') that forms the tail of tensorObj_restrict_iso f. This is exactly the δ-conjugation identity pushforward_mu_appIso_collapse (694), which expresses the strong tensorator $\delta\, G_\beta$ as the H_1-conjugate of the oplax δ(pullback φ') — the very abstract conjugation deltaConjOfMuComparison (772) specialised to the pullback adjunction — combined with uniqueness of left adjoints (leftAdjointUniq_trans, 760, and naturality of leftAdjointUniq) to match the two transports of the presheaf comparison. Here all monoidal data genuinely *exists* at the presheaf level: the category of presheaves of modules carries a symmetric monoidal structure, and no monoidal structure on the sheaf categories Scheme.Modules is invoked.

Finally, for the open immersion f the comparison pullbackTensorMap f is invertible (695), so asIso(pullbackTensorMap f) is defined; the per-leg comparisons $\rho_M =$ (restrictFunctorIsoPullback f).app M (769) reassemble the displayed conjugate factorisation tensorObj_restrict_iso $f = \rho_{M \otimes N}$; asIso(pullbackTensorMap f) ; tensorObjIsoOfIso $\rho_M^{-1}\, \rho_N^{-1}$. ☐

Lemma 777 (S2: tensor-restriction comparison commutes with further restriction)**.** *Source: [Stacks Project], Modules on Ringed Spaces, Lemma* lemma-tensor-product-pullback*: pullback commutes with tensor product* functorially*, the functoriality on which the naturality square below* *rests. Let $M, N \in$ Scheme.Modules X be arbitrary $\mathcal{O}_X$-modules and let $j : V \hookrightarrow U$ be the chart morphism above. Write $T(f) :=$ `tensorObj_restrict_iso` $f : (M \otimes_X N)|_f \cong M|_f \otimes N|_f$ (646). The tensor-restriction comparison is* natural under further restriction along j: *the square*

$$\begin{array}{ccc} \big((M \otimes_X N)|_{\iota_U}\big)\big|_j & \xrightarrow{(\texttt{restrict}\, j)\, T(\iota_U)} & \big(M|_{\iota_U} \otimes_U N|_{\iota_U}\big)\big|_j \\ \big\downarrow \varrho & & \big\downarrow \varrho\otimes\varrho \\ (M \otimes_X N)|_{\iota_V} & \xrightarrow{T(\iota_V)} & M|_{\iota_V} \otimes_V N|_{\iota_V} \end{array}$$

commutes, where ϱ is the equality-of-opens reindexing of 741. Equivalently, modulo ϱ, $(\texttt{restrict}\, j)\, T(\iota_U) = T(\iota_V)$.

Proof. This square is *not* a bespoke leg-by-leg chase: it is base-change bookkeeping built on Bridge B1 (776). Substitute B1 to rewrite both $T(\iota_U) =$ `tensorObj_restrict_iso` ι_U and $T(\iota_V) =$ `tensorObj_restrict_iso` ι_V as `restrictFunctorIsoPullback`-conjugates of the base-change tensor maps `asIso`(`pullbackTensorMap` ι_U) and `asIso`(`pullbackTensorMap` ι_V). After this substitution the comparison factors `restrictFunctorIsoPullback` cancel against the reindexing ϱ by the reindex coherence (Bridge B2, used here only through `restrictFunctorIsoPullback`'s pseudonaturality), and the displayed square reduces to the corresponding equation between the *pullback*-world tensor maps. That equation is the already-proven composition law `pullbackTensorMap_restrict` (685): for the factorisation $j ; \iota_U = \iota_V$ it identifies (`pullback` j)(`pullbackTensorMap` ι_U) with `pullbackTensorMap` ι_V, conjugated by the pullback pseudofunctoriality isomorphism `pullbackComp` $j\, \iota_U$. Naturality of `pullbackTensorMap` in the two module arguments (672) carries the per-leg comparisons ρ_M, ρ_N of B1 through the square. Finally the reindexing $\varrho =$ `restrictCompReindex` j (749) threads the equality of opens $j ; \iota_U = \iota_V$ (741) on both flanks — its tensor image $\varrho \otimes \varrho$ on the right — absorbing the congruence `restrictFunctorCongr` and yielding the displayed square. No sheaf-level monoidal structure intervenes. □

Definition 778 (Cone B / presheaf dual base-change comparison θ_M (Step 4 of `dual_restrict_iso`))**.** *Source: internal categorical construction; no external reference.* Let $f : Y \hookrightarrow X$ be an open immersion of schemes and $M \in$ Scheme.Modules X. Write $\varphi = f$.`toRingCatSheafHom` for the induced structure-ring map and β for the open-immersion structure-ring isomorphism whose pushforward is left-adjoint-equivalent to `pullback` φ. The *presheaf dual base-change comparison* is the canonical presheaf isomorphism

$$\theta_M : (\texttt{pullback}\, \varphi).\mathrm{obj}\,(\texttt{dual}\, M_{\mathrm{val}}) \xrightarrow{\sim} \texttt{dual}((\texttt{pullback}\, \varphi).\mathrm{obj}\, M_{\mathrm{val}}),$$

defined as the Step-4 residual of `dual_restrict_iso`: the composite

$$\theta_M \;=\; \left(H1_{\texttt{dual}\, M_{\mathrm{val}}}\right)^{-1} \;;\; \texttt{isoMk}(V \mapsto \texttt{sliceDualTransport}\, f\, M\, V),$$

where $H1$ is the adjunction-uniqueness isomorphism re-presenting `pullback` φ as the pushforward `pushforward` β (Stage H1 of 737), and the second factor assembles the per-slice dual transport `sliceDualTransport` (715) sectionwise over the thin poset Opens Y; the `isoMk` naturality square is the poset-thin (subsingleton) coherence of that family. Packaging Step 4 as a named isomorphism in the `pullback` φ-form is what its downstream consumers (779, 784) require: the parent `dual_restrict_iso` $f\, M$ (737) is refactored to consume it as `restrictFunctorIsoPullback` ; `sheafificationCompPullback` ; `sheafification.mapIso` θ_M (Steps 1–3 wrapping this residual).

Lemma 779 (Cone B / L1: the dual base-change comparison is sectionwise an internal-hom evaluation)**.** *Source: internal categorical construction; no external reference. This is an alternative eval-based decomposition retained for completeness; it is* not *on the critical path, as the* *immersion-naturality crux 784 closes directly by the built-in naturality of θ. Let $j : V \hookrightarrow U$ be an open immersion with $j ; \iota_U = \iota_V$ and $M \in \mathrm{Scheme}.\texttt{Modules}\,X$. On each open section the presheaf-level dual base-change comparison θ_M (778) acts as the internal-hom* evaluation *of 709: a pulled-back dual section φ — an $\mathcal{O}$-linear functional on the restricted module — is carried by θ_M to the functional on the pulled-back module whose value at a section s is φ evaluated at s, reindexed across the immersion's direct-image j.*`opensFunctor`. *In short, the* `app` *of θ_M is, sectionwise, an evaluation map of the internal hom (the immersion analogue of the eval-at-1 presentation of* `presheafDualUnitIso` *in 844).*

Proof. Unfolding θ_M through its construction (778), the forward component at an open $W \leq V$ is the categorical image of the dual-section component of φ at the j-image of W, i.e. the evaluation map of the internal hom of 709 reindexed across j.`opensFunctor`. This is a direct read-off of the definition, sectionwise; no naturality is required for this step. □

Definition 780 (Cone B / morphism-level dual precomposition operator `dualPrecompHom`)**.** *Source: internal categorical construction; no external reference.* Let R be a presheaf of commutative rings and $g : M \to M'$ a morphism of presheaves of R-modules — *not* required to be invertible. The presheaf dual carries g to a morphism of duals in the reverse direction,

$$\texttt{dualPrecompHom}\,g : \; M'^{\vee} \longrightarrow M^{\vee},$$

defined sectionwise on a fixed open U as *precomposition* of a dual section by g:

$$(\texttt{dualPrecompHom}\,g)\,(\varphi) \;=\; (\texttt{pushforward}_0(\texttt{Over.forget}\,U)).\mathrm{map}\,g\,;\varphi, \qquad \varphi \in (M'|_U \to R|_U).$$

This is the *forward leg* of the precomposition equivalence `dualPrecompEquiv` (712) stripped of any appeal to an inverse: where `dualPrecompEquiv` e needs e^{-1} to package the two one-sided maps into an $R(U)$-linear equivalence, `dualPrecompHom` g keeps only the g-precomposition map and is therefore defined for an arbitrary, possibly non-invertible, g. When $g = e.\mathrm{hom}$ for an isomorphism e, it agrees with the hom leg of `dualIsoOfIso` e (713). It is the contravariant functoriality of the presheaf dual $M \mapsto M^{\vee}$ (708) in arbitrary morphisms — the operator needed to conjugate the base-change comparison θ by the non-invertible restriction map.

Lemma 781 (Cone B / L2: morphism-level dual transport is precomposition by the restriction map)**.** *Source: internal categorical construction; no external reference. With $j : V \hookrightarrow U$ an open immersion as above, the morphism-level dual transport* `dualPrecompHom (restrict` j`)` *(780) is, on each section,* precomposition *of a dual section (functional) by the underlying section map of the restriction morphism* `(restrict` j`)`*: for a dual section φ,*

$$(\texttt{dualPrecompHom}\,(\texttt{restrict}\,j))(\varphi) \;=\; (\texttt{restrict}\,j)\,;\varphi$$

sectionwise. The restriction morphism is not *invertible, and* `dualPrecompHom` *(780) is precisely the precomposition forward leg that applies in this non-invertible regime — no inverse of* `(restrict` j`)` *is used. This is the immersion analogue of the precomposition read-off of the dual transport that, for a unit automorphism, underlies 867.*

Proof. Immediate from the definition of `dualPrecompHom` (780): its sectionwise value at φ is $(\texttt{pushforward}_0(\texttt{Over.forget}\,U)).\mathrm{map}\,(\texttt{restrict}\,j);\varphi$, i.e. precomposition of φ by the underlying section map of `(restrict` j`)`. No invertibility is invoked. □

Lemma 782 (Cone B / L3a: internal-hom evaluation commutes with the immersion restriction map (genuine base-change atom))**.** *Source: internal categorical construction; no external* *reference. This is an alternative eval-based decomposition retained for completeness; it is* not *on the critical path, as the immersion-naturality crux 784 closes directly by the built-in naturality of θ. Fix a dual section φ and a section s. Internal-hom evaluation (709) commutes with the immersion's restriction map: evaluating then restricting equals restricting both arguments then evaluating,*

$$\varphi(s)\big|_V \;=\; (\varphi\big|_V)(s\big|_V),$$

now against the non-invertible *restriction map* (`restrict` j) *in place of the multiplication by a unit scalar that governs the unit-automorphism case (844). This is the genuine internal-hom base-change content — the dual-flank analogue of Bridge B1 (776) — and is the immersion analogue of the evaluate/restrict commutation core of 844. The down-set fact $j \mathbin{;} \iota_U = \iota_V$ (741) pins the open along which both sides are restricted.*

Proof. This is exactly the evaluate/restrict-commutation identity already established in the proof of 709: the restriction $\varphi|_V$ of a functional inside the dual is its restriction to the open V, restriction is functorial, and evaluating the restricted functional on the restricted section agrees with restricting the value of the evaluation ($\varphi(s)|_V = (\varphi|_V)(s|_V)$). Because the restriction map is not invertible on sections, this is a genuine naturality of evaluation against (`restrict` j), not a scalar commutation; the open V is pinned by $j \mathbin{;} \iota_U = \iota_V$ (741). □

Lemma 783 (Cone B / L3b: pointwise naturality square of the dual base-change comparison (sectionwise form))**.** *Source: internal categorical construction; no external reference. With $j : V \hookrightarrow U$ as above, the pointwise naturality square of the dual base-change comparison holds: for a section and an element, conjugating θ by the morphism-level transport* `dualPrecompHom` (`restrict` j) *on the source equals transporting along* (`restrict` j) *on the target. Evaluating both composites at a dual section φ and a section s, the two sides agree because the underlying presheaf-of-modules morphism θ_M (778) commutes with the immersion's restriction map by construction. This is the immersion analogue of the eval-core of 844; the down-set factorisation $j \mathbin{;} \iota_U = \iota_V$ (741) pins the open along which both sides are restricted.*

Proof. The underlying presheaf-of-modules morphism θ_M (778) commutes with the immersion's restriction map by construction — this is the built-in naturality of a morphism of presheaves of modules, read off here at the immersion j and the dual section φ. Evaluating that naturality identity against the section s — functoriality of evaluation in its functional argument — yields the pointwise square directly, with no intervening internal-hom evaluation or precomposition decomposition. The open V along which both sides are restricted is pinned by $j \mathbin{;} \iota_U = \iota_V$ (741). □

Lemma 784 (Immersion-naturality of the presheaf dual base-change comparison (Cone B))**.** *Let $j : V \hookrightarrow U$ be an open immersion with $j \mathbin{;} \iota_U = \iota_V$, and let $M \in$* Scheme.`Modules` *X. The presheaf-level dual base-change comparison*

$$\theta_M : (\texttt{pullback}\,\varphi).\mathrm{obj}\,(\texttt{dual}\,M_{\mathrm{val}}) \;\cong\; \texttt{dual}((\texttt{pullback}\,\varphi).\mathrm{obj}\,M_{\mathrm{val}})$$

— the internal-hom comparison $f^*\mathcal{H}om(M,\mathcal{O}) \to \mathcal{H}om(f^*M, f^*\mathcal{O})$ packaged as θ_M (778) — is natural in the immersion j: conjugating θ by the morphism-level transport `dualPrecompHom` *(780) along j on the source matches its transport along j on the target. This is the immersion analogue of 844, with the unit automorphism there replaced by the restriction map* (`restrict` j)*.*

Proof. By element extensionality on dual sections. Testing the asserted equality of morphisms against an arbitrary dual section φ reduces the immersion-naturality square immediately to the *built-in* naturality of the underlying presheaf-of-modules morphism θ_M (778): a morphism of presheaves of modules commutes with all restriction maps by construction, so its naturality at the immersion j, read off at φ, is exactly the required square. No Iso.ext/sectionwise-`hom_ext` reduction ladder and no eval/transport decomposition is needed; the pointwise statement 783 packages this naturality of θ_M directly. ☐

Lemma 785 (Bridge dual-B1: `dual_restrict_iso` is the conjugated dual base-change comparison)**.** *Source: internal categorical construction; no external reference. Let $f : Y \hookrightarrow X$ be an open immersion and $M \in \mathrm{Scheme}.\mathtt{Modules}\, X$, with $\varphi = f.\mathtt{toRingCatSheafHom}$. The dual-restriction comparison* `dual_restrict_iso` $f\, M$ *(737) factors as the shared transport prefix followed by the sheafified Step-4 residual θ:*

$$\mathtt{dual\_restrict\_iso}\, f\, M = (\mathtt{restrictFunctorIsoPullback}\, f).\mathrm{app}\,(\mathtt{dual}\, M) \mathbin{;} (\mathtt{sheafificationCompPullback}\,\varphi).\mathrm{a}$$

where $\theta_M =$ `presheafDualPullbackComparison` $f\, M$ *(778) is the presheaf dual base-change comparison. This is the dual analogue of Bridge B1 (776): both sides are the* same *three-factor composite by construction, so the identity is definitional.*

Proof. Definitional. The parent `dual_restrict_iso` $f\, M$ (737) was refactored to *consume* the Step-4 comparison θ_M (778) as a named residual: by construction it is the composite of the restriction-to-pullback transport `restrictFunctorIsoPullback` f, the sheafification–pullback reconciliation `sheafificationCompPullback` φ at the presheaf dual `dual` M_{val}, and the sheafification of θ_M. The displayed equation therefore holds by reflexivity. ☐

Lemma 786 (Module-naturality of the presheaf dual base-change comparison θ (Cone B))**.** *Source: internal categorical construction; no external reference. Let $f : Y \hookrightarrow X$ be an open immersion and $\varphi = f.\mathtt{toRingCatSheafHom}$. The presheaf dual base-change comparison θ_M (778) is natural in the* module *argument M: for a morphism $\phi : M \to M'$ of* $\mathrm{Scheme}.\mathtt{Modules}\, X$*, the contravariance of the dual produces commuting transport, namely*

$$(\mathtt{pullback}\,\varphi).\mathrm{map}(\mathtt{dualPrecompHom}\,\phi_{\mathrm{val}});\theta_M \;=\; \theta_{M'};\mathtt{dualPrecompHom}((\mathtt{pushforward}\,\beta).\mathrm{map}\,\phi_{\mathrm{val}}),$$

where `dualPrecompHom` *(780) is the contravariant morphism-level dual transport (so that, on an isomorphism $\phi = e$, it specialises to the* hom *leg of* `dualIsoOfIso` e*, 713). This is the module-variant naturality of θ, distinct from the immersion-variant 784 (natural in j).*

Proof. The comparison factors (778) as $\theta_M = (H1_{\mathtt{dual}\, M_{\mathrm{val}}})^{-1};\mathtt{isoMk}(V \mapsto \mathtt{sliceDualTransport}\, f\, M\, V)$, a composite of two pieces each natural in M. The first factor $H1 = \mathtt{leftAdjointUniq}$ re-presents `pullback` φ as the pushforward `pushforward` β; being a natural isomorphism of functors, its component at `dual` M_{val} is natural in `dual` M_{val}, hence — by contravariance of the dual — natural in M through `dualPrecompHom`. The second factor is the `isoMk` of the per-slice dual transport `sliceDualTransport` $f\, M$, whose sectionwise component (731) is natural in M: on each slice it reindexes the dual-section component across the immersion's direct image and conjugates by the codomain unit ring-swap, both of which commute with the precomposition transport `dualPrecompHom` ϕ_{val}. Composing the two naturality squares — the $H1$ square and the slice-transport square — yields the displayed module-naturality square for θ. ☐

Lemma 787 (Module-naturality of `dual_restrict_iso` (contravariant analogue of S2's tensor naturality))**.** *Source: internal categorical construction; no external reference. Let $j : V \hookrightarrow U$ be an open immersion and $e : M \cong M'$ an isomorphism in* $\mathrm{Scheme}.\mathtt{Modules}\, U$*. The dual-restriction comparison* `dual_restrict_iso` j *(737) is natural in the module isomorphism: the* `restrict` j*-image of* `dualIsoOfIso` e *(714) equals* `dualIsoOfIso` *of the* `restrict` j*-image of e, conjugated by* *the* $\texttt{dual\_restrict\_iso}\, j$ *comparisons on both flanks (the dual being contravariant, the flanks are taken at* M' *and* M *respectively),*

$$(\texttt{restrictFunctor}\, j).\mathrm{mapIso}\,(\texttt{dualIsoOfIso}\, e) = \texttt{dual\_restrict\_iso}\, j\, M' \ggg \texttt{dualIsoOfIso}((\texttt{restrictFunctor}$$

This is the contravariant analogue of the covariant tensor naturality 788; it is the missing ingredient the Seam-1 split (838) wraps around the dual constituent, letting $\texttt{restrict}\, j$ *commute past the* $(\texttt{dualIsoOfIso}\, e^M)^{-1}$ *factor.*

Proof. By the dual-B1 factorisation (785), $\texttt{dual\_restrict\_iso}\, j$ is the shared transport prefix $\texttt{restrictFunctorIsoPullback}\, j ; \texttt{sheafificationCompPullback}$ followed by the sheafified Step-4 residual θ. The prefix — the restriction-to-pullback comparison and the sheafification–pullback reconciliation — is a natural transformation, so it commutes with the dual transport of e by built-in naturality. The residual leg is the sheafification of the presheaf comparison θ, whose module-naturality is 786; sheafification is a functor, so it carries that presheaf naturality square to the sheaf level. Composing the prefix and residual naturality squares, and rewriting the morphism-level dual transports as the hom legs of $\texttt{dualIsoOfIso}$ (714) on the isomorphism e, yields the displayed conjugated identity. □

Lemma 788 (Module-naturality of $\texttt{tensorObj\_restrict\_iso}$ (S2's bifunctorial naturality))**.** *Source: internal categorical construction; no external reference. Let* $f : Y \hookrightarrow X$ *be an open immersion and* $a : M \cong M'$, $b : N \cong N'$ *isomorphisms in* $\mathrm{Scheme}.\texttt{Modules}\, X$*. The tensor-restriction comparison* $\texttt{tensorObj\_restrict\_iso}\, f$ *(646) is natural in the two module isomorphisms: the* $\texttt{restrict}\, f$*-image of* $\texttt{tensorObjIsoOfIso}\, a\, b$ *(862) equals* $\texttt{tensorObjIsoOfIso}$ *of the* $\texttt{restrict}\, f$*-images of* a *and* b*, conjugated by the* $\texttt{tensorObj\_restrict\_iso}\, f$ *comparisons,*

$$(\texttt{restrictFunctor}\, f).\mathrm{mapIso}\,(\texttt{tensorObjIsoOfIso}\, a\, b) = \texttt{tensorObj\_restrict\_iso}\, f\, M\, N \ggg \texttt{tensorObjIsoOfIs}$$

This is the covariant ingredient the Seam-1 split (838) wraps around the trivialisation step, letting $\texttt{restrict}\, j$ *commute past the* $\texttt{tensorObjIsoOfIso}\, e^M\,(\cdots)$ *of constituent (2).*

Proof. The naturality square is established at the hom level and the iso identity then follows by cancelling the comparison at (M', N'). Promote both tensor-restriction comparisons to the pullback world via Bridge B1 (776), reducing the square to the naturality of the pullback–tensor comparison δ_{sheaf} (672) sandwiched between two naturality squares of the restriction–pullback comparison $\texttt{restrictFunctorIsoPullback}\, f$ (769) — one for $\texttt{restrictFunctorIsoPullback}.\mathrm{hom}$ on the leading leg and one for its inverse on each per-factor tail. The shared prefix $\texttt{restrictFunctorIsoPullback}.\mathrm{app};$ δ_{sheaf} cancels and the two per-leg tails match by inverse-naturality of $\texttt{restrictFunctorIsoPullback}$, closing the square. □

Lemma 789 (Bridge dual-(b): composition law of $\texttt{sheafificationCompPullback}$ over an immersion factorisation)**.** *Source: internal categorical construction; no external reference. Let* $j : V \hookrightarrow U$ *be the chart morphism above, with* $j ; \iota_U = \iota_V$*, and let* P *be a presheaf of modules over* X*. Write* $\texttt{SCP}\, g = \texttt{sheafificationCompPullback}\,(\texttt{toRingCatSheafHom}\, g)$ *for the sheafification–pullback comparison,* a_X *for sheafification on* X*, and* $\texttt{pullbackComp}\, j\, \iota_U$ *for the pullback pseudo-functoriality iso* $\iota_V^* \cong j^* \circ \iota_U^*$*. Then the comparison for the composite immersion* ι_V *factors through those for* ι_U *and for* j*, conjugated by* $\texttt{pullbackComp}\, j\, \iota_U$*:*

$$(\texttt{SCP}\,\iota_V).\mathrm{app}\, P = \left((\texttt{pullbackComp}\, j\, \iota_U).\mathrm{app}\,(a_X P)\right)^{-1} ; (\texttt{restrict}\, j)((\texttt{SCP}\,\iota_U).\mathrm{app}\, P) ; (\texttt{SCP}\, j).\mathrm{app}((\texttt{pullback}\,\iota_U)\, P) ;$$

where a_V *is sheafification on* V *and* $\texttt{pullbackComp}_{\mathrm{pre}}\, \iota_U\, j$ *is the* presheaf*-level pullback pseudofunctoriality isomorphism* $\iota_V^* \cong \iota_U^* \circ j^*$ *on presheaves of modules. Its sheafification supplies a* *fourth, dual-independent leg: this leg, not a* `restrictCompReindex` *factor, is what threads the equality of opens* $j\,;\iota_U = \iota_V$ *through the comparison. The statement is* independent of the dual*: P is an arbitrary presheaf (here instantiated at $P = \texttt{dual}\,M_{\mathrm{val}}$).*

Proof. This is the immersion-factorisation specialisation of the generic composition coherence Sq1 (884), which for composable h, f and *any* presheaf already identifies $(\texttt{SCP}\,(h \circ f))$.app with the `pullbackComp`-conjugate of the f- and h-instances. Apply Sq1 with $h = j$, $f = \iota_U$ and the presheaf P. The composite ring map of $\iota_V = j\,;\iota_U$ is reconciled with the whiskered composite of the ι_U- and j-maps by `toRingCatSheafHom_comp` (880), so Sq1 specialises to the displayed equation. Finally the reindexing $\varrho = \texttt{restrictCompReindex}\,j$ (749) threads the equality of opens $j\,;\iota_U = \iota_V$ (741) on the restriction flank, matching the $(\texttt{restrict}\,j)$-image presentation used at the seam. No dual-specific input enters: this is the `sheafificationCompPullback`-portion of the tensor base-change law 685, the analogue of the `pullbackValIso`-composition 682. ☐

Definition 790 (Cone B / c.1: object-identification iso $(\texttt{pushforward}\,\beta).\mathrm{obj}\,M_{\mathrm{val}} \cong (M|_f)_{\mathrm{val}}$)**.** *Source: internal categorical construction; no external reference.* Let $f : Y \hookrightarrow X$ be an open immersion and $M \in \mathrm{Scheme}.\texttt{Modules}\,X$, and let β be the whiskered direct-image natural isomorphism whose pushforward is left-adjoint-equivalent to $\texttt{pullback}\,\varphi$ — the data fixed inside 778. There is a canonical isomorphism of presheaves of modules over Y

$$(\texttt{pushforward}\,\beta).\mathrm{obj}\,M_{\mathrm{val}} \;\cong\; \left(M|_f\right)_{\mathrm{val}},$$

identifying the direct image of the underlying presheaf M_{val} with the underlying presheaf of the restricted module $M|_f = \texttt{restrict}\,f\,M$. It is the foundational object-level identification through which the pushforward presentation of θ is reconciled with the restriction-world form consumed downstream.

Proof. Compose two canonical identifications. First, the adjunction-uniqueness isomorphism built inside 778 re-presents the abstract presheaf pullback $\texttt{pullback}\,\varphi$ as the pushforward $\texttt{pushforward}\,\beta$ on objects, giving $(\texttt{pushforward}\,\beta).\mathrm{obj}\,M_{\mathrm{val}} \cong (\texttt{pullback}\,\varphi).\mathrm{obj}\,M_{\mathrm{val}}$. Second, the val-component of the restriction–pullback comparison $\texttt{restrictFunctorIsoPullback}\,f$ (769) followed by the sheafify-then-pullback identification (677) relates $(\texttt{pullback}\,\varphi).\mathrm{obj}\,M_{\mathrm{val}}$ to $(M|_f)_{\mathrm{val}}$. Both ingredients are canonical isomorphisms already in hand; their composite is the asserted iso, and poset-thinness of $\mathrm{Opens}\,Y$ makes the naturality square of the assembled family automatic. ☐

Lemma 791 (The presheaf pullback pseudofunctoriality isomorphism `pullbackComp`)**.** *Provided by Mathlib. For composable ring-presheaf maps φ, ψ, the composition of the two presheaf pullback functors is canonically isomorphic to the pullback along the composite:*

$$\texttt{pullback}\,\varphi \circ \texttt{pullback}\,\psi \;\cong\; \texttt{pullback}\,(\varphi\,;\psi).$$

By construction it is *the left-adjoint composition isomorphism* `leftAdjointCompIso` *of the three pullback–pushforward adjunctions, reconciled by the pushforward composition* `pushforwardComp`*; this definitional identity is the content of 763. It is leg 1 of the cocycle 799.*

Lemma 792 (Restriction along a composite as a composition of restrictions `restrictFunctorComp`)**.** *Provided by Mathlib. For composable open immersions $h : Z \hookrightarrow Y$ and $f : Y \hookrightarrow X$, restriction of sheaves of modules along the composite is canonically isomorphic to the composition of the two restrictions:*

$$\texttt{restrictFunctor}\,(h\,;f) \;\cong\; \texttt{restrictFunctor}\,f \circ \texttt{restrictFunctor}\,h.$$

Its contravariant image under the dual — `dualIsoOfIso` *(714) of the forgetful image of* $(\texttt{restrictFunctorComp}\,h\,f).\mathrm{ap}$ *— is the dual-target pseudofunctoriality leg 4 of the cocycle 799.*

Lemma 793 (Mate of a `leftAdjointUniq` comparison is the identity)**.** *Source: internal categorical construction; no external reference. Let $F, F' : C_0 \to C_1$ be two left adjoints of one and the same right adjoint $G : C_1 \to C_0$, via $\mathrm{adj}_1 : F \dashv G$ and $\mathrm{adj}_2 : F' \dashv G$. Then the mate (conjugate) of the comparison isomorphism* (`leftAdjointUniq` $\mathrm{adj}_1\,\mathrm{adj}_2$)*.*hom *under the conjugation bijection* `conjugateEquiv` $\mathrm{adj}_2\,\mathrm{adj}_1$ *is the identity natural transformation $\mathbf{1}_G$. This is the abstract content behind every* `leftAdjointUniq` *cocycle: the comparison transports the unit of one adjunction to the other and is therefore mate to $\mathbf{1}_G$.*

Proof. Unfold `leftAdjointUniq` as the symmetrisation of the conjugate-iso bijection and apply the Mathlib computation `conjugateIsoEquiv_symm_apply_inv` followed by the `Equiv.apply_symm_apply` round-trip, which collapses the mate to $\mathbf{1}_G$. □

Lemma 794 (Mate of the `leftAdjointCompIso` hom is the reconciling inverse)**.** *Source: internal categorical construction; no external reference. Given three adjunctions $\mathrm{adj}_{01} : F_{01} \dashv G_{10}$, $\mathrm{adj}_{12} : F_{12} \dashv G_{21}$, $\mathrm{adj}_{02} : F_{02} \dashv G_{20}$ and a right-adjoint composition isomorphism $e : G_{21} \circ G_{10} \cong G_{20}$, the mate (conjugate) of* (`leftAdjointCompIso` $\mathrm{adj}_{01}\,\mathrm{adj}_{12}\,\mathrm{adj}_{02}\,e$)*.*hom *under* `conjugateEquiv` $\mathrm{adj}_{02}\,(\mathrm{adj}_{01}.\mathrm{comp}\,\mathrm{adj}_{12})$ *equals e.inv. This is the companion of Mathlib's* `conjugateEquiv_leftAdjointCompIso_inv` *(which computes the conjugate of the* inv*).*

Proof. By `conjugateEquiv_leftAdjointCompIso_inv` the conjugate of the inv is e.hom; rewriting e.hom as that conjugate and using functoriality of the conjugation bijection under composition (`conjugateEquiv_comp`) together with `conjugateEquiv_id` shows the conjugate of the hom post-composed with e.hom is the identity. Cancelling the mono e.hom identifies the conjugate with e.inv. □

Lemma 795 (Explicit component of the `leftAdjointCompIso` hom)**.** *Provided by Mathlib* (`Mathlib/CategoryTheory/Adjunction/CompositionIso.lean`)*. Given three adjunctions $\mathrm{adj}_{01} : F_{01} \dashv G_{10}$, $\mathrm{adj}_{12} : F_{12} \dashv G_{21}$, $\mathrm{adj}_{02} : F_{02} \dashv G_{20}$ and a right-adjoint composition isomorphism $e : G_{21} \circ G_{10} \cong G_{20}$, the hom of the left-adjoint composition isomorphism is* fully explicit *at each object X:*

$$(\texttt{leftAdjointCompIso}\,\mathrm{adj}_{01}\,\mathrm{adj}_{12}\,\mathrm{adj}_{02}\,e).\mathrm{hom.app}\,X \;=\; F_{12}.\mathrm{map}(F_{01}.\mathrm{map}\,(\mathrm{adj}_{02}.\mathrm{unit.app}\,X))\;;\;F_{12}.\mathrm{map}(F_{01}.\mathrm{map}\,(e$$

i.e. the unit of adj_{02} pushed through F_{01} and F_{12}, the reconciling e.inv, and the two counits of adj_{01} and adj_{12}. When the reconciling iso is the identity ($e =$ `Iso.refl`*) the middle e.inv factor drops. This is the section-level component companion of the mate characterisation 794.*

Lemma 796 (Abstract $H1$ cocycle: `leftAdjointUniq` intertwines two `leftAdjointCompIso`s)**.** *Source: internal categorical construction; no external reference. Let $F_\bullet$ and $P_\bullet$ be two families of left adjoints sharing the right adjoints $G_{\bullet\bullet}$ level by level: $\mathrm{adjF}_{ij} : F_{ij} \dashv G_{ji}$ and $\mathrm{adjP}_{ij} : P_{ij} \dashv G_{ji}$ for $ij \in \{01, 12, 02\}$, together with a single shared right-adjoint composition isomorphism $e : G_{21} \circ G_{10} \cong G_{20}$. Write $H1_{ij} =$* `leftAdjointUniq` $\mathrm{adjF}_{ij}\,\mathrm{adjP}_{ij} : F_{ij} \cong P_{ij}$ *for the comparison isomorphisms and FC, PC for the two* `leftAdjointCompIso`*s built from the* same *e. Then the $H1$s intertwine the two composition isomorphisms:*

$$FC.\mathrm{hom}\;;\;(H1_{02}).\mathrm{hom} \;=\; ((H1_{01}).\mathrm{hom} \rhd F_{12})\;;\;(P_{01} \lhd (H1_{12}).\mathrm{hom})\;;\;PC.\mathrm{hom}.$$

This is the abstract $H1$ cocycle instantiated in the Cone B/c.2 lemma 797, and the generic engine behind `conjugateEquiv_restrictFunctorComp_inv` *(768): both reduce a composite comparison to a chain over the two composition isomorphisms.*

Proof. Apply the conjugation bijection `conjugateEquiv`, which is injective, to both sides. On the left, functoriality under composition turns the mate of FC.hom ; $(H1_{02})$.hom into the mate of $H1_{02}$ composed with the mate of FC.hom; by 793 the first is **1** and by 794 the second is e.inv, so the left mate is e.inv. On the right, the threefold composite splits under `conjugateEquiv_comp`; the two whiskered $H1$ factors collapse via `conjugateEquiv_whiskerRight` and `conjugateEquiv_whiskerLeft` to identities by 793, and the PC.hom factor becomes e.inv by 794. Both mates equal e.inv, so the two morphisms agree. ☐

Lemma 797 (Cone B / c.2(a): cocycle law of the adjunction-uniqueness iso $H1$ over the immersion factorisation)**.** *Source: internal categorical construction; no external reference. Let $h : Z \hookrightarrow Y$ and $f : Y \hookrightarrow X$ be composable open immersions, write $\varphi_g = g.\texttt{toRingCatSheafHom}.\mathrm{hom}$, and let*

$$H1_g \;=\; (\texttt{pushforwardPushforwardAdj}\ \cdots).\texttt{leftAdjointUniq}\,(\texttt{pullbackPushforwardAdjunction}\,\varphi_g) \;:\; \texttt{pushforwar}$$

be the left-adjoint-uniqueness isomorphism re-presenting the presheaf pullback as the pushforward (Stage H1 of 778). Then $H1$ satisfies a cocycle (composition) law over the factorisation $h\,;f$: on the pullback (source) flank it is threaded by the presheaf pseudofunctoriality iso `pullbackComp` $\varphi_f\,\varphi_h$ *(791) and on the pushforward (target) flank by the restriction-composition iso* `restrictFunctorComp` $h\,f$ *(792, equivalently the pushforward composition* `pushforwardComp`*). Writing $FC =$* `leftAdjointCompIso` *for the left-adjoint composition isomorphism of the three pushforward adjunctions* `pushforward` $\beta_g \dashv$ `pushforward` φ_g*, reconciled by the presheaf pushforward composition* `pushforwardComp` $\varphi_f\,\varphi_h$*, the realized cocycle is the* hom*-form identity*

$$FC.\mathrm{hom}\;;\;(H1_{h;f}).\mathrm{hom} \;=\; ((H1_f).\mathrm{hom} \triangleright \texttt{pushforward}\,\beta_h)\;;\;(\texttt{pullback}\,\varphi_f \triangleleft (H1_h).\mathrm{hom})\;;\;(\texttt{pullbackComp}\,\varphi_f\,\varphi_h).\mathrm{ho}$$

stated as a faithful $\forall$-strengthening over the three pushforward adjunctions `pushforward` $\beta_g \dashv$ `pushforward` φ_g *($g \in \{f,\ h,\ h;f\}$); the consumer instantiates it with the matching* `pushforwardPushforwardAdj` *adjunctions. Here $\triangleright$ is right-whiskering ($\alpha \triangleright F =$* `whiskerRight` $\alpha\,F$*) and $\triangleleft$ left-whiskering ($F \triangleleft \alpha =$* `whiskerLeft` $F\,\alpha$*). Inverting the identity (swap f and h in the two middle legs and replace* (`pullbackComp` $\varphi_f\,\varphi_h$).hom *by its* inv*) yields the form in which θ's $H1$-prefix is recombined inside 799. This is the dual-flank analogue of 763 (*`conjugateEquiv_pullbackComp_inv`*) and of* `conjugateEquiv_restrictFunctorComp_inv` *(768). This is the $H1$-flank of the cocycle 799.*

Proof. Each $H1_g$ is a `leftAdjointUniq` comparison of two left adjoints of the shared right adjoint `pushforward`. The composite-immersion comparison $H1_{h;f}$ decomposes by transitivity of `leftAdjointUniq` (760) once the composite pullback `pullback` $\varphi_{h;f}$ is identified with the iterated pullback `pullback` $\varphi_h \circ$ `pullback` φ_f through `pullbackComp` $\varphi_f\,\varphi_h$ (791), and the composite pushforward `pushforward` $\beta_{h;f}$ with the iterated pushforward through `restrictFunctorComp` $h\,f$ (equivalently `pushforwardComp`, 792). Because `pullbackComp` is *defined* as the left-adjoint composition isomorphism `leftAdjointCompIso`, the reconciliation of these two pseudofunctor isos on the conjugate is exactly 763, instantiated at the present adjunctions; substituting it collapses the conjugate to the displayed three-factor chain. Concretely, this is a direct instantiation of the abstract $H1$ cocycle 796 at the three pushforward/pullback adjunction pairs, with $F_\bullet =$ `pushforward` β, $P_\bullet =$ `pullback` φ and the shared composition iso $e =$ `pushforwardComp` $\varphi_f\,\varphi_h$; the P-side `leftAdjointCompIso` is definitionally `pullbackComp` $\varphi_f\,\varphi_h$. ☐

Lemma 798 (Generic $H1$-cancellation skeleton for the c.2 cocycle)**.** *Source: internal categorical construction; no external reference. Work in an arbitrary category C with objects $A_1, \dots, A_9$ and the morphisms appearing in the c.2 reduction. Given the eight structural facts of that reduction*

— *the two iso cancellations* $\texttt{aHinv};\texttt{aH} = \mathbf{1}$ *(of* $H1_{h;f}$*) and* $\texttt{fcinv};\texttt{fc} = \mathbf{1}$ *(of* FC*), the abstract* $H1$ *cocycle* $\texttt{fc}\,;\texttt{aH} = \texttt{phf}\,;\texttt{hh}\,;\texttt{pc}$ *(instantiating 797), the* `pullbackComp` *cancellation* $\texttt{pc}\,;\texttt{p0} = \mathbf{1}$*, the single* $H1_h$.hom *naturality slide* $\texttt{hh}\,;\texttt{Pfhif} = \texttt{Pushhif}\,;\texttt{hhdmf}$*, the* $\texttt{pushforward}\,\beta_h$*-functoriality fold* $\texttt{phf}\,;\texttt{Pushhif} = \texttt{pushSDTf}$*, the* $H1_h$ *round-trip* $\texttt{hhdmf}\,;\texttt{Hhinv} = \mathbf{1}$*, and the residual* $(**)$ $\texttt{fc}\,;\texttt{s} = \texttt{pushSDTf}\,;\texttt{sDTh}\,;\texttt{p3}$ — *the c.2 goal identity* $\texttt{aHinv}\,;\texttt{s} = \texttt{p0}\,;\texttt{Pfhif}\,;(\texttt{Hhinv}\,;\texttt{sDTh})\,;\texttt{p3}$ *holds. This is the instance-agnostic engine that performs the entire* $H1$*-cancellation of 799; the consumer supplies its eight hypotheses at the concrete pushforward/pullback morphisms (the cocycle hypothesis being 797, itself 796 at the three pushforward adjunctions, reconciled by 794).*

Proof. Pure associativity and identity rewriting. Pre-compose the cocycle goal with `fc` and use the cocycle hypothesis together with the residual $(**)$; the `pullbackComp` cancellation, the $H1_h$ naturality slide, the $H1_h$ round-trip and the functoriality fold then collapse the middle in turn, yielding $\texttt{fc}\,;(\text{RHS}) = \texttt{fc}\,;\texttt{s}$. Cancelling the iso `fc` (via `fcinv`) and then the iso `aH` (via `aHinv`) gives the stated identity. ☐

Lemma 799 (Bridge dual-(c): cocycle law of the presheaf dual base-change comparison θ over `pullbackComp`)**.** *Source: internal categorical construction; no external reference. Let* $j : V \hookrightarrow U$ *be the chart morphism,* $j\,;\iota_U = \iota_V$*, and* $M \in \text{Scheme}.\texttt{Modules}\,X$*. The presheaf dual base-change comparison* θ *(778) satisfies a* cocycle *(composition) law over the immersion factorisation, in the* exact dual shape *of the tensor composition coherence 685: the* ι_V*-instance* $\theta_{\iota_V,M}$ *factors through the* j*-pullback of the* ι_U*-instance and the* j*-instance* at the pushforward object $(\texttt{pushforward}\,\beta_{\iota_U}).\text{obj}\,M_{\text{val}}$ — *the base-change image of* M_{val}*,* not *the restricted module* $M|_{\iota_U}$ — *conjugated by the presheaf pullback pseudofunctoriality iso* $\texttt{pullbackComp}\,j\,\iota_U$ *on the source and by its contravariant transport on the dual target:*

$$\theta_{\iota_V,M} = (\texttt{pullbackComp}\,j\,\iota_U).\text{inv}_{\texttt{dual}\,M_{\text{val}}}\,;(\texttt{pullback}\,j)(\theta_{\iota_U,M})\,;\theta_{j,\;M|_{\iota_U}}\,;\texttt{dualIsoOfIso}(\texttt{forget}.\text{mapIso}\,((\texttt{restrictF}$$

Stating the middle factor at the restricted module $M|_{\iota_U}$ *is type-correct because the object-identification iso* `pushforwardObjValRestrictIso` *(790, bridge c.1) is* `Iso.refl`*: the underlying presheaf* $(M|_{\iota_U})_{\text{val}}$ *is* definitionally *the pushforward object* $(\texttt{pushforward}\,\beta_{\iota_U}).\text{obj}\,M_{\text{val}}$ *(the presheaf-level pushforward composition being the identity), so the codomain of* $(\texttt{pullback}\,j)(\theta_{\iota_U,M})$ — *namely* $(\texttt{pullback}\,\varphi_j).\text{obj}(\texttt{dual}((\texttt{pullback}\,\varphi_{\iota_U}).\text{obj}\,M_{\text{val}}))$ — *already matches the domain of* $\theta_{j,\;M|_{\iota_U}}$ *with no intervening transport. This is the form consumed by 802. On the dual (target) flank the contravariant transport is realised by* `dualIsoOfIso` *(714) of the forgetful image of the restriction-composition iso* $(\texttt{restrictFunctorComp}\,j\,\iota_U).\text{app}\,M$ *(792); this pseudofunctoriality leg appears in place of the* $\varrho\otimes\varrho$ *of the tensor side. This is the dual analogue of the composition role of 672 inside 685.*

Proof. The cocycle *factors cleanly*: every $H1$ factor cancels against the abstract $H1$ cocycle (a) (797) together with a single ordinary naturality square, leaving one isolated pushforward-flank identity that touches `sliceDualTransport`'s internals. Working sectionwise (each θ.hom is a presheaf morphism), abbreviate $dM = \texttt{dual}\,M_{\text{val}}$, $\texttt{sDT}_g = \texttt{isoMk}(\texttt{sliceDualTransport}\,g).\text{hom}$, $FC = \texttt{pushforwardComp}$, and $\varrho = (\texttt{restrictFunctorComp}\,j\,\iota_U).\text{app}\,M$, and expand each instance via 778 as $\theta_g = (H1_g)^{-1}\,;\texttt{sDT}_g$. The reduction runs in six steps.

1. **Strip the leading inverse.** Left-multiply both sides by $H1_{\iota_V}$.hom (using the isomorphism inverse-composition identity); the left side becomes $\texttt{sDT}_{\iota_V}$ and the right side acquires the prefix $H1_{\iota_V}$.hom.
2. **Expose the cocycle prefix.** Left-multiply by the `pushforwardComp` counit $FC.\text{hom}_{dM}$, legitimate because it is an isomorphism (hence an epimorphism); the left flank now reads $FC.\text{hom}_{dM}\,;H1_{\iota_V}.\text{hom}_{dM}$.

3. **Substitute cocycle (a).** By the abstract $H1$ cocycle 797 this prefix equals $(\texttt{pushforward}\,\beta_j).\mathrm{map}\,(H1_{\iota_U}.\mathrm{hom})$; $H1_j.\mathrm{hom}_{(\texttt{pushforward}\,\beta_{\iota_U})\,dM}$; $\texttt{pullbackComp}.\mathrm{hom}_{dM}$, regrouping the $H1_{\iota_U}$ factor across the pushforward middle functor on the source flank (`pullbackComp`, 791).

4. **Cancel `pullbackComp`.** The trailing $\texttt{pullbackComp}.\mathrm{hom}_{dM}$ meets the leading $(\texttt{pullbackComp}\,j\,\iota_U).\mathrm{inv}_{dM}$ of the right side and the two cancel (as the hom-inv identity of an isomorphism).

5. **Slide $H1_j$ by one naturality.** The factor $H1_j.\mathrm{hom}$ is moved past $(\texttt{pullback}\,\varphi_j).\mathrm{map}\,(\theta_{\iota_U}.\mathrm{hom})$ by a *single* ordinary naturality square of the natural transformation $H1_j.\mathrm{hom}$ at $\theta_{\iota_U}.\mathrm{hom}$ — this is the only "slide", and it is plain naturality of $H1_j.\mathrm{hom}$ as a natural transformation, not a bespoke multi-step calculation.

6. **Collapse the $H1$ round-trips.** The hom-inv identities of $H1_{\iota_U}$ and $H1_j$ annihilate the remaining $H1$ factors, leaving the sole residual

$$FC.\mathrm{hom}_{dM};\texttt{sDT}_{\iota_V} \;=\; (\texttt{pushforward}\,\beta_j).\mathrm{map}(\texttt{sDT}_{\iota_U});\texttt{sDT}_j(M|_{\iota_U});\texttt{dualIsoOfIso}(\texttt{forget}.\mathrm{mapIso}\,\varrho).\mathrm{hom}. \tag{$**$}$$

The residual $(**)$ is the pushforward-flank pseudofunctoriality of `sliceDualTransport` — the only step that touches the transport's internals — and it plays the dual role of 672 inside the proven tensor analogue 685.

The leading factor $FC.\mathrm{hom}$ is explicit, not opaque. Here $FC = \texttt{leftAdjointCompIso}$ of the three pushforward adjunctions. It is *correct* that $FC.\mathrm{hom}$ is *not* definitionally equal to the concrete presheaf pushforward-composition iso `pushforwardComp`: the domain $(\texttt{pushforward}\,\beta_j \circ \texttt{pushforward}\,\beta_{\iota_U}).\mathrm{obj}\,dM$ and the codomain $(\texttt{pushforward}\,\beta_{\iota_V}).\mathrm{obj}\,dM$ are themselves not defeq, because the `opensFunctor` of a composite open immersion is not the composite of the `opensFunctor`s on the nose. This is *not*, however, an obstruction: $FC.\mathrm{hom}$ is *fully explicit* via the Mathlib component formula `leftAdjointCompIso_hom_app` (795), which expands $FC.\mathrm{hom.app}\,X$ as the unit of the composite adjunction pushed through the two left adjoints, the reconciling $e.\mathrm{inv}$, and the two counits. With $e = \texttt{pushforwardComp} = \texttt{Iso.refl}$ the middle $e.\mathrm{inv}$ factor drops, leaving an explicit unit/counit composite; the residual $(**)$ is reachable via this formula.

The close route. Expose $FC.\mathrm{hom}$ through the component formula 795. (The mate characterisation 794 is an alternative reading — it identifies the conjugate $\texttt{conjugateEquiv}\,FC.\mathrm{hom} = e.\mathrm{inv} = \mathbf{1}$, but delivers only the mate, not the section-level component directly.) After substituting the explicit unit/counit form of $FC.\mathrm{hom}$ and cancelling all `appIso`/`dualUnitRingSwap` change-of-rings layers — the structure-ring layers reconciling through the cocycle `comp_appIso` ($\texttt{Scheme.Hom.comp\_appIso}$, $(h;f).\texttt{appIso}\,U = X.\mathrm{presheaf.mapIso}(\texttt{eqToHom})^{\mathrm{op}};f.\texttt{appIso}(h.\texttt{opensFunctor}\,U)$; $h.\texttt{appIso}\,U$) — the residual $(**)$ reduces to a single *thin-poset* / `opensFunctor`*-composition coherence*. The leading $FC.\mathrm{hom}$ is handled sectionwise via 795: it is the composite of the pushforward adjunction's unit and two counits, and *each* of these, evaluated at a section, is — by the triangle-component identities 688 and 689 — precisely a presheaf restriction map $X.\mathrm{presheaf.map}(-)^{\mathrm{op}}$ of an $\mathrm{Opens}\,X$ morphism. This form is reached *definitionally* (the units/counits are restriction maps on the nose); direct unfolding of `pushforwardPushforwardAdj` would reintroduce the `restrictScalars` carrier diamond. Folding the three restriction maps by functoriality yields one $X.\mathrm{presheaf.map}(\mathrm{composite})^{\mathrm{op}}$; the two resulting slice objects share domain ($(j;\iota_U).\texttt{opensFunctor} = \iota_U.\texttt{opensFunctor} \circ j.\texttt{opensFunctor}$, `comp_image`) and codomain, hence are `Subsingleton`-equal in the thin poset $\mathrm{Opens}\,X$, and the residual closes. The slice-side split 734 supplies the matching $\texttt{sDT}_{j;\iota_U}$ factorisation on that flank. *The mate FC must remain the* `leftAdjointCompIso`: it cannot be replaced by the concrete $\texttt{pushforwardComp}\,\beta_j\,\beta_{\iota_U}$ — their codomain functors differ by the `comp_appIso` reindex (so the two are not even type-compatible), and the swap would break the proven cocycle (a) 797, which needs both flanks to be `leftAdjointCompIso` of the *same e*. The sectionwise content, supplied by 734, runs through the forward apply lemma 731: on each open W of V and each dual section it rewrites the composite `sDT` factor into its explicit `dualUnitRingSwap` $(j\,;\iota_U)$-conjugated reindex; by the action 732 that swap is the hom-direction of the composite structure-ring iso, and the cocycle `comp_appIso` (733) splits it into the two single-immersion swaps of ι_U and j, reindexed by the direct-image composition $(j\,;\iota_U)$.`opensFunctor` $= \iota_U$.`opensFunctor` $\circ\, j$.`opensFunctor` (`comp_image`). The module-argument naturality of θ (784) together with poset-thinness of Opens — the consumer reads everything over the single open $j;\iota_U = \iota_V$ (741) — makes the assembly canonical, and the target-flank reindexing is realised by `dualIsoOfIso` (714) of the forgetful image of ϱ (792). The middle factor already sits at $M|_{\iota_U}$ because the object identification `pushforwardObjValRestrictIso` (790, c.1) is `Iso.refl`: the underlying presheaf of $M|_{\iota_U}$ is definitionally the pushforward object, so no intervening transport appears.

The entire six-step $H1$-cancellation above is carried out by a single generic, instance-agnostic skeleton lemma `c2_assemble` (798), stated over an arbitrary ambient category. It takes the structural facts of the reduction as hypotheses — the two iso cancellations of $H1_{h;f}$ and FC at dM, the abstract $H1$ cocycle (a) (797), the `pullbackComp` cancellation, the single $H1_h$.hom naturality, the `pushforward` β_h-functoriality fold, the $H1_h$ round-trip at $d(M|_{\iota_U})$, and the residual $(**)$ — and assembles the c.2 identity from them by associativity and identity rewriting alone. The abstract identity transfers across the definitional boundary between `PresheafOfModules` and `SheafOfModules` because the skeleton is stated for abstract morphisms in an arbitrary category: instantiating at the concrete pullback/pushforward morphisms specialises the identity directly to the c.2 goal, and the definitional equality between the two categories is absorbed into the abstract hypothesis statements rather than appearing as an additional verification condition. ☐

Lemma 800 (L1: folded composition cocycle for `dual_restrict_iso` along a composite)**.** *Source: internal categorical construction; no external reference (dual mirror of the tensor-flank composition law 777). Let $f : Z \hookrightarrow Y$ and $g : Y \hookrightarrow X$ be composable open immersions (*general *— not tied to the chart j) and $M \in$ Scheme.`Modules` X. Write $D(h) :=$ `dual_restrict_iso` h : $($`dual` $A)|_h \cong$ `dual`$(A|_h)$ (737). The dual-restriction comparison along the composite $f\,;g$ factors* — entirely at the folded Scheme-`dual` level, *so that every junction is a Scheme-dual ↔ Scheme-dual identification and no* `sheafification`*-colimit seam appears in the statement — as*

$$D(f;g) \;=\; (\texttt{restrictFunctorComp}\, f\, g).\mathrm{app}\,(\texttt{dual}\, M)\;;\,(\texttt{restrict}\, f)\, D(g)\;;\,D(f)\,(M|_g)\;;\,\texttt{dualIsoOfIso}\,((\texttt{restrictFu}$$

where `restrictFunctorComp` $f\,g$ *(792) is the restriction-composition pseudofunctoriality iso $A|_{f;g} \cong (A|_g)|_f$ and* `dualIsoOfIso` *(714) is its contravariant dual image. Choosing the* `restrictFunctorComp`*-induced head (rather than a* `pullbackComp` *one) keeps both flanks in the restriction world, hence syntactically folded.*

Proof. This is the dual mirror of the tensor-flank composition law (777), routed through Cone B (internal-hom base-change); no "closed functor preserves duals" principle applies, since pullback along an open immersion is neither faithful nor an equivalence. The seam between the Scheme-`dual` world and the sheafified presheaf-dual world is crossed *inside* this proof exactly as bridges (b), (c) and the S2 route do — never across the stated junction.

By the dual analogue of Bridge B1 (785) each $D(\cdot)$ factors as `restrictFunctorIsoPullback` ; `sheafificationCompPullback` ; `sheafification`(θ). Substituting this for the three instances $D(f\,;g)$, $D(g)$ and $D(f)$, the leading `restrictFunctorIsoPullback` factors cancel against the `restrictFunctorComp`-reindex by Bridge B2 (752), reducing the identity to its `sheafificationCompPullback`- and θ-legs. The `sheafificationCompPullback`-cocycle over the composite is closed by bridge (b) (789); the θ-cocycle by bridge (c) (799, itself through the immersion-naturality of θ, 784).

Bridge (c) delivers its middle θ-factor at the pushforward object (`pushforward` β_g).obj M_{val}; the object-identification iso `pushforwardObjValRestrictIso` (790, bridge c.1), which is `Iso.refl`, converts that factor to the $D(f)\,(M|_g)$-form appearing here, so the substitution is type-correct. The contravariant reindexing on the dual flank is the dual image `dualIsoOfIso` (714) of (`restrictFunctorComp` $f\,g$).ap (792). Assembling the leg-for-leg identities yields the displayed composite. ☐

Lemma 801 (L2: folded `restrictFunctorCongr` coherence for `dual_restrict_iso`)**.** *Source: internal categorical construction; no external reference. Let $f, g : Y \hookrightarrow X$ be open immersions with $f = g$ (witnessed by hf), and $M \in$ Scheme.*`Modules`*X. The dual-restriction comparison is coherent with the restriction congruence* `restrictFunctorCongr` *$hf : A|_f \cong A|_g$, at the folded Scheme-*`dual` *level (both junctions Scheme-dual ↔ Scheme-dual, seam-free):*

$$(\texttt{restrictFunctorCongr}\,hf).\mathrm{app}\,(\texttt{dual}\,M)\,;\,\texttt{dual\_restrict\_iso}\,g\,M \;=\; \texttt{dual\_restrict\_iso}\,f\,M\,;\,\texttt{dualIsoOfIso}$$

This is the dual mirror of `pullbackTensorMap_pullbackCongr`*, used to convert the ι_V-leg of S3 to the composite $j;\iota_U$.* (This brick is already verified in Lean: `dual_restrict_iso_pullbackCongr`, private, axiom-clean.)

Proof. Substitute $g := f$ along hf. Then the restriction congruence collapses, `restrictFunctorCongr` (rfl) = Iso.refl (section-level, since its hom is rfl-valued on each section), so the left factor on the source flank is the identity; and on the dual flank `dualIsoOfIso` (Iso.refl) = Iso.refl (870, 714). Both sides reduce to `dual_restrict_iso` $f\,M$, and the equation holds. ☐

Lemma 802 (S3: dual-restriction comparison commutes with further restriction (clean core))**.** *Let $M \in$ Scheme.*`Modules` *X and let $j : V \hookrightarrow U$ be the chart morphism above. Write $D(f) :=$* `dual_restrict_iso` *$f : (\texttt{dual}\,M)|_f \cong \texttt{dual}(M|_f)$ (737). The bare dual-restriction comparison is natural under further restriction along j: modulo the reindexing $\varrho =$* `restrictCompReindex` *j (749), with its contravariant image* `dualIsoOfIso` *ϱ (714) on the dual flank,*

$$D(\iota_V) \;=\; \varrho_{\texttt{dual}\,M}\,;\,(\texttt{restrict}\,j)\,D(\iota_U)\,;\,\texttt{dual\_restrict\_iso}\,j\,(M|_{\iota_U})\,;\,\texttt{dualIsoOfIso}\,\varrho_M.$$

This square proves only *the* `dual_restrict_iso` *immersion-naturality; the trivialisation step's* (`dualIsoOfIso` e^M)$^{-1}$ *transport and the datum refinement $e^M \mapsto$* `restrictIsoUnitOfLE` *$h_{VU}\,e^M$ are* not *part of the claim — they telescope in the parent (841).*

Proof. The square is now a short assembly of the folded composition cocycle L1 (800) and the folded congruence coherence L2 (801); the seam-crossing work (dual-B1, B2, bridges (b), (c), c.1) has been moved into L1's proof, where the unfolding to `restrictFunctorIsoPullback` ; `sheafificationCompPullback` ; `sheafification`(θ) is confined. The proof is the dual-flank mirror of S2, routed through Cone B (internal-hom base-change), not through any sheaf-level monoidal structure.

Specialise L1 (800) at $(f, g) = (j, \iota_U)$. The equality of opens $j\,;\iota_U = \iota_V$ ($hj\iota$, 741) converts the ι_V-leg $D(\iota_V)$ on the left of the target into the composite leg $D(j\,;\iota_U)$ via L2 (801). Combining the `restrictFunctorCongr` $hj\iota$ factor that L2 produces with the `restrictFunctorComp` $j\,\iota_U$ head of L1 reconstitutes $\varrho =$ `restrictCompReindex` j, which is by definition (`restrictFunctorCongr` $hj\iota$)$^{-1}$; `restrictFunctorComp` $j\,\iota_U$ (749), giving the head $\varrho_{\texttt{dual}\,M}$. On the dual flank the two contravariant `dualIsoOfIso` transports fuse, by contravariant functoriality (869, 714), into the single factor `dualIsoOfIso` ϱ_M. This gives the displayed square. The trivialisation step's (`dualIsoOfIso` e^M)$^{-1}$ transport and the datum refinement are not part of this square; they telescope in the parent (841). ☐

Definition 803 (Oplax-monoidal unit map η of the presheaf pullback — conceptual anchor)**.** *Provided by Mathlib.* For any oplax monoidal functor F, Mathlib supplies the *unit comparator* $\eta_F : F(\mathbf{1}) \to \mathbf{1}$, the oplax counterpart of the unit map. Applied to the oplax-monoidal presheaf pullback $\texttt{PresheafOfModules.pullback}\,\varphi$ (666) along the chart immersion $j : V \hookrightarrow U$, $\varphi = j.\texttt{toRingCatSheafHom}$, it furnishes the bare oplax unit comparison

$$\eta(\texttt{pullback}\,\varphi) : (\texttt{pullback}\,\varphi).\mathrm{obj}\,\mathbf{1}_U \longrightarrow \mathbf{1}_V.$$

This is the *conceptual* pullback-flank unit map. Its invertibility is *not* asserted here. The actual pullback-side comparator p used in the dual base-change theory is built instead as the H1 transport of the pushforward comparator q; see 804.

Definition 804 (Pullback-side unit comparator p — the H1 realization)**.** *Source: internal categorical construction; no external reference.* Let $j : V \hookrightarrow U$ be the chart immersion, $\varphi = j.\texttt{toRingCatSheafHom}$ the pullback ring map, and β the corresponding (sectionwise-bijective) pushforward ring map. The pullback-side unit comparator used throughout the dual base-change theory is the isomorphism

$$p := \texttt{presheafPullbackUnitIso}\,j : (\texttt{pullback}\,\varphi).\mathrm{obj}\,\mathbf{1}_U \;\cong\; \mathbf{1}_V,$$

realized *not* as the bare oplax η (803) but through the *H1 reconciliation.* Both $\texttt{pullback}\,\varphi$ and the open-immersion pushforward $\texttt{pushforward}\,\beta$ are left adjoints of the same right adjoint $\texttt{pushforward}\,\varphi$; the left-adjoint-uniqueness isomorphism

$$H_1 := \texttt{leftAdjointUniq}(\texttt{pushforwardPushforwardAdj}, \texttt{pullbackPushforwardAdjunction}\,\varphi) : \texttt{pushforward}\,\beta \xrightarrow{\sim}$$

identifies the two left adjoints (764, 688). Evaluated at the structure-sheaf unit $\mathbf{1}_U$, it carries the pullback flank to the pushforward flank, on which the pushforward comparator $q = \texttt{presheafPushforwardUnitIso}\,j$ (807) contracts to $\mathbf{1}_V$. Composing,

$$p \;=\; (H_1.\mathrm{app}\,\mathbf{1}_U)^{-1} \;;\; q,$$

the H1-transport of q. This is exactly the identification by which the two flank comparators are matched at the structure-sheaf unit in the θ-at-unit bridge 811.

Definition 805 (Lax-monoidal unit map ε of the presheaf pushforward — conceptual anchor)**.** *Provided by Mathlib.* For any lax monoidal functor G, Mathlib supplies the *unit comparator* $\varepsilon_G : \mathbf{1} \to G(\mathbf{1})$. Applied to the lax-monoidal presheaf pushforward $\texttt{PresheafOfModules.pushforward}\,\beta$ (665) along the chart immersion, it gives the bare lax unit comparison

$$\varepsilon(\texttt{pushforward}\,\beta) : \mathbf{1}_V \longrightarrow (\texttt{pushforward}\,\beta).\mathrm{obj}\,\mathbf{1}_U,$$

the presheaf analogue of $\texttt{SheafOfModules.unitToPushforwardObjUnit}$. This is the *conceptual* pushforward-flank unit map; its invertibility is supplied not by the lax structure but by the strong monoidality of $\texttt{pushforward}\,\beta$ for an open immersion (806), and the resulting comparator q is recorded in 807.

Lemma 806 (Strong monoidality of the open-immersion presheaf pushforward $\texttt{pushforward}\,\beta$)**.** *Source: internal categorical construction; no external reference. Let $f : Y \to X$ be an open immersion and $\beta = \texttt{whiskerRight}\,\alpha\,(\texttt{forget}_2)$ the induced structure-ring comparison, where $\alpha.\mathrm{app}\,U = (f.\texttt{appIso}\,U)^{-1}$. Since f is an open immersion each $\alpha.\mathrm{app}\,U$ is invertible, hence β is sectionwise bijective. Consequently the presheaf pushforward factors as the strong-monoidal composite*

$$\texttt{pushforward}\,\beta \;=\; \texttt{pushforward}_0^{\mathrm{Comm}} \circ \texttt{restrictScalars}\,\beta,$$

in which the topological pushforward $\texttt{pushforward}_0^{\mathrm{Comm}}$ *is strong monoidal and* $\texttt{restrictScalars}\,\beta$ *is strong monoidal because* β *is sectionwise bijective (*`restrictScalarsMonoidalOfBijective`*). Hence* $\texttt{pushforward}\,\beta$ *carries a* strong-*monoidal structure, upgrading the ambient lax instance (665); in particular its unit comparator* ε *and tensorator* μ *are isomorphisms.*

Definition 807 (Pushforward-side unit comparator q — the strong-monoidal ε inverse)**.** *Source: internal categorical construction; no external reference.* In the open-immersion setting above, the strong monoidality of $\texttt{pushforward}\,\beta$ (806) makes the unit comparator an isomorphism $\varepsilon_{\mathrm{iso}}(\texttt{pushforward}\,\beta) : \mathbf{1}_V \cong (\texttt{pushforward}\,\beta).\mathrm{obj}\,\mathbf{1}_U$. The pushforward-side unit comparator is its inverse,

$$q := \varepsilon_{\mathrm{iso}}(\texttt{pushforward}\,\beta)^{-1} : (\texttt{pushforward}\,\beta).\mathrm{obj}\,\mathbf{1}_U \;\cong\; \mathbf{1}_V,$$

the strong-monoidal $\varepsilon_{\mathrm{iso}}$ inverse (not the bare lax ε of 805).

Lemma 808 (Carrier-value of the pushforward comparator: q is the $f.\texttt{appIso}$-conjugation on carriers)**.** *Source: internal categorical construction; no external reference. This is the unit-side analogue of the tensorator collapse 694. With* $f : Y \to X$ *an open immersion,* β *the (sectionwise-bijective) structure-ring comparison, and* $q = \texttt{presheafPushforwardUnitIso}\,f = \varepsilon_{\mathrm{iso}}(\texttt{pushforward}\,\beta)^{-1}$ *(807), the sectionwise carrier action of* $q.\mathrm{hom} = \varepsilon_{\mathrm{iso}}(\texttt{pushforward}\,\beta)^{-1}$ *at an* arbitrary *carrier element* y *is, over a section* V*,*

$$(q.\mathrm{hom})_V(y) \;=\; (f.\texttt{appIso}\,V).\mathrm{hom}\,(y),$$

the $f.\texttt{appIso}$*-conjugation of* y*. The unit case* $y = 1$ *is the motivating instance; the general form is the one the L1′ consumer 811 requires (it instantiates at a non-unit carrier).*

Proof. We compute the carrier action of $q.\mathrm{hom} = \varepsilon_{\mathrm{iso}}(\texttt{pushforward}\,\beta)^{-1}$ at an arbitrary carrier element y by splitting the strong-monoidal structure of $\texttt{pushforward}\,\beta$ along the factorization $\texttt{pushforward}\,\beta = \texttt{pushforward}_0^{\mathrm{Comm}} \circ \texttt{restrictScalars}\,\beta$ furnished by 806. The argument is cleanest run on the *inverse* $q.\mathrm{inv} = \varepsilon(\texttt{pushforward}\,\beta)$: under `Functor.Monoidal.comp` the lax unit comparator ε of the composite is the composite of the two factors' ε's, and the scalar-forgetting pushforward $\texttt{pushforward}_0^{\mathrm{Comm}}$ has $\varepsilon = \mathrm{id}$, so $q.\mathrm{inv}$ reduces to the ε-value of $\texttt{restrictScalars}\,\beta$. That value is the sectionwise structure-ring transport $(f.\texttt{appIso}\,V).\mathrm{inv}$, by the abstract ε-template 809 (the strong-monoidal companion of the oplax η-template 690; $\varepsilon = \eta^{-1}$ since β is sectionwise bijective). Hence $(q.\mathrm{inv})_V(w) = (f.\texttt{appIso}\,V).\mathrm{inv}\,(w)$ (810). Now both $(q.\mathrm{hom})_V(y)$ and $(f.\texttt{appIso}\,V).\mathrm{hom}\,(y)$ are sent to y by the injective $(q.\mathrm{inv})_V$ — the former by the round-trip $q.\mathrm{hom}\,;q.\mathrm{inv} = \mathrm{id}$, the latter by `appIso`'s hom-inv identity — so they are equal: $(q.\mathrm{hom})_V(y) = (f.\texttt{appIso}\,V).\mathrm{hom}\,(y)$, the $f.\texttt{appIso}$-conjugation of y. This isolates the same structure-ring transport that the tensorator collapse 694 isolates on the μ-side. □

Lemma 809 (Abstract ε-value of `restrictScalars` on carriers)**.** *Source: internal categorical construction; no external reference. For ring-presheaves* $R, S : \mathcal{C}^{\mathrm{op}} \to \texttt{CommRingCat}$ *and a natural transformation* $\alpha : R \to S$*, the lax unit comparator* ε *of* $\texttt{PresheafOfModules.restrictScalars}\,\alpha$ *acts sectionwise on carriers by the ring map* α*: over a section* W*,* $\varepsilon.\mathrm{app}\,W(r) = (\alpha.\mathrm{app}\,W).\mathrm{hom}\,(r)$*. This is the abstract* $\{R, S : \mathcal{C}^{\mathrm{op}} \to \texttt{CommRingCat}\}$ *companion of the unit-value template 690; the abstract framing makes the ambient* `CommRing` *instance native (the concrete post-*$\texttt{forget}_2$ *Scheme carrier exposes only* `Ring`*).*

Proof. Unfold `ModuleCat.restrictScalars`'s lax-monoidal ε: it is, by construction, the scalar-action map induced by $\alpha.\mathrm{app}\,W$ on the unit module, i.e. multiplication by $(\alpha.\mathrm{app}\,W).\mathrm{hom}\,(r)$. Reading off its value at a carrier element gives the claim directly. □

Lemma 810 (Carrier-value of the pushforward comparator inverse $q.\mathrm{inv} = \varepsilon$)**.** *Source: internal categorical construction; no external reference. In the open-immersion setting, with* $q = \texttt{presheafPushforwardUnitIso}\, f$ *(807), the inverse* $q.\mathrm{inv} = \varepsilon(\texttt{pushforward}\,\beta)$ *acts sectionwise on carriers as the inverse structure-ring transport: over a section* V*,* $(q.\mathrm{inv})_V(w) = (f.\texttt{appIso}\,V).\mathrm{inv}\,(w)$*. This is the* ε*-side companion of 808; it is consumed both inside that lemma and directly by the L1′ flank 811.*

Proof. Split $\texttt{pushforward}\,\beta = \texttt{pushforward}_0^{\mathrm{Comm}} \circ \texttt{restrictScalars}\,\beta$ (806). Under $\texttt{Functor.Monoidal.comp}$ the composite ε is the composite of the factors'; $\texttt{pushforward}_0^{\mathrm{Comm}}$ contributes $\varepsilon = \mathrm{id}$, leaving the ε-value of $\texttt{restrictScalars}\,\beta$, which by 809 is $(\beta.\mathrm{app}\,V)$-multiplication, i.e. $(f.\texttt{appIso}\,V).\mathrm{inv}$ on carriers. Hence $(q.\mathrm{inv})_V(w) = (f.\texttt{appIso}\,V).\mathrm{inv}\,(w)$. ☐

Lemma 811 (S4a / L1′: θ-at-unit bridge — the presheaf dual base-change comparison at the structure sheaf is the dual-of-unit iso)**.** *Source: internal categorical construction; no external reference. Let* $j : V \hookrightarrow U$ *be the chart immersion with* $j\,;\iota_U = \iota_V$*,* $\varphi = j.\texttt{toRingCatSheafHom}$ *the pullback ring map, and* β *the corresponding pushforward ring map. Evaluate the presheaf dual base-change comparison* $\theta_M = \texttt{presheafDualPullbackComparison}\, j\, M$ *(778) at the structure-sheaf unit* $M = \mathbf{1}_U = \texttt{SheafOfModules.unit}\,U.\texttt{ringCatSheaf}$*. Because* θ *lands the dual on the pushforward flank, this gives*

$$\theta_{\mathbf{1}} : \texttt{pullback}_\varphi(\texttt{dual}\,\mathbf{1}_U) \;\cong\; \texttt{dual}((\texttt{pushforward}_\beta)\,\mathbf{1}_U).$$

Two distinct *unit comparators enter, one per flank:*

- *the pullback-side comparator* $p := \texttt{presheafPullbackUnitIso}\,j : \texttt{pullback}_\varphi\,\mathbf{1}_U \cong \mathbf{1}_V$ *(804), the H1-transport* $(H_1.\mathrm{app}\,\mathbf{1}_U)^{-1}\,; q$ *of the pushforward comparator;*
- *the pushforward-side comparator* $q := \varepsilon_{\mathrm{iso}}(\texttt{pushforward}\,\beta)^{-1} : (\texttt{pushforward}_\beta)\,\mathbf{1}_U \cong \mathbf{1}_V$ *(807), the inverse of the strong-monoidal unit comparator of the pushforward.*

*Modulo the eval-at-*1 *dual-of-unit isomorphism* $\texttt{presheafDualUnitIso}$ *(727) on the* U*- and* V*-flanks,* $\theta_{\mathbf{1}}$ is *the dual-of-unit identification, with the two comparators on opposite flanks:* q *is transported contravariantly through* $\texttt{dualIsoOfIso}$ *onto the pushforward codomain of* $\theta_{\mathbf{1}}$*, while* p *closes the pullback domain on the right:*

$$\theta_{\mathbf{1}}\;;\,(\texttt{dualIsoOfIso}\,q)^{-1}\,;\texttt{presheafDualUnitIso}_V \;=\; \texttt{pullback}_\varphi(\texttt{presheafDualUnitIso}_U)\,;p,$$

both sides being isomorphisms $\texttt{pullback}_\varphi(\texttt{dual}\,\mathbf{1}_U) \cong \mathbf{1}_V$*. This is the unit-specialisation of the immersion-naturality 784 and the immersion analogue of the unit-automorphism core 844.*

Proof. By element extensionality on dual sections, exactly as in 784, of which this is the unit specialisation. Both $\theta_{\mathbf{1}}$ and $\texttt{presheafDualUnitIso}$ are morphisms of presheaves of modules: $\theta_{\mathbf{1}}$ is the internal-hom base-change comparison $f^*\mathcal{H}om(\mathbf{1},\mathcal{O}) \to \mathcal{H}om(f^*\mathbf{1}, f^*\mathcal{O})$ (778), and $\texttt{presheafDualUnitIso}$ is the sectionwise evaluation-at-1 of the internal hom of the unit into itself (727). Testing the displayed equality against an arbitrary dual section reduces, as in 784, to the built-in naturality of θ (its $\texttt{.hom}$ naturality), now read through the eval-at-1 identification, where the internal-hom evaluation (709) collapses both composites to evaluation of the pulled-back section at the image of 1.

The two flanks carry the two *different* comparators. On the pushforward codomain of $\theta_{\mathbf{1}}$ the comparator $q = \varepsilon_{\mathrm{iso}}(\texttt{pushforward}\,\beta)^{-1}$ (807) is transported contravariantly by $\texttt{dualIsoOfIso}$; on the pullback domain the comparator $p = \texttt{presheafPullbackUnitIso}\,j$ (804) closes the square.

Because p is, by definition, the H1-transport $(H_1.\mathrm{app}\,\mathbf{1}_U)^{-1}\,;q$, the two leading H1 factors — the one hidden inside p and the one produced by unfolding $\theta_{\mathbf{1}}$ at the dual unit — are telescoped away by the naturality of the natural isomorphism $H_1 : \texttt{pushforward}\,\beta \xrightarrow{\sim} \texttt{pullback}\,\varphi$ at the dual-unit contraction $\texttt{presheafDualUnitIso}_U$ (764). After this cancellation the entire claim reduces to a single *pushforward-flank* sectionwise identity (†): writing sDT for the slice-dual transport $\texttt{sliceDualTransport}\,f\,\mathbf{1}_U$,

$$\texttt{sDT}\,;(\texttt{dualIsoOfIso}\,q)^{-1}\,;\texttt{presheafDualUnitIso}_V \;=\; (\texttt{pushforward}_\beta)(\texttt{presheafDualUnitIso}_U)\,;q.$$

This identity is proved sectionwise, as the immersion analogue of 844. Fix a section V and a dual section φ.

- *Left-hand side.* The verified forward transport formula (731) evaluates sDT at the terminal slice as the reindexing across $f.\texttt{opensFunctor}$ conjugated by $f.\texttt{appIso}$ and the dual-unit ring swap; precomposing by q and evaluating at 1 (the dual-unit iso $\texttt{presheafDualUnitIso} = \texttt{evalLin}\,(\cdot)\,1$, 727) exposes $\texttt{evalLin}\,\mathbf{1}_U\,(\mathrm{op}\,fV)\,\varphi\,1$ conjugated by $f.\texttt{appIso}$ and the dual-unit ring swap.

- *Right-hand side.* The pushforward of the eval-at-1 contraction $(\texttt{pushforward}_\beta)(\texttt{presheafDualUnitIso}_U)$ followed by the unit comparator $q.\mathrm{hom}$ is reduced by the unit-value sub-lemma 808: the carrier action of $q.\mathrm{hom}$ at 1 is the $f.\texttt{appIso}$-conjugation of 1, so the right side reduces to the same $f.\texttt{appIso}$-conjugated $\texttt{evalLin}\,\mathbf{1}_U\,(\mathrm{op}\,fV)\,\varphi\,1$.

Both sides being the same eval-at-1 value conjugated by $f.\texttt{appIso}$, equality is the commutativity of $\mathcal{O}_V$-linear endomorphisms of the (commutative) structure ring — the same linearEndo_apply_comm-style step that closes 844, now with p and q in place of the unit automorphism. Hence both sides of (†) agree sectionwise, and with the H1 telescope this gives the displayed square. □

Lemma 812 (S4a / L2′: unit assembly — the dual-restriction comparison at the unit intertwines the dual-unit contractions). *Source: internal categorical construction; no external reference. With $j : V \hookrightarrow U$ as above, write $\mathbf{1}_U, \mathbf{1}_V$ for the structure sheaves, $u_j = \texttt{unitRestrictIso}\,j$ (750) for the chart-scheme unit identification $\mathbf{1}_U|_j \cong \mathbf{1}_V$, and $\texttt{dual\_unit\_iso}_U, \texttt{dual\_unit\_iso}_V$ for the dual-unit contractions (738). The* bare *dual-restriction comparison $\texttt{dual\_restrict\_iso}\,j\,(\mathbf{1}_U) : (\texttt{dual}\,\mathbf{1}_U)|_j \cong \texttt{dual}(\mathbf{1}_U|_j)$ (737) intertwines the two contractions with the unit-restriction identification:*

$$(\texttt{restrict}\,j)\,\texttt{dual\_unit\_iso}_U\,;u_j \;=\; \texttt{dual\_restrict\_iso}\,j\,(\mathbf{1}_U)\,;(\texttt{dualIsoOfIso}\,u_j)^{-1}\,;\texttt{dual\_unit\_iso}_V,$$

both sides being isomorphisms $(\texttt{dual}\,\mathbf{1}_U)|_j \cong \mathbf{1}_V$.

Proof. Drop both sides to their underlying morphisms (Iso.ext) and flatten every ;, mapIso and inverse into a single right-associated composite. Expanding the bare comparison $\texttt{dual\_restrict\_iso}\,j\,(\mathbf{1}_U)$ by the dual-B1 factorisation (785) into the three-factor composite $\texttt{restrictFunctorIsoPullback}\,j$; $\texttt{sheafificationCompPullback}\,\varphi$; $\texttt{sheafification}(\theta_{\mathbf{1}})$, and each contraction by its definition $\texttt{dual\_unit\_iso} = \texttt{sheafification}(\texttt{presheafDualUnitIso})\,;\texttt{counit}$ (738), turns the square into a purely syntactic identity of two right-associated ;-chains.

This flattened identity is discharged by a verified *category-algebra assembly*: an abstract reassociation lemma over a bare category, proven by associativity bookkeeping alone, that crosses the SheafOfModules ; seam without any unfolding. The assembly splits the left-hand unit-restriction map m as a prefix m_{p} followed by a core m_{c} and consumes four hypotheses — a split identity, a flank identity, an scp-prefix identity, and a tail identity. The split, scp-prefix and tail data are folded ;-bookkeeping (the `restrictFunctorIsoPullback`;`sheafificationCompPullback` prefix is matched, exactly as in S3, by the composition law of bridge (b), 789, threading the equality of opens $j \,;\, \iota_U = \iota_V$, 741; and the sheafification-counit naturality moves the counits past the restriction while contravariant functoriality of `dualIsoOfIso`, 714, collects the dual transport into $(\texttt{dualIsoOfIso}\, u_j)^{-1}$). The whole θ-at-unit identification supplied by L1′ (811) is consumed inside this fold. The substantive content of the four hypotheses reduces to exactly two residual reconciliations:

- the *dual flank* (813): the forgotten sheaf-level unit-restriction comparator equals the presheaf pushforward-unit comparator q, which discharges the dual-flank leg of the assembly;
- the *pullback (unit-only) flank* (816): the sheaf pullback-unit comparator sheafifies the presheaf pullback-unit comparator p, threaded through `sheafificationCompPullback` and the two sheafification counits. After reduced-image/SCP-naturality and the common-prefix cancel, the residual is unit-only — no dual factor remains.

Modulo these two reconciliations the square is closed by the folded assembly; the keystone is therefore sorry-free up to them. The genuine open work is precisely these two adjunction-coherence reconciliations, both of which consume the already-proven left-adjoint-uniqueness and hom-transpose infrastructure (677, 679, 676, 764, 808). □

Lemma 813 (RES1 (dual-flank reconciliation): the forgotten sheaf unit-restriction iso equals the presheaf pushforward-unit comparator)**.** *Source: internal categorical construction; no external reference. Let $f : Y \hookrightarrow X$ be an open immersion with pullback ring map $\varphi = f.$*`toRingCatSheafHom` *and pushforward ring map β. Applying the forgetful functor* `forget` $:$ Scheme.`Modules` $Y \to$ `PresheafOfModules` *to the composite of the chart-scheme unit identification* `restrictFunctorIsoPullback` f *(769) at the structure-sheaf unit $\mathbf{1}_X$ with the sheaf pullback-unit iso* `pullbackUnitIso` f *yields, on the nose, the presheaf pushforward-unit comparator $q =$* `presheafPushforwardUnitIso` f *(807, the inverse strong-monoidal unit comparator $\varepsilon(\texttt{pushforward}\,\beta)^{-1}$):*

$$\texttt{forget}.\mathrm{mapIso}((\texttt{restrictFunctorIsoPullback}\, f).\mathrm{app}\,(\mathbf{1}_X) \,;\, \texttt{pullbackUnitIso}\, f) \;=\; \texttt{presheafPushforwardUnitIs}$$

This is the leg of the unit assembly (812) that closes its dual flank.

Proof. Reduce to a sectionwise carrier identity at an open V and a carrier element y (drop both isomorphisms to their homs, then test presheaf-of-modules morphisms section by section and the resulting $\mathcal{O}_V$-linear maps element by element).

On the right, the carrier action of q.hom at V is the structure-ring isomorphism $(f.\texttt{appIso}\, V)$.hom by the proven linchpin 808.

On the left, the sectionwise carrier is the composite of the two sheaf-level unit comparators $(\texttt{restrictFunctorIsoPullback}\, f).\mathrm{app}\,\mathbf{1}_X$ followed by `pullbackObjUnitToUnit` φ. The first comparator is the left-adjoint-uniqueness isomorphism `leftAdjointUniq`(`restrictAdjunction` f) (`pullbackPushfor` (769); the second is the adjunction transpose (`pullbackPushforwardAdjunction` f).homEquiv^{-1} of the canonical `unitToPushforwardObjUnit` (670). Collapsing the composite

$$(\texttt{leftAdjointUniq}\,\mathrm{adj}_1\,\mathrm{adj}_2).\mathrm{hom}.\mathrm{app}\,x \;;\; \mathrm{adj}_2.\mathrm{homEquiv}^{-1}(g) \;=\; \mathrm{adj}_1.\mathrm{homEquiv}^{-1}(g)$$

by the uniqueness coherences 765, 766 and the unit triangle 764 rewrites the left composite as the single `restrictAdjunction`-unit transpose at V. is again $(f.\texttt{appIso}\, V)$.hom — the structure-sheaf specialisation, exactly mirroring the eval-core template 844. Both sides equal $(f.\texttt{appIso}\, V)$.hom, closing the identity. □

Lemma 814 (RES2 L1: the unit pullback identification factors through `sheafificationCompPullback` and the X-sheafification counit)**.** *Source: internal categorical construction; no external reference. Let* $f : Y \hookrightarrow X$ *be an open immersion with pullback ring map* $\varphi = f.\texttt{toRingCatSheafHom}$*, let* $M_0 = \texttt{SheafOfModules.unit}\,\mathcal{O}_X$ *be the structure-sheaf unit over* X *with underlying presheaf* $\mathbf{1}^{\mathrm{p}}_X = M_0.\mathrm{val}$*, and write* $\texttt{cuX} = (\texttt{sheafificationAdjunction}\,\mathbf{1}_X).\mathrm{counit}$ *for the* X*-sheafification counit. The per-object pullback identification* $\texttt{pullbackValIso}\,f\,M_0$ *(878) factors, on the nose, as the inverse* $\texttt{sheafificationCompPullback}\,\varphi$ *comparison at* $\mathbf{1}^{\mathrm{p}}_X$ *followed by the sheaf pullback of* `cuX` *at* M_0*:*

$$(\texttt{pullbackValIso}\,f\,M_0).\mathrm{hom} = (\texttt{sheafificationCompPullback}\,\varphi).\mathrm{inv}.\mathrm{app}\,\mathbf{1}^{\mathrm{p}}_X \; ; (\texttt{pullback}\,f).\mathrm{map}(\texttt{cuX}.\mathrm{app}\,M_0).$$

Proof. Immediate from the definition $\texttt{pullbackValIso}\,f\,M_0 = ((\texttt{sheafificationCompPullback}\,\varphi).\mathrm{app}\,\mathbf{1}^{\mathrm{p}}_X)^{-1};$ $(\texttt{pullback}\,f).\mathrm{mapIso}\,((\texttt{asIso}\,\texttt{cuX}).\mathrm{app}\,M_0)$ (878): reading off the homs of the two composed isomorphisms gives the displayed identity by reflexivity. This is the unit instance of the template factorisation `hpbv` of 775. ☐

Lemma 815 (RES2 L2: the sheafified presheaf pullback-unit comparator reconciles with the sheaf pullback identification)**.** *Source: internal categorical construction; no external reference. With notation as above, let* $\texttt{cuY} = (\texttt{sheafificationAdjunction}\,\mathbf{1}_Y).\mathrm{counit}$ *be the* Y*-sheafification counit,* $a_Y = \texttt{sheafification}\,\mathbf{1}_Y$*, and* $p = \texttt{presheafPullbackUnitIso}\,f$ *(804, the presheaf pullback-unit comparator* $p = (H_1\,\mathbf{1}^{\mathrm{p}}_X)^{-1} \; ; q$ *with* H_1 *the* `leftAdjointUniq` *factor and* $q = \texttt{presheafPushforwardUnitIso}\,f$*, 807). Then sheafifying* p *and cancelling the* Y*-sheafification counit reproduces the sheaf pullback identification followed by the sheaf pullback-unit iso* $\texttt{pullbackUnitIso}\,f$ *(671):*

$$a_Y.\mathrm{map}\,(p.\mathrm{hom}) \;\; ; \;\; \texttt{cuY}.\mathrm{app}\,\mathbf{1}_Y = (\texttt{pullbackValIso}\,f\,M_0).\mathrm{hom} \;\; ; \;\; (\texttt{pullbackUnitIso}\,f).\mathrm{hom}.$$

Both sides are sheaf morphisms $a_Y((\texttt{pullback}\,\varphi').\mathrm{obj}\,\mathbf{1}^{\mathrm{p}}_X) \to \mathcal{O}_Y$*.*

Proof. Transpose-free, by a (†)-factorisation of the presheaf comparator followed by the generic unit/counit collapse. Unfold $p = (H_1\,\mathbf{1}^{\mathrm{p}}_X)^{-1} \,;q$ (804), with $q = \texttt{presheafPushforwardUnitIso}\,f$ (807). The H1-reconciliation `H1inv_app_eq_pullbackVal_restrict` (774) writes H_1^{-1} as the sheafification unit η_Y, the forgotten sheaf pullback identification $\texttt{pullbackValIso}\,f\,\mathbf{1}_X$ (878), and the forgotten inverse `restrictFunctorIsoPullback`; RES1 `forgetUnitRestrict_eq_pushforwardUnit` (813) writes q as the forgotten `restrictFunctorIsoPullback` ; $\texttt{pullbackUnitIso}\,f$ (671). The two `restrictFunctorIsoPullback` legs cancel, leaving the on-the-nose factorisation

$$p.\mathrm{hom} \;=\; \eta_Y\,;\texttt{forget}(\texttt{pullbackValIso}\,f\,\mathbf{1}_X\,;\texttt{pullbackUnitIso}\,f). \qquad (\dagger)$$

Sheafifying (†) and discharging the Y-sheafification counit `cuY` is exactly the generic adjunction unit/counit collapse `adj_unit_counit_collapse` (767) at the sheafification adjunction over Y and the morphism $g = \texttt{pullbackValIso}\,f\,\mathbf{1}_X \; ; \texttt{pullbackUnitIso}\,f$, which returns g — the asserted right-hand side. ☐

Lemma 816 ((∗) / RES2 (pullback unit-only reconciliation): the sheaf pullback-unit comparator sheafifies the presheaf pullback-unit comparator)**.** *Source: internal categorical construction; no external reference. Let* $f : Y \hookrightarrow X$ *be an open immersion. Pulling back the sheafification counit over* X *at the structure-sheaf unit and then applying the sheaf pullback-unit iso* $\texttt{pullbackUnitIso}\,f$ *equals: the* `sheafificationCompPullback` *comparison at the unit presheaf, then the sheafification of the presheaf pullback-unit comparator* $p = \texttt{presheafPullbackUnitIso}\,f$ *(804), then the sheafification counit over* Y *at the unit:*

$$(\texttt{pullback}\,f).\mathrm{map}(\texttt{sheafCounit}_X.\mathrm{app}\,\mathbf{1}_X) \; ; (\texttt{pullbackUnitIso}\,f).\mathrm{hom} \;=\; \texttt{sheafificationCompPullback}\,\varphi.\mathrm{hom.ap}$$

$$; \texttt{sheafification}.\mathrm{map}((\texttt{presheafPullbackUnitIso}\, f).\mathrm{hom})\ ;\ \texttt{sheafCounit}_Y.\mathrm{app}\, \mathbf{1}_Y,$$

where $\texttt{sheafCounit}_\bullet$ *is the counit of the sheafification adjunction over* $\bullet$ *and* $\mathbf{1}_X^{\mathrm{p}}$ *the underlying presheaf unit. This is the genuine unit-side reconciliation of the dual-unit naturality square; no dual factor remains.*

Proof. Rewrite the right-hand comparator leg $a_Y.\mathrm{map}(p.\mathrm{hom})\ ;\ \texttt{cuY}.\mathrm{app}\,\mathbf{1}_Y$ by L2 (815), then expand $(\texttt{pullbackValIso}\, f\, M_0).\mathrm{hom}$ by L1 (814); the leading $\texttt{sheafificationCompPullback}\,\varphi$ hom/inv legs cancel (`Iso.hom_inv_id_app`), leaving the left-hand side $(\texttt{pullback}\, f).\mathrm{map}(\texttt{cuX}.\mathrm{app}\, M_0)$; $(\texttt{pullbackUnitIso}\, f).\mathrm{hom}$. All seam-crossing mate-calculus is confined to L2. ☐

Lemma 817 (S4a: dual-unit contraction commutes with further restriction)**.** *With* $j : V \hookrightarrow U$ *as above, the unit-duality contraction* $\texttt{dual\_unit\_iso} : \mathcal{H}om(\mathcal{O}_U, \mathcal{O}_U) \cong \mathcal{O}_U$ *(738) commutes with further restriction along* j*. Writing* $\texttt{dual\_unit\_iso}_U$, $\texttt{dual\_unit\_iso}_V$ *for the contraction over* U, V *and* $u_j = \texttt{unitRestrictIso}\, j$ *(750) for the chart-scheme unit identification, the compatibility is the conjugated four-leg identity*

$$\texttt{dual\_unit\_iso}_V\ =\ \texttt{dualIsoOfIso}\,(u_j)\ ;\ (\texttt{dual\_restrict\_iso}\, j\,(\mathbf{1}_U))^{-1}\ ;\ (\texttt{restrictFunctor}\, j).\mathrm{mapIso}\,\texttt{dual\_uni}$$

where $\mathbf{1}_U$ *is the structure-sheaf unit over* U*,* $\texttt{dual\_restrict\_iso}\, j$ *(737) is the dual-restriction comparison along* j*, and* `dualIsoOfIso` *(714) is the contravariant transport of* u_j *on the dual flank.*

Proof. Pure folded iso-algebra from the unit assembly square L2′ (812), which records

$$(\texttt{restrict}\, j)\,\texttt{dual\_unit\_iso}_U\,;u_j = \texttt{dual\_restrict\_iso}\, j\,(\mathbf{1}_U)\,;(\texttt{dualIsoOfIso}\, u_j)^{-1}\,;\texttt{dual\_unit\_iso}_V.$$

Solving this square for $\texttt{dual\_unit\_iso}_V$ — cancelling the invertible factors $(\texttt{dualIsoOfIso}\, u_j)^{-1}$ and $\texttt{dual\_restrict\_iso}\, j\,(\mathbf{1}_U)$ on the right by the self-inverse cancellations of an isomorphism with its inverse — yields the conjugated four-leg identity verbatim, with $(\texttt{dual\_restrict\_iso}\, j\,(\mathbf{1}_U))^{-1}$ the middle leg and $\texttt{dualIsoOfIso}\, u_j$, u_j the two outer transports. The seam-crossing θ-at-unit identification is confined to L2′ (via L1′, 811); no unfolding is needed here. ☐

Lemma 818 (The unit contraction is the left unitor at the unit (Cone A coverage))**.** *The structure-sheaf unit contraction* $\texttt{tensorObj\_unit\_iso} : \mathcal{O}_X \otimes_X \mathcal{O}_X \cong \mathcal{O}_X$ *coincides with the left unitor* `tensorObj_left_unitor` *(654) evaluated at the module argument* $M = \mathcal{O}_X$*.*

Proof. Both sides unfold to the same sheafified presheaf left unitor followed by the sheafification-adjunction counit; the identity is definitional. ☐

Lemma 819 (Naturality of the left unitor (Cone A coverage))**.** *For a morphism* $g : M \to N$ *of* $\mathcal{O}_X$*-modules, the left unitor (654) is natural:*

$$(\mathbf{1}_{\mathcal{O}_X} \otimes g)\,;(\texttt{tensorObj\_left\_unitor}\, N).\mathrm{hom} = (\texttt{tensorObj\_left\_unitor}\, M).\mathrm{hom}\,; g.$$

Proof. The left unitor is the sheafification of the presheaf-level left unitor composed with the sheafification counit; both are natural transformations, so the square commutes by naturality of each leg. ☐

Lemma 820 (η mate-identification: the unit iso is the sheafified presheaf unit comparison (Cone A))**.** *Let* $f : Y \to X$ *be a morphism of schemes and* φ' *its associated change-of-rings map. The* *sheaf-level unit comparison* `pullbackUnitIso` f *(671) is the sheafification transport of the presheaf oplax unit comparison* $\eta\,($`pullback` $\varphi')$ *(666):*

$$(\texttt{pullbackUnitIso}\,f).\mathrm{hom} = (\texttt{pullbackValIso}\,f\,\mathcal{O}_X).\mathrm{inv};a_Y.\mathrm{map}(\eta\,(\texttt{pullback}\,\varphi'));\texttt{sheafifyUnitIso}.\mathrm{hom},$$

where `pullbackValIso` *(878) reconciles the sheaf-level pullback value with the sheafified presheaf pullback and* `sheafifyUnitIso` *— the unit-specific sheafification reconciliation isomorphism — closes the unit* (**1**) *seam.*

Proof. This is the unit square already isolated in the proof that the tensor comparison on the unit pair is an isomorphism (675): there the right-hand composite is shown to equal `pullbackObjUnitToUnit` φ, which by definition is `pullbackUnitIso` f (671). Reading that square as a defining equation for the sheafified η yields the displayed identity. □

Lemma 821 (δ mate-identification: the tensor comparison is the sheafified cotensorator (Cone A))**.** *Let* $f : Y \to X$ *and* $M, N \in \mathrm{Scheme}.\texttt{Modules}\,X$*. The sheaf-level tensor comparison* `pullbackTensorMap` $f\,M\,N$ *(667) is the sheafification transport of the presheaf oplax cotensorator* $\delta\,($`pullback` $\varphi')\,M_{\mathrm{val}}\,N_{\mathrm{val}}$ *(666):*

$$\texttt{pullbackTensorMap}\,f\,M\,N = (\texttt{pullbackValIso}\,f\,(M \otimes_X N)).\mathrm{inv};a_Y.\mathrm{map}(\delta\,(\texttt{pullback}\,\varphi')\,M_{\mathrm{val}}\,N_{\mathrm{val}});\beta_{M,N},$$

where $\beta_{M,N}$ *is the tensor-of-*`pullbackValIso` *reconciliation (878, 642) on the* δ*-codomain, identifying the* a_Y*-image of the presheaf tensor with* $f^*M \otimes_Y f^*N$*.*

Proof. This is the definitional unfolding of `pullbackTensorMap` (667): the sheaf comparison was constructed precisely as the sheafified presheaf cotensorator, pre- and post-composed with the `pullbackValIso` reconciliations on source and target. The stated equation is that construction read back. □

Lemma 822 (Cone A sub-lemma 1: `pullbackValIso` naturality at the sheaf left unitor (RHS reconciliation))**.** *Let* $f : Y \to X$ *and* $M \in \mathrm{Scheme}.\texttt{Modules}\,X$*; write* $F =$ `pullback` φ' *for the presheaf-level change-of-rings pullback,* a_Y *for the sheafification functor on* Y*, and* $\texttt{pbv}_{(-)} = \texttt{pullbackValIso}\,f\,(-)$ *(878). The reconciliation isomorphism* `pullbackValIso` f *is natural; instantiated against the sheaf-level left unitor* $(\texttt{tensorObj\_left\_unitor}\,M).\mathrm{hom} : \mathcal{O}_X \otimes M \to M$*, its naturality square identifies the* f^**-image of that unitor — the right-hand side of the identity of 829 — as the* a_Y*-sheafification of the* F*-image of the unitor's underlying presheaf morphism, conjugated by* `pbv` *on source and target:*

$$(\texttt{pbv}_{\mathcal{O}_X \otimes M}).\mathrm{hom};f^*((\texttt{tensorObj\_left\_unitor}\,M).\mathrm{hom}) = a_Y.\mathrm{map}(F.\mathrm{map}\,(\texttt{tensorObj\_left\_unitor}\,M).\mathrm{hom}_{\mathrm{val}});($$

where $(-)_{\mathrm{val}}$ *denotes the underlying presheaf morphism (the forget-image) of a sheaf morphism.*

Proof. Object-wise, `pullbackValIso` f is the inverse `sheafificationCompPullback` comparison post-composed with the f^*-image of the sheafification counit (878); both legs are natural transformations. Hence the displayed equation is precisely the naturality of `pullbackValIso` f against the sheaf morphism $(\texttt{tensorObj\_left\_unitor}\,M).\mathrm{hom} : \mathcal{O}_X \otimes M \to M$. Unfolding the sheaf-level unitor as the sheafified presheaf unitor $a_Y.\mathrm{map}\,\lambda$ followed by the counit (654) and cancelling that counit against the counit leg inside `pullbackValIso` reduces the square to the naturality of the `sheafificationCompPullback` comparison, which is structural. □

Lemma 823 (Localization-monoidal left-unitor collapse)**.** *Provided by Mathlib. Let* $L : \mathcal{C} \to \mathcal{D}$ *be a localization functor for a monoidal morphism property* W*, so that* $\mathcal{D} = \texttt{LocalizedMonoidal}\,L\,W\,\varepsilon$

carries the transported (Gabriel–Zisman / Day-style) monoidal structure, with unit comparison ε and tensor comparison $\mu_{X,Y} : LX \otimes LY \cong L(X \otimes Y)$. For every object Y of $\mathcal{C}$, the localized left unitor at the image object LY collapses to the L-image of the source left unitor, modulo the unit and tensor comparisons:

$$(\lambda_{LY}).\mathrm{hom} = \varepsilon^{-1} \triangleright LY \,;\, (\mu_{\mathbf{1},Y}).\mathrm{hom} \,;\, L.\mathrm{map}\,(\lambda_Y).\mathrm{hom},$$

where $\mathbf{1}$ is the source tensor unit. The proof is definitional: the localized left unitor is constructed as a `Localization.liftNatIso` of the source unitor, so on an image object it reduces by 824 to the source datum conjugated by the lifting isomorphisms, with no further coherence obligations.

Lemma 824 (Evaluation of a lifted natural transformation on an image object)**.** *Provided by Mathlib. Let $L : \mathcal{C} \to \mathcal{D}$ be a localization functor for W, let $F_1, F_2 : \mathcal{C} \to \mathcal{E}$ be functors that lift to $F_1', F_2' : \mathcal{D} \to \mathcal{E}$ along L, and let $\tau : F_1 \to F_2$. The unique lifted natural transformation* `liftNatTrans` $\tau : F_1' \to F_2'$*, evaluated at an image object LX, is the source datum conjugated by the lifting isomorphisms:*

$$(\texttt{liftNatTrans}\,\tau)_{LX} = (\texttt{Lifting.iso}\,F_1')_X.\mathrm{hom} \,;\, \tau_X \,;\, (\texttt{Lifting.iso}\,F_2')_X.\mathrm{inv}.$$

Thus a coherence between lifted transformations, tested on objects in the image of L, reduces definitionally to the corresponding coherence of the source data pinned by the lifting iso — the mechanism underlying the localization-monoidal unitor collapse above.

Lemma 825 (The sheafification of presheaves of modules is a localization)**.** *Provided by Mathlib. Let $\alpha : R_0 \to R.\mathrm{obj}$ be the comparison from a presheaf of rings to its sheafification (locally injective and locally surjective). The sheafification reflector $a =$* `PresheafOfModules.sheafification` α *:* `PresheafOfModules` $R_0 \to$ `SheafOfModules` R *is a localization functor for the morphism class $W = J.W$*`.inverseImage` (`toPresheaf` R_0) *of presheaf morphisms inverted by sheafification. Consequently the entire* `Localization.Monoidal` *API — 823 and 824 in particular — is applicable to a.*

Lemma 826 (STU-collapse: the sheaf left unitor at a pullback image object)**.** *Let $f : Y \to X$, write $a = a_Y =$* `PresheafOfModules.sheafification` *for the sheafification reflector on Y, $\varepsilon =$* `sheafifyUnitIso` *for its unit comparison and $\mu =$* `sheafifyTensorUnitIso` *for its tensor comparison, and $F =$* `pullback` φ' *for the presheaf-level change-of-rings pullback. For the image object $a.\mathrm{obj}\,(F.\mathrm{obj}\,M_{\mathrm{val}})$ — the sheafification of the presheaf pullback of M — the sheaf-level left unitor collapses, modulo ε and μ, to the sheafified presheaf left unitor:*

$$(\texttt{tensorObj\_left\_unitor}\,(a.\mathrm{obj}\,(F.\mathrm{obj}\,M_{\mathrm{val}}))).\mathrm{hom} = \varepsilon^{-1} \triangleright a.\mathrm{obj}\,(F.\mathrm{obj}\,M_{\mathrm{val}}); (\mu_{\mathbf{1},\,F.\mathrm{obj}\,M_{\mathrm{val}}}).\mathrm{hom}; a.\mathrm{map}\,(\lambda_{F.\mathrm{obj}\,M_{\mathrm{val}}}$$

This is the shared structural residual of 827 and of the central-leg sub-step of 828.

Proof. The proof is a by-hand presheaf reduction across the sheafification adjunction; it does *not* use the localization-monoidal API. The genuine content is a tensor-bifunctoriality plus adjunction-triangle collapse over the `restrictScalars` $(\mathbf{1})$ / $\mathbf{1}_{\texttt{Functor}}$ defeq seam, not a definitional pass-through.

Step 1: single-tensor presentation. Present $\mu =$ `sheafifyTensorUnitIso` by its single-tensor form (877, 875): $\mu.\mathrm{hom} = a.\mathrm{map}(\eta_{\mathbf{1}} \otimes \eta_P)$, where $\eta =$ (`sheafificationAdjunction`).unit is the sheafification unit and a the reflector; and expand `tensorObj_left_unitor` at the image object as the sheafified presheaf left unitor (654). All factors are kept inside the one `PresheafOfModules` monoidal instance on the $\circ$ $\texttt{forget}_2$ carrier.

Step 2: presheaf collapse identity. Reduce the goal to the pure presheaf equation

$$(\eta_{\mathbf{1}} \otimes \eta_P) ; (\varepsilon \otimes \mathbf{1}) ; \lambda_{aP} = \lambda_P ; \eta_P,$$

obtained from tensor bifunctoriality (the interchange law $(f_1 \otimes f_2) \circ (g_1 \otimes g_2) = (f_1 \circ g_1) \otimes (f_2 \circ g_2)$), the sheafification-adjunction *right-triangle* $\eta_{\mathbf{1}} ; \varepsilon = \mathbf{1}$ on the unit factor, and presheaf left-unitor naturality of η_P.

Step 3: merge the functor-image legs and cancel the counit. Both sides are a.map-images; merge them by functoriality $F(g) \circ F(h) = F(g \circ h)$, apply the Step-2 identity inside a.map, and cancel the trailing counit by the sheafification adjunction *left-triangle*. This leaves exactly the displayed equation. The merge, the triangle rewrites, and the unit-factor whisker are all forced to be term-mode ($\texttt{congrArg}_2$/`Eq.trans`) because the `restrictScalars` ($\mathbf{1}$) wrapper on η's codomain makes the connecting objects defeq-but-not-syntactic, so `rw`/`erw` miss or `whnf`-bomb. □

Lemma 827 (Cone A sub-lemma 2: the pullback-object left unitor is the sheafified presheaf unitor)**.** *With notation as above, write* $\texttt{pbv}_{(-)} = \texttt{pullbackValIso}\, f\,(-)$ *and let*

$$W_\lambda = (\texttt{sheafifyTensorUnitIso}_{\mathbf{1},\, F.\mathrm{obj}\, M_{\mathrm{val}}}).\mathrm{hom} ; a_Y.\mathrm{map}(\texttt{sheafifyUnitIso} \otimes (\texttt{pbv}\, M))$$

be the unit-reconciled $(\mathbf{1}, f^*M)$*-wrapper: its tensor unit is the presheaf unit* $\mathbf{1} = \mathbf{1}_{\texttt{PSh}}$ *and its left tensor leg is the unit comparison* `sheafifyUnitIso` *(*not *the δ-wrapper W of 828, whose unit is $F\mathbf{1}$ and whose left leg is* $\texttt{pbv}\, \mathcal{O}_X$*). Post-composed with the sheaf-level left unitor at the pullback object f^*M, W_λ equals the sheafified presheaf left unitor of $F.\mathrm{obj}\, M_{\mathrm{val}}$, conjugated by* `pullbackValIso` *on the target:*

$$(\texttt{sheafifyTensorUnitIso}_{\mathbf{1},\, F.\mathrm{obj}\, M_{\mathrm{val}}}).\mathrm{hom} ; a_Y.\mathrm{map}(\texttt{sheafifyUnitIso} \otimes (\texttt{pbv}\, M)) ; (\texttt{tensorObj\_left\_unitor}\, (f^*M))$$

where $\mathbf{1} = \mathbf{1}_{\texttt{PSh}}$ *is the presheaf tensor unit. The unit-reconciled wrapper W_λ is precisely the right-hand tail produced by the δ-side η-whisker identity 828; that lemma performs the $W \to W_\lambda$ handoff that converts the δ-wrapper into the unit-reconciled wrapper consumed here.*

Proof. Split the wrapper's tensor leg $a_Y.\mathrm{map}(\texttt{sheafifyUnitIso} \otimes (\texttt{pbv}\, M))$ by functoriality of the sheaf tensor into the unit-factor whisker $\texttt{sheafifyUnitIso} \triangleright (-)$ followed by the right-factor whisker $(-) \triangleleft (\texttt{pbv}\, M)$. Naturality of the sheaf-level left unitor against the $\texttt{pbv}\, M$-leg (819) slides that right-factor whisker past the unitor, producing the trailing $(\texttt{pbv}_M).\mathrm{hom}$ on the right-hand side and reducing the unitor to its value at the image object $a_Y.\mathrm{obj}\,(F.\mathrm{obj}\, M_{\mathrm{val}})$. The residual — the `sheafifyTensorUnitIso` reconciliation μ and the `sheafifyUnitIso` whisker ε composed against the sheaf left unitor at that image object — is exactly the localization-image collapse 826: the sheaf left unitor at the image object collapses, modulo ε and μ, to $a_Y.\mathrm{map}(\lambda_{F.\mathrm{obj}\, M_{\mathrm{val}}})$. Cancelling the ε-whisker against the collapse's ε^{-1}-whisker leaves the displayed identity. No new sheaf-level monoidal structure is introduced — only unitor naturality and the image-collapse 826, whose own proof is the by-hand bifunctoriality + adjunction-triangle reduction recorded there. □

Lemma 828 (Cone A sub-lemma 3: the whiskered unit comparison is the sheafified η-whisker)**.** *With notation as above, write* $\texttt{pbv}_{(-)} = \texttt{pullbackValIso}\, f\,(-)$, $\eta F = \texttt{Functor.OplaxMonoidal}.\eta\, (\texttt{pullback}\, \varphi')$ *for the presheaf oplax unit (666), and let*

$$W = (\texttt{sheafifyTensorUnitIso}_{F.\mathrm{obj}\,\mathbf{1},\, F.\mathrm{obj}\, M_{\mathrm{val}}}).\mathrm{hom} ; a_Y.\mathrm{map}((\texttt{pbv}\, \mathcal{O}_X) \otimes (\texttt{pbv}\, M)),$$

$$W_\lambda = (\texttt{sheafifyTensorUnitIso}_{\mathbf{1},\, F.\mathrm{obj}\, M_{\mathrm{val}}}).\mathrm{hom} ; a_Y.\mathrm{map}(\texttt{sheafifyUnitIso} \otimes (\texttt{pbv}\, M))$$

be respectively the δ-identification right wrapper of 821 (tensor unit $F\mathbf{1}$, left leg $\texttt{pbv}\,\mathcal{O}_X$*) and the unit-reconciled wrapper of 827 (tensor unit* $\mathbf{1} = \mathbf{1}_{\texttt{PSh}}$*, left leg* `sheafifyUnitIso`*). Then the wrapper W, post-composed with the unit comparison whiskered into the left tensor factor, equals the sheafified presheaf η-whisker followed by the unit-reconciled wrapper W_λ:*

$$W\,;\,(\texttt{tensorObjIsoOfIso}\,(\texttt{pullbackUnitIso}\,f)\,\mathbf{1}_{f^*M}).\text{hom} = a_Y.\text{map}(\eta\,F \triangleright F.\text{obj}\,M_{\text{val}})\,;\,W_\lambda.$$

This is the $W \to W_\lambda$ handoff: it converts the δ-wrapper into the unit-reconciled wrapper consumed by 827, at the cost of the η-whisker prefix.

Proof. The proof composes two reconciliations.

Bridge-1 substitution and whisker expansion. The whisker $(\texttt{tensorObjIsoOfIso}\,(\texttt{pullbackUnitIso}\,f)\,\mathbf{1}_{f^*M}).\text{hom}$ is the bifunctorial action of the sheaf tensor on $(\texttt{pullbackUnitIso}\,f).\text{hom}$ and the identity. Substituting bridge 1 (820), $(\texttt{pullbackUnitIso}\,f).\text{hom} = (\texttt{pbv}\,\mathcal{O}_X).\text{inv};a_Y.\text{map}(\eta\,F);\texttt{sheafifyUnitIso}.\text{hom}$, and splitting by functoriality of the tensor product expresses the whisker as the three whiskered legs $(\texttt{pbv}\,\mathcal{O}_X).\text{inv}$, $a_Y.\text{map}(\eta\,F)$, `sheafifyUnitIso`.hom, each whiskered by $\mathbf{1}_{f^*M}$ into the left factor.

Central-leg STU naturality. The central leg — $a_Y.\text{map}(\eta\,F)$ whiskered by f^*M, transported across the tensor reconciliation `sheafifyTensorUnitIso` identifying the sheaf tensor with the sheafification of the presheaf tensor $F.\text{obj}\,\mathbf{1} \otimes F.\text{obj}\,M_{\text{val}}$ — becomes the bare sheafified presheaf η-whisker $a_Y.\text{map}(\eta\,F \triangleright F.\text{obj}\,M_{\text{val}})$. This is *not* a definitional pass-through: it is exactly the argument-1 specialization, at $q = \mathbf{1}_{F.\text{obj}\,M_{\text{val}}}$, of the *already-proven* two-argument naturality 673 of `sheafifyTensorUnitIso`:

$$a_Y.\text{map}(p \otimes \mathbf{1})\,;\,\texttt{STU}(P', Q) = \texttt{STU}(P, Q)\,;\,a_Y.\text{map}((a_Y.\text{map}\,p)_{\text{val}} \otimes \mathbf{1}),$$

instantiated at $p = \eta\,F$. The general lemma's right-hand whisker $(a_Y.\text{map}\,(\mathbf{1}))_{\text{val}}$ collapses to the identity by `Functor.map_id` / `tensorHom_id`, so no new coherence core has to be built — the proven section-level naturality discharges this leg directly.

Left-factor device cancellation. The flanking legs $(\texttt{pbv}\,\mathcal{O}_X).\text{inv}$ and `sheafifyUnitIso`.hom introduced by the substitution, whiskered into the left factor, recombine with the wrapper W. The $(\texttt{pbv}\,\mathcal{O}_X)$-component of W's tensor leg is inverse to $(\texttt{pbv}\,\mathcal{O}_X).\text{inv}$, so it cancels and rewrites W's tensor unit $F\mathbf{1}$ to the presheaf unit $\mathbf{1}$ and its left leg to `sheafifyUnitIso`, i.e. converts W into the unit-reconciled wrapper W_λ (878). Reassociating exposes the cancelling pairs (`Iso.hom_inv_id`/`inv_hom_id`), and the composite is exactly $a_Y.\text{map}(\eta\,F \triangleright F.\text{obj}\,M_{\text{val}})\,;\,W_\lambda$, the displayed identity. This is the unit-side analogue of the whiskering for the unit pair in 675, now retaining the whisker rather than discarding it as an isomorphism. ☐

Lemma 829 (Sheaf-level left unitality of the pullback tensor comparison (Cone A core))**.** *Let $f : Y \to X$ and $M \in$ Scheme.*`Modules` *X. The sheaf-level pullback functor $f^* =$ Scheme.*`Modules.pullback` *f satisfies left unitality: the composite of the tensor comparison on the unit pair, the unit comparison whiskered into the left factor, and the target left unitor equals the f^*-image of the source left unitor,*

$$\texttt{pullbackTensorMap}\,f\,\mathcal{O}_X\,M;((\texttt{pullbackUnitIso}\,f)\otimes\mathbf{1}_{f^*M});(\texttt{tensorObj\_left\_unitor}\,(f^*M)).\text{hom} = f^*((\texttt{tensorOb}$$

Proof. Sheafify the presheaf oplax left-unitality coherence `Functor.OplaxMonoidal.left_unitality_hom` (`pullback` (666; pentagon-free): applying a_Y.map and splitting by functoriality gives a three-leg identity with right-hand side $a_Y.\text{map}(F.\text{map}\,\lambda_{M_{\text{val}}})$. Rewrite each leg by its reconciliation: the δ-leg to $\texttt{pullbackTensorMap}\,f\,\mathcal{O}_X\,M$ by 821 (which also pins the shared right wrapper), the η-whisker leg to the whiskered $\texttt{pullbackUnitIso}\,f$ by sub-lemma 3 (828), the λ-leg to $\texttt{tensorObj\_left\_unitor}\,(f^*M)$

by sub-lemma 2 (827), and the right-hand side to the f^*-image of the sheaf left unitor via the assembly-form reconciliation — the `sheafificationCompPullback`-headed naturality of `pullbackValIso` f — covered by sub-lemma 1 (822). The η-whisker leg itself — inside sub-lemma 3 (828), via the STU-collapse (826) — decomposes into the bridge-1 substitution with η-whisker expansion and the left-factor device cancellation against the `sheafifyTensorUnitIso`/`pbv` wrapper. The interior reconciliation pairs cancel, and the displayed identity follows. □

Lemma 830 (S4b inner seam: pullback-world left-unitor coherence at the chart unit (Cone A))**.** *With $j : V \hookrightarrow U$ the chart morphism ($j;\iota_U = \iota_V$), the* inner *seam of S4b is the pullback-world image of the left-unitor coherence at the chart unit. Transporting the restriction* (`restrict` j) (`tensorObj_unit_iso`) *through the chart comparison* `restrictFunctorIsoPullback` j *(769) and promoting the tensor leg by Bridge B1 (776), the square reduces to the sheaf-level left unitality* `pullbackTensorMap_left_unitality` j *(829) evaluated at $M = \mathcal{O}_U$, with the unit leg supplied by* `unitRestrictIso` j *(750) and* `pullbackUnitIso` j *(671).*

Proof. After bridging the chart restriction functor to the abstract pullback via `restrictFunctorIsoPullback` j and rewriting the tensor leg by Bridge B1, both flanks of the inner square are expressed in the pullback world: the left flank is the pullback image of the left unitor, the right flank is the cotensorator-then-unit-then-unitor composite. Their equality is precisely 829 at $M = \mathcal{O}_U$, with the unit whiskering supplied by `unitRestrictIso` j (750); iso-algebra then closes the seam. No sheaf-level monoidal instance is required — the coherence is the free presheaf-level oplax left unitality transported through sheafification. □

Lemma 831 (S4b: unit left-unitor commutes with further restriction)**.** *With $j : V \hookrightarrow U$ as above, the unit-tensor contraction* `tensorObj_unit_iso` : $\mathcal{O}_U \otimes_U (-) \cong (-)$ *(the left unitor at the unit, 654) commutes with further restriction along j: modulo the reindexing ϱ,*

$$(\texttt{restrict}\, j)\,(\texttt{tensorObj\_unit\_iso}\ \textit{over}\ U) \;=\; \texttt{tensorObj\_unit\_iso}\ \textit{over}\ V.$$

Unlike the tensor-restriction comparison `tensorObj_restrict_iso`*, the contraction* `tensorObj_unit_iso` *is* not *built from the abstract pullback: by definition it is the* sheafification of the presheaf-level left unitor at the unit*, followed by the sheafification-adjunction counit (654). Bridge B1 does not apply to the contraction directly; instead the square reduces, via the chart comparison* `restrictFunctorIsoPullback`*, to a pullback-world left-unitality coherence (Cone A, 829), established below.*

Proof. This is the *unit* companion of S2, routed through *Cone A*. Because the unit object is fixed — there is no module argument — the square has *no* naturality-in-M component, which makes it strictly simpler than B1.

Peeling the contraction. Write `tensorObj_unit_iso` = `sheafification.mapIso`($\lambda_{\mathbf{1}}$) ; ε, where $\lambda_{\mathbf{1}}$ is the presheaf-level left unitor of `PresheafOfModules` at the tensor unit $\mathbf{1} = \mathcal{O}$ and ε is the sheafification-adjunction counit. By 818 the contraction is the sheaf-level left unitor `tensorObj_left_unitor` (654) evaluated at $M = \mathcal{O}$; restriction along j commutes through the outer `Functor.mapIso`/`toSheafify`/counit legs by functoriality and naturality of the unitor (819), so the square reduces to a single *inner* seam.

Inner seam (Cone A). The inner seam is the pullback-world left-unitor coherence at the chart unit, 830. Crucially, this is *not* a missing monoidal instance: the presheaf change-of-rings pullback `pullback` φ' is *already* oplax monoidal (666, via `Adjunction.leftAdjointOplaxMonoidal`), so its left-unitality coherence `Functor.OplaxMonoidal.left_unitality_hom` holds for free. The inner seam is the *sheafification transport* of that coherence to the sheaf-level maps `pullbackTensorMap` (sheafified δ) and `pullbackUnitIso` (sheafified η); it is discharged by the sheaf-level left unitality

829 at $M = \mathcal{O}_U$, through the chart comparison `restrictFunctorIsoPullback` j and the unit conjugator `unitRestrictIso` j (750). The pentagon plays no role.

Outer seam (sheafify and counit). The remaining legs are formal. The sheafification functor acts on the inner unitor square by `Functor.mapIso`, and the sheafification unit `toSheafify` is natural, so the sheafified squares commute whenever the inner one does. The counit ε of the sheafification adjunction is a natural transformation, hence its naturality square against the restriction-induced morphism is free. Composing the inner agreement with these two formal squares yields the equality of the two contractions.

Reindexing. Finally the reindexing $\varrho =$ `restrictCompReindex` j (749) threads the equality of opens $j\,;\iota_U = \iota_V$ (741) through both flanks, absorbing the `restrictFunctorCongr` congruence exactly as in S2, and giving the displayed identity. No sheaf-level monoidal structure is involved: the structure used lives at the presheaf level and is transported through sheafification. ☐

Lemma 832 (S4c: global-unit comparison u_ι commutes with further restriction)**.** *With $j : V \hookrightarrow U$ as above, the global-unit comparison*

$$u_\iota(f) := (\texttt{restrictFunctorIsoPullback}\, f).\mathrm{app}\,(\texttt{SheafOfModules.unit}\, X);\texttt{pullbackUnitIso}\, f : \texttt{restrict}\,(\mathcal{O}_X)\, f \cong$$

(671), whose inverse $u_\iota(f)^{-1}$ is the final leg of the trivialisation chain, commutes with further restriction along j: modulo the reindexing ϱ,

$$(\texttt{restrict}\, j)\, u_\iota(\iota_U)^{-1} \;=\; u_\iota(\iota_V)^{-1}.$$

Proof. This is the cheapest constituent: it is the project's proven *unit composition law* `pullbackObjUnitToUnit_comp` (670) re-wrapped by Bridge B2 alone, with no deep content. By definition $u_\iota(f) =$ `unitRestrictIso` $f =$ `restrictFunctorIsoPullback` f;`pullbackUnitIso` f (750), and `pullbackUnitIso` $=$ `pullbackObjUnitToUnitIso` (671). Strip the `restrictFunctorIsoPullback` factor across $j;\iota_U = \iota_V$ by Bridge B2 (752); what remains is exactly the pullback-world identity `pullbackObjUnitToUnit`$(j;\iota_U) =$ `pullbackComp`$^{-1}$; (`pullback` j)(`pullbackObjUnitToUnit` ι_U) ; `pullbackObjUnitToUnit` j of 670. Taking inverses and threading the ϱ reindexing of 741 gives the stated equation for u_ι^{-1}. ☐

Lemma 833 (RFIP pseudonaturality across an equality of open immersions)**.** *Source: internal categorical construction; no external reference. Let $f, f' : Y \hookrightarrow X$ be open immersions with $h : f = f'$, and let $A \in$ Scheme.`Modules` X. The restriction–pullback comparison* `restrictFunctorIsoPullback` *(769) is pseudonatural across the equality h: the comparison at f followed by the* `pullbackCongr` *h reindex equals the* `restrictFunctorCongr` *h reindex followed by the comparison at f',*

$$(\texttt{restrictFunctorIsoPullback}\, f).\mathrm{hom.app}\, A;(\texttt{pullbackCongr}\, h).\mathrm{hom.app}\, A = (\texttt{restrictFunctorCongr}\, h).\mathrm{hom.ap}$$

This is the leaf reindex helper threaded by the Seam-1 keystone (834) to push the leading comparison onto the literal composite chart morphism.

Proof. Substituting h (`subst`) identifies f' with f and collapses both congruence reindexings `restrictFunctorCongr` h and `pullbackCongr` h to the identity: each is an `eqToHom`-reindex along a reflexivity equality, forced to be the identity by thin-poset uniqueness of the underlying opens-morphisms. Both sides reduce to (`restrictFunctorIsoPullback` f).hom.app A by the identity laws. ☐

Lemma 834 (Seam 1 keystone: the unit-of-$\le$ restriction iso is the chart restriction of e^M)**.** *Source: internal categorical construction; no external reference. Let $j : V \hookrightarrow U$ be the chart* *morphism* $(j\,;\iota_U = \iota_V)$ *and* $e^M : M|_{\iota_U} \cong \mathcal{O}_U$ *a trivialisation. Then the unit-of-*$\leq$ *restriction isomorphism factors through the chart restriction of* e^M*:*

$$\texttt{restrictIsoUnitOfLE}\, h_{VU}\, e^M \;=\; \texttt{restrictCompReindex}\, j\, M ; (\texttt{restrictFunctor}\, j).\mathrm{mapIso}\, e^M ; \texttt{unitRestrictIso}\, j,$$

where $\texttt{restrictCompReindex}\, j\, M$ *is the restriction-composite reindexing* ϱ_M *(749) and* $\texttt{unitRestrictIso}\, j$ *the unit-restriction identification* $u_\iota(j)$ *(750).*

Proof. Unfold $\texttt{restrictIsoUnitOfLE}\, h_{VU}\, e^M$ to its pullback-world chart-chase

$$\texttt{RFIP}(\iota_V);\texttt{pullbackCongr}(h_{j\iota});(\texttt{pullbackComp}\, j\, \iota_U)^{-1};(\texttt{pullback}\, j)(\texttt{RFIP}(\iota_U)^{-1});(\texttt{pullback}\, j)\, e^M;\texttt{pullbackObjUnitTo}$$

where $\texttt{RFIP} = \texttt{restrictFunctorIsoPullback}$. On the right-hand side, $\texttt{restrictCompReindex}\, j$ unfolds (749) as $(\texttt{restrictFunctorCongr}\, h_{j\iota})^{-1};\texttt{restrictFunctorComp}\, j\, \iota_U$, and $\texttt{unitRestrictIso}\, j$ unfolds (750) as $\texttt{RFIP}(j).\mathrm{app}\, \mathcal{O}_U\, ;\, \texttt{pullbackUnitIso}\, j$. Pushing the two restriction-world factors `restrictFunctorComp` and `restrictFunctorCongr` into the pullback world by the general pseudonaturality bridge (751) turns the right-hand side into the same pullback-world chart-chase as the left, the trailing $\texttt{pullbackObjUnitToUnit}\, j$ matching $\texttt{pullbackUnitIso}\, j$ definitionally. The two presentations therefore agree. ☐

Lemma 835 (Telescope seam 1, Step A: e^M-bifunctoriality split of constituent (2))**.** *Source: internal categorical construction; no external reference. Let* $j : V \hookrightarrow U$ *be the chart morphism* $(j\,;\iota_U = \iota_V)$ *and* $e^M : L|_{\iota_U} \cong \mathcal{O}_U$ *a trivialisation. Write* $c = \texttt{dual\_restrict\_iso}\, \iota_U\, ;\, (\texttt{dualIsoOfIso}\, e^M)^{-1}\, ;\, \texttt{dual\_unit\_iso}$ *for the wrapped U-side dual chain. Then* $\texttt{restrict}\, j$ *carries the constituent-(2) comparison* $\texttt{tensorObjIsoOfIso}\, e^M\, c$ *past the tensor-restriction comparison* $\texttt{tensorObj\_restrict\_iso}\, j$ *on both flanks:*

$$(\texttt{restrictFunctor}\, j).\mathrm{mapIso}\, (\texttt{tensorObjIsoOfIso}\, e^M\, c) = \texttt{tensorObj\_restrict\_iso}\, j\, (L|_{\iota_U})\, ((\texttt{dual}\, L)|_{\iota_U});\texttt{tensorO}$$

Proof. Immediate specialisation of the bifunctorial module-naturality of the tensor-restriction comparison (788) at $f = j$, $a = e^M$ and $b = c$: the source objects are $L|_{\iota_U}$, $(\texttt{dual}\, L)|_{\iota_U}$ and the target objects are $M' = N' = \mathcal{O}_U$, so the trailing comparison is $(\texttt{tensorObj\_restrict\_iso}\, j\, \mathcal{O}_U\, \mathcal{O}_U)^{-1}$. No further identification of the two inner mapIso images is performed here; they are handled by Steps B and C. ☐

Lemma 836 (Telescope seam 1, Step B: the e^M-leg refined to the unit-of-$\leq$ restriction iso)**.** *Source: internal categorical construction; no external reference. With* $j : V \hookrightarrow U$*,* $h_{VU} : V \leq U$ *and* $e^M : L|_{\iota_U} \cong \mathcal{O}_U$*, write* $\varrho_L = \texttt{restrictCompReindex}\, j\, L$ *(749) and* $u_j = \texttt{unitRestrictIso}\, j$ *(750). The* $\texttt{restrict}\, j$*-image of the trivialisation* e^M *is the unit-of-*$\leq$ *restriction iso conjugated by* ϱ_L *and* u_j*:*

$$(\texttt{restrictFunctor}\, j).\mathrm{mapIso}\, e^M = \varrho_L^{-1}\, ;\, \texttt{restrictIsoUnitOfLE}\, h_{VU}\, e^M\, ;\, u_j^{-1}.$$

Proof. The Seam-1 keystone (834) reads $\texttt{restrictIsoUnitOfLE}\, h_{VU}\, e^M = \varrho_L;(\texttt{restrictFunctor}\, j).\mathrm{mapIso}\, e^M; u_j$. Cancelling the invertible ϱ_L on the left and u_j on the right — composing with their inverses and applying the iso laws $\varrho_L^{-1}\, ;\, \varrho_L = \mathbf{1}$, $u_j\, ;\, u_j^{-1} = \mathbf{1}$ — solves for the middle factor, giving the displayed identity. ☐

Lemma 837 (Telescope seam 1, Step C: the dual-chain leg, distributed and identified leg by leg)**.** *Source: internal categorical construction; no external reference. Keep* j, h_{VU}, e^M *as above.*

Write $c = \texttt{dual\_restrict\_iso}\,\iota_U\,;(\texttt{dualIsoOfIso}\,e^M)^{-1}\,;\texttt{dual\_unit\_iso}_U$ *for the* U*-built dual chain and*

$$c_V = \texttt{dual\_restrict\_iso}\,\iota_V;\big(\texttt{dualIsoOfIso}\,(\texttt{restrictIsoUnitOfLE}\,h_{VU}\,e^M)\big)^{-1};\texttt{dual\_unit\_iso}_V$$

for its V*-built refinement (with* e^M *replaced by* $\texttt{restrictIsoUnitOfLE}\,h_{VU}\,e^M$*). Writing* $\varrho_{\texttt{dual}\,L} =$ $\texttt{restrictCompReindex}\,j\,(\texttt{dual}\,L)$ *and* $u_j = \texttt{unitRestrictIso}\,j$*, the* $\texttt{restrict}\,j$*-image of the dual chain is its* V*-built refinement conjugated by* $\varrho_{\texttt{dual}\,L}$ *and* u_j*:*

$$(\texttt{restrictFunctor}\,j).\mathrm{mapIso}\,c = \varrho_{\texttt{dual}\,L}^{-1}\,;c_V\,;u_j^{-1}.$$

Proof. As $\texttt{restrict}\,j$ is a functor it carries the composite c to the composite of the $\texttt{restrict}\,j$-images of its three legs; identify each in turn (abbreviating $D_j(X) = \texttt{dual\_restrict\_iso}\,j\,X$).

- *Leg 1*, $\texttt{dual\_restrict\_iso}\,\iota_U$: S3 (802) reads $\texttt{dual\_restrict\_iso}\,\iota_V = \varrho_{\texttt{dual}\,L};(\texttt{restrict}\,j)(\texttt{dual\_restrict\_}$ $D_j(L|_{\iota_U});\texttt{dualIsoOfIso}\,\varrho_L$; solving for the middle factor gives $(\texttt{restrict}\,j)(\texttt{dual\_restrict\_iso}\,\iota_U) =$ $\varrho_{\texttt{dual}\,L}^{-1}\,;\texttt{dual\_restrict\_iso}\,\iota_V\,;(\texttt{dualIsoOfIso}\,\varrho_L)^{-1}\,;D_j(L|_{\iota_U})^{-1}$.
- *Leg 2*, $(\texttt{dualIsoOfIso}\,e^M)^{-1}$: module-naturality of $\texttt{dual\_restrict\_iso}$ (T2, 787) reads $(\texttt{restrict}\,j)(\texttt{dualIsoOfIso}\,e^M) = D_j(\mathcal{O}_U)\,;\texttt{dualIsoOfIso}\,((\texttt{restrict}\,j)\,e^M)\,;D_j(L|_{\iota_U})^{-1}$; taking inverses, the $\texttt{restrict}\,j$-image of leg 2 is $D_j(L|_{\iota_U});(\texttt{dualIsoOfIso}\,((\texttt{restrict}\,j)\,e^M))^{-1};$ $D_j(\mathcal{O}_U)^{-1}$.
- *Leg 3*, $\texttt{dual\_unit\_iso}_U$: S4a (817) reads $\texttt{dual\_unit\_iso}_V = \texttt{dualIsoOfIso}\,u_j;D_j(\mathbf{1}_U)^{-1};$ $(\texttt{restrict}\,j)\,\texttt{dual\_unit\_iso}_U;u_j$; solving gives $(\texttt{restrict}\,j)\,\texttt{dual\_unit\_iso}_U = D_j(\mathbf{1}_U);$ $(\texttt{dualIsoOfIso}\,u_j)^{-1}\,;\texttt{dual\_unit\_iso}_V\,;u_j^{-1}$.

Composing the three legs in order, the two internal D_j comparisons cancel telescopically: the tail $D_j(L|_{\iota_U})^{-1}$ of leg 1 meets the head $D_j(L|_{\iota_U})$ of leg 2, and the tail $D_j(\mathcal{O}_U)^{-1}$ of leg 2 meets the head $D_j(\mathbf{1}_U) = D_j(\mathcal{O}_U)$ of leg 3 (the structure-sheaf unit $\mathbf{1}_U$ being $\mathcal{O}_U$). What survives is $\varrho_{\texttt{dual}\,L}^{-1}\,;\texttt{dual\_restrict\_iso}\,\iota_V\,;(*)\,;\texttt{dual\_unit\_iso}_V\,;u_j^{-1}$, where the middle factor $(*) =$ $(\texttt{dualIsoOfIso}\,\varrho_L)^{-1}\,;(\texttt{dualIsoOfIso}\,((\texttt{restrict}\,j)\,e^M))^{-1}\,;(\texttt{dualIsoOfIso}\,u_j)^{-1}$ is, by contravariant functoriality of $\texttt{dualIsoOfIso}$ (869, 714), $\big(\texttt{dualIsoOfIso}\,(\varrho_L;(\texttt{restrict}\,j)\,e^M;u_j)\big)^{-1}$. Since $\varrho_L\,;(\texttt{restrict}\,j)\,e^M\,;u_j = \texttt{restrictIsoUnitOfLE}\,h_{VU}\,e^M$ is exactly the Seam-1 keystone (834), $(*)$ is the middle leg $(\texttt{dualIsoOfIso}\,(\texttt{restrictIsoUnitOfLE}\,h_{VU}\,e^M))^{-1}$ of c_V, and the surviving expression is $\varrho_{\texttt{dual}\,L}^{-1}\,;c_V\,;u_j^{-1}$ as claimed. ☐

Lemma 838 (Telescope seam 1: e^M-bifunctoriality split of the wrapped dual constituent)**.** *With* $j : V \hookrightarrow U$ *the chart morphism* $(j\,;\iota_U = \iota_V)$ *and* $e^M : L|_{\iota_U} \cong \mathcal{O}_U$ *a trivialisation, applying* $\texttt{restrict}\,j$ *to the* U*-side constituent* $\texttt{tensorObjIsoOfIso}\,e^M\,(\texttt{dual\_restrict\_iso}\,\iota_U\,;$ $(\texttt{dualIsoOfIso}\,e^M)^{-1}\,;\texttt{dual\_unit\_iso})$ *splits, by module-naturality of the tensor iso-of-iso comparison (788), into:*

- *the* e^M*-leg, whose* V*-refinement is the identification* $\texttt{restrictIsoUnitOfLE}\,h_{VU}\,e^M = (\texttt{restrict}\,j)\,e^M$ *(834, 750); and*
- *the dual-chain leg, distributed leg by leg across* $\texttt{restrict}\,j$*: the bare dual-restriction comparison covered by S3 (802), the contravariant* $(\texttt{dualIsoOfIso}\,e^M)^{-1}$ *(714) transported by module-naturality of* $\texttt{dual\_restrict\_iso}$ *(787) into* $(\texttt{dualIsoOfIso}\,((\texttt{restrict}\,j)\,e^M))^{-1}$*, and* $\texttt{dual\_unit\_iso}$ *covered by S4a (817).*

Concretely, modulo the reindexing ϱ = `restrictCompReindex` j *(749), the* `restrict` j*-image of constituent (2) equals the* V*-built* `tensorObjIsoOfIso` (`restrictIsoUnitOfLE` $h_{VU}\, e^M$) $(\cdots_V)$ *conjugated by* ϱ *on both flanks.*

Proof. Assemble the three steps of this seam. Step A (835) splits the `restrict` j-image of constituent (2), by bifunctorial naturality of `tensorObjIsoOfIso`, into the flanking `tensorObj_restrict_iso` comparisons around `tensorObjIsoOfIso` ((`restrict` j) e^M) ((`restrict` j) c). Step B (836) identifies the first inner argument as ϱ_L^{-1};`restrictIsoUnitOfLE` $h_{VU}\, e^M$;u_j^{-1} (keystone, 834, 750) and Step C (837) the second as $\varrho_{\texttt{dual}\, L}^{-1}$;$c_V$;$u_j^{-1}$, with c_V the V-built dual chain whose (`dualIsoOfIso` e^M)$^{-1}$ leg is refined to (`dualIsoOfIso` (`restrictIsoUnitOfLE` $h_{VU}\, e^M$))$^{-1}$ — the same refinement as Step B, keeping the two e^M occurrences in sync. Regrouping the two substituted arguments by the bifunctoriality law (863), the leading ϱ^{-1} and trailing u_j^{-1} reindexings split off as `tensorObjIsoOfIso` factors that merge with the flanking `tensorObj_restrict_iso` comparisons into the conjugating ϱ on both flanks, leaving the V-built `tensorObjIsoOfIso` (`restrictIsoUnitOfLE` $h_{VU}\, e^M$) (c_V) at the centre — the asserted identity. ☐

Lemma 839 (Telescope seam 2: generic ϱ-cancellation assembly of the five squares)**.** *Source: internal categorical construction; no external reference. Work in an arbitrary category* C*. Given the five* U*-built constituents* $c_1^U, \dots, c_5^U$ *of the trivialisation chain and their* V*-built counterparts* $c_1^V, \dots, c_5^V$*, the four internal reindexings* $\varrho_1, \dots, \varrho_4$ *(the* `restrictCompReindex` j/`unitRestrictIso` *factors of 749), and the two outer reindexings* ϱ_0, ϱ_5*, suppose each constituent satisfies its restriction-naturality square*

$$(\texttt{restrict}\, j)\, c_k^U \;=\; \varrho_{k-1}^{-1} \,;\, c_k^V \,;\, \varrho_k \qquad (k = 1, \dots, 5),$$

so that the target ϱ_k *of square* k *is the source* ϱ_k *of square* $k+1$*. Then, composing the five squares in order, the internal reindexings* $\varrho_1, \dots, \varrho_4$ *cancel telescopically and*

$$(\texttt{restrict}\, j)(c_1^U \,;\cdots;\, c_5^U) \;=\; \varrho_0^{-1} \,;\, (c_1^V \,;\cdots;\, c_5^V) \,;\, \varrho_5,$$

leaving only the two outer reindexings ϱ_0 = `hobjU` *and* ϱ_5 = `hobjV`*.*

Proof. Pure `Category.assoc` cocycle collapse, identical in form to the sibling assembles `dual_scp_assemble`, `unit_assemble` and `hcore_assemble`. As `restrict` j is a functor it carries the U-composite to the composite of the `restrict` j-images; substitute each square, and for every adjacent pair the factor $\varrho_k \,;\, \varrho_k^{-1} = \mathbf{1}$ $(k = 1, \dots, 4)$ cancels by the iso law. What survives is ϱ_0^{-1} on the left and ϱ_5 on the right — the displayed identity. No `SheafOfModules`; defeq seam is crossed: the statement lives over an abstract [`Category` C] and is applied to the concrete chain by `exact`, never `rw`/`ext` on a conjugate-headed goal. ☐

Lemma 840 (Telescope seam 3: sectionwise evaluation of the two outer reindexings)**.** *Let* `hobjU` = `image_preimage_of_le` $U\, h_{VU}$ *and* `hobjV` = `image_preimage_of_le` Vle_rfl *be the two direct-image identities of 741, with images* $\iota_U{}''^U(\iota_U^{-1}(V)) = V$ *and* $\iota_V{}''^V(\iota_V^{-1}(V)) = V$*. Threading the corresponding* `eqToHom` *transports through the two flanks of the telescoped chain (839) and evaluating* `.val.presheaf.map`/`.val.app` *sectionwise over the preimage open* $\iota_U^{-1}(V)$ *yields precisely the asserted equality of 841.*

Proof. The telescoped identity (839) leaves only the outer reindexings ϱ_0 and ϱ_5; these are the `eqToHom` images of `hobjU` and `hobjV` at the presheaf level. Pushing them onto the section functors `.val.presheaf.map` and evaluating the chain's underlying morphism with `.val.app` at the preimage open $\iota_U^{-1}(V)$ discharges the bookkeeping and produces the stated section equation. ☐

Lemma 841 (Trivialisation iso-chain compatible with further restriction (cocycle residual A, step 1))**.** *Let $V \subseteq U$ be opens of X (an inclusion of trivialising opens, e.g. $V \subseteq U_i \cap U_j$). The trivialisation iso-chain that builds the local contraction — the restriction-compatibility iso 646 carrying $(L \otimes_X L^{-1})|_U$ to $\mathcal{O}_U \otimes_U \mathcal{O}_U$, composed with the unit contraction 654 — is* natural *in the open: its further restriction along $V \hookrightarrow U$ equals the same chain built directly over V. Consequently the sectionwise overlap equation of two local contractions f_i, f_j restricted to V lifts to an equation of* restricted sheaf morphisms *over V, the form on which the iso-level functoriality laws (863, 869) and 846 can act.*

Proof. The trivialisation chain carrying $(L \otimes_X L^{-1})|_U$ to $\mathcal{O}_U$, then the global-unit comparison $u_\iota(\iota_U)^{-1}$, is — in order — the five constituents S2–S4c, each carrying its own restriction-naturality square (777, 802, 817, 831, 832). The assembly is the three-seam chain of this section. Seam 1 (838) handles the $(\texttt{dualIsoOfIso}\, e^M)^{-1}$ transport the trivialisation step wraps around constituent (2): bifunctoriality of `tensorObjIsoOfIso` splits it into the e^M-leg $(\texttt{restrict}\, j)\, e^M$ and the dual-chain leg covered by S3/S4a. Seam 2 (839) is the generic abstract-[`Category`] cocycle collapse: composing the five squares, the internal ϱ reindexings cancel telescopically, leaving only the outer $\texttt{hobjU} = \texttt{image\_preimage\_of\_le}\, U\, h_{VU}$ and $\texttt{hobjV} = \texttt{image\_preimage\_of\_le}\, V \text{le\_rfl}$. Seam 3 (840) threads those two outer `eqToHoms` and evaluates the chain sectionwise over the preimage open $\iota_U^{-1}(V)$, which is the asserted equality. Restricting the sectionwise overlap equation of two local contractions f_i, f_j through this compatibility lifts it to an equation of restricted sheaf morphisms over V, the form on which the iso-level functoriality laws (863, 869) and 846 act. $\square$

Lemma 842 (Mutually-inverse unit endomorphisms cancel against the left unitor)**.** *In any monoidal category $\mathcal{C}$, if $s, s' : \mathbf{1} \to \mathbf{1}$ are endomorphisms of the monoidal unit with $s \,;\, s' = \mathbf{1}$, then*

$$(s \otimes_{\mathrm{m}} s') \;;\; \lambda_{\mathbf{1}} = \lambda_{\mathbf{1}},$$

where $\lambda_{\mathbf{1}} : \mathbf{1} \otimes \mathbf{1} \to \mathbf{1}$ is the left unitor at the unit.

Proof. Factor the tensor product $s \otimes_{\mathrm{m}} s'$ through the interchange law. Sliding the right factor past the left unitor by its naturality, and the left factor past the right unitor (which coincides with the left unitor at the unit), the composite becomes $\lambda_{\mathbf{1}}$ precomposed with $s;s' = \mathbf{1}$; cancelling that identity leaves $\lambda_{\mathbf{1}}$. $\square$

Lemma 843 (Pairing mutually-inverse unit autos through `tensorObjIsoOfIso` contracts (cocycle residual A, step 2b))**.** *Let t, s be self-isomorphisms of the trivial line bundle $\mathcal{O}_V$ with $t.\mathrm{hom} \,;\, s.\mathrm{hom} = \mathbf{1}$. Then pairing them through the iso-transport bifunctor and contracting via the unit comparison is trivial:*

$$\texttt{tensorObjIsoOfIso}\, t\, s \;;\; \texttt{tensorObj\_unit\_iso} = \texttt{tensorObj\_unit\_iso}.$$

Proof. Both $\texttt{tensorObjIsoOfIso}\, t\, s$ and `tensorObj_unit_iso` are sheafifications of presheaf-level constructions, so the claim is the image under the sheafification functor of a presheaf-level coherence. At the presheaf level the pairing is the monoidal tensor $(\,\mathrm{forget}\, t \otimes_{\mathrm{m}} \mathrm{forget}\, s\,)$ of the two underlying unit endomorphisms, whose hom-composite is the identity, and the contraction is the left unitor at the unit; 842 collapses the composite to the bare unitor. Functoriality of sheafification transports this to the stated sheaf-level identity. $\square$

Lemma 844. *[Naturality of the presheaf dual-of-unit iso against a unit automorphism (B1 eval-core)] Let $\hat{s}$ be a presheaf-level automorphism of the unit* $\mathbf{1}$ *(the underlying map of a sheaf-level unit* *automorphism s). The presheaf dual-of-unit isomorphism* `presheafDualUnitIso` $:$ `dual` $\mathbf{1} \cong \mathbf{1}$ *is natural in $\hat{s}$ for the contravariant transport on the source:*

$$\texttt{dualIsoOfIso}\,\hat{s}\ ;\ \texttt{presheafDualUnitIso} = \texttt{presheafDualUnitIso}\ ;\ \hat{s}.$$

Proof. By construction (726, 727) `presheafDualUnitIso` is the sectionwise evaluation-at-1 map of the internal hom of the unit into itself. On sections, `dualIsoOfIso` $\hat{s}$ is precomposition by the underlying map of $\hat{s}$ (867), so the left-hand composite evaluates a dual section φ at 1 after precomposing by $\hat{s}$, i.e. it computes $\varphi(\hat{s}(1))$. Since $\hat{s}$ is a unit automorphism — multiplication by a unit scalar — it commutes with evaluation, so this equals $\hat{s}$ applied to $\varphi(1)$, the right-hand composite. Equality of the two evaluation-at-1 maps is the naturality of the internal-hom evaluation (709) specialised at the unit. $\square$

Lemma 845 (Dual of a unit automorphism is itself (cocycle residual A, step 2a / B1))**.** *Let s be a self-isomorphism of the trivial line bundle $\mathcal{O}_V$. Conjugating its contravariant dual-transport by the unit-self-duality iso recovers s:*

$$\texttt{dual\_unit\_iso}^{-1}\ ;\ \texttt{dualIsoOfIso}\,s\ ;\ \texttt{dual\_unit\_iso}\ =\ s.$$

Equivalently, the sheaf-level dual of a unit automorphism, read through `dual` $\mathcal{O}_V \cong \mathcal{O}_V$*, is the automorphism itself.*

Proof. By pure iso-algebra the claim is equivalent to the single naturality square

$$(\mathrm{N}):\qquad \texttt{dualIsoOfIso}\,s\ ;\ \texttt{dual\_unit\_iso} = \texttt{dual\_unit\_iso}\ ;\ s,$$

the naturality of `dual_unit_iso` $:$ `dual` $\mathcal{O}_V \cong \mathcal{O}_V$ with respect to the unit automorphism s, acting contravariantly through `dualIsoOfIso` s on the source. This is the genuine open obligation. It decomposes through the two layers of the construction `dual_unit_iso` $=$ `sheafify`(`presheafDualUnitIso`); `counit`:

- the presheaf eval-core naturality `dualIsoOfIso` $\hat{s}$;`presheafDualUnitIso` $=$ `presheafDualUnitIso`; $\hat{s}$ (844, with $\hat{s}$ the underlying presheaf map of s), pushed through the sheafification functor; and
- the sheafification-counit naturality `sheafify`$(\hat{s})$; `counit` $=$ `counit`; s, the naturality of the sheafification counit at the unit automorphism.

Composing the two squares gives (N), and the iso-algebra reduction then yields the stated identity. $\square$

Lemma 846 (Unit self-duality evaluation collapse (cocycle residual A, step 2))**.** *Let t be a self-isomorphism of the trivial line bundle $\mathcal{O}_V$ on a trivialising open V. Pairing t on the M-leg with the dual-transported transition on the N-leg — which lands at the node* `dual` $\mathcal{O}_V$ *and so must be conjugated back to $\mathcal{O}_V$ through the unit-self-duality iso* `dual_unit_iso` $:$ `dual` $\mathcal{O}_V \cong \mathcal{O}_V$ *(738) — and then contracting via the unit comparison cancels:*

$$\texttt{tensorObjIsoOfIso}\,t\,(\texttt{dual\_unit\_iso}^{-1}\ ;\ (\texttt{dualIsoOfIso}\,t)^{-1}\ ;\ \texttt{dual\_unit\_iso})\ ;\ \texttt{tensorObj\_unit\_iso}\ =\ \texttt{tens}$$

This is the g_{ij} (inside e_x^M) meeting g_{ij}^{-1} = `dualIsoOfIso` g_{ij} *(inside e_x^N) cancellation: under the canonical evaluation of the free rank-one module against its dual, the transition iso and its inverse-transpose annihilate, leaving the trivialisation-independent unit contraction. The* `dual_unit_iso` *conjugation on the N-leg is what makes both factors live on $\mathcal{O}_V$ so the bifunctor* `tensorObjIsoOfIso` *can pair them; under the eval-at-1 semantics of* `presheafDualUnitIso` *the conjugated dual leg reduces to t^{-1}, so the pair is the symmetric $t \otimes t^{-1}$ that the multiplication kills.*

Proof. The collapse factors through the two cocycle helpers rather than a direct sectionwise computation. First, the N-leg — the dual-conjugated transition $\texttt{dual\_unit\_iso}^{-1}$;$(\texttt{dualIsoOfIso}\,t)^{-1}$; `dual_unit_iso` — is exactly t^{-1}: inverting the dual-of-a-unit identity 845 (applied to t, read off its inverse) turns the conjugated transition into t^{-1}. The collapse thus reduces to pairing t with t^{-1} on the two legs. Second, since $t.\mathrm{hom} \,;\, (t^{-1}).\mathrm{hom} = \mathbf{1}$, the pairing contracts against the unit comparison by 843, leaving `tensorObj_unit_iso` alone. This is the g_{ij}-meets-g_{ij}^{-1} cancellation, now realised as the algebraic identity "the dual of a unit-automorphism is itself, and a unit-automorphism paired with its inverse contracts to the unit". □

Remark 847 (Alternative route via object-level descent (fallback))**.** Should the slice-and-sheafify route of 706 and 710 bottom out, the dual can instead be assembled by *object-level descent*: glue the local trivial duals $\mathcal{O}_{U_i}$ along the inverse-transpose transition isomorphisms $g_{ij}^{-\mathsf{T}}$ of L into a global L^{-1} on X. Abstractly this is effective descent for sheaves of modules on the small Zariski site — the essential-surjectivity of the descent-data functor of an `IsStack` pseudofunctor $U \mapsto (\mathcal{O}_U\text{-modules})$. This is the more principled and reusable construction but the larger build (it requires constructing the restriction pseudofunctor and proving the stack/effective-descent property), and is recorded only as a fallback; the slice internal-hom route above is the intended one. It is the same morphism-/object-level descent family on which the associator re-route of 653 also waits.

16.13.2 Internal-hom and restrict-scalars helper declarations

The following are the internal Lean helper declarations assembling the morphism module 704, the slice internal hom 706, its restriction maps 707, the presheaf dual 708, the evaluation morphism 709, the lax-monoidal restrict-scalars structure 645, the ring-iso base-change equivalence 647 and the pushforward adjunction `presheaf_pushforward_adj_substrate`. Each is proved directly in Lean and carries the dependency edges of the construction.

Definition 848 (Canonical ring map from a terminal object)**.** For a presheaf of commutative rings R on a base category with terminal object T and an object Y, the unique morphism $Y \to T$ induces the ring map $R(T) \to R(Y)$ along which a global scalar acts on the value at Y.

Proof. Proved directly in Lean, as R applied to the opposite of the terminal morphism. □

Lemma 849 (Naturality of the terminal ring map)**.** *The restriction map $R(g) : R(X) \to R(Y)$ along $g : X \to Y$ carries $\texttt{termRingMap}_X\, f$ to $\texttt{termRingMap}_Y\, f$, so the images of a global scalar $f \in R(T)$ are compatible with restriction.*

Proof. Proved directly in Lean, from uniqueness of morphisms into the terminal object and functoriality of R. □

Definition 850 (Multiplication by a global scalar as an endomorphism)**.** For $f \in R(T)$, the endomorphism $m_f^N : N \to N$ of a presheaf of modules N that at each object Y is multiplication by $\texttt{termRingMap}_Y\, f \in R(Y)$; its naturality is the heart of the global-scalar action on morphism modules.

Proof. Proved directly in Lean: the naturality square commutes by `termRingMap_naturality` and the semilinearity of the restriction maps of N. □

Definition 851 (Evaluation functional of a dual section)**.** At an object X, the $R(X)$-linear map $M(X) \to R(X)$ obtained by evaluating a dual section φ at its terminal component.

Proof. Proved directly in Lean, as the underlying linear map of the terminal component of φ. □

Definition 852 (Lax-monoidal unit of restrict-scalars)**.** The lax-monoidal unit ε of `restrictScalars` α, assembled sectionwise from the **ModuleCat**-level lax units.

Proof. Proved directly in Lean, sectionwise from `ModuleCat.restrictScalars`'s lax unit. □

Definition 853 (Lax-monoidal tensorator of restrict-scalars)**.** The lax-monoidal tensorator μ of `restrictScalars` α, assembled sectionwise from the **ModuleCat**-level lax tensorators.

Proof. Proved directly in Lean, sectionwise from `ModuleCat.restrictScalars`'s lax tensorator. □

Definition 854 (Underlying Ab-presheaf of the internal hom)**.** *Source: [Stacks Project], Modules on Ringed Spaces, §Internal Hom.* The underlying abelian-group-valued presheaf of the internal hom $\mathcal{H}om(M, N)$: to each open U it assigns the group of $\mathcal{O}|_U$-linear maps $M|_U \to N|_U$, with further-restriction as the presheaf maps. This is the Ab-level part of 706; the $R(U)$-module refinement is supplied by the global scalar action 850.

Proof. Proved directly in Lean. Functoriality is `restrictionMap_id`/`restrictionMap_comp`; the group structure on each value is the pointwise Hom-addition, packaged as an AddCommGrp-valued functor. □

Definition 855 (Open-by-open evaluation pairing of the internal hom)**.** *Source: [Stacks Project], Modules on Ringed Spaces, §Internal Hom.* At each object X the $R(X)$-bilinear contraction $M(X) \otimes_{R(X)} M^\vee(X) \to R(X)$, $s \otimes \varphi \mapsto \varphi(s)$, evaluating a dual section on a module section. Assembling these over all opens gives the evaluation pairing $M \otimes M^\vee \to \mathcal{O}$ of 709; it is the substrate of the tensor–hom counit used to invert an invertible module.

Proof. Proved directly in Lean. Bilinearity over $R(X)$: linearity in s is linearity of the evaluation functional `evalLin` (851); linearity in φ is `evalLin_add`/`evalLin_smul`, the latter using that the terminal ring map is the identity. The map is the `TensorProduct.lift` of this bilinear datum. □

Lemma 856 (Extension of scalars is left adjoint to restriction of scalars)**.** *Source: [Stacks Project], Algebra, Lemma `lemma-adjoint-tensor-restrict` (Tag 05DQ); the sheaf/ringed-space-level form of the same adjunction is [Stacks Project, Tag 0096,* Sheaves on Spaces*, §Pullback and pushforward of modules]. For a morphism of presheaves of rings φ, the extension-of-scalars functor* `extendScalars` φ *on presheaves of modules is left adjoint to restriction of scalars along φ. When φ comes from a ring isomorphism, restriction of scalars is moreover strong monoidal (*`restrictScalarsMonoidalOfRingEquiv`*).*

Proof. Proved directly in Lean, sectionwise from the **ModuleCat**-level extension/restriction adjunction of [Stacks Project, Tag 05DQ], whose unit and counit are natural in the base presheaf; strong monoidality for a ring-iso base change follows because the lax unit and tensorator then become sectionwise isomorphisms. □

Lemma 857 (Pullback is left adjoint to pushforward on module sheaves)**.** *Source: [Stacks Project], Sheaves on Spaces, Lemma `lemma-adjoint-pullback-pushforward-modules` (Tag 0096). For an adjunction $F \dashv G$ of base sites together with compatible morphisms of the structure presheaves, the presheaf-of-modules pushforwards along F and G are adjoint; applied to the open-immersion adjunction this yields $f^* \dashv f_*$ on module sheaves. The right-adjoint witnesses* `pushforward₀IsRightAdjoint`, `restrictScalarsIsRightAdjoint` *and the assembled* `pullback0Adjunction` *package the scheme-level form.*

Proof. Proved directly in Lean as a de-sheafification of the **SheafOfModules**-level pushforward–pushforward adjunction of [Stacks Project, Tag 0096]: the unit and counit are built from the base adjunction's unit/counit whiskered into the pushforward, and the triangle identities reduce to those of the base adjunction via functoriality of the structure presheaf. □

16.13.3 Vestigial / off-path helper declarations

The following are the off-critical-path Lean helper declarations supporting the flat-whiskering localizer stability `flat_whisker_localizer`, the flatness-free whiskering 652, the R_x-linear stalk map 650, the stalkwise localizer characterisation `W_implies_stalkwise_iso` and the shared slice-site equivalence `open_immersion_slice_sheaf_equiv`. Each is proved directly in Lean.

Lemma 858 (A module morphism is a stalkwise isomorphism iff its stalk maps are bijective)**.** *Source: [Stacks Project], Sheaves on Spaces, Lemma* `lemma-points-exactness` *(Tag 007T), part (3). If a morphism $g\colon M \to N$ of presheaves of R-modules on* $\texttt{Opens}\, X$ *induces an isomorphism on the* Ab*-stalk at x, then the R_x-linear stalk map 650 is bijective and hence upgrades to an R_x-linear equivalence $M_x \xrightarrow{\sim} N_x$ (*`stalkLinearEquivOfIsIso`*).*

Proof. Proved directly in Lean. The forgetful functor to Ab sends an isomorphism to a bijection, and the R_x-linear structure of the stalk map 650 shares the same underlying additive map; bundling the bijection with that linear structure gives the linear equivalence. This is the varying-ring, presheaf-of-modules packaging of the "check on stalks" criterion of [Stacks Project, Tag 007T]. □

Lemma 859 (Local injectivity is equivalent to injectivity on stalks)**.** *Source: [Stacks Project], Sheaves on Spaces, Lemma* `lemma-points-exactness` *(Tag 007T), part (1); local injectivity is the sheaf-theoretic monomorphism condition spelled out at the level of the covering (canonical) topology. A morphism $g\colon M \to N$ of presheaves of R-modules on* $\texttt{Opens}\, X$ *is locally injective (for the canonical topology) if and only if the induced stalk map $g_x\colon M_x \to N_x$ is injective at every point x.*

Proof. Proved directly in Lean. A section killed on a stalk is killed on some neighbourhood, so injectivity of all stalk maps forces local injectivity; conversely local injectivity passes to the filtered colimit computing the stalk. This is a stalkwise-property criterion in the same family as 858, matching the monomorphism case of [Stacks Project, Tag 007T]. □

Lemma 860 (Tensoring with a flat module preserves the weak-equivalence class)**.** *Source: [Stacks Project], Modules on Ringed Spaces, §Flat modules, Definition* `definition-flat`*. If F is a sectionwise-flat presheaf of R-modules and g lies in the sheafification localizer $J.W$, then both whiskerings $F \triangleleft g$ and $g \triangleright F$ again lie in $J.W$; in particular $F \triangleleft g$ is J-locally injective. This is the flatness-dependent predecessor of the unconditional 659.*

Proof. Proved directly in Lean. Flatness of F makes $F \otimes (-)$ exact, hence it preserves local injectivity and, combined with the localizer's two-out-of-three behaviour, membership in $J.W$; the right-whiskering case is the braiding conjugate of the left-whiskering one. □

Lemma 861 (Sheaf-category equivalence induced by an over-slice dense subsite)**.** *Source: [Stacks Project], Sites and Sheaves, Lemma* `lemma-localize-special-cocontinuous` *(Tag 03CH). The Stacks Project's vocabulary for this comparison is "special cocontinuous functor" rather than "dense subsite"; the two describe the same phenomenon, and this is the closest verbatim match to the over-slice statement below. For an open U of a scheme, the topological site* $\texttt{Opens}\, U$ *embeds as a dense subsite of the over-slice site, inducing an equivalence between the categories of A-valued* *sheaves on U and on the slice; the sieve-cover comparison* `overEquiv_image_cover_iff` *and the dense-subsite instance* `overEquivInverseIsDenseSubsite` *are its ingredients.*

Proof. Proved directly in Lean on top of Mathlib's dense-subsite API: the over-slice inclusion is cover-dense and cover-preserving (a sieve is covering upstairs iff its image is covering downstairs), so the induced restriction of sheaves is an equivalence. This is project-local categorical glue over the over-slice specialisation [Stacks Project, Tag 03CH] of the site-comparison lemma [Stacks Project, Tag 03A0]. ☐

16.14 Substrate helper declarations (wired lean-aux blocks)

This section gives a blueprint entry to each of the project-local helper declarations of `TensorObjSubstrate.lean` that the substrate construction and the pullback–tensor monoidality machinery invoke internally. Each had been written only as an inline mid-proof `\lean` reference of an existing block, which leaves it unregistered as a graph node; here each receives its own `\label` and its `\uses` edges are read off its Lean body, and the consumers above are wired to depend on them. All but one are sorry-free; the single residual is 885, whose informal proof is given in full.

Definition 862 (Substrate $\otimes$ respects a pair of isomorphisms)**.** For $M \cong M'$ and $N \cong N'$ in Scheme.`Modules` X, sheafifying the presheaf-of-modules tensor of the underlying presheaf isomorphisms produces an isomorphism $M \otimes_X N \cong M' \otimes_X N'$. This is the functor-of-isomorphisms form of the substrate tensor, used to build the middle-four interchange and the well-definedness of the multiplication on iso-classes.

Proof. Proved directly in Lean: it is `sheafification`.mapIso applied to the monoidal `tensorIso` of the two forgetful images of the given isomorphisms. ☐

Lemma 863 (`tensorObjIsoOfIso` is bifunctorial in composition)**.** *For composable isomorphisms $M \xrightarrow{e_1} M' \xrightarrow{e_2} M''$ and $N \xrightarrow{e'_1} N' \xrightarrow{e'_2} N''$ in* Scheme.`Modules` X*,*

$$\texttt{tensorObjIsoOfIso}\,(e_1 \ggg e_2)\,(e'_1 \ggg e'_2) = \texttt{tensorObjIsoOfIso}\,e_1\,e'_1 \ggg \texttt{tensorObjIsoOfIso}\,e_2\,e'_2.$$

Tensor-transport of a composite of isomorphisms is the composite of the transports.

Proof. Both sides are sheafifications of presheaf tensor isomorphisms; the identity reduces, after applying the forgetful functor, to the interchange $(e_2 \circ e_1) \otimes (e'_2 \circ e'_1) = (e_2 \otimes e'_2) \circ (e_1 \otimes e'_1)$ of the monoidal tensor of morphisms, together with functoriality of sheafification (mapIso of a composite is the composite of mapIsos). ☐

Lemma 864 (`tensorObjIsoOfIso` preserves identities)**.** `tensorObjIsoOfIso` (Iso.refl M) (Iso.refl N) = Iso.refl $(M \otimes_X N)$*: the tensor-transport of a pair of identities is the identity.*

Proof. The monoidal tensor of two identity morphisms is the identity, and sheafification sends the identity isomorphism to the identity isomorphism. ☐

Lemma 865 (Threefold tensor/composition interchange)**.** *In a monoidal category C, for morphisms $a_i : X_i \to Y_i \to Z_i \to W_i$ $(i = 1, 2)$ presented as three composable legs on each tensor factor,*

$$(a_1^{(1)} \ggg a_1^{(2)} \ggg a_1^{(3)}) \otimes (a_2^{(1)} \ggg a_2^{(2)} \ggg a_2^{(3)}) = (a_1^{(1)} \otimes a_2^{(1)}) \ggg (a_1^{(2)} \otimes a_2^{(2)}) \ggg (a_1^{(3)} \otimes a_2^{(3)}).$$

The tensor of two threefold composites is the threefold composite of the tensors.

Proof. Two applications of the bifunctoriality of the monoidal tensor product on morphisms (`tensorHom_comp_tensorHom`). ☐

Lemma 866 (Functor image of the threefold interchange)**.** *For a functor $F : C \to D$ out of a monoidal category, the F-image of the threefold tensor/composition interchange (865) splits as the threefold composite of the F-images of the per-leg tensors.*

Proof. Apply 865 inside F, then functoriality of F on composition ($F(g \circ f) = F(g) \circ F(f)$) twice. ☐

Lemma 867 (Presheaf dual-transport is contravariantly functorial in composition)**.** *For composable presheaf-of-modules isomorphisms $L \xrightarrow{e_1} L' \xrightarrow{e_2} L''$,*

$$\texttt{dualIsoOfIso}\,(e_1 \ggg e_2) = \texttt{dualIsoOfIso}\,e_2 \ggg \texttt{dualIsoOfIso}\,e_1,$$

the order reversing because the dual is built by pre*composition (it is contravariant).*

Proof. $\texttt{dualIsoOfIso}\,e$ is precomposition of internal-hom sections by the underlying map of e. Precomposition by a composite is the composite of precompositions in reversed order, so the claim is the contravariant functoriality of $\texttt{Hom}(-, \mathcal{O})$ applied sectionwise; established by Iso.ext followed by sectionwise extensionality. ☐

Lemma 868 (Presheaf dual-transport preserves identities)**.** $\texttt{dualIsoOfIso}\,(\mathrm{Iso.refl}\,L) = \mathrm{Iso.refl}\,(L^\vee)$ *at the presheaf level.*

Proof. Precomposition by the identity is the identity on internal-hom sections. ☐

Lemma 869 (Sheaf-level dual-transport is contravariantly functorial in composition)**.** *For composable isomorphisms $L \xrightarrow{e_1} L' \xrightarrow{e_2} L''$ in* Scheme.`Modules` *X,*

$$\texttt{dualIsoOfIso}\,(e_1 \ggg e_2) = \texttt{dualIsoOfIso}\,e_2 \ggg \texttt{dualIsoOfIso}\,e_1.$$

Proof. The sheaf-level dual-transport is the presheaf-level one (867) transported through the sheaf-equivalence and the forgetful functor; functoriality of mapIso carries the presheaf identity to the sheaf identity. ☐

Lemma 870 (Sheaf-level dual-transport preserves identities)**.** $\texttt{dualIsoOfIso}\,(\mathrm{Iso.refl}\,L) = \mathrm{Iso.refl}\,(L^\vee)$ *at the sheaf level.*

Proof. Immediate from the presheaf-level statement 868 and functoriality of mapIso on the identity. ☐

Definition 871 ($\mathcal{O}_X \otimes_X \mathcal{O}_X \cong \mathcal{O}_X$)**.** The substrate tensor of the structure sheaf with itself is the structure sheaf: $\texttt{tensorObj}\,\mathcal{O}_X\,\mathcal{O}_X \cong \mathcal{O}_X$, where $\mathcal{O}_X = \texttt{SheafOfModules.unit}\,X\texttt{.ringCatSheaf}$. It is the unit isomorphism consumed by the invertibility constructions.

Proof. Proved directly in Lean: the sheafification of the presheaf left unitor $\lambda_{\mathbf{1}}$, composed with the sheafification-adjunction counit at the already-sheaf unit. ☐

Definition 872 (Middle-four interchange for $\otimes_X$)**.** For arbitrary $A, B, C, D \in$ Scheme.`Modules` X there is an isomorphism $(A \otimes B) \otimes (C \otimes D) \cong (A \otimes C) \otimes (B \otimes D)$, reassociating the four factors. It is the bookkeeping isomorphism used to show $\otimes$-invertibility is closed under tensor product.

Proof. Proved directly in Lean: an assembly of the unconditional associator 653 and the braiding 655 through `tensorObjIsoOfIso`; no coherence pentagon is consumed. □

Lemma 873 (Unit comparison is an iso from a finality hypothesis)**.** *For $g : Y \to X$ with* (`Opens.map` g.`base`).`Final`*, the Mathlib unit comparison* `pullbackObjUnitToUnit` g *is an isomorphism.*

Proof. Proved directly in Lean by `inferInstance`: the lone finality hypothesis is isolated at a clean site so the Mathlib `OfFinal` instance synthesises without colliding with other in-scope `Final`/`IsIso` hypotheses. □

Definition 874 (Bundled iso form of the unit comparison)**.** For $g : Y \to X$ with (`Opens.map` g.`base`).`Final`, the isomorphism $g^*\mathcal{O}_X \cong \mathcal{O}_Y$ bundling the unit comparison `pullbackObjUnitToUnit` g of 873. Handing out the bundled `Iso` keeps downstream reasoning at the `Iso` level.

Proof. Proved directly in Lean: `asIso` of the comparison with the explicit witness 873. □

Definition 875 (Sheafification–monoidality reconciliation brick)**.** For presheaves of modules P, Q on X, sheafifying the presheaf tensor $P \otimes Q$ agrees with sheafifying the tensor of the underlying presheaves of their sheafifications. This is the "sheafification is monoidal" reconciliation bridging a presheaf-level tensorator with the substrate $\otimes_X$.

Proof. Proved directly in Lean, exactly as in 653: whisker the sheafification unit η (a $J.W$-morphism) on each side and invert under sheafification via 649 together with the flatness-free whiskering stability 652. □

Lemma 876 (`hom` of the reconciliation brick as a whisker composite)**.** *The* `hom` *of 875 equals the sheafification of $\eta_P \triangleright Q$ followed by the sheafification of $(aP).\mathrm{val} \triangleleft \eta_Q$, stated on the common* $\circ$ `forget`$_2$ *carrier so the two whisker factors can be merged.*

Proof. Proved directly in Lean by `rfl`, stripping the carrier cast. □

Lemma 877 (`hom` of the reconciliation brick as one `tensorHom`)**.** *The* `hom` *of 875 equals the sheafification of the single* `tensorHom` $\eta_P \otimes \eta_Q$ *of the two sheafification-unit components — keeping every term in one monoidal instance so naturality reduces to plain bifunctoriality.*

Proof. Proved directly in Lean: merge the two whisker factors of 876 by ← `Functor.map_comp` and identify with `tensorHom_def`. □

Definition 878 (Sheafified presheaf-pullback ≅ abstract pullback on a value)**.** For $f : Y \to X$ and $M \in$ Scheme.`Modules` X, sheafifying the presheaf-level pullback of the underlying presheaf M.val recovers the abstract $\mathcal{O}$-module pullback (`pullback` f).obj M. It reconciles the right-hand side of `pullbackTensorMap` with the substrate tensor of the pullbacks.

Proof. Proved directly in Lean: the inverse of `sheafificationCompPullback` (677) at M.val, composed with the pullback of the sheafification counit at the already-sheaf M. □

Lemma 879 (The sheafified $\otimes$ of the two `pullbackValIso` is an iso)**.** *The fourth factor of* `pullbackTensorMap` *— the sheafification of the* `tensorHom` *of the two* `pullbackValIso` *comparisons (one for M, one for N) — is an isomorphism.*

Proof. Proved directly in Lean: it is the `hom` of the sheafification `mapIso` of the monoidal `tensorIso` of the two `pullbackValIso` isomorphisms. □

Lemma 880 (Composite ring-map reconciliation)**.** *For composable $h : Z \to Y$, $f : Y \to X$, the structure ring-presheaf map of the composite $h \circ f$ factors as φ'_f followed by the whiskered φ'_h. This feeds the presheaf* `pullbackComp` *into the oplax* `comp_`δ *decomposition (Sq2 of 685).*

Proof. Proved directly in Lean by `rfl`: the two sides hold up to definitional equality at default transparency. □

Lemma 881 (Sq2b — monoidality of the presheaf `pullbackComp`)**.** *The oplax cotensorator δ of the pullback for a composite ring map transports, through the pullback pseudofunctor coherence* `pullbackComp`*, into the* `Functor.OplaxMonoidal.comp` *cotensorator of the composite pullback. This is the $\eta \to \delta$ analogue, at the* `PresheafOfModules` *level, of the unit coherence 670.*

Proof. Proved directly in Lean by the adjunction-mate calculus: transpose under the pullback–pushforward adjunction, rewrite the oplax δ as the mate of the lax μ of the pushforward (665), and use the conjugate identity for `pullbackComp`.inv (here `pushforwardComp` = `Iso.refl`); the sole residual is the lax-μ composition coherence `pushforwardComp_lax_`μ, discharged by a sectionwise pure-tensor collapse. □

Lemma 882 (Sheafified split of the composite-pullback tensorator (step (ii) sheafified))**.** *For composable $h : Z \to Y$, $f : Y \to X$ and $M, N \in \mathrm{Scheme}.$*`Modules` *$X$, the sheafification $a_Z.\mathrm{map}$ of the oplax tensorator δ_{comp} of the* composite *presheaf pullback* `pullback` $\varphi'_f \circ$ `pullback` φ'_h *splits as*

$$a_Z.\mathrm{map}\,\delta_{\mathrm{comp}} \;=\; a_Z.\mathrm{map}((\texttt{pullback}\,\varphi'_h).\mathrm{map}\,\delta_f(M.\mathrm{val}, N.\mathrm{val}))\;;\;a_Z.\mathrm{map}\,\delta_h(F^{\mathrm{p}}_f M.\mathrm{val},\ F^{\mathrm{p}}_f N.\mathrm{val}).$$

This is the sheafified form of step (ii) of the residual chain in the proof of 685.

Proof. The oplax-monoidal structure on a composite functor *defines* its tensorator δ_{comp} to be precisely the composite (`pullback` φ'_h).map δ_f ; δ_h (`Functor.OplaxMonoidal.comp_`δ), so the underlying equality holds by definition; applying a_Z.map and distributing it over the composite by functoriality yields the displayed split. □

Lemma 883 (Sheaf-level conjugate of `pullbackComp`.inv)**.** *For composable $h : Z \to Y$, $f : Y \to X$ and $Q \in \mathrm{Scheme}.$*`Modules` *X, the unit of the composite-pullback adjunction post-composed with the pushforward of* `pullbackComp`.inv *equals the unit of the composite of the f- and h-adjunctions post-composed with* `pushforwardComp`.hom*. It is the cheap, sheafification-free R0-peel piece of the Sq1 mate calculus.*

Proof. Proved directly in Lean: the sheaf-level instance of `unit_conjugateEquiv` combined with `conjugateEquiv_pullbackComp_inv` (the mate of `pullbackComp`.inv is `pushforwardComp`.hom). □

Lemma 884 (Sq1 — composition coherence of `sheafificationCompPullback`)**.** *For composable $h : Z \to Y$, $f : Y \to X$ and a presheaf P over X, the sheafification–pullback comparison of $h \circ f$ factors through the comparisons of f and h, conjugated by the sheaf-level pullback pseudofunctor iso* `pullbackComp` *$h\,f$ and the sheafified presheaf pullback pseudofunctor iso. This is the* `sheafificationCompPullback` *twin of 670; it is the S1-foundational coherence consumed by 685.*

Proof. Proved directly in Lean: both `sheafificationCompPullback` isos are `leftAdjointUniq` of composite adjunctions (677); transpose the identity under the composite adjunction for $h \circ f$, evaluate the left-hand side by `homEquiv_leftAdjointUniq_hom_app`, peel the leading `pullbackComp`.inv factor by 883, and discharge the residual unit identity by 885. □

Lemma 885 (Sq1 tail — the R1/R5 unit collapse)**.** *Source: internal categorical construction; no external reference. After the R0-peel of 884, the residual obligation is the unit identity for the two-layer composite adjunction (presheaf pullback–pushforward composed with sheafification). Writing* B_φ *for that sheaf-level composite adjunction,* $\texttt{PresheafPullback}_f$ *for the presheaf-level pullback along* f*,* $R_1 = h^*((\texttt{sheafCompPb}\, f).\mathrm{app}\, P)$ *and* $R_5 = (\texttt{sheafCompPb}\, h).\mathrm{app}\,(\texttt{PresheafPullback}_f\, P)$ *for the two sub-comparison factors, the tail asserts that the composite unit* $B_{h\circ f}.\mathrm{unit.app}\, P$ *equals the sheafification-unit of* X *followed by the (forgetful, change-of-rings-along-identity) image of the merged factor* $((B_f.\mathrm{comp} B_h).\mathrm{unit} ; \texttt{pushforwardComp}.\mathrm{hom}) ; (\textit{pushforward of } R_1 ; R_5 ; a_Z.\mathrm{map}\, \delta_{\mathrm{pre}})$.

Proof. Source: internal categorical construction; no external reference.

This is the sheafification-laden analogue of the unit composition coherence 670, one sheafification layer up. There the adjunctions are single transposes and the coherence is sectionwise definitional; here B_f, B_h are *composite* adjunctions (presheaf pullback–pushforward) .comp (sheafification), so the coherence is a genuine mate-calculus identity rather than a reflexivity.

Both `sheafificationCompPullback` isos are `Adjunction.leftAdjointUniq` of these composite adjunctions (677). The strategy is to keep the left-hand composite unit $B_{h\circ f}.\mathrm{unit.app}\, P$ intact and *assemble* the right-hand side into it, in the following ordered steps.

(a) Strip the identity wrapper. The tail goal carries an outer `restrictScalars(`**1**`)` change-of-rings factor from the identity ring map; it is removed first (restriction of scalars along the identity is the identity functor), exposing the genuine comparison factors.

(b) Distribute the right-hand sheaf composite under `forget`*.* Pushing the forgetful functor `forget` (sheaf → presheaf) through the right-hand composite separates the two sub-comparison factors R_1 (the f-layer) and R_5 (the h-layer), confined to the right-hand side so that the left-hand composite unit $B_{h\circ f}.\mathrm{unit}$ is left intact for the final reassembly. (Distributing without confinement would unfold the left-hand sheafification unit into its `toSheafify`;`restrictHomEquiv` expansion, the wrong normal form.)

(c) Recover the sub-comparison units. Apply the P-general uniqueness-of-left-adjoints brick 679 (`homEquiv_leftAdjointUniq_hom_app` in its P-general form) to rewrite R_1 as the composite-adjunction unit $B_f.\mathrm{unit.app}\, P$ and R_5 as $B_h.\mathrm{unit.app}\,(\texttt{PresheafPullback}_f\, P)$. This is the load-bearing step: it identifies each `sheafCompPb` factor with a unit of the appropriate composite adjunction.

(d) Slide `pushforwardComp` *past the recovered units.* The sheaf-level comparison `pushforwardComp` $h\, f$ is moved across the two recovered units $B_f.\mathrm{unit}$, $B_h.\mathrm{unit}$ by its naturality.

(e) Reassemble the composite unit. Expanding $B_{h\circ f}.\mathrm{unit}$ by the composite-adjunction unit formula `Adjunction.comp_unit_app` and applying `Adjunction.unit_naturality` identifies the slid expression of step (d) with the left-hand composite unit, closing the tail.

The single non-mechanical bridge is the presheaf↔sheaf pushforward compatibility $\texttt{forget} \circ \texttt{pushforward}^{\mathrm{sheaf}} = \texttt{pushforward}^{\mathrm{pre}} \circ \texttt{forget}$ (a definitional identity, since the sheaf pushforward is built sectionwise from the presheaf pushforward and `forget` is the underlying-presheaf projection), interacting with the recovered B_f- and B_h-units of step (c); once it is in hand, steps (a)–(e) are a mechanical paste. This identity reduces, exactly as the unit twin 670 does, to `Iso.inv_hom_id` / unit-coherence at the unit presheaf. *Warning:* transposing the *whole* tail back under the composite adjunction — instead of recovering the two sub-units individually in step (c) — is circular: it returns the very identity one set out to prove. □

16.14.1 Change-of-rings, lax multiplication, and extension-of-scalars helpers

This subsection blueprints the change-of-base-ring cluster of `TensorObjSubstrate.lean`: the extension-of-scalars functor and its adjunction, the lax-monoidal structure maps μ of `restrictScalars` and `pushforward`, and the monoidality of the pushforward composition cell (`pushforwardComp`) — the change-of-rings coherence that is the sole residual of the presheaf-level Sq2b lemma 881. Each helper is sorry-free; its `\uses` edges are read off its Lean body.

Lemma 886 (`restrictScalars`($\mathbf{1}$) on morphisms). *For a morphism g of presheaves of modules over R,* (`restrictScalars` ($\mathbf{1}\,R$))$.\mathrm{map}\, g = g$*: restriction of scalars along the identity ring map acts as the identity on morphisms.*

Proof. Proved directly in Lean by `rfl`: `restrictScalars` ($\mathbf{1}$) is definitionally the identity functor, stated as a propositional rewrite so the wrapper can be stripped syntactically without a whole-term `whnf`. ☐

Lemma 887 (Sectionwise value of the `restrictScalars` tensorator μ). *The lax tensorator μ of* `restrictScalars` α*, evaluated at a section W, is the* `ModuleCat` *lax μ of* `restrictScalars` (α_W) *on the sections. It exposes the ambient presheaf μ in a form on which the pure-tensor rule matches syntactically.*

Proof. Proved directly in Lean by `rfl`: the presheaf tensorator `restrictScalarsLax`μ (853) is defined sectionwise as the `ModuleCat` tensorator. ☐

Lemma 888 (Pure-tensor value of the `ModuleCat` `restrictScalars` tensorator). *On a pure tensor $m \otimes_t n$, the* `ModuleCat` *lax tensorator μ of* `restrictScalars` f *(with* `forget`$_2$*-carrier rings coming from presheaves of* commutative *rings) is the identity: $\mu(m \otimes_t n) = m \otimes_t n$.*

Proof. Proved directly in Lean: it is `ModuleCat.restrictScalars_`μ`_tmul`, the Mathlib pure-tensor formula for the change-of-rings tensorator, with the `CommRing` instances supplied by the `CommRingCat` carriers. ☐

Lemma 889 (Pure-tensor value of the presheaf `restrictScalars` tensorator). *On a pure tensor,* (μ(`restrictScalars` α) $M_1\, M_2)_W(m \otimes_t n) = m \otimes_t n$*: the presheaf* `restrictScalars` *tensorator is the identity on pure tensors.*

Proof. Proved directly in Lean: expose the `ModuleCat` tensorator by 887, then apply the Mathlib pure-tensor rule `ModuleCat.restrictScalars_`μ`_tmul`. ☐

Lemma 890 (Pure-tensor value of the presheaf `restrictScalars` *oplax* tensorator δ). *The δ-twin of 889. For a sectionwise-bijective ring-functor map α,* `restrictScalars` α *is strong monoidal, so its comonoidal (oplax) structure map δ is the two-sided inverse of the lax μ. On a pure tensor,* (δ(`restrictScalars` α) $M_1\, M_2)_W(m \otimes_t n) = m \otimes_t n$*: δ sends a pure tensor to the same underlying tensor, viewed through* `restrictScalars`. Remark (R-module instance). *The input $m \otimes_t n$ lives in the S-tensor section, while the output $m \otimes_t n$ must be read in the codomain object* ((`restrictScalars` α)$.\mathrm{obj}\, M_1 \otimes$ (`restrictScalars` α)$.\mathrm{obj}\, M_2)_W$*, whose R-structure exists only via* `restrictScalars`*; the formalizer must ascribe the right-hand side to that codomain.*

Proof. Proved directly in Lean: since $\delta = \mu^{-1}$ (`Functor.Monoidal.`μ`_`δ), rewrite the goal through the inverse and discharge the resulting μ-value on the pure tensor by 889. ☐

Lemma 891 (Pure-tensor value of the `pushforward`-mapped `restrictScalars` tensorator). *The outer leg of 893: applied to a pure tensor,* ((`pushforward` φ)$.\mathrm{map}$ (μ(`restrictScalars` ψ) $N_1\, N_2))_W$ *is the identity.*

Proof. Proved directly in Lean: reindex the section map through `pushforward_map_app_apply` (`pushforward` φ is `pushforward`$_0$ $\cdot$ `restrictScalars` φ, so the section map at W is the μ at $F^{\mathrm{op}}.\mathrm{obj}\, W$), then collapse by 889. ☐

Lemma 892 (Reduction of the `pushforward` tensorator to the `restrictScalars` μ)**.** *The lax tensorator μ of a single* `pushforward` φ *equals the lax tensorator of the change-of-rings* `restrictScalars` φ' *on the (strongly-monoidal, $\mu_{\mathrm{Iso}} = \mathbf{1}$) reindexed objects* $\texttt{pushforward}_0^{\mathrm{Comm}}\, F R_0$*. This unfolds the opaque pushforward lax-monoidal μ into the directly computable* `restrictScalars` *μ.*

Proof. Proved directly in Lean by `rfl`: the pushforward lax-monoidal structure (665) is the `Functor.LaxMonoidal.comp` of the (identity) $\texttt{pushforward}_0$ tensorator with the `restrictScalars` tensorator (853), which is definitionally the right-hand side. ☐

Lemma 893 (Sq2b residual — monoidality of the `pushforward` composition cell)**.** *The change-of-rings coherence "*`pushforwardComp` *is monoidal": the lax tensorator μ of the composite pushforward* `pushforward` $\psi \cdot$ `pushforward` φ *(built by* `Functor.LaxMonoidal.comp`*) agrees with the lax tensorator of the single pushforward* `pushforward` $(\varphi\,;\, F^{\mathrm{op}} \triangleleft \psi)$*. It is the sole genuine residual of the presheaf-level Sq2b lemma 881.*

Proof. Proved directly in Lean by a sectionwise pure-tensor collapse: `hom_ext` then `tensor_ext`, unfold the composite μ by `Functor.LaxMonoidal.comp_`μ, lighten each μ(`pushforward _`) to a `restrictScalars` μ by 892, expose the `ModuleCat` μ by 887, and collapse the inner leg by 888 and the outer (`pushforward` φ).map leg by 891; the two collapsed pure tensors are then definitionally equal. (The equality is genuinely not `rfl` at the presheaf level — the `restrictScalars` μ on a pure tensor is real base-change content — so the atomic-object helpers are applied as explicit arguments and matched by `erw` to avoid a `whnf`-explosion.) ☐

Lemma 894 (Forgetful image of a sheaf-pushforward map)**.** *The forgetful functor* `SheafOfModules.forget` *intertwines the sheaf-level* `SheafOfModules.pushforward` φ *with the presheaf-level* `PresheafOfModules.pushforward` φ *for a morphism g of sheaves of modules,* `forget`.map ((`pushforward` φ).map g) = (`PresheafOfModules.pushforward` φ *This is the STEP-1 binding reconciliation of the Sq1 tail.*

Proof. Proved directly in Lean by `rfl`: the sheaf pushforward is built sectionwise from the presheaf pushforward and `forget` is the underlying-presheaf projection, so the two sides are definitionally equal. ☐

16.14.2 The Picard quotient: setoid, multiplication and inverse

This subsection blueprints the three quotient-level helpers underlying the invertibility-carrier Picard group 700 and its abelian-group law `pic_commgroup`: the isomorphism setoid, the tensor-induced multiplication, and the dual-induced inverse on iso-classes.

Definition 895 (The isomorphism setoid of invertible modules)**.** The setoid on $\otimes$-invertible $\mathcal{O}_X$-modules with $M \sim M'$ iff there is an isomorphism $M \cong M'$. The quotient of this setoid is the carrier of the Picard group $\operatorname{Pic} X$.

Proof. Proved directly in Lean: reflexivity, symmetry and transitivity of the `Nonempty` $(M \cong M')$ relation are witnessed by `Iso.refl`, `Iso.symm` and `Iso.trans`. ☐

Definition 896 (Multiplication on $\operatorname{Pic} X$ induced by $\otimes_X$)**.** The multiplication $[M] \cdot [M'] := [M \otimes_X M']$ on $\operatorname{Pic} X$, well-defined on iso-classes.

Proof. Proved directly in Lean by $\texttt{Quotient.lift}_2$: the value lands in $\operatorname{Pic} X$ because the tensor of invertibles is invertible (701), and it respects the setoid because `tensorObjIsoOfIso` (862) turns a pair of isomorphisms into an isomorphism of tensors. ☐

Definition 897 (Inverse on $\mathrm{Pic}\, X$ induced by the tensor-inverse)**.** The inverse class $[M] \mapsto [N]$ on $\mathrm{Pic}\, X$, where N is a chosen tensor-inverse of M.

Proof. Proved directly in Lean by `Quotient.lift`: a chosen inverse witness (`Classical.choose` of the invertibility) is itself invertible, with the braiding (655) supplying the membership isomorphism; the choice is independent of the representative because any two tensor-inverses of M are isomorphic (702). $\square$

Chapter 17

The sheaf-of-modules over-equivalence (shared slice root)

17.1 The equivalence

Definition 898 (Sheaf-of-modules over-equivalence). Let X be a scheme and $U \subseteq X$ an open subset, with associated open immersion $\iota : \widetilde{U} \hookrightarrow X$. The reindexing site equivalence

$$e \ := \ \texttt{Opens.overEquivalence}\, U \ : \ \texttt{Over}\, U \ \simeq \ \texttt{Opens}\, \widetilde{U}$$

— between the slice of $\texttt{Opens}\, X$ over U (carrying the slice topology $(\texttt{Opens.grothendieckTopology}\, X).\texttt{over}\, U$) and the opens of the subspace $\widetilde{U}$ (carrying $\texttt{Opens.grothendieckTopology}\, \widetilde{U}$) — lifts to an *equivalence of sheaf-of-modules categories*

$$\texttt{overEquivalence} \ : \ \texttt{SheafOfModules}\,((\widetilde{U}).\texttt{ringCatSheaf}) \ \simeq \ \texttt{SheafOfModules}\,(X.\texttt{ringCatSheaf.over}\, U),$$

the modules-level lift of e that transports the structure ring sheaf. It is obtained by instantiating `SheafOfModules.pushforwardPushforwardEquivalence` at e, with the open-immersion structure-sheaf ring morphism

$$\varphi \ : \ X.\texttt{ringCatSheaf.over}\, U \ \longrightarrow \ (e.\texttt{functor.sheafPushforwardContinuous}).\mathrm{obj}\,((\widetilde{U}).\texttt{ringCatSheaf})$$

— sectionwise at an object $V \in \texttt{Over}\, U$ the open-immersion comparison $\mathcal{O}_X(V.\mathrm{left}) \to \mathcal{O}_{\widetilde{U}}(e\, V)$, i.e. the section-level isomorphism $\iota.\texttt{appIso}$ of the structure sheaves — together with its inverse ψ and the two coherences H_1, H_2 expressing that φ and ψ are mutually inverse.

Proof. The construction is the assembly of three existing Mathlib primitives; the only step carrying mathematical content is the ring morphism φ.

Continuity (no work). The equivalence `pushforwardPushforwardEquivalence` requires both legs of the site equivalence e to be continuous: $e.\texttt{functor}$ and $e.\texttt{inverse}$. For a dense-subsite inclusion continuity is automatic — it is a typeclass instance — and for an equivalence the density of one leg propagates to the other. The declaration `overEquivInverseIsDenseSubsite` exhibits $e.\texttt{inverse}$ as a dense subsite of the slice site, from which both continuity legs follow by inference. The Mathlib TODO at `Topology/Sheaves/Over.lean` (lines 19–22) is discharged for this case.

The ring morphism φ (the genuine content). The two structure ring sheaves are different objects on different sites: X.`ringCatSheaf`.`over` U over `Over` U, and $(\widetilde{U})$.`ringCatSheaf` over `Opens` $\widetilde{U}$. The open immersion ι identifies them: at an object $V \in$ `Over` U the sections $\mathcal{O}_X(V.\mathrm{left})$ and $\mathcal{O}_{\widetilde{U}}(\iota^{-1}(V.\mathrm{left}))$ agree through the open-immersion section isomorphism ι.`appIso`. This is verbatim the datum that `Scheme.Modules.restrictFunctor` already uses inline (its structure map is $(\iota$.`appIso` $\cdot).\mathrm{inv}$). Packaging this section-level isomorphism naturally in V yields φ; its inverse is ψ, and the coherences H_1, H_2 record that the ι.`appIso` round-trips are identities, proved by the inverse-iso laws (`Sheaf.hom_ext` reduces them sectionwise).

Feeding e, φ, ψ, H_1, H_2 into `pushforwardPushforwardEquivalence` produces the asserted equivalence. Its underlying functor is `SheafOfModules.pushforward` φ, which is what the two consumer isomorphisms below unfold. ☐

17.1.1 Substrate helpers for the equivalence

Definition 899 (Continuity of the inverse leg of the over-equivalence)**.** The inverse leg e.`inverse` : `Opens` $\widetilde{U} \to$ `Over` U of the reindexing site equivalence $e =$ `Opens.overEquivalence` U is continuous from the subspace topology `Opens.grothendieckTopology` $\widetilde{U}$ to the slice topology (`Opens.grothendieckTopology` X).

This is the continuity prerequisite, recorded for the scheme-carrier presentation of the open subscheme, that `pushforwardPushforwardEquivalence` demands of the inverse leg.

Proof. Proved directly in Lean: it converts to the bare-subspace form $\widetilde{U}$, where the project's dense-subsite instance for the inverse leg and the priority instance `IsDenseSubsite` $\Rightarrow$ `IsContinuous` discharge it by inference. ☐

Definition 900 (Continuity of the functor leg of the over-equivalence)**.** The functor leg e.`functor` : `Over` $U \to$ `Opens` $\widetilde{U}$ of the reindexing site equivalence is continuous from the slice topology (`Opens.grothendieckTopology` X).`over` U to the subspace topology `Opens.grothendieckTopology` $\widetilde{U}$. It is the second continuity prerequisite of `pushforwardPushforwardEquivalence`.

Proof. Proved directly in Lean. Converting to the bare-subspace form, the density of the inverse leg (899) propagates, for an equivalence of sites, to density — hence continuity — of the functor leg. ☐

Lemma 901 (Reindexed open recovers its left component)**.** *For an object $V \in$ `Over` U, the open-immersion image of the reindexed open e.`functor`.obj $V = \iota^{-1}(V.\mathrm{left})$ is $V.\mathrm{left}$ itself,*

$$\iota_*\,(e.\texttt{functor}.\mathrm{obj}\,V) \;=\; V.\mathrm{left},$$

because $V.\mathrm{left} \le U$. This is the equality of opens that identifies the two structure-sheaf presheaves underlying the ring morphism φ.

Proof. Proved directly in Lean: a point y lies in $\iota_*(\iota^{-1}(V.\mathrm{left}))$ iff it lies in $V.\mathrm{left}$, using $V.\mathrm{left} \le U$ so that the preimage and image cancel. ☐

Lemma 902 (Inverse-reindexed open is its open-immersion image)**.** *For an open $W \subseteq \widetilde{U}$, the left component of the inverse-reindexed slice object e.`inverse`.obj W is the open-immersion image of W,*

$$(e.\texttt{inverse}.\mathrm{obj}\,W).\mathrm{left} \;=\; \iota_*\,W.$$

This is the symmetric counterpart of 901.

Proof. Proved directly in Lean: the two opens have equal underlying sets by definitional unfolding of the inverse reindexing functor. ☐

Definition 903 (The structure-sheaf ring morphism φ)**.** The ring morphism φ underlying the over-equivalence: at an object $V \in \texttt{Over}\, U$ it is the identification $\mathcal{O}_X(V.\mathrm{left}) \xrightarrow{\sim} \mathcal{O}_{\widetilde{U}}(\iota^{-1}(V.\mathrm{left}))$ of the two structure sheaves, realised on the nose as $X.\texttt{ringCatSheaf}$ applied to the open equality 901 (the two presheaves agreeing definitionally). Naturality in V follows from functoriality of $X.\texttt{ringCatSheaf}$.

Proof. Proved directly in Lean. It is a sub-step of 898. ☐

Definition 904 (The inverse structure-sheaf ring morphism ψ)**.** The inverse ring morphism ψ, symmetric to 903: at an open $W \subseteq \widetilde{U}$ it is $X.\texttt{ringCatSheaf}$ applied to the open equality 902, providing the section-level inverse of φ. It is the datum ψ fed into `pushforwardPushforwardEquivalence`.

Proof. Proved directly in Lean. It is a sub-step of 898. ☐

17.2 The equivalence carries restriction and unit

17.2.1 Substrate helpers for the restriction comparison

Definition 905 (The restriction ring morphism along the open-immersion functor)**.** The ring morphism along the open-immersion functor $\iota_{\texttt{opensFunctor}} : \texttt{Opens}\, \widetilde{U} \to \texttt{Opens}\, X$ underlying Scheme.`Modules.restrictFunctor` ι: at an open $W \subseteq \widetilde{U}$ it is the inverse open-immersion section comparison $(\iota.\texttt{appIso}\, W)^{-1}$, promoted to a ring-sheaf morphism. It is reconstructed so that the identification 906 holds definitionally.

Proof. Proved directly in Lean: it repackages the structure map of `restrictFunctor` ι (whose component at W is $(\iota.\texttt{appIso}\, W)^{-1}$) as a morphism of ring sheaves. ☐

Lemma 906 (Restriction functor as a pushforward)**.** *The restriction functor along the open immersion is, on the nose, a sheaf-of-modules pushforward along the morphism 905:*

$$\text{Scheme.}\texttt{Modules.restrictFunctor}\, \iota \;=\; \texttt{SheafOfModules.pushforward}\, (\texttt{psiRestrict}).$$

Proof. Proved directly in Lean: the two functors are definitionally equal, since `psiRestrict` was built to reproduce the internal structure map of `restrictFunctor` ι verbatim. ☐

Definition 907 (Index reconciliation natural isomorphism)**.** The natural isomorphism

$$\texttt{Over.forget}\, U \xrightarrow{\sim} e.\texttt{functor} \circ \iota_{\texttt{opensFunctor}}$$

reconciling the two index functors $\texttt{Over}\, U \to \texttt{Opens}\, X$ entering the composite pushforward; both send $V \mapsto V.\mathrm{left}$, and the component at V is the `eqToIso` of the open equality 901.

Proof. Proved directly in Lean: the components are `eqToIso` of 901, and naturality is automatic since the opens form a thin category. ☐

Lemma 908 (The over-equivalence carries restriction to over)**.** *Let $M \in$ Scheme.`Modules` X. Under the over-equivalence of 898 there is a canonical isomorphism*

$$\texttt{overEquivalence.functor}.\mathrm{obj}\, (M|_\iota) \xrightarrow{\sim} M.\texttt{over}\, U,$$

i.e. the equivalence carries the restriction $M|_\iota$ of M along the open immersion to the slice object $M.\texttt{over}\, U$.

Proof. The restriction $M|_\iota$ is *itself* a pushforward of sheaves of modules: by construction $M|_\iota =$ `SheafOfModules.pushforward` applied along ι.`opensFunctor` (the open-immersion functor `Opens` $\widetilde{U} \to$ `Opens` X). The functor underlying the over-equivalence is `SheafOfModules.pushforward` φ. Their composite is again a single pushforward, identified by `SheafOfModules.pushforwardComp`, which here is the identity isomorphism `Iso.refl`. It remains to identify the two underlying index functors `Over` $U \to$ `Opens` X entering the two pushforwards: both send an object V to its left component V.left, so they are equal, and `pushforwardNatIso` along the `eqToIso` of that equality transports the composite to M.`over` U. The whole argument mirrors, step for step, the existing `Scheme.Modules.restrictFunctorAdjCounitIso`. ∎

Lemma 909 (The over-equivalence carries unit to unit)**.** *Under the over-equivalence of 898 there is a canonical isomorphism*

$$\texttt{overEquivalence.functor}.\mathrm{obj}\,(\texttt{SheafOfModules.unit}\,(\widetilde{U}).\texttt{ringCatSheaf}) \xrightarrow{\sim} \texttt{SheafOfModules.unit}\,(X.\texttt{ringCat}$$

i.e. the equivalence carries the unit module (the structure sheaf-as-module) of the subspace site to the unit module of the slice site.

Proof. The functor underlying the over-equivalence is `SheafOfModules.pushforward` φ. The pushforward of a unit module along a morphism of ringed sites is the unit module of the target ring sheaf, transported along the ring identification φ: the structure sheaf pushes forward to the structure sheaf because φ is precisely the open-immersion identification of the two structure ring sheaves. Concretely the pushforward-of-unit computation produces `unit` of the codomain ring sheaf up to φ, and φ being an isomorphism makes this a genuine isomorphism.

Concretely, the canonical comparison map realising this transport is the unit-to-pushforward-of-unit morphism

$$\texttt{SheafOfModules.unitToPushforwardObjUnit}\,\varphi \;:\; \texttt{unit}\,(X.\texttt{ringCatSheaf.over}\,U) \longrightarrow (\texttt{pushforward}\,\varphi).\mathrm{obj}\,(\texttt{unit}$$

Its section at an open W is exactly the section-level ring map φ.app W: the morphism acts on a section by multiplication through φ, so on W it is the additive-group map underlying φ.app W (the characterisation `unitToPushforwardObjUnit_val_app_apply`). Since φ is an isomorphism of ring sheaves, each component φ.app W is a ring isomorphism, hence its underlying additive-group map is an isomorphism.

It follows that `unitToPushforwardObjUnit` φ is an isomorphism. One reflects the sectionwise isomorphism upward through the two forgetful functors — first `SheafOfModules.forget` (down to presheaves of modules), then `PresheafOfModules.toPresheaf` (down to presheaves of additive groups) — so that being an isomorphism reduces to the componentwise isomorphism check `NatTrans.isIso_iff_isIso_app` on the underlying natural transformation, which is the sectionwise statement just established. The desired isomorphism `unitOverIso` is then the inverse of the resulting isomorphism, (`asIso` (`unitToPushforwardObjUnit` φ))$^{-1}$. This is the slice-site analogue of `pullbackObjUnitToUnitIso`. ∎

17.3 The engine corollary

Lemma 910 (Over–restrict trivialisation bridge)**.** *Let* $M \in$ Scheme.`Modules` X *and* $U \subseteq X$ *open, and suppose given a trivialisation of the restriction along the open immersion* $\iota : \widetilde{U} \hookrightarrow X$*,*

$$e \;:\; M|_\iota \xrightarrow{\sim} \texttt{SheafOfModules.unit}\,(\widetilde{U}).\texttt{ringCatSheaf}.$$

Then there is a canonical isomorphism in the slice category

$$\texttt{chartOverIso}\,M\,U\,e \;:\; M.\texttt{over}\,U \xrightarrow{\sim} \texttt{SheafOfModules.unit}\,(X.\texttt{ringCatSheaf.over}\,U).$$

Proof. Transport the trivialisation e across the over-equivalence and reconcile the two ends with the consumer isomorphisms. Explicitly, `chartOverIso` $M\,U\,e$ is the composite

$$M.\texttt{over}\,U \xrightarrow[\sim]{(\texttt{restrictOverIso})^{-1}} \texttt{overEquivalence.functor}.\mathrm{obj}\,(M|_\iota) \xrightarrow[\sim]{\texttt{overEquivalence.functor.mapIso}\,e} \texttt{overEquivalence.f}$$

The first arrow is the inverse of 908; the middle arrow is the image of e under the (functor of the) over-equivalence, which is functorial in isomorphisms hence sends the iso e to an iso; the last arrow is 909. Each factor is an isomorphism, so the composite is.

This lemma addresses the site mismatch in the chart-presentation of 917: the given trivialisation e lives in `SheafOfModules` $(\widetilde{U})$.`ringCatSheaf` (the open subspace site), whereas the finite-presentation transport `Presentation.ofIsIso` must act in `SheafOfModules` $(X.$`ringCatSheaf.over` $U)$ (the slice site); `chartOverIso` crosses that gap. It is stated in full generality in M, U, e for use in 18. The over-equivalence of 898 likewise underlies the internal-Hom theory: the commutativity of the internal Hom with the slice-site change of base (737) passes through the same equivalence. ☐

Definition 911 (Engine bridge: line-bundle over–restrict trivialisation)**.** The line-bundle engine's local over–restrict trivialisation bridge: given $M \in$ Scheme.`Modules` X, an open $U \subseteq X$, and a trivialisation $e : M|_\iota \xrightarrow{\sim}$ `SheafOfModules.unit` $(\widetilde{U})$.`ringCatSheaf`, it produces the slice-level isomorphism $M.$`over` $U \xrightarrow{\sim}$ `SheafOfModules.unit` $(X.$`ringCatSheaf.over` $U)$. It is the consumer-facing alias used by the finite-presentation engine, defined to be precisely the general construction 910.

Proof. Proved directly in Lean: the declaration is definitionally the general Scheme.`Modules.chartOverIso` of 910 applied to the same data M, U, e. ☐

Chapter 18

Coherence of locally trivial line bundles

18.1 Setup and motivation

Fix a scheme X and write X.`Modules` for the category of sheaves of $\mathcal{O}_X$-modules (Mathlib's `SheafOfModules` on the ringed-space site of X, with structure sheaf X.`ringCatSheaf`). A *line bundle* on X is an invertible $\mathcal{O}_X$-module; in the sense of 14 we model line bundles by the predicate `IsLocallyTrivial`: a module $M : X$.`Modules` is locally trivial of rank one if every point $x \in X$ has an *affine* open neighbourhood U on which the restriction $M|_U$ is isomorphic, as an $\mathcal{O}_U$-module, to the structure sheaf-as-module $\mathcal{O}_U$:

$$\texttt{IsLocallyTrivial}(M) \;:=\; \forall x \in X,\ \exists U \ni x \text{ affine open}, \quad M|_U \;\cong\; \mathcal{O}_U.$$

This is the rank-one case of Stacks Definition 17.25.1 (Tag 01CS, invertible $\mathcal{O}_X$-modules), and agrees with the classical "locally free of rank one" description by Stacks Lemma 17.25.4 (Tag 0B8M).

The goal of the chapter is to record that such a module is *coherent*, namely finitely presented: M.`IsFinitePresentation` in Mathlib's `SheafOfModules.IsFinitePresentation` (built from a `QuasicoherentData`). The mathematics is elementary: on each trivialising chart $M|_U \cong \mathcal{O}_U$ is finite free of rank one, hence has the trivial finite presentation, and finite presentation is by definition a local condition. No stalk computation and no spreading-out are involved. The content is the bookkeeping that turns the pointwise trivialisation into the indexed `QuasicoherentData` Mathlib expects.

18.2 Site instances for quasi-coherent data

The Mathlib predicates `SheafOfModules.IsFinitePresentation` and `SheafOfModules.QuasicoherentData` are stated relative to a Grothendieck topology J on a base category C and a sheaf of rings $R : \mathrm{Sheaf}(J, \mathrm{RingCat})$. They require, for every object $x : C$, three instances on the slice topology J/x (Mathlib J.`over` x):

$$(J/x).\texttt{HasSheafCompose}(\mathrm{forget}_{\mathrm{RingCat}\to\mathrm{AddCommGrpCat}}), \quad \texttt{HasWeakSheafify}(J/x, \mathrm{AddCommGrpCat}), \quad (J/x).\texttt{WEqu}$$

For X.`Modules` = `SheafOfModules` X.`ringCatSheaf` these are the ambient site instances of the ringed space underlying X. We record their availability as a standing assumption for the chapter;

verifying it for X.`ringCatSheaf` is the single instance-resolution check the formalisation must clear at entry, and is independent of the mathematical content below.

Remark 912. For the ringed-space site underlying X with sheaf of rings $R = X$.`ringCatSheaf`, every slice topology J/x carries the three instances `HasSheafCompose`, `HasWeakSheafify` and `WEqualsLocallyBijective` (for the forgetful functor RingCat $\to$ AddCommGrpCat) demanded by `SheafOfModules.QuasicoherentData`. Under this assumption the predicate M.`IsFinitePresentation` is well-formed for every $M : X$.`Modules`.

18.3 From pointwise triviality to a trivialising cover

Lemma 913 (Trivialising affine cover)**.** *Source: [Stacks Project], Tag 01CS (Modules, Definition 17.25.1). Let M : X.*`Modules` *be locally trivial,* `IsLocallyTrivial`(M)*. Then there exist an index set I, a family of affine open subschemes $\{U_i\}_{i \in I}$ of X covering X (i.e. $\bigcup_i U_i = X$), and for each i an isomorphism of $\mathcal{O}_{U_i}$-modules*

$$e_i \ : \ M|_{U_i} \xrightarrow{\sim} \mathcal{O}_{U_i}$$

to the structure sheaf-as-module on U_i.

Proof. Apply the hypothesis `IsLocallyTrivial`(M) at every point $x \in X$: it produces an affine open $U_x \ni x$ and an isomorphism $e_x : M|_{U_x} \cong \mathcal{O}_{U_x}$. Take the index set $I := X$ (the underlying set of points), and to the index x assign the chosen neighbourhood U_x and the chosen isomorphism e_x. The family $\{U_x\}_{x \in X}$ covers X because each x lies in its own U_x. This is exactly the data the definition supplies, packaged as an indexed family rather than a pointwise existential. □

18.4 The canonical presentation of the unit module

The chart presentation of 18.5 is obtained by transporting a fixed canonical presentation of the unit module $\mathcal{O}$ = `SheafOfModules.unit` R along the trivialising isomorphism. We record that canonical presentation here. The unit module is free of rank one, so it has a presentation with a single generator and no relations.

Definition 914 (Free–unit isomorphism)**.** Over a sheaf of rings R on a site, the free R-module on a single generator is canonically isomorphic to the unit module:

$$\texttt{free}\,(\mathrm{PUnit}) \xrightarrow{\sim} \texttt{unit}\,R,$$

the coproduct of a one-element family of copies of `unit` R being `unit` R itself.

Proof. Proved directly in Lean. □

Definition 915 (Unit generating section)**.** The canonical generating family of the unit module `unit` R: the single section indexed by PUnit, namely the image of the free generator under the free–unit isomorphism 914. The associated map `unit` $R^{\oplus 1} \to$ `unit` R is an epimorphism, so the family generates.

Proof. Proved directly in Lean. □

Definition 916 (Canonical unit presentation)**.** The canonical finite free presentation of the unit module `unit` R: one generator (915) and no relations (indexed by the empty type). Since the generating map `unit` $R^{\oplus 1} \to$ `unit` R is an isomorphism, its relation kernel is the zero object, so the empty family of relations suffices. This presentation is finite.

Proof. Proved directly in Lean. □

18.5 The trivial presentation on a chart

Lemma 917 (Finite free presentation on a chart)**.** *Source: [Stacks Project], Modules, "Modules of finite presentation". On each chart U_i of the trivialising cover of 913, the restriction $M|_{U_i}$ admits the trivial finite free presentation*

$$0 \longrightarrow \mathcal{O}_{U_i} \longrightarrow M|_{U_i} \longrightarrow 0,$$

i.e. $M|_{U_i}$ is the cokernel of the zero map $\bigoplus_{j\in\emptyset} \mathcal{O}_{U_i} \to \mathcal{O}_{U_i}$ (the case $m = 0$, $n = 1$ of the finite-presentation pattern). The unit sheaf-as-module $\mathcal{O}_{U_i}$ (Mathlib's `SheafOfModules.unit`*) carries a canonical free presentation on one generator with no relations; this presentation is finite.*

Proof. The structure sheaf-as-module $\mathcal{O}_{U_i}$ = `SheafOfModules.unit` (U_i)`.ringCatSheaf` is free of rank one on the single generator 1: the identity map $\mathcal{O}_{U_i} \to \mathcal{O}_{U_i}$ exhibits it as $\bigoplus_{i\in\{*\}} \mathcal{O}_{U_i}$, so the surjection $\mathcal{O}_{U_i}^{\oplus 1} \twoheadrightarrow \mathcal{O}_{U_i}$ is an isomorphism and its kernel is zero, generated by the empty family. Thus $\mathcal{O}_{U_i}$ has the canonical presentation with $n = 1$ generators and $m = 0$ relations; this canonical `unit`-presentation is the rank-one free presentation, and Mathlib records it as *finite* (the `SheafOfModules.unit` finite-presentation instance).

The chart presentation is obtained by transporting the canonical finite presentation of `unit` (U_i)`.ringCatSheaf` along the isomorphism e_i, using `SheafOfModules.Presentation.ofIsIso`. This construction carries any finite presentation of a module to a finite presentation of any isomorphic module, so the transported presentation of $M|_{U_i}$ is again finite. □

18.6 Finite presentation of a locally trivial line bundle

Theorem 918 (Locally trivial ⇒ finitely presented)**.** *Source: [Stacks Project], Tag 0B8M (Modules, Lemma 17.25.4). Let X be a scheme satisfying 912 and let M :* X`.Modules` *be locally trivial, IsLocallyTrivial(M). Then M is finitely presented:* M`.IsFinitePresentation` *holds in Mathlib's* `SheafOfModules.IsFinitePresentation`*.*

Proof. By 913 choose a trivialising affine cover $\{U_i\}_{i\in I}$ with isomorphisms $e_i : M|_{U_i} \cong \mathcal{O}_{U_i}$. Assemble these into a `QuasicoherentData` for M: the index type is I, the covering objects are the affine opens U_i (which cover X), and the presentation attached to index i is the finite free presentation of $M|_{U_i}$ (equivalently, of the restriction M`.over` U_i) produced in 917. Each such presentation is finite by 917; the assembled `QuasicoherentData` therefore satisfies `QuasicoherentData.IsFinitePresentation`, and the constructor `SheafOfModules.IsFinitePresentation.mk` yields M`.IsFinitePresentation`.

The only non-formal input is the trivialising cover; everything else is the definitional unwinding of finite presentation as a local property (Stacks, "Modules of finite presentation"), since a rank-one free module is its own finite presentation. No stalk computation and no neighbourhood-spreading argument is used: the hypothesis already provides the local free models. □

Corollary 919 (Locally trivial ⇒ finite type)**.** *Under the hypotheses of 918, the locally trivial module M is of finite type (*M`.IsFiniteType`*). Quasi-coherence (*M`.IsQuasicoherent`*) follows from finite presentation via* `SheafOfModules.instIsQuasicoherentOfIsFinitePresentation`*.*

Proof. Finite presentation implies finite type by Mathlib's `SheafOfModules.instIsFiniteTypeOfIsFinitePresentat` immediate from 918. Quasi-coherence is immediate from finite presentation by `SheafOfModules.instIsQuasicoheren`
□

18.7 Rank one and flatness record

Lemma 920 (Chart-local rank one and flatness)**.** *Source: [Stacks Project], Modules, "Locally free sheaves" (finite locally free of rank r). Let M : X*.`Modules` *be locally trivial. The lemma records, at each point $x \in X$, a chart $U \ni x$ together with the trivialising iso $M|_U \cong \mathcal{O}_U$ (to* `SheafOfModules.unit`*). This iso is the rank-one free* model*: rank-one freeness and flatness are* consequences *of this iso, because $\mathcal{O}_U$ is the free $\mathcal{O}_U$-module of rank one and a flat module over itself (a ring is flat over itself), and both properties transport along e. In the sense of 610 this exhibits M as finite locally free of rank one and flat on the cover.*

Proof. Immediate from the trivialisation: on U_i, e_i identifies $M|_{U_i}$ with $\mathcal{O}_{U_i} = \bigoplus_{i \in \{*\}} \mathcal{O}_{U_i}$, the free module of rank one. The structure sheaf-as-module is flat over itself, and an isomorphic module is flat; the index set $\{*\}$ has cardinality one, so the rank is one. The properties are recorded chart-locally because rank-one freeness and flatness are local conditions, so the chart-local data determine them. ☐

Chapter 19

Flattening Stratification of a Coherent Sheaf

19.1 Setup and motivation

This chapter establishes the existence of the *flattening stratification* of a coherent sheaf $\mathcal{F}$ on a projective S-scheme: a finite collection of locally-closed subschemes of S on each of which the restriction of $\mathcal{F}$ becomes flat, together with the universal property that a base change makes $\mathcal{F}$ flat exactly when it factors through the stratification.

The statement is a theorem of commutative algebra and scheme theory independent of any moduli construction. It is the technical input by which Nitsure's exposition of Grothendieck's existence theorem ([1], §5) cuts the locus of *quotients with flat fibres of the prescribed Hilbert polynomial* out of a Grassmannian, once the functor of points of $\mathrm{Quot}^{\Phi,L}_{\mathcal{F}/X/S}$ has been embedded there via Castelnuovo–Mumford regularity.

We work throughout with a noetherian base scheme S, a projective morphism $\pi : X \to S$, and a coherent $\mathcal{O}_X$-module $\mathcal{F}$. The conclusion of Nitsure's theorem covers an arbitrary noetherian base with a projective $X \to S$, which applies in particular to $X = C \times_k T$, where C is a smooth proper curve over a field k, T is a noetherian k-scheme, and $\mathcal{F}$ is a coherent sheaf on the relative curve.

19.2 Coherent flatness over a base

Definition 921 (Coherent sheaf flat over the base)**.** Let S be a noetherian scheme, $f : X \to S$ a morphism of finite type, and $\mathcal{F}$ a coherent $\mathcal{O}_X$-module. We say $\mathcal{F}$ is *flat over S* (equivalently, $\mathcal{F}$ is *f-flat*, or *S-flat*) if for every point $x \in X$ the stalk $\mathcal{F}_x$ is a flat $\mathcal{O}_{S,f(x)}$-module, where the $\mathcal{O}_{S,f(x)}$-module structure is given by the composite $\mathcal{O}_{S,f(x)} \to \mathcal{O}_{X,x} \to \mathrm{End}(\mathcal{F}_x)$. *Source: cf. [Stacks Project], tag 00HB (flat modules); [Nitsure], §4 passim.* Equivalently (after pulling back to an affine open $U = \operatorname{Spec} A \subseteq S$ and an affine open $V = \operatorname{Spec} B \subseteq f^{-1}(U)$ with $\mathcal{F}|_V = \widetilde{M}$ for M a finite B-module), $\mathcal{F}|_V$ is $\mathcal{O}_U$-flat in the sense that M is a flat A-module (with the A-module structure coming from the ring map $A \to B$).

Remark 922. The condition is stable under base change: for any morphism $g : S' \to S$, pulling back along the cartesian square

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow{g} & S \end{array}$$

yields a coherent $\mathcal{O}_{X'}$-module $\mathcal{F}' := (g')^*\mathcal{F}$ which is S'-flat whenever $\mathcal{F}$ is S-flat. Indeed, at each stalk the pullback is an extension of scalars of the original flat module, and extension of scalars preserves flatness. The converse fails in general.

19.3 Generic flatness

The construction of the flattening stratification proceeds by Noetherian induction on S. The induction step relies on the following classical *generic flatness* theorem of Grothendieck: over a noetherian integral base, a coherent sheaf is flat over a non-empty open subscheme. Algebraically, this is the assertion that a finitely generated module over a finite-type algebra over a noetherian domain is generically free after inverting a single non-zero element of the base ring.

19.3.1 The dévissage layer

The proof of the algebraic generic-freeness theorem has two layers: a *dévissage* reducing an arbitrary finite module over the (noetherian) finite-type algebra B to the case of a quotient domain $B/\mathfrak{p}$, and a *normalisation core* handling that case. We first develop the dévissage layer as standalone module theory.

Definition 923 (Generically free module)**.** Let A be a commutative ring and M an A-module. We say M is *generically free over* A if there exists $f \in A$, $f \neq 0$, such that the localisation M_f is a free A_f-module.

Lemma 924 (Transport of generic freeness along further localisation)**.** *Let A be a commutative ring, M an A-module, and $f, g \in A$. If M_f is a free A_f-module, then M_{fg} is a free A_{fg}-module.*

Proof. Localisation is base change: $M_h \cong A_h \otimes_A M$ as A_h-modules for any $h \in A$. Since fg maps to a unit multiple of g in A_{fg}, the ring A_{fg} is canonically an A_f-algebra, and base-change associativity gives

$$M_{fg} \;\cong\; A_{fg} \otimes_A M \;\cong\; A_{fg} \otimes_{A_f} (A_f \otimes_A M) \;\cong\; A_{fg} \otimes_{A_f} M_f.$$

The base change of a free module along any ring map is free (a basis of M_f over A_f tensors to a basis over A_{fg}), so M_{fg} is free over A_{fg}. ☐

Lemma 925 (Generic freeness is invariant under isomorphism)**.** *Let A be a commutative ring and $M \cong N$ isomorphic A-modules. If M is generically free over A, so is N.*

Proof. An isomorphism $e : M \to N$ localises to an isomorphism $M_f \to N_f$ of A_f-modules for the witness f, and freeness transports along it. ☐

Lemma 926 (Splicing: generic freeness is stable under extensions)**.** *Let A be a domain and $0 \to M_1 \to M_2 \to M_3 \to 0$ a short exact sequence of A-modules. If M_1 and M_3 are generically free over A, then so is M_2.*

Proof. Let $f_1, f_3 \neq 0$ be witnesses for M_1, M_3 and set $f := f_1 f_3 \neq 0$ (here A is a domain). Localisation at f is exact, so $0 \to (M_1)_f \to (M_2)_f \to (M_3)_f \to 0$ is a short exact sequence of A_f-modules. By 924, $(M_1)_f$ and $(M_3)_f$ are free A_f-modules. A short exact sequence with free (hence projective) quotient splits, so $(M_2)_f \cong (M_1)_f \oplus (M_3)_f$ is free. □

Lemma 927 (Generic freeness of the zero module)**.** *Let A be a non-trivial commutative ring. The zero A-module is generically free over A, with witness $f = 1$.*

Proof. The localisation of the zero module is zero, which is free (on the empty basis), and $1 \neq 0$ since A is non-trivial. □

Lemma 928 (Torsion base case)**.** *Let A be a domain and M a finite A-module every element of which is annihilated by some non-zero element of A. Then M is generically free over A: indeed some localisation M_f with $f \neq 0$ vanishes.*

Proof. Choose a finite generating set $m_1, \dots, m_r$ of M and non-zero annihilators $a_i m_i = 0$. The product $f := a_1 \cdots a_r$ is non-zero (A is a domain) and annihilates every generator (extract the factor a_i and commute), hence — the annihilator of f being a submodule containing the generators — annihilates all of M. Every element of M_f is then zero, and the zero module is free. □

Lemma 929 (Generic freeness: finite modules)**.** *Let A be a noetherian domain and M a finite A-module. Then M is generically free over A.*

Proof. A finite module over a noetherian ring is finitely presented, and $M \otimes_A \mathrm{Frac}(A)$ is a finite-dimensional $\mathrm{Frac}(A)$-vector space, hence free. Freeness at the generic point of a finitely presented module spreads out to a basic open: there is $f \neq 0$ in A with M_f free over A_f. □

Lemma 930 (Generic flatness: finite modules)**.** *Let A be a noetherian domain and M a finite A-module. Then there is $f \neq 0$ in A with M_f flat over A_f.*

Proof. Immediate from 929: free modules are flat. □

Lemma 931 (Generic freeness: the module-finite case)**.** *Let A be a noetherian domain, B a module-finite A-algebra, and M a finite B-module. Then M is generically free over A. (This is the case of 934 where the morphism $\operatorname{Spec} B \to \operatorname{Spec} A$ is finite.)*

Proof. A finite module over a module-finite A-algebra is a finite A-module, so 929 applies. □

Lemma 932 (Generic-rank comparison for a finite module over a domain)**.** *Let D be a domain and M a finite D-module. Then there exist $r \in \mathbb{N}$, an injective D-linear map $\Phi : D^{\oplus r} \to M$, and a non-zero $d \in D$ annihilating the cokernel $M/\Phi(D^{\oplus r})$; that is, M sits in an exact sequence $0 \to D^{\oplus r} \to M \to T \to 0$ with $dT = 0$.*

Proof. Localise at $S := D \setminus \{0\}$: the localisation $S^{-1}M$ is a finite-dimensional vector space over $\mathrm{Frac}(D) = S^{-1}D$; let r be its dimension. Every element of $S^{-1}M$ has the form m/s, so scaling each vector of a basis by the (unit) image of its denominator produces a basis $(\bar{c}_1, \dots, \bar{c}_r)$ whose members are images of elements $c_1, \dots, c_r \in M$; let $\Phi(e_i) := c_i$. A D-linear relation among the c_i localises to a $\mathrm{Frac}(D)$-linear relation among the basis vectors $\bar{c}_i$, whose coefficients — images of the original ones — must vanish, and $D \to \mathrm{Frac}(D)$ is injective, so Φ is injective. For the cokernel: given $y \in M$, expand its image in the basis with coefficients in $\mathrm{Frac}(D)$ and clear their denominators by a single $s \in S$; then $s \cdot y - \Phi(u)$ is in the kernel of $M \to S^{-1}M$ for some $u \in D^{\oplus r}$, hence is killed by some $t \in S$, so $ts \neq 0$ annihilates the class of y in the cokernel T. Finally T is a finite D-module, so the product of such annihilators over a finite generating set is a single non-zero uniform annihilator d. □

Lemma 933 (Normalisation core: generic freeness of quotient domains)**.** *Let A be a noetherian domain, B a finite-type A-algebra, and $\mathfrak{p} \subset B$ a prime ideal. Then the quotient domain $B/\mathfrak{p}$ is generically free as an A-module.*

Proof. Write $C := B/\mathfrak{p}$, a finite-type A-algebra which is a domain, and let $K := \mathrm{Frac}(A)$. If the kernel of $A \to C$ is non-zero, every element of C is annihilated by any non-zero $a \in \ker(A \to C)$ (as $a \cdot c = (a \cdot 1)\, c = 0$), and 928 concludes. So assume $A \subseteq C$. Then $C_K := K \otimes_A C$ is a non-zero finite-type K-algebra domain; let m be its Krull dimension. We induct on m, the statement for all smaller values (over all noetherian domains A) being granted.

By Noether normalisation there are $b_1, \dots, b_m \in C_K$, algebraically independent over K, with C_K module-finite over $K[b_1, \dots, b_m]$; scaling by non-zero elements of A (which does not affect either property) we may take $b_i \in C$. Fix a finite generating set of C as an A-algebra; each generator satisfies a monic equation over $K[b_1, \dots, b_m]$, and clearing the finitely many denominators of all coefficients produces a single $g \neq 0$ in A such that each generator of C_g is integral over $A_g[b_1, \dots, b_m]$ (multiply the monic equation for x through by $g^{\deg}$ and rewrite it as a monic equation for a unit multiple of x). A finite-type algebra which is integral is module-finite, so C_g is a finite module over the polynomial ring $D := A_g[b_1, \dots, b_m]$.

Let r be the dimension of $C_g \otimes_D \mathrm{Frac}(D)$ as a $\mathrm{Frac}(D)$-vector space and choose $c_1, \dots, c_r \in C_g$ mapping to a basis. The resulting map $D^{\oplus r} \to C_g$ is injective (a D-linear relation among the c_i becomes a $\mathrm{Frac}(D)$-linear relation), and its cokernel T is a finite D-module with $T \otimes_D \mathrm{Frac}(D) = 0$, i.e. torsion; this generic-rank packaging, including the uniform annihilator below, is 932. A non-zero $d \in D$ annihilating T exhibits $T_K := K \otimes_{A_g} T$ as a finite module over $K[b_1, \dots, b_m]/(d')$ for the non-zero image d' of d, a ring of Krull dimension $< m$; presenting that ring as a quotient of a polynomial ring and filtering as in 934's dévissage, the inductive hypothesis (over the noetherian domain A_g) together with 926 yields $h \neq 0$ in A_g, which we may take in A, with T_h free over $(A_g)_h$. Localising the exact sequence $0 \to D^{\oplus r} \to C_g \to T \to 0$ at h and splitting off the free (hence projective) quotient T_h, we get

$$C_{gh} \;\cong\; \left(A_{gh}[b_1, \dots, b_m]\right)^{\oplus r} \oplus T_h,$$

a direct sum of free A_{gh}-modules (a polynomial ring is free on its monomials), so $f := gh$ is a generic-freeness witness for C (924 normalising the witnesses). □

Theorem 934 (Generic flatness, algebraic form)**.** *Source: [Nitsure], §4, "Lemma on Generic Flatness". Let A be a noetherian domain, B a finite-type A-algebra, and M a finite B-module. Then M is generically free over A: there exists $f \in A$, $f \neq 0$, such that the localisation M_f is a free A_f-module.*

Proof. The ring B is noetherian (it is of finite type over the noetherian A), so the finite B-module M admits a finite prime filtration

$$0 = M_0 \subset M_1 \subset \cdots \subset M_r = M$$

with successive quotients $M_i/M_{i-1} \cong B/\mathfrak{p}_i$ for prime ideals $\mathfrak{p}_i \subset B$. By induction on the filtration length it suffices that generic freeness over A holds for the zero module (927), holds for each quotient domain $B/\mathfrak{p}$ (933, transported along the isomorphism $M_i/M_{i-1} \cong B/\mathfrak{p}_i$ by 925), and is stable under extensions (the splicing lemma, 926, applied to $0 \to M_{i-1} \to M_i \to M_i/M_{i-1} \to 0$). □

19.4 Geometric glue: from generic freeness to generic flatness

The algebraic statement concerns one module over one algebra; the geometric statement quantifies over all affine pairs below the witness open. The glue consists of a mixed-base stability lemma for flatness under module localization, two restriction-compatibility lemmas for the induced module structures, and a two-layer basic-open reduction.

Lemma 935 (Mixed-base stability of flatness under module localization)**.** *Let B be an algebra over a commutative ring R, let N be a B-module which is flat as an R-module, and let $g : N \to N'$ be a localization of B-modules at a submonoid $p \subseteq B$. Then N' is flat as an R-module.*

Proof. Let $f : Q \hookrightarrow Q'$ be an injective map of R-modules. The square

$$\begin{array}{ccc} N \otimes_R Q & \longrightarrow & N \otimes_R Q' \\ \downarrow & & \downarrow \\ N' \otimes_R Q & \longrightarrow & N' \otimes_R Q' \end{array}$$

identifies the bottom map $\mathrm{id}_{N'} \otimes f$ with the localization at p of the top map $\mathrm{id}_N \otimes f$: indeed $N' \otimes_R Q$ is the localization of the B-module $N \otimes_R Q$ at p (localization commutes with $\otimes_R Q$, because $S^{-1}(N \otimes_R Q) = S^{-1}B \otimes_B (N \otimes_R Q) = (S^{-1}B \otimes_B N) \otimes_R Q$), and localizing the top map yields the bottom one. The top map is injective by flatness of N over R, and localization of modules preserves injectivity; hence $\mathrm{id}_{N'} \otimes f$ is injective. ☐

Lemma 936 (Fibre-side restriction compatibility of appLE)**.** *Let $p : X \to S$ be a morphism of schemes, $U \subseteq S$ open, $W' \subseteq W \subseteq p^{-1}(U)$ opens of X, and $u \in \Gamma(S, U)$. Then restricting $p^\sharp_{U,W}(u) \in \Gamma(X, W)$ to W' yields $p^\sharp_{U,W'}(u)$.*

Proof. Both sides are the composite of $p^\sharp$ with restriction along $W' \subseteq W \subseteq p^{-1}(U)$; this is the naturality of $p^\sharp$ in the target open. ☐

Lemma 937 (Base-side restriction compatibility of appLE)**.** *Let $p : X \to S$ be a morphism of schemes, $U' \subseteq U \subseteq S$ opens, $W \subseteq p^{-1}(U')$ an open of X, and $u \in \Gamma(S, U)$. Then $p^\sharp_{U',W}(u|_{U'}) = p^\sharp_{U,W}(u)$.*

Proof. Naturality of $p^\sharp$ in the source open. ☐

Lemma 938 (Section flatness transports along equal opens)**.** *Let $p : X \to S$, $\mathcal{F}$ a module sheaf on X, $U_0 \subseteq S$ open, and $O_1 = O_2 \subseteq p^{-1}(U_0)$ equal opens of X. If $\Gamma(\mathcal{F}, O_1)$ is flat over $\Gamma(S, U_0)$ (via $p^\sharp$), then so is $\Gamma(\mathcal{F}, O_2)$.*

Proof. Immediate: the two data coincide. ☐

Lemma 939 (Chart-basic core of generic flatness)**.** *Let $p : X \to S$, let $\mathcal{F}$ be quasi-coherent on X, let $U_0 \subseteq S$ be open, and let $W_j \subseteq p^{-1}(U_0)$ be an affine chart. Let $g \in A := \Gamma(S, U_0)$ be such that the localized section module $\Gamma(\mathcal{F}, W_j)_g$ is free over A_g. Then for every $c \in \Gamma(X, W_j)$ with $D(c) \subseteq p^{-1}(D(g))$, the module $\Gamma(\mathcal{F}, D(c))$ is flat over A.*

Proof. Write $\beta := p^\sharp_{U_0,W_j}(g) \in B_j := \Gamma(X, W_j)$, so that $D(\beta) = W_j \cap p^{-1}(D(g))$. By the quasi-compact–quasi-separated section-localization theorem (1259, applied to the affine W_j), the restriction $\Gamma(\mathcal{F}, W_j) \to \Gamma(\mathcal{F}, D(\beta))$ is the localization of B_j-modules at the powers of β; since

β^k acts on these modules as g^k acts through $p^\sharp$, it is also the localization of A-modules at the powers of g. Hence $\Gamma(\mathcal{F}, D(\beta)) \cong \Gamma(\mathcal{F}, W_j)_g$ is free, in particular flat, over A_g, and therefore flat over A (flatness over A and over the localization A_g agree for A_g-modules).

Since $D(c) \subseteq D(\beta)$, the further restriction $\Gamma(\mathcal{F}, D(\beta)) \to \Gamma(\mathcal{F}, D(c))$ is again a section localization (1259 on the affine $D(\beta)$, at the restricted element $c|_{D(\beta)}$, whose basic open is $D(c)$) — a localization of $\Gamma(X, D(\beta))$-modules. Mixed-base stability (935, with base A and fibre algebra $\Gamma(X, D(\beta))$) shows $\Gamma(\mathcal{F}, D(c|_{D(\beta)}))$ is flat over A; the induced-structure compatibilities are 936, and 938 transports the conclusion along $D(c|_{D(\beta)}) = D(c)$. □

Lemma 940 (Two-layer basic-open reduction)**.** *Let $p : X \to S$, $\mathcal{F}$ quasi-coherent on X, $U_0 \subseteq S$ an affine open with $A := \Gamma(S, U_0)$, and let $\{W_c^{(j)}\}_j$ be affine charts contained in and covering $p^{-1}(U_0)$. Let $f \in A$ be such that each localized section module $\Gamma(\mathcal{F}, W_c^{(j)})_f$ is free over A_f. Then for every affine pair $U \subseteq D(f)$, $W \subseteq p^{-1}(U)$, the module $\Gamma(\mathcal{F}, W)$ is flat over $\Gamma(S, U)$ via $p^\sharp$.*

Proof. For each $x \in W$: choose (affine interchange on S, applied to the affines U and U_0 at the point $p(x) \in U \cap U_0$) sections $a \in \Gamma(S, U)$, $a' \in A$ with $D(a) = D(a') \ni p(x)$; choose a chart $W_c^{(j)} \ni x$; shrink to a basic open of W inside $W \cap p^{-1}(D(a)) \cap W_c^{(j)}$ containing x; and apply affine interchange on X to that basic open and the affine $W_c^{(j)}$ to obtain $b \in \Gamma(X, W)$, $c \in \Gamma(X, W_c^{(j)})$ with $x \in D(b) = D(c) \subseteq p^{-1}(D(a))$.

The basic opens $D(b)$ obtained this way cover W, so the chosen elements b generate the unit ideal of $\Gamma(X, W)$. Since each restriction $\Gamma(\mathcal{F}, W) \to \Gamma(\mathcal{F}, D(b))$ is a localization of $\Gamma(X, W)$-modules at the powers of b (1259), flatness of $\Gamma(\mathcal{F}, W)$ over $\Gamma(S, U)$ may be checked on the pieces $\Gamma(\mathcal{F}, D(b))$ (flatness over the base is local with respect to a spanning family of the fibre algebra).

Fix a piece, with its data a, a', j, c. Freeness of $\Gamma(\mathcal{F}, W_c^{(j)})_f$ localizes further to fa' (924), and $D(c) = D(b) \subseteq p^{-1}(D(a)) = p^{-1}(D(a')) \subseteq p^{-1}(D(fa'))$ since $D(a) \subseteq U \subseteq D(f)$. The chart-basic core (939, at the element fa' and the chart section c) gives flatness of $\Gamma(\mathcal{F}, D(b))$ over A. Finally exchange the base ring twice along the localizations $A \to \Gamma(S, D(a')) = \Gamma(S, D(a))$ and $\Gamma(S, U) \to \Gamma(S, D(a))$ (for a module over a localized ring, flatness over the ring and over its localization agree; the compatibility of the induced structures is 937), obtaining flatness of $\Gamma(\mathcal{F}, D(b))$ over $\Gamma(S, U)$. □

Theorem 941 (Generic flatness, geometric form)**.** *Source: [Nitsure], §4, Theorem "generic flatness". Let S be an integral locally noetherian scheme, $p : X \to S$ a finite-type morphism (that is, quasi-compact and locally of finite type — quasi-compactness is essential: it bounds the number of denominators to clear, and the statement fails for the locally-of-finite-type morphism $\coprod_q \operatorname{Spec} \mathbb{F}_q \to \operatorname{Spec} \mathbb{Z}$), and $\mathcal{F}$ a coherent $\mathcal{O}_X$-module. Then there exists a non-empty open subscheme $U \subseteq S$ such that the restriction of $\mathcal{F}$ to $X_U = p^{-1}(U)$ is $\mathcal{O}_U$-flat.*

Proof. Choose a non-empty affine open $U_0 \subseteq S$; its section ring $A := \Gamma(S, U_0)$ is a noetherian domain because S is integral and locally noetherian. The preimage $p^{-1}(U_0)$ is quasi-compact (p is quasi-compact and U_0 is), and every point of it has an affine open neighbourhood inside $p^{-1}(U_0)$ on which the sections of $\mathcal{F}$ are a finite module over the sections of $\mathcal{O}_X$ (1270); extract a finite subcover by affine charts $W^{(1)}, \dots, W^{(r)}$. Each $B_i := \Gamma(X, W^{(i)})$ is of finite type over A (as p is locally of finite type and both opens are affine) and each $M_i := \Gamma(\mathcal{F}, W^{(i)})$ is a finite B_i-module, so algebraic generic freeness (934) produces $f_i \in A \setminus \{0\}$ with $(M_i)_{f_i}$ free over A_{f_i}. Set $f := f_1 \cdots f_r \neq 0$; freeness localizes from each f_i to f, so every $(M_i)_f$ is free over A_f. The witness open is $V := D(f) \subseteq U_0$, non-empty because S is reduced and $f \neq 0$. The required flatness on every affine pair $U \subseteq V$, $W \subseteq p^{-1}(U)$ is exactly the two-layer basic-open reduction, 940. □

19.5 Existence of the flattening stratification

Theorem 942. *[Existence of the flattening stratification] Source: [Nitsure], §4, Theorem "flattening stratification"; cf. [Stacks Project], tag 052H. Let S be a noetherian scheme and $\mathcal{F}$ a coherent sheaf on the relative projective space $\mathbb{P}^n_S$ over S. Then:*

(i) The set I of Hilbert polynomials $P_s(m) = \chi(\mathbb{P}^n_s, \mathcal{F}_s(m))$ that arise as s ranges over S is finite.

(ii) For each $f \in I$ there exists a locally-closed subscheme $S_f \hookrightarrow S$ whose underlying set $|S_f|$ consists of exactly those $s \in S$ with $P_s = f$. The collection $\{|S_f|\}_{f\in I}$ is pairwise disjoint with set-theoretic union $|S|$. Forming the coproduct $S' := \coprod_{f\in I} S_f$ and the morphism $i : S' \to S$ assembled from the inclusions $S_f \hookrightarrow S$, the pullback $i^*\mathcal{F}$ on $\mathbb{P}^n_{S'}$ is flat over S', and $i : S' \to S$ is universal with this property: for any morphism $\varphi : T \to S$ such that $\varphi^*\mathcal{F}$ on $\mathbb{P}^n_T$ is flat over T, φ factors uniquely through i. Each S_f is uniquely determined by f.

(iii) Order I by $f < g \iff f(n) < g(n)$ for all $n \gg 0$. Then the closure of $|S_f|$ in $|S|$ is contained in $\bigcup_{g\geq f} |S_g|$.

Proof. Following [Nitsure], §4, reduce to the special case $n = 0$ and assemble the resulting strata over the relative twists $\pi_*\mathcal{F}(N+i)$. The construction is local on S, since the strata glue by their universal property. We establish the three conclusions in order.

Part (i): finiteness of the Hilbert-polynomial set. By Lemma 944, Noetherian induction on S (whose inductive step invokes generic flatness, Theorem 941, on the integral locally-closed strata cut out by the irreducible-component decomposition of S) produces finitely many reduced, pairwise disjoint, locally-closed subschemes $V_i \hookrightarrow S$ set-theoretically covering $|S|$ on each of which $\mathcal{F}|_{\mathbb{P}^n_{V_i}}$ is flat over $\mathcal{O}_{V_i}$ in the sense of Definition 921. On a flat family the Hilbert polynomial $P_s(m) = \chi(\mathbb{P}^n_s, \mathcal{F}_s(m))$ is locally constant, so it takes finitely many values on each quasi-compact V_i; the finite union over the finitely many V_i shows that the set I of Hilbert polynomials occurring on S is finite.

Part (ii): existence of the strata and their universal property. Working over the flat pieces V_i of part (i), semicontinuity (cohomology-and-base-change) yields a uniform integer N such that, for all $s \in S$ and $m \geq N$, the higher cohomologies $H^r(\mathbb{P}^n_s, \mathcal{F}_s(m))$ vanish for $r \geq 1$ and the base-change homomorphism $(\pi_*\mathcal{F}(m))|_s \to H^0(\mathbb{P}^n_s, \mathcal{F}_s(m))$ is an isomorphism. With this N fixed, Lemma 945 applies the $n = 0$ stratification mechanism (Lemma 943) successively to the coherent sheaves $E_i = \pi_*\mathcal{F}(N+i)$ on S for $i = 0, \dots, n$, refining stratum by stratum. After $n+1$ steps this produces a finite family of locally-closed subschemes $W_{e_0,\dots,e_n} \subseteq S$, universal for the property that each E_i pulls back to a locally-free sheaf of constant rank e_i; re-indexing the tuple $(e_0, \dots, e_n)$ by the unique numerical polynomial f of degree $\leq n$ with $f(N+i) = e_i$ gives the family $\{W_f\}_{f\in I}$. At any $s \in W_f$ the base-change isomorphism and the cohomology vanishing force $P_s = f$ (since P_s has degree $\leq n$ and is determined by its values at $N, \dots, N+n$), so $|W_f| = \{s : P_s = f\}$; these subsets are disjoint with union $|S|$. The required locally-closed subscheme $S_f \subseteq W_f$ is the closed subscheme cut out by the further local-freeness conditions "$\pi_*\mathcal{F}(N+i)|_{W_f}$ is locally free of rank $f(N+i)$ for all $i \geq 0$" (a chain of coherent ideal sheaves that stabilises by Noetherianity), and $\mathcal{F}|_{\mathbb{P}^n_{S_f}}$ is flat over S_f (Definition 921) because the graded module $\bigoplus_{i\gg 0} \pi_*\mathcal{F}(i)|_{S_f}$ is locally free. Forming $S' = \coprod_{f\in I} S_f$ and the assembled morphism $i : S' \to S$, the pullback $i^*\mathcal{F}$ is S'-flat; and any $\varphi : T \to S$ with $\varphi^*\mathcal{F}$ flat over T has locally constant Hilbert polynomial, hence decomposes $T = \coprod_f T_f$ into the open-closed loci where the polynomial is f, each factoring uniquely through the corresponding S_f by the local-freeness universal property of Lemma 943. This gives the unique factorisation through i, and shows each S_f is uniquely determined by f.

Part (iii): closure of strata. Order I by $f < g \iff f(n) < g(n)$ for all $n \gg 0$. Since each S_f is a stratum of the $n = 0$ flattening stratification applied to a single direct image $\pi_* \mathcal{F}(p)$ for $p \gg 0$ separating all polynomials in I, the closure-of-strata assertion reduces to the upper-semicontinuity statement already established in the $n = 0$ case (Lemma 943): the closure of $|S_f|$ in $|S|$ is contained in $\bigcup_{g \geq f} |S_g|$. ☐

The construction begins with the special case $n = 0$, where $\mathbb{P}^n_S = S$ and $\mathcal{F}$ is a coherent sheaf on S. Its remaining ingredients are the Noetherian-induction reduction to the flat case, uniform vanishing of the higher cohomology of $\mathcal{F}(m)$ for $m \gg 0$ on the flat pieces, and base-change isomorphisms for the direct images $\pi_* \mathcal{F}(m)$.

Lemma 943 (Flat locus on S for a coherent sheaf is a locally-closed stratification)**.** *Source: [Nitsure], §4 (special case $n = 0$ of the main theorem). Let S be a noetherian scheme and let $\mathcal{F}$ be a coherent $\mathcal{O}_S$-module. For each integer $e \geq 0$, there exists a locally-closed subscheme $S_e \hookrightarrow S$ whose underlying set is $\{s \in S : \dim_{\kappa(s)} \mathcal{F}|_s = e\}$. The family $\{S_e\}_e$ is a finite, mutually disjoint, set-theoretically covering stratification of S, and a morphism $f : T \to S$ pulls back $\mathcal{F}$ to a locally-free $\mathcal{O}_T$-module of rank e if and only if f factors through S_e. The function $s \mapsto \dim_{\kappa(s)} \mathcal{F}|_s$ is upper-semicontinuous on S.*

Proof. By [Nitsure], §4, for $s \in S$, Nakayama produces a neighbourhood V of s and a right-exact local presentation $\mathcal{O}_V^{\oplus m} \xrightarrow{\psi} \mathcal{O}_V^{\oplus e} \to \mathcal{F}|_V \to 0$ where $e = \dim_{\kappa(s)} \mathcal{F}|_s$. The closed subscheme $V_e \hookrightarrow V$ defined by the ideal sheaf generated by the matrix entries of ψ represents the locus where $\mathcal{F}$ becomes locally free of rank e: by right-exactness of tensor product, a base change $f : T \to V$ has $f^* \mathcal{F}$ locally free of rank e iff $f^* \psi = 0$, iff f factors through V_e. Glueing the V_e over the cover of S by such neighbourhoods (and intersecting with $V \setminus \bigcup_{e' > e} V_{e'}$ to get the strictly-equal-to-e stratum) produces the locally-closed S_e. Upper-semicontinuity of e follows because the rank of ψ is lower-semicontinuous. ☐

Lemma 944 (Noetherian induction reduction to the flat case)**.** *Source: [Nitsure], §4 (general case, reduction by Noetherian induction). Let S be a noetherian scheme, $\pi : \mathbb{P}^n_S \to S$ the relative projective space, and $\mathcal{F}$ a coherent sheaf on $\mathbb{P}^n_S$. Then there exist finitely many reduced, locally-closed, pairwise disjoint subschemes $V_i \hookrightarrow S$ whose set-theoretic union is $|S|$ and such that $\mathcal{F}|_{\mathbb{P}^n_{V_i}}$ is flat over $\mathcal{O}_{V_i}$ for each i. Consequently, only finitely many Hilbert polynomials $P_s(m) = \chi(\mathbb{P}^n_s, \mathcal{F}_s(m))$ occur as s varies over S.*

Proof. By [Nitsure], §4: S being noetherian, decompose $|S| = \bigcup Y_j$ into finitely many irreducible components, each closed. For an irreducible component Y, let $U \subseteq Y$ be the open subset not lying on any other component, given the reduced subscheme structure (an integral, locally-closed subscheme of S). By 941 applied to $\mathcal{F}|_{\mathbb{P}^n_U}$, there is a non-empty open $V \subseteq U$ over which $\mathcal{F}$ is flat. Replace S by the reduced closed complement $S \setminus V$ and repeat. Noetherianity of S terminates the induction in finitely many steps, producing the V_i. On each V_i the Hilbert polynomial is locally constant (flat families have locally constant Hilbert polynomial), hence takes finitely many values on the noetherian (hence quasi-compact) scheme V_i; the finite union of finite sets is finite. ☐

Lemma 945 (Assembly of the strata by Noetherian induction)**.** *Source: [Nitsure], §4 (assembly step combining the $n = 0$ special case applied to direct images $\pi_* \mathcal{F}(N + i)$). Let S, $\pi : \mathbb{P}^n_S \to S$, $\mathcal{F}$ be as in 942. There exists an integer $N \geq 0$ such that for every $s \in S$ the higher cohomologies $H^r(\mathbb{P}^n_s, \mathcal{F}_s(m))$ vanish for all $r \geq 1$, $m \geq N$, and the base-change homomorphism $(\pi_* \mathcal{F}(m))|_s \to H^0(\mathbb{P}^n_s, \mathcal{F}_s(m))$ is an isomorphism for $m \geq N$, $s \in S$. Iterating the $n = 0$ stratification (943) on the coherent sheaves $E_i := \pi_* \mathcal{F}(N + i)$ for $i = 0, \dots, n$ produces a finite collection of locally-closed* *subschemes $W_{e_0,\dots,e_n} \subseteq S$ indexed by tuples $(e_0,\dots,e_n) \in \mathbb{Z}^{n+1}$. Re-indexing each tuple by the unique numerical polynomial $f \in \mathbb{Q}[\lambda]$ of degree $\leq n$ with $f(N+i) = e_i$ gives a finite stratification $\{W_f\}$ of S, and the locally-closed subschemes $S_f \subseteq W_f$ of 942 are obtained by further intersecting with the closed-subscheme conditions that $\pi_*\mathcal{F}(N+i)|_{W_f}$ be locally free of rank $f(N+i)$ for all $i \geq 0$ (terminating by Noetherianity).*

Proof. By 944, the family of Hilbert polynomials $\{P_s\}$ on S is finite, and there is a finite stratification $\{V_i\}$ of S on each of which $\mathcal{F}$ is flat with locally-constant Hilbert polynomial. Apply semicontinuity (cohomology and base change, [Hartshorne], III.12 / [Stacks Project], tag 02KH for the H^0-version) on the noetherian V_i to obtain a uniform integer N_1 with $R^r\pi_*\mathcal{F}(m) = 0$ for $r \geq 1$, $m \geq N_1$, and $H^r(\mathbb{P}^n_s, \mathcal{F}_s(m)) = 0$ for all $s \in S$, $r \geq 1$, $m \geq N_1$ (statement (B) of Nitsure's proof). Choose $r_i \geq N_1$ on each V_i such that the base-change comparison $(\pi_*\mathcal{F}(m))|_{V_i} \to (\pi_i)_*\mathcal{F}_{V_i}(m)$ is an isomorphism for $m \geq r_i$; composing with the cohomology-and-base-change isomorphism $((\pi_i)_*\mathcal{F}_{V_i}(m))|_s \xrightarrow{\sim} H^0(\mathbb{P}^n_s, \mathcal{F}_s(m))$ for the flat family on V_i yields uniform $N = \max r_i$ such that $(\pi_*\mathcal{F}(m))|_s \to H^0(\mathbb{P}^n_s, \mathcal{F}_s(m))$ is an isomorphism for $m \geq N$ and all $s \in S$ (statement (C)).

Apply 943 sequentially to the coherent sheaves $E_i = \pi_*\mathcal{F}(N+i)$ on S for $i = 0,\dots,n$: E_0 produces a stratification $\{W_{e_0}\}$ of S; on each W_{e_0} apply the lemma to $E_1|_{W_{e_0}}$ to refine to $\{W_{e_0,e_1}\}$, and continue. After $n+1$ steps the $W_{e_0,\dots,e_n}$ form a finite locally-closed stratification of S universal for the property "each E_i pulls back to a locally-free $\mathcal{O}_T$-module of constant rank e_i". Re-index $(e_0,\dots,e_n)$ by the unique numerical polynomial f of degree $\leq n$ with $f(N+i) = e_i$ to get the family $\{W_f\}$ indexed by polynomials. At any $s \in W_f$, statement (C) and vanishing (B) force $P_s = f$ (since P_s has degree $\leq n$ and is determined by its values at $N, N+1, \dots, N+n$).

The Hilbert-polynomial-coincidence locus $|W_f|$ may have non-flat scheme structure; refine to $S_f \subseteq W_f$ by intersecting with the closed-subscheme conditions "$\pi_*\mathcal{F}(N+i)|_{W_f}$ is locally-free of rank $f(N+i)$" for all $i \geq 0$. These conditions are cut out by an increasing chain of coherent ideal sheaves $I_0 \subseteq I_0 + I_1 \subseteq \cdots$ on W_f; Noetherianity terminates the chain in finitely many steps, defining a coherent ideal sheaf I and a closed subscheme $S_f \subseteq W_f$ with $|S_f| = |W_f|$ on which $\pi_*\mathcal{F}(N+i)$ is locally-free of rank $f(N+i)$ for all $i \geq 0$. The base-change-without-flatness lemma (cf. [Nitsure], §3) yields some $N' \geq N$ such that for $i \geq N'$, $(\pi_*\mathcal{F}(i))|_{S_f} \to (\pi_{S_f})_*\mathcal{F}_{S_f}(i)$ is an isomorphism, hence the graded A-module $\bigoplus_{i \geq N'} (\pi_{S_f})_*\mathcal{F}_{S_f}(i)$ is locally-free on S_f, hence $\mathcal{F}_{S_f}$ is flat over S_f. The closure-of-strata property (iii) reduces to the corresponding property in the $n = 0$ case for $\pi_*\mathcal{F}(p)$ on S at any $p \geq N$ separating all Hilbert polynomials. $\square$

19.5.1 Consequences of the flattening construction

The flattening construction also yields a countably indexed flat stratification. Further transfer lemmas give a direct Noetherian-induction construction for proper morphisms and allow the resulting finite family to be indexed injectively by integer-valued functions.

Lemma 946 (Flat-locus stratification, special case $n = 0$)**.** *Let S be a noetherian scheme and $\mathcal{F}$ a coherent $\mathcal{O}_S$-module. Then there is a family of immersions $\iota_e : S_e \to S$ indexed by $e \in \mathbb{N}$ whose images are pairwise disjoint and set-theoretically cover $|S|$, and such that for each e the pullback of $\mathcal{F}$ along ι_e is flat over S_e in the sense of 921. The index e is merely an enumeration; unlike the canonical strata of 943, it is not required to equal $\dim_{\kappa(s)} \mathcal{F}|_s$.*

Proof. Apply the existence theorem 942 to the identity morphism $\pi = \mathrm{id}_S$ and $\mathcal{F}$: it yields a finite set I, immersions $\iota_f : V_f \to S$ with pairwise disjoint images covering $|S|$, and flatness of the pullback of $\mathcal{F}$ to $S \times_S V_f$ over V_f. The second projection $p_f : S \times_S V_f \to V_f$ of the fibre product along the identity is an isomorphism, and the first projection factors as $\mathrm{pr}_1 = p_f \circ \iota_f$; take $S \times_S V_f$ itself as the stratum with pr_1 as its immersion, so disjointness and covering of images transfer along the surjection p_f. Flatness of the pullback over the stratum itself follows because the predicate of 921 is insensitive to composing the base leg with an isomorphism: an affine open of the target corresponds under the isomorphism to an affine open of the source, the two section-ring actions on an unchanged section module differ by composition with a bijective ring map, and flatness passes along such a factorization since the target ring is a free module of rank one over the source. Finally choose a bijection of I with an initial segment of $\mathbb{N}$ and pad the family at the remaining indices by the empty scheme: its section rings are trivial, so every section module over them is flat, and the empty image preserves disjointness and covering. $\square$

19.5.2 Flatness transfer to strata

The Noetherian-induction reduction is assembled from four transfer lemmas: flatness under pushout base change, the affine-piece analysis of a fibre product, affine-locality of flatness over the base, and the existence of a flat stratum over one irreducible component.

Lemma 947 (Flatness under pushout base change)**.** *Let $R \to S$ and $R \to A$ be commutative ring maps with pushout $B = S \otimes_R A$, let M be an A-module which is flat over R, and let N be a B-module which is a base change of M along $A \to B$, i.e. $N \cong B \otimes_A M$ as B-modules. Then N is flat over S.*

Proof. It suffices to prove that $B \otimes_A M$ is flat over S, and for this we show that the canonical R-linear map $\eta : M \to B \otimes_A M$, $m \mapsto 1 \otimes m$, exhibits $B \otimes_A M$ as the base change $S \otimes_R M$; flatness over S then follows from flatness of M over R since base change along $R \to S$ preserves flatness. We verify the universal property: let Q be an S-module and $g : M \to Q$ an R-linear map. The family $\beta : A \to \mathrm{Hom}_R(M, Q)$, $\beta(a)(m) = g(am)$, is R-linear into an S-module, so by the universal property of the ring base change $B = S \otimes_R A$ it lifts to an S-linear map $\Lambda : B \to \mathrm{Hom}_R(M, Q)$ with $\Lambda(1 \otimes a) = \beta(a)$. The pairing $(b, m) \mapsto \Lambda(b)(m)$ is additive, S-linear in b, and A-balanced: on generators $b = s(1 \otimes a')$ of B one has $\Lambda((1 \otimes a)b)(m) = s\, g(aa'm) = \Lambda(b)(am)$, and the general case follows since such elements generate B additively. Hence the pairing descends to an S-linear map $h : B \otimes_A M \to Q$ with $h \circ \eta = g$. Uniqueness: any S-linear h' with $h' \circ \eta = g$ agrees with h on elements $(s(1 \otimes a)) \otimes m = s\,(1 \otimes am)$, hence on all of $B \otimes_A M$. $\square$

Lemma 948 (Flat sections on an affine piece of a fibre product)**.** *Let*

$$\begin{tikzcd} Y \arrow[r, "g"] \arrow[d, "i_Y"] & X \arrow[d, "i_X"] \\ T \arrow[r, "f"] & S \end{tikzcd}$$

be a cartesian square of schemes, and let $U_S \subseteq S$, $U_X \subseteq i_X^{-1}(U_S)$, $U_T \subseteq f^{-1}(U_S)$ be affine opens. Consider the piece *$U_Y := g^{-1}(U_X) \cap i_Y^{-1}(U_T) \subseteq Y$. Then U_Y is an affine open, the section-ring square $\Gamma(S, U_S) \to \Gamma(X, U_X)$, $\Gamma(S, U_S) \to \Gamma(T, U_T)$ with corner $\Gamma(Y, U_Y)$ is a pushout of commutative rings, and for every quasi-coherent $\mathcal{O}_X$-module $\mathcal{F}$ such that $\Gamma(\mathcal{F}, U_X)$ is flat over $\Gamma(S, U_S)$, the sections $\Gamma(g^*\mathcal{F}, U_Y)$ are flat over $\Gamma(T, U_T)$. If moreover $\Gamma(\mathcal{F}, U_X)$ is a finite $\Gamma(X, U_X)$-module, then $\Gamma(g^*\mathcal{F}, U_Y)$ is a finite $\Gamma(Y, U_Y)$-module.*

Proof. Restricting the cartesian square over the affine cospan $U_X \to U_S \leftarrow U_T$ exhibits U_Y as the fibre product $U_X \times_{U_S} U_T$ of affine schemes, which is affine. Applying global sections to this pullback and using the anti-equivalence between affine schemes and commutative rings yields the pushout of section rings. Since $U_Y \subseteq g^{-1}(U_X)$ with U_X, U_Y affine and $\mathcal{F}$ quasi-coherent, the base-change section formula (1255, Stacks 01HQ/01I8) identifies $\Gamma(g^*\mathcal{F}, U_Y) \cong$

$\Gamma(Y, U_Y) \otimes_{\Gamma(X,U_X)} \Gamma(\mathcal{F}, U_X)$ as $\Gamma(Y, U_Y)$-modules, i.e. exhibits $\Gamma(g^*\mathcal{F}, U_Y)$ as a base change of $\Gamma(\mathcal{F}, U_X)$ along $\Gamma(X, U_X) \to \Gamma(Y, U_Y)$. Flatness over $\Gamma(T, U_T)$ is now 947 applied to the section-ring pushout square; finiteness is preserved because a base change of a finite module is finite. $\square$

Lemma 949 (Flatness over the base is affine-local)**.** *Let $q : Y \to T$ be a morphism of schemes and $\mathcal{G}$ a quasi-coherent $\mathcal{O}_Y$-module. Suppose given affine opens $U_j \subseteq T$ and $W_j \subseteq q^{-1}(U_j)$ ($j \in J$) such that the W_j cover Y and each $\Gamma(\mathcal{G}, W_j)$ is flat over $\Gamma(T, U_j)$. Then for* every *pair of affine opens $U \subseteq T$, $W \subseteq q^{-1}(U)$, the sections $\Gamma(\mathcal{G}, W)$ are flat over $\Gamma(T, U)$.*

Proof. This is the two-layer basic-open reduction of 940, with the freeness input replaced by chart flatness. Fix an affine pair (U, W). For each $x \in W$ choose a chart $W_j \ni x$; since $q(x) \in U \cap U_j$ and both are affine, there is a basic open $D(a) = D(a')$ of T common to U and U_j containing $q(x)$, and then a basic open $D(b) = D(c) \ni x$ common to W and W_j with $D(b) \subseteq q^{-1}(D(a))$. The elements b generate the unit ideal of $\Gamma(Y, W)$, and each restriction $\Gamma(\mathcal{G}, W) \to \Gamma(\mathcal{G}, D(b))$ is a localization at the powers of b (1259), so flatness of $\Gamma(\mathcal{G}, W)$ over $\Gamma(T, U)$ may be checked on the pieces $\Gamma(\mathcal{G}, D(b))$. On a piece: flatness of $\Gamma(\mathcal{G}, W_j)$ over $\Gamma(T, U_j)$ passes to the fibre-side localization $\Gamma(\mathcal{G}, D(c))$ (mixed-base stability of flatness under localization of the module at a submonoid of the fibre algebra), and then the base ring is exchanged twice along $\Gamma(T, U_j) \to \Gamma(T, D(a')) = \Gamma(T, D(a))$ and $\Gamma(T, U) \to \Gamma(T, D(a))$ (for a module over a localized ring, flatness over the ring and over its localization agree; the compatibility of the induced module structures is 937). $\square$

Lemma 950 (Reduced subscheme of an irreducible closed subset is integral)**.** *Let S be a scheme and $Z \subseteq S$ a closed subset. The subscheme cut out by the vanishing ideal sheaf of Z (the reduced induced structure) is reduced; if Z is irreducible (in particular non-empty), it is integral.*

Proof. The vanishing ideal sheaf is radical: it equals the vanishing ideal of its own support (Galois connection between supports and vanishing ideals), which is the radical. The subscheme is covered by the affine charts $\mathrm{Spec}(\Gamma(S, U)/I(U))$ for U ranging over affine opens of S, and each $I(U)$ is a radical ideal, so each chart is the spectrum of a reduced ring; hence the subscheme is reduced. Its underlying space is the subspace Z, which is irreducible by hypothesis, and a reduced scheme with irreducible non-empty underlying space is integral. $\square$

Lemma 951 (Flat stratum over an irreducible component)**.** *Let S be a noetherian scheme, $\pi : X \to S$ proper, $\mathcal{F}$ a coherent $\mathcal{O}_X$-module, and $Z_c \subseteq S$ an irreducible closed subset with reduced subscheme $T \to S$. Then there is a non-empty open $\Omega \subseteq T$ such that for* every *open $St \subseteq \Omega$, regarded as a locally closed subscheme $St \to S$, the pullback of $\mathcal{F}$ to $X \times_S St$ is flat over St in the sense of 921.*

Proof. T is integral (950) and noetherian (a closed subscheme of a noetherian scheme), and the base change $p : X \times_S T \to T$ is proper, hence quasi-compact and locally of finite type. The pullback of $\mathcal{F}$ to $X \times_S T$ is quasi-coherent, and it admits affine charts with finite section modules: around any point, choose affine opens $U_S \subseteq S$, $U_T \subseteq T$ above it, and an affine chart $W_X \subseteq X$ with finite sections (Stacks 01PC); the corresponding piece is an affine chart of $X \times_S T$ whose sections are the base change of $\Gamma(\mathcal{F}, W_X)$, hence finite (948), and this finiteness localizes to a basic open inside any given neighbourhood. Generic flatness (941, in its quasi-coherent form with this chart supply) yields a non-empty open $\Omega \subseteq T$ with flat sections on all affine pairs below Ω.

Now let $St \subseteq \Omega$ be open. The stratum family $X \times_S St \to St$ maps to the component family by the comparison morphism $\kappa : X \times_S St \to X \times_S T$, and the square formed by κ and the two projections over $St \hookrightarrow T$ is cartesian (cancellation of pasted pullback squares). Cover $X \times_S St$ by pieces $g^{-1}(W_X) \cap q^{-1}(U')$ with $U' \subseteq St$ affine lying over an affine $U_1 \subseteq \Omega$; on each such piece, applying 948 to the cartesian κ-square with flatness input from the generic-flatness output on the corresponding piece of $X \times_S T$ (an affine open below Ω) yields flat sections over U'. By affine-locality (949) all affine pairs of the stratum family have flat sections; finally the pulled-back sheaf along κ composed with the projection is identified with the pullback of $\mathcal{F}$ along the projection of the stratum family (functoriality of pullback, $g \circ \kappa = g'$), so the flatness predicate transports along this isomorphism of module sheaves. $\square$

Lemma 952 (Noetherian-induction reduction to the flat case)**.** *Let S be a noetherian scheme, $\pi : X \to S$ a proper morphism, and $\mathcal{F}$ a coherent $\mathcal{O}_X$-module. Then there exist finitely many immersions $\iota_i : V_i \to S$ with pairwise disjoint images set-theoretically covering $|S|$, such that for each i the pullback of $\mathcal{F}$ to $X \times_S V_i$ is flat over V_i in the sense of 921. The strata are obtained by successively removing non-empty flat open patches supplied by generic flatness (941).*

Noetherianity of S (rather than mere local noetherianity) is necessary for the finiteness *of the family: on $S = \coprod_{n \geq 1} \mathbb{A}^n$, equip the n-th component with the coherent sheaf $\bigoplus_{k \leq n} (i_{V_k})_* \mathcal{O}_{V_k}$ for a strictly nested flag $V_1 \supset \cdots \supset V_n$ of integral closed subvarieties. A locally-closed stratum on which this sheaf is flat (hence locally free) meets the flag levels in relatively clopen subsets, so a stratum through the generic point of V_k cannot also contain the generic point of $V_{k'}$ for $k' \neq k$: each neighbourhood of the latter meets the dense lower level. The n-th component therefore requires at least $n + 1$ strata, and no finite family covers all components.*

Proof. By well-founded induction on the closed subsets of S (the noetherian hypothesis makes strict inclusion of closed subsets well-founded), we prove: for every closed $Z \subseteq S$ there is a finite family of immersions with pairwise disjoint images whose union is exactly Z and with flat pullback of $\mathcal{F}$ on each member. For $Z = \emptyset$ take the empty family.

For $Z \neq \emptyset$, let $Z_1, \ldots, Z_r \subseteq S$ be the images of the (finitely many, by noetherianity) irreducible components of the subspace Z; each Z_c is an irreducible closed subset of S contained in Z, and together they cover Z. 951 applied to the reduced subscheme $T_c \to S$ of Z_c yields a non-empty open $\Omega_c \subseteq T_c$ all of whose open subsets are flat strata. Let $E_c \subseteq S$ be the open complement of the union of the other components, and take as stratum the open $St_c := \Omega_c \cap (T_c \to S)^{-1}(E_c) \subseteq T_c$, a locally closed subscheme of S via $St_c \hookrightarrow T_c \to S$, with flat pullback by the choice of Ω_c.

The images of the St_c are pairwise disjoint: the image of St_c is contained in $Z_c \setminus \bigcup_{c' \neq c} Z_{c'}$. Their union V is relatively open in Z: realizing the open image of each St_c inside Z_c by an open $W_c \subseteq S$, one has $V = Z \cap \bigcup_c (W_c \cap E_c)$, because a point of $Z \cap W_c \cap E_c$ avoids all components other than Z_c, hence lies in Z_c, hence in the image of St_c. Moreover V is non-empty: for any component Z_c, irredundancy of the component decomposition shows $Z_c \nsubseteq \bigcup_{c' \neq c} Z_{c'}$, so the open $(T_c \to S)^{-1}(E_c)$ is non-empty, and two non-empty opens of the irreducible space T_c intersect, so $St_c \neq \emptyset$.

Therefore $Z' := Z \setminus V$ is a closed subset strictly smaller than Z; the induction hypothesis provides a finite disjoint flat family with union Z', and adjoining the strata St_c gives the required family for Z (disjointness across the two groups holds because the recursive family lies in Z', which is disjoint from V). Applying the statement to $Z = S$ proves the lemma. $\square$

Lemma 953 (Assembly with injectively labelled strata)**.** *Let S be a noetherian scheme, $\pi : X \to S$ a proper morphism, and $\mathcal{F}$ a coherent $\mathcal{O}_X$-module. Then there exist a finite index set I, immersions $\iota_f : S_f \to S$ ($f \in I$) with pairwise disjoint images set-theoretically covering $|S|$, and an injective label map $P : I \to (\mathbb{N} \to \mathbb{Z})$ (so $f = g \iff P_f = P_g$), such that for each f the pullback of $\mathcal{F}$ to $X \times_S S_f$ is flat over S_f in the sense of 921.*

Proof. Since the statement requires of P only injectivity, it follows from the reduction 952: take its stratification and label the finitely many strata by any injection $I \hookrightarrow \mathbb{N} \to \mathbb{Z}$ (for instance constant functions with pairwise distinct values). Requiring these labels to be the fibrewise Hilbert polynomials gives the refinement of 945. $\square$

19.6 The entry ideal of a matrix presentation

The rank strata of 943 carry a canonical — in general non-reduced — scheme structure, cut out locally by the ideal of entries of the relation matrix of a presentation. This section develops the commutative-algebra substrate: presentations of a module by e generators, the entry ideal, its base-change behaviour, Nitsure's local universal property, independence of the chosen presentation, and the Nakayama prolongation producing e-generator presentations near a point of fiber rank e. Throughout, R is a commutative ring and M an R-module.

Definition 954 (Matrix presentation)**.** A *matrix presentation* of M by e generators and m relations consists of a matrix $\psi \in \mathrm{Mat}_{e\times m}(R)$ and a surjection $\pi : R^{\oplus e} \to M$ making the sequence

$$R^{\oplus m} \xrightarrow{\psi} R^{\oplus e} \xrightarrow{\pi} M \to 0$$

exact.

Lemma 955 (Finite presentation)**.** *A module admitting a matrix presentation is finitely presented.*

Proof. π is a surjection from a finite free module whose kernel, the image of ψ, is generated by the m columns of ψ, hence finitely generated. $\square$

Definition 956 (Entry ideal)**.** The *entry ideal* $I(\psi) \subseteq R$ of a matrix presentation is the ideal generated by the $e \cdot m$ entries ψ_{ij} of the relation matrix. It is the ideal of 1×1 minors of ψ; for an e-generator presentation this is the Fitting ideal $\mathrm{Fitt}_{e-1}(M)$.

Lemma 957 (Vanishing criterion)**.** *For an R-algebra A, the entrywise image matrix $\psi \otimes A \in \mathrm{Mat}_{e\times m}(A)$ vanishes if and only if $I(\psi) \subseteq \ker(R \to A)$.*

Proof. Both sides assert that every entry of ψ maps to 0 in A. $\square$

Definition 958 (Base change of a presentation)**.** Let A be an R-algebra. The matrix $\psi \otimes A$ obtained by applying $R \to A$ entrywise, together with the tensored projection $A^{\oplus e} \cong A \otimes_R R^{\oplus e} \xrightarrow{1\otimes\pi} A \otimes_R M$, is a matrix presentation of $A \otimes_R M$ by e generators and m relations: the tensored sequence $A^{\oplus m} \to A^{\oplus e} \to A \otimes_R M \to 0$ remains exact by right-exactness of the tensor product.

Lemma 959 (Entry ideal under base change)**.** *The entry ideal of the base-changed presentation is the image ideal: $I(\psi \otimes A) = I(\psi) \cdot A$.*

Proof. The generators of $I(\psi \otimes A)$ are exactly the images of the generators of $I(\psi)$. $\square$

Lemma 960 (Vanishing relations give an isomorphism)**.** *If $\psi = 0$, then $\pi : R^{\oplus e} \to M$ is an isomorphism.*

Proof. $\ker \pi = \operatorname{im} \psi = 0$ by exactness, and π is surjective. $\square$

Lemma 961 (Vanishing relations give a free module)**.** *If $\psi = 0$, then M is free.*

Proof. Transport freeness along the isomorphism of 960. □

Lemma 962 (Vanishing relations give constant rank)**.** *If $\psi = 0$, then the localization $M_{\mathfrak{p}}$ is free of rank e over $R_{\mathfrak{p}}$ for every prime $\mathfrak{p} \subseteq R$.*

Proof. By 960, $M \cong R^{\oplus e}$; a ring with a prime ideal is nontrivial, so $R^{\oplus e}$ has constant rank e at every prime. □

Theorem 963 (Flat everywhere-rank-e modules have vanishing relation matrix)**.** *Let M carry a matrix presentation (ψ, π) by e generators. If M is flat and $M_{\mathfrak{p}}$ has rank e over $R_{\mathfrak{p}}$ for every prime $\mathfrak{p}$, then $\psi = 0$. Equivalently, a surjection $R^{\oplus e} \twoheadrightarrow M$ onto a locally free module of everywhere-rank e has zero kernel.*

Proof. By 955, M is finitely presented, hence — being flat — projective. The surjection π therefore splits: choose $s : M \to R^{\oplus e}$ with $\pi \circ s = \mathrm{id}$. The splitting yields an isomorphism $R^{\oplus e} \cong M \times \ker \pi$, $x \mapsto (\pi x, x - s(\pi x))$, so $K := \ker \pi$ is a direct summand of a finite free module, hence itself finite projective (in particular flat). Localizing at any prime $\mathfrak{p}$ and counting ranks, $e = \mathrm{rk}_{\mathfrak{p}} R^{\oplus e} = \mathrm{rk}_{\mathfrak{p}} M + \mathrm{rk}_{\mathfrak{p}} K = e + \mathrm{rk}_{\mathfrak{p}} K$, so K has rank 0 at every prime; a finite flat module of everywhere-rank 0 is zero. Hence $\operatorname{im} \psi = \ker \pi = 0$ by exactness, and every entry $\psi_{ij} = (\psi(\delta_j))_i$ vanishes. □

Lemma 964 (Comparison of entry ideals)**.** *If (ψ, π) and (ψ', π') are two matrix presentations of M by e generators (with possibly different numbers of relations), then $I(\psi') \subseteq I(\psi)$.*

Proof. Base change both presentations to $A := R/I(\psi)$. The entries of $\psi \otimes A$ vanish by construction (957), so by 961 and 962 the module $A \otimes_R M$ is free (hence flat) of rank e at every prime of A. Applying 963 to the base-changed presentation $(\psi' \otimes A, \pi' \otimes A)$ gives $\psi' \otimes A = 0$, i.e. every entry of ψ' lies in $\ker(R \to R/I(\psi)) = I(\psi)$. □

Theorem 965 (Presentation-independence of the entry ideal)**.** *Any two matrix presentations of M by e generators have the same entry ideal. (This is the presentation-independence of the Fitting ideal Fitt_{e-1}, and is what lets the local entry-ideal subschemes of 943 glue.)*

Proof. Apply 964 in both directions. □

Definition 966 (Fiber rank)**.** For a prime $\mathfrak{p} \subseteq R$ with residue field $\kappa(\mathfrak{p})$, the *fiber rank* of M at $\mathfrak{p}$ is $\dim_{\kappa(\mathfrak{p})}(\kappa(\mathfrak{p}) \otimes_R M)$.

Theorem 967 (Point-set of the entry-ideal locus)**.** *Let (ψ, π) be a matrix presentation of M by e generators and $\mathfrak{p} \subseteq R$ a prime. Then $I(\psi) \subseteq \mathfrak{p}$ if and only if the fiber rank of M at $\mathfrak{p}$ equals e.*

Proof. Base change the presentation to the residue field $\kappa(\mathfrak{p})$, whose structure map has kernel exactly $\mathfrak{p}$. If $I(\psi) \subseteq \mathfrak{p}$, the fiber presentation has vanishing relation matrix (957), so the fiber is isomorphic to $\kappa(\mathfrak{p})^{\oplus e}$ (960) and has dimension e. Conversely, if the fiber has dimension e, then — every module over a field being free, and free modules over a field having rank equal to their dimension at the unique relevant localization — the fiber presentation satisfies the hypotheses of 963, so its relation matrix vanishes and every entry of ψ lies in $\ker(R \to \kappa(\mathfrak{p})) = \mathfrak{p}$. □

Theorem 968 (Nakayama prolongation of a fiber basis)**.** *Let M be finitely presented over R, let $\mathfrak{p} \subseteq R$ be a prime, and let e be the fiber rank of M at $\mathfrak{p}$. Then there exists $g \notin \mathfrak{p}$ such that every localization of M away from g — that is, every module N over an R-algebra R' realizing the localizations of R and M at the powers of g — admits a matrix presentation by exactly e generators.*

Proof. Every element of the fiber $\kappa(\mathfrak{p})\otimes_R M$ becomes, after scaling by some $r \notin \mathfrak{p}$ (a unit of $\kappa(\mathfrak{p})$), a pure tensor $1 \otimes s$ with $s \in M$: this holds for pure tensors $c \otimes s$ because $\kappa(\mathfrak{p})$ is a localization of $R/\mathfrak{p}$-fractions (clear the denominator of c), and sums are handled by scaling both summands by the product of their two scalars. Scaling each vector of a basis of the fiber this way produces $m_1, \dots, m_e \in M$ whose images $1 \otimes m_i$ again form a basis of the fiber (they differ from the original basis by unit scalars).

Let $\alpha : R^{\oplus e} \to M$ send the standard basis to the m_i, and let $Q := M/\operatorname{im}\alpha$; it is finitely generated. Since the $1 \otimes m_i$ span the fiber and the tensor product is right exact, $\kappa(\mathfrak{p}) \otimes_R Q = 0$; as $\kappa(\mathfrak{p}) \otimes_R Q$ is the fiber of the finite $R_{\mathfrak{p}}$-module $Q_{\mathfrak{p}}$ at the maximal ideal, Nakayama gives $Q_{\mathfrak{p}} = 0$. Hence $\mathfrak{p}$ lies outside the support of the finite module Q, which is the zero locus of its annihilator, so there is $g \in \operatorname{Ann}(Q)$ with $g \notin \mathfrak{p}$; thus $g \cdot M \subseteq \operatorname{im}\alpha$.

Now let R' and N realize the localizations of R and M away from g, with structure map $l : M \to N$. The images $l(m_i)$ generate N over R': any $n \in N$ satisfies $g^k n = l(x)$ for some $x \in M$ and some k, the element gx lies in $\operatorname{im}\alpha$, and g and g^k are units in R'. This gives a surjection $\alpha' : R'^{\oplus e} \to N$. Localization is a base change, so N is finitely presented over R' (955 for M, transported along the base change); hence $\ker\alpha'$ is finitely generated, and any finite generating family of $\ker\alpha'$, arranged as the columns of a matrix, completes α' to a matrix presentation of N by e generators. ☐

19.7 Rank strata as closed subschemes

The entry-ideal commutative algebra of 19.6 is local to an affine chart. This section globalises it: for a quasi-coherent module $\mathcal{G}$ on a scheme X it assembles the local entry ideals into a quasi-coherent ideal sheaf whose closed subscheme — the *rank-e stratum* — has support exactly the locus where the fibre of $\mathcal{G}$ has dimension e. This is the geometric substrate of Nitsure's $n = 0$ construction (943). Throughout X is a scheme and $\mathcal{G}$ a quasi-coherent $\mathcal{O}_X$-module; $\Gamma(\mathcal{G}, U)$ denotes its sections over an open U, a module over $\Gamma(X, U) = \mathcal{O}_X(U)$.

19.7.1 Base change of the fibre rank

Theorem 969 (Base-change invariance of the fibre rank)**.** *Let A be an R-algebra, M an R-module, and $\mathfrak{q}$ a prime of A with contraction $\mathfrak{p} = \mathfrak{q} \cap R$. Then the fibre rank of $A \otimes_R M$ at $\mathfrak{q}$ equals the fibre rank of M at $\mathfrak{p}$. No flatness or finiteness hypotheses are needed.*

Proof. The residue-field map $R \to \kappa(\mathfrak{p}) \to \kappa(\mathfrak{q})$ makes $\kappa(\mathfrak{q})$ a $\kappa(\mathfrak{p})$-algebra. Cancelling base changes gives $\kappa(\mathfrak{p})$-linear, then $\kappa(\mathfrak{q})$-linear, isomorphisms

$$\kappa(\mathfrak{q}) \otimes_A (A \otimes_R M) \;\cong\; \kappa(\mathfrak{q}) \otimes_R M \;\cong\; \kappa(\mathfrak{q}) \otimes_{\kappa(\mathfrak{p})} (\kappa(\mathfrak{p}) \otimes_R M).$$

Extension of scalars along a field extension preserves dimension, so $\dim_{\kappa(\mathfrak{q})} \kappa(\mathfrak{q}) \otimes_{\kappa(\mathfrak{p})} (\kappa(\mathfrak{p}) \otimes_R M) = \dim_{\kappa(\mathfrak{p})}(\kappa(\mathfrak{p}) \otimes_R M)$, which is the claim. ☐

Lemma 970 (Fibre rank is a linear invariant)**.** *If $\sigma : M \cong N$ is an isomorphism of R-modules, then M and N have the same fibre rank at every prime $\mathfrak{p}$.*

Proof. Base change σ along $R \to \kappa(\mathfrak{p})$ to obtain a $\kappa(\mathfrak{p})$-linear isomorphism $\kappa(\mathfrak{p}) \otimes_R M \cong \kappa(\mathfrak{p}) \otimes_R N$; isomorphic vector spaces have equal dimension. ☐

Corollary 971 (Fibre rank of a localized module)**.** *Let $R' = S^{-1}R$ be a localization, $N = S^{-1}M$ the corresponding localized module, and $\mathfrak{q}$ a prime of R'. Then the fibre rank of N at $\mathfrak{q}$ equals the fibre rank of M at the contraction $\mathfrak{q} \cap R$.*

Proof. Localization is a base change: the canonical map $M \to N$ induces an R'-linear isomorphism $R' \otimes_R M \cong N$. By 970 the fibre rank of N at $\mathfrak{q}$ equals that of $R' \otimes_R M$, which by 969 equals the fibre rank of M at $\mathfrak{q} \cap R$. □

19.7.2 Presentation charts and the point rank

Definition 972 (Presentation chart)**.** An affine open $V \subseteq X$ is an *e-presentation chart* for $\mathcal{G}$ if the section module $\Gamma(\mathcal{G}, V)$ admits a matrix presentation by e generators over $\Gamma(X, V)$, that is, a right-exact sequence $\mathcal{O}_V^{\oplus m} \xrightarrow{\psi} \mathcal{O}_V^{\oplus e} \to \mathcal{G}|_V \to 0$.

Definition 973 (Restriction of a presentation to a basic open)**.** Let $P = (\psi, \pi)$ be a matrix presentation of $\Gamma(\mathcal{G}, V)$ by e generators over an affine open V, and $f \in \Gamma(X, V)$. Since $\mathcal{G}$ is quasi-coherent, restriction exhibits $\Gamma(\mathcal{G}, D(f))$ as the localization $\Gamma(\mathcal{G}, V) \otimes_{\Gamma(X,V)} \Gamma(X, D(f))$. Base change of P along $\Gamma(X, V) \to \Gamma(X, D(f))$ (958), transported along this localization isomorphism, is a matrix presentation of $\Gamma(\mathcal{G}, D(f))$ by e generators, the *restriction of P to $D(f)$*; its relation matrix is the entrywise restriction $\psi|_{D(f)}$.

Lemma 974 (Entry ideal of a restricted presentation)**.** *With the notation of 973, the entry ideal of the restricted presentation is the image of $I(\psi)$ under restriction: $I(\psi|_{D(f)}) = I(\psi) \cdot \Gamma(X, D(f))$.*

Proof. The restricted presentation is a base change of P, so its entry ideal is the image ideal by 959. □

Lemma 975 (Presentation charts pass to basic opens)**.** *If V is an e-presentation chart for $\mathcal{G}$ and $f \in \Gamma(X, V)$, then the basic open $D(f) \subseteq V$ is again an e-presentation chart.*

Proof. Restrict a presentation of $\Gamma(\mathcal{G}, V)$ to $D(f)$ by 973. □

Definition 976 (Chart fibre rank)**.** For an affine open chart V and a point $x \in V$, the *chart fibre rank* of $\mathcal{G}$ at x computed on V is the fibre rank of the section module $\Gamma(\mathcal{G}, V)$ at the prime $\mathfrak{p}_x \subseteq \Gamma(X, V)$ corresponding to x.

Lemma 977 (Chart fibre rank under localization)**.** *The chart fibre rank is unchanged by passing to a basic open of the chart: for $f \in \Gamma(X, V)$ and $x \in D(f)$, the chart fibre rank computed on V equals that computed on $D(f)$.*

Proof. $\Gamma(\mathcal{G}, D(f))$ is the localization of $\Gamma(\mathcal{G}, V)$ at f, and the prime of x in $\Gamma(X, D(f))$ contracts to the prime of x in $\Gamma(X, V)$. Apply 971. □

Theorem 978 (Chart-independence of the fibre rank)**.** *Any two affine charts V, W containing a point x give the same chart fibre rank of $\mathcal{G}$ at x.*

Proof. Choose a basic open of V and a basic open of W that coincide as an open $D \ni x$ contained in $V \cap W$. By 977 the chart fibre rank computed on V and the one computed on W both equal the chart fibre rank computed on D, hence agree. □

Definition 979 (Point rank)**.** The *point rank* of $\mathcal{G}$ at a point $x \in X$ is the dimension of the fibre of $\mathcal{G}$ at x, computed as the chart fibre rank on any affine chart through x; it is well defined by 978.

Lemma 980 (Point rank on a chart)**.** *For any affine chart $V \ni x$, the point rank of $\mathcal{G}$ at x equals the chart fibre rank computed on V.*

Proof. Both are chart fibre ranks at x; apply 978. □

19.7.3 The strata ideal

Lemma 981 (Locality of entry-ideal membership)**.** *Let $P = (\psi, \pi)$ be a matrix presentation of $\Gamma(\mathcal{G}, V)$ by e generators over an affine open V, and $r \in \Gamma(X, V)$. If every point of V has a basic open $D(g) \ni x$ on which the restriction of r lies in the entry ideal of the restricted presentation $\psi|_{D(g)}$, then $r \in I(\psi)$.*

Proof. Choose for each point such a g; the basic opens $D(g)$ cover V, so the corresponding g generate the unit ideal of $\Gamma(X, V)$. By 974, membership of $r|_{D(g)}$ in $I(\psi|_{D(g)}) = I(\psi) \cdot \Gamma(X, D(g))$ means a power $g^n r$ lies in $I(\psi)$. A section that, after multiplication by a power of each member of a family generating the unit ideal, lands in a submodule, already lies in that submodule; hence $r \in I(\psi)$. □

Lemma 982 (Chart-to-chart transfer of the entry-ideal condition)**.** *Let $r \in \Gamma(X, U)$, let $V, W \subseteq U$ be affine opens with e-generator presentations P_V, P_W, and suppose the restriction of r to W lies in $I(\psi_W)$. Then for every point $x \in V \cap W$ there is a basic open $D(g) \subseteq V$ with $x \in D(g)$ on which the restriction of r lies in $I(\psi_V|_{D(g)})$.*

Proof. Choose a common basic open $D \ni x$ contained in $V \cap W$, realised as a basic open $D(g)$ of V and a basic open $D(f)$ of W. On D both $\psi_V|_{D(g)}$ and $\psi_W|_{D(f)}$ present $\Gamma(\mathcal{G}, D)$ by e generators, so by 965 their entry ideals coincide. By 974 the restriction of $r|_W$ to D lies in $I(\psi_W|_{D(f)}) = I(\psi_V|_{D(g)})$. □

Definition 983 (Strata ideal)**.** The rank-e *strata ideal* of $\mathcal{G}$ on an affine open U is the ideal of $\Gamma(X, U)$

$$I_e(U) \;=\; \bigcap_{V \leq U} \bigcap_{P} (\text{restriction } \Gamma(X, U) \to \Gamma(X, V))^{-1}(I(\psi_P)),$$

the intersection over all affine opens $V \subseteq U$ and all matrix presentations $P = (\psi_P, \pi_P)$ of $\Gamma(\mathcal{G}, V)$ by e generators, of the preimages of the entry ideals under restriction.

Theorem 984 (Strata ideal on a chart is the entry ideal)**.** *If V is an e-presentation chart with presentation P, then the strata ideal on V is the entry ideal of P: $I_e(V) = I(\psi_P)$.*

Proof. For $\subseteq$: the pair (V, P) itself occurs in the defining intersection, and restriction along $V \leq V$ is the identity, so every element of $I_e(V)$ lies in $I(\psi_P)$. For $\supseteq$: let $r \in I(\psi_P)$ and let (W, Q) be any presentation chart $W \subseteq V$. By 981 it suffices to show that each point of W has a basic open on which $r|_W$ lands in the entry ideal of the restricted Q; this is 982 applied to the two charts W and V (using $r|_V = r \in I(\psi_P)$). Hence $r|_W \in I(\psi_Q)$ for every chart, i.e. $r \in I_e(V)$. □

Lemma 985 (Strata membership from a covering family of charts)**.** *Let $r \in \Gamma(X, U)$ and let (V_i) be a family of affine presentation charts $V_i \subseteq U$, with presentations P_i, whose underlying opens cover U. If the restriction of r to each V_i lies in $I(\psi_{P_i})$, then $r \in I_e(U)$.*

Proof. Let (W, Q) be any presentation chart $W \subseteq U$. By 981 it suffices to check local membership at each $x \in W$; choosing an index i with $x \in V_i$, 982 applied to W and V_i (using $r|_{V_i} \in I(\psi_{P_i})$) supplies a basic open of W on which r lands in the entry ideal of the restricted Q. Thus $r|_W \in I(\psi_Q)$ for every chart, i.e. $r \in I_e(U)$. □

Definition 986 (Charts-cover hypothesis)**.** The module $\mathcal{G}$ *admits e-presentation charts* if every point of X lies in an e-presentation chart. This is the standing hypothesis of the rank-e stratum construction; it holds on the union of all e-presentation charts.

Lemma 987 (Finite chart covers)**.** *If $\mathcal{G}$ admits e-presentation charts, then every affine open $U \subseteq X$ is covered by finitely many e-presentation charts contained in U.*

Proof. Each point of U lies in some e-presentation chart; refining to a basic open contained in U keeps it a presentation chart (975). These charts cover U, and U is quasi-compact, being affine, so finitely many of them suffice. □

Theorem 988 (The strata ideal is quasi-coherent)**.** *Suppose $\mathcal{G}$ admits e-presentation charts. For an affine open U and $f \in \Gamma(X, U)$, restriction along $\Gamma(X, U) \to \Gamma(X, D(f))$ carries the strata ideal to the strata ideal: the image of $I_e(U)$ generates $I_e(D(f))$. Consequently the assignment $U \mapsto I_e(U)$ is quasi-coherent.*

Proof. For $\subseteq$: any presentation chart $W \subseteq D(f)$ is also a chart in U, so a section of $I_e(U)$ restricts into $I(\psi_W)$; hence its restriction to $D(f)$ lies in $I_e(D(f))$.

For $\supseteq$: let $r' \in I_e(D(f))$ and write $r' = y/f^n$ with $y \in \Gamma(X, U)$. By 987 choose finitely many presentation charts $V_1, \ldots, V_N$ covering U, with presentations P_i. On each V_i, the basic open $D(f|_{V_i}) \subseteq D(f)$ is a chart, and the strata condition on r' there says y/f^n restricts into $I(\psi_{P_i}|_{D(f|_{V_i})})$, the image ideal (974); clearing the denominator yields an exponent m_i with $f^{m_i} y|_{V_i} \in I(\psi_{P_i})$. Setting $m = \max_i m_i$, multiplying up gives $f^m y|_{V_i} \in I(\psi_{P_i})$ for all i, so $f^m y \in I_e(U)$ by 985. Therefore $r' = (f^m y)/f^{m+n}$ lies in the image of $I_e(U)$ under restriction. (984 identifies the strata ideal with the entry ideal on each chart, licensing the denominator clearing.) □

19.7.4 The rank stratum

Definition 989 (Strata ideal-sheaf datum)**.** Suppose $\mathcal{G}$ admits e-presentation charts. The rank-e *strata ideal-sheaf datum* on X is the assignment $U \mapsto I_e(U)$ of 983, which is a quasi-coherent ideal sheaf by 988.

Definition 990 (Rank-e stratum)**.** The rank-e *stratum* of $\mathcal{G}$ is the closed subscheme of X cut out by the strata ideal-sheaf datum of 989.

Definition 991 (Closed immersion of the stratum)**.** The canonical morphism from the rank-e stratum to X is a closed immersion.

Theorem 992 (Support of the rank-e stratum)**.** *A point $x \in X$ lies in the image of the closed immersion of the rank-e stratum if and only if the point rank of $\mathcal{G}$ at x equals e; that is, the support of the stratum is exactly the locus where the fibre of $\mathcal{G}$ has dimension exactly e.*

Proof. Choose an e-presentation chart $V \ni x$ with presentation P. The image of the closed immersion is the support of the ideal sheaf, which meets V in the zero locus of $I_e(V)$; and $I_e(V) = I(\psi_P)$ by 984. A point of an affine open lies in the zero locus of an ideal of sections iff its prime contains that ideal, so x lies in the stratum iff $I(\psi_P) \subseteq \mathfrak{p}_x$. By 967 this holds iff the fibre rank of $\Gamma(\mathcal{G}, V)$ at $\mathfrak{p}_x$ — the chart fibre rank at x, equal to the point rank by 980 — equals e. □

19.8 Representability of the rank strata

The rank-e stratum of 19.7 is a closed subscheme with the correct support. This section establishes that it also has the correct functor of points — Nitsure's local universal property: the pullback of $\mathcal{G}$ along a morphism is locally free of rank e exactly when the pulled-back relation matrices vanish, exactly when the morphism factors through the stratum — and assembles the strata over the chart loci into the locally closed pieces of the flattening stratification of a finitely presented module on a locally noetherian scheme. Throughout, "flat over T" for an $\mathcal{O}_T$-module means flat over T via the identity morphism of T, in the affine-local sense of 921.

19.8.1 Pullback presentations and flatness of sections

Lemma 993 (Finitely presented modules admit matrix presentations)**.** *Every finitely presented module M over a commutative ring R admits a matrix presentation: there are natural numbers e, m and a matrix presentation of M by e generators and m relations.*

Proof. By finite presentation there is a surjection $\pi : R^{\oplus e} \to M$ whose kernel K is finitely generated; choose generators $t_1, \dots, t_m$ of $K \subseteq R^{\oplus e}$ and let ψ be the $e \times m$ matrix with columns t_j. The image of $\psi : R^{\oplus m} \to R^{\oplus e}$ is the submodule generated by the columns, namely $K = \ker \pi$, so $R^{\oplus m} \xrightarrow{\psi} R^{\oplus e} \xrightarrow{\pi} M \to 0$ is exact. □

Lemma 994 (Pullback of a presentation to section modules)**.** *Let $g : Y \to X$ be a morphism of schemes, $\mathcal{G}$ a quasi-coherent $\mathcal{O}_X$-module, $V \subseteq X$ an affine open, $W \subseteq g^{-1}(V)$ an affine open, and $P = (\psi, \pi)$ a matrix presentation of $\Gamma(\mathcal{G}, V)$ by e generators and m relations over $\Gamma(X, V)$. Then $\Gamma(g^*\mathcal{G}, W)$ admits a matrix presentation by e generators and m relations over $\Gamma(Y, W)$ whose relation matrix is the entrywise image $g^\sharp(\psi)$ of ψ under the induced ring map $g^\sharp : \Gamma(X, V) \to \Gamma(Y, W)$.*

Proof. For the affine pair (V, W) the sections of the pullback module satisfy the base-change formula $\Gamma(g^*\mathcal{G}, W) \cong \Gamma(Y, W) \otimes_{\Gamma(X,V)} \Gamma(\mathcal{G}, V)$ as $\Gamma(Y, W)$-modules (1255). Base change of P along $g^\sharp$ (958) is a matrix presentation of the right-hand side — the pulled-back sequence stays exact by right-exactness of the tensor product — with relation matrix $g^\sharp(\psi)$, and transporting it along the base-change isomorphism changes neither the generator count nor the relation matrix. □

Lemma 995 (Flat sections from flatness over the identity)**.** *Let $\mathcal{G}$ be an $\mathcal{O}_T$-module that is flat over T. Then for every affine open $W \subseteq T$ the section module $\Gamma(\mathcal{G}, W)$ is flat over $\Gamma(T, W)$.*

Proof. Specialize the affine-local form of flatness over the identity to the pair of affine opens $U := W$ and $W \subseteq \mathrm{id}^{-1}(U) = U$: the module $\Gamma(\mathcal{G}, W)$ is flat over $\Gamma(T, W)$ acting through the ring map on sections induced by the identity morphism, which is the identity of $\Gamma(T, W)$, so the action is the canonical one. □

Lemma 996 (Flatness over the identity from an affine cover)**.** *Let $\mathcal{G}$ be a quasi-coherent $\mathcal{O}_T$-module and $(W_j)_j$ an affine open cover of T such that each $\Gamma(\mathcal{G}, W_j)$ is flat over $\Gamma(T, W_j)$. Then $\mathcal{G}$ is flat over T.*

Proof. Apply the affine-locality of flatness over the base (949) to the identity morphism of T with the charts $U_j := W_j$, $W_j \subseteq \mathrm{id}^{-1}(U_j)$: flatness of the section modules on the covering charts gives flatness of $\Gamma(\mathcal{G}, W)$ over $\Gamma(T, U)$ for every pair of affine opens $W \subseteq U$, which is flatness over the identity. □

Lemma 997 (Flatness over the base is stable under pullback)**.** *Let $\mathcal{G}$ be a quasi-coherent $\mathcal{O}_T$-module flat over T, and $g : T' \to T$ a morphism of schemes. Then $g^*\mathcal{G}$ is flat over T'.*

Proof. Cover T' by affine opens W, each contained in the preimage of some affine open $V \subseteq T$. For such a pair the section module $\Gamma(g^*\mathcal{G}, W) \cong \Gamma(T', W) \otimes_{\Gamma(T,V)} \Gamma(\mathcal{G}, V)$ (1255) is a base change of the flat module $\Gamma(\mathcal{G}, V)$ (995), hence flat over $\Gamma(T', W)$. The pullback $g^*\mathcal{G}$ is quasi-coherent, so 996 upgrades the cover of flat section modules to flatness over T'. $\square$

19.8.2 The rank dictionary

Theorem 998 (Stalk rank of an affine section module)**.** *Let $\mathcal{G}$ be a quasi-coherent $\mathcal{O}_T$-module and $W \subseteq T$ an affine open such that $M := \Gamma(\mathcal{G}, W)$ is flat and finite over $R := \Gamma(T, W)$. Let $\mathfrak{p}$ be a prime of R and $x \in W$ the corresponding point under the canonical identification of W with the prime spectrum of R. Then the localization $M_\mathfrak{p}$ is a free $R_\mathfrak{p}$-module whose rank equals the point rank of $\mathcal{G}$ at x.*

Proof. $M_\mathfrak{p}$ is finite and flat over the local ring $R_\mathfrak{p}$, hence free of some rank r (a finitely generated flat module over a local ring is free). Reducing modulo the maximal ideal, $M \otimes_R \kappa(\mathfrak{p}) \cong M_\mathfrak{p} \otimes_{R_\mathfrak{p}} \kappa(\mathfrak{p})$ has dimension r over $\kappa(\mathfrak{p})$, so r is the fiber rank of M at $\mathfrak{p}$ (966). The prime of R attached to the point x is $\mathfrak{p}$ itself, so this fiber rank is the chart fibre rank of $\mathcal{G}$ at x computed on W (976), which equals the point rank by 980. $\square$

Theorem 999 (Point rank of a pullback)**.** *Let $g : Y \to X$ be a morphism of schemes and $\mathcal{G}$ a quasi-coherent $\mathcal{O}_X$-module. For every $y \in Y$, the point rank of $g^*\mathcal{G}$ at y equals the point rank of $\mathcal{G}$ at $g(y)$. No flatness or finiteness hypotheses are needed.*

Proof. Choose an affine open $V \ni g(y)$ and an affine open $W \ni y$ with $W \subseteq g^{-1}(V)$. By the base-change formula for sections (1255), $\Gamma(g^*\mathcal{G}, W) \cong \Gamma(Y, W) \otimes_{\Gamma(X,V)} \Gamma(\mathcal{G}, V)$. The prime $\mathfrak{q} \subseteq \Gamma(Y, W)$ of y contracts along the induced ring map to the prime $\mathfrak{p} \subseteq \Gamma(X, V)$ of $g(y)$. By 970 and base-change invariance of the fiber rank (969), the fiber rank of $\Gamma(g^*\mathcal{G}, W)$ at $\mathfrak{q}$ equals the fiber rank of $\Gamma(\mathcal{G}, V)$ at $\mathfrak{p}$. These are the chart fibre ranks of $g^*\mathcal{G}$ at y on W and of $\mathcal{G}$ at $g(y)$ on V, i.e. the two point ranks (980). $\square$

19.8.3 Factorization through the stratum

Theorem 1000 (The strata ideal dies under a flat constant-rank pullback)**.** *Let $\mathcal{G}$ be a quasi-coherent $\mathcal{O}_X$-module admitting e-presentation charts, and let $q : T \to X$ be a morphism such that $q^*\mathcal{G}$ is flat over T and has point rank e at every point of T. Then the strata ideal sheaf is contained in the kernel of q: for every affine open $U \subseteq X$, every $r \in I_e(U)$ pulls back to $0 \in \Gamma(T, q^{-1}(U))$.*

Proof. A section of a sheaf vanishes if it vanishes on a cover, so it suffices to find, around each point $t \in q^{-1}(U)$, an open on which $q^\sharp(r)$ restricts to 0. Choose finitely many e-presentation charts $V_1, \dots, V_N \subseteq U$ covering U, with presentations $P_i = (\psi_i, \pi_i)$ (987). Given t, pick i with $q(t) \in V_i$, and an affine open $W \ni t$ with $W \subseteq q^{-1}(V_i)$.

By 994 the module $\Gamma(q^*\mathcal{G}, W)$ has a matrix presentation by e generators with relation matrix $q^\sharp(\psi_i)$; in particular it is finite over $\Gamma(T, W)$, and it is flat by 995. At every prime of $\Gamma(T, W)$ its localization is free of rank equal to the point rank of $q^*\mathcal{G}$ at the corresponding point (998), which is e by hypothesis. By 963 the relation matrix vanishes: $q^\sharp(\psi_i) = 0$. Hence the entry ideal $I(\psi_i)$ is contained in the kernel of $q^\sharp : \Gamma(X, V_i) \to \Gamma(T, W)$ (957). The strata-ideal condition (983) gives $r|_{V_i} \in I(\psi_i)$, and the maps induced by q on sections commute with restriction (937), so $q^\sharp(r)|_W = q^\sharp(r|_{V_i}) = 0$, as required. $\square$

Theorem 1001 (Unique factorization through the rank-e stratum)**.** *Let $\mathcal{G}$ be a quasi-coherent $\mathcal{O}_X$-module admitting e-presentation charts, and let $q : T \to X$ be a morphism such that $q^*\mathcal{G}$ is* *flat over T and has point rank e at every point of T. Then q factors uniquely through the closed immersion of the rank-e stratum $Z_e \hookrightarrow X$.*

Proof. The stratum Z_e is the closed subscheme cut out by the strata ideal-sheaf datum (989), and the kernel of its immersion is exactly that ideal sheaf. A morphism into X factors through the closed subscheme cut out by a quasi-coherent ideal sheaf if and only if that ideal sheaf is contained in the kernel of the morphism, and then the factorization is unique because a closed immersion is a monomorphism. By 1000 the strata ideal sheaf is contained in the kernel of q. $\square$

Theorem 1002 (The rank-e stratum flattens $\mathcal{G}$)**.** *Let $\mathcal{G}$ be a quasi-coherent $\mathcal{O}_X$-module admitting e-presentation charts, and $\iota : Z_e \to X$ the closed immersion of the rank-e stratum. Then $\iota^*\mathcal{G}$ is flat over Z_e.*

Proof. The preimages $\iota^{-1}(V)$ of the e-presentation charts V form an affine open cover of Z_e (a closed immersion is an affine morphism, and the charts cover X). Fix a chart V with presentation $P = (\psi, \pi)$. By 994 the section module $\Gamma(\iota^*\mathcal{G}, \iota^{-1}(V))$ has a matrix presentation with relation matrix $\iota^\sharp(\psi)$. Every entry of ψ lies in the entry ideal $I(\psi)$ (956), which equals the strata ideal $I_e(V)$ (984); and the kernel of $\iota^\sharp : \Gamma(X, V) \to \Gamma(Z_e, \iota^{-1}(V))$ is exactly $I_e(V)$, the ideal cutting out the subscheme. Hence $\iota^\sharp(\psi) = 0$, so the section module is free (961), in particular flat, over $\Gamma(Z_e, \iota^{-1}(V))$. Conclude by 996. $\square$

19.8.4 The chart locus

Theorem 1003 (Chart supply on a locally noetherian scheme)**.** *Let S be a locally noetherian scheme and $\mathcal{F}$ a finitely presented $\mathcal{O}_S$-module. Every point $s \in S$ lies in an e-presentation chart for $\mathcal{F}$ with e the point rank of $\mathcal{F}$ at s.*

Proof. Choose an affine open $V \ni s$ on which $M := \Gamma(\mathcal{F}, V)$ is a finite module over $R := \Gamma(S, V)$ (1270). Since S is locally noetherian, R is a noetherian ring, so the finite module M is finitely presented. Let $\mathfrak{p}$ be the prime of s in R and e the fiber rank of M at $\mathfrak{p}$; by 980, e is the point rank of $\mathcal{F}$ at s. The Nakayama prolongation (968) supplies $g \notin \mathfrak{p}$ such that every localization of M away from g admits a matrix presentation by e generators; by quasi-coherence $\Gamma(\mathcal{F}, D(g))$ is such a localization, so the basic open $D(g) \subseteq V$ is an e-presentation chart, and it contains s because $g \notin \mathfrak{p}$. $\square$

Definition 1004 (Chart locus)**.** The *e-chart locus* $W_e \subseteq S$ of an $\mathcal{O}_S$-module $\mathcal{F}$ is the union of all e-presentation charts for $\mathcal{F}$; it is an open subset of S.

Lemma 1005 (The chart locus contains the rank-e locus)**.** *Let S be locally noetherian and $\mathcal{F}$ finitely presented. Every point of point rank e lies in W_e.*

Proof. 1003 puts such a point inside an e-presentation chart. $\square$

Lemma 1006 (Rank bound on the chart locus)**.** *Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_S$-module. Every point of W_e has point rank at most e.*

Proof. A point $s \in W_e$ lies in some e-presentation chart V, so $\Gamma(\mathcal{F}, V)$ is generated by e elements. Its fiber $\kappa(\mathfrak{p}_s) \otimes \Gamma(\mathcal{F}, V)$ at the prime of s is then spanned by e vectors, so the fiber rank at $\mathfrak{p}_s$ (966) — the point rank of $\mathcal{F}$ at s, by 980 — is at most e. $\square$

Lemma 1007 (The restricted module is chart-covered)**.** *Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_S$-module. The restriction of $\mathcal{F}$ to the open subscheme W_e — its pullback along the open immersion $W_e \hookrightarrow S$ — admits e-presentation charts.*

Proof. Every point of W_e lies in some e-presentation chart $V \subseteq W_e$. The preimage of V in the open subscheme is an affine open mapped isomorphically onto V, and by 994 the section module of the restricted sheaf over it inherits a matrix presentation by e generators. □

19.8.5 The rank stratum over the chart locus

Definition 1008 (Rank-e stratum over the base)**.** The *rank-e stratum* S_e of a quasi-coherent $\mathcal{O}_S$-module $\mathcal{F}$ is the rank-e stratum (990) of the restriction of $\mathcal{F}$ to the e-chart locus W_e, which admits e-presentation charts by 1007: a closed subscheme of the open subscheme W_e, hence a locally closed subscheme of S.

Definition 1009 (Immersion of the rank-e stratum)**.** The canonical morphism $\iota_e : S_e \to S$ is the composite of the closed immersion $S_e \hookrightarrow W_e$ (991) with the open immersion $W_e \hookrightarrow S$; it is an immersion.

Theorem 1010 (Support of the rank-e stratum over the base)**.** *Let S be locally noetherian and $\mathcal{F}$ finitely presented. A point $s \in S$ lies in the image of $\iota_e : S_e \to S$ if and only if the point rank of $\mathcal{F}$ at s equals e.*

Proof. Write $\mathcal{F}' := \mathcal{F}|_{W_e}$ for the restriction to the chart locus; by 999 (applied to the open immersion) the point rank of $\mathcal{F}'$ at a point of W_e equals the point rank of $\mathcal{F}$ at its image in S.

If $s = \iota_e(z)$, then $s \in W_e$ and s, viewed in W_e, lies in the support of the stratum of $\mathcal{F}'$, so the point rank of $\mathcal{F}'$ there is e (992); hence the point rank of $\mathcal{F}$ at s is e. Conversely, if the point rank of $\mathcal{F}$ at s is e, then $s \in W_e$ (1005); the point rank of $\mathcal{F}'$ at s is e, so s lies in the support of the stratum (992), i.e. in the image of ι_e. □

Theorem 1011 (The rank-e stratum flattens $\mathcal{F}$)**.** *Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_S$-module. The pullback $\iota_e^*\mathcal{F}$ is flat over S_e.*

Proof. Since ι_e is the composite of the closed immersion $\iota : S_e \to W_e$ with the open immersion $W_e \to S$, pullback functoriality gives $\iota_e^*\mathcal{F} \cong \iota^*(\mathcal{F}|_{W_e})$. The restriction $\mathcal{F}|_{W_e}$ admits e-presentation charts (1007), so $\iota^*(\mathcal{F}|_{W_e})$ is flat over S_e by 1002; flatness is invariant under isomorphism of modules. □

19.8.6 Rank bound, open rank fibers, and the single-stratum factorization

Lemma 1012 (The point rank is bounded on a noetherian scheme)**.** *Let S be a noetherian scheme and $\mathcal{F}$ a finitely presented $\mathcal{O}_S$-module. There is an N such that the point rank of $\mathcal{F}$ is at most N at every point of S.*

Proof. For each $s \in S$ choose an $e(s)$-presentation chart $V_s \ni s$ with $e(s)$ the point rank at s (1003). The noetherian space S is quasi-compact, so finitely many charts $V_{s_1}, \dots, V_{s_k}$ cover S; set $N := \max_i e(s_i)$. Any $x \in S$ lies in some V_{s_i}, hence in the $e(s_i)$-chart locus, so its point rank is at most $e(s_i) \leq N$ (1006). □

Theorem 1013 (Openness of the rank fibers of a flat pullback)**.** *Let S be locally noetherian, $\mathcal{F}$ a finitely presented $\mathcal{O}_S$-module, and $\varphi : T \to S$ a morphism with $\varphi^*\mathcal{F}$ flat over T. For every e, the locus of points of T at which the point rank of $\varphi^*\mathcal{F}$ equals e is open in T.*

Proof. Let t_0 be a point of the locus. Choose an affine open $V \ni \varphi(t_0)$ on which $\Gamma(\mathcal{F}, V)$ is finite over the noetherian ring $\Gamma(S, V)$ (1270), hence finitely presented, and an affine open $W \ni t_0$ with $W \subseteq \varphi^{-1}(V)$. Set $M := \Gamma(\varphi^*\mathcal{F}, W)$ and $R := \Gamma(T, W)$. Then M is flat over R (995) and finitely presented: a matrix presentation of $\Gamma(\mathcal{F}, V)$ (993) pulls back to a matrix presentation of M (994), and a module with a matrix presentation is finitely presented (955).

A flat, finitely presented module is locally free, so the function $\mathfrak{p} \mapsto \operatorname{rk} M_{\mathfrak{p}}$ is locally constant on the prime spectrum of R. Under the canonical identification of W with that spectrum, this function is the point rank of $\varphi^*\mathcal{F}$ (998). Hence the set of points of W of point rank e is open in W and contains t_0; as W is open in T, the locus is a neighbourhood of each of its points. $\square$

Theorem 1014 (Unique factorization through the rank-e stratum of the base)**.** *Let S be locally noetherian, $\mathcal{F}$ a finitely presented $\mathcal{O}_S$-module, and $\varphi : T \to S$ a morphism such that $\varphi^*\mathcal{F}$ is flat over T and the point rank of $\mathcal{F}$ at $\varphi(t)$ equals e for every $t \in T$. Then φ factors uniquely through $\iota_e : S_e \to S$.*

Proof. Every $\varphi(t)$ has point rank e, hence lies in the chart locus W_e (1005), so φ factors through the open immersion $W_e \hookrightarrow S$ as $\varphi = (W_e \hookrightarrow S) \circ q$ with $q : T \to W_e$. Write $\mathcal{F}' := \mathcal{F}|_{W_e}$; it admits e-presentation charts (1007). By functoriality of pullback, $q^*\mathcal{F}' \cong \varphi^*\mathcal{F}$ is flat over T, and by 999 (applied twice, to q and to the open immersion) its point rank at t equals the point rank of $\mathcal{F}$ at $\varphi(t)$, namely e. Now 1001 gives a unique $l : T \to S_e$ with $\iota \circ l = q$ for the closed immersion $\iota : S_e \to W_e$; composing, $\iota_e \circ l = \varphi$.

For uniqueness, let $l' : T \to S_e$ satisfy $\iota_e \circ l' = \varphi$. Then $\iota \circ l'$ and q are two factorizations of φ through the open immersion $W_e \hookrightarrow S$, which is a monomorphism, so $\iota \circ l' = q$; by the uniqueness in 1001, $l' = l$. $\square$

19.9 Universal property of the strata

Theorem 1015 (Universal property of the flat-locus stratification (case $n = 0$))**.** *Let S be a noetherian scheme and $\mathcal{F}$ a coherent $\mathcal{O}_S$-module. Then there exist a finite index set I and locally-closed immersions $\iota_f : S_f \to S$ ($f \in I$) with pairwise disjoint images set-theoretically covering $|S|$, such that the pullback of $\mathcal{F}$ along ι_f is flat over S_f for every f (in the sense of 921), and such that, writing $S' := \coprod_{f \in I} S_f$ and $i : S' \to S$ for the morphism assembled from the ι_f, the pair (S', i) has the universal property: every morphism $\varphi : T \to S$ for which $\varphi^*\mathcal{F}$ is flat over T factors* uniquely *through i.*

For a finitely presented module, flatness is locally equivalent to local freeness, and the rank decomposes T into clopen pieces. Conversely, every morphism factoring through i has flat pullback by stability of flatness under base change (997). For a relative projective family, the analogous universal property additionally uses the cohomology-and-base-change layer of 945. Reduced strata alone do not give such a universal property for an arbitrary proper family: over $S = \operatorname{Spec} k[\varepsilon]$ with the flat module $\mathcal{F} = \mathcal{O}_S$, the identity must factor through the flattening stratum, which forces its canonical non-reduced scheme structure.

Proof. **The strata.** By 1012 there is an N bounding the point rank of $\mathcal{F}$ on S. Take $I := \{0, 1, \dots, N\}$ and, for $e \in I$, the rank-e stratum $\iota_e : S_e \to S$ (1008 and 1009); each ι_e is an immersion. By 1010 the image of ι_e is exactly the locus of point rank e; every point of S has a point rank, which is at most N, so the images cover $|S|$, and they are pairwise disjoint because the point rank is a single natural number. Flatness of $\iota_e^*\mathcal{F}$ over S_e is 1011.

Factorization. Let $\varphi : T \to S$ have $\varphi^*\mathcal{F}$ flat over T. For $e \in I$ let $T_e \subseteq T$ be the locus where the point rank of $\varphi^*\mathcal{F}$ equals e. By 999 the point rank of $\varphi^*\mathcal{F}$ at t is the point rank of $\mathcal{F}$ at

$\varphi(t)$, hence at most N; so the T_e cover T, they are pairwise disjoint, and each is open by 1013. Thus T is the disjoint union of the open subschemes T_e, and a morphism out of T is the same as a family of morphisms out of the T_e.

On a piece, the composite $T_e \hookrightarrow T \xrightarrow{\varphi} S$ has flat pullback of $\mathcal{F}$ — it is the restriction of $\varphi^*\mathcal{F}$ to the open subscheme T_e, flat by 997 — and, by 999 again, the point rank of $\mathcal{F}$ at the image of any $t \in T_e$ is e. By 1014 there is a unique $l_e : T_e \to S_e$ with $\iota_e \circ l_e = \varphi|_{T_e}$. Define $\psi : T \to \coprod_{f\in I} S_f$ on T_e as l_e followed by the coprojection of S_e; then $i \circ \psi = \varphi$, because this holds on each piece of the cover.

Uniqueness. Let $\psi' : T \to \coprod_f S_f$ also satisfy $i \circ \psi' = \varphi$. Fix e and $t \in T_e$, and let e' be the index of the summand containing $\psi'(t)$, say $\psi'(t) = z \in S_{e'}$. Then $\varphi(t) = i(\psi'(t)) = \iota_{e'}(z)$ lies in the image of $\iota_{e'}$, so the point rank of $\mathcal{F}$ at $\varphi(t)$ is e' (1010); it is also e, since $t \in T_e$ (999). Hence $e' = e$: the morphism ψ' maps T_e into the open summand S_e of the coproduct, so $\psi'|_{T_e}$ factors through the coprojection of S_e as some $l'_e : T_e \to S_e$, and $\iota_e \circ l'_e = i \circ \psi'|_{T_e} = \varphi|_{T_e}$. The uniqueness in 1014 forces $l'_e = l_e$; therefore ψ' and ψ agree on every piece T_e, and $\psi' = \psi$. ☐

19.10 Application: relative curves

Let C be a smooth proper curve over a field k and T a noetherian k-scheme. To apply 942 to $C \times_k T \to T$, it remains to pass from properness to projectivity.

Lemma 1016 (Smooth proper curve over k is projective)**.** *Source: classical; cf. [Hartshorne], II.6.7 & II.7.6 (curves). Let C be a smooth proper curve over a field k. Then $C \to \operatorname{Spec} k$ is projective. Consequently, for any noetherian k-scheme T, the relative curve $C \times_k T \to T$ is projective in the sense of Grothendieck (there exists a relatively very ample line bundle on $C \times_k T$ over T).*

Proof. Projectivity of a smooth proper curve over a field is classical ([Hartshorne], II.6.7 / II.7.6; alternatively: every divisor of large positive degree on a smooth proper curve C is very ample, providing the embedding $C \hookrightarrow \mathbb{P}^N_k$ for some N). Pulling the ample line bundle back along $C \times_k T \to C$ produces a relatively very ample line bundle on $C \times_k T$ over T. ☐

Corollary 1017. *[Flattening stratification for a coherent sheaf on a relative curve] Let k be a field, C a smooth proper curve over k, and T a noetherian k-scheme. Let $\mathcal{F}$ be a coherent sheaf on $C \times_k T$. Then the flattening-stratification conclusion of 942 applies: there is a finite set I of Hilbert polynomials and locally-closed subschemes $T_f \hookrightarrow T$ for $f \in I$, set-theoretically covering T disjointly, such that (a) $\mathcal{F}|_{C\times_k T_f}$ is flat over T_f for each f, and (b), after setting $T'' := \coprod_{f\in I} T_f$, any morphism $\varphi : T' \to T$ such that $\varphi^*\mathcal{F}$ on $C \times_k T'$ is T'-flat factors uniquely through the canonical morphism $T'' \to T$.*

Proof. By 1016, $C \times_k T \to T$ is projective, hence factors through a closed immersion into $\mathbb{P}^n_T$ for some $n \geq 0$. Push $\mathcal{F}$ forward along the closed immersion (extending by zero outside the image) to a coherent sheaf $\widetilde{\mathcal{F}}$ on $\mathbb{P}^n_T$; this operation preserves T-flatness in both directions because the closed immersion is a finite morphism. Apply 942 to $\widetilde{\mathcal{F}}$ on $\mathbb{P}^n_T$ to obtain $\{T_f\}$ with the stated universal property. ☐

19.11 Mathematical ingredients

The construction combines four kinds of input. First, module flatness is stable under localization and arbitrary base change, and flatness of a quasi-coherent module over a base may be checked on affine charts. Second, generic freeness over a noetherian domain supplies the dense flat open used in the Noetherian induction of 952. Third, entry ideals of finite presentations define the rank strata and their universal property, as developed in 19.6– 19.8. Finally, for a projective family, Serre vanishing and cohomology and base change compare the fibre Hilbert polynomial with the ranks of the finitely many direct images $\pi_*\mathcal{F}(N), \dots, \pi_*\mathcal{F}(N+n)$. These inputs produce the canonical Hilbert-polynomial indexing in 942.

19.12 Relation to adjacent constructions

Castelnuovo–Mumford regularity supplies the uniform twists used in the projective cohomology argument, while flattening stratification supplies the locally closed condition in the construction of Quot schemes: a family of quotients has prescribed Hilbert polynomial precisely after its classifying morphism factors through the corresponding stratum. For a smooth proper curve, 1017 gives this input after base change to an arbitrary noetherian scheme. Passing from Quot schemes to the representability of the relative Picard functor is the quotient route of 25.1, a separate argument resting on a descent step the formalisation instead avoids by the Milne–Kollár route; either way it does not alter the flattening construction proved here.

Chapter 20

The Quot scheme

20.1 Setup and motivation

Throughout this chapter, let S be a noetherian scheme, $\pi : X \to S$ a projective morphism, L a relatively very ample line bundle on X over S, and E a coherent sheaf on X. The Quot construction packages the T-flat coherent quotients $E_T \twoheadrightarrow \mathcal{F}$ of the pull-back E_T to $X_T = X \times_S T$, modulo the equivalence $\langle \mathcal{F}, q \rangle = \langle \mathcal{F}', q' \rangle$ defined by $\ker(q) = \ker(q')$. Fixing the Hilbert polynomial $\Phi(\lambda) = \chi(\mathcal{F}|_{X_t}(\lambda)) \in \mathbb{Q}[\lambda]$ (constant on connected T by flatness) stratifies the Quot functor into pieces $\mathrm{Quot}^{\Phi,L}_{E/X/S}$, each of which Grothendieck proved is representable by a projective S-scheme. This chapter develops the foundations of that picture: the Hilbert polynomial (1018), the Quot functor (1138), and the Grassmannian scheme (1176) with its representability (1183).

20.2 Hilbert polynomial of a coherent sheaf

Definition 1018 (Hilbert polynomial)**.** *Source: [Nitsure], §1 (FGA Explained Ch. 5); cf. [Hartshorne], III.5.2.* Let $X \to S$ be a finite-type morphism of noetherian schemes and let L be a line bundle on X. For a coherent sheaf $\mathcal{F}$ on X whose schematic support is proper over S and a point $s \in S$, write X_s for the fibre over s, $\mathcal{F}_s = \mathcal{F}|_{X_s}$ and $L_s = L|_{X_s}$. The *Hilbert polynomial of $\mathcal{F}$ at s relative to L* is the unique polynomial $\Phi_{\mathcal{F},s} \in \mathbb{Q}[\lambda]$ that agrees, for all $m \gg 0$, with the graded Hilbert function (1023)

$$m \longmapsto \dim_{\kappa(s)} \Gamma(X_s, \mathcal{F}_s \otimes L_s^{\otimes m})$$

of the section module $\bigoplus_{m\geq 0} \Gamma(X_s, \mathcal{F}_s \otimes L_s^{\otimes m})$ (1022), viewed as a finitely generated graded module over the homogeneous section ring $\bigoplus_{m\geq 0} \Gamma(X_s, L_s^{\otimes m})$ (1021). Uniqueness of $\Phi_{\mathcal{F},s}$ is unconditional (1038: two polynomials agreeing at all sufficiently large integers coincide); existence, for X_s proper over $\kappa(s)$, L_s ample and $\mathcal{F}_s$ coherent, is the content of 1043.

This is the classical Hilbert polynomial. When $\mathcal{F}_s$ is coherent and L_s is ample on the proper $\kappa(s)$-scheme X_s, Serre vanishing gives $H^i(X_s, \mathcal{F}_s \otimes L_s^{\otimes m}) = 0$ for all $i > 0$ and $m \gg 0$, so for $m \gg 0$

$$\dim_{\kappa(s)} \Gamma(X_s, \mathcal{F}_s \otimes L_s^{\otimes m}) \;=\; \chi(X_s, \mathcal{F}_s \otimes L_s^{\otimes m}) \;=\; \sum_{i\geq 0} (-1)^i \dim_{\kappa(s)} H^i(X_s, \mathcal{F}_s \otimes L_s^{\otimes m}),$$

and hence $\Phi_{\mathcal{F},s}$ coincides with the Euler-characteristic Hilbert polynomial of Snapper's Lemma – no numerical invariant is lost. The construction above, however, uses only the global sections

H^0: the higher cohomology enters solely through the classical statement that the two functions agree for $m \gg 0$, never through the definition of $\Phi_{\mathcal{F},s}$ itself. When $\mathcal{F}$ is S-flat the function $s \mapsto \Phi_{\mathcal{F},s}$ is locally constant on S, so Φ is well-defined on connected components.

20.3 Graded Hilbert polynomial

This section gives the graded-Hilbert-function construction that underlies 1018. It first records how a fibre X_s of a family $\pi : X \to S$ carries the data the construction consumes: the restrictions $\mathcal{F}_s$ and L_s (1019) and the $\kappa(s)$-vector-space structure on their sections (1020). Working over the fibre X_s with its ample line bundle L_s and coherent sheaf $\mathcal{F}_s$, it builds the section graded ring (1021) and module (1022) together with the graded Hilbert function (1023), records their finite generation (1024), and extracts the Hilbert polynomial (1043): the graded Hilbert–Serre rationality of the Hilbert series (1028) combines with Mathlib's polynomial-extraction lemma (1025) in 1039, and the extracted polynomial is unique by 1038. The route uses only H^0 (global sections), so it is independent of higher coherent cohomology.

Definition 1019 (Restriction of a sheaf of modules to a fibre)**.** *Source: [Nitsure], §1 (FGA Explained Ch. 5).* Let $\pi : X \to S$ be a morphism of schemes and $s \in S$ a point with residue field $\kappa(s)$. The *fibre* of π over s is the scheme

$$X_s \;=\; X \times_S \operatorname{Spec} \kappa(s),$$

the base change of π along the canonical morphism $\operatorname{Spec} \kappa(s) \to S$; write $\iota_s : X_s \to X$ for the projection to X (the *fibre embedding*). For a sheaf of $\mathcal{O}_X$-modules $\mathcal{F}$ on X, the *restriction of $\mathcal{F}$ to the fibre* is the module pullback

$$\mathcal{F}_s \;=\; \mathcal{F}|_{X_s} \;:=\; \iota_s^* \mathcal{F},$$

a sheaf of $\mathcal{O}_{X_s}$-modules. In particular a line bundle L on X restricts to a line bundle $L_s = L|_{X_s}$ on X_s.

Definition 1020 ($\kappa(s)$-structure on sections over a fibre)**.** *Source: [Nitsure], §1 (FGA Explained Ch. 5).* Let $\pi : X \to S$ be a morphism of schemes and $s \in S$. The fibre $X_s = X \times_S \operatorname{Spec} \kappa(s)$ (1019) carries a structural morphism $p_s : X_s \to \operatorname{Spec} \kappa(s)$, the second projection of the fibre product. Taking global sections of p_s and identifying $\Gamma(\operatorname{Spec} \kappa(s), \mathcal{O}) = \kappa(s)$ yields the *structural ring homomorphism of the fibre*

$$\kappa(s) \;\longrightarrow\; \Gamma(X_s,\, \mathcal{O}_{X_s}).$$

For any sheaf of $\mathcal{O}_{X_s}$-modules $\mathcal{G}$ on the fibre, the global sections $\Gamma(X_s, \mathcal{G})$ form a module over $\Gamma(X_s, \mathcal{O}_{X_s})$, hence, by restriction of scalars along the structural homomorphism, a $\kappa(s)$-vector space. All dimensions $\dim_{\kappa(s)}$ of section spaces over a fibre are taken with respect to this $\kappa(s)$-structure.

Definition 1021 (Section graded ring)**.** *Source: [Nitsure], §1 (FGA Explained Ch. 5); the section graded ring is the homogeneous coordinate ring of [Hartshorne], I.7.* Let $s \in S$ and let L_s be a line bundle on the fibre X_s, a scheme of finite type over the residue field $\kappa(s)$. The *section graded ring* of L_s is the graded commutative ring

$$R(X_s, L_s) \;=\; \bigoplus_{m \geq 0} \Gamma(X_s,\, L_s^{\otimes m}),$$

with degree-m component $\Gamma(X_s, L_s^{\otimes m})$ and multiplication the tensor product of sections $\Gamma(X_s, L_s^{\otimes m}) \otimes_{\kappa(s)} \Gamma(X_s, L_s^{\otimes m'}) \to \Gamma(X_s, L_s^{\otimes (m+m')})$, $\sigma \otimes \tau \mapsto \sigma \cdot \tau$. Its degree-zero component is $\Gamma(X_s, \mathcal{O}_{X_s})$, which contains $\kappa(s)$. When X_s is proper over $\kappa(s)$ and L_s is ample, $R(X_s, L_s)$ is a finitely generated – hence Noetherian – graded $\kappa(s)$-algebra (a suitable Veronese power of L_s is very ample and embeds X_s into a projective space, exhibiting $R(X_s, L_s)$ up to a finite extension as a quotient of a polynomial ring).

Definition 1022 (Section graded module)**.** *Source: [Nitsure], §1 (FGA Explained Ch. 5); cf. [Hartshorne], I.7.* Let $\mathcal{F}_s$ be a coherent sheaf on the fibre X_s and L_s a line bundle on X_s. The *section graded module* of $\mathcal{F}_s$ twisted by L_s is the graded $R(X_s, L_s)$-module (1021)

$$M(X_s, \mathcal{F}_s, L_s) \;=\; \bigoplus_{m\geq 0} \Gamma(X_s,\, \mathcal{F}_s \otimes L_s^{\otimes m}),$$

with degree-m component the $\kappa(s)$-vector space $\Gamma(X_s, \mathcal{F}_s \otimes L_s^{\otimes m})$ and scalar action given by the section tensor product $\Gamma(X_s, L_s^{\otimes m}) \otimes_{\kappa(s)} \Gamma(X_s, \mathcal{F}_s \otimes L_s^{\otimes m'}) \to \Gamma(X_s, \mathcal{F}_s \otimes L_s^{\otimes (m+m')})$. Its graded Hilbert function is $m \mapsto \dim_{\kappa(s)} \Gamma(X_s, \mathcal{F}_s \otimes L_s^{\otimes m})$.

Definition 1023 (Graded Hilbert function along a fibre)**.** *Source: [Nitsure], §1 (FGA Explained Ch. 5).* Let S be a locally noetherian scheme, $\pi : X \to S$ a morphism locally of finite type, L a line bundle and $\mathcal{F}$ a sheaf of $\mathcal{O}_X$-modules on X, and $s \in S$. The *graded Hilbert function of $\mathcal{F}$ at s relative to L* is the function

$$h_{\mathcal{F},s} : \mathbb{N} \longrightarrow \mathbb{N}, \qquad h_{\mathcal{F},s}(m) \;=\; \dim_{\kappa(s)} \Gamma(X_s,\, \mathcal{F}_s \otimes L_s^{\otimes m}),$$

where $\mathcal{F}_s$ and L_s are the restrictions to the fibre X_s (1019), the twist $\mathcal{F}_s \otimes L_s^{\otimes m}$ is the m-twist of $\mathcal{F}_s$ by L_s on X_s (1282), and the dimension is taken with respect to the $\kappa(s)$-vector-space structure on sections over the fibre (1020). This is exactly the graded Hilbert function of the section graded module $M(X_s, \mathcal{F}_s, L_s)$ (1022); it uses only global sections H^0.

Lemma 1024 (Finite generation of the section graded module)**.** *Source: [Nitsure], §1 (FGA Explained Ch. 5); Serre's theorem on ample line bundles, [Hartshorne], I.7 / II.5. Let X_s be projective over the field $\kappa(s)$ carrying the very ample line bundle L_s (1198: a closed immersion $X_s \hookrightarrow \mathbb{P}^N_{\kappa(s)}$ with $L_s \cong \mathcal{O}(1)|_{X_s}$), and let $\mathcal{F}_s$ be a coherent sheaf on X_s. Then the section graded ring $R(X_s, L_s)$ (1021) is a Noetherian graded ring, and the section graded module $M(X_s, \mathcal{F}_s, L_s)$ (1022) is a finitely generated graded $R(X_s, L_s)$-module. In particular each component $\Gamma(X_s, \mathcal{F}_s \otimes L_s^{\otimes m})$ is a finite-dimensional $\kappa(s)$-vector space. (For an ample L_s the same conclusions follow by passing to a very ample Veronese power; the very ample form stated here is the one consumed by the Quot-scheme construction.)*

Proof. The closed immersion $i : X_s \hookrightarrow \mathbb{P}^N_{\kappa(s)}$ with $L_s \cong i^*\mathcal{O}(1)$ reduces the claims to projective space: pushing forward along i identifies $\Gamma(X_s, \mathcal{F}_s \otimes L_s^{\otimes m})$ with $\Gamma(\mathbb{P}^N, (i_*\mathcal{F}_s)(m))$, and $i_*\mathcal{F}_s$ is coherent. By Serre's theorem the section graded ring $R(X_s, L_s)$ is a finitely generated graded $\kappa(s)$-algebra (a quotient of the polynomial ring in $N+1$ variables in large degrees), so it is Noetherian. For the coherent sheaf $\mathcal{F}_s$, Serre's theorem yields that $\bigoplus_{m\geq 0} \Gamma(X_s, \mathcal{F}_s \otimes L_s^{\otimes m})$ is a finitely generated graded module over $R(X_s, L_s)$: the twists $\mathcal{F}_s \otimes L_s^{\otimes m}$ are globally generated for $m \gg 0$, and the finitely many generating sections in low degrees together with the multiplication maps generate $M(X_s, \mathcal{F}_s, L_s)$ over $R(X_s, L_s)$. Finite dimensionality of each component over $\kappa(s)$ follows from coherence and properness (finiteness of cohomology of coherent sheaves on a proper scheme over a field). □

Lemma 1025 (Existence and uniqueness of the Hilbert polynomial of a rational graded series)**.** *Provided by Mathlib (`Mathlib.RingTheory.Polynomial.HilbertPoly`). Let F be a field of characteristic zero, let $p \in F[X]$, and let $d \in \mathbb{N}$. Then there is a unique polynomial $h \in F[X]$*

such that for some $N \in \mathbb{N}$ and all $n > N$ the coefficient of X^n in the power-series expansion of $p \cdot (1-X)^{-d} \in F[\![X]\!]$ equals $h(n)$:

$$\exists!\, h \in F[X],\ \exists N,\ \forall n > N:\ [X^n]\big(p \cdot (1-X)^{-d}\big) \;=\; h(n).$$

Equivalently, the eventual coefficient function of any rational power series with denominator a power of $1-X$ is, for $n \gg 0$, given by a unique polynomial h (Mathlib's $\mathrm{hilbertPoly}\, p\, d$*). The hypothesis* $[\mathrm{CharZero}\, F]$ *is satisfied by $F = \mathbb{Q}$, which is the instance used to obtain $\Phi_{\mathcal{F},s} \in \mathbb{Q}[\lambda]$.*

Lemma 1026 (Additivity of $\dim_\kappa$ on a short exact sequence (rank–nullity))**.** *Provided by Mathlib (`Mathlib.LinearAlgebra.Dimension.RankNullity`). Let κ be a field, V a finite-dimensional κ-vector space and $W \subseteq V$ a subspace. Then*

$$\dim_\kappa(V/W) + \dim_\kappa W \;=\; \dim_\kappa V.$$

Equivalently, $\dim_\kappa = \mathrm{finrank}_\kappa$ *is additive on a short exact sequence $0 \to W \to V \to V/W \to 0$ of finite-dimensional κ-vector spaces. This is the additivity that converts the degreewise short exact sequences of graded components into additive relations among the coefficients of the Hilbert series.*

Lemma 1027 (The rational power series $(1-X)^{-d}$)**.** *Provided by Mathlib (`Mathlib.RingTheory.PowerSeries.WellK` Let S be a commutative ring and $d \in \mathbb{N}$. Then $(1-X)^d$ is a unit in the power series ring $S[\![X]\!]$, and Mathlib names its inverse*

$$\mathrm{invOneSubPow}(S, d) \;=\; (1-X)^{-d} \;\in\; S[\![X]\!]^\times.$$

Together with the coercion $\mathrm{toPowerSeries} : F[X] \to F[\![X]\!]$ *of a polynomial to its power series, this is the data in which a rational Hilbert series $p \cdot (1-X)^{-d}$ is expressed; it is the same rational-series shape whose eventual coefficients are extracted by 1025.*

Lemma 1028 (Graded Hilbert–Serre: rationality of the Hilbert series)**.** *Source: [Stacks Project], "Commutative Algebra", Tag 00K1 (proposition on graded Hilbert polynomials); cf. [Nitsure], §1 (Snapper's Lemma), [Hartshorne], I.7. Let R be a Noetherian graded ring whose degree-zero part is a field κ and which is generated as a κ-algebra by finitely many elements, and let M be a finitely generated graded R-module with finite-dimensional graded components M_n. Then there exist $p \in \mathbb{Q}[X]$ and $d \in \mathbb{N}$ and an $N \in \mathbb{N}$ such that for all $n > N$*

$$\dim_\kappa M_n \;=\; [X^n]\big(p \cdot (1-X)^{-d}\big);$$

that is, the Hilbert series $\sum_{n \geq 0} (\dim_\kappa M_n)\, X^n$ is, up to a polynomial correction in low degrees, the rational function $p \cdot (1-X)^{-d}$, expressed in the power-series form $\mathrm{toPowerSeries}(p) \cdot \mathrm{invOneSubPow}(\mathbb{Q}, d)$ *(1027).*

Status. *This is the substantive half of the graded Hilbert–Serre theorem (rationality of the Hilbert series of a finitely generated graded module). It is not covered by Mathlib: 1025 supplies only the converse polynomial-extraction step from a rational series, not the production of the rational series from the module. The argument is realised by an* ambient subquotient induction *(1101): the whole induction ranges over pairs of homogeneous submodules $N' \leq N$ of the single fixed graded module M, and the Hilbert function is the ambient difference $n \mapsto \dim_\kappa(N \cap M_n) - \dim_\kappa(N' \cap M_n)$. The class of such pairs is closed under the kernel and cokernel of multiplication by a degree-one element (each presented again as an ambient subquotient pair), so the inductive step never forms a graded quotient ring $R/(x)$, a graded quotient module M/xM, or a regrading of a derived carrier. The genuine Mathlib gap is thereby* avoided *rather than filled; the residual obligation is the short ambient bookkeeping of the subquotient blocks (1058, 1077, 1079, 1092, 1101), each of which manipulates only ambient families $n \mapsto N_{\mathrm{aux}} \cap M_n$ of submodules of M.*

Proof. We argue with the abstract graded κ-algebra $R = \bigoplus_{n \geq 0} R_n$, whose degree-zero part $R_0 = \kappa$ is a field, and the finitely generated graded R-module $M = \bigoplus_{n \geq 0} M_n$ with finite-dimensional components M_n; there is no appeal to any geometric realisation of R or M.

Reduction to generation in degree 1. Since R is Noetherian and finitely generated as a κ-algebra, it is generated by finitely many homogeneous elements of positive degree. Replacing R by a Veronese subalgebra $R^{(e)} = \bigoplus_n R_{en}$ for a suitable e and re-grading (and M by the corresponding Veronese module), we may assume R is generated by its degree-one part, $\kappa[R_1] = R$. The degree-one part R_1 is then a finite-dimensional κ-vector space whose κ-basis $x_1, \dots, x_r$ generates R as a κ-algebra; $r = \dim_\kappa R_1$ is the number on which we induct.

Set-up of the ambient data. Regard M as the fixed ambient graded κ-module with internal decomposition $n \mapsto \mathcal{M}_n := M_n$ (1022); the components are finite-dimensional. For each basis vector $x_i \in R_1$ let $t_i : M \to M$ be multiplication by x_i, a κ-linear endomorphism that raises degree by one (it sends M_n into M_{n+1}, 1048). Since R is commutative the t_i pairwise commute, and they trivially preserve the pair of homogeneous submodules $(N, N') = (\top, \bot)$. Because R is generated by $x_1, \dots, x_r$ and M is finitely generated over R, M is finitely generated as a module over the polynomial ring $\kappa[x_1, \dots, x_r]$ acting through the t_i; this is the finiteness condition required below.

Invoking the ambient subquotient induction. Apply 1101 to the pair $(N, N') = (\top, \bot)$ with the r commuting degree-$(+1)$ endomorphisms $t_1, \dots, t_r$. It yields IsRatHilb of order r for the ambient subquotient Hilbert function of this pair (1058),

$$h(n) \;=\; \dim_\kappa(\top \cap \mathcal{M}_n) - \dim_\kappa(\bot \cap \mathcal{M}_n) \;=\; \dim_\kappa M_n - 0 \;=\; \dim_\kappa M_n.$$

Hence IsRatHilb($n \mapsto \dim_\kappa M_n$, r) (1031): there are $p \in \mathbb{Q}[X]$, $d := r \in \mathbb{N}$ and $N \in \mathbb{N}$ with

$$\dim_\kappa M_n \;=\; [X^n](p \cdot (1 - X)^{-d}) \qquad (n > N),$$

which is the claim, with pole order $d = r$. The internal mechanics of 1101 – the base case via 1032, and the inductive step assembling the kernel/cokernel closure (1077), the degreewise difference identity (1079, built on D_5, 1097) and the finiteness transfer (1092) into 1037 and 1033 – are the Stacks 00K1 induction realised entirely with ambient homogeneous submodules of M. □

20.3.1 The rationality toolkit IsRatHilb

The proof of 1028 runs the Stacks 00K1 induction on the number of degree-one generators. Each inductive step manipulates the *shape* of the Hilbert series – a numerator polynomial over a power of $(1 - X)$ – rather than the module itself. We isolate that bookkeeping in a single predicate IsRatHilb on functions $f : \mathbb{N} \to \mathbb{Q}$, together with the closure lemmas the induction consumes. These are project-internal helpers realising standard steps of the Stacks 00K1 argument at the level of power series; Mathlib supplies only the converse extraction (1025).

Lemma 1029 (Coefficients of $(1 - X)^{-1}$ times a series are partial sums)**.** *Let $F \in \mathbb{Q}[[X]]$ be a power series. Then for every $n \in \mathbb{N}$ the n-th coefficient of* invOneSubPow$(\mathbb{Q}, 1) \cdot F = (1 - X)^{-1} \cdot F$ *(1027) is the partial sum of the coefficients of F:*

$$[X^n]((1 - X)^{-1} \cdot F) \;=\; \sum_{k=0}^{n} [X^k] F.$$

Equivalently, multiplication by $(1 - X)^{-1} = \sum_{m \geq 0} X^m$ is the summation operator on coefficient sequences. Proved directly in Lean from the identity invOneSubPow$(\mathbb{Q}, 1) = \sum_{m \geq 0} X^m$ *and the Cauchy product.*

Lemma 1030 (Antidifference of a rational Hilbert series)**.** *Let $H, \delta : \mathbb{N} \to \mathbb{Q}$, let $q \in \mathbb{Q}[X]$ and $e, N \in \mathbb{N}$. Suppose that for all $n > N$ the sequence δ is the coefficient sequence of the rational series $q \cdot (1-X)^{-e}$,*

$$\delta(n) \;=\; [X^n]\big(q \cdot (1-X)^{-e}\big),$$

and that δ is the first difference of H beyond N, i.e. $H(n+1) - H(n) = \delta(n+1)$ for all $n > N$. Then there is a numerator polynomial $p \in \mathbb{Q}[X]$ such that for all $n > N$

$$H(n) \;=\; [X^n]\big(p \cdot (1-X)^{-(e+1)}\big).$$

That is, summing a coefficient sequence (anti-differencing) raises the pole order by one: the partial-sum series of $q \cdot (1-X)^{-e}$ equals $p \cdot (1-X)^{-(e+1)}$ up to a polynomial correction in low degrees. This is the power-series heart of the Stacks 00K1 inductive step.

Proof. Write $F = q \cdot (1-X)^{-e}$. Multiplying by $(1-X)^{-1}$ and using the factorisation $(1-X)^{-(e+1)} = (1-X)^{-1} \cdot (1-X)^{-e}$ (1027), 1029 gives $[X^m]\big(q \cdot (1-X)^{-(e+1)}\big) = \sum_{k=0}^{m} [X^k]F$, the partial sums of F. On the other hand, telescoping the first-difference hypothesis from $N+1$ expresses $H(N+1+j)$ as $H(N+1)$ plus a window sum of the coefficients of F. Splitting the partial sum $\sum_{k=0}^{n} [X^k]F$ at $N+2$ and absorbing the resulting constant discrepancy into a numerator $C\,(1-X)^e$ (a constant function is the order-$(e+1)$ coefficient sequence of $C\,(1-X)^e \cdot (1-X)^{-(e+1)} = C\,(1-X)^{-1}$), the explicit numerator $p = C\,(1-X)^e + q$ realises $H(n)$ as $[X^n]\big(p \cdot (1-X)^{-(e+1)}\big)$ for all $n > N$. ☐

Definition 1031 (Rational Hilbert function)**.** *Source: Stacks 00K1.* For a function $f : \mathbb{N} \to \mathbb{Q}$ and $d \in \mathbb{N}$, say that f is a *rational Hilbert function of order d*, written $\mathrm{IsRatHilb}(f, d)$, when there exist a numerator polynomial $p \in \mathbb{Q}[X]$ and an $N \in \mathbb{N}$ such that for all $n > N$

$$f(n) \;=\; [X^n]\big(p \cdot (1-X)^{-d}\big) \;=\; [X^n]\big(\mathrm{toPowerSeries}(p) \cdot \mathrm{invOneSubPow}(\mathbb{Q}, d)\big)$$

(1027). This is the "eventually equal to the coefficient sequence of a rational series $p \cdot (1-X)^{-d}$" predicate that the graded Hilbert–Serre induction tracks; its conclusion shape is exactly the one consumed by 1025.

Lemma 1032 (Base case: eventually-zero functions are rational of order 0)**.** *If $f : \mathbb{N} \to \mathbb{Q}$ is eventually zero – there is N with $f(n) = 0$ for all $n > N$ – then* $\mathrm{IsRatHilb}(f, 0)$ *(1031). Indeed f is then the coefficient sequence of the numerator polynomial* 0 *over* $(1-X)^0 = 1$*. This is the base case of the Stacks 00K1 induction (no degree-one generators). Proved directly in Lean.*

Lemma 1033 (Raising the pole order)**.** *If* $\mathrm{IsRatHilb}(f, d)$ *(1031) then* $\mathrm{IsRatHilb}(f, d+1)$*: multiplying the numerator by* $(1-X)$ *cancels one factor of* $(1-X)^{-1}$*, so the same coefficient sequence is realised at pole order $d+1$. This lets two rational Hilbert functions of different orders be brought to a common denominator. Proved directly in Lean using* $(1-X) \cdot (1-X)^{-(d+1)} = (1-X)^{-d}$ *(1027).*

Lemma 1034 (Closure under difference)**.** *If* $\mathrm{IsRatHilb}(f, d)$ *and* $\mathrm{IsRatHilb}(g, d)$ *(1031) at the same order d, then* $\mathrm{IsRatHilb}(n \mapsto f(n) - g(n),\, d)$*: the numerator of the difference is the difference of the numerators (over the common denominator* $(1-X)^{-d}$*). Proved directly in Lean.*

Lemma 1035 (Closure under right shift)**.** *If* $\mathrm{IsRatHilb}(f, d)$ *(1031) then* $\mathrm{IsRatHilb}(n \mapsto f(n-1),\, d)$*: the right shift on coefficient sequences is multiplication of the series by X, so the numerator p is replaced by $X \cdot p$ at the same pole order. Proved directly in Lean.*

Lemma 1036 (Antidifference step for the predicate)**.** *Let $H, g : \mathbb{N} \to \mathbb{Q}$ and $e, N \in \mathbb{N}$. If* $\mathrm{IsRatHilb}(g, e)$ *(1031) and g is the first difference of H beyond N ($H(n+1) - H(n) = g(n+1)$ for all $n > N$), then* $\mathrm{IsRatHilb}(H, e+1)$. *This is the predicate-level packaging of 1030: a finite difference raises the pole order by one, the inductive heart of graded Hilbert–Serre.*

Proof. Unfolding $\mathrm{IsRatHilb}(g, e)$ supplies a numerator q and a threshold N_g with $g(n) = [X^n](q \cdot (1-X)^{-e})$ for $n > N_g$. Applying 1030 with this q and threshold $\max(N, N_g)$ – whose two hypotheses are exactly the rational form of g and the first-difference identity for H – yields a numerator p with $H(n) = [X^n](p \cdot (1-X)^{-(e+1)})$ for $n > \max(N, N_g)$, i.e. $\mathrm{IsRatHilb}(H, e+1)$. □

Lemma 1037 (Inductive-step engine from a degreewise short exact sequence)**.** *Let $h_M, h_C, h_K : \mathbb{N} \to \mathbb{Q}$ and $d, N \in \mathbb{N}$. Suppose* $\mathrm{IsRatHilb}(h_C, d)$ *and* $\mathrm{IsRatHilb}(h_K, d)$ *(1031), and that for all $n > N$*

$$h_M(n+1) - h_M(n) \ = \ h_C(n+1) - h_K(n).$$

Then $\mathrm{IsRatHilb}(h_M, d+1)$. *This packages the entire power-series side of the Stacks 00K1 inductive step: h_M is the Hilbert function of M, and h_C, h_K are the Hilbert functions of the cokernel $C = M/xM$ and kernel $K = \ker(x : M \to M(1))$ of multiplication by a degree-one element x; the displayed identity is the alternating-sum consequence of the degreewise short exact sequence $0 \to K_n \to M_n \xrightarrow{x} M_{n+1} \to C_{n+1} \to 0$. Given the inductive hypothesis that h_C, h_K are rational of order d, the conclusion is rationality of h_M at order $d+1$.*

Proof. Set $g(n) = h_C(n) - h_K(n-1)$. By closure under right shift (1035) $n \mapsto h_K(n-1)$ is rational of order d, and by closure under difference (1034) so is g. The hypothesis rewrites as $h_M(n+1) - h_M(n) = g(n+1)$ for $n > N$ (since $g(n+1) = h_C(n+1) - h_K(n)$). Applying the antidifference step (1036) to $H = h_M$ and this g gives $\mathrm{IsRatHilb}(h_M, d+1)$. □

Lemma 1038 (Uniqueness of the Hilbert polynomial)**.** *Let $h_{\mathcal{F},s}$ be the graded Hilbert function of 1023. If $\Phi, \Psi \in \mathbb{Q}[\lambda]$ both agree with $h_{\mathcal{F},s}$ at all sufficiently large integers – there are $N_1, N_2 \in \mathbb{N}$ with $\Phi(m) = h_{\mathcal{F},s}(m)$ for all $m > N_1$ and $\Psi(m) = h_{\mathcal{F},s}(m)$ for all $m > N_2$ – then $\Phi = \Psi$. In particular a polynomial agreeing with $h_{\mathcal{F},s}$ for all $m \gg 0$ is unique when it exists.*

Proof. The difference $\Phi - \Psi \in \mathbb{Q}[\lambda]$ vanishes at every integer $m > \max(N_1, N_2)$, hence at infinitely many points of $\mathbb{Q}$. A nonzero polynomial over a field has at most $\deg(\Phi - \Psi)$ roots, so $\Phi - \Psi = 0$, that is $\Phi = \Psi$. □

Lemma 1039 (Hilbert polynomial from a rational Hilbert function)**.** *Let $h_{\mathcal{F},s}$ be the graded Hilbert function of 1023, and suppose $h_{\mathcal{F},s}$ is a rational Hilbert function of some order d (1031): there are $p \in \mathbb{Q}[X]$ and $N \in \mathbb{N}$ with $h_{\mathcal{F},s}(m) = [X^m](p \cdot (1-X)^{-d})$ for all $m > N$. Then there exists a unique polynomial $\Phi \in \mathbb{Q}[\lambda]$ such that $\Phi(m) = h_{\mathcal{F},s}(m)$ for all $m \gg 0$.*

Proof. Unfolding 1031 supplies $p \in \mathbb{Q}[X]$, $d \in \mathbb{N}$ and N with $h_{\mathcal{F},s}(m) = [X^m](p \cdot (1-X)^{-d})$ for all $m > N$. By 1025 over the characteristic-zero field $\mathbb{Q}$, applied to the pair (p, d), there are a polynomial $h \in \mathbb{Q}[X]$ and an N' with $h(n) = [X^n](p \cdot (1-X)^{-d})$ for all $n > N'$. Hence $\Phi := h$ satisfies $\Phi(m) = h_{\mathcal{F},s}(m)$ for all $m > \max(N, N')$, proving existence. Uniqueness is 1038: any two polynomials agreeing with $h_{\mathcal{F},s}$ at all sufficiently large integers are equal. □

Corollary 1040 (Defining property of the Hilbert polynomial)**.** *Let $h_{\mathcal{F},s}$ be the graded Hilbert function of 1023. If* some *polynomial in $\mathbb{Q}[\lambda]$ agrees with $h_{\mathcal{F},s}$ at all sufficiently large integers, then the Hilbert polynomial $\Phi_{\mathcal{F},s}$ of 1018 does: $\Phi_{\mathcal{F},s}(m) = h_{\mathcal{F},s}(m)$ for all $m \gg 0$.*

Proof. Let $\Psi \in \mathbb{Q}[\lambda]$ agree with $h_{\mathcal{F},s}$ at all $m > N$. By 1018 the Hilbert polynomial $\Phi_{\mathcal{F},s}$ is a polynomial agreeing with $h_{\mathcal{F},s}$ for all $m \gg 0$ whenever one exists, and by 1038 any such polynomial is unique, so $\Phi_{\mathcal{F},s} = \Psi$; hence $\Phi_{\mathcal{F},s}(m) = h_{\mathcal{F},s}(m)$ for all $m > N$. □

Corollary 1041 (The Hilbert polynomial under a rational Hilbert function)**.** *If the graded Hilbert function $h_{\mathcal{F},s}$ is a rational Hilbert function of some order d (1031), then the Hilbert polynomial $\Phi_{\mathcal{F},s}$ of 1018 satisfies $\Phi_{\mathcal{F},s}(m) = h_{\mathcal{F},s}(m)$ for all $m \gg 0$.*

Proof. By 1039 some polynomial agrees with $h_{\mathcal{F},s}$ at all sufficiently large integers; conclude by 1040. □

Corollary 1042 (The Hilbert polynomial computes as the extracted numerator polynomial)**.** *Suppose there are $p \in \mathbb{Q}[X]$ and $d, N \in \mathbb{N}$ with $h_{\mathcal{F},s}(m) = [X^m](p \cdot (1-X)^{-d})$ for all $m > N$ (1031). Then the Hilbert polynomial of 1018 is the polynomial extracted from the pair (p, d) by 1025: $\Phi_{\mathcal{F},s} = \mathrm{hilbertPoly}\, p\, d$.*

Proof. By 1025 there is an N' with $(\mathrm{hilbertPoly}\, p\, d)(n) = [X^n](p \cdot (1-X)^{-d})$ for all $n > N'$, so $\mathrm{hilbertPoly}\, p\, d$ agrees with $h_{\mathcal{F},s}$ at all $m > \max(N, N')$. Both $\Phi_{\mathcal{F},s}$ and $\mathrm{hilbertPoly}\, p\, d$ therefore agree with $h_{\mathcal{F},s}$ at all sufficiently large integers, and 1038 gives $\Phi_{\mathcal{F},s} = \mathrm{hilbertPoly}\, p\, d$. □

Theorem 1043 (Hilbert polynomial from the section module)**.** *Source: [Nitsure], §1 (FGA Explained Ch. 5). Let X_s be proper over the field $\kappa(s)$, let L_s be an ample line bundle on X_s, and let $\mathcal{F}_s$ be a coherent sheaf on X_s. Then there is a unique polynomial $\Phi_{\mathcal{F},s} \in \mathbb{Q}[\lambda]$ such that for all $m \gg 0$*

$$\Phi_{\mathcal{F},s}(m) \;=\; \dim_{\kappa(s)} \Gamma(X_s,\, \mathcal{F}_s \otimes L_s^{\otimes m}).$$

This $\Phi_{\mathcal{F},s}$ is the Hilbert polynomial of 1018.

Status. *The polynomial-extraction half of this theorem is formalized: uniqueness (1038) and the extraction of the unique polynomial from a rational Hilbert function (1039, `Scheme.existsUnique_hilbertPolyno`) are sorry-free. The remaining input is the geometric finite-generation theorem 1024 (Serre), which feeds the rationality hypothesis through 1028.*

Proof. By 1024 the section graded module $M(X_s, \mathcal{F}_s, L_s)$ is a finitely generated graded module over the Noetherian graded ring $R(X_s, L_s)$, with finite-dimensional components and degree-zero part the field $\kappa(s)$. Applying the abstract graded Hilbert–Serre rationality lemma (1028) to the graded $\kappa(s)$-algebra $R = R(X_s, L_s)$ and the finitely generated graded module $M = M(X_s, \mathcal{F}_s, L_s)$ exhibits the graded Hilbert function $m \mapsto \dim_{\kappa(s)} \Gamma(X_s, \mathcal{F}_s \otimes L_s^{\otimes m})$ (1023) as a rational Hilbert function of some order d (1031): there are $p \in \mathbb{Q}[X]$, $d \in \mathbb{N}$ and an N with $\dim_{\kappa(s)} \Gamma(X_s, \mathcal{F}_s \otimes L_s^{\otimes m}) = [X^m]\,(p \cdot (1-X)^{-d})$ for $m > N$. The extraction lemma 1039 then yields a unique polynomial $\Phi_{\mathcal{F},s} \in \mathbb{Q}[\lambda]$ with $\Phi_{\mathcal{F},s}(m) = \dim_{\kappa(s)} \Gamma(X_s, \mathcal{F}_s \otimes L_s^{\otimes m})$ for all $m \gg 0$, as claimed. □

20.3.2 The ambient subquotient induction for the Stacks 00K1 step

The power-series side of 1028 is fully isolated in the IsRatHilb toolkit (20.3.1). What remains is the *graded-module* side of the Stacks 00K1 induction. Rather than build the kernel and cokernel of multiplication by a degree-one element as graded modules over a smaller quotient ring $R/(x)$ – which would force a graded structure on a derived carrier – we range the induction over *pairs of homogeneous submodules of a single fixed ambient graded module M.* A pair (N, N') with $N' \leq N$ homogeneous submodules of M stands for the subquotient N/N'; its ambient Hilbert function is the difference $n \mapsto \dim_\kappa(N \cap M_n) - \dim_\kappa(N' \cap M_n)$. The class of such pairs, equipped with finitely many pairwise-commuting degree-one endomorphisms preserving N and N', is *closed* under the kernel and cokernel of one of those endomorphisms, both again presented as ambient subquotient pairs. Consequently no graded quotient ring, graded quotient module, or regrading of a subtype/quotient carrier is ever formed: every object that appears is an ambient family $n \mapsto N_{\mathrm{aux}} \cap M_n$ of submodules of the fixed M.

This subsection records that machinery as a thin layer over existing Mathlib scaffold: Mathlib already provides the notion of a homogeneous submodule (1044), the homogeneity of a span of homogeneous elements (1045), the degree-shift of multiplication by a homogeneous element (1048), the rank–nullity identity (1046), and the finite-generation transfer along a surjection (1047). What is supplied here is the internal direct-sum decomposition of a homogeneous submodule as an ambient family (the split G_1), the subquotient Hilbert data, the kernel/cokernel closure, the degreewise difference identity, the ambient finiteness transfer, and the induction itself.

Setting for this subsection. Let $R = \bigoplus_{n\geq 0} R_n$ be a graded ring whose degree-zero part $R_0 = \kappa$ is a field and which is generated as a κ-algebra in degree one, and let $M = \bigoplus_{n\geq 0} M_n$ be a graded κ-vector space (the fixed ambient module) with finite-dimensional components M_n. All submodules $N, N', \ldots$ below are homogeneous submodules of M, and the relevant endomorphisms are κ-linear maps $M \to M$ raising degree by one – typically multiplication by a homogeneous element $x \in R_1$.

Mathlib dependency anchors

Lemma 1044 (Homogeneous submodule)**.** *Provided by Mathlib (`Mathlib.RingTheory.GradedAlgebra.Homogeneou`* *Let $R = \bigoplus_n R_n$ be a graded ring and $M = \bigoplus_n M_n$ an internally graded R-module. A submodule $p \subseteq M$ is* homogeneous *when it is closed under taking homogeneous components: $m \in p$ implies each graded piece* $\mathrm{proj}_n(m) \in p$*. Equivalently $p = \bigoplus_n (p \cap M_n)$ as a set. Mathlib bundles this into a homogeneous submodule type providing order and extensionality structure; the ambient internal direct-sum decomposition $p = \bigoplus_n (M_n \cap p)$ is supplied by the split G_1 (1056, 1057).*

Lemma 1045 (A span of homogeneous elements is homogeneous)**.** *Provided by Mathlib (`Mathlib.RingTheory.Grade`* *Let $R = \bigoplus_n R_n$ be a graded ring and $s \subseteq R$ a set each of whose elements is homogeneous. Then the ideal* $\mathrm{span}\, s$ *is a homogeneous ideal. In particular, for $x \in R_1$ the principal ideal (x) is homogeneous, hence xM and the auxiliary submodule $N' + x \cdot N$ are homogeneous; this homogeneity feeds the kernel/cokernel closure (1077).*

Lemma 1046 (Rank–nullity)**.** *Provided by Mathlib (`Mathlib.LinearAlgebra.FiniteDimensional.Lemmas`). Let κ be a field, V a finite-dimensional κ-vector space and $\varphi : V \to W$ a κ-linear map. Then*

$$\dim_\kappa(\operatorname{range}\varphi) + \dim_\kappa(\ker\varphi) \;=\; \dim_\kappa V.$$

This is the rank–nullity identity underlying the degreewise difference identity D_5 (1097).

Lemma 1047 (Finite generation transfers along a surjection)**.** *Provided by Mathlib (`Mathlib.RingTheory.Finitenes`* *Let $\varphi : R \to A$ be a surjective ring homomorphism and M an A-module that is finitely generated when viewed as an R-module via φ. Then M is finitely generated as an A-module. Applied to the surjection $\kappa[t_0, \ldots, t_{r-1}] \twoheadrightarrow \kappa[t_0, \ldots, t_{r-2}]$ killing the endomorphism $x = t_{r-1}$ that annihilates the two subquotients, this transfers their finite generation down one generator (1092).*

Lemma 1048 (Multiplication by a homogeneous element shifts degree)**.** *Provided by Mathlib (`Mathlib.Algebra.GradedMulAction`). Let $\mathcal{A}$ be a graded family acting on a graded family $\mathcal{M}$ via a graded scalar multiplication. If $a \in R$ is homogeneous of degree i – in particular $x \in R_1$ – then for $m \in M_n$ one has $a \cdot m \in M_{i+n}$. This is the degree-shifting property that makes xM a* *homogeneous submodule (with degree-$(n+1)$ part $x \cdot M_n$) and that underlies the ambient identity $x \cdot N \cap M_{n+1} = x \cdot (N \cap M_n)$ used in the subquotient closure (1077) and difference identity (1079).*

Lemma 1049 (Commutative subalgebra generated by commuting elements)**.** *Provided by Mathlib (`Mathlib.Algebra.Algebra.Subalgebra.Lattice`). Let R be a commutative semiring and A an R-algebra, possibly noncommutative. If $s \subseteq A$ is a set of pairwise-commuting elements ($ab = ba$ for all $a, b \in s$), then the R-subalgebra* $\mathrm{adjoin}_R(s)$ *generated by s has commutative multiplication; the commutative-ring structure on it follows. Applied to the r pairwise-commuting degree-raising endomorphisms $x_0, \dots, x_{r-1} \in \mathrm{End}_\kappa(M)$, it yields a* commutative *κ-subalgebra $A = \mathrm{adjoin}_\kappa\{x_0, \dots, x_{r-1}\} \subseteq \mathrm{End}_\kappa(M)$; the polynomial action of $\kappa[t_0, \dots, t_{r-1}]$ on M factors through this commutative A (1092).*

Lemma 1050 (Finiteness transfers along a surjective linear map)**.** *Provided by Mathlib (`Mathlib.RingTheory.Finite` Let A be a ring, M, N two A-modules and $\varphi : M \to N$ a surjective A-linear map. If M is a finite A-module, then so is N. This is the finiteness-transfer step at the heart of the ambient finiteness transfer (1092) and of the numerator/denominator finiteness lemmas (1090, 1091).*

Lemma 1051 (Finiteness descends along an injective linear map into a Noetherian module)**.** *Provided by Mathlib (`Mathlib.RingTheory.Noetherian.Defs`). Let A be a ring, M, N two A-modules with N Noetherian (in particular, finite over a Noetherian ring A), and $\varphi : M \to N$ an injective A-linear map. Then M is a finite A-module. This is the submodule-finiteness step in 1090.*

Lemma 1052 (The lift of a linear map through a quotient)**.** *Provided by Mathlib (`Mathlib.LinearAlgebra.Quotient` Let A be a ring, M, N two A-modules, $P \subseteq M$ a submodule and $\varphi : M \to N$ an A-linear map with $P \subseteq \ker \varphi$. Then φ factors uniquely through a linear map $M/P \to N$ on the quotient. This realises the descent of a (σ-semilinear) map to the subquotient quotient in the ambient finiteness transfer (1092) and the denominator lemma (1091).*

Lemma 1053 (Only finitely many members of an independent family are nonzero)**.** *Provided by Mathlib (`Mathlib.LinearAlgebra.Dimension.Finite`). Let κ be a field (more generally a division ring) and V a finite-dimensional κ-vector space. If $(p_i)_{i \in \iota}$ is an* iSupIndep *family of subspaces of V, then the set $\{\, i : p_i \neq \bot \,\}$ is finite. This is the finiteness input that turns degreewise independence into eventual vanishing in the base case (1100).*

Lemma 1054 (The polynomial ring in finitely many variables is Noetherian)**.** *Provided by Mathlib (`Mathlib.RingTheory.Polynomial.Basic`). Let R be a Noetherian commutative ring and $r \in \mathbb{N}$. Then the multivariate polynomial ring* $\mathrm{MvPolynomial}(\{0, \dots, r-1\}, R)$ *is a Noetherian ring (Hilbert's basis theorem in finitely many variables). With $R = \kappa$ a field this makes $\kappa[t_0, \dots, t_{r-1}]$ Noetherian, the base ring over which the subquotient finiteness is measured (1090).*

Lemma 1055 (A finite module over a Noetherian ring is Noetherian)**.** *Provided by Mathlib (`Mathlib.RingTheory.Noetherian.Defs`). Let A be a Noetherian ring and M a finite A-module. Then M is a Noetherian A-module. Combined with 1054 this makes a finite $\kappa[t_0, \dots, t_{r-1}]$-module Noetherian, so its submodules are again finite (1090).*

The internal grading of a homogeneous submodule (split G_1)

Lemma 1056 (G_1 (independence) – the family $n \mapsto M_n \cap p$ is independent)**.** *Let $M = \bigoplus_n M_n$ be an internally graded κ-module and let $p \subseteq M$ be a homogeneous submodule (1044). Then the family $n \mapsto M_n \cap p$ of κ-subspaces of M is independent: for each n,*

$$(M_n \cap p) \;\cap\; \Big(\bigsqcup_{m \neq n} (M_m \cap p) \Big) \;=\; 0.$$

The whole statement lives in the ambient module M: the subspaces $M_n \cap p$ are submodules of M, never of a derived carrier $\hat{p}$.

Proof. The components M_n of the internal decomposition of M are independent. The subfamily $n \mapsto M_n \cap p$ consists of submodules with $M_n \cap p \leq M_n$, so independence is inherited: a relation among the $M_n \cap p$ is in particular a relation among the M_n, forcing each term to vanish. $\square$

Lemma 1057 (G_1 (supremum) – $\bigsqcup_n (M_n \cap p) = p$ for p homogeneous)**.** *Let $p \subseteq M$ be a homogeneous submodule (1044). Then*

$$\bigsqcup_n (M_n \cap p) \;=\; p.$$

Together with the independence of 1056, this is the ambient internal direct sum $p = \bigoplus_n (M_n \cap p)$, stated entirely inside M without requiring a direct-sum decomposition of the subtype $\hat{p}$.

Proof. The inclusion $\bigsqcup_n (M_n \cap p) \leq p$ is clear, since each $M_n \cap p \leq p$. For the reverse inclusion, take $m \in p$ and write $m = \sum_n \mathrm{proj}_n(m)$ with $\mathrm{proj}_n(m) \in M_n$ using the internal decomposition of M. Homogeneity of p gives $\mathrm{proj}_n(m) \in p$, hence $\mathrm{proj}_n(m) \in M_n \cap p$; thus m is a finite sum of elements of the $M_n \cap p$, i.e. $m \in \bigsqcup_n (M_n \cap p)$. $\square$

The ambient subquotient class and its Hilbert function

Definition 1058 (Subquotient Hilbert data)**.** Fix a graded κ-module $\mathcal{M}$, i.e. M together with an internal decomposition $M = \bigoplus_n \mathcal{M}_n$ whose components $\mathcal{M}_n$ are finite-dimensional over κ. A *subquotient datum of length r* consists of:

- a pair of homogeneous submodules $N' \leq N \subseteq M$; and
- a family $t : \{0, \dots, r-1\} \to (M \to_\kappa M)$ of pairwise commuting κ-linear endomorphisms, each raising degree by one ($t_i(\mathcal{M}_n) \subseteq \mathcal{M}_{n+1}$), each preserving N and N' ($t_i N \subseteq N$, $t_i N' \subseteq N'$); together with
- a finiteness condition: the subquotient N/N' is a finite module over the polynomial ring $\kappa[t_0, \dots, t_{r-1}]$ acting through the t_i (via the ambient t-stable carriers polySubmodule N, polySubmodule N' of 1086).

The pair (N, N') represents the subquotient N/N'. Its *ambient Hilbert function* is

$$\mathrm{hilb}(n) \;=\; \Big(\dim_\kappa(N \cap \mathcal{M}_n) \;-\; \dim_\kappa(N' \cap \mathcal{M}_n)\Big) \;\in\; \mathbb{Q},$$

the difference of the (finite) κ-dimensions of the ambient intersections, computed in $\mathbb{Z}$ and cast to $\mathbb{Q}$. Since $N' \leq N$ one has $N' \cap \mathcal{M}_n \leq N \cap \mathcal{M}_n$, so $\mathrm{hilb}(n) = \dim_\kappa((N \cap \mathcal{M}_n)/(N' \cap \mathcal{M}_n))$ is the dimension of the degree-n piece of N/N'. The split G_1 (1056, 1057) guarantees that these ambient intersections assemble the homogeneous submodules N, N', so the data depends only on the ambient families $n \mapsto N \cap \mathcal{M}_n$, $n \mapsto N' \cap \mathcal{M}_n$.

Ambient homogeneity calculus

The kernel/cokernel closure and the degreewise difference identity rest on a small calculus of how a degree-raising endomorphism interacts with the internal grading and with the lattice of homogeneous submodules. Throughout, $M = \bigoplus_n \mathcal{M}_n$ is the fixed internally graded κ-module and $x : M \to M$ is a κ-linear endomorphism raising degree by one. None of these statements forms a derived carrier: every object is a submodule of the fixed M.

Definition 1059 (Degree-raising endomorphism)**.** Let $M = \bigoplus_n \mathcal{M}_n$ be an internally graded κ-module. A κ-linear endomorphism $x : M \to M$ *raises degree by one* when it carries each graded piece into the next: $x(\mathcal{M}_n) \subseteq \mathcal{M}_{n+1}$ for every n. This is the grading-ring-free form of "multiplication by a degree-one element": it abstracts each of the r commuting degree-one generators $x \in R_1$ acting on M into a single endomorphism, with no reference to the grading ring R.

Lemma 1060 (Membership form of degree-raising)**.** *If x raises degree by one (1059) and $m \in \mathcal{M}_n$, then $xm \in \mathcal{M}_{n+1}$.*

Proof. Immediate from the definition: $xm \in x(\mathcal{M}_n) \subseteq \mathcal{M}_{n+1}$. $\square$

Lemma 1061 (Degree-shift commutation)**.** *Let x raise degree by one. Then for every $m \in M$ and every i the degree-$(i+1)$ homogeneous component of xm equals x applied to the degree-i component of m:*

$$\operatorname{proj}_{i+1}(xm) \;=\; x(\operatorname{proj}_i(m)).$$

This load-bearing commutation says a degree-raising endomorphism shifts the internal decomposition by exactly one slot; it is what makes preimages and images of homogeneous submodules under x homogeneous.

Proof. Decompose $m = \sum_j \operatorname{proj}_j(m)$ into its finitely many homogeneous components and apply x: $xm = \sum_j x(\operatorname{proj}_j(m))$ with $x(\operatorname{proj}_j(m)) \in \mathcal{M}_{j+1}$ by 1060. Reading off the degree-$(i+1)$ component, all terms with $j+1 \neq i+1$ drop out, leaving $x(\operatorname{proj}_i(m))$. $\square$

Lemma 1062 (Degree-raising kills the bottom)**.** *Let x raise degree by one. Then the degree-0 component of xm vanishes for every m:* $\operatorname{proj}_0(xm) = 0$.

Proof. Writing $xm = \sum_j x(\operatorname{proj}_j(m))$ with each summand in $\mathcal{M}_{j+1}$, no summand has degree 0 (every $j+1 \geq 1$), so the degree-0 component is 0. $\square$

Lemma 1063 (Preimage of a homogeneous submodule is homogeneous)**.** *Let x raise degree by one and let $N' \subseteq M$ be a homogeneous submodule (1044). Then the preimage $x^{-1}(N') = \{m : xm \in N'\}$ is a homogeneous submodule of M.*

Proof. Let $m \in x^{-1}(N')$ and fix a degree i. By the degree-shift commutation (1061), $x(\operatorname{proj}_i(m)) = \operatorname{proj}_{i+1}(xm)$, which lies in N' because $xm \in N'$ and N' is homogeneous. Hence $\operatorname{proj}_i(m) \in x^{-1}(N')$, so $x^{-1}(N')$ is homogeneous. $\square$

Lemma 1064 (Image of a homogeneous submodule is homogeneous)**.** *Let x raise degree by one and let $N \subseteq M$ be a homogeneous submodule. Then the image $x \cdot N = \{xm : m \in N\}$ is a homogeneous submodule of M, with degree-$(n+1)$ piece $x(N \cap \mathcal{M}_n)$ and zero degree-0 piece.*

Proof. Take $xm \in x \cdot N$ with $m \in N$ and fix a degree. In degree 0 the component $\operatorname{proj}_0(xm) = 0 \in x \cdot N$ (1062). In degree $i+1$, $\operatorname{proj}_{i+1}(xm) = x(\operatorname{proj}_i(m))$ (1061); since N is homogeneous, $\operatorname{proj}_i(m) \in N$, so this component lies in $x \cdot N$. Thus every homogeneous component of xm lies in $x \cdot N$, which is therefore homogeneous. $\square$

Lemma 1065 (Intersection of homogeneous submodules is homogeneous)**.** *If $p, q \subseteq M$ are homogeneous submodules (1044) then so is their intersection $p \cap q$. Mathlib provides no lattice-closure lemmas for homogeneous submodules, so this is recorded here.*

Proof. For $m \in p \cap q$ and any degree i, homogeneity of p and of q gives $\operatorname{proj}_i(m) \in p$ and $\operatorname{proj}_i(m) \in q$, hence $\operatorname{proj}_i(m) \in p \cap q$. $\square$

Lemma 1066 (Sum of homogeneous submodules is homogeneous). *If $p, q \subseteq M$ are homogeneous submodules then so is their sum $p + q$.*

Proof. Write an element of $p + q$ as $a + b$ with $a \in p$, $b \in q$. For any degree i, $\mathrm{proj}_i(a + b) = \mathrm{proj}_i(a) + \mathrm{proj}_i(b)$ with $\mathrm{proj}_i(a) \in p$ and $\mathrm{proj}_i(b) \in q$ by homogeneity, so $\mathrm{proj}_i(a+b) \in p+q$. □

Lemma 1067 (Ambient image identity). *Let x raise degree by one and let $N \subseteq M$ be homogeneous. Then the degree-$(n + 1)$ part of the image $x \cdot N$ is exactly x of the degree-n part of N:*

$$(x \cdot N) \cap \mathcal{M}_{n+1} \;=\; x \cdot (N \cap \mathcal{M}_n).$$

Proof. $\supseteq$: for $m \in N \cap \mathcal{M}_n$, $xm \in x \cdot N$ and $xm \in \mathcal{M}_{n+1}$ since x raises degree. $\subseteq$: an element $xm \in x \cdot N$ lying in $\mathcal{M}_{n+1}$ equals its own degree-$(n + 1)$ component $\mathrm{proj}_{n+1}(xm) = x(\mathrm{proj}_n(m))$ (1061); homogeneity of N puts $\mathrm{proj}_n(m) \in N \cap \mathcal{M}_n$, so $xm \in x \cdot (N \cap \mathcal{M}_n)$. The image $x \cdot N$ is homogeneous by 1064. □

Lemma 1068 (Ambient distributive law). *For homogeneous submodules $P, Q \subseteq M$ and any degree k, intersecting the sum with a graded piece distributes:*

$$(P + Q) \cap \mathcal{M}_k \;=\; (P \cap \mathcal{M}_k) + (Q \cap \mathcal{M}_k).$$

Proof. $\supseteq$ is the general lattice inequality. For $\subseteq$, take $z = p + q \in (P + Q) \cap \mathcal{M}_k$ with $p \in P$, $q \in Q$. Since $z \in \mathcal{M}_k$, $z = \mathrm{proj}_k(z) = \mathrm{proj}_k(p) + \mathrm{proj}_k(q)$; homogeneity of P and Q (1044) puts $\mathrm{proj}_k(p) \in P \cap \mathcal{M}_k$ and $\mathrm{proj}_k(q) \in Q \cap \mathcal{M}_k$, so $z \in (P \cap \mathcal{M}_k) + (Q \cap \mathcal{M}_k)$. The intersection and sum stay in the homogeneous lattice by 1065 and 1066. □

Closure under kernel and cokernel

The kernel/cokernel closure of the subquotient class is realised by eight ambient component facts: for a degree-raising endomorphism x and a homogeneous pair $N' \leq N$, the kernel pair is $(N \cap x^{-1}(N'),\, N')$ and the cokernel pair is $(N,\, N' + x \cdot N)$. The lemmas below record that both new pairs are homogeneous, nest correctly, are annihilated by x, and are preserved by any endomorphism t commuting with x that preserves the original pair.

Lemma 1069 (Homogeneity of the kernel pair's lower module). *Let x raise degree by one (1059) and let N, N' be homogeneous submodules of M. Then the kernel pair's lower module $N \cap x^{-1}(N')$ is homogeneous. Indeed it is the intersection (1065) of the homogeneous N with the homogeneous preimage $x^{-1}(N')$ (1063).*

Lemma 1070 (Homogeneity of the cokernel pair's upper module). *Under the same hypotheses, the cokernel pair's upper module $N' + x \cdot N$ is homogeneous: it is the sum (1066) of the homogeneous N' with the homogeneous image $x \cdot N$ (1064).*

Lemma 1071 (The kernel pair nests). *If $N' \leq N$ and x preserves N' ($x \cdot N' \subseteq N'$), then $N' \leq N \cap x^{-1}(N')$. Indeed $N' \leq N$ by hypothesis, and $x \cdot N' \subseteq N'$ is exactly $N' \subseteq x^{-1}(N')$; so N' is below the intersection. Hence the kernel pair $(N \cap x^{-1}(N'),\, N')$ is a valid subquotient pair (1058).*

Lemma 1072 (The cokernel pair nests). *If $N' \leq N$ and x preserves N ($x \cdot N \subseteq N$), then $N' + x \cdot N \leq N$: both summands lie in N. Hence the cokernel pair $(N,\, N' + x \cdot N)$ is a valid subquotient pair (1058).*

Lemma 1073 (x annihilates the kernel subquotient)**.** *The endomorphism x carries the kernel pair's lower module into N': $x \cdot (N \cap x^{-1}(N')) \subseteq N'$. Indeed $N \cap x^{-1}(N') \subseteq x^{-1}(N')$, and applying x to a preimage of N' lands in N'. Thus x acts as zero on the kernel subquotient.*

Lemma 1074 (x annihilates the cokernel subquotient)**.** *The endomorphism x carries N into the cokernel pair's upper module: $x \cdot N \subseteq N' + x \cdot N$, trivially (the right summand). Thus x acts as zero on the cokernel subquotient $N/(N' + x \cdot N)$.*

Lemma 1075 (Commuting endomorphisms preserve the preimage)**.** *Let t commute with x and preserve N' ($t \cdot N' \subseteq N'$). Then t preserves the preimage: $t \cdot x^{-1}(N') \subseteq x^{-1}(N')$. Indeed for m with $xm \in N'$ one has $x(tm) = t(xm) \in t \cdot N' \subseteq N'$, so $tm \in x^{-1}(N')$. This is what shows the remaining t_i preserve the kernel pair's lower module $N \cap x^{-1}(N')$.*

Lemma 1076 (Commuting endomorphisms preserve the image)**.** *Let t commute with x and preserve N ($t \cdot N \subseteq N$). Then t preserves the image: $t \cdot (x \cdot N) \subseteq x \cdot N$. Indeed for $m \in N$ one has $t(xm) = x(tm)$ with $tm \in N$, so $t(xm) \in x \cdot N$. This is what shows the remaining t_i preserve the cokernel pair's upper module $N' + x \cdot N$.*

Lemma 1077 (Kernel/cokernel closure of the subquotient class)**.** *Let (N, N') be a subquotient datum (1058) with endomorphisms $t_0, \dots, t_{r-1}$, and set $x = t_{r-1}$. Form the two new pairs of homogeneous submodules of M*

$$K = (\, N \cap x^{-1}(N'),\ N' \,), \qquad C = (\, N,\ N' + x \cdot N \,),$$

where $x^{-1}(N') = \{\, m \in M : x \cdot m \in N' \,\}$ is the preimage and $x \cdot N$ is the image of N under x. Then:

1. *$N \cap x^{-1}(N')$ and $N' + x \cdot N$ are homogeneous submodules of M, with $N' \leq N \cap x^{-1}(N')$ and $N' + x \cdot N \leq N$, so both K and C are subquotient pairs;*
2. *the remaining endomorphisms $t_0, \dots, t_{r-2}$ pairwise commute, raise degree by one, and preserve both members of K and of C; and*
3. *x annihilates both subquotients: $x \cdot (N \cap x^{-1}(N')) \subseteq N'$ and $x \cdot N \subseteq N' + x \cdot N$.*

Consequently K and C are again subquotient data, now of length $r - 1$ (the endomorphisms $t_0, \dots, t_{r-2}$), and $K = \ker x$ and $C = \operatorname{coker} x$ on the subquotient N/N': K represents $\ker(x : N/N' \to N/N')$ and C represents $\operatorname{coker}(x : N/N' \to N/N')$. No quotient or subtype carrier is constructed; K and C are ambient pairs of homogeneous submodules of the fixed M.

Proof. Homogeneity. The preimage $x^{-1}(N')$ is homogeneous: if $m = \sum_n \operatorname{proj}_n(m)$ with $x \cdot m \in N'$, then since x raises degree by one (1048) the components $x \cdot \operatorname{proj}_n(m) \in \mathcal{M}_{n+1}$ lie in distinct degrees, and homogeneity of N' (split G_1, 1057) forces each $x \cdot \operatorname{proj}_n(m) \in N'$, i.e. $\operatorname{proj}_n(m) \in x^{-1}(N')$. Thus $N \cap x^{-1}(N')$ is an intersection of homogeneous submodules, hence homogeneous. The image $x \cdot N$ is homogeneous with degree-$(n+1)$ piece $x \cdot (N \cap \mathcal{M}_n)$ by 1048, and a sum of homogeneous submodules is homogeneous (1045 applied at the module level), so $N' + x \cdot N$ is homogeneous. The inclusions $N' \leq N \cap x^{-1}(N')$ (as $x \cdot N' \subseteq N'$) and $N' + x \cdot N \leq N$ (as $N' \leq N$ and $x \cdot N \subseteq N$) are immediate. In the language of the ambient homogeneity calculus, this paragraph is exactly: $x^{-1}(N')$ is homogeneous (1063), $x \cdot N$ is homogeneous (1064), and the homogeneous lattice is closed under intersection (1065) and sum (1066).

Preservation. Each t_i with $i \leq r-2$ commutes with x and preserves N, N'; hence it preserves $x^{-1}(N')$ (if $x t_i m = t_i x m \in N'$ whenever $xm \in N'$), so it preserves $N \cap x^{-1}(N')$, and it preserves

$x \cdot N$ (as $t_i x N = x t_i N \subseteq xN$), so it preserves $N' + x \cdot N$. Degree-raising and commutativity are inherited.

Annihilation by x. For $m \in N \cap x^{-1}(N')$ one has $x \cdot m \in N'$ by definition, so x carries the first member of K into its second; and $x \cdot N \subseteq N' + x \cdot N$ trivially. Thus x acts as zero on N/N' restricted to K and on the cokernel quotient C.

Identification with $\ker x$ and $\operatorname{coker} x$. On the subquotient N/N', the class of $m \in N$ is killed by x iff $x \cdot m \in N'$, i.e. iff $m \in N \cap x^{-1}(N')$; so $\ker(x : N/N' \to N/N')$ is exactly $(N \cap x^{-1}(N'))/N'$, represented by K. The image of x on N/N' is $(N' + x \cdot N)/N'$, so the cokernel $(N/N')/\operatorname{im} x = N/(N' + x \cdot N)$ is represented by C. ☐

The subquotient degreewise difference identity (D_6)

Lemma 1078 (preimage dimension equals ambient-intersection dimension)**.** *For submodules p, S of a κ-vector space M, the dimension of the preimage of S under the inclusion $p \hookrightarrow M$ equals the dimension of the ambient intersection: $\dim_\kappa(\iota_p^{-1} S) = \dim_\kappa(p \cap S)$, where $\iota_p : p \to M$ is the canonical injection. Project-local helper consumed by the two kernel-dimension computations of 1079.*

Proof. The inclusion ι_p is injective, so it restricts to a κ-linear isomorphism from $\iota_p^{-1} S$ onto its image $\iota_p(\iota_p^{-1} S) = p \cap S$ (image of a comap under an injective map). An isomorphism preserves dimension, giving the claimed equality. ☐

Lemma 1079 (D_6 – subquotient degreewise difference)**.** *Source: [Stacks Project], Tag 00K1 (degreewise short exact sequence). Let (N, N') be a subquotient datum (1058) with Hilbert function h_M, and let $x = t_{r-1}$. Let h_K and h_C be the Hilbert functions of the kernel and cokernel subquotients $K = (N \cap x^{-1}(N'),\ N')$ and $C = (N,\ N' + x \cdot N)$ of 1077. Then for every n*

$$h_M(n+1) - h_M(n) \ = \ h_C(n+1) - h_K(n).$$

Writing $Q_n = (N \cap \mathcal{M}_n)/(N' \cap \mathcal{M}_n)$ for the degree-n piece of N/N' (so $\dim_\kappa Q_n = h_M(n)$), multiplication by x is a κ-linear map $x : Q_n \to Q_{n+1}$ with $\ker(x : Q_n \to Q_{n+1})$ the degree-n piece of K and $Q_{n+1}/\operatorname{im} x$ the degree-$(n+1)$ piece of C; the identity is then the degreewise rank–nullity difference D_5 (1097). Specialising to $(N, N') = (\top, \bot)$ recovers the classical $0 \to \mathcal{M}_n \xrightarrow{x} \mathcal{M}_{n+1} \to \mathcal{M}_{n+1}/x\mathcal{M}_n \to 0$.

Proof. Since $N' \leq N$ are homogeneous, the split G_1 (1057) identifies the degree-n piece of N/N' with $Q_n = (N \cap \mathcal{M}_n)/(N' \cap \mathcal{M}_n)$, a finite-dimensional κ-vector space of dimension $h_M(n)$. Because x raises degree by one and preserves N and N' (1048), it induces a κ-linear map $x : Q_n \to Q_{n+1}$.

Kernel. A class $[m] \in Q_n$ (with $m \in N \cap \mathcal{M}_n$) lies in $\ker(x : Q_n \to Q_{n+1})$ iff $x \cdot m \in N' \cap \mathcal{M}_{n+1}$, i.e. iff $m \in (N \cap x^{-1}(N')) \cap \mathcal{M}_n$. Hence $\ker(x : Q_n \to Q_{n+1}) = ((N \cap x^{-1}(N')) \cap \mathcal{M}_n)/(N' \cap \mathcal{M}_n)$ is exactly the degree-n piece of K, of dimension $h_K(n)$ (1077).

Cokernel. The image of $x : Q_n \to Q_{n+1}$ is $(x \cdot (N \cap \mathcal{M}_n) + (N' \cap \mathcal{M}_{n+1}))/(N' \cap \mathcal{M}_{n+1})$. Using the ambient identity $(N' + x \cdot N) \cap \mathcal{M}_{n+1} = (N' \cap \mathcal{M}_{n+1}) + x \cdot (N \cap \mathcal{M}_n)$ – which is the ambient distributive law (1068) applied to $P = N'$, $Q = x \cdot N$, combined with the ambient image identity $(x \cdot N) \cap \mathcal{M}_{n+1} = x \cdot (N \cap \mathcal{M}_n)$ (1067) – the cokernel $Q_{n+1}/\operatorname{im} x = (N \cap \mathcal{M}_{n+1})/((N' + x \cdot N) \cap \mathcal{M}_{n+1})$ is the degree-$(n+1)$ piece of C, of dimension $h_C(n+1)$.

Difference. Apply the degreewise rank–nullity difference D_5 (1097) to $\varphi = (x : Q_n \to Q_{n+1})$, with $V = Q_n$, $W = Q_{n+1}$:

$$h_M(n+1) - h_M(n) = \dim_\kappa Q_{n+1} - \dim_\kappa Q_n = \dim_\kappa(Q_{n+1}/\operatorname{im}\varphi) - \dim_\kappa(\ker\varphi) = h_C(n+1) - h_K(n).$$

This is the hdiff hypothesis consumed by 1037. ☐

The free polynomial-ring module structure

The finiteness condition of a subquotient datum (1058) and its transfer down one generator (1092) are phrased over the *free* polynomial ring $\kappa[t_0, \dots, t_{r-1}] = \mathrm{MvPolynomial}(\{0, \dots, r-1\}, \kappa)$. The following blocks build the action of that ring on the ambient module M from the r pairwise-commuting degree-raising endomorphisms t_i, keeping every carrier an ambient submodule of M. The free polynomial ring – rather than the subalgebra $\mathrm{adjoin}_\kappa\{t_0, \dots, t_{r-1}\} \subseteq \mathrm{End}_\kappa(M)$ – is used precisely so the inductive transfer has the ring surjection $\kappa[t_0, \dots, t_{r-1}] \twoheadrightarrow \kappa[t_0, \dots, t_{r-2}]$ available (1092).

Definition 1080 (Polynomial evaluation ring homomorphism)**.** Let $t : \{0, \dots, r-1\} \to \mathrm{End}_\kappa(M)$ be a family of pairwise-commuting κ-linear endomorphisms. The *polynomial evaluation ring homomorphism*

$$\mathrm{polyEndHom}(t) : \kappa[t_0, \dots, t_{r-1}] \longrightarrow \mathrm{End}_\kappa(M)$$

sends each variable X_i to t_i. It is constructed by evaluating into the *commutative* κ-subalgebra $A = \mathrm{adjoin}_\kappa(\{t_i\}) \subseteq \mathrm{End}_\kappa(M)$ – commutative by 1049 because the generators commute – and then including $A \hookrightarrow \mathrm{End}_\kappa(M)$; evaluation into the noncommutative $\mathrm{End}_\kappa(M)$ directly is not an algebra homomorphism, which is why the commutative A is interposed.

Lemma 1081 (Evaluation sends X_i to t_i)**.** $\mathrm{polyEndHom}(t)(X_i) = t_i$ *for each* i. *Immediate from the construction (1080), as evaluation sends* X_i *to the chosen image* t_i*.*

Lemma 1082 (Evaluation sends a constant to a scalar)**.** $\mathrm{polyEndHom}(t)(C\,c) = c \cdot \mathrm{id}_M$ *for each* $c \in \kappa$*: the evaluation homomorphism (1080) is* κ*-linear and carries the constant polynomial* $C\,c$ *to the scalar endomorphism* $c \cdot 1$*.*

Definition 1083 (Polynomial-ring module structure on M)**.** The *polynomial-module structure* is the $\kappa[t_0, \dots, t_{r-1}]$-module structure on M obtained by restricting scalars along $\mathrm{polyEndHom}(t)$ (1080): a polynomial p acts on $m \in M$ by $p \cdot m = \mathrm{polyEndHom}(t)(p)(m)$. This is the module over the free polynomial ring on which the ambient subquotient induction's finiteness condition is measured.

Lemma 1084 (X_i acts as t_i)**.** *In the polynomial-module structure (1083),* $X_i \cdot m = t_i(m)$ *for all* $m \in M$*. Immediate from* $\mathrm{polyEndHom}(t)(X_i) = t_i$ *(1081).*

Lemma 1085 (A constant acts as a scalar)**.** *In the polynomial-module structure (1083),* $(C\,c) \cdot m = c \cdot m$ *for all* $c \in \kappa$*,* $m \in M$*. Immediate from* $\mathrm{polyEndHom}(t)(C\,c) = c \cdot \mathrm{id}_M$ *(1082).*

Definition 1086 (Polynomial-ring submodule from a t-stable submodule)**.** Let $N \subseteq M$ be a κ-submodule that is stable under each t_i ($t_i \cdot N \subseteq N$). Then N, with the *same* ambient carrier, is a $\kappa[t_0, \dots, t_{r-1}]$-submodule of M for the polynomial-module structure (1083). Stability under the action of an arbitrary polynomial follows by induction on monomials: constants act as scalars (1085) and multiplication by X_i acts as t_i (1084), both of which preserve N.

Finiteness-transfer infrastructure

The ambient finiteness transfer (1092) factors the polynomial action of the larger ring through a surjection onto the smaller one. The following blocks build that surjection and the semilinearity identity it rests on, and record the two elementary finiteness manipulations (shrinking a numerator, enlarging a denominator) used to feed the kernel and cokernel constructors.

Definition 1087 (The last-variable surjection of polynomial rings)**.** For a field κ and $r \in \mathbb{N}$, the *last-variable homomorphism* is the κ-algebra map

$$\mathrm{lastVarAlgHom} : \kappa[t_0, \dots, t_r] = \mathrm{MvPolynomial}(\{0, \dots, r\}, \kappa) \longrightarrow \mathrm{MvPolynomial}(\{0, \dots, r-1\}, \kappa) = \kappa[t_0, \dots, t_{r-1}]$$

obtained by evaluation: it sends the last variable X_r to 0 and each earlier variable X_i (for $0 \le i < r$) to X_i, and fixes constants. It is a left inverse of the algebra map $\kappa[t_0, \dots, t_{r-1}] \to \kappa[t_0, \dots, t_r]$ induced by the canonical inclusion $\{0, \dots, r-1\} \hookrightarrow \{0, \dots, r\}$ on variable indices, and is therefore surjective. This is the concrete ring surjection $\kappa[t_0, \dots, t_r] \twoheadrightarrow \kappa[t_0, \dots, t_{r-1}]$ along which finiteness is transferred (1047).

Lemma 1088 (A t-stable submodule is closed under the polynomial action)**.** *Let $t_0, \dots, t_{r-1}$ be pairwise-commuting κ-linear endomorphisms of M and let $P' \subseteq M$ be a κ-submodule stable under each t_i ($t_i \cdot P' \subseteq P'$). Then P' is closed under the action of every polynomial: for all $p \in \kappa[t_0, \dots, t_{r-1}]$ and $m \in P'$,* $\mathrm{polyEndHom}(t)(p)(m) \in P'$ *(1080). The proof is induction on p: a constant acts as a scalar (1082), multiplication by a variable acts as the stabilising t_i (1081), and sums are handled additively.*

Lemma 1089 (Mod-P' semilinearity of the polynomial action)**.** *Let $t_0, \dots, t_r$ be pairwise-commuting κ-linear endomorphisms of M, and let $P' \le P$ be κ-submodules each stable under every t_i, with the last endomorphism $x = t_r$ carrying P into P' ($x \cdot P \subseteq P'$). Then for every polynomial $s \in \kappa[t_0, \dots, t_r]$ and every $m \in P$,*

$$\mathrm{polyEndHom}(t)(s)(m) \; - \; \mathrm{polyEndHom}(t_0, \dots, t_{r-1})(\mathrm{lastVarAlgHom}(s))(m) \; \in \; P'.$$

In words: acting by s through the full family t agrees, modulo P', with first projecting away the last variable via lastVarAlgHom *(1087) and acting through the reduced family $(t_0, \dots, t_{r-1})$. This is the semilinearity identity that makes the σ-semilinear quotient map of 1092 well-defined. The proof is induction on s: the case of an earlier variable $s = X_i$ ($i < r$) is the inductive hypothesis after* lastVarAlgHom *fixes it; the case of the last variable $s = X_r$ annihilates the reduced term, and the full term lands in P' because $x = t_r$ carries P into the t-stable P' (1088).*

Lemma 1090 (Shrinking the numerator preserves finiteness)**.** *Let $N_1 \le N_2$ and P' be κ-submodules of M, each stable under the commuting family $t_0, \dots, t_{r-1}$, and suppose the subquotient N_2/P' is finite over $\kappa[t_0, \dots, t_{r-1}]$ (1083, 1086). Then N_1/P' is also finite over $\kappa[t_0, \dots, t_{r-1}]$. Indeed the inclusion $N_1 \subseteq N_2$ induces an injective $\kappa[t_0, \dots, t_{r-1}]$-linear map $N_1/P' \hookrightarrow N_2/P'$; since $\kappa[t_0, \dots, t_{r-1}]$ is Noetherian (1054), the finite module N_2/P' is Noetherian (1055), so its submodule N_1/P' is finite (1051). This supplies the finiteness witness of the kernel constructor (1095).*

Lemma 1091 (Enlarging the denominator preserves finiteness)**.** *Let N and $P_1 \le P_2$ be κ-submodules of M, each stable under the commuting family $t_0, \dots, t_{r-1}$, and suppose the subquotient N/P_1 is finite over $\kappa[t_0, \dots, t_{r-1}]$ (1083, 1086). Then N/P_2 is also finite over $\kappa[t_0, \dots, t_{r-1}]$. Indeed the inclusion $P_1 \subseteq P_2$ gives a surjection $N/P_1 \twoheadrightarrow N/P_2$ (1052), and finiteness transfers along it (1050). This supplies the finiteness witness of the cokernel constructor (1096).*

Ambient finiteness transfer

Lemma 1092 (Abstract single-pair finiteness transfer down one variable)**.** *Let $t_0, \dots, t_r$ be pairwise-commuting κ-linear endomorphisms of M, and let $P' \le P$ be κ-submodules of M, each stable under every t_i. Suppose the last endomorphism $x = t_r$ carries P into P' ($x \cdot P \subseteq P'$), and that the subquotient P/P' is finite over the free polynomial ring $\kappa[t_0, \dots, t_r]$*

for the polynomial-module structure of 1083 (with carriers the ambient t-stable submodules polySubmodule P, polySubmodule P', *1086). Then P/P' is finite over $\kappa[t_0, \dots, t_{r-1}]$ for the reduced family $(t_0, \dots, t_{r-1})$.*

This is the abstract single-pair transfer: it does not mention the kernel/cokernel pairs, but is the engine applied to each of them by the kernel and cokernel constructors (1095, 1096). Because the last variable annihilates the subquotient, the $\kappa[t_0, \dots, t_r]$-action factors through the last-variable surjection $\kappa[t_0, \dots, t_r] \twoheadrightarrow \kappa[t_0, \dots, t_{r-1}]$, which is what makes the transfer drop one variable.

Proof. Write $\sigma = \mathrm{lastVarAlgHom} : \kappa[t_0, \dots, t_r] \twoheadrightarrow \kappa[t_0, \dots, t_{r-1}]$ for the surjective ring homomorphism sending the last variable to 0 and each earlier variable X_i $(i < r)$ to X_i (1087). Consider the map on numerators

$$g : \mathrm{polySubmodule}\, P \longrightarrow (\mathrm{polySubmodule}\, P)\big/(\mathrm{polySubmodule}\, P'), \qquad y \longmapsto [\, y\,],$$

where the source carries the $\kappa[t_0, \dots, t_r]$-module structure and the target the reduced $\kappa[t_0, \dots, t_{r-1}]$-module structure (both with the same ambient carrier P, 1086). The mod-P' semilinearity identity (1089) – acting by s through the full family and acting by $\sigma(s)$ through the reduced family agree modulo P', precisely because $x = t_r$ carries P into P' – says exactly that g is σ-semilinear. The map g is surjective (it is the quotient projection on the common carrier), and it kills the denominator polySubmodule P', so it descends through that submodule (1052) to a surjective $\kappa[t_0, \dots, t_{r-1}]$-linear map

$$(\mathrm{polySubmodule}\, P)\big/(\mathrm{polySubmodule}\, P') \twoheadrightarrow (\mathrm{polySubmodule}\, P)\big/(\mathrm{polySubmodule}\, P')$$

from the $\kappa[t_0, \dots, t_r]$-finite subquotient onto the reduced one. Finiteness transfers along this surjection (1050), giving finiteness of P/P' over $\kappa[t_0, \dots, t_{r-1}]$.

Why the free polynomial ring. The smaller ring must be the *free* polynomial ring $\kappa[t_0, \dots, t_{r-1}]$, not the subalgebra $\mathrm{adjoin}_\kappa\{t_0, \dots, t_{r-1}\} \subseteq \mathrm{End}_\kappa(M)$: relations among the endomorphisms inside $\mathrm{End}_\kappa(M)$ would obstruct the polynomial-ring surjection σ that the scalar transfer requires (1047). Keeping free polynomial rings at every step keeps σ – and hence the transfer – available throughout. Everything is finite generation of ambient submodules of the fixed M and their quotients. ☐

Subquotient constructors

The kernel/cokernel closure (1077) and the abstract finiteness transfer (1092) are bundled into two constructors that take a length-$(r+1)$ subquotient datum to the length-r kernel and cokernel data consumed by the induction. The two stability lemmas below first record that the new pairs are stable under the *whole* endomorphism family (not merely the reduced one), which is what lets the over-$\kappa[t_0, \dots, t_r]$ finiteness inputs be stated before the variable is dropped.

Lemma 1093 (The kernel pair's lower module is stable under the full family)**.** *Let (N, N') be a length-$(r + 1)$ subquotient datum (1058) with endomorphisms $t_0, \dots, t_r$ and $x = t_r$. Then the kernel pair's lower module $N \cap x^{-1}(N')$ is stable under* every *t_i $(0 \le i \le r)$, not only the reduced family: each t_i preserves N and preserves the preimage $x^{-1}(N')$ because it commutes with x and preserves N' (1075). This full stability is what allows the kernel subquotient's finiteness to be measured over $\kappa[t_0, \dots, t_r]$ before the transfer drops the last variable.*

Lemma 1094 (The cokernel pair's upper module is stable under the full family)**.** *Under the same hypotheses, the cokernel pair's upper module $N' + x \cdot N$ is stable under every t_i $(0 \le i \le r)$: each t_i preserves N' and preserves the image $x \cdot N$ because it commutes with x and preserves*

N (1076), and a map preserves a sum of preserved submodules. This full stability supports the over-$\kappa[t_0, \dots, t_r]$ finiteness input of the cokernel constructor.

Definition 1095 (Kernel constructor of a subquotient datum)**.** From a length-$(r+1)$ subquotient datum (N, N') with endomorphisms $t_0, \dots, t_r$ and $x = t_r$ (1058), the *kernel constructor* produces the length-r datum

$$\ker D \;=\; \big(\, N \cap x^{-1}(N'),\;\, N' \,\big)$$

on the reduced family $(t_0, \dots, t_{r-1})$. Its non-finiteness fields are supplied by the ambient kernel calculus: homogeneity of $N \cap x^{-1}(N')$ (1069), the nesting $N' \leq N \cap x^{-1}(N')$ (1071), and stability under the reduced family (1093). The finiteness witness is obtained by first shrinking the numerator from N to $N \cap x^{-1}(N')$ (1090) to get $\kappa[t_0, \dots, t_r]$-finiteness of the kernel subquotient, then transferring down one variable (1092), valid because x annihilates the kernel subquotient (1073). This realises the kernel half of 1077 as a genuine length-r datum.

Definition 1096 (Cokernel constructor of a subquotient datum)**.** From the same length-$(r+1)$ datum, the *cokernel constructor* produces the length-r datum

$$\operatorname{coker} D \;=\; \big(\, N,\;\, N' + x \cdot N \,\big)$$

on the reduced family. Its non-finiteness fields come from the ambient cokernel calculus: homogeneity of $N' + x \cdot N$ (1070), the nesting $N' + x \cdot N \leq N$ (1072), and stability under the reduced family (1094). The finiteness field is obtained by enlarging the denominator from N' to $N' + x \cdot N$ (1091) to get $\kappa[t_0, \dots, t_r]$-finiteness of the cokernel subquotient, then transferring down one variable (1092), valid because x annihilates the cokernel subquotient (1074). This realises the cokernel half of 1077 as a genuine length-r datum.

The degreewise difference identity

Lemma 1097 (D_5 – degreewise rank–nullity difference)**.** *Let $\varphi : V \to W$ be a κ-linear map between finite-dimensional κ-vector spaces. Then*

$$\dim_\kappa W - \dim_\kappa V \;=\; \dim_\kappa(W/\operatorname{range}\varphi) - \dim_\kappa(\ker\varphi).$$

Applied degreewise with $\varphi = (x : \mathcal{M}_n \to \mathcal{M}_{n+1})$, $V = \mathcal{M}_n$, $W = \mathcal{M}_{n+1}$ – so that $\ker\varphi$ is the degree-n piece K_n of the kernel subquotient and $W/\operatorname{range}\varphi = \mathcal{M}_{n+1}/x\mathcal{M}_n$ the degree-$(n+1)$ piece C_{n+1} of the cokernel subquotient (1079, via the split G_1 1057 and the degree shift 1048) – this gives, for every n,

$$\dim_\kappa \mathcal{M}_{n+1} - \dim_\kappa \mathcal{M}_n \;=\; \dim_\kappa C_{n+1} - \dim_\kappa K_n,$$

which is the difference identity hdiff *consumed by 1037. No graded structure is used in proving the underlying vector-space identity – it is pure rank–nullity.*

Proof. By rank–nullity (1046), $\dim_\kappa V = \dim_\kappa(\ker\varphi) + \dim_\kappa(\operatorname{range}\varphi)$, so $\dim_\kappa(\operatorname{range}\varphi) = \dim_\kappa V - \dim_\kappa(\ker\varphi)$. By additivity on the short exact sequence $0 \to \operatorname{range}\varphi \to W \to W/\operatorname{range}\varphi \to 0$ (1026), $\dim_\kappa(W/\operatorname{range}\varphi) = \dim_\kappa W - \dim_\kappa(\operatorname{range}\varphi) = \dim_\kappa W - \dim_\kappa V + \dim_\kappa(\ker\varphi)$. Rearranging gives $\dim_\kappa W - \dim_\kappa V = \dim_\kappa(W/\operatorname{range}\varphi) - \dim_\kappa(\ker\varphi)$. The degreewise application identifies $\ker(x : \mathcal{M}_n \to \mathcal{M}_{n+1})$ and $\mathcal{M}_{n+1}/\operatorname{range}(x : \mathcal{M}_n \to \mathcal{M}_{n+1})$ with the degree-n piece of the kernel subquotient and the degree-$(n+1)$ piece of the cokernel subquotient (1079), the latter because the degree-$(n+1)$ graded piece of the homogeneous submodule $x\mathcal{M}$ is exactly $x\mathcal{M}_n$ (1048). ∎

The base case ($r = 0$)

Lemma 1098 (Base-case finiteness over the empty polynomial ring)**.** *Let Q be a κ-vector space that is also a module over* MvPolynomial(σ, κ) *for an* empty *index set σ – in particular $\sigma = \{0, \ldots, r-1\}$ with $r = 0$ – in a way compatible with the κ-action (scalar tower). If Q is module-finite over* MvPolynomial(σ, κ)*, then Q is finite-dimensional over κ. Indeed for empty σ the polynomial ring is* MvPolynomial$(\sigma, \kappa) \cong \kappa$*, a finite κ-module, so finiteness of Q over it transfers to finiteness over κ along the scalar tower. This is the length-zero input to the ambient subquotient induction (1101): when $r = 0$ the subquotient N/N' is a finite-dimensional κ-vector space, hence* hilb *is eventually zero.*

Lemma 1099 (independence of images modulo the kernel)**.** *Let $\pi : A \to Q$ be a κ-linear map and $(g_i)_{i \in \iota}$ a family of subspaces of A that is* independent modulo $\ker \pi$*: for each i, every $a \in g_i$ that also lies in $\ker \pi + \sum_{j \neq i} g_j$ already lies in $\ker \pi$. Then the image family $(\pi(g_i))_{i \in \iota}$ is independent (*iSupIndep*) in Q. Ring-agnostic lattice core of the base-case independence step below; applied to the quotient map $N \twoheadrightarrow N/N'$.*

Proof. By the disjointness criterion for independence it suffices to show, for each i, that $\pi(g_i) \cap \sum_{j \neq i} \pi(g_j) = 0$. Take q in that intersection; write $q = \pi(a)$ with $a \in g_i$, and (pushing the sum through π, since $\pi(\sum_{j \neq i} g_j) = \sum_{j \neq i} \pi(g_j)$) $q = \pi(b)$ with $b \in \sum_{j \neq i} g_j$. Then $a - b \in \ker \pi$, so $a = (a - b) + b \in \ker \pi + \sum_{j \neq i} g_j$. By the hypothesis applied at i, $a \in \ker \pi$, whence $q = \pi(a) = 0$. $\square$

Lemma 1100 (Base case: the length-zero subquotient Hilbert function vanishes eventually)**.** *Source: [Stacks Project], Tag 00K1 (base case of the induction). Let (N, N') be a subquotient datum of length 0 (1058): a homogeneous pair $N' \leq N$ of the fixed ambient graded module $M = \bigoplus_n \mathcal{M}_n$ with* no *endomorphisms, whose finiteness witness says the subquotient $Q_0 := N/N'$ is module-finite over* MvPolynomial$(\emptyset, \kappa)$*. Then the ambient Hilbert function vanishes eventually: there is $K \in \mathbb{N}$ with* hilb$(n) = 0$ *for all $n > K$. Consequently* IsRatHilb(hilb, 0) *(1032), which is the base case of the ambient subquotient induction (1101).*

Proof. Since the index set of variables is empty, MvPolynomial$(\emptyset, \kappa) \cong \kappa$, so the finiteness witness makes the subquotient $Q_0 = N/N'$ a *finite-dimensional* κ-vector space (1098). For each degree n, composing the inclusion of the homogeneous piece $N \cap \mathcal{M}_n \hookrightarrow N$ with the quotient projection $N \twoheadrightarrow Q_0$ gives a κ-linear map $\psi_n : N \cap \mathcal{M}_n \to Q_0$, $m \mapsto [m]$, whose image is the class of the degree-n homogeneous piece inside the subquotient.

Step 1: independence of the images. The family $\left(\mathrm{range}(\psi_n)\right)_n$ is iSupIndep in Q_0. The reason is the ambient direct-sum grading of M: the homogeneous pieces $\mathcal{M}_n$ form an independent family (1056). An element of $\bigsqcup_{j \neq n} \mathrm{range}(\psi_j)$ is a finite sum of classes of homogeneous elements of degrees $j \neq n$; since N' is homogeneous, the degree-n homogeneous component of any representative of such a class lies in N', so the class has zero degree-n component in Q_0. A nonzero class in range(ψ_n), by contrast, is represented by a homogeneous element of degree n not in N', so has nonzero degree-n component. Comparing degree-n components forces the intersection $\mathrm{range}(\psi_n) \cap \bigsqcup_{j \neq n} \mathrm{range}(\psi_j) = 0$; independence follows.

Step 2: finiteness of the support. Because Q_0 is finite-dimensional and the ranges range(ψ_n) are independent, only finitely many of them are nonzero: the set $\{ n : \mathrm{range}(\psi_n) \neq 0 \}$ is finite (1053). In particular it is bounded above, by some $K \in \mathbb{N}$.

Step 3: eventual vanishing. For $n > K$ we have range$(\psi_n) = 0$, i.e. every homogeneous element of $N \cap \mathcal{M}_n$ has class 0 in $Q_0 = N/N'$; equivalently $N \cap \mathcal{M}_n \leq N'$, hence $N \cap \mathcal{M}_n = N' \cap \mathcal{M}_n$

and

$$\mathrm{hilb}(n) \;=\; \dim_\kappa(N \cap \mathcal{M}_n) - \dim_\kappa(N' \cap \mathcal{M}_n) \;=\; 0.$$

Thus hilb is eventually zero, and 1032 gives $\mathrm{IsRatHilb}(\mathrm{hilb}, 0)$. ☐

The ambient subquotient induction

Lemma 1101 (Rationality of the ambient subquotient Hilbert function)**.** *Source: [Stacks Project], Tag 00K1 (induction on the number of degree-one generators). Let (N, N') be a subquotient datum of length r (1058) with ambient Hilbert function* hilb*. Then* $\mathrm{IsRatHilb}(\mathrm{hilb}, r)$ *(1031): the difference $n \mapsto \dim_\kappa(N \cap \mathcal{M}_n) - \dim_\kappa(N' \cap \mathcal{M}_n)$ is, for $n \gg 0$, the coefficient sequence of a rational series $p \cdot (1 - X)^{-r}$ with $p \in \mathbb{Q}[X]$. The induction is on the length r, and every object that appears is an ambient pair of homogeneous submodules of the fixed graded module M.*

Proof. We induct on the length r.

Base case $r = 0$. A length-zero datum has a finiteness witness making the subquotient N/N' a finite-dimensional κ-vector space (1098); its degreewise images form an independent family inside that finite-dimensional space, so only finitely many are nonzero and $\mathrm{hilb}(n) = 0$ for $n \gg 0$ (1100). By 1032 an eventually-zero function is rational of order 0, giving $\mathrm{IsRatHilb}(\mathrm{hilb}, 0)$.

Inductive step $r \geq 1$. Set $x = t_{r-1}$. By the kernel/cokernel closure (1077) the kernel subquotient $K = (N \cap x^{-1}(N'),\ N')$ and the cokernel subquotient $C = (N,\ N' + x \cdot N)$ are subquotient data of length $r - 1$, assembled by the kernel and cokernel constructors (1095, 1096) whose finiteness witnesses come from the abstract finiteness transfer (1092). The induction hypothesis applied to K and C gives $\mathrm{IsRatHilb}(h_K, r-1)$ and $\mathrm{IsRatHilb}(h_C, r-1)$ (1031). By the degreewise difference identity (1079),

$$\mathrm{hilb}(n+1) - \mathrm{hilb}(n) \;=\; h_C(n+1) - h_K(n) \qquad \text{for all } n.$$

Feeding $\mathrm{IsRatHilb}(h_C, r-1)$, $\mathrm{IsRatHilb}(h_K, r-1)$ and this difference identity into the inductive-step engine (1037) yields $\mathrm{IsRatHilb}(\mathrm{hilb}, r)$. (The common order $r-1$ for h_C and h_K is arranged, if necessary, by raising the lower one with 1033.) This completes the induction. ☐

20.4 Support and freeness predicates

The Quot functor (1138) and the Grassmannian (1176) classify quotient sheaves subject to two geometric conditions: a coherent quotient must have *schematic support proper over the base*, and the Grassmannian's quotients must be *locally free of a fixed rank*. This section introduces both predicates.

Definition 1102 (Annihilator ideal sheaf of a sheaf of modules)**.** *Source: [Nitsure], §1 (FGA Explained Ch. 5).* Let X be a scheme and let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$-modules that is *quasi-coherent and of finite type* (over a locally noetherian base, so that on each affine open U the section module $\mathcal{F}(U)$ is a finitely generated $\mathcal{O}_X(U)$-module). The *annihilator ideal sheaf* of $\mathcal{F}$ is the ideal sheaf $\mathrm{Ann}(\mathcal{F})$ defined on affine opens: for each affine open $U \subseteq X$,

$$\mathrm{Ann}(\mathcal{F})(U) \;=\; \mathrm{Ann}_{\mathcal{O}_X(U)}(\mathcal{F}(U)) \;\subseteq\; \mathcal{O}_X(U),$$

the annihilator of the $\mathcal{O}_X(U)$-module $\mathcal{F}(U)$. These local data patch to a coherent ideal sheaf on X whose vanishing locus is the set-theoretic support of $\mathcal{F}$.

Well-definedness on basic opens. The coherence condition for an ideal sheaf requires that, for an affine open U and $f \in \mathcal{O}_X(U)$, the section $\mathrm{Ann}(\mathcal{F})(U)$ restricts correctly to the basic open $D(f) \subseteq U$, i.e. that $\mathrm{Ann}_{\mathcal{O}_X(D(f))}(\mathcal{F}(D(f)))$ is the extension of $\mathrm{Ann}_{\mathcal{O}_X(U)}(\mathcal{F}(U))$ along the localization map $\mathcal{O}_X(U) \to \mathcal{O}_X(D(f))$. This is exactly where the two hypotheses enter. The bridge 1259 supplies, for a quasi-coherent finite-type $\mathcal{F}$, that $\mathcal{O}_X(D(f)) = (\mathcal{O}_X(U))_f$ and that the restriction $\mathcal{F}(U) \to \mathcal{F}(D(f))$ exhibits $\mathcal{F}(D(f))$ as the localized module $(\text{powers } f)^{-1}\mathcal{F}(U)$; the algebra engine 1106 then yields, using that $\mathcal{F}(U)$ is finitely generated, the required identity $\mathrm{Ann}_{(\mathcal{O}_X(U))_f}((\text{powers } f)^{-1}\mathcal{F}(U)) = \mathrm{Ann}_{\mathcal{O}_X(U)}(\mathcal{F}(U)) \cdot (\mathcal{O}_X(U))_f$. Thus the local annihilators are compatible with basic-open restriction, which is the patching datum that makes $\mathrm{Ann}(\mathcal{F})$ a well-defined ideal sheaf. The finite-type hypothesis is genuinely required: without finite generation of $\mathcal{F}(U)$ the annihilator need not commute with localization, and the local data fail to patch.

Note. Mathlib does not carry an annihilator for sheaves of modules on a scheme.

Lemma 1103 (Annihilator ideal bounded by the module annihilator on affine opens)**.** *Let X be a scheme, $\mathcal{F}$ a sheaf of $\mathcal{O}_X$-modules and $U \subseteq X$ an affine open. Then the section over U of the annihilator ideal sheaf $\mathrm{Ann}(\mathcal{F})$ (1102) is contained in the module-annihilator of the sections of $\mathcal{F}$ over U:*

$$\mathrm{Ann}(\mathcal{F})(U) \;\subseteq\; \mathrm{Ann}_{\mathcal{O}_X(U)}(\mathcal{F}(U)).$$

This is the inclusion direction available directly from the definition of $\mathrm{Ann}(\mathcal{F})$; the reverse inclusion, under finiteness of the section modules, is 1104. Proved directly in Lean. It supports the well-definedness of the schematic support (1133).

Lemma 1104 (Annihilator ideal sheaf equals the module annihilator on affine opens)**.** *Source: Stacks, Algebra, Lemma* `annihilator-flat-base-change` *(localization instance); Stacks 01IB. Let X be a scheme and $\mathcal{F}$ a quasi-coherent sheaf of $\mathcal{O}_X$-modules such that $\Gamma(\mathcal{F}, V)$ is a finitely generated $\Gamma(X, V)$-module for every affine open $V \subseteq X$. Then for every affine open $U \subseteq X$,*

$$\mathrm{Ann}(\mathcal{F})(U) \;=\; \mathrm{Ann}_{\mathcal{O}_X(U)}(\mathcal{F}(U)).$$

Proof. The family $I(V) := \mathrm{Ann}_{\mathcal{O}_X(V)}(\mathcal{F}(V))$, indexed by the affine opens of X, satisfies the basic-open coherence required of an ideal sheaf: for an affine open V and $f \in \mathcal{O}_X(V)$, the restriction $\mathcal{O}_X(V) \to \mathcal{O}_X(D(f))$ is the localization away from f (1105), the section restriction $\mathcal{F}(V) \to \mathcal{F}(D(f))$ is the corresponding module localization (1259), and — because $\mathcal{F}(V)$ is finitely generated — the annihilator commutes with this localization (1106): $I(D(f)) = I(V) \cdot \mathcal{O}_X(D(f))$. Hence I is itself an ideal sheaf. Since $\mathrm{Ann}(\mathcal{F})$ is by definition the largest ideal sheaf whose affine-open values are contained in the family I (1102), and I is such an ideal sheaf, maximality gives $I(U) \subseteq \mathrm{Ann}(\mathcal{F})(U)$ for every affine U; the reverse inclusion is 1103. ∎

Lemma 1105 (Sections on a basic open localize the affine ring)**.** *Provided by Mathlib* (`Mathlib.AlgebraicGeometry.A` *Let X be a scheme, $U \subseteq X$ an affine open, and $f \in \Gamma(X, U)$. Then the restriction map $\Gamma(X, U) \to \Gamma(X, D(f))$ on the basic open $D(f) = X.\mathrm{basicOpen}\, f$ exhibits $\Gamma(X, D(f))$ as the localization of $\Gamma(X, U)$ away from f; that is, it is $\mathrm{IsLocalization.Away}\, f$, so $\Gamma(X, D(f)) = (\Gamma(X, U))_f$ as a $\Gamma(X, U)$-algebra.*

Lemma 1106 (Annihilator of a finitely generated module commutes with localization)**.** *Source: Stacks, Algebra, Lemma* `annihilator-flat-base-change` *(localization instance for finite modules). Let R be a commutative ring, $S \subseteq R$ a multiplicative submonoid, and R_S the localization of R at S. Let M be a* finitely generated *R-module and let $g : M \to M_S$ exhibit M_S as the localization of M at S (an R_S-module whose R-action is the restriction of the R_S-action along $R \to R_S$). Then*

$$\mathrm{Ann}_{R_S}(M_S) \;=\; \mathrm{Ann}_R(M) \cdot R_S,$$

the extension of the annihilator ideal along the canonical map $R \to R_S$. Finite generation is essential: for a general module only the inclusion $\supseteq$ holds.

Proof. Fix a finite generating set $m_1, \dots, m_n$ of M. An element of R annihilating every m_i annihilates all of M: the elements it kills form a submodule (they are closed under sums and scalar multiples), and that submodule contains the generators.

($\supseteq$) It suffices that the image of $\mathrm{Ann}_R(M)$ annihilates M_S. Every element of M_S has the form $g(m)/s$ with $m \in M$, $s \in S$; for $a \in \mathrm{Ann}_R(M)$, $(a/1) \cdot (g(m)/s) = g(a \cdot m)/s = 0$.

($\subseteq$) Let $y \in \mathrm{Ann}_{R_S}(M_S)$ and write $y = a/s$ with $a \in R$, $s \in S$. For each i the hypothesis gives $y \cdot (g(m_i)/1) = g(a \cdot m_i)/s = 0$ in M_S, so there is $u_i \in S$ with $u_i \cdot a \cdot m_i = 0$ in M. Put $u := u_1 \cdots u_n \in S$; since each u_i divides u in S, $(ua) \cdot m_i = 0$ for every i, hence $ua \in \mathrm{Ann}_R(M)$ by the generating-set criterion. Finally

$$\frac{a}{s} = \frac{ua}{1} \cdot \frac{1}{us}$$

exhibits y as an R_S-multiple of the image of $ua \in \mathrm{Ann}_R(M)$, so $y \in \mathrm{Ann}_R(M) \cdot R_S$. $\square$

Lemma 1107 (Sheaf Mayer–Vietoris: the pullback cone over a two-open cover is a limit)**.** *Provided by Mathlib (`Mathlib.Topology.Sheaves.SheafCondition.PairwiseIntersections`). For a sheaf F on a topological space and two opens U, V, the square relating $F(U \cup V)$, $F(U)$, $F(V)$, $F(U \cap V)$ is a pullback (limit). This is the generic module Mayer–Vietoris used in the inductive step of the descent.*

Lemma 1108 (Separatedness: a section is determined by its restrictions to a cover)**.** *Provided by Mathlib (`Mathlib.Topology.Sheaves.SheafCondition.UniqueGluing`). Let F be a sheaf on a topological space, V an open, and $\{U_i\}$ a family of opens with $\bigsqcup_i U_i = V$. If two sections $s, t \in F(V)$ have equal restrictions $s|_{U_i} = t|_{U_i}$ for every i, then $s = t$. This separatedness over a cover is the uniqueness half of the sheaf condition; it discharges the* exists_of_eq *obligation (a global section vanishing on $D(f)$ is killed by a power of f) and the final "conclude on $D(f)$" step of the cover-form descent (1124).*

Lemma 1109 (Pullback of sheaves of modules along a morphism of schemes)**.** *Provided by Mathlib (`Mathlib.AlgebraicGeometry.Modules.Sheaf`). For a morphism of schemes $f : X \to Y$, Mathlib supplies the pullback functor $f^* : Y.\mathrm{Modules} \to X.\mathrm{Modules}$ on sheaves of modules, left adjoint to pushforward, hence colimit-preserving, and carrying the unit sheaf to the unit sheaf. The project already uses this functor in the definition of local freeness. It is the geometric restriction through which a presentation is transported to an open subscheme.*

Lemma 1110 (Restriction of sheaves of modules along an open immersion)**.** *Provided by Mathlib (`Mathlib.AlgebraicGeometry.Modules.Sheaf`). For an open immersion $f : X \to Y$, Mathlib supplies the restriction functor $f^{\mathrm{res}} : Y.\mathrm{Modules} \to X.\mathrm{Modules}$, with $\Gamma(f^{\mathrm{res}} M, U) = \Gamma(M, f(U))$ definitionally; it is a left adjoint (colimit preserving) and carries the unit sheaf to the unit sheaf. This is the geometric "restrict to open U" functor on* $(\mathrm{Spec}\, R).\mathrm{Modules}$.

Lemma 1111 (Restriction is isomorphic to pullback)**.** *Provided by Mathlib (`Mathlib.AlgebraicGeometry.Modules.` For an open immersion f, the restriction functor (1110) is naturally isomorphic to the pullback functor (1109): $f^{\mathrm{res}} \cong f^*$. This identifies the restriction functor with the universal-property pullback, so a presentation transported along one transports along the other.*

Lemma 1112 (Slice of opens equivalent to opens of the open subspace)**.** *Provided by Mathlib (`Mathlib.Topology.Sheaves.Over`). For a topological space X and an open $U \subseteq X$, the slice category of opens of X over U (the opens of X contained in U, with inclusions) is equivalent to the opens of the subspace U: $(\mathrm{Opens}\, X)_{/U} \simeq \mathrm{Opens}(U)$. Mathlib supplies this underlying equivalence of sites; it leaves the lift to sheaf categories as an explicit TODO.*

Lemma 1113 (Sheaf categories of equivalent sites are equivalent)**.** *Provided by Mathlib (*`Mathlib.CategoryTheory.S`*... Let $e : C \simeq D$ be an equivalence of categories carrying Grothendieck topologies J on C and K on D, and suppose e.*inverse *is a dense subsite for (K, J). Then for any target category A the sheaf categories are equivalent:* $\mathrm{Sheaf}(J, A) \simeq \mathrm{Sheaf}(K, A)$.

Lemma 1114 (The over-equivalence functor is cocontinuous)**.** *Topological layer of bridge C; fills the* `Topology/Sheaves/Over.lean` *TODO. The functor of* Opens.overEquivalence U *(1112) is cocontinuous (cover-lifting) from the U-slice $J_X|_U$ of the opens Grothendieck topology of X to the opens Grothendieck topology J_U of the subspace U.*

Proof. A sieve covers in the slice topology $J_X|_U$ iff its image sieve covers in J_X, and covering for the opens topology is pointwise: a sieve covers iff its members' opens cover the base set-theoretically. Given a point x of a slice object and a cover member in J_U containing the image of x, transport it across the open embedding $U \hookrightarrow X$, which is injective: the set-theoretic image of an open $V \subseteq U$ is an open of X contained in the base, a neighbourhood of x lying in the image sieve. This witnesses the cover-lifting condition. □

Lemma 1115 (The over-equivalence inverse is cocontinuous)**.** *Topological layer of bridge C; fills the* `Topology/Sheaves/Over.lean` *TODO. Symmetrically, the inverse of* Opens.overEquivalence U *is cocontinuous from the opens topology J_U of the subspace U to the U-slice $J_X|_U$.*

Proof. As in 1114, reduce to the pointwise covering criterion and the over-sieve image criterion; given a point and a slice-cover member, pull the member back along the embedding $U \hookrightarrow X$ (preimage of an open of X is an open of the subspace) to obtain a cover member of J_U witnessing the lift. □

Lemma 1116 (The over-equivalence inverse is a dense subsite)**.** *Topological layer of bridge C; the dense-subsite witness assembled from the two cocontinuities. The inverse of* Opens.overEquivalence U *is a dense subsite for $(J_U, J_X|_U)$. This is formal: an equivalence both of whose legs are cocontinuous (1114, 1115) yields a dense subsite on its inverse.*

Lemma 1117 (The over-equivalence functor is continuous)**.** *Topological layer of bridge C; needed for the module-level lift of step 3. The functor of* Opens.overEquivalence U *is continuous (cover-preserving) for $(J_X|_U, J_U)$. It is obtained from the cocontinuity of the inverse (1115) through the equivalence adjunction: the left adjoint of a cocontinuous functor is continuous.*

Lemma 1118 (The over-equivalence inverse is continuous)**.** *Topological layer of bridge C; needed for the module-level lift of step 3. Symmetrically, the inverse of* Opens.overEquivalence U *is continuous for $(J_U, J_X|_U)$, obtained from the cocontinuity of the functor (1114) through the equivalence adjunction.*

Definition 1119 (The over-site/open-subspace sheaf equivalence)**.** *Keystone of the topological layer of bridge C.* For any category A, the site equivalence Opens.overEquivalence U (1112) lifts to an equivalence of sheaf categories

$$\mathrm{Sheaf}(J_X|_U,\ A) \ \simeq\ \mathrm{Sheaf}(J_U,\ A),$$

obtained by feeding the dense-subsite witness (1116) to the general sheaf-congruence construction (1113). This is exactly the equivalence left as a TODO in Mathlib's `Topology/Sheaves/Over.lean`, and it is the object across which 1121 transports the structure sheaf and the module M.

Definition 1120 (The slice-to-geometric equivalence of module categories (gap1, C, step 3)). Let X be a scheme and $U \subseteq X$ an open. The category of sheaves of modules over the sliced structure sheaf $\mathcal{O}_X.\mathrm{over}\, U$ on the slice site $J_X|_U$ is equivalent to the category of sheaves of modules on the open subscheme $U.\mathrm{toScheme}$:

$$\mathrm{SheafOfModules}(\mathcal{O}_X.\mathrm{over}\, U) \;\simeq\; U.\mathrm{toScheme}.\mathrm{Modules}.$$

The equivalence is obtained by lifting the site equivalence Opens.overEquivalence U (1112) — both of whose legs are continuous (1117, 1118) — to module categories through the ringed-site transport of 253. The two ring-sheaf comparison morphisms that transport consumes are the unit of the site equivalence whiskered into the sliced structure presheaf, and the identity: the structure sheaf of the open subscheme is precisely the transport of the sliced structure sheaf across the site equivalence.

Lemma 1121 (The Grothendieck-slice restriction equals the geometric restriction (gap1, C)). *The one lemma that must touch the Grothendieck slice site; everything downstream of it is geometric. Let X be a scheme and $U \subseteq X$ an open. The* abstract Grothendieck-slice *restriction* $M.\mathrm{over}\, U$ *— an object of the category of sheaves of modules over the sliced structure sheaf* $\mathcal{O}_X.\mathrm{over}\, U$ *on the slice site* $J.\mathrm{over}\, U$ *— is canonically isomorphic to the* geometric *restriction* $(\mathrm{restrictFunctor}\, U.\iota).\mathrm{obj}\, M$ *(1110), equivalently to the pullback* $U.\iota^* M$ *(1111), as a sheaf of modules on the small Zariski site of the open subscheme* $U.\mathrm{toScheme}$*. The isomorphism is induced by the equivalence of sites* $J.\mathrm{over}\, U \simeq \mathrm{Opens}(U.\mathrm{toScheme})$ *coming from the opens functor of the open immersion* $U.\iota$*, under which the sliced sheaf of rings* $\mathcal{O}_X.\mathrm{over}\, U$ *corresponds to the structure sheaf of* $U.\mathrm{toScheme}$*. For* $X = \mathrm{Spec}\, R$ *and* $U = D(r)$ *a basic open, the target category is* $(\mathrm{Spec}\, R_r).\mathrm{Modules}$ *under the affine identification* $D(r) \cong \mathrm{Spec}\, R_r$ *(1105).*

Proof. The isomorphism is built in four steps, ascending from the purely topological layer to the module-level statement.

Step 1 (topological layer — done). The opens functor of the open immersion $U.\iota$ realizes the slice of the opens site of X over U as the opens site of the subspace U; on sheaf categories this is the equivalence $\mathrm{Sheaf}(J_X|_U, A) \simeq \mathrm{Sheaf}(J_U, A)$ of 1119, for every target category A. This is the topos-theoretic layer, assembled from the cocontinuity/continuity of both legs of Opens.overEquivalence U (1112).

Step 2 (geometric ring-sheaf identification — the current obstacle). Identify the sliced structure sheaf $\mathcal{O}_X.\mathrm{over}\, U$ with the structure sheaf $\mathcal{O}_U$ of the open subscheme $U.\mathrm{toScheme}$, transported across the RingCat-valued instance of 1119; this is an isomorphism of sheaves of rings. The bridge is the factorization of the opens functor of the immersion,

$$U.\iota.\mathrm{opensFunctor} \;=\; (\mathrm{Opens.overEquivalence}\, U).\mathrm{inverse} \circ \mathrm{Over.forget}\, U,$$

under which the structure-sheaf comparison reduces to the standard open-immersion structure-sheaf identifications; recall that restrictFunctor $U.\iota$ (1110) is precisely pushforward along $U.\iota.\mathrm{opensFunctor}$. This ring-sheaf identification is the remaining geometric obstacle.

Step 3 (lift to modules). Given the step-2 identification of the underlying ring sheaves and the continuity of both legs (1117, 1118), lift the sheaf-of-rings comparison to an equivalence of categories of sheaves of modules via pushforwardPushforwardEquivalence (253). Under this equivalence the sliced module $M.\mathrm{over}\, U$ corresponds to the geometric restriction $(\mathrm{restrictFunctor}\, U.\iota).\mathrm{obj}\, M$.

Step 4 (compose). Compose with restrictFunctorIsoPullback (1111) to land the isomorphism $M.\mathrm{over}\, U \cong M.\mathrm{restrict}\, U.\iota \cong U.\iota^* M$. The two sides live a priori in different module categories, so the literal Lean statement of overRestrictIso is phrased through the step-3 equivalence functor; for $X = \mathrm{Spec}\, R$ and $U = D(r)$ the target is $(\mathrm{Spec}\, R_r).\mathrm{Modules}$ under $D(r) \cong \mathrm{Spec}\, R_r$ (1105). ☐

Lemma 1122 (The slice-to-geometric isomorphism in pullback form (gap1, C, step 4')). *The pullback packaging of the slice-touching bridge; the form consumed by the P1 transport. With X and $U \subseteq X$ as above and M a sheaf of modules on X, the transport of the abstract Grothendieck-slice restriction $M.\mathrm{over}\, U$ under the step-3 equivalence functor (1120) is canonically isomorphic to the inverse-image (pullback) of M along the open immersion $U.\iota$:*

$$(\mathrm{overRestrictEquiv}\, U).\mathrm{functor.obj}\,(M.\mathrm{over}\, U) \;\cong\; (U.\iota^*)\, M$$

(1109). This is the pullback packaging of the slice-touching bridge 1121; it is exactly the form in which a presentation of $M.\mathrm{over}\, U$ is transported into a presentation of the geometric pullback $U.\iota^* M$, and it is the ingredient consumed by the per-element local-tilde transport 1123.

Proof. Compose the slice-to-geometric isomorphism 1121, which identifies the transported slice restriction with the geometric restriction $(\mathrm{restrictFunctor}\, U.\iota).\mathrm{obj}\, M$, with the component at M of the natural isomorphism $\mathrm{restrictFunctorIsoPullback}\, U.\iota$ (1111) identifying the geometric restriction with the pullback $U.\iota^* M$. □

Lemma 1123 (Quasi-coherent modules restrict to a tilde on basic opens of the cover (gap1, P1)). *Let M be a sheaf of modules on $\mathrm{Spec}\, R$ equipped with quasi-coherence data: an open cover $\{X_i\}$ of $\mathrm{Spec}\, R$ together with, for each i, a presentation of the slice restriction $M.\mathrm{over}\, X_i$ — an exhibition of it as the cokernel of a map between coproducts of copies of the unit module. Let $r \in R$ with $D(r) \subseteq X_i$ for some i, and let $W := \iota_{X_i}^{-1}(D(r))$ be the corresponding affine open of the subscheme $Z := X_i$. Then the restriction of M to $D(r)$ — the pullback of M to Z, then to W, transported across the affine identification $W \cong \mathrm{Spec}\, \Gamma(Z, W)$ — has invertible tilde–Γ counit (1256); that is, $M|_{D(r)}$ is the tilde of its section module over $\Gamma(Z, W) \cong R_r$ (1105).*

Proof. Step 1 (geometrize the datum). The quasi-coherence datum supplies a presentation of the slice restriction $M.\mathrm{over}\, X_i$. The slice-to-geometric equivalence (1120) preserves colimits and carries the unit module to the unit module, so the image of this presentation, transported along the isomorphism of 1122, is a presentation of the geometric pullback $N := \iota_{X_i}^* M$ on the scheme Z.

Step 2 (restrict to W). Restriction to the slice site over W again preserves colimits and the unit, so the global presentation of N slices to a presentation of $N.\mathrm{over}\, W$; geometrizing across the slice-to-geometric equivalence at W (1120, 1122) yields a presentation of the geometric restriction $\iota_W^* N$ on the open subscheme W.

Step 3 (affine identification). W is affine: $D(r)$ is a basic open of the affine scheme $\mathrm{Spec}\, R$, and its preimage under the open immersion ι_{X_i} is again affine because $D(r)$ is contained in the immersion's range. Pull the presentation back across the isomorphism $W \cong \mathrm{Spec}\, \Gamma(Z, W)$: the pullback of sheaves of modules along an isomorphism of schemes (1109) preserves colimits, being a left adjoint, and carries the unit module to the unit module, since the induced functor on opens sites is an equivalence. This lands a global presentation of the transported module on the affine scheme $\mathrm{Spec}\, \Gamma(Z, W)$.

Step 4 (a presented module on $\mathrm{Spec}\, S$ is a tilde). Let P be a sheaf of modules on $\mathrm{Spec}\, S$ with a presentation $\mathcal{O}^{\oplus J} \to \mathcal{O}^{\oplus K} \to P \to 0$. The map of coproducts of units corresponds under the tilde–Γ adjunction (1256) to a map of free modules $S^{\oplus J} \to S^{\oplus K}$, because $\mathcal{O}^{\oplus K}$ is the tilde of $S^{\oplus K}$ and the unit of the adjunction is an isomorphism. Since the tilde functor preserves cokernels (it is a left adjoint), P is isomorphic to the tilde of the cokernel of that module map. Finally, a module isomorphic to a tilde has invertible counit: on a tilde $\widetilde{N}$ the triangle identity of the adjunction shows the counit is inverse to the tilde of the unit isomorphism $N \cong \Gamma(\widetilde{N}, \top)$, and invertibility of the counit transports across any isomorphism of modules by naturality. Applying this to the transported module of Step 3 concludes. □

Lemma 1124 (Section-localization descent from per-cover localization data (gap1, D, cover form))**.** *Let M be a sheaf of modules on* $\operatorname{Spec} R$*, $f \in R$, and $t \subseteq R$ a finite subset generating the unit ideal, so that the basic opens $\{D(r)\}_{r\in t}$ cover* $\operatorname{Spec} R$*. Assume the localization datum: for every open U contained in some $D(r)$ with $r \in t$, the section restriction*

$$\Gamma(M, U) \longrightarrow \Gamma(M, D(f) \cap U)$$

exhibits the target as the localization of $\Gamma(M, U)$ at the powers of f, over R. Then the global section restriction $\Gamma(M, \top) \to \Gamma(M, D(f))$ exhibits $\Gamma(M, D(f))$ as the localization $\Gamma(M, \top)[1/f]$ over R.

Proof. We verify the three conditions characterizing the localization map.

Units. Every power of f acts invertibly on $\Gamma(M, D(f))$: the action of f on sections over $D(f)$ is through the restriction of f to $\Gamma(\operatorname{Spec} R, D(f))$, which is a unit there (f is invertible on its own basic open), and multiplication by a unit of the acting ring is bijective. This holds for an arbitrary M, without the covering hypothesis.

Torsion. Let $x \in \Gamma(M, \top)$ restrict to 0 on $D(f)$; we find n with $f^n \cdot x = 0$ (applied to a difference $x_1 - x_2$, this is the uniqueness condition). For each $r \in t$, the restriction $x|_{D(r)}$ restricts to 0 on $D(f) \cap D(r)$ (functoriality of restriction through $D(f)$); the localization datum at $U = D(r)$ then provides k_r with $f^{k_r} \cdot x|_{D(r)} = 0$. Take $n := \max_{r\in t} k_r$; then $(f^n \cdot x)|_{D(r)} = f^{n-k_r} \cdot (f^{k_r} \cdot x|_{D(r)}) = 0$ for every r, and since the $D(r)$ cover $\operatorname{Spec} R$, sheaf separatedness (1108) gives $f^n \cdot x = 0$.

Surjectivity. Let $y \in \Gamma(M, D(f))$; we find a global x and n with $f^n \cdot y = x|_{D(f)}$. For each $r \in t$, the localization datum at $U = D(r)$ writes $f^{m_r} \cdot y|_{D(f)\cap D(r)}$ as the restriction of a section $b_r \in \Gamma(M, D(r))$; normalizing with the common exponent $N := \max_r m_r$ (replace b_r by $f^{N-m_r} \cdot b_r$) we may assume $b_r|_{D(f)\cap D(r)} = f^N \cdot y|_{D(f)\cap D(r)}$ for every r. On a pairwise overlap $D(r) \cap D(r')$ — an open contained in $D(r)$ — the difference $b_r - b_{r'}$ restricts to 0 on $D(f) \cap D(r) \cap D(r')$, since both terms restrict to $f^N \cdot y$ there; the localization datum at $U = D(r) \cap D(r')$ provides a power $f^{P_{r,r'}}$ killing the difference, and taking $P := \max P_{r,r'}$ over the finitely many pairs, the family $\{f^P \cdot b_r\}_{r\in t}$ agrees on all overlaps. By the sheaf axiom (unique gluing over the finite cover $\{D(r)\}_{r\in t}$ of $\operatorname{Spec} R$) the family glues to a global section $x \in \Gamma(M, \top)$. Finally $x|_{D(f)} = f^{N+P} \cdot y$: the two sides agree on each member of the cover $\{D(f) \cap D(r)\}_{r\in t}$ of $D(f)$, so they are equal by separatedness (1108). ☐

Lemma 1125 (Section-localization descent, basic-open hypothesis form (gap1, D))**.** *Let M, f, and t be as in 1124, but assume the localization datum only for* basic *opens: for every $s \in R$ with $D(s) \subseteq D(r)$ for some $r \in t$, the restriction $\Gamma(M, D(s)) \to \Gamma(M, D(f) \cap D(s))$ is the localization of $\Gamma(M, D(s))$ at the powers of f over R. Then the conclusion of 1124 still holds: the global restriction $\Gamma(M, \top) \to \Gamma(M, D(f))$ is the localization at the powers of f over R.*

Proof. The proof of 1124 consults the localization datum only at the cover members $U = D(r)$ ($r \in t$) and at their pairwise intersections $U = D(r) \cap D(r') = D(rr')$ — all of them *basic* opens contained in a cover member. The datum restricted to basic opens therefore suffices, and the same three-part verification applies verbatim. (The restriction of the hypothesis matters: the geometric producer below furnishes the localization datum only at basic opens, a general open of $\operatorname{Spec} R$ being neither affine nor quasi-compact in general.) ☐

20.4.1 Section-transport producer for the basic-open Hfr

This subsection builds the geometric input of the gap1 keystone descent 1128: the geometric *producer* 1127 that manufactures, for a quasi-coherent M on $\operatorname{Spec} R$ and a basic open $D(s)$ inside a quasi-coherence cover member, the per-cover-element localization datum Hfr the basic-open cover form 1125 consumes. The construction transports the R_s-level localization living on the affine slice $D(s) \cong \operatorname{Spec} R_s$ back to an R-level statement through the combiner 1126; the geometric inputs — the iterated-pullback tilde–Γ isomorphism, the identification of the immersion's image opens, and the semilinear section comparisons — enter in the proof of the producer.

Lemma 1126 (Localization transport across a semilinear section comparison)**.** *Let R be a commutative ring, A an R-algebra, S a commutative ring, and $\sigma : S \xrightarrow{\sim} A$ a ring isomorphism; let $f \in R$ and $f' \in S$ with $\sigma(f')$ equal to the image of f in A. Let $g : M_1 \to M_2$ be an S-linear map exhibiting M_2 as the localization of M_1 at the powers of f'; let N_1, N_2 be A-modules, regarded also as R-modules through $R \to A$; let $e_1 : M_1 \to N_1$ and $e_2 : M_2 \to N_2$ be additive isomorphisms that are σ-semilinear, i.e. $e_i(a \cdot x) = \sigma(a) \cdot e_i(x)$ for $a \in S$; and let $h : N_1 \to N_2$ be an A-linear map intertwining g, i.e. $h \circ e_1 = e_2 \circ g$. Then h, regarded as an R-linear map, exhibits N_2 as the localization of N_1 at the powers of f.*

Proof. We check the three localization conditions in two stages.

Stage I: crossing σ. The map h is a localization at the powers of $\sigma(f')$ over A. *Units:* for $k \geq 0$, the semilinearity of e_2 gives that the action of $\sigma(f')^k = \sigma(f'^k)$ on N_2 equals $e_2 \circ (\text{action of } f'^k \text{ on } M_2) \circ e_2^{-1}$, a composite of bijections because f'^k acts invertibly on the localization M_2. *Surjectivity:* given $y \in N_2$, choose $x \in M_1$ and k with $f'^k \cdot e_2^{-1}(y) = g(x)$; applying e_2 and semilinearity, $\sigma(f')^k \cdot y = e_2(g(x)) = h(e_1(x))$. *Uniqueness:* if $h(y_1) = h(y_2)$ then, by the intertwining relation, $e_2(g(e_1^{-1}y_1)) = e_2(g(e_1^{-1}y_2))$, so $g(e_1^{-1}y_1) = g(e_1^{-1}y_2)$; hence $f'^k \cdot e_1^{-1}y_1 = f'^k \cdot e_1^{-1}y_2$ for some k, and applying e_1, $\sigma(f')^k \cdot y_1 = \sigma(f')^k \cdot y_2$.

Stage II: descending to R. By hypothesis $\sigma(f')$ is the image of f under $R \to A$, and by compatibility of the two module structures the action of f^k through R coincides with the action of the image of f^k through A, on N_1 and on N_2. Each of the three conditions at the powers of f over R is therefore literally the corresponding Stage-I condition at the powers of $\sigma(f')$ over A: the units condition transfers because the two actions coincide; surjectivity reads $f^k \cdot y = \sigma(f')^k \cdot y = h(x)$; and uniqueness likewise. ∎

Lemma 1127 (Section-transport producer: basic-open localization inside a cover member (gap1, Hfr))**.** *Let M be a sheaf of modules on $\operatorname{Spec} R$ with quasi-coherence data (an open cover $\{X_i\}$ with per-member presentations, as in 1123), let $f, s \in R$, and suppose $D(s) \subseteq X_i$ for some i. Then the section restriction*

$$\Gamma(M, D(s)) \longrightarrow \Gamma(M, D(f) \cap D(s))$$

exhibits the target as the localization of $\Gamma(M, D(s))$ at the powers of f, over R.

Proof. Let $W := \iota_{X_i}^{-1}(D(s))$ be the affine open of the subscheme $Z := X_i$ corresponding to $D(s)$, put $S := \Gamma(Z, W)$, and let $j : \operatorname{Spec} S \to \operatorname{Spec} R$ be the composite open immersion $\operatorname{Spec} S \cong W \hookrightarrow Z \hookrightarrow \operatorname{Spec} R$ (the affine identification of W followed by the two canonical immersions). Its image is $D(s)$: the first leg is an isomorphism onto W, and $\iota_{X_i}(W) = X_i \cap D(s) = D(s)$.

Write $M' := j^* M$ (1109). The per-element transport 1123 gives invertibility of the tilde–Γ counit for the iterated pullback of M along the three legs of j; pullback along a composite agrees with the composite of the pullbacks up to canonical natural isomorphism, and invertibility of the counit is invariant under isomorphism of modules (the essential image of the tilde functor is closed under isomorphism). Hence the counit of M' is invertible, and by 1129, for every $f' \in S$ the restriction $\Gamma(M', \top) \to \Gamma(M', D(f'))$ is the localization at the powers of f' over S.

Let $\sigma : S \xrightarrow{\sim} \Gamma(\operatorname{Spec} R, D(s))$ be the ring comparison of the open immersion j on top sections ($S \cong \Gamma(\operatorname{Spec} S, \top) \cong \Gamma(\operatorname{Spec} R, j(\top))$, and $j(\top) = D(s)$), and choose $f' := \sigma^{-1}(f|_{D(s)})$.

The image $j(D(f'))$ is the locus of $D(s)$ where $\sigma(f') = f|_{D(s)}$ is invertible, that is, $j(D(f')) = D(f) \cap D(s)$.

Since j is an open immersion, applying the section functor to the identification of the pullback with the restriction (1111) gives additive isomorphisms $e_1 : \Gamma(M', \top) \cong \Gamma(M, D(s))$ and $e_2 : \Gamma(M', D(f')) \cong \Gamma(M, D(f) \cap D(s))$ which intertwine the two restriction maps (naturality of the identification in the open) and are σ-semilinear over the structure-sheaf comparisons: the action on the sections of the pullback is through the immersion's section-ring isomorphism.

Finally apply the combiner 1126 with $A := \Gamma(\operatorname{Spec} R, D(s))$ — an R-algebra through restriction of global sections — the isomorphism σ, the elements f and f' (so that $\sigma(f')$ is the image of f in A), the localization $g := (\Gamma(M', \top) \to \Gamma(M', D(f')))$, the comparisons e_1, e_2, and $h := (\Gamma(M, D(s)) \to \Gamma(M, D(f) \cap D(s)))$: the conclusion is that h is the localization at the powers of f over R. ☐

Lemma 1128 (Section-localization descent for quasi-coherent modules (gap1 keystone))**.** *For a quasi-coherent sheaf of modules M on* $\operatorname{Spec} R$ *and $f \in R$, the section restriction* $\Gamma(M, \top) \to \Gamma(M, D(f))$ *exhibits the target as the localization* $\Gamma(M, \top)[1/f]$ *over R. This is the statement of 1259 at the affine open $\top$ of* $\operatorname{Spec} R$*, proved here by finite-cover descent — not through the affine equivalence of quasi-coherent modules with modules, which is instead* derived *from this lemma in 1131.*

Proof. Quasi-coherence provides a cover $\{X_i\}$ of $\operatorname{Spec} R$ with per-member presentations. Refine it to a finite basic-open cover: the basic opens form a basis of $\operatorname{Spec} R$, so every point has a basic-open neighbourhood contained in some X_i; quasi-compactness of $\operatorname{Spec} R$ extracts a finite family $t \subseteq R$ with each $D(r)$, $r \in t$, contained in some X_i, and covering is equivalent to t generating the unit ideal. For every basic open $D(u)$ contained in some $D(r)$ with $r \in t$ — hence contained in a cover member X_i — the producer 1127 supplies the localization datum at $D(u)$. The basic-open cover form 1125 then yields the conclusion. ☐

20.4.2 G1-assemble: from per-basic-open section localization to the $\widetilde{(-)}$-counit

This subsection records the bridge *section localization on every basic open ⇒ invertible tilde–Γ counit*: given that the section restriction localizes on every basic open, the tilde–Γ counit is an isomorphism. Together with its converse 1129 it assembles the equivalence 1130, which is exactly what makes G1-core (1132) and gap1 (1131) interderivable – the route by which G1-core is obtained as a corollary of gap1. The supporting Mathlib facts enter through the tilde–Γ adjunction anchor (1256).

Lemma 1129 (Basic-open section localization for a module with invertible tilde–Γ counit)**.** *Source: Stacks, Schemes, Lemma `widetilde-pullback` (sections of a tilde over a basic open are the localized module). Let M be a sheaf of modules on* $\operatorname{Spec} R$ *whose tilde–Γ counit* $\Gamma\widetilde{(M, \top)} \to M$ *(1256) is an isomorphism. Then for every $f \in R$ the section restriction* $\Gamma(M, \top) \to \Gamma(M, D(f))$ *exhibits the target as the localization of* $\Gamma(M, \top)$ *at the powers of f, over R.*

Proof. First let $M = \widetilde{N}$ be a tilde. The unit $N \to \Gamma(\widetilde{N}, \top)$ of the adjunction (1256) is an isomorphism, and the canonical map $N \to \Gamma(\widetilde{N}, D(f))$ exhibits the target as the localized module N_f — the sections of a tilde over a basic open are the localization. The canonical map factors as the unit followed by the section restriction, so the restriction $\Gamma(\widetilde{N}, \top) \to \Gamma(\widetilde{N}, D(f))$ is the composite of the inverse unit isomorphism with a localization map, hence itself a localization at the powers of f.

For general M, put $N := \Gamma(M, \top)$. The counit isomorphism $\widetilde{N} \cong M$ induces, naturally in the open, R-linear isomorphisms $\Gamma(\widetilde{N}, \top) \cong \Gamma(M, \top)$ and $\Gamma(\widetilde{N}, D(f)) \cong \Gamma(M, D(f))$ intertwining the two restriction maps. Pre- and post-composing a localization map with R-linear isomorphisms compatible in this way preserves the localization property. $\square$

Lemma 1130 (Characterization of the invertible tilde–Γ counit by section localization)**.** *For a sheaf of modules M on $\operatorname{Spec} R$, the tilde–Γ counit $\widetilde{\Gamma(M, \top)} \to M$ is an isomorphism if and only if for every $f \in R$ the section restriction $\Gamma(M, \top) \to \Gamma(M, D(f))$ exhibits the target as the localization of $\Gamma(M, \top)$ at the powers of f over R.*

Proof. The forward direction is 1129.

For the converse, write $N := \Gamma(M, \top)$ and let $\varepsilon : \widetilde{N} \to M$ be the counit. Fix $f \in R$. The $D(f)$-component of ε is a map between two localizations of N at the powers of f: the source $\Gamma(\widetilde{N}, D(f))$ receives the canonical map from N (sections of a tilde over a basic open, as in the forward direction), and the target $\Gamma(M, D(f))$ receives the restriction $N = \Gamma(M, \top) \to \Gamma(M, D(f))$, a localization by hypothesis. The component intertwines these two maps out of N: the counit composed with the canonical section map is the section restriction, by the compatibility of the counit with the adjunction unit (1256) and naturality. A map between two localizations of the same module at the same multiplicative set commuting with the two localization maps is the canonical isomorphism between them, hence bijective. Thus ε is an isomorphism on every basic open.

Since the basic opens form a basis of $\operatorname{Spec} R$, ε is an isomorphism on every stalk: injective, because a germ in the kernel is represented by a section over some basic neighbourhood killed by ε there, hence zero; surjective, because every germ of M is the germ of a section over some basic open, which is hit by the corresponding section of $\widetilde{N}$. A morphism of sheaves that is an isomorphism on all stalks is an isomorphism, and a morphism of sheaves of modules whose underlying morphism of sheaves is an isomorphism is itself an isomorphism. $\square$

Lemma 1131 (A quasi-coherent module on an affine scheme is the tilde of its sections (gap1, Stacks 01I8))**.** *Source: Stacks 01I8 (`lemma-quasi-coherent-affine`). For a quasi-coherent sheaf of modules M on $\operatorname{Spec} R$, the tilde–Γ counit $\widetilde{\Gamma(M, \top)} \to M$ is an isomorphism; equivalently, M lies in the essential image of the tilde functor, i.e. M is isomorphic to the tilde of an R-module.*

Proof. By the keystone descent 1128, every basic-open section restriction $\Gamma(M, \top) \to \Gamma(M, D(f))$ is the localization at the powers of f. The reverse implication of 1130 concludes. $\square$

Lemma 1132 (Affine section localization for quasi-coherent modules (G1-core))**.** *For a quasi-coherent sheaf of modules M on $\operatorname{Spec} R$ and $f \in R$, the section restriction $\Gamma(M, \top) \to \Gamma(M, D(f))$ exhibits the target as the localization $\Gamma(M, \top)[1/f]$ over R. This is the statement of 1128 in its clean named form — obtained here as a corollary of gap1 rather than by re-running the descent — and the affine instance ($X = \operatorname{Spec} R$, $U = \top$) of the general-scheme keystone 1259.*

Proof. Gap1 (1131) makes the tilde–Γ counit of M an isomorphism, and 1129 applies. $\square$

Definition 1133 (Schematic support of a sheaf of modules)**.** *Source: [Nitsure], §1 (FGA Explained Ch. 5).* Let $\mathcal{F}$ be a sheaf of modules on a scheme X. The *schematic support* of $\mathcal{F}$ is the closed subscheme of X cut out by its annihilator ideal sheaf (1102):

$$\operatorname{schematicSupport}(\mathcal{F}) \;=\; V(\operatorname{Ann}(\mathcal{F})).$$

This realises Nitsure's "schematic support of $\mathcal{F}$" as a closed subscheme of X.

Definition 1134 (Schematic-support immersion)**.** Let $\mathcal{F}$ be a sheaf of modules on a scheme X. The *schematic-support immersion* is the canonical closed immersion

$$\mathrm{schematicSupport}(\mathcal{F}) \hookrightarrow X$$

of the schematic support (1133) into the ambient scheme, namely the subscheme immersion of the annihilator ideal sheaf $\mathrm{Ann}(\mathcal{F})$ (1102). It is a quasi-compact preimmersion onto the support, and it is the morphism whose composite with a base morphism $f : X \to S$ defines proper support (1136).

Lemma 1135 (Properness as a morphism property)**.** *Provided by Mathlib (`Mathlib.AlgebraicGeometry.Morphisms` A morphism $f : X \to S$ of schemes is* proper *iff it is separated, universally closed, and locally of finite type. Properness is stable under base change: if $f : X \to S$ is proper and $g : T \to S$ is any morphism then the base change $X \times_S T \to T$ is proper. Consequently the base-change clause that Nitsure invokes ("properness … preserved by base-change") applies directly to the proper-support condition below. Moreover properness is stable under composition; closed immersions are proper (they are finite morphisms); and cancellation holds along separated morphisms: if a composite $g \circ f$ is proper and g is separated, then f is proper.*

Definition 1136 (Proper support over the base)**.** *Source: [Nitsure], §1 (FGA Explained Ch. 5).* Let $\mathcal{F}$ be a sheaf of modules on a scheme X and let $f : X \to S$ be a morphism. We say $\mathcal{F}$ *has proper support over S (along f)* when the composite morphism

$$\mathrm{schematicSupport}(\mathcal{F}) \hookrightarrow X \xrightarrow{f} S$$

is proper (1135). Since properness is stable under base change (1135), this condition is preserved by pull-back along any $S' \to S$, which is exactly what the well-definedness of the Quot functor's pullback action requires.

Definition 1137 (Locally free of rank d)**.** *Source: [Nitsure], §1, Exercise (2) (FGA Explained Ch. 5).* Let $\mathcal{M}$ be a sheaf of modules on a scheme X and let $d \in \mathbb{N}$. We say $\mathcal{M}$ is *locally free of rank d* when there exists an open cover $\{U_i\}$ of X together with isomorphisms $\mathcal{M}|_{U_i} \cong \mathcal{O}_{U_i}^{\oplus d}$ for each i.

Note. Mathlib does not provide this predicate.

20.5 The Quot functor

Definition 1138 (Quot functor)**.**

Source: [Nitsure], §1 (FGA Explained Ch. 5). Let S be a locally noetherian scheme, $\pi : X \to S$ a morphism locally of finite type, L a line bundle on X, E a coherent sheaf on X, and $\Phi \in \mathbb{Q}[\lambda]$. The *Quot functor with Hilbert polynomial* Φ

$$\mathrm{Quot}^{\Phi,L}_{E/X/S} : (\mathrm{Sch}/S)^{op} \longrightarrow \mathbf{Set}$$

sends an S-scheme $T \to S$ to the set of equivalence classes $\langle \mathcal{F}, q\rangle$ of pairs $(\mathcal{F}, q)$ where

1. $\mathcal{F}$ is a coherent sheaf on $X_T = X \times_S T$ whose schematic support is proper over T (1136) and which is flat over T;

2. $q : E_T \twoheadrightarrow \mathcal{F}$ is a surjective $\mathcal{O}_{X_T}$-linear homomorphism, where E_T is the pull-back of E along $X_T \to X$;

3. for every $t \in T$ the Hilbert polynomial of $\mathcal{F}|_{X_t}$ with respect to $L|_{X_t}$ (1018) equals Φ.

Two pairs $(\mathcal{F}, q)$ and $(\mathcal{F}', q')$ are equivalent iff $\ker(q) = \ker(q')$. On morphisms $f : T' \to T$ over S, the functor acts by pulling back the quotient along $X_{T'} \to X_T$ (well-defined since tensor product is right-exact and preserves both flatness and properness). The Hilbert scheme is the special case $\mathrm{Hilb}^{\Phi,L}_{X/S} := \mathrm{Quot}^{\Phi,L}_{\mathcal{O}_X/X/S}$.

20.5.1 Base-change well-definedness of the Quot functor

The pullback action of the Quot functor (1138) along an S-morphism $T' \to T$ is well defined because each of the four conditions on a family of quotients is preserved by base change; this is the content of the following four lemmas. Throughout, write $g' : X_{T'} \to X_T$ for the induced morphism on relative products, which is cartesian over $T' \to T$.

Lemma 1139 (Pullback preserves finite presentation)**.** *Source: [Nitsure], §1 (FGA Explained Ch. 5). Let $g : Y \to X$ be a morphism of schemes and $\mathcal{F}$ a finitely presented sheaf of $\mathcal{O}_X$-modules (locally the cokernel of a map of finite free modules). Then $g^*\mathcal{F}$ is a finitely presented sheaf of $\mathcal{O}_Y$-modules.*

Proof. Choose an open cover $\{U_i\}$ of X on whose members $\mathcal{F}$ admits presentations $\mathcal{O}_{U_i}^{\oplus b_i} \to \mathcal{O}_{U_i}^{\oplus a_i} \to \mathcal{F}|_{U_i} \to 0$ with a_i, b_i finite. The preimages $g^{-1}(U_i)$ cover Y. The pullback functor is a left adjoint, hence right exact, and it sends $\mathcal{O}_X$ to $\mathcal{O}_Y$ and finite free modules to finite free modules of the same rank; moreover restriction to $g^{-1}(U_i)$ of the pullback is the pullback of the restriction along $g^{-1}(U_i) \to U_i$. Applying the restricted pullback to the chosen presentation therefore yields a presentation $\mathcal{O}^{\oplus b_i} \to \mathcal{O}^{\oplus a_i} \to (g^*\mathcal{F})|_{g^{-1}(U_i)} \to 0$ with the same finite index sets. □

Lemma 1140 (Flatness over the base is stable under base change)**.** *Source: [Nitsure], §1 (FGA Explained Ch. 5); Stacks 01U9. Let*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ \downarrow f' & & \downarrow f \\ S' & \xrightarrow{g} & S \end{array}$$

be a cartesian square of schemes and $\mathcal{F}$ a quasi-coherent sheaf of $\mathcal{O}_X$-modules that is flat over S in the sense of 921. Then $g'^*\mathcal{F}$ is flat over S'.

Proof. The pullback $g'^*\mathcal{F}$ is quasi-coherent (507). For a quasi-coherent module, flatness over the base in the sense of 921 is an affine-local condition: if every point of X' has an affine chart $W \subseteq f'^{-1}(U_t)$ (with $U_t \subseteq S'$ affine) on which $\Gamma(g'^*\mathcal{F}, W)$ is flat over $\Gamma(S', U_t)$, then the same holds at *every* affine pair $V' \subseteq f'^{-1}(U')$. Indeed, given such a pair, cover V' by opens that are simultaneously basic in V' and in one of the charts W; on such an open the sections of the quasi-coherent module $g'^*\mathcal{F}$ are a localization of $\Gamma(g'^*\mathcal{F}, W)$ along the algebra $\Gamma(X', W) \to \Gamma(X', W_b)$, and flatness over the base ring is preserved by such localizations of the module and by the base exchanges $\Gamma(S', U_t) \rightsquigarrow \Gamma(S', U')$ along basic-open localizations of the base; the local-to-global flatness criterion along a spanning family of $\Gamma(X', V')$ then yields flatness of $\Gamma(g'^*\mathcal{F}, V')$ over $\Gamma(S', U')$ (Stacks 00HT-style affine locality).

It remains to produce the charts. Given $x' \in X'$, choose an affine $U \subseteq S$ around the common image $f(g'(x')) = g(f'(x'))$, an affine $V \subseteq f^{-1}(U)$ around $g'(x')$, and an affine $U_t \subseteq g^{-1}(U)$ around $f'(x')$. The piece $W := g'^{-1}(V) \cap f'^{-1}(U_t)$ is an affine open of X': it is the fibre product

$V \times_U U_t$ of the restricted cartesian square. Taking sections, $\Gamma(X', W) = \Gamma(X, V) \otimes_{\Gamma(S,U)} \Gamma(S', U_t)$ is a pushout of rings, and by quasi-coherence of $\mathcal{F}$ the sections of the pullback on the piece are the base change

$$\Gamma(g'^*\mathcal{F}, W) \;=\; \Gamma(X', W) \otimes_{\Gamma(X,V)} \Gamma(\mathcal{F}, V) \;=\; \Gamma(\mathcal{F}, V) \otimes_{\Gamma(S,U)} \Gamma(S', U_t).$$

The hypothesis at the affine pair (U, V) gives $\Gamma(\mathcal{F}, V)$ flat over $\Gamma(S, U)$, and flatness is stable under base change of the ring, so $\Gamma(g'^*\mathcal{F}, W)$ is flat over $\Gamma(S', U_t)$, as required. □

Lemma 1141 (Module finiteness is local along a spanning family)**.** *Provided by Mathlib (`Mathlib.RingTheory.Local*
Let R be a commutative ring, M an R-module, and $t \subseteq R$ a subset generating the unit ideal. If for every $g \in t$ the localized module M_g is a finite R_g-module, then M is a finite R-module.

Lemma 1142 (Finitely presented modules have finite sections on every affine open)**.** *Source: Stacks 01PC (finite-type half). Let $\mathcal{F}$ be a finitely presented sheaf of $\mathcal{O}_X$-modules on a scheme X. Then for every affine open $V \subseteq X$ the section module $\Gamma(\mathcal{F}, V)$ is a finite $\Gamma(X, V)$-module.*

Proof. Let $t \subseteq \Gamma(X, V)$ be the set of all f such that $D(f) \subseteq V$ and $\Gamma(\mathcal{F}, D(f))$ is a finite $\Gamma(X, D(f))$-module.

The basic opens of t cover V. Fix $x \in V$. By 1270 there is an affine open chart W with $x \in W \subseteq V$ and $\Gamma(\mathcal{F}, W)$ finite over $\Gamma(X, W)$. Since the basic opens of the affine V form a basis of its topology, there is $f \in \Gamma(X, V)$ with $x \in D(f) \subseteq W$. Writing $f|_W$ for the restriction of f to W, one has $D(f) = D(f|_W)$ as opens contained in W; the restriction $\Gamma(X, W) \to \Gamma(X, D(f))$ is the localization away from $f|_W$ (1105), and $\Gamma(\mathcal{F}, W) \to \Gamma(\mathcal{F}, D(f))$ is the corresponding module localization (1259, applicable because a finitely presented module is quasi-coherent). The images of a finite generating set of $\Gamma(\mathcal{F}, W)$ generate the localized module over the localized ring, so $\Gamma(\mathcal{F}, D(f))$ is finite over $\Gamma(X, D(f))$ and $f \in t$, with $x \in D(f)$.

Conclusion. The basic opens $D(f)$, $f \in t$, cover the affine V, so t generates the unit ideal of $\Gamma(X, V)$. For each $f \in t$ the restriction $\Gamma(\mathcal{F}, V) \to \Gamma(\mathcal{F}, D(f))$ is the localization of $\Gamma(\mathcal{F}, V)$ away from f (1259), with target a finite $\Gamma(X, D(f)) = \Gamma(X, V)_f$-module by membership in t; 1141 glues these into finiteness of $\Gamma(\mathcal{F}, V)$ over $\Gamma(X, V)$. □

Lemma 1143 (Annihilators are stable under pullback: section level)**.** *Source: Stacks 056H; [Nitsure], §1. Let $g' : X' \to X$ be a morphism of schemes, $\mathcal{F}$ a quasi-coherent sheaf of $\mathcal{O}_X$-modules, $U \subseteq X$ an affine open, and $V' \subseteq g'^{-1}(U)$ any open of X'. If $r \in \mathcal{O}_X(U)$ annihilates $\Gamma(\mathcal{F}, U)$, then its image in $\mathcal{O}_{X'}(V')$ under $g'^\sharp$ and restriction annihilates $\Gamma(g'^*\mathcal{F}, V')$.*

Proof. For a sheaf of modules $\mathcal{M}$ on a scheme Z and a global section $w \in \mathcal{O}_Z(Z)$, scalar multiplication by w is an endomorphism $w\cdot : \mathcal{M} \to \mathcal{M}$: the componentwise multiplications by the restrictions of w are linear because the section rings are commutative, and they commute with the restriction maps because restriction is semilinear. The proof transports the vanishing of this endomorphism.

Pullback intertwines the multiplication endomorphisms. For a morphism $h : Z' \to Z$ one has $h^*(w\cdot) = (h^\sharp w)\cdot$ as endomorphisms of $h^*\mathcal{M}$. Indeed, under the adjunction between h^* and h_* (1109) both sides transpose to morphisms $\mathcal{M} \to h_*h^*\mathcal{M}$, and these transposes agree: the adjunction unit $\eta : \mathcal{M} \to h_*h^*\mathcal{M}$ is a morphism of $\mathcal{O}_Z$-modules, and the $\mathcal{O}_Z$-module structure of $h_*h^*\mathcal{M}$ is restriction of scalars along $h^\sharp$, so both transposes send a section m over $O \subseteq Z$ to $(h^\sharp(w|_O)) \cdot \eta(m) = (w|_O) \cdot \eta(m)$.

On an affine scheme, killing global sections kills all sections. Let Z be affine, $\mathcal{M}$ quasi-coherent, and suppose $w \cdot \Gamma(\mathcal{M}, Z) = 0$. On a basic open $D(b)$ the restriction $\Gamma(\mathcal{M}, Z) \to \Gamma(\mathcal{M}, D(b))$ is the localization away from b (1259); for $y \in \Gamma(\mathcal{M}, D(b))$ there are n and $m \in$

$\Gamma(\mathcal{M}, Z)$ with $b^n y = m|_{D(b)}$, whence $b^n(wy) = w|_{D(b)} \cdot m|_{D(b)} = (w \cdot m)|_{D(b)} = 0$; since b acts invertibly on the localized sections, $wy = 0$. A general open of Z is covered by basic opens and the abelian sheaf underlying $\mathcal{M}$ is separated, so multiplication by (the restriction of) w kills all sections of $\mathcal{M}$.

Conclusion. Restrict $\mathcal{F}$ to the affine open subscheme U, i.e. form $\iota_U^* \mathcal{F}$ for the open immersion $\iota_U : U \to X$; it is quasi-coherent (507), and the global section $r_0 \in \mathcal{O}_U(U)$ corresponding to r kills $\Gamma(\iota_U^* \mathcal{F}, U) = \Gamma(\mathcal{F}, U)$. By the affine step, $r_0 \cdot$ is the zero endomorphism of $\iota_U^* \mathcal{F}$. Let $h : V' \to U$ be the restriction of g', defined since $V' \subseteq g'^{-1}(U)$; by the intertwining step $(h^\sharp r_0)\cdot$ is the zero endomorphism of $h^* \iota_U^* \mathcal{F}$, which is identified with $\iota_{V'}^*(g'^* \mathcal{F})$ by pseudofunctoriality of pullback along $\iota_U \circ h = g' \circ \iota_{V'}$ (1109). Reading off global sections over V' — the identification $\Gamma(\iota_{V'}^* g'^* \mathcal{F}, V') = \Gamma(g'^* \mathcal{F}, V')$ is equivariant for the ring identification $\mathcal{O}_{V'}(V') = \mathcal{O}_{X'}(V')$, which carries $h^\sharp r_0$ to the image of r — the image of r annihilates $\Gamma(g'^* \mathcal{F}, V')$. ☐

Lemma 1144 (The annihilator ideal sheaf pulls back)**.** *Source: Stacks 056H. Let $g' : X' \to X$ be a morphism of schemes and $\mathcal{F}$ a finitely presented sheaf of $\mathcal{O}_X$-modules. Write $\iota' : Z' \to X'$ for the schematic-support immersion of $g'^* \mathcal{F}$ (1134). Then* $\operatorname{Ann}(\mathcal{F})$ *is contained in the kernel ideal sheaf of $\mathcal{O}_X \to (\iota' \circ g')_* \mathcal{O}_{Z'}$; equivalently, $g' \circ \iota'$ factors through the schematic support $Z \subseteq X$ of $\mathcal{F}$ by a morphism $l : Z' \to Z$ over g'.*

Proof. The kernel ideal sheaf of $(\iota' \circ g')^\sharp$ is the largest ideal sheaf whose value on each affine open $U \subseteq X$ is contained in $\ker((\iota' \circ g')^\sharp_U)$, so it suffices to prove, for each affine open U and each $r \in \operatorname{Ann}(\mathcal{F})(U)$, that $(\iota' \circ g')^\sharp_U(r) = 0$ in $\Gamma(Z', (\iota' \circ g')^{-1}(U))$. By 1103, r annihilates $\Gamma(\mathcal{F}, U)$.

The open $(\iota')^{-1}(g'^{-1}(U))$ of Z' is covered by the preimages $(\iota')^{-1}(V')$ of the affine opens $V' \subseteq g'^{-1}(U)$ of X', and the structure sheaf of Z' is separated, so it suffices that the restriction of $(\iota' \circ g')^\sharp_U(r)$ to each $(\iota')^{-1}(V')$ vanishes. By naturality of $\iota'^\sharp$, that restriction is $\iota'^\sharp_{V'}(\rho)$, where $\rho \in \mathcal{O}_{X'}(V')$ is the image of r. By 1143, ρ annihilates $\Gamma(g'^* \mathcal{F}, V')$. The pullback $g'^* \mathcal{F}$ is finitely presented (1139), hence quasi-coherent with finite section modules on all affine opens (1142); therefore $\operatorname{Ann}_{\mathcal{O}_{X'}(V')} \Gamma(g'^* \mathcal{F}, V') = \operatorname{Ann}(g'^* \mathcal{F})(V')$ (1104), and on affine opens the subscheme immersion of an ideal sheaf kills exactly that ideal, so $\iota'^\sharp_{V'}(\rho) = 0$, as required. Finally, a morphism to X under whose structure map the defining ideal sheaf of the closed subscheme Z dies factors through Z; this produces $l : Z' \to Z$ with $\iota \circ l = g' \circ \iota'$. ☐

Lemma 1145 (Proper support is stable under base change)**.** *Source: [Nitsure], §1 (FGA Explained Ch. 5); Stacks 01W4, 056H. In the cartesian square of 1140, let $\mathcal{F}$ be a finitely presented sheaf of $\mathcal{O}_X$-modules whose schematic support is proper over S (1136). Then the schematic support of $g'^* \mathcal{F}$ is proper over S'.*

Proof. Write $\iota : Z \to X$ and $\iota' : Z' \to X'$ for the schematic-support immersions (1134) of $\mathcal{F}$ and of $g'^* \mathcal{F}$. By 1144 there is a morphism $l : Z' \to Z$ with $\iota \circ l = g' \circ \iota'$.

Form the fibre product $W := Z \times_X X'$ with projections $p : W \to Z$ and $q : W \to X'$. Pasting its defining cartesian square vertically with the given cartesian square exhibits W, together with p and $f' \circ q$, as the fibre product of $Z \xrightarrow{f \circ \iota} S \xleftarrow{g} S'$. The hypothesis says $f \circ \iota$ is proper, and properness is stable under base change (1135), so $f' \circ q : W \to S'$ is proper.

The pair (l, ι') induces $c : Z' \to W$ with $q \circ c = \iota'$. The immersion ι' is a closed immersion, hence proper; and q is a closed immersion (the base change of the closed immersion ι along g'), hence separated. By cancellation of properness along separated morphisms applied to $q \circ c = \iota'$ (1135), c is proper. Finally

$$f' \circ \iota' \;=\; (f' \circ q) \circ c$$

is a composite of proper morphisms, hence proper (1135); that is, the schematic support of $g'^* \mathcal{F}$ is proper over S'. ☐

Lemma 1146 (Pullback commutes with tensor product and twists)**.** *Source: [Stacks Project], "Sheaves of Modules", Tag 01CD (pullback and tensor products); the stalkwise description is Tags 01CB and 0098. Let $f : Y \to X$ be a morphism of schemes and let $\mathcal{F}$, L be quasi-coherent sheaves of $\mathcal{O}_X$-modules. For every $m \in \mathbb{N}$ there is an isomorphism of $\mathcal{O}_Y$-modules*

$$f^*(\mathcal{F} \otimes_{\mathcal{O}_X} L^{\otimes m}) \;\cong\; f^*\mathcal{F} \otimes_{\mathcal{O}_Y} (f^*L)^{\otimes m}.$$

(The fact holds for arbitrary sheaves of modules on ringed spaces, with a canonical comparison isomorphism; the quasi-coherent case stated here is the one consumed by 1158.)

Proof. The twist $\mathcal{F} \otimes L^{\otimes m}$ is built from the binary tensor product (1279) and the unit $\mathcal{O}_X$ by the recursion $L^{\otimes 0} = \mathcal{O}_X$, $L^{\otimes(m+1)} = L^{\otimes m} \otimes L$ (1280), so by induction on m it suffices that the pullback functor carries $\mathcal{O}_X$ to $\mathcal{O}_Y$ — the unitality of 1109 — and commutes with the binary tensor product up to natural isomorphism, which is 1147. The induction is carried out in Lean (the base and step assembled from those two isomorphisms and the tensor congruence `sheafTensorObjCongr`); only the binary comparison of 1147 remains as an open leaf. □

Lemma 1147 (The canonical pullback–tensor comparison is an isomorphism)**.** *Source: [Stacks Project], "Sheaves of Modules", Tag 01CD (pullback and tensor products); the stalkwise description is Tags 01CB and 0098. Let $f : Y \to X$ be a morphism of schemes and let $\mathcal{F}$, $\mathcal{G}$ be sheaves of $\mathcal{O}_X$-modules. The canonical comparison morphism*

$$f^*(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G}) \;\longrightarrow\; f^*\mathcal{F} \otimes_{\mathcal{O}_Y} f^*\mathcal{G}$$

(the sheafified mate of the lax monoidal structure of f_*) is an isomorphism. The statement holds for arbitrary sheaves of modules on ringed spaces; no quasi-coherence hypothesis is needed.

Proof. For every $\mathcal{O}_Y$-module $\mathcal{H}$ there are isomorphisms, natural in all three variables:

$$\begin{aligned}\mathrm{Hom}_{\mathcal{O}_Y}(f^*(\mathcal{F} \otimes \mathcal{G}), \mathcal{H}) &\cong \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{F} \otimes \mathcal{G}, f_*\mathcal{H}) \cong \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{F}, \mathcal{H}om_{\mathcal{O}_X}(\mathcal{G}, f_*\mathcal{H}))\\ &\cong \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{F}, f_*\, \mathcal{H}om_{\mathcal{O}_Y}(f^*\mathcal{G}, \mathcal{H})) \cong \mathrm{Hom}_{\mathcal{O}_Y}(f^*\mathcal{F}, \mathcal{H}om_{\mathcal{O}_Y}(f^*\mathcal{G}, \mathcal{H}))\\ &\cong \mathrm{Hom}_{\mathcal{O}_Y}(f^*\mathcal{F} \otimes f^*\mathcal{G}, \mathcal{H}),\end{aligned}$$

by, in order: the pullback–pushforward adjunction; the tensor–hom adjunction for $\mathcal{O}_X$-modules; the isomorphism $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{G}, f_*\mathcal{H}) \cong f_*\, \mathcal{H}om_{\mathcal{O}_Y}(f^*\mathcal{G}, \mathcal{H})$ — over an open $V \subseteq X$ both sides compute $\mathrm{Hom}_{\mathcal{O}_{f^{-1}V}}((f^*\mathcal{G})|_{f^{-1}V}, \mathcal{H}|_{f^{-1}V})$ by the pullback–pushforward adjunction for the restricted morphism $f^{-1}V \to V$, since pullback commutes with restriction to opens, and these identifications are compatible with further restriction; the pullback–pushforward adjunction again; and the tensor–hom adjunction for $\mathcal{O}_Y$-modules. By the Yoneda lemma the composite isomorphism of functors in $\mathcal{H}$ is induced by a unique isomorphism $f^*(\mathcal{F} \otimes \mathcal{G}) \cong f^*\mathcal{F} \otimes_{\mathcal{O}_Y} f^*\mathcal{G}$, and it is inverse to the canonical comparison map. □

The following bricks are algebraic building blocks for 1157 below (its proof consumes them only through the two general facts already cited there, 1145 and flat base change for cohomology; the bricks below are project infrastructure for a from-scratch Lean reduction of that same statement along a different route, factoring through the schematic support of the twisted fibre module and its sections, rather than through the pole-divisor style argument summarised there). Both are pure module algebra, universe-monomorphic and axiom-clean.

Lemma 1148 (Annihilator monotonicity under tensoring)**.** *Let R be a commutative ring and M, N two R-modules. Then*

$$\mathrm{Ann}_R(M) \;\subseteq\; \mathrm{Ann}_R(M \otimes_R N), \qquad \mathrm{Ann}_R(N) \;\subseteq\; \mathrm{Ann}_R(M \otimes_R N).$$

Proof. A scalar $a \in \mathrm{Ann}_R(M)$ kills every elementary tensor via $a \cdot (m \otimes n) = (a \cdot m) \otimes n = 0$, hence the whole tensor product by additivity; symmetrically for N. $\square$

Remark 1149. On an affine open, the section module of a twist $\mathcal{F} \otimes L^{\otimes m}$ of quasi-coherent sheaves is the tensor product of the section modules, so 1148 is the sections-level algebraic content behind the monotonicity $\mathrm{Ann}(\mathcal{F}) \subseteq \mathrm{Ann}(\mathcal{F} \otimes L^{\otimes m})$ of the annihilator ideal sheaves (1102) — and hence, via IdealSheafData.ofIdeals_mono, the containment of schematic supports $\mathrm{Supp}(\mathcal{F} \otimes L^{\otimes m}) \subseteq \mathrm{Supp}(\mathcal{F})$ that would give properness of the twisted fibre's support from properness of the support of $\mathcal{F}$. Lifting this pure-algebra monotonicity to the sheaf-level inclusion needs an affine *tensor-section comparison* $\Gamma(\mathcal{F} \otimes \mathcal{G}, U) \cong \Gamma(\mathcal{F}, U) \otimes_{\Gamma(X,U)} \Gamma(\mathcal{G}, U)$ for quasi-coherent $\mathcal{F}, \mathcal{G}$ on an affine open U (Stacks 01CC); this comparison is the subject of the tensor-section bricks below, and its affine-open case (the genuinely open step) is recorded in 1156.

Lemma 1150 (Dimension transport across a base-change linear equivalence)**.** *Let $\kappa \to \kappa'$ be a field extension, V a κ-vector space, and W a κ'-vector space. Every κ'-linear equivalence $\kappa' \otimes_\kappa V \simeq_{\kappa'} W$ forces $\dim_{\kappa'} W = \dim_\kappa V$.*

Proof. Over fields the base-change identity $\dim_{\kappa'}(\kappa' \otimes_\kappa V) = \dim_\kappa V$ holds unconditionally (with the junk value 0 shared correctly in the infinite-dimensional case), and a linear equivalence transports dimension. $\square$

Remark 1151. This is the terminal dimension-count packaging of 1157's alternative route: flat base change over the residue field extension $\kappa(t) \to \kappa(t')$ supplies a $\kappa(t')$-linear equivalence between $\kappa(t') \otimes_{\kappa(t)} \Gamma(Z, \mathcal{N})$ and $\Gamma(Z', v^*\mathcal{N})$ for the schematic support Z of the twisted fibre module and its base change Z', and 1150 converts this directly into the dimension equality, without any finiteness case split.

20.5.2 The affine tensor-section comparison

The map-half of the affine tensor-section comparison flagged in 1149 is recorded here: for sheaves of $\mathcal{O}_X$-modules A, B, the sheafification unit of the presheaf-of-modules tensor product supplies, on every open V, a canonical $\Gamma(X, V)$-linear comparison map into the sections of the substrate tensor tensorObj(A, B) (642).

Definition 1152 (The presheaf-of-modules tensor and its section comparison)**.** Let A, B be sheaves of $\mathcal{O}_X$-modules. Write $A \otimes^p B$ for the presheaf-of-modules tensor product of the underlying presheaves of A and B; objectwise, $(A \otimes^p B)(V) = \Gamma(A, V) \otimes_{\Gamma(X,V)} \Gamma(B, V)$ for every open V. The *section comparison* at V is the $\Gamma(X, V)$-linear map

$$\tau_{A,B,V} \,:\, \Gamma(A, V) \otimes_{\Gamma(X,V)} \Gamma(B, V) \longrightarrow \Gamma(\mathrm{tensorObj}(A, B), V)$$

given by the V-component of the sheafification-adjunction unit at $A \otimes^p B$ (tensorObj(A, B) being, by definition, the sheafification of $A \otimes^p B$).

Proof. The presheaf-of-modules tensor product and its objectwise formula are Mathlib's monoidal structure on presheaves of modules; the section comparison is the sheafification-adjunction unit, restricted to the open V. $\square$

Lemma 1153 (Naturality and normalisation of the section comparison)**.** *The section comparison $\tau_{A,B,-}$ of 1152 is natural in restriction: for opens $V \subseteq W$ it commutes with the restriction maps of $A \otimes^p B$ and of* tensorObj(A, B)*. Over the total space, $\tau_{A,B,\top}$ is the section multiplication* sectionsMul *of 1314.*

Proof. Naturality is naturality of the sheafification-adjunction unit as a morphism of presheaves. At $V = \top$ the section comparison is by definition the global-sections component of that same unit evaluated at $A \otimes^p B$, which is exactly how sectionsMul is constructed in 1314. ☐

Lemma 1154 (The substrate tensor and the section-graded tensor coincide)**.** *The substrate tensor* tensorObj(A, B) *(642) and the section-graded tensor $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G}$ of 1279 are the same object: both are the sheafification of the presheaf-of-modules tensor of the underlying presheaves.*

Proof. Definitional: both constructions sheafify $A \otimes^p B$. ☐

Lemma 1155 (Sheafifying the tensor comparison unit is an isomorphism)**.** *The sheafification of the presheaf comparison unit $A \otimes^p B \to$* tensorObj(A, B) *is an isomorphism of sheaves of modules.*

Proof. The sheafification functor is a reflective localization, so it inverts the localization unit of any presheaf — here, of $A \otimes^p B$. ☐

Remark 1156 (The remaining affine step, and a narrowing recommendation)**.** What is *not* yet established is that $\tau_{A,B,V}$ (1152) is an isomorphism for A, B quasi-coherent and V affine — the Stacks 01CC affine tensor-section formula flagged in 1149. The route is: identify $\Gamma(A, V)_f \otimes_{\Gamma(X,V)_f} \Gamma(B, V)_f$ with $(\Gamma(A, V) \otimes_{\Gamma(X,V)} \Gamma(B, V))_f$ on a basic open $D(f) \subseteq V$ (a localization-tensor identity), transport this through the tilde-comparison engine for quasi-coherent sheaves, and globalize over an affine cover; 1155 is the categorical fact that makes the sheafification step in this route exact rather than merely a comparison map. This affine case is also all that is needed by the sole consumer of the general pullback–tensor comparison 1147 (1146, stated for quasi-coherent sheaves), so narrowing 1147 to the quasi-coherent case would let this affine formula suffice in place of the general ringed-space stalk argument sketched there.

Lemma 1157 (Sections of the twisted fibre module under residue field extension)**.** *Source: [Nitsure], §1 (FGA Explained Ch. 5); [Stacks Project], Tag 02KH (flat base change of cohomology), degree* 0*. Let $\pi : X \to S$ be a morphism of schemes, L a quasi-coherent sheaf of $\mathcal{O}_X$-modules, $\psi : T' \to T$ a morphism of S-schemes, and $\mathcal{F}$ a finitely presented sheaf on X_T whose schematic support is proper over T (1136). Let $t' \in T'$ with image $t \in T$, let*

$$u : (X_{T'})_{t'} \longrightarrow (X_T)_t$$

be the morphism exhibiting the fibre of $X_{T'} \to T'$ at t' as the base change of the fibre of $X_T \to T$ at t along the residue field extension $\kappa(t) \to \kappa(t')$, and let $m \in \mathbb{N}$. Then

$$\dim_{\kappa(t')} \Gamma\big((X_{T'})_{t'},\, u^*(\mathcal{F}_t \otimes L_t^{\otimes m})\big) \;=\; \dim_{\kappa(t)} \Gamma\big((X_T)_t,\, \mathcal{F}_t \otimes L_t^{\otimes m}\big),$$

where $\mathcal{F}_t$ and L_t denote the restrictions of $\mathcal{F}$ and of the pullback of L to the fibre $(X_T)_t$, and the right-hand side is the graded Hilbert function of 1023 at (t, m).

Proof. Write $\mathcal{M} := \mathcal{F}_t \otimes L_t^{\otimes m}$; it is quasi-coherent ($\mathcal{F}_t$ is finitely presented and L_t quasi-coherent, both being restrictions of such along the fibre embedding), and its pullback $u^*\mathcal{M}$ is quasi-coherent as well. Write $i : Z \hookrightarrow (X_T)_t$ for the schematic support of $\mathcal{M}$, the closed subscheme cut out by the annihilator ideal $\mathrm{Ann}(\mathcal{M})$ (1102).

The support Z is proper over $\kappa(t)$. The fibre $(X_T)_t = X_T \times_T \mathrm{Spec}\,\kappa(t)$ is a base change of X_T, so by 1145 the schematic support of $\mathcal{F}_t$ is proper over $\mathrm{Spec}\,\kappa(t)$. A local section annihilating $\mathcal{F}_t$ annihilates $\mathcal{M} = \mathcal{F}_t \otimes L_t^{\otimes m}$ (it annihilates every elementary tensor of the tensor product presheaf, hence the sheafification), so $\mathrm{Ann}(\mathcal{F}_t) \subseteq \mathrm{Ann}(\mathcal{M})$ and Z is a closed subscheme of the schematic support of $\mathcal{F}_t$; a closed immersion followed by a proper morphism is proper, so $Z \to \operatorname{Spec}\kappa(t)$ is proper — in particular Z is quasi-compact and separated.

Reduction to the support and flat base change. Since $\operatorname{Ann}(\mathcal{M})$ annihilates $\mathcal{M}$, the module $\mathcal{M}$ is i_* of a quasi-coherent module $\mathcal{N}$ on Z. Let $i' : Z' \hookrightarrow (X_{T'})_{t'}$ be the base change of i along $\operatorname{Spec}\kappa(t') \to \operatorname{Spec}\kappa(t)$, where $Z' = Z \times_{\operatorname{Spec}\kappa(t)} \operatorname{Spec}\kappa(t')$ with projection $v : Z' \to Z$; pushforward along a closed immersion commutes with flat base change of quasi-coherent modules, so $u^*\mathcal{M} = u^*i_*\mathcal{N} = i'_*(v^*\mathcal{N})$. Since $\Gamma \circ i_* = \Gamma$ and $\Gamma \circ i'_* = \Gamma$, both displayed section spaces are global sections over Z, respectively over its base change Z'. The extension $\kappa(t) \to \kappa(t')$ of fields is flat and Z is quasi-compact and separated over $\kappa(t)$, so flat base change for the cohomology of quasi-coherent modules ([Stacks Project], Tag 02KH, in degree 0) gives a $\kappa(t')$-linear isomorphism

$$\Gamma(Z', v^*\mathcal{N}) \;\cong\; \Gamma(Z, \mathcal{N}) \otimes_{\kappa(t)} \kappa(t').$$

Dimension count. For any $\kappa(t)$-vector space V, a $\kappa(t)$-basis of V induces a $\kappa(t')$-basis of $V \otimes_{\kappa(t)} \kappa(t')$, so $\dim_{\kappa(t')}(V \otimes_{\kappa(t)} \kappa(t')) = \dim_{\kappa(t)} V$ — in the infinite-dimensional case both sides are infinite. Applying this to $V = \Gamma(Z, \mathcal{N}) = \Gamma\big((X_T)_t, \mathcal{M}\big)$ yields the claim. $\square$

Lemma 1158 (The fibrewise Hilbert function is invariant under base change)**.** *Source: [Nitsure], §1 (FGA Explained Ch. 5). Let $\pi : X \to S$ be a morphism of schemes, L a quasi-coherent sheaf of $\mathcal{O}_X$-modules, $T' \to T$ a morphism of S-schemes, and $\mathcal{F}$ a finitely presented sheaf on X_T whose schematic support is proper over T. For $t' \in T'$ with image $t \in T$ and every $m \in \mathbb{N}$,*

$$\dim_{\kappa(t')} \Gamma\big((X_{T'})_{t'},\, (g'^*\mathcal{F})_{t'} \otimes L_{t'}^{\otimes m}\big) \;=\; \dim_{\kappa(t)} \Gamma\big((X_T)_t,\, \mathcal{F}_t \otimes L_t^{\otimes m}\big),$$

where $g' : X_{T'} \to X_T$ is the induced morphism. In particular the graded Hilbert function of 1023 is invariant under base change of the parameter scheme, and so is the Hilbert polynomial of 1018.

Proof. Fibre pasting. Since $X_{T'} = X_T \times_T T'$, the fibre of $X_{T'} \to T'$ at t' is the base change of the fibre of $X_T \to T$ at t along the residue field extension $\kappa(t) \to \kappa(t')$:

$$(X_{T'})_{t'} = X_{T'} \times_{T'} \operatorname{Spec}\kappa(t') = X_T \times_T \operatorname{Spec}\kappa(t') = (X_T)_t \times_{\operatorname{Spec}\kappa(t)} \operatorname{Spec}\kappa(t'),$$

by pasting the cartesian square $X_{T'} = X_T \times_T T'$ with the squares defining the two fibres, using the factorisation $\operatorname{Spec}\kappa(t') \to \operatorname{Spec}\kappa(t) \to T$ of $\operatorname{Spec}\kappa(t') \to T' \to T$. Write $u : (X_{T'})_{t'} \to (X_T)_t$ for the top morphism of the pasted cartesian square.

Matching the twisted modules. Restriction to the fibre and pullback along g' (resp. along the projections to X) are pullbacks along two factorisations of one and the same composite morphism, so $(g'^*\mathcal{F})_{t'} = u^*(\mathcal{F}_t)$ and $L_{t'} = u^*(L_t)$. All modules in sight are quasi-coherent ($\mathcal{F}$ is finitely presented, L is quasi-coherent, and quasi-coherence is preserved by module pullback), so by 1146 applied to u,

$$(g'^*\mathcal{F})_{t'} \otimes L_{t'}^{\otimes m} \;\cong\; u^*(\mathcal{F}_t) \otimes \big(u^*(L_t)\big)^{\otimes m} \;\cong\; u^*(\mathcal{F}_t \otimes L_t^{\otimes m}).$$

An isomorphism of sheaves of modules induces a $\kappa(t')$-linear isomorphism on global sections (the $\kappa(t')$-action on either side is restriction of the $\Gamma((X_{T'})_{t'}, \mathcal{O})$-action along the structural map $\kappa(t') \to \Gamma((X_{T'})_{t'}, \mathcal{O})$), so the left-hand side of the claim equals $\dim_{\kappa(t')} \Gamma\big((X_{T'})_{t'}, u^*(\mathcal{F}_t \otimes L_t^{\otimes m})\big)$.

Base change of sections. By 1157,

$$\dim_{\kappa(t')} \Gamma\big((X_{T'})_{t'},\, u^*(\mathcal{F}_t \otimes L_t^{\otimes m})\big) = \dim_{\kappa(t)} \Gamma\big((X_T)_t,\, \mathcal{F}_t \otimes L_t^{\otimes m}\big),$$

which is the claim. $\square$

20.6 The relative-divisor functor

In Grothendieck's quotient route to the Picard scheme — the route set beside the Milne–Kollár one in 25.1, and not the one the formalisation follows — the Picard scheme arises from a family of effective divisors modulo linear equivalence. This section defines the parameter functor of that family: the relative-effective-divisor functor $\mathrm{Div}_{X/S}$, encoded inside the Hilbert functor $\mathrm{Hilb}_{X/S} = \mathrm{Quot}_{\mathcal{O}_X/X/S}$ (before the Hilbert-polynomial decomposition) as the quotients of $\mathcal{O}_{X_T}$ whose kernel ideal is invertible. The functor is a piece of mathematics in its own right, and the divisor substrate it rests on is shared with the committed route.

Definition 1159 (Family of relative effective divisors)**.** *Source: [Kleiman], §3, Defs. df:red and df:div (FGA Explained Ch. 9).* Let $\pi : X \to S$ be a morphism of schemes and $T \to S$ an S-scheme; write $X_T := X \times_S T$. A *family of relative effective divisors on X_T/T* is a pair $(\mathcal{F}, q)$ where

1. $\mathcal{F}$ is a finitely presented sheaf of $\mathcal{O}_{X_T}$-modules, flat over T (921) and with schematic support proper over T (1136);
2. $q : \mathcal{O}_{X_T} \twoheadrightarrow \mathcal{F}$ is a surjective $\mathcal{O}_{X_T}$-linear homomorphism;
3. the kernel ideal $\mathcal{I} := \ker q$ is invertible, i.e. locally trivial of rank one (609).

Two families $(\mathcal{F}, q)$ and $(\mathcal{F}', q')$ are *equivalent* iff $\ker q = \ker q'$, i.e. iff they cut out the same closed subscheme.

Such a pair is precisely the structure-sheaf quotient $q : \mathcal{O}_{X_T} \to \mathcal{O}_D$ of a relative effective divisor $D \subseteq X_T$ in the sense of [Kleiman] df:red — a closed subscheme, flat over T, whose ideal is invertible — subject to two conditions that are automatic in the regime of interest: finite presentation of $\mathcal{O}_D$ always holds ($\mathcal{O}_D$ is the cokernel of $\mathcal{I} \hookrightarrow \mathcal{O}_{X_T}$ with $\mathcal{I}$ locally free of rank one), and properness of the support D over T holds whenever π is proper (a closed subscheme of the T-proper X_T is proper over T).

Lemma 1160 (Invertibility is invariant under isomorphism)**.** *Let $M \cong N$ be isomorphic sheaves of $\mathcal{O}_X$-modules on a scheme X. If M is locally trivial of rank one (609), then so is N.*

Proof. Fix $x \in X$ and choose an affine open $U \ni x$ with an isomorphism $M|_U \cong \mathcal{O}_U$. Restriction along the open immersion $U \subseteq X$ is functorial, so the given isomorphism $M \cong N$ restricts to $N|_U \cong M|_U \cong \mathcal{O}_U$. ☐

Lemma 1161 (A trivialising chart restricts to any smaller open)**.** *Source: Stacks 01HH (locally trivial modules), specialised to a chart shrink. Let $\mathcal{M}$ be an $\mathcal{O}_X$-module, $V \subseteq U$ opens of X, and $t : \mathcal{M}|_U \cong \mathcal{O}_U$ a trivialisation. Then $\mathcal{M}|_V \cong \mathcal{O}_V$.*

Proof. Restriction to V factors as restriction to U followed by restriction along the open immersion $V \subseteq U$; applying t and using that the restriction of the structure sheaf along an open immersion is again the structure sheaf gives $\mathcal{M}|_V \cong \mathcal{O}_U|_V \cong \mathcal{O}_V$. ☐

Lemma 1162 (Affine trivialising charts are cofinal)**.** *Let $\mathcal{M}$ be locally trivial of rank one (609), $x \in X$, and W an open neighbourhood of x. Then there is an affine open V with $x \in V \subseteq W$ and a trivialisation $\mathcal{M}|_V \cong \mathcal{O}_V$.*

Proof. By local triviality there is an affine open $U \ni x$ with a trivialisation of $\mathcal{M}|_U$. Choose an affine open V with $x \in V \subseteq U \cap W$ (affine opens form a basis of the topology of a scheme); restricting the trivialisation to V via 1161 gives the claim. ☐

Lemma 1163 (Flat cokernel keeps a tensored kernel inclusion injective)**.** *Source: Stacks 00HL. Let R be a commutative ring and $0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0$ an exact sequence of R-modules with C flat. Then for every R-module N the map $f \otimes \mathrm{id}_N : A \otimes_R N \to B \otimes_R N$ is injective.*

Proof. Choose a free presentation $0 \to K \xrightarrow{\iota} F_0 \xrightarrow{\pi} N \to 0$ of N; F_0 is free, hence flat. Tensoring f and g with K and with F_0, and tensoring ι and π with A, B, C, assembles a commutative 3×3 diagram with exact rows $A \otimes K \to A \otimes F_0 \to A \otimes N \to 0$ (and likewise for B, C), by right-exactness of $- \otimes_R -$; the two right-hand columns $A \otimes K \to B \otimes K \to C \otimes K$ and $A \otimes F_0 \to B \otimes F_0 \to C \otimes F_0$ are exact for the same reason, and moreover their first arrow is injective because F_0 and C are flat.

Given $x \in A \otimes N$ with $(f \otimes \mathrm{id}_N)(x) = 0$, surjectivity of π lifts x to some $y \in A \otimes F_0$ with $(\mathrm{id}_A \otimes \pi)(y) = x$. Commutativity sends the image of y in $B \otimes F_0$ to $(f \otimes \mathrm{id}_N)(x) = 0$ in $B \otimes N$, so by exactness of the middle row at $B \otimes F_0$ this image equals the image of some $z \in B \otimes K$. Pushing z down to $C \otimes K$ and then right to $C \otimes F_0$ gives 0 (the image of y in $C \otimes F_0$ vanishes because $g \circ f = 0$), so injectivity of $C \otimes K \to C \otimes F_0$ (flatness of C) forces the image of z in $C \otimes K$ to already vanish, and exactness of the left column at $B \otimes K$ produces $w \in A \otimes K$ mapping to z. Replacing y by $y - (\iota \otimes \mathrm{id}_A)(w)$ leaves its image x under $\mathrm{id}_A \otimes \pi$ unchanged and now maps to 0 in $B \otimes F_0$ (it differed from an element mapping to z exactly by the image of w), so injectivity of $A \otimes F_0 \to B \otimes F_0$ (flatness of F_0) forces this corrected y to be 0, whence $x = (\mathrm{id}_A \otimes \pi)(y) = 0$. ☐

Lemma 1164 (Flat cokernel keeps a tensored kernel inclusion injective, left-tensor form)**.** *Source: Stacks 00HL. In the situation of 1163, the left-tensored map $\mathrm{id}_N \otimes f : N \otimes_R A \to N \otimes_R B$ is injective for every R-module N.*

Proof. The left-tensored map is conjugate to the right-tensored map $f \otimes \mathrm{id}_N$ of 1163 by the two tensor-commutativity isomorphisms $N \otimes A \cong A \otimes N$ and $N \otimes B \cong B \otimes N$, hence injective as a composite of injective maps. ☐

Lemma 1165 (Base change of the ideal of a relative effective divisor)**.** *Source: [Kleiman], §3, the functoriality argument following Def. df:div. Let*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ \downarrow f' & & \downarrow f \\ S' & \xrightarrow{g} & S \end{array}$$

be a cartesian square of schemes, $q : E \twoheadrightarrow F$ an epimorphism of $\mathcal{O}_X$-modules with E quasi-coherent and F finitely presented and flat over S (921), and suppose the kernel $\mathcal{K} := \ker q$ is locally trivial of rank one (609). Then $\ker(g'^*q)$ is locally trivial of rank one; in fact the natural map $g'^*\mathcal{K} \to \ker(g'^*q)$ is an isomorphism.

Proof. All three modules in the short exact sequence $0 \to \mathcal{K} \to E \to F \to 0$ (exact because q is an epimorphism with kernel $\mathcal{K}$) are quasi-coherent: $\mathcal{K}$ is locally isomorphic to the structure sheaf, F is finitely presented, and E is quasi-coherent by hypothesis.

Charts. The claim is local on X'. Given $x' \in X'$, choose an affine open $U \subseteq S$ containing the common image $f(g'(x')) = g(f'(x'))$, a trivialising chart of $\mathcal{K}$ around $g'(x')$ and inside it an affine open $V \subseteq f^{-1}(U)$ around $g'(x')$ (a basic open of the chart; $\mathcal{K}|_V \cong \mathcal{O}_V$ persists under shrinking, 1161), and an affine open $U_t \subseteq g^{-1}(U)$ around $f'(x')$. As in 1140, the piece $W := g'^{-1}(V) \cap f'^{-1}(U_t)$ is an affine open neighbourhood of x', namely the fibre product $V \times_U U_t$ of the restricted cartesian square; writing $R := \Gamma(S, U)$, $R' := \Gamma(S', U_t)$ and $A := \Gamma(X, V)$, it satisfies $\Gamma(X', W) = A \otimes_R R'$, and for every quasi-coherent module M on X, $\Gamma(g'^*M, W) =$

$\Gamma(M,V)\otimes_A \Gamma(X',W) = \Gamma(M,V)\otimes_R R'$ (the last identification by associativity of the tensor product).

Exactness after base change. Sections over the affine V are exact on quasi-coherent modules (higher cohomology of a quasi-coherent module on an affine scheme vanishes), so $0 \to \Gamma(\mathcal{K},V) \to \Gamma(E,V) \to \Gamma(F,V) \to 0$ is an exact sequence of A-modules. Tensoring over R with R' and splicing the long exact Tor sequence gives the exact sequence

$$\mathrm{Tor}_1^R(\Gamma(F,V),R') \to \Gamma(\mathcal{K},V)\otimes_R R' \to \Gamma(E,V)\otimes_R R' \to \Gamma(F,V)\otimes_R R' \to 0,$$

and the Tor term vanishes, equivalently the left-hand map stays injective after tensoring (1163), because $\Gamma(F,V)$ is R-flat (921 at the affine pair (U,V)). By the chart identification, this says that $0 \to \Gamma(g'^*\mathcal{K},W) \to \Gamma(g'^*E,W) \to \Gamma(g'^*F,W) \to 0$ is exact.

Kernel identification. The pullback functor g'^* is right exact, so $g'^*\mathcal{K} \to g'^*E \to g'^*F \to 0$ is exact, and the induced map $g'^*\mathcal{K} \to \ker(g'^*q)$ is an epimorphism. On each chart W both sides are quasi-coherent modules on an affine scheme, and on sections over W the map is the isomorphism $\Gamma(g'^*\mathcal{K},W) \cong \ker(\Gamma(g'^*E,W) \to \Gamma(g'^*F,W)) = \Gamma(\ker(g'^*q),W)$ from the previous paragraph (the second identification is left exactness of sections). A morphism of quasi-coherent modules that is an isomorphism on sections over each member of an affine open cover is an isomorphism, so $g'^*\mathcal{K} \cong \ker(g'^*q)$.

Finally $g'^*\mathcal{K}$ is locally trivial of rank one by 611, hence so is $\ker(g'^*q)$ by 1160. □

Remark 1166. The hypothesis that E is quasi-coherent in 1165 is removable: E is an extension of the quasi-coherent F by the quasi-coherent $\mathcal{K} = \ker q$ ($\mathcal{K}$ is locally isomorphic to the structure sheaf, hence quasi-coherent), and quasi-coherent modules form a weak Serre subcategory of all $\mathcal{O}_X$-modules, in particular one closed under extensions (Stacks 01LA). Once extension-closure of quasi-coherence is available, the hypothesis on E can be dropped in favour of this derivation.

Definition 1167 (Kernel–pullback comparison map)**.** Let $g' : X' \to X$ be a morphism of schemes and $q : E \to F$ a morphism of $\mathcal{O}_X$-modules. Write $\iota_q : \ker q \to E$ for the kernel inclusion. The *kernel–pullback comparison map*

$$\kappa_{g',q} : g'^*(\ker q) \longrightarrow \ker(g'^*q)$$

is the unique morphism with $\kappa_{g',q} \circ \iota_{g'^*q} = g'^*(\iota_q)$; it exists because g'^* preserves the zero morphism, being a left adjoint (1109).

Lemma 1168 (Local triviality transports along an isomorphic kernel comparison)**.** *Let $g' : X' \to X$ and $q : E \to F$ be as in 1167, and suppose the comparison map $\kappa_{g',q}$ is an isomorphism. If $\ker q$ is locally trivial of rank one, then so is $\ker(g'^*q)$.*

Proof. By 611, $g'^*(\ker q)$ is locally trivial of rank one; transporting along the isomorphism $\kappa_{g',q}$ via 1160 gives the same for $\ker(g'^*q)$. □

Lemma 1169 (Basis-local criterion for a monomorphism of sheaves of modules)**.** *Let B be a basis of opens of a scheme X and let $\varphi : M \to N$ be a morphism of sheaves of $\mathcal{O}_X$-modules such that φ is injective on sections over every member of B. Then φ is a monomorphism.*

Proof. Injectivity on a basis of opens gives injectivity of φ on stalks, hence the underlying morphism of sheaves of abelian groups is a monomorphism; the faithful forgetful functor from sheaves of modules to sheaves of abelian groups reflects monomorphisms. □

Lemma 1170 (The kernel–pullback comparison is an epimorphism when q is)**.** *Let $g' : X' \to X$ and $q : E \twoheadrightarrow F$ be as in 1167, with q an epimorphism. Then the comparison map $\kappa_{g',q} : g'^*(\ker q) \to \ker(g'^*q)$ is an epimorphism.*

Proof. q epi identifies F with the cokernel of the kernel inclusion $\ker q \hookrightarrow E$; the pullback functor g'^* is a left adjoint (1109), hence preserves cokernels, so $g'^*(\ker q) \to g'^*E \to g'^*F \to 0$ stays exact. Exactness at g'^*E says exactly that the induced kernel-comparison lift $\kappa_{g',q}$ is an epimorphism. ☐

Lemma 1171 (The kernel–pullback comparison is an isomorphism once it is monic)**.** *Let $g' : X' \to X$ and $q : E \twoheadrightarrow F$ be as in 1167, with q an epimorphism, and suppose the pullback $g'^*(\iota_q) : g'^*(\ker q) \to g'^*E$ of the kernel inclusion is a monomorphism. Then the comparison map $\kappa_{g',q}$ is an isomorphism.*

Proof. The comparison satisfies $\kappa_{g',q} \circ \iota_{g'^*q} = g'^*(\iota_q)$ (1167); since the right-hand side is monic by hypothesis and $\iota_{g'^*q}$ is (as a kernel inclusion) automatically monic, $\kappa_{g',q}$ is monic. Combined with the epimorphism half of 1170, monic + epic = iso in the abelian category of $\mathcal{O}_{X'}$-modules. ☐

Remark 1172. The conclusion of 1165 factors through the comparison map of 1167: local triviality of $\ker(g'^*q)$ follows from 1168 as soon as $\kappa_{g',q}$ is known to be an isomorphism, and this factoring is now itself proved rather than merely anticipated: $\kappa_{g',q}$ is automatically an epimorphism whenever q is (1170, right exactness of g'^*), so by 1171 the entire remaining obligation is the single monomorphism fact that g'^* keeps the kernel inclusion $\ker q \hookrightarrow E$ injective — exactly the Tor-vanishing content isolated in the "Exactness after base change" paragraph of the proof of 1165 (1163 and 1164). That monomorphism fact is established scheme-theoretically via the basis-local criterion 1169: the affine pieces $W = g'^{-1}(V) \cap f'^{-1}(U_t)$ of the proof of 1165 cover X' but need not form a topological basis (the scheme-theoretic fibre product carries a topology finer than $|X| \times_{|S|} |S'|$); descending the per-piece section injectivity to the basic opens of each piece — which do form a basis of X', by flatness of the localisation at a section — completes the reduction, so $\kappa_{g',q}$ is an isomorphism in general (1173).

Lemma 1173 (The kernel–pullback comparison is an isomorphism under base-flatness)**.** *In the situation of 1165, the kernel–pullback comparison map $\kappa_{g',q} : g'^*(\ker q) \to \ker(g'^*q)$ (1167) is itself an isomorphism.*

Proof. The comparison map $\kappa_{g',q}$ is an epimorphism whenever q is (1170, right exactness of g'^*); the proof of 1165 shows that g'^* keeps the kernel inclusion $\ker q \hookrightarrow E$ monic, so 1171 upgrades the epimorphism $\kappa_{g',q}$ to an isomorphism. This is the isomorphism form of 1165 consumed by the Abel map and by the pullback of families of relative effective divisors (1175). ☐

Lemma 1174 (The structure sheaf is quasi-coherent)**.** *For every scheme X, the unit module $\mathcal{O}_X$ (the unit object of X.*Modules*) is quasi-coherent.*

Proof. The unit module carries the canonical one-generator, no-relation presentation given by the identity on $\mathcal{O}_X$ over the one-member cover $\{X\}$; this is a quasi-coherence datum. ☐

Definition 1175 (Relative-divisor functor)**.** *Source: [Kleiman], §3, Def. df:div (FGA Explained Ch. 9).* Let $\pi : X \to S$ be a morphism of schemes. The *relative-divisor functor*

$$\mathrm{Div}_{X/S} : (\mathrm{Sch}/S)^{op} \longrightarrow \mathbf{Set}$$

sends an S-scheme $T \to S$ to the set of equivalence classes of families of relative effective divisors on X_T/T (1159) — equivalently, the set of relative effective divisors on X_T/T — and an S-morphism $T' \to T$ to the pullback of families along the induced morphism $g' : X_{T'} \to X_T$. The action is well defined: pullback of modules preserves surjectivity (it is right exact), finite presentation (1139), flatness over the base (1140) and proper support (1145), and it preserves the invertibility of the kernel ideal because the pulled-back kernel is the kernel of the pulled-back quotient (1165, applicable since the source of q is always a pullback of the unit module $\mathcal{O}_X$, quasi-coherent by 1174 together with pullback-stability of quasi-coherence (507)). Functoriality holds because the pullback of modules is pseudofunctorial in the scheme morphism.

20.7 The Grassmannian scheme

The Quot construction proceeds by embedding $\mathrm{Quot}^{\Phi,L}_{E/X/S}$ as a locally closed subfunctor of a Grassmannian over S. Since Mathlib carries no Grassmannian *scheme* (only a finite-rank-quotient variety in linear algebra), we package the construction here.

Definition 1176 (Grassmannian scheme)**.** *Source: [Nitsure], §1, Exercise (2) and "Construction of Grassmannian" (FGA Explained Ch. 5).* Let S be a locally noetherian scheme and let V be a locally free $\mathcal{O}_S$-module of rank r. For an integer $1 \le d \le r$, the *Grassmannian of rank-d quotients of* V is the functor

$$\mathrm{Grass}(V,d) : (\mathrm{Sch}/S)^{op} \longrightarrow \mathbf{Set}, \qquad T \longmapsto \{\langle \mathcal{F}, q\rangle : q : V_T \twoheadrightarrow \mathcal{F},\ \mathcal{F} \text{ locally free of rank } d\}$$

with equivalence $\ker(q) = \ker(q')$; the rank-d local freeness of $\mathcal{F}$ is as in 1137. Over a locally noetherian base this functor agrees with the Quot functor (1138) in the special case $X = S$, $E = V$, constant Hilbert polynomial $\Phi = d$: a flat finitely presented module whose fibre dimension is d at every point is locally free of rank d.

Lemma 1177 (Representability predicate)**.** *Provided by Mathlib. Let $\mathcal{C}$ be a category and $F : \mathcal{C}^{op} \to$ **Set** a presheaf of sets. The predicate $F.\mathrm{IsRepresentable}$ asserts that F is representable, i.e. that there exists an object Y of $\mathcal{C}$ together with a natural bijection $\mathrm{Hom}_{\mathcal{C}}(-,Y) \cong F$ (the Yoneda bijection).*

Lemma 1178 (Local representability of Zariski sheaves)**.** *Provided by Mathlib (Stacks tag 01JJ). Let G be a set-valued sheaf on the big Zariski site of schemes, $\{X_i\}$ a family of schemes and $f_i : h_{X_i} \to G$ a family of relatively representable open immersions of presheaves that is jointly locally surjective. Then G is representable, by the scheme glued from the X_i along the pairwise pullbacks $h_{X_i} \times_G h_{X_j}$.*

Lemma 1179 (Representability of the Grassmannian, trivialised case)**.** *Source: [Nitsure], §1, Exercise (2). Let S be a locally noetherian scheme, V an $\mathcal{O}_S$-module equipped with a global trivialisation $V \cong \mathcal{O}_S^{\oplus r}$, and $1 \le d \le r$. Then the functor $\mathrm{Grass}(V,d)$ of 1176 is representable by the base change $S \times_{\mathrm{Spec}\,\mathbb{Z}} \mathrm{Gr}(d,r)$ of the absolute Grassmannian, with structure morphism the first projection.*

Proof. Since $\mathrm{Spec}\,\mathbb{Z}$ is the terminal scheme, the base change $S \times_{\mathrm{Spec}\,\mathbb{Z}} \mathrm{Gr}(d,r)$ is the binary product, and for an S-scheme $T \to S$ the S-morphisms $T \to S \times \mathrm{Gr}(d,r)$ (over the first projection) correspond, by the universal property of the product, to the bare scheme morphisms $T \to \mathrm{Gr}(d,r)$; these are in natural bijection with $\mathrm{Grass}(r,d)(T)$ by 1541. It therefore suffices to identify $\mathrm{Grass}(V,d)(T)$ with $\mathrm{Grass}(r,d)(T)$, naturally in T. A T-point of $\mathrm{Grass}(V,d)$ is a rank-d locally free quotient $q : V_T \twoheadrightarrow \mathcal{F}$; composing with the pullback of the trivialisation and with the canonical identification $(\mathcal{O}_S^{\oplus r})_T \cong \mathcal{O}_T^{\oplus r}$ of the pullback of a free module re-presents it as a rank-d quotient of $\mathcal{O}_T^{\oplus r}$, i.e. a T-point of $\mathrm{Grass}(r,d)$; the inverse composes with the inverse identifications, and both directions preserve the equivalence $\ker q = \ker q'$. Naturality in T is the compatibility of the free-module identification with composite pullbacks (the pseudofunctor coherence of the module pullback on free modules), and the equivalence classes match because the re-presentation composes with isomorphisms only. ∎

Definition 1180 (Zariski sheaf axiom on Sch/S)**.** A presheaf $F : (\mathrm{Sch}/S)^{op} \to \mathbf{Set}$ *satisfies the Zariski sheaf axiom* when for every S-scheme $T \to S$ and every open cover $\{W_k\}$ of T, every family of sections $x_k \in F(T|_{W_k})$ that agrees on the pairwise intersections ($x_k|_{W_k \cap W_l} = x_l|_{W_k \cap W_l}$ for all k, l) glues to a unique section $x_0 \in F(T)$ with $x_0|_{W_k} = x_k$. This is the sheaf condition for the big Zariski site of S, stated against open covers of the total space; every representable presheaf satisfies it.

Theorem 1181 (Zariski descent of representability)**.** *Let S be a scheme, $F : (\mathrm{Sch}/S)^{op} \to \mathbf{Set}$ a presheaf satisfying the Zariski sheaf axiom (1180), and $S = \bigcup_i U_i$ an open cover such that for every i the restriction of F to Sch/U_i (along $T \mapsto (T \to U_i \hookrightarrow S)$) is representable by a U_i-scheme Y_i. Then F is representable by an S-scheme.*

Proof. Form the *total functor* G of F on the absolute category of schemes: a point of $G(T)$ is a morphism $a : T \to S$ together with a family of morphisms $\gamma_i : T|_{a^{-1}(U_i)} \to Y_i$ over U_i whose classified sections $x_i \in F(T|_{a^{-1}(U_i)})$ agree on pairwise overlaps. By the sheaf axiom of F applied to the cover $\{a^{-1}(U_i)\}$ of T, such a datum is the same as a pair (a, x) with $x \in F(T, a)$ (the compatible classified sections glue to a unique x, and x restricts back to the x_i); the encoding by classifying morphisms realises G as a set-valued functor, restriction acting by composition on a and by restriction of the γ_i.

G is a sheaf for the big Zariski topology: a covering family may be replaced by the family of image opens (both generate the same sieve), and for a cover of T by opens $\{W_k\}$ with compatible points $(a_k, x_k) \in G(W_k)$, the base morphisms glue to $a : T \to S$ because morphisms of schemes glue along open covers, and the sections glue by the sheaf axiom of F at (T, a); uniqueness follows from locality of morphism equality and the uniqueness half of the sheaf axiom.

Each Y_i maps to G by the *chart* $c_i : h_{Y_i} \to G$ sending $t : T \to Y_i$ to the point with base $T \to Y_i \to U_i \subseteq S$ classified by the section that t represents. The pullback of c_i against an arbitrary point $(a, x) : h_T \to G$ is the open subscheme $a^{-1}(U_i) \subseteq T$ (a morphism $V \to T$ lifts along c_i iff its base lands in U_i, and then the lift is the unique classifying morphism of the restricted section), so the c_i are relatively representable open immersions; they are jointly locally surjective because every point of $G(T)$ restricts on the cover $\{a^{-1}(U_i)\}$ to the chart image of its classifying morphisms. By 1178 the functor G is representable by a scheme $\hat{Y}$.

Finally, the tautological point of $G(\hat{Y})$ corresponding to $\mathrm{id}_{\hat{Y}}$ has base $\hat{a} : \hat{Y} \to S$, and for $T \to S$ the identification $G(T) \cong \coprod_a F(T, a)$ restricts to a natural bijection $\mathrm{Hom}_{\mathrm{Sch}/S}(T, (\hat{Y}, \hat{a})) \cong F(T)$: a morphism $u : T \to \hat{Y}$ over S corresponds to a point of $G(T)$ with base $T \to S$, i.e. to a section of $F(T)$, and naturality is the restriction-compatibility of the glued sections. Hence F is representable by the S-scheme $(\hat{Y}, \hat{a})$. $\square$

Lemma 1182 (The Grassmannian functor is a Zariski sheaf)**.** *Let S be locally noetherian, V an $\mathcal{O}_S$-module and $d \geq 0$. The functor* $\mathrm{Grass}(V, d)$ *of 1176 satisfies the Zariski sheaf axiom (1180).*

Proof. Let $T \to S$, let $\{W_k\}$ be an open cover of T, and let $\langle \mathcal{F}_k, q_k \rangle$ be rank-d locally free quotients of $V_{T|_{W_k}}$ agreeing on overlaps. Agreement means: there are isomorphisms $f_{kl} : \mathcal{F}_k|_{W_k \cap W_l} \cong \mathcal{F}_l|_{W_k \cap W_l}$ commuting with the restricted quotient maps. Such an isomorphism is *unique* when it exists (it is determined on the image of the epimorphism q_k); consequently the f_{kl} automatically satisfy the cocycle condition on triple overlaps – pulled back to the scheme-theoretic overlaps $W_k \times_T W_l$ of the glue datum attached to the cover, both composites of transition transports commute with the restricted quotient maps, and gluing isomorphisms are unique. Descent of sheaves of modules along the glue datum of the cover (1479) glues the $\mathcal{F}_k$ to a sheaf $\mathcal{F}$ on the glued scheme, whose chart restrictions recover the $\mathcal{F}_k$ (1480); the transposed quotient maps satisfy the descent-equalizer compatibility, so they lift to $q : V \to \mathcal{F}$ out of the pulled-back test module, with chart restrictions the q_k. Epimorphy is detected on the chart cover by joint faithfulness of the chart restrictions (1470), and rank-d local freeness is local, so transporting along the canonical isomorphism between the glued scheme and T produces the glued point of $\mathrm{Grass}(V,d)(T)$; it restricts to the given classes by the chart recovery. Uniqueness: if $\langle \mathcal{F}, q\rangle$ and $\langle \mathcal{F}', q'\rangle$ both restrict to $\langle \mathcal{F}_k, q_k\rangle$ up to equivalence, the two kernels annihilate each other's quotient map locally on the cover, hence globally (the sections of a sheaf are detected on a cover), and the two epimorphism descents $\mathcal{F} \to \mathcal{F}'$, $\mathcal{F}' \to \mathcal{F}$ are mutually inverse by epi-cancellation, so the two glued points are equivalent. □

Theorem 1183 (Representability of the Grassmannian)**.**
Source: [Nitsure], §1, "Construction of Grassmannian" (FGA Explained Ch. 5). Let S be a noetherian scheme, V a locally free $\mathcal{O}_S$-module of rank r, and $1 \le d \le r$. The functor $\mathrm{Grass}(V,d)$ *of 1176 is representable by a smooth projective S-scheme* $\mathrm{Gr}_S(V,d) \to S$ *of relative dimension $d(r-d)$, equipped with a tautological quotient $\pi^*V \twoheadrightarrow \mathcal{U}$ of rank d. The determinant line bundle* $\det(\mathcal{U})$ *is relatively very ample, and the associated linear system gives a closed embedding* $\mathrm{Gr}_S(V,d) \hookrightarrow \mathbb{P}_S\big(\bigwedge^d V\big)$ *– the* Plücker embedding.

Proof. Over $S = \operatorname{Spec}\mathbb{Z}$ and $V = \mathcal{O}_S^{\oplus r}$ the Grassmannian $\mathrm{Gr}(r,d)$ is constructed by gluing $\binom{r}{d}$ affine charts $U^I = \operatorname{Spec}\mathbb{Z}[X^I]$ indexed by size-d subsets $I \subseteq \{1,\dots,r\}$. The chart U^I carries the matrix X^I of size $d \times r$ whose I-th $d \times d$ minor is the identity, with the remaining $d(r-d)$ entries free variables (so $U^I \cong \mathbb{A}_{\mathbb{Z}}^{d(r-d)}$). The transition maps $\theta_{I,J} : U^I \cap U^J \to U^J \cap U^I$ defined by $X^J \mapsto (X^I_J)^{-1}X^I$ on the locus where $\det X^I_J$ is invertible satisfy the cocycle condition $\theta_{I,K} = \theta_{I,J} \circ \theta_{J,K}$. The glued $\mathrm{Gr}(r,d) \to \operatorname{Spec}\mathbb{Z}$ is smooth of relative dimension $d(r-d)$. Separatedness is checked from the diagonal $\Delta_{I,J} \subset U^I \times U^J$ cut out by $X^J_I X^I - X^J = 0$. Properness follows from the valuative criterion for DVRs: given $\varphi : \operatorname{Spec} K \to \mathrm{Gr}(r,d)$ over a DVR R with fraction field K, factoring through some chart U^I via a ring map $f : \mathbb{Z}[X^I] \to K$, choose J minimising the valuation $\nu(f(P^I_J))$ of the minor; on the chart U^J all entries have non-negative valuation, so f extends uniquely to $\operatorname{Spec} R$.

The universal quotient $u : \mathcal{O}^{\oplus r}_{\mathrm{Gr}(r,d)} \twoheadrightarrow \mathcal{U}$ is given on U^I by the matrix X^I and glued by $g_{I,J} = (X^I_J)^{-1} \in GL_d(U^I \cap U^J)$. The determinant line bundle $\det\mathcal{U}$ has transition functions $\det g_{I,J} = 1/P^I_J$; the global sections $\sigma_I \in \Gamma(\mathrm{Gr}(r,d), \det\mathcal{U})$ defined by $\sigma_I|_{U^J} = P^J_I$ form a base-point-free linear system separating points over $\operatorname{Spec}\mathbb{Z}$, giving an embedding into $\mathbb{P}_{\mathbb{Z}}^{\binom{r}{d}-1}$, which is closed by properness. Representability of $\mathrm{Grass}(r,d)$ is then verified on the universal pair $(\mathrm{Gr}(r,d), u)$: a quotient $q : \mathcal{O}_T^{\oplus r} \twoheadrightarrow \mathcal{F}$ of rank d on T is everywhere locally trivialised; on the locus where the columns indexed by I are a basis the matrix of q has the form X^I, defining the classifying morphism $T \to U^I \subset \mathrm{Gr}(r,d)$. Base change from $\mathbb{Z}$ to a general noetherian base S with a vector bundle V is direct: trivialise V on an open cover and glue. For non-trivial V the same construction produces $\mathrm{Gr}_S(V,d)$ with Plücker embedding into $\mathbb{P}_S(\bigwedge^d V)$. □

20.8 Projective space, the Serre twist, and projective morphisms

Representability of the Quot functor holds for a *projective* structural morphism carrying a relatively very ample line bundle ([Nitsure] §5); properness alone does not suffice. This section builds the ambient vocabulary: relative projective space, the Serre twisting sheaf, and the projective-morphism predicate together with its stability properties.

Definition 1184 (Relative projective space)**.** Let S be a scheme and let n be an index set. The *relative projective space* $\mathbb{P}(n; S)$ is the fibre product

$$\mathbb{P}(n; S) \;=\; S \times_{\operatorname{Spec}\mathbb{Z}} \operatorname{Proj}\mathbb{Z}[X_i : i \in n],$$

where the polynomial ring carries the total-degree grading. Its *structural morphism* $\mathbb{P}(n; S) \to S$ is the first projection. For $n = \{0, \dots, d\}$ this is the usual projective d-space $\mathbb{P}^d_S$.

Lemma 1185 (Degree-zero part of the homogeneous coordinate ring)**.** *The degree-zero part of the total-degree grading on $\mathbb{Z}[X_i : i \in n]$ is the subring of constants: the structural map $\mathbb{Z} \to (\mathbb{Z}[X_i : i \in n])_0$ is a ring isomorphism. Consequently* Spec *of the degree-zero part is a terminal scheme.*

Proof. A polynomial is homogeneous of degree 0 exactly when it is a constant, so the degree-zero submodule is the image of $\mathbb{Z}$ under the (injective) constants map. ☐

Lemma 1186 (Properness of the integral model)**.** *For a $\mathbb{N}$-graded ring A of finite type over its degree-zero part A_0, the structural morphism* $\operatorname{Proj} A \to \operatorname{Spec} A_0$ *is proper (separated, universally closed, and of finite type).*

Lemma 1187 (Properness of relative projective space)**.** *For a finite index set n, the structural morphism $\mathbb{P}(n; S) \to S$ is proper (hence separated, universally closed, quasi-compact, and locally of finite type).*

Proof. For finite n, the polynomial ring $\mathbb{Z}[X_i : i \in n]$ is of finite type over its degree-zero part, which is $\mathbb{Z}$ by 1185; hence the integral model $\operatorname{Proj}\mathbb{Z}[X_i]$ is proper over $\operatorname{Spec}\mathbb{Z}$ by 1186, which is the terminal scheme. Properness is stable under base change, and the structural morphism of $\mathbb{P}(n; S)$ is the base change of $\operatorname{Proj}\mathbb{Z}[X_i] \to \operatorname{Spec}\mathbb{Z}$ along $S \to \operatorname{Spec}\mathbb{Z}$. ☐

Lemma 1188 (Base change of relative projective space)**.** *For every morphism $g : S' \to S$ the canonical square*

$$\begin{array}{ccc} \mathbb{P}(n; S') & \longrightarrow & \mathbb{P}(n; S) \\ \downarrow & & \downarrow \\ S' & \xrightarrow{g} & S \end{array}$$

is cartesian.

Proof. Both $\mathbb{P}(n; S')$ and $\mathbb{P}(n; S) \times_S S'$ are fibre products of $S' \to \operatorname{Spec}\mathbb{Z} \leftarrow \operatorname{Proj}\mathbb{Z}[X_i]$, by the pasting law for cartesian squares applied to the defining square of $\mathbb{P}(n; S)$ over the terminal scheme. ☐

Lemma 1189 (The transition units of the Serre twist)**.** *In the degree-zero part of the localization $\mathbb{Z}[X_i : i \in n]_{X_i X_j}$, the fraction $X_i/X_j = X_i^2/(X_i X_j)$ is a unit, with inverse $X_j/X_i = X_j^2/(X_i X_j)$.*

Proof. The product of the two fractions is $X_i^2 X_j^2/(X_i X_j)^2 = 1$. ☐

Lemma 1190 (Cocycle property of the Serre-twist transitions)**.** *On $\operatorname{Proj}\mathbb{Z}[X_i : i \in n]$, for each $m \geq 0$ the multiplication maps by $(X_i/X_j)^m$ on the trivial line bundles over the basic opens $D_+(X_i)$, transported to the pairwise scheme-theoretic overlaps of the cover, satisfy the descent cocycle condition on triple overlaps: over $D_+(X_i) \cap D_+(X_j) \cap D_+(X_k)$,*

$$\left(\frac{X_j}{X_k}\right)^m \left(\frac{X_i}{X_j}\right)^m \;=\; \left(\frac{X_i}{X_k}\right)^m,$$

compatibly with the identifications of the restricted trivial bundles.

Proof. In the degree-zero part of the localization $\mathbb{Z}[X_i]_{X_iX_jX_k}$, the three fractions satisfy $(X_i/X_j)(X_j/X_k) = X_i/X_k$ since $X_i^2X_j^2X_k^2 = (X_iX_j)(X_jX_k)(X_iX_k)$; raising to the m-th power preserves the identity. The multiplication maps by these units on the trivial line bundles commute with restriction to smaller opens, because multiplication by a section commutes with the restriction maps of the structure sheaf; and the trivialization comparisons between the restricted bundles compose transitively. Hence the transported transition isomorphisms compose as claimed. □

Definition 1191 (The Serre twisting sheaf on the integral model)**.** For $m \geq 0$, the *Serre twisting sheaf* $\mathcal{O}(m)$ on $\operatorname{Proj}\mathbb{Z}[X_i : i \in n]$ is the sheaf of modules glued from the trivial line bundles $\mathcal{O}_{D_+(X_i)}$ on the basic opens of the standard affine cover along the transition isomorphisms "multiplication by $(X_i/X_j)^m$" of 1189, which satisfy the descent cocycle condition by 1190.

Remark 1192 (Sign convention)**.** The gluing datum realises $\mathcal{O}(m)$, not $\mathcal{O}(-m)$. On $D_+(X_i)$ the trivialisation identifies a local section s of the twist with s/X_i^m; the nowhere-vanishing local frame is thus X_i^m, and its expression in the j-frame is $X_i^m/X_j^m = (X_i/X_j)^m$ — exactly the transition "multiplication by $(X_i/X_j)^m$" of the definition. Hence a global section is a compatible family (f_i) with $f_i = s/X_i^m$ and $f_iX_i^m = f_jX_j^m =: s$ homogeneous of degree m, so $\Gamma(\operatorname{Proj}\mathbb{Z}[X_i], \mathcal{O}(m))$ is the group of degree-m homogeneous polynomials; in particular $\mathcal{O}(1)$ has the X_i as sections and is very ample. The special case $m = 0$, where the factor $(X_i/X_j)^0 = 1$ makes the descent datum that of the structure sheaf, is recorded in Lean as `twistTransition_zero`.

Definition 1193 (The Serre twist on relative projective space)**.** The *Serre twist* $\mathcal{O}(m)$ on $\mathbb{P}(n; S)$ is the pullback of the twisting sheaf of 1191 along the projection $\mathbb{P}(n; S) \to \operatorname{Proj}\mathbb{Z}[X_i]$.

Lemma 1194 (Base change of the Serre twist)**.** *For $g : S' \to S$, the pullback of $\mathcal{O}(m)$ on $\mathbb{P}(n; S)$ along the base-change morphism $\mathbb{P}(n; S') \to \mathbb{P}(n; S)$ is canonically isomorphic to $\mathcal{O}(m)$ on $\mathbb{P}(n; S')$.*

Proof. Both sheaves are pullbacks of the twisting sheaf on the integral model, along the two legs of the commuting triangle formed by the base-change morphism and the two projections to $\operatorname{Proj}\mathbb{Z}[X_i]$. □

Lemma 1195 (The Serre twist is a locally trivial line bundle)**.** *For every $m \geq 0$, the Serre twisting sheaf $\mathcal{O}(m)$ on the integral model* $\operatorname{Proj}\mathbb{Z}[X_i : i \in n]$ *and its base change $\mathcal{O}(m)$ on $\mathbb{P}(n; S)$ (1193) are locally trivial line bundles (609): rank-one free on each member of the standard affine cover $\{D_+(X_i)\}$.*

Proof. On the basic-open chart $D_+(X_i)$ the gluing datum of 1191 trivialises the twist by construction: the chart restriction of the twist is the trivial line bundle $\mathcal{O}_{D_+(X_i)}$, realised through the descent-gluing comparison isomorphism of 1476. Since $\{D_+(X_i)\}$ covers $\operatorname{Proj}\mathbb{Z}[X_i]$ and each $D_+(X_i)$ is affine (a basic open of a homogeneous element of positive degree), $\mathcal{O}(m)$ is locally trivial. Local triviality is stable under pullback of sheaves of modules, so the twisting sheaf on $\mathbb{P}(n; S)$ — the pullback of $\mathcal{O}(m)$ along the projection to the integral model (1193) — is locally trivial as well. □

Corollary 1196 (Finite presentation and quasi-coherence of the Serre twist)**.** *The Serre twisting sheaf $\mathcal{O}(m)$ on the integral model and its base change on $\mathbb{P}(n; S)$ are finitely presented, hence quasi-coherent.*

Proof. Immediate from 1195 and 918 (a locally trivial line bundle is finitely presented); quasi-coherence then follows from finite presentation by 919. □

Theorem 1197 (H^0 base change of the Serre twist over a flat affine base)**.** *Let S be an affine scheme flat over* $\operatorname{Spec}\mathbb{Z}$ *and let n be a finite index set. For every $m \geq 0$ there is a $\Gamma(S, \mathcal{O}_S)$-linear equivalence*

$$\Gamma(S, \mathcal{O}_S) \otimes_{\mathbb{Z}} \Gamma(\operatorname{Proj}\mathbb{Z}[X_i : i \in n], \mathcal{O}(m)) \xrightarrow{\sim} \Gamma(\mathbb{P}(n; S), \mathcal{O}(m)).$$

Proof. Apply 1243 (H^0 flat base change) to the cartesian square defining $\mathbb{P}(n; S)$ (1188 for the base-change map $S \to \operatorname{Spec}\mathbb{Z}$), at $V = U = \top$: the structural morphism $\operatorname{Proj}\mathbb{Z}[X_i] \to \operatorname{Spec}\mathbb{Z}$ is proper (1186), hence quasi-compact and quasi-separated; the base-change morphism $S \to \operatorname{Spec}\mathbb{Z}$ is flat by hypothesis; and $\mathcal{O}(m)$ is quasi-coherent (1196). This identifies $\Gamma(S, \mathcal{O}_S) \otimes_{\mathbb{Z}} \Gamma(\operatorname{Proj}\mathbb{Z}[X_i], \mathcal{O}(m))$ with $\Gamma(\mathbb{P}(n; S), \mathcal{O}(m))$, $\Gamma(S, \mathcal{O}_S)$-linearly. ☐

Definition 1198 (Projective morphism carrying a very ample line bundle)**.** A morphism of schemes $\pi : X \to S$ is *projective carrying the line bundle L* if there are $d \geq 0$ and a closed immersion $i : X \hookrightarrow \mathbb{P}(\{0, \dots, d\}; S)$ over S with $L \cong i^*\mathcal{O}(1)$. In this situation L is relatively very ample and π is projective in the sense of [Nitsure] §5.

Lemma 1199 (Projective morphisms are proper)**.** *A projective morphism carrying a line bundle is proper.*

Proof. A closed immersion is proper, the structural morphism of projective space is proper (1187), and properness is stable under composition. ☐

Lemma 1200 (Structural consequences of projectivity)**.** *A projective morphism carrying a line bundle is locally of finite type, separated, and universally closed.*

Proof. Immediate from 1199: properness of a morphism entails that it is locally of finite type, separated, and universally closed. ☐

Lemma 1201 (Invariance under isomorphism of the bundle)**.** *If $\pi : X \to S$ is projective carrying L and $L \cong L'$, then π is projective carrying L'.*

Proof. The witnessing closed immersion and its comparison $L \cong i^*\mathcal{O}(1)$ are unchanged; precompose the comparison with the given isomorphism $L' \cong L$ to obtain $L' \cong i^*\mathcal{O}(1)$. ☐

Lemma 1202 (Composition with closed immersions)**.** *If $\pi : X \to S$ is projective carrying L and $j : Y \to X$ is a closed immersion, then $\pi \circ j$ is projective carrying j^*L.*

Proof. Compose the given closed immersion $i : X \hookrightarrow \mathbb{P}(\{0, \dots, d\}; S)$ with j: a composite of closed immersions is a closed immersion, the triangle over S still commutes, and $j^*L \cong j^*i^*\mathcal{O}(1) \cong (i \circ j)^*\mathcal{O}(1)$. ☐

Lemma 1203 (Base change of projective morphisms)**.** *If $\pi : X \to S$ is projective carrying L and $g : S' \to S$ is any morphism, then the base change $X \times_S S' \to S'$ is projective carrying the pullback of L along $X \times_S S' \to X$.*

Proof. Base-change the closed immersion i along $\mathbb{P}(\{0, \dots, d\}; S') \to \mathbb{P}(\{0, \dots, d\}; S)$ (1188): closed immersions are stable under base change, the induced comparison morphism $X \times_S S' \to \mathbb{P}(\{0, \dots, d\}; S')$ lies over S', and the pulled-back bundle is identified with $\mathcal{O}(1)$ of the base-changed projective space by 1194. ☐

Lemma 1204 (Projective space is projective over its base)**.** *The structural morphism $\mathbb{P}(\{0, \dots, d\}; S) \to S$ is projective carrying the Serre twist $\mathcal{O}(1)$; the witnessing closed immersion is the identity. In particular the projective-with predicate is non-vacuous.*

Proof. Take the identity closed immersion $\mathbb{P}(\{0, \dots, d\}; S) \to \mathbb{P}(\{0, \dots, d\}; S)$; the triangle over S is trivial and $\mathcal{O}(1) \cong \mathrm{id}^*\mathcal{O}(1)$. □

Lemma 1205 (Projectivity descends to scheme-theoretic fibres)**.** *Let $\pi : X \to S$ be projective carrying L and let $s \in S$. Then the structural morphism of the scheme-theoretic fibre $X_s = X \times_S \operatorname{Spec} \kappa(s)$, namely $X_s \to \operatorname{Spec} \kappa(s)$, is projective carrying the restriction $L_s = L|_{X_s}$.*

Proof. The fibre X_s is the base change of π along the canonical morphism $\operatorname{Spec} \kappa(s) \to S$, and $X_s \to \operatorname{Spec} \kappa(s)$ is the second projection. Apply 1203 to that morphism: the base change is projective carrying the pullback of L along the fibre embedding $X_s \to X$, which is exactly L_s. □

Lemma 1206 (Finite presentation descends to fibres)**.** *If F is a finitely presented sheaf of modules on X, then its restriction $F_s = F|_{X_s}$ to the fibre over $s \in S$ (1019) is finitely presented.*

Proof. F_s is the pullback of F along the fibre embedding $X_s \to X$, and finite presentation of a sheaf of modules is stable under pullback along an arbitrary morphism of schemes. □

Lemma 1207 (Serre finiteness along a fibre)**.** *Let $\pi : X \to S$ be projective carrying L, let F be finitely presented, and let $s \in S$. Then the section graded ring $R(X_s, L_s)$ (1021) is Noetherian and the section graded module $M(X_s, F_s, L_s)$ (1022) is a finitely generated $R(X_s, L_s)$-module.*

Proof. The fibre X_s is projective over the field $\kappa(s)$ carrying L_s (1205), and F_s is coherent (1206); apply 1024 on X_s, retaining its first two conclusions. □

Lemma 1208 (The graded Hilbert function is a genuine dimension on projective fibres)**.** *Let $\pi : X \to S$ be projective carrying L, let F be finitely presented, and let $s \in S$. Then for every m the space of global sections $\Gamma(X_s, F_s \otimes L_s^{\otimes m})$ is a finite-dimensional $\kappa(s)$-vector space. In particular the graded Hilbert function $m \mapsto \dim_{\kappa(s)} \Gamma(X_s, F_s \otimes L_s^{\otimes m})$ (1018) takes finite values, so it is not the* finrank *junk value at any m.*

Proof. This is the third conclusion of 1024 applied on the fibre X_s, which is projective over $\kappa(s)$ carrying L_s (1205) with F_s coherent (1206): finite dimensionality over the residue field of each graded component $\Gamma(X_s, F_s \otimes L_s^{\otimes m})$. □

20.8.1 Global sections of a glued sheaf of modules

The Serre-twist sheaf of 1191 is built by gluing (1479); the following computes the global sections of any glued sheaf of modules as an explicit submodule of compatible chart sections, and specialises it to the Serre twist.

Definition 1209 (Compatible families of chart sections)**.** Let D be a glue datum of schemes with chart family $(\mathcal{M}_i)_{i \in J}$ of sheaves of $\mathcal{O}_{U_i}$-modules and transition isomorphisms (g_{ij}), as in 1479. The $\Gamma(D.\mathrm{glued}, X)$-submodule of *compatible families*

$$\mathrm{Compat}(D, M, g) \subseteq \prod_{i \in J} \Gamma(U_i, \mathcal{M}_i)$$

consists of the families $(s_i)_i$ such that for every pair (i, j), the restriction of s_i to the overlap V_{ij} along f_{ij} agrees, through the transition isomorphism g_{ij}, with the restriction of s_j to V_{ij} along $t_{ij} \circ f_{ji}$.

Lemma 1210 (Global sections of a product of sheaves of modules)**.** *Let $(\mathcal{N}_i)_{i\in I}$ be a family of sheaves of $\mathcal{O}_X$-modules on a scheme X and let $\prod_i^c \mathcal{N}_i$ be their categorical product in* X.Modules*. The family of projections $\pi_i : \prod_i^c \mathcal{N}_i \to \mathcal{N}_i$ induces, on global sections, a bijection*

$$\Gamma\Big(\prod_i^c \mathcal{N}_i,\, X\Big) \xrightarrow{\ \sim\ } \prod_i \Gamma(\mathcal{N}_i, X), \qquad x \mapsto \big(\pi_i(x)\big)_i;$$

in particular the projections are jointly injective, and every family $(s_i)_i \in \prod_i \Gamma(\mathcal{N}_i, X)$ is the family of projections of some global section of the product.

Proof. Evaluation of sheaves of modules at the maximal open computes global sections and preserves all limits, so it carries the categorical product to the product, in $\Gamma(\mathcal{O}_X, X)$-modules, of the global sections, with components the section-level projections; the resulting comparison map is therefore invertible. Injectivity of the family of projections and surjectivity onto arbitrary families of sections are the two halves of this bijection. □

Lemma 1211 (Global sections of a glued sheaf are the compatible families)**.** *With the data and module cocycle conditions (C1)(C2) of 1479, restriction to the charts induces a $\Gamma(D.\mathrm{glued}, X)$-linear isomorphism*

$$\Gamma(\mathrm{glue}(D, M, g),\, X) \xrightarrow{\ \sim\ } \mathrm{Compat}(D, M, g)$$

(1209), sending a global section to the family of its restrictions to the charts U_i, with inverse sending a compatible family to the unique global section restricting to it on every chart. In particular global sections of the glued sheaf are jointly detected by their restrictions to the charts.

Proof. The glued sheaf is by construction the equalizer of the two parallel maps

$$\prod_i^c (\iota_i)_*\mathcal{M}_i \ \rightrightarrows\ \prod_{(i,j)}^c (f_{ij} \circ \iota_i)_*(f_{ij}^*\mathcal{M}_i)$$

encoding the two descent legs of 1479. Taking global sections preserves this equalizer (it is a limit-preserving functor), yielding, in $\Gamma(D.\mathrm{glued}, X)$-modules, the equalizer of the corresponding section-level maps; by 1210 the section functor identifies the two outer products with the products of sections, and global sections of a pushforward at the ambient open agree with the global sections of the original sheaf, so the section-level equalizer is exactly the submodule of families on which the two descent legs agree componentwise, i.e. $\mathrm{Compat}(D, M, g)$. Under this identification the chart-restriction map corresponds to the equalizer inclusion, giving the stated isomorphism; joint detection by chart restrictions is injectivity of the equalizer inclusion. □

Lemma 1212 (Pullback along an isomorphism is pushforward of the inverse)**.** *Let $\varphi : X \to Y$ be an isomorphism of schemes. Then the pullback functor $\varphi^* :$ Y.Modules $\to$ X.Modules is naturally isomorphic to the pushforward functor $(\varphi^{-1})_* :$ Y.Modules $\to$ X.Modules along the inverse isomorphism.*

Proof. Pushforward along φ and pushforward along φ^{-1} are mutually inverse equivalences between X.Modules and Y.Modules, by pseudofunctoriality of pushforward: the composite pushforward, in either order, is naturally isomorphic to the identity, since $\varphi \circ \varphi^{-1}$ and $\varphi^{-1} \circ \varphi$ are identities. In particular pushforward along φ^{-1} is left adjoint to pushforward along φ; since φ^* is also left adjoint to pushforward along φ, uniqueness of left adjoints up to natural isomorphism identifies the two functors. □

Lemma 1213 (The first descent leg on unit modules is restriction along the overlap immersion)**.** *Let D be a glue datum of schemes and instantiate the chart family of 1479 at the unit modules $\mathcal{M}_i = \mathcal{O}_{U_i}$. Then, through the canonical trivialisation of the pulled-back structure sheaf, the*

(i,j)-component of the first descent leg acts on a section $x \in \Gamma(U_i, \mathcal{O})$ as its restriction along the overlap immersion $f_{ij} : V_{ij} \to U_i$:

$$a_{ij}(x) = x|_{V_{ij}} \in \Gamma(V_{ij}, \mathcal{O}).$$

Proof. The first descent leg is the pushforward, along f_{ij} followed by ι_i, of the unit of the pullback–pushforward adjunction applied to $\mathcal{O}_{U_i}$, post-composed with the pushforward-composition comparison. Composed with the trivialisation of the pulled-back unit sheaf, naturality of the pushforward-composition comparison and the identification of the adjunction unit at the unit sheaf with the canonical comorphism of f_{ij} rewrite this composite as the pushforward of that comorphism; on global sections the pushforward of a comorphism acts by the ring-restriction map, giving the stated formula. □

Lemma 1214 (Global sections of the glued Serre twist)**.** *For the descent datum of 1191 (the standard affine cover of* Proj $\mathbb{Z}[X_i : i \in n]$ *by the* $D_+(X_i)$*, unit chart modules, and transition isomorphisms "multiplication by* $(X_i/X_j)^m$*"), 1211 specialises to a* $\Gamma(\text{Proj}\,\mathbb{Z}[X_i], \mathcal{O})$*-linear isomorphism between the global sections of the glued twist and the compatible families of chart sections for that datum.*

Proof. Apply 1211 with $M_i = \mathcal{O}_{D_+(X_i)}$ and g_{ij} multiplication by $(X_i/X_j)^m$; the module cocycle conditions (C1)(C2) required by 1479 hold: (C1) is the identity of the transition at $i = j$, since $(X_i/X_i)^m = 1$, and (C2) is exactly 1190. □

Lemma 1215 (Global sections of the Serre twist are the compatible chart families)**.** *The Serre twisting sheaf* $\mathcal{O}(m)$ *on* Proj $\mathbb{Z}[X_i : i \in n]$ *(1191) is isomorphic to the pushforward, along the canonical comparison from the abstractly glued model of 1214 to* Proj*, of the glued twist (1212, applied to the inverse of that comparison isomorphism); consequently its global sections form an additive group isomorphic to the compatible families of 1214:*

$$\Gamma(\text{Proj}\,\mathbb{Z}[X_i],\, \mathcal{O}(m)) \;\cong\; \text{Compat}(D, \mathcal{O}, ((X_i/X_j)^m)).$$

(The sign convention of 1192 governs the orientation of the transition isomorphisms entering this identification.)

Proof. The gluing construction realises Proj $\mathbb{Z}[X_i]$ as the target of an isomorphism from the abstractly glued scheme of the standard affine cover; pulling the twisting sheaf back along that isomorphism and applying 1212 identifies it with a pushforward of the glued twist, and global sections of a pushforward at the ambient open agree definitionally with those of the original sheaf. Composing this identification of global sections with 1214 gives the stated isomorphism. □

20.8.2 The compatible-family condition, concretely

1215 identifies global sections of $\mathcal{O}(m)$ with the abstract compatible-family submodule $\text{Compat}(D, \mathcal{O}, ((X_i/X_j)^m))$ (1209) of the descent datum instantiated at the transition isomorphisms "multiplication by $(X_i/X_j)^m$". The next lemmas make this compatibility condition concrete: through the chart identification with the degree-zero localisation $\text{Away}(X_i)$, a compatible family becomes a family of away-fractions related on overlaps by the transition unit, and a single degree-m form gives such a family.

Lemma 1216 (The second descent leg of the Serre-twist datum, concretely)**.** *Instantiate the chart family of 1479 at the unit modules and the descent transition of 1191. Through the i-chart* *trivialisation, the second ("b") descent leg sends a j-chart section $y \in \Gamma(U_j, \mathcal{O})$ to its restriction along the overlap immersion $t_{ij} \circ f_{ji} : V_{ij} \to U_j$, scaled by the inverse transition unit $(X_j/X_i)^m$:*

$$b_{ij}(y) = y|_{V_{ij}} \cdot (X_j/X_i)^m \in \Gamma(V_{ij}, \mathcal{O}).$$

This is the "b-side" companion of the a-side restriction formula of 1213.

Proof. The second descent leg is, before trivialisation, the pushforward of the canonical comorphism of $t_{ij} \circ f_{ji}$ followed by the scalar automorphism "multiplication by $(X_j/X_i)^m$" and a reindexing along the glue-datum's cocycle identity; composing with the i-chart trivialisation and cancelling it via the shared abstract unit-leg trivialisation identity (used identically for the a-leg in 1213) leaves exactly the comorphism of $t_{ij} \circ f_{ji}$ composed with the scalar action. On global sections the pushforward of a comorphism acts by ring restriction and the pushforward of a scalar automorphism acts by multiplication (both through the definitional identification $\Gamma(\rho_*\mathcal{O}, \top) = \Gamma(\mathcal{O}, \top)$), and the reindexing comparison acts as the identity on sections; composing these gives the stated formula. ☐

Lemma 1217 (The concrete Serre-twist compatible-family condition)**.** *A chart-section family $(s_i)_i$, $s_i \in \Gamma(D_+(X_i), \mathcal{O})$, is a compatible family for the descent datum of $\mathcal{O}(m)$ (1209) if and only if, on every overlap $D_+(X_i) \cap D_+(X_j)$,*

$$s_i|_{V_{ij}} = s_j|_{V_{ij}} \cdot (X_j/X_i)^m.$$

This is the section-level realisation of the sign convention of 1192: the compatible-family condition scales by $(X_j/X_i)^m$, not $(X_i/X_j)^m$.

Proof. Unfold membership in Compat (1209) and transport the a-leg and b-leg conditions through the i-chart trivialisation via 1213 and 1216 respectively; the trivialisation is injective on sections (being the global-section evaluation of an isomorphism of sheaves of modules), so the transported condition is equivalent to the original one. ☐

Definition 1218 (Chart sections as the degree-zero localisation)**.** The structure-sheaf sections $\Gamma(D_+(X_i), \mathcal{O})$ of the i-th chart of $\operatorname{Proj} \mathbb{Z}[X_i : i \in n]$ are identified with the degree-zero part of the localisation at X_i, $\operatorname{Away}(X_i) := (\mathbb{Z}[X_i]_{X_i})_0$, via Mathlib's chart isomorphism for Proj (valid since X_i is homogeneous of positive degree 1), transported to the chart's global sections through the open-subscheme identification of the chart with its own affine spectrum.

Proof. Composite of Mathlib's Proj basic-open-versus-away isomorphism with the canonical global-sections identification of an open subscheme with its $\top$-sections. ☐

20.8.3 The away-fraction bridge and the degree-m forms

Through the chart identification 1218, the two descent legs of 1217 become explicit maps between localisation rings; this section records that comparison (the "away-fraction bridge"), the resulting compatibility condition on a single homogeneous form, its consequence that the degree-m forms embed into $\Gamma(\mathcal{O}(m))$, and the coordinate sections of $\mathcal{O}(1)$.

Lemma 1219 (The away-fraction bridge)**.** *Through the chart identification of 1218, the two descent legs restricting to the overlap $D_+(X_iX_j)$ are the localisation-algebra maps $\operatorname{Away}(X_i) \to \operatorname{Away}(X_iX_j)$ and $\operatorname{Away}(X_j) \to \operatorname{Away}(X_iX_j)$ sending a fraction to its image under the further localisation at the other variable, followed by the canonical identification of $\operatorname{Away}(X_iX_j)$ with itself along the two orderings of the product.*

Proof. On global sections, restriction of a chart section along the overlap immersion into $D_+(X_iX_j) \subseteq D_+(X_i)$ (resp. $D_+(X_j)$) is, after the chart identification, exactly the further-localisation (away-map) comparison of homogeneous localisations at the common degree-zero ring $\mathrm{Away}(X_iX_j)$. $\square$

Lemma 1220 (A single form gives a compatible family)**.** *For a degree-m homogeneous form F, the chart family $(F/X_i^m)_i$ — the away-fraction of F at each chart, through 1218 — is a compatible family for the Serre-twist descent datum (1217).*

Proof. In $\mathrm{Away}(X_iX_j)$, the two away-map images of F/X_i^m and F/X_j^m differ by exactly $(X_j/X_i)^m$ (a direct fraction identity: both sides equal $F/(X_iX_j)^m$ up to the appropriate unit power). Transporting through the away-fraction bridge (1219) turns this into the compatibility condition of 1217. $\square$

Lemma 1221 (The degree-m forms embed as global sections of $\mathcal{O}(m)$)**.** *There is an additive homomorphism from the degree-m homogeneous forms of $\mathbb{Z}[X_i : i \in n]$ to $\Gamma(\mathrm{Proj}\,\mathbb{Z}[X_i], \mathcal{O}(m))$, sending F to the global section corresponding, under 1215, to the compatible chart family of 1220. When the index set n is nonempty, this homomorphism is injective: a global section is determined by any one of its chart fractions, and the i-chart fraction of the image of F recovers F since X_i is a nonzerodivisor of the domain $\mathbb{Z}[X_i]$.*

Proof. Compose 1220 (additively in F, since the chart away-fraction $F \mapsto F/X_i^m$ and the chart identification are additive) with the inverse of the compatible-family isomorphism of 1215. For injectivity, fix an index i (using nonemptiness of n): if two forms have equal images, their i-chart fractions agree by naturality of the isomorphisms involved, and equality of away-fractions $F/X_i^m = F'/X_i^m$ in a domain forces $F = F'$ after clearing the nonzerodivisor X_i^m. $\square$

Definition 1222 (Coordinate global sections of $\mathcal{O}(1)$)**.** The *coordinate global section* $x_j \in \Gamma(\mathrm{Proj}\,\mathbb{Z}[X_i], \mathcal{O}(1))$ is the image of the degree-one form X_j under the embedding of 1221. On the i-th chart it restricts to the away-fraction X_j/X_i.

Proof. Direct instantiation of 1221 at $m = 1$, $F = X_j$; the chart restriction identity is the i-chart value of 1220 applied to that same form. $\square$

Lemma 1223 (Global sections of $\mathcal{O}(m)$ are the degree-m forms)**.** *For n nonempty, the embedding of 1221 of the degree-m homogeneous forms of $\mathbb{Z}[X_i : i \in n]$ into $\Gamma(\mathrm{Proj}\,\mathbb{Z}[X_i], \mathcal{O}(m))$ is onto: every global section of $\mathcal{O}(m)$ arises from a (unique) degree-m form, i.e. $\Gamma(\mathrm{Proj}\,\mathbb{Z}[X_i], \mathcal{O}(m)) \cong (\mathbb{Z}[X_i])_m$.*

Proof. Injectivity is 1221; it remains to prove surjectivity. A global section corresponds, via 1215 and the away-fraction bridge 1219, to a compatible family $(a_i)_{i \in n}$ of chart fractions, each a_i an element of the degree-zero part of $\mathrm{Away}(X_i)$, compatible in the sense that the images of a_i and a_j in $\mathrm{Away}(X_iX_j)$ agree. Fix any index i_0 (possible since n is nonempty) and write $a_{i_0} = N_0/X_{i_0}^{k_0}$ in lowest terms, with N_0 homogeneous of degree $k_0 + m$. The ring $\mathbb{Z}[X_i : i \in n]$ is a unique factorisation domain in which every variable X_i is a prime element; comparing a_{i_0} against every other a_i on the pairwise overlaps forces $X_{i_0}^{k_0}$ to divide $X_{i_0}^m N_0$, and primality of X_{i_0} upgrades this to the existence of a homogeneous form F of degree m with $X_{i_0}^{k_0} F = X_{i_0}^m N_0$, i.e. $F/X_{i_0}^m = a_{i_0}$. For every other index i, the same cross relation applied to the pair (i_0, i), followed by cancellation of the nonzerodivisor $X_{i_0}^{k_0}$, gives $FX_i^{k_i} = N_i X_i^m$, i.e. $F/X_i^m = a_i$. Hence the compatible family (a_i) is exactly the chart family of F (1220), so the section it represents lies in the image of the embedding of 1221. $\square$

20.8.4 The graded Hilbert–Serre bridge

The rationality engine 1028 consumes an abstract finitely generated $\mathbb{N}$-graded κ-vector space carrying r commuting degree-one endomorphisms. We record its input as a package GradedHilbert, factor Serre finiteness in this engine-ready form as a single leaf (1226, a companion to 1024), and derive the rationality of the fibre graded Hilbert function — hence the genuineness of the Hilbert polynomial (1018) — as sorry-free glue. This is the DirectSum-decomposition bridge of inbox `I-0109`.

Definition 1224 (Engine-ready packaging of a Hilbert function)**.** Fix a field κ and a function $f : \mathbb{N} \to \mathbb{N}$. A GradedHilbert datum for (κ, f) is a natural number r, an $\mathbb{N}$-graded κ-vector space $M = \bigoplus_n M_n$ (an internal grading $\mathcal{M} : \mathbb{N} \to \mathrm{Submodule}_\kappa M$ with DirectSum.Decomposition) whose graded pieces M_n are all finite-dimensional, together with r pairwise-commuting κ-linear endomorphisms $g_0, \dots, g_{r-1}$ that each raise the grading degree by one, such that $\dim_\kappa M_n = f(n)$ for all n and M is a finite module over the polynomial ring $\kappa[X_0, \dots, X_{r-1}]$ acting through the g_i. This bundles exactly the data and hypotheses consumed by the graded Hilbert–Serre rationality engine 1028.

Lemma 1225 (Rationality from a GradedHilbert datum)**.** *If (κ, f) admits a* GradedHilbert *datum of order r (1224), then f is a rational Hilbert function of order r (1031): there are $p \in \mathbb{Q}[X]$ and N with $f(n) = [X^n](p \cdot (1 - X)^{-r})$ for all $n > N$.*

Proof. Apply the graded Hilbert–Serre rationality engine (1028) to the graded module M, its grading $\mathcal{M}$, and the degree-one generators g_i of the datum; its hypotheses (finite-dimensional components, finiteness over $\kappa[X_0, \dots, X_{r-1}]$) are the datum's fields. This yields that $n \mapsto \dim_\kappa M_n$ is a rational Hilbert function of order r, and $\dim_\kappa M_n = f(n)$ identifies it with f. $\square$

Lemma 1226 (Serre finiteness in engine-ready form)**.** *Source: [Nitsure], §1 (FGA Explained Ch. 5); Serre's theorem, as in 1024. Let $\pi : X \to S$ be projective carrying the very ample line bundle L, let F be finitely presented, and let $s \in S$. Then the graded Hilbert function $m \mapsto \dim_{\kappa(s)} \Gamma(X_s, F_s \otimes L_s^{\otimes m})$ (1018) admits a* GradedHilbert *datum over the residue field $\kappa(s)$ (1224).*

Status. *This is 1024 (Serre) restated in engine-ready form; its Lean statement is a named leaf. Taking M to be the fibre section module $M(X_s, F_s, L_s)$ and the $d+1$ generators to be the homogeneous coordinate sections of the very ample embedding $X_s \hookrightarrow \mathbb{P}^d_{\kappa(s)}$, the content is the finite generation of $M(X_s, F_s, L_s)$ over the homogeneous coordinate ring $\kappa(s)[x_0, \dots, x_d]$ — the finiteness of 1024 together with the degree-one coordinate action.*

Proof. The fibre X_s is projective over $\kappa(s)$ carrying L_s (1205), so it admits a closed immersion $i : X_s \hookrightarrow \mathbb{P}^d_{\kappa(s)}$ with $L_s \cong i^*\mathcal{O}(1)$, and F_s is coherent. By Serre (1024) the section module $M(X_s, F_s, L_s)$ is a finitely generated graded module over $R(X_s, L_s)$ with finite-dimensional components. The homogeneous coordinates $x_0, \dots, x_d$ restrict along i to $d+1$ sections of L_s generating $R(X_s, L_s)$ in large degrees, so $M(X_s, F_s, L_s)$ is finite over the polynomial ring $\kappa(s)[x_0, \dots, x_d]$, with x_j acting by multiplication (raising degree by one). Its m-th graded piece $\Gamma(X_s, F_s \otimes L_s^{\otimes m})$ has $\kappa(s)$-dimension the graded Hilbert function, so this data is a GradedHilbert datum. $\square$

Lemma 1227 (Rationality of the fibre graded Hilbert function)**.** *Let $\pi : X \to S$ be projective carrying L, let F be finitely presented, and let $s \in S$. Then the graded Hilbert function $m \mapsto \dim_{\kappa(s)} \Gamma(X_s, F_s \otimes L_s^{\otimes m})$ is a rational Hilbert function of some order d (1031).*

Proof. By 1226 the graded Hilbert function admits a GradedHilbert datum over $\kappa(s)$; apply 1225. $\square$

Lemma 1228 (The Hilbert polynomial computes the Hilbert function on projective fibres)**.** *Let S be locally Noetherian, let $\pi : X \to S$ be projective carrying L and locally of finite type, let F be finitely presented, and let $s \in S$. Then for all $m \gg 0$ the Hilbert polynomial of 1018 satisfies*

$$\Phi_{F,s}(m) \;=\; \dim_{\kappa(s)} \Gamma(X_s,\, F_s \otimes L_s^{\otimes m}).$$

This is the projective-fibre realisation of 1043: it is the point at which $\Phi_{F,s}$ ceases to be a formal choice and becomes the genuine Snapper/Hilbert polynomial of the fibre.

Proof. By 1227 the graded Hilbert function is a rational Hilbert function of some order d; conclude by the polynomial-extraction corollary 1041. □

20.9 Representability of the Quot scheme

Theorem 1229 (Representability of the Quot functor)**.**

Source: [Nitsure], §5, Theorem (Grothendieck), Theorem (Altman–Kleiman) (FGA Explained Ch. 5); cf. Grothendieck, FGA TDTE-IV. Let S be a noetherian scheme, $\pi : X \to S$ a projective morphism, L a relatively very ample line bundle on X, E a coherent $\mathcal{O}_X$-module, and $\Phi \in \mathbb{Q}[\lambda]$. Then the functor $\mathrm{Quot}^{\Phi,L}_{E/X/S}$ of 1138 is representable by a projective S-scheme $\mathrm{Quot}^{\Phi,L}_{E/X/S}$, equipped with a universal quotient

$$q^{\mathrm{univ}} : \pi^*_{\mathrm{Quot}} E \twoheadrightarrow \mathcal{F}^{\mathrm{univ}} \quad \textit{on } X \times_S \mathrm{Quot}^{\Phi,L}_{E/X/S},$$

flat over $\mathrm{Quot}^{\Phi,L}_{E/X/S}$ with constant Hilbert polynomial Φ on every fiber. The Hilbert scheme is the special case $E = \mathcal{O}_X$: $\mathrm{Hilb}^{\Phi,L}_{X/S} = \mathrm{Quot}^{\Phi,L}_{\mathcal{O}_X/X/S}$ represents the functor of T-flat closed subschemes $Y \subset X_T$ with fiberwise Hilbert polynomial Φ.

Proof. The proof has four steps; each step is a sub-lemma in the next sub-sections.

1. **(Boundedness, 1231.)** Pick the Castelnuovo–Mumford regularity bound $m = m(\operatorname{rank} V, \operatorname{rank} W, \Phi)$ uniformly across all fibers (by Mumford's m-regularity theorem applied to E_s and to all coherent quotients of E_s with Hilbert polynomial Φ). For $r \geq m$ and any $T \to S$ and any T-flat quotient $q : E_T \twoheadrightarrow \mathcal{F}$ with Hilbert polynomial Φ, the higher direct images $R^i(\pi_T)_*\mathcal{F}(r)$ and $R^i(\pi_T)_*E_T(r)$ and $R^i(\pi_T)_*\mathcal{G}(r)$ vanish for $i \geq 1$ (where $\mathcal{G} = \ker q$), and the direct images $(\pi_T)_*\mathcal{F}(r)$, $(\pi_T)_*E_T(r)$, $(\pi_T)_*\mathcal{G}(r)$ are locally free of ranks fixed by the data; the canonical maps $(\pi_T)^*(\pi_T)_*(-)(r) \to (-)(r)$ are surjective. Cohomology-and-base-change (1234, Stacks tag 02KH) propagates these properties across T-base change.

2. **(Grassmannian embedding, 1232.)** Reduce by 1230 to the case $X = \mathbb{P}_S(V)$, $E = \pi^*W$ with V, W locally free on S, $L = \mathcal{O}_{\mathbb{P}(V)}(1)$. Fix $r \geq m$. The Grassmannian embedding

$$\alpha : \mathrm{Quot}^{\Phi,L}_{E/X/S} \longrightarrow \mathrm{Grass}\big(W \otimes_{\mathcal{O}_S} \mathrm{Sym}^r V,\ \Phi(r)\big)$$

sends $\langle \mathcal{F}, q\rangle$ on T to the locally free quotient $(\pi_T)_*(q(r)) : (\pi_T)_*E_T(r) \to (\pi_T)_*\mathcal{F}(r)$. This is injective on T-valued points: from the map $T \to \mathrm{Gr}_S(W \otimes \mathrm{Sym}^r V, \Phi(r))$ one recovers the kernel-pull-back $(\pi_T)^*(\pi_T)_*\mathcal{G}(r) \to E_T(r)$, whose cokernel is $q(r) : E_T(r) \to \mathcal{F}(r)$; twisting back by $-r$ recovers q.

3. **(Locally closed in the Grassmannian via flattening stratification.)** Let $T_0 = \mathrm{Gr}_S(W \otimes \mathrm{Sym}^r V, \Phi(r))$ with universal quotient $f : (W \otimes \mathrm{Sym}^r V)_{T_0} \twoheadrightarrow \mathcal{J}$ and kernel

$\mathcal{K}$. Form the composite $h : \pi_{T_0}^*\mathcal{K} \hookrightarrow \pi_{T_0}^*(W \otimes \mathrm{Sym}^r V) = \pi_{T_0}^*(\pi_{T_0})_* E_{T_0} \twoheadrightarrow E_{T_0}$ and let $q := \mathrm{coker}(h) : E_{T_0} \to \mathcal{F}$. By the flattening stratification (942 of 19) applied to $\mathcal{F}$ on X_{T_0}, there exists a locally closed subscheme $T_0^\Phi \subseteq T_0$ such that for any $\phi : Y \to T_0$, the pulled-back cokernel $\mathcal{F}_Y$ is Y-flat with fiberwise Hilbert polynomial Φ iff ϕ factors through T_0^Φ. The pair $(T_0^\Phi, \text{universal cokernel})$ represents $\mathrm{Quot}_{E/X/S}^{\Phi,L}$.

4. **(Closed, hence projective, 1233.)** The functor $\mathrm{Quot}_{E/X/S}^{\Phi,L}$ satisfies the valuative criterion of properness with respect to DVRs (Exercise (7) of §1, proved using that flatness over a DVR is torsion-freeness). Combined with the locally closed embedding of step 3 into the projective $\mathrm{Gr}_S \hookrightarrow \mathbb{P}_S(\bigwedge^{\Phi(r)}(W \otimes \mathrm{Sym}^r V))$, this upgrades to a *closed* embedding. Hence $\mathrm{Quot}_{E/X/S}^{\Phi,L}$ is a projective S-scheme.

Finally, by 1230, the general case (arbitrary X projective over S and arbitrary coherent E) follows from the case $X = \mathbb{P}(V)$, $E = \pi^*W$: pick a closed embedding $X \subset \mathbb{P}(V)$ and write E as a coherent quotient of $\pi^*(W)(\nu)$ for some vector bundle W on S and integer ν; the Quot scheme of the original data is then a closed subscheme of the Quot scheme of the universal data, by the closed-embedding clause of 1230. ☐

20.10 Sub-lemmas of the construction

Lemma 1230 (Reduction to the universal case)**.**
Source: [Nitsure], §5 (FGA Explained Ch. 5). Let S, X, L, E, Φ be as in 1138.

1. *(Twist-shift.) For $\nu \in \mathbb{Z}$, tensoring by $L^{\otimes\nu}$ gives an isomorphism of functors $\mathrm{Quot}_{E/X/S}^{\Phi,L} \xrightarrow{\sim} \mathrm{Quot}_{E(\nu)/X/S}^{\Psi,L}$ with $\Psi(\lambda) = \Phi(\lambda + \nu)$.*

2. *(Surjection induces closed embedding.) Given a surjection $\phi : E \twoheadrightarrow G$ of coherent sheaves on X, the induced natural transformation $\mathrm{Quot}_{G/X/S}^{\Phi,L} \to \mathrm{Quot}_{E/X/S}^{\Phi,L}$ is a closed embedding of functors.*

Consequently, if $\mathrm{Quot}_{\pi^*W/\mathbb{P}(V)/S}^{\Phi,L}$ is representable for all V, W vector bundles on S, then so is $\mathrm{Quot}_{E/X/S}^{\Phi,L}$ for every $X \subset \mathbb{P}(V)$ closed and every coherent E on X that is a quotient of $\pi^*(W)(\nu)|_X$ for some W and ν.

Proof. (i) is direct from the definitions: tensoring with $L^{\otimes\nu}$ is an autoequivalence of coherent sheaves on X (and on every X_T), preserving surjectivity, T-flatness, and properness of support, and shifting the Hilbert polynomial by ν in the variable.

(ii) Given a family $\langle \mathcal{F}, q\rangle$ on T in $\mathrm{Quot}_{E/X/S}^{\Phi,L}$, the locus where q factors through $\phi : E \twoheadrightarrow G$ is cut out by the vanishing of the composite $\ker(\phi)|_{X_T} \hookrightarrow E_T \xrightarrow{q} \mathcal{F}$. Because $\mathcal{F}$ is T-flat, the schematic image of this composite descends to a closed subscheme $T' \subset T$ such that $f : U \to T$ factors through T' iff q_U factors through ϕ_U. Hence $\mathrm{Quot}_{G/X/S}^{\Phi,L} \hookrightarrow \mathrm{Quot}_{E/X/S}^{\Phi,L}$ is a closed embedding of subfunctors. ☐

Lemma 1231 (Uniform Castelnuovo–Mumford regularity bound)**.**
Source: [Nitsure], §5, "Use of m-Regularity" (FGA Explained Ch. 5). Let $X = \mathbb{P}_S(V)$, $E = \pi^*W$ with V, W vector bundles on S of ranks $n+1$, p, $L = \mathcal{O}_{\mathbb{P}(V)}(1)$, and $\Phi \in \mathbb{Q}[\lambda]$. There exists an integer $m = m(n, p, \Phi)$, depending only on n, p, and Φ, such that for every geometric point s of S and every coherent quotient $q : E_s \twoheadrightarrow \mathcal{F}$ with Hilbert polynomial Φ, the sheaves

E_s, $\mathcal{F}$, and $\ker q$ on X_s are all m-regular in the sense of Castelnuovo–Mumford. Consequently, for every $T \to S$ and every T-flat quotient $q : E_T \twoheadrightarrow \mathcal{F}$ on X_T with Hilbert polynomial Φ and every $r \geq m$, the conclusions **(*)** *and* **(**)** *of [1, §5, "Use of m-Regularity"] hold: the direct images $(\pi_T)_*\mathcal{F}(r)$, $(\pi_T)_*E_T(r)$, $(\pi_T)_*\mathcal{G}(r)$ are locally free of fixed ranks; the canonical surjections $(\pi_T)^*(\pi_T)_*(-)(r) \twoheadrightarrow (-)(r)$ hold; and the higher direct images $R^i(\pi_T)_*(-)(r)$ vanish for $i \geq 1$.*

Proof. Each geometric fiber E_s is a trivial bundle on $\mathbb{P}^n_{\kappa(s)}$ of rank p, so E_s is 0-regular. The Castelnuovo–Mumford m-regularity theorem of [Nitsure] §2 produces, for any fixed Hilbert polynomial Φ, a uniform integer $m = m(n, p, \Phi)$ above which every coherent quotient of E_s with Hilbert polynomial Φ – and its kernel – is m-regular. The Castelnuovo Lemma of [Nitsure] §2 converts m-regularity into vanishing of $H^i((-)(r))$ for $i \geq 1$ and $r \geq m$, and into global generation of $H^0((-)(r))$. To propagate the fiber-wise statement to families over T, apply the flat-base-change theorem (1234, Stacks tag 02KH) along $T \to S$: cohomology of flat families commutes with base change, so the direct images $(\pi_T)_*(-)(r)$ are locally free of the rank dictated by the fiber Hilbert polynomials, and the canonical maps $(\pi_T)^*(\pi_T)_*(-)(r) \to (-)(r)$ inherit fiberwise surjectivity. $\square$

Lemma 1232 (Grassmannian embedding α is injective)**.**
Source: [Nitsure], §5, "Embedding Quot into Grassmannian" (FGA Explained Ch. 5). In the universal setup $X = \mathbb{P}_S(V)$, $E = \pi^*W$, fix $r \geq m$ from 1231. The natural transformation

$$\alpha : \mathrm{Quot}^{\Phi,L}_{E/X/S} \longrightarrow \mathrm{Grass}\Big(W \otimes_{\mathcal{O}_S} \mathrm{Sym}^r V,\ \Phi(r)\Big), \qquad \langle \mathcal{F}, q\rangle \longmapsto \langle (\pi_T)_*\mathcal{F}(r),\ (\pi_T)_*(q(r))\rangle$$

is well-defined (the target is locally free of rank $\Phi(r)$, and the canonical isomorphism $\pi_*E(r) \cong W \otimes_{\mathcal{O}_S} \mathrm{Sym}^r V$ gives the source) and injective *on T-points: the data of the locally free quotient at level r determines $q : E_T \to \mathcal{F}$.*

Proof. Suppose $\alpha(T)$ sends both $\langle \mathcal{F}_1, q_1\rangle$ and $\langle \mathcal{F}_2, q_2\rangle$ to the same T-point of the Grassmannian. Then the pulled-back kernel $(\pi_T)^*(\pi_T)_*\mathcal{G}_i(r) \to (\pi_T)^*(\pi_T)_*E_T(r) = \pi^*_{T_0}(W \otimes \mathrm{Sym}^r V)_T$ is the same. The diagram **(**)** of 1231 gives a right-exact sequence

$$(\pi_T)^*(\pi_T)_*\mathcal{G}_i(r) \xrightarrow{h} E_T(r) \xrightarrow{q_i(r)} \mathcal{F}_i(r) \to 0,$$

so $q_i(r)$ is recovered as the cokernel of h; twisting by $-r$ recovers q_i, giving $q_1 = q_2$ and hence $\langle \mathcal{F}_1, q_1\rangle = \langle \mathcal{F}_2, q_2\rangle$. $\square$

Lemma 1233 (Valuative criterion of properness for Quot)**.** *Source: [Nitsure], §5, "Valuative Criterion for Properness" (FGA Explained Ch. 5); EGA IV(2) 2.8.1. The functor $\mathrm{Quot}^{\Phi,L}_{E/X/S}$ satisfies the valuative criterion for properness over S: for every discrete valuation ring R with fraction field K and every morphism $\mathrm{Spec}\, R \to S$, the restriction map*

$$\mathrm{Quot}^{\Phi,L}_{E/X/S}(\mathrm{Spec}\, R) \longrightarrow \mathrm{Quot}^{\Phi,L}_{E/X/S}(\mathrm{Spec}\, K)$$

is bijective.

Proof. Given a $\mathrm{Spec}\, K$-point $\langle \mathcal{F}_K, q_K\rangle$ on X_K, let $j : X_K \hookrightarrow X_R$ be the open inclusion. Define $\overline{\mathcal{F}}$ as the image of the composite $E_R \to j_*E_K \to j_*\mathcal{F}_K$, with induced surjection $\overline{q} : E_R \to \overline{\mathcal{F}}$. The sheaf $\overline{\mathcal{F}}$ is torsion-free over R (being a subsheaf of $j_*\mathcal{F}_K$), hence R-flat (flatness over a DVR = torsion-freeness), and restricts on the generic fiber to $\mathcal{F}_K$. The pair $(\overline{\mathcal{F}}, \overline{q})$ is the unique extension of $\langle \mathcal{F}_K, q_K\rangle$ to a $\mathrm{Spec}\, R$-point (uniqueness: any other extension agrees on the dense open $X_K \subset X_R$ and is torsion-free, hence equals the image construction). The Hilbert polynomial is constant on connected $\mathrm{Spec}\, R$ by flatness. $\square$

20.11 Cohomology and base change

The Quot construction uses cohomology-and-base-change in two places: the boundedness step (1231) and the Grassmannian embedding (1232). We record the relevant statement as a named theorem so the Lean encoding can cite it directly.

Theorem 1234 (Degree-zero flat base change)**.** *Source: [Stacks Project], tag 02KH (lemma-flat-base-change-cohomology). Let*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ \downarrow f' & & \downarrow f \\ S' & \xrightarrow{g} & S \end{array}$$

be a cartesian diagram of schemes with g flat and f quasi-compact and quasi-separated. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module with pull-back $\mathcal{F}' = (g')^*\mathcal{F}$. Then the canonical base-change map

$$g^* f_* \mathcal{F} \longrightarrow f'_* \mathcal{F}'$$

is an isomorphism.

Proof. The displayed morphism is the canonical base-change map of 1237. Its invertibility under the stated hypotheses is 1249; taking the associated isomorphism proves the claim. ☐

20.11.1 Project-side base-change substrate

The Lean encoding of 1234 (and the Quot construction's iso between pushforward and tensor-pullback for the universal section pattern) relies on the project-side declarations introduced in `AlgebraicJacobian/Picard/QuotScheme.lean` below. The whole chain is proved, including its former last leaf, the H^0 flat-base-change heart over a compatible affine pair (1243, Stacks tag 02KE), from which every isomorphism statement below follows by the affine-pair workhorse 1244 and basis locality. The abstract algebraic heart of that leaf is recorded first, in the pure module-theoretic form of 1236 below.

Lemma 1235 (Injectivity along a finite cover, by flat base change)**.** *Let $A \to B$ be a flat homomorphism of commutative rings, let M be an A-module and P a B-module, and let $(M_i)_{i\in\iota}$, $(P_i)_{i\in\iota}$ be finite families of modules with A-linear maps $\mathrm{res}_i : M \to M_i$ jointly injective and B-linear maps $\mathrm{res}'_i : P \to P_i$. Suppose $s : B \otimes_A M \to P$ is B-linear and, for each i, there is an injective B-linear comparison $\varepsilon_i : B \otimes_A M_i \to P_i$ with $\mathrm{res}'_i(s(1 \otimes m)) = \varepsilon_i(1 \otimes \mathrm{res}_i(m))$ for all $m \in M$. Then s is injective.*

Proof. An element z of $\ker s$ has, for every i, image 0 under $\varepsilon_i \circ (\mathrm{id}_B \otimes \mathrm{res}_i)$ (by B-linearity, the intertwining hypothesis extends from the generators $1 \otimes m$ to all of $B \otimes_A M$), hence lies in the kernel of $\mathrm{id}_B \otimes \mathrm{res}_i$ by injectivity of ε_i. Joint injectivity of the res_i on M transports, via flatness of B over A, to injectivity of the induced map $B \otimes_A M \to \prod_i (B \otimes_A M_i)$ (flatness of B means tensoring with B preserves injective A-linear maps, applied to $\prod_i \mathrm{res}_i$); since z lies in the kernel of every component of this map, $z = 0$. ☐

Lemma 1236 (The Stacks 02KE equalizer ladder, abstract form)**.** *Source: Stacks 02KE, module dialect. Let $A \to B$ be a flat homomorphism of commutative rings. Let M be an A-module equipped with a finite sheaf-style gluing datum: finite index sets ι (pieces) and κ (overlaps), A-linear maps $\mathrm{res}_i : M \to M_i$ jointly injective, and A-linear maps $l_k, r_k : \prod_i M_i \to M_k$ such that $l_k(\mathrm{res}(m)) = r_k(\mathrm{res}(m))$ for all $m \in M$, $k \in \kappa$ (compatibility), and every family $(s_i) \in \prod_i M_i$*

with $l_k(s) = r_k(s)$ for all k equals $\mathrm{res}(m)$ for some $m \in M$ (gluing). Let P be a B-module with an analogous gluing datum $(P_i, P_k, \mathrm{res}', l', r')$. Suppose given, for each i, a bijective B-linear comparison $\varepsilon_i : B \otimes_A M_i \to P_i$, and for each k, an injective B-linear comparison $\mu_k : B \otimes_A M_k \to P_k$, both intertwining the restriction and overlap maps on tensor generators. Then there is a B-linear equivalence

$$\Phi : B \otimes_A M \xrightarrow{\sim} P$$

compatible with all the restrictions: $\mathrm{res}'_i(\Phi(z)) = \varepsilon_i((\mathrm{id}_B \otimes \mathrm{res}_i)(z))$ for every z and i.

Proof. Assemble the maps res_i into $T : M \to \prod_i M_i$ and the overlap maps into $L, R : \prod_i M_i \to \prod_k M_k$; the gluing hypothesis on M says T is an isomorphism onto the equalizer of L and R, and likewise $T' : P \to \prod_i P_i$ is an isomorphism onto the equalizer of L' and R'. Tensoring the equalizer presentation of M with the flat A-algebra B preserves it (a flat module is one for which tensoring preserves finite limits of modules, in particular equalizers), giving $B \otimes_A M \cong \mathrm{eq}(B \otimes_A L,\ B \otimes_A R)$. Composing the bijective piece comparisons ε_i into a bijection $\prod_i (B \otimes_A M_i) \cong \prod_i P_i$, and using injectivity of the overlap comparisons μ_k to check that this bijection carries the equalizer condition for (L, R) exactly onto the equalizer condition for (L', R') (1235 supplies the injectivity input for this check), transports the equalizer presentation of $B \otimes_A M$ isomorphically onto the equalizer presentation of P, i.e. onto P itself via T'. The resulting equivalence Φ is compatible with the restrictions by construction; this compatibility, together with the fact that the generators $1 \otimes m$ generate $B \otimes_A M$ as a B-module, determines Φ uniquely. $\square$

Definition 1237 (Canonical base-change map on sheaves of modules)**.** Given a cartesian square of schemes $\square = (g' : X' \to X,\ f' : X' \to S',\ f : X \to S,\ g : S' \to S)$ with $g \circ f' = f \circ g'$, the *canonical base-change map* on sheaves of modules is the natural transformation

$$g^* f_* \mathcal{F} \longrightarrow f'_* g'^* \mathcal{F}.$$

It is the mate, under the pullback–pushforward adjunctions, of the canonical identification $f'^* g^* \cong g'^* f^*$ supplied by the cartesian square. Its construction requires no flatness hypothesis.

Theorem 1238 (Section-wise base-change isomorphism)**.** *Let $\square$ be a cartesian square as in 1237 with f quasi-compact and quasi-separated, g flat, and $\mathcal{F}$ quasi-coherent. For every open $U \subseteq S'$, the section over U of the canonical base-change map $g^* f_* \mathcal{F} \to f'_* g'^* \mathcal{F}$ is an isomorphism (Stacks 02KH(1) at $i = 0$).*

Proof. Combine 1246 (the section map is an isomorphism over every affine open of S') with the basis-locality reduction 1247. $\square$

Lemma 1239 (Base map at a reflexive inclusion)**.** *For a morphism $g : Y \to X$, a sheaf of $\mathcal{O}_X$-modules $\mathcal{N}$, an open $V \subseteq X$, and the reflexive inclusion $g^{-1}(V) \subseteq g^{-1}(V)$, the section-level base map (1252) coincides with the V-component of the adjunction unit (1251):* $\mathrm{baseMap}_g(\mathcal{N}, g^{-1}V \leq g^{-1}V) = \Gamma(V, \eta_{\mathcal{N}})$.

Proof. The base map is by construction the unit component followed by the presheaf restriction along the inclusion $g^{-1}(V) \subseteq g^{-1}(V)$; that inclusion is the identity of $g^{-1}(V)$, and a presheaf sends the identity to the identity. $\square$

Theorem 1240 (Elementwise Beck–Chevalley compatibility of the canonical mate)**.** *Let $\square = (g', f', f, g)$ be a cartesian square as in 1237 (no flatness, compactness or affineness hypotheses), $\mathcal{F}$ a sheaf of $\mathcal{O}_X$-modules, and $V \subseteq S$, $U \subseteq S'$ opens with $U \subseteq g^{-1}(V)$, so that $f'^{-1}(U) \subseteq$*

$g'^{-1}(f^{-1}(V))$. *For every section $t \in \Gamma(f_*\mathcal{F}, V) = \Gamma(\mathcal{F}, f^{-1}V)$, the U-section of the canonical base-change map sends the base-map image of t along g to the base-map image of t along g' (1252):*

$$\Gamma(U, (\mathrm{canonicalBaseChangeMap}\,\square)_{\mathcal{F}})(\mathrm{baseMap}_g(f_*\mathcal{F})(t)) \;=\; \mathrm{baseMap}_{g'}(\mathcal{F})(t) \quad \textit{in } \Gamma(g'^*\mathcal{F},\ f'^{-1}(U)) = \Gamma(f'_*g'^*\mathcal{F},$$

Proof. The component at $\mathcal{F}$ of the canonical base-change map is, by definition of the mate of the pseudofunctor 2-isomorphism $f'^*g^* \cong (f'g)^* = (g'f)^* \cong g'^*f^*$ under the two adjunctions $f^* \dashv f_*$ and $f'^* \dashv f'_*$, the composite

$$g^*f_*\mathcal{F} \xrightarrow{\eta_{f'}} f'_*f'^*g^*f_*\mathcal{F} \xrightarrow{f'_*(\alpha_{f_*\mathcal{F}})} f'_*g'^*f^*f_*\mathcal{F} \xrightarrow{f'_*g'^*(\varepsilon_f)} f'_*g'^*\mathcal{F}.$$

Evaluate on the section $\mathrm{baseMap}_g(f_*\mathcal{F})(t)$ over U and follow the stages. (1) The $\eta_{f'}$-stage produces the base-map image along f' at the reflexive inclusion (1239); (2) the $f'^*g^* \cong (f'g)^*$ stage composes the two nested base maps into $\mathrm{baseMap}_{f'g}(f_*\mathcal{F})(t)$ (1272); (3) the transport along $f'g = g'f$ turns it into $\mathrm{baseMap}_{g'f}(f_*\mathcal{F})(t)$ (1273); (4) the inverse of the $(g'f)^* \cong g'^*f^*$ stage splits it back as $\mathrm{baseMap}_{g'}(f^*f_*\mathcal{F})(\mathrm{baseMap}_f(f_*\mathcal{F})(t))$ (1272 again, cancelled against its inverse); (5) naturality of the base map in the sheaf argument (1271) moves the counit stage inside: the result is $\mathrm{baseMap}_{g'}(\mathcal{F})(\Gamma(f^{-1}V, \varepsilon_f)(\mathrm{baseMap}_f(f_*\mathcal{F})(t)))$; and (6) by 1239 and the right-triangle identity $f_*(\varepsilon_f) \circ \eta_f|_{f_*} = \mathrm{id}$ of the adjunction $f^* \dashv f_*$, the inner section is t itself. $\square$

Lemma 1241 (Affineness of the fibre-product cover pieces)**.** *Let $\square = (g', f', f, g)$ be a cartesian square as in 1237 and let $V \subseteq S$, $U \subseteq S'$, $V' \subseteq X$ be affine opens with $U \subseteq g^{-1}(V)$ and $V' \subseteq f^{-1}(V)$. Then the open $g'^{-1}(V') \cap f'^{-1}(U) \subseteq X'$ is affine.*

Proof. Restricting the cartesian square to the opens V, U, V' and $g'^{-1}(V') \cap f'^{-1}(U)$ yields a cartesian square of schemes exhibiting $g'^{-1}(V') \cap f'^{-1}(U)$ as the fibre product $V' \times_V U$. A fibre product of affine schemes over an affine scheme is affine, and affineness transports along the comparison isomorphism with the categorical pullback. $\square$

Lemma 1242 (Section algebras of a flat morphism are flat)**.** *Let $g : S' \to S$ be a flat morphism of schemes and $V \subseteq S$, $U \subseteq S'$ affine opens with $U \subseteq g^{-1}(V)$. Then $\Gamma(S', U)$ is a flat $\Gamma(S, V)$-algebra via the induced ring map.*

Proof. Flatness of a morphism of schemes is affine-local: it holds if and only if for every pair of affine opens $U \subseteq g^{-1}(V)$ the induced ring map $\Gamma(S, V) \to \Gamma(S', U)$ is flat. $\square$

Theorem 1243 (H^0 flat base change over a compatible affine pair)**.** *Source: [Stacks Project], tags 02KE and 02KH. Let $\square = (g', f', f, g)$ be a cartesian square as in 1237 with f quasi-compact and quasi-separated, g flat, and $\mathcal{F}$ quasi-coherent on X. Let $V \subseteq S$ and $U \subseteq S'$ be affine opens with $U \subseteq g^{-1}(V)$, and write $A := \Gamma(S, V)$, $B := \Gamma(S', U)$, a flat A-algebra. Then there is a B-linear equivalence*

$$B \otimes_A \Gamma(\mathcal{F},\ f^{-1}(V)) \xrightarrow{\sim} \Gamma(g'^*\mathcal{F},\ f'^{-1}(U))$$

sending $1 \otimes x$ to the canonical base-map image $\mathrm{baseMap}_{g'}(\mathcal{F})(x)$ (1252); this condition determines the equivalence on the B-module generators, hence uniquely.

Proof. Since V is affine and f is quasi-compact, $Y := f^{-1}(V)$ is a quasi-compact open of X; choose finitely many affine opens $V_1, \ldots, V_n \subseteq X$ with $Y = \bigcup_i V_i$. Because f is quasi-separated, each intersection $V_i \cap V_j$ is quasi-compact, hence a finite union of affine opens V_{ijk}. The sheaf axiom for $\mathcal{F}$ presents $\Gamma(\mathcal{F}, Y)$ as the equalizer of the two restriction maps $\prod_i \Gamma(\mathcal{F}, V_i) \rightrightarrows \prod_{i,j,k} \Gamma(\mathcal{F}, V_{ijk})$, an equalizer of A-modules with finitely many factors. Since B is flat over A and the products are finite, applying $B \otimes_A (-)$ preserves this equalizer: $B \otimes_A \Gamma(\mathcal{F}, Y)$ is the equalizer of $\prod_i B \otimes_A \Gamma(\mathcal{F}, V_i) \rightrightarrows \prod_{i,j,k} B \otimes_A \Gamma(\mathcal{F}, V_{ijk})$.

On the primed side set $W_i := g'^{-1}(V_i) \cap f'^{-1}(U)$. Then W_i, as a scheme, is the fibre product $V_i \times_V U$ of affine schemes over the affine V, hence is an affine open of X' (1241); the W_i cover $f'^{-1}(U)$ because the V_i cover $f^{-1}(V)$ and $f'^{-1}(U) \subseteq g'^{-1}(f^{-1}(V))$. The sheaf axiom for $g'^*\mathcal{F}$ presents $\Gamma(g'^*\mathcal{F}, f'^{-1}(U))$ as the equalizer of $\prod_i \Gamma(g'^*\mathcal{F}, W_i) \rightrightarrows \prod_{i,j,k} \Gamma(g'^*\mathcal{F}, W_{ijk})$ with $W_{ijk} := g'^{-1}(V_{ijk}) \cap f'^{-1}(U)$ (affine opens covering $W_i \cap W_j$).

Per index i: since $W_i = V_i \times_V U$ is an affine fibre product, $\Gamma(X', W_i) = \Gamma(X, V_i) \otimes_A B$ (global sections of an affine fibre product form the pushout of the section rings), and the affine section formula for the module pullback (1250, applied to g' and the affine pair $W_i \subseteq g'^{-1}(V_i)$) gives

$$\Gamma(g'^*\mathcal{F}, W_i) \cong \Gamma(X', W_i) \otimes_{\Gamma(X,V_i)} \Gamma(\mathcal{F}, V_i) \cong (\Gamma(X, V_i) \otimes_A B) \otimes_{\Gamma(X,V_i)} \Gamma(\mathcal{F}, V_i) \cong B \otimes_A \Gamma(\mathcal{F}, V_i),$$

the last step by tensor cancellation; the composite sends $b \otimes x \mapsto b \cdot \mathrm{baseMap}_{g'}(x|_{V_i})$, so it commutes with the restriction maps on both sides (the base map is natural in the open pair; 1271, 1239). The same computation applies verbatim to each affine W_{ijk} over V_{ijk}, giving on the overlap columns *injective* comparisons $B \otimes_A \Gamma(\mathcal{F}, V_i \cap V_j) \to \Gamma(g'^*\mathcal{F}, W_i \cap W_j)$: indeed $\Gamma(\mathcal{F}, V_i \cap V_j)$ embeds by sheaf separatedness into $\prod_k \Gamma(\mathcal{F}, V_{ijk})$, flatness of B over A preserves the injection, and on each V_{ijk}-factor the comparison is the isomorphism above.

The two equalizer presentations are therefore connected by a commuting ladder whose product-column rungs are isomorphisms and whose discriminating-column rungs are injections; hence the induced map of equalizers $B \otimes_A \Gamma(\mathcal{F}, Y) \to \Gamma(g'^*\mathcal{F}, f'^{-1}(U))$ is an isomorphism (an equalizer of two diagrams compared by isomorphisms on the product column and injections on the target column is preserved: surjectivity of the equalizer comparison uses injectivity of the overlap rung to check the equalizing condition upstairs). Finally, chasing the generator $1 \otimes x$ through the ladder: its image in the i-th factor is $\mathrm{baseMap}_{g'}(x|_{V_i})$, which is the W_i-restriction of $\mathrm{baseMap}_{g'}(x)$ by compatibility of the base map with restrictions; since the equalizer includes into the product, the equivalence sends $1 \otimes x$ to $\mathrm{baseMap}_{g'}(\mathcal{F})(x)$, as claimed. $\square$

Theorem 1244 (Affine-pair workhorse for the base-change isomorphism)**.** *Let $\square$ be a cartesian square as in 1237 with f quasi-compact and quasi-separated, g flat, and $\mathcal{F}$ quasi-coherent. For every pair of affine opens $V \subseteq S$, $U \subseteq S'$ with $U \subseteq g^{-1}(V)$, the section over U of the canonical base-change map $g^* f_* \mathcal{F} \to f'_* g'^* \mathcal{F}$ is an isomorphism.*

Proof. Write $A := \Gamma(S, V)$, $B := \Gamma(S', U)$. The pushforward $f_*\mathcal{F}$ is quasi-coherent (1265), so the affine section formula (1255, applied to g and $f_*\mathcal{F}$) provides a B-linear equivalence $e_L : B \otimes_A \Gamma(\mathcal{F}, f^{-1}V) \cong \Gamma(g^* f_* \mathcal{F}, U)$ with $e_L(1 \otimes t) = \mathrm{baseMap}_g(f_*\mathcal{F})(t)$, and the H^0 flat-base-change equivalence (1243) provides $e_R : B \otimes_A \Gamma(\mathcal{F}, f^{-1}V) \cong \Gamma(f'_* g'^* \mathcal{F}, U)$ with $e_R(1 \otimes t) = \mathrm{baseMap}_{g'}(\mathcal{F})(t)$. The section over U of the canonical base-change map is B-linear, and by 1240 it carries $e_L(1 \otimes t)$ to $e_R(1 \otimes t)$ for every t; since the elements $1 \otimes t$ generate $B \otimes_A \Gamma(\mathcal{F}, f^{-1}V)$ as a B-module and all three maps are B-linear, it equals the composite $e_R \circ e_L^{-1}$, hence is bijective. $\square$

Theorem 1245 (Affine-base case of the section-wise base-change isomorphism)**.** *Assume in addition that the base S is affine, that f is quasi-compact and quasi-separated, that g is flat, and that $\mathcal{F}$ is quasi-coherent. Then for every affine open $U \subseteq S'$ the section of the canonical base-change map (1237) over U is an isomorphism.*

Proof. Take $V := S$, affine by hypothesis; every affine open $U \subseteq S'$ satisfies $U \subseteq g^{-1}(S)$, so the affine-pair workhorse 1244 applies to the pair (V, U). $\square$

Theorem 1246 (Affine-open case of the section-wise base-change isomorphism). *With f quasi-compact and quasi-separated, g flat, and $\mathcal{F}$ quasi-coherent (but S now arbitrary), the section of the canonical base-change map (1237) over an affine open $U \subseteq S'$ is an isomorphism.*

Proof. The affine opens $W \subseteq S'$ that are *compatible* with some affine open $V \subseteq S$ (that is, $W \subseteq g^{-1}(V)$) form a basis of the topology of S': given $x \in O$ with O open, pick an affine open $V \ni g(x)$ of S, then an affine open $W \subseteq O \cap g^{-1}(V)$ containing x. On every such pair the section of the canonical base-change map is an isomorphism by the affine-pair workhorse (1244); by basis locality for morphisms of sheaves of modules (113) the map $g^*f_*\mathcal{F} \to f'_*g'^*\mathcal{F}$ is then an isomorphism of sheaves, and in particular its section over the given affine U (in fact over every open) is an isomorphism. $\square$

Theorem 1247 (Basis-locality reduction for the section-wise base-change isomorphism). *Suppose the section of the canonical base-change map (1237) is an isomorphism over* every *affine open of S'. Then it is an isomorphism over every open $U \subseteq S'$. It is the reduction half of the section-wise statement 1238, whose proof feeds the affine-open case 1246 as the affine-open hypothesis.*

Proof. The affine opens form a basis of the topology of S'. A morphism of sheaves of $\mathcal{O}_{S'}$-modules whose section map is an isomorphism over every member of a basis is an isomorphism of sheaves (113, via stalks: injectivity and surjectivity of each stalk map are checked on germs represented on basic opens); its section over every open U is then an isomorphism. $\square$

Lemma 1248 (Definitional section identity for pushforward of a pullback). *Let $f' : X' \to S'$ and $g' : X' \to X$ be morphisms of schemes, $\mathcal{F}$ a sheaf of $\mathcal{O}_X$-modules, and $U \subseteq S'$ an open. The sections over U of the pushforward along f' of the pullback along g' of $\mathcal{F}$ coincide* definitionally *with the sections of the pullback $g'^*\mathcal{F}$ over the preimage $f'^{-1}(U)$:*

$$\Gamma(U,\ f'_*(g'^*\mathcal{F})) \ = \ \Gamma(f'^{-1}(U),\ g'^*\mathcal{F}).$$

This records step (3) of the affine-open construction (1246): the target of the canonical base-change map (1237) is read through this identity.

Proof. By definition of pushforward, the sections of $f'_*\mathcal{G}$ over U are the sections of $\mathcal{G}$ over $f'^{-1}(U)$. Apply this identity to $\mathcal{G} = g'^*\mathcal{F}$. $\square$

Theorem 1249 (Canonical base-change map is a sheaf isomorphism). *With f quasi-compact and quasi-separated, g flat, and $\mathcal{F}$ quasi-coherent, the canonical base-change map $g^*f_*\mathcal{F} \to f'_*g'^*\mathcal{F}$ of 1237 is an isomorphism of sheaves of modules. This is the canonical-morphism form of the degree-zero isomorphism in 1234.*

Proof. A morphism of sheaves of modules is an isomorphism if and only if its section map over every open is an isomorphism; apply 1238. $\square$

Definition 1250 (Tensor-presentation of the pullback-of-sections functor). Given a morphism $g : X \to Y$ of schemes, affine opens $U \subseteq Y$, $V \subseteq X$ with $g(V) \subseteq U$, and a quasi-coherent sheaf of modules $\mathcal{N}$ on Y, the *tensor-presentation iso* packages the canonical iso $\Gamma(V,\, g^*\mathcal{N}) \cong \Gamma(V,\, \mathcal{O}_X) \otimes_{\Gamma(U,\mathcal{O}_Y)} \Gamma(U,\, \mathcal{N})$ as a linear equivalence. It is induced by the adjunction-unit base map: 1253 identifies the scalar extension of that map with an isomorphism, and hence with the displayed tensor-presentation equivalence.

Definition 1251 (Adjunction-unit base linear map at a section)**.** Given a morphism $g : Y \to X$ of schemes, a sheaf of $\mathcal{O}_X$-modules $\mathcal{N}$, and an open $V \subseteq X$, the unit of the pullback–pushforward adjunction yields a morphism of $\mathcal{O}_X$-modules $\mathcal{N} \to g_*(g^*\mathcal{N})$. Evaluating its V-component gives the $\Gamma(X, V)$-linear *unit-at-section* map

$$\Gamma(\mathcal{N}, V) \longrightarrow \Gamma(g_*(g^*\mathcal{N}),\ V).$$

This is the adjunction-unit map underlying the tensor-presentation isomorphism of 1250.

Proof. Evaluate the unit natural transformation $\mathcal{N} \to g_*g^*\mathcal{N}$ on the open V. Naturality with respect to multiplication by sections of $\mathcal{O}_X$ gives $\Gamma(X, V)$-linearity. □

Definition 1252 (Section-level base map for the pullback)**.** For a morphism $g : Y \to X$, a sheaf $\mathcal{N}$ on X, and compatible affine opens $U \subseteq Y$, $V \subseteq X$ with $g(U) \subseteq V$, composing the unit-at-section map (1251) with the presheaf restriction from $g^{-1}(V)$ down to U produces the $\Gamma(X, V)$-linear *base map*

$$\Gamma(\mathcal{N}, V) \longrightarrow \Gamma(g^*\mathcal{N},\ U),$$

where the $\Gamma(X, V)$-action on the target is restriction of scalars along the ring map $\Gamma(X, V) \to \Gamma(Y, U)$ induced by g. This is the candidate linear map whose base-change property (1253) yields the tensor presentation.

Proof. Restrict the section supplied by the adjunction unit from $g^{-1}(V)$ to U. The compatibility of the unit with the structure sheaf map makes the composite semilinear for $\Gamma(X, V) \to \Gamma(Y, U)$, hence linear after restriction of scalars. □

Theorem 1253 (The base map is a base change)**.** *For compatible affine opens $U \subseteq Y$, $V \subseteq X$ of a morphism $g : Y \to X$ and a quasi-coherent sheaf $\mathcal{N}$ on X, the base map of 1252 exhibits $\Gamma(g^*\mathcal{N}, U)$ as the base change of $\Gamma(\mathcal{N}, V)$ along the ring map $\Gamma(X, V) \to \Gamma(Y, U)$; i.e. the induced $\Gamma(Y, U) \otimes_{\Gamma(X,V)} \Gamma(\mathcal{N}, V) \to \Gamma(g^*\mathcal{N}, U)$ is an isomorphism. The proof feeds the packaged section-level linear equivalence and its intertwining property (1255) into the universal characterization of a base change.*

Proof. Choose the linear equivalence and its intertwining identity from 1255. The universal property of the tensor product identifies the scalar-extension map induced by the base map with this equivalence, since they agree on every pure tensor $1 \otimes x$. It is therefore an isomorphism, which is precisely the asserted base-change property. □

Theorem 1254 (Existence of the tensor-presentation equivalence)**.** *In the same affine setup, there exists a $\Gamma(Y, U)$-linear isomorphism*

$$\Gamma(g^*\mathcal{N},\ U) \ \cong\ \Gamma(Y, U) \otimes_{\Gamma(X,V)} \Gamma(\mathcal{N}, V),$$

obtained from the base-change property (1253) by passing to the induced equivalence. This is the existence statement consumed by the tensor-presentation iso 1250.

Proof. By 1253, the scalar-extension map induced by the base map is an isomorphism. Its inverse, viewed as a linear equivalence, gives the displayed tensor-presentation equivalence. □

Definition 1255 (Σ-pair packaging: section-level LinearEquiv + intertwining)**.** Write $A = \Gamma(X, V)$ and $B = \Gamma(Y, U)$. The packaged data consist of a B-linear equivalence

$$f : B \otimes_A \Gamma(\mathcal{N}, V) \xrightarrow{\sim} \Gamma(g^*\mathcal{N}, U)$$

together with the identity $f(1 \otimes x) = \mathrm{baseMap}(x)$ for every $x \in \Gamma(\mathcal{N}, V)$. To construct f, identify the restriction of $\mathcal{N}$ to $\operatorname{Spec} A$ with its associated tilde sheaf, pull this sheaf back along $\operatorname{Spec} B \to \operatorname{Spec} A$, and apply the tilde base-change isomorphism. Transporting through the affine-open identifications gives the displayed equivalence. The naturality, composition, and open-immersion identities of 20.11.6 show that this composite sends $1 \otimes x$ to the canonical base-map section.

20.11.2 Adjunction substrate for pullback of tilde sheaves

The affine pullback formula rests on the tilde–global-sections adjunction, the extension–restriction-of-scalars adjunction, and the pullback–pushforward adjunction. The following comparison identifies global sections of a pushforward with restricted scalars; uniqueness of left adjoints then yields the pullback formula for tilde sheaves used above.

Lemma 1256 (Tilde–global-sections adjunction on an affine scheme)**.** *Provided by Mathlib (`Mathlib.AlgebraicGeometry.Modules.Tilde`). For a commutative ring R, the tilde functor $\widetilde{(\cdot)} : \mathrm{Mod}_R \to (\operatorname{Spec} R)$.Modules is left adjoint to the global-sections functor $\Gamma(\operatorname{Spec} R, -) : (\operatorname{Spec} R)$.Modules $\to \mathrm{Mod}_R$ (`moduleSpecΓFunctor`); the unit is the isomorphism $M \to \Gamma(\widetilde{M}, \top)$ and the counit is the tilde–Γ comparison $\widetilde{\Gamma(\mathcal{N}, \top)} \to \mathcal{N}$ (`fromTildeΓ`). Mathlib also supplies the two companion adjunctions used with it below: extension–restriction of scalars $B \otimes_A - \dashv \mathrm{res}_\varphi$ (`ModuleCat.extendRestrictScalarsAdj`) and pullback–pushforward $\pi^* \dashv \pi_*$ for sheaves of modules over schemes (`Scheme.Modules.pullbackPushforwardAdjunction`).*

Lemma 1257 (Global sections of the pushforward as restricted scalars)**.** *For a ring homomorphism $\varphi : A \to B$ with $\pi = \operatorname{Spec} \varphi : \operatorname{Spec} B \to \operatorname{Spec} A$, the two functors $(\operatorname{Spec} B)$.Modules $\to \mathrm{Mod}_A$*

$$\mathcal{N} \longmapsto \Gamma(\operatorname{Spec} A,\, \pi_* \mathcal{N}) \qquad \textit{and} \qquad \mathcal{N} \longmapsto \mathrm{res}_\varphi\, \Gamma(\operatorname{Spec} B,\, \mathcal{N})$$

(global sections of the pushforward, with A acting through $\Gamma(\pi^\sharp)$; respectively global sections, with A acting through φ) are naturally isomorphic.

Proof. On carriers the component at $\mathcal{N}$ is restriction of sections along the equality of opens $\pi^{-1}(\operatorname{Spec} A) = \operatorname{Spec} B$, a bijection. It is A-linear: for $a \in A$ the action on the source is by $\Gamma(\pi^\sharp)(\iota_A(a))$ and on the target by $\iota_B(\varphi(a))$, where $\iota_R : R \cong \Gamma(\operatorname{Spec} R, \mathcal{O})$ is the canonical identification; these agree by naturality of ι in R applied to φ (that is, $\Gamma(\pi^\sharp) \circ \iota_A = \iota_B \circ \varphi$), combined with semilinearity of section restriction for the module sheaf $\mathcal{N}$. Naturality in $\mathcal{N}$ is naturality of section restriction against module-sheaf morphisms. □

Lemma 1258 (Pullback of tilde is tilde of base change (Stacks 01HQ))**.** *Source: Stacks 01HQ / 0BJ8 (pullback of quasi-coherent modules along a Spec morphism). For a ring homomorphism $\varphi : A \to B$ and an A-module M, there is a canonical iso between the pullback of the tilde-sheaf and the tilde of the base-changed module:*

$$(\texttt{Scheme.Modules.pullback}\,(\mathrm{Spec.map}\,\varphi)).\mathrm{obj}\,(\widetilde{M}) \;\cong\; \widetilde{B \otimes_A M}.$$

Proof. Write $\pi = \operatorname{Spec} \varphi : \operatorname{Spec} B \to \operatorname{Spec} A$. Both composite functors

$$\mathrm{Mod}_A \xrightarrow{\widetilde{(\cdot)}} (\operatorname{Spec} A).\mathrm{Modules} \xrightarrow{\pi^*} (\operatorname{Spec} B).\mathrm{Modules} \qquad \text{and} \qquad \mathrm{Mod}_A \xrightarrow{B \otimes_A -} \mathrm{Mod}_B \xrightarrow{\widetilde{(\cdot)}} (\operatorname{Spec} B).\mathrm{Modules}$$

are left adjoints: the first of $\mathcal{N} \mapsto \Gamma(\operatorname{Spec} A, \pi_* \mathcal{N})$ (compose the tilde–Γ adjunction on $\operatorname{Spec} A$, 1256, with $\pi^* \dashv \pi_*$), the second of $\mathcal{N} \mapsto \mathrm{res}_\varphi \Gamma(\operatorname{Spec} B, \mathcal{N})$ (compose extension–restriction of scalars along φ with the tilde–Γ adjunction on $\operatorname{Spec} B$). By 1257 the two right adjoints are naturally isomorphic, so uniqueness of left adjoints yields a natural isomorphism of the two composites; its component at M is the required isomorphism $\pi^*\widetilde{M} \cong \widetilde{B \otimes_A M}$.

For the section-level identity, apply the unit compatibility of the left-adjoint-uniqueness isomorphism at $m \in M$: the unit of the first composed adjunction sends m to the canonical pullback base map on global sections of $\widetilde{M}$, while the unit of the second sends m to the global section of $\widetilde{B \otimes_A M}$ attached to $1 \otimes m$; the compatibility square states precisely that the isomorphism carries the former to the latter. □

20.11.3 qcqs section localization and Stacks 01XJ

The proof of 1265 generalizes the affine section-localization keystone to quasi-compact quasi-separated opens by a Mayer–Vietoris induction, then runs the tilde–Γ comparison backwards at each affine open of the base.

Lemma 1259 (Affine basic-open section localization for quasi-coherent modules)**.** *Source: Stacks 01IB / Hartshorne II.5.3 (`lemma-invert-f-sections`, affine case). Let X be a scheme, $\mathcal{M}$ a quasi-coherent $\mathcal{O}_X$-module, $U \subseteq X$ an affine open, and $f \in \Gamma(X, U)$. Then the section restriction $\Gamma(\mathcal{M}, U) \to \Gamma(\mathcal{M}, D(f))$ exhibits the target as the localization $\Gamma(\mathcal{M}, U)[1/f]$ over $\Gamma(X, U)$.*

Proof. Pull $\mathcal{M}$ back along the affine immersion $\operatorname{Spec}\Gamma(X, U) \to X$; the pullback is quasi-coherent (pullback along an open immersion preserves quasi-coherence), so its tilde–Γ counit is an isomorphism (1131), identifying it with $\widetilde{\Gamma(\mathcal{M}, U)}$; the sections of a tilde over a basic open are the localization (1129), and the section comparison transports the statement back to X. □

Lemma 1260 (qcqs section localization, torsion half)**.** *Source: Stacks 01P0 (first half). Let X be a scheme, $\mathcal{M}$ a quasi-coherent $\mathcal{O}_X$-module, $W \subseteq X$ open, $g \in \Gamma(X, W)$, and $U \leq W$ a quasi-compact open. If $x \in \Gamma(\mathcal{M}, U)$ restricts to zero on $U \cap D(g)$, then $(g|_U)^n \cdot x = 0$ for some $n \in \mathbb{N}$.*

Proof. Induction on the quasi-compact open U, adding one affine open at a time. For $U = \emptyset$ the section vanishes by sheaf separation over the empty cover. For $U = S \cup V$ with V affine and the claim known on the quasi-compact S: the restriction of x to S is killed by a power of $g|_S$ by induction, and the restriction to V is killed by a power of $g|_V$ by the injectivity half of 1259 at the affine V (note $D(g|_V) = V \cap D(g)$). Taking the maximum n of the two exponents, $(g|_U)^n \cdot x$ restricts to zero on both members of the cover $\{S, V\}$, hence vanishes by sheaf separation. □

Lemma 1261 (qcqs section localization, surjectivity half)**.** *Source: Stacks 01P0 (second half). Let X, $\mathcal{M}$, W, g be as above, with W quasi-separated, and let $U \leq W$ be quasi-compact. For every $y \in \Gamma(\mathcal{M}, U \cap D(g))$ there are $n \in \mathbb{N}$ and $x \in \Gamma(\mathcal{M}, U)$ with $x|_{U \cap D(g)} = (g|_{U \cap D(g)})^n \cdot y$.*

Proof. Induction on the quasi-compact U as in 1260. In the step $U = S \cup V$ (V affine), the induction hypothesis and the affine surjectivity half of 1259 produce candidate sections x_S on S and x_V on V with a common normalization exponent n. Their difference on the overlap $S \cap V$ — quasi-compact because W is quasi-separated — restricts to zero on $(S \cap V) \cap D(g)$, so by the torsion half a further power g^m kills it; after multiplying both candidates by g^m they agree on the overlap and glue to a section x over U with $x|_{U \cap D(g)} = g^{n+m} \cdot y$, checked by sheaf separation over the cover $\{S \cap D(g), D(g|_V)\}$ of $U \cap D(g)$. □

Lemma 1262 (qcqs section localization)**.** *Source: Stacks 01P0. Let X be a scheme, $\mathcal{M}$ a quasi-coherent $\mathcal{O}_X$-module, and $W \subseteq X$ a quasi-compact, quasi-separated open. For every $g \in$*

$\Gamma(X,W)$ *the section restriction* $\Gamma(\mathcal{M},W) \to \Gamma(\mathcal{M},D(g))$ *exhibits the target as the localization* $\Gamma(\mathcal{M},W)[1/g]$ *over* $\Gamma(X,W)$.

Proof. The element g restricts to a unit of $\Gamma(X,D(g))$, so every power of g acts invertibly on $\Gamma(\mathcal{M},D(g))$; the surjectivity and torsion (injectivity) conditions of the localization are 1261 and 1260 at $U := W$. ☐

Lemma 1263 (Converse tilde–Γ transport at an affine open)**.** *Let Y be a scheme, $\mathcal{N}$ an $\mathcal{O}_Y$-module, and $U \subseteq Y$ an affine open with canonical immersion $j : \operatorname{Spec}\Gamma(Y,U) \to Y$. If for every $f' \in \Gamma(Y,U)$ the restriction $\Gamma(\mathcal{N},U) \to \Gamma(\mathcal{N},D(f'))$ is a localization at the powers of f' over $\Gamma(Y,U)$, then the tilde–Γ counit of the pullback $j^*\mathcal{N}$ is an isomorphism.*

Proof. The sections of $j^*\mathcal{N}$ over an open of the spectrum are the sections of $\mathcal{N}$ over its image (j is an open immersion), with $j''(\top) = U$ and $j''(D(f')) = D(f')$; under these identifications – which are semilinear over the canonical section-ring comparisons, by the naturality of the immersion's application isomorphisms – the hypothesis says exactly that all basic-open section restrictions of $j^*\mathcal{N}$ on the spectrum are localizations, which characterizes the invertibility of the tilde–Γ counit. ☐

Lemma 1264 (Pushforward basic-open section localization)**.** *Let $\pi : X \to S$ be a quasi-compact, quasi-separated morphism of schemes, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$-module, $U \subseteq S$ an affine open, and $f' \in \Gamma(S,U)$. Then the section restriction $\Gamma(\pi_*\mathcal{F},U) \to \Gamma(\pi_*\mathcal{F},D(f'))$ is a localization at the powers of f' over $\Gamma(S,U)$.*

Proof. The preimage $W := \pi^{-1}(U)$ is quasi-compact and quasi-separated (π is qcqs and U is affine), and $\pi^{-1}(D_S(f')) = D_X(g)$ for $g := \pi^\sharp f' \in \Gamma(X,W)$. The sections of the pushforward are by definition the sections of $\mathcal{F}$ over these preimages, with the $\Gamma(S,U)$-action given through $\pi^\sharp$, so the statement is 1262 for $\mathcal{F}$ at g over W, read through $\pi^\sharp$ (a localization over the base ring restricts to a localization along any ring map matching the multiplicative sets). ☐

Lemma 1265 (Pushforward of quasi-coherent modules is quasi-coherent (Stacks 01XJ))**.** *Source: Stacks 01XJ (preservation of quasi-coherence under pushforward). Let $f : X \to Y$ be a quasi-compact, quasi-separated morphism of schemes, and let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Then $f_*\mathcal{F}$ is a quasi-coherent $\mathcal{O}_Y$-module. Mathlib `b80f227` ships no `IsQuasicoherent.pushforward` instance at the `Scheme.Modules` layer.*

Proof. Quasi-coherence is local on Y, so it suffices to produce a presentation of the slice of $f_*\mathcal{F}$ over each affine open $U \subseteq Y$. By 1264 all basic-open section restrictions of $f_*\mathcal{F}$ at U are localizations, so by 1263 the pullback of $f_*\mathcal{F}$ along $\operatorname{Spec}\Gamma(Y,U) \to Y$ is isomorphic to the tilde of its global sections. A tilde admits a global presentation; transporting it along $U \hookrightarrow Y = (\operatorname{Spec}\Gamma(Y,U) \cong U) \hookrightarrow Y$ yields a presentation of the geometric restriction of $f_*\mathcal{F}$ to U, hence of the slice. (The previously sketched "right adjoints preserve quasi-coherence" argument is not a proof: adjointness yields colimit preservation only.) ☐

20.11.4 Affine-open comparison isomorphisms

The tensor-presentation equivalence uses three comparison isomorphisms: a quasi-coherent sheaf on an affine open is the tilde sheaf associated to its sections, tilde sheaves commute with affine base change, and pullback along the spectrum presentation of an affine open agrees with restriction. The following results record these ingredients separately.

Lemma 1266 (Quasi-coherent module on an affine open is `tilde` of its sections (Stacks 01I8))**.** *Source: Stacks 01I8 (a quasi-coherent module on an affine scheme is $\widetilde{M}$ for some module). Let X be a scheme, $\mathcal{N}$ a quasi-coherent $\mathcal{O}_X$-module, and $V \subset X$ an affine open. Then the pullback of $\mathcal{N}$ along `hV.fromSpec` is canonically isomorphic to `tilde` of its sections:*

$$(\texttt{Scheme.Modules.pullback}\,\texttt{hV.fromSpec}).\mathrm{obj}\,\mathcal{N} \;\cong\; \widetilde{\Gamma(\mathcal{N}, V)} \quad (\textit{on } \operatorname{Spec}\Gamma(X,V)).$$

Mathlib `b80f227` ships the analog for free *quasi-coherent modules (`isIso_fromTildeΓ_of_presentation` partial), but the general statement "$[N.IsQuasicoherent] \Rightarrow N|_V \cong \widetilde{\Gamma(N,V)}$ on an affine open V" (Stacks 01I8* lemma-affine-qc-module-iff*) is unowned at the `Scheme.Modules` layer.*

Proof. Write $j = \texttt{hV.fromSpec} : \operatorname{Spec}\Gamma(X,V) \to X$ (an open immersion with image V) and $G = j^*\mathcal{N}$. Pullback along an open immersion preserves quasi-coherence, so G is a quasi-coherent module on the affine scheme $\operatorname{Spec}\Gamma(X,V)$; by the affine structure theorem (1131, Stacks 01I8) the tilde–Γ counit $\widetilde{\Gamma(G,\top)} \to G$ is an isomorphism.

The canonical base map $b : \Gamma(\mathcal{N}, V) \to \Gamma(G, \top)$ (the V-sections of the pullback–pushforward adjunction unit followed by restriction along $\top \le j^{-1}V$) is bijective. Indeed the two adjunctions restrict $\dashv j_*$ and $j^* \dashv j_*$ share the right adjoint, so their units are identified through the canonical natural isomorphism restrict $\cong j^*$ (1111). Under this identification, the V-component of the unit of the restriction adjunction is the presheaf restriction of $\mathcal{N}$ along the *equality* of opens $j(j^{-1}V) = V$, hence bijective; the component of the natural isomorphism is bijective; and the final restriction is along the equality $\top = j^{-1}V$. Thus b is a composite of three bijections.

Since b is a bijective $\Gamma(X,V)$-linear map (the $\Gamma(X,V)$-action on $\Gamma(G,\top)$ being through the canonical ring isomorphism $\Gamma(X,V) \cong \Gamma(\operatorname{Spec}\Gamma(X,V), \top)$), it induces an isomorphism $\tilde{b} : \widetilde{\Gamma(\mathcal{N}, V)} \cong \widetilde{\Gamma(G,\top)}$. The required isomorphism is the inverse of $\tilde{b}$ followed by the counit. The characterizing section identity holds because $\widetilde{(-)}$-functoriality intertwines the canonical maps $M \to \Gamma(\widetilde{M}, U)$ (naturality of `toOpen`), while the counit composed with `toOpen` at $\top$ is the identity restriction. ☐

Lemma 1267 (Section-level transport for pullback along an affine open's `fromSpec`)**.** *Source: Project-bespoke transport, formed from Stacks 01HH + `tilde.isoTop`. Let Y be a scheme, $\mathcal{N}$ an $\mathcal{O}_Y$-module, and $U \subset Y$ an affine open. Equip $\Gamma((\operatorname{Spec}\Gamma(Y,U)), \top)$ with the $\Gamma(Y,U)$-algebra structure inherited from `Scheme.ΓSpecIso` (the canonical $\Gamma(\operatorname{Spec} R, \top) \cong R$ for any commutative ring R), and equip $\Gamma((pullback\,\texttt{hU.fromSpec}).\mathrm{obj}\,\mathcal{N}, \top)$ with the $\Gamma(Y,U)$-module structure inherited via `Module.compHom`. Then there is a canonical $\Gamma(Y,U)$-linear isomorphism between this pullback-section module and the original section module $\Gamma(\mathcal{N}, U)$:*

$$\Gamma((\textit{pullback}\,\texttt{hU.fromSpec}).\mathrm{obj}\,\mathcal{N}, \top) \simeq_{\Gamma(Y,U)} \Gamma(\mathcal{N}, U).$$

The map identifies a section of the pulled-back sheaf at $\top$ of $\operatorname{Spec}\Gamma(Y,U)$ with the corresponding section of $\mathcal{N}$ on the affine open $U \subset Y$.

Proof. The underlying additive equivalence is the composite of the $\top$-section of the natural isomorphism $j^* \cong \operatorname{restrict} j$ for the open immersion $j = \texttt{hU.fromSpec}$ (1111) with the restriction of $\mathcal{N}$ along the equality of opens $j(\top) = U$; its $\Gamma(Y,U)$-linearity reduces (through `Hom.app_smul` and semilinearity of section restriction) to the ring identity $\iota^{-1} \circ (j^\sharp\top)^{-1} \circ \mathrm{res}_{U=j(\top)} = \mathrm{id}$, which follows from `fromSpec_app_self`.

The additional Σ-pair characterization (T12, 2026-07-03) — the inverse of the equivalence is the canonical base map — is the unit-compatibility of `Adjunction.leftAdjointUniq` for the restriction adjunction versus the pullback–pushforward adjunction, as spelled out in 1274. ☐

20.11.5 Finite generation of affine sections (Stacks 01PC, finite-type half)

The chart-level input to generic flatness (Nitsure §4): over an affine open contained in a presenting-cover member, the sections of a finitely presented module sheaf form a finite module. The route runs through the tilde–Γ adjunction on the affine model.

Lemma 1268 (Finite generation from a finite generating family of the tilde)**.** *Source: Stacks 01PC (finite-type modules on affines). Let R be a commutative ring and N an R-module. If the associated sheaf $\widetilde{N}$ on $\operatorname{Spec} R$ admits a generating family of sections indexed by a finite type I — that is, an epimorphism of $\mathcal{O}_{\operatorname{Spec} R}$-modules $\pi : \mathcal{O}^{\oplus I} \twoheadrightarrow \widetilde{N}$ — then N is a finite R-module.*

Proof. Precomposing π with the canonical isomorphism $\widetilde{R^{\oplus I}} \cong \mathcal{O}^{\oplus I}$ gives an epimorphism $\pi' : \widetilde{R^{\oplus I}} \twoheadrightarrow \widetilde{N}$. Transpose π' through the adjunction $\widetilde{(\cdot)} \dashv \Gamma$: writing $t : R^{\oplus I} \to \Gamma(\widetilde{N})$ for the transpose, the counit factorization reads $\tilde{t}\,; \varepsilon_{\widetilde{N}} = \pi'$. The counit ε of the tilde–Γ adjunction is invertible at every tilde: the triangle identity $\widetilde{\eta_N}\,; \varepsilon_{\widetilde{N}} = \mathrm{id}$ exhibits it as inverse to the tilde of the (invertible) unit. Hence $\tilde{t} = \pi'\,; \varepsilon_{\widetilde{N}}^{-1}$ is an epimorphism; since $\widetilde{(\cdot)}$ is faithful, t is an epimorphism of R-modules, i.e. surjective. Composing with the inverse of the unit isomorphism $N \cong \Gamma(\widetilde{N})$ exhibits N as a quotient of the finite free module $R^{\oplus I}$. □

Lemma 1269 (Affine sections of a finitely presented module sheaf are finitely generated, chart form)**.** *Source: Stacks 01PC. Let X be a scheme, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$-module, and q a quasi-coherence datum for $\mathcal{F}$ all of whose presentations are finite. For every affine open $V \subseteq X$ contained in a cover member $q.X_i$, the section module $\Gamma(\mathcal{F}, V)$ is a finite $\Gamma(X, V)$-module.*

Proof. The finite presentation of the slice restriction $\mathcal{F}|_{q.X_i}$ provides a generating family of sections indexed by a finite type. Generating families transport along colimit-preserving functors carrying the structure sheaf to the structure sheaf, and along epimorphisms; apply this three times: across the slice-to-geometric equivalence onto the open subscheme $q.X_i$; along the pullback by the open immersion $k : \operatorname{Spec}\Gamma(X, V) \to q.X_i$ through which $\mathrm{fromSpec}_V$ factors (it factors because $V \subseteq q.X_i$); and along the composite isomorphism identifying the result with the pullback of $\mathcal{F}$ along $\mathrm{fromSpec}_V$, which by 1266 is $\widetilde{\Gamma(\mathcal{F}, V)}$. The tilde of $\Gamma(\mathcal{F}, V)$ thus has a finite generating family, and 1268 concludes. □

Lemma 1270 (Charts with finitely generated sections)**.** *Source: Stacks 01PC. Let $\mathcal{F}$ be a module sheaf of finite presentation on a scheme X. Every point x of every open $O \subseteq X$ has an affine open neighbourhood V with $x \in V \subseteq O$ such that $\Gamma(\mathcal{F}, V)$ is a finite $\Gamma(X, V)$-module.*

Proof. Choose a quasi-coherence datum q with finite presentations. The cover members $q.X_i$ cover X, so x lies in some $q.X_i$; affine opens form a basis of the topology, so there is an affine $V \subseteq O \cap q.X_i$ containing x. Conclude by 1269. □

20.11.6 Beck–Chevalley identities for the base-change map

The base map is obtained from the unit of the pullback–pushforward adjunction. Its compatibility with the affine tilde comparisons therefore follows from four identities: naturality in the module, compatibility with composition of pullbacks, invariance under equality of the underlying morphisms, and inversion of the comparison for an affine open immersion. Together these identities identify the composite used in 1255 with the canonical base map on every section.

Lemma 1271 ((N1) baseMap naturality in input sheaf)**.** *Let $g : X \to Y$ be a morphism of schemes, let $\mathcal{N}$, $\mathcal{N}'$ be $\mathcal{O}_Y$-modules, and let $\eta : \mathcal{N} \to \mathcal{N}'$ be a morphism of $\mathcal{O}_Y$-modules. For affine opens $U \subseteq Y$ and $V \subseteq X$ with $g(V) \subseteq U$ and any $x \in \Gamma(\mathcal{N}, U)$, the section-level component of 1237 satisfies*

$$((\mathit{pullback}\, g).\mathrm{map}\, \eta).\mathrm{app}\, V\, (\mathit{baseMap}\, g\, \mathcal{N}\, e\, x) \;=\; \mathit{baseMap}\, g\, \mathcal{N}'\, e\, (\eta.\mathrm{app}\, U\, x),$$

i.e. `baseMap` *is natural in its module argument. Used in Stage 2 of the 1255 decomposition to transport step1*.inv *(a morphism of $\mathcal{O}_Y$-modules) across* `baseMap` *of Spec.map φ.*

Proof. Direct from the naturality square of `pullbackPushforwardAdjunction.unit` at the morphism $\eta : \mathcal{N} \to \mathcal{N}'$ in the input category (giving the unit half), combined with the naturality of the section restriction against (pullback g).map η (giving the restriction half); the two squares chain elementwise. □

Lemma 1272 ((N2) baseMap composition via pullbackComp)**.** *Let $f : X \to Y$ and $g : Y \to Z$ be morphisms of schemes and let $\mathcal{N}$ be an $\mathcal{O}_Z$-module. The Mathlib-canonical iso pullbackComp $f\, g$: $(\mathit{pullback}\, f) \circ (\mathit{pullback}\, g) \cong \mathit{pullback}\, (f \circ g)$ intertwines* `baseMap`*: for affine opens $W \subseteq Z$, $V \subseteq X$ with $f(g(V)) \subseteq W$ and any $x \in \Gamma(\mathcal{N}, W)$,*

$$((\mathit{pullbackComp}\, f\, g)\, \mathcal{N}).\mathrm{hom.app}\, V\, (\mathit{baseMap}\, f\, ((\mathit{pullback}\, g).\mathrm{obj}\, \mathcal{N})\, e_f\, (\mathit{baseMap}\, g\, \mathcal{N}\, e_g\, x)) \;=\; \mathit{baseMap}\, (f \circ g)\, \mathcal{N}\, e\, x.$$

In words: applying `baseMap` *along g then along f, routed through the canonical composition iso, yields the* `baseMap` *along the composite $f \circ g$. Used in Stages 3 and 5 of the 1255 decomposition.*

Proof. The unit of a composition of left adjoints $F \circ G \dashv R \circ R'$ satisfies the canonical composition identity $\eta_{FG} = R'(\eta_F) \circ \eta_G$ (Mathlib `Adjunction.comp`). The composed unit is identified with the pullbackComp-twist of the unit of the composite by the conjugation identity `unit_conjugateEquiv` applied to Mathlib's `conjugateEquiv_pullbackComp_inv` (the components of `pushforwardComp` are identities). Together with the naturality of the f-unit component against the restriction $V \leq g^{-1}U$, the collapse of consecutive section restrictions, the naturality of pullbackComp.inv against the restriction $T \leq (f \circ g)^{-1}U$, and the hom–inv cancellation of pullbackComp at T-sections, this yields the stated formula elementwise. Used twice (Stages 3 and 5). □

Lemma 1273 ((N3) baseMap transport via pullbackCongr)**.** *Let $f_1, f_2 : X \to Y$ be morphisms of schemes with $f_1 = f_2$ as a propositional equality h_{eq}, and let $\mathcal{N}$ be an $\mathcal{O}_Y$-module. The Mathlib-canonical equality-transport iso pullbackCongr h_{eq} : pullback $f_1 \cong$ pullback f_2 intertwines* `baseMap`*: for affine opens $U \subseteq Y$, $V \subseteq X$ with $f_1(V) \subseteq U$ (equivalently $f_2(V) \subseteq U$ via h_{eq}) and any $x \in \Gamma(\mathcal{N}, U)$,*

$$((\mathit{pullbackCongr}\, h_{eq})\, \mathcal{N}).\mathrm{inv.app}\, V\, (\mathit{baseMap}\, f_2\, \mathcal{N}\, e_2\, x) \;=\; \mathit{baseMap}\, f_1\, \mathcal{N}\, e_1\, x.$$

I.e. `baseMap` *is invariant under the equality-transport iso of the morphism argument. Used in Stage 4 of the 1255 decomposition to re-associate the composite Spec.map $\varphi \circ$* `_hV.fromSpec` $=$ `_hU.fromSpec` $\circ\, g$ *under the Σ-pair packaging.*

Proof. By propositional-equality elimination (`subst` on h_{eq}): after substitution pullbackCongr rfl is the identity isomorphism by iota-reduction of `eqToHom`, and the two $\leq$-witnesses are proof-irrelevant, so both sides are definitionally equal. □

Lemma 1274 ((N4) step3 inversion identity for an open immersion)**.** *Let Y be a scheme, $U \subseteq Y$ an affine open with canonical open immersion* `_hU.fromSpec` $:$ *Spec* $\Gamma(Y, U) \to Y$*, and* $\mathcal{N}$ *an* $\mathcal{O}_Y$*-module. Then the canonical section-transport isomorphism of 1267 is the inverse of* `baseMap` *along* `_hU.fromSpec` *at the* $\top$*-section:*

$$\mathit{step3}(\mathit{baseMap}\texttt{\_hU.fromSpec}\,\mathcal{N}\,(\mathit{le\_of\_eq}\texttt{\_hU.fromSpec\_preimage\_self.symm})\,y) \;=\; y \qquad \text{for every } y \in \Gamma(\mathcal{N}, U).$$

This identifies the inverse of `step3` *with the natural* `baseMap` *action of the open immersion* `_hU.fromSpec`*. Used in Stage 6 of the 1255 decomposition, closing the intertwining identity.*

Proof. Realized as the Σ-pair characterization carried by 1267: the inverse of its equivalence IS the canonical base map. The proof is the unit-compatibility of `Adjunction.leftAdjointUniq` for `restrictAdjunction` versus `pullbackPushforwardAdjunction` (whose `leftAdjointUniq` is Mathlib's `restrictFunctorIsoPullback`), combined with the fact that the unit of the restriction adjunction is a plain presheaf restriction (definitional in Mathlib), the naturality of the comparison against the restriction $\top \leq j^{-1}U$, and the collapse of consecutive restrictions along the equality $j(\top) = U$. The stated inversion identity then follows by applying the equivalence to both sides of the characterization. □

20.12 Construction overview

The construction proceeds in three phases.

Phase 1: Hilbert polynomial (1018). On a locally-of-finite-type morphism $\pi : X \to S$ with S locally noetherian, the fibers determine a function $s \mapsto \Phi_{\mathcal{F},s} \in \mathbb{Q}[\lambda]$ (1019, 1020, 1023, 1038, 1039). The H^0-only route through the graded Hilbert function of the section module reduces polynomiality to finite generation (1024) and the Hilbert–Serre theorem (1028), yielding 1043.

Phase 2: Grassmannian scheme (1176, 1183). The Grassmannian $\mathrm{Gr}_S(V, d)$ is obtained by gluing $\binom{r}{d}$ affine charts U^I (each isomorphic to a relative affine space $\mathbb{A}_S^{d(r-d)}$) via the Plücker cocycle $\theta_{I,J}$. The universal quotient sheaf $\mathcal{U}$ is glued by the transition cocycle $g_{I,J} = (X_J^I)^{-1}$, and the determinant line bundle $\det \mathcal{U}$ gives the Plücker closed embedding into $\mathbb{P}_S(\bigwedge^d V)$. The universal pair $(\mathrm{Gr}_S(V, d), \mathcal{U})$ represents rank-d locally free quotients.

Phase 3: Quot scheme (1229). The Quot functor is constructed as a locally closed subfunctor of the Grassmannian $\mathrm{Grass}(W \otimes_{\mathcal{O}_S} \mathrm{Sym}^r V, \Phi(r))$, via:

1. the uniform m-regularity bound from 1231, using Castelnuovo–Mumford regularity;
2. the Grassmannian embedding α from 1232;
3. the flattening stratification (942 of 19, A.2.a) applied to the universal quotient on the Grassmannian, producing the locally closed stratum T_0^Φ;
4. the valuative-criterion upgrade (1233) to closed embedding, hence projective S-scheme.

The reduction 1230 from the universal case to the general case is a closed embedding of representable subfunctors via the kernel-vanishing scheme.

Cohomology and base change. 1234 proves the degree-zero flat-base-change isomorphism $g^* f_* \mathcal{F} \cong f'_* g'^* \mathcal{F}$ (Stacks tag 02KH). It is assembled from the canonical base-change map using the affine tensor/base-map comparison developed above; higher direct images are not claimed here.

Dependencies summary. Relative-spectrum gluing from 13 constructs the affine charts, while 942 from 19 provides the flattening stratum. Downstream, the special case $E = \mathcal{O}_C$ of 1229 on a smooth proper curve C/k gives the Hilbert scheme $\mathrm{Hilb}_{C/k} = \mathrm{Quot}^{\Phi, \mathcal{O}_C(1)}_{\mathcal{O}_C/C/k}$ and its open subscheme $\mathrm{Div}_{C/k}$ of relative effective divisors: the engine of the quotient route to the Picard scheme, which 25.1 sets beside the Milne–Kollár route the formalisation follows. The substrate developed in this chapter — Hilbert polynomial, Grassmannian scheme, graded algebra — is consumed by either route.

20.13 Out of scope

This chapter treats the Hilbert polynomial, the Quot and relative-divisor functors, the Grassmannian, and the representability of the Quot functor. The following applications and refinements are treated elsewhere:

- **Hilbert scheme as a separate chapter**. The Hilbert scheme $\mathrm{Hilb}_{X/S}$ is the special case $\mathrm{Hilb}_{X/S} = \mathrm{Quot}_{\mathcal{O}_X/X/S}$ of the Quot functor and is introduced here through 1138.
- **Snapper's Lemma** – the polynomial-eventually property of Euler characteristics. Cited via Nitsure §1 and (textbook reference) Hartshorne III.5.2. The H^0-only definition 1018 does not consume it: Snapper's Lemma enters only the comparison identifying $\Phi_{\mathcal{F},s}$ with the Euler-characteristic Hilbert polynomial for $m \gg 0$.
- **Identification with $\mathbb{P}^n$ and relative Grassmannian variants** (Nitsure §1, Exercises (1)–(4)) — optional applications, consumed by nothing in this development.
- **Group-scheme structure on** $\mathrm{Pic}_{C/k}$. The abelian-group-scheme refinement belongs to the FGA Picard-representability construction.
- **Identity-component** Pic^0 **and abelian-variety structure**. These are consequences of the later Picard and Albanese constructions.

Chapter 21

Section graded ring infrastructure: tensor powers and graded sections

The Hilbert-polynomial development of 20 represents the degree-m graded component of a fibre by the global sections $\Gamma(X_s, \mathcal{F}_s \otimes L_s^{\otimes m})$ of a twist of a coherent sheaf by a tensor power of a line bundle, and it organises these components into a graded ring $R(X_s, L_s) = \bigoplus_{m \geq 0} \Gamma(X_s, L_s^{\otimes m})$ (1021) and a graded module $M(X_s, \mathcal{F}_s, L_s) = \bigoplus_{m \geq 0} \Gamma(X_s, \mathcal{F}_s \otimes L_s^{\otimes m})$ (1022). Three pieces of that picture have no direct Mathlib counterpart and must be built here, each in its own section:

1. **Tensor powers of a sheaf of modules** (21.1). Mathlib equips the category PresheafOfModules of *presheaves* of modules over a presheaf of commutative rings with a symmetric monoidal structure (1275) and provides the sheafification functor PresheafOfModules $\to$ SheafOfModules (1276); but the category SheafOfModules of *sheaves* of modules carries *no* monoidal structure. The tensor product of sheaves of modules $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G}$ must therefore be built as the sheafification of the pointwise tensor presheaf, and the iterated tensor power $\mathcal{L}^{\otimes m}$ recursively from it.

2. **Lax-monoidal global sections** (21.2). The global-sections functor Γ is only *lax* monoidal: from the pointwise-strict monoidality of the presheaf tensor product and the sheafification unit one obtains a natural multiplication $\Gamma(\mathcal{F}) \otimes_{\Gamma(\mathcal{O}_X)} \Gamma(\mathcal{G}) \to \Gamma(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G})$, together with the associativity and unit coherence those products must satisfy.

3. **Graded-ring / graded-module assembly** (21.3). The section multiplications and the tensor-power comparison isomorphisms assemble, through Mathlib's external-direct-sum graded API (1334, 1335, 1336), into the graded *ring* structure on $\bigoplus_{m \geq 0} \Gamma(X_s, L_s^{\otimes m})$ and the graded *module* structure on $\bigoplus_{m \geq 0} \Gamma(X_s, \mathcal{F}_s \otimes L_s^{\otimes m})$. For a *general* sheaf L_s the ring is only a non-commutative graded semiring (the free tensor algebra on $\Gamma(L_s)$, 1338); it becomes *commutative* exactly when L_s is an invertible sheaf / line bundle (1281), which is the case relevant to the application (1341). Stacks defines $\Gamma_*(X, \mathcal{L})$ only for invertible $\mathcal{L}$ for this reason.

21.1 Tensor powers of a sheaf of modules

We work over a fixed scheme X (in the application $X = X_s$, a fibre over a field $\kappa(s)$). Its category of sheaves of modules is X.Modules $= \mathrm{SheafOfModules}(\mathcal{O}_X)$, with global-sections notation $\Gamma(X, \mathcal{M}) = \mathcal{M}(X)$ for the value of the underlying presheaf on the top open. The structure sheaf $\mathcal{O}_X$ is a sheaf of commutative rings, so its underlying presheaf factors through the category of commutative rings; this is the hypothesis under which Mathlib's presheaf-level monoidal structure applies.

Lemma 1275 (Monoidal structure on presheaves of modules)**.** *Provided by Mathlib (`Mathlib.Algebra.Category.Mo`* *Let $R : \mathcal{C}^{\mathrm{op}} \to \mathrm{CommRingCat}$ be a presheaf of commutative rings. Then the category* $\mathrm{PresheafOfModules}(R)$ *of presheaves of modules over R carries a symmetric monoidal structure whose tensor product is computed objectwise: for presheaves of modules $\mathcal{M}_1, \mathcal{M}_2$ and an object U,*

$$(\mathcal{M}_1 \otimes \mathcal{M}_2)(U) \;=\; \mathcal{M}_1(U) \otimes_{R(U)} \mathcal{M}_2(U),$$

with restriction maps the tensor products of the restriction maps. The unit is the presheaf R viewed as a module over itself, and the associator, unitors, and braiding are induced objectwise from those of ModuleCat*. In particular evaluation at a fixed object U is a* strict *monoidal functor* $\mathrm{PresheafOfModules}(R) \to \mathrm{ModuleCat}(R(U))$*.*

Lemma 1276 (Sheafification of presheaves of modules)**.** *Provided by Mathlib (`Mathlib.Algebra.Category.ModuleCa`* *Let R be a sheaf of (commutative) rings on a site, with underlying presheaf R_0. There is a sheafification functor*

$$(-)^{\#} \;:\; \mathrm{PresheafOfModules}(R_0) \longrightarrow \mathrm{SheafOfModules}(R),$$

left adjoint to the forgetful functor, together with the unit $\eta_{\mathcal{M}_0} : \mathcal{M}_0 \to (\mathcal{M}_0)^{\#}$ of the adjunction. On underlying sheaves of abelian groups it agrees with the ordinary sheafification, and it sends a presheaf of R_0-modules to the associated sheaf of R-modules.

Lemma 1277 (The structure sheaf as the unit module)**.** *Provided by Mathlib (`Mathlib.Algebra.Category.ModuleCa`* *For a sheaf of rings R (in the application $R = \mathcal{O}_X$) the structure sheaf, viewed as a module over itself, is a distinguished object $\mathbf{1} = \mathcal{O}_X \in$* $\mathrm{SheafOfModules}(\mathcal{O}_X)$*. It is the value at the zeroth tensor power, $\mathcal{L}^{\otimes 0} = \mathcal{O}_X$, and its global sections $\Gamma(X, \mathcal{O}_X)$ form the degree-zero component of the section graded ring.*

Definition 1278 (Unit module of a scheme)**.** For a scheme X, the *unit module* $\mathbf{1}_X \in X$.Modules is the structure sheaf viewed as a module over itself, i.e. Mathlib's SheafOfModules.unit (1277) for $\mathcal{O}_X$. It is the value of the zeroth tensor power, $\mathcal{L}^{\otimes 0} = \mathbf{1}_X$ (1280).

Definition 1279 (Tensor product of sheaves of modules)**.** *Source: [Stacks Project], "Sheaves of Modules", Tag 01CA (tensor product).* Let $\mathcal{F}, \mathcal{G}$ be sheaves of $\mathcal{O}_X$-modules. Their *tensor product* is the sheaf of $\mathcal{O}_X$-modules

$$\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G} \;=\; (\mathcal{F} \otimes_{p,\mathcal{O}_X} \mathcal{G})^{\#},$$

the sheafification (1276) of the tensor product presheaf $\mathcal{F} \otimes_{p,\mathcal{O}_X} \mathcal{G}$, whose value on an open U is the $\mathcal{O}_X(U)$-module $\mathcal{F}(U) \otimes_{\mathcal{O}_X(U)} \mathcal{G}(U)$. Concretely, the tensor product presheaf is the Mathlib presheaf-level monoidal product (1275) of the underlying presheaves of modules of $\mathcal{F}$ and $\mathcal{G}$, and $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G}$ is its sheafification. This construction is symmetric and associative because the presheaf tensor product is, and sheafification preserves the coherence isomorphisms.

Definition 1280 (Tensor power of a line bundle)**.** *Source: [Stacks Project], "Sheaves of Modules", Tag 01CU (tensor powers).* Let $\mathcal{L}$ be a sheaf of $\mathcal{O}_X$-modules (in the application an invertible sheaf / line bundle L_s). For $m \in \mathbb{N}$ the *mth tensor power* $\mathcal{L}^{\otimes m}$ is defined by recursion on m:

$$\mathcal{L}^{\otimes 0} = \mathcal{O}_X \quad \text{(the unit module, 1277)}, \qquad \mathcal{L}^{\otimes(m+1)} = \mathcal{L}^{\otimes m} \otimes_{\mathcal{O}_X} \mathcal{L} \quad (1279).$$

We restrict to non-negative powers, which is all the section graded ring uses; the Stacks definition extends $\mathcal{L}^{\otimes n}$ to all $n \in \mathbf{Z}$ using invertibility, but negative powers are not needed here. For $\mathcal{L}$ invertible each $\mathcal{L}^{\otimes m}$ is again invertible.

Definition 1281 (Invertible sheaf of modules)**.** *Source: [Stacks Project], "Sheaves of Modules", Tag 01CR (invertible modules).* A sheaf of $\mathcal{O}_X$-modules $\mathcal{L} \in X$.Modules is *invertible* when it admits a *trivializing basis*: an indexed family $\{U_i\}_{i\in\iota}$ of opens of X whose range is a basis of the topology (Opens.IsBasis of $\{U_i\}$) such that on each U_i the sections $\Gamma(\mathcal{L}, U_i)$ form an *invertible* module over the ring $\Gamma(X, U_i)$, i.e. Module.Invertible $\Gamma(X, U_i)\,\Gamma(\mathcal{L}, U_i)$ (1307; equivalently $\Gamma(\mathcal{L}, U_i)$ is finite projective of constant rank one). This is the trivializing-cover / locally-free-of-rank-one form of invertibility: on each basis open $\mathcal{L}$ restricts to a free rank-one module, hence $\mathcal{L}|_{U_i} \cong \mathcal{O}_X|_{U_i}$. Over a scheme — where every stalk $\mathcal{O}_{X,x}$ is a local ring — this is equivalent to the $\otimes$-invertible form (the existence of a sheaf $\mathcal{N}$ with $\mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{N} \cong \mathbf{1}_X$, equivalently $\mathcal{F} \mapsto \mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{F}$ an equivalence of X.Modules), Stacks Tag 01CR. The trivializing-basis form implies the triviality of the self-braiding $\beta_{\mathcal{L},\mathcal{L}} = \mathrm{id}$ (1308) by a basis-local argument; in particular $\Gamma(X, \mathcal{L})$ itself need not be an invertible $\Gamma(X, \mathcal{O}_X)$-module. This is the line-bundle hypothesis under which the section graded ring becomes commutative (1341). The structure sheaf $\mathbf{1}_X$ is invertible (it trivializes on the basis of all opens), and any tensor power $\mathcal{L}^{\otimes m}$ of an invertible sheaf is again invertible.

Definition 1282 (Twist of a coherent sheaf by a tensor power)**.** Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$-modules (in the application a coherent sheaf $\mathcal{F}_s$) and $\mathcal{L}$ a line bundle. For $m \in \mathbb{N}$ the *m-twist* of $\mathcal{F}$ by $\mathcal{L}$ is

$$\mathcal{F}(m) \;:=\; \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes m} \qquad (1279, 1280),$$

the degree-m carrier of the section graded module. By construction $\mathcal{F}(0) = \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{O}_X \cong \mathcal{F}$.

The next four isomorphisms are the parts of the (would-be) symmetric monoidal structure on X.Modules that descend through sheafification from 1275 using only *functoriality* of the sheafification functor (1283) and, for the unitors, the reflective counit isomorphism — no strong monoidality of sheafification is required. They are the structural inputs of the tensor-power comparison 1304.

The strong-monoidality comparison 1286 below is assembled from three reduction lemmas — a localization criterion, the local-isomorphism property of the unit, and their corollary — together with the coequalizer presentation of the relative module-presheaf tensor product, the single Mathlib-absent brick on which the comparison rests.

Definition 1283 (Scheme-level module sheafification functor)**.** For a scheme X, the *module sheafification functor*

$$(-)^{\#} \;:\; X.\text{PresheafOfModules} \longrightarrow X.\text{Modules}$$

is Mathlib's presheaf-of-modules sheafification (1276) for the identity morphism of the underlying presheaf of (commutative) rings of $\mathcal{O}_X$, which is locally bijective. It is left adjoint to the forgetful functor $X.\text{Modules} \to X.\text{PresheafOfModules}$, with adjunction unit $\eta_{\mathcal{M}_0} : \mathcal{M}_0 \to (\mathcal{M}_0)^{\#}$ and an invertible counit (the forgetful functor is fully faithful). Being a functor it carries isomorphisms to isomorphisms, which is precisely what lets the presheaf-level coherence isomorphisms of 1275 descend to X.Modules.

Definition 1284 (Left unitor of the sheaf tensor product)**.** For a sheaf of $\mathcal{O}_X$-modules G, the *left unitor*

$$\mathbf{1}_X \otimes_{\mathcal{O}_X} G \xrightarrow{\sim} G$$

is obtained by descending the presheaf-level left unitor of 1275 through the sheafification functor (1283) and composing with the reflective counit isomorphism $(\mathcal{M}_0)^\# \cong \mathcal{M}$ of the sheafification adjunction. It is the base case $m = 0$ of the tensor-power comparison 1304, and uses no monoidal structure on X.Modules — only functoriality of sheafification and the invertible counit.

Definition 1285 (Braiding of the sheaf tensor product)**.** For sheaves of $\mathcal{O}_X$-modules F, G, the *braiding*

$$F \otimes_{\mathcal{O}_X} G \xrightarrow{\sim} G \otimes_{\mathcal{O}_X} F$$

is the image under the sheafification functor (1283) of the symmetric braiding of the presheaf symmetric monoidal structure (1275) on the underlying presheaves of modules. As pure sheafification-functoriality of an isomorphism it is again an isomorphism, requiring no monoidal structure on X.Modules. It is the symmetry used in the inductive step of 1304.

Lemma 1286 (Sheafification inverts the whiskered localization unit)**.** *Let P, Q be presheaves of $\mathcal{O}_X$-modules and let $\eta_P : P \to P^\#$ be the unit of the sheafification adjunction (1283, 1276). Then the sheafification of the right-whiskered unit*

$$(\eta_P \triangleright Q)^\# : (P \otimes_{p,\mathcal{O}_X} Q)^\# \longrightarrow (P^\# \otimes_{p,\mathcal{O}_X} Q)^\#,$$

where $\otimes_{p,\mathcal{O}_X}$ is the presheaf monoidal product (1275), is an isomorphism of sheaves of modules. This is the strong-monoidality comparison of the module sheafification functor on a whiskered unit, the single Mathlib-absent brick on which the sheaf-level associator 1297 and the tensor-power comparison 1304 rest.

Proof. isomorphism. The unit η_P is, on underlying abelian presheaves, the ordinary sheafification unit (the canonical map from a presheaf to its associated sheaf), which is a local isomorphism. Right-whiskering by Q preserves this: the relative module-presheaf tensor product is presented as a coequalizer, and tensoring with the fixed presheaf Q carries the local isomorphism η_P to a local isomorphism. Feeding this through the localization criterion gives the claim. □

21.1.1 Monoidal structure by localization transport

The preceding isomorphisms (unitors, braiding, whiskered-unit comparison) are exactly the data Mathlib needs to *transport* the entire symmetric monoidal structure from presheaves of modules onto X.Modules along the sheafification functor, rather than reassembling associator, pentagon, triangle and hexagon by hand. The mechanism is *monoidal localization*: a localization functor whose inverted class of morphisms is compatible with the tensor product carries the monoidal structure of its source to its target, with all coherence laws inherited. The two Mathlib anchors below supply the general transport theorem and the fact that module sheafification is such a localization; the two project blocks assemble the compatibility hypothesis and read off the resulting structure.

Lemma 1287 (Monoidal localization transport)**.** *Provided by Mathlib (`Mathlib.CategoryTheory.Localization.Mo` and `….Localization.Monoidal.Braided`). Let $\mathcal{C}$ be a monoidal category and let W be a class of morphisms of $\mathcal{C}$ satisfying* MorphismProperty.IsMonoidal*: W is multiplicative and stable under left- and right-whiskering by arbitrary objects. Let $L : \mathcal{C} \to \mathcal{D}$ be a localization functor for W. Then $\mathcal{D}$ acquires a full monoidal category structure — tensor product, associator and unitors —*

for which the pentagon and triangle coherence identities hold, and L is strong monoidal, with comparison isomorphisms

$$\mu_{X,Y} : L(X) \otimes L(Y) \xrightarrow{\sim} L(X \otimes Y).$$

All of this structure is inherited from $\mathcal{C}$ through L. If moreover $\mathcal{C}$ is braided (resp. symmetric), then $\mathcal{D}$ inherits a braided (resp. symmetric) structure for which the hexagon identities hold and L is braided monoidal. The sheafification of presheaves over a site is the archetypal instance of this transport.

Lemma 1288 (Module sheafification is a monoidal localization input)**.** *Provided by Mathlib (`Mathlib.Algebra.Category.ModuleCat.Sheaf.Localization` and `….ModuleCat.Presheaf.Monoidal`). The sheafification functor for presheaves of modules (1276) is a* localization functor *for the morphism class*

$$W' \;=\; J.W.\mathrm{inverseImage}\,(\mathrm{toPresheaf}\, R_0),$$

the class of morphisms of presheaves of R_0-modules whose underlying morphism of abelian-group presheaves lies in the local-isomorphism class $J.W$ of the topology — equivalently, the morphisms that sheafification sends to isomorphisms. Moreover the source category $\mathrm{PresheafOfModules}(R_0)$ *is symmetric monoidal (1275). Thus sheafification supplies precisely a localization functor L on a symmetric monoidal source, the two standing inputs of the transport theorem 1287; the only remaining hypothesis is that W' be compatible with the tensor product. The* IsLocalization *instance quoted above is for the bare Mathlib functor; the corresponding instance for the scheme-level sheafification functor $(-)^{\#}$, re-typed onto the monoidal carrier, is 1291.*

Definition 1289 (Monoidal-carrier sheafification functor)**.** For a scheme X, sheafificationMon is the module sheafification functor (1283) re-typed so that its source is the carrier MonoidalPresheaf X of the presheaf-level symmetric monoidal structure (1275):

$$\mathrm{sheafificationMon} \;:\; \mathrm{MonoidalPresheaf}\, X \longrightarrow X.\mathrm{Modules}.$$

It is definitionally the same functor as $(-)^{\#}$; the only change is that its domain is presented as the carrier equipped with the symmetric monoidal structure of 1275, so that the localization transport 1287 applies to it. This re-typing carries no mathematical content.

Definition 1290 (The sheafification localization class W')**.** For a scheme X, sheafificationW is the morphism class

$$W' \;=\; J.W.\mathrm{inverseImage}\,(\mathrm{toPresheaf}\, R_0) \;\subseteq\; \mathrm{Mor}(\mathrm{MonoidalPresheaf}\, X),$$

for R_0 the underlying presheaf of $\mathcal{O}_X$ and J the opens topology of X: the morphisms of presheaves of modules whose underlying morphism of abelian-group presheaves is a local isomorphism, equivalently the morphisms inverted by sheafification (1288). This is the class W' referred to in 1293 and 1288.

Lemma 1291 (Sheafification is the localization at W' on the monoidal carrier)**.** *For a scheme X, the functor* sheafificationMon *(1289) is a localization functor for the class $W' =$* sheafificationW *(1290). The* IsLocalization *property of* sheafificationMon *is obtained by transporting the localization fact 1288 across the definitional identification of $(-)^{\#}$ with the Mathlib sheafification functor and the re-typing of its domain onto the monoidal carrier* MonoidalPresheaf X*. It is the instance that allows the transport theorem 1287 to be applied to* sheafificationMon*.*

Definition 1292 (Transport unit of the monoidal localization)**.** For a scheme X, localizationUnitIso is the isomorphism

$$\varepsilon\ :\ \mathrm{sheafificationMon}(\mathbf{1}_{\mathrm{MonoidalPresheaf}\,X}) \xrightarrow{\sim} \mathbf{1}_X$$

identifying the image under sheafificationMon (1289) of the presheaf-level monoidal unit with the unit module $\mathbf{1}_X$ (1278). Since the unit module is already a sheaf, this is the reflective counit isomorphism of the sheafification adjunction (1283) at the unit. It is the datum ε the transport theorem 1287 requires in order to fix the unit object of the transported monoidal structure on X.Modules to be $\mathbf{1}_X$.

Definition 1293 (Tensor-compatibility of the sheafification localization class)**.** For a scheme X, write R_0 for the underlying presheaf of the structure sheaf $\mathcal{O}_X$ and $W' = J.W.\mathrm{inverseImage}\,(\mathrm{toPresheaf}\,R_0)$ for the sheafification localization class of 1288. Then W' satisfies MorphismProperty.IsMonoidal: it is multiplicative and stable under left- and right-whiskering. This is the compatibility hypothesis that, together with 1288, feeds the transport theorem 1287.

Proof. 1286 — stated there for the unit, but established for an arbitrary morphism inverted by sheafification — shows that $f \triangleright Q$ is again inverted by sheafification, hence lies in W'; the bridge between "lies in W'" and "sheafification inverts" is the localization criterion for the sheafification functor. Left-whiskering stability follows by conjugating the right-whiskering argument with the symmetric braiding of the presheaf monoidal structure (1275): since $Q \triangleleft f$ is carried to $f \triangleright Q$ by the braiding isomorphism on both ends, and membership in W' is invariant under isomorphism of arrows, stability under left-whiskering reduces to stability under right-whiskering. □

Definition 1294 (Symmetric monoidal structure on sheaves of modules by transport)**.** For a scheme X, the category X.Modules of sheaves of $\mathcal{O}_X$-modules carries a symmetric monoidal structure obtained by transporting the symmetric monoidal structure of X.PresheafOfModules along the sheafification functor. Concretely, one instantiates the monoidal localization transport 1287 with source the symmetric monoidal category X.PresheafOfModules, localization functor the module sheafification 1288, and the tensor-compatibility of its inverted class 1293; the unit is chosen to be the unit module $\mathbf{1}_X$ (1278). The resulting tensor product agrees objectwise with the sheaf tensor product 1279, via the strong-monoidal comparison μ of 1287. In particular the associator is the canonical (Mac Lane) associator, the pentagon and triangle laws hold, and — the source being symmetric — the braiding and its hexagon laws are inherited as well, giving X.Modules the structure of a *symmetric monoidal category.*

Proof. the source being symmetric — a symmetric structure with the hexagon identities, all inherited through sheafification. What is inherited in this way is precisely the *ambient* coherence of the monoidal and braided structure: the associator, the unitors, the braiding, and their pentagon, triangle, and hexagon laws, which hold for every pair of sheaves and require no separate induction. This ambient coherence is the raw material for the section-graded-ring assembly, but it is not the same as the multiplication identities of that ring: the μ-comparison coherences 1304 (associativity) and 1312 (commutativity) are established by induction on the tensor-power index — the associativity pentagon and the commutativity hexagon respectively — with the inherited coherence supplying only the per-step canonical isomorphism. The downstream section-ring identities are thus assembled inductively *from* the inherited coherence, not read off it as a free consequence. □

Corollary 1295 (Inherited braided and symmetric structure on sheaves of modules)**.** *For a scheme X, the transported monoidal structure on X.Modules (1294) refines to a braided and a* *symmetric monoidal structure. Because the presheaf source* X.PresheafOfModules *is symmetric monoidal (1275), the braided- and symmetric-transport clauses of 1287 apply: the braiding of* X.Modules *is the localization transport of the presheaf braiding, its hexagon identities are inherited, and the symmetry* $\beta_{F,G} \circ \beta_{G,F} = \mathrm{id}$ *holds by transport. These are the intermediate* BraidedCategory *and the final* SymmetricCategory *instances on* X.Modules *whose existence the prose of 1294 asserts. What the inherited symmetric structure supplies for free — for* every *sheaf, with no hypothesis on* $\mathcal{L}$ *— is the braiding* β *together with its forward and reverse hexagon identities; this is exactly the input that 1310 descends to the tensor-power braiding and consumes.*

It does not*, by itself, make the commutativity identity of the section graded ring 1333 — equivalently the comparison law* $\mu_{m,m'} = \mu_{m',m} \circ \beta$ *of 1312 — a free consequence. That identity additionally requires the self-braiding to be trivial,* $\beta_{\mathcal{L},\mathcal{L}} = \mathrm{id}$ *(1308), which holds precisely when* $\mathcal{L}$ *is invertible (1281). For a general* $\mathcal{L}$ *the section ring is the free tensor algebra and commutativity* fails*. So commutativity is invertibility-gated and hand-proved — from the inherited hexagon together with* $\beta_{\mathcal{L},\mathcal{L}} = \mathrm{id}$ *— rather than inherited outright.*

Definition 1296 (Bridge between the inherited and the project tensor product)**.** For sheaves of $\mathcal{O}_X$-modules F, G, the *tensor-agreement bridge* is the isomorphism

$$F \otimes G \xrightarrow{\sim} F \otimes_{\mathcal{O}_X} G$$

identifying the inherited tensor product $F \otimes G$ of the transported monoidal structure (1294) with the project's sheaf tensor product $F \otimes_{\mathcal{O}_X} G$ (1279). It is built from the strong-monoidal comparison μ of 1287 — whose codomain $L(a \otimes_p b)$ is exactly the sheafification of the presheaf tensor, i.e. $F \otimes_{\mathcal{O}_X} G$ — precomposed with the reflective counit isomorphisms identifying F, G with the sheafifications of their underlying presheaves. This bridge is the structural device along which the canonical associator α, braiding β and unitors λ, ρ of the inherited symmetric monoidal structure are transported onto the project's tensorObj family: each project-level coherence isomorphism is the conjugate, by tensorObjIso, of the corresponding canonical map of X.Modules. It is the launching point for the canonical reconstruction of the associator 1297 and the tensor-power comparison 1304.

Corollary 1297 (Associator of the sheaf tensor product)**.** *For sheaves of* $\mathcal{O}_X$*-modules* A, B, C *there is a canonical associativity isomorphism*

$$(A \otimes_{\mathcal{O}_X} B) \otimes_{\mathcal{O}_X} C \xrightarrow{\sim} A \otimes_{\mathcal{O}_X} (B \otimes_{\mathcal{O}_X} C)$$

of sheaves of $\mathcal{O}_X$*-modules.*

Proof. the bridge. The required isomorphism

$$(A \otimes_{\mathcal{O}_X} B) \otimes_{\mathcal{O}_X} C \xrightarrow{\sim} A \otimes_{\mathcal{O}_X} (B \otimes_{\mathcal{O}_X} C)$$

is obtained by conjugating the canonical associator $\alpha_{A,B,C} : (A \otimes B) \otimes C \xrightarrow{\sim} A \otimes (B \otimes C)$ of X.Modules by the bridge on each bracketing: one transports both iterated project tensors to the corresponding iterated inherited tensors using Φ (whiskered appropriately on the outer and inner factors), applies α, and transports back. As a composite of isomorphisms it is an isomorphism, and being the conjugate of the canonical associator it is itself canonical — no hand-rolled double-braiding detour and no separate strong-monoidality bricks are involved. Its pentagon and triangle coherence are then inherited from those of α (1294), which is exactly what 1304 consumes. ∎

21.1.2 Bridge lemmas to the canonical coherence maps

The associator of 1297 is obtained by conjugating the canonical Mac Lane associator of X.Modules along the tensor-agreement bridge tensorObjIso (1296). The remaining coherence data of the project's tensorObj family — the unitors, the braiding, and the two whiskerings — admit the same description. The bridge lemmas of this subsection make that description explicit: each hand-built project-level coherence isomorphism is shown to equal the conjugate, by tensorObjIso, of the corresponding canonical map of the inherited symmetric monoidal structure. They are the formalization-scaffolding layer along which the inherited triangle, pentagon and hexagon are transported onto $\otimes_{\mathcal{O}_X}$; downstream, the tensorObjIso bridges introduced by adjacent factors cancel, leaving the canonical coherence laws to do the work. Throughout, write $\Phi_{F,G} : F \otimes G \xrightarrow{\sim} F \otimes_{\mathcal{O}_X} G$ for the bridge tensorObjIso (1296).

Definition 1298 (Right unitor of the sheaf tensor product)**.** For a sheaf of $\mathcal{O}_X$-modules G, the *right unitor*

$$G \otimes_{\mathcal{O}_X} \mathbf{1}_X \xrightarrow{\sim} G$$

is obtained by descending the presheaf-level right unitor of 1275 through the sheafification functor (1283) and composing with the reflective counit isomorphism $(\mathcal{M}_0)^\# \cong \mathcal{M}$ of the sheafification adjunction. Like the left unitor (1284) it uses no monoidal structure on X.Modules — only functoriality of sheafification and the invertible counit — and it is the right-unitality datum consumed by the section-level coherence of 1333.

Lemma 1299 (The right unitor is the bridge-conjugate of the canonical right unitor)**.** *For a sheaf of $\mathcal{O}_X$-modules G, the hand-built right unitor (1298) equals the conjugate of the canonical right unitor ρ of the inherited monoidal structure (1294) by the bridge:*

$$\text{tensorObjRightUnitor}_G \;=\; \rho_G \circ \Phi^{-1}_{G,\mathbf{1}},$$

where $\rho_G : G \otimes \mathbf{1} \xrightarrow{\sim} G$ and $\Phi^{-1}_{G,\mathbf{1}} : G \otimes_{\mathcal{O}_X} \mathbf{1}_X \xrightarrow{\sim} G \otimes \mathbf{1}$.

Proof. ρ exhibits exactly the descended presheaf right unitor and counit that define $\text{tensorObjRightUnitor}_G$, after the bridge $\Phi_{G,\mathbf{1}}$ is moved across by naturality of ρ. Cancelling the bridge on the unit factor gives the stated equality. □

Lemma 1300 (The left unitor is the bridge-conjugate of the canonical left unitor)**.** *For a sheaf of $\mathcal{O}_X$-modules G, the hand-built left unitor (1284) equals the conjugate of the canonical left unitor λ of the inherited monoidal structure (1294) by the bridge:*

$$\text{tensorObjUnitIso}_G \;=\; \lambda_G \circ \Phi^{-1}_{\mathbf{1},G},$$

with $\lambda_G : \mathbf{1} \otimes G \xrightarrow{\sim} G$ and $\Phi^{-1}_{\mathbf{1},G} : \mathbf{1}_X \otimes_{\mathcal{O}_X} G \xrightarrow{\sim} \mathbf{1} \otimes G$. It is an auxiliary lemma used to put the $m = 0$ base case of 1304 on canonical footing.

Proof. transported across by naturality of λ and cancelled on the unit factor. □

Lemma 1301 (The braiding is the bridge-conjugate of the canonical braiding)**.** *For sheaves of $\mathcal{O}_X$-modules F, G, the hand-built braiding (1285) equals the conjugate of the canonical braiding β of the inherited symmetric monoidal structure (1295) by the bridge on each side:*

$$\text{tensorBraiding}_{F,G} \;=\; \Phi_{G,F} \circ \beta_{F,G} \circ \Phi^{-1}_{F,G},$$

where $\beta_{F,G} : F \otimes G \xrightarrow{\sim} G \otimes F$, $\Phi^{-1}_{F,G} : F \otimes_{\mathcal{O}_X} G \xrightarrow{\sim} F \otimes G$ and $\Phi_{G,F} : G \otimes F \xrightarrow{\sim} G \otimes_{\mathcal{O}_X} F$. It is an auxiliary lemma and the template for the two whiskering bridges below.

Proof. rewrites $\Phi_{G,F} \circ \beta_{F,G} \circ \Phi_{F,G}^{-1}$ as the descended presheaf braiding, which is $\text{tensorBraiding}_{F,G}$ by definition (1285). The two bridges sit on opposite ends and are absorbed by this naturality rather than cancelling against each other. □

Lemma 1302 (Left whiskering is the bridge-conjugate of the canonical left whiskering)**.** *Let F be a sheaf of $\mathcal{O}_X$-modules and $e : G \xrightarrow{\sim} H$ an isomorphism of such sheaves. The hand-built left-whiskered isomorphism* tensorObjWhiskerLeftIso $F\,e : F \otimes_{\mathcal{O}_X} G \xrightarrow{\sim} F \otimes_{\mathcal{O}_X} H$ *equals the conjugate of the canonical left whiskering by the bridge:*

$$\text{tensorObjWhiskerLeftIso}\, F e \;=\; \Phi_{F,H} \circ (F \lhd e) \circ \Phi_{F,G}^{-1},$$

*where $F \lhd e : F \otimes G \xrightarrow{\sim} F \otimes H$ is the canonical left whiskering of X.*Modules *(1294).*

Proof. $\Phi_{F,H} \circ (F \lhd e) \circ \Phi_{F,G}^{-1}$ into the descended whiskering, which is tensorObjWhiskerLeftIso $F e$. □

Lemma 1303 (Right whiskering is the bridge-conjugate of the canonical right whiskering)**.** *Let $e : G \xrightarrow{\sim} H$ be an isomorphism of sheaves of $\mathcal{O}_X$-modules and F a further such sheaf. The hand-built right-whiskered isomorphism* tensorObjWhiskerRightIso $e\,F : G \otimes_{\mathcal{O}_X} F \xrightarrow{\sim} H \otimes_{\mathcal{O}_X} F$ *equals the conjugate of the canonical right whiskering by the bridge:*

$$\text{tensorObjWhiskerRightIso}\, e\, F \;=\; \Phi_{H,F} \circ (e \rhd F) \circ \Phi_{G,F}^{-1},$$

*where $e \rhd F : G \otimes F \xrightarrow{\sim} H \otimes F$ is the canonical right whiskering of X.*Modules *(1294).*

Proof. whiskering tensorObjWhiskerRightIso $e\,F$. □

Lemma 1304 (Tensor-power comparison isomorphism)**.** *Source: [Stacks Project], "Sheaves of Modules", Tag 01CU (canonical power isomorphisms). For a line bundle $\mathcal{L}$ and $m, m' \in \mathbb{N}$ there is a canonical isomorphism of sheaves of $\mathcal{O}_X$-modules*

$$\mu_{m,m'} : \mathcal{L}^{\otimes m} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes m'} \;\xrightarrow{\sim}\; \mathcal{L}^{\otimes(m+m')},$$

natural in $\mathcal{L}$, and these isomorphisms satisfy a commutativity constraint ($\mu_{m,m'}$ agrees with $\mu_{m',m}$ after the braiding) and an associativity constraint (the two ways of combining $\mathcal{L}^{\otimes m} \otimes \mathcal{L}^{\otimes m'} \otimes \mathcal{L}^{\otimes m''}$ into $\mathcal{L}^{\otimes(m+m'+m'')}$ agree).

Proof. All the structural isomorphisms used to assemble $\mu_{m,m'}$ are the *canonical* associator, braiding and unitors of the inherited symmetric monoidal structure on X.Modules (1294, 1295), transported to the project's tensor product $\otimes_{\mathcal{O}_X}$ along the tensor-agreement bridge tensorObjIso (1296). In particular the associator is 1297 — itself the transported Mac Lane associator α — the braiding is the transport of the canonical β (1285), and the left unitor is the transport of λ (1284). No hand-rolled associator and no separate strong-monoidality brick enters.

Construction of $\mu_{m,m'}$. Fix m and recurse on m', matching the right-growth of tensorPow (1280, $\mathcal{L}^{\otimes(n+1)} = \mathcal{L}^{\otimes n} \otimes_{\mathcal{O}_X} \mathcal{L}$) and the recursion of $\mathbb{N}$-addition on its *second* argument. For $m' = 0$, using $\mathcal{L}^{\otimes 0} = \mathbf{1}_X$ and $m + 0 = m$ (both *definitional*),

$$\mathcal{L}^{\otimes m} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes 0} = \mathcal{L}^{\otimes m} \otimes_{\mathcal{O}_X} \mathbf{1}_X \xrightarrow{\rho} \mathcal{L}^{\otimes m}$$

is the (transported) right unitor ρ of 1298. For the inductive step the recursive definition $\mathcal{L}^{\otimes(c+1)} = \mathcal{L}^{\otimes c} \otimes_{\mathcal{O}_X} \mathcal{L}$ (1280) gives, with $m + (c+1) = (m+c)+1$ again *definitional*,

$$\mathcal{L}^{\otimes m} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes(c+1)} = \mathcal{L}^{\otimes m} \otimes_{\mathcal{O}_X} (\mathcal{L}^{\otimes c} \otimes_{\mathcal{O}_X} \mathcal{L}) \xrightarrow{\alpha^{-1}} (\mathcal{L}^{\otimes m} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes c}) \otimes_{\mathcal{O}_X} \mathcal{L} \xrightarrow{\mu_{m,c} \rhd \mathcal{L}} \mathcal{L}^{\otimes(m+c)} \otimes_{\mathcal{O}_X} \mathcal{L} = \mathcal{L}^{\otimes(m+(c+1))},$$

using only the inverse associator α^{-1} (1297) and the inductive comparison $\mu_{m,c}$ whiskered by $\mathcal{L}$ on the right; the final equality is the recursive definition of tensorPow together with $m+(c+1) = (m+c)+1$. The composite is $\mu_{m,c+1}$, an isomorphism as a composite of isomorphisms. *No braiding and no $eqToIso$ reindexer enter the construction*: because both tensorPow and $\mathbb{N}$-addition grow on the right, the new $\mathcal{L}$-factor already sits at the right edge of both source and target, so it never has to be transported across $\mathcal{L}^{\otimes c}$. (This is the categorification of pow_add, proved in Mathlib by induction on the second exponent with pow_succ and associativity only, no commutativity.) Naturality in $\mathcal{L}$ holds because each constituent map is natural.

Associativity constraint. Because α and λ are the canonical coherence data of a genuine symmetric monoidal category on X.Modules (1294, 1295), the associativity diagram (the two bracketings of $\mathcal{L}^{\otimes m} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes m'} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes m''} \to \mathcal{L}^{\otimes(m+m'+m'')}$) is an instance of the pentagon identity of that inherited structure. It therefore holds by *inheritance* — Mac Lane coherence applied once, at the level of the symmetric monoidal category X.Modules — rather than by a separate induction over a hand-rolled associator. The bridge tensorObjIso (1296) transports the coherence diagram from the inherited tensor $\otimes$ to the project tensor $\otimes_{\mathcal{O}_X}$ intact, so the associativity constraint passes to the family $\{\mu_{m,m'}\}$ verbatim; this holds for an *arbitrary* sheaf of modules $\mathcal{L}$ and is 1313.

Commutativity constraint (for invertible $\mathcal{L}$ only). The commutativity square ($\mu_{m,m'}$ versus $\mu_{m',m}$ postcomposed with the braiding) does *not* follow from Mac Lane coherence alone and requires $\mathcal{L}$ to be invertible (1281). After the hexagon and bridge reductions, the per-step inductive obligation reduces to a self-braiding $\beta_{\mathcal{L},\mathcal{L}}$, which equals the identity precisely when $\mathcal{L}$ is invertible (1308); for a general sheaf of modules this identity fails. The full argument is 1312.

Concretely, the constituents α^{-1} and $\mu_{m,c} \triangleright \mathcal{L}$ of the construction are rewritten as bridge-conjugates of the canonical associator and right-whiskering of X.Modules by the whiskering bridge 1303 (the associator already canonical by 1297), so that the associativity diagram collapses to the canonical *pentagon* of 1294 — with the tensorObjIso bridges introduced by adjacent factors cancelling in pairs — and is then dischargeable by the monoidal coherence tactic. The commutativity diagram (for invertible $\mathcal{L}$ only) additionally introduces the braiding via 1301 and 1302 and collapses to the canonical hexagon. □

21.1.3 Coherence constraints of the tensor-power comparison

The commutativity and associativity constraints of the comparison family $\{\mu_{m,m'}\}$ — the content Stacks Tag 01CU leaves with "formulation omitted" in 1304 — are decomposed here into named iso-level identities, so each can be discharged against a single canonical coherence law of the inherited symmetric monoidal structure on X.Modules (1294, 1295). After re-orienting μ to recurse on its *second* index, the right-unit constraint 1305 holds *definitionally* — it is the base clause of that recursion — so the worked templates are now the left-unit constraint 1306 and the associativity constraint 1313: each is an induction mirroring the recursion of μ, whose per-step obligation collapses — after the bridge lemmas of the previous subsection rewrite every hand-built constituent to its canonical counterpart and the tensorObjIso bridges cancel in adjacent pairs — to a single canonical coherence law (the unitor coherence, resp. the pentagon) of the inherited symmetric monoidal structure on X.Modules.

Lemma 1305 (Right-unit constraint for the tensor-power comparison)**.** *For a sheaf of modules $\mathcal{L}$ and $n \in \mathbb{N}$, the degree-$(n,0)$ comparison isomorphism $\mu_{n,0}$ (1304) is the right unitor:*

$$\mu_{n,0} \;=\; \mathrm{tensorObjRightUnitor}_{\mathcal{L}^{\otimes n}} \;:\; \mathcal{L}^{\otimes n} \otimes_{\mathcal{O}_X} \mathbf{1}_X \xrightarrow{\sim} \mathcal{L}^{\otimes n}$$

(1298). Here $\mathcal{L}^{\otimes 0} = \mathbf{1}_X$ and $n+0 = n$ hold on the nose, so the two sides inhabit a common type.

Proof. holds definitionally, so no reindexing intervenes. Hence $\mu_{n,0} = \rho$ for every n, without induction. □

Lemma 1306 (Left-unit constraint for the tensor-power comparison)**.** *For a sheaf of modules $\mathcal{L}$ and $n \in \mathbb{N}$, the degree-$(0,n)$ comparison isomorphism $\mu_{0,n}$ (1304) is the left unitor, reindexed along $0 + n = n$:*

$$\mu_{0,n} \;=\; \mathrm{tensorObjUnitIso}_{\mathcal{L}^{\otimes n}} \;:\; \mathbf{1}_X \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes n} \;\xrightarrow{\sim}\; \mathcal{L}^{\otimes n}$$

(1284, transport of the left unitor λ). Unlike the right-unit case, this is not *the base clause of the second-index recursion, so it requires an induction on n.*

Proof. Induction on n. For $n = 0$, $\mu_{0,0}$ is the base clause $\mathrm{tensorObjRightUnitor}_{\mathbf{1}_X}$ (right unitor of the unit), and on the unit object the left and right unitors coincide (MonoidalCategory.unitors_equal); descending that equality through sheafification gives $\mu_{0,0} = \mathrm{tensorObjUnitIso}$. For the inductive step $n = c{+}1$, unfold $\mu_{0,c+1}$ by the succ-clause (1304), $\alpha^{-1} \circ (\mu_{0,c} \triangleright \mathcal{L})$, and substitute the inductive hypothesis $\mu_{0,c} = \lambda$. After the bridges 1300/1303 rewrite to canonical maps and the tensorObjIso bridges cancel in pairs, the step is the canonical *triangle* identity $\lambda_{A\otimes\mathcal{L}} = (\lambda_A \triangleright \mathcal{L}) \circ \alpha^{-1}_{\mathbf{1},A,\mathcal{L}}$ of the inherited monoidal structure (1294), closed by the monoidal coherence tactic. □

Lemma 1307 (Self-braiding of an invertible module is trivial (affine))**.** *Provided by Mathlib (`Mathlib.RingTheory.PicardGroup`). Let R be a commutative ring and M an invertible R-module (*Module.Invertible $R\,M$*: the evaluation map $M \otimes_R M^* \to R$ is bijective, equivalently M is finite projective of constant rank one). Then the swap involution of the tensor square is the identity:*

$$\mathrm{TensorProduct.comm}_R\, M\, M \;=\; \mathrm{id}_{M\otimes_R M},$$

i.e. $x \otimes y = y \otimes x$ in $M \otimes_R M$ for all $x, y \in M$. This is the affine kernel of self-braiding triviality. Note the abstract monoidal statement "a $\otimes$-invertible object has $\beta = \mathrm{id}$" is false *— an odd line in super vector spaces is $\otimes$-invertible yet has $\beta = -\mathrm{id}$ — so the result genuinely uses the concrete swap on a module that is invertible in the* Module.Invertible *sense, not mere categorical invertibility.*

Lemma 1308 (Trivial self-braiding of an invertible sheaf)**.** *Let $\mathcal{L}$ be an invertible sheaf (1281) on a scheme X. Then the braiding of $\mathcal{L}$ with itself is the identity:*

$$\beta_{\mathcal{L},\mathcal{L}} \;=\; \mathrm{id}_{\mathcal{L}\otimes_{\mathcal{O}_X}\mathcal{L}} \qquad (1285).$$

More generally each tensor power $\mathcal{L}^{\otimes m}$ is again invertible (1280, 1281), so its self-braiding is trivial as well, $\beta_{\mathcal{L}^{\otimes m},\mathcal{L}^{\otimes m}} = \mathrm{id}$. This is the single arithmetic input distinguishing the invertible case: it holds precisely because $\mathcal{L}$ is locally free of rank one, and fails for a general sheaf of modules — already for $\mathcal{O}_X^{\oplus 2}$ over a point, where $e_1 \otimes e_2 \neq e_2 \otimes e_1$.

Proof. Equality of morphisms of sheaves of modules is a *local* condition that can be tested on a basis of opens; we obtain $\beta_{\mathcal{L},\mathcal{L}} = \mathrm{id}$ by descent from the trivializing basis carried by the invertibility datum.

The trivializing basis. By 1281, invertibility of $\mathcal{L}$ supplies an indexed family $\{U_i\}$ of opens whose range is a basis of the topology of X, together with, on each U_i, an invertible-module structure Module.Invertible $\Gamma(X, U_i)\,\Gamma(\mathcal{L}, U_i)$ (1307).

The presheaf braiding is the identity on each basis open. By construction (1285) the braiding $\beta_{\mathcal{L},\mathcal{L}}$ is the image under sheafification (1283) of the presheaf braiding β^{pre}, whose component on an open U is the swap $\mathrm{TensorProduct.comm}_{\Gamma(X,U)}\,\Gamma(\mathcal{L}, U)\,\Gamma(\mathcal{L}, U)$ (1275). On a basis open U_i

the module $\Gamma(\mathcal{L}, U_i)$ is invertible over $\Gamma(X, U_i)$, so that swap is the identity by 1307; hence β^{pre} agrees with id on every basis open U_i.

Descent to the sheafification. Write $T = \mathcal{L} \otimes \mathcal{L}$ for the underlying presheaf tensor product and $\eta : T \to T^{\#}$ for the sheafification unit, so that $\beta_{\mathcal{L},\mathcal{L}}$ is the sheafified endomorphism of $T^{\#}$ induced by β^{pre}. Its target $T^{\#}$ is a sheaf, so two morphisms into it that agree on a basis of opens coincide (separatedness, the basis form of sheaf-morphism extensionality, applied to the underlying sheaf of abelian groups). The two presheaf morphisms $\beta^{\mathrm{pre}} \circ \eta$ and η (both $T \to T^{\#}$) agree on every basis open U_i, because there β^{pre} is the identity; therefore $\beta^{\mathrm{pre}} \circ \eta = \eta$. By naturality of η,

$$\eta \circ (\text{sheafification of } \beta^{\mathrm{pre}}) = \beta^{\mathrm{pre}} \circ \eta = \eta = \eta \circ \mathrm{id}_{T^{\#}}.$$

Precomposition with the unit η is injective on morphisms into a sheaf (the universal property of the sheafification adjunction, 1283), so the sheafified map equals $\mathrm{id}_{T^{\#}}$. An isomorphism whose forward map is the identity is the identity isomorphism, whence $\beta_{\mathcal{L},\mathcal{L}} = \mathrm{id}$. Crucially the check is performed on the trivializing basis and *never* at the global open $\top$, where $\Gamma(X, \mathcal{L})$ need not be invertible.

The generalization to $\mathcal{L}^{\otimes m}$ is the identical argument: a tensor power of an invertible sheaf is again invertible (1280), so it carries a trivializing basis on which the presheaf braiding is the identity, and the same descent gives $\beta_{\mathcal{L}^{\otimes m},\mathcal{L}^{\otimes m}} = \mathrm{id}$. □

Lemma 1309 (First-index successor recursion of the tensor-power comparison)**.** *For an* arbitrary *sheaf of $\mathcal{O}_X$-modules $\mathcal{L}$ (no invertibility needed) and $c, m \in \mathbb{N}$, the comparison*

$$\mu_{c+1,m} : \mathcal{L}^{\otimes(c+1)} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes m} \xrightarrow{\ \sim\ } \mathcal{L}^{\otimes((c+1)+m)}$$

is recovered from the lower comparison $\mu_{c,\,1+m}$ by peeling the freshly-added factor to the left:

$$\mu_{c+1,m} \ = \ \mu_{c,\,1+m} \ \circ \ \left(\mathcal{L}^{\otimes c} \triangleleft \nu_m\right) \ \circ \ \alpha_{\mathcal{L}^{\otimes c},\,\mathcal{L},\,\mathcal{L}^{\otimes m}},$$

where α is the associator (1297), $\triangleleft$ the canonical left whiskering, and $\nu_m : \mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes m} \xrightarrow{\sim} \mathcal{L}^{\otimes(1+m)}$ is $\mu_{1,m}$ precomposed with the left-unit identification $\mathcal{L} \cong \mathbf{1}_X \otimes_{\mathcal{O}_X} \mathcal{L} = \mathcal{L}^{\otimes 1} \otimes_{\mathcal{O}_X} \cdots$ (1284), modulo the reindexing $(c+1)+m = c+(1+m)$. This is the first-index dual of the definitional second-index successor clause of μ (1304); unlike the commutativity hexagon it is braiding-free *and holds for every $\mathcal{L}$, both sides realising the identity permutation of the $(c+1)+m$ identical $\mathcal{L}$-factors. This order-preserving form is correct, reusable infrastructure, but it is* not *the ingredient consumed by the commutativity proof: the glue of 1312 requires the order*-reversing *recursion 1311 (brick 1′), which exposes the lower comparison $\mu_{c,m}$ rather than the unreachable $\mu_{c,1+m}$.*

Proof. Instantiate the associativity constraint 1313 at the indices $(m, m', m'') = (c, 1, m)$, which relates the comparison $\mu_{(c+1),m} = \mu_{c+1,m}$ to lower data:

$$(\mu_{c,1} \triangleright \mathcal{L}^{\otimes m}) \circ \mu_{c+1,m} \circ (\text{reindex}) \ = \ \alpha_{\mathcal{L}^{\otimes c},\mathcal{L}^{\otimes 1},\mathcal{L}^{\otimes m}} \circ (\mathcal{L}^{\otimes c} \triangleleft \mu_{1,m}) \circ \mu_{c,\,1+m}.$$

The factor $\mu_{c,1}$ is the second-index successor of $\mu_{c,0}$, and by 1304 together with 1305 (which makes $\mu_{c,0}$ the right unitor) it is the canonical right-unit isomorphism $\mathcal{L}^{\otimes c} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes 1} \cong \mathcal{L}^{\otimes(c+1)}$; hence its right-whiskering $\mu_{c,1} \triangleright \mathcal{L}^{\otimes m}$ is invertible with a structural inverse. Solving the displayed identity for $\mu_{c+1,m}$ and absorbing the unit isomorphism $\mu_{c,1}$ and the $\mathcal{L}^{\otimes 1} = \mathbf{1}_X \otimes_{\mathcal{O}_X} \mathcal{L}$ identification into the left-unit factor ν_m — using 1300 and the whisker bridges 1302, 1303 to pass to canonical whiskerings — yields the stated formula. No braiding occurs: the entire computation stays inside the associator/unitor (pentagon–triangle) fragment, so once the hand-built whiskerings are rewritten to canonical ones it is closed by the canonical coherence tactic monoidal. □

Lemma 1310 (Descended forward hexagon for the sheaf braiding)**.** *For sheaves of $\mathcal{O}_X$-modules F, A, B, the hand-built braiding (1285) and associator (1297) satisfy the* forward hexagon *identity*

$$\alpha_{A,B,F} \circ \beta_{F,\, A\otimes_{\mathcal{O}_X} B} \circ \alpha_{F,A,B} \;=\; (A \lhd \beta_{F,B}) \circ \alpha_{A,F,B} \circ (\beta_{F,A} \rhd B),$$

where β is the braiding (1285), α the associator (1297), and $\lhd, \rhd$ the canonical whiskerings. In words: braiding F past the tensor $A \otimes_{\mathcal{O}_X} B$ factors, up to associators, as braiding F past A and then past B separately. This is the symmetric-monoidal hexagon of the inherited structure (1295) read off at the project's tensor product, the braided analogue of the right-unit/braiding coherence used in the base case of 1312.

Proof. This is the descent of the canonical forward hexagon identity through the tensor-agreement bridge, exactly mirroring how the base-case coherence $\rho = \beta_{G,\mathbf{1}} \circ \lambda$ was descended from the canonical braiding–left-unitor coherence. The inherited symmetric monoidal structure on X.Modules (1295) satisfies the canonical forward hexagon

$$\alpha'_{A,B,F} \circ \beta'_{F,A\otimes B} \circ \alpha'_{F,A,B} = (A \lhd \beta'_{F,B}) \circ \alpha'_{A,F,B} \circ (\beta'_{F,A} \rhd B)$$

for its canonical braiding β' and associator α'. By the bridge lemmas — 1301 (the hand-built braiding is the conjugate of β' by the bridge Φ of 1296), 1297 (the associator is the conjugate of α'), and the whisker bridges 1302, 1303 — every project-level construct is replaced by its canonical counterpart conjugated by Φ. The bridges introduced by adjacent factors telescope and cancel in inverse pairs (the cancellation recipe of 1305), leaving exactly the canonical hexagon, which discharges the identity. The braiding is genuine and is *not* closed by the monoidal coherence tactic (which handles only associators and unitors); the explicit canonical hexagon is what is consumed. ☐

Lemma 1311 (Order-reversing first-index successor recursion (brick 1′))**.** *Let $\mathcal{L}$ be an* invertible *sheaf (1281) and $c, m \in \mathbb{N}$. The first-index successor comparison*

$$\mu_{c+1,m} : \mathcal{L}^{\otimes(c+1)} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes m} \xrightarrow{\sim} \mathcal{L}^{\otimes((c+1)+m)}$$

is recovered from the lower *comparison $\mu_{c,m}$ by braiding the freshly-added left factor $\mathcal{L}$ past the block $\mathcal{L}^{\otimes m}$ and applying $\mu_{c,m}$ on the right, framed by associators:*

$$\mu_{c+1,m} \;=\; (\textit{reindex}) \circ (\mu_{c,m} \rhd \mathcal{L}) \circ \alpha^{-1}_{\mathcal{L}^{\otimes c},\, \mathcal{L}^{\otimes m},\, \mathcal{L}} \circ (\mathcal{L}^{\otimes c} \lhd \beta_{\mathcal{L},\, \mathcal{L}^{\otimes m}}) \circ \alpha_{\mathcal{L}^{\otimes c},\, \mathcal{L},\, \mathcal{L}^{\otimes m}},$$

where α is the associator (1297), β the braiding (1285), $\lhd, \rhd$ the canonical whiskerings, and the reindex is the definitional identification $\mathcal{L}^{\otimes(c+m)} \otimes_{\mathcal{O}_X} \mathcal{L} = \mathcal{L}^{\otimes((c+m)+1)} = \mathcal{L}^{\otimes((c+1)+m)}$. The maps are displayed in $\circ$ (right-to-left) order; the Lean realisation chains them left-to-right (diagrammatic order). In contrast with the order-preserving *1309 — which is braiding-free but supplies the wrong atom for the commutativity glue — this order*-reversing *form surfaces exactly the lower comparison $\mu_{c,m}$ demanded by the inductive hypothesis of 1312, at the cost of a genuine braiding $\beta_{\mathcal{L},\mathcal{L}^{\otimes m}}$; hence the invertibility hypothesis enters here.*

Proof. Induction on m.

Base $m = 0$. Here $\mathcal{L}^{\otimes 0} = \mathbf{1}_X$, so the only braiding occurring is $\beta_{\mathcal{L},\mathbf{1}_X}$, the unit coherence — no self-braiding $\beta_{\mathcal{L},\mathcal{L}}$ appears and no invertibility is needed. Reading $\mu_{c,0}$ and $\mu_{c+1,0}$ as the appropriate unit isomorphisms (1305, 1306) and passing the hand-built braiding and whiskerings to their canonical counterparts (1301, 1302, 1303), the residual identity lies entirely in the associator/unitor (pentagon–triangle) fragment and is closed by the canonical coherence tactic.

Successor $m \to m+1$. Expand both $\mu_{c,m+1}$ and $\mu_{c+1,m+1}$ by the second-index successor clause (1304), $\mu_{-,m+1} = \alpha^{-1} \circ (\mu_{-,m} \triangleright \mathcal{L})$. The braiding on the factored target, $\beta_{\mathcal{L},\mathcal{L}^{\otimes(m+1)}} = \beta_{\mathcal{L},\mathcal{L}^{\otimes m} \otimes_{\mathcal{O}_X} \mathcal{L}}$, splits over $\mathcal{L}^{\otimes(m+1)} = \mathcal{L}^{\otimes m} \otimes_{\mathcal{O}_X} \mathcal{L}$ by the canonical right-tensor braiding identity (the forward hexagon 1310 read on the right factor) into $(\beta_{\mathcal{L},\mathcal{L}^{\otimes m}} \triangleright \mathcal{L})$ and $(\mathcal{L}^{\otimes m} \triangleleft \beta_{\mathcal{L},\mathcal{L}})$, framed by associators. Rewrite $\beta_{\mathcal{L},\mathcal{L}^{\otimes m}}$ by the inductive hypothesis and collapse the emergent self-braiding $\beta_{\mathcal{L},\mathcal{L}}$ to the identity by 1308 (this is the sole use of invertibility). After the hand-built whiskerings are rewritten to canonical ones (1302, 1303), the remaining structural associator/unitor obligation is closed by the canonical coherence tactic. This is the braided analogue of the pentagon successor step of 1313. $\square$

Lemma 1312 (Commutativity constraint for the tensor-power comparison)**.** *Let $\mathcal{L}$ be an invertible* sheaf *(1281) and $m, m' \in \mathbb{N}$. Then the comparison family is symmetric:*

$$\mu_{m,m'} \;=\; \mu_{m',m} \circ \beta_{\mathcal{L}^{\otimes m},\mathcal{L}^{\otimes m'}},$$

i.e. $\mu_{m,m'}$ factors through the braiding $\beta_{\mathcal{L}^{\otimes m},\mathcal{L}^{\otimes m'}} : \mathcal{L}^{\otimes m} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes m'} \xrightarrow{\sim} \mathcal{L}^{\otimes m'} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes m}$ (1285) followed by $\mu_{m',m}$, after the index reindexing $m' + m = m + m'$. This is the commutativity constraint of 1304. The invertibility hypothesis is essential: for a general sheaf of modules the section ring is the free tensor algebra on $\Gamma(X, \mathcal{L})$, and $\mu_{m,m'}$ and $\mu_{m',m} \circ \beta$ realize different *permutations of the $m + m'$ identical $\mathcal{L}$-factors; already at $m = m' = 1$ the identity reduces to $\beta_{\mathcal{L},\mathcal{L}} = \mathrm{id}$, which holds exactly when $\mathcal{L}$ is invertible (1308).*

Proof. Induction mirroring the recursion of μ (now on the second index), so the statement relates two distinct inductions. The base cases $m = 0$ and $m' = 0$ are the unit constraints: the right unit 1305 and the left unit 1306 read off the two sides.

Successor $m' = c + 1$. The successor step is the symmetric-monoidal hexagon, assembled from the bricks above. Unfold the left-hand side by the second-index successor clause (1304), $\mu_{m,c+1} = \alpha^{-1} \circ (\mu_{m,c} \triangleright \mathcal{L})$, and apply the inductive hypothesis $\mu_{m,c} = \beta_{\mathcal{L}^{\otimes m},\mathcal{L}^{\otimes c}} \circ \mu_{c,m} \circ (\text{reindex})$; this exposes the lower comparison $\mu_{c,m}$ precomposed with the braiding $\beta_{\mathcal{L}^{\otimes m},\mathcal{L}^{\otimes c}}$. On the right-hand side the comparison $\mu_{c+1,m}$ is a *first-index* successor that the inductive hypothesis cannot reach (it supplies $\mu_{c,m}$, not $\mu_{c+1,m}$). Split the right-hand braiding $\beta_{\mathcal{L}^{\otimes m},\mathcal{L}^{\otimes(c+1)}}$ on $\mathcal{L}^{\otimes(c+1)} = \mathcal{L}^{\otimes c} \otimes_{\mathcal{O}_X} \mathcal{L}$ by the descended forward hexagon 1310 and distribute the whiskers; the shared prefix $\alpha^{-1} \circ (\beta_{\mathcal{L}^{\otimes m},\mathcal{L}^{\otimes c}} \triangleright \mathcal{L})$ cancels off both sides. The residual obligation is then *exactly* the order-reversing recursion of $\mu_{c+1,m}$, namely brick $1'$ (1311), whose right-hand factor $\beta_{\mathcal{L},\mathcal{L}^{\otimes m}}$ meets the surviving $\beta_{\mathcal{L}^{\otimes m},\mathcal{L}}$ coming from the hexagon. Substituting brick $1'$ and passing the hand-built braiding and whiskerings to canonical maps (1301, 1302, 1303), the two opposed braidings annihilate by the symmetry relation $\beta_{\mathcal{L}^{\otimes m},\mathcal{L}} \circ \beta_{\mathcal{L},\mathcal{L}^{\otimes m}} = \mathrm{id}$ (1295); the only remaining self-braiding $\beta_{\mathcal{L},\mathcal{L}}$ on a doubled $\mathcal{L}$-factor is the identity by 1308. With both braidings discharged and the tensorObjIso bridges cancelled pairwise (1305), the residue is a pure associator/unitor identity, closed by the canonical coherence (monoidal); the reindex $m' + m = m + m'$ matches the two ends. $\square$

Lemma 1313 (Associativity constraint for the tensor-power comparison)**.** *For an arbitrary $\mathcal{L} : \mathbf{Mod}(\mathcal{O}_X)$ (no invertibility needed — associativity holds for all $\mathcal{L}$) and $m, m', m'' \in \mathbb{N}$ the two bracketings of $\mathcal{L}^{\otimes m} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes m'} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes m''}$ into $\mathcal{L}^{\otimes(m+m'+m'')}$ agree:*

$$\mu_{m+m',m''} \circ (\mu_{m,m'} \triangleright \mathcal{L}^{\otimes m''}) \;=\; \mu_{m,m'+m''} \circ (\mathcal{L}^{\otimes m} \triangleleft \mu_{m',m''}) \circ \alpha_{\mathcal{L}^{\otimes m},\mathcal{L}^{\otimes m'},\mathcal{L}^{\otimes m''}},$$

where α is the associator 1297 and $\triangleleft, \triangleright$ the canonical whiskerings, modulo the reindexings $(m + m') + m'' = m + (m' + m'')$. This is the associativity constraint of 1304.

Proof. Induction on the *last* index m'', mirroring the second-index recursion of $\mu = \text{tensorPowAdd}$ (1304). Because that recursion is now braiding-free — each clause is built from α^{-1} and a single right-whisker — both sides of the identity are composites of canonical associators and whiskerings only; *no braiding occurs anywhere*, so the whole statement is an instance of Mac Lane's pentagon coherence and is dischargeable by the monoidal coherence tactic once the project's hand-built isos are rewritten to their canonical counterparts by 1303 (the associator is already canonical by 1297).

Base case $m'' = 0$. Both $\mu_{m+m',0}$ and $\mu_{m',0}$, $\mu_{m'+0,\ldots}$ collapse to right unitors (1305, which is now rfl) and $\mathcal{L}^{\otimes 0} = \mathbf{1}$, $\cdot + 0 = \cdot$ hold definitionally; the two sides reduce to the unit triangle $\rho_{A\otimes B} = (A \triangleleft \rho_B) \circ \ldots$, closed by monoidal.

Inductive step $m'' = c + 1$. Unfold $\mu_{-,c+1}$ on both sides by the second-index succ-clause $\alpha^{-1} \circ (\mu_{-,c} \triangleright \mathcal{L})$ (1304). The outer right-whisker distributes over the composite and, after rewriting by 1303, the inductive hypothesis (the $m'' = c$ instance, whiskered $\triangleright\mathcal{L}$) folds into the central atom via one pure-associator slide associator_inv_naturality_left (1297) — NOT a braiding. The residual structural glue between the four folded comparison atoms and the associator $\alpha_{\mathcal{L}^{\otimes m},\mathcal{L}^{\otimes m'},\mathcal{L}^{\otimes m''}}$ is a pure pentagon on canonical associators and whiskerings, closed by monoidal. The reindexings $(m + m') + c = m + (m' + c)$ etc. are Nat.add-definitional except for one add_assoc transport in the frozen statement, dispatched by a subst helper exactly as before. $\square$

21.2 Lax-monoidal global sections

The graded multiplication is supplied by the lax-monoidal structure of the global-sections functor $\Gamma(X, -) : \text{SheafOfModules}(\mathcal{O}_X) \to \text{ModuleCat}(\Gamma(X, \mathcal{O}_X))$. It is only *lax* because global sections do not commute with sheafification: a global section of a sheafified tensor product is not in general a finite sum of elementary tensors of global sections.

Definition 1314 (Section multiplication)**.** Let $\mathcal{F}, \mathcal{G}$ be sheaves of $\mathcal{O}_X$-modules. The *section multiplication* is the $\Gamma(X, \mathcal{O}_X)$-bilinear map

$$\Gamma(X, \mathcal{F}) \otimes_{\Gamma(X,\mathcal{O}_X)} \Gamma(X, \mathcal{G}) \longrightarrow \Gamma(X, \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G})$$

defined as follows. A pair of global sections (σ, τ) determines, by the objectwise formula of 1275 at the top open, the elementary tensor $\sigma \otimes \tau \in (\mathcal{F} \otimes_{p,\mathcal{O}_X} \mathcal{G})(X)$, a global section of the tensor product presheaf. Composing with the global-sections component of the sheafification unit $\eta : \mathcal{F} \otimes_{p,\mathcal{O}_X} \mathcal{G} \to (\mathcal{F} \otimes_{p,\mathcal{O}_X} \mathcal{G})^{\#} = \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G}$ (1276, 1279) yields a global section $\sigma \cdot \tau \in \Gamma(X, \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G})$. Bilinearity over $\Gamma(X, \mathcal{O}_X)$ is inherited from bilinearity of the elementary tensor and $\Gamma(X, \mathcal{O}_X)$-linearity of η.

The coherence of the section multiplication is governed by how the canonical coherence maps of X.Modules interact with sectionsMul. Because sectionsMul is the image under $\Gamma(X, -)$ of the sheafification unit η evaluated at the top open, and η is natural, each canonical coherence isomorphism — the unitors, the braiding, the associator — pushes through sectionsMul on the corresponding section factors. The four naturality helpers below record these push-through identities at the level of global sections; they are the section-level partners of the iso-level constraints 1305, 1312 and 1313. The key local fact in each case is that the MonoidalPresheaf coherence map at the top open is the corresponding ModuleCat symmetric-monoidal structure map, computed on elementary tensors of sections.

Lemma 1315 (Left unitor through the section multiplication)**.** *Let G be a sheaf of $\mathcal{O}_X$-modules and $a \in \Gamma(X, G)$. Then*

$$\Gamma(\text{tensorObjUnitIso}_G)(\text{sectionsMul}_{\mathbf{1}_X, G}(1 \otimes a)) = a,$$

i.e. applying global sections of the left unitor $\mathbf{1}_X \otimes_{\mathcal{O}_X} G \xrightarrow{\sim} G$ (1284) to the section product of the unit section $1 \in \Gamma(X, \mathcal{O}_X)$ and a (1314) returns a. This is the section-level reading of the left-unit law via η-naturality of the lax structure.

Proof. unitor). Evaluating at the top open and using the ModuleCat left-unitor formula $r \otimes m \mapsto r \cdot m$ sends $1 \otimes a$ to $1 \cdot a = a$. ☐

Lemma 1316 (Right unitor through the section multiplication)**.** *Let G be a sheaf of $\mathcal{O}_X$-modules and $a \in \Gamma(X, G)$. Then*

$$\Gamma(\text{tensorObjRightUnitor}_G)(\text{sectionsMul}_{G, \mathbf{1}_X}(a \otimes 1)) = a,$$

the right-unit analogue of 1315: global sections of the right unitor $G \otimes_{\mathcal{O}_X} \mathbf{1}_X \xrightarrow{\sim} G$ (1298) carry the section product of a with the unit section to a, via the same η-naturality and right-triangle argument and the ModuleCat *right-unitor formula $m \otimes r \mapsto r \cdot m$.*

Proof. ☐

Lemma 1317 (Braiding through the section multiplication)**.** *Let $\mathcal{F}, \mathcal{G}$ be sheaves of $\mathcal{O}_X$-modules, $a \in \Gamma(X, \mathcal{F})$ and $b \in \Gamma(X, \mathcal{G})$. Then*

$$\Gamma(\text{tensorBraiding}_{\mathcal{F}, \mathcal{G}})(\text{sectionsMul}_{\mathcal{F}, \mathcal{G}}(a \otimes b)) = \text{sectionsMul}_{\mathcal{G}, \mathcal{F}}(b \otimes a),$$

i.e. global sections of the braiding (1285) push through the section multiplication (1314), swapping the two section factors. This is the section-level partner of the commutativity constraint 1312.

Proof. presheaf braiding at the top open. The MonoidalPresheaf braiding at the top open is the ModuleCat symmetric braiding (the tensor-swap $a \otimes b \mapsto b \otimes a$), so it sends $a \otimes b$ to $b \otimes a$, giving the stated equality. ☐

The associator push-through 1326 is *not* a one-step η-naturality like the braiding 1317. The braiding is the sheafification of a single presheaf morphism, so a single naturality square moves it through sectionsMul. The associator 1297, by contrast, is the canonical Mac Lane associator α of X.Modules conjugated by five tensor-agreement bridges tensorObjIso (1296) acting through the localization comparison μ (1287) — not the image of one presheaf map — so the one-step argument does not transfer. We therefore cut the proof into a chain mirroring the bridge-telescoping construction of 1305: two whiskered-unit legs (1318, 1319) identifying each iterated section product with a sheafification unit on the raw presheaf tensor, the presheaf associator computed at the top open (1322), and a presheaf-morphism factorization (1323) assembling them; the target then evaluates that factorization on the iterated elementary tensor. Throughout, η denotes the sheafification unit (1276) and $\otimes_{p,\mathcal{O}_X}$ the presheaf monoidal product (1275).

Lemma 1318 (Right-whiskered-unit leg of the iterated section product)**.** *Let A, B, C be sheaves of $\mathcal{O}_X$-modules. As morphisms of presheaves of modules*

$$(A \otimes_{p,\mathcal{O}_X} B) \otimes_{p,\mathcal{O}_X} C \longrightarrow (A \otimes_{\mathcal{O}_X} B) \otimes_{\mathcal{O}_X} C,$$

the right-whiskered-unit leg of the iterated section product equals the sheafification unit after the right-whiskered unit:

$$\eta_{(A\otimes_{\mathcal{O}_X}B)\otimes_{p,\mathcal{O}_X}C}\circ(\eta_{A\otimes_{p,\mathcal{O}_X}B}\triangleright C),$$

where the right-whiskered unit $\eta_{A\otimes_{p,\mathcal{O}_X}B}\triangleright C$ is the comparison whose sheafification is invertible by 1286. Its component at the top open is a ModuleCat *morphism; equivalently, evaluating that component on the elementary tensor $(a\otimes b)\otimes c$ ($a\in\Gamma(X,A)$, $b\in\Gamma(X,B)$, $c\in\Gamma(X,C)$) recovers the iterated section product over the already-sheafified first factor:*

$$\mathrm{sectionsMul}_{A\otimes_{\mathcal{O}_X}B,\,C}(\mathrm{sectionsMul}_{A,B}(a\otimes b)\otimes c)\ =\ \Gamma(\eta_{(A\otimes_{\mathcal{O}_X}B)\otimes_{p,\mathcal{O}_X}C}\circ(\eta_{A\otimes_{p,\mathcal{O}_X}B}\triangleright C))((a\otimes b)\otimes c).$$

Proof. By the objectwise formula of 1275 the right-whiskering $\eta_{A\otimes_{p,\mathcal{O}_X}B}\triangleright C$ sends, at the top open, $(a\otimes b)\otimes c$ to $(\eta_{A\otimes_{p,\mathcal{O}_X}B}(a\otimes b))\otimes c$; composing the two units gives the stated equality. The whisker comparison 1286 records that this leg is the invertible right-whiskered unit, the form in which it enters the bridge telescoping of 1323. ☐

Lemma 1319 (Left-whiskered-unit leg of the iterated section product)**.** *Let A,B,C be sheaves of $\mathcal{O}_X$-modules. As morphisms of presheaves of modules*

$$A\otimes_{p,\mathcal{O}_X}(B\otimes_{p,\mathcal{O}_X}C)\ \longrightarrow\ A\otimes_{\mathcal{O}_X}(B\otimes_{\mathcal{O}_X}C),$$

the left-whiskered-unit leg of the iterated section product equals the sheafification unit after the left-whiskered unit:

$$\eta_{A\otimes_{p,\mathcal{O}_X}(B\otimes_{\mathcal{O}_X}C)}\circ(A\triangleleft\eta_{B\otimes_{p,\mathcal{O}_X}C}),$$

the left-whiskered analogue of 1318, with the left-whiskered unit $A\triangleleft\eta_{B\otimes_{p,\mathcal{O}_X}C}$ (invertible after sheafification by 1286 via the presheaf braiding) replacing the right-whiskered one. Its component at the top open is a ModuleCat *morphism; equivalently, evaluating that component on the elementary tensor $a\otimes(b\otimes c)$ ($a\in\Gamma(X,A)$, $b\in\Gamma(X,B)$, $c\in\Gamma(X,C)$) recovers the iterated section product over the already-sheafified second factor:*

$$\mathrm{sectionsMul}_{A,\,B\otimes_{\mathcal{O}_X}C}(a\otimes\mathrm{sectionsMul}_{B,C}(b\otimes c))\ =\ \Gamma(\eta_{A\otimes_{p,\mathcal{O}_X}(B\otimes_{\mathcal{O}_X}C)}\circ(A\triangleleft\eta_{B\otimes_{p,\mathcal{O}_X}C}))(a\otimes(b\otimes c)).$$

Proof. $A\triangleleft\eta_{B\otimes_{p,\mathcal{O}_X}C}$ at the top open as $a\otimes(b\otimes c)\mapsto a\otimes\eta_{B\otimes_{p,\mathcal{O}_X}C}(b\otimes c)$; composing with the outer unit gives the equality. The left-whiskered form of 1286 — its proof handles left whiskering by conjugating with the presheaf braiding — supplies the invertibility used in 1323. ☐

Lemma 1320 (Right-whisker naturality of the section product)**.** *Let $f:F\to F'$ be a morphism of sheaves of $\mathcal{O}_X$-modules and let G be a sheaf of modules. The section product* sectionsMul *is* natural in its first argument*: evaluating at the top open on $x\in\Gamma(X,F)$, $y\in\Gamma(X,G)$,*

$$\mathrm{sectionsMul}_{F',G}(\Gamma(f)(x)\otimes y)\ =\ \Gamma(f\triangleright G)(\mathrm{sectionsMul}_{F,G}(x\otimes y)),$$

where $f\triangleright G:F\otimes_{\mathcal{O}_X}G\to F'\otimes_{\mathcal{O}_X}G$ is the right-whiskering of f by G and $\Gamma(-)$ denotes the top-open component of a sheaf morphism. This is the general-morphism analogue of 1318 (which is the special case $f=\eta_{F\otimes_p\mathbf{1}}$, the sheafification unit), and is the slide used in the associativity leg of 1333 to move an inner comparison $\Gamma(\mu)$ out of the first tensor factor of an outer sectionsMul*.*

Proof. By 1314, $\mathrm{sectionsMul}_{F,G}$ is the top-open component of the sheafification unit $\eta_{F\otimes_pG}$, itself the component of a natural transformation (1276). Naturality of η along the presheaf-of-modules right-whiskering $f\triangleright_p G$ reads, as presheaf morphisms,

$$\eta_{F'\otimes_pG}\circ(f\triangleright_p G)\ =\ (f\triangleright G)\circ\eta_{F\otimes_pG}.$$

The objectwise whiskering formula of 1275 identifies $f \triangleright_p G$ at the top open as $x \otimes y \mapsto \Gamma(f)(x) \otimes y$. Evaluating the naturality square at the top open on the elementary tensor $x \otimes y$ gives the displayed equality. ☐

Lemma 1321 (Left-whisker naturality of the section product)**.** *Let F be a sheaf of $\mathcal{O}_X$-modules and let $g : G \to G'$ be a morphism of sheaves of modules. The section product* sectionsMul *is* natural in its second argument*: evaluating at the top open on $x \in \Gamma(X, F)$, $y \in \Gamma(X, G)$,*

$$\mathrm{sectionsMul}_{F,G'}(x \otimes \Gamma(g)(y)) \;=\; \Gamma(F \triangleleft g)(\mathrm{sectionsMul}_{F,G}(x \otimes y)),$$

where $F \triangleleft g : F \otimes_{\mathcal{O}_X} G \to F \otimes_{\mathcal{O}_X} G'$ is the left-whiskering of g by F. This is the left-handed analogue of 1320 (the general-morphism analogue of 1319), and is the slide used in the associativity leg of 1333 to move an inner comparison $\Gamma(\mu)$ out of the second tensor factor of an outer sectionsMul*.*

Proof. Identical to 1320 with the roles of the two tensor factors exchanged: naturality of the sheafification unit η (1276) along the left-whiskering $F \triangleleft_p g$, whose top-open component is $x \otimes y \mapsto x \otimes \Gamma(g)(y)$ by the objectwise formula of 1275, evaluated on $x \otimes y$. ☐

Lemma 1322 (Presheaf associator at the top open)**.** *Let A, B, C be presheaves of $\mathcal{O}_X$-modules. The associator $\alpha^p_{A,B,C}$ of the presheaf monoidal structure (1275) evaluated at the top open is the* ModuleCat *associator on the module sections, sending the iterated elementary tensor to its reassociation:*

$$\alpha^p_{A,B,C}\big|_\top ((a \otimes b) \otimes c) = a \otimes (b \otimes c), \qquad a \in A(\top),\ b \in B(\top),\ c \in C(\top).$$

Proof. ☐

Lemma 1323 (Presheaf-morphism factorization of the associator)**.** *Let A, B, C be sheaves of $\mathcal{O}_X$-modules. Then, as morphisms of presheaves of modules $(A \otimes_{p,\mathcal{O}_X} B) \otimes_{p,\mathcal{O}_X} C \longrightarrow A \otimes_{\mathcal{O}_X} (B \otimes_{\mathcal{O}_X} C)$,*

$$\mathrm{tensorObjAssoc}_{A,B,C} \circ \eta_{(A \otimes_{\mathcal{O}_X} B) \otimes_{p,\mathcal{O}_X} C} \circ (\eta_{A \otimes_{p,\mathcal{O}_X} B} \triangleright C) \;=\; \eta_{A \otimes_{p,\mathcal{O}_X} (B \otimes_{\mathcal{O}_X} C)} \circ (A \triangleleft \eta_{B \otimes_{p,\mathcal{O}_X} C}) \circ \alpha^p_{A,B,C},$$

where α^p is the presheaf associator (1275). This is the η-naturality-plus-bridge-telescoping identity that makes Γ(tensorObjAssoc) push the presheaf associator through the iterated sectionsMul*. It is stated as an equality of presheaf morphisms (holding component-wise at every open), the form in which the bridge-telescoping is carried out; evaluating both sides at the top open on the iterated elementary tensor is what feeds the associator push-through 1326.*

Proof. obtain it by transporting an identity of sheaf morphisms across the naturality of the sheafification unit η (1276). Each of the three constituent presheaf maps — the right-whiskered unit $\eta_{A \otimes_p B} \triangleright C$, the presheaf associator $\alpha^p_{A,B,C}$ (1275), and the left-whiskered unit $A \triangleleft \eta_{B \otimes_p C}$ — fits into a naturality square for η: composing it with the unit at its codomain equals the unit at its domain composed with $T(L(\cdot))$, the underlying presheaf map of its sheafification. Applying these three naturality squares rewrites both sides of the claim as $\eta_{(A \otimes_p B) \otimes_p C}$ post-composed with T of a single sheaf morphism, so the claim reduces to the equality of those two sheaf morphisms inside X.Modules,

$$L(\eta_{A \otimes_p B} \triangleright C) \;\circ\; \mathrm{tensorObjAssoc}_{A,B,C} \;=\; L(\alpha^p_{A,B,C}) \;\circ\; L(A \triangleleft \eta_{B \otimes_p C}),$$

which is the sheaf-level core 1325. (The shared middle object is presented two definitionally-equal but not syntactically equal ways — η's codomain carries the right-adjoint's restrictScalars($\mathbf{1}$) decoration absent from tensorObjAssoc's domain — so the three naturality rewrites are performed up to this definitional equality.) ☐

Lemma 1324 (Sheafification unit on the left triangle)**.** *Let L be the sheafification functor on presheaves of $\mathcal{O}_X$-modules, with unit η and counit ε of the sheafification adjunction (1276). For a presheaf of modules P,*

$$L(\eta_P) \;=\; \varepsilon_{LP}^{-1},$$

i.e. the sheafification of the unit at P is the inverse of the counit isomorphism at LP.

Proof. □

Lemma 1325 (Sheaf-level associator factorization (core of 1323))**.** *Let A, B, C be sheaves of $\mathcal{O}_X$-modules and L the sheafification functor. As morphisms of sheaves of $\mathcal{O}_X$-modules*

$$L(\eta_{A\otimes_{p,\mathcal{O}_X}B} \triangleright C) \;\circ\; \mathrm{tensorObjAssoc}_{A,B,C} \;=\; L(\alpha^p_{A,B,C}) \;\circ\; L(A \triangleleft \eta_{B\otimes_{p,\mathcal{O}_X}C}),$$

where η is the sheafification unit and α^p the presheaf associator (1275). Equivalently, tensorObjAssoc *(the canonical-α form 1297) coincides with the original η-conjugated associator* $(\mathrm{asIso}\, L(\eta \triangleright C))^{-1} \circ L(\alpha^p) \circ \mathrm{asIso}\, L(A \triangleleft \eta)$.

Proof. associator α of X.Modules flanked by the four tensor-agreement bridges $\Phi = \mathrm{tensorObjIso}$ (1296) whiskered on the outer and inner factors,

$$\mathrm{tensorObjAssoc} = \Phi^{-1}_{A\otimes B,\,C} \circ (\Phi^{-1}_{A,B} \triangleright C) \circ \alpha_{A,B,C} \circ (A \triangleleft \Phi_{B,C}) \circ \Phi_{A,\,B\otimes C}.$$

Each bridge Φ is built from the localization comparison μ (1287); the canonical associator α transported through the μ's is the sheafification $L(\alpha^p)$ of the presheaf associator. Substitute $L(\eta_P) = \varepsilon_{LP}^{-1}$ (1324) so each whiskered sheafification unit on the two sides becomes a whiskered counit, and rewrite the three localization-comparison segments via the strong-monoidality whisker comparison 1286 (in its right- and, via the presheaf braiding, left-whiskered forms) together with the naturality of the localization comparison μ (1287) in each of its two arguments and the comparison identifying the transported canonical associator with $L(\alpha^p)$. The tensorObjIso bridges then telescope in adjacent pairs — the bridge-cancellation recipe already carried out for 1305 — leaving exactly $L(\eta \triangleright C)$ on the left and $L(\alpha^p) \circ L(A \triangleleft \eta)$ on the right, which is the stated equality. (As in 1323, every μ-conjugation step bridges the restrictScalars($\mathbf{1}$) definitional decoration on η's codomain against tensorObjIso's undecorated factor, so the rewrites are carried out up to that definitional equality.) □

Lemma 1326 (Associator through the section multiplication)**.** *Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be sheaves of $\mathcal{O}_X$-modules and $a \in \Gamma(X, \mathcal{A})$, $b \in \Gamma(X, \mathcal{B})$, $c \in \Gamma(X, \mathcal{C})$. Applying global sections of the associator (1297) to the iterated section product reassociates the three factors:*

$$\Gamma(\mathrm{tensorObjAssoc}_{\mathcal{A},\mathcal{B},\mathcal{C}})\Big(\mathrm{sectionsMul}(\mathrm{sectionsMul}(a\otimes b)\otimes c)\Big) = \mathrm{sectionsMul}(a\otimes \mathrm{sectionsMul}(b\otimes c)),$$

i.e. $\Gamma(\alpha)$ sends $(a \otimes b) \otimes c$ to $a \otimes (b \otimes c)$ at the level of section multiplications. This is the η-naturality of the lax structure pushing the associator through sectionsMul*, the section-level partner of the associativity constraint 1313.*

Proof. 1318; its right side is the RHS of the claim by 1322 (the presheaf associator sends $(a \otimes b) \otimes c \mapsto a \otimes (b \otimes c)$) followed by 1319. □

The graded-monoid axioms below are not raw equalities between sections of two different tensor powers — on $\mathbb{N}$ the indices $(i + j) + k$ and $i + (j + k)$, or $0 + n$ and n, are equal but not definitionally so, so the two sides a priori inhabit distinct section modules. Following

`Mathlib.LinearAlgebra.TensorPower.Basic`, we phrase each coherence as an honest equality in a single section module, obtained by transporting one side along an index-equality isomorphism, and provide a bridge that repackages such a transported equality as an equality of dependent pairs.

Definition 1327 (Section component in degree m)**.** Let $\mathcal{L}$ be a sheaf of $\mathcal{O}_X$-modules. The *degree-m section component* is the $\Gamma(X, \mathcal{O}_X)$-module of global sections of the mth tensor power,

$$(\text{sectionDeg } \mathcal{L})_m \;:=\; \Gamma(X, \mathcal{L}^{\otimes m}) \;=\; \mathcal{L}^{\otimes m}(X) \qquad (1280).$$

It inherits its additive-group and $\Gamma(X, \mathcal{O}_X)$-module structure from the underlying module object $\mathcal{L}^{\otimes m}(X)$. These are the carriers $m \mapsto (\text{sectionDeg } \mathcal{L})_m$ of the section graded ring $R(X_s, L_s)$ (1341) and the domains of the index transport 1328.

Definition 1328 (Index-equality transport of section components)**.** Let $\mathcal{L}$ be an invertible sheaf on X and let $h : i = j$ be an equality of natural numbers. Applying the global-sections functor $\Gamma(X, -)$ to the canonical isomorphism $\mathcal{L}^{\otimes i} \xrightarrow{\sim} \mathcal{L}^{\otimes j}$ induced by h under the tensor-power functor (1280) yields the $\Gamma(X, \mathcal{O}_X)$-linear isomorphism

$$\text{sectionsCast}(h) : \Gamma(X, \mathcal{L}^{\otimes i}) \xrightarrow{\;\sim\;} \Gamma(X, \mathcal{L}^{\otimes j}),$$

the *section-component transport* along h. It is the section-level analogue of the tensor-power index cast TensorPower.cast of `Mathlib.LinearAlgebra.TensorPower.Basic`, which transports $\bigotimes^i M$ to $\bigotimes^j M$ along $i = j$. Because the isomorphism induced by the reflexive equality is the identity and $\Gamma(X, -)$ preserves identities, the transport along a reflexive index equality is the identity map (the reflexivity simplification sectionsCast_refl, 1329).

Lemma 1329 (Reflexivity of the section transport)**.** *For any i, the transport along the reflexive equality $i = i$ (1328) equals the identity automorphism of $\Gamma(X, \mathcal{L}^{\otimes i})$. This is the section-level counterpart of* TensorPower.cast_refl*, and serves as a simplification lemma collapsing the transport along a reflexive index equality.*

Proof. □

Lemma 1330 (Cast-mediated equality in the graded monoid)**.** *Write the carrier of the graded monoid as the dependent sum $\coprod_n \Gamma(X, \mathcal{L}^{\otimes n})$, whose elements are pairs (n, s) with $s \in \Gamma(X, \mathcal{L}^{\otimes n})$. Let $a = (i, x)$ and $b = (j, y)$ be two such pairs. If $h : i = j$ and the transported component agrees,* $\text{sectionsCast}(h)\, x = y$*, then $a = b$. That is, a component-level equality mediated by the index transport 1328 repackages into a genuine equality of dependent pairs. This is the section-level analogue of the bridge* gradedMonoid_eq_of_cast *of `Mathlib.LinearAlgebra.TensorPower.Basic` (line 123 there).*

Proof. □

Definition 1331 (Graded multiplication on section components)**.** The section components 1327 carry a graded multiplication

$$(\text{sectionDeg } \mathcal{L})_i \times (\text{sectionDeg } \mathcal{L})_j \;\longrightarrow\; (\text{sectionDeg } \mathcal{L})_{i+j}, \qquad (a, b) \longmapsto a \cdot b,$$

landing in the index-sum component (a GradedMonoid.GMul on $m \mapsto (\text{sectionDeg } \mathcal{L})_m$). It is the section multiplication $a \otimes b \mapsto a \cdot b$ (1314) on $\mathcal{L}^{\otimes i}$ and $\mathcal{L}^{\otimes j}$, followed by global sections of the tensor-power comparison isomorphism $\mu_{i,j}$ (1304),

$$\Gamma(X, \mathcal{L}^{\otimes i}) \otimes_{\Gamma(X, \mathcal{O}_X)} \Gamma(X, \mathcal{L}^{\otimes j}) \xrightarrow{\;\cdot\;} \Gamma(X, \mathcal{L}^{\otimes i} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes j}) \xrightarrow{\Gamma(\mu_{i,j})} \Gamma(X, \mathcal{L}^{\otimes (i+j)}).$$

Definition 1332 (Graded unit in degree zero)**.** The graded unit (a GradedMonoid.GOne on $m \mapsto (\text{sectionDeg } \mathcal{L})_m$) is the element

$$1 \in (\text{sectionDeg } \mathcal{L})_0 = \Gamma(X, \mathcal{L}^{\otimes 0}) = \Gamma(X, \mathcal{O}_X),$$

the image of $1 \in \Gamma(X, \mathcal{O}_X)$ under the canonical $\Gamma(X, \mathcal{O}_X)$-module identification $\Gamma(X, \mathcal{O}_X) = \Gamma(X, \mathcal{L}^{\otimes 0})$ coming from the unit module $\mathcal{L}^{\otimes 0} = \mathbf{1}_X$ (1277).

Lemma 1333 (Coherence of the section multiplication)**.** *Write $a \cdot b$ for the degreewise section multiplication on tensor powers: for $a \in \Gamma(X, \mathcal{L}^{\otimes i})$ and $b \in \Gamma(X, \mathcal{L}^{\otimes j})$ it is the elementary section multiplication (1314) followed by the tensor-power comparison isomorphism $\mu_{i,j}$ (1304), landing in $\Gamma(X, \mathcal{L}^{\otimes(i+j)})$; and let 1 denote the unit of degree 0, the image of $1 \in \Gamma(X, \mathcal{O}_X)$ under the identification $\Gamma(X, \mathcal{O}_X) = \Gamma(X, \mathcal{L}^{\otimes 0})$ (1277). Mirroring* TensorPower.{one_mul, mul_one, mul_assoc} *together with the commutativity lemma of `Mathlib.LinearAlgebra.TensorPower.Basic`, the multiplication satisfies the following four component-level identities. Each is an honest equality in a single section module, obtained by transporting one side along the relevant index equality via 1328; the subscript on* sectionsCast *records that index equality. The first three — left and right unitality and associativity — hold for an* arbitrary *sheaf of modules $\mathcal{L}$, and make the section components a non-commutative graded semiring. The fourth — commutativity — requires $\mathcal{L}$ to be* invertible *(1281); it is the upgrade to a graded* commutative *semiring and is false for general $\mathcal{L}$, the section ring then being the free tensor algebra on $\Gamma(X, \mathcal{L})$.*

- Left unitality. *For $a \in \Gamma(X, \mathcal{L}^{\otimes n})$,*

$$\text{sectionsCast}_{0+n=n}(1 \cdot a) = a.$$

- Right unitality. *For $a \in \Gamma(X, \mathcal{L}^{\otimes n})$,*

$$\text{sectionsCast}_{n+0=n}(a \cdot 1) = a.$$

- Associativity. *For $a \in \Gamma(X, \mathcal{L}^{\otimes n_a})$, $b \in \Gamma(X, \mathcal{L}^{\otimes n_b})$, $c \in \Gamma(X, \mathcal{L}^{\otimes n_c})$,*

$$\text{sectionsCast}_{(n_a+n_b)+n_c=n_a+(n_b+n_c)}((a \cdot b) \cdot c) = a \cdot (b \cdot c).$$

- Commutativity ($\mathcal{L}$ invertible, 1281). *For $a \in \Gamma(X, \mathcal{L}^{\otimes n_a})$ and $b \in \Gamma(X, \mathcal{L}^{\otimes n_b})$,*

$$\text{sectionsCast}_{n_a+n_b=n_b+n_a}(a \cdot b) = b \cdot a.$$

The transport is indispensable: the indices on the two sides of each identity are equal but not definitionally so, hence a priori live in distinct section modules, and 1328 moves one side into the other's module to produce a genuine equality there.

Proof. Each identity is obtained by unfolding the degreewise multiplication $a \cdot b = \Gamma(\mu_{i,j})(\text{sectionsMul}(a \otimes b))$ (1331, 1314, 1304) and combining two ingredients: an *iso-level constraint* on the comparison family $\{\mu_{m,m'}\}$, and the *section-level naturality* that pushes the corresponding canonical coherence map through sectionsMul. The index transport 1328 — the image under $\Gamma(X, -)$ of the reindexing isomorphism of tensor powers — matches the two sides, which a priori inhabit the distinct section modules of two equal-but-not-definitionally-equal indices, into a common module.

Left unitality. Unfolding $1 \cdot a$ and using the base case $\mu_{0,n} = \lambda$ of 1304, the inner reindexing cast pairs with the outer $\text{sectionsCast}_{0+n=n}$ and the two cancel; 1315 then closes the leg.

Right unitality. Unfolding $a \cdot 1$ and rewriting the comparison by the right-unit constraint $\mu_{n,0} = \text{tensorObjRightUnitor}$ (1305), 1316 closes the leg after the residual reflexive transport collapses.

Associativity. Unfold both bracketings to composites $\Gamma(\mu) \circ \text{sectionsMul}$. Unlike the unit legs — where the inner factor is the literal degree-0 unit and carries no comparison — here each bracketing hides an *inner* comparison nested inside one tensor factor of the outer sectionsMul, and these must first be slid out. On the left,

$$(a \cdot b) \cdot c = \Gamma(\mu_{n_a+n_b,\, n_c})\Big(\text{sectionsMul}_{\mathcal{L}^{n_a+n_b}, \mathcal{L}^{n_c}}(\Gamma(\mu_{n_a,n_b})(\text{sectionsMul}(a \otimes b)) \otimes c)\Big),$$

and the right-whisker naturality 1320 (applied to $f = \mu_{n_a,n_b}$, $G = \mathcal{L}^{n_c}$) slides $\Gamma(\mu_{n_a,n_b})$ out of the first factor, giving $\Gamma(\mu_{n_a+n_b,\, n_c} \circ (\mu_{n_a,n_b} \triangleright \mathcal{L}^{n_c}))$ applied to the bare iterated product $\text{sectionsMul}(\text{sectionsMul}(a \otimes b) \otimes c)$. Symmetrically, on the right, the left-whisker naturality 1321 (applied to $g = \mu_{n_b,n_c}$, $F = \mathcal{L}^{n_a}$) slides $\Gamma(\mu_{n_b,n_c})$ out of the second factor, giving $\Gamma(\mu_{n_a,\, n_b+n_c} \circ (\mathcal{L}^{n_a} \triangleleft \mu_{n_b,n_c}))$ applied to $\text{sectionsMul}(a \otimes \text{sectionsMul}(b \otimes c))$. Both sides are now $\Gamma(\text{comparison composite}) \circ (\text{iterated sectionsMul})$. The iso-level associativity (pentagon) constraint 1313 equates the two comparison composites along the associator α modulo the reindexing $(n_a + n_b) + n_c = n_a + (n_b + n_c)$:

$$\mu_{n_a+n_b,\, n_c} \circ (\mu_{n_a,n_b} \triangleright \mathcal{L}^{n_c}) \;=\; (\text{reindex}) \circ \mu_{n_a,\, n_b+n_c} \circ (\mathcal{L}^{n_a} \triangleleft \mu_{n_b,n_c}) \circ \alpha,$$

and the section-level associator naturality 1326 pushes the residual $\Gamma(\alpha)$ through the iterated sectionsMul, reassociating the three section factors; the transport $\text{sectionsCast}_{(n_a+n_b)+n_c = n_a+(n_b+n_c)}$ absorbs the reindexing and identifies the two sides in a single module. This is the right-unit leg with the associator triple (1313 + 1326 + the two slides 1320, 1321) in place of the right-unitor pair.

Commutativity. Symmetrically, the iso-level commutativity constraint 1312 factors μ_{n_a,n_b} through the braiding into μ_{n_b,n_a}, and the section-level naturality 1317 pushes $\Gamma(\beta)$ through sectionsMul, swapping the two section factors; the transport $\text{sectionsCast}_{n_a+n_b = n_b+n_a}$ identifies the two sides. This is again the right-unit leg with the braiding pair (1312 + 1317) in place of the right-unitor pair. □

21.3 Graded-ring and graded-module assembly

The components and multiplications of the previous two sections are assembled into the graded ring and graded module through Mathlib's *external-direct-sum* graded API: the relevant inputs are families of abelian groups indexed by $\mathbb{N}$ together with multiplication maps landing in the index-sum component, not submodules of a pre-existing ring.

Lemma 1334 (External-direct-sum graded semiring)**.** *Provided by Mathlib (`Mathlib.Algebra.DirectSum.Ring`). Let $A : \mathbb{N} \to \text{Type}$ be a family of additive commutative monoids equipped with a graded multiplication $A_i \times A_j \to A_{i+j}$ and a unit $1 \in A_0$ satisfying graded associativity, two-sided unitality and distributivity — but* not *necessarily graded commutativity (the data of a* GSemiring *on A). Then the external direct sum $\bigoplus_i A_i$ is a (not necessarily commutative) semiring, with multiplication the bilinear extension of the graded multiplication and identity the image of $1 \in A_0$. This is the general assembly; the commutative upgrade is 1335, which adds graded commutativity.*

Lemma 1335 (External-direct-sum graded commutative semiring)**.** *Provided by Mathlib (`Mathlib.Algebra.DirectS`
Let $A : \mathbb{N} \to \text{Type}$ be a family of additive commutative monoids equipped with a graded multiplication $A_i \times A_j \to A_{i+j}$ and a unit $1 \in A_0$ satisfying graded associativity, two-sided unitality,

distributivity, and graded commutativity (the data of a GCommSemiring *on* A*). Then the external direct sum* $\bigoplus_i A_i$ *is a commutative semiring, with multiplication the bilinear extension of the graded multiplication and identity the image of* $1 \in A_0$*.*

Lemma 1336 (External-direct-sum graded module)**.** *Provided by Mathlib (*`Mathlib.Algebra.Module.GradedModule`*). Let* $A : \mathbb{N} \to \mathrm{Type}$ *carry a graded semiring structure as in [1334](#) and let* $M : \mathbb{N} \to \mathrm{Type}$ *be a family of additive commutative monoids with a graded scalar action* $A_i \times M_j \to M_{i+j}$ *satisfying the graded module axioms (the data of a* Gmodule *of* M *over* A*). Then the external direct sum* $\bigoplus_j M_j$ *is a module over the semiring* $\bigoplus_i A_i$*, with scalar multiplication the bilinear extension of the graded action.*

Lemma 1337 (Graded monoid structure on the section components)**.** *Let* $\mathcal{L}$ *be an arbitrary sheaf of* $\mathcal{O}_X$*-modules. The degree family* sectionDeg $\mathcal{L}$, $m \mapsto \Gamma(X, \mathcal{L}^{\otimes m})$*, equipped with the graded multiplication* gMul *(1331) and graded unit* gOne *(1332), is a graded monoid (a* GradedMonoid.GMonoid *on* sectionDeg $\mathcal{L}$*).*

Proof. The three graded-monoid axioms — left unitality, right unitality and associativity of gMul about gOne — are the first three *hypothesis-free* clauses of 1333, stated as transport-mediated equalities of sections. Each is converted into the required equality of dependent GradedMonoid pairs by the bridge 1330, which turns a transport-mediated equality into an equality of pairs. This assembles the GMonoid on sectionDeg $\mathcal{L}$. □

Lemma 1338 (Graded semiring structure on the section components (general))**.** *Let* $\mathcal{L}$ *be an* arbitrary *sheaf of* $\mathcal{O}_X$*-modules on* X*. The family of* $\Gamma(X, \mathcal{O}_X)$*-modules* $m \mapsto \Gamma(X, \mathcal{L}^{\otimes m})$ *carries a graded* semiring *structure (a* GSemiring*, 1334) whose degree-*(m, m') *multiplication is the composite*

$$\Gamma(X, \mathcal{L}^{\otimes m}) \otimes_{\Gamma(X,\mathcal{O}_X)} \Gamma(X, \mathcal{L}^{\otimes m'}) \xrightarrow{\cdot} \Gamma(X, \mathcal{L}^{\otimes m} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes m'}) \xrightarrow{\Gamma(\mu_{m,m'})} \Gamma(X, \mathcal{L}^{\otimes (m+m')}),$$

the section multiplication (1314) followed by the tensor-power comparison isomorphism (1304), with unit $1 \in \Gamma(X, \mathcal{O}_X) = \Gamma(X, \mathcal{L}^{\otimes 0})$ *(1277). Consequently* $\bigoplus_{m \geq 0} \Gamma(X, \mathcal{L}^{\otimes m})$ *is a semiring. This structure exists for* every $\mathcal{L}$ *and is in general* non-commutative *— it is the free tensor algebra on* $\Gamma(X, \mathcal{L})$ *— because it uses only the unit and associativity clauses of 1333, which hold without hypotheses; graded commutativity (the invertible upgrade 1341) is* not *part of this layer.*

Proof. The structure is assembled field-for-field as TensorPower builds its graded semiring in `Mathlib.LinearAlgebra.TensorPower.Basic`. The degreewise multiplication $\Gamma(\mu_{m,m'}) \circ (\cdot)$ of the statement and the degree-0 unit $1 \in \Gamma(X, \mathcal{L}^{\otimes 0})$ provide the graded multiplication and unit. Their graded-monoid axioms — left and right unitality and associativity — are the first three (hypothesis-free) component-level identities of 1333, each passed through the bridge 1330, which converts a transport-mediated equality (1328) into the required equality of dependent pairs; the graded power operation is the default iterated product and contributes no further data. This yields the graded-monoid layer.

The remaining semiring axioms are the bilinearity (left and right distributivity and annihilation by 0) of the graded multiplication and the behaviour of the natural-number coercion. Bilinearity is immediate: the degreewise multiplication $\Gamma(\mu_{m,m'}) \circ (\cdot)$ is $\Gamma(X, \mathcal{O}_X)$-bilinear, being the composite of the bilinear section multiplication (1314) with the linear comparison $\Gamma(\mu_{m,m'})$ (1304), so each side vanishes on 0 and distributes over sums. The natural-number coercion sends n to $n \cdot 1$, the n-fold sum of the degree-0 unit. These are precisely the data of a GSemiring on $m \mapsto \Gamma(X, \mathcal{L}^{\otimes m})$, so 1334 endows $\bigoplus_{m \geq 0} \Gamma(X, \mathcal{L}^{\otimes m})$ with its semiring structure. No commutativity clause enters. □

Lemma 1339 (A semiring structure on the section direct sum exists)**.** *Let $\mathcal{L}$ be an arbitrary sheaf of $\mathcal{O}_X$-modules. Then the external direct sum $\bigoplus_{m\geq 0}\Gamma(X,\mathcal{L}^{\otimes m})$ admits a semiring structure; equivalently the type of semiring structures on it is nonempty.*

Proof. By 1338 the family $m\mapsto\Gamma(X,\mathcal{L}^{\otimes m})$ carries a GSemiring. Mathlib's external-direct-sum assembly (1334, via DirectSum.toSemiring) turns this graded data into a genuine semiring on $\bigoplus_{m\geq 0}\Gamma(X,\mathcal{L}^{\otimes m})$; exhibiting this one structure witnesses the nonemptiness. □

Lemma 1340 (A commutative semiring structure on the section direct sum exists)**.** *Let L be an* invertible *sheaf of $\mathcal{O}_X$-modules. Then the external direct sum $\bigoplus_{m\geq 0}\Gamma(X,L^{\otimes m})$ admits a commutative semiring structure.*

Proof. By 1341 the family $m\mapsto\Gamma(X,L^{\otimes m})$ carries a GCommSemiring when L is invertible. The external-direct-sum assembly 1335 turns this graded data into a genuine commutative semiring on $\bigoplus_{m\geq 0}\Gamma(X,L^{\otimes m})$. □

Lemma 1341 (Graded commutative semiring for an invertible sheaf)**.** *Source: [Stacks Project], "Sheaves of Modules", Tag 01CV (the graded ring structure on $\bigoplus\Gamma(X,\mathcal{L}^{\otimes n})$). Let L_s be an* invertible *sheaf — a line bundle, 1281 — on X_s. Then the graded semiring of 1338 on $m\mapsto\Gamma(X_s,L_s^{\otimes m})$ is graded* commutative *(a* GCommSemiring*, 1335), with the same degree-(m,m') multiplication*

$$\Gamma(X_s,L_s^{\otimes m})\otimes_{\Gamma(X_s,\mathcal{O}_{X_s})}\Gamma(X_s,L_s^{\otimes m'})\xrightarrow{\cdot}\Gamma(X_s,L_s^{\otimes m}\otimes L_s^{\otimes m'})\xrightarrow{\Gamma(\mu_{m,m'})}\Gamma(X_s,L_s^{\otimes(m+m')})$$

and unit $1\in\Gamma(X_s,\mathcal{O}_{X_s})=\Gamma(X_s,L_s^{\otimes 0})$ (1277). Consequently $R(X_s,L_s)=\bigoplus_{m\geq 0}\Gamma(X_s,L_s^{\otimes m})$ is a commutative semiring; this is the graded-ring structure underlying 1021. Invertibility of L_s enters only here, through the commutativity clause; for a general sheaf only the non-commutative semiring 1338 is available.

Proof. By 1338 the family $m\mapsto\Gamma(X_s,L_s^{\otimes m})$ is already a graded semiring, so it remains only to supply graded commutativity. Since L_s is invertible, the commutativity clause of 1333 holds; passing it through the bridge 1330 — which converts the transport-mediated equality (1328) into an equality of dependent pairs — yields the graded-commutativity axiom. Adjoining it to the graded-semiring data produces a GCommSemiring on $m\mapsto\Gamma(X_s,L_s^{\otimes m})$, so 1335 endows $\bigoplus_{m\geq 0}\Gamma(X_s,L_s^{\otimes m})$ with its commutative-semiring structure. □

Definition 1342 (Action comparison isomorphism for the twisted module)**.** Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$-modules and $\mathcal{L}$ a line bundle. For $i,j\in\mathbb{N}$ the *action comparison isomorphism* is the isomorphism of sheaves of $\mathcal{O}_X$-modules

$$\mathcal{L}^{\otimes i}\otimes_{\mathcal{O}_X}(\mathcal{F}\otimes_{\mathcal{O}_X}\mathcal{L}^{\otimes j})\xrightarrow{\cong}\mathcal{F}\otimes_{\mathcal{O}_X}\mathcal{L}^{\otimes(i+j)}$$

relating the left graded action of the section ring on the twisted module $\mathcal{F}(j)$ (1282) with the merge of the line-bundle factors. It is assembled from the sheaf-level associator (1297), the braiding $\beta_{\mathcal{L}^{\otimes i},\mathcal{F}}$ (1285) exchanging the two *distinct* factors $\mathcal{L}^{\otimes i}$ and $\mathcal{F}$ — a braiding of distinct objects, requiring *no* invertibility hypothesis — the whiskering isomorphisms of $\otimes$ (1279), and the tensor-power comparison $\mu_{i,j}$ merging $\mathcal{L}^{\otimes i}\otimes_{\mathcal{O}_X}\mathcal{L}^{\otimes j}\cong\mathcal{L}^{\otimes(i+j)}$ (1304, 1280). It underlies the graded-module structure of 1351.

Lemma 1343 (Associativity (hexagon) constraint for the action comparison)**.** *Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$-modules, $\mathcal{L}$ a line bundle, and $i, j, k \in \mathbb{N}$. Write $a_{p,q}$ for the action comparison isomorphism (1342) $\mathcal{L}^{\otimes p} \otimes (\mathcal{F} \otimes \mathcal{L}^{\otimes q}) \xrightarrow{\cong} \mathcal{F} \otimes \mathcal{L}^{\otimes(p+q)}$. The two ways of contracting $\mathcal{L}^{\otimes i} \otimes (\mathcal{L}^{\otimes j} \otimes (\mathcal{F} \otimes \mathcal{L}^{\otimes k}))$ down to $\mathcal{F} \otimes \mathcal{L}^{\otimes(i+j+k)}$ agree:*

$$a_{i+j,\,k} \circ (\mu_{i,j} \triangleright (\mathcal{F} \otimes \mathcal{L}^{\otimes k})) \circ \alpha^{-1}_{\mathcal{L}^{\otimes i},\,\mathcal{L}^{\otimes j},\,\mathcal{F}\otimes\mathcal{L}^{\otimes k}} \;=\; (\mathrm{reindex}) \circ a_{i,\,j+k} \circ (\mathcal{L}^{\otimes i} \triangleleft a_{j,k}),$$

where μ is the tensor-power comparison (1304), α the associator (1297), $\triangleright, \triangleleft$ the canonical whiskerings, and the reindexing is $(i + j) + k = i + (j + k)$. One side merges the two line-bundle factors first and then acts once; the other acts twice in succession. This is the module analogue of the associativity constraint 1313, with the action comparison a in place of the ring comparison μ on the two legs that touch $\mathcal{F}$.

Proof. Unlike 1313, this identity is *not* braiding-free: the action comparison $a_{p,q}$ (1342) contains the braiding $\beta_{\mathcal{L}^{\otimes p},\mathcal{F}}$ (1285) sliding the line-bundle block $\mathcal{L}^{\otimes p}$ past the module factor $\mathcal{F}$. Since $\mathcal{F}$ is *not* assumed invertible the braiding does not collapse to the identity, so the statement is a genuine symmetric-monoidal *hexagon* (Mac Lane), not a pentagon: on each side the block $\mathcal{L}^{\otimes i}$ (and then $\mathcal{L}^{\otimes j}$) must be braided across $\mathcal{F}$, and the identity records the two orders in which the merge $\mu_{i,j}$ and the two braidings may be performed.

Expand $a_{p,q}$ on both sides into its constituents — the associator (1297), the braiding $\beta_{\mathcal{L}^{\otimes p},\mathcal{F}}$ (1285), the whiskerings, and the comparison $\mu_{p,q}$ (1304). The line-bundle-merge halves of the two sides are equated by the braiding-free pentagon 1313 applied to the three factors $\mathcal{L}^{\otimes i}, \mathcal{L}^{\otimes j}, \mathcal{L}^{\otimes k}$; the residual structural glue is a hexagon among canonical associators, whiskerings and the braiding β carrying $\mathcal{L}^{\otimes i}$ (then $\mathcal{L}^{\otimes j}$) across $\mathcal{F}$, discharged by coherence for symmetric monoidal categories. ☐

Definition 1344 (Twisted-section component in degree m)**.** Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$-modules and $\mathcal{L}$ a sheaf of $\mathcal{O}_X$-modules. The *degree-m twisted-section component* is the $\Gamma(X, \mathcal{O}_X)$-module of global sections of the mth twist of $\mathcal{F}$,

$$(\mathrm{moduleSectionDeg}\ \mathcal{F}\ \mathcal{L})_m \;:=\; \Gamma(X, \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes m}) \;=\; (\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes m})(X) \qquad (1282,\ 1280).$$

It inherits its additive-group and $\Gamma(X, \mathcal{O}_X)$-module structure from the underlying module object $(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes m})(X)$. These are the carriers $m \mapsto (\mathrm{moduleSectionDeg}\ \mathcal{F}\ \mathcal{L})_m$ of the graded section module $M(X_s, \mathcal{F}_s, L_s)$ (1351) and the domains of the index transport 1345. It is the module analogue of the degree-m section component 1327, with the twisted module $\mathcal{F} \otimes \mathcal{L}^{\otimes m}$ in place of the pure power $\mathcal{L}^{\otimes m}$.

Definition 1345 (Index-equality transport of twisted-section components)**.** Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$-modules, $\mathcal{L}$ a sheaf of $\mathcal{O}_X$-modules, and let $h : i = j$ be an equality of natural numbers. Applying the global-sections functor $\Gamma(X, -)$ to the canonical isomorphism $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes i} \xrightarrow{\sim} \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes j}$ induced by h under the twist functor $m \mapsto \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes m}$ (1344) yields the $\Gamma(X, \mathcal{O}_X)$-linear isomorphism

$$\mathrm{moduleSectionsCast}(h) : \Gamma(X, \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes i}) \;\xrightarrow{\sim}\; \Gamma(X, \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes j}),$$

the *twisted-section-component transport* along h. It is the twisted-module analogue of the pure section-component transport 1328. Because the isomorphism induced by the reflexive equality is the identity and $\Gamma(X, -)$ preserves identities, the transport along a reflexive index equality is the identity map (the reflexivity simplification moduleSectionsCast_refl, 1346).

Lemma 1346 (Reflexivity of the twisted-section transport). *For any i, the transport along the reflexive equality $i = i$ (1345) equals the identity automorphism of $\Gamma(X, \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes i})$. This is the twisted-module counterpart of 1329, and serves as a simplification lemma collapsing the transport along a reflexive index equality.*

Proof. □

Lemma 1347 (Cast-mediated equality in the graded module). *Write the carrier of the graded module as the dependent sum $\coprod_n \Gamma(X, \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes n})$, whose elements are pairs (n, s) with $s \in \Gamma(X, \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes n})$. Let $a = (i, x)$ and $b = (j, y)$ be two such pairs. If $h : i = j$ and the transported component agrees,* $\mathrm{moduleSectionsCast}(h)\, x = y$*, then $a = b$. That is, a component-level equality mediated by the index transport 1345 repackages into a genuine equality of dependent pairs. This is the twisted-module analogue of the ring-side bridge 1330; it is the bridge that converts each transport-mediated clause of 1350 into the dependent-pair form required by the* Gmodule *axioms.*

Proof. □

Definition 1348 (Graded scalar action on twisted-section components). The section components 1327 act on the twisted-section components 1344 by a graded scalar multiplication

$$(\mathrm{sectionDeg}\ \mathcal{L})_i \times (\mathrm{moduleSectionDeg}\ \mathcal{F}\ \mathcal{L})_j \longrightarrow (\mathrm{moduleSectionDeg}\ \mathcal{F}\ \mathcal{L})_{i+j}, \qquad (r, x) \longmapsto r \star x,$$

landing in the index-sum component (a GradedMonoid.GSMul of moduleSectionDeg $\mathcal{F}$ $\mathcal{L}$ over sectionDeg $\mathcal{L}$; the action degree is $i + j$). It is the section multiplication $r \otimes x \mapsto r \cdot x$ (1314) of the line-bundle power $\mathcal{L}^{\otimes i}$ against the twisted module $\mathcal{F} \otimes \mathcal{L}^{\otimes j}$, followed by global sections of the action comparison isomorphism $a_{i,j}$ (1342),

$$\Gamma(X, \mathcal{L}^{\otimes i}) \otimes_{\Gamma(X, \mathcal{O}_X)} \Gamma(X, \mathcal{F} \otimes \mathcal{L}^{\otimes j}) \overset{\cdot}{\longrightarrow} \Gamma(X, \mathcal{L}^{\otimes i} \otimes (\mathcal{F} \otimes \mathcal{L}^{\otimes j})) \xrightarrow{\Gamma(a_{i,j})} \Gamma(X, \mathcal{F} \otimes \mathcal{L}^{\otimes (i+j)}).$$

It is the module analogue of the graded multiplication 1331, with the action comparison $a_{i,j}$ in place of the ring comparison $\mu_{i,j}$ and the twisted module in the right-hand factor.

Lemma 1349 (Left-unit (base-case) constraint for the action comparison). *Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$-modules, $\mathcal{L}$ a line bundle, and $k \in \mathbb{N}$. The degree-$(0, k)$ action comparison isomorphism $a_{0,k}$ (1342) is the left unitor, reindexed along $0 + k = k$:*

$$a_{0,k} \;=\; \mathrm{tensorObjUnitIso}_{\mathcal{F} \otimes \mathcal{L}^{\otimes k}} \;:\; \mathbf{1}_X \otimes_{\mathcal{O}_X} (\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes k}) \xrightarrow{\sim} \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes k}$$

(1284, transport of the left unitor λ). This is the base case of the action comparison, the module analogue of the left-unit constraint 1306, and is the key input to the Unitality clause of 1350.

Proof. Expand $a_{0,k}$ into its constituents (1342): the sheaf-level associator (1297), the braiding $\beta_{\mathcal{L}^{\otimes 0}, \mathcal{F}}$ (1285), the whiskerings, and the line-bundle merge $\mu_{0,k}$. In degree $p = 0$ the line-bundle block collapses, $\mathcal{L}^{\otimes 0} = \mathbf{1}_X$, so the merge reduces to the left-unit constraint $\mu_{0,k} = \lambda$ of 1306, and the braiding $\beta_{\mathcal{L}^{\otimes 0}, \mathcal{F}} = \beta_{\mathbf{1}_X, \mathcal{F}}$ becomes the unit-braiding, the composite $\lambda_{\mathcal{F}} \circ \rho_{\mathcal{F}}^{-1}$ of the left unitor with the inverse right unitor on the unit object.

The verification requires the unit-braiding coherence: the monoidal unit object of X.Modules is the structure sheaf $\mathbf{1}_X = \mathcal{O}_X$, and the unit-braiding identity $\beta_{\mathbf{1}, \mathcal{F}} = \lambda_{\mathcal{F}} \circ \rho_{\mathcal{F}}^{-1}$ is established over the formal monoidal unit $\mathbf{1}$ by symmetric-monoidal coherence (Mac Lane), using the equality of unitors on the unit object (MonoidalCategory.unitors_equal, descended through sheafification as in 1306), and then applied at $\mathbf{1}_X$ via the definitional equality of the two unit presentations. The bridge 1300 then identifies the left-unitor with tensorObjUnitIso, and the reindexing cast along $0 + k = k$ absorbs into it, yielding $a_{0,k} = \mathrm{tensorObjUnitIso}_{\mathcal{F} \otimes \mathcal{L}^{\otimes k}}$. □

Lemma 1350 (Coherence of the module action)**.** *Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$-modules and $\mathcal{L}$ a line bundle. For $r \in \Gamma(X, \mathcal{L}^{\otimes i})$ and $x \in \Gamma(X, \mathcal{F} \otimes \mathcal{L}^{\otimes k})$ write*

$$r \star x \;=\; \Gamma(a_{i,k})(\mathrm{sectionsMul}(r \otimes x)) \;\in\; \Gamma(X, \mathcal{F} \otimes \mathcal{L}^{\otimes(i+k)})$$

for the degreewise action: the section multiplication (1314) followed by global sections of the action comparison $a_{i,k}$ (1342). Then, as transport-mediated equalities of sections (the subscript on sectionsCast*, 1328, recording the index equality that matches the two a-priori-distinct section modules):*

- Unitality. *For $x \in \Gamma(X, \mathcal{F} \otimes \mathcal{L}^{\otimes k})$ and 1 the degree-0 unit of the section ring,*

$$\mathrm{sectionsCast}_{0+k=k}\,(1 \star x) = x.$$

- Compatibility. *For $r \in \Gamma(X, \mathcal{L}^{\otimes i})$, $r' \in \Gamma(X, \mathcal{L}^{\otimes j})$ and $x \in \Gamma(X, \mathcal{F} \otimes \mathcal{L}^{\otimes k})$,*

$$\mathrm{sectionsCast}_{(i+j)+k=i+(j+k)}\,((r \cdot r') \star x) = r \star (r' \star x),$$

  where $r \cdot r'$ is the section ring multiplication (1314 followed by $\Gamma(\mu_{i,j})$).

This is the module analogue of the unit and associativity clauses of 1333, with the action comparison a replacing the ring comparison μ on the legs touching $\mathcal{F}$.

Proof. Both identities are the section-level shadow of the iso-level constraints, pushed through sectionsMul by the naturality helpers — exactly as in 1333, but with the action comparison a in place of the ring comparison μ on the legs that touch $\mathcal{F}$.

Unitality. Unfold $1 \star x$. The base case $a_{0,k}$ of the action comparison (1342) is the left unitor $\lambda_{\mathcal{F} \otimes \mathcal{L}^{\otimes k}}$ reindexed along $0 + k = k$, by 1349 (where the unit-braiding coherence is established; see that proof). Granting that identification, the inner reindexing cast pairs with the outer $\mathrm{sectionsCast}_{0+k=k}$ and the two cancel; 1315 then closes the leg.

Compatibility. Unfold both sides to composites $\Gamma(a) \circ \mathrm{sectionsMul}$. On the left, $(r \cdot r') \star x$ hides the ring comparison $\mu_{i,j}$ inside the first factor of the outer sectionsMul; the right-whisker naturality 1320 (applied to $f = \mu_{i,j}$, $G = \mathcal{F} \otimes \mathcal{L}^{\otimes k}$) slides $\Gamma(\mu_{i,j})$ out. On the right, $r \star (r' \star x)$ hides the action comparison $a_{j,k}$ inside the second factor; the left-whisker naturality 1321 (applied to $g = a_{j,k}$, $F = \mathcal{L}^{\otimes i}$) slides $\Gamma(a_{j,k})$ out. Both sides are now $\Gamma(\text{comparison composite})$ applied to a bare iterated product, $\mathrm{sectionsMul}(\mathrm{sectionsMul}(r \otimes r') \otimes x)$ on the left and $\mathrm{sectionsMul}(r \otimes \mathrm{sectionsMul}(r' \otimes x))$ on the right. The iso-level hexagon 1343 equates the two comparison composites modulo the associator α and the braiding β. The residual $\Gamma(\alpha)$ is pushed through the iterated sectionsMul by 1326, reassociating the three section factors, and the residual $\Gamma(\beta_{\mathcal{L}^{\otimes i}, \mathcal{F}})$ is pushed through by 1317 — stated precisely for the two *distinct* factors $\mathcal{L}^{\otimes i}$ and $\mathcal{F}$ — swapping those two section factors. The transport $\mathrm{sectionsCast}_{(i+j)+k=i+(j+k)}$ absorbs the reindexing and identifies the two sides in a single module. This is the associativity leg of 1333 with the action hexagon and braiding slide in place of the pentagon and the right-unitor pair. ☐

Lemma 1351 (Graded module structure on the twisted-section components)**.** *Source: [Stacks Project], "Sheaves of Modules", Tag 01CV (the graded module $\Gamma_*(X, \mathcal{L}, \mathcal{F})$). This lemma establishes the $\mathbb{N}$-graded ($n \geq 0$) analogue of Tag 01CV: the Stacks source assembles the $\mathbb{Z}$-graded module over an* invertible *$\mathcal{L}$ (using $\mathcal{L}^\vee$ for negative degrees), whereas the action comparison $a_{i,j}$ (1342) exists for an arbitrary sheaf of modules $\mathcal{F}$ and an arbitrary $\mathcal{L}$, yielding the $n \geq 0$ truncation $\bigoplus_{m \geq 0} \Gamma(X, \mathcal{F} \otimes \mathcal{L}^{\otimes m})$ without any invertibility hypothesis. Let $\mathcal{F}_s$ be a coherent sheaf* *and L_s a line bundle on X_s. The family $m \mapsto \Gamma(X_s, \mathcal{F}_s \otimes L_s^{\otimes m})$ (1282) carries a graded module structure (a* Gmodule*, 1336) over the graded semiring of 1338, with degree-(m, m') action*

$$\Gamma(X_s, L_s^{\otimes m}) \otimes_{\Gamma(X_s, \mathcal{O}_{X_s})} \Gamma(X_s, \mathcal{F}_s \otimes L_s^{\otimes m'}) \xrightarrow{\cdot} \Gamma(X_s, L_s^{\otimes m} \otimes (\mathcal{F}_s \otimes L_s^{\otimes m'})) \xrightarrow{\sim} \Gamma(X_s, \mathcal{F}_s \otimes L_s^{\otimes (m+m')}),$$

the section multiplication followed by the comparison isomorphism that reassociates and merges the line-bundle factors (using 1304 and the symmetry of the tensor product). Consequently $M(X_s, \mathcal{F}_s, L_s) = \bigoplus_{m \geq 0} \Gamma(X_s, \mathcal{F}_s \otimes L_s^{\otimes m})$ is a graded module over $R(X_s, L_s)$; this is the graded-module structure underlying 1022.

Proof. Assemble the Gmodule field-for-field over the graded semiring of 1338, exactly as 1337 assembles the graded monoid. The action grades by ordinary addition on $\mathbb{N}$ (degree-(m, m') action landing in degree $m + m'$), with the degreewise action $r \star x$ of 1350. Its two Gmodule axioms — unitality $1 \cdot x = x$ and compatibility $r \cdot (r' \cdot x) = (rr') \cdot x$ with the ring multiplication — are the unitality and compatibility clauses of 1350, each turned from its transport-mediated form into the required equality of dependent pairs by the module bridge 1347. Bilinearity (additivity and annihilation by 0 in each variable) is free, as in 1338. By 1336 this produces the module structure of $M(X_s, \mathcal{F}_s, L_s)$ over $R(X_s, L_s)$. □

21.4 P0.5: the bridge from local triviality to invertible-graded sheaves

1281 defines invertibility of a sheaf of modules through a *trivializing basis*; the line bundles that occur in practice (the Serre twists $\mathcal{O}(m)$ of 20) are given instead as *locally trivial* sheaves in the geometric sense of IsLocallyTrivial (609: every point has an affine neighbourhood on which the sheaf restricts to the structure sheaf). This section supplies the missing bridge between the two notions, unlocking the graded-commutative section ring 1341 for every locally trivial line bundle.

Lemma 1352 (Section-level invertibility from a chart trivialisation)**.** *Let M be a sheaf of $\mathcal{O}_X$-modules and $V \subseteq X$ an open with a trivialisation $M|_V \cong \mathcal{O}_V$. Then $\Gamma(M, V)$ is an invertible $\Gamma(X, V)$-module.*

Proof. Evaluating the trivialisation at the top open of V gives a $\Gamma(X, V)$-linear isomorphism $\Gamma(M, V) \cong \Gamma(X, V)$ (using $\Gamma(\mathcal{O}_V, \top) = \Gamma(X, V)$), exhibiting $\Gamma(M, V)$ as a free rank-one, hence invertible, $\Gamma(X, V)$-module. □

Theorem 1353 (P0.5: local triviality implies invertible-graded)**.** *If M is locally trivial (609), then M is invertible in the sense of 1281.*

Proof. The trivialising opens $\{V : M|_V \cong \mathcal{O}_V\}$ form a basis of the topology of X: every neighbourhood of every point contains an affine trivialising chart (1162). On each such V, $\Gamma(M, V)$ is an invertible $\Gamma(X, V)$-module by 1352. Together these are exactly the trivializing-basis data required by 1281. □

Corollary 1354 (The Serre twist family is invertible-graded)**.** *Every member of the Serre twist family — the twist on the glued model, on the integral model* $\mathrm{Proj}\, \mathbb{Z}[X_i]$*, and on the relative projective space $\mathbb{P}(n; S)$ (1195: each is locally trivial) — is invertible in the sense of 1281. Consequently 1341 applies unconditionally to each: its section ring $\bigoplus_{m \geq 0} \Gamma(-, \mathcal{O}(m))$ is graded-commutative.*

Proof. Immediate instances of 1353 applied to 1195. □

21.5 D5: finite-dimensional graded pieces from finite generation

The Hilbert-series rationality induction 1101 (Stacks Tag 00K1) assumes each graded piece $\mathcal{M}_n$ of the ambient graded module is finite-dimensional over the base field κ. This section discharges that hypothesis from finite generation alone, for a module graded by a finite family of commuting degree-raising endomorphisms (1059) — the abstraction, used throughout 20, of the degree-one generators $X_1, \ldots, X_r$ of a polynomial grading ring acting on M.

Theorem 1355 (D5: finite generation forces finite-dimensional graded pieces)**.** *Let $M = \bigoplus_n \mathcal{M}_n$ be an $\mathbb{N}$-graded κ-vector space and $t\colon \mathrm{Fin}\, r \to \mathrm{End}_\kappa(M)$ a finite family of pairwise commuting endomorphisms, each raising degree by one (1059), making M a module over $\kappa[X_1, \ldots, X_r]$ with X_i acting as t_i (1083). If M is finitely generated over this polynomial ring, then every graded piece $\mathcal{M}_n$ is finite-dimensional over κ.*

Proof. A monomial X^α of total degree $|\alpha|$ shifts a vector of $\mathcal{M}_d$ into $\mathcal{M}_{d+|\alpha|}$ (induction on $|\alpha|$ via the degree-raising hypothesis on each t_i); by linearity, a homogeneous polynomial of degree e shifts $\mathcal{M}_d$ into $\mathcal{M}_{d+e}$. Fix a finite generating set of M over $\kappa[X_1, \ldots, X_r]$ and replace each generator by its (finitely many) homogeneous components $h \in \mathcal{M}_d$; these homogeneous pieces still generate M. For $p \in \kappa[X_1, \ldots, X_r]$ and a homogeneous generator $h \in \mathcal{M}_d$, decomposing p into its homogeneous components shows that the degree-n part of $p \cdot h$ is $(\mathrm{hc}_{n-d}\, p) \cdot h$, the action of the degree-$(n-d)$ homogeneous component of p; since a homogeneous polynomial of degree $\leq n$ ranges over a *finite-dimensional* space of coefficients (in the finitely many variables $X_1, \ldots, X_r$ of degree $\leq n$), the degree-n part of $p \cdot h$ lies in the image of a fixed finite-dimensional space under κ-linear action of h. Every $x \in \mathcal{M}_n$ is (via the finite generating set and the graded-piece projection) a finite sum of such contributions, one for each of the finitely many homogeneous generators; hence $\mathcal{M}_n$ is contained in a finite sum of finite-dimensional subspaces, so is itself finite-dimensional. ☐

Remark 1356 (Feeding the Hilbert-series induction)**.** Registering 1355's conclusion as the instance $[\forall n, \mathrm{FiniteDimensional}\, \kappa\, (\mathcal{M}_n)]$ discharges the finiteness hypothesis of the Hilbert-series rationality induction 1101 from finite generation alone, with no further geometric input.

Chapter 22

The Grassmannian over $\mathbb{Z}$: affine charts and gluing

22.1 Set-up: charts indexed by d-subsets

Fix integers $r \geq d \geq 1$. For a $d \times r$ matrix M over a ring and a subset $I \subseteq \{1, \ldots, r\}$ with $\#(I) = d$, write M_I for the *I-th minor* of M: the $d \times d$ submatrix whose columns are those indexed by I. We construct the Grassmannian scheme $\mathrm{Gr}(r, d)$ over $\mathbb{Z}$ by gluing one affine chart U^I for each such I. The construction is, conceptually, the quotient $\mathrm{GL}_{d,\mathbb{Z}} \backslash V$ of the scheme V of rank-d matrices, but we never use the language of group-scheme actions: the charts and their transition maps are elementary matrix algebra over $\mathbb{Z}$. Taking $d = 1$ recovers $\mathrm{Gr}(r, 1) = \mathbb{P}^{r-1}_{\mathbb{Z}}$.

Definition 1357 (Affine chart U^I)**.** *Source: [Nitsure], §1.* Let $I \subseteq \{1, \ldots, r\}$ with $\#(I) = d$. Let X^I denote the $d \times r$ matrix whose I-th minor X^I_I is the $d \times d$ identity matrix $1_{d \times d}$ and whose remaining $d(r-d)$ entries are independent indeterminates $x^I_{p,q}$ $(1 \leq p \leq d,\ q \in \{1, \ldots, r\} \setminus I)$ over $\mathbb{Z}$. Write $\mathbb{Z}[X^I]$ for the polynomial ring in these $d(r-d)$ variables and set

$$U^I \;:=\; \operatorname{Spec} \mathbb{Z}[X^I].$$

The choice of an ordering of the free entries gives a (non-canonical) isomorphism $U^I \cong \mathbb{A}^{d(r-d)}_{\mathbb{Z}}$; in particular U^I is a smooth affine $\mathbb{Z}$-scheme of relative dimension $d(r-d)$. Intuitively a $\mathbb{Z}$-point of U^I is a $d \times r$ matrix whose I-columns form the identity, i.e. a rank-d quotient of $\mathbb{Z}^r$ trivialised on the coordinates I.

22.2 Transition maps and the cocycle condition

Mathlib inputs for the transition map

The construction of $\theta_{I,J}$ is a Cramer-inverse computation over the localised chart ring, followed by one application of the universal property of a localisation. We record the four Mathlib facts it rests on as dependency anchors.

Lemma 1358 (Localised element is a unit)**.** *Provided by Mathlib. Let R be a commutative ring, $x \in R$, and S a localisation of R away from x (i.e. $S = R[1/x]$). Then the image of x under the structure map $R \to S$ is a unit of S.*

Lemma 1359 (Left Cramer inverse)**.** *Provided by Mathlib. For a square matrix A over a commutative ring with* $\det A$ *a unit, the nonsingular inverse A^{-1} satisfies $A^{-1}A = 1$.*

Lemma 1360 (Right Cramer inverse)**.** *Provided by Mathlib. For a square matrix A over a commutative ring with* $\det A$ *a unit, the nonsingular inverse A^{-1} satisfies $AA^{-1} = 1$.*

Lemma 1361 (Universal property of an away-localisation)**.** *Provided by Mathlib. Let R be a commutative ring, $x \in R$, and $S = R[1/x]$ a localisation of R away from x. Given any ring homomorphism $g : R \to P$ that sends x to a unit of P, there is a unique ring homomorphism $S \to P$ extending g along $R \to S$.*

The transition map by matrix algebra

We now build $\theta_{I,J}$ in seven steps over the localised chart ring $R^I_J := \mathbb{Z}[X^I, 1/P^I_J]$, splitting at the matrix-algebra seams. The ambient chart ring $\mathbb{Z}[X^I]$ is the polynomial ring of 1357 in the $d(r-d)$ free variables $x^I_{p,q}$ ($p \in \{1,\dots,d\}$, $q \notin I$).

Definition 1362 (The universal matrix X^I)**.** *Source: [Nitsure], §1.* Let $I \subseteq \{1,\dots,r\}$ with $\#(I) = d$. The *universal matrix* X^I is the $d \times r$ matrix over the chart ring $\mathbb{Z}[X^I]$ whose I-minor X^I_I is the $d \times d$ identity $1_{d\times d}$ and whose remaining entries are the free indeterminates $x^I_{p,q}$ ($1 \le p \le d$, $q \notin I$). It is the universal point of the chart U^I (1357): a ring map $\mathbb{Z}[X^I] \to A$ is the same as a $d \times r$ matrix over A with identity I-minor.

Definition 1363 (The minor determinant P^I_J)**.** *Source: [Nitsure], §1.* For $J \subseteq \{1,\dots,r\}$ with $\#(J) = d$, let X^I_J denote the J-th minor of the universal matrix X^I (1362) – the $d \times d$ submatrix whose columns are those indexed by J. Set

$$P^I_J \;:=\; \det(X^I_J) \;\in\; \mathbb{Z}[X^I].$$

Then $R^I_J := \mathbb{Z}[X^I, 1/P^I_J]$ (the localisation of the chart ring away from P^I_J) is the ring of the principal open subscheme $U^I_J = \operatorname{Spec}\mathbb{Z}[X^I, 1/P^I_J] \subseteq U^I$ on which P^I_J is invertible. One has $P^I_I = \det(1_{d\times d}) = 1$, so $R^I_I = \mathbb{Z}[X^I]$ and $U^I_I = U^I$.

Definition 1364 (The localised J-minor X^I_J over R^I_J)**.** *Source: [Nitsure], §1.* Write $X^I_J \in \mathrm{Mat}_{d\times d}(R^I_J)$ for the J-minor of the universal matrix X^I (1362) with its entries pushed forward along the structure map $\mathbb{Z}[X^I] \to R^I_J$ into the localised chart ring $R^I_J = \mathbb{Z}[X^I, 1/P^I_J]$ (1363). By construction $\det(X^I_J)$ is the image of P^I_J in R^I_J.

Lemma 1365 (The localised minor determinant is a unit)**.** *Source: [Nitsure], §1. The determinant of the localised J-minor X^I_J (1364) is a unit of R^I_J:*

$$\det(X^I_J) \in (R^I_J)^\times.$$

Proof. By 1364, $\det(X^I_J)$ is the image of P^I_J (1363) under the structure map $\mathbb{Z}[X^I] \to R^I_J$. Since R^I_J is the localisation away from P^I_J, that image is a unit by 1358. ☐

Definition 1366 (The Cramer inverse $(X^I_J)^{-1}$)**.** *Source: [Nitsure], §1.* Let $(X^I_J)^{-1} \in \mathrm{Mat}_{d\times d}(R^I_J)$ be the nonsingular (Cramer) inverse of the localised J-minor X^I_J (1364). Its entries lie in R^I_J by Cramer's rule.

Lemma 1367 (The Cramer inverse is a two-sided inverse)**.** *Because* $\det(X^I_J)$ *is a unit of R^I_J (1365), the Cramer inverse $(X^I_J)^{-1}$ (1366) is a genuine two-sided inverse:*

$$(X^I_J)^{-1}\, X^I_J = 1_{d\times d} \qquad \textit{and} \qquad X^I_J\, (X^I_J)^{-1} = 1_{d\times d}.$$

Proof. Immediate from 1365 together with 1359 (left identity) and 1360 (right identity), both of which apply once $\det(X^I_J)$ is known to be a unit. □

Definition 1368 (The image matrix $M = (X^I_J)^{-1}X^I$)**.** *Source: [Nitsure], §1.* Define the $d \times r$ matrix over R^I_J

$$M \;:=\; (X^I_J)^{-1}\, X^I \;\in\; \mathrm{Mat}_{d\times r}(R^I_J),$$

the product of the Cramer inverse (1366) with the universal matrix X^I (1362) base-changed to R^I_J. Its entries are the prospective images of the indeterminates $x^J_{p,q}$ under $\theta_{I,J}$. Its J-minor is $(X^I_J)^{-1}X^I_J = 1_{d\times d}$ (1367), so M is a point of the chart U^J; and its I-minor is $(X^I_J)^{-1}X^I_I = (X^I_J)^{-1}$.

Definition 1369 (The pre-localisation hom $\tilde\theta_{I,J} : \mathbb{Z}[X^J] \to R^I_J$)**.** *Source: [Nitsure], §1.* Let $\tilde\theta_{I,J} : \mathbb{Z}[X^J] \to R^I_J$ be the $\mathbb{Z}$-algebra homomorphism out of the chart ring of J determined, via the universal property of the polynomial ring, by sending each free indeterminate $x^J_{p,q}$ to the (p, q)-entry of the image matrix $M = (X^I_J)^{-1}X^I$ (1368); equivalently $\tilde\theta_{I,J}(X^J) = (X^I_J)^{-1}X^I$ as a matrix identity (the identity J-block of X^J matching the identity J-minor of M).

Lemma 1370 (The pre-hom sends P^J_I to a unit)**.** *Source: [Nitsure], §1. The pre-localisation hom $\tilde\theta_{I,J}$ (1369) sends P^J_I (1363, the I-minor determinant of X^J) to a unit of R^I_J. Indeed*

$$\tilde\theta_{I,J}(P^J_I) = \det((X^I_J)^{-1}X^I_I) = \det((X^I_J)^{-1}) = \det(X^I_J)^{-1} = 1/P^I_J,$$

which is a unit.

Proof. By 1369, $\tilde\theta_{I,J}(X^J) = M = (X^I_J)^{-1}X^I$, so $\tilde\theta_{I,J}$ carries the I-minor X^J_I of X^J to the I-minor of M, which is $(X^I_J)^{-1}X^I_I = (X^I_J)^{-1}$ since the I-minor of X^I is the identity. Hence $\tilde\theta_{I,J}(P^J_I) = \det((X^I_J)^{-1}) = \det(X^I_J)^{-1}$. As $\det(X^I_J)$ is a unit (1365), so is its inverse; the matrix-inverse determinant identity uses 1367. □

Definition 1371 (Transition map $\theta_{I,J}$)**.** *Source: [Nitsure], §1.* Because the pre-localisation hom $\tilde\theta_{I,J} : \mathbb{Z}[X^J] \to R^I_J$ (1369) sends P^J_I to a unit of R^I_J (1370), the universal property of the away-localisation (1361) provides a unique ring homomorphism extending it along $\mathbb{Z}[X^J] \to \mathbb{Z}[X^J, 1/P^J_I]$. This extension is the *transition map*

$$\theta_{I,J} : \mathbb{Z}[X^J, 1/P^J_I] \longrightarrow \mathbb{Z}[X^I, 1/P^I_J], \qquad \theta_{I,J}(X^J) = (X^I_J)^{-1}X^I.$$

Dually $\theta_{I,J}$ is the comorphism of a morphism of schemes $U^I_J \to U^J_I$ (1357), which we also denote $\theta_{I,J}$; it sends the universal matrix X^I to $(X^I_J)^{-1}X^I$.

Lemma 1372 ($\theta_{I,I}$ is the identity)**.** *Source: [Nitsure], §1. For every I, $\theta_{I,I} = \mathrm{id}_{U^I}$: since $P^I_I = 1$ (1363) the minor $X^I_I = 1_{d\times d}$ is its own inverse, so $\theta_{I,I}(X^I) = (X^I_I)^{-1}X^I = X^I$, i.e. $\theta_{I,I}$ (1371) is the identity ring homomorphism of $\mathbb{Z}[X^I, 1/P^I_I] = \mathbb{Z}[X^I]$.*

Proof. When $J = I$ one has $P^I_I = \det(1_{d\times d}) = 1$ (1363), so $R^I_I = \mathbb{Z}[X^I]$, the minor X^I_I is the identity, and its Cramer inverse is again the identity. Hence the image matrix is $M = 1 \cdot X^I = X^I$, the pre-hom sends each $x^I_{p,q}$ to itself, and its localisation lift (1371) is the identity. □

Project-local matrix and minor bookkeeping

The cocycle computation rests on a handful of elementary matrix-algebra identities over the (localised) chart rings. They are project-internal helpers with no external source; each is verified directly in Lean.

Lemma 1373 (Column submatrix of a product)**.** *Let $A \in \mathrm{Mat}_{d\times d}(R)$ and $B \in \mathrm{Mat}_{d\times r}(R)$ be matrices over a commutative ring, and $g : \{1, \dots, d\} \to \{1, \dots, r\}$ a column-index map. Then taking the column submatrix of the product equals multiplying by the column submatrix of the right factor:*

$$(AB)_{\bullet g} \;=\; A\,(B_{\bullet g}).$$

Proof. Entrywise from the definition of matrix multiplication (reindexing the rows by the identity leaves the contraction index untouched). Proved directly in Lean. □

Lemma 1374 (A ring map commutes with the Cramer inverse)**.** *Let $f : R \to S$ be a homomorphism of commutative rings and $A \in \mathrm{Mat}_{n\times n}(R)$ a square matrix with $\det A$ a unit. Writing f_* for the entrywise application of f, the nonsingular inverse is natural:*

$$(f_*A)^{-1} \;=\; f_*(A^{-1}).$$

Proof. Since $f_*A \cdot f_*(A^{-1}) = f_*(AA^{-1}) = f_*(1_{n\times n}) = 1_{n\times n}$ (using $AA^{-1} = 1$, 1360), the matrix $f_*(A^{-1})$ is a right inverse of f_*A, hence equals $(f_*A)^{-1}$. Proved directly in Lean. □

Lemma 1375 (Functoriality of entrywise application)**.** *Let $f : R \to A$, $g : A \to D$ and $h : R \to D$ be ring homomorphisms with $g \circ f = h$. Then for any matrix M over R, applying f then g entrywise equals applying h entrywise: $g_*(f_*M) = h_*M$.*

Proof. Immediate from functoriality of $\mathrm{Mat}(-)$ (composition of entrywise maps). Proved directly in Lean. □

Lemma 1376 (Inverse cancellation in a composite)**.** *Let $A, B \in \mathrm{Mat}_{d\times d}(R)$ and $M \in \mathrm{Mat}_{d\times e}(R)$ over a commutative ring, with $\det A$ a unit. Then*

$$(B^{-1}A)\,(A^{-1}M) \;=\; B^{-1}M.$$

Proof. By associativity and $AA^{-1} = 1_{d\times d}$ (1360). Proved directly in Lean. □

Lemma 1377 (The I-minor of X^I is the identity)**.** *The I-minor of the universal matrix X^I (1362), with its I-columns read through the order isomorphism $\{1, \dots, d\} \xrightarrow{\sim} I$, is the $d \times d$ identity: $X^I_I = 1_{d\times d}$. (This is the normalisation $P^I_I = 1$ defining the chart.)*

Proof. By 1362 the q-column with $q \in I$ is the standard basis vector e_p where p is the preimage of q under the order isomorphism $\{1, \dots, d\} \xrightarrow{\sim} I$; so the I-minor is the identity. Proved directly in Lean. □

Lemma 1378 (The J-minor of the image matrix is the identity)**.** *The J-minor of the image matrix $M = (X^I_J)^{-1}X^I$ (1368) is the identity:*

$$M_J \;=\; (X^I_J)^{-1}\,X^I_J \;=\; 1_{d\times d}.$$

Proof. The J-column submatrix of $M = (X^I_J)^{-1}X^I$ is $(X^I_J)^{-1}$ times the J-column submatrix of X^I, i.e. $(X^I_J)^{-1}X^I_J$, by 1373; this is $1_{d\times d}$ by the left-inverse identity (1367). Proved directly in Lean. □

Lemma 1379 ($\tilde{\theta}_{I,J}$ realises the matrix formula)**.** *Applying the pre-localisation hom $\tilde{\theta}_{I,J}$ (1369) entrywise to the universal matrix X^J yields the image matrix:*

$$(\tilde{\theta}_{I,J})_* X^J \ = \ M \ = \ (X^I_J)^{-1} X^I.$$

Proof. On a free entry $x^J_{p,q}$ ($q \notin J$) this holds by the definition of $\tilde{\theta}_{I,J}$ as evaluation at the entries of M (1369). On a column $q \in J$ the entry of X^J is an identity-block entry, which $\tilde{\theta}_{I,J}$ carries to the corresponding entry of $M_J = 1_{d\times d}$ (1378). Proved directly in Lean. □

Lemma 1380 (Image of a cross minor under $\tilde{\theta}_{I,J}$)**.** *For a third size-d subset K, the pre-hom $\tilde{\theta}_{I,J}$ sends the minor determinant P^J_K (1363) to the determinant of the K-minor of the image matrix $M = (X^I_J)^{-1} X^I$:*

$$\tilde{\theta}_{I,J}(P^J_K) \ = \ \det(M_K).$$

Proof. The hom $\tilde{\theta}_{I,J}$ commutes with determinants and with passing to a column submatrix, so $\tilde{\theta}_{I,J}(P^J_K) = \det\big((\tilde{\theta}_{I,J})_* X^J\big)_K$; apply 1379. Proved directly in Lean. □

Lemma 1381 (Naturality of the image matrix)**.** *Let X be a size-d subset and $\iota : R^I_X \to D$ a ring homomorphism lying over the structure map $R^I \to D$, i.e. $\iota \circ (R^I \to R^I_X) = (R^I \to D)$. Writing Y for the universal matrix X^I base-changed to D and Y_X for its X-minor, one has*

$$\iota_*((X^I_X)^{-1} X^I) \ = \ (Y_X)^{-1}\, Y.$$

Proof. Distribute ι_* across the product $M = (X^I_X)^{-1} X^I$; the factor $(X^I_X)^{-1}$ passes through ι_* as a Cramer inverse (1374, using 1365), and both factors collapse to base changes over D by functoriality (1375) since ι lies over $R^I \to D$. Proved directly in Lean. □

Triple-overlap rings and the localised transition maps

The cocycle condition $\theta_{I,K} = \theta_{I,J} \circ \theta_{J,K}$ cannot be stated as a naive composition of the landed transition maps of 1371: the codomain of $\theta_{J,K}$ (the ring $R^J[1/P^J_K]$) differs from the domain of $\theta_{I,J}$ (the ring $R^J[1/P^J_I]$). The identity therefore lives over the *triple-overlap* rings obtained by inverting both relevant minors in each chart ring:

$$S_K := R^K[1/(P^K_I P^K_J)], \quad S_J := R^J[1/(P^J_I P^J_K)], \quad S_I := R^I[1/(P^I_J P^I_K)].$$

Definition 1382 (Inclusion of an away-localisation into a double one (left))**.** For $x, y \in R$, the canonical inclusion $\iota^{\mathrm{L}}_{x,y} : R[1/x] \to R[1/(xy)]$ inverting the extra factor y, defined as the away-localisation lift (1361) of the structure map $R \to R[1/(xy)]$ along x — well-defined because x is a unit of $R[1/(xy)]$ (a factor of the unit xy, 1358).

Definition 1383 (Inclusion of an away-localisation into a double one (right))**.** For $x, y \in R$, the canonical inclusion $\iota^{\mathrm{R}}_{x,y} : R[1/y] \to R[1/(xy)]$ inverting the extra factor x, defined as the away-localisation lift (1361) of $R \to R[1/(xy)]$ along y — well-defined since y is a unit of $R[1/(xy)]$ (1358).

Lemma 1384 (Left inclusion lies over R)**.** *The inclusion $\iota^{\mathrm{L}}_{x,y}$ (1382) intertwines the two structure maps: $\iota^{\mathrm{L}}_{x,y} \circ (R \to R[1/x]) = (R \to R[1/(xy)])$.*

Proof. By the defining property of the away-localisation lift (1361). Proved directly in Lean. □

Lemma 1385 (Right inclusion lies over R)**.** *The inclusion $\iota^{\mathrm{R}}_{x,y}$ (1383) intertwines the two structure maps: $\iota^{\mathrm{R}}_{x,y} \circ (R \to R[1/y]) = (R \to R[1/(xy)])$.*

Proof. By the defining property of the away-localisation lift (1361). Proved directly in Lean. □

Lemma 1386 (Left factor is a unit in the double localisation)**.** *For $x, y \in R$, the image of x in $R[1/(xy)]$ is a unit.*

Proof. xy is a unit of $R[1/(xy)]$ (1358); a factor of a unit is a unit. Proved directly in Lean. □

Lemma 1387 (Right factor is a unit in the double localisation)**.** *For $x, y \in R$, the image of y in $R[1/(xy)]$ is a unit.*

Proof. xy is a unit of $R[1/(xy)]$ (1358); a factor of a unit is a unit. Proved directly in Lean. □

Lemma 1388 (The cross minor becomes a unit after inclusion)**.** *Let A, B, C be size-d subsets and $\iota : R^A_B \to D$ a ring homomorphism over the structure map $R^A \to D$ under which the minor determinant P^A_C becomes a unit of D. Then $\iota(\tilde{\theta}_{A,B}(P^B_C))$ is a unit of D. Indeed*

$$\tilde{\theta}_{A,B}(P^B_C) = \det((X^A_B)^{-1} X^A_C) = \det((X^A_B)^{-1}) \cdot P^A_C,$$

a product of two units once P^A_C is inverted in D.

Proof. By 1380, $\tilde{\theta}_{A,B}(P^B_C) = \det(M_C)$ with $M = (X^A_B)^{-1} X^A$; split the C-minor of the product (1373) and the determinant as $\det((X^A_B)^{-1}) \cdot P^A_C$. The first factor is a unit (1367); the second becomes a unit under ι by hypothesis. Proved directly in Lean. □

Definition 1389 (Localised transition map $\Theta_{I,J} : S_J \to S_I$)**.** The triple-overlap transition map $\Theta_{I,J} : S_J = R^J[1/(P^J_I P^J_K)] \to S_I = R^I[1/(P^I_J P^I_K)]$: the away-localisation lift (1361) along the doubly inverted minor $P^J_I P^J_K$ of the pre-hom $\tilde{\theta}_{I,J}$ (1369) post-composed with the inclusion ι^{L} into S_I (1382). It is well-defined because both inverted minors map to units of S_I: $\tilde{\theta}_{I,J}(P^J_I)$ by 1370 (then 1384), and $\tilde{\theta}_{I,J}(P^J_K)$ by 1388 (with cross factor P^I_K a unit, 1387).

Definition 1390 (Localised transition map $\Theta_{J,K} : S_K \to S_J$)**.** The triple-overlap transition map $\Theta_{J,K} : S_K = R^K[1/(P^K_I P^K_J)] \to S_J = R^J[1/(P^J_I P^J_K)]$: the away-localisation lift (1361) of $\tilde{\theta}_{J,K}$ (1369) post-composed with the inclusion ι^{R} into S_J (1383), along $P^K_I P^K_J$. Well-defined by 1370 and 1388 (cross factor P^J_I a unit, 1386).

Definition 1391 (Localised transition map $\Theta_{I,K} : S_K \to S_I$)**.** The triple-overlap transition map $\Theta_{I,K} : S_K = R^K[1/(P^K_I P^K_J)] \to S_I = R^I[1/(P^I_J P^I_K)]$: the away-localisation lift (1361) of $\tilde{\theta}_{I,K}$ (1369) post-composed with the inclusion ι^{R} into S_I (1383), along $P^K_I P^K_J$. Well-defined by 1370 and 1388 (cross factor P^I_J a unit, 1386).

Lemma 1392 (Image-matrix identity over the triple overlap)**.** *Over the triple-overlap ring S_I, the image matrix of $\theta_{I,K}$ equals $\Theta_{I,J}$ applied entrywise to the image matrix of $\theta_{J,K}$:*

$$\iota^{\mathrm{R}}_*((X^I_K)^{-1} X^I) \;=\; \left(\Theta_{I,J} \circ \iota^{\mathrm{R}}\right)_*((X^J_K)^{-1} X^J).$$

Both sides reduce to $(Y_K)^{-1} Y$, where Y is the universal matrix X^I base-changed to S_I and Y_K its K-minor.

Proof. The left side is $(Y_K)^{-1}Y$ by naturality of the image matrix (1381, via 1385). For the right side, $\Theta_{I,J} \circ \iota^{\mathrm{R}}$ carries the chart ring R^J over $\iota^{\mathrm{L}} \circ \widetilde{\theta}_{I,J}$, so the image matrix of $\theta_{J,K}$ maps to $(M^J_K)^{-1}M^J$ with $M^J = (Y_J)^{-1}Y$ (1379, 1381); the minor $M^J_K = (Y_J)^{-1}Y_K$ (1373) inverts via $(AB)^{-1} = B^{-1}A^{-1}$, and the cancellation $(Y_K)^{-1}Y_J \cdot (Y_J)^{-1}Y = (Y_K)^{-1}Y$ (1376, using $\det Y_J$ a unit by 1386) gives $(Y_K)^{-1}Y$. Proved directly in Lean. □

Lemma 1393 (Cycycle condition for the transition maps)**.** *Source: [Nitsure], §1. For any three subsets $I, J, K \subseteq \{1, \dots, r\}$ of cardinality d, the transition maps of 1371 satisfy the cocycle condition. As morphisms of schemes, on the triple overlap inside U^I,*

$$\theta_{I,K} \;=\; \theta_{J,K} \circ \theta_{I,J},$$

equivalently, as ring homomorphisms, $\theta_{I,K} = \theta_{I,J} \circ \theta_{J,K}$ (the two read the same identity through the order-reversal of Spec*). Together with $\theta_{I,I} = \mathrm{id}$, this is exactly the compatibility datum required to glue the charts.*

Proof. We verify the scheme-morphism form $\theta_{I,K} = \theta_{J,K} \circ \theta_{I,J}$ by a direct matrix computation, valid over the open locus of U^I on which all the minors involved are invertible (namely P^I_J and P^I_K, which suffices since the maps are determined by their restriction to a dense open and all rings are domains; the computation is an identity of matrices with entries in $\mathbb{Z}[X^I, 1/P^I_J, 1/P^I_K]$).

Start from the matrix X^I (the universal point of U^I). By 1371,

$$M^J \;:=\; \theta_{I,J}(X^I) \;=\; (X^I_J)^{-1}X^I,$$

a $d \times r$ matrix whose J-minor is the identity, hence a point of U^J. Its K-minor is

$$M^J_K \;=\; (X^I_J)^{-1}X^I_K, \qquad \text{so} \qquad (M^J_K)^{-1} \;=\; (X^I_K)^{-1}X^I_J,$$

using $(AB)^{-1} = B^{-1}A^{-1}$ for the invertible $d \times d$ matrices X^I_J and X^I_K. Applying $\theta_{J,K}$ to M^J gives

$$\theta_{J,K}(M^J) \;=\; (M^J_K)^{-1}M^J \;=\; (X^I_K)^{-1}X^I_J \cdot (X^I_J)^{-1}X^I \;=\; (X^I_K)^{-1}X^I,$$

the middle product $X^I_J(X^I_J)^{-1} = 1_{d\times d}$ cancelling by associativity of matrix multiplication. The right-hand side is precisely $\theta_{I,K}(X^I) = (X^I_K)^{-1}X^I$. Thus $\theta_{J,K} \circ \theta_{I,J}$ and $\theta_{I,K}$ agree on the universal matrix X^I, hence agree as morphisms $U^I \supseteq U^I_J \cap U^I_K \to U^K$. The special case $K = I$ (resp. $J = I$) together with $\theta_{I,I} = \mathrm{id}$ shows each $\theta_{I,J}$ is an isomorphism onto its image with inverse $\theta_{J,I}$, since $\theta_{J,I} \circ \theta_{I,J} = \theta_{I,I} = \mathrm{id}$. □

22.3 Scheme-level glue-data layer

The transition data of 22.2 lives at the level of rings. To feed it to a scheme-gluing machine we must re-express each piece as a morphism in the category of schemes: the charts U^I, the principal-open overlaps U^I_J, their inclusions, and the transitions $\theta_{I,J}$ all become schemes and scheme morphisms by applying the contravariant functor Spec. The lemmas of this section record those translations together with the two genuinely geometric facts needed to assemble a glue datum: that each inclusion $U^I_J \hookrightarrow U^I$ is an open immersion, and that the fibre product of two principal opens of $\operatorname{Spec} A$ is again a principal open. They are project-internal bridge results with no external source beyond the gluing construction of 1416.

Mathlib inputs for the scheme-level layer

Lemma 1394 (Spec of an away-localisation is an open immersion)**.** *Provided by Mathlib. Let R be a commutative ring and $f \in R$. The morphism* $\operatorname{Spec} R[1/f] \to \operatorname{Spec} R$ *induced by the structure map $R \to R[1/f]$ of the away-localisation is an open immersion (onto the basic open $D(f) \subseteq \operatorname{Spec} R$).*

Lemma 1395 (Fibre product of two affine spectra is Spec of the tensor product)**.** *Provided by Mathlib. Let R be a commutative ring and S, T two R-algebras. The fibre product of* $\operatorname{Spec} S \to \operatorname{Spec} R \leftarrow \operatorname{Spec} T$ *is canonically isomorphic to the spectrum of the tensor product:*

$$\operatorname{Spec} S \times_{\operatorname{Spec} R} \operatorname{Spec} T \;\cong\; \operatorname{Spec}(S \otimes_R T).$$

Lemma 1396 (An away-localisation of a tensor product is a double away-localisation)**.** *Provided by Mathlib. Let R be a commutative ring, $x, y \in R$, and $R[1/x]$ the localisation away from x. Then $R[1/x] \otimes_R R[1/y]$ is a localisation of R away from xy; equivalently it realises* $R[1/(xy)]$.

Lemma 1397 (Two away-localisations of the same element are canonically isomorphic)**.** *Provided by Mathlib. Let R be a commutative ring and $M \subseteq R$ a submonoid. Any two localisations of R at M are uniquely R-algebra isomorphic; in particular, given two rings each presented as $R[1/x]$ there is a canonical R-algebra isomorphism between them.*

Charts, overlaps, and inclusions as schemes

Theorem 1398 (The diagonal minor is the unit)**.** *For every size-d subset I, the I-minor determinant of the universal matrix X^I is the unit of the chart ring:*

$$P_I^I \;=\; 1 \;\in\; R^I.$$

Proof. The I-minor X_I^I is the $d \times d$ identity (1377), and $\det(1_{d\times d}) = 1$. Proved directly in Lean. ☐

Definition 1399 (The overlap scheme U_J^I)**.** The principal-open overlap, as a scheme, is the affine spectrum of the away-localisation of the chart ring R^I (1357) at the minor determinant P_J^I (1363):

$$U_J^I \;:=\; \operatorname{Spec} R^I[1/P_J^I].$$

It is the *V*-object of the Grassmannian glue datum.

Definition 1400 (The overlap inclusion $U_J^I \to U^I$)**.** The canonical inclusion of the overlap into the chart is the Spec of the structure map $R^I \to R^I[1/P_J^I]$:

$$U_J^I \;=\; \operatorname{Spec} R^I[1/P_J^I] \longrightarrow \operatorname{Spec} R^I \;=\; U^I.$$

It is the *f*-field of the Grassmannian glue datum.

Lemma 1401 (The overlap inclusion is an open immersion)**.** *The inclusion $U_J^I \hookrightarrow U^I$ (1400) is an open immersion, identifying U_J^I with the basic open $D(P_J^I)$ of U^I.*

Proof. This is exactly the away-localisation open-immersion fact (1394) applied to $f = P_J^I$. Proved directly in Lean. ☐

Lemma 1402 (The self-overlap inclusion is an isomorphism)**.** *For every I, the self-inclusion $U_I^I \to U^I$ (1400) is an isomorphism. This is the f_{id}-field of the glue datum.*

Proof. Since $P_I^I = 1$ is a unit (1398), the structure map $R^I \to R^I[1/P_I^I]$ is the localisation at a unit, hence an isomorphism of rings; applying Spec gives an isomorphism of schemes. Proved directly in Lean. $\square$

Definition 1403 (The scheme-level transition $U_J^I \to U_I^J$)**.** The scheme-level transition is the Spec of the ring transition map $\theta_{I,J} : R^J[1/P_I^J] \to R^I[1/P_J^I]$ (1371):

$$U_J^I \;=\; \operatorname{Spec} R^I[1/P_J^I] \longrightarrow \operatorname{Spec} R^J[1/P_I^J] \;=\; U_I^J.$$

It is the t-field of the Grassmannian glue datum.

Lemma 1404 (The self-transition is the identity)**.** *For every I, the scheme-level self-transition $U_I^I \to U_I^I$ (1403) is the identity. This is the t_{id}-field of the glue datum.*

Proof. By 1372 the ring map $\theta_{I,I}$ is the identity, and Spec sends the identity ring map to the identity morphism. Proved directly in Lean. $\square$

The away-localisation pullback isomorphism

Definition 1405 (Pullback of two principal opens of $\operatorname{Spec} A$)**.** Let A be a commutative ring and $x, y \in A$. The fibre product of the two basic-open inclusions $\operatorname{Spec} A[1/x] \to \operatorname{Spec} A \leftarrow \operatorname{Spec} A[1/y]$ is canonically isomorphic to the spectrum of the doubly-inverted localisation:

$$\operatorname{Spec} A[1/x] \times_{\operatorname{Spec} A} \operatorname{Spec} A[1/y] \;\cong\; \operatorname{Spec} A[1/(xy)].$$

The isomorphism is the composite of the tensor-product identification of the pullback (1395) with the Spec of the localisation isomorphism $A[1/x] \otimes_A A[1/y] \cong A[1/(xy)]$ (1396, 1397). It is stated over a general base ring A so that it may be instantiated at the triple-overlap rings of the Grassmannian glue datum.

Lemma 1406 (Left leg of the pullback isomorphism)**.** *Under the identification of 1405, the first projection of the pullback corresponds to the left away-inclusion: the composite* $\operatorname{Spec} A[1/(xy)] \to \operatorname{Spec} A[1/x] \times_{\operatorname{Spec} A} \operatorname{Spec} A[1/y] \xrightarrow{\mathrm{pr}_1} \operatorname{Spec} A[1/x]$ *equals* Spec *of the inclusion* $\iota^{\mathrm{L}}_{x,y}$ *(1382).*

Proof. The pullback-projection compatibility of the tensor-product identification sends pr_1 to the left tensor inclusion, which under the localisation isomorphism is the structure map $A[1/x] \to A[1/(xy)]$, i.e. $\iota^{\mathrm{L}}_{x,y}$; the verification is the universal property of the away-localisation. Proved directly in Lean. $\square$

Lemma 1407 (Right leg of the pullback isomorphism)**.** *Symmetrically, the second projection of the pullback corresponds to the right away-inclusion: the composite* $\operatorname{Spec} A[1/(xy)] \to \operatorname{Spec} A[1/x] \times_{\operatorname{Spec} A} \operatorname{Spec} A[1/y] \xrightarrow{\mathrm{pr}_2} \operatorname{Spec} A[1/y]$ *equals* Spec *of* $\iota^{\mathrm{R}}_{x,y}$ *(1383).*

Proof. As in 1406, with the right tensor inclusion in place of the left. Proved directly in Lean. $\square$

Definition 1408 (Order-swap isomorphism $A[1/(xy)] \cong A[1/(yx)]$)**.** For $x, y \in A$, the canonical ring isomorphism

$$A[1/(xy)] \;\cong\; A[1/(yx)]$$

identifying the two away-localisations, which agree because xy and yx generate the same submonoid of A (1397). It is inserted into the triple-cocycle field of the glue datum to reconcile the opposite product orders produced by the two pullback isomorphisms.

Lemma 1409 (The order-swap lies over the base ring)**.** *The order-swap isomorphism (1408) intertwines the two structure maps: for $x, y \in A$,*

$$\mathrm{swap}_{x,y} \circ (A \to A[1/(xy)]) \;=\; (A \to A[1/(yx)]).$$

Proof. Both $A[1/(xy)]$ and $A[1/(yx)]$ are localisations of A at the same submonoid (xy and yx generate it), and the canonical comparison map between two such localisations is an A-algebra isomorphism (1397); an A-algebra map commutes with the structure maps. Proved directly in Lean. □

The triple-overlap glue field t'

The scheme-gluing machine requires, beyond the pairwise transitions $t_{I,J}$, a *triple-overlap* field t' on the relevant fibre products, together with its compatibility t_{fac} with the chart projections. We record both as the next layer of bridge results; their cocycle coherence is the residual obligation discharged in 1416.

Definition 1410 (The triple-overlap transition t')**.** The t'-field of the Grassmannian glue datum is the morphism of schemes

$$t'_{I,J,K} : U^I_J \times_{U^I} U^I_K \longrightarrow U^J_K \times_{U^J} U^J_I,$$

defined as the composite

$$U^I_J \times_{U^I} U^I_K \xrightarrow{\sim} \operatorname{Spec} S_I \xrightarrow{\operatorname{Spec}\Theta_{I,J}} \operatorname{Spec} S_J \xrightarrow{\operatorname{Spec}\mathrm{swap}} \operatorname{Spec} R^J[1/(P^J_K P^J_I)] \xrightarrow{\sim} U^J_K \times_{U^J} U^J_I,$$

where the first and last arrows are the pullback identifications of 1405 – the source one for the pair (P^I_J, P^I_K) over R^I, and the target one (taken inverse) for the pair (P^J_K, P^J_I) over R^J – the middle arrow is the Spec of the localised triple-overlap transition $\Theta_{I,J} : S_J \to S_I$ (1389), and swap is the Spec of the order-swap isomorphism $R^J[1/(P^J_K P^J_I)] \cong S_J = R^J[1/(P^J_I P^J_K)]$ (1408). The order-swap is inserted precisely to reconcile the product order $P^J_K P^J_I$ carried by the target pullback identification with the order $P^J_I P^J_K$ of the codomain S_J of $\Theta_{I,J}$.

Lemma 1411 (The t' ring identity)**.** *Over the triple-overlap rings, the localised transition $\Theta_{I,J}$ (1389) pre-composed with the order-swap and the right inclusion equals the left inclusion post-composed with the plain transition $\theta_{I,J}$, as ring homomorphisms $R^J[1/P^J_I] \to S_I$:*

$$\Theta_{I,J} \circ \mathrm{swap} \circ \iota^{\mathrm{R}}_{P^J_K, P^J_I} \;=\; \iota^{\mathrm{L}}_{P^I_J, P^I_K} \circ \theta_{I,J}.$$

Proof. Both sides are ring maps out of the away-localisation $R^J[1/P^J_I]$, so by the universal property of the localisation (1361) it suffices to check equality after pre-composing with the structure map $R^J \to R^J[1/P^J_I]$, i.e. on the chart ring R^J. There the right inclusion and the order-swap collapse to structure maps (1385, 1409), and the lift defining $\Theta_{I,J}$ (1389) unwinds to $\iota^{\mathrm{L}} \circ \tilde{\theta}_{I,J}$ (1382); the right side likewise unwinds through the lift defining $\theta_{I,J}$ (1371) to the same $\iota^{\mathrm{L}} \circ \tilde{\theta}_{I,J}$. Proved directly in Lean. □

Lemma 1412 (The t' compatibility field t_{fac})**.** *The triple-overlap transition t' (1410) is compatible with the projections to the charts:*

$$\mathrm{pr}_2 \circ t'_{I,J,K} \;=\; t_{I,J} \circ \mathrm{pr}_1,$$

where pr_1 is the first projection of the source pullback $U^I_J \times_{U^I} U^I_K$, pr_2 the second projection of the target pullback $U^J_K \times_{U^J} U^J_I$, and $t_{I,J}$ the scheme-level transition (1403).

Proof. Both fibre products are affine, so the identity may be checked after applying Spec, as an identity of the underlying ring homomorphisms. Rewriting the source projection pr_1 through the pullback identification (1406) and the target projection pr_2 through 1407 turns the claim into the ring identity 1411: the three localised pieces of t' (1410) collapse to the left inclusion followed by the plain transition $t_{I,J}$ (1403). Proved directly in Lean. □

Lemma 1413 (The triple-overlap cocycle coherence)**.** *The triple-overlap transitions t' (1410) satisfy the cocycle coherence on the source triple overlap $U^I_J \times_{U^I} U^I_K$:*

$$t'_{I,J,K} \circ t'_{J,K,I} \circ t'_{K,I,J} = \mathrm{id}.$$

This is the cocycle *field of the Grassmannian glue datum – the categorical realisation of the rotated ring identity* $\Phi = \mathrm{id}$ *(1414).*

Proof. Expand each factor t' into its constituents (1410): a source pullback identification, the Spec of a localised triple-overlap transition Θ (1389) pre-composed with the Spec of an order-swap (1408), and an inverse target pullback identification. In the threefold composite the two internal pairs $\mathrm{ap}^{-1} \circ \mathrm{ap}$ of pullback identifications (1405) cancel, adjacent factors sharing the same identification. There remain the six Spec-arrows of the three Θ's and three order-swaps; these collapse into a single Spec of their composite ring endomorphism Φ. By 1414 one has $\Phi = \mathrm{id}$, so this middle Spec is the identity, and the outer conjugating pullback isomorphism cancels against its inverse, leaving the identity of the source triple overlap. Proved directly in Lean. □

Lemma 1414 (The rotated triple-overlap ring cocycle identity $\Phi = \mathrm{id}$)**.** *The ring identity underlying the* cocycle *field of the glue datum. Fix $d \le r$ and three subsets $I, J, K \subseteq \{1, \dots, r\}$ of cardinality d. On the triple-overlap chart ring $R_{IJK} :=$ Localization.Away $(P^I_J \cdot P^I_K)$ the composite ring endomorphism*

$$\Phi := \Theta_{I,J,K} \circ \mathrm{swap}_J \circ \Theta_{J,K,I} \circ \mathrm{swap}_K \circ \Theta_{K,I,J} \circ \mathrm{swap}_I$$

equals the identity RingHom.id R_{IJK}*, where each* $\Theta_{X,Y,Z} =$ `cocycleΘIJ` $d\, r\, X\, Y\, Z$ *(1389, taken for the rotated index triple) and each* swap_X *is the matching order-swap algebra equivalence* `awayMulCommEquiv` *(1408).*

Proof. Both endomorphisms are ring homomorphisms out of the localization $R_{IJK} =$ Localization.Away $(P^I_J \cdot P^I_K)$, so by Mathlib's IsLocalization.ringHom_ext for the submonoid generated by $P^I_J \cdot P^I_K$ it suffices to check equality on the chart-ring generators (the images of the universal matrix entries). On those generators Φ unwinds, via the order-swaps and the Θ definitions, to the rotated analogue of the matrix cocycle collapse $(Y_K)^{-1}Y$ already proved in 1392 — the same image-matrix identity that underlies the scheme-level cocycle condition 1393. Reusing that computation for the rotated triple shows each generator is fixed, whence $\Phi = \mathrm{id}$. □

22.4 The glued Grassmannian scheme

Definition 1415 (The Grassmannian glue datum)**.** The *Grassmannian glue datum* is the `Scheme.GlueData` bundle indexed by the set $\{ I \subseteq \{1, \dots, r\} : \#(I) = d \}$ of size-d subsets, assembling the data constructed above:

- objects $U := U^I$, the affine charts (1357);
- overlaps $V := U^I_J$ (1399);

- inclusions $f := (U^I_J \hookrightarrow U^I)$ (1400), which are open immersions (1401), with the self-inclusion an isomorphism (the f_{id} field, 1402);
- transitions $t := t_{I,J}$ (1403), with t_{id} field 1404;
- the triple-overlap field $t' := t'_{I,J,K}$ (1410), with compatibility field t_{fac} (1412) and cocycle field (1413).

The remaining structural fields f_{mono} (each inclusion is a monomorphism) and $f_{\mathrm{hasPullback}}$ (the relevant fibre products exist) are discharged by instance synthesis from the open-immersion property of the inclusions. Assembling these named fields into the `Scheme.GlueData` record yields the datum from which the glued scheme (1416) is formed.

Definition 1416 (The Grassmannian scheme $\mathrm{Gr}(r, d)$ over $\mathbb{Z}$)**.** *Source: [Nitsure], §1.* Gluing the affine charts $\{U^I\}$ (1357), one for each of the $\binom{r}{d}$ subsets $I \subseteq \{1, \dots, r\}$ of cardinality d, along the open subschemes $U^I_J \subseteq U^I$ via the transition isomorphisms $\theta_{I,J} : U^I_J \xrightarrow{\sim} U^J_I$ (1371) – which satisfy the cocycle condition $\theta_{I,K} = \theta_{J,K} \circ \theta_{I,J}$ and $\theta_{I,I} = \mathrm{id}$ of 1393 – yields a scheme

$$\mathrm{Gr}(r, d) \longrightarrow \operatorname{Spec} \mathbb{Z}.$$

It is of finite type over $\mathbb{Z}$ (a finite gluing of finitely generated $\mathbb{Z}$-algebras), and under the gluing each U^I is an open subscheme of $\mathrm{Gr}(r, d)$ with $U^I \cap U^J = U^I_J$.

Construction of the glue datum. The gluing is carried out through the Mathlib vehicle for scheme gluing, a glue datum in the sense of a category of schemes (the structure `Scheme.GlueData`, extending the abstract glue-datum interface). Its data are supplied as follows. The index set is the set $J := \{ I \subseteq \{1, \dots, r\} : \#(I) = d \}$ of size-d subsets. The objects are the charts U^I (1357); the pairwise overlaps are the schemes U^I_J (1399); the inclusions $f_{I,J} : U^I_J \to U^I$ are the open immersions 1400 (open immersions by 1401, in particular monomorphisms), with the identity field f_{id} supplied by 1402; the transition isomorphisms $t_{I,J} : U^I_J \to U^J_I$ are the scheme-level transitions 1403, with identity field t_{id} supplied by 1404.

The triple-overlap field t' and its first coherence obligation are supplied by the two preceding results. The field itself is the morphism $t'_{I,J,K} : U^I_J \times_{U^I} U^I_K \to U^J_K \times_{U^J} U^J_I$ of 1410, and the compatibility field t_{fac} – that t' is compatible with the projections to the charts – is 1412, which reduces (both fibre products being affine) to the ring identity 1411 via the leg lemmas 1406 and 1407.

Cocycle field. The t'-cocycle coherence $t'_{I,J,K} \circ t'_{J,K,I} \circ t'_{K,I,J} = \mathrm{id}$ follows from 1413 via the ring identity 1414. With this field in hand the glue datum is complete: it is the structure `Scheme.GlueData` with U := `affineChart` (1357), V := `chartOverlap` (1399), f := `chartIncl` (1400), t := `chartTransition` (1403), t_{id} := `chartTransition_self` (1404), t' := `chartTransition'` (1410), t_{fac} := `chartTransition'_fac` (1412), and the cocycle field above. The glued scheme $\mathrm{Gr}(r, d)$ is then the colimit (glue datum).glued produced by the gluing machine.

Moreover $\mathrm{Gr}(r, d) \to \operatorname{Spec} \mathbb{Z}$ is *smooth of relative dimension* $d(r-d)$. Smoothness and relative dimension are local on the source; the charts $\{U^I\}$ cover $\mathrm{Gr}(r, d)$, and each $U^I \cong \mathbb{A}^{d(r-d)}_{\mathbb{Z}}$ (1357) is affine space of dimension $d(r-d)$ over $\mathbb{Z}$, which is smooth over $\mathbb{Z}$ of relative dimension $d(r-d)$ (the polynomial ring $\mathbb{Z}[x_1, \dots, x_{d(r-d)}]$ is a smooth $\mathbb{Z}$-algebra). As the property may be checked on an open cover of the source, the conclusion follows.

22.5 Separatedness

The surjectivity of the restricted-diagonal comorphism is the ring-theoretic heart of separatedness. We isolate it through the following sequence of helper results before stating the keystone lemma.

Lemma 1417 (Two-sided inverse identity $\tilde\theta_{I,J}(P^J_I)\,P^I_J = 1$)**.** *In $R^I_J = \mathbb{Z}[X^I, 1/P^I_J]$, the pre-localisation transition hom $\tilde\theta_{I,J}$ (1369) satisfies the explicit two-sided inverse identity*

$$\tilde\theta_{I,J}(P^J_I) \cdot P^I_J \;=\; 1.$$

This refines the unit assertion of 1370 to a named inverse: $\tilde\theta_{I,J}(P^J_I)$ is the explicit inverse of P^I_J in R^I_J.

Proof. By 1380 (taking the third subset $K = I$), $\tilde\theta_{I,J}(P^J_I)$ is the determinant of the I-minor of the image matrix $M = (X^I_J)^{-1}X^I$, which by 1379 equals $\det((X^I_J)^{-1}X^I_I) = \det((X^I_J)^{-1})$. Hence $\tilde\theta_{I,J}(P^J_I) = \det(X^I_J)^{-1}$, and multiplying by $P^I_J = \det(X^I_J)$ gives $\det(X^I_J)^{-1}\cdot\det(X^I_J) = 1$ by the localisation-inverse cancellation for the unit $\det(X^I_J)$ (1370). □

Definition 1418 (Restricted-diagonal comorphism $\delta_{I,J}$)**.** The *restricted-diagonal comorphism* on the patch $U^I \times_{\mathbb{Z}} U^J$ is the $\mathbb{Z}$-algebra homomorphism

$$\delta_{I,J} : \mathbb{Z}[X^I] \otimes_{\mathbb{Z}} \mathbb{Z}[X^J] \longrightarrow R^I_J, \qquad X^I \otimes 1 \mapsto X^I, \quad 1 \otimes X^J \mapsto (X^I_J)^{-1}X^I,$$

whose surjectivity is the content of separatedness; this is the $\delta_{I,J}$ of 1429.

Proof. Construct $\delta_{I,J}$ as the tensor-product lift of the two commuting factors: the structure map $\mathbb{Z}[X^I] = R^I \to R^I_J$ on the left factor, and the pre-localisation transition hom $\tilde\theta_{I,J}$ (1369) on the right factor. Since both land in the commutative ring R^I_J, the universal property of the tensor product of $\mathbb{Z}$-algebras yields a unique $\mathbb{Z}$-algebra homomorphism out of $\mathbb{Z}[X^I]\otimes_{\mathbb{Z}}\mathbb{Z}[X^J]$ restricting to these factors; by construction its right component is $\tilde\theta_{I,J}$. □

Lemma 1419 (Left component of $\delta_{I,J}$)**.** *For $a \in \mathbb{Z}[X^I]$, the comorphism $\delta_{I,J}$ (1418) acts on the left factor by the structure map:*

$$\delta_{I,J}(a \otimes 1) \;=\; \mathrm{alg}_{R^I \to R^I_J}(a),$$

the image of a under $\mathbb{Z}[X^I] = R^I \to R^I_J$.

Proof. Immediate from the defining tensor-product lift of 1418: the value on a left elementary tensor $a \otimes 1$ is the left factor evaluated at a, i.e. the structure map. □

Lemma 1420 (Right component of $\delta_{I,J}$)**.** *For $b \in \mathbb{Z}[X^J]$, the comorphism $\delta_{I,J}$ (1418) acts on the right factor by the pre-localisation transition hom:*

$$\delta_{I,J}(1 \otimes b) \;=\; \tilde\theta_{I,J}(b)$$

(1369).

Proof. Immediate from the defining tensor-product lift of 1418: the value on a right elementary tensor $1 \otimes b$ is the right factor $\tilde\theta_{I,J}$ evaluated at b. □

Lemma 1421 (The restricted-diagonal comorphism is surjective)**.** *The comorphism $\delta_{I,J} : \mathbb{Z}[X^I]\otimes_{\mathbb{Z}} \mathbb{Z}[X^J] \to R^I_J$ (1418) is surjective.*

Proof. Let $z \in R^I_J$. Since $R^I_J = \mathbb{Z}[X^I, 1/P^I_J]$ is the localisation of $R^I = \mathbb{Z}[X^I]$ away from P^I_J, surjectivity of the localisation map provides $a \in R^I$ and $n \in \mathbb{N}$ with

$$z \cdot (P^I_J)^n \;=\; \mathrm{alg}_{R^I \to R^I_J}(a).$$

Consider the witness $a \otimes (P^J_I)^n \in \mathbb{Z}[X^I] \otimes_{\mathbb{Z}} \mathbb{Z}[X^J]$. Using multiplicativity together with 1419 and 1420,

$$\delta_{I,J}(a \otimes (P^J_I)^n) = \mathrm{alg}_{R^I \to R^I_J}(a) \cdot \tilde{\theta}_{I,J}(P^J_I)^n = z \cdot (P^I_J)^n \cdot \tilde{\theta}_{I,J}(P^J_I)^n = z \cdot \left(P^I_J \cdot \tilde{\theta}_{I,J}(P^J_I)\right)^n = z,$$

where the last equality uses $P^I_J \cdot \tilde{\theta}_{I,J}(P^J_I) = 1$ from 1417. Hence z lies in the image of $\delta_{I,J}$, so $\delta_{I,J}$ is surjective. □

Definition 1422 (Source pullback iso e_2 for the diagonal computation)**.** The *source pullback iso* e_2 identifies the categorical fibre product of two chart inclusions of the glued scheme (1416) with the overlap chart:

$$e_2 : \mathrm{pullback}\,(\iota_i)\,(\iota_j) \;\xrightarrow{\sim}\; U^I \cap U^J \;=\; \texttt{chartOverlap}\,(d, r, i, j)$$

(1399), where ι_i, ι_j are the open immersions of the charts U^I, U^J into $\mathrm{Gr}(r, d)$.

Proof. The glue data of 1416 supplies, for each pair (i, j), a pullback cone over ι_i, ι_j with apex the overlap chart $U^I \cap U^J$, and this cone is a limit cone. Comparing it with the categorical pullback limit cone via the uniqueness-up-to-isomorphism of limit cone points yields the desired isomorphism e_2. □

Definition 1423 (Structure morphism $\mathrm{Gr}(r, d) \to \operatorname{Spec} \mathbb{Z}$)**.** The *structure morphism* of the glued Grassmannian scheme (1416) is the canonical map

$$\mathrm{Gr}(r, d) \longrightarrow \operatorname{Spec} \mathbb{Z}.$$

As $\mathrm{Gr}(r, d)$ lives in the category of schemes over $\mathbb{Z}$, in which $\operatorname{Spec} \mathbb{Z}$ is the terminal object, this morphism is the unique map to that terminal object. It is the genuine base over which separatedness and properness of the Grassmannian are formulated.

Lemma 1424 (The chart map composes to the affine structure map)**.** *For each index $i = (I, \cdot)$, the chart immersion $\iota_i : U^I \to \mathrm{Gr}(r, d)$ of the glue datum (1416) composed with the structure morphism (1423) is the affine structure map of the chart:*

$$\iota_i \circ (\mathrm{Gr}(r, d) \to \operatorname{Spec} \mathbb{Z}) \;=\; \operatorname{Spec}(\mathbb{Z} \to R^I),$$

the Spec *of the canonical ring map $\mathbb{Z} \to R^I = \mathbb{Z}[X^I]$.*

Proof. Both sides are morphisms from U^I into the terminal object $\operatorname{Spec} \mathbb{Z}$, hence they coincide by uniqueness of the terminal map. □

Lemma 1425 (First projection leg of the source pullback iso)**.** *The inverse of the source pullback iso e_2 (1422), followed by the first projection of the fibre product, is the chart inclusion:*

$$e_2^{-1} \circ \mathrm{pr}_1 \;=\; \iota^I_J : U^I_J \to U^I,$$

the pr_1 leg being the projection of pullback $(\iota_i)\,(\iota_j)$ *onto U^I (1400).*

Proof. By construction e_2 is the comparison isomorphism between the categorical pullback limit cone and the glue-datum pullback cone over ι_i, ι_j. The compatibility of a limit-cone comparison isomorphism with the cone legs (uniqueness-up-to-isomorphism of limit cone points, applied to the left leg of the cospan) identifies $e_2^{-1} \circ \mathrm{pr}_1$ with the corresponding leg of the glue-datum cone, which is the chart inclusion ι_J^I. □

Lemma 1426 (Second projection leg of the source pullback iso)**.** *The inverse of the source pullback iso e_2 (1422), followed by the second projection of the fibre product, is the scheme-level transition composed with the opposite chart inclusion:*

$$e_2^{-1} \circ \mathrm{pr}_2 \;=\; t_{I,J} \circ \iota_I^J : U_J^I \to U_I^J \to U^J,$$

the transition $t_{I,J}$ of 1403 followed by the chart inclusion of 1400.

Proof. As in 1425, but using the right leg of the cospan: the glue-datum pullback cone's second leg is the composite $t \circ f$ of the glue datum (transition followed by inclusion), so the comparison isomorphism identifies $e_2^{-1} \circ \mathrm{pr}_2$ with $t_{I,J} \circ \iota_I^J$. □

Lemma 1427 (The transition-then-inclusion composite is affine)**.** *The composite of the scheme-level transition (1403) with the opposite chart inclusion (1400) is the* Spec *of the pre-localisation transition homomorphism $\tilde{\theta}_{I,J} : R^J \to R^I[1/P_J^I]$ (1369):*

$$t_{I,J} \circ \iota_I^J \;=\; \operatorname{Spec} \tilde{\theta}_{I,J} : U_J^I \to U^J.$$

Proof. Unfolding the two scheme maps, the composite is Spec of the ring map obtained by precomposing the localised transition $\theta_{I,J} : R^J[1/P_I^J] \to R^I[1/P_J^I]$ with the localisation $R^J \to R^J[1/P_I^J]$. By the universal property of the away-localisation (the lift commutes with the localisation map), this composite ring map is exactly the pre-localisation transition $\tilde{\theta}_{I,J}$; applying Spec and the contravariance of composition gives the claim. □

Lemma 1428 (The structure morphism is separated (morphism form))**.** *The structure morphism* $\mathrm{Gr}(r, d) \to \operatorname{Spec} \mathbb{Z}$ *(1423) is separated: its diagonal is a closed immersion.*

Proof. Separatedness is local on the target of the diagonal, so it may be checked on the cover of $\mathrm{Gr}(r, d) \times_{\mathbb{Z}} \mathrm{Gr}(r, d)$ by the patch products $U^I \times_{\mathbb{Z}} U^J$ (the left–right product of the chart open cover with itself). Fix a patch (i, j). On it the restricted diagonal is identified, via the source pullback iso e_2 (1422) — whose two legs are computed in 1425 and 1426, using 1424 and 1427 to read off the affine comorphisms — and the target affine iso (the pullback of two affine schemes is the Spec of the tensor product), with the affine morphism $\operatorname{Spec} \delta_{I,J}$, where $\delta_{I,J}$ is the restricted-diagonal comorphism of 1418. Since $\delta_{I,J}$ is surjective (1421), the morphism $\operatorname{Spec} \delta_{I,J}$ is a closed immersion. As every patch of the diagonal is a closed immersion, the diagonal is a closed immersion and the structure morphism is separated. □

Lemma 1429 ($\mathrm{Gr}(r, d)$ is separated over $\mathbb{Z}$)**.** *Source: [Nitsure], §1. The structure morphism* $\mathrm{Gr}(r, d) \to \operatorname{Spec} \mathbb{Z}$ *(1416) is separated; equivalently, the diagonal* $\Delta : \mathrm{Gr}(r, d) \to \mathrm{Gr}(r, d) \times_{\mathbb{Z}} \mathrm{Gr}(r, d)$ *is a closed immersion.*

Proof. The products $U^I \times_{\mathbb{Z}} U^J$, as I, J range over the size-d subsets, form an affine open cover of $\mathrm{Gr}(r, d) \times_{\mathbb{Z}} \mathrm{Gr}(r, d)$. Separatedness may be checked on this cover: it suffices to show that for each pair (I, J) the preimage $\Delta^{-1}(U^I \times_{\mathbb{Z}} U^J)$ maps by a closed immersion into $U^I \times_{\mathbb{Z}} U^J$, i.e. that the corresponding ring homomorphism is surjective.

By 1416 the diagonal preimage is the overlap $\Delta^{-1}(U^I \times_{\mathbb{Z}} U^J) = U^I \cap U^J = U^I_J = \operatorname{Spec} \mathbb{Z}[X^I, 1/P^I_J]$. The restricted diagonal corresponds to the ring map

$$\delta_{I,J} : \mathbb{Z}[X^I] \otimes_{\mathbb{Z}} \mathbb{Z}[X^J] \longrightarrow \mathbb{Z}[X^I, 1/P^I_J], \qquad X^I \otimes 1 \mapsto X^I, \quad 1 \otimes X^J \mapsto (X^I_J)^{-1} X^I,$$

the second component being the comorphism $\theta_{I,J}$ of 1371 (a point of U^I_J and its image under $\theta_{I,J}$ are the two coordinates of a point of the diagonal). We claim $\delta_{I,J}$ is surjective. Its image contains all entries of X^I (from the first factor). From the second factor it contains the entries of $(X^I_J)^{-1}X^I$; taking the I-minor of this matrix, and using $X^I_I = 1_{d\times d}$, the image contains the entries of $(X^I_J)^{-1}$, and in particular

$$\delta_{I,J}(1 \otimes P^J_I) = \det((X^I_J)^{-1}) = 1/P^I_J.$$

Hence $1/P^I_J$ lies in the image, so the image is all of $\mathbb{Z}[X^I, 1/P^I_J]$ and $\delta_{I,J}$ is surjective. Concretely the closed subscheme $\Delta_{I,J} \subseteq U^I \times_{\mathbb{Z}} U^J$ is cut out by the entries of the matrix relation

$$X^J_I \, X^I - X^J \;=\; 0,$$

which on the diagonal holds because $X^J = (X^I_J)^{-1}X^I$ forces $X^J_I = (X^I_J)^{-1}X^I_I = (X^I_J)^{-1}$ and therefore $X^J_I X^I = (X^I_J)^{-1}X^I = X^J$. As each restricted diagonal is a closed immersion, Δ is a closed immersion and $\operatorname{Gr}(r,d)$ is separated over $\mathbb{Z}$. □

22.6 Properness

Following Nitsure §1, the structure morphism $\pi : \operatorname{Gr}(r,d) \to \operatorname{Spec} \mathbb{Z}$ is proper. We reduce properness to the *existence* half of the valuative criterion: the Mathlib criterion `IsProper.of_valuativeCriterion` reduces IsProper π to four ingredients — quasi-compactness, quasi-separatedness, local finiteness of type, and the valuative criterion Existence $\sqcap$ Uniqueness. The first three and the uniqueness half are cheap (uniqueness is free from separatedness), so the only genuine content is Existence: every commutative square from a valuation ring admits a diagonal filler. This existence statement is the Nitsure chart-selection argument, which we decompose into four steps E1–E4.

Cheap ingredients and the reduction to existence

Lemma 1430 ($\operatorname{Gr}(r,d)$ is a compact space)**.** *The underlying topological space of the glued Grassmannian scheme* $\operatorname{Gr}(r,d)$ *(1416) is quasi-compact: it carries the structure of a* `CompactSpace`*.*

Proof. The glue datum (1415) is indexed by the finite set $\{ I \subseteq \{1,\dots,r\} : \#(I) = d \}$ of size-d subsets, and each chart $U^I = \operatorname{Spec} \mathbb{Z}[X^I]$ (1357) is the Spec of a ring, hence quasi-compact. A scheme glued from finitely many quasi-compact charts is quasi-compact, so $\operatorname{Gr}(r,d)$ is a compact space. □

Lemma 1431 (The structure morphism is quasi-compact)**.** *The structure morphism* $\pi : \operatorname{Gr}(r,d) \to \operatorname{Spec} \mathbb{Z}$ *(1423) is quasi-compact.*

Proof. The target $\operatorname{Spec} \mathbb{Z}$ is affine, so quasi-compactness of π is equivalent to quasi-compactness of the total space $\operatorname{Gr}(r,d)$, which is a compact space by 1430. □

Lemma 1432 (The structure morphism is locally of finite type)**.** *The structure morphism* $\pi : \operatorname{Gr}(r,d) \to \operatorname{Spec} \mathbb{Z}$ *(1423) is locally of finite type.*

Proof. Being locally of finite type is local on the source, so it may be checked on the chart open cover (1415). On the chart U^I the composite $U^I \to \mathrm{Gr}(r,d) \xrightarrow{\pi} \operatorname{Spec}\mathbb{Z}$ is, by 1424, the Spec of the structure map $\mathbb{Z} \to R^I = \mathbb{Z}[X^I]$. The polynomial ring $\mathbb{Z}[X^I]$ is a finitely generated $\mathbb{Z}$-algebra, so this map is of finite type; hence π is locally of finite type. ☐

Lemma 1433 (The structure morphism is quasi-separated)**.** *The structure morphism $\pi : \mathrm{Gr}(r,d) \to \operatorname{Spec}\mathbb{Z}$ (1423) is quasi-separated.*

Proof. The structure morphism is separated (1428), and every separated morphism is quasi-separated. ☐

Lemma 1434 (Uniqueness half of the valuative criterion)**.** *The structure morphism $\pi : \mathrm{Gr}(r,d) \to \operatorname{Spec}\mathbb{Z}$ (1423) satisfies the* uniqueness *part of the valuative criterion: for a valuation ring R with fraction field K, any two morphisms $\operatorname{Spec} R \to \mathrm{Gr}(r,d)$ over $\operatorname{Spec}\mathbb{Z}$ that agree on the generic point $\operatorname{Spec} K$ coincide.*

Proof. The structure morphism is separated (1428), and the uniqueness half of the valuative criterion is free for any separated morphism: two lifts agreeing on the schematically dense generic point of $\operatorname{Spec} R$ coincide because the diagonal is a (closed, in particular separated) immersion. ☐

Lemma 1435 (Properness from the existence half of the valuative criterion)**.** *Suppose the structure morphism $\pi : \mathrm{Gr}(r,d) \to \operatorname{Spec}\mathbb{Z}$ (1423) satisfies the* existence *part of the valuative criterion. Then π is proper.*

Proof. The Mathlib criterion `IsProper.of_valuativeCriterion` reduces properness of π to four hypotheses: π is quasi-compact (1431), quasi-separated (1433), locally of finite type (1432), and satisfies the full valuative criterion. The full valuative criterion is the conjunction of its existence and uniqueness halves; the uniqueness half holds by 1434, and the existence half is the hypothesis. Assembling the four ingredients gives properness. ☐

The existence obligation: chart selection by minimal valuation

The existence half of the valuative criterion is the genuine geometric content, and it is exactly Nitsure's chart-selection argument. A commutative square over a valuation ring presents a valuation ring R with fraction field $K = \mathrm{Frac}(R)$, valuation $v := \mathrm{val}_R : K \to \Gamma$ (written multiplicatively, so $R = \{x : v(x) \le 1\}$), a morphism $i_1 : \operatorname{Spec} K \to \mathrm{Gr}(r,d)$, and a morphism $i_2 : \operatorname{Spec} R \to \operatorname{Spec}\mathbb{Z}$ with $i_1 \circ \pi = \operatorname{Spec}(R \to K) \circ i_2$; one must construct a filler $\operatorname{Spec} R \to \mathrm{Gr}(r,d)$ making both triangles commute. We carry this out in four steps. The algebraic core — the behaviour of the transition hom on a K'-minor — is the following identity over the localised chart ring.

Lemma 1436 (The minor-ratio identity)**.** *Source: [Nitsure], §1. For size-d subsets I, J, K, the pre-localisation transition hom $\tilde{\theta}_{I,J}$ (1369) satisfies, in $R^I_J = \mathbb{Z}[X^I, 1/P^I_J]$, the minor-ratio identity*

$$\tilde{\theta}_{I,J}(P^J_K) \cdot P^I_J \;=\; P^I_K$$

(where P^I_J, P^I_K are read through the structure map $\mathbb{Z}[X^I] \to R^I_J$). Equivalently $\tilde{\theta}_{I,J}(P^J_K) = P^I_K / P^I_J$, the determinant of the K-minor of $(X^I_J)^{-1} X^I$. This generalises 1417 (the case $K = I$, where $P^I_I = 1$) and is the algebraic engine of the existence argument: pulled back through a K-point f it gives $g(P^J_K) = f(P^I_K)/f(P^I_J)$.

Proof. By 1380, $\tilde{\theta}_{I,J}(P^J_K) = \det(M_K)$ is the determinant of the K-minor of the image matrix $M = (X^I_J)^{-1}X^I$. Splitting the K-column submatrix of the product (1373) and using multiplicativity of the determinant, $\det(M_K) = \det((X^I_J)^{-1}) \cdot \det(X^I_K) = \det(X^I_J)^{-1} \cdot P^I_K$. Multiplying by $P^I_J = \det(X^I_J)$ and cancelling $\det(X^I_J)^{-1}\det(X^I_J) = 1$ (1367) gives $\tilde{\theta}_{I,J}(P^J_K) \cdot P^I_J = P^I_K$. $\square$

Lemma 1437 (E1 — the K-point factors through a single chart)**.** *Source: [Nitsure], §1. The morphism $i_1 : \operatorname{Spec} K \to \mathrm{Gr}(r,d)$ factors through a single chart: there exist a size-d subset I and a ring homomorphism $f : R^I = \mathbb{Z}[X^I] \to K$ such that*

$$i_1 \;=\; \operatorname{Spec}(f) \circ \iota_I : \operatorname{Spec} K \to U^I \to \mathrm{Gr}(r,d),$$

where $\iota_I : U^I \to \mathrm{Gr}(r,d)$ is the chart immersion of the glue datum (1415).

Proof. Since K is a field, $\operatorname{Spec} K$ is a one-point space, so its image point $x := i_1(*) \in \mathrm{Gr}(r,d)$ is a single point. The chart immersions $\{\iota_I : U^I \to \mathrm{Gr}(r,d)\}$ of the glue datum (1415) are jointly surjective (their images cover the glued scheme 1416), so x lies in the range of some ι_I. Each ι_I is an open immersion (1401 gives this for the chart inclusions, and the glue datum's immersions inherit it). A morphism out of $\operatorname{Spec} K$ whose image point lies in the range of an open immersion factors uniquely through that open immersion; hence $i_1 = j \circ \iota_I$ for a unique $j : \operatorname{Spec} K \to U^I$. As $U^I = \operatorname{Spec} R^I$ is affine, $j = \operatorname{Spec}(f)$ for a unique ring map $f : R^I \to K$, giving the claimed factorization. $\square$

Lemma 1438 (E2 — minimal-valuation chart selection)**.** *Source: [Nitsure], §1. Fix I and $f : R^I \to K$ as in 1437, and let $v := \mathrm{val}_R : K \to \Gamma$ be the (multiplicative) valuation of R. Over the finite index set $\{\, J : \#(J) = d \,\}$ choose J* maximising *$v(f(P^I_J))$ (Nitsure's minimal additive valuation ν is the maximal multiplicative v). Then I itself is a candidate and $P^I_I = 1$ (1398) gives $v(f(P^I_I)) = v(1) = 1$, so the maximum satisfies $v(f(P^I_J)) \geq 1 > 0$. In particular $f(P^I_J) \neq 0$, so the matrix $f(X^I_J)$ is invertible over K.*

Proof. The index set of size-d subsets is finite and nonempty (I is a member), so the function $J \mapsto v(f(P^I_J))$ attains a maximum at some J. Since $P^I_I = 1$ (1398) and $v(1) = 1$, the value at I is 1; hence the maximum is ≥ 1, so $v(f(P^I_J)) \neq 0$ in Γ, i.e. $f(P^I_J) \neq 0$. A square matrix over the field K with nonzero determinant $f(P^I_J) = \det f(X^I_J)$ is invertible. $\square$

Lemma 1439 (Determinant of the identity with one column replaced)**.** *For a vector $v : \operatorname{Fin} d \to R$ and a row index p, the determinant of the identity matrix with its p-th column replaced by v is the entry v_p, with* no *sign: $\det((1).\mathrm{updateCol}\, p\, v) = v_p$.*

Proof. By Cramer's rule $\det((1).\mathrm{updateCol}\, p\, v) = ((1).\mathrm{cramer}\, v)_p$ (Matrix.cramer_apply); and $(1).\mathrm{cramer}\, v = v$ since $1 \cdot \mathrm{cramer}\, v = \det(1) \cdot v = v$ (Matrix.mulVec_cramer). Proved directly in Lean. $\square$

Lemma 1440 (Free entry as a signed minor (E3 cofactor sub-step))**.** *Source: [Nitsure], §1 (the cofactor sub-step of the Properness argument). Fix a size-d subset J, a row index $p : \operatorname{Fin} d$, and a column $q \notin J$. Put $K' = (J \setminus \{j_p\}) \cup \{q\}$, again of size d. Then there is an order-iso realisation of K' under which the K'-minor of the universal matrix equals, up to sign, the free entry:*

$$\mathrm{minorDet}\, d\, r\, J\, K' \;=\; \pm X_{(p,q)}.$$

Proof. Reorder the columns of K' so that column q sits in the p-th slot and the remaining columns match J in increasing order; call σ the resulting permutation. Because the J-minor X^J_J is the identity, the so-reordered submatrix is the identity with its p-th column overwritten by the single free column carrying $X_{(p,q)}$. The determinant of an identity matrix with one column replaced by a vector v is exactly v_p, with *no* sign — via Cramer's rule, $\det((1).\mathrm{updateCol}\, p\, v) = ((1).\mathrm{cramer}\, v)_p = (1 \cdot v)_p = v_p$. Restoring the increasing column order multiplies by $\mathrm{sign}(\sigma) = \pm 1$ (det_permute′), so $\mathrm{minorDet}\, d\, r\, J\, K' = \mathrm{sign}(\sigma) \cdot X_{(p,q)} = \pm X_{(p,q)}$. ☐

Lemma 1441 (E3 — entries land in R; g factors through the valuation ring)**.** *Source: [Nitsure], §1. Define $g : R^J = \mathbb{Z}[X^J] \to K$ by the matrix formula $g(X^J) = f((X^I_J)^{-1} X^I)$; concretely $g = f' \circ \tilde{\theta}_{I,J}$, where $f' : R^I_J \to K$ is the extension of f to the away-localisation $R^I_J = \mathbb{Z}[X^I, 1/P^I_J]$ (well-defined by the universal property 1361, since $f(P^I_J)$ is a unit of K by 1438). Then every value $g(P^J_K)$ lies in R, and consequently g factors uniquely as $g = (R \hookrightarrow K) \circ g'$ for a ring homomorphism $g' : R^J \to R$. Moreover g defines the same K-point as f.*

Proof. Applying f' to the minor-ratio identity (1436), and using that f' extends f, gives for every size-d subset K'

$$g(P^J_{K'}) \cdot f(P^I_J) \;=\; f(P^I_{K'}), \qquad \text{so} \qquad g(P^J_{K'}) \;=\; \frac{f(P^I_{K'})}{f(P^I_J)}.$$

By the maximality in 1438, $v(f(P^I_{K'})) \leq v(f(P^I_J))$, hence $v(g(P^J_{K'})) = v(f(P^I_{K'}))/v(f(P^I_J)) \leq 1$, i.e. $g(P^J_{K'}) \in R$. (That $f(P^I_J)$ is a unit and the cross minors map to units is recorded by 1388, via the determinant identity of 1380.) Now the J-minor X^J_J of X^J is the identity, so cofactor expansion expresses each free entry $x^J_{p,q}$ ($q \notin J$) as, up to sign, the minor $P^J_{K'}$ with $K' = (J \setminus \{j_p\}) \cup \{q\}$ — the determinant obtained by replacing the p-th identity column of X^J_J by column q. Hence $g(x^J_{p,q}) = \pm g(P^J_{K'}) \in R$, while the entries with $q \in J$ are 0 or 1, already in R. (This entry-as-minor sign bookkeeping is the one sub-step with no pre-existing matrix-algebra scaffold; it requires expanding the determinant of a column-substituted identity matrix.) Since g sends every generator of $\mathbb{Z}[X^J]$ into R, it factors uniquely through the subring $R \subseteq K$ as $g = (R \hookrightarrow K) \circ g'$ with $g' : R^J \to R$. ☐

Lemma 1442 (E4 top-triangle K-point identity)**.** *Source: [Nitsure], §1. For charts I, J of $\mathrm{Gr}(r,d)$ and the comparison ring map $g := f' \circ \tilde{\theta}_{I,J} : R^I \to R$ (with f' the localized lift of $f : R^I \to R$ along the away-localization and $\tilde{\theta}_{I,J}$ the chart-transition comorphism), the two chart legs present the same K-point:*

$$\mathrm{Spec}(g) \circ \iota_J \;=\; \mathrm{Spec}(f) \circ \iota_I.$$

Proof. Expand $\mathrm{Spec}(g) = \mathrm{Spec}(\tilde{\theta}_{I,J}) \circ \mathrm{Spec}(f')$ and rewrite $\mathrm{Spec}(\tilde{\theta}_{I,J}) = t_{I,J} \circ \iota_{J,I}$ via 1427; the glue cocycle identity $t \circ f \circ \iota = f \circ \iota$ of the glue data (1415) together with the localized factorization f' followed by the away map $= f$ collapses the composite to $\mathrm{Spec}(f) \circ \iota_I$. ☐

Lemma 1443 (E4 — the filler and its two triangles)**.** *Source: [Nitsure], §1. The morphism*

$$\ell \;:=\; \mathrm{Spec}(g') \circ \iota_J : \mathrm{Spec}\, R \to U^J \to \mathrm{Gr}(r,d)$$

(with $g' : R^J \to R$ from 1441 and ι_J the chart immersion of 1415) is a diagonal filler of the valuative square: it restricts to i_1 over the generic point $\mathrm{Spec}\, K$*, and it lies over* $\mathrm{Spec}\, \mathbb{Z}$*. Hence the valuative square has a lift.*

Proof. Top triangle. Restricting ℓ along $\mathrm{Spec}(R \to K)$ yields $\mathrm{Spec}(g'$ followed by $R \hookrightarrow K) \circ \iota_J = \mathrm{Spec}(g) \circ \iota_J$, using $g = (R \hookrightarrow K) \circ g'$ (1441). Since $g = f' \circ \tilde{\theta}_{I,J}$ presents the same K-point as

f, and $i_1 = \mathrm{Spec}(f) \circ \iota_I$ (1437), this composite equals i_1. *Bottom triangle.* Both $\ell \circ \pi$ and the given i_2 are morphisms $\mathrm{Spec}\, R \to \mathrm{Spec}\, \mathbb{Z}$ into the terminal object (1423; 1424 identifies the chart leg with the affine structure map), hence coincide by uniqueness of the terminal map. The two triangles assemble into a lift structure for the valuative square, so a lift exists. □

Lemma 1444 (Existence half of the valuative criterion)**.** *Source: [Nitsure], §1. The structure morphism $\pi : \mathrm{Gr}(r, d) \to \mathrm{Spec}\, \mathbb{Z}$ (1423) satisfies the* existence *part of the valuative criterion: every commutative square from a valuation ring R (with fraction field K) admits a diagonal filler $\mathrm{Spec}\, R \to \mathrm{Gr}(r, d)$.*

Proof. Given a valuative square, factor the generic-point morphism through a chart (1437), select J by maximal valuation (1438), factor the transported ring map g through the valuation ring (1441), and take the filler $\mathrm{Spec}(g') \circ \iota_J$ (1443). This produces the required diagonal filler, so the existence half holds. □

Properness

Lemma 1445 ($\mathrm{Gr}(r, d)$ is proper over $\mathbb{Z}$)**.** *Source: [Nitsure], §1. The structure morphism $\pi : \mathrm{Gr}(r, d) \to \mathrm{Spec}\, \mathbb{Z}$ (1416) is proper.*

Proof. Nitsure's argument is precisely the valuative criterion, organised here so that all the topological/finiteness bookkeeping is separated from the one geometric step. The reduction 1435 packages quasi-compactness (1431), quasi-separatedness (1433), local finiteness of type (1432) and the uniqueness half (1434, free from separatedness) into the hypothesis that π satisfies only the *existence* half of the valuative criterion. That existence half is established in 1444 by the chart-selection argument of steps E1–E4: the generic-point morphism lands in a chart U^I via a ring map f (1437); one selects J with $\nu(f(P^I_J))$ minimal — equivalently $v(f(P^I_J))$ maximal — (1438), so $f(X^I_J) \in \mathrm{GL}_d(K)$; the transported map $g(X^J) = f((X^I_J)^{-1} X^I)$ then sends every generator into the valuation ring and factors as $g = (R \hookrightarrow K) \circ g'$ (1441); and $\mathrm{Spec}(g') \circ \iota_J$ is the prolonging filler (1443). Feeding this existence statement into 1435 yields that $\pi : \mathrm{Gr}(r, d) \to \mathrm{Spec}\, \mathbb{Z}$ is proper. □

22.7 Out of scope

This chapter constructs the Grassmannian *scheme* $\mathrm{Gr}(r, d)$ over the absolute base $\mathbb{Z}$ and establishes its smoothness, separatedness and properness. The following constructions are deferred to later chapters and are not treated here:

- **The tautological/universal quotient** $u : \mathcal{O}^{\oplus r}_{\mathrm{Gr}(r,d)} \twoheadrightarrow \mathcal{U}$ and the determinant line bundle $\det \mathcal{U}$ with its Plücker embedding.
- **The functor-of-points representability** of $\mathrm{Gr}(r, d)$ – that it represents $\mathrm{Grass}(r, d) = \mathrm{Quot}^{d, \mathcal{O}_{\mathbb{Z}}}_{\oplus^r \mathcal{O}_{\mathbb{Z}}/\mathbb{Z}/\mathbb{Z}}$; this is blueprinted as 1183 in 20.
- **The relative version over a general base** S with a locally free sheaf V of rank r: the scheme $\mathrm{Gr}_S(V, d)$ obtained by base change and gluing over a trivialising cover of V. The absolute-over-$\mathbb{Z}$ construction of this chapter is the input to that base change.

Chapter 23

Effective descent: restriction of the glued sheaf to a chart

The gluing primitive 1479 produces, out of a glue datum D of schemes (index set J, charts U_i, overlaps V_{ij}, overlap immersions $f_{ij} : V_{ij} \hookrightarrow U_i$, transition isomorphisms $t_{ij} : V_{ij} \xrightarrow{\sim} V_{ji}$, glued scheme X with chart immersions $\iota_i : U_i \hookrightarrow X$), a sheaf of $\mathcal{O}_X$-modules $\mathcal{M}$ as the descent equalizer

$$\mathcal{M} \;=\; \mathrm{eq}(a, b) \;\hookrightarrow\; \prod_i (\iota_i)_* \mathcal{M}_i$$

of two parallel maps into the overlap product. That construction makes $\mathcal{M}$ but does *not* yet consume the cocycle conditions (C1)/(C2) on the transition family (g_{ij}). This chapter discharges the remaining content of descent: it shows that restricting the glued sheaf back to a chart recovers the input sheaf there. Concretely, the canonical *restriction morphism* $\iota_i^* \mathcal{M} \to \mathcal{M}_i$ – the adjoint transpose along the chart immersion ι_i of the i-th equalizer projection – is an isomorphism, and it is here that (C1)/(C2) are used. The engine is that pullback along an open immersion is, up to natural isomorphism, the site-level restriction functor (1111); being a right adjoint it preserves the descent equalizer and the pushforward product, so the chart restriction of $\mathcal{M}$ is computed factor by factor. The non-formal input is the *overlap base change* β_{ij}, which identifies the restricted pushforward factor $\iota_i^*(\iota_j)_*\mathcal{M}_j$ with a pushforward of a restriction of $\mathcal{M}_j$ to the overlap, using the cartesianness of the chart–overlap square.

23.1 The descent equalizer and its overlap base change

Definition 1446 (The pushforward product carrying the descent equalizer)**.** Let D be a glue datum and $(\mathcal{M}_i)_{i \in J}$ a family of sheaves of modules with $\mathcal{M}_i$ on the chart U_i. Write

$$P \;=\; \prod_i (\iota_i)_* \mathcal{M}_i \qquad \text{and} \qquad Q \;=\; \prod_{(i,j)} (f_{ij} \circ \iota_i)_* \, (f_{ij}^* \mathcal{M}_i)$$

for the *pushforward product* and the *overlap product* of sheaves of $\mathcal{O}_X$-modules. The two descent legs of 1479,

$$P \overset{a}{\underset{b}{\rightrightarrows}} Q,$$

are: on the (i, j)-component, leg a restricts the i-th chart factor to the overlap V_{ij} using the unit of the pullback–pushforward adjunction along f_{ij} and the pushforward-composition comparison

$(\iota_i)_* \cong (f_{ij} \circ \iota_i)_* \circ f_{ij}^*$; leg b restricts the j-th chart factor, transports it across the inverse transition isomorphism g_{ij}^{-1}, and reindexes the immersion via the glue condition $(t_{ij} \circ f_{ji}) \circ \iota_j = f_{ij} \circ \iota_i$. The glued sheaf $\mathcal{M} = \mathrm{eq}(a, b)$ is the equalizer of this parallel pair; P is the object it embeds into and Q the object detecting the overlap agreement. The product P is the carrier on which the i-th projection (and, through it, the restriction morphism) is defined.

Lemma 1447 (Overlap base change of the chart square (β_{ij}))**.** *For chart indices $i, j \in J$ there is a natural isomorphism of functors* $\mathrm{SheafOfModules}(\mathcal{O}_{U_j}) \to \mathrm{SheafOfModules}(\mathcal{O}_{U_i})$,

$$\beta_{ij} \ : \ \iota_i^* \, (\iota_j)_* \ \xrightarrow{\ \sim\ } \ (f_{ij})_* \, (t_{ij} \circ f_{ji})^*,$$

identifying, for any sheaf $\mathcal{N}$ on the chart U_j, the restriction to the chart U_i of the pushforward $(\iota_j)_*\mathcal{N}$ with the pushforward along f_{ij} of the restriction of $\mathcal{N}$ to the overlap V_{ij} along $t_{ij} \circ f_{ji}$. Here the chart restriction ι_i^* and the overlap restriction $(t_{ij} \circ f_{ji})^*$ are realised as the site-level restriction functors, isomorphic to the geometric pullbacks by 1111.

Proof. The isomorphism is the cartesianness of the chart–overlap square, made site-level. The square

$$\begin{array}{ccc} V_{ij} & \xrightarrow{t_{ij} \circ f_{ji}} & U_j \\ \downarrow f_{ij} & & \downarrow \iota_j \\ U_i & \xrightarrow{\iota_i} & X \end{array}$$

is cartesian, and its cartesianness is visible already on underlying opens: for an open $V \subseteq U_i$,

$$\iota_j^{-1}(\iota_i(V)) \ = \ (t_{ij} \circ f_{ji})(f_{ij}^{-1}(V)),$$

a point of U_j lands in the image of V under ι_i precisely when it comes, via the glue relation, from a point of the overlap lying over V. Hence the two composite opens functors

$$(\iota_i)_! \circ \iota_j^{-1} \ = \ f_{ij}^{-1} \circ (t_{ij} \circ f_{ji})_!$$

agree on objects, and – morphisms of the opens poset being proof-irrelevant – as functors. Both $\iota_i^*(\iota_j)_*$ and $(f_{ij})_*(t_{ij} \circ f_{ji})^*$ are site-level pushforwards along these two (equal) opens functors, so the comparison is assembled from the pushforward-composition isomorphisms together with the natural isomorphism transporting one opens functor to the other:

$$\beta_{ij} \ = \ (\text{pushforwardComp}) \circ (\text{pushforward of the opens-functor equality}) \circ (\text{pushforwardCongr}) \circ (\text{pushforwardComp}$$

The one substantive coherence to check is that the comparison glued from the object-level opens equality is natural in the open V: on the section level the two restriction maps induced by the equal composite functors coincide because the connecting morphisms are the unique arrows of the opens poset between the identified opens. This is the pointwise compatibility that completes the pushforwardCongr factor. ☐

23.2 The restriction morphism and effective descent

Lemma 1448 (The chart restriction morphism of the glued sheaf)**.** *Let $\mathcal{M}$ be the glued sheaf of 1479. The i-th* projection *of $\mathcal{M}$,*

$$\pi_i \ : \ \mathcal{M} \longrightarrow (\iota_i)_* \, \mathcal{M}_i,$$

is the equalizer inclusion $\mathcal{M} \hookrightarrow P$ ([1446]) followed by the i-th product projection. Its adjoint transpose along the chart immersion ι_i, under the pullback–pushforward adjunction $\iota_i^* \dashv (\iota_i)_*$, is the restriction morphism

$$\mathrm{r}_i \;:\; \iota_i^* \mathcal{M} \longrightarrow \mathcal{M}_i.$$

Equivalently, r_i *is the unique morphism whose pushforward, precomposed with the unit, recovers* π_i.

Proof. The projection π_i is well-defined by 1446: the glued sheaf is the equalizer $\mathrm{eq}(a,b) \hookrightarrow P$, and composing the inclusion with the i-th projection of the product $P = \prod_i (\iota_i)_* \mathcal{M}_i$ gives a map to $(\iota_i)_*\mathcal{M}_i$. The restriction morphism is then its image under the natural bijection

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{M}, (\iota_i)_*\mathcal{M}_i) \;\cong\; \mathrm{Hom}_{\mathcal{O}_{U_i}}(\iota_i^*\mathcal{M}, \mathcal{M}_i)$$

supplied by the adjunction $\iota_i^* \dashv (\iota_i)_*$. The stated equivalent description is the unit–counit form of this transpose. $\square$

Theorem 1449 (Effective descent: the chart restriction morphism is an isomorphism)**.** *Assume the transition family (g_{ij}) satisfies the cocycle conditions (C1) and (C2) of [1479]. Then for every chart index i the restriction morphism $\mathrm{r}_i : \iota_i^*\mathcal{M} \to \mathcal{M}_i$ ([1448]) is an isomorphism of sheaves of $\mathcal{O}_{U_i}$-modules.*

Proof. The chart pullback ι_i^* is naturally isomorphic to the site-level restriction functor (1111), which is a right adjoint; consequently ι_i^* *preserves limits*. Applying it to the descent equalizer $\mathcal{M} = \mathrm{eq}(a,b)$ exhibits $\iota_i^*\mathcal{M}$ as the equalizer of the restricted legs ι_i^*a, ι_i^*b, and applying it to the product $P = \prod_j (\iota_j)_*\mathcal{M}_j$ exhibits ι_i^*P as the product $\prod_j \iota_i^*(\iota_j)_*\mathcal{M}_j$. The overlap base change β_{ij} of 1447 then identifies the j-th factor $\iota_i^*(\iota_j)_*\mathcal{M}_j$ with $(f_{ij})_*((t_{ij} \circ f_{ji})^*\mathcal{M}_j)$. Under these identifications the restricted descent data becomes the data attached to the chart U_i by the cover $\{f_{ij}\}$ of U_i, and the i-th factor (where $j = i$) is, by (C1), $\mathcal{M}_i$ itself.

We produce an inverse $\mathrm{s}_i : \mathcal{M}_i \to \iota_i^*\mathcal{M}$. Because $\iota_i^*\mathcal{M}$ is the equalizer of the restricted legs, it suffices to give a morphism $\mathcal{M}_i \to \iota_i^*P$ – equivalently a compatible family of morphisms into the factors – that equalizes ι_i^*a and ι_i^*b. The j-th component is

$$\mathcal{M}_i \xrightarrow{\eta_{f_{ij}}} (f_{ij})_* f_{ij}^*\mathcal{M}_i \xrightarrow{(f_{ij})_* g_{ij}} (f_{ij})_* (t_{ij} \circ f_{ji})^*\mathcal{M}_j \xrightarrow{\beta_{ij}^{-1}} \iota_i^*(\iota_j)_*\mathcal{M}_j,$$

the unit of the pullback–pushforward adjunction along f_{ij}, followed by the pushforward of the transition isomorphism g_{ij}, followed by the inverse of the overlap base change. That this family equalizes the two restricted legs is exactly the triple-overlap multiplicativity (C2), transported through β; and the composite $\mathrm{r}_i \circ \mathrm{s}_i = \mathrm{id}_{\mathcal{M}_i}$ reduces, via the $j = i$ component, to the self-identity (C1) together with the counit triangle identity of the adjunction. The reverse composite $\mathrm{s}_i \circ \mathrm{r}_i = \mathrm{id}$ holds because both sides are equalizer maps with the same components into P; two such agree once their factors agree, which is the separatedness of the sheaf equalizer over the chart cover. The limit computation underlying these equalities is the sheaf Mayer–Vietoris pullback square (1107), and the final identification of sections is the uniqueness-over-a-cover principle (1108). Hence r_i is an isomorphism. $\square$

23.3 Construction of the inverse: the descent obligations

This section records, as sublemmas of 1449, the internal machinery of the proof: the structure-sheaf compatibilities feeding the overlap base change β_{ij}, the bridges between the geometric and the site-level pullback–pushforward adjunctions, the explicit construction of the candidate inverse s_i as an equalizer lift, and the three obligations that consume the cocycle conditions (C1)/(C2). Every block carries a one-to-one Lean declaration.

23.3.1 Section-level compatibilities of the overlap square

Lemma 1450 (Transport of appLE along equal morphisms)**.** *Let $f, g : A \to B$ be morphisms of schemes with $f = g$, let U be an open of B and W an open of A with $W \subseteq f^{-1}(U)$ (equivalently $W \subseteq g^{-1}(U)$). Then the two induced section maps $\Gamma(B, U) \to \Gamma(A, W)$ along f and along g coincide: $f^{\mathrm{LE}}_{U,W} = g^{\mathrm{LE}}_{U,W}$.*

Proof. Substituting the equality $f = g$ reduces both sides to the same map; the two open-inclusion witnesses $W \subseteq f^{-1}(U)$ and $W \subseteq g^{-1}(U)$ are proof-irrelevant. ☐

Lemma 1451 (Structure-sheaf compatibility of the chart–overlap square)**.** *For chart indices i, j and an open $V \subseteq U_i$, the two composite section maps*

$$\Gamma(U_i, V) \longrightarrow \Gamma(U_j,\ (t_{ij} \circ f_{ji})(f_{ij}^{-1}(V)))$$

of the chart–overlap square agree. The first goes "through the glued scheme": the inverse of the comparison isomorphism of ι_i on V, the section map of ι_j over $\iota_i(V)$, and the restriction along the opens identity $\iota_j^{-1}(\iota_i(V)) = (t_{ij} \circ f_{ji})(f_{ij}^{-1}(V))$. The second goes "through the overlap": the section map of f_{ij} over V followed by the inverse comparison isomorphism of $t_{ij} \circ f_{ji}$.

Proof. Both composites are appLEs of the two composite morphisms of the square, which are equal by the glue condition $(t_{ij} \circ f_{ji}) \circ \iota_j = f_{ij} \circ \iota_i$ of 1479. Moving the two comparison isomorphisms to appLE form (using appIso as an appLE, the composition law $g^{\mathrm{LE}} \circ f^{\mathrm{LE}} = (f \circ g)^{\mathrm{LE}}$, and the absorption of presheaf restrictions into the open-inclusion witness) reduces the claim to the equality of the two appLEs of the equal composites, which is 1450. ☐

Lemma 1452 (Sections of the overlap base change, inverse side)**.** *For a sheaf $\mathcal{N}$ on U_j and an open $V \subseteq U_i$, the inverse of the overlap base change β_{ij}^{-1}, evaluated on sections over V, is the restriction map of $\mathcal{N}$ along the opens identity $\iota_j^{-1}(\iota_i(V)) = (t_{ij} \circ f_{ji})(f_{ij}^{-1}(V))$ of 1447:*

$$(\beta_{ij}^{-1})_{\mathcal{N}}\big|_V \;=\; \mathcal{N}\big(\, (t_{ij} \circ f_{ji})(f_{ij}^{-1}V) \;=\; \iota_j^{-1}(\iota_i V)\,\big)\,.$$

Proof. Each of the four functorial factors of β_{ij} (1447) acts on sections over V either as the identity or as a single presheaf restriction along the opens identity; composing them leaves exactly the named restriction map. The verification is pointwise on sections. ☐

Lemma 1453 (Sections of the overlap base change, forward side)**.** *Symmetrically, the forward overlap base change β_{ij}, evaluated on sections over an open $V \subseteq U_i$, is the restriction map of $\mathcal{N}$ along the reverse opens identity $\iota_j^{-1}(\iota_i(V)) = (t_{ij} \circ f_{ji})(f_{ij}^{-1}(V))$.*

Proof. Identical to 1452 with the roles of source and target opens exchanged: pointwise on sections, the composite of the four factors is the single presheaf restriction along the (reversed) opens identity. ☐

23.3.2 Bridges between the geometric and the site-level adjunction

The chart pullback ι^* is built by sheafification and carries the *geometric* adjunction $\iota^* \dashv \iota_*$; along an open immersion the *site-level* restriction functor ι^{res} carries a second adjunction with concrete unit and counit. They share the right adjoint ι_*, and 1111 is the uniqueness comparison of left adjoints between them. The next blocks transport units, counits and congruence casts across that comparison, so that the descent obligations can be discharged by section-level computations.

Lemma 1454 (Unit comparison across the adjunctions)**.** *For an open immersion f and a sheaf $\mathcal{N}$ on the target, the geometric unit is the site-level unit followed by the pushforward of the left-adjoint comparison isomorphism:*

$$\eta^{\mathrm{res}}_{\mathcal{N}} \;\circ\; f_*((\mathit{comparison})_{\mathcal{N}}) \;=\; \eta^{\mathrm{geo}}_{\mathcal{N}}.$$

Proof. This is the general identity relating the unit of an adjunction to the unit of any other adjunction with the same right adjoint, transported by the uniqueness isomorphism of left adjoints (1111). □

Lemma 1455 (Counit comparison across the adjunctions)**.** *Dually, the left-adjoint comparison isomorphism on $f_*\mathcal{N}$ followed by the geometric counit equals the site-level counit:*

$$(\mathit{comparison})_{f_*\mathcal{N}} \;\circ\; \varepsilon^{\mathrm{geo}}_{\mathcal{N}} \;=\; \varepsilon^{\mathrm{res}}_{\mathcal{N}}.$$

Proof. The dual of 1454: the counit of one adjunction is conjugated to the counit of the other by the uniqueness comparison of left adjoints. □

Lemma 1456 (Congruence cast at the pullback level)**.** *For equal morphisms $\varphi = \psi : X \to Y$ and a sheaf $\mathcal{N}$ on Y, the congruence isomorphism $\varphi^*\mathcal{N} \cong \psi^*\mathcal{N}$ induced by $\varphi = \psi$ is the canonical transport (the* eqToHom *of the object-level equality $\varphi^*\mathcal{N} = \psi^*\mathcal{N}$).*

Proof. Substituting $\varphi = \psi$ reduces the congruence cast to the identity, which is the canonical transport of a reflexivity. □

Lemma 1457 (Trivial site-level congruence cast)**.** *For an open immersion φ the site-level congruence cast associated to the reflexivity $\varphi = \varphi$ is the identity natural transformation.*

Proof. Pointwise on sections the cast is the presheaf restriction along a self-equality of opens, which is the identity by functoriality. □

Lemma 1458 (Comparison intertwines the two congruence casts)**.** *For equal open immersions $\varphi = \psi : X \to Y$ the left-adjoint comparison isomorphism intertwines the site-level and the pullback-level congruence casts:*

$$(\mathit{comparison}_{\varphi})_{\mathcal{N}} \;\circ\; (\varphi^* \!=\! \psi^*)_{\mathcal{N}} \;=\; (\varphi^{\mathrm{res}} \!=\! \psi^{\mathrm{res}})_{\mathcal{N}} \;\circ\; (\mathit{comparison}_{\psi})_{\mathcal{N}}.$$

Proof. Substituting $\varphi = \psi$ collapses both congruence casts to identities (1456, 1457), and the two comparison isomorphisms then coincide. □

23.3.3 The candidate inverse

Definition 1459 (Overlap base change in pullback form)**.** Conjugating the overlap base change β_{ij} (1447) on both sides by the comparison 1111 between the site-level restriction and the geometric pullback yields the isomorphism, in geometric form,

$$\gamma_{ij} \;:\; \iota_i^*((\iota_j)_* \mathcal{M}_j) \;\xrightarrow{\sim}\; (f_{ij})_*((t_{ij} \circ f_{ji})^* \mathcal{M}_j),$$

evaluated at the input sheaf $\mathcal{M}_j$.

Definition 1460 (The j-th component of the candidate inverse)**.** Given the transition isomorphisms $g_{ij} : f_{ij}^*\mathcal{M}_i \xrightarrow{\sim} (t_{ij} \circ f_{ji})^*\mathcal{M}_j$, the j-th component of the candidate inverse s_i is the composite

$$\mathcal{M}_i \xrightarrow{\eta_{f_{ij}}} (f_{ij})_* f_{ij}^*\mathcal{M}_i \xrightarrow{(f_{ij})_* g_{ij}} (f_{ij})_* (t_{ij} \circ f_{ji})^*\mathcal{M}_j \xrightarrow{\gamma_{ij}^{-1}} \iota_i^*((\iota_j)_*\mathcal{M}_j),$$

i.e. the geometric unit along f_{ij}, the pushforward of the transition g_{ij}, and the inverse of γ_{ij} (1459).

Definition 1461 (The candidate-inverse family)**.** Assembling the components 1460 through the product-preservation comparison of ι_i^* on the pushforward product $P = \prod_j (\iota_j)_*\mathcal{M}_j$ (1446) gives a single morphism

$$\mathcal{M}_i \;\longrightarrow\; \iota_i^* P,$$

the candidate inverse before it is corrected to land in the equalizer $\iota_i^*\mathcal{M}$.

Lemma 1462 (The candidate family recovers each chart component)**.** *For chart indices i, j, composing the candidate family with the pullback of the j-th product projection recovers its j-th component:*

$$\mathrm{s}_i^{\mathrm{fam}} \circ \iota_i^*(\pi_j) = \mathrm{s}_i^{(j)}.$$

This is the component formula for the product-preservation comparison used to define 1461.

23.3.4 The triple-overlap toolkit

The first descent obligation (1465) reduces, factor by factor of the restricted overlap product, to a single equation over a *triple overlap*. This subsection records that machinery. Fix three chart indices i, p, q and form the triple overlap

$$V_{ipq} \;=\; V_{ip} \times_{U_i} V_{iq},$$

with first projection $\mathrm{fst} : V_{ipq} \to V_{ip}$. Two morphisms out of V_{ipq} organise everything: the *chart leg* $q_{pq} = \mathrm{fst} \circ f_{ip} : V_{ipq} \to U_i$ (down to U_i), and the *transport leg* $\tau = t'_{ipq} \circ \mathrm{fst}_{pq} : V_{ipq} \to V_{pq}$ (across to the pair overlap, through the cocycle transport t'_{ipq}). The square

$$\begin{array}{ccc} V_{ipq} & \xrightarrow{\tau} & V_{pq} \\ \downarrow q_{pq} & & \downarrow f_{pq} \circ \iota_p \\ U_i & \xrightarrow{\iota_i} & X \end{array}$$

is the triple analogue of the chart–overlap square of 1447.

Lemma 1463 (Triple-overlap bridges of the glue datum)**.** *On the triple overlap $V_{ijk} = V_{ij} \times_{U_i} V_{ik}$ the following three identities of morphisms hold, lining up the endpoints of the three base-change transports of the cocycle equation (C2):*

1. (source) *the two projections of* V_{ijk} *to* V_{ij} *and* V_{ik}*, followed by the overlap immersions* f_{ij} *and* f_{ik} *into* U_i*, agree;*

2. (middle) *the* ij*-transport's target leg* $(t_{ij} \circ f_{ji})$ *and the* jk*-transport's source leg (through the cocycle transport* t'_{ijk} *and* f_{jk}*) agree as morphisms to* U_j*;*

3. (target) *the composite* jk*-after-*ij *transport's target leg* $(t_{jk} \circ f_{kj})$ *and the* ik*-transport's target leg* $(t_{ik} \circ f_{ki})$ *agree as morphisms to* U_k*.*

Proof. The source identity is the universal property (pullback condition) of V_{ijk}. The middle identity follows from the transition factorisation t'_{ijk} in the glue datum together with the pullback condition. The target identity – the heart of the cocycle – combines the same transition factorisation, the pullback condition, the involutivity $t_{ik} \circ t_{ki} = \mathrm{id}$ and the cocycle condition of the glue datum. □

Lemma 1464 (Pair component of the equalizing condition)**.** *For chart indices* i, p, q*, the* p*-th candidate component followed by the restricted first descent-leg factor equals the* q*-th candidate component followed by the restricted second descent-leg factor. After transposing to the triple overlap* V_{ipq}*, this is precisely the cocycle multiplicativity condition (C2), with the endpoint identifications supplied by 1463.*

23.3.5 The first descent obligation

Lemma 1465 (The candidate-inverse family equalizes the restricted legs (C2 transported))**.** *The family 1461 equalizes the two restricted descent legs* $\iota_i^* a$ *and* $\iota_i^* b$*; equivalently, it lands in the restricted equalizer* $\iota_i^* \mathcal{M}$*. This is the first descent obligation, where the triple-overlap multiplicativity (C2) of 1479 is consumed.*

Proof. It suffices to check equality after projecting to each factor (p, q) of the restricted overlap product, through the product-preservation comparison of ι_i^* (the same comparison discipline as 1467). Folding each restricted leg against the (p, q)-projection – splitting off the p- and q-th chart projections and cancelling the comparison via 1462 – turns the (p, q)-component into the equation

$$\mathrm{s}_i^{(p)} \circ \iota_i^*(a_{(p,q)}) = \mathrm{s}_i^{(q)} \circ \iota_i^*(b_{(p,q)}),$$

which is precisely the pair-component obligation 1464, where the triple-overlap multiplicativity (C2) is consumed. Assembling these componentwise equalities back through the comparison gives the equalizing condition. □

Definition 1466 (The candidate inverse s_i)**.** Because ι_i^* preserves the descent equalizer, the equalizing family 1461 (equalizing by 1465) lifts uniquely to a morphism

$$\mathrm{s}_i : \mathcal{M}_i \longrightarrow \iota_i^* \mathcal{M},$$

the candidate inverse of the chart restriction morphism.

Lemma 1467 (Comparison compatibility of the restricted projection)**.** *Through the equalizer- and product-preservation comparisons of* ι_i^**, the restricted equalizer inclusion of* $\iota_i^* \mathcal{M}$ *into* $\iota_i^* P$ *followed by the* j*-th product projection equals the chart pullback* $\iota_i^*(\pi_j)$ *of the* j*-th descent-equalizer projection (1446).*

Proof. Pure limit bookkeeping: the equalizer- and product-comparison isomorphisms associated to a limit-preserving functor send the restricted inclusion and the restricted projection to the pullbacks of the inclusion and the projection respectively, and the composite of pullbacks is the pullback of the composite, which is $\iota_i^*(\pi_j)$. □

Lemma 1468 (The candidate inverse recovers the components)**.** *The candidate inverse 1466 followed by the restricted j-th projection $\iota_i^*(\pi_j)$ recovers the j-th component 1460:*

$$\mathrm{s}_i \;\circ\; \iota_i^*(\pi_j) \;=\; \mathrm{s}_i^{(j)}.$$

Proof. Unfolding the lift 1466 and applying 1467, the equalizer- and product-preservation comparisons cancel against the lift, leaving the defining j-th projection of the family 1461, namely $\mathrm{s}_i^{(j)}$. □

Lemma 1469 (Joint detection of morphisms into the restricted glued sheaf)**.** *Two morphisms $u, v : \mathcal{Z} \to \iota_i^*\mathcal{M}$ that agree after every restricted projection $\iota_i^*(\pi_j)$ are equal.*

Proof. The restricted equalizer inclusion is a monomorphism and the restricted product projections are jointly monic, both through the preservation comparisons; so it suffices to test against $\iota_i^*(\pi_j)$ for every j, which is the hypothesis via 1467. □

Lemma 1470 (Joint faithfulness of the chart restrictions)**.** *Two morphisms $u, v : \mathcal{W} \to \mathcal{W}'$ of sheaves of modules on the glued scheme X of a glue datum (1479) that agree after pullback $\iota_i^*u = \iota_i^*v$ to every chart are equal. This is the characterisation of a glued sheaf up to unique isomorphism: a morphism out of the glued sheaf is determined by its chart-local restrictions.*

Proof. This is the separation half of the sheaf condition over the chart cover $\{\iota_i\}$: sections of $\mathcal{W}'$ are detected on the chart-image opens $\iota_i''(\iota_i^{-1}O)$, which cover any open O by the joint surjectivity of the chart immersions of the glue datum. Hence two morphisms agreeing on every chart agree on a covering family of opens, so they agree. □

23.3.6 The three cocycle obligations

Lemma 1471 (Self-component collapses to the identity (C1 + counit triangle))**.** *The (i,i)-component 1460, transposed back along the chart immersion ι_i (post-composed with the geometric counit), is the identity of $\mathcal{M}_i$:*

$$\mathrm{s}_i^{(i)} \;\circ\; \varepsilon^{\mathrm{geo}}_{\mathcal{M}_i} \;=\; \mathrm{id}_{\mathcal{M}_i}.$$

This is the second descent obligation, consuming the self-identity (C1).

Proof. Since f_{ii} factors as $t_{ii} \circ f_{ii}$ (because $t_{ii} = \mathrm{id}$), (C1) of 1479 says the transition g_{ii} is the canonical congruence cast; through 1456 and 1458 the three middle factors of the component collapse to a single site-level congruence cast. Rewriting the geometric unit by 1454 and the geometric counit by 1455 turns the whole composite into the site-level unit, the congruence cast, the section form of β_{ii}^{-1} (1452), and the site-level counit. On sections over an open this is a cycle of presheaf restrictions along opens identities whose composite open-inclusion is the identity; folding the restrictions and using proof-irrelevance of the open inclusions gives the identity. □

Lemma 1472 (Transpose of the overlap base change (mate core))**.** *The adjoint transpose along ι_i of the pullback-form overlap base change γ_{ij} (1459) is the canonical four-functor comparison of the glue square:*

$$\widehat{\gamma_{ij}} \;=\; (\iota_j)_*(\eta_{t_{ij} \circ f_{ji}}) \;\circ\; \mathrm{pushComp} \;\circ\; \mathrm{pushCongr} \;\circ\; \mathrm{pushComp}^{-1},$$

the unit along $t_{ij} \circ f_{ji}$ pushed forward along ι_j, regrouped by the pushforward-composition comparison, reindexed by the pushforward-congruence along the glue condition $(t_{ij} \circ f_{ji}) \circ \iota_j = f_{ij} \circ \iota_i$, and ungrouped by the inverse pushforward-composition. The identity is free of the cocycle hypotheses: it is pure site-level content.

Proof. Expand the adjoint transpose of γ_{ij}: the left-adjoint comparison isomorphisms (1111) conjugating the chart factor cancel against the unit/counit, leaving the bare site-level overlap base change β_{ij} (1447). The unit bridge at $t_{ij} \circ f_{ji}$ and the three naturality squares of the pushforward-composition and pushforward-congruence comparisons rewrite the transpose into the stated four-factor composite up to whiskering and reassociation. The residual site-level core is checked on sections over an open: by 1453 the section form of β_{ij} is a presheaf restriction along an opens identity, and the remaining composite is a cycle of such restrictions whose total open inclusion is the identity, folded by proof-irrelevance exactly as in the closing step of 1471. ☐

Lemma 1473 (Mate recognition of the factor transpose)**.** *For any morphism $m : \mathcal{W} \to (\iota_j)_* \mathcal{M}_j$ on the glued scheme, the conjugated transport of the inverse transpose of m along the glue square equals the inverse transpose, along f_{ij}, of $\iota_i^*(m)$ followed by γ_{ij} (1459). In other words, the factor transpose 1472 is the mate of γ_{ij} through the pseudofunctor structure of pullback.*

Proof. Both sides are computed by transposing twice. By injectivity of the two transpose bijections it suffices to compare the double transposes. Naturality of the transpose in its left argument peels m off the right-hand side; the conjugate of the pullback-composition isomorphism along the glue condition is the inverse pushforward-composition isomorphism; and folding the unit pair and reindexing along the glue condition of 1479 reduces the identity to the factor transpose 1472. ☐

Lemma 1474 (Overlap compatibility of the restriction morphisms)**.** *The (i, j)-component of the descent-equalizer condition, transposed to the pullback level, states that over the overlap V_{ij} the two restriction morphisms r_i and r_j (1448) differ exactly by the transition isomorphism g_{ij} (through the pseudofunctor congruence casts).*

Proof. The glued sheaf is the descent equalizer, so its i- and j-projections satisfy the overlap-agreement condition of 1446. Transposing that condition along ι_i and ι_j and folding the transposes back through the adjunction bijection turns it into the stated identity between the overlap pullbacks of r_i and r_j, conjugated by the glue-condition casts of 1479 and corrected by g_{ij}. ☐

Lemma 1475 (Pair condition transposed: restriction composed with the component)**.** *The chart restriction morphism r_i (1448) followed by the j-th candidate-inverse component 1460 equals the restricted j-th projection:*

$$\mathrm{r}_i \;\circ\; \mathrm{s}_i^{(j)} \;=\; \iota_i^*(\pi_j).$$

This is the third descent obligation; together with 1468 and 1469 it gives $\mathrm{r}_i \circ \mathrm{s}_i = \mathrm{id}$.

Proof. Unfold the component 1460. By the overlap compatibility 1474 the overlap pullback of r_i transported across g_{ij} is the inverse mate transpose of the restricted projection $\iota_i^*(\pi_j)$ through γ_{ij} (1473). Substituting this into the unit-naturality square of r_i along f_{ij} and folding the resulting transpose cancels the factor isomorphism, leaving exactly $\iota_i^*(\pi_j)$. ☐

23.3.7 Joint faithfulness of chart restrictions

Definition 1476 (The restriction isomorphism of the glued sheaf)**.** Let $\mathcal{M}$ be the glued sheaf of 1479, with the transition family (g_{ij}) satisfying (C1)/(C2). For each chart index i the restriction morphism r_i (1448) is promoted to the *restriction isomorphism*

$$\rho_i \; : \; \iota_i^* \mathcal{M} \xrightarrow{\sim} \mathcal{M}_i,$$

the canonical identification of the chart restriction of the glued sheaf with the i-th input sheaf, packaged from 1449. Its underlying morphism is the adjoint transpose of the i-th descent-equalizer projection, and over each overlap V_{ij} the two restrictions ρ_i, ρ_j differ exactly by the transition isomorphism g_{ij}.

Chapter 24

The tautological quotient and the universal property of $\mathrm{Gr}(r, d)$

The previous chapter constructed the Grassmannian scheme $\mathrm{Gr}(r, d)$ over $\mathbb{Z}$ by gluing the affine charts U^I (1357) along the transition isomorphisms (1371) satisfying the cocycle condition (1393), and proved that the structure morphism $\mathrm{Gr}(r, d) \to \operatorname{Spec} \mathbb{Z}$ is separated (1429) and proper (1445). This chapter adds the next layer: the *tautological rank-d quotient* $u : \mathcal{O}^r_{\mathrm{Gr}(r,d)} \twoheadrightarrow \mathcal{U}$ of the trivial rank-r sheaf, and the *universal property* that makes $\mathrm{Gr}(r, d)$ the fine moduli space of rank-d locally free quotients of $\mathcal{O}^r$ – equivalently, the representing scheme of the Quot functor of the trivial sheaf.

Throughout, $r \geq d \geq 1$ are the fixed integers of the previous chapter, and for a size-d subset $I \subseteq \{1, \dots, r\}$ we write R^I for the chart ring $\mathbb{Z}[X^I]$ of $U^I = \operatorname{Spec} R^I$ and R^I_J for the localisation along $P^I_J = \det(X^I_J)$ defining the overlap $U^I_J \subseteq U^I$.

24.1 Infrastructure: locally free sheaves and module gluing

The constructions of this chapter – the universal quotient sheaf $\mathcal{U}$, the tautological quotient u, and the functor of points – rest on two pieces of foundational machinery for sheaves of modules. The first is the predicate "locally free of rank d" (1137), which records that a sheaf of modules is, on the members of some open cover, isomorphic to the trivial bundle $\mathcal{O}^d$; it is the condition cut out in the definition of the Grassmannian functor. That predicate is the project-local SheafOfModules.IsLocallyFreeOfRank already set up in the Quot-scheme chapter (1137), which this chapter reuses verbatim; the only foundational machinery genuinely new to this chapter is the second piece: a gluing primitive that descends a sheaf of modules (and a morphism of such sheaves) from per-chart data along an open cover, given a transition cocycle compatible with the cover's own gluing; it is the vehicle that assembles the per-chart free sheaves $\mathcal{O}^d_{U^I}$ and the per-chart quotients u^I into the global $\mathcal{U}$ and u, and is set up below.

Stating the triple-overlap multiplicativity of the descent datum, however, first requires a small piece of base-change machinery for the pullback functor on sheaves of modules: the three transition isomorphisms attached to a triple of charts live a priori on three different overlaps, and to compare their composite they must be transported to a single common overlap. The next two blocks supply exactly this transport – the pseudofunctorial comparison of iterated pullbacks (1477) and its packaging along the triple-overlap data of a scheme glue datum (1478) – after which the cocycle condition can be expressed cleanly.

Definition 1477 (Comparison of iterated pullbacks of sheaves of modules)**.** *Provided by Mathlib.* Let $a : X \to Y$ and $b : Y \to Z$ be morphisms of schemes. For a sheaf of $\mathcal{O}_Z$-modules $\mathcal{M}$, pulling back first along b and then along a agrees with pulling back along the composite $b \circ a$: there is a natural isomorphism of sheaves of $\mathcal{O}_X$-modules

$$a^*(b^*\mathcal{M}) \xrightarrow{\sim} (b \circ a)^*\mathcal{M},$$

natural in $\mathcal{M}$, expressing that $\mathcal{M} \mapsto \mathcal{M}$'s pullback is a pseudofunctor on schemes. These comparison isomorphisms satisfy the pentagon (associativity) coherence for a triple of composable morphisms. This is the pullback half of the pseudofunctoriality of sheaves of modules and is supplied by Mathlib.

Lemma 1478 (Base-change transport of a transition isomorphism to a triple overlap)**.** *Let D be a glue datum of schemes (1415) with charts U_i, overlaps V_{ij}, overlap immersions $f_{ij} : V_{ij} \hookrightarrow U_i$, transitions $t_{ij} : V_{ij} \xrightarrow{\sim} V_{ji}$, and triple-overlap data t'_{ijk} with its compatibility identity t_{fac}. For an index triple (i, j, k) write*

$$V_{ijk} \;=\; V_{ij} \times_{U_i} V_{ik}$$

for the common triple overlap, with the two projections $p^{ij}_{ijk} : V_{ijk} \to V_{ij}$ and $p^{ik}_{ijk} : V_{ijk} \to V_{ik}$, and let $t'_{ijk} : V_{ijk} \xrightarrow{\sim} V_{jki}$ be the glue datum's triple transition, satisfying

$$t'_{ijk} \circ p^{jk}_{jki} \;=\; p^{ij}_{ijk} \circ t_{ij} \qquad (t_{\mathrm{fac}}).$$

Given per-chart sheaves of modules $\mathcal{M}_i$ and a transition isomorphism

$$g_{ij} \;:\; f^*_{ij}\,\mathcal{M}_i \xrightarrow{\sim} (t_{ij} \circ f_{ji})^*\,\mathcal{M}_j$$

on the overlap V_{ij} (as in 1479), the base-change transport *of g_{ij} to the triple overlap V_{ijk} is the isomorphism of sheaves of $\mathcal{O}_{V_{ijk}}$-modules*

$$\hat{g}^k_{ij} \;:\; (f_{ij} \circ p^{ij}_{ijk})^*\,\mathcal{M}_i \xrightarrow{\sim} (t_{ij} \circ f_{ji} \circ p^{ij}_{ijk})^*\,\mathcal{M}_j$$

obtained by pulling g_{ij} back along p^{ij}_{ijk} and reassociating the iterated pullbacks via 1477. With the analogous transports $\hat{g}^i_{jk}$ and $\hat{g}^j_{ik}$ (the first taken along t'_{ijk}, so that t_{fac} identifies its source and target sheaves with those of $\hat{g}^k_{ij}$ and $\hat{g}^j_{ik}$), all three transported isomorphisms become endomorphism-composable: $\hat{g}^k_{ij}$ and $\hat{g}^i_{jk}$ land on a common sheaf, and their composite is comparable to $\hat{g}^j_{ik}$ on V_{ijk}.

Proof. The pullback functor on sheaves of modules is a pseudofunctor: for composable scheme morphisms it carries the comparison isomorphisms of 1477, natural in the sheaf and coherent for triple composites. Pulling the transition isomorphism g_{ij} back along the projection $p^{ij}_{ijk} : V_{ijk} \to V_{ij}$ yields an isomorphism between $(p^{ij}_{ijk})^* f^*_{ij}\mathcal{M}_i$ and $(p^{ij}_{ijk})^*(t_{ij} \circ f_{ji})^*\mathcal{M}_j$; applying 1477 to each side reassociates these into the single-pullback form $(f_{ij} \circ p^{ij}_{ijk})^*\mathcal{M}_i$ and $(t_{ij} \circ f_{ji} \circ p^{ij}_{ijk})^*\mathcal{M}_j$, giving $\hat{g}^k_{ij}$. For the (j, k)-transition the same construction is applied along the triple transition t'_{ijk} rather than a bare projection; the glue datum's compatibility identity t_{fac}, $t'_{ijk} \circ p^{jk}_{jki} = p^{ij}_{ijk} \circ t_{ij}$, guarantees that the source of $\hat{g}^i_{jk}$ coincides with the target of $\hat{g}^k_{ij}$ (both are the pullback of $\mathcal{M}_j$ along the same morphism $V_{ijk} \to U_j$), so the two are composable as endomorphism-style isomorphisms over V_{ijk}. The reassociations introduced by 1477 are coherent (they satisfy the pentagon identity), so the composite $\hat{g}^i_{jk} \circ \hat{g}^k_{ij}$ and the third transport $\hat{g}^j_{ik}$ are isomorphisms between the same pair of sheaves on V_{ijk}, and may be compared directly. This is the module-level analogue of the scheme-level triangle $t'_{ijk}/t_{\mathrm{fac}}$ and is the shape in which the multiplicative cocycle condition is stated below. $\square$

Definition 1479 (Gluing a sheaf of modules along an open cover)**.** Let D be a glue datum of schemes (1415), with index set J, charts U_i, pairwise overlaps V_{ij}, open immersions $f_{ij} : V_{ij} \hookrightarrow U_i$, and transition isomorphisms $t_{ij} : V_{ij} \xrightarrow{\sim} V_{ji}$ satisfying the glue datum's cocycle conditions; write X for the glued scheme and $\iota_i : U_i \hookrightarrow X$ for the canonical open immersions of the charts. Suppose given:

- for each $i \in J$, a sheaf of $\mathcal{O}_{U_i}$-modules $\mathcal{M}_i$ on the chart U_i;
- for each pair (i, j), a *transition isomorphism* of sheaves of modules on the overlap V_{ij},

$$g_{ij} \;:\; f_{ij}^{*}\mathcal{M}_i \;\xrightarrow{\sim}\; (t_{ij} \circ f_{ji})^{*}\,\mathcal{M}_j,$$

  comparing the two charts' sheaves after restriction (pullback) to the common overlap V_{ij};

subject to the following module cocycle conditions, compatible with the scheme-level cocycle of D (its triple transition data t_{ij} and the overlap immersions f_{ij}), which are required as hypotheses on the family (g_{ij}):

(C1) *self-identity:* the self-transition is the identity, $g_{ii} = \mathrm{id}$ on $f_{ii}^{*}\mathcal{M}_i$ (with V_{ii} and t_{ii} the diagonal data of D);

(C2) *triple-overlap multiplicativity:* over each common triple overlap $V_{ijk} = V_{ij} \times_{U_i} V_{ik}$ the base-change transports (1478) of the three transition isomorphisms satisfy

$$\hat{g}^{i}_{jk} \circ \hat{g}^{k}_{ij} \;=\; \hat{g}^{j}_{ik},$$

where $\hat{g}^{k}_{ij}$, $\hat{g}^{i}_{jk}$, $\hat{g}^{j}_{ik}$ are the transports of g_{ij}, g_{jk}, g_{ik} to V_{ijk} through the scheme-level triple transition t'_{ijk} and its compatibility t_{fac}, reassociated by the pullback comparison 1477 so that all three land on the same sheaf – the multiplicative, GL_d-valued cocycle condition.

Conditions (C1) and (C2) are exactly the descent datum for the Zariski cover $\{\iota_i\}$; it is subject to them that the gluing is well-defined. Out of this data the construction produces a glued sheaf of $\mathcal{O}_X$-modules $\mathcal{M}$ on the glued scheme X.

Construction. The category $\mathrm{SheafOfModules}(\mathcal{O}_X)$ of sheaves of $\mathcal{O}_X$-modules has all limits, and both pushforward along a morphism and these limits preserve the sheaf condition; the glued sheaf $\mathcal{M}$ is therefore built directly as a limit – an *equalizer of pushforwards* – rather than through a hand-built presheaf of compatible families. Concretely, form the two parallel maps of sheaves of $\mathcal{O}_X$-modules

$$\prod_i (\iota_i)_*\,\mathcal{M}_i \;\overset{a}{\underset{b}{\rightrightarrows}}\; \prod_{i,j} (\iota_i \circ f_{ij})_*\,(f_{ij}^{*}\mathcal{M}_i),$$

where, on the (i, j)-component, leg a pushes the i-th factor forward along the overlap immersion f_{ij} using the unit of the pullback–pushforward adjunction together with the pushforward-composition comparison $(\iota_i)_* \cong (\iota_i \circ f_{ij})_* \circ f_{ij}^{*}$ (the dual of 1477), and leg b pushes the j-th factor forward and additionally transports it across the inverse transition isomorphism $(g_{ij})^{-1}$ and the glue datum's compatibility identity, so that both legs land on the common overlap sheaf $(\iota_i \circ f_{ij})_*(f_{ij}^{*}\mathcal{M}_i)$. The glued sheaf is the equalizer

$$\mathcal{M} \;=\; \mathrm{eq}(a, b) \;\hookrightarrow\; \prod_i (\iota_i)_*\,\mathcal{M}_i,$$

the sub-sheaf of tuples of pushed-forward chart sections whose two overlap restrictions agree – precisely the compatible families, now realised as a categorical limit rather than a bespoke presheaf. The cocycle conditions (C1)/(C2) on the family (g_{ij}) are *not* consumed in forming this equalizer object: the limit exists for any family of transition maps. They are instead what make the equalizer descend correctly, and are exactly the hypotheses pinned down downstream when the restriction isomorphisms 1480 are produced. This specialises effective descent of sheaves of modules to the concrete open cover $\{\iota_i : U_i \hookrightarrow X\}$ supplied by a scheme glue datum, and is the form in which the Grassmannian's tautological bundle and quotient are built.

The glued sheaf comes with three further structures, each recorded as its own block downstream: the *restriction isomorphisms* $\rho_i : \iota_i^*\mathcal{M} \xrightarrow{\sim} \mathcal{M}_i$ compatible with the g_{ij} over the overlaps 1480; the *characterisation up to unique isomorphism* by the restriction property 1470; and the *descent of morphisms* – a family $\phi_i : \mathcal{M}_i \to \mathcal{N}_i$ commuting with the transition isomorphisms glues to a unique $\phi : \mathcal{M} \to \mathcal{N}$ with $\rho_i^{\mathcal{N}} \circ \iota_i^*\phi = \phi_i \circ \rho_i^{\mathcal{M}}$ 1481.

Definition 1480 (Restriction isomorphisms of the glued sheaf)**.** Let $\mathcal{M}$ be the glued sheaf of 1479. For each chart index i there is a *restriction isomorphism* of sheaves of $\mathcal{O}_{U_i}$-modules

$$\rho_i \;:\; \iota_i^*\,\mathcal{M} \;\xrightarrow{\;\sim\;}\; \mathcal{M}_i,$$

compatible with the transition isomorphisms: over each overlap V_{ij} the two restrictions ρ_i and ρ_j, transported to the common overlap, differ exactly by g_{ij}.

Proof. By the equalizer-of-pushforwards description 1479, a section of $\mathcal{M}$ over an open of the form $\iota_i^{-1}V$, restricted to the chart U_i, is recovered from and recovers its i-th component $s_i \in \Gamma(\mathcal{M}_i, \cdot)$ – the i-th factor of the product $\prod_i (\iota_i)_*\mathcal{M}_i$ into which the equalizer embeds; the chart-restriction bridge 1122 identifies the pullback $\iota_i^*\mathcal{M}$ with this restriction, and the i-th projection $(s_\bullet) \mapsto s_i$ is the isomorphism ρ_i (with inverse extending a chart section by the overlap relation). The overlap compatibility is the defining relation of the compatible families: over V_{ij} the components s_i, s_j are related by g_{ij}, which is precisely the statement that ρ_i and ρ_j differ by g_{ij} after transport to V_{ij}; the triple-overlap coherence needed for this to be consistent is (C2), whose transports are aligned by 1478 and 1463. ☐

Definition 1481 (Descent of morphisms along the cover)**.** Let $(\mathcal{M}_i, g_{ij})$ and $(\mathcal{N}_i, h_{ij})$ be two glued data over the same glue datum D, with glued sheaves $\mathcal{M}, \mathcal{N}$ and restriction isomorphisms $\rho_i^{\mathcal{M}}, \rho_i^{\mathcal{N}}$. Given a family of $\mathcal{O}_{U_i}$-linear morphisms $\phi_i : \mathcal{M}_i \to \mathcal{N}_i$ commuting with the transition isomorphisms on every overlap (i.e. h_{ij} intertwines the pullbacks of ϕ_i and ϕ_j over V_{ij}), there is a unique morphism of sheaves of $\mathcal{O}_X$-modules $\phi : \mathcal{M} \to \mathcal{N}$ with

$$\rho_i^{\mathcal{N}} \circ \iota_i^*\phi \;=\; \phi_i \circ \rho_i^{\mathcal{M}} \qquad \text{for every } i.$$

Proof. On each chart the prescribed identity defines a local morphism $(\rho_i^{\mathcal{N}})^{-1} \circ \phi_i \circ \rho_i^{\mathcal{M}} : \iota_i^*\mathcal{M} \to \iota_i^*\mathcal{N}$. The commutation of the ϕ_i with the transition isomorphisms (g_{ij}, h_{ij}), together with the overlap compatibility of the ρ-families (1480), makes these local morphisms agree on every overlap V_{ij}. By the locality and gluing of sections (146) applied to the internal hom they promote to a unique global morphism $\phi : \mathcal{M} \to \mathcal{N}$ restricting to each ϕ_i through the ρ_i – the descent of Hom along the cover. ☐

24.1.1 Pullback of free sheaves along an arbitrary morphism

The functor of points (1518) acts on morphisms by pullback, so it needs the pullback of the trivial bundle to be again trivial – $\varphi^*\mathcal{O}_T^{\oplus I} \cong \mathcal{O}_{T'}^{\oplus I}$ – for an *arbitrary* scheme morphism φ. Mathlib's pullback-of-free comparison applies only when the underlying site functor $\mathrm{Opens.map}\,\varphi$ is *final*; the following three blocks remove that restriction for every morphism, and the resulting pullback machinery preserves rank-d local freeness.

Lemma 1482 (The opens-pullback functor is final)**.** *For any morphism of schemes $\varphi : T' \to T$, the inverse-image functor on opens* $\mathrm{Opens.map}\,\varphi : \mathrm{Opens}(T) \to \mathrm{Opens}(T')$*, $U \mapsto \varphi^{-1}U$, is* final.

Proof. Finality means that for every open $V \subseteq T'$ the structured-arrow category of opens $U \subseteq T$ with $V \leq \varphi^{-1}U$ is connected. That category is nonempty and has the terminal object $U = \top$ – one always has $V \leq \varphi^{-1}\top = T'$, and every object admits a unique arrow to $\top$ – and a category with a terminal object is connected. □

Lemma 1483 (Pullback of a free sheaf is free)**.** *For any scheme morphism $\varphi : T' \to T$ and index type I, there is a canonical isomorphism of sheaves of $\mathcal{O}_{T'}$-modules*

$$\varphi^*(\mathcal{O}_T^{\oplus I}) \xrightarrow{\sim} \mathcal{O}_{T'}^{\oplus I}.$$

Proof. Mathlib's pullback-of-free comparison produces this isomorphism whenever the site functor $\mathrm{Opens.map}\,\varphi$ is final; that finality is 1482, which holds for every morphism, so the comparison applies unconditionally. □

24.2 The tautological quotient on the charts

Infrastructure: scalar endomorphisms. Realising a matrix of regular functions as a morphism of free sheaves of modules has no off-the-shelf primitive in Mathlib; it is assembled from two small project-local helpers that turn a global function into a scalar endomorphism of the structure sheaf, after which the matrix is reconstituted entry-by-entry over a finite biproduct.

Definition 1484 (Global section of the unit module)**.** Let X be a scheme and $a \in \Gamma(X, \mathcal{O}_X)$ a global section of the structure sheaf. Viewing $\mathcal{O}_X$ as the unit module $\mathbf{1} \in \mathrm{SheafOfModules}(\mathcal{O}_X)$ (1277), the section a determines a global section of $\mathbf{1}$: on each open $V \subseteq X$ take the restriction $a|_V \in \Gamma(V, \mathcal{O}_X)$, the family $(a|_V)_V$ being compatible with the restriction maps because restriction is functorial. This is the section of the unit sheaf attached to a global function.

Definition 1485 (Scalar endomorphism of $\mathcal{O}_X$)**.** For $a \in \Gamma(X, \mathcal{O}_X)$ the *scalar endomorphism* $\mathrm{scalarEnd}(a) : \mathcal{O}_X \to \mathcal{O}_X$ is the $\mathcal{O}_X$-linear endomorphism of the unit module given by multiplication by a. It is obtained from the global section of 1484 under the canonical identification of endomorphisms of the unit module $\mathbf{1}$ with its global sections, $\mathrm{End}(\mathbf{1}) \cong \Gamma(X, \mathbf{1})$; concretely, on each open V it acts as multiplication by $a|_V$. These scalar endomorphisms are the matrix-entry building blocks of the chart quotient.

Lemma 1486 (Scalar endomorphism of the unit is the identity)**.** *The scalar endomorphism attached to the unit function $1 \in \Gamma(X, \mathcal{O}_X)$ is the identity of the unit module:* $\mathrm{scalarEnd}(1) = \mathrm{id}_{\mathbf{1}}$.

Proof. Multiplication by 1 is the identity on every section, so under the identification $\mathrm{End}(\mathbf{1}) \cong \Gamma(X, \mathbf{1})$ the section 1 corresponds to $\mathrm{id}_{\mathbf{1}}$. □

Lemma 1487 (Scalar endomorphism of the zero function is zero)**.** *The scalar endomorphism attached to the zero function $0 \in \Gamma(X, \mathcal{O}_X)$ is the zero endomorphism:* $\mathrm{scalarEnd}(0) = 0$.

Proof. Multiplication by 0 sends every section to 0, so the corresponding endomorphism of the unit module is the zero map. □

Lemma 1488 (Composition of scalar endomorphisms is multiplication)**.** *For $a, b \in \Gamma(X, \mathcal{O}_X)$ the scalar endomorphisms compose by multiplication:* $\mathrm{scalarEnd}(a) \circ \mathrm{scalarEnd}(b) = \mathrm{scalarEnd}(a\,b)$.

Proof. Under the identification $\mathrm{End}(\mathbf{1}) \cong \Gamma(X, \mathbf{1})$ the scalar endomorphisms act as multiplication by a and by b; their composite is multiplication by $a\,b$, because the section ring is commutative and restriction maps are ring homomorphisms. □

Lemma 1489 (Scalar endomorphism is additive)**.** *For $a, b \in \Gamma(X, \mathcal{O}_X)$ one has* $\mathrm{scalarEnd}(a + b) = \mathrm{scalarEnd}(a) + \mathrm{scalarEnd}(b)$.

Proof. Multiplication by $a + b$ equals multiplication by a plus multiplication by b on every section, since the restriction maps of $\mathcal{O}_X$ are additive; under $\mathrm{End}(\mathbf{1}) \cong \Gamma(X, \mathbf{1})$ this is the stated additivity. □

Infrastructure: matrix automorphisms of the free sheaf. To realise the GL_d bundle transitions one must promote an invertible $d \times d$ matrix of global functions to an *automorphism* of the free rank-d sheaf $\mathcal{O}_S^d$, exactly as the chart quotient 1496 realises the universal matrix. The next blocks assemble this matrix-to-morphism functor and record its two structural identities – that it carries matrix multiplication to composition and the identity matrix to the identity – which are the algebraic heart of the bundle cocycle.

Definition 1490 (Matrix endomorphism of the free sheaf)**.** Let S be a scheme, $d \in \mathbb{N}$, and $M \in \mathrm{Mat}_{d\times d}(\Gamma(S, \mathcal{O}_S))$ a matrix of global functions. The *matrix endomorphism* $\mathrm{matrixEnd}(M)$ is the endomorphism of the free rank-d sheaf $\mathcal{O}_S^d$ whose (p, q)-component, under the presentation of $\mathcal{O}_S^d$ as the finite biproduct of d copies of the unit module $\mathbf{1}$, is the scalar endomorphism $\mathrm{scalarEnd}(M_{p,q})$ (1485). It is assembled from these components by the universal property of the rank-d biproduct, exactly as the chart quotient u^I is assembled (1496). This is the substitute for the (non-existent) Mathlib “matrix $\mapsto$ endomorphism of a free sheaf” primitive.

Lemma 1491 (Matrix endomorphism carries multiplication to composition)**.** *For $M, N \in \mathrm{Mat}_{d\times d}(\Gamma(S, \mathcal{O}_S))$ the matrix endomorphisms compose according to the matrix product, with the order reversed by the contravariance of the component indexing:*

$$\mathrm{matrixEnd}(M) \circ \mathrm{matrixEnd}(N) \;=\; \mathrm{matrixEnd}(N \cdot M).$$

Proof. The composition of two biproduct-assembled morphisms is again biproduct-assembled, with (i, q)-component the sum over the middle index p of the composites of components, $\sum_p \mathrm{component}_{i,p} \circ \mathrm{component}_{p,q}$ (the categorical matrix product over the biproduct). Each composite of scalar endomorphisms is the scalar endomorphism of the product, $\mathrm{scalarEnd}(a) \circ \mathrm{scalarEnd}(b) = \mathrm{scalarEnd}(ab)$ (1488), and a finite sum of scalar endomorphisms is the scalar endomorphism of the sum (1489); hence the (i, q)-component of the composite is $\mathrm{scalarEnd}(\sum_p M_{q,p} N_{p,i}) = \mathrm{scalarEnd}((N \cdot M)_{q,i})$, which is precisely the (i, q)-component of $\mathrm{matrixEnd}(N \cdot M)$. The reversal of the factor order is the contravariance of the column/component indexing. □

Lemma 1492 (Matrix endomorphism of the identity is the identity)**.** *The matrix endomorphism of the identity matrix is the identity morphism of $\mathcal{O}_S^d$:* $\mathrm{matrixEnd}(1_{d\times d}) = \mathrm{id}_{\mathcal{O}_S^d}$.

Proof. The diagonal components are $\mathrm{scalarEnd}(1) = \mathrm{id}$ (1486) and the off-diagonal components are $\mathrm{scalarEnd}(0) = 0$ (1487); the morphism with this component matrix is the identity of the biproduct. □

Definition 1493 (Free-sheaf automorphism of an invertible matrix)**.** Let $M, N \in \mathrm{Mat}_{d\times d}(\Gamma(S, \mathcal{O}_S))$ be mutually inverse, $MN = 1$ and $NM = 1$. The *matrix automorphism* matrixToFreeIso(M, N) is the isomorphism of free sheaves $\mathcal{O}_S^d \xrightarrow{\sim} \mathcal{O}_S^d$ with forward map matrixEnd(M) and inverse map matrixEnd(N); the two composites are identities by 1491 (turning the products NM and MN into the relevant compositions) and 1492. Its forward map is matrixEnd(M) by construction (1494).

Lemma 1494 (Forward map of the matrix automorphism)**.** *The forward map of* matrixToFreeIso(M, N) *is* matrixEnd(M)*.*

Proof. Immediate from the definition 1493, whose forward field is literally matrixEnd(M). □

Lemma 1495 (Matrix automorphisms compose multiplicatively)**.** *Let A, A', B, B' be $d\times d$ matrices of global functions on S with $AA' = A'A = 1$, $BB' = B'B = 1$, so that* matrixToFreeIso(A, A') *and* matrixToFreeIso(B, B') *are automorphisms of $\mathcal{O}_S^d$. Their forward maps compose to the forward map of the product automorphism, with the order reversed by the component contravariance:*

$$\mathrm{matrixToFreeIso}(A, A')_{\mathrm{hom}} \circ \mathrm{matrixToFreeIso}(B, B')_{\mathrm{hom}} \;=\; \mathrm{matrixEnd}(B \cdot A).$$

Equivalently, composition of matrix automorphisms realises matrix multiplication of the underlying matrices. This is the linkage that turns the matrix-level cocycle identity into a composition identity of sheaf-of-module isomorphisms.

Proof. By 1494 the two forward maps are matrixEnd(A) and matrixEnd(B); their composite is matrixEnd$(A) \circ$ matrixEnd$(B) =$ matrixEnd$(B \cdot A)$ by 1491. □

Definition 1496 (Chart quotient u^I)**.** *Source: [Nitsure], §1.* Let $I \subseteq \{1, \dots, r\}$ with $\#(I) = d$. On the affine chart $U^I = \operatorname{Spec} R^I$ (1357) define the $\mathcal{O}_{U^I}$-linear homomorphism of trivial bundles

$$u^I \;:\; \mathcal{O}_{U^I}^r \longrightarrow \mathcal{O}_{U^I}^d$$

given by left multiplication by the universal $d \times r$ matrix X^I (1362): on global sections it is the R^I-linear map $(R^I)^r \to (R^I)^d$, $v \mapsto X^I v$. Since the I-th minor X_I^I of X^I is the identity matrix $1_{d\times d}$, the composite of u^I with the inclusion of the I-indexed coordinates $\mathcal{O}_{U^I}^d \hookrightarrow \mathcal{O}_{U^I}^r$ is the identity; hence u^I is a split surjection of locally free sheaves, with target the free rank-d sheaf $\mathcal{O}_{U^I}^d$.

Realisation. The free sheaves $\mathcal{O}_{U^I}^r$ and $\mathcal{O}_{U^I}^d$ are the finite biproducts (coproducts) of copies of the unit module $\mathbf{1} = \mathcal{O}_{U^I}$, indexed by $\{1, \dots, r\}$ and $\{1, \dots, d\}$. Each entry $X_{p,q}^I \in R^I$ of the universal matrix (1362) is a global function on $U^I = \operatorname{Spec} R^I$, obtained from R^I through the canonical identification $\Gamma(\operatorname{Spec} R^I, \mathcal{O}) \cong R^I$; the corresponding scalar endomorphism scalarEnd$(X_{p,q}^I)$: $\mathbf{1} \to \mathbf{1}$ (1485) is the (p, q)-component of u^I. Assembling these components over the index biproducts – the universal property of the biproduct packaging a matrix of component morphisms into a single morphism of the free sheaves – yields u^I. This entry-by-entry assembly via the finite biproduct is the substitute for a (non-existent) Mathlib "matrix $\mapsto$ morphism of free sheaves" primitive.

Lemma 1497 (The chart quotient is an epimorphism)**.** *The chart quotient $u^I : \mathcal{O}_{U^I}^r \to \mathcal{O}_{U^I}^d$ (1496) is an epimorphism of sheaves of modules.*

Proof. We exhibit a section, so that u^I is a *split* epimorphism and hence an epimorphism. Let $\iota_I : \{1, \dots, d\} \hookrightarrow \{1, \dots, r\}$ be the inclusion of the I-indexed coordinates, $k \mapsto \iota_I(k)$, the k-th element of I in increasing order. It induces a morphism of free sheaves $s_I : \mathcal{O}^d_{U^I} \to \mathcal{O}^r_{U^I}$, the coordinate inclusion sending the k-th basis section to the $\iota_I(k)$-th. We claim

$$u^I \circ s_I \;=\; \mathrm{id}_{\mathcal{O}^d_{U^I}}.$$

Both sides are morphisms out of the free rank-d sheaf, which is the coproduct of d copies of the unit module; by the universal property of that coproduct it suffices to check the equality on each coordinate, i.e. to compare component matrices. The (p,k)-component of $u^I \circ s_I$ is the entry $X^I_{p,\iota_I(k)}$ of the universal matrix at the I-indexed column, which is precisely the I-th minor X^I_I of X^I. Since that minor is the $d \times d$ identity (the columns of X^I indexed by I, read through the order isomorphism $\{1,\dots,d\} \cong I$, form the identity block – 1362), one has $X^I_{p,\iota_I(k)} = \delta_{p,k}$, so $u^I \circ s_I$ has the identity component matrix and equals id. Hence s_I splits u^I and u^I is a split epimorphism, in particular an epimorphism. □

24.3 The GL_d bundle transition cocycle

The universal quotient sheaf $\mathcal{U}$ (1510) is obtained by feeding the gluing primitive 1479 the per-chart free rank-d sheaves $\mathcal{O}^d_{U^I}$, the bundle transitions $g_{I,J} = (X^I_J)^{-1}$, and the two module cocycle hypotheses (C1)/(C2) that primitive requires. The matrix cocycle for the Cramer inverses $(X^I_J)^{-1}$ is already in hand – it is the content of the scheme-level cocycle 1393 and the minor identities 1367 – but the gluing primitive consumes its *module-level* incarnation: an honest isomorphism of sheaves of modules together with its self-identity and triple-overlap multiplicativity. The three blocks of this section package exactly that, decoupling the construction of $\mathcal{U}$ from the remaining matrix-to-module transport bookkeeping. This is the only net-new infrastructure the universal quotient requires beyond the gluing primitive itself.

Definition 1498 (The bundle transition $g_{I,J}$)**.** Let $I, J \subseteq \{1,\dots,r\}$ be size-d subsets and let $f_{IJ} : U^I_J \hookrightarrow U^I$ and $t_{IJ} : U^I_J \xrightarrow{\sim} U^J_I$ be the overlap immersion and transition of the chart cover (1371). The *bundle transition* $g_{I,J}$ is the isomorphism of sheaves of modules on the overlap U^I_J

$$g_{I,J} \;:\; f^*_{IJ}(\mathcal{O}^d_{U^I}) \;\xrightarrow{\;\sim\;}\; (t_{IJ} \circ f_{JI})^*(\mathcal{O}^d_{U^J})$$

induced by the invertible $d \times d$ matrix $g_{I,J} = (X^I_J)^{-1} \in \mathrm{GL}_d(R^I_J)$ (1366), the Cramer inverse of the localised J-minor, which is invertible because $\det(X^I_J)$ is a unit of R^I_J (1365).

Realisation. The transition is built exactly as the chart quotient 1496: each entry $(X^I_J)^{-1}_{p,q} \in R^I_J$ is a global function on the overlap, its scalar endomorphism $\mathrm{scalarEnd}((X^I_J)^{-1}_{p,q})$ (1485) is the (p,q)-component, and the universal property of the rank-d biproduct assembles these into an automorphism of the free sheaf $\mathcal{O}^d_{U^I_J}$; the source and target pullbacks of free sheaves are identified with this $\mathcal{O}^d_{U^I_J}$ through the free-pullback comparisons 1483 along f_{IJ} and $t_{IJ} \circ f_{JI}$, conjugating the matrix automorphism into the displayed isomorphism. Invertibility of the matrix makes $g_{I,J}$ an isomorphism.

Lemma 1499 (Self-identity of the bundle transition (C1))**.** *The self-transition is the identity: $g_{I,I} = \mathrm{id}$ on $f^*_{II}\mathcal{O}^d_{U^I}$. This is the module-level form of the descent condition (C1) of 1479 for the bundle data $(g_{I,J})$.*

Proof. The I-th minor of the universal matrix is the identity block, $X^I_I = 1_{d\times d}$ (1377), so its Cramer inverse is again the identity, $(X^I_I)^{-1} = 1_{d\times d}$ (1367). Hence in the biproduct presentation of 1498 the diagonal components are $\mathrm{scalarEnd}(1) = \mathrm{id}$ (1486) and the off-diagonal components are $\mathrm{scalarEnd}(0) = 0$ (1487), so the assembled automorphism is the identity of $\mathcal{O}_{U_I}^d$; conjugation by the free-pullback comparisons preserves the identity, giving $g_{I,I} = \mathrm{id}$. $\square$

Lemma 1500 (Cramer-inverse cocycle on the triple overlap)**.** *Over the common triple-overlap ring (the localisation carrying the three charts' data to V_{IJK}) the Cramer inverses of the localised minors satisfy the multiplicative cocycle identity*

$$(X_K^J)^{-1}\,(X_J^I)^{-1} \;=\; (X_K^I)^{-1}.$$

This is the pure matrix-algebra core of (C2), independent of any sheaf data.

Proof. Write all matrices over the common triple-overlap ring. The image matrix of the transition $\theta_{I,J}$ is $X^J = (X_J^I)^{-1}X^I$, and its K-minor is computed by restricting to the K-indexed columns; since taking a column submatrix commutes with left multiplication (1373),

$$X_K^J \;=\; \left((X_J^I)^{-1}X^I\right)_K \;=\; (X_J^I)^{-1}\,X_K^I.$$

Both X_J^I and X_K^I have unit determinant (1367), so this product of invertibles inverts by $(AB)^{-1} = B^{-1}A^{-1}$:

$$(X_K^J)^{-1} \;=\; (X_K^I)^{-1}\,(X_J^I).$$

Multiplying on the right by $(X_J^I)^{-1}$ and cancelling $X_J^I(X_J^I)^{-1} = 1$ via the two-sided Cramer identity (1367, the cancellation packaged in 1376) gives $(X_K^J)^{-1}(X_J^I)^{-1} = (X_K^I)^{-1}$. The agreement of the three image matrices over the triple overlap that licenses these manipulations is the content of 1392, the matrix form of the scheme-level cocycle 1393. $\square$

Lemma 1501 (Scalar endomorphism is natural under pullback)**.** *Let $p : T \to S$ be a morphism of schemes and $a \in \Gamma(S, \mathcal{O}_S)$ a global function, with comorphism on global sections $p^\sharp : \Gamma(S, \mathcal{O}_S) \to \Gamma(T, \mathcal{O}_T)$. Write $q : p^*\mathbf{1} \xrightarrow{\sim} \mathbf{1}$ for the unit-pullback comparison (1483 at a one-element index). Then the pullback of the scalar endomorphism* $\mathrm{scalarEnd}(a)$ *is conjugate, through q, to the scalar endomorphism of the base-changed function:*

$$p^*(\mathrm{scalarEnd}(a)) \;=\; q^{-1} \circ \mathrm{scalarEnd}(p^\sharp a) \circ q$$

as endomorphisms of $p^*\mathbf{1}$; equivalently $q \circ p^*(\mathrm{scalarEnd}(a)) = \mathrm{scalarEnd}(p^\sharp a) \circ q$. This is the single-entry naturality atom underlying the matrix-level statement 1502.

Proof. The scalar endomorphism $\mathrm{scalarEnd}(a)$ (1485) is multiplication by the global section a on the unit module $\mathbf{1} = \mathcal{O}_S$. Pulling back along p sends multiplication by a to multiplication by its image, and the comorphism $p^\sharp$ is precisely the action of pullback on global functions, so $p^*(\mathrm{scalarEnd}(a))$ is multiplication by $p^\sharp a$ on $p^*\mathbf{1}$. The comparison q is an isomorphism of unit modules intertwining this multiplication on $p^*\mathbf{1}$ with multiplication by $p^\sharp a$ on $\mathbf{1}$; conjugating back along q therefore yields the claimed identity. The verification is a single computation on the unit module, where multiplication by a function commutes with the natural comparison q. $\square$

Lemma 1502 (Matrix endomorphism is natural under pullback)**.** *Let $p : T \to S$ be a morphism of schemes, $d \in \mathbb{N}$, and $M \in \mathrm{Mat}_{d\times d}(\Gamma(S, \mathcal{O}_S))$. Let $p^\sharp M \in \mathrm{Mat}_{d\times d}(\Gamma(T, \mathcal{O}_T))$ be the entrywise image of M under the comorphism $p^\sharp$, and let $Q : p^*(\mathcal{O}_S^d) \xrightarrow{\sim} \mathcal{O}_T^d$ be the free-pullback comparison 1483 at the rank-d index set. Then the pullback of the matrix endomorphism* $\mathrm{matrixEnd}(M)$ *is conjugate, through Q, to the matrix endomorphism of the base-changed matrix:*

$$p^*(\mathrm{matrixEnd}(M)) \;=\; Q^{-1} \circ \mathrm{matrixEnd}(p^\sharp M) \circ Q.$$

Proof. By 1490, matrixEnd(M) is the biproduct-assembled endomorphism of $\mathcal{O}_S^d = \bigoplus_k \mathbf{1}$ whose (k,ℓ)-component is scalarEnd$(M_{k,\ell})$. The pullback functor p^* is a left adjoint, hence additive and biproduct-preserving; through the coproduct comparison underlying the free-pullback isomorphism 1483 it carries the biproduct presentation of $\mathcal{O}_S^d$ to that of $\mathcal{O}_T^d$ and the assembled morphism to the morphism reassembled from the pulled-back components, the reassociation of the iterated pullbacks being 1477. On each single (k,ℓ)-component the pulled-back scalar endomorphism is, by 1501, the comparison-conjugate of scalarEnd$(p^\sharp M_{k,\ell})$ = scalarEnd$((p^\sharp M)_{k,\ell})$. Reassembling over the biproduct, the conjugating one-element comparisons combine into the single free comparison Q, giving $p^*(\text{matrixEnd}(M)) = Q^{-1} \circ \text{matrixEnd}(p^\sharp M) \circ Q$. □

Lemma 1503 (Affine global-sections comorphism is the inducing ring homomorphism)**.** *Let $\varphi : A \to B$ be a homomorphism of commutative rings and* $\operatorname{Spec}\varphi : \operatorname{Spec} B \to \operatorname{Spec} A$ *the induced morphism of affine schemes. Under the natural identification of the global sections of an affine scheme with its coordinate ring – the counit* $\Gamma(\operatorname{Spec} A, \mathcal{O}) \xrightarrow{\sim} A$ *of the* $\Gamma \dashv \operatorname{Spec}$ *adjunction, and likewise for B – the global-sections base-change comorphism*

$$(\operatorname{Spec}\varphi)^\sharp : \Gamma(\operatorname{Spec} A, \mathcal{O}) \longrightarrow \Gamma(\operatorname{Spec} B, \mathcal{O})$$

is identified with φ itself; that is, the square relating $(\operatorname{Spec}\varphi)^\sharp$ *to φ through the two counit isomorphisms commutes.*

Proof. This is the naturality of the counit isomorphism $\Gamma \circ \operatorname{Spec} \cong \operatorname{id}$ of the global-sections / spectrum adjunction: for any ring homomorphism φ the comorphism of $\operatorname{Spec}\varphi$ on global sections is, after transport through the natural counit isomorphisms at A and B, the homomorphism φ. No Grassmannian-specific data enters; the statement is the affine-scheme naturality square applied to a single ring map. □

Lemma 1504 (First-projection bridge to awayInclLeft)**.** *Fix indices I, J, K with triple overlap $V_{IJK} = U_J^I \times_{U^I} U_K^I$ of the glue datum (1415), and let $p_{IJK}^{IJ} : V_{IJK} \to U_J^I$ be the first projection (the pullback projection of the chart inclusion* chartIncl*, 1400). Through the away-pullback identification $V_{IJK} \cong \operatorname{Spec} R^I[1/(P_J^I P_K^I)]$ (1405) the projection p_{IJK}^{IJ} is the* Spec *morphism of the left away-inclusion* awayInclLeft(P_J^I, P_K^I) *(1406, 1382). Consequently the global-sections base-change map of p_{IJK}^{IJ}, transported through the affine global-sections identification, equals the ring homomorphism* awayInclLeft(P_J^I, P_K^I)*; in particular the entrywise image of the Cramer inverse $(X_J^I)^{-1}$ under the geometric base-change agrees with its image under* awayInclLeft*.*

Proof. By 1406 the first pullback projection, precomposed with the away-pullback identification 1405, equals Spec(awayInclLeft(P_J^I, P_K^I)). The source and overlap charts are affine, so by 1503 the global-sections comorphism of this Spec morphism is, through the affine global-sections identification, the ring homomorphism awayInclLeft(P_J^I, P_K^I) (1382). Applying the comorphism entrywise to $(X_J^I)^{-1}$ gives the asserted agreement. □

Lemma 1505 (Second-projection bridge to awayInclRight)**.** *With the notation of 1504, let $p_{IJK}^{IK} : V_{IJK} \to U_K^I$ be the second projection (the pullback projection of the chart inclusion* chartIncl*, 1400). Through the away-pullback identification $V_{IJK} \cong \operatorname{Spec} R^I[1/(P_J^I P_K^I)]$ (1405) the projection p_{IJK}^{IK} is the* Spec *morphism of the right away-inclusion* awayInclRight(P_J^I, P_K^I) *(1407, 1383). Consequently the global-sections base-change map of p_{IJK}^{IK}, transported through the affine global-sections identification, equals the ring homomorphism* awayInclRight(P_J^I, P_K^I)*; in particular the entrywise image of the Cramer inverse $(X_K^I)^{-1}$ under the geometric base-change agrees with its image under* awayInclRight*.*

Proof. Identical to 1504 with the second leg in place of the first: by 1407 the second pullback projection, precomposed with the away-pullback identification 1405, equals $\mathrm{Spec}(\mathrm{awayInclRight}(P^I_J, P^I_K))$. The charts being affine, 1503 identifies its global-sections comorphism with $\mathrm{awayInclRight}(P^I_J, P^I_K)$ (1383), and the entrywise image of $(X^I_K)^{-1}$ follows. ☐

Lemma 1506 (Triple-transition bridge to Θ_{IJ})**.** *With the notation of 1504, let $t'_{IJK} : V_{IJK} \to U^J_K \times_{U^J} U^J_I$ be the triple transition of the glue datum (1410). Through the source and target away-pullback identifications (1405) and the order-swap isomorphism (1408), t'_{IJK} is the* Spec *morphism of the localised triple-overlap transition $\Theta_{IJ} : S_J \to S_I$ (1389, composed with the order-swap ring isomorphism). Consequently the global-sections base-change map of t'_{IJK}, transported through the affine global-sections identification, equals the ring homomorphism Θ_{IJ} (up to the order-swap); in particular the entrywise image of the Cramer inverse $(X^J_K)^{-1}$ under the geometric base-change agrees with its image under Θ_{IJ}.*

Proof. By 1410 the triple transition t'_{IJK} is the composite of the source away-pullback identification, $\mathrm{Spec}\,\Theta_{IJ}$ (1389), the Spec of the order-swap isomorphism (1408), and the inverse target away-pullback identification. The charts and overlaps being affine, applying 1503 to each Spec factor identifies the global-sections comorphism of t'_{IJK}, through the affine global-sections identifications and the order-swap, with the ring homomorphism Θ_{IJ}. The entrywise image of $(X^J_K)^{-1}$ under this comorphism therefore agrees with its image under Θ_{IJ}. ☐

Lemma 1507 (Base-change bridge to the localised cocycle ring homomorphisms)**.** *Fix indices I, J, K and the triple overlap V_{IJK} of the chart cover $\{U^I\}$ of* $\mathrm{Gr}(r,d)$ *(1416, 1415). The scheme-pullback base-change map on global sections $\Gamma(U^I_J, \mathcal{O}) \to \Gamma(V_{IJK}, \mathcal{O})$ induced by the first projection p^{IJ}_{IJK}, and likewise the maps induced by the second projection and by the triple transition t'_{IJK} (1371), are identified – through the natural identification of global sections of an affine scheme with its coordinate ring and the fact that the projections and triple transition are the* Spec *morphisms of the coordinate-ring homomorphisms by which the Grassmannian glue datum (1415) is assembled – with the ring homomorphisms* awayInclLeft *(1382),* awayInclRight *(1383) and Θ_{IJ} (1389) over which the matrix cocycle 1500 is stated. Consequently the entrywise image of a Cramer inverse $(X^I_J)^{-1}$ under the geometric base-change map agrees with its image under the corresponding coordinate-ring homomorphism, so 1500 applies to the base-changed matrices.*

Proof. The three identifications are exactly the three projection bridges. The first projection is handled by 1504, identifying its global-sections base-change map with awayInclLeft; the second projection by 1505, identifying it with awayInclRight; and the triple transition t'_{IJK} by 1506, identifying it with Θ_{IJ}. Each bridge already records the agreement of the entrywise image of the relevant Cramer inverse under the geometric base-change with its image under the named ring homomorphism over which 1500 is stated, so assembling the three gives the claim. ☐

Lemma 1508 (Transport and endpoint alignment of the bundle transitions)**.** *Fix indices I, J, K with triple overlap $V_{IJK} = U^I_J \times_{U^I} U^I_K$, and let $\hat{g}^K_{IJ}, \hat{g}^I_{JK}, \hat{g}^J_{IK}$ be the base-change transports (1478) of the three bundle transitions to V_{IJK}. Under the free-pullback comparisons (1483) and the endpoint bridges (1463), each transported transition is identified with the matrix automorphism of the corresponding base-changed Cramer inverse on the common free sheaf $\mathcal{O}^d_{V_{IJK}}$; granting the matrix identity 1500, the composite satisfies*

$$\hat{g}^I_{JK} \circ \hat{g}^K_{IJ} \;=\; \hat{g}^J_{IK}$$

as isomorphisms of sheaves of modules on V_{IJK}. This is the substantive transport step: the alignment of the three transports' domains and codomains.

Proof. This is pure assembly of the two infrastructure pieces over the endpoint bridges. Each bundle transition is, by 1498, matrixToFreeIso of a localised Cramer inverse conjugated by the free-pullback comparisons 1483; its base-change transport (1478) reassociates the iterated pullbacks via 1477, and the three endpoint bridges 1463 pin the source, middle and target sheaves to a single $\mathcal{O}^d_{V_{IJK}}$. By 1502 each transport then equals matrixEnd of the base-changed Cramer inverse, and by 1507 those base-changed matrices are exactly $(X^I_J)^{-1}, (X^J_K)^{-1}, (X^I_K)^{-1}$ over the common triple-overlap ring of 1500. The linkage 1495 composes the first two to matrixEnd$((X^J_K)^{-1}(X^I_J)^{-1})$, and by the matrix cocycle identity 1500 the argument equals $(X^I_K)^{-1}$, so the composite equals $\hat{g}^J_{IK}$; the bridges make both sides isomorphisms between the same pair of sheaves on V_{IJK}. □

Lemma 1509 (Triple-overlap multiplicativity of the bundle transition (C2))**.** *Over each triple overlap $V_{IJK} = U^I_J \times_{U^I} U^I_K$ the base-change transports (1478) of the three bundle transitions satisfy*

$$\hat{g}^I_{JK} \circ \hat{g}^K_{IJ} \;=\; \hat{g}^J_{IK},$$

the module-level form of the descent condition (C2) of 1479 for the bundle data $(g_{I,J})$. This is the GL_d*-valued multiplicative cocycle of* $\mathcal{U}$*.*

Proof. The statement decomposes into three independent steps. First, the matrix-algebra core 1500 establishes the Cramer-inverse cocycle $(X^J_K)^{-1}(X^I_J)^{-1} = (X^I_K)^{-1}$ over the triple overlap. Second, the linkage 1495 records that composition of matrix automorphisms realises matrix multiplication. Third, the transport step 1508 carries the three transitions to the common overlap V_{IJK}, aligns their source, middle, and target sheaves through the base-change transport and the endpoint bridges, and – feeding it the matrix identity through the linkage – yields exactly the stated equality $\hat{g}^I_{JK} \circ \hat{g}^K_{IJ} = \hat{g}^J_{IK}$. □

Definition 1510 (The universal quotient sheaf $\mathcal{U}$)**.** *Source: [Nitsure], §1.* Over each overlap $U^I_J = \mathrm{Spec}\, R^I_J$ define the bundle-transition matrix

$$g_{I,J} \;=\; (X^I_J)^{-1} \;\in\; \mathrm{GL}_d(R^I_J),$$

the Cramer inverse of the localised J-minor (1366), which is invertible because $\det(X^I_J)$ is a unit of R^I_J. The matrices $(g_{I,J})$ are compatible with the chart-gluing cocycle $(\theta_{I,J})$ (1371, 1393): over a triple overlap they satisfy $g_{I,K} = g_{J,K}\, g_{I,J}$ and $g_{I,I} = 1$, the multiplicative cocycle condition for a vector bundle. Gluing the free rank-d sheaves $\mathcal{O}^d_{U^I}$ along the isomorphisms $g_{I,J}$ over the cover $\{U^I\}$ of $\mathrm{Gr}(r,d)$ (1416) therefore produces a locally free $\mathcal{O}_{\mathrm{Gr}(r,d)}$-module $\mathcal{U}$ of rank d. Its determinant line bundle $\det \mathcal{U}$ has transition functions $\det(g_{I,J}) = 1/P^I_J$, the data underlying the Plücker embedding.

The tautological quotient u is assembled from the chart quotients u^I by the section-level descent 1481; its one remaining content is the *overlap-compatibility* of the family (u^I) with the bundle gluing $(g_{I,J})$. The next four blocks isolate that compatibility: a rectangular analogue of the square matrix-endomorphism API (1490, 1502, 1491) for the $r \to d$ chart quotient (1511, 1512, 1513), the adjunction transposition that turns the descent condition into a pullback-level identity (1514), and the matrix-core assembly that closes it (1515).

24.3.1 Matrix-calculus toolbox

The chart-cover lemma (1526) and the overlap compatibility of 1538 both rest on a small calculus of *rectangular* matrix homomorphisms of free sheaves, generalising the square API (1490, 1491, 1492). The blocks below collect that toolbox: the bridge to the square case, the identity and composition laws, injectivity of the matrix presentation, the column-restriction law, the affine splitting of epimorphisms of free sheaves and the resulting matrix right inverse, the field-level minor extraction, and the presented-matrix machinery that packages a quotient as a matrix along a morphism and proves Nitsure's change-of-basis identity. They are project-bespoke infrastructure.

Definition 1511 (Rectangular matrix homomorphism of free sheaves)**.** Let S be a scheme, $d, r \in \mathbb{N}$, and $M \in \mathrm{Mat}_{d\times r}(\Gamma(S, \mathcal{O}_S))$ a rectangular matrix of global functions. The *rectangular matrix homomorphism* matrixEndRect(M) is the morphism of free sheaves

$$\mathrm{matrixEndRect}(M) \;:\; \mathcal{O}_S^r \longrightarrow \mathcal{O}_S^d$$

whose (k, ℓ)-component – $k \in \{1, \dots, d\}$ an output index, $\ell \in \{1, \dots, r\}$ an input index – under the presentation of $\mathcal{O}_S^r$ and $\mathcal{O}_S^d$ as finite biproducts of copies of the unit module $\mathbf{1}$, is the scalar endomorphism scalarEnd$(M_{k,\ell})$ (1485). It is assembled from these components by the universal property of the two biproducts, exactly as the square matrixEnd (1490) and the chart quotient (1496). When $r = d$ it specialises to matrixEnd; and the chart quotient u^I (1496) is by construction matrixEndRect(X^I) of the universal $d \times r$ matrix X^I (1362).

Lemma 1512 (Rectangular matrix homomorphism is natural under pullback)**.** *Let $p : T \to S$ be a morphism of schemes and $M \in \mathrm{Mat}_{d\times r}(\Gamma(S, \mathcal{O}_S))$, with entrywise base-changed matrix $p^\sharp M \in \mathrm{Mat}_{d\times r}(\Gamma(T, \mathcal{O}_T))$. Write $Q_r : p^*\mathcal{O}_S^r \xrightarrow{\sim} \mathcal{O}_T^r$ and $Q_d : p^*\mathcal{O}_S^d \xrightarrow{\sim} \mathcal{O}_T^d$ for the free-pullback comparisons (1483) at the index sets $\{1, \dots, r\}$ and $\{1, \dots, d\}$. Then the pullback of* matrixEndRect(M) *is conjugate, through Q_r and Q_d, to the rectangular homomorphism of the base-changed matrix:*

$$p^*(\mathrm{matrixEndRect}(M)) \;=\; Q_d^{-1} \circ \mathrm{matrixEndRect}(p^\sharp M) \circ Q_r.$$

Proof. Identical to the square case 1502, with the source and target biproducts now indexed by $\{1, \dots, r\}$ and $\{1, \dots, d\}$ respectively. The pullback functor p^*, a left adjoint, is additive and biproduct-preserving; through the coproduct comparisons underlying the free-pullback isomorphisms 1483 it carries the biproduct presentation of $\mathcal{O}_S^r$ (resp. $\mathcal{O}_S^d$) to that of $\mathcal{O}_T^r$ (resp. $\mathcal{O}_T^d$), the reassociation of the iterated pullbacks being 1477. On each single (k, ℓ)-component the pulled-back scalar endomorphism is, by 1501, the comparison-conjugate of scalarEnd$((p^\sharp M)_{k,\ell})$. Reassembling over the two biproducts, the conjugating one-element comparisons combine into Q_r on the source and Q_d on the target, giving $p^*(\mathrm{matrixEndRect}(M)) = Q_d^{-1} \circ \mathrm{matrixEndRect}(p^\sharp M) \circ Q_r$. □

Lemma 1513 (Square-after-rectangular composition law)**.** *For $M \in \mathrm{Mat}_{d\times r}(\Gamma(S, \mathcal{O}_S))$ and $N \in \mathrm{Mat}_{d\times d}(\Gamma(S, \mathcal{O}_S))$, post-composing the rectangular homomorphism with a square one realises the matrix product, with the order reversed by the contravariance of the component indexing exactly as in 1491:*

$$\mathrm{matrixEndRect}(M) \circ \mathrm{matrixEnd}(N) \;=\; \mathrm{matrixEndRect}(N \cdot M),$$

the composite $\mathcal{O}_S^r \xrightarrow{\mathrm{matrixEndRect}(M)} \mathcal{O}_S^d \xrightarrow{\mathrm{matrixEnd}(N)} \mathcal{O}_S^d$ (here $N \cdot M$ is the $d \times r$ product).

Proof. The same biproduct-extensionality computation as 1491, the only change being that the leftmost (input) biproduct is indexed by $\{1, \dots, r\}$ rather than $\{1, \dots, d\}$. The composite of two biproduct-assembled morphisms is again biproduct-assembled, with (k, ℓ)-component the sum over the middle index m of the composites of components, $\sum_m \mathrm{scalarEnd}(N_{k,m}) \circ \mathrm{scalarEnd}(M_{m,\ell})$. Each composite of scalar endomorphisms is the scalar endomorphism of the product (1488) and a finite sum of scalar endomorphisms is the scalar endomorphism of the sum (1489); hence that component is $\mathrm{scalarEnd}(\sum_m N_{k,m}M_{m,\ell}) = \mathrm{scalarEnd}((N \cdot M)_{k,\ell})$, the (k,ℓ)-component of $\mathrm{matrixEndRect}(N \cdot M)$. ☐

Lemma 1514 (Adjunction transpose of the chart-overlap condition)**.** *Fix size-d subsets I, J and let $f_{IJ} : U^I_J \hookrightarrow U^I$, $t_{IJ} : U^I_J \xrightarrow{\sim} U^J_I$ be the overlap immersion and transition (1371). The per-chart components u^I of the tautological quotient are the adjoint transposes, along the chart immersions, of the chart quotients 1496 (precomposed with the free-pullback comparison). Their descent compatibility – the hypothesis the gluing primitive 1481 consumes, namely that the bundle transition $g_{I,J}$ (1498) intertwines the pullbacks of u^I and u^J over U^I_J – is equivalent, under the pullback–pushforward adjunction of the overlap immersion, to the pullback-level identity of morphisms $\mathcal{O}^r_{U^I_J} \to \mathcal{O}^d_{U^I_J}$*

$$g_{I,J} \circ f^*_{IJ} u^I \;=\; (t_{IJ} \circ f_{JI})^* \, u^J,$$

understood through the free-pullback comparisons (1483) on source and target.

Proof. The components are produced by applying the adjunction hom-equivalence to the chart quotients; the descent condition of 1481 is stated on those transposes. Transposing back across the adjunction is the naturality of the conjugate (mate) construction against the hom-equivalences, 1519, together with the compatibility of the conjugate with the iterated-pullback comparison 145 reassociating the two pullbacks f^*_{IJ} and $(t_{IJ} \circ f_{JI})^*$ via 1477. The free-pullback comparisons 1483 re-present both sides as morphisms of the trivial bundles $\mathcal{O}^r_{U^I_J} \to \mathcal{O}^d_{U^I_J}$, under which the transposed condition becomes the displayed pullback-level identity. ☐

Lemma 1515 (Overlap compatibility of the tautological quotient)**.** *For every pair of size-d subsets I, J, the chart quotients u^I (1496) satisfy the overlap-compatibility condition required by the section-level descent 1481 for the bundle gluing $(g_{I,J})$: the pullback-level identity $g_{I,J} \circ f^*_{IJ} u^I = (t_{IJ} \circ f_{JI})^* u^J$ of 1514 holds on each overlap U^I_J.*

Proof. By 1514 it suffices to prove the pullback-level identity $g_{I,J} \circ f^*_{IJ} u^I = (t_{IJ} \circ f_{JI})^* u^J$ on U^I_J. Both sides are morphisms of trivial bundles $\mathcal{O}^r_{U^I_J} \to \mathcal{O}^d_{U^I_J}$; write each as matrixEndRect of a $d \times r$ matrix over the overlap ring R^I_J and compare matrices.

Since $u^I = \mathrm{matrixEndRect}(X^I)$ (1496), the base-change naturality 1512 identifies $f^*_{IJ} u^I$, through the free-pullback comparisons, with matrixEndRect of the localised matrix X^I over R^I_J (the global-sections base change is the away-localisation 1504). The bundle transition $g_{I,J}$ is matrixEnd of the Cramer inverse $(X^I_J)^{-1}$ (1498, 1366); hence by the square-after-rectangular composition law 1513,

$$g_{I,J} \circ f^*_{IJ} u^I \;=\; \mathrm{matrixEndRect}((X^I_J)^{-1} \cdot X^I).$$

The transition image matrix identity $X^J = (X^I_J)^{-1} X^I$ (1379, transported across the overlap by 1381) turns the right-hand matrix into X^J. Symmetrically, $\mathrm{matrixEndRect}_J$ of X^J is, by 1512 along $t_{IJ} \circ f_{JI}$, the comparison form of $(t_{IJ} \circ f_{JI})^* u^J$. Therefore the two sides agree, which is the required overlap compatibility. ☐

Definition 1516 (The tautological quotient $u : \mathcal{O}^r \twoheadrightarrow \mathcal{U}$)**.** *Source: [Nitsure], §1.* The chart quotients $u^I : \mathcal{O}^r_{U^I} \to \mathcal{O}^d_{U^I}$ (1496) are compatible with the bundle gluing $(g_{I,J})$ of $\mathcal{U}$ (1510): over each overlap U^I_J one has the identity of $d \times r$ matrices

$$g_{I,J} X^I \;=\; (X^I_J)^{-1} X^I \;=\; X^J \qquad \text{on } U^I_J,$$

which is precisely the defining formula of the chart transition $\theta_{I,J}$ (1371, 1366). At the module level this overlap compatibility – the equalising condition consumed by the section-level descent 1481 – is exactly 1515, whose proof routes the matrix identity above through the rectangular pullback/composition API (1512, 1513) and the adjunction transpose 1514. Hence the local maps u^I, transported through the trivialisations of $\mathcal{U}$, agree on overlaps and glue to a single surjective $\mathcal{O}_{\mathrm{Gr}(r,d)}$-linear homomorphism

$$u : \mathcal{O}^r_{\mathrm{Gr}(r,d)} \twoheadrightarrow \mathcal{U},$$

the *tautological* (or *universal*) rank-d quotient of the trivial rank-r sheaf. It is surjective because each u^I is (split) surjective and surjectivity is local on $\mathrm{Gr}(r,d)$, and $\mathcal{U}$ is locally free of rank d by 1510. The pair $\langle \mathcal{U}, u\rangle$ is the family of rank-d quotients carried by $\mathrm{Gr}(r,d)$ itself.

24.4 The functor of points

The Grassmannian functor sends a scheme T to the set of *equivalence classes* of rank-d quotients of $\mathcal{O}^r_T$. The following block packages a single such quotient, the equivalence relation on them, and the pullback action through which the functor acts on morphisms; the two coherence blocks after it isolate the comparison identities that make that action functorial.

Definition 1517 (Rank-d quotient of $\mathcal{O}^r_T$)**.** A *rank-d quotient* of the trivial bundle $\mathcal{O}^r_T$ on a scheme T is the datum $x = \langle \mathcal{F}, q\rangle$ of a sheaf of $\mathcal{O}_T$-modules $\mathcal{F}$, locally free of rank d, together with an epimorphism $q : \mathcal{O}^r_T \twoheadrightarrow \mathcal{F}$. Two such quotients $\langle \mathcal{F}, q\rangle$ and $\langle \mathcal{F}', q'\rangle$ are *equivalent* when there is an isomorphism $f : \mathcal{F} \xrightarrow{\sim} \mathcal{F}'$ with $f \circ q = q'$ (equivalently, $\ker q = \ker q'$); this is the equivalence relation whose classes form the value of the Grassmannian functor.

Definition 1518 (The Grassmannian functor $\mathrm{Grass}(r,d)$)**.** The *Grassmannian functor* $\mathrm{Grass}(r,d) : \mathbf{Sch}^{\mathrm{op}} \to \mathbf{Set}$ sends a scheme T to the set of equivalence classes of rank-d quotients $\langle \mathcal{F}, q\rangle$ of $\mathcal{O}^r_T$ (1517), and a morphism $\psi : T' \to T$ to the *pullback action* $\langle \mathcal{F}, q\rangle \mapsto \langle \psi^*\mathcal{F}, \psi^* q\rangle$, the source being re-identified with $\mathcal{O}^r_{T'}$ through the free-pullback isomorphism $\psi^*\mathcal{O}^r_T \cong \mathcal{O}^r_{T'}$ (1483). Pullback preserves epimorphisms and rank-d local freeness, so the action is well defined and respects the equivalence relation, hence descends to classes. The identity and composition functoriality laws follow from the pseudofunctor coherences of ψ^*, reduced by coproduct extensionality over $\mathcal{O}^r = \bigoplus_r \mathbf{1}$ to the unit-level identities 1520 and 1521.

The two blocks below record the comparison identities, between the free-pullback isomorphism 1483 and Mathlib's pseudofunctor structure on pullback, that make the descended pullback action of 1517 functorial.

Lemma 1519 (Mate/conjugate compatibility of the adjunction hom-equivalence)**.** *Let* $\mathrm{adj}_1, \mathrm{adj}_2$ *be two adjunctions and* α *a natural transformation between their left adjoints. For a morphism* f *the hom-equivalence of* adj_2 *applied to* $\alpha_c \circ f$ *equals the hom-equivalence of* adj_1 *applied to* f*, post-composed with the component at d of the conjugate (mate) of* α*. This is the general naturality of the conjugate construction against the two hom-equivalences; it is the reusable engine behind 1521.*

Proof. Both sides unfold to the same pasting of unit/counit triangles for the two adjunctions; the equality is the standard compatibility of the mate of α with the hom-set bijections. ☐

Lemma 1520 (Free-pullback unit coherence at the identity)**.** *The free-pullback comparison morphism on the unit sheaf at the identity morphism of T agrees with the component, at the unit sheaf, of the pseudofunctor identity isomorphism* $\mathrm{id}^* \cong \mathrm{id}$ *of the pullback of sheaves of modules.*

Proof. Transpose both sides under the pullback–pushforward adjunction: the comparison becomes the canonical unit-to-pushforward-unit map, and the pseudofunctor identity becomes the conjugate (mate) of the identity adjunction, which acts as the identity on the unit section. The two transposes therefore coincide, so the original maps agree. ☐

Lemma 1521 (Free-pullback unit coherence at a composite)**.** *For composable $a : T_y \to T_x$ and $b : T_z \to T_y$, the free-pullback comparison on the unit sheaf at the composite $b \circ a$ factors, through the pseudofunctor composition isomorphism $(b \circ a)^* \cong b^* a^*$, as the pulled-back comparison of a followed by the comparison of b.*

Proof. Transpose under the composite pullback–pushforward adjunction. By 1519 the left side collapses to the unit-to-pushforward map of $b \circ a$ post-composed with the conjugate of the composition isomorphism – the inverse pushforward-composition comparison – while the right side collapses, by naturality of the hom-equivalence, to the unit map of a followed by its pushforward along b. Both reduce to the unit-section identity, so the two sides agree. ☐

Together 1520 and 1521 upgrade to the corresponding free-sheaf coherences – the free-pullback isomorphism at the identity equals the component of the pullback-identity isomorphism, and likewise for the composite – by coproduct extensionality over the decomposition $\mathcal{O}^{\oplus I} = \bigoplus_I \mathbf{1}$. These coherences discharge both functoriality laws (identity and composition) of the Grassmannian functor 1518.

24.5 The universal property

The representability theorem 1541 has two halves: the forward map $\varphi \mapsto \langle \varphi^* \mathcal{U}, \varphi^* u \rangle$ and its inverse, which manufactures a morphism $T \to \mathrm{Gr}(r, d)$ out of a rank-d quotient $\langle \mathcal{F}, q \rangle$ on T. The forward map needs the tautological quotient to actually be a quotient – that u is an epimorphism (1522). The inverse is built Zariski-locally and glued: for each size-d subset I one isolates the open locus $T_I \subseteq T$ where the I-indexed columns of q form a basis of $\mathcal{F}$, classifies $T_I \to U^I$ there, and glues over the cover $\{T_I\}$. The blocks of this section supply the four ingredients of that inverse: the iso-locus of a sheaf morphism (1523) and the fact that a morphism becomes invertible after restriction to its iso-locus (1524); the chart locus T_I (1525) and the proof that the chart loci cover T (1526); and the glued classifying morphism itself (1538).

Lemma 1522 (The tautological quotient is an epimorphism)**.** *The tautological quotient $u : \mathcal{O}^r_{\mathrm{Gr}(r,d)} \twoheadrightarrow \mathcal{U}$ (1516) is an epimorphism of sheaves of $\mathcal{O}_{\mathrm{Gr}(r,d)}$-modules.*

Proof. Being an epimorphism of sheaves of modules is local on the base: a morphism is epi precisely when its restriction to every member of an open cover is epi, since the cokernel is computed chartwise and a sheaf vanishing on a cover vanishes (the joint detection of morphisms across the chart cover is 1470). The charts $\{U^I\}$ cover $\mathrm{Gr}(r, d)$, and over U^I the tautological quotient, transported through the trivialisation $\mathcal{U}|_{U^I} \cong \mathcal{O}^d_{U^I}$ of 1510, is the chart quotient $u^I : \mathcal{O}^r_{U^I} \to \mathcal{O}^d_{U^I}$ (1516). Each u^I is a split epimorphism: its I-minor is the identity $1_{d \times d}$, so the coordinate inclusion $\mathcal{O}^d_{U^I} \hookrightarrow \mathcal{O}^r_{U^I}$ on the I-columns is a section of u^I, whence u^I is epi (1497). As u^I is epi on every chart, u is epi. ☐

Definition 1523 (Iso-locus of a morphism of module sheaves)**.** For a morphism $\varphi : \mathcal{M} \to \mathcal{N}$ of sheaves of $\mathcal{O}_X$-modules on a scheme X, the *iso-locus* $\mathrm{Iso}(\varphi) \subseteq X$ is the supremum (union) of all opens $U \subseteq X$ over which the restriction $\varphi|_U$ is an isomorphism – the largest open on which φ is invertible. A point t lies in $\mathrm{Iso}(\varphi)$ precisely when some open neighbourhood of t pulls φ back to an isomorphism.

Lemma 1524 (A morphism is invertible over its iso-locus)**.** *The restriction of $\varphi : \mathcal{M} \to \mathcal{N}$ to its iso-locus* $\mathrm{Iso}(\varphi)$ *(1523) is an isomorphism of sheaves of modules.*

Proof. Being an isomorphism of module sheaves is stalk-local. Each point of the iso-locus has an open neighbourhood on which φ pulls back to an isomorphism, and restriction along an open immersion preserves stalks; hence the stalk of the underlying abelian-sheaf morphism of φ is invertible at every point of the locus. The same stalk comparison for the locus inclusion shows $\varphi|_{\mathrm{Iso}(\varphi)}$ is stalkwise invertible, so it is an isomorphism of abelian sheaves by the stalk criterion; the forgetful functor to abelian sheaves reflects this back to module sheaves, and the restriction–pullback comparison transports it to the pullback functor. □

Definition 1525 (Chart locus T_I of a rank-d quotient)**.** For a rank-d quotient $x = \langle \mathcal{F}, q \rangle$ on T (1517) and a size-d subset $I \subseteq \{1, \dots, r\}$, the *chart composite* is $c_I = q \circ s_I : \mathcal{O}_T^d \to \mathcal{F}$ – the I-indexed coordinate inclusion s_I of free sheaves followed by the quotient map. The *chart locus* $T_I \subseteq T$ is the iso-locus $\mathrm{Iso}(c_I)$ (1523): the largest open of T on which c_I restricts to an isomorphism.

Lemma 1526 (The chart loci cover T)**.** *For a rank-d quotient x on T the chart loci $\{T_I\}_{\#I=d}$ (1525) form an open cover of T.*

Proof. Fix $t \in T$. By local freeness of $\mathcal{F}$, choose an affine open $W \ni t$ on which $\mathcal{F}|_W \cong \mathcal{O}_W^d$; the chart composite then becomes a $d \times r$ matrix of sections of $\mathcal{O}_W$. Epimorphy of q makes this matrix surjective at the residue field $\kappa(t)$, so some size-d column minor, indexed by a subset I, has nonzero determinant at t. Shrinking W to the basic open where that determinant is a unit, the minor is invertible, hence the chart composite c_I is an isomorphism there; thus $t \in T_I$. As t was arbitrary, the T_I cover T. □

24.5.1 Overlap compatibility of the chart morphisms

On the intersection $P = T_I \times_T T_J$ of two chart loci both chart composites pull back to isomorphisms, so the pulled-back quotient is presented by both chart matrices. The blocks below assemble the transports comparing the two presentations and proving the two chart morphisms agree into the glued scheme: invertibility along a composite (1528), the presented matrix along a composite (1529) and under an equality of base morphisms (1530), the chart matrix as a presented matrix (1531), its own minor (1532), the realisation $\mathrm{aeval}(M^I)(X^I) = M^I$ (1533), the image-matrix transport (1534), chart-morphism naturality (1535), the glue-condition identity (1536), and finally the overlap compatibility itself (1537).

Definition 1527 (Presented matrix M^I of a quotient along a morphism)**.** Let $x = \langle \mathcal{F}, q \rangle$ be a rank-d quotient on T and $j : P \to T$ a morphism along which the I-th chart composite c_I pulls back to an isomorphism. The *presented matrix* $M^I = \mathrm{presentedMatrix}(x, j, I)$ is the $d \times r$ matrix of sections of $\mathcal{O}_P$ whose rectangular-endomorphism realisation (1511) is the conjugated pullback $Q_r^{-1} \circ j^*q \circ (j^*c_I)^{-1} \circ Q_d$, with $Q_\bullet$ the free-pullback comparisons (1483). It generalises the chart matrix over T_I (the case $j = \iota_{T_I}$); its I-minor is the identity.

Lemma 1528 (Pullback-invertibility along a composite)**.** *If a^*c is an isomorphism of module sheaves, then so is $(p \circ a)^*c$ for any p, via the pseudofunctor comparison $(p \circ a)^* \cong p^*a^*$.*

Proof. The functor p^* carries the isomorphism a^*c to an isomorphism; transporting along the composition comparison identifies $p^*(a^*c)$ with $(p \circ a)^*c$, which is therefore an isomorphism. □

Lemma 1529 (Presented matrix along a composite)**.** *For $p : W \to V$ and $a : V \to T$, the matrix presenting x along $p \circ a$ is the entrywise Γ-restriction along p of the matrix presenting x along a:* $\mathrm{presentedMatrix}(x, p \circ a, I) = p^\sharp(\mathrm{presentedMatrix}(x, a, I))$*.*

Proof. Both sides are compared through their rectangular-endomorphism realisations (1511), which are injective on matrices. The realisation of the left side is the conjugated pullback along $p \circ a$; folding in the composition comparison $(p \circ a)^* \cong p^* a^*$ rewrites it as p^* applied to the realisation of the right side, conjugated by the free-pullback comparisons, which is exactly the realisation of the entrywise restriction $p^\sharp M^I$. □

Lemma 1530 (Presented matrix depends only on the base morphism)**.** *If $j = j'$ as morphisms $P \to T$, then* $\mathrm{presentedMatrix}(x, j, I) = \mathrm{presentedMatrix}(x, j', I)$*; the invertibility hypotheses are propositionally irrelevant.*

Proof. Substituting $j' = j$, the two presented matrices coincide by definition. □

Lemma 1531 (Chart matrix as a presented matrix)**.** *The chart matrix M^I of x over the chart locus T_I is the presented matrix of x along the locus inclusion:* $\mathrm{chartMatrix}(x, I) = \mathrm{presentedMatrix}(x, \iota_{T_I}, I)$*.*

Proof. Immediate from the definitions: the chart matrix is, entrywise, the presented matrix along the inclusion of the chart locus. □

Lemma 1532 (The own I-minor of M^I is the identity)**.** *The I-indexed column minor of* $\mathrm{presentedMatrix}(x, j, I)$ *is the $d \times d$ identity matrix.*

Proof. Through the rectangular-endomorphism realisation, the I-minor presents the inverted chart composite against itself, i.e. $(j^* c_I)^{-1} \circ (j^* c_I) = \mathrm{id}$; matching realisations and using injectivity of the realisation gives the identity matrix. □

Lemma 1533 (The classifying map realises X^I as M^I)**.** *Applying the ring map* $\mathrm{aeval}(M^I)$ *– the substitution sending each free variable $x^I_{p,q}$ to the corresponding entry of the presented matrix – entrywise to the universal matrix X^I returns M^I:* $\mathrm{aeval}(M^I)(X^I) = M^I$*.*

Proof. On the free entries $x^I_{p,q}$ (columns outside I) the substitution returns the matching entry of M^I by $\mathrm{aeval}(x^I_{p,q}) = M^I_{p,q}$. On the I-columns both X^I_I and M^I_I are the identity (1532), so the two agree there as well. □

Lemma 1534 (Transport of the image matrix along the structure map)**.** *Let I, J be size-d subsets and* $\mathrm{incl} : R^I_J \to D$ *a ring map into a commutative ring D whose restriction along $R^I \to R^I_J$ is $g : R^I \to D$. Then the image matrix $(X^I_J)^{-1} X^I$ pushed forward along* incl *equals $(Y_J)^{-1} Y$, where Y is the universal matrix X^I base-changed along g and Y_J its J-minor.*

Proof. Expand the image matrix as the product of the nonsingular inverse of the J-minor with X^I. A ring map commutes with matrix multiplication and, since the minor determinant is a unit, with the nonsingular inverse. Rewriting both factors along incl and using $\mathrm{incl} \circ (R^I \to R^I_J) = g$ yields $(Y_J)^{-1} Y$. □

Lemma 1535 (Naturality of the chart morphism)**.** *Precomposing the chart morphism $\varphi_I : T_I \to U^I$ with any $p : W \to T_I$ is the morphism $W \to U^I$ classified by the matrix presenting x along $p \circ \iota_{T_I}$; i.e. $p \circ \varphi_I$ is the* Γ–Spec *classifier of* $\mathrm{aeval}(\mathrm{presentedMatrix}(x, p \circ \iota_{T_I}, I))$*.*

Proof. By naturality of Γ–Spec, $p \circ \varphi_I$ is classified by the p-image of the ring map classifying φ_I, i.e. by $\mathrm{aeval}(p^\sharp M^I)$. By 1531 and 1529 one has $p^\sharp M^I = \mathrm{presentedMatrix}(x, p \circ \iota_{T_I}, I)$, giving the claim. □

Lemma 1536 (Chart-point identity: the glue condition in classifying form)**.** *Let K be a commutative ring and $f : R^I \to K$ a ring map under which the minor P^I_J is a unit. Then the transported point $\mathrm{lift}(f) \circ \tilde{\theta}_{I,J}$ presents the same K-point through chart J as f does through chart I; that is, $\mathrm{Spec}(\mathrm{lift}(f) \circ \tilde{\theta}_{I,J}) \circ \iota_J = \mathrm{Spec}(f) \circ \iota_I$ into the glued scheme $\mathrm{Gr}(r, d)$.*

Proof. Factor the J-classifier through the localization away from P^I_J: the transition map $\tilde{\theta}_{I,J}$ composed with the chart inclusion is the chart-transition immersion, so the glue condition $\theta_{I,J} \circ \iota_{JI} = \iota_{IJ}$ of the glue data transports $\mathrm{Spec}(\mathrm{lift}(f))$ on chart J to $\mathrm{Spec}(f)$ on chart I. The universal property of localization (that $\mathrm{lift}(f)$ restricts to f) closes the identity. □

Lemma 1537 (Overlap compatibility of the chart morphisms)**.** *On the intersection $P = T_I \times_T T_J$ of two chart loci, the two composite chart morphisms agree into the glued scheme: $\mathrm{pr}_1 \circ (\varphi_I \circ \iota_I) = \mathrm{pr}_2 \circ (\varphi_J \circ \iota_J)$.*

Proof. On P both chart composites c_I, c_J pull back to isomorphisms (1528), so the pulled-back quotient is presented by both matrices M^I_P and M^J_P. The J-minor of M^I_P is invertible – it presents c_J against c_I^{-1} (1532) – so the I-classifier sends P^I_J to a unit (1533). The change-of-basis identity $M^J_P = (M^I_{P,J})^{-1} M^I_P$, via 1534, identifies the localization-transport $\mathrm{lift}(\mathrm{aeval}\, M^I_P) \circ \tilde{\theta}_{I,J}$ of the I-classifier with the J-classifier. Hence by the chart-point identity 1536 both composites present the same point into the glued scheme, so they agree. □

Definition 1538 (The classifying morphism $T \to \mathrm{Gr}(r, d)$ of a quotient)**.** For a rank-d quotient $x = \langle \mathcal{F}, q \rangle$ on T, the *classifying morphism* $\mathrm{gr}(x) : T \to \mathrm{Gr}(r, d)$ is obtained by gluing, over the open cover $\{T_I\}$ of chart loci (1526), the composites $T_I \xrightarrow{\varphi_I} U^I \hookrightarrow \mathrm{Gr}(r, d)$ of the chart morphisms with the glue-data immersions; the overlap compatibility making the gluing legitimate is 1537. It is the inverse of the forward map of the universal property.

The two blocks below carry the chart data of a quotient through a base change $\psi : T' \to T$; they are the bridge underlying the two inverse laws of the representability theorem.

Lemma 1539 (Presented matrix of a pulled-back quotient)**.** *Source: [Nitsure], Construction of Grassmannian. For $\psi : T' \to T$, a rank-d quotient y on T, and a morphism $j : W \to T'$, presenting the pulled-back quotient $\psi^* y$ along j coincides with presenting y along the composite $j \circ \psi$:*

$$\mathrm{presentedMatrix}(\psi^* y, j, I) \;=\; \mathrm{presentedMatrix}(y, j \circ \psi, I).$$

This is the transport engine of the inverse laws of the universal property: the universal matrix X^I classifying a point of $\mathrm{Gr}(r, d)$ is, after base change, given by the same ring homomorphism out of $\mathbb{Z}[X^I]$ regardless of whether one pulls back first and then presents, or composes the base morphisms.

Proof. Compare the two matrices through their rectangular-endomorphism realisations (1511), which are injective. By definition of the pullback action (1517), $\psi^* y$ re-presents the source through the free-pullback comparison, so its pullback along j reads $j^*(\psi^* y).q = j^* P_r^{-1} \circ j^* \psi^* q$ and $j^* c_I(\psi^* y) = j^* P_d^{-1} \circ j^* \psi^* c_I(y)$, with $P_\bullet$ the comparisons for ψ. Fold the pseudofunctor composition comparison $(j \circ \psi)^* \cong j^* \psi^*$ into both the quotient map and the inverted chart composite, using the naturality of its unit and counit: the comparison square moves the conjugating isomorphisms across $j^*\psi^*q$ and $(j^*\psi^*c_I)^{-1}$, and the composite free-pullback comparisons $P_\bullet, Q_\bullet$ collapse into the single comparison for $j \circ \psi$. What remains is exactly the conjugated pullback of y along $j \circ \psi$, i.e. the realisation of presentedMatrix$(y, j \circ \psi, I)$. As the realisations agree, so do the matrices. $\square$

Lemma 1540 (Left inverse law: the classifier recovers ψ)**.** *Source: [Nitsure], Construction of Grassmannian (properness). For any morphism $\psi : T \to \mathrm{Gr}(r,d)$, classifying the pulled-back tautological quotient recovers ψ:*

$$\mathrm{gr}(\psi^*\langle \mathcal{U}, u\rangle) \;=\; \psi.$$

Proof. The preimages $\{\psi^{-1}(\iota_I(U^I))\}$ of the chart-immersion ranges cover T, so it suffices to check the equality after restriction to each member $W = \psi^{-1}(\iota_I(U^I))$. On W the composite $\iota_W \circ \psi$ factors through the chart immersion ι_I as $\rho \circ \iota_I$ for a unique $\rho : W \to U^I$, and W lies inside the chart locus T_I of $\psi^*\langle \mathcal{U}, u\rangle$ (the tautological chart composite is invertible over $\iota_I(U^I)$, and 1524 transports this), so all relevant chart composites pull back to isomorphisms. The matrix presenting $\psi^*\langle \mathcal{U}, u\rangle$ along ι_W equals, by 1539 and 1530, the matrix presenting the tautological pair along $\iota_W \circ \psi = \rho \circ \iota_I$, which by 1529 is the ρ-image of the universal matrix X^I (the tautological presentation over U^I). Hence by chart-morphism naturality 1535 the glued morphism $\mathrm{gr}(\psi^*\langle \mathcal{U}, u\rangle)$ restricted to W is the classifier of that matrix, namely $\rho \circ \iota_I = \iota_W \circ \psi$. As the members W cover T, the two morphisms agree. $\square$

Theorem 1541 ($\mathrm{Gr}(r,d)$ represents the Grassmannian functor)**.** *Source: [Nitsure], Exercises (2). The scheme* $\mathrm{Gr}(r,d)$*, together with the tautological quotient* $\langle \mathcal{U}, u\rangle$*, represents the Grassmannian functor* $\mathrm{Grass}(r,d)$ *(1518): for every scheme T there is a bijection, natural in T,*

$$\mathrm{Hom}(T, \mathrm{Gr}(r,d)) \xrightarrow{\sim} \mathrm{Grass}(r,d)(T), \qquad \psi \longmapsto \langle \psi^*\mathcal{U}, \psi^*u\rangle.$$

Proof. The forward map sends ψ to the pullback $\psi^*\langle \mathcal{U}, u\rangle$ of the tautological pair; naturality in T is the pseudofunctoriality of the pullback action (1518) evaluated at the tautological point. The inverse is the chart-by-chart classifier gr of 1538. The two are mutually inverse: the left inverse law $\mathrm{gr}(\psi^*\langle \mathcal{U}, u\rangle) = \psi$ is 1540, checked over the cover of chart-immersion preimages; the right inverse law $\langle \mathrm{gr}(x)^*\mathcal{U}, \mathrm{gr}(x)^*u\rangle \sim x$ holds because over each chart locus of x both quotients are presented by the same matrix, so the two epimorphism descents are mutually inverse, witnessing the equivalence. Surjectivity of u (1522) makes the tautological pair an honest quotient, so the forward map indeed lands in $\mathrm{Grass}(r,d)(T)$. $\square$

Chapter 25

FGA representability of the Picard scheme

25.1 Setup and motivation

Throughout this chapter, fix a base field k and a smooth proper geometrically integral curve C over k (the curve carrying the Jacobian). The relative Picard presheaf $\mathrm{Pic}^{\sharp}_{C/k} : (\mathrm{Sch}/k)^{op} \to \mathrm{Set}$ of 14 has already been packaged as a set-valued functor and refined to its abelian-group-valued étale-sheafified version $\mathrm{Pic}^{\sharp}_{(C/k)\text{ét}}$ in 15. What remains is to produce a scheme representing it: the *Picard scheme* $\mathrm{Pic}_{C/k}$. The present chapter packages that assembly, together with the abelian-group-scheme refinement.

Two routes to representability. The Picard scheme of a curve can be constructed in two ways, and this blueprint records both, because they need different inputs and only one of them is the route under construction here.

The *quotient route* is Grothendieck's, as presented by Kleiman §4: the Abel map $\mathrm{Div}_{C/k} \to \mathrm{Pic}^{\sharp}_{C/k}$ exhibits the Picard functor as a quotient of a quasi-projective scheme by a smooth proper equivalence relation, and Altman–Kleiman descent (1546) represents the quotient. Its engine is the Hilbert scheme $\mathrm{Hilb}_{C/k} = \mathrm{Quot}_{\mathcal{O}_C/C/k}$ of 20, whose open subscheme $\mathrm{Div}_{C/k}$ of relative effective divisors carries the source of the Abel map. This is the route of 1545 and its proof below.

The *Milne–Kollár route* builds $\mathrm{Pic}^{r}_{C/k}$ directly, over a separably closed field, from the open loci

$$C^{\Sigma} = \{\, t : h^0(\mathcal{O}(W - \Sigma)_t) = 1 \,\}$$

cut out inside a degree-r divisor scheme by a tuple Σ of rational points, glues them, and descends the result to k by a finite Galois quotient before assembling the degrees by a coproduct. Its inputs are a rigidified pushforward engine, uniform H^1 vanishing, and Galois descent of rigidified classes — not a Hilbert or Quot scheme. Its structure, and the parts of it that are established, are the subject of 25.4.

The decisive difference is the quotient step. Altman–Kleiman descent needs quasi-projectivity of the scheme being quotiented, which is not expressible in the formalisation at present and without which the lemma is false (1547); the Milne–Kollár route takes only finite Galois quotients under an orbit-in-affine hypothesis, and so avoids that obstruction entirely. The Lean development therefore pursues the Milne–Kollár route. The quotient-route material below is retained as the mathematics it is — the divisor and Hilbert-scheme substrate it rests on is consumed by both routes — and is not the path currently being formalised.

Comparison with the sheafified Picard functor. Kleiman §4 represents the étale-sheafified functor $\mathrm{Pic}_{(C/k)\text{ét}}$. This chapter instead works throughout under the standing hypothesis that C *has a* k*-rational point* (1559) and states representability against the *plain* relative Picard functor $\mathrm{Pic}^{\sharp}_{C/k}$ of 633. This is justified by Kleiman §2, Thm. 2.5 (`th:comp`): if $f : X \to S$ has a section and $\mathcal{O}_{S'} = f'_*\mathcal{O}_{X'}$ holds for all base changes (automatic for our proper geometrically integral curve, by cohomology and base change), then the comparison maps

$$\mathrm{Pic}_{X/S}(T) \longrightarrow \mathrm{Pic}_{(X/S)\text{ét}}(T) \longrightarrow \mathrm{Pic}_{(X/S)\text{fppf}}(T)$$

are bijective for every T. Hence under the rational-point hypothesis the plain relative functor *is* the étale sheaf, so the rational point allows us to work directly with $\mathrm{Pic}^{\sharp}_{C/k}$.

The hypothesis is not a convenience: without a section the plain functor is not even a Zariski sheaf in general, so a representability statement for $\mathrm{Pic}^{\sharp}_{C/k}$ with no rational point would be false. Two honest formulations are therefore available, and they prove different theorems.

1. Represent $\mathrm{Pic}^{\sharp}_{C/k}$ and carry $C(k) \neq \emptyset$ as a hypothesis. This is the formulation of 1545 and of every representability assertion in this chapter. It is strictly weaker than a statement about all smooth proper geometrically integral curves, since such a curve need not have a rational point — a conic without rational points, or a genus-2 curve over $\mathbb{Q}$ with no rational point, satisfies every other hypothesis.

2. Represent the étale sheafification $\mathrm{Pic}_{(C/k)\text{ét}}$ and drop the rational point. This is Kleiman's own formulation, and needs no section precisely because sheafifying supplies étale-locally what the section would have supplied globally. It requires developing the Picard functor on the étale site.

Which formulation the project should target is a decision about what theorem is being claimed, not a limitation of the platform. The two are recorded here side by side; the chapter as written realises the first.

Every representability assertion of this chapter accordingly carries the rational-point hypothesis. A pointed curve (C, P), as in 12, supplies such a section.

25.2 Line bundles and the Quot / Hilbert scheme

The bridge from line bundles on $C \times_k T$ to the Quot scheme is the Abel map: a relative effective divisor $D \subset C \times_k T$ over T (a T-flat closed subscheme whose ideal is invertible) determines a closed subscheme of $C \times_k T$ flat over T, hence a T-point of the Hilbert functor $\mathrm{Hilb}_{C/k}(T)$, which is the special case $\mathrm{Quot}_{\mathcal{O}_C/C/k}$ of the Quot functor of 20; and it determines an invertible sheaf $\mathcal{O}_{C\times_k T}(D)$ on the relative curve, representing a class in $\mathrm{Pic}^{\sharp}_{C/k}(T)$.

Definition 1542 (Abel-map witness)**.** Let C/k be a smooth proper curve. Define a natural transformation

$$A^{\mathrm{raw}}_{C/k} : \mathrm{Div}_{C/k} \longrightarrow \mathrm{Pic}^{\sharp}_{C/k}$$

as the composite $\mathrm{abelKernelNatTrans}(C)\,;\mathrm{picNeg}(C)$.

Proof. The first transformation sends the class represented by $\langle \mathcal{F}, q\rangle$ to $[\ker q]$. It is well defined because equivalent divisor families induce isomorphic kernels. It is natural under arbitrary base change because the kernel–pullback comparison is an isomorphism for an epimorphic quotient with finitely presented, base-flat target and invertible kernel. The second transformation is pointwise inversion in the group-valued presheaf $\mathrm{Pic}^{\sharp}_{C/k}$. Their composite is therefore natural and sends $\langle \mathcal{F}, q\rangle$ to $-[\ker q]$. □

Lemma 1543 (Line bundle / Quot correspondence via the Abel map)**.** *Source: [Kleiman], "The Picard scheme", §3, Thm. 3.7 + Def. 4.6 (cf. [Nitsure], "Construction of Hilbert and Quot Schemes", §1, where* $\mathrm{Hilb}_{X/S} = \mathrm{Quot}_{\mathcal{O}_X/X/S}$*). Let C/k be a smooth proper geometrically integral curve. There is a natural transformation of presheaves on* $(\mathrm{Sch}\,/k)^{op}$

$$A_{C/k} = \mathrm{abelMap} : \mathrm{Div}_{C/k} \longrightarrow \mathrm{Pic}^{\sharp}_{C/k}$$

($\mathrm{Div}_{C/k}$ *the relative-divisor functor of 1562,* $\mathrm{Pic}^{\sharp}_{C/k}$ *the relative Picard functor of 1560), with the following* defining property*: at a T-point represented by a family $\langle \mathcal{F}, q\rangle$ (1159: a finitely presented, T-flat quotient $q\colon \mathcal{O}_{C\times_k T} \twoheadrightarrow \mathcal{F}$ whose kernel is an invertible ideal sheaf cutting out a relative effective divisor $D \subseteq C \times_k T$),*

$$A_{C/k}(T)(\langle \mathcal{F}, q\rangle) \;=\; -[\ker q] \;=\; [(\ker q)^{-1}] \;=\; [\mathcal{O}_{C\times_k T}(D)] \;\in\; \mathrm{Pic}^{\sharp}_{C/k}(T),$$

the class of the associated invertible sheaf $\mathcal{O}_{C\times_k T}(D)$, i.e. the additive inverse of the class of the ideal sheaf $\ker q$ *(Kleiman §3, Def. dfn:Abel). This transformation is the required witness for* $\mathrm{HasAbelMap}(C)$ *(1564).*

Proof. The instance of 1565 identifies $A_{C/k}$ with the witness of 1542. Evaluating its two factors at $\langle \mathcal{F}, q\rangle$ first gives the kernel class $[\ker q]$ and then its inverse. Since the kernel is the ideal sheaf of D, its inverse is $\mathcal{O}_{C\times_k T}(D)$, which gives the stated formula. □

Remark 1544 (Representability of $\mathrm{Div}_{C/k}$)**.** Kleiman §3, Thm. th:repDiv, shows that $\mathrm{Div}_{C/k}$ is representable by an open subscheme $\mathrm{Div}_{C/k} \subseteq \mathrm{Hilb}_{C/k}$ of the Hilbert scheme ($\mathrm{Hilb}_{C/k} = \mathrm{Quot}_{\mathcal{O}_C/C/k}$, 20): letting $W \subseteq C \times_k \mathrm{Hilb}_{C/k}$ be the universal closed subscheme and $q\colon W \to \mathrm{Hilb}_{C/k}$ the projection, the locus $V \subseteq W$ where W is locally an effective Cartier divisor is open, so $Z := q(W \setminus V)$ is closed (q proper) and $\mathrm{Div}_{C/k} := \mathrm{Hilb}_{C/k} \setminus Z$ is the open subscheme representing relative effective divisors. This is the relative-divisor functor appearing in 1562.

25.3 The FGA representability theorem

We now state and prove the specialisation of Grothendieck's existence theorem for the Picard scheme, in the specialisation needed for the Jacobian: $S = \mathrm{Spec}\,k$, $X = C$ a smooth proper geometrically integral curve. The general theorem (Kleiman §4, Thm. 4.8) covers any X/S projective Zariski-locally over S, flat, with integral geometric fibers; we record the general statement and then specialise.

Theorem 1545 (FGA representability of the Picard scheme)**.** *Source: [Kleiman], "The Picard scheme", §4, Main Thm. 4.8 + Cor. 4.18.3. Let k be a field and C a smooth proper geometrically integral curve over k* with a k-rational point *(1559). Then the relative Picard functor* $\mathrm{Pic}^{\sharp}_{C/k}$ *of 633 — which under the rational-point hypothesis coincides with its étale sheafification* $\mathrm{Pic}_{(C/k)\text{ét}}$*, by Kleiman §2, Thm. 2.5 — is representable by a k-scheme* $\mathrm{Pic}_{C/k}$*, separated and locally of finite* *type over k, which is a disjoint union of open quasi-projective k-subschemes, each an increasing union of open quasi-projective subschemes. The scheme $\mathrm{Pic}_{C/k}$ is called the* Picard scheme *of C/k.*

Proof. We follow Kleiman §4, Thm. 4.8, specialised to $S = \operatorname{Spec} k$ and $X = C$. The proof proceeds in four conceptual steps, preceded by a comparison step; the underlying algebraic engine is the Quot scheme of 20 and the relative Picard functor of 15, together with the line-bundle / Quot correspondence of 1543.

Step 0: reduce to the étale sheaf via the rational point. By Kleiman §2, Thm. 2.5, since $C \to \operatorname{Spec} k$ has a section (1559) and $\mathcal{O}_T = \pi_{T*}\mathcal{O}_{C\times_k T}$ holds for every k-scheme T (cohomology and base change for the proper geometrically integral fibers of $C \times_k T/T$), the comparison map $\mathrm{Pic}^{\sharp}_{C/k}(T) \to \mathrm{Pic}_{(C/k)\text{ét}}(T)$ is bijective for all T. It therefore suffices to represent the étale sheaf $\mathrm{Pic}_{(C/k)\text{ét}}$, which is what Kleiman's theorem produces; the same scheme then represents $\mathrm{Pic}^{\sharp}_{C/k}$.

Step 1: stratify by Hilbert polynomial. For each polynomial $\phi \in \mathbb{Q}[n]$, let $P^{\phi} \subseteq \mathrm{Pic}^{\sharp}_{(C/k)\text{ét}}$ be the étale subsheaf whose T-points are represented by invertible sheaves $\mathcal{L}$ on $C \times_k T$ with $\chi(C_t, \mathcal{L}_t^{-1}(n)) = \phi(n)$ for all $t \in T$ (the Hilbert polynomial relative to the chosen polarisation $\mathcal{O}_C(1)$ on the projective curve C). The P^{ϕ} form a disjoint open cover of $\mathrm{Pic}^{\sharp}_{(C/k)\text{ét}}$ (openness via Kleiman's flat-base-change argument, 480 / Stacks Tag 02KH; cf. EGA III 7.9.11). It suffices to represent each P^{ϕ} by an increasing union of open quasi-projective k-schemes.

Step 2: bound twists by m-regularity. Given ϕ, fix m large enough that the T-points of P^{ϕ} are representable by $\mathcal{L}$ on $C \times_k T$ satisfying $R^i\pi_{T*}\mathcal{L}(n) = 0$ for all $i \geq 1$, $n \geq m$. Twisting by $\mathcal{O}_C(m)$ defines an automorphism of $\mathrm{Pic}^{\sharp}_{(C/k)\text{ét}}$ that carries P^{ϕ} to $P_0^{\phi_0}$ for $\phi_0(n) := \phi(m+n)$. Hence it suffices to represent each $P_0^{\phi_0}$ by a quasi-projective k-scheme.

Step 3: factor through the Abel map. Consider the Abel map $A_{C/k} : \mathrm{Div}_{C/k} \to \mathrm{Pic}^{\sharp}_{(C/k)\text{ét}}$ of 1543. The source $\mathrm{Div}_{C/k}$ is representable by an open subscheme of $\mathrm{Hilb}_{C/k}$ (and hence by an open subscheme of the Quot scheme of 1229), and is quasi-projective over k. Pulling back along $P_0^{\phi_0} \hookrightarrow \mathrm{Pic}^{\sharp}_{(C/k)\text{ét}}$ yields an open subscheme $Z := P_0^{\phi_0} \times_{\mathrm{Pic}^{\sharp}_{(C/k)\text{ét}}} \mathrm{Div}_{C/k}$ of $\mathrm{Div}_{C/k}$, which is again quasi-projective. The projection $\alpha : Z \to P_0^{\phi_0}$ is a surjection of étale sheaves: given a T-point λ of $P_0^{\phi_0}$ represented by an étale cover $T' \to T$ and an invertible sheaf $\mathcal{L}'$ on $C \times_k T'$ with $H^1(C_{t'}, \mathcal{L}'_{t'}) = 0$, the projective bundle $\mathbb{P}(\mathcal{Q}) \to T'$ associated to $\mathcal{Q} := \pi_{T'*}\mathcal{L}'$ is smooth over T'; an étale cover $T_1 \to T'$ admits a T'-map $T_1 \to \mathbb{P}(\mathcal{Q})$ (EGA IV 17.16.3 (ii)) whose composition with the universal divisor map $\mathbb{P}(\mathcal{Q}) \to Z$ produces the required T_1-point of Z. Moreover, the first projection $Z \times_{P_0^{\phi_0}} Z \to Z$ is smooth and proper.

Step 4: descend by the smooth-proper quotient lemma. Apply Kleiman §4, Lem. 4.9: given a surjection $\alpha : Z \to P$ of étale sheaves with Z quasi-projective, $R := Z \times_P Z$ representable, and the first projection $R \to Z$ smooth and proper, P is representable by a quasi-projective scheme. In our case $P = P_0^{\phi_0}$, Z and R as above, so $P_0^{\phi_0}$ is quasi-projective. Reassembling over $\phi \in \mathbb{Q}[n]$ and over m, the disjoint union of representable P^{ϕ} pieces represents $\mathrm{Pic}^{\sharp}_{(C/k)\text{ét}}$. The total scheme is separated (as a disjoint union of separated quasi-projectives) and locally of finite type. This is Kleiman §4, Cor. 4.18.3 in the $S = \operatorname{Spec} k$, $X = C$ complete case. □

25.3.1 The smooth-proper quotient lemma

The structural lemma underlying step 4 above is recorded separately because it is the abstract content allowing descent from $Z = \mathrm{Div}_0^{\phi_0}$ to the representing Picard piece.

Lemma 1546 (Smooth-proper quotient of étale sheaves)**.** *Source: [Kleiman], "The Picard scheme", §4, Lem. 4.9. Let* $\alpha : Z \to P$ *be a morphism of étale sheaves on* (Sch/k)*. Set* $R := Z \times_P Z$*. Suppose:*

1. *α is a surjection of étale sheaves;*

2. *Z is representable by a quasi-projective k-scheme;*

3. *R is representable by a k-scheme;*

4. *the first projection $R \to Z$ is representable by a smooth and proper map.*

Then P is representable by a quasi-projective k-scheme, and α is representable by a smooth map.

Proof. By (ii) the structure map $Z \to \operatorname{Spec} k$ is quasi-projective, hence separated, so the first projection $Z \times_k Z \to Z$ is separated. By (iv) the first projection $R \to Z$ is proper; it factors through $h : R \to Z \times_k Z$ over the diagonal embedding (the second component is the second projection of $R \to Z \times_P Z$). The map h is therefore proper, and a monomorphism (because R is the graph of an equivalence relation on Z, hence injective on T-points). By EGA IV 8.11.5, a proper monomorphism is a closed immersion: h is a closed embedding.

Hence $R \hookrightarrow Z \times_k Z$ is a flat (by (iv)) and proper equivalence relation on the quasi-projective k-scheme Z. By the theory of flat-and-proper effective equivalence relations on quasi-projective schemes (Altman–Kleiman; the construction places the graph $R \subseteq Z \times_k Z$ inside the universal subscheme of the Hilbert scheme $\mathrm{Hilb}_{Z/k}$ and descends to a closed subscheme $Q \subseteq \mathrm{Hilb}_{Z/k}$ representing the quotient), there exist a quasi-projective k-scheme Q and a faithfully flat, projective map $Z \to Q$ with $R = Z \times_Q Z$.

The map $Z \to Q$ is flat and proper; its fibers, up to field extension, coincide with those of $R \to Z$, which are smooth by hypothesis. Hence $Z \to Q$ is smooth.

Finally, $Z \to Q$ is a surjection of étale sheaves (an étale cover of any T-point of Q pulls back to a T'-point of Z, by EGA IV 17.16.3 (ii) applied to $Z \times_Q T \to T$). In the category of étale sheaves, both Q and P are coequalisers of the pair $R \rightrightarrows Z$; the coequaliser is unique up to unique isomorphism, so Q represents P. Smoothness of $\alpha : Z \to P$ is the smoothness of $Z \to Q$ just shown. □

Remark 1547 (Quasi-projectivity is load-bearing)**.** Hypothesis (ii) of 1546 — that Z be representable by a *quasi-projective* k-scheme — cannot be weakened to bare representability. A Hironaka-type free $\mathbb{Z}/2$-action on a smooth proper non-projective threefold gives a smooth proper equivalence relation satisfying (i), (iii) and (iv) whose quotient étale sheaf is an algebraic space and not a scheme.

This is what separates the two routes of 25.1. The quotient route must supply quasi-projectivity of the Abel-map slice; the Milne–Kollár route quotients only by a finite Galois group, under the hypothesis that every finite orbit lies in an affine open, which is a condition the Hironaka example fails.

25.4 The Milne–Kollár route

This section records the route of 25.1 that avoids the quasi-projectivity obstruction of 1547, in the three parts it decomposes into, and states the two of its inputs that are established.

Over a separably closed field k' the construction is Milne §4. Fix $r \geq 2g + 1$ and a degree-r divisor scheme Z. For a tuple Σ of $r - g$ rational points of $C' = C \times_k k'$, the locus

$$C^{\Sigma} = \{\, t \in Z : h^0(\mathcal{O}(W - \Sigma)_t) = 1 \,\}$$

is open, because h^0 is upper semicontinuous and $h^0 \geq \chi = 1$ in degree g; adding Σ back to the canonical divisor family of the rigidified pushforward gives a second map $C^\Sigma \to Z$, and the equaliser J^Σ of the two represents the subfunctor of degree-r classes $\mathcal{L}$ with $h^0(\mathcal{L}(-\Sigma)_t) = 1$ at every point. These subfunctors cover, so gluing them represents $\mathrm{Pic}^r_{C'/k'}$. Descent to k is a finite Galois quotient, and the degrees assemble as a coproduct over $\mathbb{Z}$ through the translation isomorphisms $\mathrm{Pic}^d \cong \mathrm{Pic}^r$ given by tensoring with $\mathcal{O}((r-d)x_0)$.

The two inputs the route rests on that the quotient route does not are the rigidified pushforward and the finite Galois quotient. The first is local freeness of $q_*\mathcal{L}$ together with its compatibility with base change, and it holds for every curve satisfying the hypotheses of this chapter (1551). The second is Speiser descent together with the existence of Γ-stable affine neighbourhoods.

Theorem 1548 (Local freeness of the rigidified pushforward)**.** *Source: [Mumford],* Abelian Varieties*, II §5. Let C be a smooth proper geometrically integral curve over a field k, let A be a finitely generated k-algebra and $q : C_A \to \operatorname{Spec} A$ the constant family. Let $\mathcal{L}$ be an invertible sheaf on C_A such that for every point t of $\operatorname{Spec} A$ some two-chart affine cover of the fibre C_t has surjective Čech difference map on $\mathcal{L}_t$ — for a separated fibre with affine chart intersection this computes $H^1(C_t, \mathcal{L}_t) = 0$. Then every point of $\operatorname{Spec} A$ has an open neighbourhood on which $q_*\mathcal{L}$ is free of rank $h^0(C_t, \mathcal{L}_t)$.*

Remark 1549 (Why the vanishing hypothesis is stated Čech-locally)**.** The hypothesis of 1548 asks for *some* two-chart cover of each fibre with surjective difference map, rather than for vanishing of the sheaf cohomology $H^1(C_t, \mathcal{L}_t)$. On a separated fibre whose two charts have affine intersection the Čech complex of that cover computes the sheaf cohomology, so the two agree; the existential form is the weaker-looking one and is implied by any form of $h^1 = 0$, so a consumer supplying sheaf vanishing supplies this. The identification is cover-independent by Mayer–Vietoris, but that independence is not itself established here, which is why the hypothesis is stated in the form that does not presuppose it.

Proof. Choose a finite morphism $\pi : C \to \mathbb{P}^1_k$ (1876) and base change it to $\pi_A : C_A \to \mathbb{P}^1_A$. Since π_A is finite, it is affine and π_{A*} is exact, so $q_*\mathcal{L} = p_*(\pi_{A*}\mathcal{L})$ for $p : \mathbb{P}^1_A \to \operatorname{Spec} A$, and the fibrewise cohomology of $\pi_{A*}\mathcal{L}$ agrees with that of $\mathcal{L}$.

On $\mathbb{P}^1_A$ the two standard charts form an affine cover with affine intersection, so the Čech complex of any quasi-coherent module is a two-term complex d of $\Gamma(\operatorname{Spec} A, \mathcal{O})$-modules with $H^0 = \ker d$ and $H^1 = \operatorname{coker} d$. Vanishing of H^1 at every point makes d surjective and $\ker d$ finitely generated projective, hence flat, and makes the formation of $\ker d$ commute with arbitrary base change; this is Mumford's two-term finite-free replacement.

Global sections of the pushforward are $\ker d$ by the sheaf condition, so $q_*\mathcal{L}$ is finite locally free, and it remains to identify its rank. Flatness gives $\mathrm{rank}_t = \dim_{\kappa(t)} \kappa(t) \otimes_A \ker d$, base change moves the residue field inside the kernel, and comparing the $\kappa(t)$-base-changed Čech square with the Čech square of the fibre identifies that dimension with $h^0(C_t, \mathcal{L}_t)$. A finite locally free module of locally constant rank is free on a neighbourhood of each point. ☐

Lemma 1550 (Global sections of the pushforward commute with base change)**.** *In the situation of 1548, for every ring map $A \to A'$ — with no finiteness, flatness or noetherian hypothesis on A' — the canonical map*

$$A' \otimes_A \Gamma(C_A, \mathcal{L}) \longrightarrow \Gamma(C_{A'}, g'^*\mathcal{L})$$

is bijective.

Proof. Push forward along the finite $\pi_A : C_A \to \mathbb{P}^1_A$ as in 1548 and work with the two-term Čech complex d of the preimage cover. Fibrewise vanishing of H^1 makes d surjective, and for a surjective two-term complex the formation of $\ker d$ is universally exact: base changing the short exact sequence $0 \to \ker d \to K^0 \to K^1 \to 0$ along $A \to A'$ stays exact because K^1 is flat, so $A' \otimes_A \ker d \to \ker(d \otimes A')$ is bijective. Global sections of the pushforward are $\ker d$ on both sides by the sheaf condition, which gives the claim. □

Theorem 1551 (The rigidified pushforward exists for every curve). *Let C be a smooth proper geometrically integral curve over a field k. Then C admits the rigidified pushforward: for every finitely generated k-algebra A and every invertible sheaf on C_A with fibrewise vanishing H^1, the pushforward $q_*\mathcal{L}$ is locally free of rank h^0 of the fibre and its formation commutes with arbitrary base change.*

Proof. The two conclusions are 1548 and, by affine-target descent from the module-level statement of 1550, the base-change conclusion: over an affine base both ends of the adjunction mate are quasi-coherent, and a morphism of quasi-coherent modules on an affine scheme is an isomorphism as soon as it is bijective on global sections. □

Lemma 1552 (The projective line over a field is integral). *Let k be a field. Then $\mathbb{P}^1_k$ is an integral scheme.*

Proof. Each standard chart $V_i \subseteq \mathbb{P}^1_k$ is affine with section ring the polynomial ring $k[T]$ in the chart coordinate, hence an integral domain, so each chart is an integral scheme. Reducedness of $\mathbb{P}^1_k$ follows because the two charts cover and reducedness is local. For irreducibility, the overlap $W = V_0 \cap V_1$ is a nonempty open subset of the irreducible V_0, hence irreducible and dense in V_0, and symmetrically dense in V_1; as $V_0 \cup V_1 = \mathbb{P}^1_k$, the set W is a dense irreducible subset of $\mathbb{P}^1_k$, which is therefore the closure of an irreducible set and so irreducible. □

Theorem 1553 (Speiser descent). *Let L/K be a finite Galois extension with group Γ, and let A be an L-algebra carrying an action of Γ by ring automorphisms which is semilinear over the action of Γ on L. Then the canonical map*

$$L \otimes_K A^\Gamma \longrightarrow A, \qquad a \otimes w \mapsto aw,$$

is an isomorphism of L-algebras: the invariants form a K-form of A. No finiteness hypothesis on A is required.

Proof. The same statement for semilinear Γ-modules is the classical normal-basis argument: writing $L = K[\Gamma] \cdot \theta$ for a normal basis element θ, the trace $\sum_\gamma \gamma(\theta a)\gamma$ exhibits every element of A as an L-combination of invariants, and L-independence of the invariant part follows from linear independence of the characters γ. The map above is the L-algebra map extending the inclusion $A^\Gamma \hookrightarrow A$, and its underlying additive map is the module-level descent map, which the module statement shows bijective. Bijectivity of an algebra homomorphism makes it an algebra isomorphism. □

Theorem 1554 (Points of the finite Galois quotient). *In the situation of 1553, for every K-algebra B the map $g \mapsto (\mathrm{id}_L \otimes g)$ is a bijection*

$$\mathrm{Hom}_{K\text{-}alg}(A^\Gamma, B) \;\simeq\; \{\, f \in \mathrm{Hom}_{L\text{-}alg}(A, L \otimes_K B) : f(\gamma \cdot x) = (\gamma \otimes \mathrm{id})(f(x)) \,\}.$$

Equivalently, for an affine K-scheme $T = \operatorname{Spec} B$, the T-points of $\operatorname{Spec} A^\Gamma$ are the Γ-equivariant T_L-points of $\operatorname{Spec} A$: the affine scheme $\operatorname{Spec} A^\Gamma$ is the quotient of $\operatorname{Spec} A$ by Γ.

Proof. Given g, transport $\mathrm{id}_L \otimes g$ through the isomorphism $A \cong L \otimes_K A^\Gamma$ of 1553; equivariance is the identity $\gamma \otimes \mathrm{id}$ transported along the same isomorphism, which is exactly the compatibility recorded there. Conversely, an equivariant f carries invariants to Γ-fixed elements of $L \otimes_K B$, and for the left-factor action on a base change the fixed points are $1 \otimes B$; so f restricts to a K-algebra map $A^\Gamma \to B$. The two constructions are mutually inverse: one direction is the computation of the restriction on invariants, and the other holds because the invariants span A over L, so two L-algebra maps agreeing on them agree. ☐

Lemma 1555 (Γ-stable affine neighbourhoods)**.** *Let L/K be finite Galois with group Γ and let X be a scheme over L with a semilinear Γ-action all of whose orbits are contained in affine opens. Then every point of X has a Γ-stable affine open neighbourhood.*

Proof. Let $x \in X$ and let U be an affine open containing the orbit of x. The orbit is finite and lies in the open $W = \bigcap_\gamma \gamma^{-1}U$, so by prime avoidance inside the affine U there is a section $s \in \Gamma(X, U)$ with the whole orbit contained in $D(s) \subseteq W$. Put $N = \prod_\gamma \gamma(s)|_W$. Then $D(N) = \bigcap_\gamma \gamma^{-1}D(s)$, which contains x because the orbit lies in $D(s)$, and is Γ-stable because translating the intersection permutes its factors. Finally $D(N) \subseteq D(s)$ by the factor $\gamma = 1$, so $D(N)$ is a basic open of the affine $D(s)$ and hence affine. ☐

Remark 1556 (The orbit hypothesis is load-bearing)**.** The hypothesis of 1555 that every orbit lie in an affine open cannot be dropped, and it is the same phenomenon as 1547: the Hironaka free $\mathbb{Z}/2$-action on a smooth proper non-projective threefold has an orbit contained in no affine open, and its quotient is an algebraic space rather than a scheme. The difference between the two routes is not that one needs a hypothesis and the other does not, but that this hypothesis is available for the schemes the Milne–Kollár route quotients — glued from opens carrying locally closed immersions into a Grassmannian — whereas quasi-projectivity of the Abel-map slice is not currently expressible.

25.5 Group-scheme structure

The Picard scheme $\mathrm{Pic}_{C/k}$ inherits an abelian group structure from the tensor product of invertible sheaves on $C \times_k T$. The relative Picard presheaf $T \mapsto \mathrm{Pic}(C \times_k T)/H_T$ is already an abelian group at every T (the quotient of an abelian group by a subgroup), functorially in T; this lifts to the étale sheafification, and then to a group-scheme structure on the representing scheme by Yoneda.

Theorem 1557 ($\mathrm{Pic}_{C/k}$ is a k-group scheme)**.**
Source: [Kleiman], "The Picard scheme", §2, Def. 2.2 + §4, Def. 4.1 (the Picard scheme). Let $\mathrm{Pic}_{C/k}$ be the Picard scheme of 1545. Then $\mathrm{Pic}_{C/k}$ is canonically an abelian group scheme over k: the tensor product $[\mathcal{L}] + [\mathcal{M}] := [\mathcal{L} \otimes \mathcal{M}]$ and inverse $-[\mathcal{L}] := [\mathcal{L}^{-1}]$ on $\mathrm{Pic}^\sharp_{C/k}(T)$ (well defined because both operations preserve isomorphism classes and descend through the subgroup $H_T = \pi_T^*\mathrm{Pic}(T)$ of 616) are natural in T, hence by Yoneda are represented by morphisms $m : \mathrm{Pic}_{C/k} \times_k \mathrm{Pic}_{C/k} \to \mathrm{Pic}_{C/k}$, $i : \mathrm{Pic}_{C/k} \to \mathrm{Pic}_{C/k}$, and $e : \mathrm{Spec}\, k \to \mathrm{Pic}_{C/k}$ (the trivial line bundle class) satisfying the abelian-group axioms. In particular, $\mathrm{Pic}_{C/k}$ is a k-group scheme locally of finite type.

Proof. The relative Picard functor $\mathrm{Pic}^\sharp_{C/k}$ of 633 is abelian-group valued, with sum $[\mathcal{L}] + [\mathcal{M}] = [\mathcal{L} \otimes \mathcal{M}]$, inverse $-[\mathcal{L}] = [\mathcal{L}^{-1}]$, and unit $0 = [\mathcal{O}_{C \times_k T}]$; these operations descend through the quotient by $H_T = \pi_T^*\mathrm{Pic}(T)$ because π_T^* is itself a group homomorphism (703).

By 1545, $\mathrm{Pic}^{\sharp}_{C/k}$ is represented by the scheme $\mathrm{Pic}_{C/k}$. The Yoneda embedding is fully faithful and preserves limits, so a representing object of a group-valued presheaf is canonically a group object: the natural transformations $\mathrm{Pic}^{\sharp}_{C/k} \times \mathrm{Pic}^{\sharp}_{C/k} \to \mathrm{Pic}^{\sharp}_{C/k}$ (multiplication), $\mathrm{Pic}^{\sharp}_{C/k} \to \mathrm{Pic}^{\sharp}_{C/k}$ (inverse), and the unit map $h_{\mathrm{Spec}\,k} \to \mathrm{Pic}^{\sharp}_{C/k}$ pick out unique morphisms of representing schemes m, i, e as displayed, and the abelian-group axioms hold at the level of T-points functorially in T, hence on the representing scheme. The local-finite-type property is part of 1545. $\square$

25.6 The Picard scheme

The two preceding theorems determine the following named scheme.

Definition 1558 (The Picard scheme)**.**
Source: [Kleiman], "The Picard scheme", §4, Def. 4.1. Let C/k be a smooth proper geometrically integral curve over a field k, with a k-rational point. The *Picard scheme* $\mathrm{Pic}_{C/k}$ is the k-scheme representing the relative Picard functor $\mathrm{Pic}^{\sharp}_{C/k}$ (equivalently, by Kleiman §2 Thm. 2.5 under the rational point, its étale sheafification), supplied by 1545, equipped with the abelian-group-scheme structure of 1557. The scheme is separated, locally of finite type over k, and a disjoint union of open quasi-projective k-subschemes.

25.7 Construction interfaces

The constructions above fit together through the following interfaces. The relative Picard presheaf is obtained from the quotient of invertible sheaves on $C \times_k T$ by pullbacks from T, and its set-valued version is used when applying representability. The relative-divisor functor is the specialisation of the effective Cartier-divisor functor to $C \to \mathrm{Spec}\,k$. A divisor family $\langle \mathcal{F}, q\rangle$ is sent by the Abel map to the inverse class of its kernel:

$$\langle \mathcal{F}, q\rangle \longmapsto -[\ker q].$$

The FGA theorem applies Hilbert-polynomial stratification, a regularity bound, the Abel-map factorisation, and the smooth-proper quotient lemma. Tensor product and dualisation of line bundles then induce addition and inversion on the representing scheme.

- The Picard scheme 1558 is selected by the existence assertion 1567.
- The functor 1560 is the set-valued form of the relative Picard presheaf of 15.
- The Abel map of 1543 is the natural transformation $\mathrm{Div}_{C/k} \to \mathrm{Pic}^{\sharp}_{C/k}$ sending a relative divisor family $\langle \mathcal{F}, q\rangle$ to $-[\ker q]$, the class of its associated invertible sheaf.
- The Altman–Kleiman lemma 1546 gives descent of a flat-and-proper equivalence relation $R \subseteq Z \times_k Z$ on a quasi-projective k-scheme Z to a quasi-projective quotient Q representing the coequalising sheaf.
- The theorem 1545 supplies the representing scheme.
- Yoneda transport of the abelian-group structure on $\mathrm{Pic}^{\sharp}_{C/k}$ gives the group-scheme structure of 1557.

The effective-quotient theorem for flat and proper equivalence relations on quasi-projective schemes is recorded as 1546. All other interfaces are supplied by the relative Picard, divisor, and Quot constructions.

25.8 Auxiliary existence predicates

The public constructions use a small collection of existence predicates. Each predicate records one mathematical object or property needed by the preceding representability argument, and the associated definitions select the object when the predicate is inhabited.

25.8.1 Basic functors

Definition 1559 (k-rational point)**.** The predicate HasRationalPoint(C) asserts that the structural morphism $C \to \operatorname{Spec} k$ admits a section $\sigma : \operatorname{Spec} k \to C$, i.e. that C has a k-rational point. This is the standing hypothesis of the chapter (see 25.1): by Kleiman §2, Thm. 2.5, it makes the plain relative Picard functor $\operatorname{Pic}^{\sharp}_{C/k}$ coincide with its étale (and fppf) sheafification, so representability can be stated against 633 directly. For a pointed curve (C, P), the pointing P supplies this section.

Proof. This is a definition. □

Definition 1560 (Set-valued relative Picard functor)**.** The set-valued shadow $\operatorname{Pic}^{\sharp}_{C/k} : (\mathrm{Sch}/k)^{op} \to$ Set of the group-valued relative Picard presheaf of 633 (composition with the forgetful functor Ab $\to$ Set).

A representing object therefore carries natural bijections

$$(T \to \operatorname{Pic}_{C/k}) \simeq \operatorname{Pic}(C \times_k T)/\pi_T^* \operatorname{Pic}(T).$$

Proof. Immediate from 633: apply the forgetful functor to the group-valued presheaf. □

Definition 1561 (Existence of the relative-divisor functor)**.** The predicate HasDivFunctor(C) asserts that the category of presheaves of types on $(\mathrm{Sch}/k)^{op}$ contains the relative-effective-divisor functor $\operatorname{Div}_{C/k}$, sending a k-scheme T to the set of relative effective Cartier divisors on $C \times_k T$ flat over T. It is the existence assertion for $\operatorname{Div}_{C/k}$.

Proof. This is the defining predicate. □

Definition 1562 (The relative-divisor functor)**.** The presheaf $\operatorname{Div}_{C/k} : (\mathrm{Sch}/k)^{op} \to$ Set is the relative-divisor functor $\operatorname{Div}_{C/k}$ of 1175, instantiated at the structural morphism $C \to \operatorname{Spec} k$: it sends a k-scheme T to the set of relative effective Cartier divisors on $C \times_k T$ flat over T.

Proof. Immediate from 1175: instantiate at $X = C$, $S = \operatorname{Spec} k$, π the structural morphism. □

Definition 1563 (Relative-divisor functor exists)**.** The relative-divisor functor of 1562 witnesses 1561 for every such curve C/k.

Proof. The functor $\operatorname{Div}_{C/k}$ of 1562 is an object of the presheaf category, witnessing the existential. □

25.8.2 Abel-map predicate

Definition 1564 (Abel-map data)**.** The predicate HasAbelMap(C) carries a natural transformation of presheaves abel: $\operatorname{Div}_{C/k} \to \operatorname{Pic}^{\sharp}_{C/k}$ between the relative-divisor functor of 1562 and the relative Picard functor of 1560; abelMap := HasAbelMap.abel denotes this transformation.

Proof. This is a definition. □

Definition 1565 (Existence of the Abel map)**.** For every such curve C/k, the transformation $A^{\mathrm{raw}}_{C/k}$ from 1542 supplies 1564.

Proof. The natural transformation constructed in 1542 is a term of the required type $\mathrm{Div}_{C/k} \to \mathrm{Pic}^{\sharp}_{C/k}$, and hence supplies the data of 1564. ☐

25.8.3 Smooth-proper quotient predicate

Definition 1566 (Representability of the smooth-proper quotient)**.** For a morphism $\alpha : Z \to P$ of presheaves of types on $(\mathrm{Sch}\,/k)^{op}$, the predicate HasSmoothProperQuotient(α) asserts that the target P is representable. It records the conclusion of the Altman–Kleiman lemma 1546 for the quotient occurring in the FGA construction.

Proof. This is a definition. ☐

25.8.4 Picard-scheme predicates

Definition 1567 (Existence of the Picard scheme)**.** The predicate HasPicScheme(C) asserts that there exists a k-scheme X, *locally of finite type over* k, carrying a representability witness for the relative Picard functor $\mathrm{Pic}^{\sharp}_{C/k}$ of 1560; in other words, that the Picard scheme $\mathrm{Pic}_{C/k}$ exists and is locally of finite type. It is the existence assertion from which 1558 extracts the representing object. The local-finiteness clause belongs to the same existence package as representability: Kleiman §4, Thm. th:main exhibits $\mathrm{Pic}_{C/k}$ as a disjoint union of open quasi-projective k-subschemes. The witness 1568 is conditional on the k-rational point of 1559.

Proof. This is the defining existence predicate. ☐

Definition 1568 (Picard scheme exists)**.** Let C/k be a smooth proper geometrically integral curve with a k-rational point. Then the relative Picard functor of C is represented by a scheme locally of finite type over k.

Proof. The rational point rigidifies line bundles along the section, which is what makes the plain relative functor a sheaf at all. Either route of 25.1 then produces the representing scheme. Along the Milne–Kollár route the degree pieces Pic^r are glued from the loci J^{Σ} over a separably closed field, descended to k by a finite Galois quotient, and assembled over $\mathbb{Z}$ as a coproduct, which is locally of finite type because each piece is. Along the quotient route the bounded Hilbert-polynomial strata of 1545 glue to the same conclusion. ☐

Remark 1569 (What the representability statements of this chapter assert)**.** Every representability assertion of this chapter is an assertion *about* the existence predicate HasPicScheme of 1567, not a construction of the Picard scheme. 1558 extracts a scheme from that predicate, 1545 extracts the representing property, and 1557 equips the extracted object with its group structure; each is a theorem about what follows once the predicate holds. The predicate itself is 1568, and it is the one statement of this chapter that is asserted rather than proved.

The distinction is easy to lose because a statement quantifying over the predicate is unconditional as a statement — its hypothesis is discharged by whoever applies it. It becomes visible one step later, at a curve where the predicate must be supplied: there every consequence above depends on 1568. So the correct reading of this chapter is that the passage *from* representability to the Picard scheme and its group structure is complete, and representability is not.

Definition 1570 (Representability of $\mathrm{Pic}^{\sharp}_{C/k}$)**.** The predicate PicSharpRepresentable(C) asserts that the specific scheme $\mathrm{Pic}_{C/k}$ of 1558 is a representing object for $\mathrm{Pic}^{\sharp}_{C/k}$. It is the assertion from which the representability witness 1545 is extracted.

Proof. This is a definition. □

Definition 1571 (Representability identification)**.** The scheme $\mathrm{Pic}_{C/k}$ is the chosen witness of 1567; the first component of that witness identifies it as a representing object for $\mathrm{Pic}^{\sharp}_{C/k}$.

Proof. This is the representing component of the existence statement in 1567. □

Definition 1572 (Local finite type of $\mathrm{Pic}_{C/k}$)**.** The predicate

$$\mathrm{PicSchemeLocallyOfFiniteType}(C)$$

asserts that the structural morphism of the Picard scheme $\mathrm{Pic}_{C/k}$ of 1558 is locally of finite type — part (1) of Kleiman §4, Thm. th:main ($\mathrm{Pic}_{C/k}$ is a disjoint union of open quasi-projective k-subschemes). It is the hypothesis the identity-component construction of 26 uses to specialise G^0 to $G = \mathrm{Pic}_{C/k}$. It follows from the local-finiteness component of 1567.

Proof. This is a definition. □

Definition 1573 (Local-finiteness assertion)**.** Whenever 1567 holds, the second component of its chosen witness gives local finiteness of $\mathrm{Pic}_{C/k}$. Thus representability and local finiteness are supplied by one existence package, as in Kleiman §4, Thm. th:main.

Proof. This is the local-finiteness component of the existence statement in 1567. □

25.9 Dependencies of the representability theorem

The construction of $\mathrm{Pic}_{C/k}$ separates into three mathematical inputs: the relative-divisor functor, its Abel map to the relative Picard functor, and the FGA existence theorem. The first two supply concrete functors and a natural transformation. The third combines them with regularity, cohomology and base change, and effective descent to produce a scheme locally of finite type representing $\mathrm{Pic}^{\sharp}_{C/k}$.

25.9.1 The Picard-scheme existence theorem

For a smooth proper geometrically integral curve C/k with a k-rational point, the relative Picard functor $\mathrm{Pic}^{\sharp}_{C/k}$ (1560) is representable by a k-scheme. Either route of 25.1 establishes it, and they consume different inputs.

Along the Milne–Kollár route of 25.4 — the one being formalised — the inputs are: the rigidified pushforward 1551, which holds for every curve satisfying this chapter's hypotheses and supplies both the canonical divisor family and the semicontinuity of h^0 cutting out the loci C^{Σ}; uniform H^1 vanishing in high degree, which makes those loci cover; representability of the degree-r divisor functor, which supplies the ambient Z together with its Grassmannian immersion; and the finite Galois quotient 1554 with 1555, which descends the result from a separably closed field to k. No Hilbert or Quot scheme occurs among them.

Along the quotient route the proof first applies Kleiman §2, Thm. 2.5, using the section to identify the plain functor with its étale sheafification, and then uses Kleiman's four steps

(1545): Hilbert-polynomial stratification, m-regularity bound, Abel-map factorisation through $\mathrm{Div}_{C/k}$, and the smooth-proper quotient (1546). Its inputs are the Quot/Hilbert construction of 20, representability of $\mathrm{Div}_{C/k}$ as in 1544, the regularity result 1017, the base-change result 480, and Altman–Kleiman effective descent. Both routes rest on the same divisor and comparison substrate, and the rational-point hypothesis enters both in the same way.

Either way, the resulting representing object gives natural bijections

$$(T \to \mathrm{Pic}_{C/k}) \simeq \mathrm{Pic}(C \times_k T)/\pi_T^* \mathrm{Pic}(T),$$

including the dual-number case $T = \operatorname{Spec} k[\epsilon]$ used in the tangent-space calculation.

25.9.2 The relative-divisor input

The relative-divisor functor $\mathrm{Div}_{X/S}$ of 1175 is the subfunctor of the Hilbert functor $\mathrm{Hilb}_{C/k} = \mathrm{Quot}_{\mathcal{O}_C/C/k}$ of relative effective Cartier divisors, in the invertible-kernel quotient encoding of 1159. The presheaf $\mathrm{Div}_{C/k}$ of 1562 is its instantiation at $C \to \operatorname{Spec} k$. Its divisor condition is stable under base change by 1165, while representability of $\mathrm{Div}_{C/k}$ by an open subscheme of the Hilbert scheme (Kleiman §3, Thm. th:repDiv) is the content of 1544.

25.9.3 The Abel-map input

The class 1564 carries the natural transformation whose value on a divisor family is

$$\mathrm{abelMapWitness}(C)\langle \mathcal{F}, q\rangle = -[\ker q].$$

It is the composite of the kernel-class natural transformation abelKernelNatTrans(C) (naturality via the kernel–flat-base-change comparison isomorphism) with the group-presheaf negation picNeg(C). Thus the Abel map has the defining property in 1543; it is a natural transformation of the presheaves in 1562 and 1560.

25.9.4 Assembly of the inputs

The FGA construction consumes the relative-divisor functor and Abel map, then applies the regularity, base-change, and effective-quotient results of 1545. A pointing $P : \mathbf{1} \to C$ supplies the rational-point hypothesis used to identify the plain relative Picard functor with its sheafification. The resulting representability and local-finiteness statements feed the group-scheme and Jacobian constructions.

25.10 Related constructions

The representable Picard scheme and its group structure interact with the following constructions, treated in the indicated companion chapters.

- **Identity component** $\mathrm{Pic}^0_{C/k}$ — the open and closed subgroup scheme of $\mathrm{Pic}_{C/k}$ containing the unit and consisting of those classes lying in the connected component of $\mathrm{Pic}_{C/k}$. Its dimension, smoothness, and abelian-variety structure are treated in 26 and 27.
- **Quot scheme representability** — consumed by 1545 via `thm:quot_representable` (20), not re-proved here. It is an input of the quotient route only; the Milne–Kollár route of 25.4 does not use it, although it does use the Grassmannian and flattening substrate of the same chapters.

- **Étale sheafification.** The relative Picard presheaf and its topology-parametric sheafification are constructed in 15. Under the standing rational-point hypothesis, Kleiman §2, Thm. 2.5 identifies the plain relative functor with the étale sheafification.
- **Universal Poincaré sheaf** (Kleiman §4, Ex. 4.3) — exists when $\mathrm{Pic}_{C/k}$ represents $\mathrm{Pic}_{C/k}$ itself and C/k has a section. Under the chapter's standing rational-point hypothesis both conditions in fact hold (Thm. 2.5 identifies the plain functor with the represented sheaf), so the Poincaré sheaf is then available as an associated construction.
- **Mumford's example** (Kleiman §4, Eg. 4.14), counter-examples to representability for non-integral geometric fibers — not relevant to a smooth proper geometrically integral curve.

25.11 Dependency structure

The principal constructions form a connected dependency chain rooted in 14, 20, and 15:

- 1548 uses 1552, and 1555 uses 1554, which uses 1553. These are the inputs specific to the Milne–Kollár route (25.4).
- 1543 uses 610 and 617.
- 1546 is the structural quotient statement used by the quotient-route representability argument.
- 1545 uses 1543, `thm:quot_representable`, `def:rel_pic_sharp`, 1559, 616, and 1546.
- 1557 uses 1558, 1545, `def:rel_pic_sharp`, and 616.
- 1558 uses 1567 and 1568.

The Quot-scheme theorem used above is proved in 20, while the relative Picard functor is defined in 15.

Chapter 26

The identity component of the Picard scheme

26.1 Setup and motivation

Throughout this chapter, fix a base field k and a smooth proper geometrically integral curve C over k of genus $g = g(C)$ (537). By 1545 and 1557, the étale-sheafified relative Picard functor of C/k is represented by the k-group scheme $\mathrm{Pic}_{C/k}$ (1558), which is separated, locally of finite type over k, and a disjoint union of open quasi-projective k-subschemes. The disjoint-union structure of $\mathrm{Pic}_{C/k}$ stratifies its T-points by the Hilbert polynomial of the representing invertible sheaf relative to a fixed polarisation, and on the geometric points of the curve this stratification reduces to the degree of a line bundle.

The identity component $\mathrm{Pic}^0_{C/k}$ singled out by Kleiman is the open and closed subgroup scheme consisting of the connected component of $\mathrm{Pic}_{C/k}$ containing the trivial line bundle. It is the locus of degree-zero classes, and on a smooth proper curve it is itself an abelian variety of dimension g — the Jacobian variety of C. This chapter packages those four statements as the A.3 specification.

26.2 The identity component of a group scheme

The identity component is an abstract feature of group schemes locally of finite type over a field. We record the definition and its open-and-closed subgroup-scheme structure separately, so that the Lean substrate `AlgebraicGeometry.GroupScheme.IdentityComponent` is reusable outside the Picard context.

Definition 1574 (Identity component of a group scheme)**.** *Source: [Kleiman], "The Picard scheme", §5, Lem. 5.1.* Let k be a field and G a k-group scheme locally of finite type whose identity section $e : \mathrm{Spec}\, k \to G$ factors through G. The *identity component* of G, denoted G^0, is the locally closed subscheme of G whose underlying topological space is the connected component of $|G|$ containing the image of e, equipped with the reduced-induced (a priori) subscheme structure inherited from G. In characteristic zero, or more generally when G has a geometrically reduced open subscheme, the substructure agrees with the open subscheme structure of 1575.

Theorem 1575 (Identity component is a clopen subscheme)**.** *Source: [Kleiman], "The Picard scheme", §5, Lem. 5.1 (3). Let k be a field and G a k-group scheme locally of finite type. Then* *the identity component G^0 ([1574](#)) is an open and closed subscheme of G: the inclusion $G^0 \hookrightarrow G$ is both an open immersion and a closed immersion of schemes.*

Proof. Following Kleiman's proof of Lem. 5.1 (3) (clopen-subscheme part). Since e is a k-rational point, the singleton $\{e\}$ is a closed point of G. The connected component G^0 of $|G|$ at e is the largest connected subspace containing e; the closure of a connected subspace is connected, hence G^0 is closed. On the other hand, a locally Noetherian topological space has open connected components (EGA I 6.1.9), so $G^0 \subseteq |G|$ is also open. Equip G^0 with the open-subscheme structure; the inclusion $G^0 \hookrightarrow G$ is then both an open and a closed immersion. □

Theorem 1576 (Identity component inclusion is a group-scheme homomorphism)**.**

Source: [Kleiman], "The Picard scheme", §5, Lem. 5.1 (3); the verbatim source quote of (3) is at 1575. Let k be a field and G a k-group scheme locally of finite type. The clopen inclusion $G^0 \hookrightarrow G$ (1575) is a homomorphism of k-group schemes: G^0 inherits a k-group-scheme structure from G, and the inclusion morphism in the over-category over Spec k *is compatible with the group laws on source and target.*

Proof. Following Kleiman's proof of Lem. 5.1 (3) (subgroup part); the verbatim source quote is reproduced at 1575. The product $G^0 \times_k G^0$ is connected (EGA IV_2 4.5.8): G^0 is geometrically connected (1598) and every morphism to Spec k is universally open, so the fibre product over k of the connected G^0 with itself is connected. Its image under the multiplication $\mu : G \times_k G \to G$ is therefore a connected subset of $|G|$ containing $\mu(e,e) = e$, hence contained in the clopen identity component G^0 (1575). Likewise the image of the connected set G^0 under the inversion of G is connected and contains $e^{-1} = e$, hence lies in G^0. Consequently, for every k-scheme T, the subset of $\mathrm{Hom}_k(T, G)$ consisting of morphisms whose set-theoretic image lies in G^0 is a subgroup, functorially in T; and since a morphism lands in the open subscheme G^0 exactly when its set-theoretic image lies in G^0, this subgroup presheaf is represented by G^0. By the Yoneda characterisation of group objects, G^0 inherits a k-group-scheme structure from G, for which the inclusion $G^0 \hookrightarrow G$ is a homomorphism of k-group schemes. □

Theorem 1577 (Identity component is of finite type and geometrically irreducible)**.** *Source: [Kleiman], "The Picard scheme", §5, Lem. 5.1 (3); the verbatim source quote of (3) is at 1575. Let k be a field and G a k-group scheme locally of finite type. Then the identity component G^0 is of finite type over k — equivalently, locally of finite type* and *quasi-compact — and geometrically irreducible: after base change to the algebraic closure $\bar{k}$, the underlying scheme $G^0_{\bar{k}}$ is an irreducible $\bar{k}$-scheme.*

Proof. Following Kleiman's proof of Lem. 5.1 (3) (finite-type and geometric-irreducibility part); the verbatim source quote is reproduced at 1575. Base-change to $\bar{k}$ is permitted because formation of G^0 commutes with field extension (1578), and over $\bar{k}$ the identity component G^0 carries the group structure inherited from G (1576).

Irreducibility over $\bar{k}$. Inside a Noetherian affine open of G^0 (Noetherian because G is locally of finite type over a field), removing the union of all irreducible components but one leaves a nonempty irreducible open (Stacks 0052 (3)); its image under the inclusion of the affine open is a nonempty irreducible open $U \subseteq G^0$. Pick a closed point $g \in U$ (G^0 is a Jacobson space, being locally of finite type over a field). For every closed point z of G^0, $g^{-1}z$ is again a closed — hence, $\bar{k}$ being algebraically closed, rational — point, and right translation $x \mapsto x \cdot (g^{-1}z)$ is an automorphism of G^0 carrying g to z; it therefore carries U to an irreducible open neighbourhood of z. Consequently the union of all irreducible opens of G^0 contains every closed point; since this union is open, its complement is closed, and a nonempty closed subset of a Jacobson space contains a closed point (Jacobson density) — which cannot lie outside every constructed neighbourhood — the complement is empty, so the union of irreducible opens is all of G^0. By EGA I 6.1.10 (a connected space in which every point has an irreducible open neighbourhood is irreducible) and connectedness of G^0 (1575), G^0 is irreducible.

Quasi-compactness over $\overline{k}$. Fix an affine open $W \ni e$. For a closed point z, the "division" morphism $\chi_z : x \mapsto x^{-1}z$ satisfies $\chi_z(z) = e \in W$, so $W \cap \chi_z^{-1}(W)$ is a nonempty open subset of the irreducible G^0, hence contains a closed point g by Jacobson density. Then $u := g^{-1}z$ is a rational point of W and $z = g \cdot u$ lies in the image of the multiplication map $m : W \times_{\overline{k}} W \to G^0$. Thus every closed point of G^0 lies in the image of m, which is quasi-compact as the image of the quasi-compact scheme $W \times_{\overline{k}} W$; a finite affine subcover of this image, together with Jacobson density, covers all of G^0, so G^0 is quasi-compact.

Descent to k. Both conjuncts descend along the surjective projection $(G^0)_{\overline{k}} \to G^0$ coming from the base-change identification $(G_{\overline{k}})^0 \cong (G^0)_{\overline{k}}$ (1578): a scheme admitting a surjection from a quasi-compact (resp. irreducible) scheme is itself quasi-compact (resp. irreducible). Quasi-compactness together with the standing locally-of-finite-type hypothesis on G makes G^0 of finite type over k. For geometric irreducibility at an arbitrary field extension K/k, apply the same descent along the surjective projection from the algebraic closure of K. □

Theorem 1578 (Formation of the identity component commutes with base change)**.**
Source: [Kleiman], "The Picard scheme", §5, Lem. 5.1 (3); the verbatim source quote of (3) is at 1575. Let k be a field, G a k-group scheme locally of finite type, and K/k a field extension. The natural map $(G^0)_K \to (G_K)^0$, induced by base-changing the inclusion of 1575 along $\operatorname{Spec} K \to \operatorname{Spec} k$ *and using the universal property of the identity component on G_K, is an isomorphism of K-schemes.*

Proof. Following Kleiman's proof of Lem. 5.1 (3) (base-change part); the verbatim source quote is reproduced at 1575. Geometric connectedness of G^0 follows from the existence of the k-rational point $e \in G^0(k)$: the identity component is connected and carries a k-rational section, so it is geometrically connected by 1587 (the Stacks 037Q criterion; cf. EGA IV_2 4.5.14). In particular for any field extension K/k the base change $(G^0)_K$ is connected, contains the image of the identity section of G_K, and is open and closed in G_K (since open- and closed-immersion both descend under base change, 1575). Hence $(G^0)_K$ coincides, as a subscheme of G_K, with the identity component $(G_K)^0$ of G_K; the natural comparison map $(G^0)_K \to (G_K)^0$ is therefore an isomorphism. □

26.2.1 Descent of connectedness along field extensions

The bridge from connectedness of $|G^0|$ to geometric connectedness of $G^0 \to \operatorname{Spec} k$ is the classical statement that a connected k-scheme with a k-rational point is geometrically connected (Stacks 04KV, EGA IV_2 4.5.14). We prove it from first principles. The transcendental heart is the fact that over an algebraically closed field the tensor product of two field extensions is a domain (Stacks 05P3); the geometric assembly combines universal openness of morphisms to the spectrum of a field (Stacks 0383, available in mathlib), closedness of integral base changes, and a clopen-image argument at the rational point.

Lemma 1579 (Evaluation at a maximal ideal over an algebraically closed field)**.** *Source: Zariski's lemma; Stacks, tag 00FV. Let k be an algebraically closed field, R a finite-type k-algebra, and $\mathfrak{m} \subseteq R$ a maximal ideal. Then there is a k-algebra homomorphism $\varphi : R \to k$ with* $\ker \varphi = \mathfrak{m}$.

Proof. The quotient $R/\mathfrak{m}$ is a field and a finite-type k-algebra, hence a finite (in particular integral) extension of k by Zariski's lemma. Since k is algebraically closed, the structure map

$k \to R/\mathfrak{m}$ is bijective. Composing the quotient map $R \to R/\mathfrak{m}$ with its inverse yields φ, whose kernel is $\mathfrak{m}$. $\square$

Lemma 1580 (Tensor product with a finite-type domain over an algebraically closed field)**.** *Source: Stacks, tag 05P3 (finite-type case). Let k be an algebraically closed field, R a finite-type k-algebra which is a domain, and A/k a field extension. Then $R \otimes_k A$ is a domain.*

Proof. Nontriviality holds because R and A are nontrivial and embed their fraction fields compatibly. Fix a k-basis (a_i) of A; then $R \otimes_k A$ is a free R-module with basis $(1 \otimes a_i)$, so every element x has unique coordinates $c_i(x) \in R$. Suppose $xy = 0$. For every maximal ideal $\mathfrak{m} \subseteq R$, choose $\varphi : R \to k$ with kernel $\mathfrak{m}$ by 1579, and apply the ring homomorphism $\varphi \otimes \mathrm{id} : R \otimes_k A \to A$: it sends $x \mapsto \sum_i \varphi(c_i(x))a_i$. Since A is a field, $(\varphi \otimes \mathrm{id})(x) = 0$ or $(\varphi \otimes \mathrm{id})(y) = 0$; by linear independence of (a_i) over k this means all $c_i(x) \in \mathfrak{m}$ or all $c_j(y) \in \mathfrak{m}$. In either case $c_i(x)\, c_j(y) \in \mathfrak{m}$ for all i, j and all maximal $\mathfrak{m}$. A finite-type algebra over a field is a Jacobson ring, and in a Jacobson domain the intersection of all maximal ideals is zero; hence $c_i(x)\, c_j(y) = 0$ for all i, j. If $x \neq 0$, some $c_{i_0}(x) \neq 0$, and since R is a domain every $c_j(y) = 0$, i.e. $y = 0$. $\square$

Lemma 1581 (Tensor product of field extensions over an algebraically closed field)**.** *Source: Stacks, tag 05P3. Let k be an algebraically closed field and A/k, B/k field extensions. Then $A \otimes_k B$ is a domain.*

Proof. Suppose $xy = 0$ with $x, y \in A \otimes_k B$. Writing x and y as finite sums of elementary tensors, let $R \subseteq A$ be the k-subalgebra generated by the finitely many first components; R is a finite-type k-algebra and a domain (a subring of the field A). Since B is free, hence flat, as a k-module, the natural map $R \otimes_k B \to A \otimes_k B$ is injective. The elements x, y lift to $x', y' \in R \otimes_k B$ with $x'y' = 0$; by 1580 (applied with the field extension B) we get $x' = 0$ or $y' = 0$, hence $x = 0$ or $y = 0$. Nontriviality is as in 1580. $\square$

Lemma 1582 (The spectrum of a domain is connected)**.** *Let R be a commutative domain. Then* $\operatorname{Spec} R$ *is a connected (indeed irreducible) nonempty topological space.*

Proof. In a domain the nilradical (0) is prime, so $\operatorname{Spec} R$ is irreducible with generic point (0); irreducible spaces are connected and nonempty. $\square$

Lemma 1583 (Projections from constant-field base changes are geometrically connected)**.** *Let k be an algebraically closed field, L/k a field extension, and $f : X \to \operatorname{Spec} k$ a morphism of schemes. Then the projection $q : X \times_k \operatorname{Spec} L \to X$ is geometrically connected: for every field K and morphism $\operatorname{Spec} K \to X$, the fiber product $(X \times_k \operatorname{Spec} L) \times_X \operatorname{Spec} K$ is connected.*

Proof. By the pasting law for fiber products, $(X \times_k \operatorname{Spec} L) \times_X \operatorname{Spec} K \cong \operatorname{Spec} K \times_{\operatorname{Spec} k} \operatorname{Spec} L = \operatorname{Spec}(K \otimes_k L)$, where K is a field extension of k via $\operatorname{Spec} K \to X \to \operatorname{Spec} k$. By 1581 the ring $K \otimes_k L$ is a domain, so its spectrum is connected by 1582. $\square$

Lemma 1584 (Connectedness is stable under base change from an algebraically closed field)**.** *Source: Stacks, tag 0363. Let k be an algebraically closed field, L/k a field extension, and $f : X \to \operatorname{Spec} k$ a morphism with $|X|$ connected. Then $X \times_k \operatorname{Spec} L$ is connected.*

Proof. The projection $q : X \times_k \operatorname{Spec} L \to X$ is open: it is the base change of $\operatorname{Spec} L \to \operatorname{Spec} k$, which is universally open because every morphism to the spectrum of a field is universally open (Stacks 0383; available in mathlib). By 1583 the fibers of q over points of X are connected. An open surjective map with connected fibers onto a connected space has connected total space (mathlib, `GeometricallyConnected.connectedSpace`). $\square$

Lemma 1585 (Base change to the algebraic closure preserves connectedness given a section)**.** *Source: Stacks, tag 04KV (algebraic step); EGA IV$_2$ 4.5.13. Let k be a field, $f : X \to \operatorname{Spec} k$ a morphism with $|X|$ connected admitting a section $s : \operatorname{Spec} k \to X$, and $\bar{k}$ an algebraic closure of k. Then $X \times_k \operatorname{Spec} \bar{k}$ is connected.*

Proof. The projection $p : X_{\bar{k}} := X \times_k \operatorname{Spec} \bar{k} \to X$ is: open (base change of the universally open $\operatorname{Spec} \bar{k} \to \operatorname{Spec} k$, as in 1584); closed (base change of $\operatorname{Spec} \bar{k} \to \operatorname{Spec} k$, which is integral because $\bar{k}/k$ is algebraic, and integral morphisms are universally closed); and surjective (base change of a surjection). Let $x_0 = s(*)$ be the image of the section. The section induces a k-algebra retraction $\kappa(x_0) \to k$ of the field map $k \to \kappa(x_0)$; a field homomorphism admitting a retraction is bijective, so $\kappa(x_0) = k$. Hence the fiber $p^{-1}(x_0) = \operatorname{Spec}(\kappa(x_0) \otimes_k \bar{k}) = \operatorname{Spec} \bar{k}$ is a single point ζ.

$X_{\bar{k}}$ is nonempty (it contains ζ, or equivalently the image of the base-changed section). Let $T \subseteq X_{\bar{k}}$ be clopen with $T \neq \emptyset$ and $T^{\mathsf{c}} \neq \emptyset$. Since p is both open and closed, $p(T)$ and $p(T^{\mathsf{c}})$ are nonempty clopen subsets of the connected space $|X|$, hence both equal $|X|$. In particular $x_0 \in p(T) \cap p(T^{\mathsf{c}})$, so the fiber $p^{-1}(x_0)$ meets both T and T^{c}; but that fiber is the single point ζ — a contradiction. So $X_{\bar{k}}$ admits no nontrivial clopen partition and is connected. □

Theorem 1586 (Connected with a rational section implies geometrically connected)**.** *Source: Stacks, tag 04KV; EGA IV$_2$ 4.5.14. Let k be a field and $f : X \to \operatorname{Spec} k$ a morphism with $|X|$ connected, admitting a section $s : \operatorname{Spec} k \to X$. Then f is geometrically connected: for every field extension L/k the base change $X \times_k \operatorname{Spec} L$ is connected.*

Proof. Let L/k be arbitrary, $\bar{L}$ an algebraic closure of L, and $\bar{k}$ an algebraic closure of k. Since $\bar{k}/k$ is algebraic and $\bar{L}$ is algebraically closed, there is a k-embedding $\bar{k} \hookrightarrow \bar{L}$. By 1585, $X_{\bar{k}}$ is connected; by 1584 applied over the algebraically closed field $\bar{k}$ to the extension $\bar{L}/\bar{k}$, the scheme $X_{\bar{L}} = X_{\bar{k}} \times_{\bar{k}} \operatorname{Spec} \bar{L}$ is connected. Finally the projection $X_{\bar{L}} \to X_L$ is surjective (base change of the surjection $\operatorname{Spec} \bar{L} \to \operatorname{Spec} L$) and continuous, so X_L is the continuous image of a connected space, hence connected. □

Lemma 1587 (Stacks 037Q iff-direction: connected + section geometrically connected)**.** *Source: Stacks, tag 037Q (iff-direction (2)⇒(1)). Let k be a field and let $f : X \to \operatorname{Spec} k$ be a morphism of schemes such that the underlying topological space $|X|$ is connected. Suppose there is a k-rational section $s : \operatorname{Spec} k \to X$ of f (so $s \circ f = \mathrm{id}_X$ topologically and the image of s meets every irreducible component of X through which s factors). Then $f : X \to \operatorname{Spec} k$ is geometrically connected, in the sense that for every field extension K/k the base change X_K is connected.*

Proof. Immediate from 1586 applied to f and the section s. □

26.2.2 Topological and connectedness substrate (helper lemmas)

The construction of 1574 and the proofs of its open-and-closed and base-change conclusions rest on a small chain of helper results: a purely topological core about connected components of Noetherian spaces, the identity point and its connected component (the carrier of G^0), the local connectedness of $|G|$, the identity section landing in G^0, and the connectedness of G^0. We record each helper so the dependency graph of the chapter is faithful to the Lean development.

Lemma 1588 (Finitely many connected components on a Noetherian space)**.** *Let α be a Noetherian topological space. Then the set of connected components of α is finite.*

Proof. The assignment sending an irreducible component C of α to the connected component of an arbitrarily chosen point of C is well-defined (an irreducible component is preconnected, hence lies in a single connected component) and surjective (every point lies in the irreducible component through it, which is contained in its connected component). A Noetherian space has finitely many irreducible components, so the connected components are finite as the image of a finite set. Proved directly in Lean. □

Lemma 1589 (Connected components are open on a Noetherian space)**.** *Let α be a Noetherian topological space and $x \in \alpha$. Then the connected component* $\mathrm{Comp}(x)$ *of x is open in α.*

Proof. The space of connected components of α is totally disconnected, hence T_1, and finite by 1588; a finite T_1 space carries the discrete topology, so each singleton is open. The preimage of the singleton $\{[x]\}$ under the continuous quotient map $\alpha \to \pi_0(\alpha)$ is exactly $\mathrm{Comp}(x)$, which is therefore open. Proved directly in Lean. □

Lemma 1590 (The underlying space of a locally-finite-type k-scheme is locally connected)**.** *Let k be a field and G an object of the over-category over* $\operatorname{Spec} k$ *whose structural morphism* $G \to \operatorname{Spec} k$ *is locally of finite type. Then the underlying topological space $|G|$ is locally connected.*

Proof. Being locally of finite type over the field k, the scheme G is locally Noetherian. Each point $y \in |G|$ has an affine open neighbourhood $W = \operatorname{Spec} R$ with R Noetherian, so $|W|$ is a Noetherian space; by 1589 the connected component of y inside W is open, and its image under the open inclusion $W \hookrightarrow |G|$ is an open connected neighbourhood of y contained in any prescribed open set through y. This is precisely the local-connectedness criterion. Proved directly in Lean. □

Definition 1591 (The identity point)**.** Let k be a field and G a k-group scheme. The *identity point* of G is the image $e(*) \in |G|$ of the unique point of $\operatorname{Spec} k$ under the identity section $e : \operatorname{Spec} k \to G$ (well-defined because $|\operatorname{Spec} k|$ is a one-point space).

Proof. Proved directly in Lean (a one-point evaluation of the identity section). □

Definition 1592 (The carrier of the identity component)**.** Let k be a field and G a k-group scheme locally of finite type. The *carrier of the identity component* is the open subset of $|G|$ given by the connected component $\mathrm{Comp}(e(*))$ of the identity point (1591), packaged as an open of G; its openness is supplied by the local connectedness of $|G|$ (1590).

Proof. The carrier set is $\mathrm{Comp}(e(*))$; a connected component of a locally connected space is open (1590). Proved directly in Lean. □

Lemma 1593 (The carrier of the identity component is connected)**.** *Let k be a field and G a k-group scheme locally of finite type. The carrier of the identity component (1592), with its subspace topology, is a connected topological space.*

Proof. As a subspace, the carrier $\mathrm{Comp}(e(*))$ is preconnected (a connected component is preconnected) and nonempty (it contains the identity point $e(*)$, 1591), hence connected. Proved directly in Lean. □

Lemma 1594 (The identity component is connected)**.** *Let k be a field and G a k-group scheme locally of finite type. The underlying topological space of the identity component G^0 (1574) is connected.*

Proof. By construction the underlying space of G^0 coincides with the carrier of 1592, whose connectedness is 1593. Proved directly in Lean. □

Lemma 1595 (The identity section lands in the identity-component carrier)**.** *Let k be a field and G a k-group scheme locally of finite type. The image of the identity section $e : \operatorname{Spec} k \to G$ is contained in the carrier of the identity component (1592).*

Proof. The image of e is the single point $e(*)$ (1591), which lies in its own connected component, i.e. in the carrier. Proved directly in Lean. $\square$

Definition 1596 (The identity section into the identity component)**.** Let k be a field and G a k-group scheme locally of finite type. The identity section $e : \operatorname{Spec} k \to G$ factors through the open immersion $G^0 \hookrightarrow G$ (1592); the resulting morphism $\operatorname{Spec} k \to G^0$ is the *identity section into the identity component*, obtained by the universal property of the open immersion using the range containment of 1595.

Proof. The lift exists and is unique by the universal property of the open immersion $G^0 \hookrightarrow G$, the hypothesis being the range containment of 1595. Proved directly in Lean. $\square$

Lemma 1597 (The lifted identity section is a section)**.** *Let k be a field and G a k-group scheme locally of finite type. The morphism $\operatorname{Spec} k \to G^0$ of 1596 is a section of the structural morphism $G^0 \to \operatorname{Spec} k$: composing it with $G^0 \to \operatorname{Spec} k$ recovers the identity on $\operatorname{Spec} k$.*

Proof. Composing through the open immersion recovers the original identity section e (the defining factorisation of 1596), and e is a section of $G \to \operatorname{Spec} k$; hence the lift is a section of $G^0 \to \operatorname{Spec} k$. Proved directly in Lean. $\square$

Lemma 1598 (The identity component is geometrically connected)**.** *Let k be a field and G a k-group scheme locally of finite type. The structural morphism $G^0 \to \operatorname{Spec} k$ of the identity component is geometrically connected.*

Proof. The identity component G^0 is connected (1594) and carries the k-rational section of 1596, which is a section of $G^0 \to \operatorname{Spec} k$ by 1597. The criterion "connected + k-rational section $\Rightarrow$ geometrically connected" (1587) then yields the conclusion. $\square$

26.3 The identity component of the Picard scheme

We now specialise the abstract identity-component construction to $G = \operatorname{Pic}_{C/k}$.

Definition 1599 (The identity component of the Picard scheme)**.**
Source: [Kleiman], "The Picard scheme", §5, opening + Prp. 5.3. Let C/k be a smooth proper geometrically integral curve over a field k. The *identity component of the Picard scheme* $\operatorname{Pic}^0_{C/k}$ is the identity component $(\operatorname{Pic}_{C/k})^0$ of the k-group scheme $\operatorname{Pic}_{C/k}$ of 1558, in the sense of 1574. By 1575, $\operatorname{Pic}^0_{C/k}$ is an open and closed subgroup scheme of $\operatorname{Pic}_{C/k}$ of finite type over k, geometrically irreducible, and its formation commutes with extension of the base field.

26.4 The degree map

The disjoint-union structure of $\operatorname{Pic}_{C/k}$ (1545: a disjoint union of open quasi-projective k-subschemes) is indexed by the Hilbert polynomial $\Phi \in \mathbb{Q}[\lambda]$ of a representing invertible sheaf (1018). For a smooth proper geometrically integral curve C/k of genus g with a fixed degree-one polarisation $\mathcal{O}_C(1)$ on C, the Hilbert polynomial of an invertible sheaf $\mathcal{L}$ on C is, by Riemann–Roch, $\Phi_{\mathcal{L}}(n) = n \cdot \deg \mathcal{L} + 1 - g$, so it is linear in n with leading coefficient $\deg \mathcal{L} \in \mathbb{Z}$. This identifies the connected components of $\operatorname{Pic}_{C/k}(\overline{k})$ with the integers, the index being the degree.

Definition 1600 (Degree of a Picard class)**.**

Source: [Milne], "Abelian Varieties" (course notes, 2008), §III.1, p. 88. Let C/k be a smooth proper geometrically integral curve of genus g admitting a Picard scheme $\mathrm{Pic}_{C/k}$ as in 1558 (run-0008: the Lean statement carries the `[HasPicScheme C]` hypothesis, supplied under a k-rational point via 1559). The *degree map*

$$\deg : \ \mathrm{Pic}_{C/k}(k) \longrightarrow \mathbb{Z}$$

is defined as follows: a k-point $\lambda \in \mathrm{Pic}_{C/k}(k)$ represents (after étale base change) an invertible sheaf $\mathcal{L}$ on C; its Hilbert polynomial with respect to a fixed degree-one polarisation $\mathcal{O}_C(1)$ on C is, by 1018 and Riemann–Roch,

$$\Phi_{\mathcal{L}}(n) \ = \ \chi(C, \mathcal{L} \otimes \mathcal{O}_C(n)) \ = \ n \cdot \deg \mathcal{L} + 1 - g,$$

a polynomial in n linear in n. The integer $\deg(\lambda) := \deg \mathcal{L}$ is the leading coefficient of this polynomial, well-defined on the isomorphism class $[\mathcal{L}]$ and (because $\mathrm{Pic}_{C/k}$ represents the étale-sheafified relative Picard functor) on the k-point λ. The degree map is a group homomorphism: for $\mathcal{L}, \mathcal{M}$ invertible sheaves on C one has $\deg(\mathcal{L} \otimes \mathcal{M}) = \deg \mathcal{L} + \deg \mathcal{M}$, because the Euler characteristic is additive on the additive groups of twists. Equivalently, the degree on k-points is the index of the connected component of $\mathrm{Pic}_{C/k}$ containing λ under the Hilbert-polynomial decomposition of 1545: the disjoint union $\mathrm{Pic}_{C/k} = \coprod_{d \in \mathbb{Z}} \mathrm{Pic}^d_{C/k}$ is indexed by the degree, with $\mathrm{Pic}^d_{C/k}$ the component representing classes of degree d.

26.5 The Pic-zero abelian variety

The terminal statement of the chapter identifies $\mathrm{Pic}^0_{C/k}$ with an abelian variety of dimension g — the Jacobian variety of C. Recall (Milne §I.1, p. 8) that a k-group scheme is an *abelian variety* iff it is a complete (proper) connected group variety over k; commutativity and projectivity then follow automatically.

Theorem 1601 ($\mathrm{Pic}^0_{C/k}$ is an abelian variety)**.**

Source: [Kleiman], "The Picard scheme", §5, Ex. 5.23 + Rmk. 5.26; cf. [Milne], "Abelian Varieties", §I.1 (def. of abelian variety, p. 8). Let C/k be a smooth proper geometrically integral curve of genus g over a field k. The identity component $\mathrm{Pic}^0_{C/k}$ *of 1599 is an* abelian variety *over k: it is proper, smooth, and geometrically irreducible as a k-scheme, and it carries a k-group-scheme structure. (In the Lean encoding this is the conjunction of `[IsProper]`, `[Smooth]`, and `[GeometricallyIrreducible]` on the structural morphism `(Pic0Scheme C).hom` together with the existence of a `[GrpObj]` structure on* $\mathrm{Pic}^0_{C/k}$*. Commutativity follows automatically from Milne §I.1, Cor. 1.4 and is not separately stated.)*

Proof. We assemble the four conjuncts of the abelian-variety structure on $\mathrm{Pic}^0_{C/k}$ (1599) from four inputs.

(i) *Clopen subgroup of finite type, geometrically irreducible.* By 1575, 1576, 1577, and 1578 applied to $G = \mathrm{Pic}_{C/k}$ (a k-group scheme locally of finite type by 1557 and 1545), $\mathrm{Pic}^0_{C/k}$ is an open and closed subgroup scheme of $\mathrm{Pic}_{C/k}$, of finite type over k, geometrically irreducible, and its formation commutes with extending k. In particular $\mathrm{Pic}^0_{C/k}$ is separated and locally of finite type as a subscheme of $\mathrm{Pic}_{C/k}$, and connected.

(ii) *Quasi-projectivity.* The disjoint-union structure of $\mathrm{Pic}_{C/k}$ (1545: a disjoint union of open quasi-projective k-subschemes) restricted to $\mathrm{Pic}^0_{C/k}$ expresses the identity component as a quasi-projective k-scheme (Kleiman §5, Thm. 5.4: X/k projective and geometrically integral $\Longrightarrow \mathrm{Pic}^0_{X/k}$ is quasi-projective).

(iii) *Projectivity (properness).* For a smooth proper geometrically integral curve C/k of genus $g > 0$, C/k is also geometrically normal (the smoothness implies geometric regularity, hence geometric normality). Kleiman §5, Thm. 5.4 then upgrades the quasi-projective conclusion to projective: $\mathrm{Pic}^0_{C/k}$ is projective over k, in particular proper. The proof of Theorem 5.4 uses the Chevalley–Rosenlicht structure theorem to reduce to ruling out non-constant maps $\mathbb{G}_m \to \mathrm{Pic}^0_{C/k}$, which on a geometrically normal X becomes a divisor-theoretic computation.

(iv) *Smoothness.* For a smooth proper curve C/k, the Picard scheme is smooth at 0: this is Exercise 5.23 of Kleiman §5, which asserts that for X/S projective and flat with integral geometric fibres of arithmetic genus p_a, the generalised Jacobians $\mathrm{Pic}^0_{X_s/k_s}$ form a smooth quasi-projective family of relative dimension p_a; specialising to $S = \mathrm{Spec}\, k$ and $X = C$ a smooth proper curve of (geometric = arithmetic) genus g gives smoothness of $\mathrm{Pic}^0_{C/k}$.

Combining (i)–(iv): $\mathrm{Pic}^0_{C/k}$ is a smooth, projective, geometrically irreducible k-group scheme. A proper connected k-group variety is by definition an *abelian variety* (Milne §I.1, p. 8), and is automatically commutative (Milne §I.1, Cor. 1.4). This establishes 1601. ☐

Theorem 1602 (Dimension of $\mathrm{Pic}^0_{C/k}$ equals the genus of C)**.**

Source: [Milne], "Abelian Varieties" (course notes, 2008), §III.1, Rmk. 1.4(e), p. 86; cf. [Kleiman], "The Picard scheme", §5, Cor. 5.13 + Ex. 5.23. Let C/k be a smooth proper geometrically integral curve of genus $g = g(C)$ (537) over a field k. The dimension of the abelian variety $\mathrm{Pic}^0_{C/k}$ (1601) equals the genus:

$$\dim_k \mathrm{Pic}^0_{C/k} \;=\; g(C).$$

Proof. By Kleiman §5, Cor. 5.13, the inequality $\dim \mathrm{Pic}_{C/k} \le \dim_k H^1(C, \mathcal{O}_C)$ holds, with equality if and only if $\mathrm{Pic}_{C/k}$ is smooth at 0, in which case $\mathrm{Pic}_{C/k}$ is smooth of dimension $\dim_k H^1(C, \mathcal{O}_C)$ everywhere. The abelian-variety structure of 1601 (step (iv) of its proof) shows $\mathrm{Pic}^0_{C/k}$ is smooth at 0; equality therefore holds on the open identity component, and $\dim_k \mathrm{Pic}^0_{C/k} = \dim_k H^1(C, \mathcal{O}_C) = g(C)$ by 537. This is Milne §III.1, Rmk. 1.4(e): "The dimension of J is the genus of C". ☐

Theorem 1603 (k-points of $\mathrm{Pic}^0_{C/k}$ are the kernel of the degree map)**.**

Source: [Milne], "Abelian Varieties" (course notes, 2008), §III.1, p. 88 ("$\mathrm{Pic}^0(C)$… degree zero"); the verbatim source quote is at 1600. Let C/k be a smooth proper geometrically integral curve of genus g over a field k, admitting a Picard scheme (run-0008: the Lean statement carries the `[HasPicScheme C]` hypothesis, supplied under a k-rational point via 1559). A k-point $\lambda \in \mathrm{Pic}_{C/k}(k)$ lies in the image of the inclusion of the identity component $\mathrm{Pic}^0_{C/k} \hookrightarrow \mathrm{Pic}_{C/k}$ (1575 applied to $G = \mathrm{Pic}_{C/k}$, specialised in 1599) if and only if the degree $\deg(\lambda) \in \mathbb{Z}$ of 1600 vanishes. Equivalently, $\mathrm{Pic}^0_{C/k}(k) = \ker(\deg : \mathrm{Pic}_{C/k}(k) \to \mathbb{Z})$.

Proof. A class $\lambda \in \mathrm{Pic}_{C/k}(k)$ lies in the connected component of the identity — i.e. in the image of the open immersion $\mathrm{Pic}^0_{C/k} \hookrightarrow \mathrm{Pic}_{C/k}$ (1575, specialised to $G = \mathrm{Pic}_{C/k}$ in 1599) — precisely when the Hilbert polynomial of a representing invertible sheaf coincides with that of $\mathcal{O}_C$, equivalently when its degree (the leading coefficient of the Hilbert polynomial of 1600) is zero. Since $\mathrm{Pic}^0_{C/k}$ is the abelian variety of 1601, this image is exactly the degree-zero locus on k-points. Hence $\mathrm{Pic}^0_{C/k}(k) = \ker(\deg : \mathrm{Pic}_{C/k}(k) \to \mathbb{Z})$, which is the content of Milne §III.1, p. 88 (the verbatim quote at 1600 explicitly writes "$\mathrm{Pic}^0(C)$ for the group of isomorphism classes of invertible sheaves of degree zero on C"). ☐

26.6 Lean encoding

The Lean target file is `AlgebraicJacobian/Picard/IdentityComponent.lean`, sibling to `AlgebraicJacobian/Picar`
(25). The blueprint declarations of this chapter correspond to Lean targets as follows.

- `AlgebraicGeometry.GroupScheme.IdentityComponent` (1574) is a structural constant naming the open-and-closed identity-component subscheme of a k-group scheme G locally of finite type. The construction is the abstract substrate of Kleiman §5 Lem. 5.1, independent of the Picard context; it is new project material (*not* yet in Mathlib).

- `AlgebraicGeometry.GroupScheme.IdentityComponent.isOpenSubgroupScheme` (1575) packages the clopen-subscheme inclusion $G^0 \hookrightarrow G$ (open immersion *and* closed immersion) — conclusion (a) of Kleiman §5 Lem. 5.1 (3).

- `AlgebraicGeometry.GroupScheme.IdentityComponent.isSubgroupHomomorphism` (1576) packages the group-homomorphism property of the inclusion of 1575: G^0 inherits a k-group-scheme structure from G compatible with the inclusion. Conclusion (b) of Kleiman §5 Lem. 5.1 (3).

- `AlgebraicGeometry.GroupScheme.IdentityComponent.isFiniteTypeGeometricallyIrreducible` (1577) packages the finite-type-ness and geometric-irreducibility conclusions of Kleiman §5 Lem. 5.1 (3) (conclusion (c)). In Lean these are the instances `[LocallyOfFiniteType (IdentityComponent G).hom]`, `[QuasiCompact (IdentityComponent G).hom]`, and `[GeometricallyIrreduc` `(IdentityComponent G).hom]`.

- `AlgebraicGeometry.GroupScheme.IdentityComponent.baseChangeIso` (1578) packages the base-change-commutation conclusion (d) of Kleiman §5 Lem. 5.1 (3): for any field extension K/k, the natural map $(G^0)_K \to (G_K)^0$ (in Lean, `(IdentityComponent G)_K →` `IdentityComponent (G_K)`) is an isomorphism of K-schemes.

- `AlgebraicGeometry.Scheme.Pic0Scheme` (1599) is the open subscheme of $\mathrm{Pic}_{C/k}$ obtained by applying the `IdentityComponent` substrate to $G = \mathrm{Pic}_{C/k}$, inheriting the k-group-scheme structure from 1557.

- `AlgebraicGeometry.Scheme.PicScheme.degree` (1600) is the group homomorphism $\mathrm{Pic}_{C/k}(k) \to \mathbb{Z}$ extracting the leading coefficient of the Hilbert polynomial of a representing invertible sheaf relative to a fixed degree-one polarisation $\mathcal{O}_C(1)$ on C. On the disjoint-union decomposition of 1545 the degree is the index of the component.

- `AlgebraicGeometry.Scheme.Pic0Scheme.isAbelianVariety` (1601) is the bundled abelian-variety structure on `Pic0Scheme`: the four-conjunct conjunction `[IsProper]`, `[Smooth]`, `[GeometricallyIrreducible]` on the structural morphism `(Pic0Scheme C).hom` together with `Nonempty (GrpObj (Pic0Scheme C))`.

- `AlgebraicGeometry.Scheme.Pic0Scheme.finrank_eq_genus` (1602) is the dimension formula $\dim_k \mathrm{Pic}^0_{C/k} = g(C)$ (Milne §III.1, Rmk. 1.4(e)). In Lean this is the equation `Module.finrank k (Pic0Scheme C).left = AlgebraicGeometry.genus C` (precise statement via the project's Krull-dimension API on the underlying scheme).

- `AlgebraicGeometry.Scheme.Pic0Scheme.kPoints_iff_kerDegree` (1603) is the k-point-characterisation $\mathrm{Pic}^0_{C/k}(k) = \ker(\deg)$ (Milne §III.1, p. 88). In Lean, `  : Spec (.of k)  (PicScheme C).left, (  : Spec (.of k)  (Pic0Scheme C).left,  pic0Inclusion =  )  PicScheme.degree C  = 0`.

The abstract identity-component substrate (`GroupScheme.IdentityComponent` and its four conclusion-specific theorems `IdentityComponent.isOpenSubgroupScheme`, `IdentityComponent.isSubgroupHomomo`
`IdentityComponent.isFiniteTypeGeometricallyIrreducible`, `IdentityComponent.baseChangeIso`)
is new project material not present in Mathlib; all four theorems are fully proved (`isFiniteTypeGeometricallyIrredu`
closed by Kleiman's translation argument over the algebraic closure — closed-point translation, Jacobson density, EGA I 6.1.10, then descent along the base-change isomorphism — with no need for EGA IV_2 4.6.1-type input). The downstream `Pic0Scheme.isAbelianVariety`, `Pic0Scheme.finrank_eq_genus`, and `Pic0Scheme.kPoints_iff_kerDegree` consume Kleiman §5 Cor. 5.13 + Ex. 5.23 and Milne §III.1; the latter two remain open, blocked on the typed-sorry FGA representability foundation (25, `AJC.picrep`).

26.7 Out of scope

This chapter packages *only* the identity component $\mathrm{Pic}^0_{C/k}$ as an open-and-closed subgroup scheme of $\mathrm{Pic}_{C/k}$, its degree-zero characterisation, and its abelian-variety structure. The following downstream constructions live in separate sibling chapters and are explicitly out of scope here:

- **Albanese universal property of** $\mathrm{Pic}^0_{C/k}$ (A.4.d.ii) — the universal property of $f_P : C \to \mathrm{Pic}^0_{C/k}$ is the content of 32, gated on this chapter via 1601.
- **The autoduality** $\mathrm{Pic}^0_{C/k} \cong (\mathrm{Pic}^0_{C/k})^\vee$ (Milne §III.6, Thm. 6.6) — not needed for the Jacobian's existence as an abelian variety, hence out of scope.
- **The torsion component** $\mathrm{Pic}^\tau_{C/k}$ (Kleiman §6) — for a smooth proper curve $\mathrm{Pic}^\tau_{C/k} = \mathrm{Pic}^0_{C/k}$, so the finer torsion-component finiteness statements of Kleiman §6 are not load-bearing on Route A.
- **Construction of the degree map as a morphism of group schemes** $\mathrm{Pic}_{C/k} \to \underline{\mathbb{Z}}_k$. This chapter defines the degree on k-points only; the upgrade to a morphism of group schemes (with the constant group scheme $\underline{\mathbb{Z}}_k$) is the natural assembly statement but is not required for any downstream consumer in this project.

26.8 Internal-consistency check

The ten declaration blocks form a connected `\uses` chain rooted in the sibling chapters 25, 20, and 9:

- 1574 is a structural Archon-internal abstract substrate; no cross-chapter `\uses` pointers.
- 1575, 1576, 1577, 1578 all use 1574. 1576 additionally uses 1575 (the clopen inclusion whose homomorphism status it asserts).
- 1599 uses 1574 and 1558 (from 25).
- 1600 uses 1558 (from 25) and 1018 (from 20).
- 1601 uses 1599, 1575, 1576, 1577, 1578, and 1545 (from 25).
- 1602 uses 1601 and 537 (from 9).

- 1603 uses 1601, 1575, 1599, and 1600.

The cross-chapter labels `def:pic_scheme`, `thm:fga_pic_representability`, `thm:pic_is_group_scheme`, `def:hilbert_polynomial`, and `def:genus` all exist in sibling chapters (verified against `Picard_FGAPicRepresentabil` `Picard_QuotScheme.tex`, and `Genus.tex` at writing time).

Chapter 27

The identity component $\mathrm{Pic}^0_{C/k}$ as an abelian variety

27.1 Tangent space at the identity

For a k-group scheme G locally of finite type, the tangent space at the identity $\mathrm{T}_0 G$ is the k-vector space of k-derivations of its local ring at e into k, or equivalently (and functorially) the set $G(k[\varepsilon]/(\varepsilon^2))_e$ of k_ε-points whose underlying k-point is e. For the Picard scheme of a geometrically integral projective k-scheme, this tangent space is identified with the first sheaf cohomology of the structure sheaf via the truncated exponential sequence and the étale-sheaf comparison theorem; on a smooth proper curve, this is the genus-many-dimensional k-vector space $H^1(C, \mathcal{O}_C)$.

We first record the local-algebra core of the "tangent space via dual numbers" identification: for a local k-algebra R with maximal ideal $\mathfrak{m}$ and residue field κ, and with the k-rational-point hypothesis that the structure map $k \to \kappa$ is an isomorphism, the k-derivations of R into κ, the local k-algebra maps $R \to k[\varepsilon]/(\varepsilon^2)$, and the linear functionals on the cotangent space $\mathfrak{m}/\mathfrak{m}^2$ all agree. Applied to the local ring of $\mathrm{Pic}_{C/k}$ at the identity, this is the identification $\mathrm{T}_0 \mathrm{Pic}_{C/k} \cong \mathrm{Hom}(\mathfrak{m}_0/\mathfrak{m}_0^2, k)$ used in the main theorem.

Lemma 1604 (Derivations descend to the cotangent space)**.** *Let k be a field and R a local k-algebra with maximal ideal $\mathfrak{m}$ and residue field κ. Every k-derivation $d : R \to \kappa$ vanishes on $\mathfrak{m}^2$, and its restriction to $\mathfrak{m}$ is R-linear; consequently d induces an R-linear map*

$$\bar{d} : \mathfrak{m}/\mathfrak{m}^2 \longrightarrow \kappa.$$

Proof. Every element of $\mathfrak{m}$ acts by zero on the R-module $\kappa = R/\mathfrak{m}$, since the action factors through the residue class. For $x, y \in \mathfrak{m}$ the Leibniz rule gives $d(xy) = x \cdot d(y) + y \cdot d(x) = 0$, so d vanishes on $\mathfrak{m}^2$. For $r \in R$ and $x \in \mathfrak{m}$, $d(rx) = r \cdot d(x) + x \cdot d(r) = r \cdot d(x)$, so $d|_{\mathfrak{m}}$ is R-linear; it therefore descends to the quotient $\mathfrak{m}/\mathfrak{m}^2$. □

Lemma 1605 (Derivations are the cotangent dual)**.** *Let k be a field and R a local k-algebra whose structure map $k \to \kappa$ to the residue field is an isomorphism. Then descent $d \mapsto \bar{d}$ (1604) is an isomorphism of abelian groups*

$$\mathrm{Der}_k(R, \kappa) \;\cong\; \mathrm{Hom}_R(\mathfrak{m}/\mathfrak{m}^2, \kappa).$$

Proof. Write $\sigma : \kappa \to k$ for the inverse of the structure map. Each $r \in R$ splits canonically as $r = c_r + x_r$ with constant part $c_r := \sigma(\bar{r}) \cdot 1 \in k \cdot 1$ and $x_r := r - c_r \in \mathfrak{m}$ (its residue class vanishes by construction).

Injectivity. A k-derivation kills the constants $k \cdot 1$, so $d(r) = d(x_r)$ for every r: the derivation is determined by its restriction to $\mathfrak{m}$, i.e. by $\bar{d}$.

Surjectivity. Given an R-linear $\varphi : \mathfrak{m}/\mathfrak{m}^2 \to \kappa$, set $d(r) := \varphi(x_r \bmod \mathfrak{m}^2)$. Additivity and k-linearity follow since $r \mapsto c_r$ is k-linear. For the Leibniz rule, expand

$$x_{rs} \;=\; c_r\, x_s + c_s\, x_r + x_r x_s,$$

where $x_r x_s \in \mathfrak{m}^2$ dies in the cotangent space; since r and c_r have the same residue class, they act identically on κ, whence $d(rs) = c_r \varphi(\bar{x}_s) + c_s \varphi(\bar{x}_r) = r \cdot d(s) + s \cdot d(r)$. On elements of $\mathfrak{m}$ the constant part vanishes, so the two constructions are mutually inverse. $\square$

Lemma 1606 (Dual-number points are derivations)**.** *Let k be a field and R a local k-algebra whose structure map $k \to \kappa$ is an isomorphism. Sending a k-algebra homomorphism $f : R \to k[\varepsilon]/(\varepsilon^2)$ to the ε-coefficient of its values is a bijection*

$$\{\, f : R \to k_\varepsilon \text{ local } k\text{-algebra maps} \,\} \;\cong\; \mathrm{Der}_k(R, \kappa),$$

where locality means $f(\mathfrak{m}) \subseteq (\varepsilon)$, the maximal ideal of the local ring k_ε.

Proof. Write $f(r) = a(r) + b(r)\varepsilon$. Splitting $r = c_r + x_r$ into constant part and maximal-ideal part as in 1605, locality and k-linearity force $a(r) = a(c_r) = \sigma(\bar{r})$: the constant component of f is the residue map read through $\kappa \cong k$. Multiplicativity of f and $\varepsilon^2 = 0$ give $b(rs) = a(r)\, b(s) + a(s)\, b(r)$, which — transported through $\kappa \cong k$ — is exactly the Leibniz rule for the map $D := \bar{b} : R \to \kappa$; k-linearity of D is inherited from f. Conversely, a derivation $D : R \to \kappa$ assembles with the residue map into $f(r) := \sigma(\bar{r}) + \sigma(D(r))\,\varepsilon$, a local k-algebra homomorphism by the same computation. The two constructions are mutually inverse: the constant component of any local f is forced as above, and the ε-component is preserved by both directions. $\square$

Corollary 1607 (The tangent space is the cotangent dual)**.** *Let k be a field and R a local k-algebra whose structure map $k \to \kappa$ is an isomorphism. Then the set of local k-algebra maps $R \to k_\varepsilon$ — the dual-number points of* $\operatorname{Spec} R$ *lifting the closed point — is in bijection with* $\mathrm{Hom}_R(\mathfrak{m}/\mathfrak{m}^2, \kappa)$*, the dual of the cotangent space.*

Proof. Compose the bijections of 1606 and 1605. $\square$

Lemma 1608 (The cotangent dual is κ-linear)**.** *Let R be a local ring with maximal ideal $\mathfrak{m}$ and residue field κ. Every R-linear map $\mathfrak{m}/\mathfrak{m}^2 \to \kappa$ is κ-linear for the natural κ-vector-space structures on both sides:*

$$\mathrm{Hom}_R(\mathfrak{m}/\mathfrak{m}^2, \kappa) \;=\; \mathrm{Hom}_\kappa(\mathfrak{m}/\mathfrak{m}^2, \kappa) \;=\; (\mathfrak{m}/\mathfrak{m}^2)^\vee.$$

Proof. Both $\mathfrak{m}/\mathfrak{m}^2$ and κ are modules over $\kappa = R/\mathfrak{m}$, and their R-actions factor through the surjection $R \twoheadrightarrow \kappa$. Hence for an R-linear φ and a scalar $c = \bar{r} \in \kappa$, $\varphi(c \cdot x) = \varphi(r \cdot x) = r \cdot \varphi(x) = c \cdot \varphi(x)$. $\square$

Corollary 1609 (Tangent space as the dual vector space of $\mathfrak{m}/\mathfrak{m}^2$)**.** *In the situation of 1607, the dual-number points of* $\operatorname{Spec} R$ *lifting the closed point are in bijection with the κ-linear dual vector space* $(\mathfrak{m}/\mathfrak{m}^2)^\vee$*. In particular their κ-dimension equals* $\dim_\kappa \mathfrak{m}/\mathfrak{m}^2$ *when the latter is finite.*

Proof. Compose 1607 with the identification of 1608; the dimension statement is the equality $\dim_\kappa V^\vee = \dim_\kappa V$ for finite-dimensional V. $\square$

The local-algebra bijections above consume local homomorphisms out of the stalk $\mathcal{O}_{X,x}$. The next three statements supply the geometric input: morphisms $\operatorname{Spec} k_\varepsilon \to X$ landing at a fixed point x are exactly such stalk data.

Lemma 1610 (Local maps to the dual numbers)**.** *Let k be a field, R a local ring, and $f : R \to k_\varepsilon = k[\varepsilon]/(\varepsilon^2)$ a ring homomorphism. Then f is a local homomorphism if and only if the constant component of f kills the maximal ideal:*

$$f(\mathfrak{m}) \subseteq (\varepsilon) \quad \iff \quad \forall\, x \in \mathfrak{m},\ \operatorname{fst}(f(x)) = 0.$$

Proof. The dual numbers k_ε form a local ring whose maximal ideal is (ε), the kernel of the constant-component projection fst; an element of k_ε is a unit iff its constant component is a nonzero element of the field k. If f is local and $x \in \mathfrak{m}$, then x is a nonunit, so $f(x)$ is a nonunit, i.e. $\operatorname{fst}(f(x)) = 0$. Conversely, if the constant component kills $\mathfrak{m}$ and $f(a)$ is a unit, then $\operatorname{fst}(f(a)) \neq 0$, so $a \notin \mathfrak{m}$, i.e. a is a unit. ☐

Lemma 1611 (Spec of a local ring at a fixed point)**.** *Let R be a local ring, X a scheme, and $x \in X$ a fixed point. Morphisms $\operatorname{Spec} R \to X$ carrying the closed point of $\operatorname{Spec} R$ to x are in bijection with the local ring homomorphisms $\mathcal{O}_{X,x} \to R$:*

$$\{\, f : \operatorname{Spec} R \to X \mid f(\mathfrak{m}_R) = x \,\} \;\cong\; \{\, \varphi : \mathcal{O}_{X,x} \to R \text{ local} \,\}.$$

Proof. This is the fiber at x of the standard description of morphisms out of the spectrum of a local ring (Stacks 01J6; in Mathlib, `SpecToEquivOfLocalRing`): $\operatorname{Hom}(\operatorname{Spec} R, X)$ is the set of pairs (y, φ) of a point $y \in X$ and a local homomorphism $\varphi : \mathcal{O}_{X,y} \to R$. Forward, a morphism f with $f(\mathfrak{m}_R) = x$ induces on stalks a local homomorphism $\mathcal{O}_{X,f(\mathfrak{m}_R)} \to \mathcal{O}_{\operatorname{Spec} R,\mathfrak{m}_R} = R$, transported to $\mathcal{O}_{X,x}$ along $f(\mathfrak{m}_R) = x$. Backward, a local φ yields $\operatorname{Spec}(\varphi)$ composed with the canonical map $\operatorname{Spec} \mathcal{O}_{X,x} \to X$, which carries the closed point to x precisely because φ is local. The two constructions are mutually inverse by the same computation as for the unfibered bijection, the transport along $f(\mathfrak{m}_R) = x$ cancelling on both sides. ☐

Corollary 1612 (Dual-number points at a fixed point)**.** *Let k be a field, X a scheme, and $x \in X$. The dual-number points of X at x — morphisms $\operatorname{Spec} k_\varepsilon \to X$ carrying the closed point to x — are in bijection with the ring homomorphisms $\mathcal{O}_{X,x} \to k_\varepsilon$ whose constant component kills the maximal ideal. This is exactly the interface consumed by 1609, once the k-algebra structure on the stalk at a k-rational point of a k-scheme is threaded through.*

Proof. Instantiate 1611 at $R = k_\varepsilon$ (a local ring, its maximal ideal being (ε)) and rewrite the locality condition by 1610. ☐

The stalk data above is so far only ring-theoretic. The structure morphism of a k-scheme endows each stalk with a canonical k-algebra structure, and morphisms *over* $\operatorname{Spec} k$ are exactly those whose stalk data is k-linear. This is the missing thread between 1612 and 1609.

Definition 1613 (Structure homomorphism into a stalk)**.** Let k be a field, X a scheme, $f : X \to \operatorname{Spec} k$ a morphism, and $x \in X$. The *structure homomorphism* $\sigma_x : k \to \mathcal{O}_{X,x}$ is the composite

$$k \;\cong\; \Gamma(\operatorname{Spec} k, \mathcal{O}) \;\longrightarrow\; \mathcal{O}_{\operatorname{Spec} k,\, f(x)} \;\longrightarrow\; \mathcal{O}_{X,x}$$

of the global-sections identification, the germ map at $f(x)$, and the stalk map of f at x.

Definition 1614 (The stalk as a k-algebra)**.** In the situation of 1613, the ring homomorphism σ_x makes the stalk $\mathcal{O}_{X,x}$ a k-algebra, with algebraMap $= \sigma_x$.

Lemma 1615 (Canonical map from the spectrum of a stalk, over $\operatorname{Spec} k$)**.** *In the situation of 1613, the canonical morphism $c_x : \operatorname{Spec} \mathcal{O}_{X,x} \to X$ followed by f is the morphism of affine schemes induced by the structure homomorphism:*

$$f \circ c_x \;=\; \operatorname{Spec}(\sigma_x).$$

Proof. By naturality of the canonical map out of the spectrum of a stalk, $f \circ c_x$ factors as $c_{f(x)} \circ \operatorname{Spec}(f_x^\sharp)$, where $f_x^\sharp : \mathcal{O}_{\operatorname{Spec} k, f(x)} \to \mathcal{O}_{X,x}$ is the stalk map of f and $c_{f(x)} : \operatorname{Spec} \mathcal{O}_{\operatorname{Spec} k, f(x)} \to \operatorname{Spec} k$ is the canonical map for the affine scheme $\operatorname{Spec} k$. The latter is Spec applied to the composite of the global-sections identification $k \cong \Gamma(\operatorname{Spec} k, \mathcal{O})$ with the germ map at $f(x)$. By contravariant functoriality of Spec, the composite is Spec of $f_x^\sharp \circ (\text{germ at } f(x)) \circ (\text{identification}) = \sigma_x$. ☐

Lemma 1616 (Over-triangles are structure-compatibilities)**.** *Let k be a field, $f : X \to \operatorname{Spec} k$ a morphism of schemes, S a local ring, $g : \operatorname{Spec} S \to X$, and $\psi : k \to S$ a ring homomorphism. Write $x = g(\mathfrak{m}_S)$ for the image of the closed point and $\varphi_g : \mathcal{O}_{X,x} \to S$ for the local homomorphism induced by g on stalks (1611). Then*

$$f \circ g = \operatorname{Spec}(\psi) \quad \iff \quad \varphi_g \circ \sigma_x = \psi.$$

Proof. By 1611 the morphism g factors as $g = c_x \circ \operatorname{Spec}(\varphi_g)$ with $c_x : \operatorname{Spec} \mathcal{O}_{X,x} \to X$ the canonical map. Hence, using 1615 and contravariant functoriality,

$$f \circ g = (f \circ c_x) \circ \operatorname{Spec}(\varphi_g) = \operatorname{Spec}(\sigma_x) \circ \operatorname{Spec}(\varphi_g) = \operatorname{Spec}(\varphi_g \circ \sigma_x).$$

Since Spec is faithful on ring homomorphisms (it is fully faithful onto affine schemes), the equality $f \circ g = \operatorname{Spec}(\psi)$ holds if and only if $\varphi_g \circ \sigma_x = \psi$. ☐

Corollary 1617 (Over-$\operatorname{Spec} k$ dual-number points are local k-algebra maps)**.** *Let k be a field, X a scheme over $\operatorname{Spec} k$, and $x \in X$. The dual-number points of X at x* over $\operatorname{Spec} k$ *— morphisms $g : \operatorname{Spec} k_\varepsilon \to X$ over $\operatorname{Spec} k$ carrying the closed point to x — are in bijection with the k*-algebra *homomorphisms $\mathcal{O}_{X,x} \to k_\varepsilon$ whose constant component kills the maximal ideal. (A plain ring homomorphism $\mathcal{O}_{X,x} \to k_\varepsilon$ need not be k-linear; the over-structure is what enforces k-linearity.)*

Proof. Under the bijection of 1612, a morphism g with $g(\mathfrak{m}) = x$ corresponds to the ring homomorphism $\varphi_g : \mathcal{O}_{X,x} \to k_\varepsilon$, and by 1616 (applied with $S = k_\varepsilon$ and ψ the canonical inclusion $k \to k_\varepsilon$) the over-triangle condition $f \circ g = \operatorname{Spec}(k \to k_\varepsilon)$ holds if and only if $\varphi_g \circ \sigma_x$ is the canonical inclusion, i.e. precisely when φ_g is a homomorphism of k-algebras for the structure of 1614. The side condition on the constant component is untouched by this refinement. ☐

Theorem 1618 (Tangent space of a k-scheme at a rational point)**.** *Let k be a field, X a scheme over $\operatorname{Spec} k$, and $x \in X$ a k-rational point (the structure map $k \to \kappa(x)$ to the residue field is bijective). Then the dual-number points of X at x over $\operatorname{Spec} k$ are in bijection with the $\kappa(x)$-linear dual of the cotangent space:*

$$\{\, g : \operatorname{Spec} k_\varepsilon \to X \textit{ over } \operatorname{Spec} k \mid g(\mathfrak{m}) = x \,\} \;\cong\; \operatorname{Hom}_{\kappa(x)}(\mathfrak{m}_x/\mathfrak{m}_x^2,\, \kappa(x)).$$

Proof. Compose the bijection of 1617 with the local-algebra identification of 1609, applied to the local k-algebra $R = \mathcal{O}_{X,x}$ (1614), whose residue field is k by the rationality hypothesis. ☐

Lemma 1619 (A section retracts the structure homomorphism)**.** *Let k be a field, $f : X \to \operatorname{Spec} k$ a morphism of schemes, and $e : \operatorname{Spec} k \to X$ a section of f (that is, $f \circ e = \mathrm{id}$). Write $x = e(*)$ for the image of the unique point of $\operatorname{Spec} k$, $\sigma_x : k \to \mathcal{O}_{X,x}$ for the structure homomorphism (1613), and $\varphi_e : \mathcal{O}_{X,x} \to k$ for the local homomorphism induced by e on stalks. Then $\varphi_e \circ \sigma_x = \mathrm{id}_k$.*

Proof. Apply 1616 with $S = k$, $g = e$ and $\psi = \mathrm{id}_k$: the over-triangle $f \circ e = \mathrm{id} = \mathrm{Spec}(\mathrm{id}_k)$ holds by hypothesis, and it is equivalent to $\varphi_e \circ \sigma_x = \mathrm{id}_k$. □

Lemma 1620 (The image of a section is a rational point)**.** *Let k be a field, X a scheme over* $\mathrm{Spec}\, k$*, $e : \mathrm{Spec}\, k \to X$ a section of the structure morphism, and $x = e(*)$. Then x is a k-rational point: the structure map $k \to \kappa(x)$ to the residue field is bijective.*

Proof. Injectivity is automatic: a ring homomorphism out of a field into a nonzero ring is injective. For surjectivity, the retraction $\varphi_e : \mathcal{O}_{X,x} \to k$ of 1619 is a local homomorphism into a field, so it kills the maximal ideal $\mathfrak{m}_x$ and descends to a field homomorphism $\bar{\varphi} : \kappa(x) \to k$ with $\bar{\varphi} \circ \iota = \mathrm{id}_k$, where $\iota : k \to \kappa(x)$ is the structure map. Given $a \in \kappa(x)$, the element $a - \iota(\bar{\varphi}(a))$ lies in the kernel of $\bar{\varphi}$; but $\bar{\varphi}$ is a homomorphism of fields, hence injective, so $a = \iota(\bar{\varphi}(a))$. Thus ι is surjective. □

Corollary 1621 (Tangent space at the image of a section)**.** *Let k be a field, X a scheme over* $\mathrm{Spec}\, k$*, $e : \mathrm{Spec}\, k \to X$ a section of the structure morphism, and $x = e(*)$. Then the dual-number points of X at x over* $\mathrm{Spec}\, k$ *form the $\kappa(x)$-linear dual of the cotangent space:*

$$\{\, g : \mathrm{Spec}\, k_\varepsilon \to X \ \textit{over}\ \mathrm{Spec}\, k \mid g(\mathfrak{m}) = x \,\} \ \cong\ \mathrm{Hom}_{\kappa(x)}(\mathfrak{m}_x/\mathfrak{m}_x^2,\, \kappa(x)).$$

Proof. By 1620 the point x is k-rational, so 1618 applies. □

Definition 1622 (Identity section of a group scheme)**.** Let k be a field and G a monoid (in particular, group) scheme over k: a scheme over $\mathrm{Spec}\, k$ equipped with a monoid-object structure in the category of schemes over $\mathrm{Spec}\, k$. The *identity section* $e : \mathrm{Spec}\, k \to G$ is the underlying scheme morphism of the unit $\eta : \mathbf{1} \to G$, where the monoidal unit $\mathbf{1}$ is $\mathrm{Spec}\, k$ with structure morphism the identity.

Lemma 1623 (The identity section is a section)**.** *In the situation of 1622, the identity section satisfies $f \circ e = \mathrm{id}_{\mathrm{Spec}\, k}$, where $f : G \to \mathrm{Spec}\, k$ is the structure morphism.*

Proof. The unit η is a morphism of schemes over $\mathrm{Spec}\, k$ out of the monoidal unit $(\mathrm{Spec}\, k, \mathrm{id})$; its compatibility triangle over $\mathrm{Spec}\, k$ is precisely the equation $f \circ e = \mathrm{id}$. □

Theorem 1624 (Tangent space of a group scheme at the identity)**.** *Let k be a field and G a monoid (in particular, group) scheme over k, with identity section $e :$ $\mathrm{Spec}\, k \to G$ (1622) and $x = e(*)$. Then the dual-number points of G at x over* $\mathrm{Spec}\, k$ *— the Zariski tangent space of G at the identity in its functor-of-points form — form the $\kappa(x)$-linear dual of the cotangent space* $\mathfrak{m}_x/\mathfrak{m}_x^2$.

Proof. By 1623 the identity section is a section of the structure morphism, so 1621 applies at $x = e(*)$. □

Theorem 1625 ($\mathrm{Pic}^0_{C/k}$ is a k-group scheme)**.** *Let C/k be a smooth proper geometrically integral curve over a field k. The identity component $\mathrm{Pic}^0_{C/k}$ (1599) carries a k-group-scheme structure, inherited from the group scheme $\mathrm{Pic}_{C/k}$ via the clopen inclusion $\mathrm{Pic}^0_{C/k} \hookrightarrow \mathrm{Pic}_{C/k}$.*

Proof. The Picard scheme $\mathrm{Pic}_{C/k}$ is a k-group scheme (1557) locally of finite type over k (1545), and $\mathrm{Pic}^0_{C/k}$ is by definition its identity component. By 1576 the identity component of a k-group scheme locally of finite type is a subgroup scheme; in particular $\mathrm{Pic}^0_{C/k}$ inherits a k-group-scheme structure compatible with the inclusion. □

Theorem 1626 (Dual-number points of $\mathrm{Pic}^0_{C/k}$ at the identity)**.** *Let C/k be a smooth proper geometrically integral curve over a field k, and equip $\mathrm{Pic}^0_{C/k}$ (1599) with its k-group-scheme structure (1625). Then there is an identity section $e : \operatorname{Spec} k \to \mathrm{Pic}^0_{C/k}$ of the structure morphism such that the dual-number points of $\mathrm{Pic}^0_{C/k}$ at $e(*)$ over $\operatorname{Spec} k$ form the $\kappa(e(*))$-linear dual of the cotangent space:*

$$\mathrm{T}_0 \,\mathrm{Pic}^0_{C/k} \;=\; \{\, g : \operatorname{Spec} k_\varepsilon \to \mathrm{Pic}^0_{C/k} \ \textit{over}\ \operatorname{Spec} k \mid g(\mathfrak{m}) = e(*) \,\} \;\cong\; \mathrm{Hom}_{\kappa(e(*))}(\mathfrak{m}_{e(*)}/\mathfrak{m}^2_{e(*)},\, \kappa(e(*))).$$

This is the geometric half of 1627; what remains is the identification of the right-hand side with $H^1(C, \mathcal{O}_C)$.

Proof. The group-scheme structure (1625) provides the identity section, and 1624 applied to $G = \mathrm{Pic}^0_{C/k}$ gives the identification. □

Theorem 1627 (Tangent space at the identity: $\mathrm{T}_0 \,\mathrm{Pic}^0_{C/k} \cong H^1(C, \mathcal{O}_C)$)**.**

Source: [Kleiman], "The Picard scheme", §5, Thm. 5.11; cf. [Hartshorne], "Algebraic Geometry", GTM 52, IV.1.1 (genus def. $g(C) = \dim_k H^1(C, \mathcal{O}_C)$). Let C/k be a smooth proper geometrically integral curve of genus $g = g(C)$ over a field k, and let $\mathrm{Pic}^0_{C/k}$ (1599) be the identity component of the Picard scheme of C/k. There is a canonical isomorphism of k-vector spaces

$$\mathrm{T}_0 \,\mathrm{Pic}^0_{C/k} \;\cong\; H^1(C, \mathcal{O}_C).$$

In particular, the k-dimension of $\mathrm{T}_0 \,\mathrm{Pic}^0_{C/k}$ equals the genus $g(C) = \dim_k H^1(C, \mathcal{O}_C)$ (537).

Proof. We follow Kleiman §5 Thm. 5.11. By 1545, the Picard scheme $\mathrm{Pic}_{C/k}$ exists and represents the étale-sheafified relative Picard functor $\mathrm{Pic}_{(C/k)\text{ét}}$; the identity component $\mathrm{Pic}^0_{C/k}$ is an open subscheme of $\mathrm{Pic}_{C/k}$ (1575), so the tangent spaces at the identity of $\mathrm{Pic}^0_{C/k}$ and of $\mathrm{Pic}_{C/k}$ agree; we may therefore work with $\mathrm{Pic}_{C/k}$ throughout.

Tangent space via dual numbers. Let $k_\varepsilon := k[\varepsilon]/(\varepsilon^2)$ be the ring of dual numbers. For any k-scheme P locally of finite type and k-rational point $e \in P(k)$, the Zariski tangent space $\mathrm{T}_e P$ is in functorial bijection with the set $P(k_\varepsilon)_e$ of k_ε-valued points whose underlying k-point is e: such points are, by 1612, the local k-algebra maps $\mathcal{O}_{P,e} \to k_\varepsilon$, which by 1607 (via 1605) form the dual of the cotangent space $\mathfrak{m}_e/\mathfrak{m}_e^2$. When P is a k-group scheme, the natural homomorphism $k_\varepsilon \to k$ ($\varepsilon \mapsto 0$) induces a group homomorphism $P(k_\varepsilon) \to P(k)$ whose kernel coincides with $P(k_\varepsilon)_e$; under this identification the vector-space addition on $\mathrm{T}_e P$ corresponds to the group multiplication on the kernel. Specialising to $P = \mathrm{Pic}_{C/k}$,

$$\mathrm{T}_0 \,\mathrm{Pic}_{C/k} \;=\; \ker(\mathrm{Pic}_{C/k}(k_\varepsilon) \to \mathrm{Pic}_{C/k}(k)).$$

The identification of $\mathrm{T}_0 \,\mathrm{Pic}^0_{C/k}$ — the dual-number points over the identity section — with the $\kappa(0)$-linear dual of the cotangent space $\mathfrak{m}_0/\mathfrak{m}_0^2$ is 1626.

Computing the kernel via the truncated exponential sequence. Set $C_\varepsilon := C \otimes_k k_\varepsilon$. On the étale site of C the truncated exponential sequence of sheaves of abelian groups

$$0 \to \mathcal{O}_C \to \mathcal{O}^\times_{C_\varepsilon} \to \mathcal{O}^\times_C \to 1,$$

whose first map sends a local section b to $1 + b\varepsilon$, is split by $a \mapsto a + 0 \cdot \varepsilon$. Taking sheaf cohomology produces a split exact sequence

$$0 \to H^1(C, \mathcal{O}_C) \to H^1(C, \mathcal{O}^\times_{C_\varepsilon}) \to H^1(C, \mathcal{O}^\times_C) \to 1.$$

Identification with the tangent space. Because $\mathrm{Pic}_{C/k}$ represents the étale-sheafification of $T \mapsto H^1(C_T, \mathcal{O}_{C_T}^\times)$, there is a commutative diagram of abelian groups

$$\begin{array}{ccc} H^1(C, \mathcal{O}_{C_\varepsilon}^\times) & \longrightarrow & H^1(C, \mathcal{O}_C^\times) \\ \downarrow & & \downarrow \\ \mathrm{Pic}_{C/k}(k_\varepsilon) & \longrightarrow & \mathrm{Pic}_{C/k}(k), \end{array}$$

whose vertical maps are isomorphisms after base change to the algebraic closure $\bar{k}$ (Kleiman §2, Exercise 2.3; invocation legitimised by 1578). Taking horizontal kernels yields a k-linear comparison map $v : H^1(C, \mathcal{O}_C) \to \mathrm{T}_0 \mathrm{Pic}_{C/k}$ which is an isomorphism after base change to $\bar{k}$, hence already an isomorphism over k (the source and target are finite-dimensional k-vector spaces, and an isomorphism of k-vector spaces is detected after a faithfully flat extension).

Compatibility with the additive structure. The diagram of the preceding paragraph is functorial in k-linear endomorphisms of k_ε; in particular scalar multiplication by $a \in k$ on the dual numbers corresponds, on $H^1(C, \mathcal{O}_C)$, to scalar multiplication by a in the natural k-module structure. Hence v is a homomorphism of k-vector spaces, as required.

Finally, by the genus definition (Hartshorne IV.1.1, 537), $\dim_k H^1(C, \mathcal{O}_C) = g(C)$; thus $\dim_k \mathrm{T}_0 \mathrm{Pic}^0_{C/k} = g(C)$. □

27.1.1 The truncated-exponential splitting, ring level

The truncated-exponential sequence of sheaves used in the proof of 1627 is, on each ring of sections, the following split short exact sequence of groups attached to the dual numbers $R[\varepsilon] := R[\varepsilon]/(\varepsilon^2)$ of a commutative ring R:

$$1 \longrightarrow (R, +) \xrightarrow{b \mapsto 1 + b\varepsilon} (R[\varepsilon])^\times \xrightarrow{\varepsilon \mapsto 0} R^\times \longrightarrow 1,$$

split by $a \mapsto a + 0 \cdot \varepsilon$. This subsection develops the sequence and its naturality in R; the sheaf-level upgrade on the dual-number thickening C_ε is the remaining input to 1627.

Lemma 1628 (Reduction mod ε is a ring homomorphism)**.** *Let R be a commutative ring. The first-component projection $a + b\varepsilon \mapsto a$ is a ring homomorphism $R[\varepsilon] \to R$.*

Proof. The projection is additive and unital, and multiplicative because $(a + b\varepsilon)(a' + b'\varepsilon) = aa' + (ab' + a'b)\varepsilon$ has first component aa'. □

Lemma 1629 (The truncated exponential is a unit)**.** *Let R be a commutative ring and $b \in R$. Then $1 + b\varepsilon$ is a unit of $R[\varepsilon]$, with inverse $1 - b\varepsilon$.*

Proof. $(1 + b\varepsilon)(1 - b\varepsilon) = 1 - b^2\varepsilon^2 = 1$ since $\varepsilon^2 = 0$. □

Definition 1630 (The truncated exponential)**.** Let R be a commutative ring. The *truncated exponential* is the group homomorphism $(R, +) \to (R[\varepsilon])^\times$, $b \mapsto 1 + b\varepsilon$; it is a homomorphism because $(1 + b\varepsilon)(1 + c\varepsilon) = 1 + (b + c)\varepsilon$ ($\varepsilon^2 = 0$).

Lemma 1631 (The truncated exponential is injective)**.** *The truncated exponential $b \mapsto 1 + b\varepsilon$ is injective.*

Proof. The ε-component of $1 + b\varepsilon$ is b. □

Definition 1632 (Reduction mod ε on unit groups)**.** Let R be a commutative ring. Reduction mod ε (1628) induces a group homomorphism $(R[\varepsilon])^\times \to R^\times$ on unit groups.

Definition 1633 (The constant section of the unit groups)**.** Let R be a commutative ring. The inclusion $a \mapsto a + 0 \cdot \varepsilon$ (a ring homomorphism $R \to R[\varepsilon]$) induces a group homomorphism $R^\times \to (R[\varepsilon])^\times$ on unit groups.

Lemma 1634 (The unit-group sequence is split)**.** *The composite $R^\times \to (R[\varepsilon])^\times \to R^\times$ of 1633 followed by 1632 is the identity. In particular reduction mod ε on unit groups is surjective.*

Proof. The first component of $a + 0 \cdot \varepsilon$ is a. □

Lemma 1635 (First component of an inverse unit)**.** *For a unit $u \in (R[\varepsilon])^\times$, the first components of u^{-1} and of u multiply to 1; that is, reduction mod ε carries u^{-1} to the inverse of the reduction of u.*

Proof. Apply the ring homomorphism of 1628 to $u^{-1}u = 1$. □

Lemma 1636 (Exactness of the truncated-exponential sequence)**.** *The range of the truncated exponential $(R, +) \to (R[\varepsilon])^\times$ equals the kernel of reduction mod ε on unit groups: a unit of $R[\varepsilon]$ reduces to 1 if and only if it is of the form $1 + b\varepsilon$ for a (unique) $b \in R$.*

Proof. A unit $1 + b\varepsilon$ visibly reduces to 1. Conversely a unit $u = a + m\varepsilon$ with $a = 1$ equals $1 + m\varepsilon$, the truncated exponential of m; uniqueness of m is 1631. □

Theorem 1637 (Splitting of the dual-number unit group)**.** *Let R be a commutative ring. There is a group isomorphism*

$$(R[\varepsilon])^\times \;\cong\; R^\times \times (R, +),$$

sending a unit $u = a + m\varepsilon$ (so $a \in R^\times$) to $(a, a^{-1}m)$, with inverse $(a, b) \mapsto a(1 + b\varepsilon) = a + ab\,\varepsilon$.

Proof. The two maps are mutually inverse: composing them in either order and comparing components uses only $a^{-1}a = 1$ (1635) and $\varepsilon^2 = 0$. Multiplicativity of the forward map: for units $u = a+m\varepsilon$ and $v = a'+m'\varepsilon$, $uv = aa'+(am'+a'm)\varepsilon$, and $(aa')^{-1}(am'+a'm) = a'^{-1}m'+a^{-1}m$, which in the target $R^\times \times (R, +)$ is exactly the product of the images $(a, a^{-1}m)$ and $(a', a'^{-1}m')$; commutativity of R is used here. □

Definition 1638 (Functoriality of the dual numbers)**.** A ring homomorphism $f : R \to S$ of commutative rings induces a ring homomorphism $R[\varepsilon] \to S[\varepsilon]$, $a + b\varepsilon \mapsto f(a) + f(b)\varepsilon$.

Lemma 1639 (Functoriality: identity)**.** *The ring homomorphism $R[\varepsilon] \to R[\varepsilon]$ induced by the identity of R is the identity.*

Proof. Immediate from the componentwise formula. □

Lemma 1640 (Functoriality: composition)**.** *For ring homomorphisms $f : R \to S$ and $g : S \to T$ of commutative rings, the map on dual numbers induced by $g \circ f$ is the composite of the maps induced by f and by g.*

Proof. Immediate from the componentwise formula. □

Lemma 1641 (Naturality of the truncated exponential)**.** *For a ring homomorphism $f : R \to S$ of commutative rings, the induced map on dual-number units carries $1 + b\varepsilon$ to $1 + f(b)\varepsilon$.*

Proof. Immediate from the componentwise formula. □

Lemma 1642 (Naturality of reduction mod ε)**.** *For a ring homomorphism $f : R \to S$ of commutative rings, the square*

$$\begin{array}{ccc} (R[\varepsilon])^\times & \longrightarrow & (S[\varepsilon])^\times \\ \downarrow & & \downarrow \\ R^\times & \longrightarrow & S^\times \end{array}$$

of reduction mod ε on unit groups against the maps induced by f commutes.

Proof. Both composites send $a + b\varepsilon$ to $f(a)$, on first components. □

27.2 Smoothness of $\mathrm{Pic}^0_{C/k}$

By the tangent-space dimension bound for a Noetherian local ring, $\dim_0 \mathrm{Pic}^0_{C/k} \leq \dim_k \mathrm{T}_0 \mathrm{Pic}^0_{C/k} = g$, with equality if and only if $\mathrm{Pic}^0_{C/k}$ is regular at 0. For a smooth proper curve, this inequality becomes an equality: smoothness at the identity propagates to smoothness everywhere by translation, hence $\mathrm{Pic}^0_{C/k}$ is a smooth k-group scheme of dimension g.

Theorem 1643 (Smoothness of $\mathrm{Pic}^0_{C/k}$)**.**
Source: [Kleiman], "The Picard scheme", §5, Cor. 5.13 + Cor. 5.14; cf. [Milne], "Abelian Varieties" (course notes, 2008), §III.1, Rmk. 1.4(e), p. 86 (dimension of J equals the genus). Let C/k be a smooth proper geometrically integral curve of genus g over a field k. The identity component $\mathrm{Pic}^0_{C/k}$ (1599) is smooth over k of dimension g: the structural morphism $(\mathrm{Pic}^0_{C/k}) \to \mathrm{Spec}\, k$ is smooth, and every closed point of $\mathrm{Pic}^0_{C/k}$ lies in a smooth open neighbourhood.

Proof. We follow Kleiman §5 Cor. 5.13, with the curve-specific dimension input supplied by 1627.

Since smoothness of a scheme over k descends along faithfully flat extensions, we may replace k by its algebraic closure $\bar{k}$ (using 1578 to identify $(\mathrm{Pic}^0_{C/k})_{\bar{k}}$ with $\mathrm{Pic}^0_{C_{\bar{k}}/\bar{k}}$). Over $\bar{k}$ every closed point of $\mathrm{Pic}^0_{C/k}$ is $\bar{k}$-rational, and the group law translates the identity 0 onto an arbitrary closed point: the translation morphism $t_\lambda : \mathrm{Pic}^0_{C/k} \to \mathrm{Pic}^0_{C/k}$, $x \mapsto x + \lambda$, is an automorphism by 1576. Consequently $\mathrm{Pic}^0_{C/k}$ has the same local dimension at λ as at 0, and is smooth at λ if and only if it is smooth at 0.

By the standard tangent-space dimension bound, for any Noetherian local ring $(A, \mathfrak{m})$ with residue field k one has $\dim A \leq \dim_k(\mathfrak{m}/\mathfrak{m}^2)$, with equality if and only if A is regular. Over an algebraically closed field, regularity at a $\bar{k}$-rational point is equivalent to smoothness. Hence $\dim_0 \mathrm{Pic}^0_{C/k} \leq \dim_k \mathrm{T}_0 \mathrm{Pic}^0_{C/k} = g(C)$ by 1627, with equality iff $\mathrm{Pic}^0_{C/k}$ is regular (equivalently, smooth) at 0.

It remains to show that for C/k a smooth proper curve, equality holds (so smoothness of $\mathrm{Pic}^0_{C/k}$ follows). For a smooth proper curve there is no obstruction: as Kleiman §5 Ex. 5.23 notes, projectivity and flatness of C/k together with the smoothness of C over k imply that $\mathrm{Pic}^0_{C/k}$ is a smooth quasi-projective family of relative dimension $p_a(C) = g(C)$. In characteristic zero this reduces to Cartier's theorem (Kleiman §5 Cor. 5.14): every k-group scheme locally of finite type over a field of characteristic zero is smooth; in positive characteristic the smoothness uses the curve-specific Hodge-theoretic vanishing $H^2(C, \mathcal{O}_C) = 0$ (the source-target of the obstruction to lifting tangent vectors to first-order deformations), which holds for any curve since C has dimension 1. In either case $\mathrm{Pic}^0_{C/k}$ is smooth at 0; by the translation argument above, it is smooth everywhere.

Thus $\mathrm{Pic}^0_{C/k} \to \mathrm{Spec}\, k$ is smooth and $\dim \mathrm{Pic}^0_{C/k} = g(C)$, completing the proof. □

27.3 Properness of $\mathrm{Pic}^0_{C/k}$

A smooth proper geometrically integral curve is geometrically normal, so Kleiman's Pic^0-projectivity theorem applies and upgrades the quasi-projective conclusion to projective: $\mathrm{Pic}^0_{C/k}$ is projective over k, in particular proper.

Theorem 1644 (Properness of $\mathrm{Pic}^0_{C/k}$)**.**

Source: [Kleiman], "The Picard scheme", §5, Thm. 5.4; cf. [Milne], "Abelian Varieties" (course notes, 2008), §I.6 (abelian varieties are projective, p. 27). Let C/k be a smooth proper geometrically integral curve over a field k. The identity component $\mathrm{Pic}^0_{C/k}$ (1599) is projective over k, in particular proper: the structural morphism $(\mathrm{Pic}^0_{C/k}) \to \mathrm{Spec}\, k$ is a projective morphism, hence satisfies the universal closedness and separatedness conditions of properness.

Proof. We follow Kleiman §5 Thm. 5.4, specialised to $X = C$ a smooth proper geometrically integral curve.

Quasi-projectivity. By 1545, $\mathrm{Pic}_{C/k}$ exists, represents $\mathrm{Pic}_{(C/k)\text{ét}}$, and is locally of finite type; its identity component $\mathrm{Pic}^0_{C/k}$ is of finite type by 1577. Kleiman's quasi-projectivity statement (§5 Thm. 5.4, first conclusion) then gives that $\mathrm{Pic}^0_{C/k}$ is a quasi-projective k-scheme, since C/k is projective and geometrically integral.

Reduction to algebraically closed k. Properness descends through faithfully flat extensions of the base field (EGA IV_2 2.7.1(vii)), and formation of $\mathrm{Pic}^0_{C/k}$ commutes with extension of k (1578). Hence it suffices to prove that $(\mathrm{Pic}^0_{C/k})_{\bar{k}} = \mathrm{Pic}^0_{C_{\bar{k}}/\bar{k}}$ is proper, and we may assume $k = \bar{k}$ is algebraically closed throughout.

Reduction via Chevalley–Rosenlicht. A quasi-projective scheme is projective if and only if it is proper, so it suffices to show that $\mathrm{Pic}^0_{C/k}$ is proper over $\bar{k}$. By the Chevalley–Rosenlicht structure theorem (Conrad, "A modern proof of Chevalley's theorem on algebraic groups", Thm. 1.1, p. 3), every reduced connected algebraic $\bar{k}$-group is an extension of an abelian variety by a linear (affine) algebraic group. The multiplicative group $\mathbb{G}_m$ and the additive group $\mathbb{G}_a$ generate every connected solvable linear group (Lie–Kolchin); since $\mathrm{Pic}^0_{C/k}$ is commutative, hence solvable, properness will follow once we show that every $\bar{k}$-morphism $t : \mathbb{G}_m \to (\mathrm{Pic}^0_{C/k})_{\mathrm{red}}$ is constant.

No non-constant map from $\mathbb{G}_m$ (uses geometric normality of C). The smooth proper curve $C/\bar{k}$ is in particular geometrically normal. Suppose $t : T \to \mathrm{Pic}^0_{C/\bar{k}}$, where $T = \mathbb{A}^1 \setminus \{0\}$ is the multiplicative group; by the comparison theorem (Kleiman §2 Thm. 2.5) t is represented by an invertible sheaf $\mathcal{L}$ on $C \times T$. Since $C \times T$ is normal and integral, every invertible sheaf is the sheaf of a Cartier divisor: write $\mathcal{L} = \mathcal{O}_{C\times T}(D)$ for some divisor D (Hartshorne II Ex. 6.15). Restricting $\mathcal{L}$ to the generic fibre of the projection $p : C \times T \to C$: the generic fibre is T localised at the generic point of C, and an invertible sheaf on an open subscheme of $\mathbb{A}^1$ is trivial. Hence there is a rational function ϕ on $C \times T$ such that $(\phi) + D$ restricts to the trivial divisor on the generic fibre. Pick any section $s : C \to C \times T$ of p (e.g. at a closed point of T), and set $E := s^*((\phi) + D)$; then E is a divisor on C with the property that p^*E and $(\phi) + D$ coincide as cycles, hence as divisors (Altman–Kleiman AK70, Prp. 3.10, p. 139, valid since $C \times T$ is normal). Therefore $\mathcal{L} \cong p^*\mathcal{O}_C(E)$, so the associated map $t : T \to \mathrm{Pic}^0_{C/\bar{k}}$ factors through the generic fibre and is constant. This rules out non-constant maps $\mathbb{G}_m \to \mathrm{Pic}^0_{C/k}$; the additive case $\mathbb{G}_a \to \mathrm{Pic}^0_{C/k}$ is analogous (and follows from the multiplicative case via the standard embedding $\mathbb{G}_a \hookrightarrow \mathbb{G}_m$ on solvable subgroups). Hence the linear part in the Chevalley–Rosenlicht decomposition of $\mathrm{Pic}^0_{C/k}$ is trivial, so $\mathrm{Pic}^0_{C/k}$ is an abelian variety; in particular, it is proper. Combined with quasi-projectivity, this yields projectivity of $\mathrm{Pic}^0_{C/k}$ over $\bar{k}$, hence over k by faithfully flat descent of properness. $\square$

27.4 Geometric irreducibility of $\mathrm{Pic}^0_{C/k}$

Geometric irreducibility is delivered directly by the abstract identity-component substrate (1577) applied to the k-group scheme $G = \mathrm{Pic}_{C/k}$. We restate the specialisation here for the assembly downstream.

Theorem 1645 (Geometric irreducibility of $\mathrm{Pic}^0_{C/k}$)**.**

Source: [Kleiman], "The Picard scheme", §5, Prp. 5.3 (geometric irreducibility of the connected component of the identity). Let C/k be a smooth proper geometrically integral curve over a field k. The identity component $\mathrm{Pic}^0_{C/k}$ (1599) is geometrically irreducible: after base change to the algebraic closure $\bar{k}$, the scheme $(\mathrm{Pic}^0_{C/k})_{\bar{k}}$ is irreducible.

Proof. Apply the abstract identity-component substrate to the k-group scheme $G = \mathrm{Pic}_{C/k}$, which is locally of finite type by 1545. By 1577, the identity component $(\mathrm{Pic}_{C/k})^0 = \mathrm{Pic}^0_{C/k}$ is of finite type and geometrically irreducible over k. The base-change compatibility of 1578 identifies $(\mathrm{Pic}^0_{C/k})_{\bar{k}}$ with $\mathrm{Pic}^0_{C_{\bar{k}}/\bar{k}}$, closing the proof. □

27.5 Assembly: $\mathrm{Pic}^0_{C/k}$ is an abelian variety

A k-group scheme is an *abelian variety* iff its underlying k-scheme is a complete (proper) connected group variety (Milne §I.1, p. 8). Combining the smoothness, properness, and geometric irreducibility statements of this chapter with the identity-component group-scheme structure of 26 packages $\mathrm{Pic}^0_{C/k}$ as an abelian variety of dimension g. This is the load-bearing A.3.vii gate for Route A.

Theorem 1646 ($\mathrm{Pic}^0_{C/k}$ is an abelian variety)**.**

Source: [Milne], "Abelian Varieties" (course notes, 2008), §I.1, p. 8 (definition of abelian variety); [Kleiman], "The Picard scheme", §5, Rmk. 5.26 (the Jacobian $J := \mathrm{Pic}^0_{X/S}$ of a smooth proper family of curves is a projective abelian S-scheme). Let C/k be a smooth proper geometrically integral curve of genus g over a field k. The identity component $\mathrm{Pic}^0_{C/k}$ (1599) is an abelian variety *over k of dimension g: the underlying k-scheme is smooth (1643), proper (1644), and geometrically irreducible (1645), and it carries a k-group-scheme structure inherited from $\mathrm{Pic}_{C/k}$ via the inclusion $\mathrm{Pic}^0_{C/k} \hookrightarrow \mathrm{Pic}_{C/k}$ (1576). In particular, $\mathrm{Pic}^0_{C/k}$ is automatically commutative (Milne §I.1, Cor. 1.4).*

Proof. We assemble Milne's defining axioms of an abelian variety from the four conjuncts established earlier in this chapter and in 26.

(i) Group-variety structure. By 1576 applied to $G = \mathrm{Pic}_{C/k}$ (a k-group scheme locally of finite type by 1545), the identity component $\mathrm{Pic}^0_{C/k}$ inherits a k-group-scheme structure from $\mathrm{Pic}_{C/k}$ compatible with the clopen inclusion.

(ii) Smoothness. By 1643, the structural morphism $\mathrm{Pic}^0_{C/k} \to \operatorname{Spec} k$ is smooth (of relative dimension g). In particular $\mathrm{Pic}^0_{C/k}$ is a smooth k-variety: it is reduced (Milne's standing convention bundles reducedness into "variety"), geometrically reduced (by smoothness over k), and locally of finite type.

(iii) Properness. By 1644, the structural morphism $\mathrm{Pic}^0_{C/k} \to \operatorname{Spec} k$ is proper (in fact projective). This is Milne's "complete" axiom.

(iv) Connectedness (geometric irreducibility). By 1645, the underlying scheme $\mathrm{Pic}^0_{C/k}$ is geometrically irreducible; in particular it is geometrically connected, hence connected.

Combining (i)–(iv): $\mathrm{Pic}^0_{C/k}$ is a smooth, proper, geometrically irreducible k-scheme equipped with a k-group-scheme structure. By Milne §I.1, p. 8, "a complete connected group variety is called an abelian variety". Thus $\mathrm{Pic}^0_{C/k}$ is an abelian variety over k; commutativity is automatic (Milne §I.1, Cor. 1.4). Its dimension is the relative dimension of the smooth morphism $\mathrm{Pic}^0_{C/k} \to \mathrm{Spec}\, k$, which equals $g(C) = \dim_k H^1(C, \mathcal{O}_C)$ by 1627 and 1643.

This is the content of Kleiman §5 Rmk. 5.26: for X/S projective and smooth with connected curve fibres of genus $g > 0$ (specialise to $S = \mathrm{Spec}\, k$, $X = C$), the Jacobian $J := \mathrm{Pic}^0_{X/S}$ exists and is a projective abelian S-scheme. The genus-zero case is degenerate ($\mathrm{Pic}^0_{C/k} = 0$, Milne §III.1, Rmk. 1.4(e)) but is consistent with the assembly: a point scheme $\mathrm{Spec}\, k$ is trivially an abelian variety of dimension 0. □

27.6 Lean encoding

The Lean target file is `AlgebraicJacobian/Picard/Pic0AbelianVariety.lean`, sibling to `AlgebraicJacobian/Pica`
(26). The five theorems of this chapter correspond to Lean targets as follows.

- `AlgebraicGeometry.Scheme.Pic0.tangentSpaceIso` (1627) packages the canonical isomorphism $\mathrm{T}_0 \,\mathrm{Pic}^0_{C/k} \cong H^1(C, \mathcal{O}_C)$, the curve-specific content of Kleiman §5 Thm. 5.11. In Lean this is an isomorphism of k-modules between the Zariski tangent space of $\mathrm{Pic}^0_{C/k}$ at the identity 0 and the first sheaf-cohomology k-module $H^1(C, \mathcal{O}_C)$.
- `AlgebraicGeometry.Scheme.Pic0.smooth` (1643) packages the smoothness of $\mathrm{Pic}^0_{C/k}$ as a `[Smooth]` instance on the structural morphism, of relative dimension g. Kleiman §5 Cor. 5.13 + Cor. 5.14.
- `AlgebraicGeometry.Scheme.Pic0.proper` (1644) packages the properness of $\mathrm{Pic}^0_{C/k}$ as a `[IsProper]` instance on the structural morphism. Kleiman §5 Thm. 5.4 (projectivity upgrade for geometrically normal C/k).
- `AlgebraicGeometry.Scheme.Pic0.geometricallyIrreducible` (1645) packages the geometric irreducibility of $\mathrm{Pic}^0_{C/k}$ as a `[GeometricallyIrreducible]` instance on the structural morphism. Kleiman §5 Prp. 5.3 (specialisation of the abstract substrate `IdentityComponent.isFiniteTyp`
of 26 to $G = \mathrm{Pic}_{C/k}$).
- `AlgebraicGeometry.Scheme.Pic0.isAbelianVariety` (1646) packages the assembled abelian-variety structure: the conjunction of `[Smooth]`, `[IsProper]`, `[GeometricallyIrreducible]` on the structural morphism together with a k-group-scheme structure (`Nonempty (GrpObj (Pic0 C))`) inherited via the clopen inclusion of 1576. Milne §I.1, p. 8 (defining axioms) + Kleiman §5 Rmk. 5.26.

The five Lean declarations `tangentSpaceIso`, `smooth`, `proper`, `geometricallyIrreducible`, and `isAbelianVariety` are new project material: they are not currently in Mathlib, and the Lean skeleton is owed in a follow-up iteration. The downstream `isAbelianVariety` consumes the four conjunct theorems together with the abstract identity-component substrate of 26, and is the load-bearing A.3.vii gate of Route A for `thm:nonempty_jacobianWitness`.

27.7 Out of scope

This chapter packages *only* the five A.3.iii–vii steps. The following downstream constructions live in separate sibling chapters and are explicitly out of scope here:

- **Albanese universal property of** $\mathrm{Pic}^0_{C/k}$ (A.4.d.ii) — the universal property of $f_P : C \to \mathrm{Pic}^0_{C/k}$ is the content of 32, gated on this chapter via 1646.

- **Autoduality** $\mathrm{Pic}^0_{C/k} \cong (\mathrm{Pic}^0_{C/k})^\vee$ (Milne §III.6, Thm. 6.6; Kleiman §5 Rmk. 5.26, Abel map) — not needed for the Jacobian's existence as an abelian variety, hence out of scope.

- **Torsion component** $\mathrm{Pic}^\tau_{C/k}$ (Kleiman §6) — for a smooth proper curve $\mathrm{Pic}^\tau_{C/k} = \mathrm{Pic}^0_{C/k}$, so the finer torsion-component finiteness statements of Kleiman §6 are not load-bearing on Route A.

- **Genus-zero degenerate case.** When $g(C) = 0$, $\mathrm{Pic}^0_{C/k}$ is the trivial group scheme $\mathrm{Spec}\, k$ (Milne §III.1, Rmk. 1.4(e)); this is consistent with the abelian-variety assembly (a point is trivially an abelian variety of dimension 0) and is handled uniformly here, but is not separately stated.

27.8 Internal-consistency check

The five declaration blocks of this chapter form a connected `\uses`-chain rooted in 26, 25, and 9:

- 1627 uses 1599, 1575, 537 (statement); proof additionally uses 1545 and 1578.
- 1643 uses 1627, 1599, and 1576 (statement); proof additionally uses 1578.
- 1644 uses 1599 and 1577 (statement); proof additionally uses 1545 and 1578.
- 1645 uses 1599 and 1577 (statement); proof additionally uses 1578 and 1545.
- 1646 uses 1643, 1644, 1645, and 1576 (statement); proof additionally uses 1577.

The cross-chapter labels `def:pic_zero_subscheme`, `thm:identity_component_open_subgroup`, `thm:identity_component_is_subgroup_homomorphism`, `thm:identity_component_finite_type_geom_irreducib` `thm:identity_component_base_change_commutes`, `thm:fga_pic_representability`, and `def:genus` all exist in sibling chapters (`Picard_IdentityComponent.tex`, `Picard_FGAPicRepresentability.tex`, and `Genus.tex` at writing time).

Chapter 28

Codimension-1 indeterminacy extension (A.4.a)

This chapter isolates two codimension-one inputs to the extension theorem for rational maps. A rational map from a normal variety to a complete variety is defined away from a closed subset of codimension at least two, while a map to a group variety has indeterminacy that is either empty or purely codimension one. We also record the equivalent Weil-divisor obstruction in terms of orders along prime divisors. The depth-≥ 2 reduction uses 1765 from 30, and the two codimension-one outputs feed the extension theorem in 31.

28.1 Setup and motivation

Fix an algebraically closed field $\bar{k}$ throughout. By a *variety* over $\bar{k}$ we mean a geometrically integral, separated scheme of finite type over $\bar{k}$ (Milne's convention, *Abelian Varieties*, p. v); a variety is *nonsingular* (or *smooth*) if its local ring at every closed point is regular. A variety is *complete* if it is proper over $\bar{k}$.

A *rational map* $f\colon X \dashrightarrow Y$ between varieties is the equivalence class of a regular morphism $\varphi_U\colon U \to Y$ defined on a dense open subset $U \subseteq X$, where two pairs (U, φ_U), $(U', \varphi_{U'})$ are equivalent if φ_U and $\varphi_{U'}$ agree on $U \cap U'$ (Milne §I.3, p. 16). The map f is *defined at* a point $x \in X$ if there is some representative (U, φ_U) with $x \in U$; the set of such x is an open subset $\mathrm{Dom}(f) \subseteq X$, the *domain of definition* of f.

The motivating question of this chapter, following Milne §I.3, is:

When does a rational map $f\colon X \dashrightarrow Y$ extend to a regular morphism on all of X?

Without hypotheses on X or Y the question has no positive answer (Milne lists three counterexamples on p. 16: the inclusion of a proper open into the ambient variety; the rational map to the cuspidal cubic whose inverse fails to extend; the inverse of a blow-up at a smooth point). The two structural inputs that buy unconditional extension in our setting are:

- *Target completeness:* if Y is complete (proper over $\bar{k}$), the valuative criterion of properness forces the rational map to extend across every codim-1 point of a normal source X (Milne Theorem 3.1).
- *Target group structure:* if Y is a group variety, the difference map $\Phi(x, y) := f(x) \cdot f(y)^{-1}$ on $X \times X$ has a vanishing/pole divisor along the diagonal that controls the codimension of

the indeterminacy locus (Milne Lemma 3.3).

For Y an abelian variety both conditions hold, and combining them yields Milne's Theorem 3.2 (the headline of 31). The present chapter packages the two inputs separately: 1705 is Milne 3.1, and 1711 is Milne 3.3. The Weil-divisor reformulation 1713 translates the codim-1 extension obstruction in Milne 3.1's proof into the order-at-a-prime-divisor language pinned in 33.

28.2 The indeterminacy locus of a rational map

Definition 1647 (Indeterminacy locus of a rational map)**.** *Source: Milne,* Abelian Varieties*, §I.3, p. 16.* Let $f\colon X \dashrightarrow Y$ be a rational map between varieties over $\bar{k}$. The *domain of definition* of f is the open set

$$\mathrm{Dom}(f) \;:=\; \bigcup_{(U,\varphi_U)\in f} U \;\subseteq\; X,$$

i.e. the union of the dense opens of all representatives of f. The *indeterminacy locus* of f is the closed complement

$$Z(f) \;:=\; X \smallsetminus \mathrm{Dom}(f) \;\subseteq\; X.$$

The map f is regular (i.e. defined on all of X) if and only if $Z(f) = \emptyset$.

Lemma 1648 (Indeterminacy locus is closed)**.** *For any rational map $f\colon X \dashrightarrow Y$ of schemes, the indeterminacy locus $Z(f) = X \smallsetminus \mathrm{Dom}(f)$ (1647) is a closed subset of X: it is the complement of the open domain of definition* $\mathrm{Dom}(f)$*.*

Proof. The domain of definition is open, and the complement of an open subset is closed. □

Definition 1649 (Codimension-1-indeterminacy-free rational maps)**.** *Source: Milne,* Abelian Varieties*, Theorem 3.1, §I.3, p. 16 (the conclusion).* Let $f\colon X \dashrightarrow Y$ be a rational map between varieties over $\bar{k}$. We say f is *codim-1-indeterminacy-free* (or has *codimension-≥ 2 indeterminacy*) if every irreducible component of the closed subset $Z(f) \subseteq X$ has codimension ≥ 2 in X. Equivalently, no prime divisor of X is contained in $Z(f)$; equivalently, every point $\eta \in X$ with $\operatorname{codim}_X \overline{\{\eta\}} = 1$ lies in $\mathrm{Dom}(f)$.

Why the abstraction. Milne's Theorem 3.1 (1705 below) and Lemma 3.3 (1711 below) both produce conclusions about the codimension of $Z(f)$. Phrasing the codim-≥ 2 condition as a named predicate keeps the cited statements concise and lets the consumer 31 read off the combination ($Z(f)$ has codim ≥ 2 from 1705 *and* $Z(f)$ is empty or pure codim 1 from 1711 $\Rightarrow Z(f)$ is empty) as a single line.

28.3 Smoothness yields a DVR at every codim-1 point

The codim-1 extension argument in Milne's proof of Theorem 3.1 hinges on the local ring at the generic point of a prime divisor being a discrete valuation ring (so that the valuative criterion of properness applies). On a normal variety this is the defining property of regularity in codimension one; on a smooth variety it is automatic.

The smooth-implies-regular-local-stalk implication itself is the Jacobian-criterion direction of Stacks tag `00TT`; we package it as a separate sub-lemma so that 1685 below reads cleanly as "regular local of dimension 1 is a DVR", with the regularity prerequisite supplied by the sub-lemma.

28.3.1 Kähler differentials under localisation

The Jacobian criterion for smooth algebras is local. Near a point of a smooth scheme one chooses a standard-smooth affine presentation, identifies the stalk with a localisation of its section ring, and localises the module of Kähler differentials. The next lemmas record these reductions and the preservation of freeness and rank under localisation. The dimension and cotangent-space comparisons needed for regularity are treated in 28.3.2.

Lemma 1650 (Smooth implies flatness on stalks). *Let $X \to \operatorname{Spec}(\bar{k})$ be a smooth morphism over an algebraically closed field $\bar{k}$. Then for every point $z \in X$ the induced ring map on stalks is flat.*

Proof. Every smooth morphism is flat, and flatness of a morphism is equivalent to flatness of all induced maps on local rings. □

Lemma 1651 (A smooth morphism is locally standard smooth). *For a smooth morphism $X \to \operatorname{Spec}(\bar{k})$ and a point $z \in X$, there exist affine open neighbourhoods $U \subseteq \operatorname{Spec}(\bar{k})$ and $V \ni z$ in X with $V \subseteq X^{-1}(U)$ such that the induced ring map $\Gamma(\operatorname{Spec}\bar{k}, U) \to \Gamma(X, V)$ is standard smooth (Stacks 00T7).*

Proof. This is the local standard-smooth presentation theorem for smooth morphisms (Stacks tag 00T7). □

Lemma 1652 (A standard-smooth presentation of a smooth stalk). *For a smooth morphism $X \to \operatorname{Spec}(\bar{k})$ and a point $z \in X$, there is an affine open neighbourhood $V \ni z$ and an $\Gamma(\operatorname{Spec}\bar{k}, U)$-algebra structure on $\Gamma(X, V)$ making it a standard-smooth algebra, together with a witness that the stalk $\mathcal{O}_{X,z}$ is the localisation of $\Gamma(X, V)$ at the prime ideal corresponding to z.*

Proof. Choose the affine neighbourhoods supplied by 1651. Their map on section rings is standard smooth, and 1676 identifies the stalk at z with the localisation of the target section ring at the prime of z. □

Lemma 1653 (Kähler differentials of a standard-smooth algebra are free). *For a standard-smooth R-algebra S, the module of Kähler differentials $\Omega_{S/R}$ is free as an S-module (Stacks 00T7(2)), with basis the family $\{ds_i\}$ of the standard-smooth presentation.*

Proof. The invertible Jacobian minor of a standard-smooth presentation makes the differentials of the free variables a basis of $\Omega_{S/R}$. □

Lemma 1654 (Kähler-differential freeness is preserved by localisation). *Let R and S be commutative rings equipped with an R-algebra structure on S, and assume that the module of Kähler differentials $\Omega_{S/R}$ is free as an S-module. Let $M \subseteq S$ be a submonoid and let S_M be a localisation of S at M, so that $R \to S \to S_M$ is a tower of R-algebras. Then the module of Kähler differentials $\Omega_{S_M/R}$ is free as an S_M-module.*

Proof. The canonical comparison map $S_M \otimes_S \Omega_{S/R} \to \Omega_{S_M/R}$ is an isomorphism. Localising a basis of $\Omega_{S/R}$ therefore gives a basis of $\Omega_{S_M/R}$ over S_M. □

Lemma 1655 (The rank of localised Kähler differentials equals the relative dimension). *With notation as in 1654, assume in addition that the R-algebra S is* standard smooth of relative dimension n *(in the sense of Stacks tag* `00T7`*(2)) and that the localisation S_M is non-trivial as a ring. Then the rank of $\Omega_{S_M/R}$ over S_M equals n.*

Proof. A standard-smooth presentation of relative dimension n gives $\Omega_{S/R}$ a basis indexed by an n-element set. By 1654, localising this basis gives a basis of $\Omega_{S_M/R}$ with the same index set. Its rank is therefore n. □

28.3.2 Dimension and cotangent criteria for smooth local rings

Let k be a perfect field and let $k \to S$ be a standard-smooth presentation near a point z, and write $S_{\mathfrak{q}} = \mathcal{O}_{X,z}$ for the corresponding localisation. The regularity of this local ring follows by comparing three dimensions. The finite-type dimension formula (Stacks tag 00OE) and the cotangent exact sequence (Stacks tag 02JK) give

$$\dim S_{\mathfrak{q}} = n - \operatorname{trdeg}_k \kappa(\mathfrak{q}) = \dim_{\kappa(\mathfrak{q})} \mathfrak{m}/\mathfrak{m}^2,$$

where $n = \operatorname{rank}_{S_{\mathfrak{q}}} \Omega_{S_{\mathfrak{q}}/k}$, which is the cotangent-space characterisation of a regular local ring. The next three lemmas state the dimension formula, the cotangent comparison, and their combination.

Lemma 1656 (Smooth-algebra Krull-dimension formula (Stacks 00OE))**.** *Source: [Stacks Project], tag 00OE (=* `lemma-dimension-at-a-point-finite-type-field`*). Let k be a field, let S be a finite-type k-algebra, and let $\mathfrak{q} \subset S$ be a prime ideal corresponding to a point $x \in \operatorname{Spec}(S)$. Then*

$$\dim_x(\operatorname{Spec} S) \;=\; \dim S_{\mathfrak{q}} + \operatorname{trdeg}_k \kappa(\mathfrak{q}).$$

In particular, if $\mathfrak{q}$ is a maximal ideal (so x is a closed point) and k is algebraically closed, then $\kappa(\mathfrak{q}) = k$ (Nullstellensatz), so $\operatorname{trdeg}_k \kappa(\mathfrak{q}) = 0$ and $\dim S_{\mathfrak{q}} = \dim_x(\operatorname{Spec} S)$.

Geometric application. *Let $k = \bar{k}$, let S be a standard-smooth $\bar{k}$-algebra of relative dimension n, and let $\mathfrak{q}$ be the prime defining $z \in \operatorname{Spec} S$. The formula yields $\dim(\mathcal{O}_{X,z}) = \dim S_{\mathfrak{q}} = n - \operatorname{trdeg}_{\bar{k}} \kappa(z)$. For z a closed point this is simply n; for the generic point of a codim-1 subvariety ($\operatorname{trdeg} = n-1$) it is 1.*

Proof. This is exactly Stacks tag `00OE` (`lemma-dimension-at-a-point-finite-type-field`); the proof is a citation of the dimension theory of finitely generated algebras over a field. By Noether normalisation the finite-type k-algebra S is finite over a polynomial subalgebra, so the going-up / catenary dimension theory of such algebras applies. Write $r = \operatorname{trdeg}_k \kappa(\mathfrak{q})$; this transcendence degree is exactly the dimension of the closed subvariety $\overline{\{x\}} = V(\mathfrak{q})$. A maximal chain of primes through $\mathfrak{q}$ therefore splits into a length-$\dim S_{\mathfrak{q}}$ part below $\mathfrak{q}$ (recording the local codimension, i.e. the height of $\mathfrak{q}$ in the relevant irreducible component) and a length-r part above it (recording the dimension of $\overline{\{x\}}$). Taking the maximum over the minimal primes contained in $\mathfrak{q}$ yields

$$\dim_x(\operatorname{Spec} S) = \dim S_{\mathfrak{q}} + r = \dim S_{\mathfrak{q}} + \operatorname{trdeg}_k \kappa(\mathfrak{q}),$$

which is the asserted formula.

For the geometric specialisation, take $k = \bar{k}$ algebraically closed and S standard smooth of relative dimension n, so that $\dim_x(\operatorname{Spec} S) = n$ via 1655 (the free Kähler module $\Omega_{S_{\mathfrak{q}}/R}$ has rank n, which pins the relative dimension). The formula then gives $\dim S_{\mathfrak{q}} = n - \operatorname{trdeg}_{\bar{k}} \kappa(\mathfrak{q})$. At a closed point $\mathfrak{q}$ is maximal, so by the Nullstellensatz $\kappa(\mathfrak{q}) = \bar{k}$ and $\operatorname{trdeg}_{\bar{k}} \kappa(\mathfrak{q}) = 0$; hence $\dim S_{\mathfrak{q}} = \dim_x(\operatorname{Spec} S) = n$. □

Lemma 1657 (Cotangent and Kähler differentials over a field)**.** *Source: [Stacks Project], tag 02JK. The right-exact half is* `lemma-differential-seq`*; the separable-residue refinement occurs in* `lemma-separable-smooth`*. Let k be a field and let R be a Noetherian local k-algebra with maximal ideal $\mathfrak{m}$ and residue field $\kappa = R/\mathfrak{m}$. The canonical exact sequence*

$$\mathfrak{m}/\mathfrak{m}^2 \;\longrightarrow\; \Omega_{R/k} \otimes_R \kappa \;\longrightarrow\; \Omega_{\kappa/k} \;\longrightarrow\; 0$$

identifies the cotangent space $\mathfrak{m}/\mathfrak{m}^2$ (a κ-vector space) with the kernel of the right map. When κ/k is separable (in particular when k is algebraically closed and $\kappa = k$, so $\Omega_{\kappa/k} = 0$) the leftmost map is injective and the sequence is short exact — in fact split short exact when there is a k-algebra section $\kappa \to R$ (e.g. at a closed point of a $\bar{k}$-variety). In particular, over an algebraically closed field $k = \bar{k}$ and at a closed point $\mathfrak{m}$,

$$\Omega_{R/k} \otimes_R \kappa \;\simeq\; \mathfrak{m}/\mathfrak{m}^2$$

as κ-vector spaces; hence $\dim_\kappa(\Omega_{R/k} \otimes_R \kappa) = \dim_\kappa \mathfrak{m}/\mathfrak{m}^2$.

The cotangent–Kähler comparison can be separated into the following finite-dimensional statements. They compute the dimension after residue-field base change and then transfer it across the isomorphism of 1657.

Lemma 1658 (Residue-field tensor dimension from free rank)**.** *Let $R \to S_\mathfrak{q}$ be an algebra with $S_\mathfrak{q}$ a nontrivial local ring whose Kähler differentials $\Omega_{S_\mathfrak{q}/R}$ are $S_\mathfrak{q}$-free of rank n. Then the residue-field base change $\kappa \otimes_{S_\mathfrak{q}} \Omega_{S_\mathfrak{q}/R}$ has κ-dimension n, where κ is the residue field of $S_\mathfrak{q}$.*

Proof. A basis of the free module $\Omega_{S_\mathfrak{q}/R}$ base-changes to a basis of $\kappa \otimes_{S_\mathfrak{q}} \Omega_{S_\mathfrak{q}/R}$ with the same n-element index set. □

Lemma 1659 (Residue-tensor dimension for a standard-smooth algebra)**.** *If $S_\mathfrak{q}$ is a nontrivial local R-algebra that is standard smooth of relative dimension n, then the residue-field base change of its Kähler differentials has κ-dimension n. This is the standard-smooth-tied packaging of 1658, discharging the freeness and rank hypotheses directly from the relative-dimension instance.*

Proof. Standard smoothness of relative dimension n makes $\Omega_{S_\mathfrak{q}/R}$ free of rank n, so apply 1658. □

Lemma 1660 (Cotangent isomorphism in maximal-ideal form)**.** *Under the closed-point hypotheses of 1657 ($S_\mathfrak{q}$ formally smooth over R, residue field κ formally smooth over R, and $\Omega_{\kappa/R} = 0$), there is an $S_\mathfrak{q}$-linear isomorphism between the cotangent module of the maximal ideal $\mathfrak{m}$ and the residue-field base change $\kappa \otimes_{S_\mathfrak{q}} \Omega_{S_\mathfrak{q}/R}$. This restates the kernel-side cotangent isomorphism of 1657 with the canonical maximal-ideal cotangent space as its domain.*

Proof. Transport the isomorphism of 1657 along the canonical equality $\ker(S_\mathfrak{q} \to \kappa) = \mathfrak{m}$. □

Lemma 1661 (Cotangent-space dimension with formally smooth residue field)**.** *With the closed-point hypotheses of 1660 and $\Omega_{S_\mathfrak{q}/R}$ free of $S_\mathfrak{q}$-rank n, the cotangent space $\mathfrak{m}/\mathfrak{m}^2$ has κ-dimension n.*

Proof. Combine the cotangent isomorphism 1660 with the residue-tensor dimension formula 1658. □

Lemma 1662 (Cotangent-space dimension at a $\bar{k}$-rational point)**.** *Bundled closed-point form of 1661: if $S_\mathfrak{q}$ is formally smooth over R, $\Omega_{S_\mathfrak{q}/R}$ is free of rank n, and the structure map $R \to \kappa$ to the residue field is bijective (as at a $\bar{k}$-rational closed point, by the Nullstellensatz), then* $\dim_\kappa \mathfrak{m}/\mathfrak{m}^2 = n$.

Proof. A bijective structure map identifies the residue field with R; it is therefore formally smooth over R, with zero module of relative differentials. The result follows from 1661. □

Lemma 1663 (A standard-smooth stalk is regular local)**.** *Let $\bar{k}$ be an algebraically closed field, and let $\bar{k} \to S$ be a standard-smooth algebra presentation of relative dimension n on an affine chart $\operatorname{Spec} S \subseteq X$, and let $z \in \operatorname{Spec} S$ be a point corresponding to a prime $\mathfrak{q} \subset S$. Then the stalk $S_{\mathfrak{q}}$ is a regular local ring of Krull dimension $n - \operatorname{trdeg}_{\bar{k}} \kappa(\mathfrak{q})$. In the closed-point case ($\kappa(\mathfrak{q}) = \bar{k}$, $\operatorname{trdeg} = 0$) the dimension equals n; in the codim-1 generic-point case ($\operatorname{trdeg} = n - 1$) the dimension equals 1.*

Proof. Combine the three sub-lemmas:

1. By 1655, the localised Kähler module $\Omega_{S_{\mathfrak{q}}/\bar{k}}$ is free of rank n over $S_{\mathfrak{q}}$.

2. By 1656, the Krull dimension of $S_{\mathfrak{q}}$ equals $\dim_z \operatorname{Spec} S - \operatorname{trdeg}_{\bar{k}} \kappa(\mathfrak{q})$, which for a standard-smooth presentation of relative dimension n equals $n - \operatorname{trdeg}_{\bar{k}} \kappa(\mathfrak{q})$. (For the standard-smooth presentation the dimension $\dim_z \operatorname{Spec} S = n$ follows from the global-complete-intersection structure: n variables modulo $n - \dim$ equations, see Stacks `lemma-relative-global-complete-intersection-smooth`.)

3. By 1657, the cotangent exact sequence is short exact because $\bar{k}$ is perfect and $\kappa(\mathfrak{q})/\bar{k}$ is separably generated. Its middle term has dimension n by (1), while $\Omega_{\kappa(\mathfrak{q})/\bar{k}}$ has dimension $\operatorname{trdeg}_{\bar{k}} \kappa(\mathfrak{q})$. Hence
$$\dim_{\kappa(\mathfrak{q})} \mathfrak{m}/\mathfrak{m}^2 = n - \operatorname{trdeg}_{\bar{k}} \kappa(\mathfrak{q}).$$
At a closed point the transcendence degree is zero, and the cotangent sequence reduces to the displayed isomorphism in 1657.

4. Compare (2) and (3): both $\dim S_{\mathfrak{q}}$ and $\dim_{\kappa(\mathfrak{q})} \mathfrak{m}/\mathfrak{m}^2$ equal $n - \operatorname{trdeg}_{\bar{k}} \kappa(\mathfrak{q})$; the regularity criterion for Noetherian local rings then gives $S_{\mathfrak{q}}$ regular. The scheme-level transfer $S_{\mathfrak{q}} = \mathcal{O}_{X,z}$ on the affine chart completes the conclusion.

□

28.3.3 Krull dimension of standard-smooth localisations

For a prime $\mathfrak{p}$ of a Noetherian ring, the Krull dimension of $R_{\mathfrak{p}}$ is the height of $\mathfrak{p}$. If $R = k[x_i]_{i \in \iota}$ is a polynomial ring in finitely many variables over a field and $\mathfrak{m}$ is maximal, then $\operatorname{ht}(\mathfrak{m}) = \#\iota$. The lower bound follows by induction on the number of variables and the principal ideal theorem; the upper bound follows from the dimension of the polynomial ring. Consequently $\dim R_{\mathfrak{m}} = \#\iota$.

If $f_1, \dots, f_c$ is a regular sequence in a Noetherian local ring A, then
$$\dim A/(f_1, \dots, f_c) + c = \dim A.$$
Thus a standard-smooth presentation gives the desired local dimension once its defining relations are known to form a regular sequence after localisation.

For a submersive presentation, the invertible Jacobian minor makes the cotangent classes of the defining relations linearly independent. This independence is preserved after localisation. Matsumura's criterion then shows that the localised relations form a regular sequence. More explicitly:

- On a regular local Noetherian ring $(A, \mathfrak{m})$, a sequence $f_1, \dots, f_c \in \mathfrak{m}$ whose images in $\mathfrak{m}/\mathfrak{m}^2$ are κ-linearly independent forms a regular sequence in A. Indeed, $f_1 \notin \mathfrak{m}^2$, so it is a non-zero-divisor and $A/(f_1)$ is again regular local; the remaining cotangent classes stay independent in the quotient, and induction applies.

- The conormal isomorphism

$$(I/I^2) \otimes_S S_{\mathfrak{m}} \simeq (IS_{\mathfrak{m}})/(IS_{\mathfrak{m}})^2$$

  transports linear independence from I/I^2 over S to $\mathfrak{m}_A/\mathfrak{m}_A^2$ over $\kappa(\mathfrak{m}_A)$ on the localisation A.

- Applying the preceding regular-sequence criterion and the dimension-drop formula gives the dimension of the quotient of the localised polynomial ring by the relations.

28.3.4 Algebraic ingredients and scheme bridges

The following lemmas make the preceding dimension argument explicit. They feed the Stacks-00OE Krull-dimension formula 1656 and the regular-stalk assembly, either directly or through the Jacobian regular-sequence witness.

Stacks 00OE Krull-dimension chain.

Lemma 1664 (Localised Krull dimension equals height)**.** *For a prime ideal $\mathfrak{m}$ of a commutative ring R and any localisation A of R at $\mathfrak{m}$, the Krull dimension of A equals the height of $\mathfrak{m}$.*

Proof. Prime ideals of $R_{\mathfrak{m}}$ correspond order-preservingly to prime ideals of R contained in $\mathfrak{m}$. Thus chains of primes in the localisation have the same maximal lengths as chains ending at $\mathfrak{m}$, whose supremum is its height. □

Lemma 1665 (Polynomial-ring maximal ideal height lower bound)**.** *For a field k, every maximal ideal of the polynomial ring* $\mathrm{MvPolynomial}\,(\mathrm{Fin}\,n)\,k$ *has height at least n.*

Proof. Induct on n, transporting the ideal along the one-variable-splitting isomorphism and using that adjoining a polynomial variable raises height by one, together with the Jacobson contraction of the maximal ideal. □

Lemma 1666 (Polynomial-ring ideal height upper bound)**.** *For a field k and a finite index set ι, every proper ideal of* $\mathrm{MvPolynomial}\,\iota\,k$ *has height at most $\#\iota$.*

Proof. The height of a proper ideal is bounded by the Krull dimension of its ring, and the polynomial ring in $\#\iota$ variables over a field has Krull dimension $\#\iota$. □

Lemma 1667 (Maximal-ideal height in a polynomial ring indexed by $\mathrm{Fin}\,n$)**.** *For a field k, every maximal ideal of* $\mathrm{MvPolynomial}\,(\mathrm{Fin}\,n)\,k$ *has height exactly n.*

Proof. Combine the lower bound 1665 and the upper bound 1666 by antisymmetry. □

Lemma 1668 (Maximal-ideal height in a finitely generated polynomial ring)**.** *For a field k and a finite index set ι, every maximal ideal of* $\mathrm{MvPolynomial}\,\iota\,k$ *has height exactly $\#\iota$.*

Proof. Choose a bijection $\iota \simeq \mathrm{Fin}(\#\iota)$ and transport 1667 along the renaming isomorphism that it induces. Ring isomorphisms preserve heights of ideals. □

Lemma 1669 (Krull dimension of a localised polynomial ring)**.** *For a field k, a finite index set ι, and any localisation A of* $\mathrm{MvPolynomial}\,\iota\,k$ *at a maximal ideal $\mathfrak{m}$, the Krull dimension of A equals $\#\iota$.*

Proof. By 1664 (Krull dimension equals height), the dimension is the height of $\mathfrak{m}$, which is $\#\iota$ by 1668. □

Lemma 1670 (Dimension drop by a regular sequence). *Let R' be a Noetherian local ring of dimension a, and let rs be a regular sequence in R' of length b. Then* $\dim(R'/(rs)) + b = a$.

Proof. Quotienting a Noetherian local ring by one non-zero-divisor lowers its dimension by one. Apply this successively to the elements of the regular sequence. ☐

Lemma 1671 (Dimension of a localised polynomial quotient). *Let A be a Noetherian localisation of* MvPolynomial $\iota\, k$ *at a maximal ideal, and let rs be a regular sequence in A of length b. Then* ringKrullDim $(A/(rs)) + b = \#\iota$.

Proof. The localised polynomial ring has dimension $\#\iota$ by 1669; now apply 1670. ☐

Stacks 00SW submersive-presentation cotangent independence.

Lemma 1672 (00SW — relations are cotangent-independent). *For a submersive presentation P of an R-algebra S, the family of cotangent classes of the relations $i \mapsto [\,P$*.relation $i\,]$ *in the cotangent module of P is S-linearly independent.*

Proof. In a submersive presentation, the cotangent classes of the relations are the distinguished cotangent basis, hence are linearly independent. ☐

Lemma 1673 (00SW — cotangent independence transported to a localisation). *With P a submersive presentation of S and p a prime ideal of S, the image of the cotangent classes of the relations under the canonical localisation map of the cotangent module at p is* Localization.AtPrime p*-linearly independent.*

Proof. Transport 1672 through the localisation map, using that linear independence is preserved by a localised-module map. ☐

Relative dimension and scheme-to-algebra bridges.

Lemma 1674 (Existence of a relative dimension for a standard-smooth algebra). *Every standard-smooth R-algebra S is standard smooth of some specific relative dimension $n \in \mathbb{N}$.*

Proof. Take the difference between the number of polynomial variables and the number of relations in a submersive presentation of S. ☐

Lemma 1675 (Submersive presentation of a relative-dimension-n algebra). *An R-algebra S that is standard smooth of relative dimension n admits an explicit submersive presentation P with* P.dimension $= n$.

Proof. By definition, a standard-smooth algebra of relative dimension n is equipped with a submersive presentation of dimension n. ☐

Lemma 1676 (An affine-open stalk is a localisation). *For an affine open $V \ni z$ of a scheme X, the stalk $\mathcal{O}_{X,z}$ is the localisation of the section ring $\Gamma(X, V)$ at the prime ideal of z.*

Proof. Identify V with Spec $\Gamma(X, V)$. Under this identification, the stalk at z is the localisation at the corresponding prime ideal. ☐

Lemma 1677 (A nonempty open in a one-point space is the whole space). *On a scheme whose underlying space is a subsingleton (at most one point), every nonempty open subset equals the whole space.*

Proof. A nonempty open contains the unique point and therefore equals the whole underlying space. ☐

Lemma 1678 (Sections of Spec $\bar{k}$ over a nonempty open)**.** *For a field $\bar{k}$ and any nonempty open U of* $\mathrm{Spec}(\bar{k})$*, the section ring* $\Gamma(\mathrm{Spec}\,\bar{k}, U)$ *is ring-isomorphic to $\bar{k}$. This gives the algebraically-closed base ring of the standard-smooth presentation a clean $\bar{k}$-identification.*

Proof. The spectrum of a field has a single point, so 1677 forces $U = \top$ and the sections collapse to the global sections $\Gamma(\mathrm{Spec}\,\bar{k}, \top) \simeq \bar{k}$. ☐

Matsumura regular-sequence bridge (Stacks 00NQ).

Lemma 1679 (Module quotient equals ring quotient)**.** *For r in a commutative ring A, there is a canonical $(A/(r))$-linear isomorphism between the module quotient $A/(r \cdot A)$ and the ring quotient $A/(r)$.*

Proof. Both quotients identify elements of A exactly when their difference lies in the principal ideal (r). The identity on representatives therefore induces the required linear isomorphism. ☐

Lemma 1680 (Ring-quotient cons rule for regular sequences)**.** *Let $r \in A$ be such that multiplication by r is injective on A, and suppose the images of a list rs form a regular sequence over the ring quotient $A/(r)$. Then $r :: rs$ is a regular sequence over A.*

Proof. Injectivity of multiplication by r says that r is a non-zero-divisor. The regularity of the tail in the module quotient then gives regularity of the full sequence; use 1679 to identify this module quotient with $A/(r)$. ☐

Lemma 1681 (Cotangent independence descends to a quotient)**.** *Let $(A, \mathfrak{m})$ be a local ring and $f_0, \dots, f_m \in \mathfrak{m}$ with $\kappa(A)$-linearly independent cotangent classes. Then the tail $f_1, \dots, f_m$, pushed into the quotient $A' = A/(f_0)$, has $\kappa(A')$-linearly independent cotangent classes in $\mathfrak{m}'/\mathfrak{m}'^2$.*

Proof. The cotangent map $A \to A'$ is surjective with kernel spanned by the class of f_0; a kernel-disjointness argument reduces a $\kappa(A')$-relation to a $\kappa(A)$-relation killed by the full independence hypothesis. ☐

Lemma 1682 (Matsumura: cotangent-independent elements form a regular sequence)**.** *Let $(A, \mathfrak{m})$ be a Noetherian regular local ring and $f_1, \dots, f_n \in \mathfrak{m}$ whose cotangent classes in $\mathfrak{m}/\mathfrak{m}^2$ are $\kappa(A)$-linearly independent. Then $f_1, \dots, f_n$ form a regular sequence in A (Matsumura,* Commutative Ring Theory*, Thm. 14.2; Stacks 00NQ).*

Proof. Induct on n. The head f_1 is a non-zero-divisor (a regular local ring is a domain), the cotangent independence descends to the quotient $A/(f_1)$ by 1681, and the ring-quotient cons rule 1680 reassembles the regular sequence. In the standard-smooth application the linear-independence input is supplied by the relations of a submersive presentation via 1673. ☐

Lemma 1683 (Smooth over $\bar{k} \Rightarrow$ regular local stalk (Stacks 00TT))**.** *Source: [Stacks Project], tag 00TT. Let X be a variety over an algebraically closed field $\bar{k}$ (geometrically integral, separated, locally of finite type) and assume the structure morphism $X \to \mathrm{Spec}(\bar{k})$ is* smooth*. Then for every point $z \in X$ the stalk $\mathcal{O}_{X,z}$ is a regular local ring.*

The algebraic-closure hypothesis ensures that every closed point is $\bar{k}$-rational. Regularity at an arbitrary point may then also be obtained by choosing a closed specialization and localizing its regular local ring. The arbitrary-field form follows directly from the general smooth-implies-regular criterion of tag `00TT`*.*

Proof. Source: [Stacks Project], tag 00TT. The cited lemma packages the Jacobian-criterion equivalence and concludes "in this case the local ring $S_{\mathfrak{q}}$ is regular". Translated to the scheme level: a smooth finite-type morphism $X \to \operatorname{Spec}(\bar{k})$ restricted to an affine chart $\operatorname{Spec}(S) \subseteq X$ gives an affine algebra $\bar{k} \to S$ that is smooth at every prime $\mathfrak{q} \subset S$; tag 00TT then yields regularity of the localisation $S_{\mathfrak{q}} = \mathcal{O}_{X,z}$ at every point $z \in X$ lying over $\mathfrak{q}$. □

Lemma 1684 (Smooth at a codim-1 point $\Rightarrow$ principal nonzero maximal ideal)**.** *Let X be a smooth, geometrically-irreducible, separated, integral variety over an algebraically closed field $\bar{k}$, and let $z \in X$ be a point with* coheight $z = 1$ *(a codimension-one point). Then the maximal ideal of the stalk $\mathcal{O}_{X,z}$ is principal and nonzero.*

Proof. Smoothness gives a regular local stalk via 1683; the coheight bridge 1719 gives Krull dimension 1; the cotangent-space characterisation of regularity then forces $\dim_\kappa \mathfrak{m}/\mathfrak{m}^2 = 1$, i.e. a principal maximal ideal, which is nonzero since the stalk is not a field. This is the geometric core consumed by 1685. □

Lemma 1685 (Smooth $\Rightarrow$ DVR at every codim-1 point)**.** *Source: Hartshorne, II.6, p. 130 (definition of regular in codimension one; the DVR consequence at a codim-1 point). Let X be a nonsingular variety over $\bar{k}$ and let $\eta \in X$ be a point with* $\operatorname{codim}_X \overline{\{\eta\}} = 1$*. Then the local ring $\mathcal{O}_{X,\eta}$ is a discrete valuation ring with fraction field the function field $K(X)$ of X.*

Proof. By 1683 (Stacks tag `00TT`, the Jacobian-criterion direction), the stalk $\mathcal{O}_{X,x}$ is regular for every point $x \in X$ — in particular at every closed point (the case Hartshorne I.5.1 states classically). Localising a regular local ring at a prime ideal yields a regular local ring; in particular, for $\eta \in X$ with $\operatorname{codim}_X \overline{\{\eta\}} = 1$, the local ring $\mathcal{O}_{X,\eta}$ is regular of Krull dimension 1. A regular local ring of Krull dimension 1 is exactly a discrete valuation ring (Atiyah–Macdonald, Proposition 9.2; or Stacks tag `00PD`). Its fraction field is the function field $K(X)$ of X by irreducibility of X (the unique generic point of X lies above η, and the stalk at the generic point of an integral scheme is the function field). □

28.4 Milne Theorem 3.1: extension across codimension at least two

Before the extension theorem itself we record the function-field functoriality $K(Y) \to K(X)$ attached to a dominant rational map, which the codim-1 step of the extension proof uses to view a representative as a morphism $\operatorname{Spec} K(X) \to Y$.

Lemma 1686 (Image of the closed point of $\operatorname{Spec} K(X)$)**.** *Let X be an integral scheme and Y an irreducible scheme, and let $f\colon X \dashrightarrow Y$ be a dominant rational map. The induced morphism* $\operatorname{Spec} K(X) \to Y$ *(the restriction of f to the generic point of X) sends the closed point of* $\operatorname{Spec} K(X)$ *to the generic point of Y.*

Proof. Choose a representative (U, φ_U) of f; its underlying morphism $\varphi_U\colon U \to Y$ is dominant. The morphism $\operatorname{Spec} K(X) \to Y$ factors as $\operatorname{Spec} \mathcal{O}_{X,\eta_X} \to U \xrightarrow{\varphi_U} Y$, where the first map sends the closed point to the generic point η_U of U (checked after the open immersion $U \hookrightarrow X$, which identifies η_U with η_X). A dominant morphism carries the generic point of its irreducible source to a generic point of Y; as Y is a T_0 space its generic point is unique, so the image is η_Y. □

Definition 1687. For a dominant rational map $f\colon X \dashrightarrow Y$ from an integral scheme X to an irreducible scheme Y, the **function-field pullback** is the ring homomorphism $K(Y) \to K(X)$

obtained from the germ pullback $\mathcal{O}_{Y,f(\eta_X)} \to K(X)$ of f at the generic point, identified along 1686 with the stalk $\mathcal{O}_{Y,\eta_Y} = K(Y)$.

The difference map of Milne's Lemma 3.3 (28.5 below) is built by precomposing f with the two projections $X \times X \to X$ and pairing the results. The precomposition operation is the following; it is the left-hand analogue of the right-composition of a rational map with a morphism, $f \mapsto f \circ g$.

Definition 1688. Let $p\colon W \to X$ be a morphism of schemes whose underlying continuous map is *open*, and let $f\colon X \dashrightarrow Y$ be a rational map. The **precomposition** $f \circ p\colon W \dashrightarrow Y$ is the rational map represented, for any partial-map representative (V, φ_V) of f, by

$$(p^{-1}(V),\ \ p^{-1}(V) \xrightarrow{p|_{p^{-1}(V)}} V \xrightarrow{\varphi_V} Y).$$

The domain $p^{-1}(V)$ is dense because the preimage of a dense open set under an open continuous map is dense; the same fact makes the construction independent of the chosen representative, so it descends to a well-defined rational map.

Lemma 1689. *Precomposition on the left commutes with right-composition by a morphism: for $p\colon W \to X$ open, $f\colon X \dashrightarrow Y$, and $g\colon Y \to Z$,*

$$(f \circ p) \circ g \;=\; (f \circ g) \circ p \qquad \textit{in} \qquad W \dashrightarrow Z.$$

Furthermore, if p and f are defined over a base scheme S, then so is $f \circ p$.

Proof. Both sides act on a chosen partial-map representative (V, φ_V) of f by restriction along p and composition with g; the resulting equality is the associativity of morphism composition, $(p|_{p^{-1}(V)} \circ \varphi_V) \circ g = p|_{p^{-1}(V)} \circ (\varphi_V \circ g)$. For the over-$S$ claim, $p|_{p^{-1}(V)}$ is an S-morphism because it commutes with the structure morphisms via the restriction identity $p|_{p^{-1}(V)} \circ (V \hookrightarrow X) = (p^{-1}(V) \hookrightarrow W) \circ p$, and a composite of S-morphisms is an S-morphism. $\square$

Lemma 1690. *Precomposing the rational map of an honest morphism $g\colon X \to Y$ with an open morphism $p\colon W \to X$ is the rational map of the composite:*

$$(g \textit{ as a rational map}) \circ p \;=\; (p \circ g) \textit{ as a rational map.}$$

Proof. Both sides are everywhere-defined; on the whole of W the left-hand side is $p \circ g$ by the restriction identity $p|_W \circ (X \hookrightarrow X)_{\mathrm{id}} = p$, i.e. the two partial maps have the same (dense, in fact total) domain and equal underlying morphism. $\square$

To pair the two precompositions $f \circ \mathrm{pr}_1$ and $f \circ \mathrm{pr}_2$ into a single rational map $X \times X \dashrightarrow G \times_{\bar{k}} G$, we use the function-field correspondence: for an integral scheme X over S and a scheme Y locally of finite type over S, a rational map $X \dashrightarrow Y$ over S is the same datum as an S-morphism $\operatorname{Spec} K(X) \to Y$ (restriction to the generic point, spread out again by local finite type). Morphisms out of the single scheme $\operatorname{Spec} K(X)$ pair freely through the universal property of the fibre product.

Definition 1691. Let $s_X\colon X \to S$, $s_Y\colon Y \to S$, $s_Z\colon Z \to S$ be structure morphisms with X integral and s_Y, s_Z locally of finite type, and let $a\colon X \dashrightarrow Y$, $b\colon X \dashrightarrow Z$ be rational maps over S (i.e. a restricted to the generic point commutes with the structure morphisms, and likewise for b). The **pairing** $a \times_S b\colon X \dashrightarrow Y \times_S Z$ is the rational map corresponding, under the function-field correspondence, to the morphism

$$\operatorname{Spec} K(X) \xrightarrow{\langle a_\eta, b_\eta \rangle} Y \times_S Z$$

obtained from the generic-point morphisms $a_\eta\colon \operatorname{Spec} K(X) \to Y$ and $b_\eta\colon \operatorname{Spec} K(X) \to Z$ by the universal property of the fibre product (their composites with the structure morphisms both equal $\operatorname{Spec} K(X) \to \operatorname{Spec} \mathcal{O}_{X,\eta} \to X \to S$).

Lemma 1692. *The two projections of the pairing recover the factors, and the pairing is again over S:*

$$(a \times_S b) \circ \mathrm{pr}_1 = a, \qquad (a \times_S b) \circ \mathrm{pr}_2 = b, \qquad (a \times_S b) \circ (\mathrm{pr}_1 \circ s_Y) = s_X,$$

where $\mathrm{pr}_1\colon Y \times_S Z \to Y$, $\mathrm{pr}_2\colon Y \times_S Z \to Z$ are the projections (the last equation being the statement that $a \times_S b$ is over S).

Proof. Right-composition with a morphism is functorial on generic-point morphisms: $(\varphi \circ g)_\eta = \varphi_\eta \circ g$ for any rational map φ and morphism g. Hence the generic-point morphism of $(a \times_S b) \circ \mathrm{pr}_1$ is $\langle a_\eta, b_\eta \rangle \circ \mathrm{pr}_1 = a_\eta$, which is the generic-point morphism of a; since a rational map out of an integral scheme is determined by its generic-point morphism, $(a \times_S b) \circ \mathrm{pr}_1 = a$. The second identity is symmetric. The over-S identity follows by combining the first identity with associativity of right-composition, $(a \times_S b) \circ (\mathrm{pr}_1 \circ s_Y) = ((a \times_S b) \circ \mathrm{pr}_1) \circ s_Y = a \circ s_Y = s_X$. □

The two precompositions and their pairing are assembled into Milne's difference map by right-composing with the *difference morphism* of the group object G, the honest group law $(g, h) \mapsto g \cdot h^{-1}$.

Definition 1693. Let G be a group object in a cartesian-monoidal category. The **difference morphism** $\mathrm{diff}_G\colon G \otimes G \to G$ is fst / snd, the division in the group structure on the hom-set $\mathrm{Hom}(G \otimes G, G)$; pointwise it is $(g, h) \mapsto g \cdot h^{-1}$, i.e. the group law composed with inversion on the second factor. Equivalently, it is built from the group structure on the hom-set. When G lives over $\operatorname{Spec} \bar{k}$, its underlying scheme morphism $\mathrm{diff}^{\mathrm{left}}_G\colon G \times_{\bar{k}} G \to G$ is Milne's $m \circ (\mathrm{id} \times \mathrm{inv})$.

Lemma 1694. *The difference morphism is over the base:* $\mathrm{diff}^{\mathrm{left}}_G \circ (G \to \operatorname{Spec} \bar{k}) = \mathrm{pr}_1 \circ (G \to \operatorname{Spec} \bar{k})$.

Proof. This is the triangle identity $\varphi_{\mathrm{left}} \circ g_{\mathrm{hom}} = f_{\mathrm{hom}}$ of the over-category morphism diff_G, together with the identification of the structure morphism of $G \times_{\bar{k}} G$ as $\mathrm{pr}_1 \circ (G \to \operatorname{Spec} \bar{k})$. □

Definition 1695. Let X be smooth over an algebraically closed field $\bar{k}$ with $X \times_{\bar{k}} X$ integral, let G be a group variety over $\bar{k}$, and let $f\colon X \dashrightarrow G$ be a rational map over $\bar{k}$. Milne's **difference rational map**

$$\Phi \;=\; ((f \circ \mathrm{pr}_1) \times_{\bar{k}} (f \circ \mathrm{pr}_2)) \circ \mathrm{diff}^{\mathrm{left}}_G\colon\; X \times_{\bar{k}} X \;\dashrightarrow\; G$$

is $(x, y) \mapsto f(x) \cdot f(y)^{-1}$. The projections pr_i are open (being base changes of the smooth structure morphism), so the precompositions $f \circ \mathrm{pr}_i$ are defined; both are over $\bar{k}$ (1689, 1690) with common structure composite $\mathrm{pr}_1 \circ (X \to \operatorname{Spec} \bar{k})$, since pr_1 and pr_2 agree after $X \to \operatorname{Spec} \bar{k}$.

Lemma 1696. *The difference map is a $\bar{k}$-rational map:* $\Phi \circ (G \to \operatorname{Spec} \bar{k}) = \mathrm{pr}_1 \circ (X \to \operatorname{Spec} \bar{k})$*, the structure morphism of $X \times_{\bar{k}} X$.*

Proof. By associativity of right-composition, $\Phi \circ (G \to \operatorname{Spec} \bar{k}) = ((f \circ \mathrm{pr}_1) \times (f \circ \mathrm{pr}_2)) \circ (\mathrm{diff}^{\mathrm{left}}_G \circ (G \to \operatorname{Spec} \bar{k}))$; by 1694 the inner composite is $\mathrm{pr}_1 \circ (G \to \operatorname{Spec} \bar{k})$, and the over-$S$ identity of the pairing (1692) evaluates the result to the structure morphism of $X \times_{\bar{k}} X$. □

The following elementary fact is the workhorse for Milne's "Φ is defined at (x, x) if f is defined at x" step: post-composing a rational map with an honest morphism never destroys definedness.

Lemma 1697. *Right-composition of a rational map $F\colon X \dashrightarrow Y$ with a morphism $g\colon Y \to Z$ enlarges the domain of definition:* $\mathrm{Dom}(F) \subseteq \mathrm{Dom}(F \circ g)$.

Proof. A partial-map representative (V, F_V) of F yields the representative $(V, F_V \circ g)$ of $F \circ g$ with the same domain V. $\square$

Lemma 1698. ***(Milne Lemma 3.3, Sub-step 1 domain bound / Sub-step 2 easy direction.)*** *The difference rational map Φ is defined on* $\mathrm{Dom}(f) \times_{\bar{k}} \mathrm{Dom}(f)$*: every point of $X \times_{\bar{k}} X$ projecting into* $\mathrm{Dom}(f)$ *under both projections lies in* $\mathrm{Dom}(\Phi)$*. In particular Φ is defined at the diagonal point (x, x) whenever f is defined at x.*

Proof. Right-composition with diff_G only enlarges the domain of definition (1697), so it suffices to show the pairing $(f \circ \mathrm{pr}_1) \times (f \circ \mathrm{pr}_2)$ is defined on $\mathrm{Dom}(f) \times_{\bar{k}} \mathrm{Dom}(f)$. The maximal representative (U, φ_U) of f (on its full domain of definition) is defined on $U = \mathrm{Dom}(f)$ and over $\bar{k}$; pairing $\varphi_U \circ \mathrm{pr}_i$ into $G \times_{\bar{k}} G$ over the common domain $\mathrm{pr}_1^{-1} U \cap \mathrm{pr}_2^{-1} U$ gives Milne's explicit morphism $U \times_{\bar{k}} U \to G \times_{\bar{k}} G$, a partial map whose induced function-field morphism on the integral source $X \times_{\bar{k}} X$ matches that of the pairing (1691) factor-by-factor. Hence it represents the pairing, and its domain $\mathrm{pr}_1^{-1} U \cap \mathrm{pr}_2^{-1} U$ lies in the pairing's domain of definition. $\square$

The converse direction of Milne's slice argument rests on the reconstruction formula $f(x) = \Phi(x, u) \cdot f(u)$. The next four lemmas establish its group-theoretic content and the resulting domain bound. Together with openness of $\mathrm{Dom}(\Phi)$, irreducibility of the fibre $X_{\kappa(x)}$, and descent of definedness along the smooth projection, this proves that definedness of Φ at (x, x) forces definedness of f at x.

Lemma 1699. *For a rational map $f\colon X \dashrightarrow Y$ and an open morphism $p\colon W \to X$, the precomposition is defined on the preimage of the domain:* $p^{-1}(\mathrm{Dom}(f)) \subseteq \mathrm{Dom}(f \circ p)$.

Proof. If $p(w) \in \mathrm{Dom}(f)$, choose a partial-map representative (V, f_V) of f with $p(w) \in V$. Then $(p^{-1}V, f_V \circ (p\,|_V))$ is a representative of $f \circ p$ defined at w, so $w \in \mathrm{Dom}(f \circ p)$. $\square$

Lemma 1700. *Let X be integral, let S, Y, Z be separated schemes with structure morphisms $s_Y\colon Y \to S$, $s_Z\colon Z \to S$ locally of finite type, and let $a\colon X \dashrightarrow Y$, $b\colon X \dashrightarrow Z$ be rational maps over S. Then the pairing is defined on the intersection of the two domains:* $\mathrm{Dom}(a) \cap \mathrm{Dom}(b) \subseteq \mathrm{Dom}(a \times_S b)$.

Proof. The maximal representatives of a and b are defined on $\mathrm{Dom}(a)$, resp. $\mathrm{Dom}(b)$, and are S-morphisms; over the common domain $\mathrm{Dom}(a) \cap \mathrm{Dom}(b)$ they agree after the structure morphisms to S, hence pair through the fibre product into a partial map defined on $\mathrm{Dom}(a) \cap \mathrm{Dom}(b)$. Its induced function-field morphism restricts, on each fibre-product factor, to a's, resp. b's; so it represents $a \times_S b$, and its domain lies in the domain of definition of the pairing. $\square$

Lemma 1701. *Let G be a group object in a cartesian-monoidal category and $a, b\colon T \to G$ two T-points. Then $\langle \langle a, b\rangle \circ \mathrm{diff}_G,\ b\rangle \circ m = a$, i.e. $(a \cdot b^{-1}) \cdot b = a$.*

Proof. Precomposition by $\langle a, b\rangle$ is a group homomorphism on T-points, so $\langle a, b\rangle \circ \mathrm{diff}_G = \langle a, b\rangle \circ (\mathrm{fst} \,/\, \mathrm{snd}) = a/b$. Then $\langle a/b, b\rangle \circ m = (a/b) \cdot b = a$ by the group cancellation law. $\square$

Lemma 1702. *Let G be a group variety over $\bar{k}$ and $A, B\colon T \to G$ two $\bar{k}$-morphisms sharing a structure map ($A \circ (G \to \mathrm{Spec}\,\bar{k}) = B \circ (G \to \mathrm{Spec}\,\bar{k})$). Then the scheme-level reconstruction holds:*

$$m^{\mathrm{left}} \circ \langle \mathrm{diff}_G^{\mathrm{left}} \circ \langle A, B\rangle,\ B\rangle \ = \ A.$$

Proof. The forgetful functor $(-)^{\mathrm{left}}\colon \mathbf{Sch}/\bar{k} \to \mathbf{Sch}$ sends the cartesian lift, first and second projections, and composition to pullback-lift, $\mathrm{pr}_1, \mathrm{pr}_2$ and composition. Lifting A, B to $\bar{k}$-morphisms $T \to G$ in the over-category and applying $(-)^{\mathrm{left}}$ to 1701 yields the claim. ☐

Lemma 1703. *The reconstruction $f(x) = \Phi(x,y)\cdot f(y)$ holds as an identity of rational maps on $X \times_{\bar{k}} X$:*

$$f \circ \mathrm{pr}_1 \;=\; m^{\mathrm{left}} \circ (\Phi \times_{\bar{k}} (f \circ \mathrm{pr}_2)).$$

Proof. Both sides are $\bar{k}$-rational maps out of the integral scheme $X \times_{\bar{k}} X$, so it suffices to match their function-field morphisms $\operatorname{Spec} K(X \times X) \to G$. The left side is $A = (f \circ \mathrm{pr}_1)^{\mathrm{ff}}$; the right side unfolds, via the function-field formula for the pairing and difference map, to $m^{\mathrm{left}} \circ \langle \mathrm{diff}_G^{\mathrm{left}} \circ \langle A, B\rangle,\ B\rangle$ with $B = (f \circ \mathrm{pr}_2)^{\mathrm{ff}}$. These agree by 1702. ☐

Lemma 1704. ***(Milne Lemma 3.3, Sub-step 2 hard direction, algebraic content.)*** *$f \circ \mathrm{pr}_1$ is defined wherever Φ is defined and $f \circ \mathrm{pr}_2$ is defined:*

$$\mathrm{Dom}(\Phi) \cap \mathrm{pr}_2^{-1}(\mathrm{Dom}(f)) \;\subseteq\; \mathrm{Dom}(f \circ \mathrm{pr}_1).$$

This is the group-theoretic half of Milne's converse: at a point where Φ is defined and f is defined on the second coordinate, the reconstruction $f(x) = \Phi(x,u)\cdot f(u)$ exhibits $f \circ \mathrm{pr}_1$ as regular.

Proof. Rewriting $f \circ \mathrm{pr}_1$ by 1703, it suffices to bound $\mathrm{Dom}(m^{\mathrm{left}} \circ (\Phi \times (f \circ \mathrm{pr}_2)))$. Right-composition only enlarges the domain (1697), and the pairing is defined on $\mathrm{Dom}(\Phi) \cap \mathrm{Dom}(f \circ \mathrm{pr}_2)$ (1700), which contains $\mathrm{Dom}(\Phi) \cap \mathrm{pr}_2^{-1}(\mathrm{Dom}(f))$ by 1699. ☐

The first input to Milne's Theorem 3.2 is the following: a rational map from a normal variety to a *complete* variety automatically extends across every codim-1 point.

Theorem 1705 (Milne Theorem 3.1: codim-≥ 2 extension to a complete target)**.** *Source: Milne,* Abelian Varieties*, Theorem 3.1, §I.3, p. 16. Let X be a nonsingular variety over $\bar{k}$, let Y be a complete variety over $\bar{k}$, and let $f\colon X \dashrightarrow Y$ be a rational map. Then the indeterminacy locus $Z(f)$ (1647) has codimension ≥ 2 in X; equivalently, f is codim-1-indeterminacy-free (1649).*

Proof. Fix an indeterminacy point $z \in Z(f)$; we must show its coheight is ≥ 2. Suppose not, so $\mathrm{coheight}_X(z) \leq 1$, and argue by cases.

Coheight 0. Then z is a maximal point, hence $z = \eta_X$, the generic point of the irreducible X. But η_X specializes into the dense open $\mathrm{Dom}(f)$, so $z \in \mathrm{Dom}(f)$, contradicting $z \in Z(f)$.

Coheight 1. Then $\overline{\{z\}}$ has codimension 1, so by 1685 the local ring $\mathcal{O}_{X,z}$ is a discrete valuation ring with fraction field the function field $K(X)$. The rational map f determines a morphism $\operatorname{Spec} K(X) \to Y$ at the generic point, and the over-$\operatorname{Spec} \bar{k}$ condition $f \circ Y_{\mathrm{hom}} = X_{\mathrm{hom}}$ makes the resulting square commute. Since Y is proper, the valuative criterion of properness (Milne cites Hartshorne II.4.7) lifts $\operatorname{Spec} K(X) \to Y$ across $\operatorname{Spec} K(X) \to \operatorname{Spec} \mathcal{O}_{X,z}$ to a morphism $\operatorname{Spec} \mathcal{O}_{X,z} \to Y$. This spreads to a regular representative of f on a neighbourhood of z, so $z \in \mathrm{Dom}(f)$, again a contradiction.

Hence every point of $Z(f)$ has coheight ≥ 2; equivalently f is codim-1-indeterminacy-free. ☐

Remark 1706. Milne 3.1 is valuative-only; Auslander–Buchsbaum is not used here. The proof above uses only normality (via the codim-1 DVR of 1685) and the valuative criterion of properness (Hartshorne II.4.7), and is characteristic-free. There is *no* "extend across a codim-≥ 2 point" step: the theorem asserts only that $Z(f)$ has codimension ≥ 2, and the actual extension of f to a morphism happens later, once $Z(f)$ is shown to be empty (1705 together with Milne 3.3

gives 1770, $Z(f) = \emptyset$, and then 1766 extends across the empty locus). The depth-≥ 2/local-cohomology (Stacks `0AVF`) route for a group-variety-free target is *not* taken and would in any case be false for a general complete target (*cf.* $\mathbb{P}^2 \dashrightarrow \mathbb{P}^1$, whose single codim-2 indeterminacy point admits no regular extension).

Where 1765 and the Auslander–Buchsbaum machinery of 30 *do* enter is the Milne 3.3 analysis of a group-variety target below (regularity of the diagonal and of local quotients), not this codim-≥ 2 statement.

28.5 Milne Lemma 3.3: indeterminacy locus of a map to a group variety is codim 1

The second input to Milne's Theorem 3.2 is structural: when the target Y is a group variety, the indeterminacy locus is either empty or of *pure* codimension 1. Combined with 1705's "codimension ≥ 2" conclusion, this forces the locus to be empty.

The following four lemmas isolate two ingredients of Milne's proof: the topological existence statement completing the slice argument and Krull's Hauptidealsatz transported to a scheme-theoretic vanishing statement. The first supplies a point where reconstruction applies; the other three show that a nonzero local function vanishing at a point also vanishes along a codimension-one generisation.

Lemma 1707 (Slice existence for the fibre of the self-product, and its difference-map corollary)**.**

(Milne Lemma 3.3, Sub-step 2, topological existence half.) *Let X be smooth and geometrically irreducible over $\bar{k}$, let $x \in X$, let $V \subseteq X \times_{\bar{k}} X$ be an open set meeting the fibre $\mathrm{pr}_1^{-1}\{x\}$ (at some point p_0), and let $D \subseteq X$ be a dense open subset. Then there is a point p of the fibre $\mathrm{pr}_1^{-1}\{x\}$ with $p \in V$ and $\mathrm{pr}_2(p) \in D$.*

Consequently, if Milne's difference rational map Φ (1695) is defined at a point p_0 lying over x — in particular at the diagonal point (x, x) — then there is a point p over x, with $p \in \mathrm{Dom}(\Phi)$, at which the precomposition $f \circ \mathrm{pr}_1$ is defined.

Proof. The fibre $\mathrm{pr}_1^{-1}\{x\}$ is the base change $X_{\kappa(x)}$, hence is irreducible because X is geometrically irreducible. The set V meets the fibre at p_0 by hypothesis; the set $\mathrm{pr}_2^{-1}(D)$ meets the fibre too. Indeed, for any $u \in D$, the tensor product $\kappa(x) \otimes_{\bar{k}} \kappa(u)$ is nonzero and therefore has a prime ideal, which determines a point of $X \times_{\bar{k}} X$ lying over (x, u). Preirreducibility of the fibre forces V and $\mathrm{pr}_2^{-1}(D)$ to meet it *simultaneously*, producing the point p.

For the corollary, apply the existence statement with $V := \mathrm{Dom}(\Phi)$ (which meets the fibre at p_0) and $D := \mathrm{Dom}(f)$, dense since f is a rational map; the resulting point p satisfies $p \in \mathrm{Dom}(\Phi)$ and $\mathrm{pr}_2(p) \in \mathrm{Dom}(f)$, so 1704 places p in the domain of $f \circ \mathrm{pr}_1$. □

Lemma 1708 (Krull's Hauptidealsatz, transported to a height-one prime below a given prime)**.** ***(Milne Lemma 3.3, Sub-step 4, ring-theoretic core.)*** *Let R be a Noetherian domain, let $t \in R$ be nonzero, and let $\mathfrak{p}$ be a prime ideal of R with $t \in \mathfrak{p}$. Then there is a prime ideal $\mathfrak{q}$ of height exactly 1 with $\mathfrak{q} \subseteq \mathfrak{p}$ and $t \in \mathfrak{q}$.*

Proof. The principal ideal (t) is contained in $\mathfrak{p}$, so it has a minimal prime $\mathfrak{q}$ lying below $\mathfrak{p}$; minimality of $\mathfrak{q}$ over (t) gives $t \in \mathfrak{q}$. Krull's principal ideal theorem bounds the height of a minimal prime over a principal ideal by 1. Since R is a domain and $t \neq 0$, the unique height-0 prime (0) does not contain t, so $\mathfrak{q} \neq (0)$ and hence $\mathrm{height}(\mathfrak{q}) \geq 1$. The two bounds give $\mathrm{height}(\mathfrak{q}) = 1$. □

Lemma 1709 (Regularity propagates to a generisation)**.** *On an irreducible scheme X, a function-field element regular at a point P is regular at every generisation z of P (a point z with $z \rightsquigarrow P$, i.e. P a specialisation of z): the germ pullback $\mathcal{O}_{X,P} \to K(X)$ factors through the stalk map $\mathcal{O}_{X,P} \to \mathcal{O}_{X,z}$ induced by the specialisation, so its image is contained in the image of $\mathcal{O}_{X,z} \to K(X)$.*

Proof. Immediate from functoriality of stalks along the specialisation map: for $s \in \mathcal{O}_{X,P}$, the germ of s in the function field equals the germ, under $\mathcal{O}_{X,z} \to K(X)$, of the image of s under $\mathcal{O}_{X,P} \to \mathcal{O}_{X,z}$. ☐

Lemma 1710 (Zero loci of regular functions are pure codimension one: the diagonal transport)**.**

(Milne Lemma 3.3, Sub-step 4, scheme wrapper.) *Let X be an integral, locally Noetherian scheme with regular local rings at every point (e.g. X smooth over a field). Let $P \in X$ and let s be an element of the maximal ideal of $\mathcal{O}_{X,P}$ whose image t in the function field $K(X)$ is nonzero. Then there is a generisation $z \rightsquigarrow P$ of coheight 1 and an element s' of the maximal ideal of $\mathcal{O}_{X,z}$ whose image in $K(X)$ is again t.*

Geometrically: the zero locus of a nonzero regular function on X is of pure codimension 1, passing through P.

Proof. Since s is a non-unit and $t \neq 0$, the reciprocal t^{-1} is not regular at P: if it were the image of a unit $u \in \mathcal{O}_{X,P}$, then $1 - us$ would be a unit mapping to 0 in $K(X)$, which is impossible. Purity of the pole locus of the nonzero rational function t^{-1} — the codimension-1 structure of the zero/pole locus of a regular function on a locally Noetherian integral scheme with regular stalks, an instance of 1708 applied to a local equation of the locus at P — produces a coheight-1 generisation $z \rightsquigarrow P$ at which t^{-1} is still not regular. By 1709, t itself remains regular at z, witnessed by some $s' \in \mathcal{O}_{X,z}$ with image t; if s' were a unit, its inverse would exhibit t^{-1} as regular at z, a contradiction, so s' lies in the maximal ideal of $\mathcal{O}_{X,z}$. ☐

Lemma 1711 (Milne Lemma 3.3: pure-codim-1 structure of the indeterminacy locus into a group variety)**.** *Source: Milne,* Abelian Varieties*, Lemma 3.3, §I.3, p. 17. Let X be a nonsingular variety over $\bar{k}$, let G be a group variety over $\bar{k}$ (a smooth group scheme of finite type), and let $f\colon X \dashrightarrow G$ be a rational map. Then either f is defined on all of X, or the indeterminacy locus $Z(f) \subseteq X$ is a closed subset of pure codimension 1 (i.e. a finite union of prime divisors).*

Equivalently: it is never *the case that f has a non-empty indeterminacy locus all of whose irreducible components have codimension ≥ 2.*

Proof. Following Milne's proof of Lemma 3.3. The argument has four sub-steps. Throughout we fix a representative (U, φ_U) of f.

Sub-step 1: the difference rational map. Define

$$\Phi\colon X \times X \dashrightarrow G, \qquad \Phi(x, y) \;:=\; \varphi(x) \cdot \varphi(y)^{-1},$$

via the composition

$$U \times U \xrightarrow{\varphi_U \times \varphi_U} G \times G \xrightarrow{\mathrm{id} \times \mathrm{inv}} G \times G \xrightarrow{m} G,$$

where $\mathrm{inv}\colon G \to G$ is the group-variety inverse and $m\colon G \times G \to G$ is the group law. The map Φ is rational by composition: precomposing f with each projection $\mathrm{pr}_i\colon X \times X \to X$ (which is open, being a base change of the flat, locally-finitely-presented structure morphism $X \to \operatorname{Spec} \bar{k}$) yields rational maps $f \circ \mathrm{pr}_i\colon X \times X \dashrightarrow G$ (1688, whose over-$\bar{k}$ part is 1689, using 1690 to identify $(s_X \circ \mathrm{pr}_i)$ with the structure morphism of $X \times X$); pairing them into $X \times X \dashrightarrow G \times G$ (1691, whose projection and over-$\bar{k}$ properties are 1692) and composing on the right with the difference morphism $\mathrm{diff}_G^{\mathrm{left}} = m \circ (\mathrm{id} \times \mathrm{inv})$ (1693) gives the named difference map Φ (1695); it is again over $\bar{k}$ (1696, using 1694). Its domain is at least $U \times U \subseteq X \times X$ (1698), which is dense because U is dense in X.

Sub-step 2: Φ-defined at $(x, x) \Leftrightarrow \varphi$-defined at x. If φ is defined at x, then $(x, x) \in U \times U \subseteq \mathrm{Dom}(\Phi)$ (1698), so Φ is defined at (x, x), with value $\Phi(x, x) = e$ (the identity of G). Conversely, if Φ is defined at (x, x), openness of $\mathrm{Dom}(\Phi)$ in $X \times X$ produces an open set $W \ni (x, x)$ on which Φ is defined; by projection (taking the slice $\{x\} \times W' \subseteq W$ for some open $W' \subseteq X$, possibly shrinking W' so φ is defined on W' as well — W' need not contain x), the identity

$$\varphi(x) \;=\; \Phi(x, u) \cdot \varphi(u), \qquad u \in W',$$

exhibits a regular representative of φ at x. Hence $x \in \mathrm{Dom}(\varphi)$ if and only if $(x, x) \in \mathrm{Dom}(\Phi)$.

The reconstruction identity $\varphi(x) = \Phi(x, u) \cdot \varphi(u)$ is 1703, and its domain consequence $\mathrm{Dom}(\Phi) \cap \mathrm{pr}_2^{-1}(\mathrm{Dom}(\varphi)) \subseteq \mathrm{Dom}(\varphi \circ \mathrm{pr}_1)$ is 1704. A suitable u exists by openness of $\mathrm{Dom}(\Phi)$ and irreducibility of the fibre $X_{\kappa(x)}$, which follows from geometric irreducibility of X; this is 1707. Finally, the smooth-descent reflection $\mathrm{Dom}(\varphi \circ \mathrm{pr}_1) \subseteq \mathrm{pr}_1^{-1}(\mathrm{Dom}(\varphi))$ transports definedness back to X.

Sub-step 3: rephrase via pullback of local rings. The rational map Φ pulls $\mathcal{O}_{G,e}$ back to a $\bar{k}$-subalgebra of the function field $K(X \times X)$:

$$\Phi^* \colon \mathcal{O}_{G,e} \;\longrightarrow\; K(X \times X).$$

Because Φ sends (x, x) to e wherever it is defined, Φ is defined at (x, x) if and only if $\Phi^*(\mathcal{O}_{G,e}) \subseteq \mathcal{O}_{X \times X,(x,x)}$ (i.e. every local function on G near e has no pole at (x, x) after pullback).

Sub-step 4: pole locus is pure codimension 1. The variety $X \times X$ is nonsingular (smoothness is stable under products over $\bar{k}$), so the Weil-divisor apparatus of 1797 and the order map of 1816 apply on $X \times X$. For any nonzero rational function $f \in K(X \times X)^*$ decompose its divisor as

$$\mathrm{div}(f) \;=\; \mathrm{div}(f)_0 - \mathrm{div}(f)_\infty,$$

with $\mathrm{div}(f)_0$ and $\mathrm{div}(f)_\infty$ effective and $\mathrm{div}(f)_\infty = \mathrm{div}(f^{-1})_0$ (Milne's notation, matching the project's $\div(f) = \sum \mathrm{ord}_Y(f) \cdot Y$ of 1839: $\mathrm{div}(f)_0$ collects the prime divisors with $\mathrm{ord}_Y(f) > 0$ and $\mathrm{div}(f)_\infty$ those with $\mathrm{ord}_Y(f) < 0$). The local ring at (x, x) can be characterised as

$$\mathcal{O}_{X \times X,(x,x)} \;=\; \big\{\, f \in K(X \times X) \;\big|\; \mathrm{div}(f)_\infty \text{ does not contain } (x, x) \,\big\} \cup \{0\}.$$

Suppose $f \in K(X \times X)^*$ is not in $\mathcal{O}_{X \times X,(x,x)}$; equivalently, $(x, x) \in \mathrm{supp}(\mathrm{div}(f)_\infty)$. The pole divisor $\mathrm{div}(f)_\infty$ is a finite sum of prime divisors of $X \times X$ (i.e. a closed subset of pure codimension 1 in $X \times X$). Its intersection with the diagonal $\Delta_X \subseteq X \times X$ is a closed subset of pure codimension 1 in Δ_X (this is Milne's reference to AG 9.2, the standard fact that a prime divisor of $X \times X$ either contains Δ_X or meets it in pure codimension 1; on a nonsingular variety this is an instance of Krull's principal-ideal theorem applied to the local equation of Δ_X inside $X \times X$ — the scheme-theoretic transport of this purity statement along a generisation of a point of Δ_X is recorded as 1710, built from the ring-theoretic core 1708).

Conclusion. Suppose $f \colon X \dashrightarrow G$ is not defined on all of X. Pick $x \in Z(f)$. By Sub-steps 2–3, there is some $f_0 \in \Phi^*(\mathcal{O}_{G,e})$ with $(x, x) \in \mathrm{supp}(\mathrm{div}(f_0)_\infty)$. By Sub-step 4 the intersection $\mathrm{supp}(\mathrm{div}(f_0)_\infty) \cap \Delta_X$ is pure codimension 1 in Δ_X, passing through (x, x). Identifying Δ_X with X via the diagonal isomorphism, the corresponding closed subset of X has pure codimension 1 and passes through x. By Sub-step 2, every point of this codim-1 subset is a point where Φ is undefined on the diagonal, hence where φ is undefined. Thus $Z(\varphi)$ contains a pure codim-1 closed subset passing through x; since $x \in Z(\varphi)$ was arbitrary, every irreducible component of $Z(\varphi)$ has codimension exactly 1, i.e. $Z(\varphi)$ is of pure codimension 1. $\square$

Remark 1712. Characteristic-free. The above proof uses only: (i) the group law on G and its inverse (formal smoothness of G as a group object); (ii) the Weil-divisor apparatus on the nonsingular variety $X \times X$ (Hartshorne II.6, or 33 in the project's notation); (iii) the decomposition of a divisor on a nonsingular variety into effective pieces, and Krull's principal-ideal theorem on $X \times X$. None of these has a characteristic restriction, so 1711 holds over every algebraically closed base field, in arbitrary characteristic. (Milne states the lemma in this generality at §I.3, p. 17.)

28.6 Weil-divisor obstruction to codim-1 extension

The proof of 1705's Step 1 (codim-1 ruling-out) goes through the valuative criterion. In the project's Weil-divisor language (33) the same obstruction has a cleaner phrasing: the extension fails at a prime divisor W if and only if some affine coordinate of f around the image $\overline{f(\eta_W)}$ has a strictly negative order $\mathrm{ord}_W(\cdot) < 0$. This is the valuation-theoretic form of the obstruction recorded by 1816 and 1839.

Theorem 1713. *[Weil-divisor criterion for codimension-one extension] Source: Hartshorne, II.6, pp. 130–131 (valuation v_Y and the order map). Let X be a nonsingular variety over $\bar{k}$, let Y be a variety over $\bar{k}$, let $f\colon X \dashrightarrow Y$ be a rational map, and let $W \subseteq X$ be a prime divisor with generic point $\eta_W \in W$ (so $\mathrm{codim}_X \overline{\{\eta_W\}} = 1$). Let (U, φ_U) be any representative of f with $U \cap (X \setminus W) \neq \emptyset$. The following are equivalent:*

1. *f is defined at η_W (i.e. $\eta_W \in \mathrm{Dom}(f)$, equivalently $W \not\subseteq Z(f)$).*
2. *For some (equivalently, every) affine open neighbourhood $V \subseteq Y$ of the generic image of f near W, every regular function $g \in \mathcal{O}_Y(V)$ pulls back to a rational function $\varphi_U^*(g) \in K(X)$ satisfying*
$$\mathrm{ord}_W(\varphi_U^*(g)) \ \geq \ 0.$$

Consequently: f is codim-1-indeterminacy-free (1649) if and only if, for every prime divisor $W \subseteq X$ and every affine open $V \subseteq Y$ as above, the pullback $\varphi_U^*(g)$ has $\mathrm{ord}_W \geq 0$ for all $g \in \mathcal{O}_Y(V)$.

Proof. By 1685, the local ring $\mathcal{O}_{X,\eta_W}$ is a DVR with fraction field $K(X)$ and uniformiser determined by W; by 1816 its valuation is exactly ord_W. The local ring at η_W is therefore characterised inside $K(X)$ as the subring

$$\mathcal{O}_{X,\eta_W} \ = \ \{\, h \in K(X) \mid \mathrm{ord}_W(h) \geq 0 \,\} \cup \{0\}.$$

The rational map f is defined at η_W if and only if, for any affine open $V = \operatorname{Spec} A \subseteq Y$ around the image $\overline{f(\eta_W)}$, the pullback $\varphi_U^*\colon A \to K(X)$ lands in $\mathcal{O}_{X,\eta_W}$: this is the standard local characterisation of where a rational map is regular at a codim-1 point of a smooth variety (it is the "regular = no pole" criterion). By the displayed identity, the latter is equivalent to $\mathrm{ord}_W(\varphi_U^*(g)) \geq 0$ for every $g \in A$, which is condition (2). The equivalence (1) $\Leftrightarrow$ (2) is therefore established.

For the consequence, f is codim-1-indeterminacy-free iff no prime divisor W is contained in $Z(f)$, which by the equivalence above is iff the $\mathrm{ord}_W \geq 0$ condition holds for every prime divisor W. $\square$

Remark 1714. Role of the valuation form. In 31, the abstract codimension statement 1705 suffices and 1713 is not needed. The Weil-divisor form instead isolates an equivalent local criterion for regularity at every prime divisor; this criterion is also useful for the symmetric-product arguments of 32.

Lemma 1715. *[Definedness at a prime divisor, repackaged as a through-the-point partial map] Let X and Y be varieties over $\bar{k}$ and let $f\colon X \dashrightarrow Y$ be a rational map. For a prime divisor $W \subseteq X$ with generic point $\eta_W = W.\mathrm{point}$, the following are equivalent:*

1. *$\eta_W \in \mathrm{Dom}(f)$ (f is defined at the generic point of W).*
2. *There exists a partial-map representative (U, φ_U) of f — i.e. a regular morphism $\varphi_U\colon U \to Y$ on a dense open $U \subseteq X$ whose induced rational map equals f — whose domain U contains η_W.*

Proof. By definition, $\eta_W \in \mathrm{Dom}(f)$ precisely when some representative of the rational map is defined on an open neighbourhood of η_W. Such a representative gives (2), and conversely the domain of the representative in (2) witnesses membership in $\mathrm{Dom}(f)$. □

Remark 1716. Relationship to 1713. 1715 captures the definitional half of the equivalence: a partial-map representative exists through the generic point of the prime divisor. The additional mathematical input in 1713 is the identification of the local ring with the nonnegative part of the associated discrete valuation, applied to the pullbacks of regular functions on an affine neighbourhood of the image.

28.7 The codimension-one method

The results above form a short chain of local and global arguments:

1. Smoothness makes every codimension-one local ring a discrete valuation ring (1685).
2. Properness and the valuative criterion then rule out codimension-one indeterminacy (1705).
3. For a group-variety target, the difference map and its pole divisors force every nonempty indeterminacy locus to be pure of codimension one (1711).
4. For a prime divisor, regularity is equivalently the absence of negative order in every pulled-back affine coordinate (1713).

Thus a rational map from a nonsingular variety to a complete group variety has empty indeterminacy locus: it has no codimension-one component by (2), whereas (3) says that every nonempty indeterminacy locus has such a component.

28.8 Relations with neighbouring results

The regular-implies-Cohen–Macaulay theorem 1765, obtained from Auslander–Buchsbaum in 30, gives a depth-theoretic description of the same regular local rings. The valuative proof of 1705 does not require that stronger description.

Combining 1705 and 1711 gives Milne's Theorem 3.2, 1766. Its later rigidity consequences use the group structure after the extension has been obtained.

The divisor argument needs only the decomposition

$$\div(h) = \div(h)_0 - \div(h)_\infty$$

and the codimension-one support of principal divisors. It does not use the degree of a divisor on a curve or intersection numbers on a surface. Cartier divisors could be substituted for Weil divisors on a smooth variety, where the two theories agree.

The algebraically closed base guarantees that closed points have residue field $\bar{k}$. Over a general field, the local statements persist after accounting for residue-field extensions, while descent of the resulting global morphism is a separate question.

Chapter 29

Coheight–Krull dim bridge for scheme points

29.1 Setup and motivation

A point z of a scheme X has two natural "dimension" invariants attached to it. The *Krull dimension of the local ring at* z, written $\dim \mathcal{O}_{X,z}$, is an algebraic quantity: the supremum of the lengths of chains of prime ideals in the stalk $\mathcal{O}_{X,z}$. The *coheight* of z in X, written $\mathrm{coheight}_X(z)$, is a topological quantity attached to the specialisation preorder on the underlying topological space of X: the supremum of the lengths of strict-generisation chains $z = a_0 \prec a_1 \prec \cdots \prec a_n$ in X, where $\prec$ is the strict specialisation order $x \prec y \iff y \rightsquigarrow x \wedge y \neq x$.

These two invariants agree:

$$\dim \mathcal{O}_{X,z} \;=\; \mathrm{coheight}_X(z),$$

which is the headline theorem 1719 of this chapter. The identification is folklore in algebraic geometry — it is the statement that the dimension of the local ring at z counts irreducible closed subsets of X containing z — but at the project's pinned Mathlib commit (`b80f227`) the equation is not packaged as a single declaration and must be assembled project-side from four Mathlib API pieces.

Why the bridge is needed. Two downstream files in the project currently work around the missing bridge by threading explicit `[Ring.KrullDimLE 1 (X.presheaf.stalk _)]` instance arguments at every codim-1 use site:

- 28 carries an internal conjunction

  $$\mathrm{hreg\_dim} : \mathrm{IsRegularLocalRing}\ \mathcal{O}_{X,z} \;\wedge\; \mathrm{ringKrullDim}(\mathcal{O}_{X,z}) = 1$$

  inside the private helper `smooth_codim_one_maximalIdeal_isPrincipal_and_ne_bot`; the Krull-dim conjunct is supplied by *this chapter's* 1720, leaving only the irreducible `IsRegularLocalRing` half (which is the separate Stacks tag `00TT` sub-project, out of scope here).

- 33's order-at-a-prime-divisor map `Scheme.RationalMap.order` currently carries an explicit `[Ring.KrullDimLE 1 (X.presheaf.stalk Y.point)]` instance argument; after 1720 lands, the instance is synthesisable from the codim-1 hypothesis on Y alone, so the explicit argument can be dropped.

Chapter structure. 29.2 proves the topology-only lemma that coheight is preserved by open immersions (1717). 29.3 identifies coheight on a Spec with the height of the underlying prime ideal (1718). 29.4 assembles the bridge 1719. The closing 29.5 packages the codim-1 specialisation 1720 that downstream files consume.

29.2 Coheight is preserved by open immersions

Lemma 1717 (Coheight on opens equals coheight in the ambient space)**.** *Let X be a topological space carrying the specialisation preorder, and let $U \subseteq X$ be an open subset. For any $z \in U$,*

$$\mathrm{coheight}_X(z) \;=\; \mathrm{coheight}_U(\langle z,\, z \in U\rangle),$$

where on the right U carries the subspace topology and $\langle z,\, z \in U\rangle$ denotes z viewed as a point of U.

Proof. Both sides are by definition the supremum, over natural numbers n, of the set of n for which there exists a strictly increasing chain $z = a_0 \prec a_1 \prec \cdots \prec a_n$ in the relevant specialisation preorder (with $\prec$ denoting strict specialisation). It therefore suffices to give an order-preserving bijection between the strict-generisation chains starting at z in X and those starting at the corresponding point of U.

Chains in U embed into X. The inclusion $U \hookrightarrow X$ is continuous, so it preserves specialisation: if $y \rightsquigarrow x$ in U then $y \rightsquigarrow x$ in X, and distinct points of U remain distinct in X. Hence every strict-generisation chain in U starting at z lifts to one in X of the same length.

Chains in X starting at z all lie in U. Conversely, suppose $z = a_0 \prec a_1 \prec \cdots \prec a_n$ is a strict-generisation chain in X with $a_0 = z \in U$. For each $i \geq 0$, the relation $a_i \succeq z$ (in the specialisation preorder) is $z \rightsquigarrow a_i$, i.e. a_i is in the closure of $\{z\}$ *viewed from above*; in our convention the strict order $a \prec b$ records $b \rightsquigarrow a$ (the generic point is the strict maximum), so $a_i \succeq z$ in X means $a_i \rightsquigarrow z$ in X. By the Mathlib lemma `Topology.Inseparable.Specializes.mem_open` (any specialisation $a \rightsquigarrow z$ with z in an open set U forces $a \in U$, since U is open and contains z, hence contains every point whose closure contains z — equivalently every a with $a \rightsquigarrow z$), every a_i lies in U. Thus the chain descends to a strict-generisation chain $\langle z\rangle = \langle a_0\rangle \prec \langle a_1\rangle \prec \cdots \prec \langle a_n\rangle$ in U of the same length.

These two operations are mutually inverse on the set of strict-generisation chains starting at z (resp. at $\langle z,\, z \in U\rangle$) and preserve length, so the suprema agree. ☐

29.3 Coheight on Spec equals height in PrimeSpectrum

Lemma 1718 (Coheight on $\mathrm{Spec}\,R$ equals height in $\mathrm{PrimeSpectrum}\,R$)**.** *Let R be a commutative ring and let $p \in \mathrm{Spec}\,R$ (viewed as a point of the scheme spectrum, with the specialisation preorder). Then*

$$\mathrm{coheight}_{\mathrm{Spec}\,R}(p) \;=\; \mathrm{height}_{\mathrm{PrimeSpectrum}(R)}(p),$$

where the right-hand side is the supremum of the lengths of strict prime-ideal chains $\mathfrak{p}_0 \subsetneq \cdots \subsetneq \mathfrak{p}$ ending at the prime $\mathfrak{p}$ corresponding to p.

Proof. The specialisation preorder on $\mathrm{Spec}\,R$ is the dual of the natural inclusion preorder on $\mathrm{PrimeSpectrum}(R)$. Concretely, for $p, q \in \mathrm{Spec}\,R$ with underlying primes $\mathfrak{p}, \mathfrak{q}$, the relation $p \leq q$ in the specialisation preorder (which is $q \rightsquigarrow p$ in topological notation) corresponds to the prime containment $\mathfrak{q} \subseteq \mathfrak{p}$. This is the content of the Mathlib lemma `spec_le_iff` (located in

`Mathlib.AlgebraicGeometry.AffineSpace`, lines 438–447 at commit `b80f227`, with the explicit remark that the preorder on $\operatorname{Spec} R$ equals the order-dual of the preorder on PrimeSpectrum R).

Under this order anti-isomorphism, coheight of p in $\operatorname{Spec} R$ (= supremum of strict-generisation chain lengths starting at p, i.e. ascending chains $p \prec a_1 \prec \cdots \prec a_n$ in the specialisation preorder) translates to height of $\mathfrak{p}$ in PrimeSpectrum(R) (= supremum of strict descending prime-ideal chains $\mathfrak{p} \supsetneq \mathfrak{q}_1 \supsetneq \cdots \supsetneq \mathfrak{q}_n$, equivalently strict ascending chains $\mathfrak{q}_n \subsetneq \cdots \subsetneq \mathfrak{q}_1 \subsetneq \mathfrak{p}$). The transport is mechanised by the Mathlib lemmas `Order.coheight_orderIso` (coheight is invariant under order isomorphisms) and `Order.height_toDual` (height in the order-dual equals coheight in the original order), applied to the order-dual identification $\operatorname{Spec} R \simeq_o (\text{PrimeSpectrum}\, R)^{\text{op}}$ furnished by `spec_le_iff`. $\square$

29.4 The coheight = Krull-dim-of-stalk bridge

Theorem 1719 (Coheight-to-Krull-dim bridge for a scheme point)**.**

Let X be a scheme and let $z \in X$ be any point. Then

$$\text{ringKrullDim}(\mathcal{O}_{X,z}) \;=\; \text{coheight}_X(z),$$

i.e. the Krull dimension of the stalk at z equals the coheight of z in the underlying topological space of X (with respect to the specialisation preorder).

Proof. The proof proceeds in five steps, assembling Mathlib API rather than arguing from a single textbook reference.

Step 1: pick an affine open neighbourhood of z. By `AlgebraicGeometry.exists_isAffineOpen_mem_and_subset` applied to the open set X itself, there exists an affine open $U \subseteq X$ with $z \in U$. Write $\langle z, z \in U\rangle \in U$ for z viewed as a point of the subscheme U.

Step 2: name the prime corresponding to z inside U. Since $U \cong \operatorname{Spec}\Gamma(U, \mathcal{O}_X)$ via the affine equivalence $U \simeq \operatorname{Spec} R$ (with $R := \Gamma(U, \mathcal{O}_X)$), the point $\langle z, z \in U\rangle$ corresponds to a prime ideal $\mathfrak{p} \subseteq R$. In Mathlib this prime is named `IsAffineOpen.primeIdealOf` $\langle z, z \in U\rangle$.

Step 3: the stalk is the localisation of $\Gamma(U)$ at $\mathfrak{p}$. The classical affine-stalk identification says $\mathcal{O}_{X,z} \cong R_{\mathfrak{p}}$. In Mathlib this is the instance `IsAffineOpen.isLocalization_stalk`, which witnesses $\mathcal{O}_{X,z}$ as a localisation of R at $\mathfrak{p}$ in the `IsLocalization.AtPrime` sense.

Step 4: Krull dim of the localisation equals the height of $\mathfrak{p}$. The Mathlib lemma `IsLocalization.AtPrime.ringKru` gives

$$\text{ringKrullDim}(\mathcal{O}_{X,z}) \;=\; \text{ringKrullDim}(R_{\mathfrak{p}}) \;=\; \text{height}(\mathfrak{p}),$$

where $\text{height}(\mathfrak{p})$ is the prime-ideal height in PrimeSpectrum(R). By `Ideal.height_eq_primeHeight` this matches the $\text{height}_{\text{PrimeSpectrum}(R)}$ used in 1718.

Step 5: lift back from U to X. By 1718 applied to R and the affine identification $U \simeq \operatorname{Spec} R$,

$$\text{height}_{\text{PrimeSpectrum}(R)}(\mathfrak{p}) \;=\; \text{coheight}_{\operatorname{Spec} R}(\langle\text{image of } z\rangle) \;=\; \text{coheight}_U(\langle z, z \in U\rangle).$$

Then 1717 applied to the open $U \subseteq X$ promotes the right-hand side back to $\text{coheight}_X(z)$. Concatenating the equalities from Steps 4–5 gives the conclusion. $\square$

29.5 Codim-1 specialisation: synthesising KrullDimLE 1

Lemma 1720 (Coheight = 1 forces stalk Krull dim ≤ 1)**.**

Let X be a scheme and let $z \in X$ be a point with $\text{coheight}_X(z) = 1$. *Then the stalk $\mathcal{O}_{X,z}$ has Krull dimension at most 1, equivalently* Ring.KrullDimLE 1 $\mathcal{O}_{X,z}$ *holds.*

Proof. By 1719, $\operatorname{ringKrullDim}(\mathcal{O}_{X,z}) = \operatorname{coheight}_X(z) = 1$. The Mathlib predicate $\operatorname{Ring.KrullDimLE} 1\, R$ unfolds to $\operatorname{ringKrullDim}(R) \leq 1$, so the inequality $1 \leq 1$ discharges the goal. □

29.6 Consumer wiring and the Stacks 00TT carve-out

Consumer 1: the `hreg_dim` refactor in 28. The private helper `smooth_codim_one_maximalIdeal_isPrincipal_` inside that chapter's Lean target currently carries an internal `sorry` of the form

$$\text{hreg\_dim} : \operatorname{IsRegularLocalRing} \mathcal{O}_{X,z} \;\wedge\; \operatorname{ringKrullDim} \mathcal{O}_{X,z} = 1.$$

After this chapter's 1720 (and its underlying 1719) land, the conjunction splits into two independent goals:

- The right conjunct $\operatorname{ringKrullDim}(\mathcal{O}_{X,z}) = 1$ is discharged by $\operatorname{coheight}_X(z) = 1$ (the standing codim-1 hypothesis on z) plus 1719; the antisymmetric pair $\operatorname{ringKrullDim} \leq 1$ (1720) and $\operatorname{ringKrullDim} \geq 1$ (also from the bridge, since $1 \leq 1$) gives equality.
- The left conjunct $\operatorname{IsRegularLocalRing} \mathcal{O}_{X,z}$ is the genuinely-irreducible *Stacks tag 00TT* statement "smooth ring map over a field forces regular stalks". It is a separate, out-of-scope sub-project (Mathlib does *not* ship the implication `Algebra.Smooth k A` → `IsRegularLocalRing` $A_{\mathfrak{p}}$ at `b80f227`; the project-side build is ∼ 200–300 LOC of cotangent-space and pure-dimension machinery, not packaged in this chapter).

Thus the present chapter *halves* the `hreg_dim` obligation in 28: one conjunct closes via the bridge, one remains as a named Mathlib gap.

Consumer 2: the `Scheme.RationalMap.order` threading hygiene in 33. The order-at-a-prime-divisor map currently carries an explicit instance argument

$$[\operatorname{Ring.KrullDimLE} 1\, (X.\text{presheaf.stalk}\, Y.\text{point})]$$

at every use site. Because Y is a prime divisor of X (so $\operatorname{coheight}_X(Y.\text{point}) = 1$ by the definition of a prime divisor as a codim-1 irreducible closed subset), 1720 synthesises the instance automatically from the codim-1 hypothesis on Y, so the explicit instance argument can be dropped from `Scheme.RationalMap.order`'s signature.

Consumer 3: future codim-1 infrastructure. Any subsequent project lemma that needs to discharge "the stalk at a codim-1 point has Krull dim ≤ 1" — the implicit hypothesis behind every DVR-at-codim-1 argument (cf. 1685 of 28) — now has a one-liner consequence. The instance form 1720 is the right shape for typeclass-search to find it without ceremony.

29.7 Lean encoding

The Lean target file is the new `AlgebraicJacobian/Albanese/CoheightBridge.lean`. The declarations to introduce, in order:

1. `Order.coheight_eq_of_isOpenEmbedding` — 1717. Signature (informal): for X a topological space, $U \subseteq X$ open, and $z \in U$, $\operatorname{coheight}_X(z) = \operatorname{coheight}_U(\langle z, hz\rangle)$. Tools: `Topology.Inseparable.Special` for the chain-lifting direction; `Subtype.val` continuity for the embedding direction. Estimated ∼ 20–30 LOC.

2. `Order.coheight_spec_eq_height_primeSpectrum` — 1718. Signature (informal): for R a `CommRingCat` and p : Spec R a point, coheight of p in Spec R equals height in PrimeSpectrum R. Tools: `spec_le_iff` (`Mathlib.AlgebraicGeometry.AffineSpace` L438–447), `Order.coheight_orderIso`, `Order.height_toDual` (`Mathlib.Order.KrullDimension`). Estimated $\sim$ 10–15 LOC.

3. `AlgebraicGeometry.Scheme.ringKrullDim_stalk_eq_coheight` — 1719. Signature (informal): for X : `Scheme` and z : X, ringKrullDim(X.presheaf.stalk z) = coheight z. Five-step assembly per the proof above. Tools: `exists_isAffineOpen_mem_and_subset`, `IsAffineOpen.primeIdeal` `IsAffineOpen.isLocalization_stalk`, `IsLocalization.AtPrime.ringKrullDim_eq_height`, `Ideal.height_eq_primeHeight`, plus the two lemmas above. Estimated $\sim$ 30–50 LOC.

4. `AlgebraicGeometry.Scheme.ringKrullDimLE_of_coheight_eq_one` — 1720. Signature (informal): given X : `Scheme`, z : X, and coheight $z = 1$, derive `Ring.KrullDimLE 1 (X.presheaf.stalk z)`. Trivial corollary of (3). Estimated $\sim$ 10 LOC. Lemma vs. instance phrasing: the `coheight z = 1` hypothesis is not a typeclass, so this lands as a plain lemma. Downstream files that have the codim-1 hypothesis in hand invoke it explicitly; downstream typeclass infrastructure that bundles "z is a codim-1 point" into a typeclass (if any is added later) can re-export as an instance.

Mathlib readiness audit. At commit `b80f227` the prover should verify:

- `Order.coheight`, `Order.height`, `Order.coheight_orderIso`, `Order.height_toDual` all live in `Mathlib.Order.KrullDimension` and are imported transitively by `Mathlib.AlgebraicGeometry.Stalk.`
- `Topology.Inseparable.Specializes.mem_open` lives in `Mathlib.Topology.Inseparable`; the file already imports it via `Mathlib.Topology.Defs.Filter`.
- `IsAffineOpen.isLocalization_stalk` and `IsAffineOpen.primeIdealOf` live in `Mathlib.AlgebraicGeometry` (lines 806–813 and surrounding, at the pinned commit).
- `IsLocalization.AtPrime.ringKrullDim_eq_height` and `Ideal.height_eq_primeHeight` live in `Mathlib.RingTheory.Ideal.Height` (lines 341–348 and 45 at the pinned commit).
- `Ring.KrullDimLE` is defined and unfolds to ringKrullDim(R) $\leq n$ at `Mathlib.RingTheory.KrullDimension.Ba`

No new core axiom is required. The chapter introduces no new macro.

29.8 Out of scope

This chapter packages *only* the coheight-to-Krull-dim bridge and its codim-1 specialisation. The following adjacent content is explicitly out of scope:

- **Stacks tag `00TT` (smooth $\Rightarrow$ regular stalks).** This is a separate sub-project of $\sim$ 200–300 LOC, tracked in `analogies/stacks-00tt-coheight.md` Decision 1. The directive flags it as iter-200+ work. The natural project landing site, when attacked, is a standalone file `AlgebraicJacobian/Albanese/SmoothToRegular.lean` (not part of this chapter).
- **The codim-1 valuative criterion / Milne Lemma 3.3 difference-map construction.** This is the second multi-iter sub-project flagged in `analogies/stacks-00tt-coheight.md` Decision 3 ($\sim$ 300–500 LOC of project-side new infrastructure); see 1711 in 28 for the chapter that owns it.

- **Refactoring `Scheme.RationalMap.order` to drop its explicit `[Ring.KrullDimLE 1 _]` instance argument.** This is a downstream consequence of 1720 that lives in 33; the present chapter only *enables* the refactor, it does not perform it.
- **Mathlib upstream PRs.** The two lemmas 1717 and 1719 are natural candidates for upstreaming (their natural Mathlib homes are `Mathlib.Order.KrullDimension` and a new `Mathlib.AlgebraicGeometry.Coheight` respectively); the chapter flags this as an observation but does not pursue it.
- **General codim-n specialisations.** Only the codim-1 instance is packaged here, because that is the consumer-facing case. A general `ringKrullDimLE_of_coheight_le` or `ringKrullDimLE_of_coheight_eq` family of lemmas can be added iteratively if downstream demand emerges, but is not required for any A.4.* sub-row at the time of writing.

Chapter 30

Auslander–Buchsbaum

30.1 Setup and motivation

The Auslander–Buchsbaum formula relates the projective dimension of a finite module over a Noetherian local ring to its depth. For a regular local ring, a regular system of parameters also shows directly that the ring is Cohen–Macaulay. On a regular surface this gives depth two at every closed point and hence the familiar Hartogs extension theorem for regular functions across such a point.

This local-algebra consequence must be distinguished from extension of rational maps to a complete target: depth two alone does not imply that a rational map extends. In particular, the Albanese rational-map extension used later follows the separate codimension-one and group-scheme indeterminacy argument; it does not depend on the Cohen–Macaulay corollary proved here.

This chapter packages:

1. the regular-sequence definition of depth and its Ext reformulation (1721 and 1722),
2. the projective-dimension invariant of a finitely-generated module (1723),
3. the depth-on-a-short-exact-sequence lemma (1724),
4. the Auslander–Buchsbaum formula proper (1725), and
5. the corollary regular ⇒ Cohen–Macaulay (1765).

A short closing application (30.8) records the Hartogs consequence for regular functions and its precise separation from the Albanese extension argument.

30.2 Depth of a module over a local ring

Definition 1721 (Depth of a module)**.** *Source: [Stacks Project], tag 00LF (definition-depth).* Let R be a commutative ring, let $I \subseteq R$ be an ideal, and let M be a finitely generated R-module. The I-*depth* of M, written $\operatorname{depth}_I(M)$, is the supremum in $\{0, 1, 2, \dots, \infty\}$ of the lengths of M-regular sequences contained in I, provided $IM \neq M$; if $IM = M$ we set $\operatorname{depth}_I(M) = \infty$. When $(R, \mathfrak{m})$ is a Noetherian local ring we abbreviate $\operatorname{depth}(M) := \operatorname{depth}_{\mathfrak{m}}(M)$ and call this simply the *depth* of M. The depth of the ring R itself is $\operatorname{depth}(R)$, the depth of R as a module over itself.

Recall that a sequence $f_1, \dots, f_r \in R$ is called *M-regular* if $M/(f_1, \dots, f_r)M \neq 0$ and for each i, the element f_i is a non-zero-divisor on $M/(f_1, \dots, f_{i-1})M$. The empty sequence is regular on M iff $M \neq 0$ (it is *not* regular on the zero module, but the convention $\mathrm{depth}(0) = \infty$ above is the usual one).

The two practical computations of depth are the regular-sequence definition (used to establish a *lower* bound) and the Ext characterisation (used to establish an *upper* bound and to prove the depth-on-a-short-exact-sequence inequality).

Lemma 1722 (Depth via Ext)**.**

Source: [Stacks Project], tag 00LP (lemma-depth-ext). Let $(R, \mathfrak{m})$ be a Noetherian local ring with residue field $\kappa = R/\mathfrak{m}$ and let M be a nonzero finitely generated R-module. Then

$$depth(M) \;=\; \inf\{\, i \in \mathbb{N} \;\big|\; Ext^i_R(\kappa, M) \neq 0 \,\}.$$

Proof. Write $\delta := \mathrm{depth}(M)$ and $i := \inf\{j : \mathrm{Ext}^j_R(\kappa, M) \neq 0\}$. The base case $\delta = 0$ is exactly $\mathrm{Hom}_R(\kappa, M) \neq 0$ (some element of M is annihilated by $\mathfrak{m}$, i.e. $\mathfrak{m} \in \mathrm{Ass}(M)$, i.e. $\mathfrak{m}$ contains a zero-divisor on M), which is $i = 0$. For the inductive step pick $x \in \mathfrak{m}$ a non-zero-divisor on M with $\mathrm{depth}(M/xM) = \delta - 1$. The short exact sequence $0 \to M \xrightarrow{x} M \to M/xM \to 0$ induces a long exact sequence of $\mathrm{Ext}^*_R(\kappa, -)$. Multiplication by x annihilates each $\mathrm{Ext}^j_R(\kappa, M)$ (because $x \in \mathfrak{m}$ and κ is annihilated by $\mathfrak{m}$), so the long exact sequence breaks into short pieces giving $\mathrm{Ext}^{j-1}_R(\kappa, M/xM) \cong \mathrm{Ext}^j_R(\kappa, M)$ at the relevant indices, and the smallest non-vanishing index drops by 1. By induction $i(M/xM) = i - 1 = \delta - 1$, so $i = \delta$. □

30.3 Projective dimension

Definition 1723 (Projective dimension)**.** Let R be a commutative ring and let M be an R-module. The *projective dimension* of M, written $\mathrm{pd}_R(M)$, is the infimum in $\{0, 1, 2, \dots, \infty\}$ of the lengths n of projective resolutions

$$0 \longrightarrow P_n \longrightarrow P_{n-1} \longrightarrow \cdots \longrightarrow P_1 \longrightarrow P_0 \longrightarrow M \longrightarrow 0$$

with each P_i a projective R-module. We say M has *finite projective dimension* if $\mathrm{pd}_R(M) < \infty$, and *infinite projective dimension* otherwise. This chapter adopts the convention $\mathrm{pd}_R(0) = 0$, represented by the empty resolution $0 \to 0$; some references instead use $-\infty$. The Auslander–Buchsbaum formula in 1725 concerns only nonzero M. In the local Noetherian setting a finitely generated R-module of finite projective dimension admits a *minimal finite free resolution*: a resolution with each P_i free of finite rank and each transition map $P_{i+1} \to P_i$ landing in $\mathfrak{m}P_i$.

The two basic facts about $\mathrm{pd}_R(M)$ needed downstream are:

- $\mathrm{pd}_R(M) = 0$ iff M is projective (in the local Noetherian setting, iff M is free);
- if $0 \to K \to P \to M \to 0$ is a short exact sequence with P projective, then $\mathrm{pd}_R(M) = \mathrm{pd}_R(K) + 1$ provided $K \neq 0$.

Both statements follow by splitting or splicing finite projective resolutions.

30.4 Depth on a short exact sequence

Lemma 1724 (Depth on a short exact sequence)**.**

Source: [Stacks Project], tag 00LE (lemma-depth-in-ses). Let $(R, \mathfrak{m})$ be a Noetherian local ring and let $0 \to N' \to N \to N'' \to 0$ be a short exact sequence of nonzero finitely generated R-modules. Then

1. *depth*$(N) \geq \min(\mathit{depth}(N'),\ \mathit{depth}(N''))$;
2. *depth*$(N'') \geq \min(\mathit{depth}(N),\ \mathit{depth}(N') - 1)$;
3. *depth*$(N') \geq \min(\mathit{depth}(N),\ \mathit{depth}(N'') + 1)$.

Proof. By 1722, $\operatorname{depth}(N')$ (resp. $\operatorname{depth}(N)$, $\operatorname{depth}(N'')$) is the smallest index at which $\operatorname{Ext}^*_R(\kappa, -)$ is nonzero on N' (resp. N, N''). Apply the long exact $\operatorname{Ext}^*_R(\kappa, -)$ sequence to $0 \to N' \to N \to N'' \to 0$:

$$\cdots \to \operatorname{Ext}^{i-1}_R(\kappa, N'') \to \operatorname{Ext}^i_R(\kappa, N') \to \operatorname{Ext}^i_R(\kappa, N) \to \operatorname{Ext}^i_R(\kappa, N'') \to \operatorname{Ext}^{i+1}_R(\kappa, N') \to \cdots$$

Each of the three inequalities is then a direct read-off of the fact that if two of the three modules in a length-3 chunk vanish then the middle vanishes (for (1)) / the boundary terms vanish (for (2),(3)). Compare with the diagonal vanishing argument in the proof of 1722. ☐

30.5 The Auslander–Buchsbaum formula

Theorem 1725 (Auslander–Buchsbaum formula)**.** *Source: [Stacks Project], tag 090V (proposition-Auslander-Buchsbaum); cf. Matsumura,* Commutative Ring Theory*, Theorem 19.1; Auslander–Buchsbaum,* Homological dimension in local rings*, 1957. Let $(R, \mathfrak{m})$ be a Noetherian local ring and let M be a nonzero finitely generated R-module of finite projective dimension. Then*

$$\mathit{pd}_R(M) \ +\ \mathit{depth}(M) \ =\ \mathit{depth}(R).$$

Proof. We induct on $\operatorname{depth}(M)$.

Base case depth$(M) = 0$. Let $e := \operatorname{pd}_R(M)$ and pick a minimal finite free resolution

$$0 \longrightarrow R^{n_e} \longrightarrow R^{n_{e-1}} \longrightarrow \cdots \longrightarrow R^{n_0} \longrightarrow M \longrightarrow 0$$

with all transition maps having matrix coefficients in $\mathfrak{m}$ (this is the “minimal-resolution-after-trimming-trivial-summands” normalisation, cf. Stacks `lemma-add-trivial-complex`). On the one hand, the “what is exact” criterion (Stacks tag 00MF, `proposition-what-exact`, expressing exactness of a sequence of finite free modules in terms of depths of ideals of r-minors) gives $\operatorname{depth}(R) \geq e$. On the other hand, splitting the resolution into short exact sequences

$$0 \to R^{n_e} \to R^{n_{e-1}} \to K_{e-2} \to 0, \quad 0 \to K_{e-2} \to R^{n_{e-2}} \to K_{e-3} \to 0, \quad \ldots, \quad 0 \to K_0 \to R^{n_0} \to M \to 0,$$

and applying 1724 iteratively, yields the chain of inequalities $\operatorname{depth}(K_{e-2}) \geq \operatorname{depth}(R) - 1$, $\operatorname{depth}(K_{e-3}) \geq \operatorname{depth}(R) - 2$, ..., $\operatorname{depth}(K_0) \geq \operatorname{depth}(R) - (e-1)$, and finally $\operatorname{depth}(M) \geq \operatorname{depth}(R) - e$. Since $\operatorname{depth}(M) = 0$ we deduce $\operatorname{depth}(R) \leq e$. Combined with the previous inequality, $\operatorname{depth}(R) = e = \operatorname{pd}_R(M) + 0 = \operatorname{pd}_R(M) + \operatorname{depth}(M)$.

Inductive step depth$(M) > 0$. We claim there exists $x \in \mathfrak{m}$ which is a non-zero-divisor on both R and M. Indeed: either $\operatorname{pd}_R(M) > 0$, in which case $\operatorname{depth}(R) > 0$ by the “what is exact” criterion applied to the start of a minimal resolution, or $\operatorname{pd}_R(M) = 0$ in which case M is finite free and $\operatorname{depth}(R) = \operatorname{depth}(M) > 0$. Either way both $\operatorname{depth}(R) > 0$ and $\operatorname{depth}(M) > 0$, hence $\operatorname{depth}(R \oplus M) > 0$ by 1724, so a common non-zero-divisor $x \in \mathfrak{m}$ exists. Take a minimal finite free resolution of M as above; the snake lemma applied to multiplication by x gives a minimal finite free resolution

$$0 \to (R/xR)^{n_e} \to (R/xR)^{n_{e-1}} \to \cdots \to (R/xR)^{n_0} \to M/xM \to 0$$

of M/xM over R/xR, of the same length e. So $\mathrm{pd}_{R/xR}(M/xM) = \mathrm{pd}_R(M)$. By Stacks `lemma-depth-drops-by-one`, both $\mathrm{depth}(R/xR) = \mathrm{depth}(R) - 1$ and $\mathrm{depth}(M/xM) = \mathrm{depth}(M) - 1$, and one verifies $\mathrm{depth}_R(M/xM) = \mathrm{depth}_{R/xR}(M/xM)$ (regular sequences in $\mathfrak{m}$ correspond to regular sequences in $\mathfrak{m}/(x)$). Apply the inductive hypothesis to M/xM over R/xR:

$$\mathrm{pd}_{R/xR}(M/xM) + \mathrm{depth}(M/xM) = \mathrm{depth}(R/xR).$$

Substituting yields $\mathrm{pd}_R(M) + (\mathrm{depth}(M) - 1) = \mathrm{depth}(R) - 1$, i.e. $\mathrm{pd}_R(M) + \mathrm{depth}(M) = \mathrm{depth}(R)$, as required. ☐

Lemma 1726 (Auslander–Buchsbaum: inductive step pd $= k + 1$)**.** *Source: [Stacks Project], tag 090V (proposition-Auslander-Buchsbaum), induction-on-pd re-organisation; cf. Matsumura, Theorem 19.1.*

Let $(R, \mathfrak{m})$ be a Noetherian local ring, let M be a nonzero finite R-module of projective dimension $k + 1$, and assume the Auslander–Buchsbaum formula for modules of projective dimension at most k. Then $\mathrm{pd}_R(M) + \mathrm{depth}(M) = \mathrm{depth}(R)$.

Proof. Choose a minimal surjection $f : R^n \twoheadrightarrow M$, and write $K = \ker f$. The syzygy sequence

$$0 \longrightarrow K \longrightarrow R^n \xrightarrow{f} M \longrightarrow 0$$

gives $\mathrm{pd}_R(K) = k$ by 1745.

Suppose first that $k > 0$. By the inductive hypothesis, $k + \mathrm{depth}(K) = \mathrm{depth}(R)$, so $\mathrm{depth}(K) < \mathrm{depth}(R)$. The depth inequalities for the syzygy sequence then force $\mathrm{depth}(K) = \mathrm{depth}(M) + 1$. Substituting this equality into the inductive hypothesis gives the desired formula.

It remains to treat $k = 0$. Then K is a nonzero finite free module, hence $\mathrm{depth}(K) = \mathrm{depth}(R) = d$. Minimality of f means that every matrix entry of $K \to R^n$ belongs to $\mathfrak{m} = \mathrm{Ann}_R(R/\mathfrak{m})$. Therefore the induced map $\mathrm{Ext}^d_R(R/\mathfrak{m}, K) \to \mathrm{Ext}^d_R(R/\mathfrak{m}, R^n)$ is zero. The source is nonzero by the characterization of depth, so exactness of the long Ext sequence supplies a nonzero class in $\mathrm{Ext}^{d-1}_R(R/\mathfrak{m}, M)$. Thus $\mathrm{depth}(M) \le d - 1$, while the depth inequality for the same short exact sequence gives the reverse inequality. Hence $\mathrm{depth}(M) = d - 1$, and $1 + \mathrm{depth}(M) = \mathrm{depth}(R)$. ☐

Lemma 1727. *[Depth drops by one under a regular element] Source: [Stacks Project], tag 00LL (`lemma-depth-drops-by-one`).*

For a Noetherian local ring $(R, \mathfrak{m})$, a nonzero finite R-module M, and an M-regular element $x \in \mathfrak{m}$,

$$\mathrm{depth}_{\mathfrak{m}}(M/xM) + 1 \;=\; \mathrm{depth}_{\mathfrak{m}}(M).$$

Proof. Reduces to the Ext characterisation (1722) on both sides; the short exact sequence $0 \to M \xrightarrow{\cdot x} M \to M/xM \to 0$ gives a long exact sequence in $\mathrm{Ext}^*_R(\kappa, -)$ on which multiplication by x is zero (since $x \in \mathrm{Ann}_R \kappa$), breaking the LES into short exact pieces $0 \to \mathrm{Ext}^i(\kappa, M) \to \mathrm{Ext}^i(\kappa, M/xM) \to \mathrm{Ext}^{i+1}(\kappa, M) \to 0$. The translation to the $\mathbb{N}_\infty$ depth equality goes through `ENat.forall_natCast_le_iff_le`. ☐

30.5.1 The syzygy step in the induction

Let M have projective dimension $k+1$, and choose a minimal surjection $f : R^n \twoheadrightarrow M$ with kernel K. The short exact sequence

$$0 \longrightarrow K \longrightarrow R^n \xrightarrow{f} M \longrightarrow 0$$

reduces the projective dimension by one: $\mathrm{pd}_R(K) = k$. The depth inequalities for this sequence compare $\mathrm{depth}(K)$, $\mathrm{depth}(M)$, and $\mathrm{depth}(R^n) = \mathrm{depth}(R)$. For $k > 0$, the inductive hypothesis and these inequalities give $\mathrm{depth}(K) = \mathrm{depth}(M) + 1$. When $k = 0$, the minimality of f makes the matrix of $K \to R^n$ vanish after reduction modulo the maximal ideal; the induced map on $\mathrm{Ext}^*_R(R/\mathfrak{m}, -)$ is therefore zero, and the same depth equality follows from the long exact sequence. Substitution into the inductive hypothesis yields

$$(k + 1) + \mathrm{depth}(M) = \mathrm{depth}(R).$$

30.5.2 Minimal finite-free surjections

Lemma 1728 (Per-syzygy minimal surjection from R^n for finite modules over a local ring)**.** *Source: [Stacks Project], tag 00LK (`lemma-add-trivial-complex`). Let R be a local ring (Noetherian-ness NOT required at this step), $\mathfrak{m} \subseteq R$ the maximal ideal, $\kappa = R/\mathfrak{m}$ the residue field, and M a finite R-module. Then there exist $n \in \mathbb{N}$ and an R-linear surjection $f : R^n \twoheadrightarrow M$ such that:*

1. *$n = \dim_\kappa(\kappa \otimes_R M)$ (the minimal generator count of M, via Nakayama);*
2. *$\ker f \subseteq \mathfrak{m} \cdot R^n$ (the "minimality" condition: every entry of every relation lives in the maximal ideal, so the chosen lift has no trivial summands).*

Proof. Pick a κ-basis $b : \mathrm{Fin}\, n \to \kappa \otimes_R M$ via `Module.finBasis` with $n = \dim_\kappa(\kappa \otimes_R M)$. Surjectivity of the canonical map $M \twoheadrightarrow \kappa \otimes_R M$ (composition of `TensorProduct.mk_surjective` and `Ideal.Quotient.mk_surjective`) gives lifts $m_i \in M$ with $1 \otimes m_i = b_i$. Define $f : R^n \to M$ by $f x := \sum_i x_i \cdot m_i$ via `(Pi.basisFun R (Fin n)).constr R m`.

Surjectivity: $\mathrm{range}\, f = \mathrm{span}_R(m_i)$ by `Module.Basis.constr_range`, and $\mathrm{span}_R(m_i) = M$ by Nakayama via `IsLocalRing.span_eq_top_of_tmul_eq_basis`.

Kernel containment: for $x \in \ker f$ the relation $\sum_i x_i m_i = 0$ tensors to $\sum_i (1 \otimes x_i \cdot m_i) = 0$; using `TensorProduct.tmul_smul` and the linear independence of b (`Module.Basis.linearIndependent`) we obtain $\mathrm{residue}_R(x_i) = 0$ for every i, i.e. $x_i \in \mathfrak{m}$. Decomposing $x = \sum_i x_i \cdot \mathrm{Pi.single}\, i\, 1$ then yields $x \in \mathfrak{m} \cdot R^n$ by `Submodule.smul_mem_smul` and `Submodule.sum_mem`. ☐

The resulting short exact sequence is the syzygy sequence used in the induction on projective dimension.

30.5.3 Projective-dimension descent along a syzygy

The following lemmas express the two directions of projective-dimension control in the syzygy sequence and the corresponding lower bound on depth.

Lemma 1729 (Bridge: `projectiveDimension` $M = n \Rightarrow$ `HasProjectiveDimensionLT` $M\,(n{+}1)$)**.** *Source: Mathlib's `CategoryTheory.projectiveDimension_lt_iff`. For a Noetherian local ring R and a finite R-module M with `Module.projectiveDimension` $R\,M = ((n : \mathbb{N}) :$ `WithBot` $\mathbb{N}_\infty)$, one has `HasProjectiveDimensionLT`(`ModuleCat.of` $R\,M$) $(n + 1)$. This is the strict-inequality formulation of the assumed projective-dimension equality.*

Lemma 1730 (Syzygy descent of `HasProjectiveDimensionLT`)**.** *Source: Mathlib's* `ShortComplex.ShortExact.hasP` *+* `ModuleCat.projective_of_free`*. Let $f : R^n \twoheadrightarrow M$ be a surjection of R-modules with* `HasProjectiveDimensionLT`*(*`ModuleCat.of` *R M) $(k{+}2)$. Then* `HasProjectiveDimensionLT`*(*`ModuleCat.of` *R (ker f 1). Proof: build the SES $0 \to \ker f \to R^n \to M \to 0$ via* `LinearMap.shortExact_shortComplexKer`*; discharge projectivity of R^n via* `ModuleCat.projective_of_free`*(*`Pi.basisFun` *R (*`Fin` *n)); lift* `HasProjectiveDimensionLT` *1 on R^n up to $k+1$ via* `hasProjectiveDimensionLT_of_ge`*; apply* `ShortExact.hasProjectiveDimensionLT_X_1` *to close.*

Lemma 1731 (Ascent companion: ker f-pd-control $\Rightarrow$ M-pd-control)**.** *Companion ascent direction: with the same SES,* `HasProjectiveDimensionLT`*(ker f) $(k{+}1)$ gives* `HasProjectiveDimensionLT` *M $(k{+}2)$ via* `ShortExact.hasProjectiveDimensionLT_X_3`*. Together with 1730 this enables the contradiction argument that pins* `pd` *$\ker f = k$* exactly *(rather than merely $\leq k$) under* `pd` *$M = k+1$ exactly.*

Lemma 1732 (Depth lower bound on the kernel of a finite-free surjection)**.** *Source:* `depth_of_short_exact` *part (3) +* `depth_pi_const_eq_depth_of_nonempty`*. Let R be a Noetherian local ring, M a nonzero finite R-module, $n \geq 1$, and $f : R^n \twoheadrightarrow M$ a surjection with $\ker f$ nontrivial. Then* min(`depth` R, `depth` $M+1$) $\leq$ `depth`(ker f)*. Proof: assemble the SES $0 \to \ker f \to R^n \to M \to 0$ via (ker f).*`subtype`*; rewrite* `depth`*(R^n) =* `depth` *R via* `depth_pi_const_eq_depth_of_nonempty`*; apply* `depth_of_short_exact` *part (3) for the lower bound.*

Together, the preceding lemmas identify the projective dimension of the syzygy and supply the depth comparison used in 1726. The base case is completed by the Ext-annihilation lemmas recorded below.

30.6 Auxiliary homological lemmas

This section records the elementary Ext, depth, and regular-sequence lemmas used in the proofs above.

30.6.1 Ext-annihilation and depth-bound substrate

Lemma 1733 (Annihilator kills Ext)**.** *Let R be a commutative ring, let N, M be R-modules, fix $i \in \mathbb{N}$, and let $e \in Ext^i_R(N, M)$. If $x \in R$ annihilates N (i.e. $x \in Ann_R(N)$), then the R-action $x \cdot e = 0$. Concretely, $x \cdot e = (mk_0(x \cdot \mathbf{1}_N)) \circ e$, and $x \cdot \mathbf{1}_N = 0$ since x annihilates N.*

Proof. The scalar action of x on $\mathrm{Ext}^i_R(N, M)$ is induced by precomposition with multiplication by x on N. Since $xN = 0$, that endomorphism is zero, and functoriality sends it to the zero map. □

Lemma 1734 (Ext-vanishing below the depth)**.** *Let $(R, \mathfrak{m})$ be a Noetherian local ring with residue field κ, and let M be a nonzero finitely generated R-module. If $(i : \mathbb{N}_\infty) < depth(M)$, then every element of $Ext^i_R(\kappa, M)$ is zero. This is the index-wise repackaging of 1722 convenient for the long-exact-sequence chases.*

Proof. If $\mathrm{Ext}^i_R(\kappa, M)$ were nonzero, the characterization in 1722 would give $\mathrm{depth}(M) \leq i$, contrary to the hypothesis. □

Lemma 1735 ($\mathbb{N}_\infty$ successor bridge)**.** *For $d \in \mathbb{N}_\infty$ and $a \in \mathbb{N}$ with $1 \leq a$: if $(a : \mathbb{N}_\infty) \leq d - 1$ then $(a + 1 : \mathbb{N}_\infty) \leq d$. A truncated-subtraction arithmetic lemma used for the $depth(N') - 1$ shift in the second inequality of 1724.*

Proof. If $d = \infty$, the conclusion is immediate. Otherwise write $d = m$. Truncated subtraction agrees with ordinary subtraction under $1 \le a \le m-1$, so $a+1 \le m = d$. $\square$

Lemma 1736 (Depth is a linear-equivalence invariant)**.** *Let R be a commutative ring, $I \subseteq R$ an ideal, and M, M' two R-modules related by an R-linear equivalence $e : M \xrightarrow{\sim} M'$. Then $depth_I(M) = depth_I(M')$: both the side condition $I \cdot M = M$ and the regular-sequence supremum are preserved under e.*

Proof. The equivalence carries IM onto IM', so the exceptional case $IM = M$ is preserved. It also carries each regular sequence on M to a regular sequence of the same length on M', and conversely via the inverse equivalence. The two suprema are therefore equal. $\square$

30.6.2 Depth of a constant product module

Lemma 1737 (Ideal action on a constant product)**.** *For a commutative ring R, an ideal I, a finite index type ι, and an R-module M, the submodule $I \cdot \top_{\iota \to M}$ equals the product submodule $\prod_\iota (I \cdot \top_M)$ of fibrewise ideal actions.*

Proof. For one inclusion, take coordinates in a finite sum of elements am with $a \in I$; every coordinate lies in IM. Conversely, decompose a tuple into its finitely many coordinate vectors. Each vector lies in the ideal action because its unique nonzero coordinate does. $\square$

Lemma 1738 (Triviality of the ideal action on a constant product)**.** *For a nonempty finite index type ι: $I \cdot \top_{\iota \to M} = \top$ if and only if $I \cdot \top_M = \top$. The forward direction reads off the fibre witness from a Pi.single projection; the backward direction lifts a fibre witness through Pi.single.*

Proof. Apply 1737. If $IM = M$, the product equality follows coordinatewise. Conversely, evaluate the product equality at any index, which exists by nonemptiness, to obtain $IM = M$. $\square$

Definition 1739 (Quotient of a constant product by a scalar)**.** For a commutative ring R, a finite index type ι, a scalar $r \in R$, and an R-module M, the canonical R-linear equivalence

$$(\iota \to M)\big/ r \cdot \top \xrightarrow{\sim} \iota \to (M/r \cdot \top),$$

obtained by rewriting $r \cdot \top = \mathrm{span}\{r\} \cdot \top$ and applying the quotient-of-a-product identification.

Lemma 1740 (Regularity on a constant product)**.** *For a nonempty finite index type ι and a list rs of elements of R, the sequence rs is $(\iota \to M)$-regular if and only if it is M-regular. The proof inducts on rs: the empty case reduces to Nontrivial$(\iota \to M) \leftrightarrow$ Nontrivial(M), and the cons case combines pointwise SMul-regularity with 1739 to feed the inductive hypothesis.*

Proof. Induct on the sequence. Nontriviality of a nonempty finite product is equivalent to nontriviality of one factor. For the inductive step, scalar regularity is coordinatewise, and 1739 identifies the quotient product with the product of the quotients; apply the inductive hypothesis. $\square$

Lemma 1741 (Depth of a constant product)**.** *For a commutative ring R, an ideal I, an R-module M, and a nonempty finite index type ι, one has $depth_I(\iota \to M) = depth_I(M)$. In particular a finite free module $M \cong R^k$ has $depth(M) = depth(R)$, which closes the $pd_R(M) = 0$ base case of 1725.*

Proof. The exceptional condition in the definition of depth is equivalent by 1738. The regular sequences, with their lengths, agree by 1740. Hence their suprema agree. $\square$

30.6.3 Surjection and projective-dimension substrate for the inductive step

Lemma 1742 (A regular element exists when depth is positive)**.** *For a Noetherian local ring $(R, \mathfrak{m})$ and a nonzero finitely generated R-module M, if $1 \le depth(M)$ then there exists $x \in \mathfrak{m}$ that is M-regular. (Nakayama rules out the trivialisation $\mathfrak{m} \cdot M = M$, so the depth supremum exhibits a length-1 witness.)*

Proof. Nakayama gives $\mathfrak{m}M \neq M$. Since depth is at least one, its defining supremum contains a regular sequence of positive length. The first element of that sequence lies in $\mathfrak{m}$ and is M-regular. □

Lemma 1743 (Kernel-SES depth inequalities for a finite free surjection)**.** *Let R be a Noetherian local ring, M a nonzero finite R-module, $n \ge 1$, and $f : R^n \twoheadrightarrow M$ a surjection with $\ker f$ nontrivial. Then*

$$\min\big(depth(R),\ depth(\ker f){-}1\big) \le depth(M) \quad and \quad \min\big(depth(R),\ depth(M){+}1\big) \le depth(\ker f).$$

These are parts (2) and (3) of 1724 for the SES $0 \to \ker f \to R^n \to M \to 0$, after identifying $depth(R^n) = depth(R)$.

Proof. Apply parts (2) and (3) of 1724 to $0 \to \ker f \to R^n \to M \to 0$, and replace $\operatorname{depth}(R^n)$ by $\operatorname{depth}(R)$ using 1741. □

Lemma 1744 (A nonzero Ext class at the depth index)**.** *For a nonzero finite R-module M over a Noetherian local ring $(R, \mathfrak{m})$ with $depth(M) = D$ finite, there is a nonzero element of $Ext^D_R(\kappa, M)$. (All Ext^i vanish for $i < D$, and vanishing fails at index D.) This exhibits the nonzero class $Ext^{depth(R)}(\kappa, R^k) \neq 0$ in the base case of the formula.*

Proof. By 1722, D is the least index at which $\operatorname{Ext}^i_R(\kappa, M)$ does not vanish. Thus the group at index D is nonzero, and any nonzero group contains the required class. □

Lemma 1745 (Exact projective dimension of a syzygy)**.** *For a surjection $f : R^n \twoheadrightarrow M$ with $pd_R(M) = k+2$, the kernel satisfies $pd_R(\ker f) = k+1$ exactly. The upper bound is the syzygy descent of 1730; the lower bound is the contrapositive of the ascent 1731. This is the exact-pd input that licenses the inductive hypothesis on $\ker f$.*

Proof. Syzygy descent gives $\operatorname{pd}_R(\ker f) \le k+1$. If it were at most k, the ascent statement applied to the same short exact sequence would give $\operatorname{pd}_R(M) \le k+1$, contradicting the assumed value $k+2$. □

Lemma 1746 ($\mathbb{N}_\infty$ combine for the inductive step)**.** *A purely arithmetic statement in $\mathbb{N}_\infty$: given $j{+}d_K = d$, $\min(d, d_K{-}1) \le d_M$, $\min(d, d_M{+}1) \le d_K$, and $1 \le j$, one concludes $(j{+}1){+}d_M = d$. This packages the inductive hypothesis $j + depth(\ker f) = depth(R)$ with the two kernel-SES depth inequalities into the next instance of the Auslander–Buchsbaum equation.*

Proof. The equality $j + d_K = d$ and $1 \le j$ exclude $d_K = \infty$ unless $d = \infty$, where the conclusion is immediate. In the finite case the two minimum inequalities give $d_M + 1 \le d_K \le d_M + 1$; hence $d_K = d_M + 1$, and substitution yields $(j+1) + d_M = d$. □

30.6.4 Matrix-collapse on Ext

Definition 1747 (Elementary linear map)**.** For a commutative ring R and indices $i \in \{1, \dots, n\}$, $j \in \{1, \dots, m\}$, the *elementary map* $E_{i,j} : R^m \to R^n$ is the R-linear map sending the standard basis vector Pi.single j 1 to Pi.single i 1 and every other standard basis vector to 0.

Lemma 1748 (Evaluation of the elementary map)**.** *For $x \in R^m$ one has $E_{i,j}(x) = Pi.single\, i\, (x_j)$: the elementary map reads off the j-th coordinate and places it in the i-th slot.*

Proof. Expand x in the standard basis. By linearity, $E_{i,j}$ kills every basis term except the j-th and sends that coefficient to the i-th coordinate, giving the displayed formula. ☐

Lemma 1749 (Matrix decomposition into elementary maps)**.** *Every R-linear map $A : R^m \to R^n$ decomposes as $A = \sum_{i,j} A_{i,j} \cdot E_{i,j}$, where $A_{i,j} = (A\,(Pi.single\, j\, 1))_i$ is the corresponding matrix entry.*

Proof. Both sides are linear maps, so it suffices to evaluate them on each standard basis vector. The j-th column of the sum is $\sum_i A_{i,j} e_i = A(e_j)$, proving equality. ☐

Lemma 1750 (Matrix-collapse on Ext)**.** *Let $A : R^m \to R^n$ be an R-linear map every matrix entry of which lies in $Ann_R(N)$. Then the postcomposition map $Ext^p_R(N, R^m) \to Ext^p_R(N, R^n)$ induced by A is identically zero. The proof decomposes $A = \sum_{i,j} A_{i,j} \cdot E_{i,j}$ (1749) and kills each scalar summand via 1733.*

Proof. Use the elementary-map decomposition of A. The induced Ext map is additive and linear in each matrix coefficient. Every coefficient annihilates N, so 1733 kills every summand; their sum is zero. ☐

30.6.5 Regular-sequence and domain substrate for regular local rings

Lemma 1751 (Regular sequences are bounded by the Krull dimension)**.** *For a Noetherian local ring R, every R-regular sequence rs has length at most $\dim(R)$. This is the upper-bound half of 1765, obtained from the dimension identity $\dim(R/(rs)) + |rs| = \dim(R)$ together with $\dim(R/(rs)) \geq 0$.*

Proof. Quotient successively by the elements of the regular sequence. Each regular element lowers the Krull dimension by one, while the final quotient has nonnegative dimension. Thus $\dim(R) \geq |rs|$. ☐

Lemma 1752 (Cotangent image of $x \in \mathfrak{m} \setminus \mathfrak{m}^2$)**.** *For a local ring $(R, \mathfrak{m})$ and $x \in \mathfrak{m}$ with $x \notin \mathfrak{m}^2$, the image of x in the cotangent space $\mathfrak{m}/\mathfrak{m}^2$ is nonzero. This is the positivity input for the cotangent dimension-drop.*

Proof. The kernel of the quotient map $\mathfrak{m} \to \mathfrak{m}/\mathfrak{m}^2$ is exactly $\mathfrak{m}^2$. Hence the class of x vanishes precisely when $x \in \mathfrak{m}^2$. ☐

Lemma 1753 (Cotangent dimension drops by one modulo x)**.** *For a Noetherian local ring $(R, \mathfrak{m})$ and $x \in \mathfrak{m} \setminus \mathfrak{m}^2$, the cotangent space of $R/(x)$ has dimension one less than that of R:*

$$\dim_{\kappa'}(CotangentSpace(R/(x))) + 1 = \dim_{\kappa}(CotangentSpace(R)),$$

where κ, κ' are the residue fields of R and $R/(x)$. The kernel of the induced surjection on cotangent spaces is the line spanned by the image of x, which is one-dimensional by 1752.

Proof. The quotient map induces a surjection from the cotangent space of R onto that of $R/(x)$. Its kernel is the line generated by the nonzero class of x, so rank-nullity lowers the dimension by exactly one. $\square$

Lemma 1754 (Nakayama witness for a positive cotangent dimension)**.** *For a Noetherian local ring $(R, \mathfrak{m})$ with positive minimal-generator count of $\mathfrak{m}$ (i.e. spanFinrank$(\mathfrak{m}) \geq 1$), there exists $x \in \mathfrak{m}$ with $x \notin \mathfrak{m}^2$. By Nakayama, $\mathfrak{m} \subseteq \mathfrak{m}^2$ would force $\mathfrak{m} = 0$, contradicting positivity.*

Proof. If every element of $\mathfrak{m}$ lay in $\mathfrak{m}^2$, then $\mathfrak{m} = \mathfrak{m}^2$. Nakayama's lemma would force $\mathfrak{m} = 0$, making the cotangent dimension zero, a contradiction. $\square$

Lemma 1755 (Zero cotangent dimension forces a field)**.** *For a Noetherian local ring R with spanFinrank$(\mathfrak{m}) = 0$, the maximal ideal collapses to 0, so R is a field and hence an integral domain. This is the base case of 1759.*

Proof. Vanishing cotangent dimension gives $\mathfrak{m}/\mathfrak{m}^2 = 0$, hence $\mathfrak{m} = \mathfrak{m}^2$. Nakayama yields $\mathfrak{m} = 0$. A local ring whose maximal ideal is zero is a field, and therefore a domain. $\square$

Lemma 1756 (Regularity descends to $R/(x)$)**.** *For a regular Noetherian local ring R with spanFinrank$(\mathfrak{m}) = k + 1$ and $x \in \mathfrak{m} \setminus \mathfrak{m}^2$, the quotient $R/(x)$ is again a regular local ring, with spanFinrank$(\mathfrak{m}') = k$. The cotangent dimension drop (1753) plus a Krull-height bound give the matching dimension equality, so $R/(x)$ is regular. Notably this avoids assuming x is a non-zero-divisor, so it does not depend on 1759.*

Proof. By 1753, the embedding dimension of $R/(x)$ is k. The principal ideal theorem gives $\dim(R/(x)) \geq \dim(R) - 1 = k$, while Krull dimension is bounded above by embedding dimension. Equality follows, which is the regularity criterion for the local quotient. $\square$

Lemma 1757 (Members of a minimal prime are zero-divisors)**.** *For a commutative ring R and a minimal prime $\mathfrak{p}$ of R, every $x \in \mathfrak{p}$ is a zero-divisor: there exists $y \neq 0$ with $xy = 0$. This follows from the disjointness of a minimal prime from the non-zero-divisors.*

Proof. A minimal prime is disjoint from the set of non-zero-divisors. Hence x is not a non-zero-divisor, so multiplication by x is not injective. Thus there is $y \neq 0$ with $xy = 0$. $\square$

Lemma 1758 ((x) is not a minimal prime in the inductive step)**.** *Let R be a regular Noetherian local ring with spanFinrank$(\mathfrak{m}) = k+1$ and $x \in \mathfrak{m} \setminus \mathfrak{m}^2$, and suppose every k-dimensional regular local ring is a domain. Then span$\{x\}$ is* not *a minimal prime of R. Combined with the regular descent (1756), this is the residual case of the regular-local domain argument.*

Proof. Suppose that (x) is minimal. Since a Noetherian ring has finitely many minimal primes, prime avoidance supplies $x' \in \mathfrak{m}$ lying neither in $\mathfrak{m}^2$ nor in any minimal prime: indeed, $\mathfrak{m} \nsubseteq \mathfrak{m}^2$ by positive cotangent dimension, and $\mathfrak{m}$ cannot lie in a minimal prime without forcing $\dim(R) = 0$.

The quotient $R/(x')$ is regular of dimension k, hence a domain by the inductive hypothesis, so (x') is prime. Choose a minimal prime $\mathfrak{p} \subseteq (x')$. Since $x' \notin \mathfrak{p}$, every $y \in \mathfrak{p}$, written $y = x'z$, has $z \in \mathfrak{p}$ by primality. Thus $\mathfrak{p} \subseteq \mathfrak{m}\mathfrak{p}$, and the reverse inclusion is automatic. The ideal $\mathfrak{p}$ is finitely generated, so Nakayama gives $\mathfrak{p} = 0$. Consequently R is a domain. Its only minimal prime is then 0, so the assumed minimality of (x) gives $x = 0$, contradicting $x \notin \mathfrak{m}^2$. $\square$

Lemma 1759 (Regular local rings are domains)**.** *Every regular Noetherian local ring R is an integral domain. The proof is a strong induction on spanFinrank$(\mathfrak{m})$: the base case is the field case (1755); the inductive step picks $x \in \mathfrak{m} \setminus \mathfrak{m}^2$ (1754), uses that $R/(x)$ is regular local of smaller dimension (1756) and a domain by induction, and rules out the obstruction case via 1758.*

Proof. Induct on the cotangent dimension. The zero-dimensional case is 1755. In positive dimension choose $x \in \mathfrak{m} \setminus \mathfrak{m}^2$. The quotient is regular of smaller dimension and hence a domain by induction. The only possible obstruction to R being a domain would put (x) among the minimal primes, contrary to 1758. □

Lemma 1760 (A regular element outside $\mathfrak{m}^2$)**.** *For a regular local ring R of dimension $k+1$, there exists $x \in \mathfrak{m} \setminus \mathfrak{m}^2$ that is moreover an R-regular element. The Nakayama witness (1754) is nonzero, and in a domain (1759) every nonzero element is a non-zero-divisor.*

Proof. Choose $x \in \mathfrak{m} \setminus \mathfrak{m}^2$ by 1754. It is nonzero, and 1759 makes multiplication by x injective. Thus x is regular. □

Lemma 1761 (Regular element with regular quotient)**.** *For a regular local ring R of dimension $k+1$, there exists $x \in \mathfrak{m}$ that is R-regular and such that $R/(x)$ is again a regular local ring of dimension k (its maximal ideal has spanFinrank $= k$). This consolidates the "pick an R-regular system-of-parameters element and drop dimension" step.*

Proof. Take the element supplied by 1760. It is regular, and its quotient is regular of dimension k by 1756. □

Lemma 1762 (Inductive step for the regular sequence)**.** *Let R be a regular local ring of dimension $k+1$, and suppose that every regular local ring of dimension k admits a length-k regular sequence inside its maximal ideal. Then R admits a length-$(k+1)$ regular sequence inside $\mathfrak{m}$: extract the regular element x with regular quotient (1761), lift a length-k regular sequence of $R/(x)$ back to R, and prepend x.*

Proof. Choose a regular x whose quotient is regular of dimension k. By the inductive hypothesis, $R/(x)$ has a regular sequence $\bar{x}_1, \dots, \bar{x}_k$. Lift these elements to $\mathfrak{m}$; the definition of regularity in successive quotients shows that $x, x_1, \dots, x_k$ is regular on R. □

Lemma 1763 (Regular local rings have a full regular sequence)**.** *For a regular Noetherian local ring R, there is an R-regular sequence inside $\mathfrak{m}$ whose length equals spanFinrank$(\mathfrak{m}) = \dim(R)$. The proof is a strong induction on the dimension: the base case is the empty sequence, and the inductive step is 1762. This is the lower-bound half of 1765.*

Proof. Use strong induction on $\dim(R)$. The empty sequence proves the zero-dimensional case. In positive dimension, 1762 extends the regular sequence from the quotient of dimension one less, producing a sequence of length $\dim(R)$. □

30.7 Corollary: regular local rings are Cohen–Macaulay

Definition 1764 (Cohen–Macaulay local ring)**.** *Source: [Stacks Project], tag 00N4 (definition-local-ring-CM).* A Noetherian local ring $(R, \mathfrak{m})$ is called *Cohen–Macaulay* if $\operatorname{depth}(R) = \dim(R)$ (equivalently, if R is Cohen–Macaulay as a module over itself, where Cohen–Macaulay for a finite R-module M means $\dim(\operatorname{Supp}(M)) = \operatorname{depth}(M)$, cf. Stacks tag 00N3).

Corollary 1765 (Regular local rings are Cohen–Macaulay)**.** *Source: [Stacks Project], tag 00OD (lemma-regular-ring-CM). Every regular Noetherian local ring is Cohen–Macaulay: if $(R, \mathfrak{m})$ is regular local of Krull dimension d, then depth$(R) = d = \dim(R)$.*

Proof. Let $(R, \mathfrak{m})$ be regular of dimension d and pick a minimal generating set $x_1, \ldots, x_d$ of $\mathfrak{m}$. By Stacks tag 00NQ (lemma-regular-domain), R is an integral domain, so x_1 is a non-zero-divisor. The quotient R/x_1R is a Noetherian local ring of dimension $\geq d-1$ (by Krull's principal ideal theorem, `lemma-one-equation`) with maximal ideal generated by $x_2, \ldots, x_d$, hence has dimension exactly $d-1$ and is again regular of dimension $d-1$. Inducting on d, $x_1, \ldots, x_d$ is an R-regular sequence, so $\operatorname{depth}(R) \geq d$. The reverse inequality $\operatorname{depth}(R) \leq \dim(R) = d$ always holds for any nonzero finite module over a Noetherian local ring (Stacks tag 00LK, `lemma-bound-depth`). Thus $\operatorname{depth}(R) = d = \dim(R)$, i.e. R is Cohen–Macaulay.

Remark on the Auslander–Buchsbaum route. The same conclusion can be obtained from 1725 as follows: take $M := R$ itself. Then M is finitely generated and projective (free of rank 1), so $\operatorname{pd}_R(R) = 0$. The formula gives $\operatorname{depth}(R) = 0 + \operatorname{depth}(R)$, a tautology. The non-tautological use of Auslander–Buchsbaum is rather: for a regular local ring of dimension d, every finitely generated module M has $\operatorname{pd}_R(M) \leq d$ (Stacks tag 00O2, `proposition-finite-gl-dim-regular`); applying the formula to such an M then relates $\operatorname{depth}(M)$ to $d - \operatorname{pd}_R(M)$, which is the form most often used in practice. The direct regular-sequence argument above is the cleanest proof of R Cohen–Macaulay itself. □

30.8 Application: Hartogs extension on a regular surface

Let S be a regular Noetherian surface and let $x \in S$ be a closed point. The local Hartogs mechanism for the structure sheaf is:

1. The local ring $\mathcal{O}_{S,x}$ is a regular Noetherian local ring of Krull dimension 2 (because S is regular and $\dim S = 2$).
2. By 1765 it is Cohen–Macaulay, so $\operatorname{depth}(\mathcal{O}_{S,x}) = 2$. Equivalently, the maximal ideal contains an $\mathcal{O}_{S,x}$-regular sequence of length 2.
3. Sections of the structure sheaf $\mathcal{O}_S$ on $U = S \setminus \{x\}$ extend uniquely across x because $H^0_x(\mathcal{O}_S) = 0$ and $H^1_x(\mathcal{O}_S) = 0$ (these local-cohomology vanishings are equivalent to depth ≥ 2 at x by Stacks tag 0AVF; see Hartshorne, III.3.5, for the surface case).

Thus 1765 supplies the depth-two input needed to extend regular functions across a codimension-two point on a regular surface. It does not extend a rational map by choosing affine coordinates on the target: such an implication is false for general complete targets. The Albanese development instead combines the codimension-one valuative result with the group-scheme theorem on indeterminacy loci; that geometric argument is independent of the commutative-algebra result proved here.

30.9 Algebraic scope

The Auslander–Buchsbaum formula and its Cohen–Macaulay corollary depend only on the homological algebra of finite modules over Noetherian local rings. They are independent of the geometric constructions used later for Albanese varieties and may therefore be applied wherever the corresponding local rings are regular.

30.10 Further consequences

The local result above also leads to the usual global Cohen–Macaulay criterion and to local-cohomology vanishing below the depth. Those consequences, as well as refinements involving dualizing complexes and Gorenstein rings, are not needed for the Auslander–Buchsbaum formula or its application to regular surfaces.

Chapter 31

Milne Theorem 3.2: Rational maps into abelian varieties

31.1 Setup and statement

Fix an algebraically closed field $\bar{k}$, a nonsingular variety X over $\bar{k}$, and an abelian variety A over $\bar{k}$. Thus X is integral and smooth, while A is a complete smooth group variety. A rational map $f\colon X \dashrightarrow A$ is represented by a regular morphism $\varphi_U\colon U \to A$ on a dense open subset $U \subseteq X$, two such pairs (U, φ_U) and $(U', \varphi_{U'})$ being equivalent if φ_U and $\varphi_{U'}$ agree on $U \cap U'$.

Theorem 1766 (Milne Theorem 3.2: a rational map from a nonsingular variety to an abelian variety extends)**.** *Let X be a nonsingular variety over an algebraically closed field $\bar{k}$, let A be an abelian variety over $\bar{k}$, and let $f\colon X \dashrightarrow A$ be a rational map. Then f is defined on the whole of X: there is a unique regular morphism $\tilde{f}\colon X \to A$ whose restriction to some (equivalently, every) dense open set of definition $U \subseteq X$ of f agrees with φ_U.*

Proof. Let U be the maximal open subset on which f is defined, and put $Z = X \setminus U$. By 1770, $Z = \emptyset$. Hence $U = X$, and the representative of f on U is the desired morphism $\tilde{f}\colon X \to A$.

If two morphisms $X \to A$ represent f, they agree on a dense open subset of X. Their equalizer is closed because A is separated, and it contains that dense open subset. Since X is reduced, the two morphisms are equal. Thus the extension is unique. ☐

Remark 1767. The Auslander–Buchsbaum theorem is not needed for 1766. The codimension-one argument uses regularity only to obtain normality and discrete valuation rings at prime divisors; properness then supplies extensions by the valuative criterion. It does not extend across a codimension-two locus by a depth argument. The stronger Cohen–Macaulay consequences of Auslander–Buchsbaum in 30 describe the same regular local rings but are logically separate from Milne's proof.

31.2 Why the proof is characteristic-free

Both ingredients are independent of the characteristic:

- *Theorem 3.1 / codim-≥ 2 extension* (1705): a codimension-one local ring on a normal variety is a discrete valuation ring, and the valuative criterion of properness extends the map over that local ring.

- *Lemma 3.3 / pure-codim-*1 *indeterminacy for maps into a group variety* (1711): Milne's proof forms the difference map $\Phi(x, y) := \varphi(x)\, \varphi(y)^{-1} : X \times X \dashrightarrow A$ and examines its pole divisors along the diagonal. This uses the group law, regularity, and the codimension-one support of principal divisors, none of which imposes a characteristic restriction.

Consequently 1766 holds over every algebraically closed field, in arbitrary characteristic.

31.3 Application to the symmetric-product Albanese map

Let C be a complete nonsingular curve of genus $g > 0$, let J be its Jacobian, and let $\varphi\colon C \to A$ send a chosen point P to zero. The symmetric map

$$(P_1, \dots, P_g) \longmapsto \varphi(P_1) + \dots + \varphi(P_g)$$

descends to a morphism $\varphi^{(g)} : C^{(g)} \to A$. The Abel–Jacobi morphism $C^{(g)} \to J$ is birational, so composing $\varphi^{(g)}$ with its rational inverse gives a rational map $J \dashrightarrow A$. Since J is nonsingular, 1766 makes this map regular. It factors φ through $C \to J$, and the rigidity theorem shows that it is a homomorphism. This is the existence step in the Albanese universal property of J proved in 32.

31.4 Structure of the proof

The argument separates into three elementary reductions. Smoothness makes the abelian variety reduced; geometric irreducibility then makes it integral. Milne's two indeterminacy results force the indeterminacy locus to be empty. Finally, a rational map with empty indeterminacy locus has a global representative, unique because its source is reduced and its target is separated.

31.5 Auxiliary lemmas

We record the reductions concerning the target and the indeterminacy locus.

Lemma 1768 (A scheme smooth over a field is reduced)**.** *Let A be a scheme over an algebraically closed field $\bar{k}$. If the structure morphism $A \to \operatorname{Spec} \bar{k}$ is smooth, then A is reduced.*

Proof. An algebraically closed field is perfect. A scheme smooth over a perfect field is regular, since its local rings are geometrically regular over the field. Regular local rings are reduced, and reducedness is local; hence A is reduced. ☐

Lemma 1769 (Integrality of the abelian variety)**.** *An abelian variety A over an algebraically closed field $\bar{k}$ is integral. Indeed, it is geometrically irreducible and smooth over $\bar{k}$, hence irreducible and reduced.*

Proof. The spectrum of $\bar{k}$ consists of one point. Geometric irreducibility of the structure morphism therefore implies that the underlying topological space of A is irreducible. By 1768, A is also reduced. A nonempty scheme is integral precisely when it is irreducible and reduced, so A is integral. ☐

Lemma 1770 (Emptiness of the indeterminacy locus of a rational map into an abelian variety)**.** *For a rational map $f\colon X \dashrightarrow A$ from a nonsingular variety X to an abelian variety A over an algebraically closed field $\bar{k}$, the indeterminacy locus $Z(f)$ is empty.*

Proof. The abelian variety A is integral (1769), so Milne's Lemma 3.3 (1711) applies and gives the dichotomy: either $Z(f) = \emptyset$, or every point of $Z(f)$ specialises from a point $z \in Z(f)$ of coheight 1. In the second branch, applying the codim-$\geq$ 2 indeterminacy theorem (1705, Milne 3.1) to such a $z \in Z(f)$ forces coheight$(z) \geq 2$, contradicting coheight$(z) = 1$. Hence $Z(f) = \emptyset$. $\square$

31.6 Relations with neighbouring results

Milne's codimension estimate and pure-codimension-one theorem are proved in 28 as 1705 and 1711. Their combination here is special to group-variety targets: a rational map to a general complete variety can have a nonempty codimension-two indeterminacy locus.

The extension theorem feeds both the Albanese construction discussed above and the usual rigidity consequences for rational maps between abelian varieties. Over a non-algebraically-closed field the local extension arguments remain available after suitable residue-field bookkeeping, but descent of the resulting morphism is an additional problem.

Chapter 32

The Albanese universal property of $\mathrm{Pic}^0_{C/k}$

Let k be an algebraically closed field, let C be a smooth proper geometrically irreducible curve over k, and let $P_0 \in C(k)$. Write g for the genus of C and assume $g > 0$. This chapter proves that the Abel–Jacobi map

$$\iota_{P_0} : C \longrightarrow \mathrm{Pic}^0_{C/k}$$

is the Albanese morphism of the pointed curve (C, P_0). The proof uses the canonical map from the g-th symmetric power of C to its Jacobian. In positive characteristic, generic injectivity of this map does not by itself imply birationality: a purely inseparable map may also have one geometric point in its generic fibre. The differential calculation that excludes this possibility is therefore included explicitly.

32.1 The Jacobian and the Abel–Jacobi map

The construction of the identity component of the Picard scheme gives

$$J := \mathrm{Pic}^0_{C/k}.$$

It is a group scheme by 1625, smooth by 1643, proper by 1644, and geometrically irreducible by 1645.

Definition 1771 (The Abel–Jacobi morphism)**.**

The relative degree-zero line bundle associated with the rigidified diagonal on $C \times C$ defines a classifying morphism $\iota_{P_0} : C \to J$, called the *Abel–Jacobi morphism.*

Lemma 1772 (Formula and basepoint of the Abel–Jacobi morphism)**.**

There is a regular morphism

$$\iota_{P_0} : C \longrightarrow J, \qquad Q \longmapsto [\mathcal{O}_C(Q - P_0)],$$

and $\iota_{P_0}(P_0) = 0_J$.

Proof. On $C \times C$, viewed over the second factor, consider the rigidified relative degree-zero line bundle represented by

$$\mathcal{O}_{C\times C}(\Delta - P_0 \times C - C \times P_0).$$

Its restriction over $Q \in C(k)$, modulo a line bundle pulled back from the base, is $\mathcal{O}_C(Q - P_0)$. The moduli property of $\mathrm{Pic}^0_{C/k}$ therefore gives the displayed morphism. At $Q = P_0$ the fibre is the trivial line bundle, whose class is the identity of J. ☐

32.2 Symmetric powers and their canonical maps

Definition 1773 (Symmetric power of a curve)**.** For $r \geq 1$, the *r-th symmetric power* $C^{(r)}$ is the quotient of C^r by permutation of the factors, with quotient morphism $\pi_r \colon C^r \to C^{(r)}$. Every symmetric morphism from C^r to a k-scheme factors uniquely through π_r.

Lemma 1774 (Properties of the symmetric-power projection)**.**
The quotient morphism $\pi_r \colon C^r \to C^{(r)}$ is finite, surjective, and separable.

Proof. On an invariant affine chart the target ring is the ring of S_r-invariants. Every element of the source ring is integral over this invariant ring, so the quotient is finite; the topological quotient description gives surjectivity. Finally, the induced extension of function fields is the finite invariant-field extension for a group of automorphisms, and is therefore separable. ☐

Definition 1775 (The symmetric-power carrier)**.**
The underlying k-scheme of $C^{(r)}$ is denoted $\mathrm{SymmetricPower}(C, r)$.

Lemma 1776 (Nonsingularity of a symmetric power of a curve)**.**
If C is nonsingular, then $C^{(r)}$ is nonsingular.

Proof. At the image of $(Q, \dots, Q)$, choose a parameter X at Q. The completed local ring of C^r is $k[[X_1, \dots, X_r]]$, and the completed local ring of the quotient is its invariant subring. The fundamental theorem on symmetric functions identifies this with $k[[\sigma_1, \dots, \sigma_r]]$, which is regular. At a divisor with several distinct support points, the completed local ring is the completed tensor product of the corresponding invariant power-series rings, and is again regular. ☐

Definition 1777 (The symmetric sum morphism)**.**
For a morphism $\varphi \colon C \to A$ to an abelian variety, denote the induced morphism from the symmetric-power carrier by $\varphi^{(r)} \colon C^{(r)} \to A$.

Lemma 1778 (Symmetrisation of a morphism to an abelian variety)**.**
Let A be an abelian variety and let $\varphi \colon C \to A$ be a morphism. There is a unique morphism $\varphi^{(r)} \colon C^{(r)} \to A$ satisfying

$$\varphi^{(r)}(P_1 + \dots + P_r) = \varphi(P_1) + \dots + \varphi(P_r).$$

Proof. The morphism $C^r \to A$ obtained by applying φ in every coordinate and then adding is invariant under every permutation, because the group law of A is commutative. The universal property of $C^{(r)}$ gives the unique factorisation. ☐

Definition 1779 (The canonical symmetric-power morphism to the Jacobian)**.**
The Lean construction obtained by symmetrising ι_{P_0} is denoted

$$f^{(r)} \colon C^{(r)} \longrightarrow J.$$

Lemma 1780 (Divisor description of the canonical symmetric-power morphism)**.**
On effective divisors, the canonical morphism is given by

$$f^{(r)}(D) = [\mathcal{O}_C(D - rP_0)].$$

Denote its reduced closed image by W^r. The fibre through an effective divisor D is its complete linear system $|D|$, a projective space of dimension $h^0(D) - 1$.

Proof. Pulling back along π_r gives the sum of the r Abel–Jacobi classes, hence the displayed divisor class. Two effective divisors have the same class precisely when they are linearly equivalent. The functor of effective divisors linearly equivalent to D is represented by the projectivisation of $H^0(C, \mathcal{O}_C(D))$, giving the fibre statement. □

Lemma 1781 (General divisors of degree at most the genus)**.**
If $1 \le r \le g$, there is a dense open subset of $C^{(r)}$ on which $h^0(D) = 1$. Thus the fibres of $f^{(r)}$ over this open consist of one geometric point.

Proof. If $h^1(D) > 0$, Serre duality identifies

$$H^1(C, \mathcal{O}_C(D + Q))^\vee$$

with the subspace of $H^0(C, \Omega^1_C(-D))$ consisting of differentials that vanish at Q. Outside the common zero locus of a basis of this latter space, adding Q lowers h^1 by exactly one. Starting with the zero divisor, for which $h^1 = g$, and repeating this argument r times gives a dense open subset of C^r on which $h^1(P_1 + \cdots + P_r) = g - r$. Riemann–Roch then gives

$$h^0(P_1 + \cdots + P_r) = r + 1 - g + (g - r) = 1.$$

Intersecting its finitely many translates under S_r gives a dense symmetric open with the same property. Its image in the finite quotient $C^{(r)}$ is the complement of the image of its closed invariant complement, hence is open and dense. Since the fibre of $f^{(r)}$ through D is $|D|$, it is a single point there. □

Lemma 1782 (Invariant one-forms on the Jacobian)**.** *Evaluation at the identity is an isomorphism*

$$H^0(J, \Omega^1_J) \xrightarrow{\sim} T_0 J^\vee.$$

Proof. A cotangent vector at the identity extends uniquely to a translation-invariant one-form on the smooth group scheme J. Conversely, for a global one-form η, the coefficients of $t_a^*\eta - \eta$, expressed in an invariant frame, vary regularly with $a \in J$. They are regular functions on the proper geometrically irreducible variety J, hence constant; at $a = 0$ they vanish. Thus every global one-form is invariant, and evaluation at the identity gives the asserted inverse pair. □

Lemma 1783 (Differential of the Abel–Jacobi map)**.**
Under the tangent-space identification $T_0 J \simeq H^1(C, \mathcal{O}_C)$ and Serre duality, the transpose of the translated differential of ι_{P_0} at Q is evaluation

$$H^0(C, \Omega^1_C) \longrightarrow T_Q^* C, \qquad \omega \longmapsto \omega(Q).$$

Proof. Moving Q to first order changes the class $[\mathcal{O}_C(Q - P_0)]$ by the connecting homomorphism associated with the principal-parts sequence at Q. The Serre-duality transpose of this connecting homomorphism is the restriction of a differential to its value at Q, which is the displayed evaluation map. □

Lemma 1784 (One-forms pulled back by the Abel–Jacobi map)**.**
Pullback by ι_{P_0} *is an isomorphism*

$$H^0(J,\Omega^1_J) \xrightarrow{\sim} H^0(C,\Omega^1_C).$$

Proof. By 1782, a form on J is determined by its value at the identity. Under 1783, pulling it back along ι_{P_0} recovers at every Q the corresponding canonical differential on C. Under the Serre-duality identification this map is the identity, hence an isomorphism. □

Lemma 1785 (Dimension of the Jacobian)**.** *The Jacobian has dimension* g*.*

Proof. Since J is smooth, its dimension is the dimension of its cotangent space at the identity. Translation identifies that cotangent space with the global one-forms on J. By 1784, this space is isomorphic to $H^0(C,\Omega^1_C)$, whose dimension is the genus g. □

Proposition 1786 (Differentials on a symmetric power)**.**
Pullback by $f^{(r)}$ *is an isomorphism*

$$H^0(J,\Omega^1_J) \xrightarrow{\sim} H^0(C^{(r)},\Omega^1_{C^{(r)}}).$$

Under the canonical identification of either side with $H^0(C,\Omega^1_C)$*, the form corresponding to* ω *vanishes at an effective divisor* D *if and only if* $\operatorname{div}(\omega) \geq D$*.*

Proof. A global one-form on C^r is a sum of pullbacks from the factors. Hence the invariant forms are exactly $\sum_i p_i^*\omega$, with $\omega \in H^0(C,\Omega^1_C)$. The quotient map π_r is separable, so pullback embeds $H^0(C^{(r)},\Omega^1)$ into this invariant subspace. The composite

$$H^0(J,\Omega^1_J) \longrightarrow H^0(C^{(r)},\Omega^1) \longrightarrow H^0(C^r,\Omega^1)^{S_r}$$

is the pullback by the sum of the r copies of the Abel–Jacobi map. By 1784, it sends a form to $\sum_i p_i^*\omega$ and is an isomorphism. Consequently both arrows in the display are isomorphisms.

It remains to identify vanishing at a divisor. First take $D = rQ$, choose a parameter X at Q, and write $\omega = (a_0 + a_1X + \cdots)dX$. In the completed local ring of $C^{(r)}$ at D, use the elementary symmetric functions $\sigma_1,\dots,\sigma_r$ in local parameters $X_1,\dots,X_r$, and put $\tau_j = \sum_i X_i^j\,dX_i$. The identities

$$\sigma_m\tau_0 - \sigma_{m-1}\tau_1 + \cdots + (-1)^m\tau_m = d\sigma_{m+1} \qquad (0 \leq m < r)$$

show that $\tau_0,\dots,\tau_{r-1}$ form a basis of the cotangent module. The descended form is $\sum_{j\geq 0} a_j\tau_j$, so it vanishes at D exactly when $a_0 = \cdots = a_{r-1} = 0$, equivalently when $\operatorname{ord}_Q(\omega) \geq r$. The same calculation in the completed local product for $D = \sum_Q n_Q Q$ gives the stated criterion. □

Lemma 1787 (Tangent sequence of the canonical symmetric-power map)**.**
Let $F = |D|$ *be the fibre of* $f^{(r)}$ *through an effective divisor* D*, and put* $a = f^{(r)}(D)$*. The sequence*

$$0 \longrightarrow T_DF \longrightarrow T_DC^{(r)} \xrightarrow{df_D^{(r)}} T_aJ$$

is exact. In particular, if $h^0(D) = 1$*, then* $df_D^{(r)}$ *is injective.*

Proof. The first arrow is injective because the fibre is a closed subscheme, and its image lies in the kernel because $f^{(r)}$ is constant on the fibre. By 1786, the annihilator in $T_aJ^\vee$ of the image of $df_D^{(r)}$ is $H^0(C,\Omega^1_C(-D))$. Serre duality therefore gives

$$\dim\ker(df_D^{(r)}) = r - g + h^1(D).$$

The fibre F is the projective space of effective divisors in the complete linear system, so it is smooth of dimension $h^0(D)-1$. Thus $\dim T_D F = \dim |D| = h^0(D)-1$, and Riemann–Roch says $h^0(D)-1 = r-g+h^1(D)$. The included subspaces have the same dimension, proving exactness. When $h^0(D)=1$, this dimension is zero. $\square$

Lemma 1788 (Birationality of the canonical map)**.**
For $1 \le r \le g$, the morphism

$$f^{(r)}: C^{(r)} \longrightarrow W^r$$

is birational. In particular, $W^g = J$, and $f^{(g)}: C^{(g)} \to J$ is a birational surjective morphism.

Proof. The product C^r has dimension r, and the finite surjective map $C^r \to C^{(r)}$ shows that $C^{(r)}$ also has dimension r. By 1781, the map is injective on geometric points over a dense open subset. It is therefore generically quasi-finite, so its image W^r has dimension r. Thus the induced extension of function fields has separable degree one. In positive characteristic this still leaves open the possibility of a nontrivial purely inseparable extension. Choose D in that open. Then $h^0(D)=1$, so 1787 makes $df_D^{(r)}$ injective. Since source and image both have dimension r, the differential has full rank, and the function-field extension is generically separable. Its inseparable degree is therefore also one. The function fields of $C^{(r)}$ and W^r are equal, proving birationality.

For $r=g$, the image W^g is a closed irreducible subvariety of the irreducible variety J and has dimension $g = \dim J$, the latter equality being 1785. Hence $W^g = J$. Surjectivity follows because $C^{(g)}$ is proper and its image is exactly W^g. $\square$

32.3 Descent and the universal property

Lemma 1789 (Descent through the birational symmetric power)**.**
Let A be an abelian variety and let $\varphi: C \to A$ be pointed at P_0. There is a unique regular morphism $\psi: J \to A$ such that

$$\varphi^{(g)} = \psi \circ f^{(g)}.$$

Proof. On dense opens $U \subseteq C^{(g)}$ and $V \subseteq J$, the map $f^{(g)}|_U: U \to V$ is an isomorphism. Thus $\varphi^{(g)}|_U \circ (f^{(g)}|_U)^{-1}$ defines a rational map $J \dashrightarrow A$. The Jacobian is smooth and A is an abelian variety, so 1766 extends this rational map uniquely to a regular morphism $\psi: J \to A$. The two morphisms $\varphi^{(g)}$ and $\psi \circ f^{(g)}$ agree on the dense open U; since $C^{(g)}$ is nonsingular by 1776, hence reduced, and A is separated, they agree everywhere. The same dense-open agreement proves uniqueness. $\square$

Definition 1790 (The basepoint section of the symmetric power)**.**
Define

$$s_{P_0}: C \longrightarrow C^{(g)}, \qquad Q \longmapsto Q + (g-1)P_0.$$

Lemma 1791 (The basepoint section followed by the canonical map)**.**
One has $f^{(g)} \circ s_{P_0} = \iota_{P_0}$.

Proof. At $Q \in C$, the left side represents

$$\mathcal{O}_C(Q + (g-1)P_0 - gP_0) = \mathcal{O}_C(Q - P_0),$$

which is the defining class of $\iota_{P_0}(Q)$. $\square$

Lemma 1792 (The basepoint section followed by a symmetric sum)**.**
If $\varphi(P_0) = 0_A$, then $\varphi^{(g)} \circ s_{P_0} = \varphi$.

Proof. At $Q \in C$, the left side is $\varphi(Q) + (g-1)\varphi(P_0) = \varphi(Q)$. □

Lemma 1793 (Equivalence of the two factorisation equations)**.**
Let $\varphi\colon C \to A$ satisfy $\varphi(P_0) = 0_A$. For every regular morphism $\psi\colon J \to A$,

$$\varphi = \psi \circ \iota_{P_0} \iff \varphi^{(g)} = \psi \circ f^{(g)}.$$

Proof. Suppose first that $\varphi = \psi \circ \iota_{P_0}$. Evaluating at P_0 gives $\psi(0_J) = 0_A$, so 554 shows that ψ is a homomorphism. Pulling both sides of the desired symmetric-power equation back to C^g, both send $(P_1, \dots, P_g)$ to $\sum_i \psi(\iota_{P_0}(P_i))$. The uniqueness clause in the universal property of $C^{(g)}$ gives the equation on $C^{(g)}$.

Conversely, compose the symmetric-power equation with s_{P_0}. The two basepoint-section identities 1792 and 1791 identify the resulting equation with $\varphi = \psi \circ \iota_{P_0}$. Hence $\varphi = \psi \circ \iota_{P_0}$. □

Theorem 1794 (Albanese factorisation through $\mathrm{Pic}^0_{C/k}$)**.**
For every abelian variety A and every morphism $\varphi\colon C \to A$ with $\varphi(P_0) = 0_A$, there exists a unique regular morphism $\psi\colon J \to A$ such that

$$\varphi = \psi \circ \iota_{P_0}.$$

Proof. By 1789, there is a unique $\psi\colon J \to A$ satisfying $\varphi^{(g)} = \psi \circ f^{(g)}$. The equivalence in 1793, applied to every candidate morphism, transports both existence and uniqueness to the equation $\varphi = \psi \circ \iota_{P_0}$. □

Corollary 1795 (Albanese universal property)**.**
In the situation of 1794, the unique factor $\psi\colon J \to A$ is a homomorphism of group schemes. Consequently ι_{P_0} is initial among pointed morphisms from C to abelian varieties.

Proof. Since $\iota_{P_0}(P_0) = 0_J$ and $\varphi(P_0) = 0_A$, the factorisation gives $\psi(0_J) = 0_A$. The pointed rigidity theorem 554 therefore makes ψ a group homomorphism. Uniqueness among all regular factors implies uniqueness among homomorphisms. □

Chapter 33

Weil divisors on a regular curve

Let X be a Noetherian integral separated scheme that is regular in codimension one. Thus every codimension-one local ring of X is a discrete valuation ring. This chapter constructs the group of Weil divisors, the divisor of a rational function, and linear equivalence. It also records the coefficient-degree formalism used for curves over an algebraically closed field.

The final degree-zero theorem for principal divisors requires the finite-morphism geometry of complete nonsingular curves. Its two geometric inputs are stated separately below. The later Riemann–Roch argument is developed in 34.

33.1 Prime divisors and Weil divisors

Definition 1796 (Prime divisor)**.** A prime divisor of X is a codimension-one integral closed subscheme. Equivalently, it is determined by a point $x \in X$ whose closure has codimension one. We encode the latter condition by

$$\mathrm{coheight}(x) = 1.$$

The reduced closed subscheme on $\overline{\{x\}}$ is integral, so this point and its codimension witness determine the prime divisor.

Definition 1797 (Weil divisor group)**.**

The group of Weil divisors of X is the free abelian group on its prime divisors:

$$\mathrm{Div}(X) = \bigoplus_{Y \in X^{(1)}} \mathbb{Z} \cdot Y.$$

Thus a divisor is a finite formal sum $D = \sum_Y n_Y Y$. It is *effective* when every $n_Y \geq 0$.

Definition 1798 (Regularity in codimension one)**.**

An integral scheme is *regular in codimension one* if $\mathcal{O}_{X,Y}$ is a discrete valuation ring for every prime divisor $Y \in X^{(1)}$.

Lemma 1799 (The stalk at a prime divisor is a DVR)**.**

If X is regular in codimension one and $Y \in X^{(1)}$, then $\mathcal{O}_{X,Y}$ is a discrete valuation ring.

Proof. This is the defining property of regularity in codimension one. □

Lemma 1800 (Dimension of a prime-divisor stalk)**.**

If X is regular in codimension one and $Y \in X^{(1)}$, then $\dim \mathcal{O}_{X,Y} \leq 1$.

Proof. A discrete valuation ring is a principal ideal domain of Krull dimension at most one. □

33.2 Restriction to an open subscheme

Let $U \subseteq X$ be open.

Lemma 1801 (Extensionality of prime divisors)**.**
Two prime divisors represented by the same point of X are equal.

Proof. Their codimension witnesses are propositions, hence proof-irrelevant. □

Definition 1802 (Restriction of a prime divisor)**.**
If $Y \in X^{(1)}$ and its generic point lies in U, restriction gives a prime divisor $Y|_U \in U^{(1)}$.

Definition 1803 (Extension from an open subscheme)**.**
A prime divisor $Z \in U^{(1)}$ determines the prime divisor of X represented by the image of its generic point.

Lemma 1804 (Prime divisors on an open subscheme)**.**
Restriction and extension are inverse bijections

$$\{Y \in X^{(1)} \mid \eta_Y \in U\} \;\simeq\; U^{(1)}.$$

Proof. Both composites preserve the representing point. The result follows from 1801. □

Lemma 1805 (Point of a restricted prime divisor)**.**
The point representing $Y|_U$ is η_Y, regarded as a point of U.

Proof. This follows directly from the definition of restriction. □

Lemma 1806 (Point of an extended prime divisor)**.**
The point representing the extension of $Z \in U^{(1)}$ is the image of η_Z in X.

Proof. This follows directly from the definition of extension. □

Lemma 1807 (Stalks under restriction to an open)**.**
If $\eta_Y \in U$, restriction induces a canonical isomorphism

$$\mathcal{O}_{U,Y|_U} \;\cong\; \mathcal{O}_{X,Y}.$$

Proof. Stalks of the structure sheaf are unchanged after restriction to an open neighbourhood of the point. □

Theorem 1808 (Regularity in codimension one is open-local)**.**
If X is regular in codimension one, then every integral open subscheme $U \subseteq X$ is regular in codimension one.

Proof. Extend a prime divisor of U to X. Its stalk in X is a DVR, and 1807 transports this property to the corresponding stalk in U. □

33.3 Orders of rational functions

33.3.1 Naturality under isomorphisms

Lemma 1809 (Order length is invariant under a ring isomorphism). *Let $e\colon R \cong S$ be an isomorphism of commutative rings. For $x \in R$, the order lengths of x and $e(x)$ agree.*

Proof. The isomorphism e induces an equivalence $R/(x) \cong S/(e(x))$, and module length is invariant under this equivalence. □

Lemma 1810 (Non-zero-divisors are invariant under isomorphism). *For a ring isomorphism $e\colon R \cong S$, an element $r \in R$ is a non-zero-divisor if and only if $e(r)$ is a non-zero-divisor.*

Proof. Multiplication by $e(r)$ is conjugate, through e, to multiplication by r. □

Lemma 1811 (Naturality of the order monoid homomorphism).
The order monoid-with-zero homomorphism is invariant under a ring isomorphism.

Proof. Split according to whether the element is a non-zero-divisor. The non-zero-divisor branch is 1809; the other branch is sent to zero on both sides. □

Lemma 1812 (Naturality of fraction-field order).
Let $R \cong S$ be an isomorphism of Noetherian integral domains of dimension at most one, and let $K_R \cong K_S$ be a compatible isomorphism of fraction fields. Then

$$\mathrm{ord}_S(e_K(x)) = \mathrm{ord}_R(x) \qquad (x \in K_R).$$

Proof. The assertion is immediate for $x = 0$. Otherwise write $x = a/b$ with a and b non-zero-divisors. Compatibility identifies $e_K(x)$ with $e(a)/e(b)$, and the result follows by applying 1811 to numerator and denominator. □

Definition 1813 (Function field of a nonempty open subscheme). If X is integral and $U \subseteq X$ is nonempty and open, restriction induces a canonical isomorphism

$$K(U) \;\cong\; K(X).$$

Lemma 1814 (Compatibility of the function-field isomorphism).
Let $Y \in X^{(1)}$ have generic point in a nonempty integral open U. The diagram

$$\begin{tikzcd} \mathcal{O}_{U,Y|_U} \arrow[r] \arrow[d, "\sim"'] & K(U) \arrow[d, "\sim"] \\ \mathcal{O}_{X,Y} \arrow[r] & K(X) \end{tikzcd}$$

commutes.

Proof. Both composites send a germ represented on an open neighbourhood of η_Y to the same germ at the generic point. □

Lemma 1815 (Fraction-field order across an open immersion).
Suppose the local rings in the preceding square are Noetherian domains of dimension at most one. Any compatible isomorphism $K(U) \cong K(X)$ preserves their fraction-field orders.

Proof. Apply 1812 to the stalk isomorphism and the compatible function-field isomorphism. □

Definition 1816 (Order along a prime divisor)**.**
Let X be integral and locally Noetherian, and let $Y \in X^{(1)}$ have local ring of dimension at most one. The *order of vanishing* of $f \in K(X)$ along Y is the integer

$$\mathrm{ord}_Y(f) = v_Y(f).$$

We use the harmless total convention $\mathrm{ord}_Y(0) = 0$.

Lemma 1817 (Order is invariant under restriction to an open)**.**
Let X be integral, locally Noetherian, and regular in codimension one. If $Y \in X^{(1)}$ meets a nonempty integral open U, then for every $f \in K(U)$,

$$\mathrm{ord}_Y(f) = \mathrm{ord}_{Y|_U}(f),$$

where the left side uses $K(U) \cong K(X)$.

Proof. The compatibility square of 1814 supplies the hypothesis of 1815. ☐

33.3.2 Valuation identities

Lemma 1818 (Order of zero)**.** *For every prime divisor Y, $\mathrm{ord}_Y(0) = 0$.*

Proof. This is the total convention in 1816. ☐

Lemma 1819 (Order of one)**.** *For every prime divisor Y, $\mathrm{ord}_Y(1) = 0$.*

Proof. The valuation of a multiplicative unit is zero. ☐

Lemma 1820 (Order of a product)**.** *If $f, g \in K(X)$ are nonzero, then*

$$\mathrm{ord}_Y(fg) = \mathrm{ord}_Y(f) + \mathrm{ord}_Y(g).$$

Proof. This is multiplicativity of the discrete valuation, written additively. ☐

Lemma 1821 (Order of an inverse)**.** *For $f \in K(X)$,*

$$\mathrm{ord}_Y(f^{-1}) = -\,\mathrm{ord}_Y(f).$$

Proof. For $f \neq 0$, additivity applied to $ff^{-1} = 1$ gives the result. The total convention makes the equality true also for $f = 0$. ☐

Lemma 1822 (Order of a unit inverse)**.** *For $u \in K(X)^\times$, $\mathrm{ord}_Y(u^{-1}) = -\,\mathrm{ord}_Y(u)$.*

Proof. This is 1821 applied to the underlying field element. ☐

Lemma 1823 (Order is insensitive to sign)**.** *For $f \in K(X)$, $\mathrm{ord}_Y(-f) = \mathrm{ord}_Y(f)$.*

Proof. Since $(-f)^2 = f^2$, additivity gives twice each side; cancellation in $\mathbb{Z}$ proves the equality. ☐

Lemma 1824 (Order of a power)**.** *If $f \neq 0$ and $n \in \mathbb{N}$, then*

$$\mathrm{ord}_Y(f^n) = n\,\mathrm{ord}_Y(f).$$

Proof. Induct on n, using 1819 at zero and 1820 at the successor step. ☐

33.4 Closed points and coefficient degree

Definition 1825 (Divisor associated with a closed point)**.**
Let P be a closed point of a scheme C. Define

$$[P] = \begin{cases} 1 \cdot \langle P, h \rangle, & \text{if } h : \mathrm{coheight}(P) = 1, \\ 0, & \text{otherwise.} \end{cases}$$

Lemma 1826 (Closed points of a nonsingular curve are prime divisors)**.**
Let C be a nonsingular integral curve over an algebraically closed field. Every closed point $P \in C$ has coheight one, and the divisor $[P]$ of 1825 is the corresponding prime divisor with multiplicity one.

Proof. A closed point of an integral curve is distinct from the generic point. Since the curve has dimension one, the specialization chain from the closed point to the generic point has length one. Thus $\mathrm{coheight}(P) = 1$, and the positive branch of 1825 gives the asserted divisor. ☐

Lemma 1827 (Closed point in the codimension-one branch)**.**
If $h : \mathrm{coheight}(P) = 1$, then $[P] = 1 \cdot \langle P, h \rangle$.

Proof. Select the positive branch in 1825. ☐

Lemma 1828 (Closed point outside the codimension-one branch)**.**
If $\mathrm{coheight}(P) \neq 1$, then $[P] = 0$.

Proof. Select the negative branch in 1825. ☐

Definition 1829 (Coefficient degree)**.**
For any scheme X, define the coefficient degree of $D = \sum_Y n_Y Y$ by

$$\deg(D) = \sum_Y n_Y.$$

If X is a nonsingular curve over an algebraically closed field, this is the usual degree because every closed point has residue degree one.

Theorem 1830 (Coefficient degree is additive)**.**
Coefficient degree defines a homomorphism

$$\deg \colon \mathrm{Div}(X) \longrightarrow \mathbb{Z}.$$

Proof. The sum of coefficients is additive under addition of finite formal sums. ☐

Lemma 1831 (Bundled degree agrees with coefficient degree)**.** *For every divisor D, the homomorphism of 1830 takes D to $\deg(D)$.*

Proof. Both sides are the sum of the coefficients of D. ☐

Lemma 1832 (Degree of zero)**.** *One has $\deg(0) = 0$.*

Proof. The empty finite sum is zero. ☐

Lemma 1833 (Degree of a sum)**.** *For divisors D_1, D_2, $\deg(D_1 + D_2) = \deg(D_1) + \deg(D_2)$.*

Proof. This is additivity in 1830. ☐

Lemma 1834 (Degree of a negation)**.** *For a divisor D,* $\deg(-D) = -\deg(D)$.

Proof. Apply the homomorphism of 1830 to $-D$. □

Lemma 1835 (Degree of a difference)**.** *For divisors D_1, D_2,* $\deg(D_1 - D_2) = \deg(D_1) - \deg(D_2)$.

Proof. Combine 1833 and 1834. □

Lemma 1836 (Degree of a one-point divisor)**.** *For $Y \in X^{(1)}$ and $n \in \mathbb{Z}$,* $\deg(nY) = n$.

Proof. The divisor nY has the single coefficient n. □

33.5 Principal divisors

Lemma 1837 (Finite support of the order function)**.**
Let X be a Noetherian integral scheme regular in codimension one. For every $f \in K(X)$, the set

$$\{Y \in X^{(1)} \mid \operatorname{ord}_Y(f) \neq 0\}$$

is finite.

Proof. If $f = 0$, the support is empty by convention. Otherwise choose a finite affine open cover $X = \bigcup_i \operatorname{Spec} R_i$. On one chart, write $f = a/b$. If the height-one prime $\mathfrak{p}_Y$ contains neither a nor b, both become units in $(R_i)_{\mathfrak{p}_Y}$, so $\operatorname{ord}_Y(f) = 0$. Hence every point of the support is a height-one prime containing a or b. Such a prime is minimal over (a) or (b), and a Noetherian ring has only finitely many minimal primes over an ideal. Taking the finite union over the cover proves the claim. □

Remark 1838. Global Noetherianness cannot be replaced by local Noetherianness. The affine line with infinitely many origins is integral, locally Noetherian, and regular in codimension one, but its coordinate function has order one at every origin. The scheme is not quasi-compact.

Definition 1839 (Principal divisor)**.**
For a nonzero $f \in K(X)$, its principal divisor is

$$\operatorname{div}(f) = \sum_{Y \in X^{(1)}} \operatorname{ord}_Y(f)\, Y.$$

The sum is finite by 1837.

Lemma 1840 (Coefficient of a principal divisor)**.** *For $Y \in X^{(1)}$,*

$$\operatorname{div}(f)(Y) = \operatorname{ord}_Y(f).$$

Proof. This is the coefficient formula in the defining finite sum. □

Lemma 1841 (Principal divisor of one)**.** *One has* $\operatorname{div}(1) = 0$.

Proof. Every coefficient is $\operatorname{ord}_Y(1) = 0$. □

Theorem 1842 (The principal-divisor homomorphism)**.**
The assignment

$$K(X)^\times \longrightarrow \operatorname{Div}(X), \qquad f \longmapsto \operatorname{div}(f)$$

is a homomorphism from the multiplicative group to the additive group.

Proof. Compare coefficients at every Y. The product formula for the order gives $\operatorname{div}(fg)(Y) = \operatorname{div}(f)(Y) + \operatorname{div}(g)(Y)$, and 1819 treats the identity. □

33.6 Degree under finite morphisms

Definition 1843 (Pullback of a divisor along a finite morphism of curves)**.**

Let $\varphi\colon C \to D$ be a finite morphism of nonsingular curves over an algebraically closed field. For a closed point $Q \in D$, choose a local parameter t at Q and set

$$\varphi^*[Q] = \sum_{\varphi(P)=Q} \operatorname{ord}_P(t)\,[P].$$

This is independent of the local parameter and extends linearly to $\varphi^*\colon \operatorname{Div}(D) \to \operatorname{Div}(C)$.

Theorem 1844 (Degree of a finite pullback)**.**

If $\varphi\colon C \to D$ is a finite morphism of complete nonsingular curves over an algebraically closed field and $E \in \operatorname{Div}(D)$, then

$$\deg(\varphi^* E) = \deg(\varphi)\deg(E).$$

Proof. It suffices to take $E = [Q]$. On an affine neighbourhood $\operatorname{Spec} B$ of Q, let A be the integral closure of B in $K(C)$, localize at Q, and write the resulting finite torsion-free $\mathcal{O}_{D,Q}$-module as a free module of rank $[K(C) : K(D)] = \deg(\varphi)$. If t is a local parameter at Q, the dimension of A/tA is this rank. Decomposing A/tA at the points P over Q identifies the same dimension with $\sum_{\varphi(P)=Q} \operatorname{ord}_P(t)$, which is $\deg(\varphi^*[Q])$. □

Lemma 1845 (A rational function gives a finite map to the projective line)**.**

Let C be a complete nonsingular curve over an algebraically closed field k, and let $f \in K(C)\setminus k$. There is a finite morphism $\varphi_f\colon C \to \mathbb{P}^1_k$ inducing the inclusion $k(f) \subseteq K(C)$, and

$$\operatorname{div}(f) = \varphi_f^*([0] - [\infty]).$$

Proof. The rational function defines a morphism to $\mathbb{P}^1$ because C is complete and nonsingular. It is nonconstant, hence dominant; the induced extension $K(C)/k(f)$ is finite, so the morphism is finite. At every closed point, the pullback multiplicity of 0 or ∞ is respectively the order of the zero or the pole of f. Equality of all coefficients gives the divisor formula. □

Theorem 1846 (Principal divisors have degree zero)**.**

Let C be a complete nonsingular curve over an algebraically closed field. For every nonzero $f \in K(C)$,

$$\deg(\operatorname{div}(f)) = 0.$$

Proof. If f is constant, its order at every point is zero. Otherwise 1845 gives a finite morphism $\varphi_f\colon C \to \mathbb{P}^1$ with $\operatorname{div}(f) = \varphi_f^*([0] - [\infty])$. Therefore

$$\deg(\operatorname{div}(f)) = \deg(\varphi_f)\deg([0] - [\infty]) = \deg(\varphi_f)(1-1) = 0$$

by 1844. □

33.7 Positive parts

Definition 1847 (Positive part of a Weil divisor)**.**

For $D = \sum_Y n_Y Y$, define

$$D_+ = \sum_Y \max(n_Y, 0)\,Y.$$

Lemma 1848 (Positive part of zero)**.** *One has* $0_+ = 0$.

Proof. Every coefficient is $\max(0,0) = 0$. □

Lemma 1849 (Positive part of a one-point divisor)**.** *For* $Y \in X^{(1)}$ *and* $n \in \mathbb{Z}$,

$$(nY)_+ = \max(n,0)Y.$$

Proof. This follows coefficientwise from the definition. □

Lemma 1850 (Degree of a positive part)**.** *If* $D = \sum_Y n_Y Y$, *then*

$$\deg(D_+) = \sum_Y \max(n_Y, 0).$$

Proof. Expand the two definitions and compare the finite sums. □

Lemma 1851 (Sum over positive coefficients)**.** *For a finitely supported function* $a \mapsto n_a \in \mathbb{Z}$,

$$\sum_a \max(n_a, 0) = \sum_{a:\, n_a > 0} n_a.$$

Proof. Split each coefficient according to whether it is positive. Positive coefficients contribute themselves to both sides and all other coefficients contribute zero. □

Lemma 1852 (A simple zero contributes positive degree)**.**
If $f \neq 0$ *and* $\operatorname{ord}_{Y_0}(f) = 1$, *then*

$$1 \leq \deg(\operatorname{div}(f)_+).$$

Proof. In the sum of nonnegative coefficients, the Y_0-term is $\max(\operatorname{ord}_{Y_0}(f), 0) = 1$; every other term is nonnegative. □

33.8 Linear equivalence and the divisor class group

Definition 1853 (Linear equivalence)**.**
Two divisors $D, D' \in \operatorname{Div}(X)$ are linearly equivalent, written $D \sim D'$, if there is a nonzero $f \in K(X)$ such that

$$D - D' = \operatorname{div}(f).$$

Lemma 1854 (Linear equivalence is an equivalence relation)**.** *Linear equivalence is reflexive, symmetric, and transitive.*

Proof. Reflexivity uses $\operatorname{div}(1) = 0$. If $D - D' = \operatorname{div}(f)$, then $D' - D = \operatorname{div}(f^{-1})$, proving symmetry. Adding the equations for $D \sim D'$ and $D' \sim D''$, and using $\operatorname{div}(fg) = \operatorname{div}(f) + \operatorname{div}(g)$, proves transitivity. □

Definition 1855 (Divisor class group)**.** The divisor class group is

$$\operatorname{Cl}(X) = \operatorname{Div}(X) / \operatorname{im}\big(K(X)^\times \xrightarrow{\operatorname{div}} \operatorname{Div}(X)\big).$$

Theorem 1856 (Degree on the divisor class group of a curve)**.** *On a complete nonsingular curve over an algebraically closed field, coefficient degree descends to a homomorphism*

$$\deg\colon \operatorname{Cl}(C) \longrightarrow \mathbb{Z}.$$

Proof. By 1846, the degree homomorphism vanishes on the subgroup of principal divisors, hence factors uniquely through the quotient. □

Chapter 34

Adelic Riemann–Roch: the repartition cokernel (RR.A)

34.1 The two-chart model

Let C be a smooth proper geometrically integral curve over a field k, and let $K = k(C)$. The divisor theory of 33 assigns to every prime divisor P an order $v_P\colon K^\times \to \mathbb{Z}$. The Mayer–Vietoris theory of 4 computes the first cohomology of the structure sheaf from two affine charts.

The concrete-repartition idea. Classically (Weil), for the function field $K = k(C)$ one forms the *adele* (repartition) space $\mathbb{A}_K = \{(x_P)_P \in \prod_P K : v_P(x_P) \geq 0 \text{ for almost all } P\}$, its bounded subspaces $\mathbb{A}_K(D) = \{(x_P) : v_P(x_P) \geq -D(P)\ \forall P\}$, and the two Riemann–Roch spaces

$$L(D) = K \cap \mathbb{A}_K(D) = \{f \in K : \operatorname{div} f + D \geq 0\}, \qquad H^1(D) = \mathbb{A}_K/(\mathbb{A}_K(D) + K).$$

The genus is $g = \dim_k H^1(0)$, and the index of speciality is $i(D) = \dim_k H^1(D)$. A two-affine cover gives a finite presentation of the same cohomology without first constructing the restricted product.

The 2-affine-cover cokernel. Fix affine opens U_0, U_1 covering C, with affine intersection, and write

$$A_i = \Gamma(U_i, \mathcal{O}_C), \qquad A_{01} = \Gamma(U_0 \cap U_1, \mathcal{O}_C).$$

The degree-1 Čech cohomology of the two-element cover is the cokernel of the difference map $(f_0, f_1) \mapsto f_0|_{01} - f_1|_{01}$:

$$\check{H}^1 \;=\; A_{01}\big/(A_0 + A_1).$$

For a Weil divisor D the *twisted* version replaces each ring of sections by the Riemann–Roch sheaf of sections $\Gamma(U, \mathcal{O}_C(D)) = \{f \in K : v_P(f) \geq -D(P) \text{ for all prime divisors } P \in U\}$, giving

$$L(D) = \Gamma(U_0, \mathcal{O}_C(D)) \cap \Gamma(U_1, \mathcal{O}_C(D)).$$

$$\check{H}^1(D) = \Gamma(U_0 \cap U_1, \mathcal{O}_C(D))\big/\big(\Gamma(U_0, \mathcal{O}_C(D)) + \Gamma(U_1, \mathcal{O}_C(D))\big).$$

The bridge of 34.3 identifies $\check{H}^1(0)$ with $H^1(C, \mathcal{O}_C)$.

Residue-degree bookkeeping. Because k need not be algebraically closed, every place carries a *residue degree*. For a prime divisor P with residue field $\kappa(P) = \mathcal{O}_{C,P}/\mathfrak{m}_P$, set

$$\deg P \;=\; [\kappa(P) : k] \;=\; \dim_k \kappa(P) \;<\; \infty, \qquad \deg_k D \;=\; \sum_P D(P) \deg P.$$

Over an algebraically closed field this agrees with the coefficient degree of 1829; over a general field it does not. In this chapter $\deg D$ always means the residue-weighted degree $\deg_k D$.

We write $\ell(D) = \dim_k L(D)$, $i(D) = \dim_k \check{H}^1(D)$, and $\chi(D) = \ell(D) - i(D)$.

34.2 Valuations and divisor sections

Proposition 1857 (Affine chart is a Dedekind domain with fraction field K)**.** *Let $U \subseteq C$ be a nonempty affine open with coordinate ring $A_U = \Gamma(U, \mathcal{O}_C)$. The canonical map $A_U \hookrightarrow K$ exhibits K as its fraction field. Under the hypothesis that every nonempty affine chart of C is Dedekind, A_U is a Dedekind domain.*

Proof. The fraction-field assertion is localization at the generic point. The Dedekind assertion is the specialization of the stated chart hypothesis to U. □

Definition 1858 (Point valuation)**.**
For a prime divisor P, the discrete valuation ring $\mathcal{O}_{C,P}$ defines its adic valuation

$$\nu_P \colon K \longrightarrow \mathbb{Z}^0_{\mathrm{m}}.$$

Lemma 1859 (Order is the additive normalization of the point valuation)**.**
For every $f \in K$,

$$\mathrm{ord}_P(f) = -\log(\nu_P(f)).$$

Proof. This is the comparison between the height-one-stalk adic valuation and the ordFrac normalization used in 1816. □

Definition 1860 (Residue field and residue degree)**.**
If P lies in a nonempty affine open U, it determines a height-one prime $\mathfrak{p}_P$ of the Dedekind chart $A_U = \Gamma(U, \mathcal{O}_C)$. Its residue field and residue degree are

$$\kappa(P) = A_U/\mathfrak{p}_P = \mathcal{O}_{C,P}/\mathfrak{m}_P, \qquad \deg P = [\kappa(P) : k] = \dim_k \kappa(P).$$

The stalk $\mathcal{O}_{C,P}$ is the localisation $(A_U)_{\mathfrak{p}_P}$ because P lies in U. The definition is independent of the chart, and $\deg P$ is finite by Zariski's lemma because P is a closed point of the finite-type k-scheme C.

Definition 1861 (Residue-weighted degree)**.**
For a Weil divisor $D = \sum_P n_P[P]$ on a curve over k, define its residue-weighted degree by

$$\deg_k(D) = \sum_P n_P[\kappa(P) : k].$$

The sum is finite because a Weil divisor has finite support. If k is algebraically closed, every closed point has residue field k, so $\deg_k$ agrees with the coefficient degree of 1829.

Theorem 1862 (Weighted degree of a principal divisor)**.** *For every $f \in K^\times$,*

$$\deg_k \mathrm{div}(f) = 0.$$

Proof. A constant function has zero order everywhere. Otherwise f defines a finite morphism $\varphi_f\colon C \to \mathbb{P}^1_k$, and $\operatorname{div}(f) = \varphi_f^*([0] - [\infty])$. The residue-weighted degree of a finite pullback is multiplied by $\deg \varphi_f$; since both $[0]$ and $[\infty]$ have degree one, the displayed divisor has degree zero. □

Definition 1863 (Sections bounded by a divisor; the Riemann–Roch space)**.** For an open $U \subseteq C$ and a Weil divisor D (1797) define the additive subgroup of K

$$\Gamma(U, \mathcal{O}_C(D)) \;=\; \{f \in K : f = 0 \text{ or } v_P(f) \geq -D(P) \text{ for every prime divisor } P \text{ with point in } U\}$$

(the disjunct $f = 0$ encodes $v_P(0) = +\infty$). This is an additive subgroup of K. Its global instance gives the *Riemann–Roch space* $L(D) = \Gamma(U_0, \mathcal{O}_C(D)) \cap \Gamma(U_1, \mathcal{O}_C(D))$, equal to $\{f \in K : \operatorname{div} f + D \geq 0\}$ since $U_0 \cup U_1 = C$.

Proof. The set contains 0 ($v_P(0) = +\infty \geq -D(P)$). It is closed under addition by the ultrametric inequality of the discrete valuation $v_P(f + h) \geq \min(v_P f, v_P h) \geq -D(P)$. It is closed under negation because $v_P(-f) = v_P(f)$. The intersection identity for $L(D)$ holds because a prime divisor of C has its point in U_0 or in U_1; imposing $v_P(f) \geq -D(P)$ on the union of the two chart conditions is imposing it at every prime divisor, i.e. $\operatorname{div} f + D \geq 0$. □

Lemma 1864 (Finiteness of the non-integral places on a Dedekind chart)**.** *Let R be a Dedekind domain with fraction field K. For a nonzero $k \in K$ the set $\{v \in \operatorname{Spec}^1 R : v(k) \neq 1\}$ of height-one places at which k is a zero or a pole is finite.*

This is the affine-chart core of the finite-support statement: a curve is covered by finitely many Dedekind charts, and applying the lemma on each chart shows that for $f \in K^\times$ only finitely many prime divisors have $v_P(f) \neq 0$, so $\operatorname{div} f = \sum_P v_P(f)\, P$ is a well-defined Weil divisor. The general scheme-level assembly (finite affine cover plus the height-one/minimal-primes bound) is 1837.

Proof. Apply the finiteness of the support of a nonzero fractional ideal in a Dedekind domain to both k and k^{-1}. A place v with $v(k) \neq 1$ satisfies either $v(k) > 1$ (a zero of k) or $v(k) < 1$, equivalently $v(k^{-1}) > 1$ (a pole of k); each of the two sets of such places is finite, and their union contains every place with $v(k) \neq 1$. □

34.3 The cover cokernel and first cohomology

Definition 1865 (The two-cover cokernel of a sheaf)**.** For a sheaf F of k-modules and an affine Mayer–Vietoris square, define

$$\mathrm{H1Cok}_S(F) = \Gamma(U_0 \cap U_1, F)/\operatorname{im}(\Gamma(U_0, F) \oplus \Gamma(U_1, F) \xrightarrow{f_0 - f_1} \Gamma(U_0 \cap U_1, F)).$$

Definition 1866 (The function-field twist cokernel)**.** Independently of a sheaf structure, the divisor section subgroups define the additive quotient

$$\check{H}^1_{\mathrm{ff}}(D) = \Gamma(U_0 \cap U_1, \mathcal{O}_C(D))/(\Gamma(U_0, \mathcal{O}_C(D)) + \Gamma(U_1, \mathcal{O}_C(D))).$$

Proposition 1867 (Comparison at the zero divisor)**.** *There is a canonical isomorphism*

$$\check{H}^1_{\mathrm{ff}}(0) \simeq_k \mathrm{H1Cok}_S(\mathcal{O}_C).$$

Proof. On every nonempty affine open of the integral normal curve, a rational function is regular exactly when it has nonnegative order at every closed point of that open. This identifies each term and each restriction map in the two difference complexes, hence their cokernels. □

Lemma 1868 (The unnormalized coface differential is the alternating coface sum)**.** *For a cosimplicial object Y in a preadditive category and $n \geq 0$, the degree-n differential of the associated (unnormalized) alternating coface cochain complex is the alternating sum of coface maps,*

$$d^n \;=\; \sum_{i=0}^{n+1} (-1)^i \, Y(\delta_i) \colon Y^n \to Y^{n+1}.$$

Proof. This is the cosimplicial dual of the standard alternating-face-map formula for the (co)chain complex of a simplicial object: the differential of the alternating coface complex is defined as this alternating sum, so the identity is definitional unfolding. □

Definition 1869 (The Čech cosimplicial object of a cover)**.** For a family of opens $\mathcal{U} = (U_i)_{i \in \iota}$ of C and a sheaf F of k-modules, the *Čech cosimplicial object* $Y_{\mathcal{U},F}$ is given in degree n by

$$Y^n_{\mathcal{U},F} = \prod_{j \colon \mathrm{Fin}(n+1) \to \iota} \Gamma\Big(\bigsqcap_a U_{j(a)}, F\Big).$$

Its coface maps $\delta_{i_0} : Y^n \to Y^{n+1}$ are induced by restriction along the shrinking of the intersection when an index is duplicated. The Čech cochain complex is by definition the alternating coface complex of $Y_{\mathcal{U},F}$: cechCochain $C\,F\,\mathcal{U}$ = the alternating coface complex of $Y_{\mathcal{U},F}$ (1868).

Proof. Immediate from the definition of cechCochain as the alternating coface complex associated to the cosimplicial object obtained by evaluating the Čech nerve of $\mathcal{U}$ against F. □

Lemma 1870 (Cofaces are restriction maps)**.** *For every finite index family $j \colon \mathrm{Fin}\, m \to \iota$, the categorical product $\prod_a U_{j(a)}$ equals the intersection $\bigsqcap_a U_{j(a)}$ of opens of C. Consequently, for $n \geq 0$, $i_0 \in \mathrm{Fin}(n+2)$, and $j \colon \mathrm{Fin}(n+2) \to \iota$, the coface map $\delta_{i_0} : Y^n_{\mathcal{U},F} \to Y^{n+1}_{\mathcal{U},F}$, projected to the j-indexed factor of the target, is the restriction map*

$$\Gamma\Big(\bigsqcap_a U_{j(\delta_{i_0}(a))}, F\Big) \longrightarrow \Gamma\Big(\bigsqcap_a U_{j(a)}, F\Big)$$

along the inclusion $\bigsqcap_a U_{j(a)} \subseteq \bigsqcap_a U_{j(\delta_{i_0}(a))}$ (dropping the index i_0 enlarges the intersection).

Proof. The abstract categorical product of a finite family of opens, computed in the poset Opens C, is its infimum, by the universal property of both constructions. Under this identification the coface map's action on the j-indexed factor — reindexing the source factor along $j \circ \delta_{i_0}$ followed by the canonical comparison map between the two products — is exactly the restriction map along the inclusion of the larger-index intersection into the smaller-index one; since Opens C is a poset, this comparison map is the unique restriction along that inclusion. □

Lemma 1871 (The degree-0 and degree-1 Čech differentials)**.** *In terms of the coface maps of $Y_{\mathcal{U},F}$, the Čech differentials in degrees 0 and 1 are*

$$d^0 = \delta_0 - \delta_1, \qquad\qquad d^1 = \delta_0 - \delta_1 + \delta_2.$$

Proof. Immediate from 1868 at $n = 0$ and $n = 1$, expanding the alternating sum over Fin 2 and Fin 3 respectively. □

Proposition 1872 (Two-element Čech H^1 is the cokernel)**.** *For any sheaf F of k-modules on C and any affine Mayer–Vietoris square S with cover family $\mathcal{U} = \{U_0, U_1\}$, there is a k-linear isomorphism*

$$\mathrm{cechCohomology}\, C\, F\, \mathcal{U}\, 1 \;\simeq_k\; \Gamma(U_0 \cap U_1, F) \big/ \big(\Gamma(U_0, F) + \Gamma(U_1, F)\big),$$

the image of $\Gamma(U_0, F) \oplus \Gamma(U_1, F)$ *under* $(f_0, f_1) \mapsto f_0|_{01} - f_1|_{01}$ *being identified with the sum* $\Gamma(U_0, F) + \Gamma(U_1, F)$ *inside* $\Gamma(U_0 \cap U_1, F)$*. Specialising* $F = \mathrm{toModuleKSheaf}\, C$ *gives*

$$\mathrm{cechCohomology}\, C\, (\mathrm{toModuleKSheaf}\, C)\, S.\mathrm{cover}\, 1 \ \simeq_k \ \check{H}^1 = A_{01}/(A_0 + A_1).$$

Proof. Unfold $\mathrm{cechCohomology}\, C\, F\, \mathcal{U}\, 1$ as the homology at the middle term of the short complex $C^0 \xrightarrow{d^0} C^1 \xrightarrow{d^1} C^2$ of the Čech cochain complex, i.e. $\ker d^1 / \operatorname{im}(C^0 \to \ker d^1)$. For the two-element index set $\mathcal{U} = \{U_0, U_1\}$ every triple of indices $x\colon \mathrm{Fin}\, 3 \to \{0,1\}$ repeats a value, so by 1871 the degree-1 differential d^1, evaluated in each of the eight index components of $C^2 = \prod_x \Gamma(\prod_a U_{x(a)}, F)$, collapses by the coface-restriction identities of 1870 (parallel restrictions cancel and diagonal restrictions round-trip). Consequently $\ker d^1 \subseteq C^1 = \Gamma(U_0, F) \times \Gamma(U_1, F) \times \Gamma(U_1, F) \times \Gamma(U_0, F)$ (indexed by the four functions $\mathrm{Fin}\, 2 \to \{0,1\}$) consists exactly of the tuples $(0, q, -q, 0)$ for $q \in \Gamma(U_0 \cap U_1, F)$, giving a k-linear isomorphism $\Gamma(U_0 \cap U_1, F) \xrightarrow{\sim} \ker d^1$, $q \mapsto (0, q, -q, 0)$. Under this identification the image of $d^0\colon \Gamma(U_0, F) \oplus \Gamma(U_1, F) \to \ker d^1$, $(f_0, f_1) \mapsto (0,\, f_0|_{01} - f_1|_{01},\, f_1|_{01} - f_0|_{01},\, 0)$, corresponds to $\{f_0|_{01} - f_1|_{01} : f_i \in \Gamma(U_i, F)\} = \Gamma(U_0, F) + \Gamma(U_1, F)$ inside $\Gamma(U_0 \cap U_1, F)$ (each $\Gamma(U_i, F)$ being closed under negation). Composing the two identifications gives the claimed k-linear isomorphism $\mathrm{cechCohomology}\, C\, F\, \mathcal{U}\, 1 \simeq_k \Gamma(U_0 \cap U_1, F)/(\Gamma(U_0, F) + \Gamma(U_1, F))$. □

Proposition 1873 (First cohomology vanishes on the affine pieces)**.** *If* $U \subseteq C$ *is affine, then* $H^1(U, \mathcal{O}_C|_U) = 0$*. In particular this holds for both members of an affine Mayer–Vietoris square.*

Proof. This is affine vanishing for the quasi-coherent structure sheaf. □

Proposition 1874 (The degree-one Mayer–Vietoris quotient)**.** *Let* F *be a sheaf of* k*-modules. If* $H^1(U_0, F) = H^1(U_1, F) = 0$*, then the connecting homomorphism in the Mayer–Vietoris sequence induces an isomorphism*

$$\Gamma(U_0 \cap U_1, F)/\operatorname{im}(\Gamma(U_0, F) \oplus \Gamma(U_1, F)) \simeq_k H^1(U_0 \cup U_1, F).$$

Proof. Exactness identifies the kernel of the connecting map with the image of the difference of restrictions. Vanishing on the two pieces makes the connecting map surjective, so the first isomorphism theorem gives the displayed quotient. □

Theorem 1875 (The structure-sheaf H^1 is the cover cokernel)**.** *Every affine Mayer–Vietoris square on* C *gives a* k*-linear isomorphism*

$$\mathrm{HModule}\, k\, (\mathrm{toModuleKSheaf}\, C)\, 1 \ \simeq_k \ \check{H}^1 \ = \ A_{01}/(A_0 + A_1).$$

Proof. Apply 1874 to the structure sheaf; its two vanishing hypotheses are 1873. Since $U_0 \cup U_1 = C$, its target is the global first cohomology. □

34.4 Finiteness via the projective line

By 1875, it remains to prove that $A_{01}/(A_0 + A_1)$ is finite-dimensional. We pull back the two standard charts of $\mathbb{P}^1$ along a finite morphism.

Proposition 1876 (A nonconstant map to $\mathbb{P}^1$ is finite)**.** *If a* k*-morphism* $\pi\colon C \to \mathbb{P}^1_k$ *takes two distinct values, then* π *is finite.*

Proof. Let $\pi\colon C \to \mathbb{P}^1_k$ be a nonconstant k-morphism (1877), with $\pi(x_1) \neq \pi(x_2)$ for some $x_1, x_2 \in C$. Since $\mathbb{P}^1_k$ is separated and of finite type over k, 1887 applied to π shows π is a finite morphism. □

Definition 1877 (A nonconstant map to $\mathbb{P}^1$)**.** The class ExistsNonconstantMapToP1 C asserts the existence of a k-morphism $\pi\colon C \to \mathbb{P}^1_k$ taking two distinct values.

Proposition 1878 (A nonconstant map to the integral projective line)**.** *A smooth proper geometrically integral curve admits a morphism $f\colon C \to \operatorname{Proj} \mathbb{Z}[X_0, X_1]$ taking two distinct values.*

Proof. A transcendental element of K/k defines a rational map to the projective line. Properness and regularity in dimension one extend it across every closed point, and transcendence makes the map nonconstant. □

Proposition 1879 (Integral-model bridge to $\mathbb{P}^1_k$)**.** ExistsNonconstantMapToProjInt C *implies* ExistsNonconstantMapToP1 C.

Proof. The concrete $\mathbb{P}^1_k = \mathbb{P}(\mathrm{ULift}(\mathrm{Fin}\,2); \operatorname{Spec} k)$ is, by construction, the fibre product $\operatorname{Spec} k \times_\top \operatorname{Proj} \mathbb{Z}[X_0, X_1]$ over the *terminal* scheme $\top$. Since any two morphisms into $\top$ agree, the compatibility square required by the universal property of this pullback is automatic: given $f\colon C \to \operatorname{Proj} \mathbb{Z}[X_0, X_1]$ and the ambient structure map $C \to \operatorname{Spec} k$, the two assemble into a unique $g\colon C \to \mathbb{P}^1_k$ with g over $\operatorname{Spec} k$ and g composed with the projection $\mathbb{P}^1_k \to \operatorname{Proj} \mathbb{Z}[X_0, X_1]$ equal to f. If $f(x_1) \neq f(x_2)$ then, composing with the projection, $g(x_1) \neq g(x_2)$; hence g is a nonconstant k-morphism $C \to \mathbb{P}^1_k$. □

Lemma 1880 (Quasi-finite algebras do not raise Krull dimension)**.** *For a quasi-finite R-algebra A,* ringKrullDim $A \leq$ ringKrullDim R*.*

Proof. Quasi-finiteness gives incomparability of primes over the contraction map $\operatorname{Spec} A \to \operatorname{Spec} R$: if two primes of A contract to the same prime of R and one is contained in the other, they are equal. Hence the contraction map is strictly monotone on chains of primes, so it strictly increases no chain length in the wrong direction and ringKrullDim $A \leq$ ringKrullDim R. □

Lemma 1881 (Dimension one for standard-smooth curves)**.** *Let S be standard-smooth of relative dimension* 1 *over a field k, and let q be a prime of S. Every localisation S_q of S at q satisfies* ringKrullDim $S_q \leq 1$*.*

Proof. Standard smoothness of relative dimension one makes $\Omega_{S/k}$ free of rank one; writing the spanning vector as a finite combination of exact differentials, some dt (for a generator t of S) has coefficient $a \notin q$ in this basis. Over S_q the element a is a unit, so dt spans $\Omega_{S_q/k}$; the k-algebra map $k[X] \to S_q$, $X \mapsto t$, then makes S_q formally unramified over $k[X]$ ($\Omega_{S_q/k[X]} = 0$) and essentially of finite type, i.e. *quasi-finite* over $k[X]$. Since $k[X]$ is a PID, ringKrullDim $k[X] = 1$, and 1880 gives ringKrullDim $S_q \leq 1$. □

Lemma 1882 (Stalks of the curve have dimension ≤ 1)**.** *Every stalk $\mathcal{O}_{C,z}$ of the curve C (smooth of relative dimension one over* $\operatorname{Spec} k$*) has* ringKrullDim $\mathcal{O}_{C,z} \leq 1$*.*

Proof. As $\operatorname{Spec} k$ is a single point, z has a standard smooth chart V of relative dimension one over all of $\operatorname{Spec} k$; the stalk $\mathcal{O}_{C,z}$ is the localisation of $\Gamma(V, \mathcal{O}_C)$ at the prime of z, so 1881 applies directly. □

Lemma 1883 (Coheight ≤ 1 on the curve)**.** *Every point z of C has topological coheight* coheight $z \leq 1$*.*

Proof. Immediate from 1882 and the stalk–coheight bridge 1719:

$$\operatorname{coheight} z = \operatorname{ringKrullDim} \mathcal{O}_{C,z}.$$

□

Lemma 1884 (Non-generic points of the curve are closed)**.** *On an irreducible scheme all of whose points have coheight ≤ 1, every point z other than the generic point is closed.*

Proof. If $\{z\}$ were not closed, its closure would contain a point $z' \neq z$ with $z \rightsquigarrow z'$, giving a strict specialisation chain $z' \prec z \prec \eta$ (η the generic point, since $z \neq \eta$); each strict step adds at least one to coheight, so $\operatorname{coheight} z' \geq 2$, contradicting the hypothesis. □

Lemma 1885 (Proper closed subsets of the curve are finite)**.** *On an irreducible Noetherian sober scheme all of whose points have coheight ≤ 1, every closed subset $F \neq C$ is finite.*

Proof. A Noetherian space has finitely many irreducible components; a proper closed F does not contain the generic point, so each of its (finitely many) irreducible components has a generic point $\zeta \neq \eta$, which is closed by 1884. A sober irreducible closed set with closed generic point is that singleton, so each component is a single point and F is finite. □

Lemma 1886 (Fibres of a nonconstant map are finite)**.** *Let $\pi\colon C \to Y$ be a k-morphism to a locally-of-finite-type k-scheme Y taking two distinct values. Every fibre $\pi^{-1}(y)$ is finite.*

Proof. If $\{y\}$ is closed, $\pi^{-1}(y)$ is a closed subset of C that is not all of C (since π is nonconstant), hence finite by 1885. If $\{y\}$ is not closed, then y is not a closed point of the Jacobson scheme Y; since closed points of C map to closed points of Y, the fibre $\pi^{-1}(y)$ avoids every closed point of C (1884: every non-generic point of C is closed), so $\pi^{-1}(y) \subseteq \{\eta\}$ is finite. □

Proposition 1887 (Nonconstant morphisms from the curve are finite)**.** *Let Y be a separated, locally-of-finite-type k-scheme. Any k-morphism $\pi\colon C \to Y$ taking two distinct values is a* finite *morphism.*

Proof. π is proper: π composed with the separated structure map $Y \to \operatorname{Spec} k$ is the proper structure map $C \to \operatorname{Spec} k$, and properness cancels along a separated morphism, so π itself is proper. π is locally quasi-finite by finiteness of its fibres (1886). A proper, locally quasi-finite morphism is finite (Zariski's main theorem), so π is finite. □

Proposition 1888 (Charts are module-finite over the $\mathbb{P}^1$ charts)**.** *Take x as in 1876 and the cover $U_0 = \pi^{-1}(\mathbb{A}^1_x)$, $U_1 = \pi^{-1}(\mathbb{A}^1_{1/x})$; both are affine, since the preimage of an affine open under a finite (hence affine) morphism is affine. Then $A_0 = \Gamma(U_0, \mathcal{O}_C)$ is a finite module over $k[x]$, and $A_1 = \Gamma(U_1, \mathcal{O}_C)$ is a finite module over $k[1/x]$: for a finite morphism ρ and an affine open V of its target Y, $\Gamma(\rho^{-1}V, \mathcal{O}_X)$ is always a finite module over $\Gamma(V, \mathcal{O}_Y)$ (the integral-closure finiteness of a finite morphism, applied to $\rho = \pi$, $Y = \mathbb{P}^1_k$). Packaging U_0, U_1 and their affine overlap as a 2-affine cover of C is the pulled-back cover square consumed by the keystone (1890). (The sharper classical description of A_0, A_1, and A_{01} as integral closures, generically free of rank $n = [K : k(x)]$, is not needed for this argument.)*

Proof. Affineness of U_0 and U_1 is preimage-stability of affine opens under a finite morphism. Module-finiteness of A_0 over $k[x]$ (and symmetrically of A_1 over $k[1/x]$) is the general fact that for a finite morphism $\rho\colon X \to Y$ and an affine open V of Y, the induced ring map $\Gamma(V, \mathcal{O}_Y) \to \Gamma(\rho^{-1}V, \mathcal{O}_X)$ makes the target a finite module over the source. □

Lemma 1889 (The $\mathbb{P}^1$ base cokernel vanishes)**.** *The untwisted cokernel of the standard cover of $\mathbb{P}^1_k$ vanishes:*

$$k[x,x^{-1}]\big/\big(k[x]+k[x^{-1}]\big) \;=\; 0, \qquad \textit{i.e.} \qquad H^1(\mathbb{P}^1_k,\mathcal{O}_{\mathbb{P}^1})=0.$$

Proof. As a k-vector space $k[x,x^{-1}]$ has basis the Laurent monomials $\{x^m : m\in\mathbb{Z}\}$. Every monomial x^m with $m\geq 0$ lies in $k[x]$, and every monomial with $m\leq 0$ lies in $k[x^{-1}]$; since each integer is ≥ 0 or ≤ 0, the sum $k[x]+k[x^{-1}]$ contains every basis monomial and hence equals $k[x,x^{-1}]$. Therefore the quotient is zero. This is the genus-zero statement for $\mathbb{P}^1_k$. □

Theorem 1890 (The two-cover cokernel of the curve is finite-dimensional)**.** *Let $\pi\colon C\to Y$ be a finite k-morphism of* Spec k*-schemes whose target carries* Laurent chart data*: affine opens $V_0\cup V_1=Y$, coordinates $x\in\Gamma(V_0)$, $y\in\Gamma(V_1)$ with $V_0\cap V_1=D(x)=D(y)$, $x\cdot y=1$ on the overlap, and $\Gamma(V_0)=\sum_n k\,x^n$, $\Gamma(V_1)=\sum_n k\,y^n$ (for $Y=\mathbb{P}^1_k$: $\Gamma(V_0)=k[x]$, $\Gamma(V_1)=k[x^{-1}]$). Set $U_i=\pi^{-1}V_i$, $A_i=\Gamma(U_i,\mathcal{O}_C)$, $M=\Gamma(U_0\cap U_1,\mathcal{O}_C)$. Then the two-cover Čech cokernel $\check{H}^1=M/(A_0+A_1)$ is a finite-dimensional k-vector space. In particular, under the gates* HasFiniteMapToP1 C *and* P1HasLaurentChartData k*, some 2-affine cover of C (the pullback of the standard charts) has finite-dimensional $\check{H}^1$.*

Proof. Write $t=\pi^\sharp x|_{U_0\cap U_1}$, $u=\pi^\sharp y|_{U_0\cap U_1}$; by naturality of $\pi^\sharp$ and $x\cdot y=1$ they satisfy $t\cdot u=1$ in M (an abstract Laurent pair). Let $N_0,N_1\subseteq M$ be the images of the restrictions $A_0\to M$, $A_1\to M$; N_0 is stable under multiplication by t and N_1 under multiplication by u, since restriction is a ring map.

Extension. $U_0\cap U_1=\pi^{-1}(V_0\cap V_1)=\pi^{-1}(D(x))=D(\pi^\sharp x)$ is the basic open of $\pi^\sharp x$ inside the affine U_0, so M is the localization $(A_0)_t$; hence for every $m\in M$ there are n and $a\in A_0$ with $t^n m=a|_{U_0\cap U_1}$, and symmetrically $u^n m\in N_1$ for $n\gg 0$.

Ladder spanning. By finiteness of π, A_0 is a finite $\Gamma(V_0)$-module; choose generators $g_1,\dots,g_r$. Since $\Gamma(V_0)$ is the k-span of the powers of x (the geometric source of the Laurent split of 1889), A_0 is the k-span of $\{x_0^j g_i\}$ where $x_0=\pi^\sharp x$; combining with the extension property and $tu=1$, every $m\in M$ lies in the k-span of the two-sided ladder $\{t^j\bar{g}_i\}\cup\{u^j\bar{g}_i\}$, $j\in\mathbb{N}$, with $\bar{g}_i\in N_0$ the images of the generators.

Middle band. The rungs $t^j\bar{g}_i$ lie in N_0 (t-stability); choosing n_i with $u^{n_i}\bar{g}_i\in N_1$ and $N=\max_i n_i$, all rungs $u^j\bar{g}_i$ with $j\geq N$ lie in N_1 (u-stability). Hence modulo N_0+N_1 the module M is spanned by the finitely many middle rungs $\{u^j\bar{g}_i : i\leq r,\ j<N\}$, so $\check{H}^1=M/(N_0+N_1)$ is finite-dimensional over k. □

34.4.1 The projective-line chart rings

The two standard charts of $\mathbb{P}^1_k$ have coordinate rings $k[x]$ and $k[x^{-1}]$, and their intersection has coordinate ring $k[x,x^{-1}]$. The following construction obtains these charts by base change from the degree-zero away rings of Proj $\mathbb{Z}[X_0,X_1]$.

Definition 1891 (The coordinate fraction X_j/X_i)**.** For $i,j\in\{0,1\}$, the coordinate fraction X_j/X_i is the element of the degree-zero away ring $(\mathbb{Z}[X_0,X_1]_{X_i})_0$ with numerator X_j and denominator X_i (both homogeneous of degree 1).

Definition 1892 (The pulled-back chart coordinates)**.** The chart coordinates $x=X_1/X_0\in\Gamma(V_0)$ and $y=X_0/X_1\in\Gamma(V_1)$ on the standard two-chart cover V_0,V_1 of the concrete model $\mathbb{P}^1_k$, obtained by pulling back the away-section fractions $X_1/X_0\in$ Away $\mathcal{A}\,X_0$ and $X_0/X_1\in$ Away $\mathcal{A}\,X_1$ (1891) along the projection $\mathbb{P}^1_k\to$ Proj $\mathbb{Z}[X_0,X_1]$.

Lemma 1893 (The monomial-to-power identity)**.** *For $a, b \in \mathbb{N}$, the degree-$(a+b)$ monomial fraction $X_0^a X_1^b / X_0^{a+b}$ in $(\mathbb{Z}[X_0, X_1]_{X_0})_0$ equals $(X_1/X_0)^b$.*

Proof. Both sides are the class of $X_0^a X_1^b$ over the denominator X_0^{a+b} in the localisation: cross-multiplying, $(X_0)^b \cdot (X_0^a X_1^b) = X_0^{a+b} \cdot X_1^b$, a monomial identity in the commutative monoid of monomials. □

Theorem 1894 (Span of the projective-line chart)**.** *The degree-zero away ring $(\mathbb{Z}[X_0, X_1]_{X_0})_0$ is spanned, as a module over its constant subring $(\mathbb{Z}[X_0, X_1])_0$, by the powers of the coordinate fraction X_1/X_0.*

Proof. Every element of the away ring is a fraction p/X_0^N with p homogeneous of degree N ($N \geq 0$). Expanding p over its monomial support $\{c_d X_0^{d_0} X_1^{d_1}\}$, homogeneity forces $d_0 + d_1 = N$ for every occurring exponent vector d, so each summand is, by 1893, the constant c_d times $(X_1/X_0)^{d_1}$. Summing over the (finite) support writes p/X_0^N as a $(\mathbb{Z}[X_0, X_1])_0$-combination of powers of X_1/X_0. □

Theorem 1895 (Laurent chart data on $\mathbb{P}^1$)**.** *For every field k, the standard two-chart cover of $\mathbb{P}^1_k$ carries Laurent chart data: its chart rings are spanned by nonnegative powers of x and x^{-1}, the overlap is both $D(x)$ and $D(x^{-1})$, and the two coordinates multiply to one there.*

Theorem 1896 (Finite-dimensionality of $H^1(C, \mathcal{O}_C)$)**.** *If C is smooth and proper with geometrically integral fibres, then*

$$\text{Module.Finite } k \text{ } (\text{HModule } k \, (\text{toModuleKSheaf} \, C) \, 1).$$

Thus $\text{genus} C$ *is the dimension of this finite-dimensional vector space.*

Proof. Pull back the standard charts along the finite morphism to $\mathbb{P}^1_k$. Their cokernel is finite-dimensional by 1890, and 1875 transports this finiteness to $H^1(C, \mathcal{O}_C)$. □

34.5 The Euler-characteristic ledger

Fix the same two-affine cover. The divisor section groups become k-subspaces of K, and their quotients give a concrete numerical Euler-characteristic calculation.

34.5.1 The twist sequence

Lemma 1897 (Antitonicity of the section subgroup in the open)**.** *For opens $U \subseteq U'$ of C and a Weil divisor D, $\Gamma(U', \mathcal{O}_C(D)) \subseteq \Gamma(U, \mathcal{O}_C(D))$: a larger open imposes the order bound $v_P(f) \geq -D(P)$ at more prime divisors, so its section subgroup is smaller.*

Proof. A nonzero $f \in \Gamma(U', \mathcal{O}_C(D))$ satisfies $v_P(f) \geq -D(P)$ at every prime divisor P with point in U', in particular at every such P with point in the smaller open $U \subseteq U'$, so $f \in \Gamma(U, \mathcal{O}_C(D))$. □

Definition 1898 (The coboundary subgroup)**.** The *coboundary* is the sum of the two chart section subgroups inside K:

$$B(D) = \Gamma(U_0, \mathcal{O}_C(D)) + \Gamma(U_1, \mathcal{O}_C(D)).$$

Lemma 1899 (The coboundary and the linear system land in the overlap)**.** *$B(D) \subseteq \Gamma(U_0 \cap U_1, \mathcal{O}_C(D))$ and $L(D) \subseteq \Gamma(U_0 \cap U_1, \mathcal{O}_C(D))$: each of the two chart sections, and every global section, restricts to the overlap.*

Proof. Immediate from the antitonicity of $\Gamma(-, \mathcal{O}_C(D))$ in the open (1897: a larger open imposes more order conditions, hence restricts into a smaller open's section group), applied to $U_0 \cap U_1 \subseteq U_0$, $U_0 \cap U_1 \subseteq U_1$, and $U_0 \cap U_1 \subseteq \top$. $\square$

Theorem 1900 (Exactness at the L-terms)**.** *Suppose $D \leq D'$ pointwise and $D' - D$ is supported on the overlap $V = U_0 \cap U_1$ (off V, D and D' agree). Then*

$$L(D') \cap \Gamma(V, \mathcal{O}_C(D)) \;=\; L(D).$$

Proof. ($\supseteq$) A global section of $\mathcal{O}_C(D)$ is a global section of $\mathcal{O}_C(D')$ ($D \leq D'$) that trivially restricts into $\Gamma(V, \mathcal{O}_C(D))$. ($\subseteq$) Let $f \in L(D')$ restrict into $\Gamma(V, \mathcal{O}_C(D))$; if $f = 0$ it lies in $L(D)$ trivially. Otherwise, to check $f \in L(D)$ it suffices to bound $v_P(f) \geq -D(P)$ at every prime divisor P: on V this is the hypothesis $f \in \Gamma(V, \mathcal{O}_C(D))$, and off V it is the bound $v_P(f) \geq -D'(P) = -D(P)$ (using $D = D'$ off V) already known from $f \in L(D')$. $\square$

Definition 1901 (The function-field quotient and the window map)**.** The function-field quotient $\check{H}^1_{\mathrm{ff}}(D) = \Gamma(V, \mathcal{O}_C(D))/B(D)$ ($B(D) \subseteq \Gamma(V, \mathcal{O}_C(D))$ by 1899). For $D \leq D'$, the inclusions $\Gamma(V, \mathcal{O}_C(D)) \subseteq \Gamma(V, \mathcal{O}_C(D'))$ and $B(D) \subseteq B(D')$ induce the twist map $\check{H}^1_{\mathrm{ff}}(D) \to \check{H}^1_{\mathrm{ff}}(D')$, and the inclusions $L(D) \subseteq \Gamma(V, \mathcal{O}_C(D))$ induce the *window map*

$$L(D')/L(D) \longrightarrow \Gamma(V, \mathcal{O}_C(D'))\big/\Gamma(V, \mathcal{O}_C(D)),$$

the left-hand map of the four-term ledger sequence of 1905.

Theorem 1902 (The window map is injective)**.** *Under the hypotheses of 1900 ($D \leq D'$, $D' - D$ supported on the overlap), the window map $L(D')/L(D) \to \Gamma(V, \mathcal{O}_C(D'))/\Gamma(V, \mathcal{O}_C(D))$ is injective: this is the exactness of the ledger sequence at the two L-terms.*

Proof. A class $[f] \in L(D')/L(D)$ mapping to zero means $f \in L(D')$ restricts into $\Gamma(V, \mathcal{O}_C(D))$; by 1900 this forces $f \in L(D)$, i.e. $[f] = 0$. $\square$

Definition 1903 (The divisorial sheaf)**.** The divisorial sheaf $\mathcal{O}_C(D)$ is the subsheaf of the constant sheaf K whose sections on a nonempty open U are the rational functions satisfying

$$v_P(f) + D(P) \geq 0 \quad \text{for every prime divisor } P \text{ meeting } U.$$

Its section over the empty open is zero. It is an invertible, hence quasi-coherent, $\mathcal{O}_C$-module.

Proposition 1904 (Comparison with the divisorial sheaf)**.** *For every divisor D, the order-bounded function-field quotient is canonically isomorphic to the first cohomology of the divisorial sheaf:*

$$\check{H}^1_{\mathrm{ff}}(D) \simeq_k H^1(C, \mathcal{O}_C(D)).$$

Proof. The valuation description of $\mathcal{O}_C(D)$ identifies its sections on U_0, U_1, and $U_0 \cap U_1$ with the three divisor section groups. The resulting difference complex has cokernel $\check{H}^1_{\mathrm{ff}}(D)$. Since $\mathcal{O}_C(D)$ is quasi-coherent and both charts are affine, Mayer–Vietoris identifies this cokernel with $H^1(C, \mathcal{O}_C(D))$. $\square$

Proposition 1905 (Twist exact sequence)**.** *For Weil divisors $D \le D'$ there is a four-term exact sequence of k-vector spaces*

$$0 \longrightarrow L(D')/L(D) \longrightarrow \mathcal{Q}(D,D') \longrightarrow \check{H}^1_{\mathrm{ff}}(D) \longrightarrow \check{H}^1_{\mathrm{ff}}(D') \longrightarrow 0,$$

where $\mathcal{Q}(D,D') = H^0(C, \mathcal{O}_C(D')/\mathcal{O}_C(D))$ is the space of global sections of the finite-support quotient sheaf.

Proof. The inclusion of divisorial sheaves has finite-support cokernel $\mathcal{T}$:

$$0 \longrightarrow \mathcal{O}_C(D) \longrightarrow \mathcal{O}_C(D') \longrightarrow \mathcal{T} \longrightarrow 0.$$

A finite-support sheaf has no higher cohomology. The associated long exact sequence therefore ends after $H^1(C, \mathcal{O}_C(D'))$, and the degree-zero terms identify with $L(D)$, $L(D')$, and $\mathcal{Q}(D,D')$. Since the cover is affine and the divisorial sheaves are quasi-coherent, their first sheaf cohomology groups are the displayed cover cokernels. Quotienting the first two terms gives the stated sequence. □

34.5.2 The one-point step

Definition 1906 (The single-point order-$\ge m$ subgroup)**.** For a prime divisor P and $m \in \mathbb{Z}$, $\mathfrak{G}_P^{\ge m} = \{f \in K : f = 0 \text{ or } v_P(f) \ge m\}$, an additive subgroup of K (for $m \le 0$ the fractional ideal $\mathfrak{m}_P^{-m} \subset \mathcal{O}_{C,P}$ viewed inside K).

Theorem 1907 (The local step identity)**.** *Let $D \le D'$ agree except at a prime divisor $P \in U$ with $D'(P) = D(P) + 1$. Then*

$$\Gamma(U, \mathcal{O}_C(D)) \;=\; \Gamma(U, \mathcal{O}_C(D')) \cap \mathfrak{G}_P^{\ge -D(P)}.$$

Proof. A nonzero $f \in \Gamma(U, \mathcal{O}_C(D'))$ lies in $\Gamma(U, \mathcal{O}_C(D))$ iff its order bound at P tightens from $\ge -D(P) - 1$ to $\ge -D(P)$ (unchanged at every other point of U, where $D = D'$), i.e. iff $f \in \mathfrak{G}_P^{\ge -D(P)}$. □

Theorem 1908 (The local step quotient injects into the valuation quotient)**.** *Under the hypotheses of 1907, the inclusion $\Gamma(U, \mathcal{O}_C(D')) \subseteq \mathfrak{G}_P^{\ge -D(P)-1}$ induces an injective map on the relative quotients,*

$$\Gamma(U, \mathcal{O}_C(D'))\big/\Gamma(U, \mathcal{O}_C(D)) \;\hookrightarrow\; \mathfrak{G}_P^{\ge -D(P)-1}\big/\mathfrak{G}_P^{\ge -D(P)}.$$

Proof. Injectivity transports 1907 (a kernel computation) into the quotient language: a class mapping to zero has a representative $f \in \Gamma(U, \mathcal{O}_C(D')) \cap \mathfrak{G}_P^{\ge -D(P)} = \Gamma(U, \mathcal{O}_C(D))$, hence is zero already in the source quotient. □

Lemma 1909 (Order-nonnegativity is stalk-integrality)**.** *For $f \in K^\times$ with $v_P(f) \ge 0$, f lifts to the discrete valuation ring $\mathcal{O}_{C,P}$, i.e. $f = a$ for some $a \in \mathcal{O}_{C,P}$ (viewed in K).*

Proof. $v_P(f) \ge 0$ exactly says the normalised valuation of f at P satisfies $|f|_P \le 1$; a discrete valuation ring is precisely the set of elements of its fraction field of valuation ≤ 1, so f lifts to $\mathcal{O}_{C,P}$. □

Theorem 1910 (The ground field has order zero)**.** *The structure morphism embeds k in K, and every nonzero $c \in k$ is a unit in every local ring of C. Hence $v_P(c) = 0$ for every prime divisor P. Consequently every divisor section group is naturally a k-subspace of K.*

Theorem 1911 (The local step is bounded by the residue degree)**.** *Put*

$$\deg_P^{\mathrm{val}} = \dim_k(\mathfrak{G}_P^{\geq 0}/\mathfrak{G}_P^{\geq 1}).$$

Assume this degree-zero valuation step is finite-dimensional over k. Multiplication by an element of prescribed order identifies every quotient $\mathfrak{G}_P^{\geq m-1}/\mathfrak{G}_P^{\geq m}$ with the quotient defining $\deg_P^{\mathrm{val}}$. Under the hypotheses of 1907,

$$\dim_k(\Gamma(U, \mathcal{O}_C(D'))/\Gamma(U, \mathcal{O}_C(D))) \ \leq\ \deg_P^{\mathrm{val}}.$$

Proof. The injection of 1908 lands in a one-step valuation quotient. Shifting by a uniformizer power identifies this target with the degree-zero step, and dimension is monotone under an injective linear map. ☐

Theorem 1912 (Residue degree one over an algebraically closed base)**.** *Let C be a smooth curve over an algebraically closed field k. Then every prime divisor P of C satisfies*

$$\deg P = [\kappa(P) : k] = 1.$$

Proof. A prime divisor of C is a closed point, and C is locally of finite type over k because it is smooth. The underlying space of a scheme locally of finite type over a field is a Jacobson space, so $\kappa(P)$ is an algebraic extension of k by Zariski's lemma; since k is algebraically closed, the structure map $k \to \kappa(P)$ is an isomorphism, giving $\dim_k \kappa(P) = 1$. ☐

Corollary 1913 (The weighted degree is the geometric degree)**.** *For a smooth curve C over an algebraically closed field k and every Weil divisor D on C,*

$$\deg_k(D) = \deg(D),$$

the right-hand side being the unweighted degree of 1829.

Proof. Both degrees are the sum over the (finite) support of D of its coefficients, weighted by $[\kappa(P) : k]$ on the left and by 1 on the right. The weights agree by 1912. ☐

Lemma 1914 (The valuation quotient is the residue field)**.** *Reduction modulo the maximal ideal induces a k-linear isomorphism*

$$\mathfrak{G}_P^{\geq 0}/\mathfrak{G}_P^{\geq 1} \simeq_k \kappa(P).$$

In particular $\deg_P^{\mathrm{val}} = [\kappa(P) : k] = \deg P$.

Proof. The subgroup $\mathfrak{G}_P^{\geq 0}$ is the valuation ring $\mathcal{O}_{C,P}$, while $\mathfrak{G}_P^{\geq 1}$ is its maximal ideal. Their quotient is therefore the residue field. ☐

Theorem 1915 (Approximation on a Dedekind chart)**.** *Let $P_1, \dots, P_r$ be pairwise distinct points in a common affine Dedekind chart. Given $a_i \in K$ and integers m_i, there exists $f \in K$ such that*

$$v_{P_i}(f - a_i) \geq m_i \qquad (1 \leq i \leq r).$$

Proof. Write the chart as $\operatorname{Spec} A$, and choose a common denominator $0 \neq d_0 \in A$ with $d_0 a_i \in A$ for every i. Choose $N_i \geq 0$ with $v_{P_i}(d_0) + N_i + m_i \geq 0$. The nonzero ideal $\prod_i \mathfrak{p}_i^{N_i}$ contains a nonzero element b; put $d = d_0 b$ and $e_i = v_{P_i}(d) + m_i$. Then $d a_i \in A$ and $e_i \geq 0$. The powers $\mathfrak{p}_i^{e_i}$ of the distinct maximal ideals are pairwise comaximal. By the Chinese remainder theorem there is $y \in A$ with $y \equiv d a_i \pmod{\mathfrak{p}_i^{e_i}}$ for every i. For $f = y/d$ one has

$$v_{P_i}(f - a_i) = v_{P_i}(y - d a_i) - v_{P_i}(d) \geq e_i - v_{P_i}(d) = m_i.$$

☐

Theorem 1916 (The local step has residue degree under approximation)**.** *Suppose every element of $\mathfrak{G}_P^{\geq -D(P)-1}$ is congruent, modulo $\mathfrak{G}_P^{\geq -D(P)}$, to a member of $\Gamma(U, \mathcal{O}_C(D+P))$. Then*

$$\dim_k(\Gamma(U, \mathcal{O}_C(D+P))/\Gamma(U, \mathcal{O}_C(D))) = \deg_P^{\mathrm{val}}.$$

Proof. The approximation hypothesis makes the local-step map surjective; it is injective by 1908. Hence it is an isomorphism onto the one-step valuation quotient. □

Proposition 1917 (Sections of a finite-support quotient)**.** *For $D \leq D'$, the quotient $\mathcal{T} = \mathcal{O}_C(D')/\mathcal{O}_C(D)$ is supported on the finite set $\operatorname{supp}(D'-D)$, and restriction to stalks induces*

$$H^0(C, \mathcal{T}) \simeq_k \bigoplus_{P \in \operatorname{supp}(D'-D)} \mathcal{T}_P.$$

Proof. Divisors are finitely supported. The coherent quotient is the pushforward of a module on a finite closed subscheme $Z \subset C$ with this support. The finite scheme Z is affine, and its Artinian coordinate ring is the product of its local Artinian factors. The corresponding module therefore decomposes as the direct sum of its localizations at the points of Z, which are precisely the stalks $\mathcal{T}_P$. Taking global sections of the finite pushforward gives the displayed isomorphism. □

Proposition 1918 (The one-point stalk quotient)**.** *If $D' = D + P$ and $\mathcal{T} = \mathcal{O}_C(D')/\mathcal{O}_C(D)$, then*

$$\mathcal{T}_P \simeq_k \mathfrak{G}_P^{\geq -D(P)-1}/\mathfrak{G}_P^{\geq -D(P)} \simeq_k \kappa(P),$$

while $\mathcal{T}_Q = 0$ for $Q \neq P$.

Proof. Localising the two divisorial sheaves at P gives two consecutive fractional ideals of the DVR $\mathcal{O}_{C,P}$. Multiplication by a uniformiser power shifts their quotient to $\mathcal{O}_{C,P}/\mathfrak{m}_P = \kappa(P)$. At every other point the two divisors, and hence the two stalks, agree. □

Proposition 1919 (Dimension of the twist quotient)**.** *For $D \leq D'$ the skyscraper section space is finite-dimensional with*

$$\dim_k \mathcal{Q}(D, D') \;=\; \deg_k(D'-D) \;=\; \sum_P (D'(P) - D(P)) \deg P.$$

Proof. By 1917, global sections split into the finitely many stalk quotients. Filter each stalk by the coefficient difference $D'(P) - D(P)$. Every consecutive quotient is $\kappa(P)$ by 1918, and therefore has k-dimension $\deg P$. Additivity of dimension along the filtration gives $\dim_k \mathcal{Q}(D, D') = \sum_P (D'(P) - D(P)) \deg P = \deg_k(D'-D)$, the residue-weighted degree of 1861. □

34.5.3 Euler characteristic and the Riemann inequality

Lemma 1920 (Alternating dimension identity of a four-term exact sequence)**.** *For a four-term exact sequence of finite-dimensional k-vector spaces $0 \to A \xrightarrow{i} B \xrightarrow{p} C \xrightarrow{q} E \to 0$ (i injective, $\operatorname{im} i = \ker p$, $\operatorname{im} p = \ker q$, q surjective),*

$$\dim A - \dim B + \dim C - \dim E = 0.$$

Proof. Rank–nullity gives $\dim(\ker p) = \dim(\operatorname{im} i) = \dim A$ and $\dim(\operatorname{im} p) + \dim(\ker p) = \dim B$, so $\dim B - \dim A = \dim(\operatorname{im} p) = \dim(\ker q)$. Rank–nullity for q gives $\dim(\ker q) + \dim(\operatorname{im} q) = \dim C$, and surjectivity gives $\operatorname{im} q = E$, so $\dim(\ker q) = \dim C - \dim E$. Combining, $\dim B - \dim A = \dim C - \dim E$. □

Definition 1921 (The χ-ledger dimensions)**.** Fix a field of constants k. Write $\ell(D) = \dim_k L(D)$, $h^1(D) = \dim_k \check{H}^1_{\mathrm{ff}}(D)$, and $\chi(D) = \ell(D) - h^1(D)$ for the Euler characteristic.

Theorem 1922 (Global regular functions are constants)**.** *The structure map induces a k-algebra equivalence*

$$\Gamma(C, \mathcal{O}_C) \simeq_k k.$$

Equivalently, the zeroth cohomology of the structure sheaf has dimension one on every affine Mayer–Vietoris square.

Proof. Proper geometric integrality identifies the global section ring with the base field; the cover kernel is the global section space. □

Corollary 1923 (The base value of the divisor ledger)**.** *The zero divisorial sheaf is the structure sheaf, its valuation-defined global sections satisfy $L(0) = k$, and therefore $\ell(0) = 1$.*

Proof. The local condition $v_P(f) \geq 0$ at every prime divisor says precisely that the rational function f is regular everywhere on the regular curve. Thus $L(0) = \Gamma(C, \mathcal{O}_C)$, and 1922 identifies this space with k. □

Theorem 1924 (Conditional overlap-ledger identity)**.** *Put $V = U_0 \cap U_1$. Suppose the four displayed spaces are finite-dimensional and the window, connecting, and twist maps form an exact sequence*

$$0 \longrightarrow L(D')/L(D) \longrightarrow \Gamma(V, \mathcal{O}_C(D'))/\Gamma(V, \mathcal{O}_C(D)) \longrightarrow \check{H}^1_{\mathrm{ff}}(D) \longrightarrow \check{H}^1_{\mathrm{ff}}(D') \longrightarrow 0.$$

Then

$$\dim_k(L(D')/L(D)) - \dim_k(\Gamma(V, \mathcal{O}_C(D'))/\Gamma(V, \mathcal{O}_C(D))) + h^1(D) - h^1(D') = 0.$$

For the canonical window map, injectivity follows from 1902 when $D' - D$ is supported on V; exactness at the other terms remains a separate hypothesis here.

Proof. 1920 applied to the four-term sequence, whose injectivity at the left is 1902. □

Theorem 1925 (One-step χ-additivity)**.** *Under the exactness and finite-dimensionality hypotheses of 1924 for $D \leq D'$,*

$$\chi(D') \;=\; \chi(D) + \dim_k(\Gamma(V, \mathcal{O}_C(D'))/\Gamma(V, \mathcal{O}_C(D))).$$

For a one-point bump $D' = D + P$ with $P \in V$, the approximation hypothesis of 1916 gives $\chi(D + P) = \chi(D) + \deg_P^{\mathrm{val}}$.

Proof. Rank–nullity identifies $\dim_k(\Gamma(\top, \mathcal{O}_C(D'))/\Gamma(\top, \mathcal{O}_C(D))) = \ell(D') - \ell(D)$; substituting into 1924 and rearranging with $\chi = \ell - h^1$ gives the first claim. The one-point formula follows from 1916. □

Theorem 1926 (Telescoping along an effective divisor)**.** *Let $P_1, \ldots, P_r$ be prime divisors and suppose the one-point identity $\chi(E + P) = \chi(E) + \deg_P^{\mathrm{val}}$ holds at every step. Then*

$$\chi(P_1 + \cdots + P_r) = \chi(0) + \sum_{j=1}^{r} \deg_{P_j}^{\mathrm{val}}.$$

Proof. Induct on r. The empty sum is immediate; the induction step applies the one-point identity to $P_1 + (P_2 + \cdots + P_r)$. □

Theorem 1927 (Euler characteristic of a divisor)**.** *The Euler characteristic $\chi(D) = \ell(D) - i(D)$ is additive in the degree:*

$$\chi(D') - \chi(D) = \deg(D' - D) \quad \textit{for } D \le D', \qquad \textit{hence} \qquad \chi(D) = \deg D + 1 - g.$$

Proof. By 1896 the base cohomology $\check{H}^1_{\mathrm{ff}}(0)$ is finite-dimensional; the twist sequence bound every $\ell(D), i(D)$ by finite numbers, so all dimensions occurring are finite and the alternating-sum identity for an exact sequence of finite-dimensional spaces applies. Apply it to the four-term sequence of 1905 for $D \le D'$:

$$\dim\bigl(L(D')/L(D)\bigr) - \dim \mathcal{Q}(D, D') + \dim \check{H}^1_{\mathrm{ff}}(D) - \dim \check{H}^1_{\mathrm{ff}}(D') = 0.$$

With $\dim(L(D')/L(D)) = \ell(D') - \ell(D)$, $\dim \mathcal{Q}(D, D') = \deg(D' - D)$ (1919), and $\dim \check{H}^1_{\mathrm{ff}}(D) = i(D)$, this rearranges to

$$\bigl(\ell(D') - i(D')\bigr) - \bigl(\ell(D) - i(D)\bigr) = \deg(D' - D), \qquad \text{i.e.} \qquad \chi(D') - \chi(D) = \deg(D' - D).$$

Every divisor D satisfies $D - D_- \le D \le D_+$ with $0 \le D_+$; applying the additivity to $0 \le D_+$ and to $D \le D + D_-$ and adding, one gets $\chi(D) = \chi(0) + \deg D$ for all D (additivity of $\chi - \deg$ as a function on the free group of divisors, which agrees at 0 and is additive on each unit step, hence is $\chi(0)$ plus the homomorphism deg). Finally $\chi(0) = \ell(0) - i(0) = 1 - g$: the global sections $L(0) = k$ have dimension $\ell(0) = 1$ by 1923. Moreover, $i(0) = \dim_k \check{H}^1_{\mathrm{ff}}(0) = g$ by 1867 and 1875. Hence $\chi(D) = \deg D + 1 - g$. ☐

Lemma 1928 (Nonnegativity of the index of speciality)**.** *$\chi(D) \le \ell(D)$, i.e. $h^1(D) = i(D) \ge 0$.*

Proof. Immediate from $\chi(D) = \ell(D) - h^1(D)$ and $h^1(D) \ge 0$. ☐

Lemma 1929 (The Riemann inequality from a telescoped χ)**.** *Given $\chi(D) = \chi(0) + \deg D$,*

$$\deg D + \chi(0) \ \le\ \ell(D).$$

Proof. Substitute $\chi(D) = \chi(0) + \deg D$ into 1928. ☐

Corollary 1930 (Riemann inequality and finiteness of the twist cokernel)**.** *For every Weil divisor D, the space $\check{H}^1_{\mathrm{ff}}(D)$ is finite-dimensional and*

$$\deg D + 1 - g \ \le\ \ell(D).$$

Proof. Finiteness: for $0 \le D''$ with $D \le D''$ and $0 \le D''$, the last map of the twist sequence (1905) makes $\check{H}^1_{\mathrm{ff}}(D) \to \check{H}^1_{\mathrm{ff}}(D'')$ surjective with kernel a quotient of $\mathcal{Q}(D, D'')$, which is finite (1919); and $\check{H}^1_{\mathrm{ff}}(0) \to \check{H}^1_{\mathrm{ff}}(D'')$ is surjective, so $\dim \check{H}^1_{\mathrm{ff}}(D'') \le \dim \check{H}^1_{\mathrm{ff}}(0) = g$. Hence $\dim \check{H}^1_{\mathrm{ff}}(D) \le g + \deg(D'' - D) < \infty$. Inequality: 1929 specialises, with $\chi(0) = 1 - g$ (1927), to $\ell(D) = \deg D + 1 - g + i(D) \ge \deg D + 1 - g$. ☐

34.6 Weil differentials and duality

The adelic evaluation pairing identifies the dual of the twist cokernel with a space of Weil differentials. A fixed nonzero differential then turns this space into an ordinary Riemann–Roch space.

Proposition 1931 (The principal-parts resolution)**.** *Let $\mathcal{K}$ be the sheaf of rational functions and let*

$$\mathcal{P}_D = \bigoplus_P i_{P,*}(K/\mathfrak{G}_P^{\geq -D(P)}).$$

There is an exact sequence

$$0 \longrightarrow \mathcal{O}_C(D) \longrightarrow \mathcal{K} \longrightarrow \mathcal{P}_D \longrightarrow 0,$$

in which $\mathcal{K}$ and $\mathcal{P}_D$ are flasque, and $\Gamma(C, \mathcal{P}_D) \simeq \mathbb{A}_K/\mathbb{A}_K(D)$.

Proof. Exactness is stalkwise. At the generic point the first arrow is the identity of K. At a closed point P, its kernel is the fractional ideal $\mathfrak{G}_P^{\geq -D(P)}$, and the quotient is the displayed principal-parts module. The rational-function sheaf is constant on nonempty opens of the integral curve and is flasque; each skyscraper summand is flasque, and a finite-support section of their direct sum extends componentwise. Finally an adele modulo $\mathbb{A}_K(D)$ records exactly its finitely many nonzero principal parts, which gives the global-section isomorphism. □

Proposition 1932 (Adelic and two-chart cohomology agree)**.** *For every divisor D,*

$$H^1_{\mathbb{A}}(D) = \mathbb{A}_K/(\mathbb{A}_K(D) + K) \simeq_k \check{H}^1_{\mathrm{ff}}(D).$$

Proof. Taking global sections of 1931 identifies its degree-one cohomology with the cokernel of $K \to \mathbb{A}_K/\mathbb{A}_K(D)$, namely $\mathbb{A}_K/(\mathbb{A}_K(D) + K)$. Thus this quotient is $H^1(C, \mathcal{O}_C(D))$, which is the two-chart quotient by 1904. □

Definition 1933 (Weil differentials)**.** A Weil differential is a k-linear form on $\mathbb{A}_K$ that vanishes on $\mathbb{A}_K(E) + K$ for some divisor E. Write Ω_K for their union and

$$\Omega_K(D) = \{\eta \in \Omega_K : \eta(\mathbb{A}_K(D) + K) = 0\}.$$

Multiplication by $x \in K$ is defined by $(x\eta)(\alpha) = \eta(x\alpha)$.

Closure under the vector-space operations. If η vanishes on $\mathbb{A}_K(E)+K$ and θ vanishes on $\mathbb{A}_K(F)+K$, then both vanish on $\mathbb{A}_K(\min(E, F)) + K$, where the minimum is taken coefficientwise. Hence so does $\eta + \theta$. If $x \neq 0$, then

$$x\mathbb{A}_K(E + \mathrm{div}(x)) = \mathbb{A}_K(E) \quad \text{and} \quad xK = K,$$

so $(x\eta)(\alpha) = \eta(x\alpha)$ vanishes on $\mathbb{A}_K(E + \mathrm{div}(x)) + K$. The zero scalar is immediate. Thus the displayed operation makes Ω_K a K-vector space. □

Proposition 1934 (The Weil-differential pairing)**.** *Evaluation induces a canonical isomorphism*

$$\Omega_K(D) \simeq_k H^1_{\mathbb{A}}(D)^*,$$

and hence a perfect pairing

$$\check{H}^1_{\mathrm{ff}}(D) \times \Omega_K(D) \longrightarrow k.$$

Proof. A functional in $\Omega_K(D)$ factors uniquely through the finite-dimensional quotient $H^1_{\mathbb{A}}(D) = \mathbb{A}_K/(\mathbb{A}_K(D) + K)$, and every linear functional on that quotient pulls back to such a Weil differential. This gives the first isomorphism; transport it across 1932 for the displayed pairing. □

Theorem 1935 (Eventual vanishing of the adelic index)**.** *There is an integer c such that $H^1_{\mathbb{A}}(D) = 0$ whenever $\deg D \geq c$.*

Proof. Choose adele representatives of a basis of the finite-dimensional space $H^1_{\mathbb{A}}(0)$. Every representative belongs to $\mathbb{A}_K(B_j)$ for some effective divisor B_j; a common upper bound B therefore kills the basis under the surjection $H^1_{\mathbb{A}}(0) \to H^1_{\mathbb{A}}(B)$, so the latter space is zero. Put $c = \deg B + g$. If $\deg D \geq c$, the Riemann inequality gives $\ell(D - B) > 0$. For nonzero $f \in L(D - B)$, the divisor $E = D + \mathrm{div}(f)$ satisfies $E \geq B$, hence $H^1_{\mathbb{A}}(E) = 0$. Multiplication by f^{-1} identifies the quotients for D and E, proving the claim. ☐

Theorem 1936 (The divisor of a Weil differential)**.** *For every nonzero $\eta \in \Omega_K$ there is a unique divisor $\mathrm{div}(\eta)$ maximal among the divisors D for which $\eta \in \Omega_K(D)$. Consequently*

$$\Omega_K(D) = \{0\} \cup \{\eta \neq 0 : \mathrm{div}(\eta) \geq D\}.$$

Proof. The set $M(\eta) = \{D : \eta \in \Omega_K(D)\}$ is nonempty by the definition of a Weil differential. If $D \in M(\eta)$, then η gives a nonzero functional on $H^1_{\mathbb{A}}(D)$; hence 1935 bounds the degrees occurring in $M(\eta)$. Choose $W \in M(\eta)$ of maximal degree. If $D_0 \in M(\eta)$ and $D_0(Q) > W(Q)$, decompose any $\alpha \in \mathbb{A}_K(W + Q)$ into the adele supported away from Q and the adele supported at Q. The first lies in $\mathbb{A}_K(W)$ and the second in $\mathbb{A}_K(D_0)$, so $\eta(\alpha) = 0$. Thus $W + Q \in M(\eta)$, contradicting maximality. Therefore every $D \in M(\eta)$ satisfies $D \leq W$, which proves existence, uniqueness, and the displayed characterization. ☐

Theorem 1937 (Scaling a differential)**.** *For $x \in K^\times$ and $0 \neq \eta \in \Omega_K$,*

$$\mathrm{div}(x\eta) = \mathrm{div}(x) + \mathrm{div}(\eta).$$

Proof. Multiplication by x carries $\mathbb{A}_K(E)$ isomorphically onto $\mathbb{A}_K(E - \mathrm{div}(x))$. Hence $x\eta \in \Omega_K(E)$ exactly when $\eta \in \Omega_K(E - \mathrm{div}(x))$. Comparing the unique maximal divisors on the two sides gives the formula. ☐

Theorem 1938 (One-dimensionality of Weil differentials)**.** *The K-vector space Ω_K is one-dimensional.*

Proof. First $\Omega_K \neq 0$: for $\deg D \ll 0$, the identity $i(D) = \ell(D) - \deg D - 1 + g$ makes $i(D) > 0$, so $\Omega_K(D) \simeq H^1_{\mathbb{A}}(D)^*$ is nonzero. Choose $0 \neq \eta_i \in \Omega_K(D_i)$ for $i = 1, 2$. For an effective divisor B of sufficiently large degree, multiplication gives injections

$$L(D_i + B) \hookrightarrow \Omega_K(-B), \qquad x \longmapsto x\eta_i.$$

The Riemann inequality bounds the two source dimensions below by $\deg(D_i + B) + 1 - g$, while negative-degree vanishing and the χ-identity give $\dim_k \Omega_K(-B) = i(-B) = \deg B + g - 1$. For $\deg B$ large, the two images therefore have positive-dimensional intersection. Thus $x_1\eta_1 = x_2\eta_2 \neq 0$ for some $x_i \in K^\times$, and $\eta_2 = (x_1/x_2)\eta_1$. Every two nonzero differentials are proportional. ☐

Theorem 1939 (Serre duality: $i(D) = \ell(K_C - D)$)**.** *For a canonical divisor $K_C = \mathrm{div}\,\omega$ of a nonzero Weil differential ω, the map $x \mapsto x\omega$ induces a k-linear isomorphism*

$$\mu\colon L(K_C - D) \xrightarrow{\sim} \Omega_K(D) \simeq_k H^1_{\mathbb{A}}(D)^* \simeq_k \check{H}^1_{\mathrm{ff}}(D)^*,$$

so that $i(D) = \dim_k \check{H}^1_{\mathrm{ff}}(D) = \ell(K_C - D)$.

Proof. By 1938, every Weil differential is uniquely $x\omega$. By 1937, $\mathrm{div}(x\omega) = \mathrm{div}(x) + K_C$; hence

$$x\omega \in \Omega_K(D) \iff \mathrm{div}(x) + K_C \geq D \iff x \in L(K_C - D).$$

This proves the first isomorphism. The second is 1934, and the third is 1932; taking dimensions gives the final equality. ☐

Theorem 1940 (Geometric degree of a principal divisor, from the χ-ledger)**.** *Let C be a smooth curve over an algebraically closed field k, and suppose the χ-ledger $\chi(D) = \chi(0) + \deg_k(D)$ holds at every Weil divisor D. Then for every $g \in K^\times$,*

$$\deg(\operatorname{div}(g)) = 0.$$

Proof. The invariant χ is unchanged along a linear equivalence, so $\chi(\operatorname{div}(g)) = \chi(0)$; the ledger then forces $\deg_k(\operatorname{div}(g)) = 0$. Over an algebraically closed base the weighted and geometric degrees coincide by 1913, giving the displayed equality. □

Lemma 1941 (Negative-degree divisors have no nonzero sections)**.** *If $\deg E < 0$, then $L(E) = 0$ and $\ell(E) = 0$.*

Proof. A nonzero $f \in L(E)$ would make $\operatorname{div}(f) + E$ effective. Its degree is $\deg E < 0$, because principal divisors have weighted degree zero by 1862; an effective divisor cannot have negative degree. □

Theorem 1942 (Degree of a canonical divisor)**.** *Every canonical divisor K_C has*

$$\ell(K_C) = g, \qquad \deg K_C = 2g - 2.$$

Proof. At $D = 0$, duality gives $i(0) = \ell(K_C)$, while $\ell(0) = 1$, $i(0) = g$, so $\ell(K_C) = g$. At $D = K_C$, duality gives $i(K_C) = \ell(0) = 1$. Substituting these two dimensions into $\ell(K_C) - i(K_C) = \deg K_C + 1 - g$ yields $\deg K_C = 2g - 2$. □

Corollary 1943 (Vanishing: $i(D) = 0$ for $\deg D > 2g-2$)**.** *If $\deg D > 2g-2$, then $\check{H}^1_{\mathrm{ff}}(D) = 0$, and consequently $H^1(C, \mathcal{O}_C(D)) = 0$. This is the curve-level replacement for Castelnuovo–Mumford boundedness in the Quot endgame.*

Proof. By 1942, $\deg(K_C - D) < 0$. Therefore 1941 and duality give $i(D) = \ell(K_C - D) = 0$. Finally 1904 supplies the sheaf-cohomology statement. □

Theorem 1944 (Riemann–Roch)**.** *For every Weil divisor D and canonical divisor K_C,*

$$\ell(D) - \ell(K_C - D) \;=\; \deg D + 1 - g.$$

Proof. Immediate from the two keystones: the χ-ledger 1927 gives $\ell(D) - i(D) = \deg D + 1 - g$, and Serre duality 1939 gives $i(D) = \ell(K_C - D)$; substituting the second into the first yields the identity. □

Bibliography

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